

UNIFORMITY, PRIME SUPPORT, AND REACHABLE LATTICES IN APPROACHES TO THE *abc* CONJECTURE

CHATGPT

ABSTRACT. We study the standard *abc* conjecture through parallel arithmetic and geometric routes, recording complete local proofs, conditional global reductions, counterfamilies to precise auxiliary claims, and scoped Lean certificates. The conjecture is neither proved nor disproved. Shared exponent generators improve all earlier disjoint-partition estimates on an actual infinite family. The second continuation refines both places to the remainder proportion alone and proves elementary moving-prime average bounds from exact shifted ranks, uniform in residual and root heights. A separate route transports actual radicals through successive norm maps and describes the second norm by a fixed quartic unit, including compatible local lifting at two primes. Formal implications connect an explicit unbounded-family condition to the repository's standard conjecture. That family and the forward signed large-prime bound remain unproved. Counterfamilies delimit precise auxiliary strategies while their unrefuted parent routes remain active. Ordinary proofs, external inputs, independent agent review, finite computations, and the exact scope of kernel verification are stated separately throughout.

RESEARCH OVERVIEW

We develop proof reductions, obstruction theorems, and formally checked finite certificates for several routes toward the standard *abc* conjecture. The conjecture is neither proved nor disproved here. Our common method is to separate a global uniformity statement from its local algebraic and combinatorial inputs, prove the latter before formalization, and test every proposed strengthening against all of its hypotheses. Exact prime-support identities and radical formulas isolate sufficient height and counting gates; explicit Pell, affine, Mersenne, and local-field examples delimit claims that those gates alone can support.

The project sharpens several active interfaces. In the IUT-facing model, primitive local factors give the exact *ef* coefficient, total-degree product-Haar normalization, and a pointed same-pilot reduction, while the source-level Ind3 comparison remains open. At the Mersenne endpoint, an exact Farey prefix–tail inequality shows that linear energy would force a polynomial swarm of distinct depth-three rows; the critical arithmetic counting exponent remains open. In the affine route, divisibility-maximal intersection tops admit ownership-preserving Cauchy aggregation and a strict non-arm energy reduction, leaving a concrete catalogue-sparsity estimate. In the Pell route, all-index polynomial norm and derivative identities and the full Taylor–Hensel digit law isolate the fixed-coefficient squarefull exclusion.

An earlier gain/packet/trace checkpoint adds five exact boundaries. For primitive nonunit triples, the canonical product gives $1/3 < A_{\text{can}} < 1/2$ and the correlated identity $q = 1 + X - Y$; a uniform defect-flag budget is proved equivalent to standard *abc*, without being assumed. An infinite primitive family refutes the uncompensated synchronized-packet bound at exponent $4/3$, while the exact *ab*-compensated implication remains a live route. Finally, adjacent signed Pell traces encode simultaneous squarefullness as fourth-power support. Two complete local collisions show that depth two need not promote to depth five, but do not decide the simultaneous two-channel gate.

A new labelled valuation-incidence complex retains the arm, prime, exponent, defect filtration, and local congruence of every selected vertex. Its top face gives an exact rational-power

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half-space for abc . A dyadic infinite family refutes its fixed absolute-budget selector, and a balanced two-prime family refutes the first coefficient-one binary-scale repair. Allowing all three arms still leaves a raw weighted-defect gate false: an infinite primitive Pythagorean-square family is a complete-premise obstruction. An ordered fractional successor gives an exact height reduction, but the odd prime family $(1, p^2 - 1, p^2)$ forces an inaccessible source of mass $\log p$ and refutes its uniform gate at $\varepsilon = 1/4$. The same family refutes the ordered endpoint-flow gate, whose conditional implication to abc remains valid. Merely deleting the order reduces optimal unmatched mass to the positive scalar height defect. The first costed bidirectional repair is also false: prime-hypotenuse Pythagorean squares force energy at least $\frac{1}{4} \log p$ while their conductor is at most $3 \log p$. We then isolate an indivisible prime-packet boundary gate which implies abc and does not collapse pointwise to the scalar relaxation. Linnik's least-prime theorem then supplies a prime-neighbour family that refutes its exclusive-ownership rule while satisfying $h \leq r$. A shared CRT-incidence successor then lets a saturated hyperedge certify several endpoint powers while charging each sink capacity once. It gives a new height bridge and is strictly non-scalar at $(343, 625, 968)$. Arbitrary partial sharing collapses back to the scalar defect, while a saturated-only child is refuted by $(2, 15^n - 2, 15^n)$ for $n \equiv 284 \pmod{310}$. The exact shared-incidence uniform estimate, weighted multi-face, and genuine homological successors remain open.

The present continuation analyzes the unused capacity of a saturated shared block. Sending it to an arbitrary target is proved to collapse exactly to the scalar defect. Requiring a compatible nonempty proper subface instead gives a once-charged flagged boundary satisfying $\Delta \leq B_{\text{FCRT}} \leq B_{\text{SCRT}} \leq B_{\text{PBT}}$. Exact points show both strict improvement and strict noncollapse; at $(1, 65024, 65025)$ all four levels are distinct. Endpoint packets form Boolean cubes over finite residue groups, where the desired flag is a nested cross-source incidence rather than a same-fibre collision. The resulting uniform flagged estimate remains open and is never inserted as a formal assumption. An anchored-prefix theorem now produces the required proper face under an explicit residue-subgroup and weight-cell entropy condition. The actual point $(1, 4715, 4716)$ refutes the weaker raw Boolean-cardinality shortcut while leaving the anchored theorem and the parent FCRT route active.

We also introduce a divisor-valued contact surface. Its standard Bloch five-term boundary gives an exact quadratic Veronese peeling and a valuation-layer expansion of logarithmic contact area. Primitive Pythagorean squares furnish full-premise infinite counterexamples to two single-cell gates and to several fixed one-move policies. These examples do not refute arbitrary multi-cell fillings or the abc conjecture. Every finite scan is recorded without asymptotic extrapolation, and every formal claim is separated from its paper-only analytic or arithmetic bridge. The remaining pointed tensor comparison, critical Farey counting, affine catalogue sparsity, fixed Pell squarefull exclusion, compensated packet compression, canonical excess/slack control, and calibrated five-term filling estimates are explicit open obligations.

1. THE TARGET AND THE ROLE OF THE RESULTS

Throughout, an abc triple means positive integers a, b, c such that $a + b = c$ and a, b, c are pairwise coprime. Write

$$r = \text{rad}(abc), \quad R = \log r, \quad h = \log c.$$

Here $\text{rad}(n)$ is the product of the distinct prime divisors of n . Pairwise coprimality is equivalent to $\gcd(a, b, c) = 1$ under the additive relation. The target is the standard assertion

$$(1) \quad \forall \varepsilon > 0 \quad \exists C_\varepsilon \in \mathbb{R} \quad \forall (a, b, c) : \quad h \leq (1 + \varepsilon)R + C_\varepsilon.$$

Exponentiation gives the usual multiplicative constant. The repository's **ABCConjecture** uses $\log \max(a, b, c)$, which equals h here. Its definition is not changed in this work.

The quantifier order in (1) is essential. Its negation requires one fixed positive ε and violations for every additive constant, hence at unbounded height. A finite set of high-quality triples cannot supply that negation: it is absorbed into C_ε . Likewise, a sparse exceptional-set bound does not imply that the exceptional set is finite.

Our results concern several specific interfaces. Section 2 uses finite prime support to test a proposed analytic construction. Section 3 bounds distinct outputs of two amplification constructions under explicit sufficient radical certificates. Section 5 records the cancellation that an endpoint approach must preserve. Section 6 supplies a local generator certificate relevant to possible-image constructions. Sections 8 and 9 examine effective arithmetic geometry and recent literature. In particular, Corollary 8.5 proves effective finiteness of a specified moving-index family. These components are partial theorems and applicability tests; none constructs the missing uniform bound in (1). Section 4 extends the amplification test from sufficient certificates to actual radicals. Section 7 uses the original admissible-arrow definitions and identifies a precise Ind2 hull equality. Section 10 supplies constructions for the full family of *abc* triples, together with their exact remaining endpoint and denominator costs. Sections 11 and 12 construct full local Galois arrows and distinguish their specified point, pre-ideal and whole-product sources. Sections 13, 14 and 15 supply explicit arithmetic realizations, simultaneous local formulas and global initial-data constructions. Section 16 proves a uniform existence theorem at unbounded height. Section 17 proves the two-prime bound and the rational-return and odd-part fibre statements. Section 18 determines an entire rational isogeny class and its intrinsic bounds. Section 19 calculates its exact minimum Weil height, retaining the contribution of the reduced denominator. Section 20 computes the point hull for every independent finite torsion representative of the standard theta values. Section 21 proves canonical local membership in one synchronized base branch, keeping the predecessor carrier and the transfer direction explicit. Section 22 constructs the arithmetic bundles and calculates their degrees and reference changes. Section 23 gives the exact three-prime reduction, the mixed-full radical theorem, and the newest Campana counting boundary. Section 24 proves the failure of every coefficient below six on one Frey family and the finite selected-place product-hull theorem. It does not prove that six is an attainable optimal coefficient. Section 25 gives the exact positive uniformity equivalence, the counting threshold, the residual-kernel and polynomial counterexample obstructions, and the conditional Pell disproof gate. Section 28 gives the corrected local-volume holonomy ledger and the latest affine, Pell, and Danilov continuations. Section 29 gives a literal asymptotic counterfamily to the remaining printed global- ω hypothesis in a claimed 2026 disproof, while retaining and broadening its valid prime-power radical-neighbour target. Section 30 lifts the same-pilot comparison to a faithful labelled prime/unit signature before taking volume, and gives complete counterexamples to three weaker coordinate interfaces. Section 31 splits a generic fixed-stage packet into primitive field factors, restores zero to the signature, and states the exact source-level covariance still needed for IUT. Section 32 audits the newest LANA conditional implication, proves the empty-set obstruction to its current local-volume interface, and isolates the missing abstract-to-integer *abc* bridge. Section 33 gives an inhabited p -adic ball repair, proves its exact orbit constraints, and supplies the uniform comparison package that would carry abstract Statement I to the integer target. Section 34 realizes that package with zero error on the actual open rational tripod, proves that Statement I for this shadow is exactly equivalent to integer *abc*, and uses full-premise models to separate the three semantic inputs still missing for an intended theory. Section 35 then matches the genuine degree-one rational points, log-canonical height, different, and conductor to their source definitions and isolates the still-missing intended arithmetic height-theory instance and upstream Statement I proof. Section 39 constructs a simultaneous two-arm CRT packet and tests the two necessary marginal excess gates against a complete canonical counterexample. Section 40 proves cubic-product separation for a full fixed three-arm certificate, closes its large-modulus boundary, and derives the quantitative certificate-entropy obstruction while retaining all adaptive and correlated affine routes. Section 41 combines all three affine congruences in the determinant, forces canonical collinearity, and derives the stronger repeated-kernel entropy requirement with its exact boundary counterexamples. Section 42 proves the sharp determinant bound for three point-adaptive certificates, improves the canonical support constants, and retains the unresolved common-kernel selection problem. Section 43 proves the line cap, large-label fibre and multiplicity bounds, and the third-gcd energy identity; an actual box-wide collinear example isolates the remaining overlap gate. Section 44 extracts the exact arithmetic period on a ray, proves cube-root and weighted cubic-energy bounds away from constant-arm directions, and

treats the vertical direction under an individual arm cap. Section 45 computes the actual Euler-totient catalogue mass and proves that coincident divisor labels can only increase its cubic energy, while preserving the separate singleton- novelty and signed-line gates. Section 46 treats arbitrary signed primitive directions, proves the strict capture squeeze and exact linear caps on all three canonical arm levels, and reduces the remaining global energy to one non-arm inverse-period sum and a singleton baseline. Section 58 separates base order blocks from the LTE lifting remainder and proves unconditional radical bounds at prime Mersenne layers. Section 59 identifies the exact divisor-average endpoint and proves why the available coarse order estimates do not control the moving weighted fibre. Sections 61– 64 resolve the super-Wieferich contribution into finite depth layers, give pointwise and divisor-average sufficient gates, and certify the odd-order counterexample at 3511 without closing the surviving route. Section 67 proves the exact divisor moment, its $O(\log \log(3m))$ moving localization, the actual cyclotomic cap, and the resulting near-diagonal exceptional-mass equivalence. Section 68 replaces that moving range by the exact family of fixed-polylogarithmic co-divisor gates, with its atomic and diffuse failure alternatives. Section 69 then reduces every surviving linear mass to deep lifts, transition clustering, or extreme small order. Section 70 removes the auxiliary exception threshold, localizes the actual block mass, and raises the three-arm coefficient to every value below the sharp algebraic ceiling $1/3$. Section 71 uses the injective exact-order multiplier to remove the localized low arm and raises the remaining two-arm coefficient to every value below $1/2$. Section 72 moves the one-copy cutoff to every fixed positive log-log slack and removes the growing low-multiplier part of the deep arm using Yamada’s pointwise valuation bound. Section 73 replaces every fixed positive power by an arbitrarily slowly divergent factor, adds the exact multiplier character table, and records the precise arithmetic and abstract endpoint counterexample boundaries. Section 49 adds the exact modulo-eight and quadratic-character restrictions in the Pell packet, while Section 50 compresses a hypothetical squarefull packet into two odd-exponent kernels and retains a third quotient digit. Section 51 splices both Pell channels into a norm-one Lucas sequence, proves the exact support-unit staircase at every multiplication order, and recovers the opposite channel signs from a paired companion correction. Section 52 constructs the companion staircase in closed form, proves its coefficientwise differential correlation, and combines coherent every-order splitters with vertexwise character incidence and certified finite exclusion through prime index 271. Section 36 audits the current public IUT source, compiles an admissible-domain quantifier repair, and proves the exact finite index of a congruence order with surjective component projections. Section 37 realizes the repaired local domain with actual Haar measure, derives the local-degree packet coefficient, and isolates the still-unconstructed tensor and same-pilot steps. Section 47 cancels the singleton incidence baseline and gives exact Euler and tail bounds for pairwise inverse-period catalogues. Section 53 joins the quotient ledger to the all-order Lucas system and computes its sharp endpoint curvature. Section 74 closes one critical endpoint arm and identifies the exact-order Farey-energy obstruction for the other. Section 38 derives the primitive-factor degree coefficient and isolates the pointed source comparison still needed by the IUT-facing route. Section 48 aggregates divisibility-maximal affine tops without double-counting their owners. Section 54 proves the polynomial Hensel specialization on paper and records the fixed-parameter squarefull boundary. Section 55 kernel-checks the all-index polynomial identities, the all-support displacement equivalence under explicit scale-unit and transversality hypotheses, finite CRT steering, and the complete moving-parameter witness. Section 56 proves those transversality premises for every actual support prime at the fixed coefficient two and identifies the still-open simultaneous-displacement gate with the prime-index squarefull exclusion. Section 57 rewrites that surviving gate as two adjacent fourth-power support packets, proves the exact projector defect and its Newton lift, and uses complete-premise depth-two collisions to retire only stronger fifth-power promotions. Section 75 converts endpoint energy into a quantitative depth-three row swarm under an explicitly open global counting hypothesis. Section 76 kernel-checks the harmonic prefix budget, frequent-subsequence quantifiers, and finite injection of actual depth-three exact-order rows into the super-Wieferich prime count. Section 77 audits the latest alternative quality metrics, proves their exact packing-coordinate transfer, and gives an abstract full-premise countermodel to divergence transfer without claiming an arithmetic

counterexample. Section 78 introduces a three-arm divisor packet spectrum directly on primitive *abc* triples, proves its exact product, height and orientation laws, and isolates an uncompensated radical-energy candidate that is tested and corrected in the following section. Section 79 proves that the original uncompensated radical-energy gate fails on an infinite primitive family at exponent $4/3$, and retains an exact *ab*-compensated implication to the required height bound. Section 80 audits the 2026 approximation/power gain proposal, constructs the canonical gain surface and its exact scale groupoid cocycles, and proves that the resulting uniform defect-flag budget is equivalent to the standard conjecture rather than a substitute for it. Section 94 develops the divisor-contact five-term complex, proves the surviving calibrated filling reduction, and uses primitive Pythagorean squares to retire only its single-cell gates. Section 95 constructs the integer exponent-gcd split and the actual finite-support divisor norm, then instantiates the filling inequality under an explicit surface-boundary equality. Section 96 constructs the algebraic and positive generated five-term relations and proves their boundary bridge; arbitrary-target filling existence and the two Gate VF costs remain open. Section 97 analyzes the exact quadratic five-term move, proves its valuation-layer flag, and delimits three fixed one-move policies without rejecting arbitrary fillings. No priority over existing literature is claimed for consequences of the cited classical theorems.

The new statements below were proved mathematically before their corresponding Lean implementations. Separate research agents reviewed the arguments, source hypotheses, and formal scope; this internal review is not external journal peer review. Classical published theorems may be used in the paper, but are not silently represented as proved Lean declarations. The formal scope is specified in Section 171. The literature and public-repository cutoff is September 3, 2026.

Six recent primary sources are useful for locating the boundary. Letendre’s short-interval bounds for ω assume a new *abc*-type Conjecture 1 that the paper presents as stronger than *abc* [24]; they are therefore test interfaces rather than unconditional inputs here. Zhou’s announced effective rational inequalities explicitly invoke the μ_6 version of IUT III, Corollary 3.12 in Proposition 1.9 [31]; this does not remove the source-faithful comparison obligation isolated below. Falk–Harrington–Jones give exact generalized Wieferich/Lucas criteria for monogenic trinomials [23], but no density bound for the fixed-base depth-three packet. Laniewski’s 29 August 2026 preprint reduces Mersenne transgression to aggregate ordinary-Wieferich order-defect growth, while explicitly leaving that growth open; it likewise supplies no critical count for the fixed-base depth-three packet [25]. Sankaran’s DGM and power-mean quality metrics have an unconditional divergent alternative-quality family, but the packing-efficiency formula $q_{\text{std}} = \eta q_{\text{DGM}}$ shows that this is not a proof or disproof of standard *abc*; the paper’s Theorem 4.15 is an exact reformulation whose missing content is control of η on actual triples [26]. Müller and Taktikos introduce approximation and power gains for *abc* equations arising from algebraic approximation [27]. Their exact factorization motivates our canonical gain surface, while fixed separate bounds on the two factors yield only their product coefficient and therefore do not by themselves imply the coefficient- $1 + \varepsilon$ quantifier of standard *abc*. Versioned PDFs, extracted text, metadata, and SHA-256 checksums for Letendre, Zhou, and Falk–Harrington–Jones are archived under `research/sources/latest_abc_proposals_2026_09_02/`; the Laniewski source record is archived under `research/sources/mersenne_farey_denominator_entropy_2026_09_02/`, and the Sankaran record under `research/sources/alternative_quality_metrics_2026_09_03/`, and the Müller–Taktikos record under `research/sources/abc_gain_surface_2026_09_03/`.

2. PRIME SUPPORT IN SHORT INTERVALS

Let $\omega(n)$ be the number of distinct prime divisors of n , and let $P^+(n)$ be its greatest prime divisor, with $P^+(1) = 1$. For integers $N, Y \geq 1$ and $w \geq 0$, define

$$\mathcal{A}(N, Y, w) = \{1 \leq n \leq N : P^+(n) \leq Y, \omega(n) \leq w\}.$$

Theorem 2.1 (Finite prime-power encoding). *With $L = \lfloor \log_2 N \rfloor$, one has*

$$(2) \quad |\mathcal{A}(N, Y, w)| \leq (1 + YL)^w.$$

Proof. Use the alphabet

$$\mathcal{Q} = \{1\} \cup \{q^e : 1 \leq q \leq Y, 1 \leq e \leq L\}.$$

It has at most $1 + YL$ members; the bases need not be prime. For $n \in \mathcal{A}(N, Y, w)$, write its actual prime factorization as $n = \prod_{j=1}^t p_j^{e_j}$, with $p_1 < \dots < p_t$ and $t \leq w$. Since $2^{e_j} \leq p_j^{e_j} \leq n \leq N$, each $e_j \leq L$. Thus every factor belongs to $\mathcal{Q}$. Append $w - t$ copies of 1 to obtain a word of length w . Its product recovers n , so different integers have different encoded words. Counting all words proves (2). The empty factorization covers $n = 1$ and $w = 0$; $N = 1$ is also included. $\square$

Corollary 2.2 (A finite first-moment bound). *Let S be a finite set of distinct positive Y -smooth integers at most N , and put $B = (1 + Y \lfloor \log_2 N \rfloor)^w$. Then*

$$(3) \quad |\{n \in S : \omega(n) \leq w\}| \leq B,$$

$$(4) \quad \sum_{n \in S} \omega(n) \geq (w + 1) \max\{|S| - B, 0\}.$$

Proof. The first set is contained in $\mathcal{A}(N, Y, w)$. Its complement in S has at least $\max\{|S| - B, 0\}$ members, each with at least $w + 1$ distinct prime factors. Summation proves the second assertion. $\square$

Suppose now that $y = y(x) \geq 2$ satisfies

$$(5) \quad \log \log x = o(\log y), \quad \log y = o(\log x), \quad u = \frac{\log x}{\log y}.$$

For every fixed $\kappa > 0$, Theorem 2.1, with $N = \lfloor 2x \rfloor$, $Y = \lfloor y \rfloor$ and $w = \lfloor \kappa u \rfloor$, gives

$$(6) \quad \#\{n \leq 2x : P^+(n) \leq y, \omega(n) \leq \kappa u\} \leq x^{\kappa + o(1)}.$$

Indeed,

$$w \log(1 + YL) \leq \kappa \frac{\log x}{\log y} (\log y + O(\log \log x)) = \kappa \log x + o(\log x).$$

More generally the upper bound is $x^{o(1)}$ whenever $w(\log y + \log \log x) = o(\log x)$.

2.1. The external analytic input. Write $\Psi(x, y) = \#\{1 \leq n \leq x : P^+(n) \leq y\}$. Younis's unconditional Theorem 1.1 [73], p. 2, states that for each fixed $17/30 < \theta \leq 1$ there is $C_Y(\theta) > 0$ such that

$$(7) \quad \frac{\Psi(x + h, y) - \Psi(x, y)}{h} = \frac{\Psi(x, y)}{x} \left(1 + O_\theta \left(\frac{\log(u + 1)}{\log y} \right) \right)$$

uniformly for

$$x^\theta \leq h \leq x, \quad \exp\left(C_Y(\theta)(\log x)^{2/3}(\log \log x)^{4/3}\right) \leq y \leq 2x.$$

Fix

$$(8) \quad \frac{17}{30} < \theta < 1, \quad C \geq C_Y(\theta), \quad h = x^\theta, \quad y = \exp\left(C(\log x)^{2/3}(\log \log x)^{4/3}\right).$$

The lower bound on C is part of the hypotheses, not a disposable constant. Put

$$S_x = \{n \in (x, x + h] : P^+(n) \leq y\}, \quad M_x = |S_x|.$$

The classical long-interval estimate recalled in [51, equations (1.8)–(1.9), p. 415] yields $\Psi(x, y) = x\rho(u)(1 + o(1))$ here. Its range $u \leq (\log y)^{3/5 - \varepsilon}$ is satisfied, for example with $\varepsilon = 1/20$: the power of $\log x$ on the right is $11/30$, larger than the power $1/3$ in u . Also $-\log \rho(u) = u \log u(1 + o(1)) = o(\log x)$. Consequently (7) gives

$$(9) \quad M_x = h\rho(u)(1 + o(1)) = x^{\theta - o(1)}.$$

Only this mass estimate is needed below; no short-interval moment formula is assumed.

Theorem 2.3 (A lower tail and a first-moment obstruction). *For (8) and every fixed $0 < \kappa < \theta$,*

$$(10) \quad \frac{\#\{n \in S_x : \omega(n) \leq \kappa u\}}{M_x} \leq x^{\kappa-\theta+o(1)} \rightarrow 0.$$

In particular,

$$(11) \quad \liminf_{x \rightarrow \infty} \frac{1}{uM_x} \sum_{n \in S_x} \omega(n) \geq \theta.$$

For every fixed $A > 0$, the proportion of $n \in S_x$ with $\omega(n) \leq (\log \log x)^A$ tends to zero.

Proof. Every member of S_x is at most $2x$. Divide (6) by (9) to obtain (10). If its numerator is B_x , then

$$\sum_{n \in S_x} \omega(n) \geq \kappa u(M_x - B_x).$$

The lower limit after division by uM_x is at least each fixed $\kappa < \theta$, and hence at least θ . This argument does not require a variable κ or a uniform error term in that variable. Finally,

$$(\log \log x)^A (\log y + \log \log x) = o(\log x).$$

The last observation after (6) bounds the corresponding population by $x^{o(1)}$. Divide again by (9). $\square$

Corollary 2.4 (A false intermediate assertion in a claimed disproof). *Lemma 4.2 of Carella [14], p. 8, equation (4.5), is false in its stated range. The same holds for the second-moment assertion in Lemma 4.4, p. 9, and the relative exceptional estimate (4.24), p. 11.*

Proof. Choose $\theta = 3/5$ and a fixed $C \geq C_Y(3/5)$. Then $h > x^{7/12}$, $y < h < x$, and $y = o(h)$, exactly as required in the first lemma. Its proposed formula would give

$$\frac{1}{M_x} \sum_{n \in S_x} \omega(n) = O(\log \log y).$$

However,

$$\frac{u}{\log \log y} \asymp \frac{(\log x)^{1/3}}{(\log \log x)^{7/3}} \rightarrow \infty,$$

contradicting (11). Changing the left endpoint of the interval from open to closed adds at most one integer; its contribution is at most $\log_2(2x)$ and is negligible relative to M_x . The proposed second moment is $O(M_x(\log \log y)^2)$, whereas (10) gives a lower bound $\kappa^2 u^2 M_x(1 - o(1))$ for any fixed $0 < \kappa < 3/5$. Finally the threshold in (4.24) is bounded by a fixed power of $\log \log x$. Theorem 2.3 shows that the proportion above it tends to one, although the claimed relative upper bound tends to zero. $\square$

Remark 2.5. This refutes those intermediate assertions, not *abc* and not the possibility of an exceptionally sparse counterexample family. In particular a set of size $x^{o(1)}$ can still be infinite. It is also important to distinguish $o(h)$ from $o(M_x)$: the whole smooth set already has size $o(h)$.

2.2. Two further transfer tests. For a fixed integer $k \geq 4$, the intervals

$$(p^k, p^k + (p^k)^{3/5}], \quad X < p^k \leq 2X,$$

are pairwise disjoint for large X . Indeed, successive distinct integer bases give gaps at least $X^{1-1/k}$, larger than $(2X)^{3/5}$. The finite encoding, using the largest thresholds in this range, shows that at most $X^{o(1)}$ centres can contain a smooth candidate with $\omega(n) \leq 2 \log \log(p^k)$ at the scale (8). The prime number theorem gives $X^{1/k+o(1)}$ centres, so their proportion is at most $X^{-1/k+o(1)}$. This rules out a positive-proportion selection of this particular kind, while leaving a sparse selection unresolved.

Recent work on exceptional sets [7, 12] and on the existence of smooth neighbours [54] addresses different quantifiers. A useful general check is the following. If a counting theorem gives $N(T) \ll T^{\eta+o(1)}$, then a seed of height H contradicts it only when it produces more than $H^{\beta\eta+o(1)}$ distinct qualifying outputs of height at most $O(H^\beta)$. Repeated encodings of one output

do not count. None of the existence results just cited supplies the required uniform low-radical neighbours or such an amplification theorem.

3. TWO EXPLICIT AMPLIFICATION FAMILIES

Let $N_\mu(T)$ count positive primitive triples $A + B = C \leq T$ with $\text{rad}(ABC) < C^\mu$, where $0 < \mu < 1$. The current estimates [7, Proposition 1.1 and Theorems 1.2–1.3] give

$$(12) \quad N_\mu(T) \ll_{\mu,\delta} T^{F(\mu)+\delta}, \quad F(\mu) = \min \left\{ \frac{2\mu}{3}, \frac{23\mu+3}{40}, \frac{3}{5} \right\}, \quad \delta > 0.$$

We test two actual constructions from a primitive seed $a + b = c$. An amplifier contradicting (12) at output height $T = c^K$, for fixed $K > 0$, would need more than $c^{KF(\mu)+o(1)}$ distinct qualifying outputs. Our conclusions concern explicit radical certificates, rather than the entire exceptional set.

3.1. Inherited congruence moduli. Fix pairwise coprime positive integers U, V, W , put $M = UVW$, and let $R_M = \text{rad}(M)$. Outputs with $U \mid A$, $V \mid B$, $W \mid C$ satisfy

$$\text{rad}(ABC) \leq \frac{R_M ABC}{M}.$$

Call an output *size-certified* if the right side is at most C^μ .

Proposition 3.1. *For $T \geq 1$, there are at most $1 + \lfloor \log_2 T \rfloor$ size-certified primitive outputs with $C \leq T$, for $0 < \mu \leq 1$. Allowing every inherited template $U \mid a$, $V \mid b$, $W \mid c$ gives at most*

$$\tau(abc)(1 + \lfloor \log_2 T \rfloor)$$

distinct outputs. For fixed K and $T \leq c^K$ this is $c^{o(1)}$.

Proof. Suppose first $M \geq 2$. Certification implies $R_M AB \leq M$, and $R_M \geq 2$. Consider two outputs in the strip $L \leq A, A' < 2L$. Their determinant $\Delta = AB' - A'B$ is divisible by UV and also by W , since $\Delta = AC' - A'C$. Thus $M \mid \Delta$. The bounds $B' \leq M/(R_M A')$ and $B \leq M/(R_M A)$ give

$$0 < AB', A'B < 2M/R_M \leq M.$$

Consequently $|\Delta| < M$, so $\Delta = 0$. Coprimality implies $A \mid A'$ and $A' \mid A$, hence equality of all coordinates. There is at most one output in each dyadic strip. The strict strip endpoint also handles $R_M = 2$. If $M = 1$, certification forces $AB \leq 1$; only $(1, 1, 2)$ can remain, and only when the exponent permits it.

The number of inherited templates is $\tau(a)\tau(b)\tau(c) = \tau(abc)$. A union bound proves the second claim. For every fixed $\eta > 0$, the elementary estimate $\tau(n) \ll_\eta n^\eta$ follows from $e + 1 \leq 2^e \leq p^{\eta e}$ for sufficiently large primes p and bounded multiplicative costs at the finitely many smaller primes. Since $abc \leq c^3$, this proves the last claim. $\square$

Outputs whose remaining factors have additional powers or smaller actual radicals are not ruled out by this proposition. A count of raw CRT solutions cannot replace the count of distinct certified primitive triples.

3.2. Square completion on a conic. Put $P = abc$ and $R_P = \text{rad}(P)$. Positive integer points on

$$(13) \quad ax^2 + by^2 = cz^2$$

give candidate outputs (ax^2, by^2, cz^2) . Retain only primitive outputs. Then $\gcd(x, y, z) = 1$, and the positive coordinates are recovered uniquely from the output. Their radical is at most $R_P xyz$.

Lemma 3.2. *For coprime integers m, n , not both zero, put*

$$Q = an^2 + bm^2, \quad J = an + bm, \quad X = Q - 2nJ, \quad Y = Q - 2mJ,$$

and set

$$D_0 = \gcd(a, m) \gcd(b, n) \gcd(c, m - n), \quad G = \gcd(X, Y, Q) > 0.$$

Then

$$aX^2 + bY^2 = cQ^2, \quad \gcd(Q, J) = D_0, \quad G = \gcd(Q, 2J), \quad D_0 \mid G \mid 2D_0.$$

Every positive primitive point of (13) is represented by $(X/G, Y/G, Q/G)$ for some such pair. Passing to absolute values only overcounts outputs, and (m, n) and $(-m, -n)$ give the same point.

Proof. Expansion proves the conic identity. Coprimality of m, n gives $\gcd(Q, 2nJ, 2mJ) = \gcd(Q, 2J)$. Here is a valuation check of the remaining gcd, including the prime two. If a prime divides both Q, J , the identities

$$aQ = J^2 - 2bmJ + bcm^2, \quad bQ = J^2 - 2anJ + acn^2$$

show that it divides abc . At a prime dividing a , the common valuation is $\min(v_p(a), v_p(m))$: compare the terms of $J = an + bm$ and $Q = an^2 + bm^2$, using that b, n are units whenever $p \mid m$. The case $p \mid b$ is identical with the roles exchanged. At $p \mid c$, set $e = v_p(c)$, $r = v_p(n - m)$. If $r = 0$ the common valuation is zero. If $r > 0$, both coordinates are units and $J = a(n - m) + cm$ has valuation $\min(e, r)$ unless $e = r$. In that exceptional case either $v_p(J) = e$, or

$$Q = 2mJ - cm^2 + a(n - m)^2$$

has valuation exactly e . Thus the common valuation is always $\min(e, r)$. This argument does not divide by two. It proves the gcd formula and hence the assertions about G .

In the chart $z = 1$, a line through $(1, 1)$ has equation $(x, y) = (1, 1) + t(n, m)$. Its intersection parameters satisfy $t(2J + tQ) = 0$, giving the displayed second point. Every other rational point determines such a rational direction, which may be chosen primitive. The base point itself comes from the tangent direction $(m, n) = (-a, b)$. Dividing homogeneous coordinates by their gcd gives every primitive point. $\square$

Theorem 3.3. *For every $T > 0$, the number of positive primitive square-completion outputs with $cz^2 \leq T$ is at most*

$$(14) \quad \tau(P) \left(1 + 4\pi\sqrt{T/P} \right).$$

The subfamily certified by $R_Pxyz \leq (cz^2)^\mu$, for $0 < \mu < 1$, has size at most

$$(15) \quad \tau(P) \left(1 + \frac{4\pi}{\sqrt{c}} T^{\mu/2} \right).$$

Proof. Fix the three divisors in D_0 from Lemma 3.2. The conditions

$$d_a \mid m, \quad d_b \mid n, \quad d_c \mid m - n$$

define a lattice in $\mathbb{Z}^2$ of index $D_0 = d_a d_b d_c$. Indeed the first two have index $d_a d_b$, and the third residue map is surjective there because these three divisors are pairwise coprime. The height bound and $G \leq 2D_0$ imply

$$an^2 + bm^2 = Q \leq 2D_0\sqrt{T/c}.$$

This ellipse has area $V = 2\pi D_0\sqrt{T/P}$.

A centred ellipse of area V contains at most $1 + 2V/D_0$ primitive integer directions in a lattice of index D_0 , modulo sign. To see this, select representatives in a fixed half-plane and order them by angle. There is at most one primitive vector on each positive ray. Consecutive distinct directions have nonzero determinant divisible by D_0 , so their triangles with the origin have area at least $D_0/2$. These triangles have disjoint interiors and lie in the ellipse. The cases of zero or one direction satisfy the bound as well. There are $\tau(P)$ divisor templates, proving (14).

For a certified output write $C = cz^2$. Equation (13) implies $\max(x, y) \geq z$, so $xyz \geq z^2$. Hence

$$R_P C / c \leq R_P xyz \leq C^\mu, \quad C^{1-\mu} \leq c / R_P.$$

All certified heights are at most $U = \min\{T, (c/R_P)^{1/(1-\mu)}\}$. Moreover, $ab \geq c - 1 \geq c/2$ and $R_P \geq 2$, giving $R_P P \geq c^2$. Therefore

$$\sqrt{U/P} = U^{\mu/2} \sqrt{U^{1-\mu}/P} \leq T^{\mu/2} \sqrt{c/(R_P P)} \leq c^{-1/2} T^{\mu/2}.$$

Apply (14) at height U to obtain (15). $\square$

For fixed $K > 0$ and $T = c^K$, (15) is $c^{\max\{0, K\mu/2-1/2\}+o(1)}$. Every term defining $F(\mu)$ is strictly greater than $\mu/2$, so this family cannot exceed the exponent required to contradict (12). The conclusion is limited to the certificate. For example, the seed $(1, 8, 9)$ and point $(7, 2, 3)$ give $(49, 32, 81)$, whose radical is 42, while $R_{Pxyz} = 252$. Since $42^2 < 3^7$, $42 < 81^{9/10}$, although the certificate fails even with exponent one. This example proves that actual exceptional descendants cannot be identified with the certified subfamily; it is not a disproof of abc.

4. ACTUAL RADICAL BUDGETS AND A NECESSARY AMPLIFICATION WINDOW

Here we count actual prime support, allowing arbitrary new primes and their repeated powers. Fix a positive primitive seed $a + b = c$ and put $P = abc$. *Only in this section*, $R = \text{rad}(P)$ denotes the integer radical, rather than its logarithm as in the introductory notation. Thus $c \geq 2$ and $R \leq P \leq c^3$. All output triples below are ordered.

Theorem 4.1 (A height-independent radical budget). *Let $R \geq 1$ be a squarefree integer and $Y > 0$ a real number. The number $\mathcal{N}(R, Y)$ of all positive primitive triples $A + B = C$ with $R \mid \text{rad}(ABC)$ and $\text{rad}(ABC) \leq Y$ is zero if $Y < R$. If $Y \geq R$, then, with $X = Y/R \geq 1$,*

$$(16) \quad \mathcal{N}(R, Y) \leq 905 \, 45^{\omega(R)} X(1 + \log X)^{44}.$$

There is no restriction on the output height C .

Proof. For each output, $D = \text{rad}(ABC)/R$ is a squarefree integer with $D \leq X$ and $\gcd(D, R) = 1$. Fix D and take S to consist of the archimedean place of $\mathbb{Q}$ and the primes dividing RD . The pair $(A/C, B/C)$ solves $u + v = 1$ in S -units. This association is injective: A/C is reduced because $\gcd(A, C) = 1$, so it recovers A, C and then B .

Theorem 1.1 of [52] bounds the number of solutions over a degree- m number field by $(3.1 + 5(3.4)^m)45^{|S|}$. Here $m = 1$ and $|S| = 1 + \omega(R) + \omega(D)$, including the infinite place. The number for this D is consequently at most $904.5 \, 45^{\omega(R)+\omega(D)}$, which is smaller than the same expression with coefficient 905.

For an integer $M \geq 1$, let $d_M(n)$ count positive ordered M -tuples of product n . Prime factorization gives

$$M^{\omega(n)} \leq d_M(n) = \prod_{p^e \parallel n} \binom{e + M - 1}{M - 1}.$$

Fixing the first $M - 1$ entries of a tuple, and then dropping their product restriction, yields for every real $X \geq 1$

$$\begin{aligned} \sum_{n \leq X} d_M(n) &\leq X \sum_{n_1 \cdots n_{M-1} \leq X} \frac{1}{n_1 \cdots n_{M-1}} \\ &\leq X \left(\sum_{1 \leq n \leq X} \frac{1}{n} \right)^{M-1} \leq X(1 + \log X)^{M-1}. \end{aligned}$$

The empty tuple covers $M = 1$, and every sum equals one when $X = 1$. Sum the fixed- D estimate, enlarge the sum by discarding coprimality and squarefreeness, and use $M = 45$. This proves the bound and finiteness. The case $Y < R$ follows from divisibility of positive integers. $\square$

Corollary 4.2 (Uniformity for a seed). *Fix $K > 0$, $0 < \mu < 1$, and $\varepsilon > 0$. The number of outputs with*

$$R \mid \text{rad}(ABC), \quad \text{rad}(ABC) \leq C^\mu, \quad C \leq T \leq c^K$$

is zero if $T^\mu < R$, and otherwise is

$$(17) \quad \ll_{\varepsilon, K, \mu} c^\varepsilon \frac{T^\mu}{R}.$$

The implied constant is independent of the seed and of T .

Proof. Apply Theorem 4.1 with $Y = T^\mu$. For any fixed $\eta > 0$, separating the finitely many primes $p < 45^{1/\eta}$ gives $45^{\omega(R)} \ll_\eta R^\eta \leq c^{3\eta}$. Choose $\eta = \varepsilon/6$ before varying the seed. Since $\log(T^\mu/R) \leq K\mu \log c$, the remaining logarithmic factor is $\mathcal{O}_{\varepsilon,K,\mu}(c^{\varepsilon/2})$. This proves the assertion. $\square$

To record exactly what repeated primes save, for positive integers P, n set

$$D_P(n) = \prod_{\substack{p|n \\ p \nmid P}} p, \quad E_P(n) = n/D_P(n) \in \mathbb{N}_{>0}.$$

The factor $D_P(n)$ divides n , is squarefree, and is coprime to R . In $E_P(n)$, an old prime contributes its entire exponent, and a new prime of exponent e contributes exponent $e - 1$.

Proposition 4.3 (Exact excess accounting). *Let x, y, z be positive integers satisfying $ax^2 + by^2 = cz^2$, and put $(A, B, C) = (ax^2, by^2, cz^2)$. Then*

$$(18) \quad \text{rad}(ABC) = RD_P(xyz), \quad \text{rad}(ABC)E_P(xyz) = Rxyz.$$

In particular, $\text{rad}(ABC) \leq C^\mu$ implies

$$(19) \quad E_P(xyz) \geq (R/c)C^{1-\mu}.$$

These statements do not require the output to be primitive.

Proof. The prime divisors of Pn^2 are exactly those of Pn , so $\text{rad}(Pn^2) = RD_P(n)$. Use $ABC = P(xyz)^2$ to obtain (18). The conic equation makes z^2 a positive weighted average of x^2, y^2 . Thus $\max(x, y) \geq z$, and $x, y \geq 1$ give $xyz \geq z^2 = C/c$. Consequently $RC/c \leq Rxyz = \text{rad}(ABC)E_P(xyz) \leq C^\mu E_P(xyz)$. $\square$

Theorem 4.4 (A necessary amplification window). *Put $\rho = \log P / \log c$ and $\sigma = \log R / \log c$. Fix $K > 0$ and $0 < \mu < 1$. Consider all positive integer square completions whose unscaled output (ax^2, by^2, cz^2) is primitive and satisfies $C \leq c^K$ and $\text{rad}(ABC) \leq C^\mu$. If nonempty, this family has size*

$$(20) \quad \ll_{\varepsilon,K,\mu} c^{\min\{\max(0, (K-\rho)/2), K\mu-\sigma\}+\varepsilon} \quad (\varepsilon > 0).$$

For the exponent before ε to exceed $KF(\mu)$ from (12), it is necessary that

$$(21) \quad K > \rho + 4\sigma, \quad \frac{3\sigma}{K} < \mu < \frac{3}{4} \left(1 - \frac{\rho}{K}\right) < \frac{3}{4}.$$

All implied constants are uniform in the seed for fixed K, μ, ε .

Proof. Integer square completion retains every prime of P , so Corollary 4.2 gives the exponent $K\mu - \sigma$; nonemptiness makes it nonnegative. The complete fibre bound (14) in Theorem 3.3, together with $\tau(P) \ll_\eta P^\eta$ and $P \leq c^3$, gives $\max(0, (K - \rho)/2)$. Taking the smaller bound proves (20). The elementary divisor estimate used here follows by splitting primes at a fixed threshold; for large primes, $e + 1 \leq p^{\eta e}$, and for each remaining prime the quotient has a finite supremum over $e \geq 0$.

The BBLT exponent is $F(\mu) = \min\{2\mu/3, (23\mu + 3)/40, 3/5\}$ [7]. If $\mu \geq 3/4$, all three entries are at least $1/2$, whereas $\max(0, (K - \rho)/2) < K/2$ because $K, \rho > 0$. Thus the desired strict exponent gain is impossible in this range, even for the complete fibre. If $0 < \mu < 3/4$, then $F(\mu) = 2\mu/3$. Both exponents in (20) must exceed $2K\mu/3$, forcing

$$K\mu - \sigma > 2K\mu/3, \quad (K - \rho)/2 > 2K\mu/3.$$

These inequalities give the two bounds on μ in (21); their interval is nonempty only if $K > \rho + 4\sigma$. $\square$

In particular, along seeds satisfying $K \leq \rho + 4\sigma$, this family cannot supply a uniform lower count c^α with a fixed gap $\alpha > KF(\mu)$. Choose ε in (20) smaller than that gap to see this directly. The conclusion is for fixed K, μ ; it asserts no uniformity for growing K . An exceptional seed bounds σ from above, and that must not be substituted as a lower bound in the counting estimate. The interval (21) remains open: no sufficient lower bound for actual excesses or distinct outputs has been proved here.

Full support retention also needs its stated domain. The primitive seed $(49, 576, 625)$, of radical 210, has the rational completion $(x, y, z) = (3/7, 1/6, 1/5)$ with primitive output $(9, 16, 25)$ of radical 30. Equivalently, the integer point $(90, 35, 42)$ loses the prime 7 after its output is divided by the common factor 44100. Thus arbitrary projective normalization is not covered by the full- R estimate. If the output itself retains the rational squareclasses ax^2, by^2, cz^2 , only primes of odd valuation in P are automatically retained: their corresponding output valuations are nonnegative odd integers.

A separate attempt uses the three quadratic tripod identities

$$((a-b)^2, 4ab, c^2), \quad (a^2, 4bc, (b+c)^2), \quad (b^2, 4ac, (a+c)^2),$$

with $a \neq b$ and each row divided by its actual common divisor. That divisor is 1 or 4: for coprime u, v , the entries $(u-v)^2, 4uv, (u+v)^2$ have no common odd prime, and their gcd is 4 exactly when both u, v are odd. Their primitive radicals are exactly

$$R \text{rad}_{\text{odd}}(|a-b|), \quad R \text{rad}_{\text{odd}}(a+2b), \quad R \text{rad}_{\text{odd}}(2a+b),$$

where rad_{odd} omits the prime 2. Every new odd prime in the displayed factor is coprime to P , while all seed odd primes survive normalization and every primitive positive triple contains 2. Thus all three branches, and their positive iterates, retain R .

Nevertheless, no branch is guaranteed to preserve quality $q = \log C / \log \text{rad}(ABC)$. The seed $(1, 8, 9)$ gives $(49, 32, 81)$, $(1, 288, 289)$, and $(16, 9, 25)$, with radicals 42, 102, 30. The first quality is smaller than $\log 9 / \log 6$ because $42 > 36$; the last is less than one. For the middle one,

$$\frac{\log 289}{\log 102} < \frac{38}{31} < \frac{\log 9}{\log 6}, \quad 17^{24} < 6^{38}, \quad 2^{38} < 3^{24},$$

where the integer comparisons prove the two logarithmic inequalities. This refutes only the proposed universal quality-preserving selection among these three branches. Neither finite example is an abc disproof, and more general maps and the remaining window are not excluded.

5. THE COUPLED SIGNED PRIME-EXPONENT DEFECT

For $n \geq 1$ put

$$E_1(n) = \sum_{\substack{p|n \\ v_p(n)=1}} \log p, \quad E_3(n) = \sum_{p|n} (v_p(n) - 2)_+ \log p.$$

Write $m = \min(a, b)$ and $M = \max(a, b)$ for an abc triple.

Proposition 5.1 (Exact signed identity). *For every positive integer n ,*

$$(22) \quad \delta_2(n) := \log n - 2 \log \text{rad}(n) = E_3(n) - E_1(n).$$

The coupled defect satisfies

$$(23) \quad \begin{aligned} \Delta &:= \log(Mc) - 2R \\ &= E_3(M) + E_3(c) - E_1(M) - E_1(c) - 2 \log \text{rad}(m), \end{aligned}$$

$$(24) \quad 2h - 2R - \log 2 \leq \Delta \leq 2h - 2R.$$

Consequently abc is equivalent to

$$(25) \quad \forall \varepsilon > 0 \quad \exists K_\varepsilon \quad \forall (a, b, c) : \quad E_3(M) + E_3(c) \leq E_1(M) + E_1(c) \\ + 2 \log \text{rad}(m) + 2\varepsilon R + K_\varepsilon.$$

Proof. For every positive exponent e , $e - 2 = (e - 2)_+ - \mathbf{1}_{e=1}$. Multiply by $\log p$ and sum over the actual prime factorization to obtain (22). Pairwise coprimality gives $r = \text{rad}(m) \text{rad}(M) \text{rad}(c)$, which proves (23). Since $c/2 \leq M \leq c$, taking logarithms of $c^2/2 \leq Mc \leq c^2$ gives (24). An abc constant C_ε supplies $K_\varepsilon = 2C_\varepsilon$. Conversely (25) and the lower corridor give $C_\varepsilon = (K_\varepsilon + \log 2)/2$. $\square$

The equivalence does not prove (25); it identifies the entire missing estimate without losing the negative exponent-one terms. Replacing the coupled inequality by separate endpoint bounds fails.

Proposition 5.2 (An unbounded separate-endpoint obstruction). *For every $0 \leq \rho < 1$ and $K \in \mathbb{R}$, there is an *abc* triple with $m = 1$ and*

$$\delta_2(M) > \log \text{rad}(m) + \rho R + K.$$

Proof. Take $(a, b, c) = (1, 2^N, 2^N + 1)$ with $N \geq 1$. Then $m = 1$, $M = 2^N$, and $\delta_2(M) = (N - 2) \log 2$. Without making any assumption about the odd neighbour,

$$r \leq 2(2^N + 1) \leq 2^{N+2}, \quad R \leq (N + 2) \log 2.$$

Therefore

$$\delta_2(M) - \rho R \geq ((1 - \rho)N - 2 - 2\rho) \log 2 \longrightarrow \infty.$$

This proves the claim. The bound uses only $\text{rad}(n) \leq n$ and $\text{rad}(2^N) = 2$. $\square$

This argument neither proves nor assumes that all odd neighbours have large radical. It only rejects a false stronger estimate that discards their possible contribution. The coupled route remains open.

6. FINITE CERTIFICATES FOR REACHABLE LOCAL LATTICES

Let K be a nonarchimedean local field, $\mathcal{O}$ its valuation ring, π a uniformizer, and $k = \mathcal{O}/\pi\mathcal{O}$ its residue field, with $|k| = q > 1$ and $v(\pi) = 1$. Fix $N \geq 1$, write $L = \mathcal{O}^N$, and normalize additive Haar measure on K^N by $\mu(L) = 1$. For any set $\mathcal{R} \subset L$, let

$$H(\mathcal{R}) = \overline{\text{span}_{\mathcal{O}}(\mathcal{R})}.$$

If nonsingular N -tuples exist in $\mathcal{R}$, define

$$d(\mathcal{R}) = \min_{\substack{r_1, \dots, r_N \in \mathcal{R} \\ \det[r_1 \ \dots \ r_N] \neq 0}} v(\det[r_1 \ \dots \ r_N]);$$

otherwise set $d(\mathcal{R}) = \infty$.

Theorem 6.1 (A finite witness for the exact hull measure). *With $q^{-\infty} = 0$, one has*

$$\mu(H(\mathcal{R})) = q^{-d(\mathcal{R})}.$$

If $d(\mathcal{R}) < \infty$, some matrix A whose columns belong to $\mathcal{R}$ satisfies

$$v(\det A) = d(\mathcal{R}), \quad H(\mathcal{R}) = A\mathcal{O}^N.$$

Proof. In the full-rank case the nonzero determinant valuations are nonnegative integers, so choose A attaining their minimum d . For $r \in \mathcal{R}$, Cramer's rule gives

$$(A^{-1}r)_i = \frac{\det A_i(r)}{\det A},$$

where $A_i(r)$ replaces column i by r . Its columns still belong to $\mathcal{R}$. Every nonzero numerator has valuation at least d ; a zero numerator causes no difficulty. Thus $A^{-1}r \in \mathcal{O}^N$ and $\mathcal{R} \subset A\mathcal{O}^N$. Since the columns of A lie in $\mathcal{R}$, their integral spans are equal. The set $A\mathcal{O}^N$ is compact, hence closed.

Integral row and column operations give $UAV = \text{diag}(\pi^{a_1}, \dots, \pi^{a_N})$, with $U, V \in \text{GL}_N(\mathcal{O})$, $a_i \geq 0$, and $\sum a_i = d$. One may obtain this diagonalization by repeatedly moving an entry of least valuation to the upper left and clearing its row and column by integral operations. The quotient $\mathcal{O}^N/A\mathcal{O}^N$ has q^d elements. Its equal-measure cosets partition L , proving $\mu(A\mathcal{O}^N) = q^{-d}$.

In the rank-deficient case a nonzero K -linear functional annihilates $\mathcal{R}$. Scale its coefficients to lie in $\mathcal{O}$ with at least one unit coefficient. It then defines a surjection $\ell : L \rightarrow \mathcal{O}$ whose closed kernel contains $H(\mathcal{R})$. For every $t \geq 1$, $\ell^{-1}(\pi^t \mathcal{O})$ has index q^t in L . Hence $0 \leq \mu(H(\mathcal{R})) \leq q^{-t}$ for all t , proving measure zero. $\square$

Corollary 6.2. *The equality $H(\mathcal{R}) = L$ holds if and only if the reductions of $\mathcal{R}$ span k^N , equivalently if some reachable column matrix has unit determinant. More generally, any measurable $\mathcal{O}$ -submodule containing all columns of a reachable matrix A with $v(\det A) = d$ has measure at least q^{-d} .*

Proof. A determinant is a unit exactly when its reduction is nonzero; a spanning set of k^N contains a basis. Apply Theorem 6.1. For the last claim, the target contains $A\mathcal{O}^N$ and measure is monotone. The target need not be contained in L . $\square$

6.1. Independent and simultaneous actions. Let $L_j = \mathcal{O}^{d_j}$, with $d_j \geq 1$, and let $T = \bigotimes_{j=1}^s L_j$, where $s \geq 1$ and $N = \prod_j d_j$. For nonzero $x_j \in L_j$, write $x_j = \pi^{m_j} u_j$ with u_j primitive.

Proposition 6.3. *Under the independent action of $\prod_j \mathrm{GL}_{d_j}(\mathcal{O})$, the orbit of $\bigotimes_j x_j$ spans $\pi^{\sum_j m_j} T$. The measure of this spanned lattice, normalized by $\mu(T) = 1$, is $q^{-N \sum_j m_j}$.*

Proof. Integral row operations send a primitive vector to any chosen standard basis vector. Choosing these operations independently in each factor gives every standard tensor basis vector multiplied by $\pi^{\sum_j m_j}$. Conversely all orbit vectors lie in this scaled lattice. Its index is $q^{N \sum_j m_j}$, giving the measure. $\square$

Proposition 6.4 (A strict simultaneous-action counterexample). *Over any commutative ring A , identify $A^2 \otimes_A A^2$ with $\mathrm{Mat}_2(A)$. The A -span of $\{v \otimes v : v \in A^2\}$ is exactly the submodule of symmetric matrices. If $A \neq 0$, it omits E_{12} . For $A = \mathcal{O}$ it has Haar measure zero in $\mathrm{Mat}_2(K)$.*

Proof. Every generator has equal off-diagonal entries, a condition preserved by linear combinations. Conversely $e_1 \otimes e_1, e_2 \otimes e_2$, and

$$(e_1 + e_2) \otimes (e_1 + e_2) - e_1 \otimes e_1 - e_2 \otimes e_2 = E_{12} + E_{21}$$

span all symmetric matrices. No division by two occurs. The functional $M \mapsto M_{12} - M_{21}$ vanishes on the span and takes value one on E_{12} . In the local-field case, the congruence $M_{12} = M_{21} \pmod{\pi^t}$ occupies a proportion q^{-t} of the integral matrix lattice for every t . Its exact equality locus therefore has measure zero. $\square$

The generating set in Proposition 6.4 is stable under every simultaneous action $g \otimes g$. Thus even the full simultaneous integral linear group does not provide Proposition 6.3. This rejects a specific abstract substitute for independent generation; it does not identify the actual IUT outputs with this diagonal example.

6.2. A tensor point inside the holomorphic hull. An important distinction changes the reachability question. The hull in [56, Section 9.10.7, p. 125] is, after the finite-etale decomposition of a tensor algebra, a product of principal local ideals. It is therefore a module over the *whole product of integer rings*. Independent linear actions are not necessary for the following containment.

Let $E_1, \dots, E_m$ be finite extensions of $\mathbb{Q}_p$, and write

$$T = \bigotimes_{i=1}^m E_i = \prod_{\alpha} F_{\alpha}, \quad B = \prod_{\alpha} \mathcal{O}_{F_{\alpha}}, \quad A = \bigotimes_{i=1}^m \mathcal{O}_{E_i} \subset T,$$

where the integral tensor product is over $\mathbb{Z}_p$. Tensoring integral bases embeds A as a full lattice. Its elements are integral over $\mathbb{Z}_p$, so $A \subset B$, of finite index $I = [B : A]$.

Proposition 6.5. *For nonzero $\tau_i \in E_i$, put $t = \bigotimes_i \tau_i$. If a B -submodule $H \subset T$ contains a set S with $t \in S$, then*

$$\bigotimes_i \tau_i \mathcal{O}_{E_i} = tA \subset tB \subset H.$$

Both displayed lattices have full rank, and the smallest B -module containing the singleton $\{t\}$ is exactly tB .

Proof. The inverse of t is $\bigotimes_i \tau_i^{-1}$, by multiplication in the tensor algebra. Closure under B gives $tB \subset H$. The inclusion $A \subset B$ and the tensor multiplication rule give the other equality and inclusion. Multiplication by the unit t is a $\mathbb{Q}_p$ -linear automorphism and preserves full rank. Finally tB itself is a B -module containing t , proving the singleton assertion. $\square$

The conclusion applies *after* taking the specified holomorphic hull. It does not put tA in the raw tensor-image set or its ordinary $\mathbb{Z}_p$ convex closure. As printed, Section 9.10.7 requires the component generators to be nonzero and integral. An arbitrary bounded nonintegral set is not covered by that definition. With fractional generators allowed, a bounded set containing a unit has a unique such hull: take the least valuation in each nonzero component. This is an explicit extension of the convention, not an assumption that every source output is integral.

Proposition 6.6 (The two closures can differ). *Let $E = \mathbb{Q}_3(\pi)$ with $\pi^2 = 3$. Under*

$$E \otimes_{\mathbb{Q}_3} E \simeq E^2, \quad a \otimes b \longmapsto (ab, \sigma(a)b), \quad \sigma(\pi) = -\pi,$$

the tensor order satisfies

$$A = \{(x, y) \in \mathcal{O}_E^2 : x - y \in \pi \mathcal{O}_E\}, \quad [\mathcal{O}_E^2 : A] = 3.$$

It is already a closed $\mathbb{Z}_3$ -module, whereas its holomorphic hull is $B = \mathcal{O}_E^2$. The singleton $(1, 1)$ also has holomorphic hull B , while its ordinary convex closure is contained in $\mathbb{Z}_3(1, 1)$.

Proof. Valuations show that $\pi \notin \mathbb{Q}_3$ and $\mathcal{O}_E = \mathbb{Z}_3 + \mathbb{Z}_3\pi$. The images of an integral tensor basis are

$$(1, 1), \quad (\pi, -\pi), \quad (\pi, \pi), \quad (3, -3).$$

Their differences are divisible by π . Conversely write $x = a + b\pi$, $y = c + d\pi$. The asserted congruence is $3 \mid a - c$, and coefficients $(a+c)/2, (b-d)/2, (b+d)/2, (a-c)/6$ express (x, y) in the displayed basis. Reduction of $x - y$ modulo π gives the quotient $\mathbf{F}_3$ and hence the index. Both A and the singleton contain $(1, 1)$; its B -multiples are all of B . The compact module $\mathbb{Z}_3(1, 1)$ is convex and omits $(1, 0)$, proving the last distinction without relying on a choice between affine and absolutely convex conventions. $\square$

This gives a counterexample to the general equality of holomorphic hull and minimum ordinary convex closure in [56, Proposition 9.10.8.1]. It does not refute IUT or *abc*. In particular the positive containment in Proposition 6.5 must be used on the holomorphic-hull side of this distinction.

6.3. Native coefficients and compatible volumes. There is a coherent marking-preserving collation of the relevant local Kummer classes. For a fixed finite extension $E/\mathbb{Q}_p$ and marked copies $i_y : E \simeq E_y$, Kummer theory identifies the integral multiplicative cohomology with

$$\mathcal{K}(E_y) = \varprojlim_n E_y^\times / (E_y^\times)^{p^n}.$$

This is the identification used in [55, Proposition 7.2.2]. At each finite level, the map $i_z i_y^{-1}$ induces an isomorphism of these quotients; their inverse limit gives a topological isomorphism $F_{y \rightarrow z}$ with

$$F_{y \rightarrow z}(\kappa_y(i_y(u))) = \kappa_z(i_z(u)), \quad F_{z \rightarrow r} F_{y \rightarrow z} = F_{y \rightarrow r}.$$

It preserves the unit-completion subgroup and can be chosen coherently every time a label is repeated. This uses one family of the factorwise topological collations in the cited source; it does not assert that every such collation preserves a marked arithmetic element.

For $b_p = p$ when p is odd and $b_2 = 4$, the coefficient in [56, equation (9.7.2.2)] is

$$\lambda_{b_p}(a) = \frac{\log(1 + b_p a)}{b_p}, \quad a \in \mathcal{O}_E.$$

For $a \neq 0$ it is nonzero and has the same valuation as a . Indeed, with $v(p) = 1$ and $x = b_p a$, every term after the first in the logarithm series has greater valuation:

$$v(x^n/n) - v(x) = (n-1)v(x) - v_p(n) > 0 \quad (n \geq 2).$$

For odd p use $v_p(n) < n - 1$, and for $p = 2$ use $v(x) \geq 2$. The terms tend to zero, so $v(\log(1+x)) = v(x)$. The field markings commute with this convergent series, yielding the same normalized coefficient in the fixed target. In particular, the background coefficient $\lambda_{b_p}(1)$ is a unit.

Proposition 6.7. Put $d_i = [E_i : \mathbb{Q}_p]$, $D_T = \prod_i d_i$, and normalize Haar measure on T by $\mu(A) = 1$. For $t = \bigotimes_i \tau_i$,

$$\mu(tB) = I \prod_i |\tau_i|_{E_i}^{D_T/d_i}, \quad |x|_{E_i} = |N_{E_i/\mathbb{Q}_p}(x)|_p.$$

For any $c_0 > 0$, the functional $\nu(U) = \mu(U)^{c_0/D_T}$ is monotone, and every measurable $H \supset tB$ satisfies

$$\nu(H) \geq I^{c_0/D_T} \prod_i |\tau_i|_{E_i}^{c_0/d_i} \geq \prod_i |\tau_i|_{E_i}^{c_0/d_i}.$$

A factor prescription $\prod_i \mu_i(V_i)^{\gamma_i}$ for the same integral tensor lattice can be independent of its presentation only if all $d_i \gamma_i$ agree. These equalities are also sufficient on full tensor lattices.

Proof. The I translates of A partition B , so $\mu(B) = I$. Multiplication by τ_i in its tensor factor has determinant $N_{E_i/\mathbb{Q}_p}(\tau_i)^{D_T/d_i}$. Multiplication by t is the composition of these maps, giving the displayed Haar factor. Monotonicity and the positive power give the inequalities. For necessity of balancing, put a scalar p in any one factor of the tensor lattice pA . The proposed values are $p^{-d_i \gamma_i}$ and must agree. If $d_i \gamma_i = c_0$, the normalized functional μ^{c_0/D_T} gives the factor prescription by the same determinant computation, proving sufficiency. $\square$

The normalization matters when using an upper estimate from another source. Denote the preceding Haar measure by μ_A , and instead let $\mu_B(B) = 1$. On a measurable region of finite positive measure, put

$$V_A(U) = \frac{\log \mu_A(U)}{D_T}, \quad V_B(U) = \frac{\log \mu_B(U)}{D_T}.$$

Uniqueness of Haar measure and $\mu_A(B) = I$ give

$$(26) \quad V_A(U) = V_B(U) + \frac{\log I}{D_T}, \quad V_B(tB) = \sum_i \frac{\log |\tau_i|_{E_i}}{d_i}.$$

The positive power of Haar measure used above need not itself be an additive measure. Only its monotonicity was asserted.

This conversion is required by the original texts: [56, Section 9.10.3.1] assigns value one to the tensor order A , whereas [65, Proposition 1.4(i), author-version p. 13] normalizes the log-volume of the maximal order B to zero. The positive index contribution cannot be carried into a B -normalized upper estimate. The second equality in (26) preserves the native lower bound without that contribution.

For example, if all m factors equal a Galois extension $E/\mathbb{Q}_p$ of degree d , then

$$E^{\otimes m} \simeq E^{d^{m-1}}, \quad \log_p[B : A] = \frac{(m-1)d^{m-1}\delta_E}{2},$$

where δ_E is the exponent of its discriminant over $\mathbb{Z}_p$. Indeed, the trace Gram matrix on a tensor basis has discriminant exponent $md^{m-1}\delta_E$. On the product maximal order that exponent is $d^{m-1}\delta_E$. Their difference is twice the exponent of the lattice index. Thus

$$V_A(U) - V_B(U) = \frac{(m-1)\delta_E}{2d} \log p.$$

For copies of a fixed selected field $E = L'_w$ over $F = L_{\text{mod},v}$, the source weight $\gamma = 1/[E : F]$ obeys $[E : \mathbb{Q}_p]\gamma = [F : \mathbb{Q}_p]$. It is thus balanced. A tuple with r theta coefficients $\lambda_{b_p}(a)$ and the remaining coefficients $\lambda_{b_p}(1)$ gives the native lower bound

$$\nu(H) \geq |a|_E^{r/[E:F]} \geq |a|_E^r \quad \text{if } 0 < |a|_E < 1.$$

For a rational Frey curve C , the field of moduli is $\mathbb{Q}$. The underlying punctured curve is defined over $\mathbb{Q}$; its absolute field of moduli, used before adjoining level structure in [56, Section 3.1(4)], is therefore $\mathbb{Q}$. Equivalently the Legendre parameter $-b/a$ gives $\mathbb{Q}(j) = \mathbb{Q}$ in [57, Section 5.3]. The later extensions $L = \mathbb{Q}(i, C[30])$ and $L' = L(C[\ell])$ are Galois over $\mathbb{Q}$ and do not redefine that unlevelled moduli field. Choosing one place of L' above each rational prime gives the place-compatible selected bijection of [56, Section 3.3(14) and Proposition 9.4.2.4(4)]. Thus there is one selected place per prime. The internal field decomposition of $E^{\otimes m}$ still remains.

For a split multiplicative place with $q \in \mathbb{Q}_p$, take a root $a^{2\ell} = q$ in the selected field E . The native module absolute value satisfies $|q|_E = |q|_p^d$, so

$$|\lambda_{b_p}(a)|_E^{1/d} = |q|_p^{1/(2\ell)}.$$

A block with one such entry and background units therefore gives

$$(27) \quad V_B(H) \geq V_B(tB) = -Q_p, \quad Q_p = -\frac{\log |q|_p}{2\ell} > 0.$$

The chosen marking-induced isomorphisms give an actual tuple in the collation, and convex closure retains it. The tuple-to-tensor map and the module hull then give the stated containment. No independent action on repeated labels is needed.

6.4. Changing the target of the whole collation. Fix a family of source groups C_y , a set of configurations of source classes, and a fixed slot set in each block. Require repeated occurrences of a source label to use the same isomorphism. Let Ψ_k be the union of all resulting tuples in a target C_k , where all allowed topological group isomorphisms are used. The same discussion applies to a groupoid of Galois-induced isomorphisms.

Proposition 6.8 (Covariance for a fixed source family). *If $c : C_0 \simeq C_1$ is an allowed target isomorphism, then*

$$\Psi_1 = c^{\text{slots}}(\Psi_0).$$

If c in logarithmic coordinates is induced by a local field marking, the same covariance holds for closed convex closure, tensor image, and the maximal-order module hull. In particular the transported hulls have equal V_A volumes, and equal V_B volumes with the respective normalizations.

Proof. Postcomposition $f \mapsto c \circ f$ is a bijection from $\text{Iso}(C_y, C_0)$ to $\text{Iso}(C_y, C_1)$, with inverse postcomposition by c^{-1} . Apply it to the whole family of maps used by a source configuration. Equal maps on repeated labels remain equal in both directions. Taking the union proves the set equality. For a groupoid the same argument uses closure under composition and inverse.

The marking induces a linear homeomorphism in logarithmic coordinates, so it carries a smallest closed convex region to the corresponding smallest closed convex region. Tensoring the field maps commutes with the map on pure tensors. The resulting algebra isomorphism preserves integrality over $\mathbb{Z}_p$ and thus carries A_0, B_0 to A_1, B_1 . Minimality of the coefficient-module hull, applied also to the inverse, proves equality of the transported hulls. Finally uniqueness of Haar measure transports the chosen normalization, and dimensions are unchanged. $\square$

The source family and its classes are hypotheses of this proposition. It justifies a change of target in the all-isomorphism construction of [56, Section 9.7.5.1]; it does not show that every operation intended by a later Frobenius comparison keeps that source family fixed. In particular a favorable native point and covariance for this family cannot be combined with an upper estimate for a different family.

To see the normalization issue directly, write a rescaled untilt norm on the marked finite field as $N_y(x) = |x|_E^{s_y}$, $s_y > 0$. The native Haar contribution is

$$|\lambda_{b_p}(a)|_E^{1/d} = N_y(\lambda_{b_p}(a))^{1/(ds_y)}.$$

Changing s_y alone leaves it unchanged. Also replacing q by q^p rebuilds the principal unit from a^p and gives coefficient $\lambda_{b_p}(a^p)$, whereas raising $1 + b_p a$ to its p th power gives $p\lambda_{b_p}(a)$. They need not agree: for $p = 3, E = \mathbb{Q}_3, a = 3$, their valuations are three and two, respectively. The

Frobenius in [55, Corollary 5.6.2] acts on a perfected multiplicative monoid, not by definition on this fixed integral Kummer lattice.

The original IUT IV computation explicitly uses a squared exponent: [65, Theorem 1.10, Step (v), author-version pp. 27–29] takes the j th theta-pilot valuation to be $j^2 v_p(q)/(2\ell)$. Replacing the coefficient above by $\lambda_{b_p}(a^{j^2})$ would give the single-point lower bound $-j^2 Q_p$, rather than $-Q_p$. For $s = (\ell - 1)/2$ the average of these coefficients is

$$\frac{1}{s} \sum_{j=1}^s j^2 = \frac{(s+1)(2s+1)}{6} = \frac{\ell(\ell+1)}{12}.$$

Proposition 6.8 can be proved for either fixed family, but it does not identify them. The required favorable lower bound for the correctly powered possible-image family is a further assertion.

6.5. The source weights for several selected places. The mixed-place weights can also be read from the original upper source, [65, Remark 1.7.1, author-version p. 17]. Let $F = L_{\text{mod}}$ have degree n , put $h_v = [F_v : \mathbb{Q}_p]$, $d_v = [L'_{w(v)} : \mathbb{Q}_p]$, and $\gamma_v = h_v/d_v$ for $v \mid p$. Then $\sum_v h_v = n$. For a word f of length m in these places put

$$D_f = \prod_i d_{f(i)}, \quad \rho_f = \frac{\prod_i h_{f(i)}}{n^m}.$$

With B_f the maximal order in the corresponding tensor algebra, the degree-normalized functional on a product of ideal hulls is

$$V_{B,p,m}(H) = \sum_f \frac{\rho_f}{D_f} \log \mu_{B_f}(H_f), \quad \mu_{B_f}(B_f) = 1.$$

The word weights sum to one and have marginal h_v/n at every slot. Applying the tensor determinant identity therefore gives

$$V_{B,p,m}(tB) = \frac{1}{n} \sum_i \sum_{v \mid p} \gamma_v \log |\tau_{i,v}|_{L'_{w(v)}}.$$

The corresponding tensor-order normalization differs by exactly $\sum_f (\rho_f/D_f) \log[B_f : A_f]$. This proves the formula on product ideal hulls. It is not a product formula for an arbitrary measurable non-product region. The extra global factor n must be coordinated with the arithmetic-degree convention; it equals one in the rational branch.

These constructions prove the local point-to-lattice implication and fixed-source target covariance with their stated normalizations. They do not yet identify the full Frobenius-shifted source families in [57, Theorem 6.10.1] with those of the later upper estimate. That estimate must concern the same hull and volume functional. Part III already introduces the volume of a containing compact region on p. 121; omission of a hull symbol in a later formula is therefore not, by itself, evidence that a different set was estimated. What is needed is the actual comparison with the region and measure in IUT IV, rather than an identification by notation.

6.6. What a source-derived application must establish. The July 2026 LANA report [59] isolates an identification between q-pilot and possible-output volume data in the passage to IUT III, Corollary 3.12 [62]. Joshi's July 30 response [58] refers to distinct arithmeticoids and his Part IV comparison [57, Theorem 6.10.1]. These are source claims to reconstruct, not assumptions used in our local theorems.

Part III [56, Proposition 9.7.5.1] allows specified topological isomorphisms in its collation. Section 6.2 constructs one marking-preserving family and proves containment after the stated holomorphic hull. It does not require unrestricted independent linear actions. The remaining identification must compare the same set of source objects, the same target region, and the same numerical normalization with the later upper estimate.

For a source-faithful finite packet of local sets $\mathcal{R}_v \subset \mathcal{O}_v^{N_v}$ and positive weights λ_v , Theorem 6.1 gives, when all defects are finite,

$$V_H = - \sum_v \frac{\lambda_v}{N_v} d(\mathcal{R}_v) \log q_v.$$

Suppose genuine local target regions Θ_v contain the respective hulls and have finite positive measure. Define, on these same carriers, $V_\Theta = \sum_v (\lambda_v/N_v) \log \mu_v(\Theta_v)$. Let $V_q \in \mathbb{R}$ be the fixed native q -pilot log-volume under the same source normalization. Then

$$\sum_v \frac{\lambda_v}{N_v} d(\mathcal{R}_v) \log q_v \leq -V_q \implies V_q \leq V_H \leq V_\Theta.$$

The final inequality follows from local measure monotonicity, followed by the logarithm and the positive weighted sum. A different volume functional on a non-product region requires a separate monotonicity theorem. This is a sufficient numerical interface. The native points and their holomorphic hulls above provide one local construction; identifying them with the region bounded in IUT IV [65], controlling the same normalized determinant sum, and deriving the uniform global comparison remain separate obligations. Archimedean factors and infinite-packet convergence are not covered by the finite local theorem.

7. ADMISSIBLE UNIT ACTIONS AND A CYCLOTOMIC TRACE INVARIANT

This section uses the actual all-open-subgroup condition in [64, Example 1.8(iv), author-version pp. 38–39]. It must be distinguished from allowing every automorphism of one fixed integral lattice. Fix a finite extension $E/\mathbb{Q}_p$, an algebraic closure $\overline{E}$, and $G = \text{Gal}(\overline{E}/E)$. For each finite K/E inside $\overline{E}$, put

$$U_K = \mathcal{O}_K^\times, \quad L_K = \log U_K \subset K, \quad I_K = p^{-1}L_K.$$

The logarithm is used only on units. Its kernel on U_K is the finite group of roots of unity in K , and L_K is a full $\mathbb{Z}_p$ -lattice in K . These assertions hold also for $p = 2$.

Logarithm induces a G -equivariant additive isomorphism

$$(28) \quad \mathcal{O}_{\overline{E}}^\times / \mu(\overline{E}) \simeq (\overline{E}, +).$$

Indeed the finite-field kernel statement proves injectivity. Given $x \in \overline{E}$, choose n large enough for $\exp(p^n x)$ to converge in a finite extension and choose a p^n -th root y of it. Then y is a unit and $\log y = x$, proving surjectivity. If $H = \text{Gal}(\overline{E}/K)$, the invariants of the quotient in (28) are K , whereas the image of the invariants of the original unit group is precisely L_K .

Under this identification, the group $\text{Ism}(G)$ specified in [64] consists of additive G -equivariant automorphisms F of $\overline{E}$ preserving L_K for every finite K/E . The source requires preservation of these lattices, not only of their rational spans. Such a map is $\mathbb{Q}_p$ -linear on each K : preservation of $p^n L_K$ implies continuity, which extends integer linearity to $\mathbb{Z}_p$ -linearity and then to $\mathbb{Q}_p$ -linearity.

7.1. The all-open condition forces scalar action.

Lemma 7.1 (Realizing logarithmic sublattices by norms). *For every finite-index $\mathbb{Z}_p$ -submodule $J \subset L_E$, there is a finite abelian extension K/E such that*

$$\text{Tr}_{K/E}(L_K) = J.$$

Proof. Let $V = \log^{-1}(J) \subset U_E$, choose a uniformizer π of E , and put $N = \pi^{\mathbb{Z}}V$. This is an open finite-index subgroup of $E^\times$. The local existence theorem of class field theory [67, Chapter I, Theorem 1.4] supplies a finite abelian K/E with $\text{Norm}_{K/E}(K^\times) = N$. For integer-normalized valuations,

$$v_E(\text{Norm}_{K/E} z) = f_{K/E} v_K(z).$$

Consequently a norm is a unit exactly when its preimage is a unit, and $\text{Norm}_{K/E}(U_K) = N \cap U_E = V$. The identity $\log \text{Norm} z = \text{Tr} \log z$ follows by summing the logarithms of the conjugates. Therefore $\text{Tr}_{K/E}(L_K) = \log V = J$. $\square$

Lemma 7.2 (Sublattice rigidity). *An automorphism of a nonzero finite free $\mathbb{Z}_p$ -module preserving every finite-index submodule is multiplication by one element of $\mathbb{Z}_p^\times$.*

Proof. Choose a basis $e_1, \dots, e_d$. Preservation of $\mathbb{Z}_p e_i + p^n L$ for every n puts the image of e_i in their intersection, which is $\mathbb{Z}_p e_i$: every other coordinate is divisible by every power of p and hence is zero. Write $F(e_i) = \lambda_i e_i$. The same argument for $e_i + e_j$ implies $\lambda_i = \lambda_j$. Thus $F = \lambda \text{id}$; surjectivity shows that λ is a unit. The rank-one case is already covered by the first step. $\square$

Theorem 7.3 (Classification of the specified Ism action). *For the group defined in [64, Example 1.8(iv)],*

$$\text{Ism}(G) = \mathbb{Z}_p^\times$$

under (28), with the right side acting by scalars.

Proof. For finite abelian K/E , equivariance gives $F_E \text{Tr}_{K/E} = \text{Tr}_{K/E} F_K$. Since $F_K(L_K) = L_K$, Lemma 7.1 implies that F_E preserves every finite-index submodule of L_E . Lemma 7.2 gives $F_E = \lambda_E \text{id}$ with $\lambda_E \in \mathbb{Z}_p^\times$. Repeat over any finite K/E : the hypothesis includes all finite extensions of K , since their Galois groups are open in G . Thus $F_K = \lambda_K \text{id}$. Restriction to the nonzero subfield E gives $\lambda_K = \lambda_E$, proving the assertion on all of $\bar{E}$. Conversely every $\mathbb{Z}_p$ -unit scalar preserves all the lattices and commutes with G . It is the image of a unit of $\hat{\mathbb{Z}}$ under the unit-power action; prime-to- p components act trivially after quotienting by all roots of unity. $\square$

Only preserving L_E would leave the full group $\text{GL}_{[E:\mathbb{Q}_p]}(\mathbb{Z}_p)$; the finite-extension hypothesis is decisive. The full-unit lifting theorem [53, Theorem 7.6] by itself does not classify automorphisms of the torsion quotient. The preceding proof supplies that additional step. For the indeterminacy called Ind2 in [62, Theorem 3.11(i)], the theorem applies to its indicated Ism factors. It does not classify the Galois arrows of Ind1 or the set operations of Ind3.

For example, if $1 < e(E/\mathbb{Q}_p) \leq p-2$, logarithm and exponential on the maximal ideal give $I_E = \pi^{1-e} \mathcal{O}_E$. The unit $u = \log(1+p)/p$ has native valuation zero, and all its Ism images still have valuation zero. None reaches the least valuation $1/e - 1$ of this larger lattice. This statement concerns Ism; it adds no unproved restriction to Ind1.

7.2. What the procession retains, and an exact Ind2 hull equality. The Ind1 source in [62, Theorem 3.11(i)] is the mono-analytic procession of [63, Proposition 6.9(ii)]. At a nonarchimedean place its retained category is $\mathcal{B}(E)^0$, the connected finite étale covers of E , rather than the earlier curve-covering category [63, Examples 3.2(i), 3.3(i), Definition 4.1(iii)–(iv)]. Its self-equivalences, up to natural isomorphism, correspond to $\text{Out}(G_E)$: pullback of finite transitive G_E -sets constructs one direction, and adjoining finite disjoint unions and recovering the fibre functor constructs the inverse.

For fixed capsule labels, the capsule-full poly-isomorphism contains all collections of constituent isomorphisms [63, §0, pp. 33–34; Definition 4.10]. Consequently every local outer Galois automorphism occurs as a representative in an allowed poly-automorphism, with the same representative used at repeated labels if required. To see this directly, identify each labelled category with $\mathcal{B}(E)^0$, conjugate the chosen pullback equivalence to each copy, and use identity maps at the other places. Keep the capsule and stage injections equal to the identity. Each resulting collection belongs to the required full set of constituent isomorphisms, and conjugation preserves the full connecting sets. There is no further requirement to lift back to the curve category. This is a statement about representatives inside poly-morphisms, not a claim that each representative is a distinct object or morphism of the coarse category. It supplies no arbitrary integral linear lift.

Proposition 7.4 (Ind2 does not enlarge a fixed module hull). *On a fixed marked nonarchimedean tensor carrier, let*

$$T = \bigotimes_{i, \mathbb{Q}_p} \left(\prod_{v \in J_i} E_{i,v} \right), \quad A = \bigotimes_{i, \mathbb{Z}_p} \left(\prod_{v \in J_i} \mathcal{O}_{E_{i,v}} \right) \subset B,$$

where B is the maximal order of the finite etale algebra T . For any subset $S \subset T$, write $\Gamma_2 S$ for the union of its images under the local Ind2 operators of [62, Theorem 3.11(i)]. Then

$$\text{span}_B(\Gamma_2 S) = \text{span}_B(S).$$

The same equality holds for closed B -module hulls.

Proof. Theorem 7.3 makes the action on each local direct summand multiplication by a scalar $\lambda_{i,v} \in \mathbb{Z}_p^\times$. In factor i , their tuple $c_i = (\lambda_{i,v})_v$ is a unit of $\prod_v \mathcal{O}_{E_{i,v}}$. The tensor product operator is multiplication by $c = \bigotimes_i c_i \in A^\times$, with inverse the tensor of the c_i^{-1} . Thus every image of S lies in $\text{span}_B(S)$. The identity Ind2 operator supplies the reverse inclusion of spans. Taking closures gives the closed version. Any extra convention forcing some scalars to coincide leaves the argument unchanged. $\square$

This removes the Ind2 step from a B -module-hull computation once the Ind1 image set and native markings are fixed. It does not remove the distinction between A and B , turn an Ind1 map into a B -linear map, or identify Ind3 with multiplication by a unit.

7.3. Integral Galois transport preserves trace depth. A coefficient-compatible Galois arrow consists of a continuous isomorphism $\alpha : G_{E'} \simeq G_E$ and an isomorphism $\beta : \alpha^* \mathbb{Z}_p(1)_E \simeq \mathbb{Z}_p(1)_{E'}$ of integral Tate modules. The cyclotomic character and inertia are group-theoretically determined [68, Propositions 1.1–1.2 and Corollary 1.3]. In particular the coefficient isomorphism can be chosen integrally; one cannot insert multiplication by p as an isomorphism.

Proposition 7.5. *The induced logarithmic map $F_{\log} : E \rightarrow E'$ satisfies*

$$(29) \quad \text{Tr}_{E'/\mathbb{Q}_p}(F_{\log} x) = c_\alpha \text{Tr}_{E/\mathbb{Q}_p}(x) \quad (x \in E)$$

for some $c_\alpha \in \mathbb{Z}_p^\times$. The statement also holds with both logarithmic coordinates divided by p .

Proof. Put $h_E = \log_p \chi_E \in H^1(G_E, \mathbb{Z}_p)$, where the latter coefficient action is trivial. Equivariance of β gives $\chi_E \alpha = \chi_{E'}$, hence $\alpha^* h_E = h_{E'}$. Local duality and the invariant map identify $H^2(G_E, \mathbb{Z}_p(1))$ with $\mathbb{Z}_p$. Integral cohomology transport is an isomorphism, so it acts there by a unit c_α , not by an arbitrary nonzero rational p -adic scalar.

The local reciprocity cup-product formula, for an integral character h and a Kummer class $\kappa(z)$, is

$$\text{inv}(h \cup \kappa(z)) = h(\text{rec}_E z).$$

It follows at finite level from the cyclic-algebra description and then by inverse limit; see [67, Chapter III, Proposition 3.6 and §4]. In the arithmetic-Frobenius convention,

$$h_E(\text{rec}_E z) = -\log_p \text{Norm}_{E/\mathbb{Q}_p} z = -\text{Tr}_{E/\mathbb{Q}_p} \log_p z \quad (z \in U_E).$$

The integral unit Kummer submodule is also preserved: it is the annihilator, under this pairing, of the unramified $\mathbb{Z}_p$ -character. Characteristic inertia carries a generator of that unramified character module to a unit multiple of a generator, and cup-product naturality preserves its annihilator.

Applying the same naturality with h_E now proves (29) on L_E . Torsion unit classes are killed by logarithm and pair to zero in the torsion-free integral H^2 ; the argument therefore descends to the actual free unit lattice. Since L_E spans E , $\mathbb{Q}_p$ -linearity proves the formula everywhere. Uniform division by p on both sides does not change it. $\square$

Composing with the Ism factors of Theorem 7.3 only changes c_α by a unit. Thus the valuation of a nonzero absolute trace is an invariant of these local Galois/Kummer/Ism arrows. For example, on $E = \mathbb{Q}_7(\pi)$ with $\pi^3 = 7$, the normalized integral log lattice I_E has basis $1, \pi/7, \pi^2/7$. Swapping its first two basis vectors is not an admissible Galois/Kummer arrow: their traces are 3 and 0. No claim that every matrix satisfying (29) is admissible is made.

Theorem 7.6 (A native root-power obstruction). *Let $p \neq \ell$ be odd primes, $b \in \mathbb{Q}_p$ with $N = v_p(b) > 0$ and $\ell \nmid N$, and let $E/\mathbb{Q}_p$ contain μ_ℓ and a with $a^\ell = b$. Write $d = [E : \mathbb{Q}_p]$ and, for $s \geq 1$ with $\ell \nmid s$, put $\tau_s = \log(1 + pa^s)/p$. Then*

$$(30) \quad \mathrm{Tr}_{E/\mathbb{Q}_p} \tau_s = \frac{d}{\ell p} \log(1 + p^\ell b^s),$$

$$(31) \quad v_p(\mathrm{Tr}_{E/\mathbb{Q}_p} \tau_s) = v_p(d/\ell) + \ell - 1 + sN.$$

In particular, distinct such s give distinct orbits under the arrows of Proposition 7.5, even after Ism action.

Proof. The field $F = \mathbb{Q}_p(\mu_\ell)$ is unramified over $\mathbb{Q}_p$. The valuation assumption makes b a non- ℓ -th-power in F , so $K = F(a)$ has degree ℓ over F . Since multiplication by s permutes the exponents modulo ℓ , and ℓ is odd,

$$\mathrm{Norm}_{K/F}(1 + pa^s) = \prod_{\zeta \in \mu_\ell} (1 + p\zeta a^s) = 1 + p^\ell b^s.$$

All factors are principal units. Taking logarithms, dividing by p , and using trace transitivity gives (30): the remaining factor is $[E : K][F : \mathbb{Q}_p] = d/\ell$. For $x = p^\ell b^s$, the term x is the unique least-valuation term in $\log(1 + x)$, since

$$v_p(x^k/k) - v_p(x) = (k-1)(\ell + sN) - v_p(k) > 0 \quad (k \geq 2).$$

This proves (31), including nonvanishing. Its values differ by $(s-t)N$ for distinct labels, which (29) cannot change. $\square$

For a concrete example take $p = 11$, $\ell = 5$, $b = 11$, and $E = \mathbb{Q}_{11}(\pi)$ with $\pi^5 = 11$. The base field contains μ_5 ; the trace depths of τ_1 and τ_4 are 5 and 8. This is a specified local example, not a construction of all the global initial theta data.

There is a direct local link with a split Tate curve having rational full 2-torsion. If q is its parameter, choose $b^2 = q$. The class of b in $\overline{\mathbb{Q}_p}^\times/q^\mathbb{Z}$ is rational, so $g(b)/b \in q^\mathbb{Z} \cap \{1, -1\}$. Valuation makes this intersection $\{1\}$; hence $b \in \mathbb{Q}_p$. Rationalizing full ℓ -torsion also rationalizes full 2ℓ -torsion and supplies the required a and μ_ℓ . The theorem consequently applies when $\ell \nmid v_p(q)/2$. It does not assert that every bad prime meets that condition.

For a pure tensor with $m-1$ factors $u = \log(1+p)/p$ and one factor τ_s , trace in $E^{\otimes m}$ is the product of the factor traces. Its valuation is

$$(m-1)v_p(d) + v_p(d/\ell) + \ell - 1 + sN.$$

Local arrows, Ism actions, and factor permutations preserve this number, including when labels require repeated use of one arrow. This separates the exponent-1 input from the exponent- j^2 input for $j > 1$, $j < \ell$. Raising $1 + pa$ to the power j^2 , which scales its logarithm, is a different operation from replacing a by a^{j^2} .

7.4. Why this is not yet a holomorphic-hull comparison. Multiplying generators by coefficients of the maximal tensor order need not preserve their trace depths. Thus distinct pure orbits do not imply distinct module hulls. The upper-semicompatibility operation Ind3 in [62, Theorem 3.11(ii)] is a further operation, and has not been replaced here by an automorphism of one fixed Kummer module. Nor does changing an untilt's real-valued norm change the native trace in a fixed marked p -adic field.

A converse test is useful: trace preservation alone permits a common least-layer displacement, without establishing an Ind1 lift.

Lemma 7.7 (Common avoidance with fixed trace). *Let k be a field of characteristic different from two, let V be a k -vector space, and let $t, l \in V^*$. Suppose $t(w) = 0$, $l(w) = 1$. For any nonzero $x, y \in V$, there is an automorphism F with $tF = t$ and $l(Fx), l(Fy) \neq 0$.*

Proof. Set $P(v) = v - l(v)w$. Choose a functional h nonzero on each nonzero vector among $P(x), P(y)$. Such a functional exists: separate the first vector; if that functional kills the second, separate the second and use either the new functional or the sum, according to its value on the

first. Put $f = hP$, so $f(w) = 0$. The maps $F_z(v) = v + zf(v)w$ have inverses F_{-z} and preserve t . For $v = x, y$, the polynomial $l(v) + zf(v)$ is nonzero: if $l(v) = 0$, then $P(v) = v \neq 0$. Two nonzero affine polynomials have at most two roots altogether. The three distinct scalars $0, 1, -1$ therefore contain a valid choice of z . $\square$

For finite Galois $E/\mathbb{Q}_p$ with $2 \leq e \leq p-2$, one has $I_E = \pi^{1-e}\mathcal{O}_E$, the inverse different. Both $\text{Tr} : I_E \rightarrow \mathbb{Z}_p$ and $\text{Tr} : \mathcal{O}_E \rightarrow \mathbb{Z}_p$ are onto; the latter follows by tracing first through the tame ramified extension and then through the unramified one. On I_E/pI_E , trace modulo p is independent of any nonzero $\mathbb{F}_p$ -coordinate of $I_E/\pi I_E$: the latter vanishes on $\mathcal{O}_E \subset \pi I_E$, whereas trace does not. The lemma gives an automorphism preserving trace modulo p and reaching the least valuation layer for two primitive vectors simultaneously. Choose a basis splitting integral trace. The residue matrix has first row $(1, 0, \dots, 0)$; lift that row exactly and the others arbitrarily. The determinant is a unit, so this is an integral invertible lift preserving trace exactly. No lift to an absolute Galois automorphism is established by this construction. Determining that image, and then performing all source-prescribed hull operations, remains necessary for a global comparison. No IUT theorem or the *abc* conjecture has been disproved by these local nonreachability statements.

8. EFFECTIVE BOUNDS AND FINITENESS IN A SPECIFIED PELL FAMILY

We now restrict to positive integers b, A, B, u, v, r_0, s, X and a prime $p \geq 37$ satisfying

$$(32) \quad b = Au^2, \quad b+1 = Bv^2, \quad b+2 = 3r_0^2, \quad b+3 = s^2,$$

$$(33) \quad D = 3AB, \quad Z = b^2 + 3b + 1 = T_p(X), \quad D+1 \leq X^2, \quad X^p \leq Z.$$

Here T_p is the Chebyshev polynomial of the first kind. These equations define the specified residual packet; no reduction of general *abc* counterexamples to it is asserted. Set

$$H = \log(b+2), \quad L = \log(D+1), \quad C_0 = 12^{331776}, \quad K_0 = \log(2C_0).$$

The packet immediately yields

$$(34) \quad (D+1)^p \leq X^{2p} \leq Z^2 < (b+2)^4, \quad pL < 4H.$$

Theorem 2.2 of Bérczes–Evertse–Györy [6], specialized to $K = \mathbb{Q}$, $S = \{\infty\}$, and exponent two, gives the following published input. If $f \in \mathbb{Z}[t]$ is squarefree of degree $n \geq 3$, with coefficient height $\hat{h} = \log \max(1, |a_0|, \dots, |a_n|)$, and $f(x) = y^2$ in integers, then

$$(35) \quad \max\{h(x), h(y)\} \leq (4n)^{4096n^4} \exp(50n^4 \hat{h}), \quad h(t) = \log \max(1, |t|).$$

Proposition 8.1 (Two effective constraints). *Every packet (32)–(33) satisfies*

$$(36) \quad H \leq \exp(4300p^5), \quad H \leq 2C_0 D^{4050}.$$

Proof. First put $f_p(t) = 4T_p(t) + 5$. The relation in (33) gives $f_p(X) = (2b+3)^2$. The identities $T'_p = pU_{p-1}$ and $T_p^2 - 1 = (t^2 - 1)U_{p-1}^2$ show that a repeated root of f_p would have $T_p = \pm 1$ and $T_p = -5/4$, a contradiction. If M_n is the sum of the absolute values of the coefficients of T_n , then $M_{n+2} \leq 2M_{n+1} + M_n$ and $M_0 = M_1 = 1$. Induction gives $M_n \leq 3^n$. Thus every coefficient of f_p has absolute value at most $4 \cdot 3^p + 5 \leq 3^{p+2}$, and $\hat{h} \leq (p+2) \log 3 \leq 4p$. Since $\log(4p) \leq p$ for $p \geq 3$, (35) implies

$$H \leq \log(2b+3) \leq \exp\left(4096p^4 \log(4p) + 50p^4 \hat{h}\right) \leq \exp(4300p^5).$$

For the second estimate, set $x = b+1$ and $Y = Duvr_0$. The first three equations in (32) give

$$(37) \quad Y^2 = D(x^3 - x).$$

The cubic $D(t^3 - t)$ is squarefree and has coefficient height $\log D$. Equation (35), with $n = 3$, gives

$$\log(b+1) \leq 12^{4096 \cdot 3^4} \exp(50 \cdot 3^4 \log D) = C_0 D^{4050}.$$

Finally $b+2 \leq (b+1)^2$ for $b \geq 1$, proving the second bound. $\square$

Proposition 8.2 (Descent to the squarefree parameter). *Write $D = d\tau^2$, where d is a positive squarefree integer and $\tau \geq 1$ is an integer. Then every packet satisfies*

$$H \leq 2C_0 d^{4050}.$$

Proof. Equation (37) implies $\tau^2 \mid Y^2$. Comparing prime valuations gives $\tau \mid Y$. Write $Y = \tau y$ and cancel the nonzero factor τ^2 to obtain

$$y^2 = d(x^3 - x), \quad x = b + 1.$$

The polynomial $d(t^3 - t)$ has distinct roots and coefficient height $\log d$. Equation (35) gives $\log(b + 1) \leq C_0 d^{4050}$, and again $H \leq 2 \log(b + 1)$. $\square$

Combining (34) and (36) gives

$$(38) \quad p \geq \left(\frac{\log H}{4300} \right)^{1/5}, \quad L < 4 \cdot 4300^{1/5} \frac{H}{(\log H)^{1/5}},$$

$$(39) \quad D \geq \left(\frac{H}{2C_0} \right)^{1/4050}, \quad p(\log H - K_0) < 16200H.$$

For example, the first upper bound follows by multiplying the lower bound for p by $L > 0$ and applying $pL < 4H$. For the final inequality use $\log H \leq K_0 + 4050 \log D$ and $\log D < L$. In a form avoiding fractional powers, the former calculation is

$$(\log H)L^5 \leq 4300(4H)^5.$$

These inequalities alone are compatible with unbounded height: they would force both the index and the squarefree kernel to grow, while $\log(D + 1)/H$ tends to zero. The following argument supplies the missing absolute index bound and excludes such an unbounded sequence in this specified family. It does not freeze a field or an index in advance.

8.1. An absolute index bound from normalized logarithmic forms. The preceding effective height estimate can be combined with a different logarithmic-form argument to prove finiteness, uniformly in the index. For this argument retain only positive integers b, r_0, s, X and an integer $p \geq 3$ satisfying

$$(40) \quad b + 2 = 3r_0^2, \quad b + 3 = s^2, \quad Z = b^2 + 3b + 1 = T_p(X), \quad X > 1.$$

Primality of p , the endpoint kernels A, B , and their congruences are not needed. Put

$$\varepsilon = 2 + \sqrt{3}, \quad \eta = s + r_0\sqrt{3}, \quad \delta = X + \sqrt{X^2 - 1}, \quad W = Z + \sqrt{Z^2 - 1}, \quad H = \log(b + 2).$$

The Pell equation gives $\eta = \varepsilon^m$ for an integer $m \geq 1$. The Chebyshev identity gives $W = \delta^p$; δ need not be a fundamental unit. In fact $b \geq 22$: below 22 the fixed Pell equations leave only $b = 1, r_0 = 1, s = 2$, whereas $Z = 5 < T_p(X)$ for $p \geq 3, X \geq 2$. Here $T_p(X) \geq X^p$ follows by induction from its recurrence and $T_{n+1}(X) \geq T_n(X) > 0$.

We use Matveev's original Corollary 2.3 [66], with the normalized coefficient parameter in his equation (1.3). For a real number field of degree d , algebraic numbers $\alpha_i > 1$, and a nonzero form $\Lambda = \sum_{i=1}^n b_i \log \alpha_i$ with integral coefficients, take

$$A_i \geq \max\{d\mathbf{h}(\alpha_i), |\log \alpha_i|, 0.16\}, \quad B_* = \max \left\{ 1, \max_i \frac{|b_i| A_i}{A_n} \right\}, \quad \Omega = \prod_i A_i.$$

Then

$$(41) \quad \log |\Lambda| > -C_1(n, 1)d^2\Omega \log(ed) \log(eB_*), \quad C_1(n, 1) \leq 2^{6n+20}.$$

Unlike Theorem 2.1 of that source, this corollary does not require the logarithms to be independent. It also does not require A_n to be the largest parameter. Both points matter when δ varies.

Lemma 8.3. *For (40),*

$$1 < \frac{\eta^4}{8W} < 1 + \frac{2}{b}, \quad W > \eta^2, \quad \log W < 3H.$$

Consequently the real linear form

$$(42) \quad \Lambda = 4m \log \varepsilon - 3 \log 2 - p \log \delta = \log \frac{\eta^4}{8W}$$

satisfies $0 < \Lambda < 2/b$ and $-\log \Lambda > H/2$.

Proof. For $t > 1$, write $f(t) = t + \sqrt{t^2 - 1}$, so $2t - 1 < f(t) < 2t$. The Pell identities give

$$\eta^2 = f(2b + 5), \quad \eta^4 = f(8b^2 + 40b + 49), \quad W = f(Z).$$

Thus

$$\eta^4 > 16b^2 + 80b + 97 > 16Z > 8W,$$

and

$$\frac{\eta^4}{8W} - 1 < \frac{32b + 90}{16b^2 + 48b + 8} < \frac{2}{b}.$$

The final comparison is equivalent to $32b^2 + 90b < 32b^2 + 96b + 16$. Also $W > 2b^2 + 6b + 1 > 4b + 10 > \eta^2$ for $b \geq 22$. Finally $W < 2Z < 2(b + 2)^2$ gives $\log W < \log 2 + 2H < 3H$. The strict inequality $\log(1 + u) < u$ for $u > 0$ proves the asserted bounds for Λ . Since $(b/2)^2 > b + 2$, $-\log \Lambda > \log(b/2) > H/2$. $\square$

Theorem 8.4 (An absolute index bound). *Every tuple (40) satisfies $p < 2^{59}$.*

Proof. Use the real field $K = \mathbb{Q}(\sqrt{3}, \sqrt{X^2 - 1})$ of degree $d \leq 4$, and in this order choose

$$(\alpha_1, \alpha_2, \alpha_3) = (\varepsilon, 2, \delta), \quad (b_1, b_2, b_3) = (4m, -3, -p),$$

$$A_1 = 2 \log \varepsilon, \quad A_2 = 4 \log 2, \quad A_3 = 2 \log \delta.$$

For integer $X \geq 2$, $X^2 - 1$ lies strictly between two consecutive squares. Hence δ is a quadratic algebraic integer with minimal polynomial $t^2 - 2Xt + 1$ and conjugate δ^{-1} . The absolute heights are

$$h(\varepsilon) = \frac{1}{2} \log \varepsilon, \quad h(2) = \log 2, \quad h(\delta) = \frac{1}{2} \log \delta.$$

These identities, $d \leq 4$, and $\delta \geq \varepsilon$ verify all the height and 0.16 conditions in (41). The fields may coincide; no independence assertion is used.

By Lemma 8.3, the normalized coefficients obey

$$\frac{4mA_1}{A_3} = \frac{4p \log \eta}{\log W} < 2p, \quad \frac{3A_2}{A_3} = \frac{6 \log 2}{\log \delta} < 4 \leq 2p.$$

For the latter strict inequality use $\delta^2 \geq \varepsilon^2 > 8$. The remaining entries are p and one, so $B_* \leq 2p$. Using the same chosen parameters, rather than changing them when estimating B_* , gives

$$\Omega = 16 \log \varepsilon \log 2 \frac{\log W}{p} < 96 \frac{H}{p}.$$

Apply (41), with $C_1(3, 1) \leq 2^{38}$, $d^2 \leq 16$, and $\log(ed) < 3$. Comparison with the upper approximation gives

$$\frac{H}{2} < -\log \Lambda < 2^{38} \cdot 16 \cdot 3 \cdot 96 \frac{H}{p} \log(2ep).$$

Cancel the positive H and multiply by p to obtain

$$(43) \quad p < 9 \cdot 2^{48} \log(2ep) < 2^{52} \log(2ep).$$

With $t = p/2^{52} > 0$, this reads $t < 53 \log 2 + 1 + \log t$. The elementary inequality $\log t \leq t/2$ gives $t < 2(53 \log 2 + 1) < 108 < 128$. Thus $p < 2^{59}$, independently of the height and all varying fields. $\square$

Corollary 8.5 (Effective finiteness of the specified family). *The tuples (40) form a finite set. Every such tuple satisfies*

$$(44) \quad H < \exp(4300 \cdot 2^{295}), \quad b + 2 < \exp(\exp(4300 \cdot 2^{295})).$$

In particular the more restrictive residual (32)–(33) is effectively finite.

Proof. The first part of the proof of Proposition 8.1 applies to $4T_p(X) + 5 = (2b + 3)^2$ for every integer $p \geq 3$, giving $H \leq \exp(4300p^5)$. Combine it with Theorem 8.4 and exponentiate. The resulting bound leaves finitely many positive integers b . The square equations determine r_0, s , the index is bounded, and $X^p \leq Z$ bounds X . If the endpoint data are also retained, then $A \mid b$, $B \mid b + 1$, and their positive square multipliers have only finitely many possibilities. $\square$

These bounds are effective but far beyond feasible enumeration. They prove finiteness without listing the exceptional tuples. They improve a fixed-index finiteness statement: the same absolute bound applies while both the index and the quadratic field vary.

Corollary 8.6 (The product Pell unit outside the finite set). *Suppose only that b, r_0, s are positive integers with $b + 2 = 3r_0^2$ and $b + 3 = s^2$. Put $Z = b^2 + 3b + 1$ and write $Z^2 - 1 = D_0 y^2$ with $D_0 > 1$ squarefree and y a positive integer. Outside the bound in (44), $Z + y\sqrt{D_0}$ is the fundamental positive norm-one unit of the integral Pell group $\mathbb{Z}[\sqrt{D_0}]$.*

Proof. Write $Z + y\sqrt{D_0} = \varepsilon_{D_0}^k$, $k \geq 1$. If $k = 2j$, then $Z = T_2(T_j(X_0))$ for the integer first coordinate $X_0 > 1$ of ε_{D_0} . Thus, with $Y = T_j(X_0)$,

$$2Y^2 = Z + 1 = (b + 1)(b + 2) = 3(b + 1)r_0^2.$$

Since $3 \nmid b + 1$, the right side has odd 3-adic valuation and the left side has even 3-adic valuation, a contradiction. Therefore k is odd. If $k > 1$, it is at least three and (40) holds with $p = k$, $X = X_0$. Corollary 8.5 bounds b , proving the assertion. $\square$

This last statement concerns the integral Pell group, without identifying it with the unit group of a larger maximal order. A fundamental moving Pell unit can still have a large regulator relative to its discriminant. Neither finiteness of the nonfundamental cases nor the absolute index bound gives the radical estimate (1), and no reduction of arbitrary abc triples to (40) has been established.

8.2. A stronger bound from a fixed-norm element. For the height and kernel estimates in this subsection use only (32), and impose the additional conditions that A, B are squarefree and

$$A \equiv 22 \pmod{24}, \quad B \equiv 23 \pmod{24}.$$

They specify the squarefree branch under consideration; the more general packet estimates above did not require them. No Chebyshev decomposition is needed except in the displayed consequence for p . In particular $b \geq 22$, $A, B > 1$, $3 \nmid AB$, $\gcd(A, B) = 1$, and $D = 3AB$ is squarefree. Let R_A, R_B denote the regulators of the maximal real quadratic orders.

We use two general tools from the proof of the new result [69, Theorem 2.4 and Lemma 2.5], without applying its irreducible-cubic theorem to a split polynomial. First, in a totally real degree-four field, for a multiplicative group generated modulo torsion by three independent elements g_1, g_2, g_3 and $g \neq 1$ in that group,

$$(45) \quad -\log |1 - g| \leq C_* \log \max\{2, h(g)\} \prod_{i=1}^3 \max\{1, h(g_i)\},$$

where $C_* > 0$ is effective and absolute. Second, for an algebraic integer ω of absolute norm $q > 1$ in a real quadratic field K , the group $\mathcal{O}_K^\times \langle \omega \rangle$ has independent elements ξ_1, ξ_2 generating a subgroup modulo torsion of index $1 \leq N \leq 2$ such that

$$(46) \quad \omega^N = \zeta \xi_1^{e_1} \xi_2^{e_2}, \quad \zeta \in \{\pm 1\}, \quad h(\xi_1)h(\xi_2) \leq \frac{3}{8} R_K \log q.$$

Here the set S consists of both real places and no finite place, so the all-or-none condition in Lemma 2.5 is satisfied and $q' = q$. The small-generator estimate follows from the exterior-product

theorem of Akhtari–Vaaler [5, Theorem 1.2]; its joint use of the unit and nonunit generators is what saves a regulator factor. Finally Landau’s fixed-degree estimate gives an effective absolute C_Q with

$$(47) \quad R_A \leq C_Q \sqrt{A} \log(2A), \quad R_B \leq C_Q \sqrt{B} \log(2B).$$

Theorem 8.7. *There is an effective absolute constant C_A such that every packet in this subsection satisfies*

$$H \leq C_A \sqrt{A} [\log(2A)]^2.$$

Proof. Put $\varphi = (1 + \sqrt{5})/2$ and $\rho = \frac{1}{2} \log \varphi$. Every nonzero algebraic number of degree at most two which is not a root of unity has height at least ρ . Indeed a rational such number has height at least $\log 2$. For a quadratic number, its primitive minimal polynomial has Mahler measure at least two if the leading or constant coefficient has absolute value at least two. Otherwise it is a quadratic unit. For norm minus one a nonzero integral trace gives a root of absolute value at least φ ; trace zero is reducible. For norm plus one, traces of absolute value at most two give roots of unity or reducibility, and the other traces give a root of absolute value at least φ^2 . The formula $2h = \log M$ proves the height lower bound.

In $K = \mathbb{Q}(\sqrt{A})$ take

$$\omega = s + u\sqrt{A} = \sqrt{b+3} + \sqrt{b}, \quad \eta = s + r_0\sqrt{3} = \sqrt{b+3} + \sqrt{b+2}.$$

Their respective quadratic norms are three and one. The latter is a positive power of $\varepsilon_3 = 2 + \sqrt{3}$. Apply (46) with $q = 3$. The field $F = \mathbb{Q}(\sqrt{A}, \sqrt{3})$ has degree four, since $A > 1$ is squarefree and $A \neq 3$. Define $\gamma = \omega/\eta$. At the chosen real embedding,

$$0 < \gamma < 1, \quad 1 - \gamma = \frac{2}{(\sqrt{b+2} + \sqrt{b})(\sqrt{b+3} + \sqrt{b+2})} < \frac{1}{2b}.$$

As $1 \leq N \leq 2$, it follows that $0 < 1 - \gamma^N < 1/b$.

The group $\langle -1, \xi_1, \xi_2, \varepsilon_3 \rangle$ has rank three modulo torsion and contains γ^N . Independence follows because $K \cap \mathbb{Q}(\sqrt{3}) = \mathbb{Q}$, and a rational power of ε_3 must equal its conjugate, forcing that power to be zero. The other two generators are independent by construction. Moreover $h(\varepsilon_3) < 1$ and $\max(1, h(\xi_i)) \leq h(\xi_i)/\rho$, giving

$$\prod_i \max(1, h(g_i)) \leq \frac{3 \log 3}{8\rho^2} R_A.$$

Both conjugates of ω, η are positive. Their smaller conjugates are $3/\omega < 1$ and $1/\eta < 1$, so their global heights are $\frac{1}{2} \log \omega$ and $\frac{1}{2} \log \eta$. Thus

$$h(\gamma^N) \leq 2(h(\omega) + h(\eta)) < \log(4(b+3)) < H + 2 \leq 3H.$$

Applying (45) gives $\log b \ll R_A \log(3H)$, with an effective absolute constant. Since $H \leq \log b + \log 3$, we have $H \leq M \log(3H)$ for $M = C_1 \max(1, R_A) \geq 1$ and an absolute C_1 .

For any positive X, M with $X \leq M \log(3X)$, one has

$$(48) \quad X \leq 2M \log(3M).$$

To prove it, write $t = X/M$ and use $\log t \leq t/2$, which follows from $\log(t/2) \leq t/2 - 1$ and $\log 2 < 1$. Then $t \leq \log(3M) + t/2$, proving (48). Equation (47) bounds M by $C_2 \sqrt{A} \log(2A)$ and $\log(3M)$ by $C_3 \log(2A)$. Applying (48) proves the theorem. $\square$

Replacing ω by $s + v\sqrt{B}$, of norm two, gives the analogous bound with $\sqrt{B} \log^2(2B)$. For this endpoint, however, a stronger published input is available and was already identified in the repository. The unit

$$\varepsilon_B = (b+2) + vs\sqrt{B}$$

is the fundamental positive norm-one unit: its first coordinate is $3r_0^2$, so Bennett–Walsh [13, Theorem 1.2] says that its unit index is the first index at which three divides the first Pell coordinate. If the fundamental first coordinate is 1 or -1 modulo three, the Chebyshev recurrence gives only 1 or alternating signs modulo three, respectively. A first coordinate divisible by three

therefore forces the fundamental first coordinate itself to be divisible by three, and the index is one. Since $B \equiv 3 \pmod{4}$, the order $\mathbb{Z}[\sqrt{B}]$ is maximal and a norm-minus-one unit is impossible modulo four. Consequently

$$(49) \quad H < R_B \leq C_Q \sqrt{B} \log(2B).$$

We use this established endpoint result, rather than presenting it as a new consequence of the recent preprint.

Corollary 8.8. *There is an effective absolute $K \geq 1$ such that these packets satisfy*

$$(50) \quad H^4 \leq KD[\log(2D)]^6.$$

In particular,

$$(51) \quad D \geq \frac{H^4}{K5^6(\log H)^6}, \quad \log D \geq 4 \log H - 6 \log \log H - \log(K5^6).$$

When the latter lower bound is positive,

$$p < \frac{4H}{4 \log H - 6 \log \log H - \log(K5^6)}.$$

Proof. Square Theorem 8.7 and (49) and multiply. Since $AB = D/3$ and $A, B \leq D$, the result is (50). If $D \geq H^4$, the first inequality in (51) is immediate because $H > 2$ and $K \geq 1$. Otherwise $\log(2D) \leq 5 \log H$, giving that inequality from (50). Taking logarithms proves the second one. Finally use $p \log(D+1) < 4H$. $\square$

The absolute index bound already excludes an unbounded sequence with $p \geq 3$. The kernel estimate (50) remains useful without a Chebyshev decomposition, including cases where the product Pell unit is fundamental. Its polynomial dependence on H does not supply the radical bound required for abc.

Proposition 8.9 (A general split-cubic estimate). *For positive squarefree d , an integral solution of $y^2 = d(x^3 - x)$ with $x \geq 2$ satisfies*

$$\log x \ll d^{2/3} [\log(2d)]^3$$

with an effective absolute implied constant.

Proof. Write $x-1 = a_0 u^2$, $x = b_0 v^2$, $x+1 = c_0 w^2$ with positive squarefree coefficients. Adjacent coefficients are coprime and $g = \gcd(a_0, c_0) \in \{1, 2\}$, so $a_0 b_0 c_0 = g^2 d \leq 4d$. No two coefficients are equal: equality for adjacent ones would force positive squares to differ by one; equality at the ends forces the common coefficient to divide two and gives the same impossibility or a difference of squares equal to two. The quadratic fields of the three products $a_0 b_0, b_0 c_0, a_0 c_0$ are therefore distinct.

The three positive norm-one units

$$U = 2x - 1 + 2uv\sqrt{a_0 b_0}, \quad V = 2x + 1 + 2vw\sqrt{b_0 c_0}, \quad W = x + uw\sqrt{a_0 c_0}$$

satisfy $1 < V/U, V/(2W), 2W/U < 1 + 2/x$. Indeed $2t - 1 < t + \sqrt{t^2 - 1} < 2t$ for $t > 1$ gives these inequalities directly at $t = 2x - 1, 2x + 1, x$. Each ratio is generated by two quadratic fundamental units and two; two may also be adjoined for the first ratio. Distinct quadratic fields give independence, and two is an independent nonunit. The global height of each ratio is at most $\log x + O(1)$. Choose the pair whose radicands share $m = \min(a_0, b_0, c_0)$. Landau's bound gives a product of regulators at most

$$C \sqrt{a_0 b_0 c_0 m} [\log(2a_0 b_0 c_0)]^2 \leq C (a_0 b_0 c_0)^{2/3} [\log(2a_0 b_0 c_0)]^2.$$

Passing to squarefree radicands can only decrease the discriminant bound. Apply (45) to the selected ratio and then (48), adjusting absolute constants for bounded x . The latter step adds one logarithmic power. Using $a_0 b_0 c_0 \leq 4d$ proves the claim. $\square$

9. APPLICABILITY OF RECENT ELLIPTIC-CURVE ESTIMATES

9.1. A denominator lower bound. For the Frey curve $E : y^2 = x(x - a)(x + b)$ attached to an abc triple, let $Q = a^2 + ab + b^2$. Its invariant is

$$j_E = \frac{256Q^3}{(abc)^2}.$$

Let n, d be the positive numerator and denominator in lowest terms.

Proposition 9.1. *For every such triple,*

$$(52) \quad c^4 \leq 1024d, \quad n \leq 256c^6.$$

Proof. Primitivity implies $\gcd(Q, abc) = 1$. Indeed, reduction modulo a prime dividing any one of a, b, c leaves in Q the square of a nonzero other coordinate. Therefore $g = \gcd(256Q^3, (abc)^2)$ divides 256, and $(abc)^2 = gd \leq 256d$. Since $(a - 1)(b - 1) \geq 0$, we have $ab \geq c - 1$; as $c \geq 2$, this gives $2ab \geq c$. Squaring $2abc \geq c^2$ proves $c^4 \leq 4(abc)^2 \leq 1024d$. Finally $Q = c^2 - ab \leq c^2$, so $n \leq 256Q^3 \leq 256c^6$. $\square$

Pasten's 2026 small-denominator theorem [70, Theorem 1.2] applies under a bound $\text{den}(j_E) \leq A_0(\log \text{num}(j_E))^{B_0}$ with fixed positive A_0, B_0 . For Frey curves, (52) would imply

$$c^4 \leq 1024A_0(\log 256 + 6 \log c)^{B_0}.$$

A positive power of c dominates every fixed power of its logarithm. Thus only bounded c , and consequently finitely many positive triples, can satisfy this hypothesis for fixed A_0, B_0 . This is a precise restriction on direct applicability, not a counterexample to the theorem or to other Szpiro strategies.

9.2. The August 2026 Mordell estimate. The preprint [69, Theorem 1.3], submitted August 24, 2026, gives an effective bound for $y^2 = x^3 + k$, $k \neq 0$, using

$$\underline{k} = \prod_{q|k} q^{\min(2, v_q(k))}.$$

The stated estimate is

$$(53) \quad \log \max(|x|, |y|) \ll \underline{k}^{1/2} \log^2(2\underline{k}) \log(2|k|) \log(\underline{k} \log(3|k|)).$$

We record this new result without treating its publication date as a substitute for checking its hypotheses or proof.

Directly substituting $c_6^2 = c_4^3 - 1728\Delta$ and $\Delta = 16(abc)^2$ gives $k = -27648(abc)^2$. Then $\underline{k}^{1/2} \ll r$, while both $\log(2|k|)$ and $\log \max(|c_4|, |c_6|)$ are comparable with h . The resulting right-hand side has the shape

$$r \log^2(2r) h \log(2r \log(3c)).$$

The height remains a multiplicative factor on the right; cancellation does not yield an upper bound for c in terms of r . The integral- j result in Theorem 1.6 of the same paper concerns the height of the *curve*, and cannot be substituted for a point-height bound.

There is a more primitive Mordell point, but it preserves the same height factor in this substitution.

Proposition 9.2. *Put $m = abc/2$ and $Y = (a - b)(2a + b)(a + 2b)/2$. These are integers, and*

$$Y^2 = Q^3 - 27m^2, \quad \gcd(Q, m) = 1, \quad c_4 = 4^2Q, \quad c_6 = 4^3Y.$$

There is no integer $z \geq 2$ with $z^2 \mid Q$ and $z^3 \mid Y$. The associated binary cubic is split over $\mathbb{Z}$:

$$U^3 - 3QUV^2 + 2YV^3 = (U - (a - b)V)(U - (a + 2b)V)(U + (2a + b)V).$$

It remains reducible after every change of variables in $\text{GL}_2(\mathbb{Z})$.

Proof. At least one of a, b, c is even, so m is integral. If one of a, b is even then one of $2a+b, a+2b$ is even; if both are odd then $a-b$ is even. This proves integrality of Y . Expansion gives

$$4Q^3 - ((a-b)(2a+b)(a+2b))^2 = 27(abc)^2$$

and the asserted cubic factorization. The standard invariant formulas give the two scalings. Coprimality follows from $\gcd(Q, abc) = 1$. If the proposed z existed, then $z^6 \mid 27m^2$ while $\gcd(z, m) = 1$; thus $z^6 \mid 27$, impossible for $z \geq 2$. An invertible linear change of variables takes each nonzero linear factor to another nonzero linear factor, preserving reducibility. $\square$

The normalized parameter is therefore $k = -27m^2$ and its prime support is still that of $3m$. Removing the fixed powers of four does not remove the factor comparable with h on the right of (53). The split cubic also does not meet an irreducible Thue-form hypothesis.

There is also a strict obstruction to one possible transfer from (37). With $U = Dx$ and $V = DY$, it becomes $E_D : V^2 = U^3 - D^2U$, of j -invariant 1728.

Proposition 9.3. *For $D, k \neq 0$, there is no nonconstant algebraic morphism over $\overline{\mathbb{Q}}$ from E_D to the nonsingular Mordell curve $M_k : v^2 = u^3 + k$.*

Proof. Their geometric endomorphism algebras are respectively $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$: the automorphisms $(x, y) \mapsto (-x, iy)$ and $(u, v) \mapsto (\zeta_3 u, v)$ give these quadratic actions, and in characteristic zero the endomorphism algebra of an elliptic curve has dimension at most two over $\mathbb{Q}$. A nonconstant morphism, translated to send the origin to the origin, is an isogeny. An isogeny induces an isomorphism of rational endomorphism algebras by conjugation. The two imaginary quadratic fields are not isomorphic, a contradiction; see also [71, Chapters III and VI]. $\square$

The proposition excludes this morphism strategy, even if k depends on D . It does not exclude other Diophantine encodings or extending the new logarithmic-form methods to a different family.

10. THREE AUXILIARY POINTS AND THE REMAINING ENDPOINT COST

The constructions here apply to every positive primitive abc triple. Write its canonical cube decompositions as

$$(54) \quad a = Au^3, \quad b = Bv^3, \quad c = Cw^3,$$

where A, B, C are positive cube-free integers. These coefficients, and the three original endpoints, are pairwise coprime within their respective triples. Put

$$R_0 = \text{rad}(ABC), \quad R_3 = \prod_{\substack{p|abc \\ 3 \nmid v_p(abc)}} p, \quad H = \log c.$$

Thus $\text{rad}(abc) = R_0 R_3$ and $\gcd(R_0, R_3) = 1$. The subscript on R_0 denotes the support left after removing cubes, not the primes whose exponents are divisible by three.

10.1. Actual integral points and their prime costs. For each omitted endpoint n_i , define the following integers:

i	n_i	d_i	r_i	x_i	y_i
a	a	vw	BC	$4vwBC$	$4BC(a+2b)$
b	b	uw	AC	$4uwAC$	$4AC(2a+b)$
c	c	uv	AB	$-4uvAB$	$4AB(a-b)$

Set $k_i = 16(n_i r_i)^2$.

Proposition 10.1. *Every row gives an integral point $y_i^2 = x_i^3 + k_i$ with $|x_i|^3 \geq 32c$. If $\rho_i = \text{rad}(2n_i r_i)$, then*

$$(55) \quad \rho_a \rho_b \rho_c = 4^{\mathbf{1}_{3 \mid v_2(abc)}} R_0^3 R_3, \quad \min_i \rho_i \leq 2^{2/3} R_0 R_3^{1/3}.$$

Proof. For the first row use $(a + 2b)^2 - a^2 = 4bc$ and $bc = (vw)^3 BC$; the second is the same identity with a, b interchanged. For the last, $(a - b)^2 - c^2 = -4ab$ gives the negative sign of x_c . The three values of $|x_i|^3$ are respectively $64bc(BC)^2$, $64ac(AC)^2$, and $64ab(AB)^2$. The first two exceed $32c$, and the last is at least $32c$ because $ab \geq c - 1 \geq c/2$.

An odd prime in R_0 occurs in all three $n_i r_i$: once in the omitted endpoint and twice in the coefficient products. A prime in R_3 occurs only in its own omitted endpoint. The prime 2 divides every ρ_i , and exactly one endpoint is even. If its exponent is divisible by three this adds a factor 4 to the preceding product; otherwise no correction is needed. This proves the exact identity; the minimum is at most the geometric mean. $\square$

These are auxiliary curves of j -invariant zero. Their standard 3-isogeny is

$$(56) \quad (x, y) \mapsto \left(\frac{x^3 + 4k}{x^2}, \frac{y(x^3 - 8k)}{x^3} \right), \quad E_k \longrightarrow E_{-27k}.$$

On the unextracted omit- b point with $k = 16(abc)^2$, it gives $(4Q, 4T)$, where

$$Q = a^2 + ab + b^2, \quad T = (a - b)(2a + b)(a + 2b).$$

These are $(c_4/4, c_6/8)$ for the original Frey curve. The assertion is an identity involving its invariants; the Frey curve itself has not become isogenous to a $j = 0$ curve. After cube extraction, the same image is generally rational rather than integral.

10.2. A small field discriminant with an exact growing order index. Form the monic cubic

$$(57) \quad G_i(Z) = Z^3 - 3x_i Z + 2y_i, \quad \text{disc } G_i = -108k_i = -1728(n_i r_i)^2.$$

For the omit- b row, put $\alpha = \sqrt[3]{A^2 C}$ and $\beta = \sqrt[3]{AC^2} = \alpha^2/A$. Then $\tau_b = -2(u\alpha + w\beta)$ is a root. The omit- a row uses (B, C, v, w) instead, while the omit- c row has root $\tau_c = -2(u\sqrt[3]{A^2 B} - v\sqrt[3]{AB^2})$.

If the selected product r_i exceeds one, its pure cubic is irreducible. At a prime of either cube-free coefficient, the valuation of the radicand is not divisible by three. Independence of $1, \alpha, \alpha^2$ then makes τ_i nonrational, so it generates the same degree-three field K_i . In particular the root fields are

$$K_a = \mathbb{Q}(\sqrt[3]{B^2 C}), \quad K_b = \mathbb{Q}(\sqrt[3]{A^2 C}), \quad K_c = \mathbb{Q}(\sqrt[3]{A^2 B}),$$

with signature $(1, 1)$. Reducible rows are exactly those with $r_i = 1$.

Proposition 10.2. *In an irreducible row, put $S_i = \text{rad}(r_i)$. There is an order $\mathcal{O}_i \subset K_i$ with*

$$(58) \quad [\mathcal{O}_i : \mathbb{Z}[\tau_i]] = 8n_i, \quad |\Delta_{K_i}| \mid 27S_i^2.$$

More precisely, there is a larger order $\mathcal{O}'_i$ such that

$$(59) \quad [\mathcal{O}_{K_i} : \mathbb{Z}[\tau_i]] = 8n_i(r_i/S_i)[\mathcal{O}_{K_i} : \mathcal{O}'_i].$$

Proof. Write the two selected coefficients as E, F , their cube bases as h, j , and $r = EF$. The lattice $\mathcal{O} = \mathbb{Z}\{1, \alpha, \beta\}$ is an integral order, because

$$\alpha^2 = E\beta, \quad \beta^2 = F\alpha, \quad \alpha\beta = r.$$

Its trace matrix is $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 3r \\ 0 & 3r & 0 \end{pmatrix}$, and hence its discriminant is $-27r^2$. The coordinate matrices of $1, \tau, \tau^2$ in this basis are

$$M_+ = \begin{pmatrix} 1 & 0 & 8hjr \\ 0 & -2h & 4Fj^2 \\ 0 & -2j & 4Eh^2 \end{pmatrix}, \quad M_- = \begin{pmatrix} 1 & 0 & -8hjr \\ 0 & -2h & 4Fj^2 \\ 0 & 2j & 4Eh^2 \end{pmatrix}.$$

Their determinants are $8(Fj^3 - Eh^3)$ and $-8(Eh^3 + Fj^3)$, respectively. The absolute value is $8n_i$ in the corresponding row. Since τ is an integral generator of a cubic field, this gives the first equality of (58).

Write $E = E_1 E_2^2$, $F = F_1 F_2^2$ with the four factors squarefree and pairwise coprime. Put $s = E_2 F_1$, $t = E_1 F_2$, $\theta = \sqrt[3]{st^2}$, and $\eta = \theta^2/t$. The lattice $\mathcal{O}' = \mathbb{Z}\{1, \theta, \eta\}$ is an order by $\theta^2 = t\eta$, $\eta^2 = s\theta$, $\theta\eta = st$. Its discriminant is $-27(st)^2 = -27S_i^2$. Furthermore $\alpha = E_2\theta$, $\beta = F_2\eta$, so $[\mathcal{O}' : \mathcal{O}] = E_2 F_2 = r_i/S_i$. The discriminant-index formula and multiplication of indices prove the remaining claims. $\square$

Thus the small discriminant concerns the field, while the original monogenic order has an index containing the unbounded endpoint n_i . Replacing its polynomial discriminant by the field discriminant in a point-height argument would discard this factor.

10.3. A full-family relative height estimate.

Theorem 10.3. *There is an effective absolute constant $C_0 > 0$ such that every positive primitive abc triple, and each of its three rows, satisfies*

$$(60) \quad H \leq C_0 S_i [\log(2S_i)]^2 L_i \log(3S_i L_i), \quad L_i = \log(2n_i r_i), \quad S_i = \text{rad}(r_i).$$

Proof. We use the stated unconditional Theorem 1.7 and the reduction in Section 4 of Pasten's 2026 preprint [69], together with its classical degree-three regulator bound. We keep the actual pure cubic root field, rather than bounding its regulator using disc G_i .

First suppose $r_i > 1$ and put $D = 1728(n_i r_i)^2$. The reduction in that section supplies $\gamma \in \text{GL}_2(\mathbb{Z})$ such that the binary cubic $F = G_i \circ \gamma$ has coefficient height at most $\sqrt{D}$. For $(u_0, v_0) = \gamma^{-1}(1, 0)$ one has $F(u_0, v_0) = 1$ and $\gcd(u_0, v_0) = 1$. The fractional linear change of root leaves the cubic field K_i unchanged. Theorem 1.7, applied with empty prime list, unit right side, and just the two archimedean places of K_i , gives, for $X = \max(3, |u_0|, |v_0|)$,

$$(61) \quad \log X \ll \log(2D) + \mathcal{R}_i \log(2D) \log(2 + 2\mathcal{R}_i \log(2D)),$$

where $\mathcal{R}_i$ is the ordinary regulator. The degrees and the place count are fixed, so the implied constant is absolute and effective. Hessian and Jacobian recovery in the same section gives

$$\log \max(2, |x_i|, |y_i|) \ll \log |k_i| + \log X.$$

These covariants recover the point, not just the auxiliary curve. Proposition 10.1, together with $\log(2D) \leq 12L_i$ and $\log |k_i| \leq 4L_i$, therefore yields

$$H \ll \mathcal{R}_i^* L_i \log(3\mathcal{R}_i^* L_i), \quad \mathcal{R}_i^* = \max(1, \mathcal{R}_i).$$

Here all logarithm arguments are bounded away from one because $L_i \geq \log 2$.

The degree-three Landau bound, with class number at least one, and Proposition 10.2 imply

$$\mathcal{R}_i^* \ll S_i [\log(2S_i)]^2.$$

Substitution loses no further logarithmic power. Indeed, for $B = \log(2S_i)$ and $z = S_i L_i \geq \log 2$, one has $B \leq 2S_i$ and $L_i \geq 1/2$, hence $S_i B^2 L_i \leq 16z^3$. Thus $\log(3CS_i B^2 L_i) \ll_C \log(3S_i L_i)$ for any fixed positive C . This proves (60) in the irreducible case.

If $r_i = 1$, the selected endpoints are cubes. In a plus row their difference is $n_i = w^3 - h^3$ with $w > h \geq 1$. Thus $n_i \geq w^2$ and $H = 3 \log w \leq 3L_i/2$. In the minus row $n_i = c$, so $H \leq L_i$. As $S_i = 1$ and $L_i \geq \log 2$, enlarging one absolute constant proves the same bound in all reducible cases. $\square$

Cube-freeness gives $r_i \leq S_i^2$, so L_i may be replaced on the right by $\log(2n_i) + 2 \log S_i$. Also

$$(62) \quad S_a S_b S_c = R_0^2, \quad \min_i S_i \leq R_0^{2/3}.$$

Choosing the omitted endpoint $m = \min(a, b)$ instead gives

$$(63) \quad H \ll R_0 [\log(2R_0)]^2 M \log(3R_0 M), \quad M = \log(2m) + 2 \log R_0.$$

It is not legitimate to cancel an unknown height from the right: L_i may be comparable to H . For fixed bounded R_0 , (63) does imply $\log m \gg_{R_0} H/\log H$ along an unbounded sequence. A stronger fixed-support statement follows next, with an ineffective threshold.

10.4. **One common curve and its three torsion translates.** Put $N = ABC$ and $t_a = u, t_b = v, t_c = w$. Since $n_i r_i = N t_i^3$, the points

$$Q_i = (X_i, Y_i) = (x_i/t_i^2, y_i/t_i^3)$$

all lie on the same nonsingular curve

$$(64) \quad E_N: \quad Y^2 = X^3 + 16N^2.$$

Primitivity gives $\gcd(t_i, d_i r_i) = 1$, so their reduced X -denominators are exactly

$$(65) \quad \text{den}(X_i) = (t_i / \gcd(t_i, 2))^2.$$

Indeed the numerator is $\pm 4d_i r_i$, and the only cancellation with t_i^2 is $\gcd(t_i^2, 4) = \gcd(t_i, 2)^2$.

Proposition 10.4. *The point $T_0 = (0, 4N)$ has order three on E_N , and*

$$Q_b + T_0 = Q_c, \quad Q_b - T_0 = -Q_a.$$

Proof. The horizontal tangent at T_0 gives $2T_0 = -T_0$. The slopes from T_0 and $-T_0$ to Q_b are respectively

$$\frac{2Au^2}{vw}, \quad \frac{2Cw^2}{uv}.$$

Squaring the first and subtracting X_b gives $-4ABuv/w^2 = X_c$; the ordinate from the line through T_0 is $4AB(a-b)/w^3 = Y_c$. The second gives $4BCvw/u^2 = X_a$ and ordinate $-4BC(a+2b)/u^3 = -Y_a$. These computations use only $Au^3 + Bv^3 = Cw^3$ and positive denominators. $\square$

Writing $V = uvw$, the three points $Q_b, Q_c, -Q_a$ have the same image $(4Q/V^2, 4T/V^3)$ under (56). Their changing denominators describe three translates of one rational point; an integral-point bound for a fixed curve does not bound them.

Theorem 10.5 (Asymptotic balance for bounded residual support). *Fix $R_* \geq 1$. For every $\varepsilon > 0$, all sufficiently large positive primitive abc triples with $R_0 \leq R_*$ satisfy*

$$\min(a, b) \geq c^{1-\varepsilon}.$$

Equivalently, along any such sequence with $c \rightarrow \infty$, $\log \min(a, b) / \log c \rightarrow 1$. The threshold may depend on R_, ε ; the proof does not make it effective.*

Proof. For a fixed elliptic curve over $\mathbb{Q}$, Siegel's local approximation theorem, with the even function x , the point at infinity, and the real place, states that

$$(66) \quad \frac{\log \max(1, |x(P)|)}{h_x(P)} \rightarrow 0 \quad \text{as } h_x(P) \rightarrow \infty, \quad P \in E(\mathbb{Q}).$$

Here $h_x(P) = \log \max(|A_x|, B_x)$ for $x(P) = A_x/B_x$ in lowest terms, $B_x > 0$. This is [71, Theorem IX.3.1 and Example IX.3.3], not just finiteness of integral points. Since $N = ABC \leq R_0^2 \leq R_*^2$, only finitely many curves (64) occur. Formula (66) is uniform over this finite list.

By symmetry suppose $a = m \leq b$, and set $g = \gcd(u, 2) \in \{1, 2\}$. Equation (65) gives

$$X_a = A_x/B_x, \quad A_x = 4vwBC/g^2, \quad B_x = u^2/g^2.$$

Moreover $X_a^3 = 64N^2bc/a^2 \geq 64$, so $X_a \geq 4$ and $h_x(Q_a) = \log A_x$. Direct substitution yields

$$\log A_x = \frac{1}{3}(\log b + \log c) + \frac{2}{3} \log(BC) + \log 4 - 2 \log g,$$

$$\log B_x = \frac{2}{3} \log a - \frac{2}{3} \log A - 2 \log g.$$

As $c/2 \leq b < c$, these imply

$$(67) \quad \frac{2}{3}H - \frac{1}{3} \log 2 \leq h_x(Q_a) \leq \frac{2}{3}H + \log 4 + \frac{2}{3} \log N,$$

$$(68) \quad 0 \leq H - \log a \leq \frac{3}{2} \log X_a + \frac{1}{2} \log 2.$$

Thus $h_x(Q_a) \rightarrow \infty$ and is comparable to H , uniformly for bounded N . Equation (66) makes the right side of (68) equal to $o(H)$. This proves the stated ratio and hence the power inequality. If $b < a$, use Q_b instead. An infinite countersequence to the eventual assertion would contradict the same finite-list application of Siegel, so the quantifier is over all triples with $R_0 \leq R_*$. $\square$

10.5. An infinite fixed-support family and the exact limitation. Bounded R_0 alone does not imply bounded height.

Proposition 10.6. *There are infinitely many positive primitive abc triples of unbounded height with $R_0 = 3$.*

Proof. On $Y^2 = X^3 - 34992$, let $P = (36, -108)$. Doubling gives

$$2P = (252, 3996), \quad X(4P) = \left(\frac{882}{37}\right)^2 - 504 = \frac{87948}{1369}.$$

The final numerator has remainder 36 modulo 37, so $4P$ is not integral. The Nagell–Lutz integrality theorem on this integral nonsingular short Weierstrass model implies that P has infinite order; see [72, Theorem 24.21]. The mutually inverse formulas

$$u = \frac{324 + Y}{6X}, \quad v = \frac{324 - Y}{6X}, \quad X = \frac{108}{u + v}, \quad Y = \frac{324(u - v)}{u + v}$$

identify its rational affine points with $u^3 + v^3 = 9$. No affine rational point has $X = 0$, and $u + v = 0$ is impossible. Nonzero multiples of P therefore supply infinitely many rational solutions. Clear denominators to obtain

$$x^3 + y^3 = 9z^3, \quad z > 0, \quad \gcd(x, y, z) = 1.$$

Neither x nor y is zero, since 9 is not a rational cube. If $3 \mid x$, reduction modulo three forces $3 \mid y$, and then $27 \mid 9z^3$ forces $3 \mid z$, a contradiction. The same holds for y . A different prime dividing two coordinates divides the third, so x, y, z are pairwise coprime.

If $x, y > 0$, take $(a, b, c) = (x^3, y^3, 9z^3)$. If one is negative, move that term to the other side; for example $x < 0$ gives $((-x)^3, 9z^3, y^3)$. Each resulting triple is positive and primitive. Every prime other than three has exponent divisible by three in its product, while the exponent at three is $2 + 3v_3(z)$. Thus $R_0 = 3$. Only finitely many choices of signs and order give the same positive triple, so infinitely many distinct triples occur. Their heights are unbounded because a bounded height admits only finitely many triples. $\square$

This family is consistent with Theorem 10.5. It refutes only a bound depending on the cubic residual support alone: the growing primes of cube-divisible exponent still contribute to $\text{rad}(abc)$. For moving R_0 , neither the finite-list threshold in Siegel’s theorem nor the field discriminant estimate removes the growing point, denominator, or order-index cost. Controlling that cost uniformly remains a global abc problem.

11. FULL LOCAL GALOIS LIFTS AND SIMULTANEOUS MINIMUM LAYERS

Proposition 7.5 gives a necessary trace constraint for local Galois transport. We now construct actual automorphisms of full local absolute Galois groups and compute their action on specified vectors. The two examples below give a common minimum layer for a background vector and, in the second example, three different square-index vectors. All tensor factors use the same local arrow. The resulting exact maximal-order hulls concern the native families defined here; their identification with a published global collation is a separate question.

Throughout this section, p is odd, $E/\mathbb{Q}_p$ is finite, and the native valuation is normalized by $v_p(p) = 1$. Write

$$G_E = \text{Gal}(\overline{E}/E), \quad L_E = \log \mathcal{O}_E^\times, \quad I_E = p^{-1}L_E.$$

The notation for free generators and local coordinate vectors in this section is independent of the global endpoints of an abc triple.

11.1. A word automorphism and the full relative presentation. Let $n = [E : \mathbb{Q}_p]$ be even. The original Jannsen–Wingberg construction [74, §1.2, Theorem 2, and §1.4(a), pp. 74–76] uses the full tame group $\mathfrak{T}$, generated by σ, θ , and the relative profinite group

$$(69) \quad V = \widehat{F}_{n+1} * \mathfrak{T}, \quad Z = \ker(V \longrightarrow \mathfrak{T}), \quad J = \ker(Z \longrightarrow Z^{(p)}), \quad \mathcal{F} = V/J.$$

Here $\widehat{F}_{n+1}$ is free profinite on $x_0, \dots, x_n$, and $Z^{(p)}$ is the maximal pro- p quotient of Z . One then imposes the additional relation

$$(70) \quad \sigma x_0 \sigma^{-1} = W_0(x_0, \theta) x_1^{p^s} [x_1, x_2] [x_3, x_4] \cdots [x_{n-1}, x_n],$$

where W_0 belongs to the closed subgroup generated by x_0, θ . The commutator convention is $[x, y] = xyx^{-1}y^{-1}$. The tame relation $\sigma\theta\sigma^{-1} = \theta^{p^f}$ and all internal relations of $\mathfrak{T}$ are retained in (69). In particular, this construction is not the maximal pro- p quotient of the whole absolute Galois group.

Lemma 11.1 (An exact cross-handle word). *In the free group on a, b, c, d_0 , put*

$$r = bab^{-1}, \quad z = d_0 r d_0^{-1}, \quad \delta = [a, b][c, d_0].$$

The substitution

$$(71) \quad \begin{aligned} F(a) &= a, & F(b) &= d_0 b, \\ F(c) &= z r^{-1} c z^{-1}, & F(d_0) &= z d_0 z^{-1} \end{aligned}$$

is an automorphism fixing a and δ exactly. Its inverse is

$$(72) \quad \begin{aligned} G(a) &= a, & G(b) &= r^{-1} d_0^{-1} r b, \\ G(c) &= r^{-1} d_0^{-1} r d_0 c r, & G(d_0) &= r^{-1} d_0 r. \end{aligned}$$

On the free abelianization it acts as

$$(73) \quad a \longmapsto a, \quad b \longmapsto b + d_0, \quad c \longmapsto c - a, \quad d_0 \longmapsto d_0.$$

Proof. All substitutions act on the rightmost expression first. Set $D = r^{-1} d_0 r$ and $s = D^{-1} r D$. From (72),

$$G(b) = D^{-1} b, \quad G(r) = s, \quad G(z) = r, \quad G(c) = s r^{-1} c r, \quad G(d_0) = D.$$

For example, $G(r) = (D^{-1} b) a (b^{-1} D) = D^{-1} r D$, and $G(z) = D(D^{-1} r D) D^{-1} = r$. Substitution in (71) now gives

$$G(F(a)) = a, \quad G(F(b)) = D D^{-1} b = b, \quad G(F(c)) = r s^{-1} (s r^{-1} c r) r^{-1} = c,$$

and $G(F(d_0)) = r D r^{-1} = d_0$. Conversely, $F(r) = z$, $F(D) = z^{-1} (z d_0 z^{-1}) z = d_0$, and $F(s) = d_0^{-1} z d_0 = r$. Applying F to the preceding expressions for $G(b), G(c), G(d_0)$ gives b, c, d_0 , respectively. Thus both compositions are the identity on all four generators.

The boundary is preserved as a word, not just up to conjugacy:

$$\begin{aligned} F([a, b]) &= a d_0 r^{-1} d_0^{-1}, \\ F([c, d_0]) &= z r^{-1} c d_0 c^{-1} r d_0^{-1} z^{-1}. \end{aligned}$$

Using $z = d_0 r d_0^{-1}$, their product reduces to $a r^{-1} c d_0 c^{-1} d_0^{-1} = [a, b][c, d_0]$. Finally r and z both have abelian image a , which gives (73) directly. $\square$

Proposition 11.2 (Descent to the full Galois group). *Suppose $n \geq 4$ is even. Apply Lemma 11.1 to $(a, b, c, d_0) = (x_1, x_2, x_3, x_4)$ and fix all other generators, including the tame factor. This defines a continuous automorphism of G_E with the action (73) on abelianization. More generally, a free-word automorphism of $x_1, \dots, x_n$ descends in this manner if it fixes x_1 and the ordered commutator product in (70) exactly.*

Proof. The substitution and its finite-word inverse extend continuously to the free profinite product V by its universal property. Both leave the projection to $\mathfrak{T}$ unchanged, and hence preserve Z . The kernel J of the canonical maximal pro- p quotient of Z is characteristic in Z . Thus both substitutions preserve J and give inverse automorphisms of $\mathcal{F}$. This verifies the wild-normal-subgroup condition in the original presentation.

They fix σ, θ, x_0 , and therefore fix W_0 and all its profinite powers. They fix $x_1^{p^s}$ literally. The first two commutators are fixed as their ordered product, and the later commutators are fixed term by term. Thus the additional relator and its closed normal closure are preserved. The induced maps on the full quotient are inverse; the identification of that quotient with G_E is Jannsen–Wingberg’s Theorem 2 with the algebraic closure as extension. The same argument proves the general assertion. $\square$

For clarity, the descended cross-handle automorphism is not inner. The character to $\mathbb{Z}_p$ sending x_4 to 1 and the other generators and tame factor to 0 satisfies the relative presentation. It has value 0 on x_2 and value 1 on $F(x_2) = x_4 x_2$; an inner automorphism would preserve every abelian character. This observation does not require any choice of logarithmic basis.

11.2. The integral JW basis and the available linear actions. We keep the same full presentation and its cyclotomic parameters when passing to local reciprocity. Let y_i be the logarithm of the principal-unit image of x_i . Hoshi–Nishio [75, Lemma 1.3 and its proof] prove that $y_1, \dots, y_n$ are a $\mathbb{Q}_p$ -basis of E and that $y_0, \dots, y_n$ generate L_E over $\mathbb{Z}_p$. The former assertion alone does not give an integral basis.

Lemma 11.3 (Integral upgrade in the small tame range). *Assume $2 \leq e(E/\mathbb{Q}_p) \leq p - 2$. Then $L_E = \mathfrak{m}_E$, and*

$$(74) \quad v_i = y_i/p \quad (1 \leq i \leq n)$$

form a $\mathbb{Z}_p$ -basis of I_E . For even n , in these same coordinates,

$$(75) \quad I_E \cap \ker \mathrm{Tr}_{E/\mathbb{Q}_p} = \mathbb{Z}_p v_1 \oplus \bigoplus_{i=3}^n \mathbb{Z}_p v_i.$$

If additionally $\mathrm{Tr} I_E = \mathbb{Z}_p$, then $\mathrm{Tr}(v_2)$ is a unit.

Proof. The inequality $1/e > 1/(p-1)$ makes logarithm and exponential inverse on $1 + \mathfrak{m}_E$ and $\mathfrak{m}_E$. The prime-to- p torsion units have zero logarithm, proving $L_E = \mathfrak{m}_E$.

Let g, h be the lifts of the tame cyclotomic actions appearing in the original presentation. With the notation of [75, Proposition 1.1], the mixed factor can be written

$$W_0 = \left(x_0^{h^{p-1}} \theta x_0^{h^{p-2}} \theta \cdots x_0^h \theta \right)^{\epsilon_p g/(p-1)},$$

where $\epsilon_p \in \widehat{\mathbb{Z}}$ is the projector onto $\mathbb{Z}_p$. The tame relation makes the image of θ torsion in abelianization, so it contributes zero to logarithm. Taking logarithms in (70) gives

$$(76) \quad \left(1 - \frac{g}{p-1} \sum_{r=1}^{p-1} h^r \right) y_0 = p^s y_1.$$

The reduction of h modulo p is not 1. Indeed, trivial inertia action on μ_p would make $E(\mu_p)/E$ unramified, forcing $p-1$ to divide $e(E/\mathbb{Q}_p)$, contrary to the assumed range. Thus the geometric sum in (76) vanishes modulo p . Its parenthesized coefficient is a unit, and $y_0 \in \mathbb{Z}_p y_1$. The remaining y_i therefore generate L_E integrally. Their known rational independence gives an integral basis and proves (74).

For even degree, Kondo’s trace-kernel theorem [76, Theorem 1.3] identifies $\ker \mathrm{Tr}$ with the rational span of $v_1, v_3, \dots, v_n$ in this JW basis. Intersecting with the proved integral basis gives (75). Consequently the trace ideal of I_E is generated by $\mathrm{Tr}(v_2)$, which proves the final assertion. $\square$

Write $a = v_1$, $b = v_2$, and $W = \bigoplus_{i=3}^n \mathbb{Z}_p v_i$. Let ω be the standard integral alternating form on these remaining handle coordinates, with $\omega(v_3, v_4) = 1$. This form records the handle action; it is not being identified with field multiplication. The covariant logarithmic action of Proposition 11.2 is

$$N_{v_4}(a) = a, \quad N_{v_4}(b) = b + v_4, \quad N_{v_4}(x) = x + \omega(v_4, x)a \quad (x \in W).$$

The twist $x_2 \mapsto x_2 x_1$ gives $C_1(b) = b + a$, fixing a, W .

The boundary-preserving surface construction in [76, pp. 19–20 and Theorem 2.17] also gives an individual full Galois lift of every integral symplectic matrix on the remaining handles, fixing $\sigma, \theta, x_0, x_1, x_2$. The construction on p. 19 explicitly includes even degree; the odd-degree restriction later on p. 20 is not used here. In particular, handle exchanges and rotations move v_4 to any displayed basis direction. Conjugating the cross-handle lift therefore realizes N_w for every such direction.

For a general integer coordinate vector $w \in \text{span}_{\mathbb{Z}}\{v_3, \dots, v_n\}$, the linear formulas are

$$(77) \quad \begin{aligned} N_w(Aa + Bb + x) &= (A + \omega(w, x))a + Bb + x + Bw, \\ C_c(Aa + Bb + x) &= (A + cB)a + Bb + x. \end{aligned}$$

The following correction makes their realization precise.

Lemma 11.4 (Ordered central correction). *As identities of the induced linear maps,*

$$(78) \quad N_w N_v = C_{\omega(w, v)} N_{w+v}, \quad C_c C_d = C_{c+d}, \quad C_c N_w = N_w C_c.$$

For $w = \sum_i n_i e_i$ in the fixed ordered integer handle basis,

$$(79) \quad N_w = C_{-\sum_{i < j} n_i n_j \omega(e_i, e_j)} N_{e_1}^{n_1} \cdots N_{e_{n-2}}^{n_{n-2}}.$$

Thus every such N_w is the canonical action of an actual full Galois automorphism.

Proof. In coordinates (A, B, x) , applying N_v and then N_w gives

$$(A + \omega(v, x) + \omega(w, x + Bv), \quad B, \quad x + Bv + Bw).$$

Bilinearity gives (78); the other two identities follow directly from (77). Alternation implies $N_w^{-1} = N_{-w}$ and $N_w^m = N_{mw}$ for every integer m . An induction in the displayed left-to-right basis order gives (79). Each basis cross and C_1 already has a full lift. Integer powers, inverses and compositions of those lifts realize the displayed linear image. $\square$

The central identities are not asserted in the full nonabelian automorphism group: lifts may differ by elements acting trivially on abelianization. Nor is a group-homomorphic section of a mapping class group into an outer Galois group required. Individual lifts and the functoriality of their linear action suffice. In particular, any prescribed vector modulo p can be realized by choosing an integer coordinate lift in (79).

11.3. Covariance and integral Kummer transport. For $\alpha \in \text{Aut}(G_E)$, let M_α be the canonical covariant action on principal-unit abelianization, extended by logarithm and $\mathbb{Q}_p$ -linearity to E . It sends each y_i to the logarithm of the image of $\alpha(x_i)$ and preserves I_E . The norm/trace compatibility of this action is the one in [75, Lemma 2.3].

Proposition 11.5 (The inverse in Kummer transport). *For a continuous automorphism α and an integral equivariant Tate-module isomorphism, let F_α be the induced logarithmic Kummer pullback. Then*

$$(80) \quad F_\alpha = c_\alpha M_\alpha^{-1} \quad \text{for some } c_\alpha \in \mathbb{Z}_p^\times.$$

Thus the native valuations attained by M_α are attained by one integral Kummer arrow belonging to α^{-1} .

Proof. The cyclotomic character is determined by the local absolute Galois group, so an equivariant integral Tate-module isomorphism exists; its choices differ by $\mathbb{Z}_p^\times$ [68, Proposition 1.1]. Local duality identifies $H^2(G_E, \mathbb{Z}_p(1))$ with $\mathbb{Z}_p$, and integral transport acts on it by a unit c_α .

The cup-product reciprocity pairing gives, for a continuous integral character χ and a Kummer class z ,

$$(81) \quad \langle \chi \circ \alpha^{\text{ab}}, F_\alpha z \rangle = c_\alpha \langle \chi, z \rangle.$$

Here the pairing is evaluation of χ on the pro- p completion of $E^\times$ under local reciprocity; see the conventions in Proposition 7.5 and [67, Chapter III, Proposition 3.6 and §4]. The unit submodule is the annihilator of the unramified character, so it is preserved. Torsion classes disappear after logarithm.

On the free unit lattice, the pullback character is $\chi \circ M_\alpha$. These characters separate the lattice after rational extension. Equation (81) therefore says $M_\alpha F_\alpha = c_\alpha \text{id}$, proving (80). Applying this to α^{-1} gives $F_{\alpha^{-1}} = c_{\alpha^{-1}} M_\alpha$. A $\mathbb{Z}_p$ -unit scalar preserves the native valuation of every vector, simultaneously. Uniform division of logarithmic coordinates by p does not alter the identity. $\square$

11.4. A degree-ten example.

Lemma 11.6 (One minimal vector). *Use the integral basis in Lemma 11.3, with even $n \geq 4$, and let π be a uniformizer. Suppose*

$$\text{Tr } I_E = \text{Tr}(\pi I_E) = \mathbb{Z}_p, \quad u_0 \in \pi I_E, \quad \text{Tr}(u_0) \in \mathbb{Z}_p^\times,$$

and $t_0 \in I_E \setminus \pi I_E$ with $\text{Tr}(t_0) \in p\mathbb{Z}_p$. Then one full Galois automorphism has canonical action M satisfying $Mu_0, Mt_0 \in I_E \setminus \pi I_E$.

Proof. Put $\beta = \text{Tr}(b) \in \mathbb{Z}_p^\times$. The respective b -coordinates are $B_u = \text{Tr}(u_0)/\beta \in \mathbb{Z}_p^\times$ and $B_t = \text{Tr}(t_0)/\beta \in p\mathbb{Z}_p$. If $a \notin \pi I_E$, use C_1 . Modulo πI_E , its image of u_0 is $B_u a \neq 0$, and its image of t_0 is unchanged, because $pI_E \subseteq \pi I_E$.

If $a \in \pi I_E$, then $I_E \cap \ker \text{Tr}$ surjects onto $I_E/\pi I_E$: subtract an element of πI_E with the same trace from any representative. Equation (75) implies that some displayed basis vector $w \in W$ is outside πI_E . The actual cross lift N_w changes u_0 by $B_u w + A_u a$ and t_0 by $B_t w + A_t a$, for integral A_u, A_t . Modulo πI_E , the first image is $B_u w \neq 0$, and the second is the original nonzero image of t_0 . The same map works for both. $\square$

Theorem 11.7 (A common minimum layer over $\mathbb{Q}_{19}$). *Let*

$$K_0 = \mathbb{Q}_{19}(\mu_5), \quad \pi^5 = 19, \quad E = K_0(\pi),$$

and put

$$u = \frac{\log 20}{19}, \quad \tau = \frac{\log(1 + 19\pi)}{19}, \quad t = \tau/19.$$

Then some full Galois automorphism has canonical action M with

$$(82) \quad v_{19}(Mu) = v_{19}(Mt) = -4/5.$$

One integral Kummer arrow attains the same two valuations.

Proof. The order of 19 modulo 5 is 2, so $K_0/\mathbb{Q}_{19}$ is unramified quadratic. The Eisenstein polynomial $X^5 - 19$ and $\mu_5 \subset K_0$ give a Galois extension with $(n, e, f) = (10, 5, 2)$. The logarithm range yields

$$I_E = \pi^{-4} \mathcal{O}_E = \bigoplus_{r=-4}^0 \mathcal{O}_{K_0} \pi^r.$$

The negative powers in this direct sum have relative trace zero, and $\text{Tr}_{E/K_0}(1) = 5$. Hence $\text{Tr } I_E = \mathbb{Z}_{19}$. Also $1/10 \in \mathcal{O}_E \subset \pi I_E$ has absolute trace 1, proving $\text{Tr}(\pi I_E) = \mathbb{Z}_{19}$.

Leading logarithm terms give $v_{19}(u) = 0$ and $v_{19}(t) = -4/5$. The norm identity

$$\prod_{\zeta \in \mu_5} (1 + 19\zeta\pi) = 1 + 19^6$$

gives the exact traces

$$\text{Tr}(u) = 10u \in \mathbb{Z}_{19}^\times, \quad \text{Tr}(t) = \frac{2}{19^2} \log(1 + 19^6) \in 19^4 \mathbb{Z}_{19}^\times.$$

Lemma 11.6 proves (82), and Proposition 11.5 gives the Kummer assertion. $\square$

The covariant action in Theorem 11.7 is not induced by a field automorphism: it changes the native valuation of u from 0 to $-4/5$. This is an assertion about a specified local vector, rather than a classification of all admissible linear maps.

11.5. A simultaneous finite-family lemma. The degree-ten argument begins with t already at the minimum layer. The square-index example below does not have this feature. We therefore need a common operation on several vectors initially lying above the minimum layer.

Lemma 11.8 (Simultaneous parabolic reachability). *Let k be a finite field with q elements, and let the finite-dimensional k -vector space*

$$V = ka \oplus kb \oplus W$$

carry the standard nondegenerate alternating form on W , and let B be the b -coordinate. Let $L \subset V$ have codimension two, and suppose $\ker B \rightarrow V/L$ is onto. Assume

$$U, T_1, \dots, T_r \in L, \quad B(U) \neq 0, \quad B(T_j) = 0, \quad T_j \notin ka,$$

with $r + 1 < q$. The linear operations C_c, N_w of (77), together with symplectic operations on W fixing a, b , contain one map sending $U, T_1, \dots, T_r$ all outside L . If $k = \mathbf{F}_p$ and the coordinates come from the preceding integral JW basis, one such map is induced by a full Galois automorphism, using integer coordinate lifts.

Proof. Write $T_j = A_j a + w_j$. The hypotheses imply $w_j \neq 0$, so each functional $w \mapsto \omega(w, w_j)$ is nonzero.

Suppose first that $a \notin L$. Fewer than q proper hyperplanes cannot cover W : in dimension D , their union has fewer than $q \cdot q^{D-1} = q^D$ elements. Choose w with $\omega(w, w_j) \neq 0$ for all j . Then

$$N_w(T_j) + L = \omega(w, w_j)(a + L) \neq 0.$$

If $N_w(U) \notin L$, the construction is complete. Otherwise compose with C_1 . It adds the nonzero class $B(U)(a + L)$ to $N_w(U)$ and leaves all the T_j images unchanged because their B -coordinates remain zero.

Suppose now that $a \in L$. Then $\lambda : W \rightarrow V/L$ is onto. Also $\lambda(w_j) = 0$ for every j . Avoid $\ker \lambda$ and the r hyperplanes $\ker \omega(-, w_j)$ simultaneously, obtaining $s \in W$ such that

$$\lambda(s) \neq 0, \quad \omega(s, w_j) \neq 0 \quad \text{for all } j.$$

There are $r + 1 < q$ proper subspaces, so the same union bound applies. The single symplectic transvection

$$S_s(x) = x + \omega(s, x)s \quad (x \in W), \quad S_s(a) = a, \quad S_s(b) = b$$

has inverse given by changing the plus sign to a minus sign. Expanding its pairing shows it preserves ω ; the cross terms cancel and $\omega(s, s) = 0$. Since the original projections vanish,

$$S_s(T_j) + L = \omega(s, w_j)\lambda(s) \neq 0$$

for every j . If $S_s(U) \notin L$, stop. Otherwise choose a displayed basis vector e of W with $\lambda(e) \neq 0$ and apply N_e . Its change on the zero projection of $S_s(U)$ is $B(U)\lambda(e) \neq 0$. It does not change any T_j projection, because $B(T_j) = 0$ and $a \in L$.

For the arithmetic assertion, lift the chosen residue vectors to integer coordinate vectors. Lemma 11.4 realizes each required cross. The lifted transvection is an element of $\mathrm{Sp}(W_{\mathbb{Z}})$ and has an individual full Galois lift by the surface construction cited above. Compose these lifts. The argument uses one map for the whole family, not a different map for each vector. $\square$

11.6. Three square labels at the prime 139. We state the next example for a general unit part of the parameter. Let $p = 139$ and choose any $b_0 \in \mathbb{Q}_p$ with $v_p(b_0) = 1$. Put

$$(83) \quad K_0 = \mathbb{Q}_p(\mu_{210}), \quad \pi^{105} = b_0, \quad E = K_0(\pi), \quad q_0 = b_0^2.$$

Thus $\pi^{210} = q_0$ and $\pi^{15} = q_0^{1/14}$ with these choices of roots. The field in (83) is exactly $\mathbb{Q}_p(\mu_{210}, q_0^{1/210})$; no global curve with a prescribed exact parameter is assumed here.

Theorem 11.9 (One full Galois arrow for three square labels). *In (83), set*

$$(84) \quad \begin{aligned} u &= \frac{\log(1+p)}{p}, & \tau_s &= \frac{\log(1+p\pi^{15s})}{p}, \\ k_s &= \left\lfloor \frac{s}{7} + \frac{104}{105} \right\rfloor, & T_s &= p^{-k_s} \tau_s \quad (s = 1, 4, 9). \end{aligned}$$

Then $k_1 = k_4 = 1$, $k_9 = 2$. One full Galois automorphism has canonical action M such that

$$(85) \quad v_p(Mu) = v_p(MT_1) = v_p(MT_4) = v_p(MT_9) = -104/105.$$

Consequently

$$(86) \quad v_p(M\tau_1) = 1/105, \quad v_p(M\tau_4) = 1/105, \quad v_p(M\tau_9) = 106/105.$$

One integral Kummer arrow attains all these values simultaneously.

Proof. Since $139^2 \equiv 1 \pmod{210}$ and $139 \not\equiv 1 \pmod{210}$, $K_0/\mathbb{Q}_p$ is unramified quadratic. The Eisenstein polynomial $X^{105} - b_0$ and $\mu_{105} \subset K_0$ give a Galois field with $(n, e, f) = (210, 105, 2)$. As $1/105 > 1/138$,

$$(87) \quad I_E = \pi^{-104} \mathcal{O}_E, \quad pI_E = \pi \mathcal{O}_E, \quad \pi I_E = \pi^{-103} \mathcal{O}_E.$$

These are equalities of fractional ideals: b_0/p is a rational unit and need not be 1.

The power basis gives $I_E = \bigoplus_{r=-104}^0 \mathcal{O}_{K_0} \pi^r$. All negative powers in this sum have relative trace zero and $\text{Tr}_{E/K_0}(1) = 105$. Hence $\text{Tr } I_E = \mathbb{Z}_p$. Also $1/210 \in \mathcal{O}_E \subset \pi I_E$ has trace 1. Subtracting $\text{Tr}(z)/210$ from $z \in I_E$ proves that $I_E \cap \ker \text{Tr}$ surjects onto $I_E/\pi I_E$. Lemma 11.3 applies.

The integer k_s is exactly the largest k for which $\tau_s \in p^k I_E$. Leading logarithm terms and the exact norm give the following data:

s	k_s	$v_p(T_s)$	$v_p(\text{Tr } T_s)$
1	1	$-90/105$	6
4	1	$-45/105$	9
9	2	$-75/105$	13

Indeed, for these s , the seven conjugates of π^{15s} occur fifteen times in the relative norm, and therefore

$$(88) \quad \begin{aligned} \text{Nm}_{E/K_0}(1 + p\pi^{15s}) &= (1 + p^7 b_0^s)^{15}, \\ \text{Tr}(T_s) &= \frac{30}{p^{k_s+1}} \log(1 + p^7 b_0^s). \end{aligned}$$

The last expression has valuation $s + 6 - k_s$, as displayed. Furthermore $\text{Tr}(u) = 210 \log(1+p)/p$ is a unit.

Pass to the $\mathbf{F}_p$ -space

$$V = I_E/pI_E, \quad L = \pi I_E/pI_E, \quad \dim V = 210, \quad \text{codim } L = 2.$$

The integral JW basis gives $V = \mathbf{F}_p a \oplus \mathbf{F}_p b \oplus W$, with $\dim W = 208$ and B its b -coordinate. Equation (75) and $\text{Tr}(b) \in \mathbb{Z}_p^\times$ imply

$$B(u) \neq 0, \quad B(T_s) = 0, \quad u, T_s \in L, \quad T_s \neq 0 \text{ in } V.$$

The trace-zero subspace surjects onto V/L by the subtraction argument above. There remains a possible fixed-line obstruction: one or more of the T_s might belong to $\mathbf{F}_p a$.

We remove it by one common field automorphism. The logarithm tail satisfies

$$(89) \quad T_s \equiv \pi^{15s}/p^{k_s} \pmod{pI_E}.$$

For its term of degree $m \geq 2$, use $v_p(m) \leq m - 2$ to obtain

$$m(1 + s/7) - (k_s + 1) - v_p(m) \geq 1 + ms/7 - k_s \geq 1 + 2s/7 - k_s.$$

The last three bounds are $2/7, 8/7, 11/7$, all larger than $1/105$, the minimum valuation of pI_E .

Let $\sigma(\pi) = \zeta_{105}\pi$, fixing K_0 . For $h = 0, \dots, 6$, equation (89) gives

$$\sigma^h(T_s) \equiv \zeta_7^{sh} \pi^{15s} / p^{k_s} \pmod{pI_E}.$$

For each s , these determine seven distinct $\mathbf{F}_{139}$ -lines. To see this, the reduction of ζ_7 has order seven and $7 \nmid 138$, so its group meets $\mathbf{F}_{139}^\times$ only in 1. Equality of two lines would force their leading coefficients to have a ratio in that intersection. A nonzero residue difference would be a unit times π^{15s}/p^{k_s} , whose negative valuation precludes membership in pI_E . Since s is prime to seven, the two exponents must be equal modulo seven.

For each of the three labels, at most one of these seven values of h puts $\sigma^h(T_s)$ in the fixed line $\mathbf{F}_p a$. Avoid the at most three excluded values. This one σ^h fixes u , preserves L and trace, and places all three labels outside the fixed line. It is an actual field automorphism, hence has a representative on the full local Galois group.

Apply Lemma 11.8 to these three resulting vectors and u . The numerical condition is $3 + 1 < 139$. More precisely, in its second case the excluded sets have at most $139^{206} + 3 \cdot 139^{207} < 139^{208}$ elements. The lemma gives one further actual lift sending all four vectors outside L . Composing with the chosen inertia representative proves (85). Linearity and $\tau_s = p^{k_s} T_s$ give (86); Proposition 11.5 supplies the common Kummer arrow. $\square$

The unit b_0/p is fixed by inertia. Thus the proof includes all parameters in (83), rather than only $b_0 = p$. The separate passage from a rational Frey curve to these local data does not enter the simultaneous-reachability argument.

11.7. Exact native maximal-order hulls. Let $E/\mathbb{Q}_p$ now be either Galois field above. For $m \geq 1$, write

$$\mathcal{T}_m = E^{\otimes_{\mathbb{Q}_p} m}, \quad B_m = \prod_{\text{Gal}(E/\mathbb{Q}_p)^{m-1}} \mathcal{O}_E,$$

using the decomposition

$$(90) \quad x_1 \otimes \cdots \otimes x_m \longmapsto (x_1 \gamma_2(x_2) \cdots \gamma_m(x_m))_{(\gamma_2, \dots, \gamma_m)}.$$

Thus B_m is the maximal order in this fixed finite etale carrier. In $\pi^r B_m$, the uniformizer is embedded through the first tensor factor, so it has the same valuation in every component. Let Γ_E be the set of integral logarithmic Kummer transports arising from automorphisms of the same G_E , with integral Tate-module identifications. Every $F \in \Gamma_E$ is $\mathbb{Q}_p$ -linear and preserves I_E and $p^k I_E$ for all integers k .

Proposition 11.10 (An attained block hull). *Suppose $I_E = \pi^{1-e} \mathcal{O}_E$, and put $\kappa = 1 - 1/e$. Let $u, t \in I_E$, $\tau = p^k t$ with integer k , and suppose one $F_0 \in \Gamma_E$ satisfies $v_p(F_0 u) = v_p(F_0 t) = -\kappa$. Define the coherent repeated-label family*

$$S_m = \{F(\tau) \otimes F(u) \otimes \cdots \otimes F(u) : F \in \Gamma_E\}, \quad H_m = \text{span}_{B_m}(S_m).$$

Then

$$(91) \quad H_m = \pi^{ek - (e-1)m} B_m.$$

The equality also holds after taking topological closure. More generally, it remains true if the set of source tuples is enlarged to any set lying between the displayed coherent tuples and $p^k I_E \times I_E^{m-1}$.

Proof. For every F , the first factor has valuation at least $k - \kappa$, and each remaining factor at least $-\kappa$. All field embeddings in (90) preserve valuation. Every component of every tensor generator therefore has valuation at least $k - m\kappa$, proving containment in the right-hand side of (91).

The single arrow F_0 gives equality in this valuation in every component. Its tensor is consequently $\pi^{ek - (e-1)m}$ times a unit of the product ring B_m . Its principal B_m -span is the entire right-hand side, giving the reverse containment. This fractional product lattice is closed. The same upper bound holds for every tuple in the stated containing product, while the original attaining tuple remains present; this proves the enlargement assertion. $\square$

The enlargement includes the closed $\mathbb{Z}_p$ -convex hull taken in the product of the source coordinate spaces, since that product lattice is closed and $\mathbb{Z}_p$ -convex. This does not identify such a convex hull with a B_m -module hull: tensor formation and the final B_m -span remain distinct operations.

Corollary 11.11 (The two exact local calculations). *For Theorem 11.7, the family with one τ factor and $m - 1$ background factors has*

$$H_m = \pi^{5-4m} B_m \quad (m \geq 1), \quad \text{in particular } H_3 = \pi^{-7} B_3.$$

For Theorem 11.9, let H_j use one τ_{j^2} factor and j background factors, for $j = 1, 2, 3$. Then one and the same arrow attains all three hulls, and

$$(92) \quad H_1 = \pi^{-103} B_2, \quad H_2 = \pi^{-207} B_3, \quad H_3 = \pi^{-206} B_4.$$

Proof. Use Proposition 11.10 with $(e, k) = (5, 1)$ in the first case. In the second case $e = 105$, $m = j + 1$, and $(k_1, k_4, k_9) = (1, 1, 2)$. Thus the three exponents are $105k_{j^2} - 104(j + 1) = -103, -207, -206$. The common witness is supplied by Theorem 11.9 and the single inverse/unit conversion of Proposition 11.5. $\square$

These nonintegral hulls are exact calculations for stated local source families. They do not establish a comparison between different native markings, a cross-Frobenius identification, or compatible global initial theta data. In particular, identifying the source sets needed to compare a published pilot container with one of these hulls remains a separate requirement.

The free-word identities, the discrete free-group automorphism, the linear central correction, and the finite-family transvection statements have independent Lean proofs. The relative profinite quotient, local class field theory, local duality, integral JW basis, and logarithmic norm calculations are mathematical source arguments, not Lean declarations of local-field theorems in this work. The global abc bound remains an additional obligation.

12. NATIVE PILOT POINTS, MULTIPLICATIVE FAMILIES, AND TRANSPORTED IDEALS

We now distinguish three local inputs: a specified point, the image of multiplication on a log lattice, and a principal maximal-order ideal before transport. There are positive comparisons between these inputs, even for nongeometric Galois arrows. In the attained examples the point and pre-ideal hulls coincide, while saturation to the whole multiplication image produces a strictly larger hull. All the constructions in this section use one native carrier and one normalization; they do not assume the cross-arithmeticoid comparison in [57, Theorem 6.10.1].

12.1. The local objects and the logarithmic scale. Let $p > 2$ and let $E/\mathbb{Q}_p$ be finite Galois, of degree d and ramification index $2 \leq e \leq p - 2$. Normalize $v(p) = 1$, choose a uniformizer π , and write

$$(93) \quad \kappa = \frac{e-1}{e}, \quad I = p^{-1} \log \mathcal{O}_E^\times = p^{-1} \mathfrak{m}_E = \pi^{1-e} \mathcal{O}_E, \quad u = p^{-1} \log(1+p).$$

Logarithm and exponential are inverse isometries on $1 + \mathfrak{m}_E$ and $\mathfrak{m}_E$, since $1/e > 1/(p-1)$. The tame different is $\mathfrak{D}_{E/\mathbb{Q}_p} = \pi^{e-1} \mathcal{O}_E$, so $I = \mathfrak{D}^{-1}$. The convention $(2p)^{-1} \log \mathcal{O}_E^\times$ differs only by a $\mathbb{Z}_p^\times$ -factor, which does not affect the ideals below.

For $m \geq 1$, set

$$T_m = E^{\otimes_{\mathbb{Q}_p} m}, \quad A_m = \mathcal{O}_E^{\otimes_{\mathbb{Z}_p} m}, \quad B_m = \text{the normalization of } A_m \text{ in } T_m.$$

The algebra T_m is a product of d^{m-1} copies of E , and B_m is the product of their integer rings. An ideal written $\pi^t B_m$ means the product ideal of valuation t/e in every component, without requiring identical component generators.

Let Γ be a family containing the identity of actual integral Kummer actions on I , or more generally any such family in $\text{Aut}_{\mathbb{Z}_p}(I)$. Each action extends $\mathbb{Q}_p$ -linearly to E . We do not suppose that the actual Galois family exhausts this general linear group. With the same arrow in each factor, put

$$(94) \quad \mathcal{H}_\Gamma(S) = \overline{\text{span}_{B_m} \{Fx_1 \otimes \cdots \otimes Fx_m : F \in \Gamma, (x_1, \dots, x_m) \in S\}} \quad (S \subset E^m).$$

This is an orbit followed by a module hull. Taking a product convex hull before tensoring is a separate operation.

Write Kum for the integral Kummer map on principal units. The standard Bloch–Kato exponential, extended by p^{-n} Kum($\exp(p^n x)$) when necessary, has inverse $\log_{\text{BK}}^{\text{std}}$ satisfying

$$\log_{\text{BK}}^{\text{std}}(\text{Kum}(1 + pa)) = \log(1 + pa).$$

The following coordinate map is different:

$$(95) \quad \rho = p^{-1} \log_{\text{BK}}^{\text{std}}, \quad \lambda_p(a) = \rho(\text{Kum}(1 + pa)) = p^{-1} \log(1 + pa).$$

The three displays [56, (9.7.1.1), (9.7.2.1)–(9.7.2.2), PDF p. 110] require this distinction. If the exponential has its standard cohomological scalar meaning and the class is the unscaled class of $1 + pa$ defined on p. 109, its inverse cannot take that class to $\lambda_p(a)$. Indeed, for nonzero integral a ,

$$\exp_{\text{BK}}(\lambda_p(a)) = p^{-1} \text{Kum}(1 + pa) \neq \text{Kum}(1 + pa).$$

The Kummer class is nonzero because its logarithm has nonzero leading term pa . Formula (95) is a consistent normalized coordinate, but it is not that inverse formula.

Passing from ρ to $\log_{\text{BK}}^{\text{std}}$ multiplies each entry, including each background entry, by p . Since integral Kummer arrows commute with this scalar, every length- m hull satisfies

$$(96) \quad H^{\text{std}} = p^m H^\rho, \quad V_B(H^{\text{std}}) = V_B(H^\rho) - m \log p, \quad V_B(H) = d^{-m} \log \mu_B(H), \quad \mu_B(B_m) = 1.$$

The determinant of multiplication by p^m on T_m is p^{md^m} , which proves the volume formula. If instead $\mu_A(A_m) = 1$, then $V_A(H) = V_B(H) + d^{-m} \log[B_m : A_m]$. This constant cancels when both sets use the same reference order.

12.2. A point bridge that survives every integral arrow.

Proposition 12.1 (Transported root and crystalline point). *For $0 \neq a \in \mathcal{O}_E$, set $r = v(a)$ and $k = \lfloor r + \kappa \rfloor$. Every $F \in \text{Aut}_{\mathbb{Z}_p}(I)$ satisfies*

$$(97) \quad v(F\lambda_p(a)) = v(Fa), \quad v(Fu) = v(F1).$$

Consequently

$$(98) \quad \mathcal{H}_\Gamma(\{(\lambda_p(a), u, \dots, u)\}) = \mathcal{H}_\Gamma(\{(a, 1, \dots, 1)\}).$$

Proof. The choice of k gives $a \in p^k I \setminus p^{k+1} I$. In the convergent expansion

$$\lambda_p(a) - a = \sum_{n \geq 2} (-1)^{n+1} \frac{p^{n-1} a^n}{n},$$

every term has valuation at least $1 + 2r$: for $p > 2$ and $n \geq 2$, $v_p(n) \leq n - 2$. Hence

$$v(\lambda_p(a) - a) \geq 1 + 2r \geq k + 1 - \kappa, \quad \lambda_p(a) - a \in p^{k+1} I.$$

As F preserves each $p^n I$ in both directions,

$$v(Fa) < k + 1 - \kappa, \quad v(F(\lambda_p(a) - a)) \geq k + 1 - \kappa.$$

The strict ultrametric inequality proves the first equality in (97); apply it to $a = 1$ for the second.

In every component of T_m , the two nonzero tensors $F\lambda_p(a) \otimes (Fu)^{\otimes(m-1)}$ and $Fa \otimes (F1)^{\otimes(m-1)}$ have the same valuation. Their componentwise ratio is a unit of B_m , so they generate the same principal B_m -ideal. Taking spans and closures proves (98). No E -linearity or multiplicativity of F was used. $\square$

The same proof allows different prescribed integral arrows in different factors. Root-of-unity ambiguities can be handled by applying the proposition separately to each permitted ζa and then taking their union; one does not assert $F(\zeta a) = \zeta F(a)$. Also, (98) is a statement about module hulls of raw orbits. The tensor map is not additive in its tuple of inputs, so equality after a product convex hull needs a further argument. An attaining-point argument below supplies it in the explicit example.

12.3. A whole multiplication image requires a whole unit family.

Proposition 12.2 (Exact saturated Kummer family). *For $0 \neq a \in \mathcal{O}_E$, the group $U_a = 1 + paI = 1 + a\mathfrak{m}_E$ satisfies, as actual sets,*

$$(99) \quad \rho(\text{Kum}(U_a)) = aI, \quad \log_{\text{BK}}^{\text{std}}(\text{Kum}(U_a)) = paI.$$

Proof. The ideal $paI = a\mathfrak{m}_E$ lies in the common strict convergence disk of logarithm and exponential. These maps preserve valuations, so $\log(1 + paI) = paI$. More explicitly, for every $z \in aI$, $\exp(pz) \in U_a$ and $\rho(\text{Kum}(\exp(pz))) = z$. The inverse series gives the reverse containment. The positive-valuation ideal also makes U_a closed under multiplication and inversion. $\square$

In particular, the background family is the whole principal-unit group U_1 , not the singleton $1 + p$. Taking products in (99) realizes $aI \times I^{m-1}$ before any arrows or hulls are applied. This equality therefore survives the same transport, product convex closure, tensor map, and final module hull, in any specified order.

Proposition 12.3 (The singleton can be strictly smaller). *Put*

$$P(a) = \mathcal{H}_\Gamma(\{(\lambda_p(a), u, \dots, u)\}), \quad S(a) = \mathcal{H}_\Gamma(aI \times I^{m-1}).$$

If $v(a) < \lfloor v(a) + \kappa \rfloor$, then $P(a) \subsetneq S(a)$. The strict inclusion persists if both product inputs first undergo closed $\mathbb{Z}_p$ -convex closure.

Proof. Writing $r = v(a)$ and $k = \lfloor r + \kappa \rfloor$, the logarithmic expansion gives $\lambda_p(a)/a \in \mathcal{O}_E^\times$. Also $u \in \mathcal{O}_E \subset I$, so the singleton input lies in the whole input. Every transported singleton lies in the closed product lattice $p^k I \times I^{m-1}$, so every tensor in $P(a)$ has component valuation at least $k - m\kappa$. The whole input at the identity contains

$$(a\pi^{1-e}, \pi^{1-e}, \dots, \pi^{1-e}).$$

Its tensor has component valuation $r - m\kappa < k - m\kappa$, proving strictness. Convex closure keeps the smaller input in its closed product container and retains the displayed larger input. $\square$

Proposition 12.2 supplies a positive cohomological realization of a specified multiplication image. It is an enlargement of a single-class recipe. Proposition 12.3 shows why equality of coefficient norms alone cannot justify omitting this enlargement.

12.4. One actual arrow for six normalized inputs. We use the field of (83): $p = 139$, $v(b_0) = 1$, $\pi^{105} = b_0$, and $E = \mathbb{Q}_p(\mu_{210}, \pi)$. Thus $(d, e, f) = (210, 105, 2)$ and $I = \pi^{-104}\mathcal{O}_E$. Put

$$r_0 = \pi^{15}, \quad \alpha_s = r_0^s, \quad \tau_s = \lambda_p(\alpha_s), \quad n_s = \lfloor s/7 \rfloor, \quad k_s = \lfloor s/7 + 104/105 \rfloor \quad (s = 1, 4, 9),$$

and define two normalized families

$$(100) \quad T_s = \tau_s/p^{k_s}, \quad X_s = \alpha_s\pi^{-104}/p^{n_s}.$$

Lemma 12.4 (A common arrow for both families). *One actual integral Kummer arrow F satisfies*

$$v(Fu) = v(FT_s) = v(FX_s) = -104/105 \quad (s = 1, 4, 9).$$

Proof. The triples (n_s) and (k_s) are $(0, 0, 1)$ and $(1, 1, 2)$. The native valuations are

s	1	4	9
$v(T_s)$	$-90/105$	$-45/105$	$-75/105$
$v(X_s)$	$-89/105$	$-44/105$	$-74/105$

Thus all six vectors are in $\pi I \setminus pI$. The Kummer conjugates show $\text{Tr}(X_s) = 0$, since $105 \nmid 15s - 104$, including for the negative powers occurring here. The direct norm computation used in Theorem 11.9 gives

$$\text{Tr}(T_s) = \frac{30 \log(1 + p^7 b_0^s)}{p^{k_s+1}}, \quad v(\text{Tr}(T_s)) = s + 6 - k_s \in \{6, 9, 13\}, \quad \text{Tr}(u) = 210u \in \mathbb{Z}_p^\times.$$

For the norm formula, the seven distinct conjugates of $1 + p\pi^{15s}$ multiply to $1 + p^7 b_0^s$, and there are fifteen repetitions over the unramified quadratic subfield.

Let $V = I/pI$ and $L = \pi I/pI$, of codimension two. Lemma 11.3 gives coordinates $V = \mathbf{F}_p a \oplus \mathbf{F}_p b \oplus W$ in which the b -coordinate is a unit multiple of the trace. The trace is onto on πI : the element $1/210$ lies in $\mathcal{O}_E \subset \pi I$ and has trace one. Consequently $\ker B \rightarrow V/L$ is onto. The vector u is in L with nonzero b -coordinate, and the six other vectors are nonzero in $L \cap \ker B$.

There is an actual field automorphism $\sigma(\pi) = \zeta_7 \pi$, fixing $\mathbb{Q}_p(\mu_{210})$. Modulo pI , the leading term of T_s is α_s/p^{k_s} . Proposition 12.1 places its logarithmic error in pI . The characters on T_s and X_s are therefore respectively ζ_7^s and $\zeta_7^{15s-104} = \zeta_7^{s+1}$. All six are nontrivial. Since $\mu_7 \cap \mathbf{F}_{139}^\times = \{1\}$, each of their projective orbits has length seven. Indeed a difference between a nontrivial such scalar and a scalar of $\mathbf{F}_{139}^\times$ is a residue-field unit and cannot annihilate a nonzero vector.

For each vector there is at most one $h \in \{0, \dots, 6\}$ placing σ^h of that vector on the line $\mathbf{F}_p a$. Avoid all six forbidden choices at once. This common inertia action fixes u and preserves L and the trace. Now the six vectors satisfy all hypotheses of Lemma 11.8, with $r = 6$ and $r + 1 < 139$. Its full-Galois lift sends them and u outside L simultaneously. Finally Proposition 11.5 converts the canonical action M into a Kummer action by taking the inverse Galois arrow: the resulting action is a $\mathbb{Z}_p^\times$ -multiple of M . This preserves all seven minimum valuations. $\square$

Theorem 12.5 (Exact point and whole-family hulls). *For $s = j^2$, $j = 1, 2, 3$, and $m = j + 1$, define*

$$P_j^\rho = \mathcal{H}_\Gamma(\{(\tau_s, u, \dots, u)\}), \quad S_j^\rho = \mathcal{H}_\Gamma(\alpha_s I \times I^{m-1}),$$

where Γ is the full integral Galois–Kummer family. Then

$$(101) \quad P_j^\rho = \pi^{105k_s-104m} B_m, \quad S_j^\rho = \pi^{105n_s-104m} B_m = p^{-1} P_j^\rho.$$

One common arrow attains both bounds in all three blocks. The same equalities hold with intermediate closed product convex hulls.

Proof. The containments $\tau_s \in p^{k_s} I$ and $\alpha_s I \subseteq p^{n_s} I$ give componentwise lower valuation bounds $k_s - m\kappa$ and $n_s - m\kappa$ for the two transported families. Lemma 12.4 attains the first bound on $(\tau_s, u, \dots, u)$, and the second on $(\alpha_s \pi^{-104}, u, \dots, u)$, using the same arrow: their first entries are respectively $p^{k_s} T_s$ and $p^{n_s} X_s$. Each attaining tensor has that valuation in every field component. Its principal B_m -span is the entire asserted product ideal, proving the reverse containments and closedness. Since $k_s = n_s + 1$, the two ideals differ by p^{-1} .

The containing product lattices $p^{k_s} I \times I^{m-1}$ and $p^{n_s} I \times I^{m-1}$ are closed and convex. Convex closure cannot violate the upper containments and retains the attaining tuples. $\square$

Proposition 12.1 gives the same point hull with $(\alpha_s, 1, \dots, 1)$; the same argument now also treats its product convex hull. The formulas remain valid for independent integral factor arrows if the diagonal witnesses are retained. For reference, the exponents of π in the two logarithmic scales are

j	m	P_j^ρ	S_j^ρ	P_j^{std}	S_j^{std}
1	2	−103	−208	107	2
2	3	−207	−312	108	3
3	4	−206	−311	214	109

The last two columns add $105m$ to the corresponding ρ columns. Nonintegrality in a ρ column is therefore not a violation of a bound stated in the standard cohomological coordinate.

12.5. The maximal-order ideal before transport. For $z = a \otimes 1 \otimes \dots \otimes 1$ and $\Phi_F = F^{\otimes m}$, put

$$(102) \quad \mathcal{M}_\Gamma(a) = \overline{\text{span}_{B_m}\{\Phi_F(b) : b \in zB_m, F \in \Gamma\}}.$$

The ideal zB_m is the input, not an ideal formed after applying Φ_F .

Proposition 12.6 (A trace-dual comparison for one source set). *For $0 \neq a \in \mathcal{O}_E$,*

$$(103) \quad \mathcal{H}_\Gamma(\{(a, 1, \dots, 1)\}) \subseteq \mathcal{M}_\Gamma(a) \subseteq \mathcal{H}_\Gamma(aI \times I^{m-1}).$$

Proof. The first containment follows from $z \in zB_m$. Take the integral trace dual of A_m in the finite etale algebra T_m . Trace factorizes on pure tensors; choosing an integral basis and its trace-dual basis in each factor therefore gives

$$A_m^\vee = (\mathfrak{D}_{E/\mathbb{Q}_p}^{-1})^{\otimes_{\mathbb{Z}_p} m} = I^{\otimes_{\mathbb{Z}_p} m}.$$

If $b \in B_m$ and $c \in A_m \subset B_m$, the product bc is integral in every component field. Its trace belongs to $\mathbb{Z}_p$, proving $B_m \subseteq A_m^\vee$. Consequently zB_m is contained in the $\mathbb{Z}_p$ -span of pure tensors with entries in $aI \times I^{m-1}$. Apply the $\mathbb{Q}_p$ -linear map Φ_F , then take the common B_m -span and closure. $\square$

In particular, the preceding native example gives bounds for the same pre-hull input:

$$(104) \quad P_j^\rho \subseteq \mathcal{M}_\Gamma(\alpha_{j^2}) \subseteq S_j^\rho = p^{-1}P_j^\rho,$$

and hence

$$(m\kappa - k_{j^2}) \log p \leq V_B(\mathcal{M}_\Gamma(\alpha_{j^2})) \leq (m\kappa - n_{j^2}) \log p.$$

This is a valid interval bound of width $\log p$. The next result sharpens it and determines the middle ideal exactly in the attained examples.

Proposition 12.7 (The sharper trace-dual bound). *Let $0 \neq a \in \mathcal{O}_E$, $r = v(a)$, and $k = \lfloor r + \kappa \rfloor$. If a $\mathbb{Q}_p$ -linear map $\Phi : T_m \rightarrow T_m$ satisfies $\Phi(A_m^\vee) \subseteq A_m^\vee$, then*

$$(105) \quad \text{span}_{B_m} \Phi(zB_m) \subseteq \pi^{ek-m(e-1)} B_m, \quad z = a \otimes 1 \otimes \cdots \otimes 1.$$

In particular this bounds $\mathcal{M}_\Gamma(a)$ for every family of integral factor arrows used above. The upper bound itself does not require Φ to factor as a tensor product.

Proof. Take all trace duals with respect to the absolute algebra trace $\text{Tr}_{T_m/\mathbb{Q}_p}$. Since $E/\mathbb{Q}_p$ is Galois, every component field of T_m is a copy of E . The idempotents of B_m isolate the components of the trace sum, so

$$(106) \quad B_m^\vee = \prod_{\alpha} I_{\alpha} \subseteq A_m^\vee = I^{\otimes_{\mathbb{Z}_p} m},$$

where I_{α} is the inverse different in the corresponding copy of E . There is no additional degree or component-count factor in this formula. The containment follows from $A_m \subseteq B_m$ by contravariance of the trace dual.

Each component of z has valuation r . Because $r - k \geq -\kappa$, one has

$$p^{-k}zB_m \subseteq B_m^\vee \subseteq A_m^\vee.$$

It follows that $\Phi(zB_m) \subseteq p^k A_m^\vee$. Every pure tensor in $I^{\otimes m}$ has component valuation at least $-m\kappa$, and $(\pi^{1-e})^{\otimes m}$ attains this value in every component. Consequently

$$\text{span}_{B_m} A_m^\vee = \pi^{-m(e-1)} B_m.$$

Multiplying by p^k proves (105). The bounding ideal is closed, so unions and closures preserve this upper bound. Integral factor arrows preserve $I^{\otimes m} = A_m^\vee$, as required. $\square$

Corollary 12.8 (Exact pre-ideal hull). *If the point orbit attains depth $k - m\kappa$ in every component, then*

$$\mathcal{M}_\Gamma(a) = \mathcal{H}_\Gamma(\{(a, 1, \dots, 1)\}) = \pi^{ek-m(e-1)} B_m.$$

In the degree-210 three-label example, this gives

$$(107) \quad \begin{aligned} \mathcal{M}_\Gamma(\alpha_{j^2}) &= P_j^\rho, & S_j^\rho &= p^{-1}\mathcal{M}_\Gamma(\alpha_{j^2}), \\ V_B(\mathcal{M}_\Gamma(\alpha_{j^2})) &= (m\kappa - k_{j^2}) \log p, & m &= j+1, \quad j = 1, 2, 3. \end{aligned}$$

The same actual full-Galois arrow can witness all three point attainments.

Proof. The point-orbit containment in Proposition 12.6 and the assumed attainment give the reverse inclusion to (105). In the stated example, Proposition 12.1 identifies each transported root-point principal ideal with its normalized crystalline-point ideal, including the background factors. Theorem 12.5 supplies the common attaining arrow and the separate whole-family formula. The measure follows from the depth of the resulting product ideal. $\square$

This comparison has the input type of the containing calculation in [65, Proposition 1.2, PDF pp. 10–11; Theorem 1.10, Step (v), pp. 27–28]. Indeed Φ_F preserves $I^{\otimes m}$, and hence $(\log \mathcal{O}_E^\times)^{\otimes m} = p^m I^{\otimes m}$, as required there. The displayed input is $\phi(p^\lambda B_m)$, with $\lambda = v(q^{j^2})$ at a bad place. Here $\underline{q} = q^{1/(2\ell)}$ is the root fixed in [63, Example 3.2(iv), PDF p. 71]. In the example $e = 105, p = 139$, its parameters are

$$d_i = 104/105, \quad a_i = 1/105, \quad b_i = -1/105.$$

Thus its displayed containing ideal has depth

$$\lfloor \lambda - d_I - a_I \rfloor - b_I = \lfloor \lambda \rfloor - m\kappa, \quad \lambda = j^2/7,$$

which is the upper endpoint in (104). By (107) it is a larger containing ideal, equal to p^{-1} times the exact pre-ideal hull in these examples. A loose upper bound is not a contradiction, and its agreement with the whole-family formula does not identify those input sets.

If all tensor coordinates are changed from ρ to the standard logarithm, the pre-ideal input changes from zB_m to $p^m zB_m$. Its hull then becomes $p^m \mathcal{M}_\Gamma(\alpha_{j^2}) = P_j^{\text{std}}$, with measure shift $-m \log p$. This conversion does not insert p^m into a printed $\phi(p^\lambda B_m)$ input without changing its coordinate description.

12.6. The source distinctions that remain. Three distinctions are necessary when using these local interfaces in the published constructions.

First, [56, §9.6.2 and Theorem-Definition 9.8.1.1, PDF pp. 109, 114–115] defines a chosen class from $1 + p q^{1/(2\ell)}$ and background classes from $1 + p$. The optional family in its Remark 9.6.2.1 is printed with exponent $j/(2\ell)$, whereas the squared-exponent pilot in [62, Theorem 3.11; Remark 3.11.1(i)] involves $\underline{q}^{j^2}$. The valuation scaling of [56, (4.2.2.2)] is additional marking data. It does not by itself replace an algebraic root by its squared power. For example, in our fixed native field $\lambda_p(r_0)$ belongs to $pI \setminus p^2 I$, while $\lambda_p(r_0^9)$ belongs to $p^2 I \setminus p^3 I$. No integral log-lattice isomorphism identifies these two specified elements. A change $v \mapsto cv$ changes $v(p)$ as well and does not change this integral content.

Second, the monoid representation in [62, Theorem 3.11(i)(b), PDF pp. 153–154] is a subset equipped with a multiplicative action; the pilot also has the fractional-ideal realization of its Proposition 3.7(v) and Definition 3.8(i). A point, an operator image, and a principal ideal before transport have different quantifiers. Proposition 12.1 supplies an equality for specified point hulls, and Proposition 12.2 supplies a set equality for a specified whole family. Neither permits a B_m -span to be moved across a non- B_m -linear arrow. Proposition 12.7 instead bounds the entire pre-ideal before the arrow is applied; together with point attainment it proves (107) without moving a module span across that arrow.

Third, the standard versus normalized logarithmic map must be the same on both sides of a volume comparison. Equation (96) is its explicit conversion, including background factors. The native fixed-source covariance of an all-isomorphism collation cannot at the same time be interpreted as a change of its source element or source principal ideal. Remark 6.10.2 of [57] itself restricts direct comparison of local quantities at different arithmeticoids.

The degree-210 calculation applies locally to a split rational Frey curve with full rational 2-torsion and $v(q) = 2$. Tate uniformization [71, Theorem C.14.1] makes a chosen square root of q rational: its class is a rational point of order two, so its Galois sign must be a power of q ; valuations force that power to be zero. Thus it supplies precisely the b_0 allowed above, without requiring $q = p^2$. This local observation does not establish all the initial data in [63, Definition 3.1].

There is also a small-parameter limitation to using the particular existence proof of [57, §5.8]. Its prime threshold must satisfy $\xi_{\text{prm}} > 10$, because $\theta(10) = \log 210 < 20/3$. With the normalized Q of its Definition 5.4.1 and Theorem 5.7.1, every candidate with $\sqrt{Q} \leq \ell = 7$ is therefore included in the proof's small- Q exceptional set, if it is in the chosen bounding domain. The opening display of Section 5.8 omits the degree divisor present in that definition. If this display is instead read literally as a different unnormalized quantity, the normalized value and numerical prime window cannot be reused for it. Neither reading turns a local example into an automatic

outside-exception global input. This statement concerns that proof-constructed set, not every possible sufficient exceptional set or a direct verification of initial data.

Finally, the actual global family, its product weights, the Ind3 operation, and the cross-Frobenius comparison are not computed in this section. The trace-dual bound, the attained exact pre-ideal equality, and the two positive Kummer interfaces are local theorems about their explicitly specified sets. Their p -adic analytic and full local-Galois hypotheses have not yet been completely formalized in Lean. They neither prove nor disprove the unconditional abc conjecture.

13. EXPLICIT FREY CURVES REALIZING THE LOCAL FIELDS

The local calculations above use actual finite extensions of $\mathbb{Q}_p$. We now realize their ramification by torsion fields of explicit rational elliptic curves. This separates a local model from a claim that every condition on global initial data has already been supplied. The latter construction is given separately in Section 15.

For positive coprime a, b , put $c = a + b$ and

$$D_{a,b}: \quad y^2 = x(x-a)(x+b), \quad S = a^2 + ab + b^2.$$

Direct calculation from the Weierstrass coefficients gives

$$(108) \quad \Delta(D_{a,b}) = 16(abc)^2, \quad c_4(D_{a,b}) = 16S, \quad j(D_{a,b}) = \frac{256S^3}{(abc)^2}.$$

Reducing S modulo a, b, c gives b^2, a^2, a^2 , respectively; thus $\gcd(S, abc) = 1$. At an odd prime $r \mid abc$, the displayed equation is minimal and multiplicative, and its Tate order is $2v_r(abc)$ after a splitting extension. Indeed c_4 is a unit, which both proves minimality and gives the reduction criterion [71, VII.5.1].

Lemma 13.1 (Two transvections and passage to a torsion field). *Let ℓ be a prime and let $G \subseteq \mathrm{GL}_2(\mathbf{F}_\ell)$ be a subgroup. If G contains a nonidentity transvection and an element preserving no $\mathbf{F}_\ell$ -line, then G contains $\mathrm{SL}_2(\mathbf{F}_\ell)$. If $H \triangleleft G$ has finite index prime to ℓ , then H also contains this special linear group.*

Proof. Let T be the transvection, with fixed line L , and let g be the second element. The lines L and gL differ. In a basis along them, T and gTg^{-1} have matrices

$$U(s) = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \quad V(t) = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}, \quad st \neq 0.$$

Their integer powers give $U(x)$ and $V(y)$ for every $x, y \in \mathbf{F}_\ell$, since the field is prime. Elementary row reduction shows that these two root groups generate $\mathrm{SL}_2(\mathbf{F}_\ell)$: for a matrix with nonzero upper-left entry, a lower row operation first clears the lower-left entry, and an upper operation clears the upper-right entry; diagonal matrices are products of elementary matrices. Explicitly, $W(t) = U(t)V(-t^{-1})U(t)$ has entries $\begin{pmatrix} 0 & t \\ -t^{-1} & 0 \end{pmatrix}$, so $W(t)W(-1) = \mathrm{diag}(t, t^{-1})$. If the upper-left entry is zero, a preliminary upper operation makes it nonzero. For the last assertion, the image of T in G/H has order dividing both ℓ and $[G:H]$, hence is trivial. Normality retains gTg^{-1} , so the same argument applies inside H . $\square$

Here and below the arithmetic use of this lemma invokes two classical facts. Good-reduction Frobenius has characteristic polynomial $T^2 - a_r T + r$, with $a_r = r + 1 - \#D(\mathbf{F}_r)$ [71, V.2; VII.4]. At a split multiplicative prime $p \neq \ell$ with $\ell \nmid v_p(q)$, the Tate description $\overline{\mathbb{Q}}_p^\times / q^\mathbb{Z}$ supplies a nontrivial inertia transvection on $D[\ell]$: adjoining $q^{1/\ell}$ has ramification index ℓ , while μ_ℓ is unramified. See [77, Theorems 1–2 and Proposition 3]. These statements about arithmetic representations are not consequences of the formal elementary-matrix lemma alone.

Proposition 13.2 (Degree 210 at the prime 139). *Each of the curves*

$$D_{139,279}, \quad D_{1,2362}$$

has split multiplicative reduction at 139 with native Tate order two. For either curve D , the mod-7 image over $F = \mathbb{Q}(i, D[30])$ contains $\mathrm{SL}_2(\mathbf{F}_7)$. The completion at 139 of $\mathbb{Q}(i, D[210])$ is

$$(109) \quad E = \mathbb{Q}_{139}(\mu_{210}, \pi), \quad \pi^{105} = b_0, \quad b_0^2 = q, \quad v_{139}(b_0) = 1.$$

It has ramification index 105, residue degree two, and degree 210. The second curve is already a rational Legendre curve, with parameter -2362 .

Proof. The factorizations are

$$279 = 3^2 \cdot 31, \quad 418 = 2 \cdot 11 \cdot 19, \quad 2362 = 2 \cdot 1181, \quad 2363 = 17 \cdot 139.$$

All displayed factors other than 3^2 are prime. For the first curve, reduction at 139 gives $y^2 = x^2(x+1)$; for the second it gives $y^2 = x(x-1)^2$. The respective nodes have two distinct rational tangent directions, of slopes ± 1 . Equation (108) gives $v_{139}(\Delta) = 2$ and unit c_4 in both cases.

Counting the actual points modulo five, including infinity, gives 8 and 4, respectively. The two Frobenius polynomials are $T^2 + 2T + 5$ and $T^2 - 2T + 5$; both have discriminant -16 , a nonsquare modulo seven. The transvection at 139 and Lemma 13.1 therefore give the special linear subgroup over $\mathbb{Q}$. The field F is Galois and

$$(110) \quad [F : \mathbb{Q}] \mid 2 \mid |\mathrm{GL}_2(\mathbb{Z}/30\mathbb{Z})| = 2 \cdot 6 \cdot 48 \cdot 480 = 276480 = 2^{11} 3^3 \cdot 5.$$

Its degree is prime to seven, so the normal-subgroup part of the lemma applies over F .

Full rational two-torsion makes a square root b_0 of q rational over $\mathbb{Q}_{139}$. In fact its Tate class is rational, so for every Galois element σ the quotient $\sigma(b_0)/b_0$ belongs to both $\{\pm 1\}$ and $q^{\mathbb{Z}}$. Its valuation is zero, forcing that quotient to be one. The order of 139 modulo 210 is two, so the cyclotomic field in (109) is unramified quadratic. The polynomial $T^{105} - b_0$ is Eisenstein there. The Tate torsion classes give $\mathbb{Q}_{139}(D[210]) = \mathbb{Q}_{139}(\mu_{210}, q^{1/210}) = E$; fixing their classes forces the root-of-unity multipliers to be one. The unramified quadratic field also contains i . Finally $D_{1,2362}$ is $y^2 = x(x-1)(x+2362)$, as asserted. No equality $q = 139^2$ has been assumed. $\square$

These small examples require care if used with the particular finite-exception construction of [57, §5.8]. Write

$$N_1 = 8105229 = 139 \cdot 279 \cdot 418/2, \quad N_2 = 2790703 = 2362 \cdot 2363/2.$$

In both cases the normalized Tate quantity is $Q_D = 2 \log N_i$. At two, $v_2(j) = 6$; adjoining the full three-torsion gives good reduction there, by the finite-torsion argument below. Summing the odd pole orders of j gives the asserted normalized value. The exact inequalities

$$3^{25} < N_i^2 < 2^{49}, \quad i = 1, 2,$$

and $2 < \exp(1) < 3$ imply $25 < Q_D < 49$. But the prime threshold in that existence proof must exceed ten, since $\theta(10) = \log 210 < 6 < 20/3$. Thus candidates with $\sqrt{Q_D} \leq 7$ belong to its proof-constructed small-height exceptional set when they lie in the chosen bounding domain. This observation uses the normalized definition of Q_D in Definition 5.4.1 and Theorem 5.7.1 of that source. The opening display of its Section 5.8 omits the degree divisor; a literal unnormalized reading must not reuse the preceding normalized values. This is a limitation of invoking that sufficient existence proof, not a failure of the local calculations or of a direct initial-data construction.

Theorem 13.3 (An exact balanced example at level 43). *Put*

$$p = 1289, \quad \ell = 43, \quad A = p(p^{16} + 428), \quad (a, b, c) = (A^2, A^2 + 1, 2A^2 + 1), \quad D = D_{a,b}.$$

*This is a primitive *abc* triple, and D is rationally isomorphic to the Legendre curve of parameter $\lambda = -1 - A^{-2}$. For $F = \mathbb{Q}(i, D[30])$, the following assertions hold:*

- (1) *D is good above two and at 43, and split multiplicative at every odd prime dividing abc .*
- (2) *The mod-43 image over both $\mathbb{Q}$ and F contains $\mathrm{SL}_2(\mathbf{F}_{43})$.*
- (3) *Every nonzero integral Tate order over F is prime to 43.*

(4) *The normalized Tate quantity satisfies the exact bounds*

$$(111) \quad Q_D = 2 \log \left(\frac{abc}{2} \right), \quad 1224 < Q_D < 1648 < 43^2.$$

(5) *At p , the completion of $K = \mathbb{Q}(i, D[1290])$ has degree 1290, ramification index 645, and residue degree two.*

Proof. The three endpoints are pairwise coprime, and $a + b = c$. The rational substitution $x = A^2X$, $y = A^3Y$ gives

$$Y^2 = X(X - 1)(X + 1 + A^{-2}),$$

so no quadratic twist is introduced. The exact residues are $A \equiv 5 \pmod{8}$ and $A \equiv 1 \pmod{5 \cdot 43}$. At an odd prime dividing a or b , the reduced equation is $y^2 = x^2(x + 1)$, with tangent slopes ± 1 . At a prime dividing c , translation of its node gives $y^2 = X^2(X + A^2)$, with slopes $\pm A$. All these slopes are nonzero. The endpoints are $(1, 2, 3)$ modulo 43, so the discriminant is a unit there.

At two, $v_2(abc) = 1$ and $3A^4 + 3A^2 + 1$ is odd, whence $v_2(j) = 6$. Here is a direct reason that the particular field F gives good reduction. At any completion F_w above two, the group $D[3] \simeq (\mathbb{Z}/3\mathbb{Z})^2$ is rational. If reduction were additive, both the formal subgroup $D_1(F_w)$ and the quotient $D_0(F_w)/D_1(F_w)$ would have no three-torsion. Thus $D[3]$ would inject into $D(F_w)/D_0(F_w)$, whose order is at most four in the additive case, a contradiction. These are [71, VII.2.1, VII.3.1, VII.5.1(c), VII.6.1], valid over finite extensions of $\mathbb{Q}_2$ as well. Multiplicative reduction would require negative j -valuation. The only remaining possibility is good reduction.

At five the reduced model has four points and Frobenius trace two. The discriminant -16 of $T^2 - 2T + 5$ is a nonsquare modulo 43. Also $v_p(A) = 1$, giving native Tate order four at p . Lemma 13.1, followed by (110), proves the image assertion.

For clarity, the assertion about all prime exponents does not use a factorization of the large endpoints. Let $P = \prod_{r \leq 511, r \text{ prime}} r$ be the actual primorial. The exact integer calculations give

$$a, b, c < 2^{377}, \quad \gcd(a, P) = 3, \quad \gcd(b, P) = 2, \quad \gcd(c, P) = 17,$$

and $v_3(a) = 2$, $v_2(b) = 1$, $v_{17}(c) = 1$. These identities can be verified by integer Euclidean division; the companion Lean proof also checks the primorial defined as the product over all primes in the indicated range. Every prime $r \geq 512$ has $r^{43} \geq 2^{387} > a, b, c$. Since the endpoints are pairwise coprime, it follows that

$$(112) \quad v_r(abc) < 43 \quad \text{for every prime } r.$$

At a place $w \mid r$ of F , the nonzero integral Tate order is $2e(w/r)v_r(abc)$. The ramification index divides the Galois degree in (110), which is prime to 43; equation (112) proves the claim.

The normalized sum over F is the sum of the rational odd pole orders, because

$$\sum_{w \mid r} e(w/r)f(w/r) = [F : \mathbb{Q}].$$

This proves the identity in (111). For its strict bounds, put $N = abc/2$. We have $p^{17} < A < 2p^{17}$ and $A^6 < N < 3A^6$. The integer inequalities $3^6 < p < (8/3)^8$ give

$$3^{1224} < N^2 < 9 \cdot 2^{12}(8/3)^{1632} < (8/3)^{1648}.$$

For the last step use $9 \cdot 2^{12} < 3^{12} < (8/3)^{16}$. Together with $8/3 < \exp(1) < 3$, these imply (111) without a floating point approximation.

Finally full rational two-torsion gives $b_0 = \sqrt{q} \in \mathbb{Q}_p$ with $v_p(b_0) = 2$, by the proof of Proposition 13.2. Put $K_0 = \mathbb{Q}_p(\mu_{1290})$ and choose $\pi^{645} = b_0$. The field K_0 is unramified quadratic, since $p \equiv -1 \pmod{1290}$. Writing $\eta = p^2/b_0 \in \mathbb{Z}_p^\times$, the genuine uniformizer is

$$(113) \quad \beta = \pi^{323}/p, \quad \beta^{645} = p\eta^{-323}, \quad \pi = \eta\beta^2.$$

The first polynomial in β is Eisenstein. Tate torsion identifies $K_v = K_0(\pi) = K_0(\beta)$, and K_0 contains i . The claimed degrees follow. In particular the native root $q^{1/86} = \pi^{15}$ has valuation $2/43$, not $1/43$. $\square$

The numerical upper bound on the level in [57, Theorem 5.7.1] is also satisfied here: with $\delta = 2^{12}3^3 \cdot 5$ and $d_{\text{mod}} = 1$,

$$\sqrt{Q_D} < 43 < 10\delta\sqrt{Q_D}\log(2\delta\log Q_D).$$

This directly verifies a window for the specified prime. It does not identify that prime with an unspecified existential choice in another theorem. The next sections construct the additional data and an unbounded family independently of that identification.

14. ALL SQUARE LABELS IN A TAME VALUATION-FOUR TATE FIELD

The preceding constructions extend to every square label for an arbitrary prime level. The new step is one inertia choice for both the point and whole-product inputs. We retain the actual full-Galois lifts of Section 11 and the distinct source types of Section 12.

Let $\ell \geq 7$ and p be primes such that

$$p \equiv -1 \pmod{30\ell}, \quad e = 15\ell, \quad d = 2e, \quad h = (\ell - 1)/2.$$

No condition $\ell \equiv 3 \pmod{4}$ is needed for this local result. Choose $b_0 \in \mathbb{Q}_p$ with $v_p(b_0) = 2$, and set

$$(114) \quad \begin{aligned} q &= b_0^2, & K_0 &= \mathbb{Q}_p(\mu_{2e}), & \pi^e &= b_0, & E &= K_0(\pi), \\ \beta &= \pi^{(e+1)/2}/p, & r &= \pi^{15} = q^{1/(2\ell)}, & \kappa &= 1 - 1/e. \end{aligned}$$

All roots are chosen once, before the common arrow below is selected.

Lemma 14.1 (The genuine uniformizer). *The extension $E/\mathbb{Q}_p$ is Galois with ramification index e , residue degree two, and degree d . The element β is a uniformizer; π has valuation $2/e$ and is not one. Writing $\eta = p^2/b_0 \in \mathbb{Z}_p^\times$, one has*

$$(115) \quad \beta^e = p\eta^{-(e+1)/2}, \quad \pi = \eta\beta^2, \quad r = \eta^{15}\beta^{30}, \quad r^\ell = b_0.$$

Moreover

$$(116) \quad I = p^{-1} \log \mathcal{O}_E^\times = \beta^{1-e} \mathcal{O}_E, \quad \text{Tr } I = \text{Tr}(\beta I) = \mathbb{Z}_p.$$

The trace kernel surjects onto $I/\beta I$.

Proof. The order of p modulo $2e$ is two, so $K_0/\mathbb{Q}_p$ is unramified quadratic. Since e is odd, the exponent $(e+1)/2$ is an integer, and direct calculation gives (115). Its first equation is Eisenstein over K_0 , and its second gives $K_0(\beta) = K_0(\pi)$. The roots of unity in K_0 make E a splitting field. These observations prove the field assertions.

One has $p \geq 2e - 1$, hence $e \leq p - 2$. Logarithm and exponential are inverse isometries on $1 + \beta\mathcal{O}_E$ and $\beta\mathcal{O}_E$. Dividing the logarithmic image by p proves the lattice identity; the unit in (115) is not discarded as an element. Tameness gives $I = \mathfrak{D}_{E/\mathbb{Q}_p}^{-1}$, so its trace is integral. Also $p \nmid d$ and $\text{Tr}(1/d) = 1$, with $1/d \in \mathcal{O}_E \subset \beta I$. This proves the trace identities. For $x \in I$, subtracting $\text{Tr}(x)/d \in \mathcal{O}_E$ kills its trace without changing its class modulo βI . $\square$

This field is the full local 30ℓ -torsion field of the split Tate curve with parameter q . Conversely, a split elliptic curve over $\mathbb{Q}_p$ with full rational 2-torsion and Tate valuation four supplies a b_0 as above. Indeed, the rationality of the Tate class of $\sqrt{q}$ makes its Galois ratio a power of q ; valuation zero forces that power to be zero. Applying the same argument to both Tate torsion generators proves equality with $\mathbb{Q}_p(\mu_{30\ell}, q^{1/(30\ell)})$, not merely containment. Here we use the Galois-equivariant Tate uniformization theorem [71, Theorem C.14.1].

For $1 \leq j \leq h$, put $s = j^2$ and define

$$(117) \quad \begin{aligned} z_j &= r^{j^2}, & n_j &= \left\lfloor \frac{2j^2}{\ell} \right\rfloor, & k_j &= n_j + 1 = \left\lceil \frac{2j^2}{\ell} \right\rceil, \\ u &= p^{-1} \log(1 + p), & \tau_j &= p^{-1} \log(1 + pz_j), & t_j &= p^{-k_j} \tau_j, & x_j &= p^{-n_j} z_j \beta^{1-e}. \end{aligned}$$

The ratio $2j^2/\ell$ is nonintegral in this range, and $k_j = \lfloor 2j^2/\ell + \kappa \rfloor$.

Lemma 14.2 (Contents and trace depths). *The elements t_j, x_j lie in $\beta I \setminus pI$, while $u \in \beta I$ and $\text{Tr } u$ is a unit. For every label,*

$$(118) \quad \begin{aligned} \text{Tr}(t_j) &= \frac{30}{p^{k_j+1}} \log(1 + p^\ell b_0^{j^2}), \\ v_p(\text{Tr}(t_j)) &= \ell - 1 + 2j^2 - k_j \geq \ell, \quad \text{Tr}(x_j) = 0. \end{aligned}$$

Modulo pI , t_j has the pure leading term z_j/p^{k_j} .

Proof. Write $2j^2 = \ell n_j + a_j$, with $1 \leq a_j \leq \ell - 1$. Then

$$e v_p(t_j) = 15a_j - e, \quad e v_p(x_j) = 15a_j - e + 1.$$

Both exponents are strictly greater than $2 - e$ and less than 1, which proves the lattice assertions. Also $\text{Tr } u = du \in \mathbb{Z}_p^\times$.

For $R = v_p(z_j) = 2j^2/\ell$, the logarithmic tail has valuation at least $1 + 2R$. Dividing by p^{k_j} leaves valuation at least $1 + R - \kappa \geq 1/e$, so the remainder belongs to pI . The relation $r^\ell = b_0$ and $\gcd(j^2, \ell) = 1$ give

$$N_{E/K_0}(1 + pr^{j^2}) = (1 + p^\ell b_0^{j^2})^{15}.$$

Taking logarithms and then the degree-two trace proves the first formula in (118). Its valuation follows from $v_p(b_0) = 2$, $p \nmid 30$, and $k_j \leq j^2$. Finally x_j is a rational unit times $\beta^{30j^2+1-e}/p^{n_j}$. Its exponent is 1 modulo 15, so summing its e conjugates over K_0 gives zero. The geometric sum applies also to a negative exponent. $\square$

Use the integral JW basis of Lemma 11.3, writing it as a, b, W , with W of rank $d - 2$. Let B be the b -coordinate. Lemma 14.1 gives $B(v) = \text{Tr}(v)/\text{Tr}(b)$ with unit denominator. Set

$$V = I/pI, \quad V_0 = \beta I/pI.$$

The quotient V/V_0 has dimension two, and $\ker B \rightarrow V/V_0$ is onto. Lemma 14.2 places the $2h$ nonzero inputs in $V_0 \cap \ker B$, while $B(u) \neq 0$.

Lemma 14.3 (One inertia exponent for both families). *There is one $\sigma^\nu \in \text{Gal}(E/K_0)$ that puts all t_j, x_j off the line $\mathbf{F}_p a$ in V , without changing u , trace, or membership in V_0 . At most*

$$(119) \quad 15h + (h - 1 + \ell) = 9\ell - 9 < 15\ell = e$$

of the exponents $0 \leq \nu < e$ are forbidden.

Proof. Choose $\sigma(\beta) = \zeta_e \beta$. Since $\gcd(e, p - 1) = \gcd(e, 2) = 1$, the roots μ_e have trivial intersection with $\mathbf{F}_p^\times$ after reduction. For a character of order $o \mid e$, its leading projective coefficients therefore run through o distinct $\mathbf{F}_p$ -lines. A fixed line is hit at most e/o times as $0 \leq \nu < e$ varies. If the valuation of the target vector differs, there is no hit. In every case equality of full vector lines in V requires this leading-line equality, which is all that the following count uses.

The leading character of t_j is $\zeta_e^{30j^2}$, of order ℓ , because $\gcd(e, 30j^2) = 15$. Thus each of the h point inputs forbids at most 15 exponents.

For x_j the exponent modulo e is $30j^2 + 1$. It is 1 modulo 15, so its gcd with e is either 1 or ℓ . The second possibility requires

$$30j^2 + 1 \equiv 0 \pmod{\ell}.$$

There is at most one such j in $1, \dots, h$: the two possible roots modulo the prime ℓ are a nonzero pair $j, -j$, with exactly one member in this half range. A nonexceptional character has order e and forbids at most one exponent. The possible exceptional character has order 15 and forbids at most ℓ . Thus these inputs forbid at most $h - 1 + \ell$ exponents, also a valid bound when no exception occurs. Adding the two bounds gives (119), leaving a common choice of ν . Field inertia preserves valuation and trace and fixes the rational element u , as required. $\square$

Theorem 14.4 (A common arrow for every square label). *There is one actual integral Kummer arrow $F \in \Gamma_E$ with*

$$(120) \quad v_p(Fu) = v_p(Ft_j) = v_p(Fx_j) = -\kappa \quad (1 \leq j \leq h).$$

The chosen arrow may depend on the field and the chosen roots; within these choices it is the same for all labels and both families.

Proof. Apply Lemma 14.3. Every resulting input has a nonzero W -component, because it lies in $\ker B$ but off $\mathbf{F}_p a$. The other hypotheses of Lemma 11.8 were verified above. Its required inequality is

$$2h + 1 = \ell < p.$$

It therefore gives one parabolic composition moving all inputs and u outside V_0 . The required residue vectors have integer coordinate lifts; Lemma 11.4 and the full-presentation handle construction realize this composition by one actual Galois-group automorphism. Compose it with the chosen field inertia. Being outside V_0 means valuation $-\kappa$. Finally Proposition 11.5 converts the covariant canonical action to the Kummer action of the inverse automorphism, up to a common $\mathbb{Z}_p^\times$ -factor. That factor changes none of the valuations in (120). $\square$

For $m = j + 1$, use the native algebra T_m and maximal order B_m of Section 12, and define

$$\begin{aligned} P_j^\rho &= \mathcal{H}_{\Gamma_E}(\{(\tau_j, u, \dots, u)\}), \\ S_j^\rho &= \mathcal{H}_{\Gamma_E}(z_j I \times I^{m-1}). \end{aligned}$$

Corollary 14.5 (Two attained endpoint hulls). *For every $1 \leq j \leq h$,*

$$(121) \quad \begin{aligned} P_j^\rho &= \beta^{ek_j - (e-1)m} B_m, \\ S_j^\rho &= \beta^{en_j - (e-1)m} B_m = p^{-1} P_j^\rho. \end{aligned}$$

The same arrow witnesses every equality, and the spans are closed.

Proof. For the point hull, apply Proposition 11.10 to $\tau_j = p^{k_j} t_j$ and (120). For the whole product, $z_j I \subset p^{n_j} I$ bounds each component valuation below by $n_j - m\kappa$. The first input $z_j \beta^{1-e} = p^{n_j} x_j$ and the repeated remaining input u attain that bound under the same arrow. The attaining tensor is a unit times the asserted power of β in every field component, so its principal B_m -span is the entire product ideal. Closedness follows from that fractional-lattice description. Since $k_j = n_j + 1$ and p is a unit times β^e , the ratio is p^{-1} . $\square$

The full principal-unit families in Proposition 12.2 realize $z_j I \times I^{m-1}$ cohomologically. Selecting those families is an explicit enlargement of the single-class input; it is not entailed by matching the coefficient norm of one point.

Proposition 14.6 (A trace-dual upper bound for the pre-transport ideal). *Let $A_m = \mathcal{O}_E^{\otimes_{\mathbb{Z}_p} m} \subset T_m$, and take trace duals using the absolute algebra trace on T_m . For any family of $\mathbb{Q}_p$ -linear maps $\Phi : T_m \rightarrow T_m$ satisfying $\Phi(A_m^\vee) \subseteq A_m^\vee$, one has*

$$(122) \quad \overline{\text{Span}_{B_m}\{\Phi(x) : x \in (z_j \otimes 1 \otimes \dots \otimes 1) B_m\}} \subseteq \beta^{ek_j - (e-1)m} B_m,$$

where the span includes every map in the specified family. No B_m -linearity or trace preservation of the maps is required.

Proof. Put $Z_j = z_j \otimes 1 \otimes \dots \otimes 1$. The product decomposition $T_m \simeq \prod_{d^{m-1}} E$ identifies its absolute algebra trace with the sum of the component field traces. Testing against the individual idempotents in B_m gives

$$B_m^\vee = \prod_{d^{m-1}} I \subseteq A_m^\vee = I^{\otimes_{\mathbb{Z}_p} m}.$$

The inclusion reverses $A_m \subseteq B_m$; the last equality follows by tensoring an integral basis and its trace-dual basis. There is no factor equal to the number of product components.

Every component of Z_j has valuation $R_j = 2j^2/\ell$. Since $k_j = \lfloor R_j + \kappa \rfloor$, one has

$$p^{-k_j} Z_j B_m \subseteq B_m^\vee \subseteq A_m^\vee.$$

Thus $Z_j B_m \subseteq p^{k_j} A_m^\vee$. Applying any allowed Φ preserves this containing lattice by $\mathbb{Q}_p$ -linearity. Its B_m -span is contained in

$$p^{k_j} \text{Span}_{B_m}(I^{\otimes_{\mathbb{Z}_p} m}) = \beta^{ek_j - (e-1)m} B_m.$$

The last equality follows from factor valuations at least $-\kappa$, attained by $\beta^{1-e} \otimes \dots \otimes \beta^{1-e}$ in every component. This product ideal is closed, completing the proof. $\square$

Corollary 14.7 (The exact pre-transport pilot and the two scales). *For the pre-transport ideal hull of (102),*

$$(123) \quad \mathcal{M}_{\Gamma_E}(z_j) = P_j^\rho \subsetneq S_j^\rho = p^{-1} P_j^\rho.$$

The equality still holds upon enlarging the allowed tensor arrows to any family of $\mathbb{Q}_p$ -linear maps preserving $A_m^\vee$ and containing the already constructed attaining diagonal arrow. Every exponent in (121) is strictly negative. Consistently multiplying all m coordinates and the corresponding source by p gives

$$(124) \quad P_j^{\text{std}} = \beta^{ek_j + m} B_m, \quad S_j^{\text{std}} = \beta^{en_j + m} B_m,$$

whose valuation exponents are strictly positive, and $p^m \mathcal{M}_{\Gamma_E}(z_j) = P_j^{\text{std}}$.

Proof. Proposition 12.1 identifies the point orbit hull with that of $(z_j, 1, \dots, 1)$, separately after every integral arrow. Since $Z_j \in Z_j B_m$, the attaining diagonal arrow supplies $P_j^\rho \subseteq \mathcal{M}_{\Gamma_E}(z_j)$. Each integral tensor arrow preserves $A_m^\vee = I^{\otimes_{\mathbb{Z}_p} m}$, so Proposition 14.6 proves the opposite inclusion. The same proof works for the larger stated map families. This strengthens Proposition 12.6 for the same pre-transport input; it never moves the B_m -span through a merely $\mathbb{Q}_p$ -linear map. The whole-product hull is strictly larger because $p^{-1} P_j^\rho \neq P_j^\rho$.

For $j \leq h$, one has $2j^2/\ell < j$, hence $k_j \leq j$. Thus

$$ek_j - (e-1)(j+1) \leq j - (e-1) < 0.$$

The whole exponent is smaller by e . The scale rule (96) adds em to each exponent, giving $ek_j + m > 0$ and $en_j + m > 0$ and proving (124). $\square$

For additive Haar measure on the entire T_m , normalized by $\mu_B(B_m) = 1$, put $V_B(H) = d^{-m} \log \mu_B(H)$. Then

$$(125) \quad \begin{aligned} V_B(\mathcal{M}_{\Gamma_E}(z_j)) &= V_B(P_j^\rho) = (m\kappa - k_j) \log p > 0, \\ V_B(S_j^\rho) &= V_B(P_j^\rho) + \log p. \end{aligned}$$

The remaining $\log p$ difference belongs to the larger whole source. After consistent standard scaling, the exact volumes of the pre-transport pilot and the whole source are respectively $-(k_j + m/e) \log p$ and $-(n_j + m/e) \log p$, both strictly negative. Thus the signs of valuation exponents and logarithmic Haar volumes are opposite. This Haar measure is not a probability measure restricted to B_m .

The equality in (123) identifies two native hulls, not their source objects, and it leaves a strict distinction from the whole-product source. This result does not supply global initial data, a compact bounding domain, escape from a finite exceptional set, the additional Ind3 unions, or the cross-Frobenius comparison. The specified native square powers and coordinate scale remain part of its statement.

The finite arithmetic of the half-range exception, the budget $9\ell - 9$, the strict floor/ceiling bracket, and the exponent identities is kernel-checked in the companion module `IUTGeneralTameSquareLabels20260830`. The local-field, logarithmic, full-Galois-lift and pilot arguments above remain paper proofs using the stated mathematical inputs; the companion does not introduce them as Lean axioms. The algebraic trace-dual and transported-span steps are separately checked in `TraceDualPreidealHull20260831` using the actual algebra trace; this does not formalize the local inverse different, tensor-basis identification, or attaining Galois arrow.

15. CONSTRUCTING THE GLOBAL INITIAL THETA DATA

The local fields of Section 13 arise from a specific rational elliptic curve. We first construct its additional global covering, cusp, and section of places required by the original definition of initial theta data, then extend the construction to every member of Theorem 16.2. The construction uses [63, Definition 3.1(a)–(f)] directly. It does not invoke an existence theorem outside a finite exceptional set, and it keeps the chosen prime and the global base field fixed.

For the first construction put

$$(126) \quad \begin{aligned} \ell &= 43, & p &= 1289, & A &= p(p^{16} + 428), \\ a &= A^2, & b &= A^2 + 1, & c &= 2A^2 + 1, \\ D : y^2 &= x(x - a)(x + b). \end{aligned}$$

The rational change $x = A^2X$, $y = A^3Y$ identifies D with the Legendre curve of parameter $-1 - A^{-2}$. Define

$$(127) \quad \begin{aligned} F &= \mathbb{Q}(i, D[30]), & K &= F(D[43]) = \mathbb{Q}(i, D[1290]), \\ X_F &= D_F \setminus \{O\}, & C_F &= [X_F / \{\pm 1\}], & C_K &= C_F \times_F K, \\ S &= \{r : r \text{ is an odd prime and } r \mid abc\}. \end{aligned}$$

Fix an algebraic closure $\overline{F}$ containing these fields. Quotients by inversion are stack quotients; in particular, their finite étale covering maps are not claims about unramified maps of coarse curves. The set S is finite and contains 1289.

We first make explicit the two conditions that cannot be replaced by genericity assertions: stable reduction above 2 and the required core.

Lemma 15.1 (The place above 2). *Let k be a finite extension of $\mathbb{Q}_2$ and let E/k be an elliptic curve with rational full 3-torsion and integral j -invariant. Then E has good reduction over k . In particular, D has good reduction at every place of F above 2.*

Proof. Use a minimal Weierstrass equation over the complete valuation ring of k to define $E_0(k)$ and $E_1(k)$. Its residue field is finite and perfect, as required in Chapter VII of [71]. Suppose that reduction is additive. The group $E_1(k)$ has no nonzero 3-torsion, by [71, VII.3.1(a)]; the quotient $E_0(k)/E_1(k)$ is the additive group of the residue field, by [71, VII.2.1 and VII.5.1(c)], and also has no 3-torsion. Hence $E_0(k)[3] = 0$. The rational group $E[3] \cong (\mathbb{Z}/3\mathbb{Z})^2$ would inject into $E(k)/E_0(k)$. The latter group has order at most four in the additive case, by the Kodaira–Néron bound [71, VII.6.1]. This is a contradiction. The cited bound applies in residue characteristic two and over each finite extension k ; no unramifiedness assumption is being made. Finally, multiplicative reduction would force $v_k(j(E)) < 0$, whereas the j -invariant was assumed integral. Only good reduction remains.

For (126), A is odd, so $A^2 \equiv 1 \pmod{8}$. Writing $S_D = 3A^4 + 3A^2 + 1$, one has

$$j(D) = \frac{256S_D^3}{(abc)^2}, \quad v_2(b) = 1, \quad v_2(S_D) = 0.$$

Thus $v_2(j(D)) = 6$. Every completion of F contains $D[3]$, and the first assertion applies. Notice that the minimal equation over the completion need not be the rational equation in (126). $\square$

The references in this argument have a useful precise location in the archived second edition of [71]: the Chapter VII local-field hypotheses are on printed page 185, and VII.2.1, VII.3.1(a), VII.5.1(c), and VII.6.1 are on printed pages 188, 192, 196, and 200, respectively. In the archived PDF these are pages 201, 204, 208, 212, and 216, respectively.

Proposition 15.2 (The required core). *The hyperbolic orbicurve C_K in (127) is a K -core in the sense of [78, Remark 2.1.1]. The same assertion holds after replacing K by any of its completions.*

Proof. Takeuchi's classification gives exactly four geometric arithmetic once-punctured elliptic curves [80, Theorem 4.1(i)]. Their j -invariants, as computed in [82, Section 3.1, Table 4], are

$$(128) \quad \frac{2^{14}31^3}{5^3}, \quad \frac{2^273^3}{3^4}, \quad 1728, \quad 0.$$

All four are integral at 1289. The curve D satisfies $v_{1289}(j(D)) = -4$, by the split multiplicative calculation in Section 13. Thus its once-punctured curve is not among the four arithmetic curves. Arithmeticity is unchanged under finite étale coverings, so the hemi-elliptic quotient is also nonarithmetic over an algebraic closure.

By [78, Proposition 2.7], a nonarithmetic punctured hemi-elliptic orbicurve over an algebraically closed field of characteristic zero is its own core. Proposition 2.3(i)–(ii) of the same paper gives invariance under algebraically closed base extension and descent from a separable closure. Apply these results first over K and then over each completion. They give the asserted cores. $\square$

The source locations for this step are author-PDF pages 9–10 and 14–15 of [78], published page 392 (PDF12) of [80], and PDF11 of the pinned arXiv version of [82]. In particular, the argument does not replace the four exclusions by a non-CM hypothesis.

Lemma 15.3 (Decorated quotient transitivity). *Let V be a two-dimensional vector space over a field, equipped with a nondegenerate alternating form. The group $\mathrm{SL}(V)$ acts transitively on pairs $(H, \bar{v})$ consisting of a line $H \subset V$ and a nonzero vector $\bar{v} \in V/H$.*

Proof. Choose a lift v of $\bar{v}$. There is a unique $h \in H$ with $\omega(h, v) = 1$. The pair (h, v) is a symplectic basis. Given a second decorated pair, construct (h', v') in the same way. The linear map $h \mapsto h', v \mapsto v'$ preserves the alternating form, has determinant one, and carries the first decorated quotient to the second. $\square$

Theorem 15.4 (Explicit initial theta data). *For every nonempty subset $S_0 \subseteq S$, there exist a hyperbolic orbicurve $\underline{C}_K$, a cusp ϵ of it, and a subset $\mathbb{V} \subseteq V(K)$ mapping bijectively to $V(\mathbb{Q})$, such that*

$$(129) \quad (\bar{F}/F, X_F, 43, \underline{C}_K, \mathbb{V}, S_0, \epsilon)$$

satisfies all of [63, Definition 3.1(a)–(f)]. More precisely, one may fix any line $H \subset D[43](K)$ and any nonzero $\bar{v} \in D[43](K)/H$ in advance, construct a single global cover and cusp from them, and then choose $\mathbb{V}$ with the required local properties. The fields F and K remain exactly those in (127).

Proof. We follow the letters of the source definition, whose full statement is on author-PDF pages 61–63 of [63].

The arithmetic requirements (a)–(c). The field F contains i and all 6-torsion, and is Galois over $\mathbb{Q}$. Section 13 gives

$$(130) \quad [F : \mathbb{Q}] \mid 2 \mid |\mathrm{GL}_2(\mathbb{Z}/30\mathbb{Z})| = 276480, \quad 43 \nmid [F : \mathbb{Q}],$$

and proves that the image of G_F on $D[43]$ contains $\mathrm{SL}_2(\mathbb{F}_{43})$. Its kernel cuts out K . Since the pointed curve has a model over $\mathbb{Q}$, every automorphism of $\bar{\mathbb{Q}}/\mathbb{Q}$ fixes its geometric isomorphism class. Consequently its field of moduli is exactly $F_{\mathrm{mod}} = \mathbb{Q}$.

Every odd bad prime of D belongs to S and is already split multiplicative over $\mathbb{Q}$. All other finite primes except possibly 2 are good. Lemma 15.1 treats 2. These properties persist after the extension to F . The zero section makes the smooth genus-one model stable as a one-pointed curve. At a multiplicative place it gives a stable nodal model, either directly from the minimal cubic or by contracting the unmarked two-valent components of the semistable polygon. Thus X_F has stable reduction at every finite place, in the pointed-curve sense needed in the definition.

For each $r \in S$, Section 13 proves

$$1 \leq v_r(abc) < 43, \quad \mathrm{ord}_w(q_w) = 2e(w/r)v_r(abc) \quad (w \mid r \text{ in } F).$$

The ramification index divides (130), so 43 is prime to all of these positive Tate orders. It is also different from every $r \in S$, since $(a, b, c) \equiv (1, 2, 3) \pmod{43}$. This verifies all the required

order and residue-characteristic conditions for S_0 , including the choice $S_0 = S$. There is no need to know the complete factorization of the large endpoints: the finite divisibility certificate and size bound in Section 13 prove the displayed inequalities.

The global covering in (d). Put $V = D[43](K)$ and fix $(H, \bar{v})$ as in the statement. Take the quotient isogeny and its dual

$$\phi : D_K \longrightarrow D_K/H, \quad \psi : D_K/H \longrightarrow D_K, \quad \psi\phi = [43].$$

Both are defined over K , since V is rational over K . Their kernels give the exact sequence of constant finite groups

$$(131) \quad 0 \longrightarrow H \longrightarrow V \xrightarrow{\phi} Q := \ker \psi \longrightarrow 0.$$

In particular, $Q \cong V/H$ is cyclic of order 43. Set

$$(132) \quad \underline{X}_K = (D_K/H) \setminus Q, \quad \underline{C}_K = [\underline{X}_K / \{\pm 1\}].$$

The restriction of ψ is a degree-43 finite étale cyclic cover $\underline{X}_K \rightarrow X_K$ with group Q . Its cusps are all K -rational. Since the cover extends étale across the origin of the completed elliptic curve, inertia at that cusp acts trivially on its fiber. The residue-field part of the cusp decomposition group also acts trivially, because every point of Q is rational. The quotient of the fundamental group is therefore trivial on the full cusp decomposition group and surjective on geometric monodromy. As an isogeny cover, it factors through the elliptic abelian quotient. This is precisely the cyclic quotient used in [79, Definition 2.1 and its preceding construction].

Inversion yields the cartesian diagram

$$(133) \quad \begin{array}{ccc} \underline{X}_K & \longrightarrow & X_K \\ \downarrow & & \downarrow \\ \underline{C}_K & \longrightarrow & C_K. \end{array}$$

The lower arrow is finite étale by descent along $X_K \rightarrow C_K$. Thus $\underline{C}_K$ has type $(1, 43\text{-tors})^\pm$ in the source terminology. It is not necessary, and generally not true, that this lower cover is Galois. By Proposition 15.2, C_K is a K -core. Finite étale commensurable orbicurves have the same category of finite étale correspondences, so this is also the K -core of $\underline{C}_K$.

The nonzero element $\bar{v} \in V/H$ gives a nonzero cusp in Q through (131); its inversion orbit gives a cusp ϵ of $\underline{C}_K$. Both the cover and ϵ are now fixed globally. They will not be changed in the construction at the different primes.

The section and the local cusp conditions in (e)–(f). For each $r \in S_0$, initially choose any place w of K above r . Split Tate uniformization over $F_{w|F}$ and then K_w gives

$$(134) \quad 0 \longrightarrow \mu_{43} \longrightarrow D[43](K_w) \longrightarrow \mathbb{Z}/43\mathbb{Z} \longrightarrow 0.$$

All these torsion points come from V . The kernel defines a line $H_w \subset V$, and the class of $q_w^{1/43}$ determines a nonzero vector $\bar{v}_w \in V/H_w$. Changing the root only multiplies by μ_{43} . Reversing the Tate orientation changes the vector's sign, which is permitted for the cusp of the inversion quotient. For the following argument choose one of the two orientations.

The decorated pairs are equivariant under $\text{Gal}(K/F)$. With the convention $(gw)(x) = w(g^{-1}x)$, the map of completions $g : K_w \rightarrow K_{gw}$ transports Tate uniformization and its multiplicative subgroup. It also transports the graph generator. Thus

$$(135) \quad (H_{gw}, \bar{v}_{gw}) = g(H_w, \bar{v}_w)$$

for compatible orientation choices; the equality modulo sign does not require those choices. Lemma 15.3, together with the actual $\text{SL}_2(\mathbb{F}_{43})$ in the Galois image, therefore gives $g_r \in \text{Gal}(K/F)$ such that $g_r(H_w, \bar{v}_w) = (H, \bar{v})$. Choose $v_r = g_r w$ as the selected place above r .

The isogeny direction is significant. At this place, write the curve as $E(q)$ and the matched line as $H = \mu_{43}$. Then

$$(136) \quad \begin{aligned} \phi : E(q) &\longrightarrow E(q^{43}), & [z] &\longmapsto [z^{43}], \\ \psi : E(q^{43}) &\longrightarrow E(q), & [z] &\longmapsto [z]. \end{aligned}$$

The kernel of ψ is generated by $[q]$, and $\phi([q^{1/43}]) = [q]$. Accordingly ψ , which defines our cover, is the graph cover obtained by reducing the natural $\Pi_X^{\text{tp}} \rightarrow \mathbb{Z}$ quotient modulo 43. Its distinguished cusp is the image of the generator, up to sign. This is the graph quotient constructed at the beginning of [79, Section 1]; its profinite completion is the $\widehat{\mathbb{Z}}$ quotient in Definition 2.5(i) of that paper. It follows that $\underline{C}_{v_r}$ has type $(1, \mathbb{Z}/43\mathbb{Z})^\pm$, and that ϵ_{v_r} is its specified ± 1 cusp.

At all remaining rational places, including the archimedean place, choose any one place of K above it. Combining these choices with the v_r gives $\mathbb{V}$. The source requires a section of $V(K) \rightarrow V(F_{\text{mod}})$, with no equivariance, continuity, or previously fixed section condition. Thus the g_r can be chosen independently at the finitely many bad primes. There is still one global H , one global cover, and one global ϵ ; the argument has not substituted different global covers at different places.

The auxiliary theta covers. We finish by checking the auxiliary construction recalled in parts (d)–(f), without extending F . For odd ℓ , inversion determines the canonical geometric subgroups of the theta cover, by [79, Proposition 2.2]. An arithmetic splitting of the indicated cusp extension gives a model over the base field; the choices form a torsor under $H^1(G_k, \mu_{43}) = k^\times / (k^\times)^{43}$. Part (d) of the initial-data definition only invokes the geometric subgroups. Part (e) then invokes their natural local models at each selected bad place; it does not prescribe a single arithmetic splitting over K compatible with all local splittings.

Let $k = K_v$ at a selected bad place. Its residue characteristic is odd and prime to 43, the curve is split multiplicative, and $i \in k$. The nonarithmeticity needed at the beginning of [79, Section 2] follows from Proposition 15.2 and its proof. Moreover rational full 2-torsion implies $q^{1/2} \in k$. Indeed, a square root b_0 represents a rational Tate 2-torsion class. For every $\sigma \in G_k$, the ratio $\sigma(b_0)/b_0$ belongs to $q^\mathbb{Z}$ and has valuation zero, so it equals one. Thus $k = k(\mu_2, q^{1/2}) = \check{k}$, the field condition of [79, Definition 2.5]. The root-of-unity condition recalled in Remark 1.10.1(ii) of that paper also holds: rational $D[3]$ and the Weil pairing give $\mu_3 \subset F$, while $i \in F$ gives $\mu_4 \subset F$. Hence $\mu_{12} \subset F \subset k$, without any further field extension.

The classical theta series in [79, Proposition 1.4] has value $2i$ plus terms of positive valuation at the point represented by i . This value is a unit, so a k -unit multiple normalizes it to one and gives a class of standard type in Definition 1.9 of that paper. Theorem 1.10(iii) there supplies the compatible $\{\pm 1\}$ -structure on the cusp torsor. Passing to the order-43 quotient gives the required splitting class: the two representatives differing by -1 agree in $k^\times / (k^\times)^{43}$ because $(-1)^{43} = -1$. Proposition 2.2 with that splitting gives exactly the local theta cover in Definition 2.5(i). All cusps used here are k -rational; also $\mu_{43} \subset k$ by the Weil pairing on the rational 43-torsion.

These are the local models invoked in [63, Definition 3.1(e)]. They use no global extension obtained by adjoining arbitrary 43rd roots of constants. The oriented covers at the end of part (f) are then determined by the fixed triple $(X_K, \underline{C}_K, \epsilon)$, as in Definition 1.1 and Remark 1.1.2 of [63]. This verifies the final auxiliary construction and completes all parts of the initial-data definition. $\square$

For checking the last part directly, the relevant author-PDF pages of [79] are: the graph quotient on page 11; $\check{k} = k(\mu_2, q^{1/2})$ on page 16; Proposition 1.4 on pages 20–21; Definition 1.9 and Theorem 1.10 on pages 27–28; and Definitions 2.1, 2.5 with Proposition 2.2 on pages 32–36. The original PDFs and their hashes are recorded separately, so these locations refer to fixed source objects.

15.1. A uniform construction along the power-free family. The covering construction uses no property specific to the number 43 beyond the arithmetic hypotheses already verified. The following criterion states precisely what can vary.

Proposition 15.5 (A rational initial-data criterion). *Let $D/\mathbb{Q}$ be an elliptic curve, let $\ell \geq 7$ be prime, and set*

$$F_D = \mathbb{Q}(i, D[30]), \quad K_D = F_D(D[\ell]) = \mathbb{Q}(i, D[30\ell]).$$

Suppose that D_{F_D} has good or multiplicative reduction at every finite place, that the image of G_{F_D} on $D[\ell]$ contains $\text{SL}_2(\mathbb{F}_\ell)$, and that $v_{p_0}(j(D)) < 0$ for some rational prime $p_0 > 5$. Let S_0 be

a nonempty finite set of odd rational primes, different from ℓ , where $D/\mathbb{Q}$ is split multiplicative. Suppose that every integral Tate order at a place of F_D above S_0 is prime to ℓ .

For any line $H \subset D[\ell](K_D)$ and any nonzero $\bar{v} \in D[\ell](K_D)/H$, the quotient isogeny and its dual construct one global cover $\underline{C}_{K_D}$ and cusp ϵ . A section $\mathbb{V} \subset V(K_D)$ of the rational place map can be chosen such that

$$(\bar{F}_D/F_D, D_{F_D} \setminus \{O\}, \ell, \underline{C}_{K_D}, \mathbb{V}, S_0, \epsilon)$$

satisfies [63, Definition 3.1(a)–(f)]. No extension of F_D or K_D is required.

Proof. The universal degree bound $[F_D : \mathbb{Q}] \mid 276480 = 2^{11}3^35$ is prime to ℓ . The field is Galois, contains i and rational full 6-torsion, and the original pointed curve has field of moduli $\mathbb{Q}$. The mod- ℓ kernel cuts out exactly K_D . The reduction hypothesis gives a stable one-pointed model at every finite place, by the same smooth or nodal construction used above. The image and Tate-order hypotheses then give all the other conditions in parts (a)–(c). The orders are imposed over F_D , as in the source, not over K_D .

All four invariants in (128) are integral at every prime greater than 5. The condition at p_0 therefore proves nonarithmeticity by the same classification and excludes no additional curves. The proof of Proposition 15.2 gives a core over K_D and over each completion.

Take the quotient $\phi : D \rightarrow D/H$ and its dual $\psi : D/H \rightarrow D$ over K_D . Their constant kernel sequence is $0 \rightarrow H \rightarrow D[\ell] \rightarrow \ker \psi \rightarrow 0$. Removing $\ker \psi$ and quotienting by inversion constructs the cover exactly as in (132). Its cusps are rational, its geometric monodromy is cyclic of order ℓ , and its full cusp decomposition group acts trivially on the fiber at the origin. It therefore has the type required in part (d), with the same core. The class $\bar{v}$ fixes one nonzero global cusp.

Lemma 15.3 applies over $\mathbb{F}_\ell$. For each $r \in S_0$, choose any place w above it and then an actual $g_r \in \text{Gal}(K_D/F_D)$ carrying its Tate decorated quotient to the fixed $(H, \bar{v})$. Formula (135) is independent of 43, so selecting $g_r w$ gives the required local interpretation of that one global cover and cusp. The dual isogeny is $E(q^\ell) \rightarrow E(q)$, $[z] \mapsto [z]$; its graph generator is $[q]$, the image of $[q^{1/\ell}]$. All remaining places may be selected arbitrarily. This verifies the section and generator requirements in parts (e)–(f).

Finally, the auxiliary-cover argument above only uses oddness of ℓ , the original full mod- ℓ torsion, and the explicit local conditions. At a selected place k , rational 2-torsion gives $k = \ddot{k}$; rational 3-torsion and i give $\mu_{12} \subset k$; and the theta value at i is a unit in odd residue characteristic. EtTh’s canonical geometric subgroups and natural local theta splittings therefore apply. The sign ambiguity disappears modulo $(k^\times)^\ell$ because $(-1)^\ell = -1$. For the oriented covers, $\ell \geq 7$ is prime to 6, and the original curve’s mod- ℓ abelian action is trivial over K_D . There is no extra global arithmetic splitting condition, and no need to adjoin the full ℓ -torsion of D/H . These are exactly the remaining assumptions checked in the preceding proof. $\square$

Corollary 15.6 (Initial data throughout the unbounded family). *For every member (ℓ, p, A) of Theorem 16.2, put*

$$F_A = \mathbb{Q}(i, D_A[30]), \quad K_A = \mathbb{Q}(i, D_A[30\ell]),$$

$$S_A = \{r : r \text{ is an odd prime, } r \mid A^2(A^2 + 1)(2A^2 + 1)\}.$$

For every nonempty $S_0 \subseteq S_A$, every line $H \subset D_A[\ell](K_A)$, and every nonzero quotient vector $\bar{v}$, Proposition 15.5 supplies initial theta data with precisely these fields and the chosen prime ℓ . In particular $S_0 = S_A$ is allowed. The construction introduces no further largeness threshold on ℓ beyond that in the family existence theorem.

Proof. Every odd bad prime is split multiplicative by the family calculation, and all other odd primes are good. Odd A gives $v_2(j(D_A)) = 6$, so Lemma 15.1 gives good reduction over each completion of F_A at 2. Thus the first hypothesis of Proposition 15.5 holds. The family theorem gives the required SL_2 image over F_A . At its prescribed prime $p \geq 30\ell - 1 > 5$, $v_p(A) = 1$ and $v_p(j(D_A)) = -4$, proving the core criterion uniformly. Also $p \in S_A$, so this set is nonempty, while $(a, b, c) \equiv (1, 2, 3) \pmod{\ell}$ gives $\ell \notin S_A$.

For the order condition one must use the precise power-free statement; it is not asserted that A^2 is ℓ -power-free. For $w \mid r$ in F_A , the three mutually exclusive cases give

$$(137) \quad \text{ord}_w(q_w) = \begin{cases} 4e(w/r)v_r(A), & r \mid A, \\ 2e(w/r)v_r(A^2 + 1), & r \mid A^2 + 1, \\ 2e(w/r)v_r(2A^2 + 1), & r \mid 2A^2 + 1. \end{cases}$$

Each valuation on the right lies between 1 and $\ell - 1$, and $e(w/r)$ divides the prime-to- ℓ degree $[F_A : \mathbb{Q}]$. Since ℓ is odd, all three orders are prime to ℓ . This verifies the last hypothesis of the proposition without the unnecessary stronger inequality $v_r(abc) < \ell$. The proposition now applies to each chosen S_0 and fixed decorated quotient. $\square$

When $p \in S_0$, changing the selected place by g_p leaves the exact local field in the family theorem unchanged up to isomorphism. It still has ramification index 15ℓ , residue degree two, and total degree 30ℓ . In the notation of (149), β remains a uniformizer and

$$v_p(q^{1/(2\ell)}) = 2/\ell.$$

Thus the compatible section does not remove the actual Tate unit or change the valuation to $1/\ell$. The cover and cusp are single fixed global objects for each family member; there is no assertion that distinct curves share one covering or one field.

Remark 15.7 (Scope of the construction). Theorem 15.4 and Corollary 15.6 verify the original global initial data, including one covering and cusp with simultaneous local interpretations for each curve. Neither identifies our prescribed ℓ with an existentially selected prime in [57, Theorem 5.7.1]. For the isolated level-43 example, no exceptional-set avoidance is claimed. For the unbounded family, avoidance of a finite set fixed before the family varies is the separate statement of Corollary 16.4; it uses the original compactly bounded domain specified there.

At the selected place above 1289, the completion is still, up to isomorphism, the exact field of Section 13, with ramification index 645, residue degree two, and total degree 1290. Selecting a Galois-conjugate place changes none of these quantities. The native root $q^{1/86}$ retains valuation $2/43$.

Initial data alone do not identify the published pilot family with the explicit native point family, its saturated unit family, or a principal maximal-order ideal formed before transport. The exact local pre-ideal/point-hull equality proved separately applies to its specified native source and integral arrows; this initial-data construction does not identify them with the whole published pilot. The standard Bloch–Kato versus p^{-1} log normalization, the j^2 -dependent absolute-value rescaling, the specified order of hull operations, and the remaining Ind3 and global Frobenius comparisons still need their own compatible proofs. Neither the initial-data theorems in this section nor the external core and theta-cover theorems invoked here have been formalized in Lean in this work.

16. AN UNBOUNDED FAMILY WITH PRESCRIBED LOCAL LEVEL DATA

The preceding local constructions can be supplied by rational elliptic curves of unbounded height while the relevant local bounding domain remains fixed. We prove this using a least-prime theorem and a finite sieve in one arithmetic progression. The result is an existence family with specified reduction, torsion-image, and level-window properties. No lower bound for its abc quality with an exponent greater than one is asserted.

16.1. A uniform finite sieve. An integer is k -power-free if it is not divisible by q^k for any prime q . We shall use the following estimate with k and the progression modulus both growing.

Lemma 16.1 (Power-free values in a progression). *Let $k \geq 5$, let M be a positive even integer, and let $X \geq 1$. Fix a residue class $r \bmod M$, and suppose that, for every integer $A \in [X, 2X]$ in this class, none of*

$$A, \quad A^2 + 1, \quad 2A^2 + 1$$

is divisible by q^k at a prime $q \mid M$. The number N_{good} of such A for which all three values are k -power-free satisfies

$$(138) \quad N_{\text{good}} \geq \frac{X}{M} \left(1 - 5 \sum_{q \text{ prime}} q^{-k} \right) - 1 - 5(9X^2)^{1/k} \geq \frac{3X}{4M} - 1 - 5(9X^2)^{1/k}.$$

Proof. The interval contains at least $X/M - 1$ integers of the prescribed class. If $q \nmid M$, then q is odd. The polynomial T has one root modulo q^k , and each of $T^2 + 1$ and $2T^2 + 1$ has at most two. For the latter assertion, there are at most two roots modulo q , and the derivative is nonzero at every root. A root $x \bmod q^h$ has lifts $x + tq^h$; reduction of

$$f(x + tq^h) \equiv f(x) + tq^h f'(x) \pmod{q^{h+1}}$$

gives exactly one admissible $t \bmod q$. This proves the root bound at every exponent without a uniformity assumption on the prime.

Each of these at most five roots, combined with $r \bmod M$, specifies one class modulo Mq^k . It contributes at most $X/(Mq^k) + 1$ integers. All three polynomial values are positive and at most $9X^2$. Thus only primes

$$q \leq Y := (9X^2)^{1/k}$$

can exclude an integer. Summing these upper bounds over $q \leq Y$ excludes at most

$$\frac{5X}{M} \sum_{q \text{ prime}} q^{-k} + 5Y$$

integers. The sum of the error terms is finite; no error of size one is summed over infinitely many primes. Subtracting gives the first inequality. Finally,

$$\sum_{q \text{ prime}} q^{-k} \leq \sum_{n=2}^{\infty} n^{-k} \leq 2^{-k} + \frac{2^{1-k}}{k-1} \leq \frac{3}{2} 2^{-k}.$$

For $k \geq 5$, five times this bound is at most $15/64 < 1/4$. □

16.2. The existence theorem. The only prime-distribution input needed below is the effective estimate of Xylouris [83, Theorem 1.1, author manuscript p. 9]. With $P(m)$ the maximum, over reduced classes modulo m , of the least prime in the class, it gives an effective absolute constant C such that

$$(139) \quad P(m) \leq Cm^{26/5}.$$

We use the exponent 5.2 of that archived author version throughout; the later improvement is unnecessary. In particular, C is independent of both the modulus and the reduced class.

Theorem 16.2 (An unbounded balanced Frey–Legendre family). *For every sufficiently large prime $\ell \equiv 43 \pmod{60}$, let p be the least prime congruent to $-1 \pmod{30\ell}$, and put*

$$(140) \quad M = 10\ell p^2, \quad X = \exp(\ell^2/16).$$

There are at least $X/(2M)$ integers $A \in [X, 2X]$ satisfying

$$(141) \quad A \equiv 1 \pmod{2}, \quad A \equiv 1 \pmod{5\ell}, \quad A \equiv p \pmod{p^2},$$

for which $A, A^2 + 1, 2A^2 + 1$ are all ℓ -power-free. Every such A gives the positive primitive triple and curve

$$(142) \quad (a, b, c) = (A^2, A^2 + 1, 2A^2 + 1), \quad D_A : y^2 = x(x - a)(x + b).$$

The curve is rationally isomorphic to the Legendre curve with

$$\lambda_A = -1 - A^{-2}.$$

For $L_A = \mathbb{Q}(i, D_A[30])$, the following assertions hold:

- (1) *At p , the curve is split multiplicative with native Tate valuation $v_p(q) = 4$.*
- (2) *The mod- ℓ image over $\mathbb{Q}$ and over L_A contains $\text{SL}_2(\mathbf{F}_\ell)$. Reduction at ℓ is good.*
- (3) *Every nonzero integral Tate order over L_A is prime to ℓ .*

(4) The Tate degree normalized by $[L_A : \mathbb{Q}]^{-1}$ is

$$(143) \quad Q_A = 2 \log \frac{A^2(A^2 + 1)(2A^2 + 1)}{2}, \quad \frac{3}{4}\ell^2 < Q_A < \ell^2.$$

The parameters lie in one fixed compactly bounded domain in the original sense specified below, and both $h(\lambda_A)$ and $h(j(D_A))$ tend to infinity with ℓ . The threshold is effective in terms of the effective constant in (139); no numerical value of that constant is extracted here.

Existence of the integers. The three moduli in (141) are pairwise coprime. By (139),

$$30\ell - 1 \leq p \leq C(30\ell)^{26/5}, \quad M \leq C_0\ell^{57/5}, \quad C_0 = 10C^2 30^{52/5}.$$

The primes dividing M cause no exclusion in Lemma 16.1. At 2, A is odd, so $v_2(A^2 + 1) = 1$ and $2A^2 + 1$ is odd. At 5 and ℓ , the three values are 1, 2, 3. At p , their valuations are 1, 0, 0. All four primes are distinct.

For $k = \ell$, the error parameter is

$$Y = 9^{1/\ell} \exp(\ell/8).$$

It suffices that $1 + 5Y \leq X/(4M)$. Since $1 + 5Y \leq 46 \exp(\ell/8)$, an explicit sufficient condition is

$$(144) \quad \frac{\ell^2}{16} - \frac{\ell}{8} - \frac{57}{5} \log \ell \geq \log(184C_0).$$

The left side tends to infinity and is increasing for $\ell \geq 43$: its derivative is $\ell/8 - 1/8 - 57/(5\ell) > 0$. Lemma 16.1 now gives the lower bound $X/(2M)$. All constants are independent of the chosen ℓ, p, A .

There are infinitely many eligible primes ℓ . Indeed, if the primes congruent to 43 modulo 60 were a finite list $\ell_1, \dots, \ell_s$, the reduced CRT class congruent to 43 modulo 60 and to 1 modulo each ℓ_i would contain a prime by (139). It would be a new prime in the original progression, a contradiction. $\square$

Arithmetic and height assertions. Pairwise coprimality in (142) follows from $\gcd(A^2, A^2 + 1) = 1$ and $2(A^2 + 1) - (2A^2 + 1) = 1$. The substitution $x = A^2 x_1$, $y = A^3 y_1$ gives

$$y_1^2 = x_1(x_1 - 1)(x_1 + 1 + A^{-2}).$$

It is an isomorphism over $\mathbb{Q}$, not a quadratic twist. The general invariants (108) give, with $S_A = 3A^4 + 3A^2 + 1$,

$$(145) \quad \Delta = 16(abc)^2, \quad c_4 = 16S_A, \quad j(D_A) = \frac{256S_A^3}{(abc)^2}, \quad \gcd(S_A, abc) = 1.$$

At an odd prime $r \mid abc$, unit c_4 makes the equation minimal and multiplicative, with Tate order $2v_r(abc)$. Splitness is also uniform in A . If $r \mid a$ or $r \mid b$, the reduced node has tangent slopes ± 1 ; if $r \mid c$, translation of the node at $x = a$ gives $y^2 = x_1^2(x_1 + a)$, with slopes $\pm A$. In each case they are distinct units. These are the reduction criteria of [71, Chapter VII, Proposition 5.1]; the split Tate description is [77, Theorems 1–2]. At the prescribed p , $v_p(A) = 1$, and hence $v_p(q) = 4$. At ℓ , the endpoints are 1, 2, 3, so reduction is good.

The good reduction at 5 is the same fixed curve $y^2 = x(x - 1)(x + 2)$ used in Section 13. It has four $\mathbf{F}_5$ -points, giving Frobenius polynomial $T^2 - 2T + 5$. Its discriminant -16 is a nonsquare modulo ℓ , as $\ell \equiv 3 \pmod{4}$; thus this image element fixes no line. At p , Tate uniformization supplies a nonidentity inertia transvection of order ℓ , because $p \neq \ell$ and $\ell \nmid v_p(q) = 4$ [77, Proposition 3]. Conjugating its fixed line by the Frobenius element gives a distinct fixed line. Lemma 13.1 therefore gives the full $\mathrm{SL}_2(\mathbf{F}_\ell)$ subgroup. The same lemma applies over L_A : the latter is Galois, and (110) gives

$$(146) \quad [L_A : \mathbb{Q}] \mid 2|\mathrm{GL}_2(\mathbb{Z}/30\mathbb{Z})| = 276480 = 2^{11} 3^3 5,$$

which is prime to ℓ . The normal image over L_A retains the order- ℓ transvection and its conjugate.

At 2, (145) gives $v_2(j) = 6$, hence potential good reduction [71, Chapter VII, Proposition 5.5]. We check why the specific field L_A already gives good reduction. Potential good reduction makes the inertia image on $T_3(D_A)$ finite. Since all 3-torsion is rational over L_A , that image lies in

$1 + 3 \operatorname{Mat}_2(\mathbb{Z}_3)$, a torsion-free group. To see the last assertion, write a nonidentity element as $1 + 3^h B$, where $h \geq 1$ and $B \not\equiv 0 \pmod{3}$. Powers prime to 3 preserve h , and

$$(1 + 3^h B)^3 = 1 + 3^{h+1}(B + 3^h B^2 + 3^{2h-1} B^3)$$

increases it by exactly one. Thus no positive finite power is the identity. Inertia is trivial, and the Neron–Ogg–Shafarevich criterion [71, Chapter VII, Theorem 7.1] gives good reduction over every completion of L_A at 2.

For $w \mid r$, where r is odd and divides abc , the integral Tate order over L_A is

$$\operatorname{ord}_w(q_w) = 2e(w/r)v_r(abc).$$

If $r \mid A$, this equals $4e(w/r)v_r(A)$; otherwise it is twice the ramification index times the valuation of one of the two quadratic factors. Each latter valuation, and $v_r(A)$, lies strictly between zero and ℓ . Also $\ell \nmid e(w/r)$, by (146). Since ℓ is odd, every nonzero order is prime to ℓ . We do not need or assert that A^2 itself is ℓ -power-free.

Sum these orders with residue-degree weights and divide by $[L_A : \mathbb{Q}]$. For each r , the identity $\sum_{w \mid r} e(w/r)f(w/r) = [L_A : \mathbb{Q}]$, good reduction at 2, and $v_2(abc) = 1$ give the exact formula for Q_A in (143). Finally, for $A > 1$,

$$A^6 < \frac{A^2(A^2 + 1)(2A^2 + 1)}{2} < 3A^6.$$

Together with $X \leq A \leq 2X$, this yields

$$(147) \quad \frac{3}{4}\ell^2 < Q_A < \frac{3}{4}\ell^2 + 12\log 2 + 2\log 3 < \frac{3}{4}\ell^2 + 16 < \ell^2.$$

The remaining domain and unbounded-height assertions follow from Proposition 16.3 below. $\square$

The prime-to- ℓ order assertion is over L_A , as stated. It is not an assertion over $L_A(D_A[\ell])$, whose ramification at p has an additional factor ℓ . Set $\delta = 2^{12}3^35 = 552960$, the constant for $d_{\text{mod}} = 1$ in [57, Theorem 5.7.1]. Equation (147) also proves both numerical endpoints

$$(148) \quad \sqrt{Q_A} < \ell < 10\delta\sqrt{Q_A}\log(2\delta\log Q_A).$$

For the second inequality, use $\sqrt{Q_A} > \ell/2$ and $\log(2\delta\log Q_A) > 1$. This directly verifies a window for our chosen ℓ ; it does not identify it with an existential choice in that source.

16.3. The full local torsion field. The local field in this family is determined with its actual Tate unit retained. Put $N = 30\ell$ and $e = 15\ell$. Full rational 2-torsion implies that a square root b_0 of q belongs to $\mathbb{Q}_p$. Indeed, its Tate class is rational, so for every Galois element the ratio of a conjugate to b_0 is a power of q . The ratio has valuation zero, hence is one. Thus $v_p(b_0) = 2$.

Let $K_0 = \mathbb{Q}_p(\mu_N)$, choose $\varpi^e = b_0$, and write $u_0 = b_0/p^2$. Since $p \equiv -1 \pmod{N}$, $K_0/\mathbb{Q}_p$ is unramified quadratic. The element ϖ has valuation $2/e$; the uniformizer is instead

$$(149) \quad \beta = \frac{\varpi^{(e+1)/2}}{p}, \quad \beta^e = p u_0^{(e+1)/2}, \quad \varpi = u_0^{-1}\beta^2.$$

The equation for β is Eisenstein over K_0 . Consequently

$$E = K_0(\varpi) = K_0(\beta), \quad [E : \mathbb{Q}_p] = 30\ell, \quad e(E/\mathbb{Q}_p) = 15\ell, \quad f(E/\mathbb{Q}_p) = 2.$$

Tate uniformization identifies

$$\mathbb{Q}_p(D_A[N]) = \mathbb{Q}_p(\mu_N, q^{1/N}) = E :$$

fixing each torsion class forces its root-of-unity multiplier to be one, because a nontrivial power of q has nonzero valuation. The unramified quadratic field K_0 contains i , so E is also the completion at p of $\mathbb{Q}(i, D_A[N])$. It is Galois, being the splitting field obtained from the cyclotomic field by adjoining all roots of $T^e - b_0$. Moreover,

$$(150) \quad e = 15\ell \leq p - 2, \quad \gcd(e, p - 1) = 1,$$

since $p \geq 30\ell - 1$ and $p - 1 \equiv -2 \pmod{15\ell}$. Thus the small tame range of the local arguments occurs at unbounded global height. No substitution $q = p^4$ or $b_0 = p^2$ has been made.

16.4. A fixed domain and the finite-set quantifier. Let $U = \mathbf{P}^1 \setminus \{0, 1, \infty\}$. Consider

$$(151) \quad \begin{aligned} \mathcal{K}_\infty &= \{z \in \mathbf{C} : |z + 1| \leq 1/2\}, \\ \mathcal{K}_2 &= \{z \in \overline{\mathbb{Q}}_2 : |z + 2|_2 \leq 1/8\}. \end{aligned}$$

For nonarchimedean places, the original compactly bounded condition of [91, Example 1.3(ii), author manuscript pp. 5–6] is formulated on intersection with each finite local extension. It does not require the entire set $\mathcal{K}_2$ to be compact in $\overline{\mathbb{Q}}_2$.

Proposition 16.3 (Fixed local bounds and unbounded rational height). *The sets (151) give one compactly bounded domain of support $\{\infty, 2\}$ in that original sense. Every member of Theorem 16.2 belongs to this domain, and*

$$(152) \quad h(\lambda_A) = \log(A^2 + 1) \longrightarrow \infty, \quad h(j(D_A)) \geq Q_A \longrightarrow \infty.$$

On all of $\mathcal{K}_2$, the Legendre invariant has valuation 6.

Proof. Both sets avoid the three removed points and are stable under the respective Galois actions. The complex set is a compact regular closed disc. For every finite extension $k/\mathbb{Q}_2$,

$$\mathcal{K}_2 \cap k = -2 + 8\mathcal{O}_k$$

is compact and open, hence is the closure of its interior in k . The set $\mathcal{K}_2$ is itself open and closed in $\overline{\mathbb{Q}}_2$, so any separate regular-closed requirement also holds. These are the stated original conditions.

For $A \geq 2$, $|\lambda_A + 1| = A^{-2} \leq 1/4$. Since A is odd, A^2 and its inverse are 1 modulo 8, giving $\lambda_A \equiv -2 \pmod{8}$. Thus the domain is fixed independently of ℓ, p, A , and the degree bound is one. If $z \in \mathcal{K}_2$, then $v_2(z) = 1$, $v_2(1 - z) = 0$, and $v_2(1 - z + z^2) = 0$. The formula

$$j(z) = \frac{256(1 - z + z^2)^3}{z^2(1 - z)^2}$$

therefore gives $v_2(j(z)) = 6$. On $\mathcal{K}_\infty$, the inequalities $1/2 \leq |z| \leq 3/2$ and $|1 - z| \geq 3/2$ also give the fixed coarse bound

$$|j(z)| \leq \frac{256(19/4)^3}{(1/2)^2(3/2)^2} = \frac{438976}{9} < 50000.$$

The numerator and denominator of $\lambda_A = -(A^2 + 1)/A^2$ are coprime. Its rational height is therefore the first expression in (152). Also $abc/2$ is odd, and (145) reduces to

$$j(D_A) = \frac{64S_A^3}{(abc/2)^2}$$

in lowest terms. Its denominator alone gives $h(j(D_A)) \geq 2 \log(abc/2) = Q_A$. $\square$

Corollary 16.4 (Avoidance of any fixed finite set). *If a finite set of parameters in $U(\overline{\mathbb{Q}})$, or a finite set of geometric elliptic moduli points, is fixed before ℓ varies, all sufficiently large members of the family avoid it.*

Proof. Take the maximum of the finitely many parameter heights, or of the heights of their j -invariants, and apply (152). In particular, bounded complex absolute values of these invariants do not bound their rational heights. $\square$

16.5. The source-interface limits. The finite-set statement has its exact order of quantifiers: the set must be fixed before the family varies. It cannot be used for an exceptional set chosen separately for each member. There is also a topological difference that prevents silently substituting one definition for another.

Proposition 16.5 (The literal ambient compactness condition). *In the usual valuation topology of $U(\overline{\mathbb{Q}}_2)$, every compact subset has empty interior. Consequently no nonempty compact subset is the closure of its ambient interior.*

Proof. Every nonempty open ball contains, after translation and a sufficiently small nonzero scaling, infinitely many algebraic integral units with distinct residues in $\overline{\mathbf{F}}_2$. For example, one may take lifts in finite unramified extensions. The distances between the scaled points are all the same positive constant. Thus the ball is not totally bounded and cannot be contained in a compact set. A compact set with nonempty interior would contain such a ball, a contradiction. $\square$

Section 5.1 of the archived v2 of [57], p. 50, literally requires nonempty compact subsets in the product of the local algebraic closures and equality with the closure of their ambient interiors. Under the usual valuation topology, the proposition applies to the nonarchimedean factor, and the same obstruction holds for a product containing it. The domain (151) satisfies the original finite-extension-section definition; it must not be described as a compact subset of that full algebraic closure. Using finite-extension sections or a different topology would require stating that interpretation explicitly. This observation does not disprove the original IUT statements.

Finally, the separate existence quantifier in [57, Theorem 5.7.1, p. 53] supplies some prime in a window. Our chosen ℓ satisfies (148) and the arithmetic properties proved above directly, but these facts do not identify it with that existential prime. The covering and marking requirements of [63, Definition 3.1] for these members are supplied separately by Corollary 15.6. They do not identify the complete possible-image source including subsequent indeterminacies. The classical least-prime, elliptic-curve, and local-field inputs in this section remain mathematical source dependencies; they are not asserted as new Lean axioms or as a formal proof of the *abc* conjecture.

17. TWO-PRIME SUPPORT AND RATIONAL-RETURN RIGIDITY

We give a complete elementary bound on one moving-support subclass of *abc* triples, and then examine a proposed passage from local logarithmic arrows to rational S -unit solutions. These are different conclusions: the first is a special-case height estimate, while the second restricts the number of outputs of a specified construction. Throughout this section, $r = \text{rad}(abc)$ and $R = \log r$, as in (1).

17.1. The complete two-prime subclass.

Theorem 17.1 (A sharp bound on two-prime support). *Let a, b, c be positive integers with $a + b = c$ and $\gcd(a, b) = 1$. If $\omega(abc) \leq 2$, then*

$$(153) \quad c \leq \frac{3}{2} \text{rad}(abc).$$

Equality holds precisely for $(a, b, c) = (1, 8, 9)$ or $(8, 1, 9)$. Every other triple in this class satisfies $c \leq \text{rad}(abc)$.

Proof. The entries are pairwise coprime. Exactly one is even: the addends cannot both be even, and their parities determine that of their sum. If all three entries exceeded 1, each would contribute a different prime divisor, contradicting the support bound. An addend is therefore 1. The case $(1, 1, 2)$ has $c = r = 2$. Otherwise, after exchanging the addends if necessary, write the triple as $(1, N, N + 1)$, with $N \geq 2$. Both consecutive integers have a prime factor, and their supports are disjoint. Thus, for an odd prime q and positive integers u, v ,

$$(154) \quad \{N, N + 1\} = \{2^u, q^v\}.$$

Suppose first that $2^u = q^v + 1$. If v is even, then $q^v \equiv 1 \pmod{4}$, so $2^u \equiv 2 \pmod{4}$ and $u = 1$, contrary to $q^v + 1 > 2$. If $v \geq 3$ is odd, the factorization

$$q^v + 1 = (q + 1)(q^{v-1} - q^{v-2} + \cdots - q + 1)$$

has an odd second factor greater than 1. Indeed, its v terms are odd modulo 2, and its value is at least $q^2 - q + 1 > 1$. Such a factor cannot divide a power of 2. This ordering therefore requires $v = 1$.

Now suppose that $q^v = 2^u + 1$. An odd $v \geq 3$ is excluded by

$$q^v - 1 = (q - 1)(1 + q + \cdots + q^{v-1}),$$

whose second factor is again odd and greater than 1. If v is even, set $Q = q^{v/2} \geq 3$. Both factors in

$$(Q - 1)(Q + 1) = 2^u$$

are positive powers of 2. Write $Q - 1 = 2^s$ and $Q + 1 = 2^t$, with $1 \leq s < t$. Subtraction gives $2^s(2^{t-s} - 1) = 2$, hence $s = 1$, $t = 2$ and $Q = 3$. Since q is prime, this forces $q = 3$, $v = 2$ and $u = 3$. The resulting pair is $\{8, 9\}$.

In every remaining case $v = 1$, the radical is $r = 2q$, and the larger endpoint is q or $q + 1$, both at most $2q$. For $(1, 8, 9)$ and its exchanged triple, $c = 9$ and $r = 6$. These observations prove both the bound and its equality statement. $\square$

The estimate is uniform as the odd prime varies. For every $\varepsilon > 0$, the entire subclass in Theorem 17.1 satisfies

$$\log c \leq R + \log(3/2) \leq (1 + \varepsilon)R + \log(3/2),$$

with no constant depending on the prime support. The proof is elementary and uses no classification theorem for general consecutive perfect powers. No priority claim is made for this classical special case, and no estimate for $\omega(abc) \geq 3$ follows from it.

17.2. Nonzero rational return under the actual trace constraint. Let E, E' be finite extensions of $\mathbb{Q}_p$. We retain the coefficient-compatible arrow of Proposition 7.5: a continuous isomorphism $\alpha : G_{E'} \simeq G_E$ together with an integral equivariant coefficient isomorphism

$$\beta : \alpha^* \mathbb{Z}_p(1)_E \simeq \mathbb{Z}_p(1)_{E'}.$$

Its induced logarithmic map $F : E \rightarrow E'$ is $\mathbb{Q}_p$ -linear and satisfies, for some $\kappa \in \mathbb{Z}_p^\times$,

$$(155) \quad \text{Tr}_{E'/\mathbb{Q}_p}(Fz) = \kappa \text{Tr}_{E/\mathbb{Q}_p}(z) \quad (z \in E).$$

This is the previously proved identity (29). Its unit multiplier comes from the integral transport of $H^2(G_E, \mathbb{Z}_p(1)) \cong \mathbb{Z}_p$, together with the reciprocity cup-product formula [67, Chapter III, Proposition 3.6 and §4]. Cyclotomic character, inertia and absolute degree are recovered from the local Galois group [68, Propositions 1.1–1.2 and Corollary 1.3]. In particular the degrees of E and E' agree for these arrows. None of these assertions requires the arrow to be induced by a field isomorphism.

Lemma 17.2 (Rational return and the scalar line). *Suppose $[E : \mathbb{Q}_p] = [E' : \mathbb{Q}_p] = d$ and a $\mathbb{Q}_p$ -linear map F satisfies (155). If $z \in \mathbb{Q}_p \subset E$ and $F(z) = z' \in \mathbb{Q}_p \subset E'$, then*

$$(156) \quad z' = \kappa z.$$

For nonzero z , the output is nonzero and $v_p(z') = v_p(z)$. Moreover, one such nonzero return implies

$$F(t) = \kappa t \quad (t \in \mathbb{Q}_p).$$

Proof. The traces of the two rational elements are dz and dz' . Equation (155) gives $dz' = \kappa dz$. The positive integer d is nonzero in $\mathbb{Q}_p$, so cancellation gives (156), even when $p \mid d$. Its valuation assertion follows from $v_p(\kappa) = 0$. If $z \neq 0$, linearity gives $F(1) = z^{-1}F(z) = \kappa$, and then $F(t) = tF(1) = \kappa t$. $\square$

Equal degree is an explicit hypothesis in this linear-algebra statement. With unequal degrees d, d' , the same calculation would instead yield $z' = \kappa(d/d')z$. Uniform division of both logarithmic coordinates by p leaves the displayed trace identity unchanged; independent rescalings by different powers of p do not. The lemma does not assert that every local Galois arrow preserves the rational line. It describes what follows if a specified point returns to that line.

Lemma 17.3 (Depth of an odd-prime principal-unit logarithm). *For an odd prime p and $0 \neq \xi \in p\mathbb{Z}_p$,*

$$(157) \quad v_p(\log_p(1 - \xi)) = v_p(\xi).$$

Proof. Put $e = v_p(\xi) \geq 1$. In the convergent series

$$\log_p(1 - \xi) = - \sum_{n \geq 1} \frac{\xi^n}{n},$$

the first term has valuation e . For $n \geq 2$ one has $v_p(n) \leq n - 2$: this is immediate for $n = 2$, and for $n \geq 3$ it follows from $v_p(n) \leq \log_3 n \leq n - 2$. Every subsequent term therefore has valuation $ne - v_p(n) \geq e + 1$, and these valuations tend to infinity. The first term is the unique term of least valuation. The logarithm is nonzero and has valuation e . $\square$

For an odd prime $p \mid abc$, choose the following vanishing coordinate and its companion rational S -unit, where S is the full prime support of abc :

$$(158) \quad \begin{array}{c|cc} & \xi_p & u_p = 1 - \xi_p \\ \hline p \mid a & a/c & b/c \\ p \mid b & b/c & a/c \\ p \mid c & c/a & -b/a. \end{array}$$

Pairwise coprimality makes every denominator a p -adic unit. Thus $\xi_p \in p\mathbb{Z}_p$, $u_p \in 1 + p\mathbb{Z}_p$, and $v_p(\xi_p)$ is exactly the exponent of p in the indicated endpoint. In the last row $1 - c/a = -b/a$; the negative real sign does not prevent p -adic congruence to 1.

Proposition 17.4 (An endpoint exponent cannot change under rational return). *Let (a, b, c) and (A, B, C) be positive primitive triples, and let an odd prime p divide the same labelled endpoint in both triples. Form u_p, u'_p using that row of (158). Suppose an equal-degree coefficient-compatible logarithmic arrow satisfies the pointwise equality*

$$(159) \quad F_p(\log_p u_p) = \log_p u'_p.$$

Then the exponent of p in the indicated endpoint is the same in the two triples.

Proof. Both logarithms are nonzero rational p -adic elements by Lemma 17.3. Lemma 17.2 equates their valuations, and (157) identifies those valuations with the corresponding endpoint exponents. $\square$

The proposition imposes no condition at $p = 2$. The depth identity used here is not valid for every $\xi \in 2\mathbb{Z}_2$. Nor can (159) be replaced by membership of the two logarithms in the same ideal or hull.

17.3. The exact arithmetic fibre bound. For $n > 0$, write $n_{\text{odd}} = n/2^{v_2(n)}$. Fix a positive primitive triple (a, b, c) and put $(a_0, b_0, c_0) = (a_{\text{odd}}, b_{\text{odd}}, c_{\text{odd}})$. Let $\mathcal{F}(a, b, c)$ be the set of all positive primitive triples (A, B, C) with

$$(A_{\text{odd}}, B_{\text{odd}}, C_{\text{odd}}) = (a_0, b_0, c_0).$$

These conditions fix all odd-prime exponents in their labelled endpoints, while allowing the even endpoint to move.

Theorem 17.5 (Two possible outputs). *For every positive primitive triple,*

$$(160) \quad |\mathcal{F}(a, b, c)| \leq 2.$$

If the position of the even endpoint is fixed to that of the seed, the fibre contains only the seed. The bound two is attained.

Proof. Every target has exactly one even endpoint, with exponent $k \geq 1$. The three possible positions give

$$(161) \quad \begin{array}{c|cc} \text{even endpoint} & (A, B, C) & \text{required equation} \\ \hline A & (2^k a_0, b_0, c_0) & 2^k a_0 + b_0 = c_0 \\ B & (a_0, 2^k b_0, c_0) & a_0 + 2^k b_0 = c_0 \\ C & (a_0, b_0, 2^k c_0) & a_0 + b_0 = 2^k c_0. \end{array}$$

Each row admits at most one k , since 2^k multiplies a fixed positive integer and is strictly increasing. Either of the first two rows implies $c_0 > a_0 + b_0$, whereas the third implies $c_0 < a_0 + b_0$. The third row cannot coexist with either of the first two, so at most two triples occur.

When the even position is fixed, only one row remains, and it already contains the seed. No division by a difference of odd parts was used. In particular, if $a_0 = b_0$, coprimality forces $a_0 = b_0 = 1$ and the same argument applies. When all three odd parts are 1, the only possibility is $(1, 1, 2)$.

Finally, the two distinct triples

$$(162) \quad (4, 3, 7), \quad (1, 6, 7)$$

are primitive and have the same labelled odd parts $(1, 3, 7)$. Thus the bound is sharp. $\square$

Corollary 17.6 (The specified rational-return construction). *Fix a seed (a, b, c) with full prime support S . Consider positive primitive targets (A, B, C) satisfying all of the following:*

- (1) *Their exact prime support is S .*
- (2) *Every odd $p \in S$ belongs to the same labelled endpoint as in the seed.*
- (3) *For each odd $p \in S$, an equal-degree coefficient-compatible local logarithmic arrow gives (159), with the companion units in (158).*

There are at most two targets. If the even endpoint is prescribed as well, the only target is the seed.

Proof. Proposition 17.4 preserves every odd exponent on the specified support. Primitivity makes its exponents on the other two endpoints zero. Exact support equality excludes any new odd primes, so all three odd parts agree. Apply Theorem 17.5. Identity arrows admit the seed itself. $\square$

The local arrows may depend on the prime and the target; the upper bound requires no common global arrow. Exact support and endpoint-label preservation are hypotheses, not consequences of trace transport. Checking the old primes alone would not exclude new prime factors.

Proposition 17.7 (Sharpness with freely chosen coefficient units). *In the class of pairs (α, β) in which an identity-Galois arrow may be accompanied by any integral Tate-module coefficient unit, the upper bound in Corollary 17.6 is attained by the seed and target in (162).*

Proof. Both triples have support $\{2, 3, 7\}$; the odd prime 3 stays on the second endpoint and 7 on the third. Their companion units are

$$(u_3, u'_3) = (4/7, 1/7), \quad (u_7, u'_7) = (-3/4, -6).$$

The vanishing coordinates at 3 are $3/7, 6/7$, and those at 7 are $7/4, 7$. All have valuation one at their indicated prime. Lemma 17.3 therefore shows that

$$\kappa_p = \frac{\log_p u'_p}{\log_p u_p} \in \mathbb{Z}_p^\times \quad (p = 3, 7).$$

Take $E = E' = \mathbb{Q}_p$ and $\alpha = \text{id}_{G_{\mathbb{Q}_p}}$. Multiplication by κ_p on $\mathbb{Z}_p(1)$ is an integral equivariant coefficient isomorphism. It acts as that same scalar on unit Kummer cohomology and its logarithmic module, so it gives exactly (159). These are principal units at odd primes, where logarithm is injective; no torsion ambiguity affects the stated points. Together with the identity arrows for the seed, this supplies two targets in the specified coefficient-pair class. $\square$

The coefficient convention in this proposition is essential. The purely arithmetic sharpness of Theorem 17.5 has no Galois hypothesis. Its additional realization just proved allows β to be multiplied by a freely chosen element of $\mathbb{Z}_p^\times$. It does not show that every resulting κ_p is induced by a strict automorphism of $G_{\mathbb{Q}_p}$ when a Tate coefficient framing has been fixed. For that possibly smaller class, the upper bound remains valid, but this example alone does not establish sharpness. No arrow condition at $p = 2$ was imposed.

17.4. Formal scope and the unproved uniform step. Theorem 17.1, including its equality classification, is formalized in the separate module `ABCTwoPrimeSupport20260831`. Its proof starts with the actual prime-factor support and the repository radical; the prime-power classification is proved rather than assumed. All 22 public theorems in that module were compiled and checked for axiom dependencies. The module `TraceCovariantRationalReturn20260831` proves the algebraic cancellation and scalar-line conclusions using the actual algebra trace and actual dimensions. Trace covariance remains an explicit hypothesis there: this does not formalize its local Galois or Kummer derivation, or the principal-unit logarithm argument. The separate module `ABCOddPartFibre20260831` now proves Theorem 17.5 for the actual natural-number odd part and the unchanged primitive-triple type. It constructs an injection of each entire fibre into a two-element type, proves that the fibre is finite, and establishes cardinality exactly two for the displayed example. This is an arithmetic fibre theorem; the logarithmic endpoint-transfer proposition and the coefficient-pair realization remain the mathematical proofs above, rather than formalized Galois-orbit assertions.

The rational-return obstruction is compatible with the minimum-layer constructions of the preceding sections. Those constructions can move a rational input outside the rational line and then form a hull. Returning to an actual rational S -unit solution is an additional condition. Multiplication by tensor-order coefficients, addition and ideal generation do not automatically preserve a point's trace, so the fibre bound cannot be applied after replacing (159) by hull membership.

No general S -unit method, IUT theorem, complete Ind3 operation or *abc* conjecture is disproved by this obstruction. Mechanisms introducing new prime support, changing prime labels, or using nonrational intermediate points followed by a proved arithmetic descent remain outside its scope. The uniform estimate (1) for $\omega(abc) \geq 3$, even for a fixed number of primes which themselves vary, is not proved here. Neither finite exceptional sets depending on the support nor density-one bounds supply that missing quantifier.

18. AN OBSTRUCTION IN THE ENTIRE RATIONAL ISOGENY CLASS

For one infinite family of primitive *abc* triples, we determine the entire rational isogeny class of the actual Frey curve and obtain matching bounds for two intrinsic quantities on that class. The largest minimal discriminant has order c^5 , while the smallest complex absolute j -invariant has order c . Thus selecting an arbitrary rationally isogenous representative cannot recover a sixth power of the endpoint from its minimal discriminant.

18.1. Odd isogenies and prime-degree paths. We use the following unconditional classification: if an elliptic curve over $\mathbb{Q}$ admits a cyclic $\mathbb{Q}$ -isogeny of degree N , then

$$(163) \quad N \in \{1, \dots, 19, 21, 25, 27, 37, 43, 67, 163\}.$$

The prime-degree theorem is due to Mazur [2, Theorem 1]. Its completion by Kenku for composite cyclic degrees is stated in [1, Theorem 2.2, p. 239]. We use this completed statement, including its CM cases. Quotients by Galois-stable finite subgroups, dual isogenies, and descent of uniquely determined factors are the standard characteristic-zero isogeny results of [71, Chapter III].

Proposition 18.1. *Let $E/\mathbb{Q}$ have full rational 2-torsion. If E admits a $\mathbb{Q}$ -isogeny of odd prime degree ℓ , then $\ell = 3$.*

Proof. Write the isogeny as $\psi : E \rightarrow E'$, and choose distinct nonzero points $T_0, T_1 \in E[2](\mathbb{Q})$. Since ℓ is odd, ψ identifies $E[2]$ with $E'[2]$. Form the rational quotient maps

$$\phi_0 : E \longrightarrow E_0 = E/\langle T_0 \rangle, \quad \phi_1 : E' \longrightarrow E_1 = E'/\langle \psi(T_1) \rangle.$$

The composite

$$\Phi = \phi_1 \circ \psi \circ \widehat{\phi_0} : E_0 \longrightarrow E_1$$

has degree 4ℓ . We check that its geometric kernel is cyclic; the degree alone is insufficient.

On $E_0[2]$, the dual $\widehat{\phi_0}$ has a kernel of order two. Its image has order two and is contained in $\ker \phi_0$, since $\phi_0 \widehat{\phi_0} = [2]$. Consequently its image is $\langle T_0 \rangle$. The subgroups $\langle \psi(T_0) \rangle$ and $\langle \psi(T_1) \rangle$ are distinct subgroups of order two. Therefore

$$\ker \Phi \cap E_0[2] = \ker \widehat{\phi_0}, \quad \#(\ker \Phi)[2] = 2.$$

The 2-primary subgroup of $\ker \Phi$ has order four. It cannot be $(\mathbb{Z}/2\mathbb{Z})^2$, so it is cyclic. The ℓ -primary subgroup has prime order and is also cyclic. Their coprime orders imply that $\ker \Phi$ is cyclic of order 4ℓ . The list (163) permits this only for $\ell = 3$. $\square$

The hypothesis here is a rational isogeny, or equivalently a Galois-stable subgroup, and does not require a rational point of order ℓ .

Lemma 18.2. *Any two elliptic curves that are isogenous over $\mathbb{Q}$ are joined by a finite path of rational prime-degree isogenies, up to $\mathbb{Q}$ -isomorphism at the endpoint.*

Proof. Let the curves be E, F . By duality there is a rational isogeny $E \rightarrow F$. Among all such maps choose θ of least positive degree, and let H be its finite geometric kernel. If H is not cyclic, one of its primary subgroups is not cyclic, and for some prime ℓ the group $H[\ell]$ has dimension two over $\mathbb{F}_\ell$. Here we used $H[\ell] \subseteq E[\ell]$ and $E[\ell] \cong (\mathbb{Z}/\ell\mathbb{Z})^2$. Thus $E[\ell] \subseteq H$. The quotient property gives a factorization

$$\theta = \theta_1 \circ [\ell], \quad \deg \theta_1 = \deg \theta / \ell^2 < \deg \theta.$$

The factor θ_1 is defined over $\mathbb{Q}$: it is unique, and both θ and $[\ell]$ are defined over $\mathbb{Q}$. This contradicts the choice of θ .

Hence H is cyclic. Its unique subgroup of each divisor order is Galois-stable. Quotienting along a subgroup chain with successive prime-order quotients, then identifying E/H with F , proves the claim. $\square$

This argument replaces the initial map by one of least degree. It does not assert that every given rational isogeny factors through rational prime-degree maps: multiplication by ℓ on a curve with irreducible $E[\ell]$ need not do so.

Lemma 18.3. *Fix an odd prime ℓ . If $E/\mathbb{Q}$ has no rational ℓ -isogeny, neither does any curve rationally isogenous to E by an isogeny of 2-power degree.*

Proof. Take such a curve F and a rational isogeny $\alpha : F \rightarrow E$ of 2-power degree, using a dual if necessary. Its restriction to $F[\ell]$ is a Galois-equivariant isomorphism onto $E[\ell]$. A Galois-stable subgroup of order ℓ in $F[\ell]$ would therefore give one in $E[\ell]$, contrary to the hypothesis. $\square$

18.2. An explicit family and its four models. For every integer $n \geq 1$, put

$$(164) \quad c = c_n = 1792n + 2, \quad (a, b, c) = (1, c_n - 1, c_n), \quad E_c : y^2 = x(x-1)(x+c-1).$$

These are positive primitive abc triples, and c_n tends to infinity. Moreover,

$$c \equiv 2 \pmod{8}, \quad c \equiv 2 \pmod{7}, \quad c \geq 1794.$$

The curve E_c is the actual Frey curve of this triple, with full rational 2-torsion.

Lemma 18.4. *The curve E_c in (164) has no rational 3-isogeny.*

Proof. Its displayed discriminant is $16(c-1)^2c^2$, a 7-adic unit, so it has good reduction at 7. Its reduction is $y^2 = x^3 - x$. The complete affine count is

x	0	1	2	3	4	5	6
$x^3 - x \pmod{7}$	0	0	6	3	4	1	0
$\# \{y : y^2 = x^3 - x\}$	1	1	0	0	2	2	1

and gives seven affine points. Adding the point at infinity yields $\#E_c(\mathbb{F}_7) = 8$, hence $a_7 = 0$. The good-reduction Frobenius theorem identifies the characteristic polynomial on $E_c[3]$ with $T^2 - a_7T + 7$, reduced modulo 3 [71, Chapters V and VII]. This is $T^2 + 1$, which has no root in $\mathbb{F}_3$. A rational 3-isogeny would supply a Galois-stable line in $E_c[3]$, on which Frobenius would have an eigenvalue in $\mathbb{F}_3$. This is impossible. $\square$

Proposition 18.1 and Lemma 18.4 now exclude every odd-prime rational isogeny from E_c .

For the quotient at $(0,0)$ of $y^2 = u^3 + Au^2 + Bu$, the usual integral model is

$$(165) \quad Y^2 = X^3 - 2AX^2 + (A^2 - 4B)X.$$

For example the formula $X = u + A + B/u$, $Y = y(1 - B/u^2)$ verifies the equation and extends to the degree-two quotient with kernel $\{O, (0,0)\}$. The discriminant of the remaining quadratic factor on the right of (165) is $16B$. Hence the quotient has extra rational 2-torsion exactly when B is a rational square.

Translate each of the three rational points of order two of E_c to $(0,0)$. The resulting values of B are, respectively,

$$-(c-1), \quad c, \quad c(c-1).$$

The first is negative, and the other two are 2 modulo 8. An integer that is a rational square is an integer square, and a square cannot be 2 modulo 8. Thus each of the three quotients has exactly one nonzero rational point of order two. Its only rational degree-two quotient is the dual edge back to E_c , up to rational isomorphism.

Denote these quotient curves by E_0, E_a, E_b , where the last subscript refers to the kernel at $x = -(c-1)$. Their equations are

$$(166) \quad \begin{aligned} E_c : \quad & y^2 = x^3 + (c-2)x^2 - (c-1)x, \\ E_0 : \quad & Y^2 = X^3 + (4-2c)X^2 + c^2X, \\ E_a : \quad & Y^2 = X^3 - 2(c+1)X^2 + (c-1)^2X, \\ E_b : \quad & Y^2 = X^3 + (4c-2)X^2 + X. \end{aligned}$$

Direct computation of their Weierstrass invariants gives

$$(167) \quad \begin{array}{c|cc} \text{curve} & c_4/16 & \Delta_{\text{disp}} \\ \hline E_c & c^2 - c + 1 & 16(c-1)^2c^2 \\ E_0 & c^2 - 16c + 16 & -256(c-1)c^4 \\ E_a & c^2 + 14c + 1 & 256c(c-1)^4 \\ E_b & 16c^2 - 16c + 1 & 256c(c-1) \end{array}$$

Here the subscript indicates the discriminant of the displayed integral model, without an assertion of minimality at 2.

Theorem 18.5. *Every elliptic curve over $\mathbb{Q}$ isogenous to E_c over $\mathbb{Q}$, by an isogeny of any degree, is $\mathbb{Q}$ -isomorphic to one of the four curves in (166).*

Proof. The quotient computation exhausts the connected rational 2-isogeny graph: E_c has precisely the three indicated subgroups of order two, and each quotient has only its dual edge back. Lemma 18.3 excludes every odd-prime isogeny from every vertex of this graph.

Any rationally isogenous curve is connected to E_c by a prime-degree path, by Lemma 18.2. Every initial 2-edge stays among the four displayed vertices. A first odd-prime edge is impossible by the preceding paragraph. Thus the entire path stays in the list. Possible isomorphisms between displayed vertices only shorten the list and do not affect the conclusion. $\square$

The theorem has no non-CM or conjectural Galois-surjectivity hypothesis. Its external inputs are the stated cyclic-isogeny classification, the basic quotient and dual theory, and the good-reduction Frobenius theorem.

18.3. Matching bounds throughout the class. Let $\mathcal{I}_c$ be the nonempty finite set of rational isomorphism classes in Theorem 18.5, and define

$$D_{\max}(c) = \max_{F \in \mathcal{I}_c} |\Delta_{\min}(F)|, \quad J_{\min, \infty}(c) = \min_{F \in \mathcal{I}_c} |j(F)|.$$

Here $\Delta_{\min}$ is the global minimal discriminant over $\mathbb{Q}$. The absolute value in $J_{\min, \infty}$ is the single complex absolute value of the rational number $j(F)$; it is not its global Weil height.

Theorem 18.6. *For every $c = c_n$ in (164),*

$$(168) \quad \frac{c^5}{32} \leq D_{\max}(c) \leq 256c^5, \quad 2c \leq J_{\min, \infty}(c) \leq 32c.$$

Proof. Every absolute displayed discriminant in (167) is at most $256c^5$. Passing to a minimal integral model cannot increase a prime valuation of the discriminant. Theorem 18.5 therefore gives the upper bound for $D_{\max}$ over the entire class.

For its lower bound use E_0 . At an odd prime $q \mid c$, its c_4 invariant is congruent to 256 modulo q . At an odd prime $q \mid c - 1$, it is congruent to 16 modulo q . Thus c_4 is a unit at every odd prime dividing the displayed discriminant, and the integral model is minimal there. We use here the usual minimality criterion for integral Weierstrass models [71, Chapter VII]: lowering the discriminant by q^{12} would require $q^4 \mid c_4$. Since $v_2(c) = 1$ and $c - 1$ is odd, the odd part of the displayed discriminant is retained, giving

$$(169) \quad |\Delta_{\min}(E_0)| \geq (c - 1)(c/2)^4 = \frac{(c - 1)c^4}{16} \geq \frac{c^5}{32}.$$

No calculation of the minimal model at 2 is needed.

For the j -bounds, $c \geq 1794 > 32$ implies that each polynomial in the $c_4/16$ column of (167) is at least $c^2/2$. In particular,

$$c^2 - 16c + 16 - \frac{c^2}{2} = \frac{c(c - 32)}{2} + 16 \geq 0;$$

the other three inequalities follow directly for $c \geq 2$. Consequently every displayed curve has $|c_4| \geq 8c^2$ and

$$|j| = \frac{|c_4|^3}{|\Delta_{\text{disp}}|} \geq \frac{512c^6}{256c^5} = 2c.$$

This ratio is unchanged when one passes to a minimal model. Finally $0 < c^2 - 16c + 16 \leq c^2$ for our values of c , so E_0 gives

$$|j(E_0)| = \frac{16(c^2 - 16c + 16)^3}{(c - 1)c^4} \leq \frac{16c^2}{c - 1} \leq 32c.$$

This proves both pairs of inequalities. □

With $\log^+ x = \max\{0, \log x\}$, the bounds imply

$$(170) \quad \lim_{n \rightarrow \infty} \frac{\log D_{\max}(c_n)}{\log c_n} = 5, \quad \lim_{n \rightarrow \infty} \frac{\min_{F \in \mathcal{I}_{c_n}} \log^+ |j(F)|}{\log c_n} = 1.$$

The second limit concerns complex j -values, not a height of a rational point on an auxiliary elliptic curve.

Corollary 18.7. *For every real $\theta > 5$ and constant $C > 0$, there are positive primitive abc triples for which no elliptic curve $F/\mathbb{Q}$ rationally isogenous to the actual Frey curve satisfies*

$$c^\theta \leq C|\Delta_{\min}(F)|.$$

Proof. Choose n so large that $c_n^{\theta-5} > 256C$, and apply (168). □

Thus a uniform $c^{6-\delta}$ recovery for $0 < \delta < 1$, or a $c^{6-o(1)}$ recovery along this family, fails even when the representative can depend arbitrarily on the triple. This refutes the specified discriminant-replacement step. It does not refute a Szpiro conjecture or abc: no sufficiently small radical has been established for these triples. Bounds retaining the archimedean contribution, or using curves outside this rational isogeny class, remain separate possibilities.

Remark 18.8 (Formalization boundary). The Lean arithmetic companion verifies the actual primitive family, the actual elliptic-curve point count over $\mathbb{F}_7$, the absence of a root of $T^2 + 1$ over $\mathbb{F}_3$, the four Weierstrass models and their invariants, and the displayed discriminant and rational absolute j -bounds. It does not formalize the cyclic-degree classification, the Frobenius interpretation of the point count, the entire isogeny-class identification, or minimal-discriminant theory. These are the external mathematical steps explicitly used above.

19. EXACT WEIL HEIGHTS WITHIN THE RATIONAL ISOGENY CLASS

The complex absolute value in Theorem 18.6 is a single local quantity. We now determine the absolute Weil height of the same four j -invariants, including every cancellation in their rational coordinates. The minimum over the entire rational isogeny class is $3 \log(c^2 - 16c + 16)$. In particular its leading term remains $6 \log c$, although the logarithm of the minimum complex absolute value has leading term $\log c$.

19.1. Reduced rational coordinates. Throughout this section put

$$(171) \quad n \in \mathbb{Z}_{\geq 1}, \quad c = 1792n + 2, \quad u = c/2 = 896n + 1.$$

Thus $c \geq 1794 > 32$, and $u, c - 1$ are positive odd integers. Let E_c, E_0, E_a, E_b be the four actual curves of Theorem 18.5, and write

$$(172) \quad \begin{aligned} P &= c^2 - c + 1, & Q &= c^2 - 16c + 16, \\ R &= c^2 + 14c + 1, & S &= 16c^2 - 16c + 1. \end{aligned}$$

For a rational number $z = N/D$ in lowest terms, with $D > 0$, we use the absolute logarithmic and multiplicative Weil heights

$$(173) \quad h(z) = \log \max\{|N|, D\}, \quad H(z) = \max\{|N|, D\}.$$

Over $\mathbb{Q}$ the relative and absolute normalizations agree.

Proposition 19.1. *The following are the signed reduced rational coordinates of the actual j -invariants. Every denominator is positive.*

curve	numerator N	denominator D
E_c	$64P^3$	$u^2(c-1)^2$
E_0	$-Q^3$	$(c-1)u^4$
E_a	$8R^3$	$u(c-1)^4$
E_b	$8S^3$	$u(c-1)$

Proof. The previously computed invariants give

$$\begin{aligned} j(E_c) &= \frac{256P^3}{c^2(c-1)^2}, & j(E_0) &= -\frac{16Q^3}{(c-1)c^4}, \\ j(E_a) &= \frac{16R^3}{c(c-1)^4}, & j(E_b) &= \frac{16S^3}{c(c-1)}. \end{aligned}$$

Substitution of $c = 2u$ gives the displayed coordinates. The sign of $j(E_0)$ follows from its negative displayed discriminant. This computation uses $j = c_4^3/\Delta$ and does not require the displayed models to be minimal at 2.

For coprimality, the exact residue identities give the table

	mod u	mod $(c-1)$
P	1	1
Q	16	1
R	1	16
S	1	1

For example, $Q = 16 + u(4u - 32) = 1 + (c-1)(c-15)$ and $R = 1 + u(4u + 28) = 16 + (c-1)(c+15)$. Because $u, c-1$ are odd, 16 is coprime to either modulus. Consequently each polynomial is coprime to both factors of every denominator. Taking powers and multiplying numerator factors 64 or 8 preserves this coprimality.

For completeness the exact 2-parts are also visible directly: P, R, S are odd, whereas

$$Q = 4(u^2 - 8u + 4), \quad v_2(Q) = 2.$$

The 2-adic valuations of the four absolute numerators are respectively 6, 6, 3, 3, and every denominator is odd. Thus no odd prime or power of 2 can cancel in any of the four displayed fractions. $\square$

19.2. The two exact minima. Let $\mathcal{I}_c$ denote the rational isogeny class of E_c , modulo rational isomorphism, and define

$$h_{\min}(c) = \min_{F \in \mathcal{I}_c} h(j(F)), \quad J_{\min, \infty}(c) = \min_{F \in \mathcal{I}_c} |j(F)|_{\infty}.$$

Theorem 18.5 supplies the entire-class identification used in these minima; the following arithmetic calculations do not provide a replacement proof of that classification.

Lemma 19.2. *The unique minimizing rational isomorphism class for the complex absolute value is represented by E_0 , and*

$$(174) \quad J_{\min, \infty}(c) = \frac{Q^3}{(c-1)u^4} = \frac{16Q^3}{(c-1)c^4} > 2c > 1.$$

In particular all four absolute numerators in Proposition 19.1 exceed their denominators.

Proof. For $c \geq 32$ one has

$$(175) \quad \begin{aligned} Q - \frac{c^2}{2} &= \frac{c(c-32)}{2} + 16 > 0, \quad Q < c^2, \\ P - Q &= 15(c-1) > 0, \\ R - Q &= 15(2c-1) > 0, \\ S - Q &= 15(c^2-1) > 0. \end{aligned}$$

Thus all four polynomials are positive. Since $Q > c^2/2$ and $c-1 < c$, the absolute value of $j(E_0)$ is greater than $2c$. The exact ratios of the other three absolute values to this one are

$$(176) \quad \begin{aligned} \frac{|j(E_c)|}{|j(E_0)|} &= \frac{16c^2}{c-1} \left(\frac{P}{Q}\right)^3 > 1, \\ \frac{|j(E_a)|}{|j(E_0)|} &= \left(\frac{cR}{(c-1)Q}\right)^3 > 1, \\ \frac{|j(E_b)|}{|j(E_0)|} &= \left(\frac{cS}{Q}\right)^3 > 1. \end{aligned}$$

These inequalities follow from (175) and $c > 1$. The entire-class theorem now gives the assertion, including uniqueness. Strictness separates E_0 from all other displayed models, regardless of any possible coincidence among those others. $\square$

Theorem 19.3. *The absolute logarithmic Weil heights of the actual four j -invariants are*

$$(177) \quad \begin{aligned} h(j(E_c)) &= \log 64 + 3 \log P, \\ h(j(E_0)) &= 3 \log Q, \\ h(j(E_a)) &= \log 8 + 3 \log R, \\ h(j(E_b)) &= \log 8 + 3 \log S. \end{aligned}$$

The unique minimizer in the entire rational isogeny class is represented by E_0 , with

$$(178) \quad h_{\min}(c) = 3 \log(c^2 - 16c + 16), \quad \min_{F \in \mathcal{I}_c} H(j(F)) = Q^3.$$

Moreover,

$$(179) \quad \begin{aligned} 6 \log c - 3 \log 2 &< h_{\min}(c) < 6 \log c, \\ c^6/8 &< \min_{F \in \mathcal{I}_c} H(j(F)) < c^6. \end{aligned}$$

Proof. The fractions in Proposition 19.1 are reduced, and Lemma 19.2 shows that their absolute numerators exceed their denominators. Therefore (173) gives exactly (177). The inequalities $P, R, S > Q > 0$, together with $64, 8 > 1$, show that each height other than that of E_0 is strictly greater than $3 \log Q$. The entire-class theorem extends this minimum and uniqueness to every rationally isogenous curve. Finally $c^2/2 < Q < c^2$ gives (179), first by cubing and then by taking logarithms. $\square$

19.3. Finite-place contribution and the bounded gain. The minimizers above are the same, but the two minimum values measure different quantities. If N/D is reduced, each prime dividing D contributes exactly $v_q(D) \log q$ to the finite-place sum in the logarithmic Weil height; primes not dividing D contribute zero. The entire finite-place contribution is therefore $\log D$. For E_0 , whose complex absolute value exceeds one, this proves the exact identity

$$(180) \quad \begin{aligned} h_{\min}(c) - \log J_{\min, \infty}(c) &= \log((c-1)u^4) \\ &= 5 \log c - \log 16 + \log(1 - 1/c). \end{aligned}$$

Equivalently, the exact formulas yield

$$(181) \quad \begin{aligned} h_{\min}(c) - 6 \log c &= 3 \log(1 - 16/c + 16/c^2) \rightarrow 0, \\ \frac{J_{\min, \infty}(c)}{c} &= 16 \frac{(1 - 16/c + 16/c^2)^3}{1 - 1/c} \rightarrow 16. \end{aligned}$$

All limits here and below are along the unbounded sequence (171). Continuity of $\log$ at positive arguments proves these limits. In particular

$$\frac{h_{\min}(c)}{\log c} \rightarrow 6, \quad \frac{\log J_{\min, \infty}(c)}{\log c} \rightarrow 1.$$

The denominator accounts for the missing term of size $5 \log c$; it cannot be discarded when using the absolute Weil height.

The gain achieved by choosing the best representative in the entire class is also exact:

$$(182) \quad h(j(E_c)) - h_{\min}(c) = \log 64 + 3 \log(P/Q) \in (6 \log 2, 9 \log 2).$$

Indeed $P > Q$ by (175), while $P < c^2$ and $Q > c^2/2$ imply $P/Q < 2$. These strict inequalities prove the stated interval, using $\log 64 = 6 \log 2$. Since $P/Q \rightarrow 1$, the gain tends to $6 \log 2$. Thus optimization over the entire class saves only a bounded additive amount in this family.

Corollary 19.4. *For every $\delta > 0$ and $C \in \mathbb{R}$, there exists an $n \geq 1$ such that every elliptic curve $F/\mathbb{Q}$ rationally isogenous to E_{c_n} satisfies*

$$h(j(F)) > (6 - \delta) \log c_n + C.$$

Equivalently, for every $\theta < 6$ and $C_0 > 0$, some $n \geq 1$ satisfies

$$H(j(F)) > C_0 c_n^\theta \quad \text{for every } F \in \mathcal{I}_{c_n}.$$

Proof. Choose n with $\delta \log c_n > C + 3 \log 2$ and apply the lower bound in (179). Such n exist because $c_n \rightarrow \infty$. For the multiplicative formulation, choose $c_n^{6-\theta} > 8C_0$ and use the corresponding lower bound. $\square$

This corollary excludes the specific proposed step of lowering the leading sixth-power j -height size by selecting a different rationally isogenous representative. It is not a counterexample to a conductor–height estimate, modified Szpiro, or *abc*. No upper bound for the radicals of these triples is asserted, and their unresolved height–radical comparison is unaffected by this calculation.

19.4. Formalization boundary. The Lean height companion reuses the four actual Weierstrass curves from the preceding section. It proves the signed rational equalities, integer coprimality, and equality with the canonical `Rat.num` and `Rat.den` coordinates. It then uses the actual multiplicative and logarithmic height functions in `mathlib` and proves compatibility with the existing normalized-height interface over $\mathbb{Q}$. This gives the exact height tables, attained minima over the actual model enumeration, uniqueness, and the logarithmic bounds (179). No artificially defined height is substituted for the library functions.

The enumeration is not asserted in Lean to exhaust a rational isogeny class. That passage uses the separately proved Theorem 18.5, including its stated external isogeny and Frobenius inputs. The strict comparison of the complex minimum, the explicit finite-place identity, the asymptotic limits, the bounded-gain formula, and Corollary 19.4 are paper proofs in this section; they are not added to the companion’s formal claims. The module adds no axioms or conjectural hypotheses.

20. STANDARD-TYPE THETA POINTS IN A DECLARED ADDITIVE CARRIER

We now restrict the local source to numerical representatives of standard-type theta values, with their finite root-of-unity ambiguity. For each independent choice of these representatives, one actual full-Galois Kummer operator attains every point-hull bound below. The operator acts on a declared native additive carrier. This does not identify its action there with an action on the original multiplicative theta Kummer class, or establish membership in the global output family of [62, Corollary 3.12].

Page references to the theta and IUT sources in this section use the archived author versions: December 2008 for [79], May 2020 for [63, 62], and December 2020 for [64].

20.1. The normalized evaluation and its finite ambiguity. Retain the field and roots of (114). Thus $\ell \geq 7$ and p are primes, $p \equiv -1 \pmod{30\ell}$, $v_p(b_0) = 2$, and

$$(183) \quad \begin{aligned} e &= 15\ell, & d &= 2e, & h &= (\ell - 1)/2, & K_0 &= \mathbb{Q}_p(\mu_{30\ell}), \\ \varpi^e &= b_0, & E &= K_0(\varpi), & q &= b_0^2, & r_0 &= \varpi^{15} = q^{1/(2\ell)}, \\ \beta &= \varpi^{(e+1)/2}/p, & \kappa &= 1 - 1/e, & I &= \beta^{1-e} \mathcal{O}_E. \end{aligned}$$

Here ϖ is the element denoted π in (114), and is not a uniformizer. Lemma 14.1 gives $I = p^{-1} \log \mathcal{O}_E^\times$, $\text{Tr } I = \mathbb{Z}_p$, and $\mu_{2\ell} \subset K_0$. In particular, $e \leq p - 2$, $p \nmid d$, and $E/\mathbb{Q}_p$ is Galois with residue degree two.

Choose $i^2 = -1$ in an algebraic closure. The series of [79, Proposition 1.4, pp. 20–21] is

$$(184) \quad \vartheta(U) = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(n+1)/2} U^{2n+1}, \quad f(U) = \vartheta(U)/\vartheta(i).$$

The standard-type theta function is the reciprocal of an ℓ -th root of this normalized function [63, Example 3.2(ii),(iv)].

Lemma 20.1 (The normalized theta coset). *For every integer a and every nonzero U in a finite extension of E ,*

$$\vartheta(q^{a/2}U) = (-1)^a q^{-a^2/2} U^{-2a} \vartheta(U), \quad f(\pm i q^{a/2}) = \pm q^{-a^2/2}.$$

The reciprocals of the ℓ -th roots of these two evaluations form exactly the coset

$$(185) \quad \mu_{2\ell} r_0^{a^2}.$$

Proof. At the indicated points the term valuations grow quadratically as $|n| \rightarrow \infty$, so reindexing is legitimate. Substitution $n \mapsto n + a$ gives the functional equation. Also $\vartheta(-U) = -\vartheta(U)$. At $U = i$ the terms $n = 0, -1$ sum to $2i$, and all other terms have positive valuation. Thus $\vartheta(i)$ is a unit. Since $i^{-2a} = (-1)^a$, the two normalized evaluations follow. Their reciprocal roots have ℓ -th powers $\pm q^{a^2/2}$, whose complete root set, for odd ℓ , is (185). $\square$

In the standard cyclotomic and Kummer marking, [64, Remark 2.5.1(i), p. 72] identifies the unperfected evaluation set at label j with the Kummer image of $\mu_{2\ell} r_0^{j^2}$, where the exponent uses the integer representative $0 \leq j \leq h$. The ambiguities at different labels are independent [64, Remark 2.5.1(iii), p. 73]. They are not arbitrary elements of $\mathcal{O}_E^\times$: standard-type normalization removes that larger constant-unit ambiguity [64, Remark 1.12.2(i),(ii)].

20.2. One arrow for independent torsion representatives. For each $1 \leq j \leq h$, choose $\zeta_j \in \mu_{2\ell}$ independently and put

$$(186) \quad \begin{aligned} x_j &= \zeta_j r_0^{j^2}, & r_j &= v_p(x_j) = 2j^2/\ell, \\ n_j &= \lfloor r_j \rfloor, & k_j &= n_j + 1, & T_j &= p^{-k_j} x_j, & V &= I/pI, & V_0 &= \beta I/pI. \end{aligned}$$

The quotient V/V_0 has dimension two over $\mathbf{F}_p$. Bars will denote reduction modulo pI .

Lemma 20.2 (Content, trace and projective inertia). *For every choice of the ζ_j ,*

$$(187) \quad \begin{aligned} k_j &= \lfloor r_j + \kappa \rfloor, & T_j &\in \beta I \setminus pI, & \text{Tr}_{E/\mathbb{Q}_p}(T_j) &= 0, \\ 1 &\in \beta I \setminus pI, & \text{Tr}_{E/\mathbb{Q}_p}(1) &= d \in \mathbb{Z}_p^\times. \end{aligned}$$

If $\sigma(\beta) = \epsilon\beta$ generates $\text{Gal}(E/K_0)$, with ϵ of order e , then

$$(188) \quad \sigma(T_j) = \epsilon^{30j^2} T_j.$$

The projective orbit of $\overline{T_j}$ over $\mathbf{F}_p$ has exactly ℓ elements.

Proof. Write $2j^2 = \ell n_j + \delta_j$, where $1 \leq \delta_j \leq \ell - 1$. Then $0 < \delta_j/\ell - 1/e < 1$, proving the formula for k_j . Moreover

$$e v_p(T_j) = 15\delta_j - e, \quad 2 - e < 15 - e \leq 15\delta_j - e \leq -15 < 1.$$

The exponents $2 - e$ and 1 are the thresholds for βI and pI . This proves the two membership assertions, including the analogous assertion for 1 , whose exponent is zero.

With $\gamma = p^2/b_0 \in \mathbb{Z}_p^\times$, (115) gives $r_0 = \gamma^{15}\beta^{30}$. Both γ and ζ_j lie in K_0 and are fixed by σ , proving (188). Its character has order $e/\gcd(e, 30j^2) = \ell$. Summing its inertia conjugates gives zero trace over K_0 , and then zero absolute trace.

These roots reduce injectively and $\mu_\ell \cap \mathbf{F}_p^\times = \{1\}$, since $p \equiv -1 \pmod{\ell}$. If two inertia images of $\overline{T_j}$ are proportional over $\mathbf{F}_p$, their character residues are therefore equal. Indeed, a nonzero difference of residues is a unit of K_0 , and multiplying $T_j \notin pI$ by such a unit cannot put it in pI . Thus proportionality forces the inertia exponents to agree modulo ℓ , while agreement gives equality already in E . This proves the projective orbit assertion. $\square$

Let Γ_E be the actual integral Kummer operators of Section 11, extended $\mathbb{Q}_p$ -linearly to E . They preserve I ; we do not identify this group of operators with every integral linear automorphism.

Theorem 20.3 (Common attainment for the finite theta source). *For every tuple $(\zeta_1, \dots, \zeta_h) \in \mu_{2\ell}^h$, there is one $F \in \Gamma_E$ such that*

$$(189) \quad v_p(F(1)) = v_p(F(T_j)) = -\kappa \quad (1 \leq j \leq h).$$

The chosen F may depend on the tuple.

Proof. The hypotheses of Lemma 11.3 hold: d is even, $2 \leq e \leq p - 2$, and $\text{Tr } I = \mathbb{Z}_p$. Use its basis

$$I = \mathbb{Z}_p a_{\text{JW}} \oplus \mathbb{Z}_p b_{\text{JW}} \oplus W, \quad I \cap \ker \text{Tr} = \mathbb{Z}_p a_{\text{JW}} \oplus W,$$

and let B be the b_{JW} -coordinate. The trace of b_{JW} is a unit, so $B = \text{Tr} / \text{Tr}(b_{\text{JW}})$.

For each j , at most one of the first ℓ inertia powers can place $\overline{T_j}$ on $\mathbf{F}_p \overline{a_{\text{JW}}}$. Since $h < \ell$, one power avoids this line for every label. It fixes 1 and preserves V_0 and trace. We now verify every remaining hypothesis of Lemma 11.8, with $k = \mathbf{F}_p$, $L = V_0$, $U = \overline{1}$ and the transformed vectors $\overline{T_j}$. All lie in L by Lemma 20.2; $B(U) \neq 0$, $B(T_j) = 0$, and $T_j \notin \mathbf{F}_p a_{\text{JW}}$. The trace kernel maps onto V/V_0 : for $y \in I$, subtract $\text{Tr}(y)/d \in \mathcal{O}_E \subset \beta I$. Finally its cardinality condition is $h + 1 < p$.

That lemma consequently gives one parabolic composition sending all these vectors outside V_0 . Its integer-coordinate crosses and symplectic operations have actual full-group lifts by Proposition 11.2 and Lemma 11.4, as used in the arithmetic conclusion of Lemma 11.8. Compose them with the chosen field inertia automorphism. The resulting canonical action M sends every input into $I \setminus \beta I$, of valuation $-\kappa$.

Proposition 11.5 gives Kummer action $c_\alpha M_\alpha^{-1}$, with $c_\alpha \in \mathbb{Z}_p^\times$. Using the inverse of the constructed group automorphism therefore gives a unit multiple of M , proving (189). $\square$

This proves $\forall(\zeta_j) \exists F \forall j$, not the stronger assertion that one F works for every tuple. It does not commute multiplication by ζ_j with F , which need not be K_0 -linear.

20.3. The additive point hull and the logarithmic bridge. For $m = m_j = j + 1$, let

$$\mathcal{T}_m = E^{\otimes_{\mathbb{Q}_p} m}, \quad A_m = \mathcal{O}_E^{\otimes_{\mathbb{Z}_p} m}, \quad B_m = \prod_{\text{Gal}(E/\mathbb{Q}_p)^{m-1}} \mathcal{O}_E$$

under the decomposition (90). Thus B_m is the integral closure of A_m . The multiplication and literal unit 1 here belong to the fixed native field E . The normalization $I = p^{-1} \log \mathcal{O}_E^\times$ specifies a lattice in that field; it does not replace its multiplication by a structure transported through ρ . In the declared additive tensor carrier put

$$(190) \quad Z_{j,\zeta_j} = 1 \otimes \cdots \otimes x_j, \quad H_{j,\zeta_j} = \overline{\text{span}_{B_m} \{F^{\otimes m}(Z_{j,\zeta_j}) : F \in \Gamma_E\}}.$$

The literal ones agree with the single-slot embedding of [62, Proposition 3.1(ii), p. 93]. Choosing this embedding does not itself prove global source membership.

Theorem 20.4 (An attained point-source hull). *For every label and every finite torsion representative,*

$$(191) \quad H_{j,\zeta_j} = P_j = \beta^{ek_j - (e-1)m_j} B_{m_j}.$$

Here the uniformizer exponent is taken in every field component. For each tuple of representatives, one F generates these ideals simultaneously for all labels. With $\mu(B_m) = 1$ and $D_m = d^m$,

$$(192) \quad D_{m_j}^{-1} \log \mu(H_{j,\zeta_j}) = (m_j \kappa - k_j) \log p.$$

Proof. Every F preserves I and $p^{k_j} I$. Since $x_j \in p^{k_j} I$ and $1 \in I$, each tensor component has valuation at least $k_j - m_j \kappa$. The component embeddings preserve valuation, giving containment in P_j . The common F from Theorem 20.3 gives $v_p(Fx_j) = k_j - \kappa$ and $v_p(F1) = -\kappa$. Its tensor has the required valuation in every component, so it is a generator of P_j times an element of $B_{m_j}^\times$. This proves the reverse inclusion before closure; the fractional ideal is closed. It is also the direct specialization $u = 1$, $t = T_j$ of Proposition 11.10.

There are d^{m-1} field components, each with residue degree two. Hence $\log \mu(\beta^\nu B_m) = -2d^{m-1}\nu \log p$. Division by d^m proves (192). No further division by m_j is made. $\square$

The entire finite orbit $\mu_{2\ell} r_0^{j^2}$ gives the same hull. The source in this proof contains no arbitrarily chosen $\mathcal{O}_E^\times$ -multiple and need not contain $r_0^{j^2} B_m$ or a product of fractional ideals.

Proposition 20.5 (A comparison for each arrow). *For $0 \neq x \in \mathcal{O}_E$, set $\lambda(x) = p^{-1} \log(1 + px)$, $r = v_p(x)$ and $k = \lfloor r + \kappa \rfloor$. For every $F \in \Gamma_E$,*

$$(193) \quad v_p(\lambda(x) - x) \geq 1 + 2r \geq k + 1 - \kappa, \quad \lambda(x) - x \in p^{k+1} I, \\ v_p(F(\lambda(x))) = v_p(F(x)).$$

Thus, for $t_j = \lambda(x_j)$ and $u = \lambda(1)$, the point tensors with factors $(x_j, 1, \dots, 1)$ and $(t_j, u, \dots, u)$ generate the same principal B_{m_j} -ideal after each individual $F^{\otimes m_j}$. Both orbit hulls are P_j , with the same common attainer.

Proof. For $n \geq 2$, the n -th term of $\lambda(x) - x$ has valuation $(n-1) + nr - v_p(n)$. For odd p , $v_p(n) \leq n-2$, so this is at least $1 + 2r$. Convergence and ultrametricity give the first inequality; the second uses $k \leq r + \kappa$ and $r \geq 0$. By the definition of k , $x \in p^k I \setminus p^{k+1} I$. Since F is an automorphism of this filtered lattice,

$$v_p(Fx) < k + 1 - \kappa \leq v_p(F(\lambda(x) - x)).$$

Strict ultrametricity proves the valuation equality. Also $u \in \mathbb{Z}_p^\times$, so $\mathbb{Q}_p$ -linearity gives $F(u) = uF(1)$. All component valuations of the two tensors therefore agree, proving the ideal assertions. Finally $(t_j - x_j)/p^{k_j} \in pI$, so the common attainer for T_j also works for t_j/p^{k_j} . $\square$

Here $\lambda(x)$ is the coefficient of the separately defined class $\text{Kum}_p(1 + px)$ in $\rho = p^{-1} \log_{\text{BK}}^{\text{std}}$. It is not the theta evaluation class. In particular, exact zero trace was proved for T_j , not for t_j . If

all m_j coordinates of this unit-cohomology packet are changed from ρ to $\log_{\text{BK}}^{\text{std}}$, the tensor is multiplied by p^{m_j} . Consequently

$$(194) \quad \begin{aligned} P_j^{\text{std}} &= p^{m_j} P_j = \beta^{ek_j+m_j} B_{m_j}, \\ D_{m_j}^{-1} \log \mu(P_j^{\text{std}}) &= -(k_j + m_j/e) \log p. \end{aligned}$$

The change is $-m_j \log p$. This rescales every coordinate of the specified cohomological source; it does not give the original theta function an additional factor p^{m_j} .

20.4. Synchronized source conditions. The constructions distinguish $\text{Kum}(x_j)$, the numerical scalar x_j , and $\text{Kum}_p(1 + px_j)$. Their valuations already separate the first and third: $v_p(x_j) = 2j^2/\ell > 0$, whereas $v_p(1 + px_j) = 0$. Under normalized logarithm the scalar x_j instead comes from the unit $\exp(px_j)$. Equal additive hulls therefore do not identify the multiplicative inputs.

The LGP construction uses the prescribed log-link pullback before the single-slot insertion [62, Propositions 3.4(ii) and 3.5, pp. 102–106]. A tautological local logarithmic marking can give this pullback the numerical coordinate x_j . It neither makes the log-link a ring homomorphism from the predecessor's original field nor permits independent choices at all labels. The theater log-link comes from one isomorphism and retains its synchronization [62, Proposition 1.3(i) and Remark 1.3.1, pp. 42–43].

Theorem 20.4 alone does not establish membership in the possible-image family of [62, Corollary 3.12, pp. 173–175]. Section 21 supplies one globally synchronized copy representative and proves the canonical local inclusion in its fixed base branch. Compatibility of the complete family of log-Kummer maps and reference data with the undetermined q -pilot, including the required Ind3 comparison, remains separate. The full local Galois constructions used here are mathematical inputs from Section 11; their local class field theory and source identifications are not newly formalized in Lean by this section. No global *abc* conclusion is asserted.

21. A SYNCHRONIZED BASE BRANCH OF THE THETA-PILOT IMAGES

We now place the local points of Section 20 in a specified base branch of the possible-image construction in [62, Corollary 3.12]. The construction uses one concrete global theta-pilot, two tagged copies of its Hodge theater, and the log-link representative induced by their single global copy-identity. The resulting local inclusion concerns the raw union of possible images in one vertically coric column. It does not identify the complete global family with the one-prime arithmetic bundles of Section 22.

We use the reconstruction results of the cited sources, with their original objects and variance conventions. We do not reprove all those results, or assume the global multiradial and horizontal compatibility assertions of [62, Theorem 3.11] in order to establish the inclusion below. References to [63, 62], [64], and [4] use respectively the archived May 2020, December 2020, and November 2015 author versions.

21.1. Fixed initial data and the standard log-field. Fix a member of Corollary 15.6. Its global initial theta data, including its core, auxiliary cover, distinguished quotient element and place section, have already been constructed. At its distinguished rational prime p , retain (183): in particular,

$$e = 15\ell, \quad d = 2e, \quad h = (\ell - 1)/2, \quad I = p^{-1} \log \mathcal{O}_E^\times = \beta^{1-e} \mathcal{O}_E, \quad r_0^{2\ell} = q = b_0^2.$$

Here $v_p(b_0) = 2$, $E/\mathbb{Q}_p$ is Galois with residue degree two, $e \leq p - 2$, and $\mu_{2\ell} \subset K_0 = \mathbb{Q}_p(\mu_{30\ell})$. For $1 \leq j \leq h$, put

$$(195) \quad \begin{aligned} m_j &= j + 1, & a_j &= r_0^{j^2}, & x_j &= \zeta_j a_j, & \zeta_j &\in \mu_{2\ell}, \\ \mathcal{T}_{m_j} &= E^{\otimes_{\mathbb{Q}_p} m_j}, & Z_{j, \zeta_j} &= 1^{\otimes j} \otimes x_j. \end{aligned}$$

The distinguished last slot is the label- j slot in the source procession; it is not an extra factor.

The field of moduli is $F_{\text{mod}} = \mathbb{Q}$. The source's selected place set V maps bijectively to the places of F_{mod} , so it has exactly one selected v above p . This is not a claim that the global

torsion field has only one prime above p . It means that the finite local packet at this rational prime has no other selected direct summands: its marked carrier is $\mathcal{T}_{m_j}$.

Let E^- denote the original predecessor field. Definition 1.1(i) of [62, pp. 23–25] equips the perfection of its multiplicative unit group with a *new field structure*, denoted here by L^- . In the concrete model, standard logarithm gives a unital field isomorphism

$$(196) \quad \mathcal{L} : L^- \xrightarrow{\sim} E.$$

It transports both field operations from E ; the new multiplication is not the predecessor's old multiplication of units.

For $x \in E$, choose N sufficiently large and write

$$(197) \quad y(x) = p^{-N}[\exp(p^N x)] \in L^-.$$

The bracket is written additively in the old unit perfection. The equality $\exp(p^{N+1}x) = \exp(p^N x)^p$ proves independence of sufficiently large N , and $\mathcal{L}(y(x)) = x$. For $x \in I$, one may take $N = 1$, since $pI = \mathfrak{m}_E$ and $1/e > 1/(p-1)$.

In the chart (196), the pre-log-shell has coordinates $\log \mathcal{O}_E^\times$; the canonical shell is its multiple $p^{-1} \log \mathcal{O}_E^\times = I$. This division by p does not change the field multiplication or its unit. The new field unit is $y(1)$, of coordinate 1, whereas the old multiplicative unit $1 \in (E^-)^\times$ has coordinate zero after logarithm. Thus the background entries of (195) are the new field units.

For comparison, changing the *entire field chart* to $u = x/p$ would instead give

$$(198) \quad \begin{aligned} u \star v &= puv, & 1_\rho &= 1/p, & \mathbb{Q}_p \ni t &\longmapsto t/p, \\ v_\rho(u) &= v_E(pu), & \mathcal{O}_\rho &= \mathcal{O}_E/p, & I_\rho &= I/p. \end{aligned}$$

The equation $a^n = b$ would become $p^{n-1}a_\rho^n = b_\rho$. In an m -fold packet, both the tensor coordinates and the reference integer module would be transported by p^{-m} . This is different from changing only the outputs of a fixed unit-cohomology source from $\rho = p^{-1} \log_{\text{BK}}^{\text{std}}$ to $\log_{\text{BK}}^{\text{std}}$, which gives the factor p^m in (194). Neither rescaling is made in (196).

21.2. A global identity and three distinct cyclotomes. Take distinct tagged copies $\mathcal{H}_{n,r}$, $n, r \in \mathbb{Z}$, of the fixed concrete Hodge theater. Retain the full vertical and horizontal poly-arrows required by [62, Definition 3.8(iii), pp. 113–114]. Inside the vertical arrow $\mathcal{H}_{0,-1} \rightarrow \mathcal{H}_{0,0}$, select the representative induced by the global copy-identity

$$(199) \quad \Xi : \mathcal{H}_{0,-1}^D \xrightarrow{\sim} \mathcal{H}_{0,0}^D.$$

Selecting a representative does not replace a full poly-arrow by a singleton.

Proposition 21.1 (Synchronized standard-log representatives). *The single isomorphism (199) gives, at every finite place and every labeled prime-strip occurrence, the log-field map*

$$(200) \quad \lambda_v = \text{copy}_v \circ \mathcal{L}_v : L_{0,-1,v} \xrightarrow{\sim} E_{0,0,v}.$$

These are simultaneously allowed representatives of the theater log-link. At the distinguished place, the pullbacks of x_j and of each background field unit have standard-log coordinates x_j and 1.

Proof. Definition 1.1(i) of [62] constructs the natural isomorphism from the tautological log-Frobenioid to the original Frobenioid. The group acts compatibly on all the monoids in that construction. Remark 1.1.2(i), pp. 29–30, identifies its underlying local fundamental groups by the tautological identity. Composing this natural isomorphism with the copy map therefore covers the D-prime-strip isomorphism induced by Ξ .

The map from F-prime-strip isomorphisms to their D-prime-strip isomorphisms is bijective by [63, Corollary 5.3(ii), p. 144]. Consequently the displayed local map is the unique lift used in [62, Proposition 1.3(i), p. 42]. That proposition applies these lifts to the one Ξ across every prime-strip of both bridges; Remark 1.3.1, p. 43, makes the synchronization requirement explicit. The construction at the archimedean constituents uses the corresponding standard tautological isomorphism.

The concrete marked theater has the synchronized constant-monoid identifications of [64, Corollary 4.6(iii), p. 138]. Its copy-identity neither permutes the labels nor changes these identifications. In those fixed coordinates (200) is the identity field map. The asserted pullbacks follow. $\square$

The proposition is about this Ξ , not every theater isomorphism. An arbitrary such isomorphism may change label markings. A unital local field isomorphism fixing $q \in \mathbb{Q}_p$ preserves the coset defined by $X^{2\ell} = q^{j^2}$, but that observation alone does not provide a synchronized global representative.

It is essential to distinguish the three coefficient objects

$$\mu_{\text{prev}}, \quad \mu_{\text{log}}, \quad \mu_{\text{current}},$$

arising respectively from multiplication in the old field E^- , the new log-field L^- , and the current field E^+ . The old *linear unit-Kummer coordinate* is

$$(201) \quad \kappa_{\text{lin}}^-(x) = p^{-N} \text{Kum}_{\mu_{\text{prev}}}(\exp(p^N x)) \in H_f^1(G_E, \mathbb{Q}_p(1)_{\text{prev}}).$$

It is the inverse standard Bloch–Kato logarithm on the unit subspace. On I it takes values in $p^{-1}H_f^1(G_E, \mathbb{Z}_p(1)_{\text{prev}})$.

There is separately the new multiplicative Kummer map

$$(L^-)^{\times, \log} \longrightarrow H^1(G_{L^-}, \mathbb{Q}_p(1)_{\log}).$$

The field isomorphism λ of (200) gives the usual field-theoretic Kummer comparison

$$(202) \quad \text{Kum}_{\mu_{\log}}(y(x)) \longmapsto \text{Kum}_{\mu_{\text{current}}}(\lambda(y(x))).$$

Its group and coefficient maps are those induced by that field isomorphism. The operation $y \mapsto \text{Kum}_{\mu_{\log}}(y)$ is not linear in the old unit-H1 coordinate (201). The new multiplication used here is already part of [62, Definition 1.1(i), p. 24], and its monoid, units and action are used explicitly in [62, Proposition 3.4(i), p. 102].

Indeed, a same-group integral-coefficient H1 arrow cannot identify the two Kummer maps just displayed. If $v_E(x) > 0$, the raw class $\text{Kum}_{\mu_{\text{current}}}(x)$ has nonzero valuation component, while $\kappa_{\text{lin}}^-(x)$ has valuation component zero. The unit subspace is the annihilator of unramified characters under local Tate duality, so a Galois-induced isomorphism with compatible integral Tate coefficients preserves it. The comparison (202) concerns the new multiplicative structure and does not assert this impossible old-H1 identification.

No global exponential is being used. The element $\exp(p^N x)$ in (197) need only belong to the completion E . It is not a global rational-function torsor generator. The source global localization is pulled back along the globally synchronized local isomorphisms [62, Proposition 3.3]. The same distinction is explicit in [4, Definition 5.4(ii), pp. 125–126]: the global log construction re-embeds the global object through its natural local isomorphisms and keeps the underlying global Galois theater. Its Remark 5.4.1, p. 129, warns that the local logarithm diagrams do not extend as a logarithm on the global number field.

21.3. The canonical core coordinate and the preceding layer. Let $\mathcal{C}_E$ denote the finite Galois-invariant part of the canonical module $k^\sim(G_E)$ in [4, Proposition 5.8(ii), p. 139]. Use the local reciprocity convention of that unit reconstruction, $\text{rec}_E : E^\times \rightarrow G_E^{\text{ab}}$. The concrete model of $\mathcal{C}_E$ is the rationalized old unit group with its finite torsion removed.

Proposition 21.2 (The specified unit comparison). *There is a standard comparison*

$$(203) \quad \sigma_E : E \xrightarrow{\sim} \mathcal{C}_E, \quad \sigma_E(x) = p^{-N}[\text{rec}_E(\exp(p^N x))],$$

and the source shell is $\sigma_E(I)$. For the unit Kummer comparison

$$(204) \quad \eta_E : H_f^1(G_E, \mathbb{Q}_p(1)_{\text{prev}}) \xrightarrow{\sim} \mathcal{C}_E, \quad \eta_E(\text{Kum}(u)) = [\text{rec}_E(u)],$$

one has

$$(205) \quad \sigma_E = \eta_E \kappa_{\text{lin}}^-.$$

Proof. The brackets in (203) refer to the unit-perfection module. Independence of sufficiently large N follows from the exponential identity used for (197). Standard logarithm on a sufficiently small unit subgroup proves additivity, $\mathbb{Q}_p$ -linearity and bijectivity. The actual unit image is the pre-log-shell, and its multiple p^{-1} is the shell. This gives $\sigma_E(I)$, with no further factor p^{-1} in (203).

The Kummer sequence and Hilbert 90 identify the rationalized torsion-free unit group with H_f^1 , so (204) is well defined. Evaluating it on $p^{-N} \text{Kum}(\exp(p^N x))$ proves (205).

This is the particular comparison specified by the reconstruction source. Parts (c) and (d) of [4, Corollary 5.10(iv), p. 148] give the natural comparison from the field-theoretic shell to the canonical mono-analytic shell, compatible with the shells and volumes. Its proof on p. 149 compares the base field inside abelianizations with its Kummer reconstruction by the fundamental-class and cyclotomic isomorphisms of Corollary 1.10(a),(c). The proof identifies the result as the reciprocity map of local class field theory. On a concrete unit u , this is (204). Thus the comparison used here is not an arbitrary linear isomorphism chosen after the point. $\square$

Write $k_-^\times, k_+^\times$ for the original constant-monoid reconstruction maps of [64, Corollary 4.6(iii)], from the two copied theaters to the common core. In the fixed concrete model they reconstruct the same labeled field. Before forgetting multiplication, the relevant diagram is

$$(206) \quad \begin{array}{ccc} L^- & \xrightarrow{\lambda} & E^+ \\ \log(k_-^\times) \downarrow & & \downarrow k_+^\times \\ L^c & \xrightarrow{\lambda_c} & E^c. \end{array}$$

The lower map is the corresponding tautological standard-log representative. Both paths send $y(x)$ to the constant-field element of coordinate x . To check this directly, first take a sufficiently small old unit: its constant-monoid reconstruction is its actual reconstructed constant, and standard logarithm commutes term by term with this copy-identity field reconstruction. Division by p^N extends the equality to every $y(x)$. The left arrow is formed from the old-unit map and then the new log-field structure; its value in the core additive carrier is $\sigma_E(x)$. The right arrow is a current multiplicative constant-monoid map. These are not two versions of the same H1 arrow.

Let

$$\mathcal{C}_{j,p} = I^{\mathbb{Q}}(S_{j+1}^\pm; {}^{0,\circ}D_p^\perp), \quad \iota_j = \sigma_E^{\otimes m_j} : \mathcal{T}_{m_j} \xrightarrow{\sim} \mathcal{C}_{j,p}.$$

The superscript $\perp$ here denotes the source's mono-analytic prime-strip, not an orthogonal complement. Its identification with the tensor of $\mathcal{C}_E$ is the natural one in [62, Proposition 3.2(i),(ii), pp. 97–99]. Denote the label- j components of the split LGP monoids at level 0 and at the core by $\mathcal{S}_j^+$ and $\mathcal{S}_j^c$. Applying the distinguished-slot recipe to (206) gives

$$(207) \quad \begin{array}{ccc} \mathcal{S}_j^+ & \hookrightarrow & \mathcal{T}_{m_j} \\ k_{\text{LGP},0} \downarrow & & \downarrow \iota_j \\ \mathcal{S}_j^c & \hookrightarrow & \mathcal{C}_{j,p}. \end{array}$$

The upper right is written in the standard-log coordinates of the *preceding* field L^- . Before choosing coordinates the right-hand map is $\iota_j \mathcal{L}^{\otimes m_j}$.

Here the upper inclusion is [62, Proposition 3.4(ii), pp. 102–103]: pull the current Gaussian monoid back to the preceding log-field, then tensor with the new units in the other slots. The lower inclusion is the same recipe on the reconstructed constant monoids. Proposition 3.5(i), pp. 103–104, explicitly uses the comparisons at *both* indices $r' = r, r - 1$. The commutativity of (206), followed by this tensor operation, verifies (207) in the selected concrete base representative. It is not obtained by replacing the preceding carrier with the current H1 space.

The same diagram at all places and the copy-identity of the global field give the corresponding base fractional-ideal comparison. Both recipes use the same local ideals and the same global rational-function monoid acting on them. This is the construction recorded in [62, Proposition 3.10(i), pp. 147–148], again with both adjacent indices. We use this fixed representative, not the later assertion of compatibility for every vertical iterate.

Proposition 21.3 (The variance of the actual core action). *For a continuous automorphism α of G_E , let M_α be its action on the unit abelianization, expressed in logarithmic coordinates:*

$$(208) \quad M_\alpha(\log u) = \log \left(\text{rec}_E^{-1} \alpha^{\text{ab}}(\text{rec}_E(u)) \right).$$

Extend this map $\mathbb{Q}_p$ -linearly to E . The actual canonical core map Φ_α satisfies

$$(209) \quad \Phi_\alpha \sigma_E = \sigma_E M_\alpha^{-1}, \quad \Phi_\alpha(\sigma_E(I)) = \sigma_E(I).$$

Proof. The unit image in the abelianization is preserved by the reconstruction algorithm. Proposition 5.8(i) of [4, p. 139] specifies its *contravariant* functoriality by transfer. For a group isomorphism this transfer is the inverse of the induced abelianization map; there is no index factor. Thus

$$\Phi_\alpha(\sigma_E(x)) = p^{-N} [\alpha^{-1, \text{ab}}(\text{rec}_E(\exp(p^N x)))] = \sigma_E(M_\alpha^{-1}x).$$

The unit image and its perfection are preserved. Dividing the unit image by p proves the shell assertion. $\square$

For comparison, a Kummer action with a chosen compatible integral Tate-module isomorphism δ is

$$F_{\alpha, \delta} = c_{\alpha, \delta} M_\alpha^{-1}, \quad c_{\alpha, \delta} \in \mathbb{Z}_p^\times,$$

by Proposition 11.5. We do not silently identify this with (209). All pointwise statements below use the actual canonical core operator $F = M_\alpha^{-1}$. In particular, if a full Galois lift α_0 has canonical matrix M , then α_0^{-1} gives core matrix $M_{\alpha_0}^{-1} = M$. No additional scalar choice is required.

21.4. Local ideal elements of one global pilot.

Lemma 21.4 (Points belonging to one pilot object). *In the synchronized representative above, there is one global theta-pilot $\mathcal{P}$, in the local fractional-ideal realization of [62, Definition 3.8(i)], whose label- j ideal at the distinguished place contains Z_{j, ζ_j} for every $\zeta_j \in \mu_{2\ell}$.*

Proof. Choose one standard normalized theta-evaluation representative. Its label- j value is a $\mu_{2\ell}$ -multiple of a_j , by Lemma 20.1 and [64, Remark 2.5.1(i), p. 72]. The Gaussian splitting monoid is generated, up to torsion, by this value-profile [64, Corollary 3.6(ii),(iii), pp. 99–100]; Corollary 4.6(iv), pp. 138–139, supplies it in the chosen theater.

Proposition 21.1 and [62, Proposition 3.4(ii)] pull that generator to the preceding log-field with the same standard coordinate. The single-slot insertion of [62, Proposition 3.1(ii), p. 93] tensors it with the new field units. The local fractional ideal generated by this element contains its generator times the unit. Multiplication by a root of unity does not change that ideal, since these are units of the new log-field. This proves the local claim for every ζ_j .

At the other bad places choose the prescribed splitting generators of the same theater, and use its standard good-place constituents. The finite collection of bad-place local ideals defines a global object by [62, Proposition 3.7(v), pp. 111–112]. Definition 3.8(i), p. 112, is precisely the theta-pilot designation for that object. Thus all the local ideal elements in the statement belong to one $\mathcal{P}$. The construction does not require the local generators to be a single element of the global number field. $\square$

The lemma uses elements of the actual local ideals. It need not assert that every independent tuple (ζ_j) arises from one strictly chosen global theta-evaluation representative. The ideals, and hence the global pilot object, are unchanged by those torsion choices.

We also record exactly which local Galois operations are permitted. The Ind1 object in [62, Theorem 3.11(i), pp. 153–154] is the output mono-analytic procession of [63, Proposition 6.9(ii)]. At the selected nonarchimedean place, its constituent category is the category of finite transitive continuous G_E -sets. Pullback along any continuous $\alpha \in \text{Aut}(G_E)$ is a self-equivalence, with inverse the pullback along α^{-1} . Use this equivalence at every repeated occurrence above p and the identity elsewhere, keeping the place and capsule indices fixed.

Definition 4.1(iii),(iv), p. 96, of [63] makes this a constituent prime-strip isomorphism. Its Section 0's capsule-full poly-isomorphisms contain the chosen collection with the identity index

map. Definition 4.10, pp. 119–120, makes these precisely the representative data permitted at the stages of the procession. The same choice at repeated occurrences is allowed, and conjugation leaves the full connecting collections unchanged. Thus the collection occurs inside an allowed procession poly-automorphism, as in the representative argument of Section 7. No extension of α to the earlier curve covering category or global Hodge theater is imposed by this output object.

Theorem 21.5 (Membership in a specified base branch). *Let $\mathcal{P}$ be the fixed global pilot of Lemma 21.4, and let ${}^{0,\circ}U_{j,p}^{\text{raw}}$ denote the raw possible-image union defined in [62, Corollary 3.12, p. 174], before its holomorphic hull is taken. For every continuous $\alpha \in \text{Aut}(G_E)$, put $F = M_\alpha^{-1}$. Then, simultaneously for all labels,*

$$(210) \quad \iota_j(F^{\otimes m_j} Z_{j,\zeta_j}) \in {}^{0,\circ}U_{j,p}^{\text{raw}} \quad (1 \leq j \leq h).$$

These images use the fixed adjacent-layer branch (207), the same local Ind1 representative at every occurrence above p , the identity elsewhere, and identity Ind2. The assertion is membership for this branch, not compatibility of all branches.

Proof. Lemma 21.4 puts Z_{j,ζ_j} in the local ideal of $\mathcal{P}$. The concrete base construction (207), including its fractional-ideal realization, therefore places $\iota_j Z_{j,\zeta_j}$ in the core realization of that same pilot. The permitted representative described above acts on this local packet by $\Phi_\alpha^{\otimes m_j}$. Proposition 21.3 gives

$$\Phi_\alpha^{\otimes m_j}(\iota_j Z_{j,\zeta_j}) = \iota_j F^{\otimes m_j} Z_{j,\zeta_j}.$$

This is thus an element of the image of an actual local ideal of the pilot under an allowed representative.

The raw set on p. 174 is the union of such possible images. The basic Kummer image is the branch specified in [62, Proposition 3.5(ii)(a)(1), pp. 104–105]; here it has not been precomposed with a positive log-iterate. Including the further vertical branches in that union does not remove the chosen image. This proves (210). In particular, we have not treated Ind3 as a group with an identity element or used a claimed commutativity of every vertical Kummer map. $\square$

Remark 21.6 (Transport of an ideal and its test module). The source gives the splitting monoid both as a subset of the packet and as a multiplicative action [62, Theorem 3.11(i)(b), pp. 153–154]. If its local ideal is presented as $J = m_z(R)$, with $1 \in R$, then any additive carrier isomorphism Φ satisfies

$$(211) \quad \begin{aligned} \Phi J &= (\Phi m_z \Phi^{-1})(\Phi R), \\ (\Phi m_z \Phi^{-1})(\Phi 1) &= \Phi z. \end{aligned}$$

Both identities follow by applying the maps to an element of R . They require no multiplicativity of Φ . Using the conjugated operator on a fixed native 1 or a fixed native integer module would be a different operation. The proof of Theorem 21.5 uses ΦJ , with the test module and vector transported as in (211). Its embedded global field is also transported; it is not required to be stabilized in its old native coordinates.

For general initial data with more than one selected place above p , the ambient carrier is $\bigotimes_{\text{slots}} (\bigoplus_{v|p} E_v)$. The point $1^{\otimes j} \otimes x_j$ in a specified all- v block is then the projection of the source's actual single-slot point. The corresponding statement is membership in the projection of the full possible-image set, with the same global pilot image as preimage. Zero extension of that block does not follow. The condition $F_{\text{mod}} = \mathbb{Q}$ above is what permits the literal membership (210) in the whole local packet.

21.5. An attained local lower inclusion and its boundary. Let

$$\Gamma_E^{\text{can}} = \{M_\alpha^{-1} : \alpha \in \text{Aut}_{\text{cont}}(G_E)\}.$$

These are the canonical core operators, expressed in the fixed standard-log chart. Retain $k_j = \lfloor 2j^2/\ell \rfloor + 1$, $\kappa = 1 - 1/e$, and the normalized tensor integer ring B_{m_j} of Section 20.

Corollary 21.7 (An attained lower ideal in the possible images). *For every tuple $(\zeta_1, \dots, \zeta_h) \in \mu_{2\ell}^h$, there is one $F \in \Gamma_E^{\text{can}}$ such that*

$$(212) \quad \begin{aligned} v_E(F(1)) &= -\kappa, \\ v_E(F(x_j)) &= k_j - \kappa \quad (1 \leq j \leq h). \end{aligned}$$

For every label and fixed torsion representative,

$$(213) \quad \begin{aligned} \overline{\text{span}_{B_{m_j}} \{F^{\otimes m_j} Z_{j, \zeta_j} : F \in \Gamma_E^{\text{can}}\}} &= P_j, \\ P_j &= \beta^{ek_j - (e-1)m_j} B_{m_j}. \end{aligned}$$

Consequently, in this fixed local chart,

$$(214) \quad P_j \subseteq \overline{\text{span}_{B_{m_j}} (\iota_j^{-1}(0, \circ U_{j,p}^{\text{raw}}))}.$$

Proof. The proof of Theorem 20.3 first constructs a canonical matrix M from a genuine full-Galois lift, by tame inertia and the explicit parabolic operations. Before making the final Tate-coefficient comparison, this M already sends 1 and every x_j/p^{k_j} to $I \setminus \beta I$. If its lift is α_0 , Proposition 21.3 shows that α_0^{-1} has canonical core action M . This proves (212) without a coefficient rescaling.

Every canonical core operator preserves I , hence each $p^k I$. Since $1 \in I$ and $x_j \in p^{k_j} I$, each field component of its tensor image has valuation at least $k_j - m_j \kappa$. The component embeddings preserve the native valuation, so every such image belongs to P_j . The common operator just constructed gives this exact valuation in every component. Its tensor is therefore a generator of P_j times a unit of B_{m_j} . This gives the reverse inclusion before closure; the fractional ideal is closed. Finally Theorem 21.5 places all these orbit points in the inverse image of the raw union, proving (214). $\square$

The quantifier order remains $\forall(\zeta_j) \exists F \forall j$. Nothing requires one operator to work for all torsion tuples. Equation (214) does not identify the whole possible-image hull with P_j . Other allowed representatives and vertical branches may enlarge it.

The difference between the pure log-field point and a one-plus-unit Kummer input can also be detected exactly.

Lemma 21.8 (Trace distinguishes two points with the same ideal behavior). *For $s = j^2$, $\zeta \in \mu_{2\ell}$, put*

$$a = r_0^s, \quad k = \lfloor 2s/\ell + \kappa \rfloor, \quad t = p^{-1} \log(1 + p\zeta a), \quad \varepsilon = \zeta^\ell \in \{1, -1\}.$$

Then

$$(215) \quad \begin{aligned} \text{Tr}_{E/\mathbb{Q}_p}(\zeta a/p^k) &= 0, \\ \text{Tr}_{E/\mathbb{Q}_p}(t/p^k) &= \frac{30}{p^{k+1}} \log(1 + \varepsilon p^\ell b_0^s) \neq 0. \end{aligned}$$

The valuation of the second line is $\ell - 1 + 2s - k \geq \ell$.

Proof. The tame character of r_0^s has order ℓ , because $\ell \nmid j^2$. Its ℓ conjugates sum to zero, each occurring fifteen times in E/K_0 ; ζ is fixed by this inertia. This proves the first trace equality. Multiplication of the same conjugates gives

$$N_{E/K_0}(1 + p\zeta r_0^s) = (1 + \zeta^\ell p^\ell b_0^s)^{15}.$$

The sign is positive because ℓ is odd. The right side lies in $\mathbb{Q}_p$, so its norm from K_0 squares it. Taking logarithms on principal units and dividing by p^{k+1} proves the second line of (215). Since $p \nmid 30$, its first nonzero logarithmic term gives valuation $\ell + 2s - k - 1$. Finally $k = \lfloor 2s/\ell \rfloor + 1 \leq s$ for $s \geq 1$ and $\ell \geq 7$. The lower bound follows. $\square$

Allowed integral-coefficient Galois/Kummer maps preserve the trace-zero kernel by Proposition 7.5. Canonical core maps differ from such maps by a unit normalization, so they preserve it as well. The two points in (215) cannot literally lie in the same such orbit. This is consistent with the per-operator principal-ideal equality of Proposition 20.5. The membership theorem

used the pure points from the actual log-field pullback, not a substitution of the one-plus-log points for them.

The remaining boundary is global. The initial curve, prime ℓ , places and q were fixed at the start. Before Ind1, the copy-identity preserves the root equation and its marked value-group generator. After Ind1, however, one cannot conclude that F fixes the native scalar q , the native ring unit or the old global field embedding. The transported multiplicative data retain their abstract value-group generator; its valuation in the old native chart is a different datum.

The q-pilot in [62, Corollary 3.12, p. 174] is explicitly left outside Ind1, Ind2 and Ind3. Keeping that original object in the notation is not a proof of the full horizontal compatibility assertion. Theorem 21.5 concerns a fixed base branch in one column, not a transport of that possible-image set through a horizontal theta-link.

One still has to match the complete globally synchronized pilot family, including the other places and all required vertical branches, with the precise holomorphic reference, global weights and region used in the subsequent upper bound. The local inclusion does not identify the one-prime bundle of Theorem 22.1 with that full pilot. Nor does it establish the restriction to prime-strip-interpretable regions made later in the proof on p. 175 of [62]. The reconstruction comparisons, actual full-Galois action and source membership in this section are mathematical arguments using the specified external reconstruction results; they are not new Lean formalizations. No global *abc* conclusion is inferred.

22. ARITHMETIC BUNDLES FOR THE SPECIFIED NATIVE HULLS

The local equalities already proved in Section 14 admit an explicit realization by arithmetic vector bundles on the actual level field. We construct these bundles, compute their normalized degrees, and descend a sufficiently powered weighted determinant to the field of moduli $\mathbb{Q}$. This realizes ordinary arithmetic objects with the required numerical normalization. It does not establish their membership in the full possible-output family of IUT III, Corollary 3.12.

The relevant source conventions are the relative weighted determinant and sufficiently divisible positive tensor powers in [62, Remark 3.9.5(vii)(Ob3)], the label average in [62, Proposition 3.9(i)–(iii)], and the normalized local weights in [65, Proposition 1.4(i), Remark 1.7.1]. The additional Ind3 comparisons and the particular global prime-strip identification in [62, Theorem 3.11(ii), Corollary 3.12] are separate requirements; none is assumed in the arithmetic arguments below.

22.1. The fixed native input and its reference. Fix a member of the family in Theorem 16.2, with primes $\ell \equiv 43 \pmod{60}$ and $p \equiv -1 \pmod{30\ell}$, and write

$$D_A : y^2 = x(x - A^2)(x + A^2 + 1), \quad K = \mathbb{Q}(i, D_A[30\ell]).$$

The field K is Galois over $\mathbb{Q}$. At the specified place over p , its completion is the field of (114):

$$E = \mathbb{Q}_p(\mu_{30\ell}, \varpi), \quad \varpi^{15\ell} = b_0, \quad b_0 \in \mathbb{Q}_p, \quad v_p(b_0) = 2.$$

Put

$$\begin{aligned} e &= 15\ell, \quad f = 2, \quad d = ef, \quad h = (\ell - 1)/2, \quad \kappa = (e - 1)/e, \\ \beta &= \varpi^{(e+1)/2}/p, \quad r_0 = \varpi^{15}, \\ m_j &= j + 1, \quad N_j = d^{m_j-1}, \quad D_j = d^{m_j}, \\ a_j &= r_0^{j^2}, \quad k_j = \left\lfloor \frac{2j^2}{\ell} \right\rfloor + 1, \\ \eta_j &= ek_j - (e - 1)m_j \quad (1 \leq j \leq h). \end{aligned} \tag{216}$$

The element β is a uniformizer by Lemma 14.1; ϖ has valuation $2/e$ and is not one. All roots are fixed before the permitted arrows are chosen.

For each label define

$$\begin{aligned} T_j &= E^{\otimes_{\mathbb{Q}_p} m_j}, \quad A_j = \mathcal{O}_E^{\otimes_{\mathbb{Z}_p} m_j}, \\ B_j &= \prod_{\text{Gal}(E/\mathbb{Q}_p)^{m_j-1}} \mathcal{O}_E \subset T_j. \end{aligned}$$

The product decomposition uses the first factor as coefficient field and has N_j copies of E . In general $A_j \neq B_j$. The native lattice is $I = p^{-1} \log \mathcal{O}_E^\times = \beta^{1-e} \mathcal{O}_E$. Let Γ_E be the already constructed family of actual native integral Kummer arrows on I , using the same arrow in repeated slots. We retain precisely the marked point inputs

$$(217) \quad \begin{aligned} \rho &= p^{-1} \log_{\text{BK}}^{\text{std}}, \\ u &= \rho(\text{Kum}(1+p)) = p^{-1} \log(1+p), \\ \tau_j &= \rho(\text{Kum}(1+pa_j)) = p^{-1} \log(1+pa_j). \end{aligned}$$

Their closed B_j -span after the permitted arrows is

$$P_j^\rho = \overline{\text{span}_{B_j} \{F(\tau_j) \otimes (Fu)^{\otimes j} : F \in \Gamma_E\}}.$$

Corollaries 14.5 and 14.7, with the per-arrow bridge of Proposition 12.1, give

$$(218) \quad M_j^\rho := \mathcal{M}_{\Gamma_E}(a_j) = P_j^\rho = \beta^{\eta_j} B_j.$$

The first term is the pre-transport principal-ideal hull already defined in (102). Equality here identifies these specified native hulls, not their input points or the full published theta-pilot family. No additional choice of a torsion representative, vertical level, or Ind3 enlargement is included in (217)–(218).

Let μ_{B_j} be ordinary additive Haar measure on T_j with $\mu_{B_j}(B_j) = 1$, and write

$$(219) \quad V_j^B(H) = D_j^{-1} \log \mu_{B_j}(H).$$

The positive root $\mu_{B_j}^{1/D_j}$ is not another additive Haar measure. For a general field of moduli F_0 , the normalized weight of an ordered word $(v_0, \dots, v_j)$ over a rational prime is

$$\frac{\prod_{i=0}^j [(F_0)_{v_i} : \mathbb{Q}_p]}{[F_0 : \mathbb{Q}]^{m_j}}.$$

Indeed the sum of the numerator over all words is the m_j -th power of $\sum_{v|p} [(F_0)_v : \mathbb{Q}_p] = [F_0 : \mathbb{Q}]$. In the rational branch there is one selected-place word and this weight is 1. The remaining procession normalization is $h^{-1} \sum_{j=1}^h$, with no further division by m_j .

22.2. Actual fractional ideals and normalized degree. At a finite place w of K , put $q_w = \#k(w)$ and $|x|_w = q_w^{-\text{ord}_w(x)}$. This is the local norm absolute value. At an infinite place σ , use ordinary real or complex modulus and multiplicity $n_\sigma = 1$ or 2 , respectively. For a metrized line $\bar{L}$, declare its finite local lattices to be the unit balls. For a nonzero rational section $s \in L \otimes_{\mathcal{O}_K} K$, define

$$(220) \quad \widehat{\deg}_K(\bar{L}) = -\frac{1}{[K : \mathbb{Q}]} \left(\sum_{w \text{ finite}} \log \|s\|_w + \sum_{\sigma|\infty} n_\sigma \log \|s\|_\sigma \right).$$

For a fractional ideal with the metrics below, the finite sum is finite. Replacing s by xs adds zero: the finite norm sum for $x \in K^\times$ is $-\log |N_{K/\mathbb{Q}}(x)|$, whereas the infinite sum is $\log |N_{K/\mathbb{Q}}(x)|$. This proves independence of the rational section with the indicated multiplicities.

For a fractional ideal $\mathfrak{a} \subset K$ with standard ambient infinite metric $\|x\|_\sigma = |\sigma(x)|$, the section 1 gives

$$(221) \quad \widehat{\deg}_K(\bar{\mathfrak{a}}) = -[K : \mathbb{Q}]^{-1} \log N(\mathfrak{a}).$$

In fact if $\mathfrak{a}_w = \pi_w^b \mathcal{O}_{K_w}$, then $\|1\|_w = q_w^b$. Thus a positive finite ideal exponent has negative degree in this lattice convention. Degree of a vector bundle means degree of its determinant, with the induced determinant metric.

Theorem 22.1 (Arithmetic realization of the native hulls). *Set*

$$(222) \quad \mathfrak{a}_j = \prod_{w|p} \mathfrak{p}_w^{\eta_j}, \quad \mathcal{E}_j = \mathfrak{a}_j^{\oplus N_j} \subset K^{N_j},$$

and give $\mathcal{E}_j$ the standard ambient Hermitian metric at every infinite place. These are projective $\mathcal{O}_K$ -modules of rank N_j . At each $w \mid p$ their local lattices identify with the conjugate of M_j^ρ over its first coefficient field. At all other finite places they are the standard lattices. Moreover

$$(223) \quad \frac{\widehat{\deg}_K(\overline{\mathcal{E}_j})}{N_j} = -\frac{\eta_j}{e} \log p = V_j^B(M_j^\rho).$$

Proof. A fractional ideal of the Dedekind domain $\mathcal{O}_K$ is invertible, hence projective of rank one. This applies also to negative η_j . Since $K/\mathbb{Q}$ is Galois, all completions above p have the same ramification index e , residue degree f , and local degree d . The product coordinates of T_j identify its first-factor lattice with N_j copies of $\mathcal{O}_{K_w}$. The depth η_j of (218) in every component is precisely the depth specified by (222). Uniformizer units do not change the lattice. Assigning the same exponent to every place over p also makes $\mathfrak{a}_j$ Galois-stable.

If g places lie above p , then $[K : \mathbb{Q}] = g e f$ and $N(\mathfrak{p}_w) = p^f$. Since $\det \mathcal{E}_j = \mathfrak{a}_j^{N_j}$, (221) gives

$$\widehat{\deg}_K(\overline{\mathcal{E}_j}) = -\frac{g f N_j \eta_j}{g e f} \log p = -\frac{N_j \eta_j}{e} \log p.$$

The rational section used here is the standard coordinate wedge, of norm 1 at each infinite place. Thus the infinite terms are exactly zero for the stated metrics, not omitted terms.

The local index calculation, valid for either sign of η_j , gives $\mu_{B_j}(M_j^\rho) = p^{-f N_j \eta_j}$. Dividing its logarithm by $D_j = e f N_j$ proves (223). $\square$

Equivalently the contribution per rank is

$$\sum_{w \mid p} \frac{[K_w : \mathbb{Q}_p]}{[K : \mathbb{Q}]} \left(-\frac{\eta_j}{e} \log p \right) = -\frac{\eta_j}{e} \log p.$$

Thus passing from one selected completion to all conjugate places introduces no further degree or tensor-length factor.

22.3. Weighted determinants, common twists, and descent.

Proposition 22.2. *Let C be any positive integer divisible by every $h N_j$. The metrized line*

$$(224) \quad \overline{\mathcal{L}} = \bigotimes_{j=1}^h (\det \overline{\mathcal{E}_j})^{\otimes C/(h N_j)}$$

is defined using positive integral tensor powers and satisfies

$$(225) \quad \frac{\widehat{\deg}_K(\overline{\mathcal{L}})}{C} = \frac{1}{h} \sum_{j=1}^h V_j^B(M_j^\rho).$$

Replacing every $\overline{\mathcal{E}_j}$ by $\overline{\mathcal{E}_j} \otimes \overline{Q}$, for one metrized line $\overline{Q}$, tensors (224) by $\overline{Q}^{\otimes C}$.

Proof. The divisibility hypothesis makes all powers integral and positive. Additivity of degree and (223) prove (225). The canonical determinant isometry

$$\det(\overline{\mathcal{E}_j} \otimes \overline{Q}) = \det \overline{\mathcal{E}_j} \otimes \overline{Q}^{\otimes N_j}$$

shows that the total exponent of $\overline{Q}$ is $\sum_j N_j C/(h N_j) = C$. $\square$

The reference determinant is present in this computation: $\det \mathcal{O}_K^{N_j}$ has the same standard metric and degree zero. Each determinant in (224) can equivalently be replaced by $\det \mathcal{E}_j \otimes (\det \mathcal{O}_K^{N_j})^{-1}$. If covolumes are instead computed after restriction of scalars to $\mathbb{Q}$, the field-discriminant and reference covolume factors appear in both equal-rank terms and cancel in this relative determinant.

Corollary 22.3 (Descent to the field of moduli). *If C is divisible by every ehN_j , put*

$$(226) \quad t = \frac{C}{eh} \sum_{j=1}^h \eta_j \in \mathbb{Z}.$$

Then $\bar{\mathcal{L}}$ is the isometric pullback from $\mathbb{Q}$ of the actual metrized line $p^t \mathbb{Z} \subset \mathbb{Q}$ with its ordinary ambient absolute-value metric. In rational tensor powers, $\bar{\mathbf{a}}_j$ is the pullback of the arithmetic line with finite exponent η_j/e at p and the standard infinite metric.

Proof. Unique ideal factorization and the common ramification index give

$$(227) \quad p\mathcal{O}_K = \prod_{w|p} \mathfrak{p}_w^e, \quad \mathfrak{a}_j^e = p^{\eta_j} \mathcal{O}_K.$$

Multiplication identifies the tensor metric on the left of the second equality with the ambient metric on the right, since absolute values multiply at every embedding. This proves the rational-tensor-power statement.

In (224), the exponent at each $w \mid p$ is

$$\sum_{j=1}^h \eta_j N_j \frac{C}{hN_j} = \frac{C}{h} \sum_{j=1}^h \eta_j = et.$$

There is no finite support elsewhere. Hence the actual lattice is $p^t \mathcal{O}_K$, and the multiplication identification preserves its standard infinite metric. This is the stated pullback. In particular its normalized degree is $-t \log p$. $\square$

An explicit simultaneous choice is $C = ehd^h$, since $N_j = d^j$ and $C/(hN_j) = e d^{h-j} \in \mathbb{Z}_{>0}$. Thus the descent concerns an actual powered determinant object, not merely an equality of degrees. Galois stability by itself would not have supplied unpowered integral descent; (227) is the additional step.

22.4. The exact procession average.

Proposition 22.4. *Let*

$$R_\ell = \sum_{j=1}^h (2j^2 \bmod \ell),$$

where residues are taken in $\{1, \dots, \ell-1\}$. Then

$$(228) \quad \frac{\widehat{\deg}_K(\bar{\mathcal{L}})}{C \log p} = \frac{\ell+1}{12} - \frac{\ell+5}{60\ell} + \frac{2R_\ell}{\ell(\ell-1)}.$$

Each summand $V_j^B(M_j^\rho)$ in (225) is strictly positive.

Proof. Since ℓ is odd prime and $1 \leq j \leq h < \ell$, the indicated residue is nonzero. The fractional part of $2j^2/\ell$ is therefore at least $1/\ell > 1/e$, which also verifies $k_j = \lfloor 2j^2/\ell + \kappa \rfloor$. Elementary summation gives

$$\sum_{j=1}^h m_j = \frac{h(\ell+5)}{4}, \quad \sum_{j=1}^h \left\lfloor \frac{2j^2}{\ell} \right\rfloor = \frac{\ell^2-1}{12} - \frac{R_\ell}{\ell}.$$

For the second identity, sum $2j^2/\ell$ and subtract the fractional parts. Substitution into the average

$$h^{-1} \sum_j (\kappa m_j - k_j)$$

gives (228). Moreover $2j^2/\ell < j$, so $k_j \leq j$ and

$$\kappa m_j - k_j \geq 1 - \frac{j+1}{e} > 0.$$

$\square$

As a numerical check at $\ell = 43$, direct integer summation gives $R_{43} = 473$, $\sum_j k_j = 164$, and $\sum_j \eta_j = -56508$. The right side of (225) evaluates to $18836 \log p / 4515$. No distributional assertion about quadratic residues is used here. In particular this positive degree is a property of (222), not a sign assertion about the complete theta-pilot quantity in the source.

22.5. The tensor-order correction.

Proposition 22.5. *Let μ_{A_j} be additive Haar measure with $\mu_{A_j}(A_j) = 1$. For every measurable region $H \subset T_j$ of finite positive volume,*

$$(229) \quad D_j^{-1} \log \mu_{A_j}(H) = D_j^{-1} \log \mu_{B_j}(H) + \frac{(m_j - 1)\kappa}{2} \log p.$$

Proof. Tameness gives the discriminant exponent $f(e - 1)$ for $\mathcal{O}_E/\mathbb{Z}_p$. On a tensor integral basis of A_j , the Gram matrix of the absolute algebra trace is the Kronecker product of the factor trace Gram matrices. Its determinant therefore has valuation

$$v_p \text{disc}(A_j) = m_j d^{m_j - 1} f(e - 1).$$

The product order B_j has $N_j = d^{m_j - 1}$ copies of $\mathcal{O}_E$, so

$$v_p \text{disc}(B_j) = d^{m_j - 1} f(e - 1).$$

Under a finite-index change of lattice, the trace determinant changes by the square of the index. Consequently

$$\log[B_j : A_j] = \frac{m_j - 1}{2} d^{m_j - 1} f(e - 1) \log p.$$

Finally $\mu_{A_j} = [B_j : A_j] \mu_{B_j}$. Dividing by $D_j = e f d^{m_j - 1}$ proves the formula. $\square$

The factor-weight prescription in [56, (9.10.3.1), (9.10.5.1)], for this rational branch, uses exponent $1/d$ on each factor. On tensor lattices this agrees with $\mu_{A_j}^{1/D_j}$, since tensoring a lattice change in one factor raises its determinant to the power D_j/d . It therefore has the correction (229) relative to the maximal-order reference. Agreeing on the factor weights alone does not remove the A_j/B_j index.

22.6. All-slot scaling and transport of the infinite metrics.

Proposition 22.6. *Write $\mathcal{E}_j^\rho = \mathcal{E}_j$ for the realization (222). Replace all m_j entries of the tuple $(\tau_j, u, \dots, u)$ specified in (217) by their standard Bloch–Kato logarithms, retaining the same cohomology classes, arrows, and ambient reference B_j . Then*

$$(230) \quad \begin{aligned} M_j^{\text{std}} &= p^{m_j} M_j^\rho = \beta^{\eta_j + e m_j} B_j = \beta^{e k_j + m_j} B_j, \\ V_j^B(M_j^{\text{std}}) &= V_j^B(M_j^\rho) - m_j \log p. \end{aligned}$$

For the global realization, making this change only at the specified rational prime p , while keeping the ambient infinite metrics fixed, gives

$$(231) \quad \begin{aligned} \mathcal{E}_j^{\text{std}} &= p^{m_j} \mathcal{E}_j^\rho, \\ \frac{\widehat{\deg}_K(\overline{\mathcal{E}_j^{\text{std}}}) - \widehat{\deg}_K(\overline{\mathcal{E}_j^\rho})}{N_j} &= -m_j \log p. \end{aligned}$$

The label-averaged change is $-(\ell + 5) \log p / 4$. If instead the infinite metrics are transported along multiplication by p^{m_j} , the map is an isometry of metrized bundles and their degrees are equal.

Proof. The two coordinates differ by the rational scalar p at every entry, including all backgrounds. The permitted arrows are $\mathbb{Q}_p$ -linear, and the tensor map is multilinear. Thus every image tensor is multiplied by p^{m_j} . Closure and B_j -span commute with this invertible scalar. Its determinant on the D_j -dimensional space has absolute value $p^{-m_j D_j}$, which proves (230).

At finite places away from p the scalar is a unit, and at every place above p its uniformizer exponent is $e m_j$. The degree calculation in Theorem 22.1 proves (231). The average of $m_j = j + 1$ is $(\ell + 5)/4$.

For the transported metric, the new infinite norm on a vector y is $p^{-m_j} \|y\|_\sigma$. The determinant norm is therefore multiplied by $p^{-m_j N_j}$. Since $\sum_{\sigma|\infty} n_\sigma = [K : \mathbb{Q}]$, its normalized infinite degree contribution increases by $m_j N_j \log p$, cancelling the finite change. This also follows from the resulting isometry. $\square$

Thus scaling a finite coordinate with fixed references differs from transporting a whole metrized global object. Neither operation allows the active root alone to change scale while the other source entries are silently left unchanged.

22.7. An almost-everywhere reference test.

Proposition 22.7. *Fix one global member and one label j . Outside a finite set of rational primes p' , take the declared background class $\text{Kum}(1 + p')$ at the selected completion, its m_j -fold tuple, and the native integral Kummer arrows. Relative to the maximal order $B_{j,p'}$, its native hull is $B_{j,p'}$, whereas its standard-coordinate hull is $(p')^{m_j} B_{j,p'}$. With the reference unchanged, the sum of standard-coordinate normalized local log-volumes is $-\infty$, and the volume product is zero. This is not a finite-support fractional-ideal modification of a fixed arithmetic bundle.*

Proof. Exclude 2, the primes ramified in K , the bad primes of the curve, and the other finitely many distinguished primes. The remaining completion $E_{p'}/\mathbb{Q}_{p'}$ is unramified. The logarithm and exponential on principal units give

$$\log \mathcal{O}_{E_{p'}}^\times = p' \mathcal{O}_{E_{p'}}, \quad I_{p'} = \mathcal{O}_{E_{p'}}, \quad u_{p'} = (p')^{-1} \log(1 + p') \in \mathbb{Z}_{p'}^\times.$$

The residue-root-of-unity part has logarithm zero. The tensor product of the unramified integral orders is finite etale over $\mathbb{Z}_{p'}$ and is already its maximal product order.

Every integral automorphism of $I_{p'}$ preserves nonzero classes modulo $p' I_{p'}$. Since the field is unramified, this is precisely the unit condition. Each permitted image of $u_{p'}$ is therefore a unit, and its repeated tensor has unit value in every field component. The identity arrow is present. The $B_{j,p'}$ -span of the native images is consequently exactly $B_{j,p'}$. The all-slot argument of Proposition 22.6 gives the standard hull $(p')^{m_j} B_{j,p'}$, with normalized log-volume $-m_j \log p'$, independently of the local degree.

Infinitely many rational primes remain, and $\sum_{p'} m_j \log p' = +\infty$, since every summand is at least $m_j \log 2 > 0$. This proves the assertions about the sum and product. A lattice in a fixed finite-dimensional K -space agrees with a fixed integral reference at all but finitely many places. Nonzero depth m_j at infinitely many distinct rational primes violates this condition. A finite contribution from fixed metrics at infinity cannot cancel the divergence. $\square$

Equivalently, the local scalars $(p')_{p'}$ do not form an idele: their inverses are nonintegral at infinitely many primes. One can replace the reference by $(p')^{m_j} B_{j,p'}$ at the same time, obtaining local volume 1 again, but that specifies a different restricted product and requires an explicit comparison with the old reference.

Remark 22.8 (The remaining identification). The almost-everywhere test concerns the declared native branch, not the complete Ind3 family. Likewise, the lattices assigned away from p in (222) and their standard infinite metrics are part of our construction. They have not been identified with the other components of the full theta-pilot output. Theorems about these actual arithmetic bundles therefore supply neither the particular membership in IUT III, Corollary 3.12, nor the required comparison across Ind3. Establishing that membership must retain the source classes, allowed arrows, reference structures, weights, and the link to the same global q-pilot. Equality of degrees of separately constructed objects does not provide it.

23. THREE-PRIME SUPPORT AND MIXED-FULL CAMPANA POINTS

This section records two complementary descriptions of the first locus not covered by the two-prime theorem. The first is an exact support decomposition. The second keeps multiplicities rather than support size and isolates the log-general-type Campana locus studied most recently by Browning–Verzobio [8].

23.1. The support-three gate. Let $a + b = c$ be a positive primitive triple with $\omega(abc) \leq 3$. Pairwise coprimality gives

$$\omega(abc) = \omega(a) + \omega(b) + \omega(c).$$

If $a, b > 1$, every term on the right is positive. Hence equality holds in the support bound, and there are distinct primes p, q, r and positive integers x, y, z such that

$$(232) \quad a = p^x, \quad b = q^y, \quad c = r^z.$$

If one input is one, the support split between the two consecutive nonunits is exactly $(1, 1)$, $(1, 2)$, or $(2, 1)$. Thus, after exchanging the inputs, the remaining equations are

$$(233) \quad 1 + p^x = q^y, \quad 1 + p^x = q^y r^z, \quad 1 + p^x q^y = r^z,$$

with distinct displayed prime bases. This proves an exhaustion rather than a density reduction: the two semiprime branches in (233) remain moving Diophantine problems.

The all-nonunit branch has the following useful rigidity.

Theorem 23.1 (Common-exponent rigidity). *Suppose that distinct primes satisfy $p^x + q^y = r^z$ and $d = \gcd(x, y, z) > 1$. Then, up to exchanging the inputs,*

$$2^4 + 3^2 = 5^2.$$

Proof. Set $X = p^{x/d}$, $Y = q^{y/d}$, and $Z = r^{z/d}$. Then $X^d + Y^d = Z^d$. Fermat's Last Theorem [84, 81] forces $d = 2$. In the resulting primitive Pythagorean triple the even leg is a power of two, say $X = 2^A$. Euclid's parametrization gives

$$X = 2mn, \quad Y = m^2 - n^2, \quad Z = m^2 + n^2,$$

where m, n are coprime and of opposite parity. Since mn is a power of two, one has $n = 1$ and $m = 2^k$. Now

$$Y = (2^k - 1)(2^k + 1)$$

is a prime power, while the two odd factors are coprime. The smaller factor must be one, so $k = 1$ and $(X, Y, Z) = (4, 3, 5)$. Squaring proves the assertion. $\square$

For $x, y, z \geq 2$, sorting the exponents shows that $1/x + 1/y + 1/z \geq 1$ occurs only for

$$(2, 2, n), (2, 3, 3), (2, 3, 4), (2, 3, 5), (3, 3, 3), (2, 4, 4), (2, 3, 6).$$

Elementary difference-of-powers arguments exclude the full signatures $(2, 3, 3)$ and $(2, 3, 6)$ in every output orientation. Theorem 23.1 excludes $(3, 3, 3)$ and $(2, 4, 4)$. For $(2, 2, n)$, the odd-exponent orientation reduces to $S^2 + 4 = R^n$; the theorem of Arif–Abu Muriefah [3] then yields the second of exactly two solutions

$$2^4 + 3^2 = 5^2, \quad 2^2 + 11^2 = 5^3.$$

The other orientations follow by factoring a difference of squares; even n follows from Theorem 23.1.

Consequently, if $c > \text{rad}(abc)$ in an all-nonunit support-three triple, then the exponent $\gcd$ is one and the all-nonlinear remainder consists only of $(2, 2, 3)$, $(2, 3, 4)$, $(2, 3, 5)$, or signatures with reciprocal sum below one. The first case is the genuine triple $2^2 + 11^2 = 5^3$, for which $125 > 110$. Linear exponents are essential:

$$7 + 181^2 = 2^{15}, \quad \frac{2^{15}}{2 \cdot 7 \cdot 181} > 12.$$

This example refutes a putative extension $c \leq \text{rad}(abc)$ to support three; it is not an abc counterexample. The unresolved branches are the two semiprime unit equations, linear exponents, the prime-base signatures $(2, 3, 4)$ and $(2, 3, 5)$, and the union over varying hyperbolic signatures. Darmon–Granville [32] gives finiteness for each fixed hyperbolic signature, not uniformly over that union.

23.2. Different fullness exponents. An integer $n > 0$ is m -full when every prime divisor occurs with valuation at least m . Let $P = (a, b, c)$ be a primitive *abc* point and suppose that a, b, c are respectively p -, q -, and r -full, where $p, q, r > 0$. Put

$$\sigma_{p,q,r} = \frac{1}{p} + \frac{1}{q} + \frac{1}{r}.$$

Theorem 23.2 (Mixed-full radical compression). *One has*

$$(234) \quad \text{rad}(abc)^{pqr} \leq c^{qr+pr+pq},$$

$$(235) \quad \log \text{rad}(abc) \leq \sigma_{p,q,r} \log c.$$

Moreover,

$$\sigma_{p,q,r} < 1 \iff qr + pr + pq < pqr.$$

Proof. Fullness gives $\text{rad}(a)^p \mid a$, and similarly for b, c . Raise these three divisibilities to qr, pr, pq , multiply them, and use radical multiplicativity across a primitive triple. Since $a, b < c$, this gives

$$\text{rad}(abc)^{pqr} \leq a^{qr} b^{pr} c^{pq} \leq c^{qr+pr+pq},$$

which proves (234). Taking the three logarithmic inequalities separately gives

$$\log \text{rad}(abc) \leq \frac{\log a}{p} + \frac{\log b}{q} + \frac{\log c}{r} \leq \sigma_{p,q,r} \log c,$$

proving (235). Clearing the positive denominator pqr proves the final equivalence. $\square$

Corollary 23.3 (A strict Campana disproof gate). *Fix $p, q, r > 0$ with $\sigma_{p,q,r} < 1$. The *abc* conjecture implies a uniform height bound for every primitive *abc* point with the three prescribed fullness conditions. Conversely, an unbounded family of such points implies the negation of the standard *abc* conjecture.*

Proof. Choose $\varepsilon > 0$ so that $(1 + \varepsilon)\sigma_{p,q,r} < 1$. If *abc* holds with constant C_ε , Theorem 23.2 gives

$$(1 - (1 + \varepsilon)\sigma_{p,q,r}) \log c \leq C_\varepsilon.$$

The coefficient is positive, so c is bounded. This proves the first claim. The same displayed inequality contradicts unbounded height and proves the second. $\square$

Browning–Verzobio [8, Theorems 1.1 and 1.2] study exactly these points. If $p = r + u$, $q = r + v$, with fixed $u \geq v \geq 0$, they prove, for all sufficiently large r ,

$$N(B) \ll B^{1/p+1/q-\eta_{u,v}(r)+\varepsilon}, \quad \eta_{u,v}(r) = r^{-2} + O_{u,v}(r^{-5/2}) > 0.$$

They also obtain a power saving for every $r \geq 2$ in the special family $(p, q, r) = (r + 2, r + 1, r)$. This is a genuine improvement over the trivial $B^{1/p+1/q}$ count and applies precisely to the low-radical locus in Corollary 23.3. Its exponent remains positive, so it proves neither the predicted boundedness nor the existence of an unbounded family. The determinant-method and function-field inputs of that paper are cited mathematical results and are not declared as Lean axioms.

24. A COEFFICIENT-BELOW-SIX OBSTRUCTION AND FINITE SELECTED-PLACE HULLS

We next close two proposed shortcuts. On the Frey route, optimization over an entire rational isogeny class cannot lower the coefficient below six. On the local IUT route, projected membership does suffice after the actual finite product-order hull, but that finite statement does not determine a global Arakelov degree.

24.1. An exact conductor and an isogeny-class obstruction. For $n \geq 1$, put $c = 1792n + 2$ and

$$E_c : y^2 = x(x-1)(x+c-1) = x^3 + (c-2)x^2 + (1-c)x.$$

The preceding sections determine the four vertices of its rational isogeny class and prove that the minimum absolute logarithmic Weil height in that class is

$$(236) \quad 3 \log(c^2 - 16c + 16).$$

Proposition 24.1 (Exact local and global conductor). *The displayed equation is minimal at every prime. At 2 it has Kodaira symbol III and conductor exponent five. Every odd prime $s \mid c(c-1)$ is multiplicative of type $I_{2v_s(c(c-1))}$ and has conductor exponent one. Hence every elliptic curve $F/\mathbb{Q}$ rationally isogenous to E_c has*

$$(237) \quad N(F) = 16 \operatorname{rad}(c(c-1)).$$

If $D = (c-1)(c/2)^4$ is the reduced denominator of the height-minimizing j -invariant, then

$$(238) \quad N(F) = 32 \operatorname{rad}(D).$$

Proof. Write $A = c - 2$ and $B = 1 - c$. Then

$$v_2(A) \geq 8, \quad v_2(B) = 0, \quad B \equiv -1 \pmod{4}.$$

The corresponding row of the complete two-torsion model table in Mulholland [50, Theorem 2.1 and Table 2.1], which specializes Papadopoulos's classification, gives type III, conductor exponent five, and minimality at two. The invariants are

$$c_4 = 16(c^2 - c + 1), \quad \Delta = 16(c-1)^2 c^2.$$

If an odd prime s divides c or $c-1$, then $c^2 - c + 1 \equiv 1 \pmod{s}$. Thus c_4 is a unit and the model is minimal multiplicative, with the asserted type and conductor exponent. All other odd primes are good. This proves (237) for E_c . Rational isogenies identify the rational Tate modules and hence preserve every local Artin conductor, so the same formula holds throughout the class.

Writing $u = c/2$, one has $D = (2u-1)u^4$ and $\gcd(u, 2u-1) = 1$. Therefore

$$\operatorname{rad}(c(c-1)) = \operatorname{rad}(2u(2u-1)) = 2 \operatorname{rad}(D),$$

which gives (238). □

The congruence modulus permits an infinite power subfamily. Put

$$c_k = 2 \cdot 897^k, \quad n_k = \frac{897^k - 1}{896} = 1 + 897 + \cdots + 897^{k-1}.$$

Then $c_k = 1792n_k + 2$ and, since $897 = 3 \cdot 13 \cdot 23$,

$$(239) \quad N(E_{c_k}) = 28704 \operatorname{rad}(2 \cdot 897^k - 1) < 28704 c_k.$$

Theorem 24.2 (Uniform failure of every coefficient below six). *For every $0 \leq \theta < 6$ and $C \in \mathbb{R}$, all sufficiently large k satisfy*

$$h(j(F)) > \theta \log N(F) + C$$

simultaneously for every $F/\mathbb{Q}$ rationally isogenous to E_{c_k} .

Proof. For $c_k > 32$, (236) gives

$$h(j(F)) > 6 \log c_k - 3 \log 2.$$

Equation (239) and $\theta \geq 0$ give

$$\theta \log N(F) < \theta \log c_k + \theta \log 28704.$$

Their difference is greater than

$$(6 - \theta) \log c_k - 3 \log 2 - \theta \log 28704,$$

which tends to infinity because $c_k = 2 \cdot 897^k$ and $\theta < 6$. □

This rules out every subcritical coefficient obtained merely by changing the representative inside the rational isogeny class. It leaves the coefficient-six and all-epsilon estimate $(6 + \varepsilon) \log N + C_\varepsilon$ untouched; in particular, it does not prove that six is an attainable or optimal coefficient. On this family the remaining input is a lower bound for $\text{rad}(2 \cdot 897^k - 1)$; the elementary upper bound in (239) cannot supply it.

24.2. Finite products of selected-place components. Fix a rational prime and one label. After decomposing the finite-etale tensor packet into field factors, write

$$T = \prod_{w \in W} T_w, \quad B = \prod_{w \in W} B_w,$$

where W is the finite set of ordered selected-place words and B_w is the product of the integer rings in the field factors of T_w .

Proposition 24.3 (Exact product-order span). *For every subset $S \subset T$,*

$$(240) \quad \text{span}_B(S) = \prod_{w \in W} \text{span}_{B_w}(\pi_w S).$$

The same identity holds after topological closure.

Proof. Projection of a B -linear combination gives the forward inclusion. For the reverse inclusion, let $e_w \in B$ be the central idempotent supported on the w -component. If $y_w = \sum_i b_i \pi_w(s_i)$, then $\sum_i (e_w b_i) s_i$ belongs to $\text{span}_B(S)$, has component y_w at w , and has every other component zero. Sum this construction over the finite set W . Closure commutes with a finite product. $\square$

Suppose now that S satisfies the first-branch hypotheses of IUT III, Remark 3.9.5(i): it is relatively compact and contains a relatively compact finite-log-volume subset. The source holomorphic hull is the smallest λB containing S , with every field component of λ nonzero. Its definition is componentwise, and therefore

$$(241) \quad \text{Hull}_B(S) = \prod_{w \in W} \text{Hull}_{B_w}(\pi_w S).$$

The compactness hypothesis is essential: in the other branch of the source definition, the hull is the entire ambient space.

If full-rank product fractional ideals P_w satisfy

$$P_w \subset \overline{\text{span}_{B_w}(\pi_w S)} \quad (w \in W),$$

then (240) gives

$$\prod_w P_w \subset \text{Hull}_B(S).$$

For an ordered word $w = (v_0, \dots, v_j)$, set

$$h_v = [(F_{\text{mod}})_v : \mathbb{Q}_p], \quad \omega_w = \frac{\prod_i h_{v_i}}{[F_{\text{mod}} : \mathbb{Q}]^{j+1}}.$$

IUT IV, Remark 1.7.1 gives precisely these normalized weights. They sum to one because

$$\sum_w \prod_i h_{v_i} = \left(\sum_{v|p} h_v \right)^{j+1} = [F_{\text{mod}} : \mathbb{Q}]^{j+1}.$$

Monotonicity of Haar measure now yields the finite weighted lower bound

$$(242) \quad \sum_w \omega_w V_w(\pi_w \text{Hull}(S)) \geq \sum_w \omega_w V_w(P_w),$$

where $V_w = D_w^{-1} \log \mu_w$ and $D_w = \dim_{\mathbb{Q}_p} T_w$. No single raw member must realize all word projections: the central idempotents combine witnesses only after taking the B -span.

Equation (242) is a finite-place result. It does not specify the archimedean metric or prove that the finite lattices localize one global projective module. Indeed, keep every finite lattice of $L = \mathbb{Z} \subset \mathbb{Q}$ fixed and set $\|x\|_{\infty, t} = e^{-t}|x|$. All finite projections, hulls, and Haar volumes are

independent of t , while $\widehat{\deg}(\overline{L}_t) = t$. Thus finite projected membership alone cannot determine a global Arakelov inequality. The remaining source problem is to construct the same globally metrized marked family and compare it, through the actual Ind3 and horizontal links, with the fixed q -pilot.

25. SUBCRITICAL UNIFORMITY AND A PELL COUNTEREXAMPLE GATE

This section runs the positive and counterexample programs against the same unchanged target. The positive result identifies the exact pointwise uniformity statement equivalent to *abc*. The counterexample analysis shows why fixed residual coefficients and polynomial identities cannot supply an unbounded strict mixed-full family, while retaining one explicit conditional Pell route. No condition introduced below is asserted to hold without proof.

25.1. The exact positive gate. For an *abc* point P , retain

$$H(P) = \log c, \quad R(P) = \log \text{rad}(abc).$$

Consider the pointwise uniformity statement

$$(243) \quad \forall \mu \in [0, 1) \quad \exists B_\mu \in \mathbb{R} \quad \forall P, \quad R(P) \leq \mu H(P) \implies H(P) \leq B_\mu.$$

Theorem 25.1 (Subcritical-locus equivalence). *The standard logarithmic abc conjecture is equivalent to (243).*

Proof. Assume *abc*, fix $0 \leq \mu < 1$, and set $\eta = (1 - \mu)/2$. Then

$$d = 1 - (1 + \eta)\mu = \frac{(1 - \mu)(2 - \mu)}{2} > 0.$$

If C_η is the *abc* constant and $R(P) \leq \mu H(P)$, then

$$dH(P) \leq C_\eta,$$

so $B_\mu = C_\eta/d$ works.

Conversely, assume (243). Given $\varepsilon > 0$, put $\mu = (1 + \varepsilon)^{-1}$ and let B_μ be its bound. If $R(P) \leq \mu H(P)$, then

$$H(P) \leq B_\mu \leq (1 + \varepsilon)R(P) + \max\{B_\mu, 0\}.$$

If the displayed premise fails, then $H(P) < (1 + \varepsilon)R(P)$. Thus the single constant $C_\varepsilon = \max\{B_\mu, 0\}$ works for every P . $\square$

This is a proved equivalence, rather than a replacement definition. It shows that density zero, an exceptional-set estimate, or a positive power saving cannot by itself prove *abc*: every point on every fixed subcritical slope must be bounded.

25.2. What counting would have to accomplish. The logical gap persists even under very rapid separation. Put

$$X_n = 2^{2^n}.$$

Then $X_{n+1} = X_n^2$, but the sequence is infinite. If $A(X) = \#\{n : X_n \leq X\}$, then for every integer $X \geq 2$,

$$(244) \quad A(X)^2 \leq X.$$

Indeed, two base-two logarithms give $A(X) \leq 1 + \lfloor \log_2 \log_2 X \rfloor$. The elementary induction $(N + 1)^2 \leq 2^{2^N}$ completes (244). Thus an actual exponent-1/2 count and an exact square-gap law are compatible with infinitely many objects. This is a counterexample to that proposed counting inference, not to *abc*.

Two upgrades do reach pointwise boundedness. If the number of objects in the k -th shell is at most $C2^{-\delta k}$ with $\delta > 0$, then the integer shell cardinality is eventually strictly below one and hence zero. There is also a useful amplification criterion.

Proposition 25.2 (Single-source amplification). *Suppose $C > 0$, $\theta, \kappa, \beta \geq 0$, and a target class satisfies $|T(Y)| \leq CY^\theta$ for $Y \geq 1$. If every bad source of height $X \geq 1$ produces a set of distinct targets F_X with*

$$F_X \subseteq T(X^\kappa), \quad |F_X| \geq X^\beta, \quad \beta > \kappa\theta,$$

then every such source satisfies

$$X \leq C^{1/(\beta - \kappa\theta)}.$$

Proof. For each individual source,

$$X^\beta \leq |F_X| \leq |T(X^\kappa)| \leq CX^{\kappa\theta}.$$

Division and the positive exponent $\beta - \kappa\theta$ give the result. No overlap estimate between fibres of different sources is used. $\square$

For the Browning–Verzobio bound in Section 23, the resulting design inequality is

$$\beta > \kappa \left(\frac{1}{p} + \frac{1}{q} - \eta_{u,v}(r) + \varepsilon \right).$$

Their present positive exponent reaches neither an empty-shell bound nor this amplification inequality without a new source-to-target construction.

25.3. Residual-kernel escape and polynomial rigidity. For $m \geq 1$ and $n > 0$, define

$$\kappa_m(n) = \prod_{\ell|n} \ell^{v_\ell(n) \bmod m}, \quad \rho_m(n) = \prod_{\ell|n} \ell^{\lfloor v_\ell(n)/m \rfloor}.$$

Primewise Euclidean division gives the unique decomposition

$$(245) \quad n = \kappa_m(n) \rho_m(n)^m,$$

with $\kappa_m(n)$ m -th-power-free.

Proposition 25.3 (Kernel escape). *Fix $p, q, r \geq 2$ with $1/p + 1/q + 1/r < 1$. In every infinite family of distinct primitive points whose coordinates are respectively p -, q -, and r -full, the triples*

$$(\kappa_p(a), \kappa_q(b), \kappa_r(c))$$

take infinitely many values.

Proof. For a fixed residual triple (A, B, C) , (245) converts $a + b = c$ into

$$Ax^p + By^q = Cz^r.$$

Primitivity of (a, b, c) implies $\gcd(x, y, z) = 1$. Darmon and Granville [32, Theorem 2] prove that this fixed equation has only finitely many proper integer solutions in the strict reciprocal range. A finite residual packet would therefore give a finite union of finite sets. $\square$

The proposition rules out a finite list of fixed generalized-Fermat coefficient triples. It does not rule out a fixed Pell or elliptic source curve whose coordinate values induce infinitely many residual kernels; the Pell gate below has exactly this form.

There is a separate obstruction to putting all multiplicities into one polynomial identity.

Proposition 25.4 (Mason–Stothers obstruction). *Let nonzero, pairwise coprime polynomials A, B, C over a characteristic-zero field satisfy $A + B = C$. If they are respectively polynomial- p -, q -, and r -full and at least one is nonconstant, then*

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1.$$

Proof. Put $D = \max\{\deg A, \deg B, \deg C\} > 0$. The polynomial *abc* theorem of Stothers [33] gives

$$D + 1 \leq \deg \text{rad}(ABC).$$

Fullness and pairwise coprimality give

$$\deg \text{rad}(ABC) \leq \frac{\deg A}{p} + \frac{\deg B}{q} + \frac{\deg C}{r} \leq D \left(\frac{1}{p} + \frac{1}{q} + \frac{1}{r} \right).$$

The reciprocal sum cannot be at most one. $\square$

This excludes polynomial-full identities. It does not exclude a sparse set of specializations at which polynomials with simple algebraic factors take full integer values.

25.4. A strict conditional Pell route. The classical consecutive-powerful-number Pell orbit studied by Walker [35]

$$x_0 = 1, \quad y_0 = 0, \quad x_{n+1} = 3x_n + 8y_n, \quad y_{n+1} = x_n + 3y_n$$

satisfies

$$(246) \quad 1 + 8y_n^2 = x_n^2.$$

Its positive coordinates are strictly increasing. Hence it gives an unbounded primitive family $(1, 8y_n^2, x_n^2)$ in which both nonunit coordinates are always squarefull.

If y is itself squarefull, then $8y^2$ is cubefull: the prime 2 occurs to exponent at least three, and every odd prime occurs in y^2 to exponent at least four. The unit is 7-full and x^2 is 2-full, so

$$(1, 8y^2, x^2)$$

has the strict signature $(7, 3, 2)$, with

$$\frac{1}{7} + \frac{1}{3} + \frac{1}{2} = \frac{41}{42} < 1.$$

Theorem 25.5 (Pell-squarefull disproof gate). *If an unbounded subsequence of the positive Pell roots y_n is squarefull, then the standard abc conjecture is false.*

Proof. The corresponding points from (246) are positive and primitive, have the fixed strict mixed-full signature $(7, 3, 2)$, and have unbounded height. The strict mixed-full disproof gate from Section 23 applies to the unchanged standard target. $\square$

The actual radical slope is stronger than $41/42$. Squarefullness gives $\text{rad}(y)^2 \leq y$; also $\text{rad}(x) \leq x$ and $y < x$. Therefore

$$\text{rad}(8y^2x^2)^2 \leq 4\text{rad}(y)^2\text{rad}(x)^2 \leq 4yx^2 < 4x^3.$$

With $H = \log(x^2)$ and $R = \log \text{rad}(8y^2x^2)$ this yields

$$(247) \quad R \leq \frac{3}{4}H + \log 2.$$

The unresolved premise is squarefullness of infinitely many y_n . Here $\alpha = 3 + \sqrt{8}$ and $\beta = 3 - \sqrt{8}$ are algebraic integers with $\alpha + \beta = 6$, $\alpha\beta = 1$, and coprime nonzero integral sum and product; moreover $\alpha/\beta = \alpha^2 > 1$ is not a root of unity and $y_n = (\alpha^n - \beta^n)/(\alpha - \beta)$. Thus this is a nondegenerate Lucas pair in the sense required by Bilu–Hanrot–Voutier. Their Theorem 1.4 says that every Lucas or Lehmer term of index greater than 30 has a primitive divisor [34]. Their theorem does not assert that this divisor has valuation one, so it alone does not negate the premise. The first twelve terms were factored only to check the recurrence and formulas; only $y_1 = 1$ is squarefull among them, and no asymptotic conclusion is drawn.

The Lean companion proves the Pell norm recurrence and unboundedness, the squarefull-to-cubefull transfer, the exact $(7, 3, 2)$ slope, and the conditional implication to the original **ABCConjecture**. It does not assume or prove an unbounded squarefull subsequence. The Darmon–Granville, Mason–Stothers, three-quarter radical, and Lucas–Wieferich arguments remain paper proofs.

26. BALANCED PERSISTENCE: AFFINE AMPLIFICATION AND EXPLICIT COUNTEREXAMPLE GATES

This section records a further simultaneous attack on the positive and negative directions. A route is kept active whenever its missing statement is merely difficult or unknown. We mark a construction mechanism as closed only when a theorem or an explicit counterexample contradicts the statement that mechanism actually needs. A local failure is never promoted to a failure of the surrounding route.

The standard conjecture remains the target throughout. No existence premise for a squarefull Pell subsequence, a squarefull Hall remainder, or a large low-radical affine subfibre is assumed.

26.1. A complete two-parameter primitive shear. Fix a primitive positive seed

$$a + b = c, \quad \gcd(a, b) = 1,$$

and put

$$P = abc, \quad R = \text{rad}(P), \quad \rho = \frac{\log P}{\log c}, \quad \sigma = \frac{\log R}{\log c}.$$

After interchanging a and b , assume $b \geq c/2$. The elementary bounds

$$(248) \quad c(c-1) \leq P \leq \frac{c^3}{4}$$

will be used repeatedly.

For integers $h, k \geq 1$ define

$$(249) \quad \begin{aligned} U &= 1 + Ph, \\ V &= 1 + P(h + ck), \\ W &= 1 + P(h + bk). \end{aligned}$$

Theorem 26.1 (Primitive affine shear). *If $\gcd(U, k) = 1$, then*

$$(aU, bV, cW)$$

is a positive primitive abc point. The ordered output determines the parameter pair (h, k) .

Proof. The three identities

$$V - U = cPk, \quad W - U = bPk, \quad V - W = aPk$$

and $a + b = c$ give

$$aU + bV = cW.$$

All three cofactors are congruent to 1 modulo P , so they avoid every prime of the seed. A common divisor of U and V divides cPk and is coprime to cP ; it must therefore divide k , contrary to $\gcd(U, k) = 1$. The same argument gives $\gcd(U, W) = 1$.

A common divisor of V and W divides aPk . It avoids aP , while $V \equiv U \pmod{k}$ and hence is coprime to k . Thus U, V, W are pairwise coprime. The seed coordinates are pairwise coprime, and all cross terms are coprime because the cofactors avoid P . This proves primitivity. Finally $U = (aU)/a$ recovers h , and $V - U = cPk$ then recovers k . $\square$

The construction is not a thin parametrization.

Lemma 26.2 (Uniform admissible count). *For all sufficiently large M ,*

$$\#\{1 \leq h, k \leq M : \gcd(1 + Ph, k) = 1\} \geq \frac{M^2}{8}.$$

Inside the upper square $\lceil M/2 \rceil \leq h, k \leq M$, the count is at least $M^2/32$.

Proof. A bad pair has a prime $\ell \leq M$ dividing both k and $1 + Ph$; such a prime does not divide P . For each ℓ , there are at most M/ℓ choices of k and at most $M/\ell + 1$ choices of h . Hence

$$\#\{\text{bad pairs}\} \leq M^2 \sum_{\substack{\ell \leq M \\ \ell \text{ prime}}} \ell^{-2} + M \sum_{2 \leq n \leq M} n^{-1} \leq \frac{3M^2}{4} + M \log M.$$

Here $\sum_{n \geq 2} n^{-2} \leq 3/4$. This proves the first assertion for large M .

For the upper square let L denote its side length. The contribution of ℓ is at most $(L/\ell + 1)^2$, so the total number of bad pairs is at most

$$\frac{3L^2}{4} + 2L \log M + \pi(M).$$

For large M the final two terms are at most $L^2/8$. Since $L \geq M/2$, at least $L^2/8 \geq M^2/32$ pairs remain. $\square$

For fixed $K > 5$, set

$$M_K = \left\lfloor \frac{c^{K-2}}{4P} \right\rfloor.$$

Equation (248), Theorem 26.1, and Lemma 26.2 give

$$(250) \quad \#\{\text{distinct outputs of height at most } c^K\} \gg M_K^2 \gg c^{2K-10}.$$

Indeed,

$$cW \leq c + cPM_K(1+b) < c^K.$$

At $K = 8$ this produces at least $c^6/32$ outputs after increasing the fixed lower threshold for c .

26.2. The actual-radical gate. The raw count does not inherit the seed's radical saving automatically. For $n \geq 1$ put

$$E(n) = \frac{n}{\text{rad}(n)}.$$

Because the cofactors in (249) are pairwise coprime and avoid P , radical multiplicativity gives the exact identity

$$(251) \quad \text{rad}(aUbVcW) = R\text{rad}(UVW) = \frac{RUVW}{E(U)E(V)E(W)}.$$

Writing $H = cW$, the condition

$$\text{rad}(aUbVcW) < H^\mu$$

is therefore equivalent to

$$(252) \quad E(U)E(V)E(W) > \frac{R}{c} UVH^{1-\mu}.$$

This equivalence is the positive route's exact remaining arithmetic object.

At $K = 8$, $\mu = 3/4$, and on the upper parameter square, every exceptional output has

$$(253) \quad E(U)E(V)E(W) > \frac{R}{8192} c^{14}.$$

To see the direction of the height inequality explicitly, the lower bounds in the upper square give $UVW \geq c^{20}/8192$, while $H < c^8$, and

$$\frac{R}{c} UVH^{1/4} = \frac{RUVW}{H^{3/4}}.$$

Define the square-root part

$$Q(n) = \prod_p p^{\lfloor v_p(n)/2 \rfloor}.$$

Primewise, $Q(n)^2 \mid n$ and $Q(n)^2 \geq E(n)$. Thus (253) places every desired output in the tail

$$(254) \quad Q(UVW) \gg R^{1/2} c^7,$$

whereas the parameter box has side $M \ll c^4$. This explains why an ordinary small-modulus squarefree sieve does not settle either direction.

There is nevertheless a uniform unconditional upper budget. Let $\mathcal{E}_c$ be the $K = 8$, $\mu = 3/4$ exceptional subfibre. Then

Theorem 26.3 (Three-way exceptional budget). *For every fixed subcritical source locus and every $\delta > 0$,*

$$(255) \quad |\mathcal{E}_c| \ll_{\delta} c^{\min\{12-2\rho, 6-\sigma, 8-\rho-\sigma/3\}+\delta},$$

uniformly in the seed.

Proof. The first term is the size of the full two-parameter fibre. The second is the actual-support S -unit estimate from Section 4, applied with target radical below c^6 and inherited radical R .

For the third term, (251) implies

$$D := \text{rad}(UVW) < \frac{c^6}{R}.$$

Since the three cofactors are pairwise coprime, one of their radicals is below

$$Z = (c^6/R)^{1/3} = c^{2-\sigma/3}.$$

The de Bruijn radical count used by Bernert–Browning–Lichtman–Teräväinen [7] bounds the number of possible values of that cofactor by $O_{\delta}(c^{2-\sigma/3+\delta})$. The exponent is uniform: for a fixed source slope $\sigma \in [0, \lambda]$, the relevant de Bruijn exponent ranges over a compact interval, so a finite mesh and monotonicity reduce the claim to finitely many fixed estimates.

For fixed U , the parameter h is unique and there are at most M choices of k . Fixed V or W has no larger representation multiplicity. Since $M \ll c^{6-\rho}$, the third exponent follows. $\square$

Theorem 26.4 (Seed-sensitive affine upper gate). *Fix a positive primitive seed $a + b = c$, and put $R = \text{rad}(abc)$. Let $X \geq 2$ and $0 < \mu < 1$. For an arbitrary subset of the admissible positive integral parameters, let $\mathcal{E}(X)$ be its set of distinct affine-shear outputs satisfying $H = cW \leq X$ and $\text{rad}(aUbVcW) < H^{\mu}$. Then, for every $\varepsilon > 0$,*

$$(256) \quad \#\mathcal{E}(X) \ll_{\mu, \varepsilon} R^{-2/3} X^{2\mu/3+\varepsilon}.$$

The implied constant is independent of the seed and of the chosen parameter subset.

Proof. Positivity in $aU + bV = cW = H$ gives $U, V, W \leq H \leq X$. Exact support avoidance and cofactor coprimality give

$$\text{rad}(aUbVcW) = R \text{rad}(U) \text{rad}(V) \text{rad}(W),$$

so every exceptional output satisfies

$$\text{rad}(U) \text{rad}(V) \text{rad}(W) < X^{\mu}/R.$$

All three pair projections (U, V) , (U, W) , and (V, W) are injective. The two-coordinate consequence of the de Bruijn radical count is

$$\#\{(m, n) : m, n \leq X, \text{rad}(m) \text{rad}(n) < Y\} \ll_{\varepsilon} Y X^{\varepsilon}.$$

Writing r_1, r_2, r_3 for the three cofactor radicals, the identity $(r_1 r_2)(r_1 r_3)(r_2 r_3) = (r_1 r_2 r_3)^2$ shows that one pair product is below $(X^{\mu}/R)^{2/3}$. Apply the pair count through the corresponding injective projection and sum over the three choices. This proves (256). $\square$

The relaxed region $\rho \leq 3$, $\sigma < 1$ leaves every exponent in (255) above 4; the infimum of the third term is $14/3$. This is a lower envelope of the available upper-budget exponents, not a pointwise upper bound $|\mathcal{E}_c| \ll c^{14/3}$. For a fixed source slope $\lambda < 1$ its corresponding relaxed envelope is $5 - \lambda/3$.

At $\mu = 3/4$, Proposition 1.1 of Bernert–Browning–Lichtman–Teräväinen gives exponent $2\mu/3 = 1/2$ in target height [7]. Their Theorem 1.3 gives the different uniform exponent $3/5$ for fixed $\mu < 1$; at $\mu = 3/4$ it is the weaker of the two applicable estimates. Hence the following statement would close the positive route through the subcritical-locus equivalence.

Corollary 26.5 (Exact conditional closing step). *Fix $\lambda < 1$. Suppose every sufficiently large seed satisfying $R < c^{\lambda}$ has at least $c^{17/4}$ members of $\mathcal{E}_c$. Then the height of that source locus is bounded.*

Proof. Every member is a distinct primitive point of height below c^8 . For every $\delta > 0$, the cited exceptional-set theorem gives

$$|\mathcal{E}_c| \ll_\delta c^{4+8\delta}.$$

Take $\delta = 1/64$. Then $4 + 8\delta = 33/8 < 17/4$, so comparison with the hypothesized lower bound bounds c . $\square$

The fixed exponent $17/4$ is sufficient but no longer optimal. By (256), the sharp conditional comparison at $X = c^8$ is

$$|\mathcal{E}_c| \geq R^{-2/3} c^{4+\eta} \quad (\eta > 0).$$

For a seed with $R = c^{\sigma+o(1)}$, the required exponent is therefore $4 - 2\sigma/3 + \eta$. This is a conditional lower target, not a consequence of the upper bound.

There is also a precise local warning. Conditional on local admissibility at $p \nmid P$, each of $p^2 \mid U, V, W$ has probability $1/(p(p+1))$, but the three events are pairwise disjoint. Thus the model asserting independence of the three same-prime events is refuted. Finite local data at distinct primes remain exactly independent by the Chinese remainder theorem. This local counterexample closes only that probabilistic model; it does not refute the affine lower-bound route.

No theorem presently supplies the lower bound in Corollary 26.5. The upper budget does not refute it. The affine route therefore remains active.

26.3. Square-root descent on the balancing Pell orbit. Let

$$u_0 = 0, \quad u_1 = 1, \quad u_{n+2} = 6u_{n+1} - u_n,$$

and define x_n by

$$x_n^2 - 8u_n^2 = 1.$$

Equivalently,

$$(3 + \sqrt{8})^n = x_n + u_n \sqrt{8}.$$

Introduce the companion coordinates

$$(1 + \sqrt{2})^n = A_n + B_n \sqrt{2}.$$

Theorem 26.6 (Coprime square-root factorization). *For every $n \geq 1$,*

$$(257) \quad x_n = A_n^2 + 2B_n^2, \quad u_n = A_n B_n, \quad A_n^2 - 2B_n^2 = (-1)^n, \quad \gcd(A_n, B_n) = 1.$$

Consequently u_n is squarefull if and only if both A_n and B_n are squarefull.

Proof. Squaring $(1 + \sqrt{2})^n$ and using $(1 + \sqrt{2})^2 = 3 + \sqrt{8}$ gives the first two identities. Taking norms gives the third. A common divisor of A_n and B_n divides the norm, hence is one. The prime supports of the two factors are therefore disjoint, which proves the squarefull equivalence. $\square$

Adding and subtracting the norm identity gives

$$(258) \quad \left(\frac{x_n - 1}{2}, \frac{x_n + 1}{2} \right) = \begin{cases} (2B_n^2, A_n^2), & n \text{ even}, \\ (A_n^2, 2B_n^2), & n \text{ odd}. \end{cases}$$

Thus the associated primitive point Q_n is obtained by inserting the unit between the two entries in (258).

Theorem 26.7 (One-half radical slope). *If u_n is squarefull and c_n is the larger nonunit entry of Q_n , then*

$$(259) \quad \log \text{rad}(Q_n) < \frac{1}{2} \log c_n + \log 2.$$

Hence an unbounded squarefull subsequence of (u_n) would disprove the standard abc conjecture.

Proof. Theorem 26.6 makes both factors squarefull, so

$$\text{rad}(Q_n) = \text{rad}(2A_n^2B_n^2) \leq 2\text{rad}(A_n)\text{rad}(B_n) \leq 2\sqrt{A_nB_n}.$$

For even n , $c_n = A_n^2$ and $B_n < A_n$; for odd n , $c_n = 2B_n^2$ and $A_n < 2B_n$. Thus $A_nB_n < c_n$ in both cases, proving (259). Applying standard *abc* with, for example, $\varepsilon = 1/2$ would bound the heights, contradicting an unbounded subsequence. $\square$

On an even squarefull index, both nonunit entries are 4-full, so the fixed signature is $(3, 4, 4)$ with reciprocal sum $5/6$. This is an additional conversion of the same unresolved squarefull premise, rather than a new source of indices.

26.4. Valuation saturation and prime-index descent. For a prime p , let $z(p)$ be its rank of apparition in (u_n) . Strong divisibility gives $p \mid u_n$ if and only if $z(p) \mid n$. For odd p , the order calculation in the quadratic residue algebra gives

$$(260) \quad z(p) \mid p - \left(\frac{2}{p}\right).$$

Sanna's exact valuation theorem [36] specializes to

$$(261) \quad v_p(u_n) = \begin{cases} v_p(u_{z(p)}) + v_p(n), & z(p) \mid n, \\ 0, & z(p) \nmid n, \end{cases} \quad (p \text{ odd}),$$

and

$$v_2(u_n) = 0 \quad (n \text{ odd}), \quad v_2(u_n) = v_2(n) \quad (n \text{ even}).$$

Equations (260)–(261) yield an exact saturation rule. If $p \parallel u_{z(p)}$, $z(p) \mid N$, and u_N is squarefull, then $p \mid N/z(p)$. Iterating only the factorizations

$$\begin{array}{l|l} 2 & 2 \cdot 3 \\ 3 & 5 \cdot 7 \\ 4 & 2^2 \cdot 3 \cdot 17 \\ 5 & 29 \cdot 41 \\ 6 & 2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \\ 7 & 13^2 \cdot 239 \end{array}$$

proves the finite obstruction

$$(262) \quad u_N \text{ squarefull and } 2 \mid N \implies 22,318,790,340 \mid N.$$

The factor 13^2 at rank seven supplies no outgoing exponent-one edge. This is a real obstruction to the naive use of primitive-divisor theorems:

$$(263) \quad u_7 = 13^2 \cdot 239.$$

Carmichael's theorem in Yabuta's formulation [40] provides a primitive divisor at every relevant index, but (263) proves that “primitive” does not mean “valuation one.”

The strongest unconditional descent obtained here reduces the full squarefull question to prime indices.

Theorem 26.8 (Largest-prime descent). *Suppose $N > 1$ and u_N is squarefull. If ℓ is the largest prime divisor of N , then ℓ is odd and u_ℓ is squarefull.*

Proof. If N were a power of two, then $z(3) = 2$ and (261) would give $v_3(u_N) = 1$, a contradiction. Thus ℓ is odd.

Let $q \mid u_\ell$. Since u_ℓ is odd and ℓ is prime, $z(q) = \ell$. Equation (260) gives $\ell \mid q \pm 1$. An odd prime with this property is larger than ℓ : the first positive representatives $\ell - 1$ and $\ell + 1$ are even. Hence $q \nmid N$ by the maximality of ℓ . Since $\ell \mid N$, valuation lifting gives

$$v_q(u_N) = v_q(u_\ell) + v_q(N) = v_q(u_\ell).$$

The left side is at least two. This holds for every prime divisor of u_ℓ , proving squarefullness. $\square$

An odd prime p for which $p^2 \mid u_{z(p)}$ is a balancing-Wieferich prime. Write $(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}$. At an odd prime index ℓ , $u_\ell = A_\ell B_\ell$ and $\gcd(A_\ell, B_\ell) = 1$. The perfect-power classifications of Cohn and Ljunggren [45, 46, 47] sharpen the prime-index alternative as follows.

Theorem 26.9 (Prime-index four-prime packet). *For every odd prime ℓ , either some prime satisfies $q \parallel u_\ell$, or $\ell \neq 7$ and u_ℓ is squarefull with at least four distinct support primes of rank exactly ℓ : at least two divide A_ℓ and at least two divide B_ℓ . In this second alternative every such prime has first valuation at least two, and each channel contains a prime whose first valuation is at least three.*

Proof. If no support valuation equals one, the term is squarefull. Coprimality makes both channels squarefull. Cohn's associated-Pell theorem says that $A_\ell > 1$ is not a perfect power. The standard Pell classifications say the same for B_ℓ , except at $\ell = 7$; there $u_7 = 13^2 \cdot 239$ and $239 \parallel u_7$. A squarefull integer which is not a perfect power has at least two support primes and an odd valuation at least three. Applying this separately to the two channels proves the packet. The rank and valuation claims follow from (260)–(261). $\square$

The channel orders also force

$$q \mid A_\ell \implies q \equiv 1 \text{ or } 2\ell + 1 \pmod{4\ell}, \quad q \mid B_\ell \implies q \equiv 1 \text{ or } 2\ell - 1 \pmod{4\ell}.$$

Combining Theorem 26.9 with the largest-prime descent shows that every hypothetical nonunit squarefull balancing term forces such a packet at an odd prime rank. The adjacent Pell point consequently has the sharpened bound

$$(264) \quad \text{rad}(2A_\ell^2 B_\ell^2) \leq 2\sqrt{\frac{u_\ell}{4\ell^2 - 1}}.$$

No theorem currently rules out all such packets or constructs infinitely many of them. The Pell route remains active.

26.5. Elliptic and Hall alternatives. The Mordell curve

$$E : y^2 = x^3 - 2, \quad P = (3, 5)$$

supplies an independent critical family. Write

$$nP = \left(\frac{X_n}{D_n^2}, \frac{Y_n}{D_n^3} \right)$$

in lowest terms. Then

$$(265) \quad Y_n^2 + 2D_n^6 = X_n^3.$$

The three entries are pairwise coprime. Since

$$2P = \left(\frac{129}{10^2}, \frac{-383}{10^3} \right),$$

the denominator elliptic divisibility sequence satisfies $D_2 \mid D_{2k}$, so D_n is even at every even index [49]. Equation (265) then has critical signature $(2, 6, 3)$. It becomes strict if Y_n or X_n is squarefull on an unbounded subsequence, or if the odd part of D_n is squarefull. Elliptic primitive-divisor theorems control new support, but not its first valuation, so none of these upgrade gates is currently refuted.

There is also a size-sensitive Hall gate.

Proposition 26.10 (Squarefull Hall gate). *Suppose positive primitive data satisfy*

$$X^3 + K = Y^2, \quad K^2 \leq X,$$

and K is squarefull. Then

$$(266) \quad \text{rad}(X^3 K Y^2)^{12} < (Y^2)^{11},$$

so an unbounded family of such data would disprove standard abc.

Proof. Put $D = \text{rad}(X^3KY^2)$. Radical submultiplicativity gives $D \leq X \text{rad}(K)Y$. Squarefullness and the size condition give

$$\text{rad}(K)^{12} \leq K^6 \leq X^3.$$

Therefore $D^{12} \leq X^{15}Y^{12}$. Since $K > 0$, $X^3 < Y^2$, whence $X^{15} < Y^{10}$. This proves (266). Its logarithmic slope is strictly below $11/12$. $\square$

Danilov's identity, as recorded by Dujella [41, 42], is

$$(267) \quad (z^2 + 6z + 4)^3 - (z^2 + 1)(z^2 + 9z + 19)^2 = -27(2z + 11).$$

Choose positive solutions

$$z^2 - 5w^2 = -1, \quad z \equiv 57 \pmod{125}.$$

There are infinitely many: one is $(682, 305)$, and multiplication by a fixed positive power of the norm-one unit $9 + 4\sqrt{5}$ preserves the congruence and gives an unbounded orbit. Put

$$\begin{aligned} A &= z^2 + 6z + 4, & B &= z^2 + 9z + 19, & L &= 2z + 11, \\ X &= A/5, & Y &= wB/5, & K &= 27L/125. \end{aligned}$$

The congruence gives $125 \mid L$, $A \equiv 20 \pmod{25}$, and $5 \mid w$. Equation (267) becomes

$$(268) \quad X^3 + K = Y^2.$$

Moreover $\gcd(A, B) = 1$ and $\gcd(A, wB) = 5$, so $\gcd(X, Y) = 1$. Thus this is an unbounded primitive Hall family with

$$X \sim \frac{z^2}{5}, \quad K \sim \frac{54z}{125} \sim \frac{54\sqrt{5}}{125} \sqrt{X}, \quad \frac{54\sqrt{5}}{125} < 1.$$

Consequently $K^2 < X$ on a sufficiently large tail of the orbit. The first point is the exact identity

$$93844^3 + 297 = 28748141^2.$$

Here $297 = 3^3 \cdot 11$ is not squarefull. This refutes only the universal claim that every remainder in the congruence orbit is squarefull; it does not refute an unbounded squarefull subsequence. Such a subsequence would satisfy Proposition 26.10 after discarding its finite initial segment and would disprove *abc*.

For each fixed $m \geq 2$, forcing the moving factor $(2z + 11)/125$ to be one exact m th power leads to the hyperelliptic equation

$$(2w)^2 = 3125t^{2m} - 550t^m + 25.$$

The polynomial on the right is squarefree of degree $2m$, so its smooth completion has genus $m - 1 \geq 1$. Siegel's theorem yields only finitely many integral points for that fixed m . This closes the fixed exact-power mechanism. It does not close the squarefull route because the square and cube kernels may both move.

Certified finite computation. A reproducible exact audit in the companion computation archive checked both open squarefull routes without extrapolating from finite data. For the balancing recurrence, every one of the 999 terms u_n with $2 \leq n \leq 1000$ has an explicit prime p and residue q such that

$$u_n \equiv pq \pmod{p^2}, \quad 1 \leq q < p.$$

Thus each of those terms is rigorously non-squarefull. The extended certified run through $n = 2000$ now settles 1997 of the 1999 nonunit terms and leaves precisely

$$\{1873, 1951\}$$

unresolved. These two indices are neither squarefull hits nor certified non-hits. The seven new exclusions use local recursive Pocklington proofs and two independent recurrences modulo p^2 .

An exhaustive scan of all 183,071 odd primes $q \leq 2,500,000$ tested $u_{q-(2/q)}$ modulo q^3 . The only balancing-Wieferich hits are 13, 31, 1546463, all of exact depth two; no depth-three hit occurs in this finite range. This is pressure against the four-prime packet, not an infinite nonexistence theorem.

On the Danilov orbit obtained by iterating $(9 + 4\sqrt{5})^{10}$ from $(682, 305)$, the audit verifies the Pell equation, integrality, primitivity, $X^3 + K = Y^2$, and $K^2 \leq X$ for $0 \leq t \leq 80$; every one of the 81 remainders also has a certified prime with $v_p(K) = 1$. The five-small-prime periodic sieve already fails to decide $t = 326$. A subsequent symbolic prime-square lift eliminates the finite-search boundary: if K_t is squarefull, then

$$(269) \quad \begin{aligned} t &\equiv 122136955032565025967809449110840347537827 \\ &\pmod{183205432548847538951714173666260521306741}. \end{aligned}$$

Hence every nonnegative index below the displayed representative is ruled out as a squarefull index. The surviving index residue class is nonempty, but the theorem asserts no squarefull value in it and does not exclude all such values. The Danilov route remains active.

The Cohn–Nitaj constructions provide genuine unbounded critical $(3, 3, 3)$ families [43, 44]. Walsh’s Theorem 1.1 supplies another infinite family for an odd prime p under the explicit hypothesis that $Y^2 = X^3 - 432p^2$ has positive rank [48]; without a proved rank statement for the selected p , that branch remains conditional. In each case the reciprocal sum equals one. A powerful-coordinate subsequence would make a strict signature, and no cited theorem either constructs or refutes that subsequence. These routes also remain active.

26.6. Formalization boundary and route ledger. Seven Lean modules formalize the elementary deterministic core after the paper proofs were written:

Module	Kernel-checked content
PellAdjacentFactor Counterexample20260831	Half-factor existence, primitive adjacent point, radical square bound, one-half conductor inequality, fourth-full transfer, and the conditional implication to $\neg \text{ABCConjecture}$.
PellSquareRoot Descent20260831	The $(1 + \sqrt{2})^n$ recurrence, norm, coprimality, exact square shapes, fullness splitting, and equivalence with the Pell coordinate.
PellPrimeIndex Dichotomy20260831	Odd squarefull exponents have depth at least three, the complete index-seven obstruction $239 \parallel u_7$, and the two-channel numerical product bound.
HallSquarefull Counterexample20260831	The primitive Hall point, the twelfth-power radical inequality, the 11/12 conductor inequality, and the unbounded-family disproof gate.
AffineShear Amplification20260831	The shear equation, avoidance of the seed support, all cofactor and endpoint coprimality statements, the actual <code>ABCPPoint</code> , and injectivity of the parameter map.
AffineExcess UpperBound20260831	Injectivity of all three pair projections, same-prime double-divisibility exclusion, and the integer geometric-mean lemma.
DanilovGlobalIndex Sieve20260831	The 11- and 89-adic stages, ten prime-square lift packets, exact CRT, and the global progression and lower-index exclusion in (269).

The analytic BBLT and S -unit estimates; the Cohn–Ljunggren perfect-power classifications; Sanna’s valuation theorem; Carmichael–Yabuta and Siegel; elliptic-divisibility and primitive-divisor results; the Danilov parametrization, primitivity, and orbit unboundedness; and the Cohn–Nitaj and conditional Walsh existence results all remain outside this Lean increment. They are proved on paper or cited, rather than inserted as project axioms.

The resulting route ledger is deliberately narrow:

- the affine repeated-prime lower bound, simultaneous squarefull Pell coordinates, exclusion or construction of the forced four-prime balancing-Wieferich packets, the surviving Danilov residue class, and powerful elliptic-coordinate subsequences are *active*;

- a fixed strict generalized-Fermat equation, a finite residual-kernel packet, a primitive polynomial-full identity, and the fixed exact-power Danilov specialization are closed only as the stated construction mechanisms;
- the example $K = 297$ refutes “every Danilov remainder is squarefull” but leaves all unbounded-subsequence variants open; the same-prime independent affine model is also refuted, without closing the affine route.

No unconditional closed term of type **ABCConjecture**, and no rigorous unconditional disproof, is obtained in this increment.

27. RADICAL-STEP FIBRES, GLOBAL PRIME PACKETS, AND THE SAME-PILOT TYPE BOUNDARY

This section continues the positive and negative searches simultaneously. Failure of a finite search, absence of a known theorem, and present technical difficulty do not retire a route. A counterexample below is used only against the fully quantified statement that it satisfies; the surrounding route is kept whenever a stronger or differently typed mechanism survives.

27.1. Replacing the product step by the seed radical. Let $a + b = c$ be primitive, put $P = abc$ and $R = \text{rad}(P)$, and let Q be any positive multiple of R . For $h, k \geq 1$ define

$$(270) \quad U = 1 + Qh, \quad V = 1 + Q(h + ck), \quad W = 1 + Q(h + bk).$$

Theorem 27.1 (Radical-step primitive shear). *If $\gcd(U, k) = 1$, then (aU, bV, cW) is a primitive *abc* triple. Each of the three pair projections (U, V) , (U, W) , and (V, W) determines (h, k) . If $K \geq 2$ and*

$$M_Q = \left\lfloor \frac{c^{K-2}}{4Q} \right\rfloor,$$

then every parameter pair $1 \leq h, k \leq M_Q$ has height below c^K . For sufficiently large M_Q , at least $M_Q^2/8$ of these pairs are admissible.

Proof. Every prime dividing a seed coordinate divides Q , while all three cofactors are congruent to one modulo Q . Thus every cofactor avoids the entire seed support. The identities

$$V - U = cQk, \quad W - U = bQk, \quad V - W = aQk$$

show that a common divisor of U and V , or of U and W , must divide k after the factors Q and the appropriate seed coordinate have been removed. The hypothesis $\gcd(U, k) = 1$ excludes both possibilities. A common divisor of V and W divides aQk ; it avoids aQ , and $V \equiv U \pmod{k}$, so it also avoids k . Hence U, V, W are pairwise coprime. Combining this with support avoidance and the pairwise coprimality of a, b, c proves primitivity. The identity $aU + bV = cW$ follows by expansion.

Any one of the three displayed differences recovers k from the indicated pair. If the pair contains U , then $U = 1 + Qh$ recovers h ; from (V, W) one instead uses $h = (V - 1)/Q - ck$. This proves the three injectivity statements. Since $b + 1 \leq c$,

$$W \leq 1 + QcM_Q \leq 1 + \frac{c^{K-1}}{4},$$

and therefore $cW < c^K$ for $c \geq 2$ and $K \geq 2$. Finally, the elementary sieve used in Lemma 26.2 applies verbatim with Q in place of P : the bad pairs are covered by primes dividing both k and $1 + Qh$. It leaves at least $M_Q^2/8$ pairs once M_Q is sufficiently large. $\square$

The minimal step $Q = R$ is therefore optimal among these multiples for the raw box. At $K = 8$ its elementary lower count has scale

$$(271) \quad \frac{c^{12}}{R^2},$$

instead of c^{12}/P^2 . This is a genuine improvement when the seed has large prime powers. It does not itself count low-radical outputs. Indeed, the pair-projection upper argument is intrinsic to the output factors and remains

$$(272) \quad \#\mathcal{E}_Q(c^8) \ll_\varepsilon R^{-2/3} c^{4+\varepsilon}$$

on the $\mu = 3/4$ locus.

There is no gain from averaging the same construction over multiples of R . If $Q = sR$, then the Q -point at (h, k) is exactly the R -point at (sh, sk) . Moreover $1 + Qh \equiv 1 \pmod{s}$, so admissibility at (h, k) implies admissibility at (sh, sk) . Thus the larger-step fibres are subfibres of the minimal one after rescaling their boxes.

The often stated “matching lower bound” should also be interpreted with care. Assume the uniform upper estimate (272). For fixed $\lambda < 1$ and $\eta > 0$, the assertion that there are constants $A > 0, c_0$ such that every seed with $R < c^\lambda$ and $c \geq c_0$ satisfies

$$(273) \quad \#\mathcal{E}_R(c^8) \geq AR^{-2/3}c^{4+\eta}$$

is equivalent to boundedness of the heights of the seeds in that subcritical locus. The forward implication compares (273) with (272) at $\varepsilon = \eta/2$. Conversely, after a height bound has already been proved, one may choose c_0 above it and the lower assertion is vacuous. Consequently (273) is an exact closing gate, but it is not by itself an explanatory distribution theorem.

A fixed congruence template cannot provide that explanation. Suppose d_1, d_2, d_3 are pairwise coprime, $D = d_1d_2d_3$, and $\gcd(D, Qabc) = 1$. The conditions

$$d_1 \mid U, \quad d_2 \mid V, \quad d_3 \mid W$$

have exactly D solutions (h, k) modulo D , hence density D^{-1} . The deterministic lower certificate for repeated-prime excess supplied solely by these divisibilities is

$$(274) \quad \Delta = \prod_{i=1}^3 \frac{d_i}{\text{rad}(d_i)} \leq D.$$

On the upper half of the $K = 8$ radical-step box, forcing the target radical inequality requires $\Delta \gg Rc^{14}$. It therefore also requires $D \gg Rc^{14}$, while the periodic main term is at most

$$\frac{M_R^2}{D} \ll \frac{c^{12}/R^2}{Rc^{14}} = \frac{1}{c^2R^3}.$$

This rules out obtaining the matching lower bound from the periodic main term of a single template alone. It leaves large unions of templates, boundary terms, and accidental high valuations open.

There are also exact finite warnings against replacing excess by a visual “square cofactor” condition. For the subcritical seed $(1, 8, 9)$, $R = 6$, the choices

$$(Q, h, k) = (6, 840, 60) \quad \text{and} \quad (72, 70, 5)$$

both give

$$(U, V, W) = (5041, 8281, 7921) = (71^2, 91^2, 89^2).$$

Nevertheless the output height is 71289, its radical is 3450174, and the desired fourth-power versus third-power radical inequality fails. A second example is

$$(Q, h, k) = (72, 1160, 798), \quad (U, V, W) = (289^2, 775^2, 737^2),$$

and it fails the same target. These are counterexamples to the assertion that three square cofactors automatically give an exceptional output. They do not refute the eventual arithmetic lower gate.

27.2. Exact order stagnation in the balancing orbit. Let

$$u_0 = 0, \quad u_1 = 1, \quad u_{n+2} = 6u_{n+1} - u_n,$$

and put $\alpha = 3 + 2\sqrt{2}$ and $\gamma = \alpha^2 = 17 + 12\sqrt{2}$. Then

$$(275) \quad u_n = \alpha^{1-n} \frac{\gamma^n - 1}{\gamma - 1}.$$

For an odd rational prime q , let $z(q)$ be its rank of apparition and put $e(q) = v_q(u_{z(q)})$.

Theorem 27.2 (Order-stagnation tower). *For every odd prime q and $k \geq 1$,*

$$(276) \quad z(q^k) = z(q)q^{\max(0, k-e(q))}.$$

In particular, depth two is exactly one failure of order growth and depth three is exactly two consecutive failures.

Proof. In $K = \mathbb{Q}(\sqrt{2})$, every odd q is unramified and $N(\gamma - 1) = -32$. At every prime ideal $\mathfrak{q} \mid q$, (275) identifies $z(q)$ with the residual order of γ and $e(q)$ with $v_{\mathfrak{q}}(\gamma^{z(q)} - 1)$. If y is a local unit with $v(y - 1) = e \geq 1$, the odd-prime binomial expansion gives

$$v(y^m - 1) = e + v_q(m).$$

Thus the least multiplier needed to raise the congruence from level one to level k is $q^{\max(0, k-e)}$. Substitution gives (276). $\square$

At an odd prime index ℓ , write $(1 + \sqrt{2})^\ell = A_\ell + B_\ell\sqrt{2}$. The previous packet theorem says that a squarefull $u_\ell = A_\ell B_\ell$ forces four distinct rank- ℓ repeated primes, two in each coprime channel, with a depth-three prime in each channel. There is also a first-order relation among all packet primes. If

$$A_\ell = 1 + 2\ell a, \quad \left(\frac{2}{\ell}\right) B_\ell = 1 + 2\ell b,$$

then the Pell norm gives the exact identity

$$(277) \quad a - 2b + \ell(a^2 - 2b^2) = 0.$$

Expanding the prime factorizations modulo $4\ell^2$ turns its reduction modulo ℓ into a coupled congruence between the quotient coordinates in the two channels. This is stronger than independent residue conditions, although the free quotient variables prevent a contradiction at present.

The latest number-field input yields a useful global alternative. Fellini and Murty state that finiteness of base- x super-Wieferich prime ideals forces infinitely many non-Wieferich prime ideals [39]. An adversarial audit of their Section 8 architecture gives a sharper prime-order consequence, but it also requires two local proof repairs and one textual correction. At a selected prime index ℓ , a prime ideal $\mathfrak{p} \mid \Phi_\ell(x)$ has residual order dividing ℓ . Order one would imply both $x \equiv 1$ and $\ell \equiv 0$ modulo $\mathfrak{p}$; the defining exclusion of primes above ℓ which divide $(x - 1)$ rules this out. Thus every such factor has order exactly ℓ , and its exponent in $\Phi_\ell(x)$ is its first-occurrence valuation. Excluding the finitely many super-Wieferich orders leaves exponents one or two. If only finitely many prime-order exponent-one ideals existed, the Section 8 localization, finite square-class decomposition, and Siegel argument would again place infinitely many distinct points on a finite family of genus-one curves.

This streamlined prime-index proof deliberately avoids the printed second case of Lemma 6.4. The displayed equality there is generally false: with $x = 2, p = 3, f = 2, i = 1$, it identifies $\Phi_6(2) = 3$ with $(2^6 - 1)/(2^2 - 1) = 21$. Its binomial proof also treats $\binom{p}{p} = 1$ as if it were divisible by p ; separating the last term repairs the valuation estimate. A contradictory sentence on printed p. 226 is avoided by allowing the localized square generator to be a unit. These counterexamples invalidate intermediate printed steps. They do not invalidate the repaired valuation conclusion or the resulting Siegel argument.

Specializing this proof extraction to $x = \gamma$ and descending prime ideals to rational primes gives the unconditional alternative. After discarding the finitely many prime ideals above 2, the local dictionary applies to odd rational primes:

$$(278) \quad \begin{aligned} &\text{either infinitely many odd primes } q \text{ satisfy } e(q) \geq 3, \\ &\text{or infinitely many odd prime indices } \ell \text{ admit } q \parallel u_\ell \text{ with } z(q) = \ell. \end{aligned}$$

This is a repaired refinement of the cited proof architecture, not a quotation of its theorem statement. In the first branch it gives neither a common rank nor opposite channels; in the second it gives infinitely many nonsquarefull prime-index terms but not all prime indices. It therefore does not settle the forced four-prime packet.

Three natural shortcuts have exact counterexamples. At index seven,

$$B_7 = 13^2, \quad 13^2 \parallel u_7, \quad z(13) = 7, \quad 13 = 2 \cdot 7 - 1.$$

Moreover

$$F_7(X) = X^6 - 5X^4 + 6X^2 - 1, \quad F_7(6) = 40391 = 13^2 \cdot 239, \quad F_7'(6) \equiv 2 \pmod{13}.$$

Thus a primitive divisor need not be simple, equality in the minimal channel bound does not force simplicity, and a nonsingular polynomial root modulo q does not force the fixed integer value to avoid q^2 . The opposite channel has the further exact example

$$1546463 = 2 \cdot 773231 + 1, \quad z(1546463) = 773231, \quad 1546463^2 \parallel A_{773231},$$

with both displayed rank parameters prime. These examples close only the three stated implications.

Two independent exact programs exhaust all 5,761,454 odd primes $q \leq 10^8$. The only repeated primes are 13, 31, 1546463, all of exact depth two; there is no depth-three hit. Hence both depth-three primes in a hypothetical squarefull prime-index packet exceed 10^8 . This is a finite certificate and is never used as an infinitude conclusion.

27.3. Recursive prime-square lifts on the Danilov progression. On the surviving Danilov progression, write $t = T + Qr$ and maintain

$$(279) \quad 3T + 1 = hQ, \quad h \in \{1, 2\}.$$

The quadratic-unit orbit gives the exact Fibonacci factorization

$$(280) \quad L_T = 5F_{10hQ}F_{10hQ-5}.$$

If $p \parallel F_{10Q}$, $p \nmid 3375Q$, then the packet modulo p^2 is nondegenerate. Squarefullness forces the unique residue

$$(281) \quad h + 3r \equiv 0 \pmod{p}.$$

For its representative $0 \leq \rho < p$, divisibility and the inequalities $0 < h + 3\rho < 3p$ give a unique $h' \in \{1, 2\}$ with $h + 3\rho = h'p$. The update

$$T' = T + Q\rho, \quad Q' = Qp$$

then preserves $3T' + 1 = h'Q'$.

Carmichael's theorem in Yabuta's form supplies a primitive divisor of F_{10Q} outside its explicit small exceptions [40]. Rank of apparition makes such a divisor fresh relative to Q . The theorem does not control its first valuation. This distinction yields the exact conditional closure.

Theorem 27.3 (Conditional simple-primitive closure). *Assume that at every adaptively generated state, F_{10Q} has a primitive prime divisor p with $p \parallel F_{10Q}$. Then no index in the initial Danilov survivor progression has squarefull remainder.*

Proof. The simple divisor supplies a nondegenerate fresh packet and (281) nests the next progression while preserving (279). Repetition constructs arbitrarily many distinct fresh primes. If one fixed index t survived as squarefull, every chosen prime would divide the one fixed nonzero integer L_t . A finite integer has only finitely many distinct prime divisors, a contradiction. $\square$

The valuation-one hypothesis is essential to this proof and is not a consequence of the cited primitive-divisor theorem. Nor does one abstract packet automatically generate another. In $\mathbb{Z}[\sqrt{5}]$, take

$$\eta = (9 + 4\sqrt{5})^8, \quad \alpha_0 = 19 + \sqrt{5}, \quad L_r = K_r = 2 \operatorname{Re}(\alpha_0 \eta^r) + 11.$$

Modulo 49, $\eta = 1 + 35\sqrt{5}$ and

$$(282) \quad L_r \equiv 7r \pmod{49}.$$

At $(T, Q, p) = (0, 1, 7)$ the slope is nonzero, the forced root is $r = 0$, and $K_0 = 49$ is squarefull. The updated modulus is 7, while every prime divisor of $L_0 = 49$ already divides it. Hence no fresh successor exists. This satisfies the full abstract local-packet mechanism, but its initial

norm is not the Danilov norm. It refutes automatic abstract continuation and does not refute the Danilov-specific simple-primitive route.

An exact adaptive computation constructs 626 distinct packets. They occur in thirteen nonempty batches. The final state has a 4398-digit modulus and exactly 638 distinct prime factors; no next eligible packet occurs below 10^8 . Every selected prime has exact valuation one, and every CRT update is independently replayed. This certifies the finite recursion, while the endpoint remains open above its bound.

27.4. The simple-primitive gate and Wall–Sun–Sun support. The missing valuation-one premise can be characterized exactly for the Fibonacci indices arising above. If $n > 5$, $5 \mid n$, and p is a primitive divisor of F_n , then the rank of apparition satisfies

$$z(p) = n \mid p - \left(\frac{5}{p}\right).$$

Reduction modulo five and quadratic reciprocity exclude the inert sign, so

$$(283) \quad p \equiv 1 \pmod{n}.$$

Sanna’s exact valuation formula then gives

$$(284) \quad v_p(F_n) = v_p(F_{p-1}).$$

Thus a primitive divisor is repeated precisely when it is a Fibonacci–Wieferich, or Wall–Sun–Sun, prime [36, 37]. No such prime is presently known; that empirical fact is not used as a nonexistence theorem.

Let Q_* be the final 4398-digit squarefree modulus in the certified recursive chain and put $n_* = 10Q_*$. Its complete factorization consists of exactly 638 primes, each below 10^8 , and $\gcd(Q_*, 30) = 1$. Let C_n denote the Fibonacci cyclotomic factor. For $n > 2$ divisible by 10, Yabuta’s correction analysis says that C_n has at most one nonprimitive prime factor; if present, it is the greatest prime factor of n and occurs once. At n_* this correction cannot occur. Indeed, squarefreeness would give $n_* = qz(q)$, whence $n_* \leq q(q+1) \leq 10^8(10^8+1)$, contrary to the number of digits.

Theorem 27.4 (Exact surviving simple-primitive failure). *If F_{n_*} has no simple primitive divisor, then C_{n_*} is powerful and every prime $p \mid C_{n_*}$ is Wall–Sun–Sun and satisfies $p \equiv 1 \pmod{n_*}$. At least one such prime satisfies*

$$p \geq 41n_* + 1.$$

Proof. The correction just excluded shows that every factor of C_{n_*} is primitive with its full F_{n_*} multiplicity. Equations (283) and (284) therefore give the powerful and Wall–Sun–Sun assertions. Hong’s explicit large primitive-divisor theorem with $\kappa = 40$ applies because $\log n_* > 4398 \log 10 > 10036$ [38]. It produces a primitive prime outside $jn_* \pm 1$ for $1 \leq j \leq 40$. The split congruence leaves only the plus sign, so its coefficient is at least 41. $\square$

This theorem sharply locates the surviving bad case but does not contradict it. A broad real-Lucas shortcut is false under its full standard hypotheses: for

$$U_{m+2} = 2U_{m+1} + 3U_m, \quad U_{10} = 14762 = 2 \cdot 11^2 \cdot 61,$$

the prime 11 is the unique primitive divisor at index ten and occurs exactly twice. The roots are 3, -1 , so the sequence is real, nondegenerate and has coprime parameters. This counterexample closes only the sequence-uniform transfer; it does not satisfy the Fibonacci-specific hypothesis. An exact search over the 252 squarefree $Q \leq 1000$ with $\gcd(Q, 30) = 1$ finds simple primitive certificates for 207 cases below $5 \cdot 10^7$. The remaining 45 cases are unresolved by that finite bound, not counterexamples.

27.5. The current LANA specification and the pointed same-pilot gate. The public Project LANA Lean repository was audited at the pinned commit `ddaddc2` [60]. Its declared Corollary 3.12 variant shares initial theta data between a q -pilot scalar and an independently supplied right-hand-side container. At that commit the right-hand-side record has a prior consistency defect.

Theorem 27.5 (Empty-set obstruction in the pinned low-resolution signature). *For every initial theta datum D , the pinned type RHSDat_D is uninhabited.*

Proof. The admissible prime satisfies $\ell \geq 5$, and the required standard procession therefore has positive length. Choose one capsule index and the rational prime $p = 2$. The record asserts that the finite sum of the weights over the p -fibre is one, so that fibre cannot be empty. A constant function from the capsule labels to a point of the fibre supplies a component c .

For this component the record further asserts, for every set U , that

$$(285) \quad \text{vol}((x \mapsto px)^{-1}U) = \text{vol}(U) + \log p.$$

Take $U = \emptyset$. Its inverse image is empty, so cancellation in (285) gives $\log 2 = 0$, contrary to $\log 2 > 0$. $\square$

Thus the assembled variant-data type, which contains an RHSDat field, is also empty. A universal theorem over that type would be vacuous, and there can be no counterexample satisfying all its fields. This result rules out the pinned low-resolution signature as a satisfiable specification. It says nothing against a corrected Haar-volume interface, the published IUT argument, or *abc*. A direct repair restricts (285) to nonempty measurable regions of finite nonzero volume and proves stability of that domain under preimages; alternatively one may use an extended-real logarithmic codomain with explicit zero-volume semantics.

After this repair, the main typing gap remains. The two sides currently share the initial datum but have no field realizing the q -pilot as a region in the right-hand container, no identity for that region's volume, no bridge between the two weight systems, no multiradial output map, no IPL/SHE/APT or Ind3 data, no determinant normalization, and no volume monotonicity theorem. Sharing a parameter supplies no order relation between two independently chosen real outputs.

The July 2026 LANA report isolates a stronger target: for a suitable output choice S , prove equality of maps

$$\eta^q = \eta_S^{\text{anab}} : \mathbb{R}^{\text{val}} \longrightarrow \mathbb{R}^{\text{ss}}.$$

The report states that this equality is not currently proved and does not declare it manifestly false [59]. The numerical conclusion needs less.

Proposition 27.6 (Minimal pointed same-pilot certificate). *Let p_q be the canonical q -pilot point, let ν be the correctly normalized volume coordinate, and put $q_* = -|\log q|$, $T = -|\log \Theta|$. Suppose a globally synchronized, source-permitted output S satisfies*

$$\nu(\eta^q(p_q)) = q_*, \quad \eta^q(p_q) = \eta_S^{\text{anab}}(p_q), \quad \nu(\eta_S^{\text{anab}}(p_q)) \leq T.$$

Then $q_ \leq T$.*

Proof. Apply ν to the pointed equality and substitute the two inequalities. $\square$

The proposition is elementary after its hypotheses are available. Its content is the construction of the middle equality before taking the volume quotient: the diagram must start with the fixed q -pilot arithmetic line, follow the theta link, IPL/SHE/APT construction, the required indeterminacies, log-Kummer correction and determinant normalization, and return to the same right-hand arithmetic holomorphic structure with compatible local and global metrics. Defining the quotient equality by equality of the desired scalar values would be circular. A source-defined witness in $\Phi(P_q)$ also suffices, but membership in that class already contains the lower volume inequality and must be constructed independently.

The pinned specification failure is therefore a narrow no-go theorem, while the pointed same-pilot construction remains an active positive route. The public report's full map equality is a

stronger active target. None of these results yields an unconditional term of type **ABCConjecture** or its negation.

28. CORRECTED VOLUME HOLONOMY, AFFINE DENSITY, AND DEEP-PRIME ESCAPE

This section records the next continuation of the four active routes. The results are deliberately asymmetric. On the positive side we construct an inhabited valuation-ball model for the local volume law, prove an affine squarefree-bulk theorem, and amplify the arithmetic consequences of a hypothetical Pell or Danilov survivor. On the negative side we give complete counterexamples to two scalar reconstruction mechanisms. Neither counterexample satisfies the stronger object-level hypotheses of the corresponding IUT or arithmetic route.

28.1. A corrected local volume model and its holonomy ledger. Fix a rational prime p , normalize $|p|_p = p^{-1}$, and for $k \in \mathbb{Z}$ put

$$B_p(k) = \{x \in \mathbb{Q}_p : |x|_p \leq p^{-k}\}.$$

These compact-open balls avoid the empty-set obstruction in Theorem 27.5 while retaining the intended scaling law.

Proposition 28.1 (Valuation-ball transport). *For every $k \in \mathbb{Z}$,*

$$(x \mapsto px)^{-1}B_p(k) = B_p(k-1).$$

Moreover, $B_p(k) \subseteq B_p(m)$ if and only if $m \leq k$. If $\text{vol}(B_p(k)) = p^{-k}$, then

$$\log \text{vol}((x \mapsto px)^{-1}B_p(k)) = \log \text{vol}(B_p(k)) + \log p.$$

The same additive shift survives every normalized finite weighted packet and every positive-length procession average.

Proof. The first assertion is the chain

$$|px|_p \leq p^{-k} \iff p^{-1}|x|_p \leq p^{-k} \iff |x|_p \leq p^{-(k-1)}.$$

The inclusion criterion follows because $p^{-k} \leq p^{-m}$ precisely when $m \leq k$; necessity may also be tested at p^k . The volume formula is $\log p^{-(k-1)} = \log p^{-k} + \log p$. Finally, a common additive shift passes through a weighted sum whose weights total one, and through division by a positive finite procession length. $\square$

Let a logarithmic transport from a pointed object x to a pointed object y with coordinate L and shift δ mean

$$L(y) = L(x) + \delta.$$

Composition adds shifts and reversal negates them.

Theorem 28.2 (Zero logarithmic holonomy). *Every transport returning to the same pointed object has total shift zero. Consequently an uncorrected loop whose edge shifts are nonnegative and whose total shift is positive cannot be a same-pilot loop. A correcting edge in a closed loop must contribute the negative of the accumulated shift.*

Proof. If $y = x$, then $L(x) = L(x) + \delta$, and cancellation gives $\delta = 0$. The remaining assertions are immediate applications to a sum of edge shifts. $\square$

For rational coefficients the scalar test can be sharpened place by place.

Theorem 28.3 (Prime-local rational ledger). *Let $p_1, \dots, p_r$ be distinct rational primes and $c_i \in \mathbb{Q}$. If*

$$\sum_{i=1}^r c_i \log p_i = 0,$$

then $c_i = 0$ for every i . Hence a closed rational-prime logarithmic transport must cancel separately at every rational place.

Proof. Clear a common denominator and write the resulting coefficients as $m_i \in \mathbb{Z}$. Exponentiation gives $\prod_i p_i^{m_i} = 1$. Moving negative powers to the other side produces an equality of positive integers, and unique factorization forces every m_i to vanish. $\square$

The coefficient field and the amount of retained local data are essential. Arbitrary real coefficients are dependent, for example $(\log 3) \log 2 - (\log 2) \log 3 = 0$. Positivity and normalization do not repair the loss of labelled information.

Proposition 28.4 (Positive normalized scalar collision). *There are two different triples of strictly positive real weights, each having sum one, whose weighted averages of $\log 2, \log 3, \log 5$ are equal.*

Proof. Put $a = \log 2 < b = \log 3 < c = \log 5$ and define

$$W = \left(\frac{c-b}{2(c-a)}, \frac{1}{2}, \frac{b-a}{2(c-a)} \right), \quad V = \left(\frac{c-b}{3(c-a)}, \frac{2}{3}, \frac{b-a}{3(c-a)} \right).$$

All six entries are positive, and each row sums to one. More generally, for $n = 2, 3$,

$$\frac{c-b}{n(c-a)}a + \left(1 - \frac{1}{n}\right)b + \frac{b-a}{n(c-a)}c = b,$$

because $(c-b)a + (b-a)c = b(c-a)$. The middle coordinates $1/2$ and $2/3$ show that the triples differ. $\square$

Proposition 28.4 is a full counterexample to recovering labelled weights from one positive normalized scalar. It does not refute a construction which transports the labelled local objects, nor the stronger same-pilot map equality discussed after Proposition 27.6.

28.2. Squarefree bulk and the coupled long-arm affine gate. Return to the minimal radical-step shear in Theorem 27.1. Thus $a + b = c$ is primitive, $R = \text{rad}(abc)$, and

$$U = 1 + Rh, \quad V = 1 + R(h + ck), \quad W = 1 + R(h + bk).$$

Proposition 28.5 (Support-closed gap recovery). *For $h, k > 0$,*

$$\gcd(V - U, W - U) = Rk, \quad (a, b, c) = \frac{1}{Rk}(V - W, W - U, V - U).$$

Conversely, let $1 < U < W < V$, put $g = \gcd(V - U, W - U)$, and divide the three consecutive gaps by g to obtain (a, b, c) . If $\text{rad}(abc) \mid g$ and $\text{rad}(abc) \mid U - 1$, then the triple is exactly a minimal positive-parameter radical-step shear with $k = g/R$ and $h = (U - 1)/R$.

Proof. The forward identities follow from $(V - W, W - U, V - U) = Rk(a, b, c)$ and $\gcd(b, c) = 1$. Conversely the definition of g makes the normalized gaps integral and primitive, while addition of the first two gives the third. The two divisibilities define integral h, k ; the strict inequality $U > 1$ gives $h > 0$, and substitution recovers the three displayed affine forms. The weaker condition $U \geq 1$ is false: $(U, W, V) = (1, 3, 5)$ gives $(a, b, c) = (1, 1, 2)$ and $R = g = 2$, but $h = 0$. $\square$

Write $E(n) = n/\text{rad}(n)$. The inherited three-quarter exceptional inequality in the upper-half canonical box is

$$E(U)E(V)E(W) > \frac{Rc^{14}}{8192},$$

while $U \leq c^6$ and $V, W \leq c^7$. It follows pointwise that

$$(286) \quad 8192E(V) > Rc, \quad 8192E(W) > Rc.$$

Thus both long arms must have large repeated-prime excess.

The same box nevertheless contains a uniformly positive squarefree bulk. Put

$$M = \left\lfloor \frac{c^6}{4R} \right\rfloor, \quad I_M = \{\lfloor M/2 \rfloor + 1, \dots, M\}, \quad N = |I_M|.$$

Theorem 28.6 (Uniform squarefree-admissible bulk). *Fix $\theta < 5/2$. For all sufficiently large primitive seeds satisfying $R < c^\theta$, at least*

$$\frac{1}{2} \left(5 - \frac{\pi^2}{2} \right) N^2$$

*pairs $(h, k) \in I_M^2$ obey $\gcd(U, k) = 1$ and have U, V, W all squarefree. Every resulting *abc* output has radical at least its height and is therefore nonexceptional for every fixed exponent below one.*

Proof. For a prime $p \nmid R$, failure of $\gcd(U, k) = 1$ prescribes one residue class for both h and k modulo p , and hence occurs for at most $(N/p + 1)^2$ pairs. For a fixed $F \in \{U, V, W\}$, the condition $p^2 \mid F$ prescribes one class of h modulo p^2 after k is fixed, and hence occurs for at most $N^2/p^2 + N$ pairs. Primes dividing R cause neither event. Since $2 \mid R$, the union bound and

$$\sum_{j \geq 1} \frac{1}{(2j+1)^2} = \frac{\pi^2}{8} - 1$$

leave at least

$$\left(5 - \frac{\pi^2}{2} \right) N^2 - 2N(1 + \log M) - M - 3Nc^{7/2}$$

good pairs. Here every possible square divisor is covered because $U, V, W \leq c^7$. The hypothesis $R < c^\theta$ gives $N \gg c^{6-\theta}$, so the three error terms are smaller than half the positive main term when $\theta < 5/2$.

For a good pair, pairwise coprimality and avoidance of the seed support give $\text{rad}(aUbVcW) = RUVW$. Also $U \geq c$ for all sufficiently large seeds in this range, and therefore $RUVW \geq cW$, the output height. $\square$

Theorem 28.6 and (286) point in opposite directions without contradiction. The desired exceptional lower count occupies a vanishing fraction of the raw box, so a positive-density generic bulk cannot be complemented into that lower bound. A proof must create simultaneous high-excess tails in both long arms or exploit the support-closed reverse description in Proposition 28.5.

28.3. Prime-rank depth and second-order Pell coupling. Let $u_0 = 0, u_1 = 1, u_{n+2} = 6u_{n+1} - u_n$ and write $(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}$. At an odd prime index ℓ , $u_\ell = A_\ell B_\ell$, and the inherited valuation dictionary identifies every odd divisor of either channel with rank ℓ .

Theorem 28.7 (Pointwise channel escape). *For every odd prime ℓ , the A channel has either a prime $q \parallel A_\ell$ or an odd-exponent divisor $p^{e(p)} \mid A_\ell$ with rank ℓ and $e(p) \geq 3$. The same holds for the B channel except for $B_7 = 13^2$. Consequently each channel separately has simple divisors at all but finitely many prime indices, or has infinitely many distinct prime-rank depth-three primes. If the global set of depth-three primes is finite, then both simple divisors occur simultaneously at every sufficiently large prime index.*

Proof. An integer which is not a perfect power either has an exponent-one prime or has an odd prime exponent at least three. Apply this observation to the perfect-power classifications of the two Pell coordinates. Different prime indices give different ranks, hence different primes in the second branch. If only finitely many depth-three primes exist, delete their finitely many ranks and the exceptional index seven; both channels must then take the simple branch at every remaining prime index [45, 46, 47]. $\square$

Factoring the channels and expanding every signed local factor through its quadratic term yields a necessary congruence one digit deeper than the first-order ledger. With the notation K_A, K_B, C_A, C_B for the linear and quadratic factor coefficients, respectively,

$$(287) \quad K_A - 2K_B + \ell(K_A^2 - 2K_B^2) + 2\ell(C_A - 2C_B) \equiv 0 \pmod{4\ell^2}.$$

Quadratic reciprocity also gives the all-pairs identity

$$\prod_{q \mid A_\ell} \prod_{r \mid B_\ell} \left(\frac{q}{r} \right)^{v_q(A_\ell)v_r(B_\ell)} = \left(\frac{2}{\ell} \right).$$

These ledgers constrain a full four-prime packet but do not yet force an inconsistency.

Applying the repaired Fellini–Murty prime-order argument separately to $1 + \sqrt{2}$ and its negative produces an alternative: either infinitely many rational balancing primes have depth at least three, or infinitely many prime indices have a simple A -divisor and infinitely many prime indices have a simple B -divisor. The latter two index sets are not asserted to intersect; the synchronized conclusion in Theorem 28.7 uses the stronger finite-depth-three hypothesis instead [39].

An exhaustive certificate checks every one of the 50,847,533 odd primes $q \leq 10^9$. Independent segmented-sieve/Lucas-doubling and dense-sieve/quotient-ring implementations agree: the only depth-two primes are

$$13, \quad 31, \quad 1546463,$$

and each has exact depth two; no depth-three prime occurs. Arbitrary-precision replay verifies the three residues modulo q^3 . Thus each depth-three prime required by an actual opposite-channel packet exceeds 10^9 . This is a finite lower bound, not a nonexistence result.

28.4. Divisor-pair amplification on the Danilov survivor. Let F_n be the Fibonacci sequence. The Danilov factorization has the form

$$K_t = \frac{27}{25} F_N F_{N-5}, \quad N = 10(3t + 1),$$

with $\gcd(F_N, F_{N-5}) = 5$. The final recursive certificate supplies a squarefree, 638-prime, 4398-digit modulus Q_* , coprime to 30, such that $Q_* \mid 3t + 1$ throughout the surviving progression.

Theorem 28.8 (Wall–Sun–Sun divisor-pair amplification). *Let $Q > 1$ be squarefree and coprime to 10, let R be a positive multiple of Q , and put $N = 10R$. If $(27/25)F_N F_{N-5}$ is squarefull and Q has s distinct prime factors, then there are at least 2^{s-1} distinct Wall–Sun–Sun primes. They may be indexed by the divisors $d \mid Q$ satisfying $d < Q/d$, and the indexed prime p_d obeys*

$$z(p_d) = N/d, \quad p_d \equiv 1 \pmod{N/d}, \quad p_d > N/\sqrt{Q}.$$

For the final Danilov progression this supplies a distinguished subfamily of at least 2^{637} such primes, each exceeding 10^{2199} .

Proof. The involution $d \mapsto Q/d$ pairs the 2^s divisors of nonsquare Q , so exactly 2^{s-1} satisfy $d < Q/d$. Put $m_d = N/d$. Carmichael–Yabuta gives a primitive divisor p_d of F_{m_d} [40]. Since $10 \mid m_d$, its rank is m_d and the split-rank congruence gives $p_d \equiv 1 \pmod{m_d}$. The inequality $d < \sqrt{Q}$ implies $m_d > d$, whence p_d divides neither m_d nor d and therefore does not divide $N = m_d d$.

Now $p_d \mid F_N$ and, because $p_d > 5$, it does not divide F_{N-5} or the rational prefactor. Squarefullness forces $v_{p_d}(F_N) \geq 2$. The exact Lucas valuation formula and $p_d \nmid N$ transfer this valuation to F_{m_d} , making p_d a Wall–Sun–Sun prime [36]. Equality of two selected primes would force equality of their ranks and hence equality of the corresponding divisors, so they are distinct. Finally $p_d \geq N/d + 1 > N/\sqrt{Q}$.

For Q_* take $s = 638$. Since Q_* has 4398 decimal digits and $R \geq Q_*$, the size bound gives $p_d > 10\sqrt{Q_*} > 10^{2199}$. $\square$

The certified upper bound for the prime factors of Q_* permits almost all divisors, rather than only one representative from each complementary pair.

Theorem 28.9 (Factor-bound amplification). *Under the hypotheses of Theorem 28.8, suppose every prime factor of Q is at most B , and put*

$$E_B(Q) = \#\{e : e \mid Q, 10e \leq B\}.$$

Then squarefullness forces at least $2^s - E_B(Q)$ distinct Wall–Sun–Sun primes. At the verified endpoint $B = 99,966,059$ and exact truncated subset-product enumeration gives $E_B(Q_) = 622$. Thus a survivor forces at least $2^{638} - 622$ distinct Wall–Sun–Sun primes. At least one forced prime is at least $41N + 1 > 10^{4399}$.*

Proof. Write $R = kQ$. For every divisor $e \mid Q$ with $10e > B$, put $d = Q/e$ and $m_e = N/d = 10ke$. A primitive divisor p_e of F_{m_e} is congruent to one modulo m_e , so $p_e > m_e > B$. It divides neither Q nor m_e , and hence does not divide $N = m_e d$. The squarefullness and valuation transfer argument from Theorem 28.8 makes it a Wall–Sun–Sun prime. Different e give different ranks and hence different primes. Exactly $E_B(Q)$ of the 2^s divisors were omitted.

For Q_* the certified factor list has maximum $B = 99,966,059$; exact enumeration below $B/10$ returns 622 divisors. The final large-prime claim uses Hong’s primitive-divisor theorem with $\kappa = 40$: since $\log N > 10036$, it supplies a primitive divisor of F_N outside $jN \pm 1$ for $1 \leq j \leq 40$. The split congruence leaves the plus sign and coefficient at least 41, and squarefullness transfers repeatedness as before [38]. $\square$

No known theorem excludes $2^{638} - 622$ Wall–Sun–Sun primes at these sizes, so Theorems 28.8 and 28.9 are strong, explicitly quantified conditional obstructions, not a contradiction. The cyclotomic-derivative shortcut also fails: the complete example

$$\Phi_{10}(-3) = 121 = 11^2, \quad \Phi'_{10}(-3) = -142, \quad \gcd(121, 142) = 1$$

shows that a squared value need not come from a multiple root. Hensel uniqueness does not prevent a fixed algebraic unit from landing on the unique lift. This counterexample closes only that derivative mechanism; the Fibonacci and Danilov routes remain active.

28.5. Formal and logical boundary. The elementary parts of Propositions 28.1 and 28.4, Theorems 28.2 and 28.3, the affine gap and excess identities, the Pell second-order ledger, and the abstract Danilov divisor-pair transfer have independent Lean implementations. Across the four companion modules these comprise 65 theorem or lemma declarations, 18 definitions, four abbreviations, and five structures, for 92 declarations; 58 embedded `#print axioms` queries report only `propext`, `Classical.choice`, and `Quot.sound`. The squarefree-bulk estimate, the perfect-power classifications and Fellini–Murty argument, the primitive-divisor and valuation theorems, Hong’s theorem, and the large finite certificates remain explicit paper, literature, or computation inputs and are not introduced as Lean axioms. The computation packages freeze source, output, byte count, and SHA-256 manifests.

At this stage the four live endpoints are: construct an object-level same-pilot transport satisfying the prime-local zero-holonomy ledger; produce or exclude the simultaneous affine high-excess tails; rule out or construct opposite-channel prime-rank depth-three Pell packets; and exclude or realize the enormous Wall–Sun–Sun population forced by a Danilov survivor. None is closed by difficulty or by a finite no-hit, and no unconditional term of type `ABCConjecture` or its negation is claimed.

29. A LITERAL DISTINCT-PRIME-FACTOR COUNTERMODEL AND THE RETAINED RADICAL TARGET

Carella’s current preprint [14], Theorem 5.1(iii), p. 12, prints the hypothesis

$$(288) \quad \#\{n \leq x : \omega(n) > w\} = o(x^{3/5}), \quad w \leq 2 \log \log x.$$

The prose following the display refers instead to smooth integers in the short interval $[x, x + x^{3/5}]$. The two sets are different. We first audit the displayed global statement literally.

Let $p_1 = 2 < p_2 < \dots$ be the primes and put $Q_r = \prod_{j=1}^r p_j$.

Theorem 29.1 (Primorial-multiple counterfamily). *For a real number $w \geq 0$, let $r = \lfloor w \rfloor + 1$. Then, for $x \geq Q_r$,*

$$(289) \quad \#\{n \leq x : \omega(n) > w\} \geq \left\lfloor \frac{x}{Q_r} \right\rfloor.$$

If $w = w(x) \leq 2 \log \log x$ for all sufficiently large x , then the left side is $x^{1-o(1)}$ and is therefore not $o(x^{3/5})$.

Proof. For every $1 \leq m \leq \lfloor x/Q_r \rfloor$, the integer mQ_r is at most x and is divisible by the r distinct primes $p_1, \dots, p_r$. Thus $\omega(mQ_r) \geq r > w$, and multiplication by the positive integer Q_r is injective. This proves (289).

Bertrand's postulate gives $p_{j+1} < 2p_j$, hence $p_j \leq 2^j$ and

$$\log Q_r \leq \frac{r(r+1)}{2} \log 2.$$

Under $w(x) \leq 2 \log \log x$, this is $O((\log \log x)^2) = o(\log x)$. Consequently $Q_r = x^{o(1)}$, and the lower bound in (289) is $x^{1-o(1)}$. Dividing it by $x^{3/5}$ gives $x^{2/5-o(1)} \rightarrow \infty$. $\square$

Thus (288) is false with all of its displayed conditions. Moreover, Theorem 5.1 is conditional on that statement, while the subsequent proof of the preprint's Theorem 1.1 treats the conditional result as available for every large x . The claimed unconditional conclusion does not follow.

This literal counterfamily is separate from the earlier local audit in Section 2. If the display is repaired by restricting to B -smooth integers in $[x, x + x^{3/5}]$, the theorem above no longer applies literally. Theorem 2.3 then shows that almost every smooth integer in the relevant Younis regime has much more than $2 \log \log x$ distinct prime divisors, contradicting the proposed majority and moment mechanism. Neither result excludes one exceptionally sparse low-radical neighbour in infinitely many prime-power-centred intervals.

The final radical calculation of the proposed construction is in fact valid under such an existence premise. The following formulation keeps the larger route without requiring smoothness or a bound on ω .

Theorem 29.2 (Three-fifths radical-neighbour gate). *Fix $k \geq 1$, $\sigma \geq 0$, and suppose*

$$(290) \quad \sigma < \frac{2}{5} - \frac{1}{k}.$$

If there are infinitely many primes p and integers c , at unbounded $X = p^k$, for which

$$(291) \quad X < c \leq X + X^{3/5}, \quad p \nmid c, \quad \text{rad}(c) \leq X^{\sigma+o(1)},$$

then the standard abc conjecture is false.

Proof. Set $a = c - p^k$ and $b = p^k$. Since $p \nmid c$, the three entries are pairwise coprime. Radical submultiplicativity gives

$$\text{rad}(abc) \leq a p \text{rad}(c) \leq X^{3/5+1/k+\sigma+o(1)}.$$

Condition (290) makes the fixed exponent $q = 3/5 + 1/k + \sigma$ strictly smaller than one. Choose $\varepsilon > 0$ with $(1 + \varepsilon)q < 1$. After absorbing the $o(1)$ term into half of the positive gap,

$$\frac{c}{\text{rad}(abc)^{1+\varepsilon}} \geq X^{(1-(1+\varepsilon)q)/2} \rightarrow \infty.$$

This contradicts the existence of the uniform abc constant for that fixed ε . $\square$

The conclusion shows that the condition $\text{rad}(c) = X^{o(1)}$ is stronger than needed. At $k = 4$, for example, every fixed $\sigma < 3/20$ suffices; the allowed upper exponent tends to $2/5$ as k grows.

There is an exact extra restriction if $k \geq 3$ and one asks for candidates at a positive proportion of prime centres. The Rankin radical count proved in the underlying research report gives

$$\#\{n \leq X : \text{rad}(n) \leq X^\sigma\} \ll_\eta X^{\sigma+\eta}.$$

Disjointness of the $X^{3/5}$ intervals and the prime number theorem then force $\sigma \geq 1/k$.

Proposition 29.3 (Exact density-compatible exponent window). *There exists a real σ such that*

$$(292) \quad \frac{1}{k} \leq \sigma < \frac{2}{5} - \frac{1}{k}$$

if and only if $k > 5$. For every integer $k \geq 6$, the uniform choice $\sigma = 1/5$ is feasible and gives

$$\frac{3}{5} + \frac{1}{k} + \frac{1}{5} \leq \frac{29}{30} < 1.$$

Proof. The interval in (292) is nonempty precisely when $1/k < 2/5 - 1/k$, equivalently $2/k < 2/5$, or $k > 5$. If $k \geq 6$, then $1/k \leq 1/6 < 1/5$ and the displayed total-slope bound follows. $\square$

For $k = 1, 2$, the target (290) is already empty because $\sigma \geq 0$. The positive-density versions with $3 \leq k \leq 5$ are rejected by the counting contradiction. A zero-density unbounded subsequence is not rejected for $3 \leq k \leq 5$. For $k \geq 6$, Proposition 29.3 provides a concrete positive target that current density estimates do not forbid. What remains is an arithmetic theorem producing the neighbours, not another manipulation of the final radical budget.

30. A FAITHFUL PRIME-UNIT-LABEL VECTOR BRIDGE

The scalar holonomy ledger in Section 28 cannot reconstruct a pilot: two different positive normalized labelled packets may have the same one-dimensional weighted value. We therefore retain the rational prime, complete unit part, and fixed label until after the object comparison. IUT III [62] distinguishes unit/coric data and mono-analytic log-shells before the final log-volume operation. The pinned Project LANA container [59, 60] likewise retains the rational place fibre and procession labels in a packet component. These sources motivate the coordinate type; they do not prove that the actual multiradial transport preserves the coordinates below.

Fix a rational prime p . For $x \in \mathbb{Q}^\times$, put

$$(293) \quad e_p(x) = v_p(x), \quad u_p(x) = \frac{x}{p^{e_p(x)}}, \quad C_p(x) = (e_p(x), u_p(x)).$$

Negative powers have their ordinary field meaning.

Theorem 30.1 (Exact prime-unit reconstruction). *For every $x \in \mathbb{Q}^\times$,*

$$(294) \quad v_p(u_p(x)) = 0, \quad p^{e_p(x)} u_p(x) = x.$$

Consequently C_p is injective. The identical assertions hold on the actual complete local field $\mathbb{Q}_p^\times$ with its normalized additive valuation.

Proof. Valuation additivity and $v_p(p) = 1$ give

$$v_p(u_p(x)) = v_p(x) - e_p(x)v_p(p) = 0.$$

The second equality in (294) is cancellation of the same nonzero field power used in the definition. Equal coordinate pairs therefore have equal reconstruction expressions. This proves injectivity. The proof uses only the corresponding field and valuation identities in $\mathbb{Q}_p$ and is unchanged there. $\square$

There is also a valuation-free field-summand template. If K is any field, $\pi \in K^\times$, and $e : K^\times \rightarrow \mathbb{Z}$ is any function, then

$$(295) \quad u_{\pi,e}(x) = \frac{x}{\pi^{e(x)}} \neq 0, \quad \pi^{e(x)} u_{\pi,e}(x) = x.$$

Thus $x \mapsto (e(x), u_{\pi,e}(x))$ is injective. When e is a normalized discrete valuation and π a uniformizer, the first calculation in Theorem 30.1 additionally proves that the second coordinate is a genuine unit. Hence the reconstruction mechanism applies separately to every actual field summand once that summand and its transport have been constructed.

Let L be a label set and let a packet be a map $P : L \rightarrow \mathbb{Q}^\times$. Define its complete labelled signature by

$$(296) \quad \Sigma_p(P) = (C_p(P(i)))_{i \in L}.$$

Theorem 30.2 (Labelled and region reconstruction). *The map Σ_p is injective. More generally, let X be a carrier with a family of exponent and unit coordinates whose joint value is faithful at each point, and let Σ be the corresponding labelled signature. For packet regions $A, B \subseteq (L \rightarrow X)$,*

$$(297) \quad \Sigma(A) \subseteq \Sigma(B) \implies A \subseteq B.$$

If μ is a monotone finite real-valued region functional, then $\mu(B) \leq T$ and (297) imply $\mu(A) \leq T$.

Proof. Equality of two signatures gives equality of both local coordinates at every fixed label. Theorem 30.1, or the assumed pointwise faithfulness, gives equality of the packet entries and then of the packets.

For the region statement, take $P \in A$. Image containment supplies $Q \in B$ with $\Sigma(Q) = \Sigma(P)$. Injectivity gives $Q = P$, hence $P \in B$. Monotonicity of μ and transitivity of the real order give the last assertion. $\square$

This is a non-circular route from vector data to a scalar bound. It does not assume equality of the regions or equality of their volumes. It requires an independently proved containment of complete signature images, followed by a monotonicity theorem on the correct finite-positive-volume domain.

The unit coordinate also completes the holonomy ledger. Suppose a labelled transport $P \rightarrow Q$ has exponent shifts $\delta_i \in \mathbb{Z}$ and unit twists $\tau_i \in \mathbb{Q}$ satisfying

$$(298) \quad e_p(Q(i)) = e_p(P(i)) + \delta_i, \quad u_p(Q(i)) = u_p(P(i))\tau_i.$$

Proposition 30.3 (Pointed vector holonomy). *Every twist in (298) is nonzero and has p -adic valuation zero. Transports compose labelwise by*

$$(\delta, \tau) \circ (\epsilon, \nu) = (\delta + \epsilon, \tau\nu).$$

If the labelled transport is a pointed loop, $Q = P$, then $\delta_i = 0$ and $\tau_i = 1$ for every fixed label i .

Proof. Both unit parts in the second equation of (298) are nonzero, so $\tau_i \neq 0$. Applying v_p and Theorem 30.1 gives $0 = 0 + v_p(\tau_i)$. Substitution of two consecutive transport equations gives the composition law. If $Q = P$, cancellation in the exponent equation gives $\delta_i = 0$, and cancellation of the nonzero unit part gives $\tau_i = 1$. $\square$

Each of the three retained layers is necessary for this particular reconstruction theorem.

Proposition 30.4 (Complete counterexamples to three weakened interfaces). *The following data satisfy all the indicated weakened hypotheses.*

- (1) *At $p = 5$, the one-label packets $P = 1$, $Q = 2$, with weight one, have the same exponent zero, the same fixed label, and the same positive normalized weight, but are unequal.*
- (2) *At $p = 5$, the one-label packets $P = 1$, $Q = 6$, with weight one, have equal exponent zero and equal first unit residue modulo 5, but are unequal.*
- (3) *At $p = 5$, take $P = (1, 2)$, $Q = (2, 1)$ and weights $(1/2, 1/2)$. Their multisets of complete coordinates agree after transposition. At the fixed labels the exponent shifts vanish and the unit twists are $(2, 1/2)$, whose product is one, but the labelled packets differ and neither twist is one.*

Proof. The integers 1, 2, 6 are 5-adic units, and $1 \equiv 6 \pmod{5}$. The stated weights are positive and sum to one. In the two-label example, transposition exchanges the two complete coordinates, while at fixed labels the quotient unit twists are 2/1 and 1/2. Their product is one and each is nontrivial. Every asserted inequality is immediate. $\square$

The first example closes only exponent-only reconstruction; the second closes only first-residue truncation; the third closes only unordered reconstruction and aggregate unit-holonomy closure. The third example does not refute a transport that records and returns an explicit label permutation.

The active IUT target is therefore a source-level theorem on the actual common carrier: through the theta link, log-Kummer correction, determinant normalization, and every required Ind1–Ind3 branch, the globally synchronized output must preserve the faithful prime-unit coordinate at every required place and every fixed label, or at least contain the input signature image. Theorem 30.2 would then transfer the output bound back to the native pilot before taking scalar log-volume. No such global preservation theorem is proved here, and none of the finite counterexamples disproves it. The IUT route remains active.

31. REFINED-FACTOR ZERO-AWARE PACKET SIGNATURES

The prime-unit signature of Section 30 must be corrected at two source-level seams. A place-tuple tensor algebra need not be a field: IUT III, Proposition 3.1, splits it into a direct sum of field factors [62]. Moreover its integral packet includes zero, so a coordinate defined only on the multiplicative group is not a faithful coordinate on the source carrier.

Let K be a field, choose $\pi \in K^\times$, and allow an arbitrary map $e : K^\times \rightarrow \mathbb{Z}$. Define

$$\mathcal{C}_{\pi,e}(x) = \begin{cases} \text{none}, & x = 0, \\ \text{some}(e(x), x/\pi^{e(x)}), & x \neq 0. \end{cases}$$

Theorem 31.1 (Zero-aware reconstruction). *The map $\mathcal{C}_{\pi,e}$ is injective. Its reconstruction map is*

$$\text{none} \mapsto 0, \quad \text{some}(n, u) \mapsto \pi^n u.$$

No valuation property of e is needed for injectivity.

Proof. The zero tag separates zero from every nonzero input. For $x \neq 0$, $\pi^{e(x)}(x/\pi^{e(x)}) = x$. Applying reconstruction to equal coordinates therefore recovers equal inputs. $\square$

If e is a normalized discrete valuation and π a uniformizer, the second component is the complete unit part. The signature is functorial only after the necessary multiplicative data have been supplied.

Theorem 31.2 (Factorwise covariance). *Let $\phi : K \simeq K'$ be a field equivalence. Suppose the target scale is $\phi(\pi)$ and*

$$e'(\phi(x)) = e(x) \quad (x \neq 0).$$

Then

$$\mathcal{C}_{\phi(\pi),e'}(\phi(x)) = \hat{\phi}(\mathcal{C}_{\pi,e}(x)),$$

where $\hat{\phi}$ fixes the exponent and applies ϕ to the second coordinate.

Proof. The zero case is immediate. For $x \neq 0$, preservation of division and integer powers gives

$$\frac{\phi(x)}{\phi(\pi)^{e(x)}} = \phi\left(\frac{x}{\pi^{e(x)}}\right).$$

$\square$

Fix one finite-dimensional stage of the source inductive system. Let C be its place-tuple index and let A_c be the associated finite étale algebra. The correct refined index at this stage is

$$D = \coprod_{c \in C} \text{MaxSpec}(A_c), \quad L_{(c,m)} = A_c/\mathfrak{m}.$$

The semisimple decomposition gives an algebra equivalence

$$\Psi : \prod_{c \in C} A_c \simeq \prod_{d \in D} L_d.$$

Choose nonzero scales and exponent maps on every L_d .

Corollary 31.3 (Faithful refined packet signature). *The map*

$$x \mapsto (\mathcal{C}_{\pi_d,e_d}(\Psi(x)_d))_{d \in D}$$

is injective. It is covariant under a reindexing $D \simeq D'$ and a family of field equivalences satisfying the scale and exponent hypotheses of Theorem 31.2.

Proof. Each factor coordinate is injective by Theorem 31.1; equality in the dependent product is componentwise. Compose with the injective algebra equivalence Ψ . Covariance is the factorwise theorem after reindexing. $\square$

If the target uniformizer is chosen independently, a unit twist must be recorded. With $\eta_d = \phi_d(\pi_d)/\pi'_{\sigma(d)}$, the unit coordinate transforms as

$$u'_{\sigma(d)}(\phi_d(x)) = \phi_d(u_d(x))\eta_d^{e_d(x)}.$$

Raw equality of units is therefore too strong unless $\eta_d = 1$.

The source currently transports less information. IUT III, Theorem 3.11(ii)(a), supplies chosen-branch poly-isomorphisms and compatible log-volumes; Proposition 3.9 supplies the relevant permutation invariance. The indeterminacy (Ind3) is described by inclusions or surjections, so its faithful abstraction is a reachability relation followed by image containment. These statements do not by themselves produce the componentwise field equivalences and valuation covariance required above. The zero-labelled singleton core from IUT I, Corollary 6.10(iii), survives as an object-level remnant [63], but it does not select a complete unit coordinate.

At the integral level, let R be a discrete valuation ring with fraction field F , let each K_i/F be finite separable, and set $T = \bigotimes_R \mathcal{O}_{K_i}$ and $B = \prod_m \mathcal{O}_{L_m}$ inside $\bigotimes_F K_i \simeq \prod_m L_m$, then

$$T \subseteq B, \quad B/T \text{ is finitely generated torsion over } R,$$

Equality can fail in a ramified quadratic example; the resulting Fitting ideal records an index correction. Relating that ideal to a different or conductor requires an additional theorem.

The counterexample audit fixes the scope of the route. The decomposition $E \otimes_F E \simeq E \times E$ refutes one-field-per-tuple indexing. A determinant-one shear preserves a lattice and Haar volume while changing a component valuation, so volume compatibility alone does not imply signature covariance. Equal determinants do not reconstruct rank-two lattices, and a proper sublattice with surjective coordinate projections need not be a product of independent exponent enlargements. These examples reject only the weakened interfaces just named. The current LANA strand remains a specification rather than a construction of the multiradial output [59, 60]; no full-premise counterexample to the source-level refined covariance target is known, so the IUT route remains active.

32. AUDIT OF THE CURRENT CONCRETE IUT IMPLICATION

After the preceding packet work was completed, the public LANA IUT repository added a conditional implication strand. We audited commit [6e963070](#). Its main declaration has the schematic type

$$(299) \quad \begin{aligned} &P(T) \rightarrow (\forall K, d, \text{C}_{22}(T, K, d)) \rightarrow \\ &(\forall K, d, \text{E}(K, d)) \rightarrow \text{Cheb} \rightarrow \text{PNT} \rightarrow \\ &(\forall D, LT, TL, QI, \text{C}_{312}(D, LT, TL, QI)) \rightarrow \\ &\text{ABC}(T), \end{aligned}$$

Here $P(T)$ abbreviates `T.ProofPackage`, $\text{C}_{22}(T, K, d)$ abbreviates `Corollary22Inputs(T, K, d)`, and `Cheb`, `PNT` denote the two stated prime bounds. The remaining abbreviations are

$$\begin{aligned} \text{E}(K, d) &:= \text{ConcreteThetaDataExistence}(I(K, d)), \\ \text{C}_{312}(D, LT, TL, QI) &:= \text{Corollary312Variant}(\text{concreteVariantData}(D, LT, TL, QI)). \end{aligned}$$

Here $LT : \text{LocalTheory}(D.K_t)$ and $TL : \text{ThetaLocalData}(D, LT)$. Moreover,

$$\text{ABC}(T) := T.\text{StatementI}.$$

Thus (299) is a proved conditional implication, not a construction of its inputs and not a closed proof of the integer statement (1).

The proof graph is nevertheless useful. Concrete theta-data existence is converted to theta-data existence restricted by a predicate; its invariants, certificate and local estimate enter Theorem 1.10; the assumed Corollary 3.12 variant and the prime bounds yield Corollary 2.2;

finite exceptional sets are absorbed to prove *T.StatementII*; finally the *genl* proof package proves

$$T.StatementII \implies T.StatementI.$$

Every source assumption remains visible in the type.

The new concrete interface has, however, a complete elementary counterexample. Its local log-volume field requires, for every finite label type ι , every prime p , every place tuple c , and every set U in the tensor packet,

$$(300) \quad V((x \mapsto px)^{-1}U) = V(U) + \log p, \quad V : \mathcal{P}(A) \longrightarrow \mathbb{R}.$$

Theorem 32.1 (Empty-set obstruction to the current *LocalTheory*). *At the audited commit, *LocalTheory*(K) is empty for every number field K .*

Proof. Take the empty finite label type, its unique place tuple, $p = 2$, and $U = \emptyset$. The preimage of the empty set under every function is empty, so (300) becomes

$$V(\emptyset) = V(\emptyset) + \log 2.$$

As $V(\emptyset)$ is a finite real number, subtraction gives $\log 2 = 0$, contradicting $\log 2 > 0$. $\square$

This counterexample satisfies every premise of the displayed volume law. It therefore rejects that exact total-real-valued specification. It does not reject the Haar scaling identity on nonempty measurable sets of finite positive measure. The upstream change that adds nonemptiness to the packet-product combination law does not repair (300), where U remains unrestricted. Each non-vacuous witness requested by *ConcreteThetaDataExistence* contains a term of *LocalTheory*; consequently the advertised concrete branch cannot currently be instantiated. The definition is implication-valued and may still hold vacuously in a degenerate model with no qualifying input point, which is not the intended theta-data construction.

There is a second independent boundary. The repository target *ABCConjecture* is the explicit integer inequality (1), whereas *T.StatementI* is an abstract bounded-discrepancy inequality for hyperbolic curves. No checked declaration in the audited upstream snapshot proves

$$(301) \quad T.StatementI \implies ABCConjecture.$$

A proof of (301) must construct the intended height theory, send every primitive positive triple to a degree-one tripod point, compare the canonical height with $\log \max(a, b, c)$, compare log-different plus log-conductor with $\log \text{rad}(abc)$, and unpack bounded discrepancy into one uniform additive constant.

We therefore retain the reproducible pin `ddaddc274281adb5674d647e24fa478745ac6d40`. A direct update would also change packet summands from fields to commutative rings, integral subrings to sets, and nonzero hull scales to units. The refined-factor signature then needs an explicit equivalence of each tensor packet with a finite product of fields. A successful compilation after an immediate pin change would not cure the empty-set contradiction.

The constructive continuation is precise:

- (1) restrict logarithmic volume to a proof-carrying domain of measurable finite positive-volume regions, and prove closure under the scaling preimages;
- (2) construct an inhabited local theory on that corrected domain;
- (3) provide the finite-product-of-fields decomposition needed by the refined zero-aware coordinates;
- (4) construct the compatible 2ℓ -th Tate-parameter roots and the remaining concrete theta data;
- (5) prove the actual Corollary 3.12 statement; and
- (6) prove the integer bridge (301).

Difficulty of these steps does not eliminate the corrected IUT route.

The companion Lean audit proves the universal empty-set theorem for any real-valued set-volume with the unrestricted prime-preimage law, specializes the already pinned public no-go, and introduces the honest proposition

$$\text{StatementIToIntegerABCBridge}(T) := T.StatementI \rightarrow ABCConjecture.$$

It proves only that a supplied term of this bridge consumes a supplied term of $T.$ **StatementI**. No bridge or IUT premise is populated by a hidden axiom.

33. AN ADMISSIBLE LOG-VOLUME MODEL AND AN EXPLICIT INTEGER BRIDGE

The empty-set obstruction in Section 32 identifies a defect in one total real-valued interface, rather than in Haar scaling. We give an explicit extensional repair and then state the exact comparison data that transfer the abstract general-position inequality to the repository's integer target.

Fix $s : X \rightarrow X$ and $\delta \in \mathbb{R}$. An admissible preimage log-volume consists of an inhabited region type $\mathcal{R}$, an injective carrier map $C : \mathcal{R} \rightarrow \mathcal{P}(X)$ with $C(U) \neq \emptyset$, a map $P : \mathcal{R} \rightarrow \mathcal{R}$, and $V : \mathcal{R} \rightarrow \mathbb{R}$ such that

$$(302) \quad C(PU) = s^{-1}C(U), \quad V(PU) = V(U) + \delta.$$

Injectivity prevents assigning several volumes to the same underlying set; the typed operation P supplies the required closure proof.

Theorem 33.1 (Valuation-ball model). *For every rational prime p , the interface (302) with $X = \mathbb{Q}_p$, $s(x) = px$, and $\delta = \log p$ is inhabited.*

Proof. For $k \in \mathbb{Z}$, let

$$B_k = \{x \in \mathbb{Q}_p : |x|_p \leq p^{-k}\}, \quad P(k) = k - 1, \quad V(k) = -k \log p.$$

Every ball contains zero. Since $|p|_p = p^{-1}$, $px \in B_k$ if and only if $x \in B_{k-1}$, proving carrier closure. The inclusions among these balls recover the integer index, so the carrier map is injective. Finally $V(k - 1) = -k \log p + \log p = V(k) + \log p$. $\square$

The shift law also imposes an exact consistency condition. Induction gives

$$(303) \quad V(P^n U) = V(U) + n\delta.$$

If $\delta \neq 0$, no positive iterate can fix U ; if $P^m U = P^n U$, applying V and cancelling δ gives $m = n$. Every orbit is therefore infinite. This supplies full-premise contradictions to proposed repairs with a fixed region, a positive-period cycle, or only finitely many regions. The valuation lattice passes all three tests. It is a model only of this local scaling layer, not a construction of LANA's tensor packets or of IUT.

We next make the abstract-to-integer bridge non-tautological. Let T be a height theory, let P_{abc} be the set of positive primitive triples, and put

$$H(a, b, c) = \log \max(a, b, c), \quad R(a, b, c) = \log \text{rad}(abc).$$

A uniform tripod comparison consists of an encoding $\iota : P_{abc} \rightarrow T.\text{Pt}(T.\text{tripod})$, membership of every $\iota(t)$ in the degree-at-most-one set, and constants $A_H, A_R \in \mathbb{R}$ such that

$$(304) \quad H(t) \leq h_{\text{can}}(\iota(t)) + A_H,$$

$$(305) \quad \log \text{Diff}(\iota(t)) + \log \text{Cond}(\iota(t)) \leq R(t) + A_R.$$

Theorem 33.2 (Bounded-discrepancy transfer). *If $T.$ **StatementI** and a uniform tripod comparison are supplied, then the repository's exact **ABCConjecture** holds.*

Proof. Fix $\epsilon > 0$. Apply $T.$ **StatementI** to the tripod, degree bound one, and ϵ . It supplies C_ϵ such that every encoded point satisfies

$$h_{\text{can}}(\iota(t)) \leq (1 + \epsilon)(\log \text{Diff}(\iota(t)) + \log \text{Cond}(\iota(t))) + C_\epsilon.$$

Since $1 + \epsilon > 0$, the two comparisons give

$$H(t) \leq (1 + \epsilon)R(t) + C_\epsilon + A_H + (1 + \epsilon)A_R.$$

The final three terms form one constant independent of t , which is exactly the required quantifier order. $\square$

Each field is logically necessary at this level. For example, on $\mathbb{N}$, $h = k = R = 0$ and $H(n) = n$ satisfy the source inequality and radical comparison but refute every uniform target without the height comparison. Taking $h = k = H = n$ and $R = 0$ does the same without the radical comparison. Restricting the source inequality to the empty set shows why degree membership is necessary, and replacing A_H by the point-dependent error $A_H(n) = n$ destroys uniformity. These are full-premise countermodels within this bare sequence-transfer schema to the corresponding weakened bridges. They are not *abc* counterexamples and do not retire the IUT/same-pilot research route. Constructing the intended degree-one encoding and proving (304)–(305) remain open arithmetic work.

34. THE EXACT RATIONAL-TRIPOD SHADOW AND ITS SEMANTIC BOUNDARY

The comparison in Section 33 has an exact arithmetic realization on the open rational tripod

$$U(\mathbb{Q}) = \{x \in \mathbb{Q} : 0 < x < 1\}.$$

For a positive primitive triple $t = (a, b, c)$, put $\lambda(t) = a/c$. Conversely, if $x = m/n$ is reduced in $U(\mathbb{Q})$, set $t(x) = (m, n - m, n)$. These maps are inverse. With H_{tr} the rational Weil height and N_{tr} the truncated counting function for the three marked points $0, 1, \infty$, one has

$$(306) \quad H_{\text{tr}}(\lambda(t)) = \log c = \log \max(a, b, c), \quad N_{\text{tr}}(\lambda(t)) = \log \text{rad}(abc).$$

The first identity follows because a/c is reduced and $0 < a < c$; the second follows because the supports of a/c and $1 - a/c = b/c$ have union exactly the prime support of abc .

Define a one-curve shadow height theory T_{rat} . Its point set is $U(\mathbb{Q})$; its degree-at-most-zero locus is empty and every point has exact degree one; and

$$h_{\text{can}} = H_{\text{tr}}, \quad \log \text{Diff} = 0, \quad \log \text{Cond} = N_{\text{tr}}.$$

The remaining structural fields have elementary witnesses. The identity map on $U(\mathbb{Q})$ therefore gives a degree-one comparison, while (306) gives both normalization errors equal to zero.

Theorem 34.1 (Exact shadow boundary). *Statement I for T_{rat} is equivalent to the standard integer *abc* conjecture:*

$$T_{\text{rat}}.\text{Statement I} \iff \text{ABCConjecture}.$$

Proof. The forward implication is the bounded-discrepancy transfer of Section 33, with both errors zero. For the reverse implication, integer *abc* transfers through the inverse reduced triple $t(x)$ to the uniform rational-tripod inequality. Since the shadow has one curve, the degree-zero case is empty and every positive degree bound contains all of $U(\mathbb{Q})$, that inequality is exactly its Statement I. $\square$

For an arbitrary height theory T , an open-rational tripod comparison is a map from $U(\mathbb{Q})$ to the degree-at-most-one points of the distinguished tripod, together with uniform height and radical errors. Composition with $t \mapsto a/c$ turns this into the integer comparison package. Composition with $x \mapsto t(x)$ gives the converse, with the same constants. Thus the rational and integer semantic comparison packages have exactly the same logical strength.

Three full-premise models delimit what the current bare structural fields can prove. First, a one-point theory with every degree locus empty satisfies Statement I vacuously but admits no comparison from the nonempty set $U(\mathbb{Q})$. Second, on the actual rational points take

$$h_{\text{can}} = 0, \quad \log \text{Diff} = 0, \quad \log \text{Cond} = N_{\text{tr}}.$$

Statement I, degree membership, and the radical identity all hold, but no uniform height error exists because $H_{\text{tr}}(1/(n+1)) = \log(n+1)$ is unbounded. Third, take

$$h_{\text{can}} = H_{\text{tr}}, \quad \log \text{Diff} = e^{2H_{\text{tr}}}, \quad \log \text{Cond} = N_{\text{tr}}.$$

Statement I follows from $e^{2H} \geq 1 + 2H \geq H$ for $H \geq 0$, and the height comparison is exact, but a uniform radical error would bound $e^{2H_{\text{tr}}}$, again impossible.

These are counterexamples to automatic degree realization and to automatic uniform normalization from the present abstract fields. They are not counterexamples to the intended

arithmetic height theory, IUT, Statement I for that theory, or *abc*. The positive route remains the source-level construction of the rational point, its degree-one membership, and the two triple-independent normalization constants in the intended objects. The Lean companion constructs the shadow, proves Theorem 34.1, and verifies all three pressure models without placeholders or new axioms.

35. A SOURCE-FAITHFUL RATIONAL DEGREE-ONE TRIPOD REALIZATION

The shadow construction of Section 34 can be compared directly with the source definitions in Mochizuki's *Arithmetic Elliptic Curves in General Position*. Put

$$P = \mathbb{P}_{\mathbb{Q}}^1, \quad C = \{0, 1, \infty\}, \quad U = P \setminus C, \quad U_{(0,1)}(\mathbb{Q}) = \{x \in \mathbb{Q} : 0 < x < 1\}.$$

The map $j(x) = [x : 1]$ is an injective map to $U(\mathbb{Q})$. Its points have minimal field of definition $\mathbb{Q}$, and hence exact degree one. The recurrence in the abstract LANA height theory already forces

$$(307) \quad \text{ptLE}(X, 1) = \text{ptEQ}(X, 1),$$

because the degree-at-most-zero locus is empty.

Let h be the normalized logarithmic Weil height and let n be the truncated count for the three marked points. For the genuine arithmetic objects, one has

$$(308) \quad \text{ht}_{\omega_P(C)} \circ j \approx h, \quad \log\text{-diff}_P \circ j = 0, \quad \log\text{-cond}_C \circ j \approx n.$$

Indeed, $\omega_P(C) \cong \mathcal{O}_{\mathbb{P}^1}(1)$, and invariance of the height BD class under isomorphism gives the first comparison. The different ideal of $\mathbb{Q}/\mathbb{Q}$ is the unit ideal, giving the middle equality. Finally, if $x = m/r$ is reduced with $0 < m < r$, pulling $0, 1, \infty$ back to $\text{Spec } \mathbb{Z}$ gives the divisors of $m, r - m, r$. Reduction of the pullback divisor therefore gives

$$\log\text{-cond}_C(j(x)) = \sum_{p|m(r-m)r} \log p = n(x)$$

for the standard model. Changing the proper flat model changes this function only by bounded discrepancy.

For an abstract height theory T , call a map $j : U_{(0,1)}(\mathbb{Q}) \rightarrow T.\text{Pt}(T.\text{tripod})$ a source-faithful degree-one realization when it has exact-degree-one image and satisfies the three relations in (308) after replacing the arithmetic functions by the corresponding fields of T . Write H_T, D_T, N_T for the three pulled-back functions. Define the restricted Statement I by

$$(309) \quad \forall \epsilon > 0 \exists C_\epsilon \forall x, \quad H_T(x) \leq (1 + \epsilon)(D_T(x) + N_T(x)) + C_\epsilon.$$

We denote this proposition by $\text{RStmtI}(T, j)$.

Theorem 35.1 (Source-faithful degree-one transport). *For every source-faithful degree-one realization,*

$$\text{RStmtI}(T, j) \iff \text{ABCConjecture}.$$

Consequently, full T .StatementI together with such a realization implies the integer abc conjecture.

Proof. Choose uniform constants A_+, A_-, B_+, B_- such that

$$H_T \leq h + A_+, \quad h \leq H_T + A_-, \quad N_T \leq n + B_+, \quad n \leq N_T + B_-.$$

If (309) holds, then, using $D_T = 0$,

$$h(x) \leq (1 + \epsilon)n(x) + C_\epsilon + A_- + (1 + \epsilon)B_+.$$

This is the uniform rational-tripod formulation of integer *abc*. Conversely, that formulation gives

$$H_T(x) \leq (1 + \epsilon)N_T(x) + C_\epsilon + A_+ + (1 + \epsilon)B_-,$$

which is (309). Finally, full Statement I restricts along j because (307) puts every image point in its degree-one domain. $\square$

This theorem closes the elementary source normalization but not the IUT argument. At the pinned versions, `Gen1.HeightTheory` is only an abstract record, no genuine arithmetic instance is constructed, and the pinned IUT repository has no connection to Statement I. A later public IUT snapshot has only a conditional connection that still takes the height theory, its proof package, and the IUT and analytic inputs as premises.

The distinction cannot be erased by adding the abstract proof package. The degree-empty, height-zero, and different-inflated pressure models all admit full `ProofPackage` instances. They show respectively that this package does not force rational degree-one points, the canonical height normalization, or the rational different. A separate conductor-inflated model preserves exact points, degree, height, and zero different while destroying every uniform upper conductor comparison. These are full-premise countermodels to their precisely stated weakened extraction claims; they do not contradict the genuine computations above, IUT, or *abc*. For clarity, the first three models retain the abstract proof package. The conductor-inflated model retains the height theory, Statement I, and the stated point, degree, height, and different data, but no proof-package instance is claimed for it.

The Lean companion proves the recurrence (307), formalizes the abstract transport theorem, constructs the rational-shadow consistency witness, and checks the pressure models without placeholders or new axioms. Constructing the genuine arithmetic height-theory instance and proving its Statement I remain active upstream tasks.

36. ADMISSIBLE SCALING AT THE CURRENT IUT INTERFACE AND AN EXACT ORDER INDEX

We now compare the preceding source-faithful bridge with Project LANA commit `c65b28c` [61]. This snapshot contains a genuine positive advance: `Iut/Concrete/Existence.lean` constructs the previously abstract `ConcreteThetaDataExistence` datum from an elliptic curve and its arithmetic, Tate, representation-theoretic, and anabelian inputs. The new theorem `CurveInputs.concreteThetaDataExistence` therefore removes one assembly gap. The end-point `cor312Variant_implies_abc_curves`, however, remains conditional. Its hypotheses still include

- the height-theory proof package `Gen1.HeightTheory.ProofPackage` and the curve inputs;
- anabelian existence, together with providers for local theory, theta-local data, and tower arithmetic; and
- the analytic prime bounds and the concrete Corollary 3.12 comparison.

It is therefore not an unconditional proof of Statement I or of integer *abc*.

The same snapshot retains a logically impossible quantifier in its local log-volume interface. Abstract the issue as follows. Let $s : X \rightarrow X$, let $V : \mathcal{P}(X) \rightarrow \mathbb{R}$, and fix $\delta \in \mathbb{R}$.

Theorem 36.1 (Total-set and nonempty-set obstructions). *If*

$$(310) \quad V(s^{-1}U) = V(U) + \delta \quad (U \subseteq X),$$

then $\delta = 0$. If X is nonempty, the same conclusion holds when (310) is required only for nonempty U . More generally, the conclusion holds on any asserted domain containing a set U with $s^{-1}U = U$.

Proof. For the total law take $U = \emptyset$; its preimage is again empty and additive cancellation gives $\delta = 0$. Under the nonempty restriction take $U = X$, whose preimage is X . The last assertion is the same cancellation with the stated fixed set. $\square$

In the current `LocalTheory.componentVol.prime_preimage` field one has $\delta = \log p$ for a prime p . Since $p \geq 2$, this shift is positive, so no inhabitant of the literal all-set structure can exist. Merely appending `U.Nonempty` would move the contradiction from the empty set to the whole set and would not repair it.

The source-level repair used in our audit makes the domain explicit. It adds a predicate $\mathcal{A}$ of admissible finite, nonzero-volume regions and requires

$$(311) \quad U \in \mathcal{A} \implies U \neq \emptyset, \quad U \in \mathcal{A} \implies s^{-1}U \in \mathcal{A}, \quad U \in \mathcal{A} \implies V(s^{-1}U) = V(U) + \log p.$$

The predicate is transported from `LocalTheory` to `LogVolumeData` and through `Concrete.Container.vol`. No numeric equality is imposed on sets outside its domain. This is the minimal quantifier correction suggested by the finite, nonzero-volume regime of IUT III, Proposition 3.9 and Remark 3.9.5 [62].

Proposition 36.2 (Admissible orbit law). *Assume (311). For $U \in \mathcal{A}$ and $r \geq 0$,*

$$V(s^{-r}U) = V(U) + r \log p.$$

If $\log p \neq 0$, the admissible preimage orbit of U is injective and in particular is not periodic.

Proof. Induction on r proves the formula: closure under preimage supplies the admissibility hypothesis needed at each successor step. If $s^{-r}U = s^{-t}U$, the formula gives $(r - t) \log p = 0$, whence $r = t$. $\square$

The patched versions of the three affected source files compile together with both upstream targets `Iut` and `Iut4Sec1`; the build finishes all 8767 tasks. This verifies that the repaired dependency shape is compatible with the current source tree. It does not construct the local Haar volume or prove the comparison theorem.

There is a second accounting seam at the passage from an integral tensor order to the product of maximal factor orders. For $n \in \mathbb{N}$ put

$$(312) \quad \mathcal{O}_n = \{(a, b) \in \mathbb{Z}^2 : n \mid a - b\}, \quad \delta_n(a, b) = a - b \pmod{n}.$$

Theorem 36.3 (Exact congruence-order quotient). *The map $\delta_n : \mathbb{Z}^2 \rightarrow \mathbb{Z}/n\mathbb{Z}$ is a surjective additive homomorphism with kernel $\mathcal{O}_n$. Consequently*

$$\mathbb{Z}^2/\mathcal{O}_n \simeq \mathbb{Z}/n\mathbb{Z}, \quad [\mathbb{Z}^2 : \mathcal{O}_n] = n.$$

For $n > 1$, $\mathcal{O}_n$ is a proper finite-index subring of $\mathbb{Z}^2$, although both coordinate projections $\mathcal{O}_n \rightarrow \mathbb{Z}$ are surjective.

Proof. Additivity is immediate, and every residue class is the image of $(a, 0)$. Moreover $\delta_n(a, b) = 0$ exactly when $n \mid a - b$, so the first isomorphism theorem gives the quotient. The kernel is a subring: it contains $(0, 0)$ and $(1, 1)$, is additively closed, and for two of its elements

$$ac - bd = a(c - d) + d(a - b),$$

so it is closed under componentwise multiplication. Its quotient cardinality is n . If $n > 1$, the point $(1, 0)$ is absent, proving properness; every (z, z) is present, proving both projection statements. $\square$

The theorem includes $n = 0$ under the `Nat.card` index convention: both the quotient and $\mathbb{Z}/0\mathbb{Z} \cong \mathbb{Z}$ are infinite and their recorded natural cardinality is zero. For positive m, n , exactness also gives

$$(313) \quad [\mathbb{Z}^2 : \mathcal{O}_{mn}] = [\mathbb{Z}^2 : \mathcal{O}_m][\mathbb{Z}^2 : \mathcal{O}_n], \quad \log[\mathbb{Z}^2 : \mathcal{O}_{mn}] = \log[\mathbb{Z}^2 : \mathcal{O}_m] + \log[\mathbb{Z}^2 : \mathcal{O}_n].$$

Thus a finite integral-order defect contributes an additive logarithmic term. Identifying the actual IUT tensor-order quotient with a local different or conductor exponent is a further arithmetic theorem, not a consequence of this model.

The empty and whole sets are full-premise counterexamples to the two stated overbroad volume laws. The orders $\mathcal{O}_n$, $n > 1$, are full-premise counterexamples to the assertion that surjective component projections force the product order. These witnesses retire precisely those shortcuts. They do not satisfy the premises of an admissible, index-aware same-pilot theorem, so that corrected route remains active. Its unresolved obligations are to construct the actual admissible Haar regions, calculate the tensor-order indices at every factor and place, prove covariance through the horizontal link and `Ind3`, and establish the concrete Corollary 3.12 comparison.

The Lean companion proves the two no-go theorems, identifies the exact kernel and quotient in (312), proves the index and its logarithmic multiplication law, and verifies properness together with both surjective projections. The source ledger separately freezes the upstream commit, exact

patch replay, and complete build log. No same-pilot theorem, Statement I, or *abc* consequence is introduced as a formal assumption.

37. ACTUAL HAAR ADMISSIBILITY AND PACKET NORMALIZATION

The preceding source audit leaves a concrete local question: can the repaired admissible-domain interface be realized by an actual measure, with the sign and coefficient required by the source? Proposition 3.9(i) of [62] takes the nonarchimedean log-volume domain to be the nonempty compact open subsets, normalizes the integral structure to log-volume zero, and uses packet normalization so that multiplication by the local rational prime subtracts its logarithm. We now construct the underlying Haar model and separate raw local volume from that packet normalization.

Let K be a nontrivially normed field with proper metric, let μ be a regular additive Haar measure, and let $a \in K^\times$. The local fields used below have this properness property. Write $m_a(x) = ax$ and let $\Delta_\mu : K^\times \rightarrow \mathbb{R}_{>0}$ be the distributive Haar character, so that

$$\mu(aU) = \Delta_\mu(a)\mu(U).$$

Define the finite-positive domain

$$\mathcal{F}_\mu = \{U : U \text{ is measurable and } 0 < \mu(U) < \infty\}, \quad L_\mu(U) = \log \mu(U).$$

The conversion from an extended nonnegative measure to a real logarithm is made only after both inequalities in the definition of $\mathcal{F}_\mu$ have been proved.

Theorem 37.1 (Actual scalar-preimage law). *For every $U \in \mathcal{F}_\mu$,*

$$(37.1.1) \quad m_a^{-1}(U) = a^{-1}U \in \mathcal{F}_\mu, \quad L_\mu(m_a^{-1}U) = L_\mu(U) + \log \Delta_\mu(a^{-1}).$$

If U is also compact and open, then $m_a^{-1}(U)$ is compact and open.

Proof. The carrier identity follows in both directions from $ax \in U \iff x = a^{-1}(ax) \in a^{-1}U$. Multiplication by a is a homeomorphism, so its inverse image preserves measurability, openness, and compactness. The Haar change-of-variables identity gives

$$\mu(m_a^{-1}U) = \mu(a^{-1}U) = \Delta_\mu(a^{-1})\mu(U).$$

The character is finite and strictly positive. The right side is therefore finite and positive, and taking real logarithms proves the formula. $\square$

Suppose now that the valuation ring $\mathcal{O}_K$ is a discrete valuation ring, the residue field has cardinality q , and $\mu(\mathcal{O}_K) = 1$. Let π be a uniformizer.

Theorem 37.2 (Uniformizer orbit and its obstruction). *Put $B_n = \pi^{-n}\mathcal{O}_K$ for $n \geq 0$. Every B_n is a nonempty compact open member of $\mathcal{F}_\mu$, and*

$$(37.2.1) \quad m_\pi^{-1}(B_n) = B_{n+1}, \quad L_\mu(B_n) = n \log q.$$

Consequently $n \mapsto B_n$ is injective, and no single $V \in \mathcal{F}_\mu$ contains every B_n .

Proof. The q additive cosets of $\pi\mathcal{O}_K$ partition $\mathcal{O}_K$. Translation invariance and $\mu(\mathcal{O}_K) = 1$ imply $\mu(\pi\mathcal{O}_K) = q^{-1}$, hence $\Delta_\mu(\pi) = q^{-1}$. Theorem 37.1 therefore adds $\log q$ under a uniformizer preimage. The unit ball is a nonempty open ball; properness makes it compact, and scalar multiplication preserves these properties. Induction gives the displayed identities. Since $q > 1$, the log-volumes are strictly increasing, proving injectivity. If every B_n lay in one finite-positive V , monotonicity would give $n \log q \leq L_\mu(V)$ for every n , contrary to the Archimedean property. $\square$

The last conclusion rules out only a bounded, finite, or periodic implementation of the complete preimage orbit. It does not conflict with the least-hull construction for one fixed relatively compact region in Remark 3.9.5(i) of [62], and it does not rule out the Haar/IUT route.

The rational-prime coefficient requires one further normalization. Let $K/\mathbb{Q}_p$ have ramification index e and residue degree f . Choose a unit u with

$$p = u\pi^e, \quad q = p^f.$$

Multiplication by u preserves Haar measure. Thus raw Haar log-volume changes under rational-prime preimage by

$$(37.2.2) \quad \log \Delta_\mu(p^{-1}) = e \log q = ef \log p.$$

Theorem 37.3 (Local-degree and packet normalization). *Define*

$$L_\mu^{\text{pkt}}(U) = \frac{L_\mu(U)}{ef}.$$

Then

$$(37.3.1) \quad L_\mu^{\text{pkt}}(p^{-1}U) = L_\mu^{\text{pkt}}(U) + \log p.$$

For finitely many components, if $\sum_i w_i = 1$, the weighted packet volume $\sum_i w_i L_{\mu_i}^{\text{pkt}}(U_i)$ also changes by exactly $\log p$ under simultaneous rational-prime preimage.

Proof. Divide the raw identity $ef \log p$ by the positive local degree ef . Every normalized component then has the same increment $\log p$, so the weighted increment is $(\sum_i w_i) \log p = \log p$. $\square$

This two-stage calculation explains which coefficient has to be realized in the actual tensor components. Ordinary weight normalization cannot perform the first stage.

Proposition 37.4 (Full-premise raw-volume counterexample). *The condition $\sum_i w_i = 1$ alone does not turn raw Haar log-volume into a $\log p$ rational-prime shift.*

Proof. Let $K/\mathbb{Q}_p$ be the unramified quadratic extension, take the one-element packet, and give it weight one. Then $e = 1$, $f = 2$, $q = p^2$, and all local field, Haar, positivity, and weight premises hold. The raw shift is $\log(p^2) = 2 \log p \neq \log p$. Division by $ef = 2$ restores the conclusion of Theorem 37.3. $\square$

Proposition 37.4 retires exactly the weaker raw-volume claim. The corrected route remains active. It must still construct $p = u\pi^e$ and $q = p^f$ uniformly in every actual tensor component, identify the component functional with $L_\mu/(ef)$, transport the compact-open admissible class through the direct-sum and least-hull operations, and then prove the horizontal same-pilot and Ind3 comparison. None of these global requirements follows from a local change-of-variables formula.

38. REFINED TENSOR HAAR VOLUME AND THE POINTED SAME-PILOT GATE

Author: ChatGPT. This section isolates an unconditional local theorem that is useful at the present Project-LANA/IUT interface. It does not assert IUT III, Corollary 3.12, or the *abc* conjecture. In particular, the finite-etale factorization, Haar-positivity, Ind3-openness, and pointed-horizontal identification hypotheses below are logically distinct.

Primary-source boundary. The terminology follows Mochizuki's May 2020 author version of *Inter-universal Teichmüller Theory III*. The exact locators are Proposition 3.9(i), Remark 3.9.5(i), Theorem 3.11(i)–(ii), and Corollary 3.12, especially proof Steps (xi-b)–(xi-f). The identification still missing after the three indeterminacies is also recorded in Sections 6–7 of the July 2026 LANA formalization report. The frozen Project LANA source is commit

c65b28c9f9631635e742294c3a5df15759e7c74c.

No source claim is imported as a premise unless it is displayed below.

38.1. Primitive factors and the prime Jacobian. Fix a rational prime p . For a place tuple c , let

$$A_c = \bigotimes_{j \in S, \mathbb{Q}_p} K_{c_j}.$$

Since the local extensions are finite and separable, A_c is a finite-etale $\mathbb{Q}_p$ -algebra. Thus

$$(314) \quad A_c \simeq \prod_{d \in D_c} L_{c,d}, \quad N_c := \dim_{\mathbb{Q}_p} A_c = \sum_{d \in D_c} [L_{c,d} : \mathbb{Q}_p].$$

For a factor $L = L_{c,d}$, write $e = e(L/\mathbb{Q}_p)$ and $f = f(L/\mathbb{Q}_p)$.

Primitive-factor theorem. For every uniformizer $\pi \in \mathcal{O}_L$, there is $u \in \mathcal{O}_L^\times$ such that

$$(315) \quad p = u\pi^e, \quad |k_L| = p^f, \quad [L : \mathbb{Q}_p] = ef.$$

Indeed, unique factorization in the discrete valuation ring gives $p = u\pi^m$. Applying the integer-valued valuation normalized by $v_L(\pi) = 1$ yields $m = v_L(p) = e$. The residue field is an f -dimensional vector space over $\mathbb{F}_p$, and the fundamental equality for finite extensions of complete discretely valued fields gives $[L : \mathbb{Q}_p] = ef$. Consequently,

$$(316) \quad N_c = \sum_{d \in D_c} e_{c,d} f_{c,d} > 0.$$

The last identity follows by taking dimensions in (314); it is not an assumption about an unsplit tensor product.

Normalize additive Haar measure $\mu_{c,d}$ by $\mu_{c,d}(\mathcal{O}_{c,d}) = 1$. For nonempty compact-open $U_{c,d} \subseteq L_{c,d}$, set

$$U_c = \prod_d U_{c,d}, \quad \mu_c = \prod_d \mu_{c,d}, \quad \lambda_c(U_c) = \frac{1}{N_c} \log \mu_c(U_c).$$

All measures in this display are finite and strictly positive. The additive Haar change-of-variables formula and (315) give

$$\mu_{c,d}(p^{-1}U_{c,d}) = p^{e_{c,d}f_{c,d}} \mu_{c,d}(U_{c,d}).$$

Taking products and using (316) proves

$$(317) \quad \lambda_c(p^{-1}U_c) = \lambda_c(U_c) + \log p.$$

Moreover, with $n_{c,d} = [L_{c,d} : \mathbb{Q}_p]$ and $\rho_{c,d} = n_{c,d}/N_c$, one has

$$(318) \quad \sum_d \rho_{c,d} = 1, \quad \lambda_c(U_c) = \sum_d \rho_{c,d} \frac{\log \mu_{c,d}(U_{c,d})}{n_{c,d}}.$$

Thus the normalization is by total local degree, rather than by the number of primitive factors.

If a tuple has weight W_c , its primitive factors must receive

$$(319) \quad \widetilde{W}_{c,d} = W_c \rho_{c,d}.$$

Then $\sum_d \widetilde{W}_{c,d} = W_c$, so normalized tuple weights remain normalized after primitive refinement.

38.2. Finite-positive orbit regions. Let X be a finite product of locally compact primitive fields with product Haar measure μ . Let $U \subseteq X$ be nonempty and open, and let $\{\phi_a : X \simeq X\}_{a \in A}$ be homeomorphisms, one of which is the identity. Suppose that every $\phi_a(U)$ lies in a common set E with compact closure. Then

$$(320) \quad \Theta(U) := \bigcup_{a \in A} \phi_a(U)$$

is nonempty and open. It therefore has positive Haar measure. Its closure is a closed subset of the compact set $\overline{E}$, so it is compact; Haar measure is finite on that closure. Hence

$$(321) \quad 0 < \mu(\Theta(U)) < \infty.$$

When A is finite and U is compact-open, $\Theta(U)$ is itself compact-open. For an infinite family the stated hypotheses yield an open region with compact closure; they do not imply that the orbit union is closed.

The same argument applies to an inductively generated Ind1–Ind3 system only under the following separate premises: ordinary branches are open and one is nonempty; Ind1 and Ind2 preserve openness; every relational Ind3 target of an open reachable region is open; and all constructors preserve the common compact-envelope containment. Structural induction then proves openness and envelope containment for every reachable region, after which (321) follows. The finite-etale theorem above does not supply any of these source-level Ind hypotheses.

38.3. The pointed same-pilot implication. Let P be the actual native q -pilot region, Θ the actual union of reachable theta outputs, and H a finite-positive hull. If the object-level construction proves the pointed inclusions

$$(322) \quad P \subseteq \Theta \subseteq H,$$

then Haar monotonicity and monotonicity of the real logarithm give

$$(323) \quad \log \mu(P) \leq \log \mu(\Theta) \leq \log \mu(H).$$

Thus any independent estimate $\log \mu(H) \leq T$ implies $\log \mu(P) \leq T$. No volume-preservation assertion for the Ind maps is needed for this conditional implication. However, naming some generated branch “native” does not establish (322). The open source-level obligation is to transport the original pointed pilot through the theta link, IPL/SHE/APT, log-Kummer correction, determinant normalization, and Ind3.

38.4. Exact counterexamples and route status. Each counterexample below satisfies all premises of the deliberately weaker claim that it refutes.

- (1) A one-factor quadratic local field has factor count 1 and total degree 2. Its raw prime-preimage shift is $2 \log p$, so division by factor count does not give (317). This refutes only factor-count normalization.
- (2) For an unramified quadratic Galois extension $K/\mathbb{Q}_p$, $K \otimes_{\mathbb{Q}_p} K \simeq K \times K$. Copying a parent weight 1 to both factors gives total weight 2; the relative-degree weights $(1/2, 1/2)$ satisfy (319). This refutes only copied parent weights.
- (3) On $\mathbb{R}$ with Lebesgue Haar measure, an Ind3 relation may send $(-1, 1)$ to $\{0\}$ while both lie in the compact envelope $[-1, 1]$. The target has measure zero. This refutes the claim that envelope preservation alone yields finite-positive Ind3 outputs; it does not refute a concrete Ind3 theorem that proves openness or positivity.
- (4) In the discrete three-point group with counting Haar measure, take $P = \{0, 1\}$ and $\Theta = H = \{0\}$. All regions are compact-open and finite-positive and $\Theta \subseteq H$, but $\log \mu(P) = \log 2 > 0 = \log \mu(H)$. This refutes only an unpointed inference.

The finite-etale factor identities and total-degree Haar normalization are proved. The relative-degree weights, bounded-open orbit lemma, and conditional pointed sandwich are also proved. The identification of the concrete IUT Ind1 and Ind2 maps, openness or positive measure of the required Ind3 targets, and the pointed IPL, SHE, and APT horizontal transport remain open gaps. No counterexample satisfying all premises of those source statements is presently supplied; therefore those routes remain active. IUT III, Corollary 3.12, IUT, and the standard *abc* conjecture remain open in this work.

38.5. Formalization boundary. The companion Lean development derives the identities in (315) and (316) on the actual primitive quotients of a `TupleFiniteEtalePacket`. It also formalizes the product-Haar formulas, the refined weights, the orbit and Ind interface lemmas and the pointed sandwich. It checks coefficient-level versions of the factor-count and copied-weight obstructions and set-level versions of the envelope and unpointed obstructions. The quadratic local-field and tensor decomposition witnesses for the first two remain paper proofs. Its axiom audit reports only `propext`, `Classical.choice`, and `Quot.sound`; there are no project axioms or admitted proofs.

The following source-specific work remains on paper: identifying each IUT capsule’s completed tensor algebra with the finite-etale packet; comparing the tensor product of integral orders with the product of maximal factor orders, including finite-index/different corrections; instantiating the abstract Ind1/Ind2 homeomorphisms and common envelope; proving the required Ind3 openness or positivity; and constructing the pointed horizontal map. Finite computation audits the rational normalization formulas only and gives no evidence for these source-specific obligations or for *abc*.

39. A TWO-ARM CRT PACKET FOR THE MINIMAL AFFINE SHEAR

Let $a + b = c$ be primitive, put $R = \text{rad}(abc)$, and consider the minimal-step affine shear

$$(324) \quad U = 1 + Rh, \quad V = 1 + R(h + ck), \quad W = 1 + R(h + bk).$$

Write $E(n) = n/\text{rad}(n)$. The density analysis in Section 28 shows that every three-quarter exception in the upper-half canonical box must satisfy

$$(325) \quad Rc < 8192E(V), \quad Rc < 8192E(W).$$

The two inequalities carry more joint information than two unrelated numerical lower bounds.

Theorem 39.1 (Coprime joint carrier). *If V, W, T, K are positive integers, $(V, W) = 1$, and $T < KE(V)$ and $T < KE(W)$, then*

$$(E(V), E(W)) = 1, \quad E(V)E(W) \mid VW, \quad T^2 < K^2 E(V)E(W).$$

Proof. Each powerful part divides its parent integer. Divisor monotonicity of coprimality gives $(E(V), E(W)) = 1$, and multiplication gives the divisibility. Multiplying the two strict positive inequalities proves the last assertion. For (325), take $T = Rc$ and $K = 8192$. $\square$

The diagonal $h = k$ is automatically admissible because $(1 + Rk, k) = 1$, and its long arms are

$$V = 1 + R(c + 1)k, \quad W = 1 + R(b + 1)k.$$

Theorem 39.2 (Diagonal two-arm CRT packet). *Let D, F be positive coprime squarefree integers such that*

$$(D, R(c + 1)) = 1, \quad (F, R(b + 1)) = 1.$$

There is a unique residue class $k_0 \pmod{D^2 F^2}$ such that

$$k \equiv k_0 \pmod{D^2 F^2} \iff D^2 \mid V \text{ and } F^2 \mid W.$$

Every member is admissible and satisfies $D \mid E(V)$ and $F \mid E(W)$.

Proof. The two coefficients $R(c + 1)$ and $R(b + 1)$ are units modulo D^2 and F^2 , respectively. Each affine congruence therefore has one root, and the Chinese remainder theorem combines them into one class modulo $D^2 F^2$. If a squarefree D satisfies $D^2 \mid n$, then each prime of D occurs in n with valuation at least two and hence occurs in $E(n)$; thus $D \mid E(n)$. Apply this observation to both arms. Admissibility was noted above. $\square$

The theorem supplies simultaneous repeated-prime mass, but the following exact calculation shows that the two marginal gates do not contain enough information to force exceptionality.

Proposition 39.3 (A counted packet and a complete insufficiency witness). *For the seed $(a, b, c) = (1, 242, 243)$ one has $R = 66$ and canonical scale*

$$M = \left\lfloor \frac{243^6}{4 \cdot 66} \right\rfloor = 779890651873.$$

The diagonal progression

$$h = k = 356 + 1225t$$

has exactly 318322715 distinct points with $M/2 < k \leq M$. Every point satisfies $5 \mid E(V)$, $7 \mid E(W)$, and both inequalities in (325). At its first point,

$$h = k = 389945326231, \quad (E(U), E(V), E(W)) = (1, 5, 7),$$

the primitive output is

$$(A, B, C) = (25736391531247, 1519682447137014050, 1519708183528545297), \quad A + B = C,$$

but $\text{rad}(ABC)^4 > C^3$. Hence the point is not a three-quarter exception.

Proof. On the progression,

$$V = 25(229321 + 789096t), \quad W = 49(116521 + 400950t).$$

The canonical inequalities are equivalent to $318322715 \leq t \leq 636645429$, an interval of exactly 318322715 integers; the affine function of t is injective. The square divisibilities and the squarefree-carrier lemma give the two excess divisibilities, while $66 \cdot 243 < 8192 \cdot 5$ and $66 \cdot 243 < 8192 \cdot 7$ give the gates.

For the first point, exact factorization gives

$$\begin{aligned} U &= 17 \cdot 1513905384191, \\ V &= 5^2 \cdot 23 \cdot 37 \cdot 139267 \cdot 2119433, \\ W &= 7^2 \cdot 17431 \cdot 7322098141. \end{aligned}$$

All residual factors displayed here are prime. A shorter certificate for nonexceptionality is the squarefree divisor

$$d = 139267 \cdot 2119433 \cdot 17431 = 5145057294975341$$

of ABC : it divides $\text{rad}(ABC)$ and direct integer comparison gives $d^4 > C^3$. $\square$

Proposition 39.3 refutes only the assertion that the two necessary long-arm gates are sufficient. It does not refute their necessity or the affine route. The remaining positive theorem must produce, uniformly over every fixed subcritical seed range $R < c^\lambda$, enough canonical admissible parameters satisfying the full pointwise inequality

$$(326) \quad E(U)E(V)E(W) > \frac{RUVW}{(cW)^{3/4}} = \frac{R}{c} UV(cW)^{1/4}.$$

Within the present density architecture, one seeks constants $\eta, \kappa > 0$ and at least $\kappa R^{-2/3} c^{4+\eta}$ such parameters for every sufficiently large seed. Growing squarefree carriers and a correlated powerful-value theorem for all three arms remain distinct active ways to reach (326).

40. LARGE-MODULUS SEPARATION AND AFFINE CERTIFICATE ENTROPY

The periodic estimate for a fixed divisibility template in the minimal affine shear has an error term too large at the modulus forced by the full three-arm excess condition. This section removes that boundary error by a separation argument. The result controls one prescribed certificate, and then quantifies how many such certificates a matching lower bound would need. It does not bound a pointwise adaptive or otherwise correlated family of certificates.

40.1. A three-form separation lemma. Let $0 < B < C$ and let $d_U, d_V, d_W, L, X_U, X_V, X_W$ be positive integers. Put $D = d_U d_V d_W$, and assume that the three d 's are pairwise coprime. For two pairs $(h, k), (h', k') \in \mathbb{Z}^2$, set

$$x = h - h', \quad y = k - k', \quad H = \max\{|x|, |y|\}.$$

Suppose

$$(327) \quad d_U \mid x, \quad d_V \mid x + Cy, \quad d_W \mid x + By,$$

and impose the six cancellation conditions

$$(328) \quad \begin{aligned} (d_V, C) &= (d_W, B) = 1, & (d_U, C) &= (d_W, C - B) = 1, \\ (d_U, B) &= (d_V, C - B) = 1. \end{aligned}$$

Theorem 40.1 (Cubic-product separation). *Let $T > 0$. In addition to (327)–(328), assume*

$$(329) \quad \begin{aligned} d_U &\leq X_U, & d_V &\leq X_V, & d_W &\leq X_W, \\ D &> T > (C + 1)^2 L^3, & LX_U &< T, & LX_V &< T, & LX_W &< T. \end{aligned}$$

If $(h, k) \neq (h', k')$, then $H > L$.

Proof. Suppose first that x , $x + Cy$, and $x + By$ are all nonzero. Pairwise coprimality and (327) give

$$D \mid x(x + Cy)(x + By).$$

Since $B < C$, a nonzero multiple of D therefore satisfies

$$(330) \quad D \leq |x| |x + Cy| |x + By| \leq (C + 1)^2 H^3.$$

The assumption $H \leq L$ would contradict the middle inequality in (329).

The zero factors require the individual modulus caps. If $x = 0$, then (328) permits cancellation of C and B from the last two divisibilities. Pairwise coprimality gives $d_V d_W \mid y$. Distinctness gives $y \neq 0$, so $d_V d_W \leq |y| \leq H$. Under $H \leq L$ this would imply

$$D = d_U d_V d_W \leq X_U L < T,$$

contrary to $D > T$. If $x + Cy = 0$, cancellation from $d_U \mid Cy$ and $d_W \mid (C - B)y$ gives $d_U d_W \mid y$; hence $H \leq L$ would give $D \leq X_V L < T$. If $x + By = 0$, the same argument gives $d_U d_V \mid y$ and then $D \leq X_W L < T$. These four cases exhaust the possibilities. $\square$

Proposition 40.2 (Product-cell packing). *Let $S \subseteq \{1, \dots, M\}^2$ and suppose distinct members of S have sup distance greater than L . Then*

$$(331) \quad |S| \leq \left\lceil \frac{M}{L+1} \right\rceil^2 \leq \left(\frac{M}{L+1} + 1 \right)^2.$$

Proof. Divide each coordinate interval into consecutive cells of length $L + 1$. Two points in one product cell have both coordinate differences at most L , so every product cell contains at most one member of S . $\square$

40.2. The canonical affine box. Let $a + b = c$ be a positive primitive seed, orient it so that $b \geq c/2$, and write $\mathcal{R} = \text{rad}(abc)$. With the radical step, the three cofactor forms are

$$(332) \quad U = 1 + \mathcal{R}h, \quad V = 1 + \mathcal{R}(h + ck), \quad W = 1 + \mathcal{R}(h + bk).$$

Take

$$(333) \quad M = \left\lfloor \frac{c^6}{4\mathcal{R}} \right\rfloor, \quad L = \left\lfloor \frac{c^4}{13} \right\rfloor.$$

Assume $c \geq 6$ and $\mathcal{R} < c$. For $1 \leq h, k \leq M$ one has

$$(334) \quad U < c^6, \quad V < c^7, \quad W < c^7.$$

Indeed $\mathcal{R}M \leq c^6/4$, and $\mathcal{R}(1 + c)M \leq (1 + c)c^6/4$; the claimed strict inequalities follow for $c \geq 6$, while $b < c$ makes the W estimate no larger.

The seed conditions also imply $\mathcal{R} \geq 6$. If c is odd, then $c \geq 7$, the product abc is even, and an odd prime divides c . If c is even, coprimality makes a, b odd, and $c \geq 6$ makes one of them at least three. In both cases distinct primes 2 and $q \geq 3$ divide abc , whence $\mathcal{R} \geq 2q \geq 6$.

For a positive integer n , write $E(n) = n/\text{rad}(n)$. If $d \mid n$, then $d/\text{rad}(d) \leq E(n)$, as follows prime by prime from $v_p(d) - 1 \leq v_p(n) - 1$ on the support of d . Fix positive moduli d_U, d_V, d_W , and let S be the parameter pairs in the box satisfying $\gcd(U, k) = 1$, $d_U \mid U$, $d_V \mid V$, and $d_W \mid W$. Put

$$(335) \quad \Delta = \prod_{Z \in \{U, V, W\}} \frac{d_Z}{\text{rad}(d_Z)}.$$

Thus $\Delta > \mathcal{R}c^{14}/8192$ is a sufficient fixed certificate for the full necessary excess inequality

$$(336) \quad E(U)E(V)E(W) > \frac{\mathcal{R}c^{14}}{8192}$$

from the upper-half three-quarter analysis.

For every point of S , the identities

$$V - U = \mathcal{R}ck, \quad W - U = \mathcal{R}bk, \quad V - W = \mathcal{R}ak$$

and $\text{rad}(abc) = \mathcal{R}$ show that U, V, W are pairwise coprime and each is coprime to $\mathcal{R}abc$. If S is nonempty, one point of S then shows that the fixed divisors are pairwise coprime and satisfy every cancellation condition in (328) with $(B, C) = (b, c)$. For two points of S , subtracting the corresponding template divisibilities gives divisibility of $\mathcal{R}x$, $\mathcal{R}(x + cy)$, and $\mathcal{R}(x + by)$. Each d_Z is coprime to $\mathcal{R}$, so cancellation gives (327).

Theorem 40.3 (One full-strength affine template). *If S is nonempty and $\Delta > \mathcal{R}c^{14}/8192$, then*

$$(337) \quad |S| < \frac{12c^4}{\mathcal{R}^2}.$$

Proof. Set $T = \mathcal{R}c^{14}/8192$. Since $D = d_U d_V d_W \geq \Delta > T$ and $6(c + 1) \leq 7c$, equations (333) and (334) give

$$(c + 1)^2 L^3 \leq \frac{49c^{14}}{36 \cdot 13^3} < T.$$

Here the strict comparison is $49 \cdot 8192 = 401408 < 474552 = 36 \cdot 6 \cdot 13^3 \leq 36\mathcal{R}13^3$. They also give $Ld_U < T$ and $Ld_V, Ld_W < T$. The weakest of these comparisons is

$$\frac{c^4}{13} c^7 < \frac{\mathcal{R}c^{14}}{8192},$$

which reduces to $8192 < 13\mathcal{R}c^3$ and holds already at the conservative endpoint $\mathcal{R} = c = 6$, where the right side is 16848. Theorem 40.1 now applies to every two points of S .

Because $L + 1 > c^4/13$ and $M \leq c^6/(4\mathcal{R})$,

$$\frac{M}{L + 1} < \frac{13c^2}{4\mathcal{R}}.$$

The last quantity exceeds one when $\mathcal{R} < c$. Also $\mathcal{R} \leq c - 1$, and the elementary factorization of $5c^2 - 36c + 36$ at its roots $6/5$ and 6 gives

$$\frac{\mathcal{R}}{c^2} \leq \frac{c - 1}{c^2} \leq \frac{5}{36}, \quad \left(\frac{13}{4} + \frac{5}{36}\right)^2 = \left(\frac{61}{18}\right)^2 = \frac{3721}{324} < 12.$$

Proposition 40.2 therefore gives

$$|S| < \left(\frac{13c^2}{4\mathcal{R}} + 1\right)^2 < \frac{12c^4}{\mathcal{R}^2}.$$

□

Corollary 40.4 (Certificate entropy). *Let $\kappa, \eta > 0$. Suppose a union of N fixed templates satisfying Theorem 40.3 contains at least $\kappa\mathcal{R}^{-2/3}c^{4+\eta}$ parameter pairs. Then*

$$(338) \quad N > \frac{\kappa}{12} \mathcal{R}^{4/3} c^\eta.$$

Proof. The union bound and (337) give $|\bigcup_{i=1}^N S_i| < N(12c^4/\mathcal{R}^2)$. Comparing this with the assumed lower bound and cancelling positive factors proves (338). □

The corollary closes the large-modulus boundary for a single full-strength template and rules out every fixed finite list as a source of the matching lower bound. It leaves active an unbounded correlated family, pointwise certificate selection, an algebraic parametrization, and excess contributed by unprescribed residual factors.

40.3. Why the individual caps cannot be removed. The determinant inequality alone does not imply separation. Take

$$B = 1, \quad C = 2, \quad L = 1, \quad T = 10, \quad d_U = 31, \quad d_V = d_W = 1, \quad X_U = X_V = X_W = 1$$

and the two points (30, 1) and (30, 2). The moduli are pairwise coprime, all six conditions in (328) hold, and the three divisibilities (327) hold because $x = 0$. Moreover

$$D = 31 > T = 10 > (C + 1)^2 L^3 = 9, \quad LX_U = LX_V = LX_W = 1 < T,$$

but the sup distance is $1 = L$. Both points have $U = 1 + h = 31$ when the step is one. This is a complete counterexample to the strengthening obtained by deleting $d_Z \leq X_Z$: it fails exactly $d_U \leq X_U$. It does not refute Theorem 40.1 or any broader affine route.

The exact constants and this counterexample are independently replayed in the companion affine template-entropy computation directory. That finite replay also stress-tests 6244 bounded abstract parameter cases and 494515 canonical pairs $(c, \mathcal{R})$. These no-hit checks are diagnostic; the general conclusions above come from the proofs, and no route is retired by finite computation.

41. DETERMINANT LAYERS AND CANONICAL-KERNEL ENTROPY

We refine the fixed-template estimate of Section 40 by using all three affine congruences simultaneously. Let $S \subseteq [1, M]^2$, put $N = M - 1$, and suppose every difference vector (x, y) in one fixed template satisfies

$$d_U \mid x, \quad d_V \mid x + Cy, \quad d_W \mid x + By,$$

where d_U, d_V, d_W are pairwise coprime. Set $D = d_U d_V d_W$.

Proposition 41.1 (Three-form determinant divisor). *For any two template difference vectors $v = (x, y)$ and $w = (x', y')$,*

$$D \mid \det(v, w).$$

Proof. The modulus d_U divides $xy' - yx'$. The identities

$$\det(v, w) = (x + Cy)y' - y(x' + Cy') = (x + By)y' - y(x' + By')$$

give divisibility by d_V and d_W . Pairwise coprimality multiplies the three divisibilities. $\square$

Choose a shortest nonzero difference vector v and put $\delta = \|v\|_\infty$. The determinant map $p \mapsto \det(v, p - p_0)$ has values in $D\mathbb{Z}$, range length at most $N(|v_1| + |v_2|)$, and level sets parallel to v . Projection to a dominant coordinate gives the following capacity.

Theorem 41.2 (Determinant-layer capacity). *If $|S| \geq 2$, then*

$$|S| \leq \left(\left\lfloor \frac{N}{\delta} \right\rfloor + 1 \right) \left(\left\lfloor \frac{N(|v_1| + |v_2|)}{D} \right\rfloor + 1 \right) \leq \frac{4N^2}{D} + \frac{N}{\delta} + 1.$$

If $D > 2N^2$, all points of S are collinear. If the complete three-form separation theorem also yields $\delta > L$, then

$$|S| \leq \left\lfloor \frac{M-1}{L+1} \right\rfloor + 1.$$

Proof. An interval of the displayed determinant range contains at most the second number of multiples of D ; a level contains at most the first number of points. Expanding and using $\delta \leq N$ gives the coarse inequality. When $D > 2N^2$, every determinant has absolute value below D and hence is zero. The final estimate is one-dimensional interval packing. $\square$

Return to a positive primitive seed $a + b = c$, oriented by $b \geq c/2$, with $R = \text{rad}(abc) < c$ and

$$M = \left\lfloor \frac{c^6}{4R} \right\rfloor, \quad T_0 = \frac{Rc^{14}}{8192}.$$

The cases $c = 6, 7, 8$ are incompatible with $R < c$, so $c \geq 9$; also $R \geq 6$. Consequently $T_0 > 2(M-1)^2$. Every complete certificate with repeated-loss product above T_0 is therefore in the zero determinant layer.

Theorem 41.3 (Canonical one-dimensional capacities). *At the fixed scale $L_{12} = \lfloor c^4/12 \rfloor$ every canonical template satisfies*

$$|S| < \frac{31c^2}{10R}.$$

More generally, if $s \geq 1$ and $s^3 \leq R$, then at $L_s = \lfloor sc^4/22 \rfloor$ one has

$$|S| < \frac{57c^2}{10Rs}.$$

Proof. The cubic separation inequality at scale 12 follows from

$$8192 \cdot 100 < 81 \cdot 12^3 \cdot 6,$$

and the weakest long-arm cap from $8192 < 12 \cdot 6 \cdot 9^3$. At scale $s/22$, use $s^3 \leq R$ and $8192 \cdot 100 < 81 \cdot 22^3$, together with $8192 < 22 \cdot 9^3$. One-dimensional packing gives respectively $|S| < 3c^2/R + 1$ and $|S| < 11c^2/(2Rs) + 1$; the inequalities $10R < c^2$ and $5Rs < c^2$ absorb the final terms. $\square$

For $n > 0$ define its repeated kernel by

$$K(n) = \prod_{p^e \parallel n, e \geq 2} p^e.$$

Primewise comparison proves

$$K(n) \mid n, \quad \frac{K(n)}{\text{rad } K(n)} = \frac{n}{\text{rad } n},$$

and $K(n)$ divides every divisor of n preserving this exact loss. Thus partitioning exceptional affine points by $(K(U), K(V), K(W))$ is the unique coarsest exact-lossless template partition.

Corollary 41.4 (Forced canonical-kernel entropy). *Let $\kappa, \eta > 0$. If the exceptional packet has size at least $\kappa R^{-2/3} c^{4+\eta}$ and N_{ker} canonical-kernel triples are realized, then, for any integer $s \geq 1$ with $s^3 \leq R$,*

$$N_{\text{ker}} > \max \left\{ \frac{10\kappa}{31} R^{1/3} c^{2+\eta}, \frac{10\kappa}{57} s R^{1/3} c^{2+\eta} \right\}.$$

Taking $s = \lfloor R^{1/3} \rfloor$ yields

$$N_{\text{ker}} > \frac{5\kappa}{57} R^{2/3} c^{2+\eta}.$$

Every realized triple additionally satisfies

$$\text{rad}(K(U)K(V)K(W)) < \frac{8192c^6}{R}.$$

Proof. The canonical-kernel triples partition the exceptional packet into disjoint fixed-template classes. Each class retains the complete loss threshold $\Delta > T_0$, so Theorem 41.3 applies to it at both scales. Summing the two strict class bounds gives

$$|\mathcal{E}| < N_{\text{ker}} \frac{31c^2}{10R}, \quad |\mathcal{E}| < N_{\text{ker}} \frac{57c^2}{10Rs}.$$

Combining these inequalities with the assumed lower bound and dividing by the positive capacities gives the displayed maximum.

For $s = \lfloor R^{1/3} \rfloor$, the standing bound $R \geq 6$ implies $s \geq 1$, $s^3 \leq R$, and $s \geq R^{1/3}/2$. Substitution in the second term of the maximum yields $N_{\text{ker}} > (5\kappa/57)R^{2/3}c^{2+\eta}$.

Finally put $P = K(U)K(V)K(W)$. The repeated-kernel identities, together with the pairwise coprimality of the primitive arms, give $P/\text{rad } P = \Delta > T_0$. Also $K(n) \mid n$ and the canonical arm bounds give $P \leq UVW < c^{20}$. Hence

$$\text{rad } P = \frac{P}{P/\text{rad } P} < \frac{c^{20}}{T_0} = \frac{8192c^6}{R},$$

as required. $\square$

The positivity of κ is essential to this exact formulation. If it is omitted, $\kappa = 0$, an empty exceptional packet, and $N_{\text{ker}} = 0$ satisfy the remaining premises but contradict the strict conclusion. This full-premise counterexample retires only the zero-density strengthening.

For a fractional cover in which each template meets $\mathcal{E}$ in at most K_s points and each point receives total nonnegative weight at least one, the finite inequality $|\mathcal{E}| \leq K_s \sum_{\tau} w_{\tau}$ handles correlated overlaps exactly. It does not produce the required adaptive weight bound. Two complete integer witnesses mark precise boundaries: one certified template has $4M^2 < D$, disproving an area-only estimate, while two adjacent points carry different complete certificates, disproving the assertion that a union of pointwise selected templates inherits fixed-template

separation. The latter witness lies outside the canonical box. Neither counterexample removes the canonical adaptive-kernel route; the unresolved task is to produce or exclude the required $R^{2/3}c^{2+\eta}$ family under the actual exception inequality.

42. ADAPTIVE THREE-POINT KERNELS IN THE AFFINE SHEAR

The fixed-template argument of Section 41 does not allow the three arm certificates to vary from point to point. We now isolate the part of the determinant mechanism that survives completely adaptive choices.

Retain the minimal-radical shear

$$U(h, k) = 1 + Rh, \quad V(h, k) = 1 + R(h + ck), \quad W(h, k) = 1 + R(h + bk),$$

and let p, q, r be three admissible parameters in $[1, M]^2$. At each point s choose arbitrary divisors

$$d_U(s) \mid U(s), \quad d_V(s) \mid V(s), \quad d_W(s) \mid W(s).$$

Define the triplewise common moduli

$$g_Z = \gcd(d_Z(p), d_Z(q), d_Z(r)) \quad (Z = U, V, W), \quad G = g_U g_V g_W.$$

Lemma 42.1 (Sharp square-triangle determinant). *If three points lie in an axis-parallel square of side N , then*

$$|\det(q - p, r - p)| \leq N^2.$$

The constant is sharp.

Proof. Translate the square to $[0, N]^2$ and order the three first coordinates as $x_1, x_1 + s, x_1 + s + t$, where $s, t \geq 0$ and $s + t \leq N$. If the corresponding second coordinates are Y_1, Y_2, Y_3 , translation gives

$$D = tY_1 - (s + t)Y_2 + sY_3.$$

The positive contribution and the absolute value of the negative contribution are each at most $(s + t)N \leq N^2$. A permutation changes only the sign. Three corners attain equality. $\square$

At any one admissible point the three arms are pairwise coprime and each is coprime to R . For a difference vector (x, y) between two of the points, cancellation of R therefore gives

$$g_U \mid x, \quad g_V \mid x + cy, \quad g_W \mid x + by.$$

Applying this to both $q - p$ and $r - p$ and using

$$\det((x, y), (x', y')) = (x + cy)y' - y(x' + cy') = (x + by)y' - y(x' + by')$$

shows that every g_Z divides the determinant. The three moduli are pairwise coprime, hence their product divides it.

Theorem 42.2 (Adaptive common-kernel determinant bound). *If $p, q, r \in [1, M]^2$ are non-collinear, then*

$$G \leq (M - 1)^2.$$

Proof. The preceding congruences give $G \mid \det(q - p, r - p)$. Noncollinearity makes the determinant nonzero, so G is at most its absolute value. Apply Lemma 42.1 to the square of side $M - 1$. $\square$

This theorem allows unrelated certificates at all three points. What it does not provide is a density theorem forcing a triple whose three triplewise gcds have large product. That is now the precise positive selection problem.

There are also useful canonical constant improvements. Put

$$M = \left\lfloor \frac{c^6}{4R} \right\rfloor, \quad c \geq 9, \quad E(n) = \frac{n}{\text{rad}(n)}.$$

For every $(h, k) \in [1, M]^2$,

$$(339) \quad 3U \leq c^6, \quad 3V \leq c^7, \quad 3W \leq c^7.$$

Indeed, $4RM \leq c^6$. The short-arm estimate follows from $12 + 12Rh \leq 12 + 3c^6 \leq 4c^6$. For a long arm,

$$4(3V) \leq 12 + 3c^6(c+1) \leq 4c^7,$$

where the last inequality is $12 \leq (c-3)c^6$; $b \leq c$ gives the same bound for W .

Suppose the affine point meets the necessary exceptional threshold

$$(340) \quad Rc^{14} < 8192E(U)E(V)E(W).$$

Since $3E(U) \leq c^6$, cancellation gives

$$3Rc^8 < 8192E(V)E(W).$$

Thus at least one long arm $Z \in \{V, W\}$ satisfies

$$(341) \quad 3Rc^8 < 8192E(Z)^2.$$

Combining $Z = \text{rad}(Z)E(Z)$, $3Z \leq c^7$, and (341) yields

$$(342) \quad 27R\text{rad}(Z)^2 < 8192c^6.$$

For the full repeated kernel

$$K(n) = \prod_{p^e \parallel n, e \geq 2} p^e, \quad K = K(U)K(V)K(W),$$

pairwise coprimality gives

$$K = \text{rad}(K)E(U)E(V)E(W).$$

Moreover, (339) gives $27K \leq 27UVW \leq c^{20}$. Multiplying (340) by $27\text{rad}(K)$ and cancelling c^{14} proves the improved support bound

$$(343) \quad 27R\text{rad}(K) < 8192c^6.$$

The geometric alternatives have been tested on actual parameters. For the primitive seed $(a, b, c) = (1, 8, 9)$ one has $R = 6$ and $M = 22143$. The admissible points $(20, 1)$ and $(20, 2)$ have arm triples

$$(121, 175, 169), \quad (121, 229, 217).$$

They share $K(U) = 121$, have distance one and vertical difference, while

$$121 > (b+1)(c+1) = 90, \quad 121 > (c+1)^2 = 100.$$

Taking divisor certificates $(121, 25, 169)$ and $(121, 1, 1)$ gives common gcds $(121, 1, 1)$ and product 121. Thus this is a full-premise in-box counterexample to the precise local implication that deletes the zero-first-coordinate branch and concludes $G \leq (c+1)^2\|p-q\|_\infty^3 = 100$.

Likewise $(160, 1), (160, 2), (160, 3)$ are admissible, collinear and share $U = K(U) = 961$. Their diameter is two and

$$961 > (c+1)^2 2^3 = 800.$$

Choosing certificates $(961, 1, 1)$ at every point gives triplewise gcds $(961, 1, 1)$ and satisfies all six difference congruences. Hence the precise full-premise local cubic implication to noncollinearity is false. This is not a counterexample to Theorem 42.2, nor to its numerical box-wide conclusion: $961 < (22143 - 1)^2$. It refutes only the local diameter-scale strengthening and the invalid step from divisibility to a size bound when the determinant may be zero.

Lean verifies the sharp determinant lemma for every point order, the derivation of the adaptive congruence interface from actual pointwise arm divisors and triplewise gcds, all three bounds in (339), the long-arm dichotomy, (342), the arithmetic seam of (343), and the negations of both exact local implications. No adaptive common-kernel density input is assumed. The affine route therefore remains active at the triple-selection gate.

43. INCIDENCE BUDGETS AT THE AFFINE TRIPLE-SELECTION GATE

The adaptive determinant theorem leaves one precise selection problem: an exceptional point may choose its own arm divisors, while the determinant bound needs three points with a large *common* divisor product. We record the finite geometry and the exact incidence budget available for this step.

Let $S \subseteq [1, M]^2 \cap \mathbb{Z}^2$, put $N = M - 1$, and retain the affine arms U, V, W of Section 42. For a fixed actual arm-divisor label

$$\lambda = (d_U, d_V, d_W), \quad D(\lambda) = d_U d_V d_W,$$

define

$$F_\lambda = \{x \in S : d_U \mid U(x), d_V \mid V(x), d_W \mid W(x)\}.$$

Lemma 43.1 (Integer-line cap). *If every triple in S is collinear, then $|S| \leq M$. Equivalently, more than M points in the square contain a noncollinear triple.*

Proof. Choose distinct $p, q \in S$ when $|S| > 1$. If $p_1 = q_1$, collinearity forces every point of S to have that first coordinate, so projection to the second coordinate is injective and has at most M values. If $p_1 \neq q_1$, and $r, s \in S$ have $r_1 = s_1$, subtracting $\det(q - p, r - p) = \det(q - p, s - p) = 0$ gives $(q_1 - p_1)(r_2 - s_2) = 0$. Hence $r = s$, so projection to the first coordinate is injective and again $|S| \leq M$. $\square$

Theorem 43.2 (Large-label fibre and incidence bounds). *Assume the three arms are pairwise coprime at every point of S . If $D(\lambda) > N^2$, then $|F_\lambda| \leq M$. Consequently, for every finite family $\mathcal{L}$ of such large labels,*

$$\sum_{\lambda \in \mathcal{L}} |F_\lambda| \leq M |\mathcal{L}|.$$

If every x in a set $E \subseteq S$ belongs to at least μ of these fibres, then

$$(344) \quad \mu |E| \leq M |\mathcal{L}|.$$

Proof. If $|F_\lambda| > M$, Lemma 43.1 produces a noncollinear triple in the fibre. Use the same label at all three points. Actual arm divisibility yields the six difference congruences, and pairwise coprimality of the arms makes d_U, d_V, d_W pairwise coprime. The adaptive determinant theorem then gives $D(\lambda) \leq N^2$, a contradiction. Summing the fibre bound proves the first displayed incidence inequality. Double-counting $\{(x, \lambda) : x \in E \cap F_\lambda\}$ proves (344). $\square$

The associated energy identity shows exactly what an analytic refinement must estimate. For positive labels $d_Z(x)$, set

$$\mathcal{E}_3 = \sum_{x, y, z \in S} \prod_{Z \in \{U, V, W\}} \gcd(d_Z(x), d_Z(y), d_Z(z))$$

and

$$n(e) = \#\{x \in S : e_U \mid d_U(x), e_V \mid d_V(x), e_W \mid d_W(x)\}.$$

Then the divisor identity $m = \sum_{e|m} \varphi(e)$ gives, by expanding three finite sums,

$$(345) \quad \mathcal{E}_3 = \sum_{e_U, e_V, e_W} \varphi(e_U) \varphi(e_V) \varphi(e_W) n(e)^3.$$

Thus total energy alone is insufficient: one must place enough of it on $e_U e_V e_W > N^2$ and control the contribution of collinear triples.

That collinear qualification cannot be removed. For the canonical seed $(a, b, c) = (1, 8, 9)$, $R = 6$ and $M = 22143$. At

$$(21480, 282), \quad (21480, 6211), \quad (21480, 12140)$$

use the common square label

$$(d_U, d_V, d_W) = (128881, 49, 121) = (359^2, 7^2, 11^2).$$

The corresponding arm rows are

	U	V	W
1	128881	144109	142417
2	128881	464275	427009
3	128881	784441	711601.

The label divides every row, all canonical admissibility and coprimality conditions hold, and the vertical increment $5929 = 49 \cdot 121$ proves all six congruences. Yet

$$128881 \cdot 49 \cdot 121 = 764135449 > 22142^2 \quad \text{and} \quad \det(q - p, r - p) = 0.$$

This is a full-premise counterexample to the extra implication “common product above the box square implies noncollinearity.” It does not meet the noncollinearity premise of the valid determinant theorem and therefore does not retire the affine route.

The remaining positive gate is now quantitative. One may close this branch by constructing a finite family $\mathcal{L}$ and a multiplicity μ with $\mu|E| > M|\mathcal{L}|$, or by proving that the large-product part of (345) exceeds every possible collinear contribution. No full-premise counterexample to either conditional criterion is known.

44. DIRECTION PERIODS AND COLLINEAR AFFINE ENERGY

The common-kernel selection in Section 43 leaves a collinear concentration problem. We now extract an exact arithmetic period along each nonnegative ray. Write

$$U(h, k) = 1 + Rh, \quad V(h, k) = 1 + R(h + Ck), \quad W(h, k) = 1 + R(h + Bk)$$

and parametrize the ray by $x(n) = (h_0 + ns, k_0 + nt)$. For a positive pairwise-coprime label $\lambda = (d_U, d_V, d_W)$ define

$$\begin{aligned} A_U &= s, & A_V &= s + Ct, & A_W &= s + Bt, \\ r_Z &= \frac{d_Z}{\gcd(d_Z, A_Z)}, & T_\lambda &= r_U r_V r_W, \\ C_\lambda &= \gcd(d_U, A_U) \gcd(d_V, A_V) \gcd(d_W, A_W). \end{aligned}$$

Theorem 44.1 (Exact ray period and capacity). *If the same label divides the corresponding three arms at $x(n)$ and $x(m)$, then*

$$T_\lambda \mid |n - m|, \quad T_\lambda C_\lambda = d_U d_V d_W.$$

Consequently, if all occurrences have indices in $[0, H]$, then

$$n_\lambda \leq \left\lfloor \frac{H}{T_\lambda} \right\rfloor + 1.$$

Proof. Each d_Z divides a number congruent to one modulo R , hence is coprime to R . Subtracting the two arm values and cancelling R gives $d_Z \mid |n - m|A_Z$. Dividing d_Z and A_Z by their gcd shows $r_Z \mid |n - m|$. The r_Z remain pairwise coprime because they divide the pairwise-coprime components d_Z , so their product divides the gap. The factorization follows from $(d_Z / \gcd(d_Z, A_Z)) \gcd(d_Z, A_Z) = d_Z$. All indices lie in one residue class modulo the positive integer T_λ , which gives the capacity. $\square$

Assume $s > 0$ and $N > 0$, put $L = \max(s, t)$ and $K = (B + 1)(C + 1)$, and suppose $HL \leq N$. Then every direction coefficient is positive and

$$C_\lambda \leq s(s + Ct)(s + Bt) \leq KL^3.$$

Theorem 44.2 (Nonconstant-ray cube-root cap). *If $N^2 < d_U d_V d_W$, then*

$$(n_\lambda - 1)^3 < KN.$$

For any finite catalogue $\mathcal{L}$ of such labels and nonnegative integer weights w_λ ,

$$\sum_{\lambda \in \mathcal{L}} w_\lambda n_\lambda^3 \leq 4(B + 1)(C + 1)N \sum_{\lambda \in \mathcal{L}} w_\lambda.$$

Proof. Put $a = n_\lambda - 1$ and $T = T_\lambda$. The period capacity gives $aT \leq H$, hence $aTL \leq N$. Since

$$N^2 < TC_\lambda \leq TKL^3,$$

squaring the first inequality and cancelling positive factors gives $a^2T < KL$. Thus $a^2 < KL$ and $aL \leq N$; multiplication yields $a^3 < KN$. For positive integers n, X , the implication $(n-1)^3 < X \Rightarrow n^3 \leq 4X$ follows from

$$4(a^3 + 1) - (a+1)^3 = 3(a-1)^2(a+1) \geq 0.$$

Apply this with $X = KN$, multiply by w_λ , and sum. The constant four is optimal at this integer level: $n = X = 2$ defeats the constant three. $\square$

Ambient positivity is indispensable. The pure ledger

$$(n, H, T, C, D, N, K, L) = (1, 0, 1, 1, 1, 0, 1, 1)$$

satisfies every other period, factorization, capture, and strict large-label premise, while the proposed conclusion becomes $0 < 0$. This refutes exactly the $N > 0$ -omitted strengthening.

Taking $w_\lambda = \varphi(d_U)\varphi(d_V)\varphi(d_W)$ converts the collinear part of the third-gcd identity to a total totient-weight problem. The remaining cost is the weight of the actual large-label catalogue and the number of supporting rays, rather than the raw length of a line.

The hypothesis $s > 0$ is essential. On a vertical ray the U component is constant and may be absorbed in full. If $d_U \leq X_U$, $B, C, t > 0$, and $Ht \leq N$, then

$$C_\lambda \leq X_U BC t^2, \quad (n_\lambda - 1)^2 < X_U BC.$$

Signed analogues for the two negative-slope arm-level directions, together with their individual component caps, remain part of the active route.

There is a full actual pressure test. For the primitive seed $(1, 8, 9)$ take

$$R = 6, \quad N = 22142, \quad x(n) = (21480, 282 + n), \quad \lambda = (128881, 49, 121)$$

and $S = \{0, 5929, 11858, 17787\}$. All four points lie in the canonical box, satisfy the admissibility conditions, and carry the displayed pairwise- coprime label, whose product exceeds N^2 . Here

$$T_\lambda = 5929, \quad C_\lambda = 128881, \quad |S| = 17787/5929 + 1 = 4,$$

whereas $\lfloor 17787/(d_U d_V d_W) \rfloor + 1 = 1$. This refutes the stronger claim that the whole label product is itself a ray period and also refutes deleting the nonconstant-direction premise. It attains the correct period capacity and does not contradict the cube-root or vertical theorems.

The formal companion proves all period, capacity, cube-root, weighted-energy, and vertical statements above and checks this actual counterexample. No full-premise counterexample is known to the remaining catalogue-weight and signed arm-level transfer problems, so those routes remain active.

45. ACTUAL AFFINE CATALOGUE WEIGHT AND CUBIC AGGREGATION

The ray-period argument reduces the remaining non-arm-level collinear term to the Euler-totient weight of the divisor labels that actually arise from canonical repeated kernels. We now compute that weight and aggregate its cubic energy without assuming that different kernel catalogues are disjoint.

For a positive kernel triple $d = (d_U, d_V, d_W)$ put $D(d) = d_U d_V d_W$ and

$$\mathcal{D}(d) = \{(e_U, e_V, e_W) : e_Z \mid d_Z\}, \quad w(e) = \varphi(e_U)\varphi(e_V)\varphi(e_W).$$

For a threshold T , define

$$L_T(d) = \sum_{\substack{e \in \mathcal{D}(d) \\ e_U e_V e_W > T}} w(e).$$

Theorem 45.1 (Actual catalogue weight and finite tail). *For every positive triple d ,*

$$\sum_{e \in \mathcal{D}(d)} w(e) = D(d)$$

and

$$D(d) \leq L_T(d) + T\tau(d_U)\tau(d_V)\tau(d_W).$$

Moreover, if $T < D(d)$, then the top label gives

$$\varphi(d_U)\varphi(d_V)\varphi(d_W) \leq L_T(d).$$

Proof. The three sums separate, and $\sum_{e|n} \varphi(e) = n$. On the complementary range $e_U e_V e_W \leq T$, one has $w(e) \leq e_U e_V e_W \leq T$, and there are at most $\tau(d_U)\tau(d_V)\tau(d_W)$ labels. Finally, $e = d$ occurs in the large range whenever $T < D(d)$. $\square$

At a canonical exceptional affine point let $d = (K(U), K(V), K(W))$, where K retains the exact prime powers of exponent at least two. With

$$M = \left\lfloor \frac{c^6}{4R} \right\rfloor, \quad N = M - 1,$$

the established arm and excess inequalities give

$$27D(d) \leq c^{20}, \quad D(d) > \frac{Rc^{14}}{8192}, \quad N^2 < \frac{c^{12}}{16R^2}.$$

The elementary bound $\tau(n) \ll_\epsilon n^\epsilon$, applied with $\epsilon = 1/40$ to the three components, therefore yields uniformly

$$\frac{D(d) - L_{N^2}(d)}{D(d)} \ll c^{-3/2}.$$

Thus asymptotically all of the actual totient weight lies above the sharp box-square determinant threshold.

Let $\mathcal{K}$ be the realized canonical kernel triples, and let m_κ be the size of the corresponding class in the exceptional-set partition. If $n(e)$ is the number of exceptional points whose canonical kernel is divisible componentwise by e , then

$$n(e) = \sum_{\kappa: e|\kappa} m_\kappa.$$

Theorem 45.2 (Monotone catalogue aggregation). *For the determinant-large part of the third-gcd energy,*

$$\sum_{\kappa \in \mathcal{K}} m_\kappa^3 L_T(\kappa) \leq \sum_{\substack{e \\ e_U e_V e_W > T}} w(e) n(e)^3.$$

Consequently, at $T = N^2$ the right side is at least

$$(1 - O(c^{-3/2})) \sum_{\kappa \in \mathcal{K}} D(\kappa) m_\kappa^3.$$

Proof. For each fixed label,

$$\sum_{\kappa: e|\kappa} m_\kappa^3 \leq \left(\sum_{\kappa: e|\kappa} m_\kappa \right)^3 = n(e)^3.$$

Multiply by $w(e)$, sum over e , and interchange the two finite sums. The asymptotic consequence follows from Theorem 45.1. $\square$

This theorem controls energy, including its diagonal part. It does not by itself force two kernel classes to share a large label. If every point introduces a new catalogue, all relevant occupancies may still be one. A bound for deduplicated catalogue weight or singleton novelty remains necessary for a genuine reuse conclusion.

The form best matched to ray capacities is shifted. For nonnegative weights and integer occupancies put

$$W = \sum_i w_i, \quad I = \sum_i w_i n_i, \quad J = \sum_i w_i (n_i - 1).$$

Weighted Hölder gives

$$J^3 \leq W^2 \sum_i w_i (n_i - 1)^3.$$

Since every occurring label has $n_i \geq 1$, the termwise identity $n_i = 1 + (n_i - 1)$ gives $I = W + J$. Thus the same estimate is the explicit overlap inequality

$$(I - W)^3 \leq W^2 \sum_i w_i (n_i - 1)^3.$$

Hence individual caps $(n_i - 1)^3 \leq X_i$ imply

$$J^3 \leq W^2 \sum_i w_i X_i.$$

Labels may use different supporting lines and different X_i ; no raw number-of-rays factor occurs. This also avoids the factor four used when the earlier paper converted shifted cubes to n_i^3 .

The finite tail cannot be strengthened by simply subtracting T . The positive pairwise-coprime powerful triple $(d_U, d_V, d_W) = (9, 25, 1)$ at $T = 5$ has total weight 225, small-label weight $1 + 2 + 4 = 7$, and therefore

$$L_5(d) = 218 < 220 = 225 - 5.$$

This is a complete counterexample only to $L_T(d) \geq D(d) - T$. It leaves the correct divisor-count tail and the affine route intact.

The Lean companion checks the exact catalogue identities, tail bounds, top-label baseline, cubic aggregation, polynomial and shifted weighted Hölder inequalities, and this precise counterexample. The active next steps are a signed period/capture theorem for every non-arm-level supporting line, the two negative-slope arm-level caps, and an arithmetic bound on deduplicated catalogue novelty. No full-premise counterexample to those corrected targets is known.

46. SIGNED RAY CAPTURE AND EXACT CANONICAL ARM CEILINGS

We close the signed-direction and constant-arm losses left by Sections 44 and 45. Retain the three affine arms

$$U(h, k) = 1 + Rh, \quad V(h, k) = 1 + R(h + Ck), \quad W(h, k) = 1 + R(h + Bk), \quad 0 < B < C.$$

A line through at least two lattice points has a primitive signed direction (s, t) and a parametrization $(h_0 + js, k_0 + jt)$. Put

$$(346) \quad A_U = s, \quad A_V = s + Ct, \quad A_W = s + Bt, \quad L = \max(|s|, |t|).$$

If the index interval has length H inside a square of side difference N , then $HL \leq N$.

For a positive pairwise-coprime divisor label $\lambda = (d_U, d_V, d_W)$, set

$$(347) \quad r_Z = \frac{d_Z}{\gcd(d_Z, |A_Z|)}, \quad T_\lambda = r_U r_V r_W, \quad C_\lambda = \prod_Z \gcd(d_Z, |A_Z|), \quad D_\lambda = d_U d_V d_W.$$

Theorem 46.1 (Signed period and strict capture squeeze). *Suppose λ occurs n_λ times on the segment, and put $a_\lambda = n_\lambda - 1$. Then*

$$(348) \quad T_\lambda \mid |i - j| \text{ for two occurrence indices,} \quad T_\lambda C_\lambda = D_\lambda, \quad a_\lambda T_\lambda L \leq N.$$

If $N > 0$ and $D_\lambda > N^2$, then

$$(349) \quad a_\lambda N L < C_\lambda,$$

$$(350) \quad a_\lambda^2 T_\lambda L^2 < C_\lambda, \quad a_\lambda^3 T_\lambda^2 L^3 < C_\lambda N,$$

$$(351) \quad (a_\lambda T_\lambda L)^3 < D_\lambda N.$$

Proof. Subtract the two arm divisibilities. Actual occurrence makes every d_Z coprime to R , so cancellation gives $d_Z \mid (i - j)A_Z$. Dividing by the gcd in (347) gives $r_Z \mid |i - j|$. The r_Z remain pairwise coprime, hence their product divides the gap; the factorization of D_λ is immediate. All occurrence indices lie in one residue class modulo T_λ , so $a_\lambda T_\lambda \leq H$ and then $a_\lambda T_\lambda L \leq N$.

Now

$$a_\lambda T_\lambda L N \leq N^2 < T_\lambda C_\lambda.$$

Cancel T_λ to obtain (349). Squaring the period capacity and using the same strict inequality gives $a_\lambda^2 T_\lambda L^2 < C_\lambda$; multiplication by the capacity proves the cubic inequality. Multiplying once more by T_λ gives (351). $\square$

This theorem includes all signs and all zero coefficients. For a non-arm direction none of A_U, A_V, A_W vanishes. With

$$(352) \quad P(s, t) = |s| |s + Ct| |s + Bt|, \quad K = (B + 1)(C + 1),$$

one has

$$(353) \quad C_\lambda \leq P(s, t) \leq KL^3.$$

The constant is sharp at $s = t = 1$. Consequently every repeated large non-arm label obeys

$$(354) \quad a_\lambda NL < P(s, t), \quad a_\lambda^3 T_\lambda^2 < KN, \quad KL^2 > N.$$

The last condition follows because $KL^2 \leq N$ would contradict the first inequality as soon as $a_\lambda \geq 1$. Thus non-arm repetitions are confined to large primitive directions and retain an inverse-square period saving.

The canonical relation is $C = B + 1$. There are then three zero-coefficient primitive directions, up to sign.

Theorem 46.2 (Exact canonical arm capture). *For the constant U, V, W directions, respectively, the absolute coefficient triples, scales, and captures are*

constant arm	(A_U , A_V , A_W)	L	C_λ
U	$(0, C, B)$	1	d_U
V	$(C, 0, 1)$	C	d_V
W	$(B, 1, 0)$	B	d_W .

Hence a large occurring label satisfies the three respective linear caps

$$(355) \quad a_\lambda N < d_U, \quad a_\lambda NC < d_V, \quad a_\lambda NB < d_W.$$

Proof. Each component of an actual label divides an arm value congruent to one modulo $R = \text{rad}(BC)$, and is therefore coprime to B and C . For the U direction,

$$C_\lambda = \gcd(d_U, 0) \gcd(d_V, C) \gcd(d_W, B) = d_U.$$

For the V direction the coefficients are $(C, 0, C - B) = (C, 0, 1)$, giving $C_\lambda = d_V$. The W direction is $(B, -1, 0)$ and gives $C_\lambda = d_W$. Substitution into (349) proves (355). $\square$

With the preceding canonical point bounds $3d_U \leq C^6$ and $3d_V, 3d_W \leq C^7$, this gives

$$3a_\lambda N < C^6, \quad 3a_\lambda NC < C^7, \quad 3a_\lambda NB < C^7.$$

These are linear in the relevant label component and improve the former square-root treatment of the vertical direction.

We next pass from a shifted occupancy to the full cubic energy. For every $n \in \mathbb{N}$, with truncated $a = n - 1$,

$$(356) \quad n^3 \leq 1 + 7a^3.$$

For $n \geq 2$, the difference of the right side and $(a + 1)^3$ is $3a(a - 1)(2a + 1) \geq 0$; the cases $n = 0, 1$ are immediate. Equality at $n = 2$ shows that seven is optimal.

Let $\mathcal{K}$ be the realized selected canonical kernel triples $\kappa = (k_U, k_V, k_W)$, with $D_\kappa = k_U k_V k_W$. Define $\mathcal{L}$ as the deduplicated union of their downward divisor catalogues, then retain only

labels whose coordinate product exceeds N^2 . Let n_λ be the number of selected canonical points realizing the label λ . For

$$w_\lambda = \varphi(d_U)\varphi(d_V)\varphi(d_W), \quad W_0 = \sum_{\lambda \in \mathcal{L}} w_\lambda, \quad E = \sum_{\lambda \in \mathcal{L}} w_\lambda n_\lambda^3, \quad E_{\text{sh}} = \sum_{\lambda \in \mathcal{L}} w_\lambda (n_\lambda - 1)^3,$$

equation (356) gives

$$(357) \quad E \leq W_0 + 7E_{\text{sh}}.$$

This charges a singleton once and introduces no number-of-lines factor.

Choose one owner $o(\lambda) \in \mathcal{K}$ for every deduplicated label. Euler's divisor identity gives, for each coordinate Z ,

$$(358) \quad \sum_{\lambda \in \mathcal{L}} w_\lambda d_Z(\lambda)^3 \leq \sum_{\kappa \in \mathcal{K}} D_\kappa k_Z(\kappa)^3, \quad W_0 \leq \sum_{\kappa \in \mathcal{K}} D_\kappa.$$

Indeed, labels owned by κ form a subset of its full downward catalogue, whose total totient weight is D_κ , and $d_Z^3 \leq k_Z^3$.

Partition repeated labels into non-arm labels and the three arm directions. Let

$$\mathcal{S}_{\text{non}} = \sum_{\substack{\lambda \text{ repeated} \\ \lambda \text{ non-arm}}} \frac{w_\lambda}{T_\lambda^2}.$$

Theorem 46.3 (Global signed affine energy ceiling). *In the canonical setup,*

$$(359) \quad \begin{aligned} E &\leq \sum_{\kappa \in \mathcal{K}} D_\kappa + 7KN\mathcal{S}_{\text{non}} \\ &\quad + \frac{7}{N^3} \sum_{\kappa \in \mathcal{K}} D_\kappa \left(k_U^3 + \frac{k_V^3}{C^3} + \frac{k_W^3}{B^3} \right). \end{aligned}$$

Every term in $\mathcal{S}_{\text{non}}$ is supported on $KL^2 > N$.

Proof. The non-arm cubic cap in (354) gives $E_{\text{non}} \leq KN\mathcal{S}_{\text{non}}$. Cubing (355), multiplying by w_λ , and summing gives the three coordinate moments divided by N^3, N^3C^3, N^3B^3 . Enlarge each arm subset to all labels and apply (358). Add the terms and use (357). $\square$

The arm-level contribution is therefore explicit. The remaining affine upper-bound task is the non-arm inverse-period sum $\mathcal{S}_{\text{non}}$, together with the comparison of the singleton baseline against catalogue-class multiplicities. These are active gates.

Several complete pressure tests determine the exact boundary. Replacing $N^2 < D_\lambda$ by $N^2 \leq D_\lambda$ fails at $(n, H, T, C_\lambda, D, N, L) = (2, 1, 1, 1, 1, 1, 1)$. The tuple

$$(n, H, T, C_\lambda, D, N, K, L) = (2, 3, 3, 6, 18, 3, 6, 1)$$

satisfies the period, span, factorization, strict-threshold, and capture premises, but refutes adding one more factor of T to either higher capture inequality. Scaling a primitive arm direction by five gives full-premise examples showing why primitivity is required, and four separate primitive examples show why each coefficient-coprimality premise is required. At occupancy two, coefficient six in (356) fails. These witnesses retire only the stated strengthenings; actual canonical labels retain the strictness, primitivity, and coprimality premises.

Catalogue membership is also essential. In the actual box $(B, C, M, N, R) = (1, 2, 10, 9, 2)$, with 82 admissible points, the totient weight of all arm-divisor labels above N^2 is 972,496, whereas the owner mass of the twelve distinct powerful-kernel triples is only 1,072. Thus deleting membership in the selected downward-catalogue union would assert the false inequality $972,496 \leq 1,072$. This full-premise counterexample retires precisely that enlarged owner statement, while leaving the selected-catalogue theorem unchanged.

An independent exact-integer audit checks 15,840 signed directions, 1,776,807 cubic ledgers, 2,390,018 normalized quadratic ledgers, 43,403 exact arm-capture cases, and four actual canonical boxes. Those boxes contain 4,307 distinct local large labels and 259 repeated labels, testing every ray-level identity. The selected powerful-kernel subcatalogues for $M = 10$ contain 110

large labels and 12 repeated labels. A separate $(B, C, M) = (4, 5, 30)$ stress box contains 113 selected large labels, nine repeated labels, and one repeated non-arm label. All owner and global energy checks pass. These finite audits are pressure tests and supply no asymptotic inference.

The Lean companion kernel-checks the signed period/capture algebra, all three canonical arm identities, the sharp constant K , the optimal coefficient seven, the finite weighted owner bounds, the global shifted-energy interface, and the full-premise boundary countermodels. It assumes no exceptional-point lower bound, no catalogue novelty theorem, and no *abc* statement.

47. AFFINE INCIDENCE CANCELLATION AND INVERSE-PERIOD CATALOGUES

We refine the selected powerful-kernel catalogue without enlarging it to all arm divisors. Let $\mathcal{K}$ be the finite set of kernel classes, let m_κ be the number of selected points in class κ , and let

$$\mathcal{L}_N(\kappa) = \{d = (d_U, d_V, d_W) : d_Z \mid k_Z(\kappa), d_U d_V d_W > N^2\}.$$

Give d the weight $w_d = \varphi(d_U)\varphi(d_V)\varphi(d_W)$ and put

$$L_\kappa = \sum_{d \in \mathcal{L}_N(\kappa)} w_d.$$

On the deduplicated union $\mathcal{L}$, write n_d for the number of selected points supporting d , and define

$$I = \sum_{d \in \mathcal{L}} w_d n_d, \quad E = \sum_{d \in \mathcal{L}} w_d n_d^3, \quad E_{\text{sh}} = \sum_{d \in \mathcal{L}} w_d (n_d - 1)^3.$$

Proposition 47.1 (Exact point-label incidence). *One has*

$$(47.1.1) \quad I = \sum_{\kappa \in \mathcal{K}} m_\kappa L_\kappa.$$

No disjointness assumption on the class tails is required.

Proof. For every label, $n_d = \sum_{\kappa: d \in \mathcal{L}_N(\kappa)} m_\kappa$. Substitute this identity in I and interchange the two finite sums. The inner sum for a fixed class is exactly L_κ . $\square$

The preceding monotone-overlap theorem gives $\sum_\kappa m_\kappa^3 L_\kappa \leq E$. The correct conversion of E uses the incidence I , rather than paying the deduplicated union weight as an unrelated error.

Lemma 47.2 (Optimal incidence shift). *For every $n \in \mathbb{N}$,*

$$(47.2.1) \quad n^3 \leq n + 6(n - 1)^3,$$

where subtraction is in $\mathbb{N}$. The coefficient six is optimal.

Proof. The cases $n = 0, 1$ are equalities. For $n \geq 2$, write $a = n - 1$; then

$$n + 6a^3 - n^3 = a(a - 1)(5a + 2) \geq 0.$$

At $n = 2$ equality reads $8 = 2 + 6$, so no smaller coefficient works. $\square$

Theorem 47.3 (Singleton cancellation). *With*

$$\Delta_{\text{mult}} = \sum_{\kappa \in \mathcal{K}} (m_\kappa^3 - m_\kappa) L_\kappa,$$

one has

$$(47.3.1) \quad \Delta_{\text{mult}} \leq 6E_{\text{sh}}.$$

Every class of multiplicity one cancels exactly from the left side. For occupancy at least three, the coefficient six may be replaced by three; the two coefficients are separately sharp at occupancies two and three.

Proof. Multiply Lemma 47.2 by w_d and sum to obtain $E \leq I + 6E_{\text{sh}}$. Combine this with the monotone-overlap inequality and Proposition 47.1, then subtract the finite common incidence. For $n \geq 3$, the refinement follows from $n^3 \leq n + 3(n - 1)^3$; equality at $n = 3$ proves sharpness. $\square$

There are two distinct sources of shifted incidence. Put

$$A_0 = \sum_{\kappa} L_{\kappa}, \quad A_1 = \sum_{\kappa} (m_{\kappa} - 1) L_{\kappa}, \quad W = \sum_{d \in \mathcal{L}} w_d, \quad \Omega = A_0 - W.$$

If r_d is the number of supporting kernel classes, then $\Omega = \sum_d w_d(r_d - 1)$. Double counting gives

$$(47.3.2) \quad J := \sum_d w_d(n_d - 1) = A_1 + \Omega.$$

Weighted Hölder therefore yields the exact normalized pressure inequality

$$(47.3.3) \quad (A_1 + \Omega)^3 \leq (A_0 - \Omega)^2 E_{\text{sh}}.$$

The term A_1 records multiplicity inside one class; Ω records overlap of distinct class catalogues. Neither can be deleted.

We next retain the inverse period. For two selected points write their difference as $q(s, t)$ with (s, t) primitive, set

$$A_U = s, \quad A_V = s + Ct, \quad A_W = s + Bt, \quad L = \max(|s|, |t|), \quad qL \leq N,$$

and assume the direction is non-arm. Let

$$g_Z = \gcd(k_Z(x), k_Z(y)), \quad c_Z = \gcd(g_Z, |A_Z|), \quad G = \prod_Z g_Z, \quad C_g = \prod_Z c_Z, \quad T_g = G/C_g.$$

Pairwise coprimality of the three arms and cancellation of $R = \text{rad}(BC)$ give

$$(47.3.4) \quad T_g \mid q, \quad G = T_g C_g \leq q C_g.$$

For $k \geq 1$ and $a \geq 0$, define

$$F(k, a) = \sum_{d \mid k} \varphi(d) \frac{\gcd(d, a)^2}{d^2}.$$

If $p^e \parallel k$ and $r = \min(e, v_p(a))$, with $r = e$ for $a = 0$, a direct prime-power summation gives

$$(47.3.5) \quad F(p^e, a) = \begin{cases} p^e, & r = e, \\ p^r + p^{r-1}(1 - p^{-(e-r)}), & r < e. \end{cases}$$

Thus F has an exact Euler product and $F(k, a) \leq \gcd(k, a) \prod_{p \mid k} (1 + p^{-1})$.

Let $Q(x, y)$ be the sum of w_d/T_d^2 over all common labels with $d_U d_V d_W > N^2$, where $T_d = \prod_Z d_Z / \gcd(d_Z, |A_Z|)$. Also put

$$H(k) = \sum_{d \mid k} \frac{\varphi(d)}{d} = \prod_{p^e \parallel k} (1 + e(1 - p^{-1})).$$

Theorem 47.4 (Hybrid pair-catalogue bound). *Every non-arm pair satisfies*

$$(47.4.1) \quad Q(x, y) \leq \min \left\{ \prod_Z F(g_Z, |A_Z|), \frac{C_g^2 G}{N^4}, \frac{C_g^2 H(G)}{N^2} \right\}.$$

If q_0 and $P_0 = \prod_Z |A_Z|^{(R)}$ denote the R -free parts, then the first term is at most $P_0 q_0 / \varphi(q_0)$ and $G \mid q_0 P_0$.

Proof. Dropping the large-product restriction makes the three divisor sums independent and gives the Euler-product term. For a retained label, $D_d > N^2$ and $C_d \leq C_g$, hence $T_d^{-2} < C_g^2 / N^4$; summing all totient weights gives G . Alternatively,

$$\frac{w_d}{T_d^2} = \frac{w_d}{D_d} \frac{C_d^2}{D_d} < \frac{C_g^2}{N^2} \frac{w_d}{D_d},$$

and the enlarged divisor sum is $H(G)$. The R -free conclusions follow from $T_g \mid q_0$, $C_g \mid P_0$, and the local Euler factors. $\square$

Because G is powerful, the same pair also obeys a useful direction filter. Writing $\mathfrak{E}(n) = n/\text{rad}(n)$, one has

$$(47.4.2) \quad T_g \mathfrak{E}(P_0) > N, \quad \mathfrak{E}(P_0) > L.$$

Indeed $G > N^2$ and powerfulness give $\mathfrak{E}(G) > N$, while $\mathfrak{E}(G) \leq T_g \mathfrak{E}(C_g) \leq T_g \mathfrak{E}(P_0)$; then use $T_g \mid q$ and $qL \leq N$. In the canonical branch $C = B + 1$, the three R -free coefficients are pairwise coprime. If one of them has powerful excess greater than $L^{1/3}$, then the corresponding original linear coefficient has at least that excess. The original coefficient, rather than the operation of removing R -factors, is injective along each side of the direction square. The elementary estimate

$$\#\{n \leq X : \mathfrak{E}(n) > Y\} = O(X/\sqrt{Y})$$

follows by putting $m = \prod_p p^{\lfloor v_p(n)/2 \rfloor}$: then $m^2 \mid n$ and $m^2 \geq \mathfrak{E}(n) > Y$, after which one sums X/m^2 over $m > \sqrt{Y}$. On the shell $\max(|s|, |t|) = L$, this counts $O(CL^{5/6})$ eligible values on the nonconstant sides. The exceptional vertical-side values of s are summed by the same square-divisor estimate in dyadic blocks. Summing the shells reduces the number of eligible primitive directions of scale at most N from $O(N^2)$ to $O(CN^{11/6})$. This is a real saving, although it does not yet count the maximal powerful intersection tops on each direction.

Finally, let $S_{\text{non}} = \sum w_d/T_d^2$ over repeated non-arm labels. A label of occupancy n_d occurs in exactly $\binom{n_d}{2}$ pair catalogues, so

$$\sum_{\{x,y\}} Q(x,y) = \sum_{d \text{ repeated, nonarm}} \binom{n_d}{2} \frac{w_d}{T_d^2}.$$

Choosing one loop for every class of multiplicity at least two and one edge for every relevant pair of singleton classes gives a smaller support skeleton $\mathcal{E}$ with

$$(47.4.3) \quad S_{\text{non}} \leq \sum_{e \in \mathcal{E}} Q_e.$$

The inverse period cannot supply a uniform saving by itself. In the actual canonical box

$$(B, C, R, M, N) = (8, 9, 6, 22143, 22142)$$

the two admissible points (3128, 10183) and (21897, 11423) lie in distinct singleton kernel classes and share the sole large label

$$d = (137^2, 173^2, 1), \quad D_d = 561737401 > N^2.$$

Their primitive direction has coefficients (18769, 29929, 28689), so $C_d = D_d$ and $T_d = 1$. Hence

$$(47.4.4) \quad Q_{\{x,y\}} = S_{\text{non}} = \varphi(137^2)\varphi(173^2) = 554413792.$$

This full-premise example has $R < C$ and refutes any strict fractional saving deduced only from the subcritical branch, singleton support, or a positive period lower bound. It does not refute the pair-skeleton theorem.

The active normalized contradiction target is

$$(47.4.5) \quad (A_1 + \Omega)^3 > (A_0 - \Omega)^2 \left[(B+1)(C+1)N \sum_{e \in \mathcal{E}} Q_e + \frac{1}{N^3} \sum_{\kappa} L_{\kappa} \left(k_U^3 + \frac{k_V^3}{C^3} + \frac{k_W^3}{B^3} \right) \right].$$

Proving this requires a global aggregation theorem for the divisibility-maximal powerful intersection tops. No counterexample to that complete statement is known, so the affine route remains active.

48. MAXIMAL-INTERSECTION OWNERSHIP AGGREGATION

We continue the affine catalogue notation of Section 47. This section proves an unconditional compression theorem for the non-arm catalogue. It does not prove the *abc* conjecture: the final strict comparison isolated below remains open.

Let X be the selected canonical parameter points in a square of side difference N . Write $\kappa(x) = (k_U(x), k_V(x), k_W(x))$ for the powerful-kernel triple of x , and order triples by coordinatewise divisibility. For distinct $x, y \in X$, set

$$\gamma(x, y) = \underset{\text{coord}}{\gcd}(\kappa(x), \kappa(y)).$$

Let Γ be the distinct values $\gamma(x, y)$ having coordinate product greater than N^2 and a non-arm supporting line, and let $\mathcal{M}$ be the divisibility-maximal elements of Γ . Put

$$\text{Supp}(\mu) = \{x \in X : \mu \preceq \kappa(x)\}, \quad r_\mu = |\text{Supp}(\mu)|.$$

Proposition 48.1 (Maximal supports and the reduced skeleton). *Every repeated non-arm large label divides a maximal top on the same unique non-arm line. For $\mu \in \mathcal{M}$, one has $r_\mu \geq 2$, and every pair of distinct points in $\text{Supp}(\mu)$ has exact top μ . Consequently distinct maximal supports meet in at most one point and*

$$(48.1.1) \quad \sum_{\mu \in \mathcal{M}} \binom{r_\mu}{2} \leq \binom{|X|}{2}.$$

Moreover, the reduced class skeleton consisting of one loop for every kernel class of multiplicity at least two and one edge for every pair of singleton classes is cofinal in Γ , and hence has exactly the same maximal top values.

Proof. A repeated label is contained in the exact gcd top of any two of its support points. Finiteness supplies a maximal top above it. The top is itself a large label, so large-label line uniqueness keeps every comparable top on the original non-arm line. If two points support a maximal top μ , their exact top lies above μ and on that line; maximality forces equality. Two common points of distinct maximal supports would therefore have two different exact tops, proving the intersection and pair count assertions.

For cofinality, consider an exact pair top omitted by the reduced skeleton. At least one endpoint belongs to a class κ of multiplicity at least two. Its retained loop has top κ , which dominates the omitted gcd. Every point in this class supports the omitted large label, so the loop is on the same unique non-arm line and is eligible. A cofinal subset of a finite poset has the same maximal elements: dominate a whole-poset maximum by a subset element, and dominate any whole-poset extension of a subset maximum once more inside the subset; antisymmetry then gives equality. $\square$

Fix a total order on $\mathcal{M}$, and assign every repeated non-arm label λ to the first maximal top $o(\lambda)$ divisible by it. Define

$$\mathcal{O}_\mu = \{\lambda : o(\lambda) = \mu\}, \quad S_\mu = \sum_{\lambda \in \mathcal{O}_\mu} \frac{w_\lambda}{T_\lambda^2}, \quad E_\mu = \sum_{\lambda \in \mathcal{O}_\mu} w_\lambda (n_\lambda - 1)^3.$$

These are exact partitions of S_{non} and E_{non} . If $\lambda \in \mathcal{O}_\mu$, then

$$(48.1.2) \quad n_\lambda \geq r_\mu, \quad T_\lambda \mid T_\mu.$$

The second relation is in this direction: for $d \mid m$, $d/\gcd(d, a) \mid m/\gcd(m, a)$ coordinatewise. It does not permit T_λ^{-2} to be bounded above by T_μ^{-2} .

On the line of μ , let

$$Q(\mu) = \sum_{\substack{d \preceq \mu \\ D_d > N^2}} \frac{w_d}{T_d^2}, \quad B_\mu \geq S_\mu, \quad \mathcal{H}_3 = \sum_{\mu \in \mathcal{M}} \frac{B_\mu}{(r_\mu - 1)^3}.$$

One valid owner-independent choice has

$$(48.1.3) \quad B_\mu = \min \left\{ F(\mu, A_\mu), \frac{C_\mu^2 G_\mu}{N^4}, \frac{C_\mu^2 H(G_\mu)}{N^2} \right\}, \quad S_\mu \leq Q(\mu) \leq B_\mu,$$

with the notation of Theorem 47.4.

Theorem 48.2 (Ownership Cauchy inequality and strict ray closure). *One has*

$$(48.2.1) \quad S_{\text{non}}^2 \leq E_{\text{non}} \mathcal{H}_3.$$

Writing $K = (B + 1)(C + 1)$, the nonempty non-arm domain satisfies the strict bounds

$$(48.2.2) \quad S_{\text{non}} < KN \mathcal{H}_3, \quad E_{\text{non}} < (KN)^2 \mathcal{H}_3.$$

For an empty domain the corresponding weak inequalities hold with both sides zero.

Proof. For each maximal top, catalogue coverage and support containment give

$$S_\mu \leq B_\mu, \quad S_\mu \leq \sum_{\lambda \in \mathcal{O}_\mu} w_\lambda \leq \frac{E_\mu}{(r_\mu - 1)^3}.$$

Multiplication and Cauchy–Schwarz over the finite index set $\mathcal{M}$, rather than over labels or point pairs, prove the first claim. The signed-ray theorem gives termwise $(n_\lambda - 1)^3 T_\lambda^2 < KN$. Positivity of w_λ / T_λ^2 makes the summed inequality $E_{\text{non}} < KN S_{\text{non}}$ strict whenever the exact summation domain is nonempty. In that case $S_{\text{non}} > 0$, and the owner cap of the nonempty owner set also gives $\mathcal{H}_3 > 0$. Combining the two inequalities and dividing by S_{non} proves the strict conclusions. $\square$

Proposition 48.3 (Full-catalogue multiplicity is paid by energy). *The maximal-top catalogues satisfy*

$$(48.3.1) \quad \sum_{\mu \in \mathcal{M}} Q(\mu) \leq E_{\text{non}}.$$

Thus the exact choice $B_\mu = Q(\mu)$ gives $\mathcal{H}_3 \leq \sum_\mu Q(\mu) \leq E_{\text{non}}$.

Proof. For a repeated non-arm label λ , let $m_\lambda = \#\{\mu \in \mathcal{M} : \lambda \preceq \mu\}$. Every divisor term of $Q(\mu)$ is an actual repeated label on the same unique line, so finite summation gives the exact identity

$$\sum_{\mu \in \mathcal{M}} Q(\mu) = \sum_{\lambda \text{ repeated, nonarm}} \frac{w_\lambda}{T_\lambda^2} m_\lambda.$$

Choose one realizing pair for every top counted by m_λ . These pairs lie in the n_λ -point support and are distinct, because one pair has one exact gcd top. Hence $m_\lambda \leq \binom{n_\lambda}{2}$. For $n_\lambda \geq 2$ and $T_\lambda \geq 1$,

$$\frac{1}{T_\lambda^2} \binom{n_\lambda}{2} \leq (n_\lambda - 1)^3.$$

Multiplying by w_λ and summing proves the claim. Finally $r_\mu \geq 2$, so the cubic support denominator only decreases $Q(\mu)$. $\square$

Let $\mathcal{A}_{\text{arm}}$ denote the three proved arm caps and let $\widehat{Q}_{\text{dir}}$ be the sum of the pairwise-coprime direction-lcm envelopes. Exact label containment in those envelopes and the preceding theorem give

$$(48.2.3) \quad E_{\text{sh}} \leq \min\{KN \widehat{Q}_{\text{dir}}, (KN)^2 \mathcal{H}_3\} + \mathcal{A}_{\text{arm}}.$$

Since $J = A_1 + \Omega$, $W = A_0 - \Omega$, and $J^3 \leq W^2 E_{\text{sh}}$, every selected configuration with $W > 0$ obeys

$$(48.2.4) \quad \left(\frac{A_1 + \Omega}{A_0 - \Omega} \right)^3 \leq \min \left\{ KN \frac{\widehat{Q}_{\text{dir}}}{A_0 - \Omega}, (KN)^2 \frac{\mathcal{H}_3}{A_0 - \Omega} \right\} + \frac{\mathcal{A}_{\text{arm}}}{A_0 - \Omega}.$$

This is the ownership-preserving normalized gate.

For completeness, in the canonical consecutive-coefficient setting $1 \leq B$, $C = B + 1$, the arithmetic divisor enumeration gives, for every $\varepsilon > 0$,

$$(48.2.5) \quad \begin{aligned} |\mathcal{M}| &\ll_{\varepsilon} (CN)^{\varepsilon} \min\{N^2, CN^{11/6}\}, \\ \sum_{\mu \in \mathcal{M}} B_{\mu} &\ll_{\varepsilon} K(CN)^{\varepsilon} \min\{N^5, CN^{29/6}\}, \end{aligned}$$

provided $S_{\mu} \leq B_{\mu} \leq F(\mu, A_{\mu})$. On $r_{\mu} - 1 \geq R_0 \geq 1$, the pressure has the further factor R_0^{-4} . Indeed a closest support pair has step q_{μ} satisfying $q_{\mu}(r_{\mu} - 1)L \leq N$; the fixed-direction candidates divide the three integers $q_{\mu}|A_Z|$; and the original, before R -free reduction, coefficient is finite-to-one on each side of the direction square. Here the three direction coefficients are pairwise coprime after removing the primes of $\text{rad}(BC)$: their pairwise gcds divide B , C , and $C - B = 1$. The square-divisor excess estimate then gives $O(CL^{5/6})$ ordinary directions on shell L , and summation gives the displayed exponents.

Certified boundaries. The following statements are refuted only with the premises stated here. All examples are exact finite certificates.

- (1) The implication $T_{\lambda} \geq T_{\mu}$ for every owned label is false under canonical admissibility, non-arm maximal ownership, and full catalogue containment. At $B = 7, C = 8, M = 400, N = 399$, the unique top $\mu = (9, 121, 529)$ has $T_{\mu} = 3$, while its proper owned label $(3, 121, 529)$ has period one; the exact catalogue has two terms and $Q(\mu) = 445280/3$.
- (2) The assertion that one kernel class belongs to at most one maximal top is false under canonical admissibility and global divisibility maximality. At $B = 5, C = 6, M = 254, N = 253$, the singleton class $(1, 841, 43681)$ lies in the two incomparable maximal supports with tops $(1, 841, 121)$ and $(1, 841, 361)$, whose intersection is exactly that one point.
- (3) Raw pair enumeration is not ownership preserving. At $B = 4, C = 5, M = 390, N = 389$, all three pairs of three canonical points have the same top $(49, 361, 9)$. Its support has size three, its period-one catalogue has mass 86184, and equality holds in the local cubic Cauchy bound after the single top is deduplicated.
- (4) Neither $Q(\mu) \leq w_{\mu}$ nor $Q(\mu) \leq 2w_{\mu}$ follows from canonical admissibility, non-arm maximal ownership, and full catalogue coverage. At $B = 55123, C = 55124, M = 3, N = 2$, the two-point top $\mu = (1, 1, 55125)$ has 34 period-one terms,

$$Q(\mu) = 55122, \quad w_{\mu} = 25200, \quad \frac{Q(\mu)}{w_{\mu}} = \frac{9187}{4200} > 2.$$

The explicit three-term cap equals 55125, with inflation $35/16 > 2$.

- (5) Maximality, ownership, cap coverage, $r_{\mu} \geq 2$, and linear support alone imply neither a strict Cauchy saving nor a linear number of tops. The abstract four-vertex construction assigns an edge-specific prime square to each pair, realizes all six pair tops, and has $S_{\text{non}} = E_{\text{non}} = \mathcal{H}_3$. This item is an abstract ledger certificate and does not claim canonical affine realizability.

The first four examples are canonical admissible local selections, but they are not asserted to satisfy the exceptional-point inequality. They refute the listed local implications and no stronger statement that adds exceptional-point geometry. The fifth omits the affine divisibility and powerful-excess filters. None of these certificates closes the affine route.

Open gate. The unresolved task is to prove the strict reverse of

$$(48.2.6) \quad (A_1 + \Omega)^3 \leq (A_0 - \Omega)^2 \left[\min\{KN\widehat{Q}_{\text{dir}}, (KN)^2\mathcal{H}_3\} + \mathcal{A}_{\text{arm}} \right]$$

for every hypothetical exceptional selection, or to obtain an actual standard *abc* counterexample. This requires a canonical low-support supersaturation or catalogue-sparsity theorem coupling the normalized incidence, $\mathcal{H}_3$, and the arm term. The finite scan found no same-line or same-direction multiplicity in its tested range, but these are explicitly finite no-hit observations and are not used as theorems.

Formalization boundary. The companion Lean module proves 24 finite or algebraic theorems and prints the axiom dependencies of all 24. It formalizes finite-poset cofinality, abstract maximal-support and codegree arguments, owner sums, Cauchy aggregation, strict-ray closure, the polynomial normalized gate, abstract tree subtraction, and the numerical cores of the boundary certificates. It does not formalize the affine realization of the maximal cover, the general period-divisibility lemma, the exact catalogue double count and its support-pair injection, the direction Euler products and tails, the spanning-tree realization, the closest-pair shell counts, the arm and weighted Hölder instantiations, or the complete canonical-kernel verification of the finite examples. The analytic bounds above remain paper proofs; the complete finite canonical premises are independently replayed by the accompanying Python certificate.

49. RESIDUE-PARITY LOCALIZATION IN THE BALANCING-PELL PACKET

Write

$$(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}.$$

For odd n these coprime coordinates satisfy $A_n^2 - 2B_n^2 = -1$. The preceding four-prime analysis proves that, at an odd prime index ℓ , the odd-exponent prime sets O_A, O_B obey

$$(360) \quad \prod_{q \in O_A} \prod_{r \in O_B} \left(\frac{q}{r}\right) = \left(\frac{2}{\ell}\right).$$

We now localize the depth-three primes by congruence class.

Multiplication by $1 + \sqrt{2}$ gives

$$A_{n+1} = A_n + 2B_n, \quad B_{n+1} = A_n + B_n.$$

Since

$$(1 + \sqrt{2})^8 = 577 + 408\sqrt{2} \equiv 1 \pmod{8},$$

the coordinate pair is periodic modulo eight. Direct evaluation gives

$$(361) \quad \begin{array}{c|cccc} n & (\text{mod } 8) & 1 & 3 & 5 & 7 \\ \hline A_n & (\text{mod } 8) & 1 & 7 & 1 & 7 \\ B_n & (\text{mod } 8) & 1 & 5 & 5 & 1. \end{array}$$

If an odd prime q divides A_n , the norm equation gives $2B_n^2 \equiv 1 \pmod{q}$, whence $q \equiv 1, 7 \pmod{8}$. If an odd prime r divides B_n , then $A_n^2 \equiv -1 \pmod{r}$, whence $r \equiv 1, 5 \pmod{8}$.

We use the following finite-product observation. If $\rho^2 = 1$ and $\rho \neq 1$ in a commutative monoid, every x_i is 1 or ρ , and

$$\prod_i x_i^{e_i} = \rho,$$

then some $x_i = \rho$ has odd exponent. Otherwise each nontrivial factor would occur to even exponent and the product would be one. If every $e_i \geq 2$, the selected exponent is at least three.

Theorem 49.1 (Forced-residue depth-three localization). *Let ℓ be an odd prime and suppose $A_\ell B_\ell$ is squarefull. Then:*

- (1) *if $\ell \equiv 3, 7 \pmod{8}$, some prime $q_A \mid A_\ell$ occurs to an odd exponent at least three and satisfies $q_A \equiv 7 \pmod{8}$;*
- (2) *if $\ell \equiv 3, 5 \pmod{8}$, some prime $q_B \mid B_\ell$ occurs to an odd exponent at least three and satisfies $q_B \equiv 5 \pmod{8}$.*

Proof. The residue table shows that both coordinates are odd. Reduce the two prime-power factorizations modulo eight. The allowed local classes are $\{1, 7\}$ and $\{1, 5\}$, respectively. In the stated index classes, (361) makes the total product the nontrivial involution 7 or 5. The finite-product observation selects an odd exponent. Since $\gcd(A_\ell, B_\ell) = 1$, squarefullness of $A_\ell B_\ell$ makes every positive exponent in either channel at least two, so the selected odd exponent is at least three. $\square$

For $\ell \equiv 3 \pmod{8}$, both conclusions hold simultaneously. Thus the two previously unspecified deep primes can be required to occupy the 7-class in the A channel and the 5-class in the B channel.

The Lean statement uses an explicit factor-list packet interface. Each entry stores a prime and its exponent; the packet records the allowed local classes and the squarefull lower bound. If the A residue product is 7 or the B residue product is 5, the corresponding list contains the prime and odd exponent asserted above. Thus the formal theorem combines the residue-product hypothesis and the depth conclusion in one statement, while passage from an actual Pell factorization to that interface is the elementary argument in this proof.

Theorem 49.2 (A forced cross-channel nonresidue). *If $\ell \equiv 3, 5 \pmod{8}$ and $A_\ell B_\ell$ is squarefull, then there are odd-exponent primes $q \mid A_\ell$ and $r \mid B_\ell$, each of exponent at least three, such that*

$$\left(\frac{q}{r}\right) = -1.$$

Proof. Here $(2/\ell) = -1$. Equation (360) is a finite product of nonzero signs with value -1 : coprimality of the two channels prevents a zero Legendre symbol. Hence at least one factor is negative. Membership in $O_A \times O_B$ gives odd exponents; squarefullness makes them at least three. $\square$

Here too the Lean statement is combined rather than left as two unrelated lemmas: a flattened factor-list packet stores every pair in $O_A \times O_B$, its two odd squarefull exponents, and its actual Legendre symbol. A negative product yields a stored pair with symbol -1 and both exponents at least three.

The existential quantifier is sharp under the ordinary actual prime-index premises. At $\ell = 11$ one has

$$A_{11} = 8119 = 23 \cdot 353, \quad B_{11} = 5741,$$

with all three factors prime, while

$$\left(\frac{23}{5741}\right) = -1, \quad \left(\frac{353}{5741}\right) = 1.$$

Indeed, $252^2 \equiv 353 \pmod{5741}$, whereas quadratic reciprocity reduces the first symbol to $\left(\frac{14}{23}\right) = -1$. The product is still $-1 = \left(\frac{2}{11}\right)$ because $11 \equiv 3 \pmod{8}$. The factorization shows that $23, 353 \in O_A$ and $5741 \in O_B$. This is a full actual counterexample to the strengthening that every cross pair is a nonresidue. It has three exponent-one factors, so it does not satisfy the squarefull premise and does not close the Pell route.

The companion Lean module checks the integral recurrence and norm, the eight-step formula, all residue classes in (361), the abstract odd-exponent and depth-three extraction, the two explicit factor-list packet theorems, and one combined index-eleven certificate containing the prime-index premise, global character, coordinates, factorization, exponent-one witnesses and Legendre values. The open point remains an actual opposite-channel depth-three exclusion; the new residue and character conditions narrow that packet without yet making it inconsistent.

50. ODD KERNELS AND A THIRD-ORDER BALANCING-PELL PACKET

Continue with $(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}$ and an odd prime index ℓ . Suppose, for pressure testing, that both A_ℓ and B_ℓ are squarefull. A finite squarefull factorization has the canonical form

$$N = D^3 C^2, \quad D = \prod_{v_p(N) \text{ odd}} p.$$

Indeed an even exponent is $2 + 2c$ and an odd exponent is $3 + 2c$. The inherited perfect-power classifications force each exponent vector in a hypothetical packet to have gcd one, so both odd kernels are nontrivial:

$$(362) \quad A_\ell = D_A^3 X^2, \quad B_\ell = D_B^3 Y^2, \quad D_A, D_B > 1.$$

Every kernel prime consequently occurs at depth at least three in its assigned channel. The negative Pell equation becomes

$$D_A^6 X^4 - 2D_B^6 Y^4 = -1.$$

Let $s_\ell = (2/\ell)$. The rank congruences, sign product, and residue orbit from the preceding Pell sections give

$$(363) \quad D_A \equiv 1 \pmod{2\ell}, \quad D_B \equiv s_\ell \pmod{2\ell},$$

$$(364) \quad X^2 \equiv 1 \pmod{2\ell}, \quad Y^2 \equiv 1 \pmod{2\ell}.$$

Because ℓ is prime and the square cores are odd, each of X, Y is congruent to one of ± 1 modulo 2ℓ . The complete mod-eight data are

$\ell \bmod 8$	1	3	5	7
$D_A \bmod 8$	1	7	1	7
$D_B \bmod 8$	1	5	5	1.

Writing $d_A = (D_A - 1)/(2\ell)$ and $d_B = (D_B - s_\ell)/(2\ell)$ further gives

$\ell \bmod 8$	1	3	5	7
$d_A \bmod 4$	0	1	0	1
$d_B \bmod 4$	0	1	3	0.

The two kernels also form a three-edge Jacobi-character triangle with the index:

$$\left(\frac{D_A}{D_B}\right) = s_\ell, \quad \left(\frac{\ell}{D_A}\right) = \left(\frac{-1}{\ell}\right), \quad \left(\frac{\ell}{D_B}\right) = \left(\frac{s_\ell}{\ell}\right).$$

Thus the three signs, in the order displayed, are

$\ell \bmod 8$	1	3	5	7
(D_A/D_B)	+1	-1	-1	+1
(ℓ/D_A)	+1	-1	+1	-1
(ℓ/D_B)	+1	-1	+1	+1.

In the 3 (mod 8) class all three aggregate edges are negative. This does not assert that every individual cross pair is negative; an actual index- eleven factorization already refutes that stronger statement.

One additional quotient digit can be retained. For a finite list T , let $E_j(T)$ be its j th elementary symmetric sum. The exact expansion

$$\prod_i (1 + xt_i) \equiv 1 + xE_1(T) + x^2E_2(T) + x^3E_3(T) \pmod{x^4}$$

follows by induction. Repeat each A -channel quotient according to its valuation to form T_A , and repeat each signed B -channel quotient to form T_B . Write $(K_A, C_A, H_A) = (E_1, E_2, E_3)(T_A)$ and similarly for B . For

$$a = \frac{A_\ell - 1}{2\ell}, \quad b = \frac{s_\ell B_\ell - 1}{2\ell},$$

cancellation of 2ℓ gives

$$a \equiv K_A + 2\ell C_A + 4\ell^2 H_A, \quad b \equiv K_B + 2\ell C_B + 4\ell^2 H_B \pmod{8\ell^3}.$$

Substitution into the exact identity $a - 2b + \ell(a^2 - 2b^2) = 0$ yields the necessary third-order ledger

$$\begin{aligned} 0 \equiv & K_A - 2K_B + \ell(K_A^2 - 2K_B^2) + 2\ell(C_A - 2C_B) \\ & + 4\ell^2(H_A - 2H_B + K_A C_A - 2K_B C_B) \\ & + 4\ell^3(C_A^2 - 2C_B^2) \pmod{8\ell^3}. \end{aligned}$$

Its reduction modulo $4\ell^2$ recovers the preceding second-order ledger. The new congruence is a necessary condition and is not asserted to be a contradiction.

A deterministic two-direction search examined all 668 odd prime indices $3 \leq \ell \leq 5000$ and all primes $q \leq 2,000,000$ in the necessary classes $q \equiv \pm 1 \pmod{2\ell}$. It certified 481 indices with an actual exponent-one divisor of A_ℓ or B_ℓ ; 187 bounded-search rows remain open. An independent verifier recomputed every witness modulo q^2 , checked primality, and, for every odd prime $\ell \leq 191$, verified a simple divisor from the exact coordinates. The largest such witness,

$$13558774610046711780701 \parallel B_{59},$$

is supported by a complete Pocklington certificate. Hence

$$\boxed{A_\ell B_\ell \text{ is not squarefull for every odd prime } \ell \leq 191.}$$

This is a finite theorem. It neither proves eventual simplicity nor treats an unresolved row as a squarefull packet.

The Lean companion checks the cubic-square factor identity, residue transport, Jacobi reciprocity steps, arbitrary-list third-order expansion, quotient cancellation, and the coupled ledger. The remaining positive gate is an actual opposite-channel depth-three exclusion satisfying all kernel, residue, character, and quotient constraints. No full-premise packet counterexample was found, so this route remains active.

51. ALL-ORDER PELL–LUCAS STAIRCASES AND A PAIRED COMPANION SPLITTER

The two balancing-Pell channels admit a useful norm-one splice. Put

$$(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}, \quad \alpha = 3 + 2\sqrt{2}, \quad \beta = 3 - 2\sqrt{2},$$

and let

$$u_0 = 0, \quad u_1 = 1, \quad u_{n+2} = 6u_{n+1} - u_n,$$

with companion $v_0 = 2, v_1 = 6, v_{n+2} = 6v_{n+1} - v_n$. Binet's formulas give

$$(365) \quad u_n = \frac{\alpha^n - \beta^n}{\alpha - \beta} = \frac{B_{2n}}{2} = A_n B_n, \quad v_n = 2A_{2n}, \quad v_n^2 - 32u_n^2 = 4.$$

Thus the norm-one Lucas sequence sees the coprime channel product rather than either channel alone.

We use Zhang's all-orders multiplication formula [90, Proposition 5.1]. This source is an arXiv v1 preprint, so we record the specialization and the additional argument in full. Fix an odd prime ℓ , set

$$A = A_\ell, \quad B = B_\ell, \quad U = u_\ell = AB, \quad \theta = \frac{\ell - 1}{2},$$

and define

$$(366) \quad c_r(\ell) = \frac{\ell \prod_{j=1}^r (\ell^2 - (2j-1)^2)}{4^r (2r+1)!}, \quad a_r = 32^r c_r(\ell) \quad (0 \leq r \leq \theta).$$

The source proves $c_r(\ell) \in \mathbf{Z}_{>0}$ and the exact identity

$$(367) \quad Q_\ell := \frac{u_{\ell^2}}{u_\ell} = \sum_{r=0}^{\theta} a_r U^{2r}.$$

The prime-index rank congruences from the preceding Pell sections imply that every prime $p \mid U$ satisfies

$$(368) \quad p \equiv \left(\frac{2}{p} \right) \pmod{2\ell}, \quad p \geq 2\ell - 1.$$

This lower bound makes every coefficient in (367) a unit on the actual support.

Lemma 51.1 (All multiplication coefficients are support units). *For $0 \leq r \leq \theta$,*

$$\gcd(a_r, U) = 1.$$

Moreover $c_\theta(\ell) = 1$.

Proof. Let $p \mid U$. For $1 \leq j \leq r$, equation (368) gives

$$0 < \ell - (2j - 1) < p, \quad 0 < \ell + (2j - 1) \leq 2\ell - 2 < p.$$

Consequently p divides none of the numerator factors in (366). Since the numerator equals the denominator times the integer $c_r(\ell)$, the prime cannot divide $c_r(\ell)$. It also cannot divide 32. This proves the coprimality.

For the final coefficient write $\ell = 2\theta + 1$. Then

$$\prod_{j=1}^{\theta} (\ell^2 - (2j - 1)^2) = 4^{\theta} (2\theta)!,$$

so the numerator and denominator of c_{θ} coincide. $\square$

For $0 \leq r \leq \theta$, subtract the first r terms and put

$$D_r = Q_{\ell} - \sum_{i < r} a_i U^{2i}, \quad E_r = \sum_{i=r}^{\theta} a_i U^{2(i-r)}.$$

Theorem 51.2 (All-order support-unit staircase). *For every $0 \leq r \leq \theta$,*

$$(369) \quad D_r = U^{2r} E_r, \quad E_r \equiv a_r \pmod{U^2}, \quad \gcd(E_r, U) = 1.$$

Hence, for $p \mid U$ and $e_p = v_p(U)$,

$$(370) \quad v_p(D_r) = 2re_p.$$

At the last step,

$$D_{\theta} = 32^{\theta} U^{\ell-1}.$$

Proof. Factoring U^{2r} from the exact tail proves the first two statements in (369). The leading coefficient is coprime to U by Lemma 51.1; adding a multiple of U^2 preserves that coprimality. This proves the exact valuation. The final formula uses $c_{\theta} = 1$. $\square$

For a hypothetical squarefull packet, every $e_p \geq 2$, and at an odd-kernel prime $e_p \geq 3$. Thus the r th deviation has exact depth at least $4r$, respectively $6r$, at those primes. This is an all-order necessary condition, not a contradiction.

The first tail also couples to the companion. Define

$$W = \frac{u_{\ell^2}/u_{\ell} - \ell}{U^2}, \quad S = \frac{v_{\ell^2} - v_{\ell}}{U^2}.$$

The fourth-order formulas in [90, Corollary 5.2] yield

$$(371) \quad W \equiv \frac{4\ell(\ell^2 - 1)}{3} \pmod{U^2},$$

$$(372) \quad S \equiv 4v_{\ell}(\ell^2 - 1) \pmod{U^2}.$$

In particular W is a support unit.

Theorem 51.3 (Paired companion splitter). *One has*

$$(373) \quad \ell S \equiv 3v_{\ell} W \pmod{U^2},$$

and therefore

$$(374) \quad \ell S \equiv 6W \pmod{A^2}, \quad \ell S \equiv -6W \pmod{B^2}.$$

The factor $6W$ is invertible modulo U^2 . The unique $Z \pmod{U^2}$ satisfying

$$6WZ \equiv \ell S \pmod{U^2}$$

obeys

$$Z \equiv A_{2\ell} \pmod{U^2}, \quad Z \equiv 1 \pmod{A^2}, \quad Z \equiv -1 \pmod{B^2}, \quad Z^2 \equiv 1 \pmod{U^2}.$$

Proof. Multiply (372) by ℓ and (371) by $3v_\ell$; their right sides coincide. Next,

$$\frac{v_\ell}{2} = A_{2\ell} = 2A^2 + 1 = 4B^2 - 1,$$

which gives the opposite channel signs in (374). Every prime divisor of U exceeds $\ell + 1$ and is therefore coprime to $6W$. Cancellation shows that the splitter residue is $A_{2\ell}$. Since $A^2 - 2B^2 = -1$, one has $\gcd(A, B) = 1$; hence its two squared signs combine modulo $A^2B^2 = U^2$ and give the last congruence. $\square$

There is also a precise negative result. A single normalized correction is not rigid. Consider the full-premise claim that every positive odd $k \equiv 1 \pmod{u_3^2}$ satisfies

$$\frac{u_{3k}/u_3 - 1}{u_3^2} \equiv 0 \pmod{5}.$$

It is false. Since $u_3 = 35$, take

$$k = 2451 = 1 + 2 \cdot 35^2.$$

Two independent recurrence computations, one linear and one by binary powering in $\mathbf{Z}[T]/(T^2 - 6T + 1)$, give

$$u_{7353} \equiv 85785 = 35 \cdot 2451 \pmod{214375 = 35(5 \cdot 35^2)}.$$

After the justified divisions, the normalized correction is 2 modulo five. This is an explicit instance of Zhang's local-surjectivity theorem [90, Theorem 5.6]. It retires only the exact fixed-zero subclaim. The all-order staircase and paired companion route remain active because that theorem does not independently prescribe both corrections.

The Lean companion kernel-checks the even-power tail algebra, coefficient support argument, paired congruence, channel signs, modular cancellation, square-root reconstruction, and the local-surjectivity logical boundary. It also checks the displayed recurrence residue by kernel reduction, without `native_decide`. The remaining sufficient target is a correlated opposite-channel depth-three exclusion satisfying this complete staircase, the earlier kernel character triangle, and the third-order factorization ledger. No full-premise packet counterexample was found, so that route is not retired.

52. CORRELATED PELL–LUCAS STAIRCASES AND OPPOSITE-CHANNEL INCIDENCE

We continue the norm-one splice of Section 51. For a norm-one Lucas pair write

$$u_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}, \quad v_n = \alpha^n + \beta^n, \quad \alpha\beta = 1,$$

and let $\delta = (\alpha - \beta)^2$. For a positive odd integer $k = 2\theta + 1$, define

$$(375) \quad d_j(k) = \binom{\theta + j}{2j}, \quad \Psi_k(W) = \sum_{j=0}^{\theta} d_j(k) W^j.$$

Proposition 52.1 (Exact companion multiplication polynomial). *For every $n \geq 1$,*

$$(376) \quad v_{nk} = v_n \Psi_k(\delta u_n^2).$$

Thus the notation v_{nk}/v_n below denotes an exact integer identity and does not assume division in a residue ring.

Proof. In a splitting ring put $x = \alpha^n$ and $W = (x - x^{-1})^2 = \delta u_n^2$. No quotient in the splitting ring is needed. Put $S_m = x^{2m+1} + x^{-(2m+1)}$. The exact Laurent-polynomial recurrence is $S_{m+1} = (W + 2)S_m - S_{m-1}$, with $S_0 = x + x^{-1}$ and $S_1 = (x + x^{-1})(1 + W)$. The polynomials $P_m(W) = \sum_{j=0}^m \binom{m+j}{2j} W^j$ have the same recurrence and initial values, by Pascal's identity. Induction therefore gives the exact factorization $S_m = (x + x^{-1})P_m(W)$ without cancelling a possibly nonunit factor. Taking $m = \theta$ is (376); being symmetric in x, x^{-1} , it descends to the integer identity. $\square$

Let $\Phi_k(W) = \sum c_j(k)W^j$ be the first-sequence multiplication polynomial from [90, Proposition 5.1], where

$$c_j(k) = \frac{k \prod_{h=1}^j (k^2 - (2h-1)^2)}{4^j (2j+1)!}.$$

Theorem 52.2 (Coefficientwise companion correlation). *For $0 \leq j \leq \theta$,*

$$(377) \quad (2j+1)c_j(k) = kd_j(k).$$

Consequently

$$(378) \quad \Phi_k(W) + 2W\Phi'_k(W) = k\Psi_k(W).$$

Proof. Since $k = 2\theta + 1$,

$$\frac{1}{4^j} \prod_{h=1}^j (k^2 - (2h-1)^2) = \frac{(\theta+j)!}{(\theta-j)!}.$$

Substitution gives $c_j(k) = \frac{k}{2^{j+1}} \binom{\theta+j}{2j}$, proving (377). Coefficient comparison then proves (378). $\square$

Specialize to the balancing-Pell pair at an odd prime index ℓ :

$$A = A_\ell, \quad B = B_\ell, \quad U = AB = u_\ell, \quad v_\ell = 2A_{2\ell}, \quad \delta = 32.$$

Put $\theta = (\ell-1)/2$, $a_j = 32^j c_j(\ell)$, $b_j = 32^j d_j(\ell)$ and, for $0 \leq s \leq (\ell-1)/2$, define

$$\mathcal{E}_s(X) = \sum_{j=s}^{\theta} a_j X^{j-s}, \quad \mathcal{F}_s(X) = \sum_{j=s}^{\theta} b_j X^{j-s}, \quad E_s = \mathcal{E}_s(U^2), \quad F_s = \mathcal{F}_s(U^2).$$

Theorem 52.3 (Paired all-order support-unit staircase). *Removing the first s terms from either multiplication polynomial leaves respectively $U^{2s}E_s$ and $U^{2s}F_s$. Moreover*

$$(379) \quad \gcd(E_s, U) = \gcd(F_s, U) = 1, \quad (2s+1)\mathcal{E}_s(X) + 2X\mathcal{E}'_s(X) = \ell\mathcal{F}_s(X),$$

and in particular

$$(380) \quad (2s+1)E_s \equiv \ell F_s \pmod{U^2}.$$

If $p \mid U$ and $e_p = v_p(U)$, both unnormalized tails have exact valuation $2se_p$ at p .

Proof. The tail factorizations are exact by construction. Every support prime $p \mid U$ satisfies $p \geq 2\ell - 1 > \ell$. All factorial arguments in $d_s(\ell) = \binom{(\ell-1)/2+s}{2s}$ are therefore below p , so d_s is a p -adic unit. The first-sequence support-unit argument from the preceding section makes c_s a p -adic unit, and $p \nmid 32$. Hence the leading terms a_s, b_s , and thus E_s, F_s modulo U^2 , are units at every support prime.

Apply (377) term by term. The coefficient of X^{j-s} on the left of the differential identity is $(2s+1+2(j-s))a_j = (2j+1)a_j = \ell b_j$. This proves (379) and its reduction (380) after setting $X = U^2$. The exact valuations follow from the unit leading tails. $\square$

Set $T_s = v_\ell F_s$ and $Z_0 = v_\ell/2 = A_{2\ell}$.

Theorem 52.4 (Coherent every-order channel splitters). *For every s , the congruence*

$$(381) \quad 2(2s+1)E_s Z_s \equiv \ell T_s \pmod{U^2}$$

has the unique solution $Z_s \equiv Z_0 \pmod{U^2}$. Consequently

$$(382) \quad Z_s \equiv 1 \pmod{A^2}, \quad Z_s \equiv -1 \pmod{B^2}, \quad Z_s^2 \equiv 1 \pmod{U^2}.$$

For two levels s, t one has

$$(383) \quad (2t+1)E_t T_s \equiv (2s+1)E_s T_t \pmod{U^2}.$$

If $q^3 \mid A$ and $r^3 \mid B$, every Z_s satisfies

$$(384) \quad Z_s \equiv 1 \pmod{q^6}, \quad Z_s \equiv -1 \pmod{r^6}.$$

Proof. Multiplying (380) by v_ℓ gives

$$\ell T_s \equiv (2s+1)v_\ell E_s = 2(2s+1)E_s Z_0 \pmod{U^2}.$$

Every prime divisor of U exceeds ℓ , so $2(2s+1)E_s$ is invertible modulo U^2 . This proves uniqueness and identifies Z_s . The Pell identities $A_{2\ell} = 2A^2 + 1 = 4B^2 - 1$ give the two signs; coprimality of A, B combines them modulo U^2 . Cross-multiplication at levels s, t and cancellation of ℓ , which is coprime to U , proves (383). Finally $q^3 \mid A$ and $r^3 \mid B$ imply $q^6 \mid A^2$ and $r^6 \mid B^2$. $\square$

The splitters therefore form one coherent projective line; higher orders are not independent tests. The character information can nevertheless be resolved vertex by vertex. In a hypothetical squarefull packet write

$$A = D_A^3 X^2, \quad B = D_B^3 Y^2,$$

with D_A, D_B squarefree, and let O_A, O_B be their prime supports.

Theorem 52.5 (Vertexwise character incidence). *For $r \in O_B$ and $q \in O_A$ one has*

$$(385) \quad \prod_{q \in O_A} \left(\frac{q}{r} \right) = \left(\frac{2}{r} \right), \quad \prod_{r \in O_B} \left(\frac{q}{r} \right) = \left(\frac{B}{q} \right).$$

Both total products equal $(2/\ell)$. If $q \equiv 1 \pmod{8}$, the second endpoint is the quartic-two sign $2^{(q-1)/4} \pmod{q}$. For $\ell \equiv 3$ or $5 \pmod{8}$, there exist $q \in O_A, r \in O_B$ such that

$$(386) \quad v_q(A) \geq 3, \quad v_r(B) \geq 3, \quad \left(\frac{q}{r} \right) = -1, \quad 2\ell \mid q+r,$$

and the every-order signs (384) hold at this same pair.

Proof. For $r \mid B$, the Pell equation gives $A^2 \equiv -1 \pmod{r}$. Every B -channel prime is $1 \pmod{4}$, so Euler's criterion yields $(A/r) = (2/r)$. Removing the square core from $A = D_A^3 X^2$ proves the first product. Removing the square core from B and using quadratic reciprocity, whose sign is trivial because $r \equiv 1 \pmod{4}$, proves the second. Multiplying all rows or all columns gives the same edge product, and the inherited kernel ledger evaluates it as $(2/\ell)$.

If $q \equiv 1 \pmod{8}$, the congruence $2B^2 \equiv 1 \pmod{q}$ implies $(B/q) = 2^{(q-1)/4}$ because this quarter-power is a sign. Finally, in the two stated index classes the inherited table gives $D_B \equiv 5 \pmod{8}$. Thus some $r \in O_B$ is $5 \pmod{8}$ and its row product is negative; one edge (q/r) is negative. The rank congruences are $q \equiv 1 \pmod{2\ell}$ and $r \equiv -1 \pmod{2\ell}$, proving the last divisibility. Squarefull odd exponents are at least three, and Theorem 52.4 supplies the depth-six signs. $\square$

This necessary packet is highly constrained but has not been contradicted. An independently replayed computation proves a separate finite theorem: for every one of the 57 odd prime indices $3 \leq \ell \leq 271$, at least one of A_ℓ, B_ℓ has a certified prime divisor of exponent exactly one. Hence $A_\ell B_\ell$ is not squarefull throughout that range. The verifier checks each divisor, its primality, and the coordinate modulo its square; the large witness at $\ell = 59$ uses a complete Pocklington certificate. It also checks 138,675 coefficient pairs, 228 recurrence evaluations, and the incidence graphs at all 13 prime indices through 43.

A bounded search of 527,352 candidate primes up to $2 \cdot 10^6$ finds $13^2 \parallel B_7$ but no depth-three opposite pair. This is only a finite no-hit and retires no assertion. At index 11, the row at $r = 5741$ contains both $(23/r) = -1$ and $(353/r) = +1$, a full-premise counterexample to the stronger claim that every edge of a negative row is negative. It is not a squarefull packet and therefore does not refute the active exclusion route.

Any surviving packet must simultaneously satisfy the earlier third-order congruence modulo $8\ell^3$, both support-unit staircases, every cross-order determinant, the depth-six signs, and the vertexwise incidence laws. The missing theorem must couple the prime-factor quotients in the third-order ledger to this all-order projective line. Its current absence is a difficult open gate, not evidence against the route.

The Lean companion checks the rational coefficient identity, exact weighted list identity, normalized-tail congruence, every-order recovery, cross-order determinant, sixth-power propagation, negative-row extraction, and the quartic-two power algebra. The Lucas multiplication

theorem, rank theorem, finite factor certificates, and the open uniform exclusion are kept at their stated mathematical or computational boundaries and are not added as axioms.

53. FACTOR-QUOTIENT JETS AND PELL PROJECTIVE CURVATURE

Write

$$(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}, \quad U = A_\ell B_\ell, \quad v = 2A_{2\ell}, \quad x = 2\ell,$$

where ℓ is an odd prime. Put

$$a = \frac{A_\ell - 1}{x}, \quad b = \frac{s_\ell B_\ell - 1}{x}, \quad s_\ell = \left(\frac{2}{\ell}\right).$$

For a prime factor $q \mid A_\ell$, write $q = 1 + xk_q$ and repeat k_q according to its valuation. For $r \mid B_\ell$, write $r = s_r + xh_r = s_r(1 + xs_r h_r)$ and repeat $s_r h_r$. If the first three elementary symmetric functions of the two resulting lists are (K_A, C_A, H_A) and (K_B, C_B, H_B) , respectively, exact finite-product expansion gives

$$(53.1) \quad a \equiv K_A + xC_A + x^2H_A \pmod{x^3}, \quad b \equiv K_B + xC_B + x^2H_B \pmod{x^3}.$$

Three copies of one quotient t have the exact fingerprint

$$(53.2) \quad \begin{aligned} E_1(t, t, t, R) &= 3t + K_0, \\ E_2(t, t, t, R) &= 3t^2 + 3tK_0 + C_0, \\ E_3(t, t, t, R) &= t^3 + 3t^2K_0 + 3tC_0 + H_0, \end{aligned}$$

where (K_0, C_0, H_0) is the residual triple. A depth-three carrier is therefore visible in the third-order ledger even when its actual depth is larger.

The factor quotients also determine a jet of the Lucas companion.

Theorem 53.1 (Two-channel companion jet). *Define*

$$\begin{aligned} V_A(K, C, H) &= 6 + 8xK + x^2(8C + 4K^2) + x^3(8H + 8KC), \\ V_B(K, C, H) &= 6 + 16xK + x^2(16C + 8K^2) + x^3(16H + 16KC). \end{aligned}$$

Then

$$(53.1.1) \quad v \equiv V_A(K_A, C_A, H_A) \pmod{x^4}, \quad v \equiv V_B(K_B, C_B, H_B) \pmod{x^4}.$$

Proof. The negative Pell identity gives $v = 4A_\ell^2 + 2 = 8B_\ell^2 - 2$. Substitute $A_\ell = 1 + xa$ in the first expression and $s_\ell B_\ell = 1 + xb$ in the second. Modulo x^4 , the square of $K + xC + x^2H$ contributes only $K^2 + 2xKC$ after multiplication by x^2 . The displayed polynomials follow by collecting coefficients. $\square$

Let $\theta = (\ell - 1)/2$ and let α_j, β_j be the correlated Lucas coefficients from the preceding section, satisfying

$$(2j + 1)\alpha_j = \ell\beta_j, \quad \alpha_\theta = \beta_\theta = 32^\theta.$$

For the tail polynomials E_j, F_j , put $T_j = vF_j(U^2)$ and define

$$\Delta_{\text{top}} = T_{\theta-1}\ell E_\theta(U^2) - T_\theta(\ell - 2)E_{\theta-1}(U^2).$$

Theorem 53.2 (Exact endpoint curvature). *For every odd prime index,*

$$(53.2.1) \quad \Delta_{\text{top}} = 2v 32^{\ell-1}U^2.$$

Moreover

$$(53.2.2) \quad \gcd(2v32^{\ell-1}, U) = 1, \quad v_p(\Delta_{\text{top}}) = 2v_p(U) \quad (p \mid U).$$

Thus U^2 always divides the determinant and U^3 never does.

Proof. The two top tails are

$$E_{\theta-1} = \alpha_{\theta-1} + \alpha_{\theta}U^2, \quad F_{\theta-1} = \beta_{\theta-1} + \alpha_{\theta}U^2, \quad E_{\theta} = F_{\theta} = \alpha_{\theta}.$$

After substituting $T_j = vF_j$, the constant part of the determinant is

$$v\alpha_{\theta}\{\ell\beta_{\theta-1} - (\ell-2)\alpha_{\theta-1}\} = 0$$

by coefficient correlation. Its U^2 coefficient is $2v\alpha_{\theta}^2 = 2v32^{\ell-1}$, proving the first identity. Odd-index Pell coordinates make U odd, while

$$v^2 - 32U^2 = 4.$$

Every common odd divisor of v and U would divide four. The quotient in the first identity is therefore a unit at every prime of U , proving the valuation statement and sharpness. $\square$

With $\kappa_{\ell} = \Delta_{\text{top}}/U^2 = 2v32^{\ell-1}$, Theorems 53.1 and 53.2 give the direct coupling

$$(53.2.3) \quad \kappa_{\ell} \equiv 232^{\ell-1}V_A(K_A, C_A, H_A) \equiv 232^{\ell-1}V_B(K_B, C_B, H_B) \pmod{16\ell^4}.$$

Thus the third-order factor-quotient ledger determines the 2ℓ -adic jet of the exact unit measuring failure of integral projective collinearity. Since the rank theorem gives $\gcd(2\ell, U) = 1$, the jet modulus and U^2 are coprime; the Chinese remainder theorem combines them without contradiction. A closing argument must relate their values through reciprocity, height, or distribution.

Two full-premise counterexamples delimit stronger local claims. First, the local-only assertion that the forced depth-three quotient data and complete third-order ledger are inconsistent is false. Take

$$\ell = 3, \quad q = 7 = 1 + 6, \quad r = 797 = -1 + 6 \cdot 133,$$

and quotient lists $[1, 1, 1]$ and $[-133, -133, -133]$. They satisfy the required congruence and character table, and the full ledger is divisible by $8\ell^3$. The synthetic values are squarefull of depth three. They do not satisfy the global Pell equation:

$$q^6 - 2r^6 + 1 = -512601560592751008 \neq 0.$$

This retires only the local inconsistency claim.

Second, the genuine Pell index $\ell = 7$ has

$$A_7 = 239, \quad B_7 = 169, \quad U = 40391, \quad v = 228486,$$

and

$$\Delta_{\text{top}} = 800495088185894730989568, \quad \Delta_{\text{top}} \bmod U^3 = 24354030047568 \neq 0.$$

It is a complete counterexample to the proposed U^3 strengthening and an instance of the exact U^2 theorem. Since $239 \parallel A_7$, it is not a counterexample to a hypothetical squarefull Pell packet.

Independent certificates exhibit a prime divisor of exponent one in $A_{\ell}B_{\ell}$ for every one of the 57 odd prime indices $3 \leq \ell \leq 271$. This proves finite nonsquarefullness in that range and nothing beyond it. The active route must still rule out or construct an unbounded family satisfying simultaneously squarefullness, the opposite depth-three nonresidue pair, both factor-quotient jets, the complete third-order ledger, all correlated Lucas tails, exact endpoint curvature, and vertexwise character incidence. No full-premise counterexample to that global statement is known.

54. TRANSVERSE HENSEL SPECIALIZATION IN THE PELL–LUCAS PACKET

Let

$$F_0(T) = 0, \quad F_1(T) = 1, \quad F_{n+2}(T) = TF_{n+1}(T) + F_n(T)$$

and

$$L_0(T) = 2, \quad L_1(T) = T, \quad L_{n+2}(T) = TL_{n+1}(T) + L_n(T).$$

For $(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}$ one has $F_n(2) = B_n$ and $L_n(2) = 2A_n$. Recent work on powerful Fibonacci and Horadam polynomials clarifies polynomial multiplicities [10], while primitive-divisor theory over polynomial rings supplies a complementary global context [11]. Polynomial

squarefreeness does not, by itself, control the valuation of a specialized integer. The results below make that distinction exact.

Put $\Delta = T^2 + 4$. The Binet formulas in the quadratic extension of $\mathbb{Q}(T)$ give

$$(387) \quad L_n(T)^2 - \Delta F_n(T)^2 = 4(-1)^n, \quad L'_n(T) = nF_n(T),$$

and

$$(388) \quad \Delta F'_n(T) = nL_n(T) - TF_n(T).$$

Indeed, for $\alpha = (T + \sqrt{\Delta})/2$ and $\beta = (T - \sqrt{\Delta})/2$, one has $\alpha\beta = -1$, $F_n = (\alpha^n - \beta^n)/(\alpha - \beta)$, and $L_n = \alpha^n + \beta^n$; subtraction and differentiation prove the three identities. At $T = 2$ they become

$$L'_n(2) = nB_n, \quad 4F'_n(2) = nA_n - B_n.$$

Consequently, if ℓ is an odd prime and an odd support prime $p \nmid \ell$ divides B_ℓ , then $F'_\ell(2) \not\equiv 0 \pmod{p}$. Likewise, if $q \nmid \ell$ divides A_ℓ , then $L'_\ell(2) \not\equiv 0 \pmod{q}$. Coprimality of A_ℓ and B_ℓ is the only additional input. Thus every actual repeated support factor is a transverse specialization coincidence rather than a multiple root of the polynomial.

Lemma 54.1 (One-digit Hensel displacement). *Let $f \in \mathbb{Z}[T]$, let p be prime, and suppose $p^e \mid f(t)$ with $e \geq 1$. Then*

$$\frac{f(t + p^e h)}{p^e} \equiv \frac{f(t)}{p^e} + hf'(t) \pmod{p}.$$

If $p \nmid f'(t)$, there is a unique $h \pmod{p}$ for which $p^{e+1} \mid f(t + p^e h)$, namely

$$h \equiv -\frac{f(t)}{p^e} f'(t)^{-1} \pmod{p}.$$

The old representative t persists precisely when $p^{e+1} \mid f(t)$.

Proof. The integral Taylor expansion has the form $f(t+z) = f(t) + zf'(t) + z^2 G(t, z)$. Substituting $z = p^e h$ and using $2e \geq e + 1$ gives the congruence after division by p^e . Its right-hand side is affine with nonzero slope modulo p , proving existence and uniqueness of the displayed digit. Taking $h = 0$ proves the final assertion. $\square$

For a simple support root write

$$\lambda_{f,p,e}(t) = -\frac{f(t)}{p^e} f'(t)^{-1} \pmod{p}.$$

The product $A_\ell B_\ell$ is squarefull exactly when every level-one displacement at the fixed parameter $T = 2$ vanishes in both channels. A support factor of exponent at least three also has zero level-two displacement. This is an exact reformulation, not a proof of the required global exclusion: the remaining problem is to prevent one fixed integer $T = 2$ from agreeing simultaneously with all of the relevant lifted roots.

Local Hensel conditions cannot supply that global prohibition when the parameter is allowed to move. For distinct primes p_i and simple roots $p_i \mid f_i(t_0)$, Lemma 54.1 gives one class modulo p_i^2 for each lifted root, and the Chinese remainder theorem gives a unique simultaneous class modulo $\prod_i p_i^2$. At index three this construction is completely explicit:

$$F_3(T) = T^2 + 1, \quad L_3(T) = T^3 + 3T.$$

The lifts of the F -channel prime 5 and the L -channel prime 7 from $T = 2$ meet at $T = 282 \pmod{1225}$. At the least positive representative,

$$F_3(282) = 79525 = 5^2 \cdot 3181, \\ \frac{L_3(282)}{2} = 11213307 = 3^2 \cdot 7^2 \cdot 47 \cdot 541,$$

and $D = (282^2 + 4)/4 = 19882 = 2 \cdot 9941$ is squarefree. Specializing (387) yields the genuine global point

$$11213307^2 - 19882 \cdot 79525^2 = -1.$$

This is a full counterexample to the precise claim that simple opposite channel roots cannot both become repeated while the parameter moves and the Pell coefficient remains squarefree. It has coefficient 19882, and both coordinates retain exponent-one factors, so it is neither a counterexample to the fixed coefficient-two packet nor to the *abc* conjecture.

There is also an actual fixed-parameter collision:

$$F_7(T) = T^6 + 5T^4 + 6T^2 + 1, \\ F_7(2) = 169 = 13^2, \quad F_7'(2) = 376 \equiv -1 \pmod{13}.$$

The next Hensel digit is one and

$$13^3 \mid F_7(171) = 25006385400373, \quad 13^4 \nmid F_7(171).$$

Thus a simple polynomial root can specialize to an integer with exponent two. This refutes exactly the stronger specialization claim; it does not give a squarefull Pell packet because $A_7 = 239$ occurs only to the first power.

Finally, an exhaustive external search over positive even $T = 2s$ with $s \leq 10^7$ uses the unique representation $N = x^2y^3$ of a positive squarefull integer with y squarefree. It checks 43,355,470 canonical representations. The sole squarefull value in the full F_3 channel is

$$F_3(682) = 465125 = 5^3 \cdot 61^2,$$

whereas

$$\frac{L_3(682)}{2} = 158608307 = 11 \cdot 13 \cdot 31 \cdot 37 \cdot 967$$

is not squarefull. Independent Python and C++ implementations agree. Lean checks the singleton identities, the nonsquarefull witness, the Pell norm, and squarefreeness of $116282 = 2 \cdot 53 \cdot 1097$; the entire 43-million-case enumeration is not yet replayed in the kernel and is therefore reported as an external finite computation.

The original formal companion proves the integer Hensel divisibility core, supplied derivative readouts, index-three Taylor and CRT certificates, and the displayed numerical points. At that stage the all-index identities, general Taylor and simultaneous-CRT theorems, all-support squarefull equivalence, and one bundled moving witness were paper-level obligations. The supplemental Section 55 now proves those exact items in Lean under their displayed hypotheses. It does not supply the fixed- $T = 2$ all-support transversality or the global exclusion described below.

The fixed $D = 2$ route remains open. Its smallest surviving target is to prove that every odd prime index has a support prime with nonzero level-one displacement at $T = 2$, or more narrowly to exclude the opposite-channel level-one/level-two packet after imposing the previously proved rank, character, factor-quotient, all-order-tail, and endpoint-curvature constraints. No counterexample meeting those complete premises is known, so difficulty does not retire the route.

55. ALL-INDEX PELL POLYNOMIALS AND COMPLETE HENSEL STEERING

Author: ChatGPT. This section closes the algebraic formalization gaps left in Section 54. It does not prove the fixed coefficient-two squarefull exclusion and does not prove or disprove the *abc* conjecture. Its purpose is to put the all-index identities, the full polynomial Hensel law, the all-support equivalence, finite simultaneous steering, and the complete moving counterexample under one exact interface.

Let $F_n, L_n \in \mathbb{Z}[T]$ be defined by

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+2} = TF_{n+1} + F_n, \\ L_0 = 2, \quad L_1 = T, \quad L_{n+2} = TL_{n+1} + L_n.$$

Theorem 55.1 (All-index polynomial identities). *For every $n \geq 0$,*

$$(389) \quad L_n^2 - (T^2 + 4)F_n^2 = 4(-1)^n,$$

$$(390) \quad L'_n = nF_n,$$

$$(391) \quad (T^2 + 4)F'_n = nL_n - TF_n.$$

Evaluation at any $t \in \mathbb{Z}$ preserves the two recurrences and all three identities.

Proof. Two-step induction gives

$$(392) \quad L_n = 2F_{n+1} - TF_n, \quad L_{n+1} = TF_{n+1} + 2F_n.$$

Ordinary induction in the F recurrence gives Cassini's identity

$$F_{n+1}^2 - TF_{n+1}F_n - F_n^2 = (-1)^n.$$

Substitution of the first equality in (392) proves (389). Differentiating the L recurrence and using the second equality in (392) proves (390) by two-step induction. For the remaining formula, differentiate the F recurrence and use its two preceding instances together with

$$(T^2 + 4)F_{n+1} = TL_{n+1} + 2L_n,$$

which follows from (392). Polynomial evaluation is a ring homomorphism, so it transports every displayed equality. $\square$

Theorem 55.2 (Polynomial Taylor–Hensel law). *Let $f \in \mathbb{Z}[T]$, let $p, t, c, h \in \mathbb{Z}$, and let $e \geq 1$. For every fixed $z \in \mathbb{Z}$ there is an integer $G = G(f, t, z)$ such that*

$$f(t + z) = f(t) + zf'(t) + z^2G.$$

In particular,

$$f(t + p^e h) \equiv f(t) + p^e h f'(t) \pmod{p^{e+1}}.$$

If $p \neq 0$ and $f(t) = p^e c$, then

$$(393) \quad p^{e+1} \mid f(t + p^e h) \iff p \mid c + h f'(t).$$

If moreover $\gcd(f'(t), p) = 1$, a digit satisfying (393) exists and is unique modulo p .

Proof. The integral Taylor expansion gives the first formula. Substitution of $z = p^e h$ and the inequality $e + 1 \leq 2e$ give the congruence. After writing $f(t) = p^e c$, cancellation of the nonzero factor p^e gives (393). Bézout's identity supplies an admissible digit, and subtracting two admissible affine congruences proves uniqueness. $\square$

Call $N \in \mathbb{N}$ squarefull when every prime $p \mid N$ satisfies $p^2 \mid N$. For a polynomial f , an integer t , and a support prime p , define zero first displacement to mean

$$(394) \quad p \mid f(t), \quad \gcd(f'(t), p) = 1, \quad p^2 \mid f(t).$$

By Theorem 55.2, the final condition is equivalent to the unique next Hensel digit being zero.

Proposition 55.3 (All-support two-channel equivalence). *Let $A, B \in \mathbb{N}$ be coprime, let $f_A(t) = u_A A$ and $f_B(t) = u_B B$, and suppose that each scale u_A, u_B is coprime to every prime in the corresponding support. Assume also that all support roots of f_A, f_B at t are simple. Then AB is squarefull if and only if (394) holds at every support prime in both channels.*

In particular, this applies formally to

$$f_A = L_\ell, \quad u_A = 2, \quad f_B = F_\ell, \quad u_B = 1, \quad t = 2,$$

once the odd-support scale condition and both all-support transversality conditions have been proved.

Proof. For coprime A, B , Euclid's lemma gives

$$AB \text{ squarefull} \iff A \text{ and } B \text{ squarefull}.$$

In each channel, a support prime square divides $u_A N$ if and only if it divides N , since it is coprime to the scale. The asserted equivalence now follows directly from (394). $\square$

Theorem 55.4 (Finite simultaneous Hensel steering). *Let $f_1, \dots, f_m \in \mathbb{Z}[T]$, let $p_1, \dots, p_m$ be distinct primes, and suppose*

$$p_i \mid f_i(t_0), \quad p_i \nmid f'_i(t_0) \quad (1 \leq i \leq m).$$

There is a unique residue class

$$t \pmod{\prod_{i=1}^m p_i^2}$$

such that

$$t \equiv t_0 \pmod{p_i}, \quad p_i^2 \mid f_i(t) \quad (1 \leq i \leq m).$$

Proof. Theorem 55.2 gives one and only one local class modulo p_i^2 . The square moduli are pairwise coprime, so the integer Chinese remainder theorem gives a joint class and uniqueness modulo their product. Polynomial evaluation respects congruence. Conversely, any integer satisfying the displayed base and divisibility conditions lies in each unique local class, hence in the joint CRT class. $\square$

At index three the recurrence gives

$$F_3(T) = T^2 + 1, \quad L_3(T) = T^3 + 3T.$$

The selected roots at $t_0 = 2$ are

$$7 \mid L_3(2), \quad 7 \nmid L'_3(2), \quad 5 \mid F_3(2), \quad 5 \nmid F'_3(2).$$

Their lifted classes are $t \equiv 37 \pmod{49}$ and $t \equiv 7 \pmod{25}$; the explicit joint representative used below is 282.

Proposition 55.5 (Complete moving witness). *Put*

$$t = 282, \quad A = 11213307, \quad B = 79525, \quad D = 19882.$$

Then

$$\begin{aligned} 2A &= L_3(t), & B &= F_3(t), & 4D &= t^2 + 4, \\ A &= 3^2 7^2 47 541, & B &= 5^2 3181, & D &= 2 \cdot 9941, \end{aligned}$$

all displayed factors are prime, D is positive and squarefree, and

$$A^2 - DB^2 = -1.$$

The selected factors satisfy $7^2 \mid A$ and $5^2 \mid B$, while

$$47^2 \nmid A, \quad 541^2 \nmid A, \quad 3181^2 \nmid B.$$

Thus this tuple satisfies every premise of the moving-parameter exclusion and refutes that exact strengthening. It is not a squarefull two-channel packet and has $D \neq 2$.

Proof. All evaluations and factorizations are direct integer identities. Trial division proves the primality assertions; in particular 9941 is prime, so $D = 2 \cdot 9941$ is squarefree. The global norm is the specialization of (389), divided by four. The three failed square divisibilities certify the stated non-squarefull scope. $\square$

The companion Lean module checks each result above. Its structure `HGlobalMoveWitness` contains the prime-index, base-root, simple-derivative, lifted-residue, repeated-divisibility, factorization, primality, positivity, squarefree-coefficient, global-norm, and non-squarefull fields in one object. The associated axiom audit reports only the standard Mathlib principles `propext`, `Classical.choice`, and `Quot.sound`.

The arithmetic route remains open at a precise point: one must discharge the all-support transversality premises for the actual fixed Pell packet and then exclude simultaneous zero displacements at $T = 2$. No counterexample meeting those full fixed-parameter premises has been found, so the route is retained.

56. A MATRIX FROBENIUS ANCHOR AT THE FIXED PELL PARAMETER

Write

$$(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}, \quad F_n(2) = B_n, \quad L_n(2) = 2A_n.$$

The preceding Hensel section left derivative transversality at all actual support primes as an explicit premise. It can be discharged without using a rank-of-aparition theorem.

Set

$$M = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}.$$

Then $M = I + K$, $K^2 = 2I$, and the first column of M^n is $(A_n, B_n)^t$. If $\ell = 2m + 1$ is prime, the intermediate binomial coefficients are divisible by ℓ , so for an integral matrix R ,

$$M^\ell = I + K^\ell + \ell KR = I + 2^m K + \ell KR.$$

Taking the first column proves the following statement.

Theorem 56.1 (Prime-index Frobenius anchor). *For every odd prime $\ell = 2m + 1$ there are integers r_A, r_B such that*

$$A_\ell = 1 + \ell r_A, \quad B_\ell = 2^m + \ell r_B.$$

In particular $\ell \nmid A_\ell B_\ell$.

Proof. Only the last assertion remains. Divisibility of the first coordinate by ℓ would give $\ell \mid 1$. Divisibility of the second would give $\ell \mid 2^m$, hence $\ell = 2$, contrary to oddness. $\square$

The all-index differential identities specialize to

$$L'_\ell(2) = \ell B_\ell, \quad 4F'_\ell(2) + B_\ell = \ell A_\ell.$$

Theorem 56.2 (All-support transversality at $T = 2$). *If $q \mid A_\ell$ is prime, then $\gcd(L'_\ell(2), q) = 1$. If $p \mid B_\ell$ is prime, then $\gcd(F'_\ell(2), p) = 1$.*

Proof. The coordinates are coprime. In the first channel, q divides neither ℓ nor B_ℓ , so the first differential identity proves the claim. In the second, p divides neither ℓ nor A_ℓ and is odd. Reducing the second identity modulo p gives $4F'_\ell(2) \equiv \ell A_\ell \not\equiv 0 \pmod{p}$. $\square$

For a simple support root let $Z_1(f, t, p)$ mean that $p \mid f(t)$, $\gcd(f'(t), p) = 1$, and $p^2 \nmid f(t)$. The one-digit Hensel law identifies this with zero first displacement.

Theorem 56.3 (Exact fixed-parameter gate). *For every odd prime ℓ ,*

$$A_\ell B_\ell \text{ is squarefull} \iff \begin{cases} Z_1(L_\ell, 2, q) & (q \mid A_\ell), \\ Z_1(F_\ell, 2, p) & (p \mid B_\ell), \end{cases}$$

where both quantifiers range over all rational support primes.

Proof. The preceding theorem supplies simplicity at every support prime. The factor 2 in $L_\ell(2) = 2A_\ell$ is a unit at every such prime, and the two coordinates are coprime. Thus the square divisibilities in Z_1 are exactly the defining square divisibilities for squarefullness of the two coordinates. $\square$

This theorem instantiates the earlier conditional all-support statement; it does not prove either side. The universal exclusion of simultaneous zero displacements is exactly the still-open prime-index squarefull exclusion.

The stronger assertion that no individual support displacement vanishes is false. At $\ell = 7$,

$$A_7 = 239, \quad B_7 = 169 = 13^2, \quad F'_7(2) = 376 \equiv -1 \pmod{13}.$$

Hence $Z_1(F_7, 2, 13)$ holds with every premise. This does not refute the all-support exclusion because 239 has exponent one.

An independent bounded replay examined all odd prime $\ell \leq 20000$ and all primes $q \leq 10^7$ in the necessary classes $q = \pm 1 \pmod{2\ell}$. Its 3,091,963 candidate tests found only the same valuation-two hit $(7, 13, B)$, no valuation-three hit, and no opposite-channel repeated pair. The

789 indices without an in-bound simple certificate remain unresolved, as do all larger support primes. The finite no-hit is not an asymptotic theorem.

Lean proves Theorems 56.1–56.3. It also proves that the surviving zero-displacement exclusion is equivalent to squarefull exclusion, together with the complete index-seven refutation of the stronger local assertion. The standard *abc* conjecture and the fixed-Pell squarefull exclusion both remain open.

57. SIGNED TRACE PROJECTORS FOR A PRIME-INDEX PELL SQUAREFULL GATE

Author: ChatGPT.

Summary. Let $(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}$. We give an exact signed-trace encoding of the still open assertion that $A_\ell B_\ell$ is not squarefull at an odd prime index. At every odd index,

$$A_{2n} - 1 = 2A_n^2, \quad A_{2n} + 1 = 4B_n^2.$$

Thus squared support in either coordinate is equivalent to fourth-power support in the corresponding adjacent trace factor. We also identify a canonical Chinese-remainder projector $E_n = (A_{2n} + 1)/2 = 2B_n^2$ whose exact idempotent defect is $2(A_n B_n)^2$, and show that one explicit Newton correction raises its precision from $(A_n B_n)^2$ to $(A_n B_n)^4$.

An exhaustive computation over odd prime $\ell \leq 800000$ and support prime $q \leq 2000000$, independently replayed by matrix powering, finds exactly two depth-two support collisions, one in each channel. Both give signed-trace depth exactly four and therefore disprove several tempting strengthenings. The index-seven row is nonsquarefull because its opposite coordinate is the exponent-one prime 239; the global squarefull status of the large row is unresolved. Neither row is a counterexample to the *abc* conjecture. All algebraic claims and the small counterexample are kernel checked in Lean.

57.1. The Pell coordinates and the unresolved gate. Define nonnegative integers A_n, B_n by

$$(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}.$$

Equivalently, $(A_0, B_0) = (1, 0)$ and

$$(A_{n+1}, B_{n+1}) = (A_n + 2B_n, A_n + B_n).$$

The norm identity is

$$(395) \quad A_n^2 - 2B_n^2 = (-1)^n.$$

For an integer $N > 0$, *squarefull* means that $p \mid N$ implies $p^2 \mid N$ for every prime p .

The present work begins with the following fixed-parameter result from the companion transversality analysis.

Proposition 57.1 (fixed-parameter gate). *Let ℓ be an odd prime. Every support prime of A_ℓ and B_ℓ is transverse at the Pell polynomial parameter $T = 2$. Furthermore, all first Hensel displacements vanish if and only if $A_\ell B_\ell$ is squarefull.*

Proposition 57.1 is recalled as an input, rather than silently being folded into a computation. Classical rank-of-apparition theory for Lucas sequences supplies the necessary candidate classes $q \equiv \pm 1 \pmod{2\ell}$ used below; primitive-divisor results such as [34] provide the broader historical setting. The exclusion of a squarefull product at every odd prime index remains open.

57.2. Depth doubling in adjacent signed trace factors.

Lemma 57.2 (addition law). *For all $m, n \geq 0$,*

$$\begin{aligned} A_{m+n} &= A_m A_n + 2B_m B_n, \\ B_{m+n} &= A_m B_n + B_m A_n. \end{aligned}$$

Proof. Multiply $(A_m + B_m\sqrt{2})(A_n + B_n\sqrt{2})$ and compare the rational and $\sqrt{2}$ coefficients. This is also an induction using the defining recurrence. $\square$

Put $Z_n = A_{2n}$. Taking $m = n$ in Lemma 57.2 gives

$$Z_n = A_n^2 + 2B_n^2, \quad B_{2n} = 2A_nB_n.$$

Theorem 57.3 (signed trace-square identities). *If n is odd, then*

$$Z_n - 1 = 2A_n^2, \quad Z_n + 1 = 4B_n^2.$$

Proof. For odd n , equation (395) gives $2B_n^2 = A_n^2 + 1$. Substitute this identity into $Z_n = A_n^2 + 2B_n^2$. $\square$

Theorem 57.4 (exact depth dictionary). *Let n be odd.*

(1) *If q is an odd prime dividing A_n , then*

$$q^4 \mid Z_n - 1 \iff q^2 \mid A_n.$$

(2) *If q is an odd prime dividing B_n , then*

$$q^4 \mid Z_n + 1 \iff q^2 \mid B_n.$$

Moreover, coordinate depth exactly two gives signed-trace depth exactly four.

Proof. Use Theorem 57.3. The factors 2 and 4 are coprime to an odd prime. Unique factorization gives both fourth-power equivalences; the same comparison at powers two, three, four, and five gives the exact depth-two-to-four assertion. $\square$

For odd integers A, B and a trace value Z satisfying the two identities of Theorem 57.3, define

$$\text{ST}_4(A, B, Z) \iff \begin{cases} q^4 \mid Z - 1 & \text{for every prime } q \mid A, \\ q^4 \mid Z + 1 & \text{for every prime } q \mid B. \end{cases}$$

Corollary 57.5 (exact signed-packet gate). *For every odd prime ℓ ,*

all fixed- $T = 2$ first Hensel displacements vanish

$$\iff A_\ell B_\ell \text{ is squarefull}$$

$$\iff \text{ST}_4(A_\ell, B_\ell, A_{2\ell}).$$

Proof. The first equivalence is Proposition 57.1. The coordinates are coprime, so their product is squarefull exactly when both coordinates are squarefull. Apply Theorem 57.4 to every support prime. $\square$

Corollary 57.6 (radical compression). *If both coordinates at an odd index are squarefull, then*

$$\text{rad}(A_n)^4 \mid Z_n - 1, \quad \text{rad}(B_n)^4 \mid Z_n + 1,$$

and

$$\text{rad}(A_n B_n)^4 \mid (Z_n - 1)(Z_n + 1) = 8(A_n B_n)^2.$$

Proof. Squarefullness gives $\text{rad}(A_n)^2 \mid A_n$ and $\text{rad}(B_n)^2 \mid B_n$. Square these divisibilities and use Theorem 57.3. Coprimality combines the radicals, while direct multiplication gives the displayed product identity. $\square$

The last divisibility is compatible with squarefullness and is not a contradiction. Its content is the placement of the two complete support packets in adjacent factors of one trace coordinate.

57.3. The exact projector and its Newton continuation. Fix an odd index and abbreviate $A = A_n$, $B = B_n$, $U = AB$, and $Z = Z_n$. Define

$$E = \frac{Z + 1}{2} = 2B^2 = A^2 + 1.$$

Theorem 57.7 (channel projector). *The integer E satisfies*

$$E \equiv 1 \pmod{A^2}, \quad E \equiv 0 \pmod{B^2},$$

and has exact defect

$$(396) \quad E^2 - E = 2U^2.$$

Thus E is idempotent modulo U^2 . If $U \geq 3$, it is not idempotent modulo U^3 .

Proof. The congruences follow from the two expressions for E . Their product gives

$$E(E - 1) = 2B^2A^2 = 2U^2.$$

If U^3 divided the last expression, then U would divide 2, which is impossible for $U \geq 3$. $\square$

The sharpness of the raw projector does not terminate the construction. For integers e, d , set

$$\mathcal{N}(e, d) = e - (2e - 1)d.$$

Theorem 57.8 (one Newton step). *If $e^2 - e = d$, then*

$$\mathcal{N}(e, d)^2 - \mathcal{N}(e, d) = d^2(4d - 3).$$

For $d = 2U^2$, the corrected projector is congruent to e modulo U^2 and is idempotent modulo U^4 .

Proof. Write $z = 2e - 1$. Since $z^2 = 4d + 1$, expansion yields

$$\begin{aligned} (e - zd)^2 - (e - zd) &= d - z^2d + z^2d^2 \\ &= d^2(4d - 3). \end{aligned}$$

The correction zd is divisible by U^2 , and the new defect contains $d^2 = 4U^4$. $\square$

57.4. Complete-premise finite counterexamples. The search enumerates every odd prime $\ell \leq 800000$ and every prime $q \leq 2000000$ in the necessary classes $q \equiv \pm 1 \pmod{2\ell}$. It computes (A_ℓ, B_ℓ) modulo q^2 , and recomputes every repeated hit modulo q^5 . The producer uses binary powering in $\mathbb{Z}[\sqrt{2}]$. A verifier independently powers

$$\begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}^\ell.$$

A separate certificate checks primality by complete trial division and recomputes each collision by a third, linear-recurrence algorithm.

Proposition 57.9 (replayed finite search). *In the stated rectangle there are 63950 prime indices, 764366 candidate-prime tests, and 12356 actual support hits. Exactly two support hits have depth at least two. They are the rows in Table 1; both have coordinate depth exactly two and signed trace depth exactly four. No depth-three hit and no index with repeated support in both channels occurs in the rectangle.*

TABLE 1. The complete repeated-support rows. The two quotient columns are nonzero residues modulo q ; C_ℓ denotes the indicated channel and $S_\ell = A_{2\ell} \mp 1$ its signed trace shift.

ℓ	q	ch.	C_ℓ/q^2	$C'_\ell(2)$	S_ℓ/q^4
7	13	B	1	12	4
773231	1546463	A	1090979	326969	1407445

Proof. The exhaustive producer and independent matrix replay agree on every count and on both complete rows. For the larger row, complete trial division through 879 and 1243 proves the primality of ℓ and q , respectively. Three orbit algorithms agree modulo $q^5 = 8844996565598309452666138088543$. The nonzero residues in Table 1 prove $q^2 \parallel C_\ell$, transversality of the fixed parameter, and $q^4 \parallel A_{2\ell} \mp 1$. $\square$

The first row can also be read without software:

$$B_7 = 169 = 13^2, \quad A_{14} + 1 = 114244 = 4 \cdot 13^4.$$

It follows that $13^5 \nmid A_{14} + 1$.

Corollary 57.10 (refuted strengthenings). *Each of the following universal statements is false:*

- (1) *no support prime repeats at an odd prime index;*

- (2) *repetition is confined to the A channel;*
- (3) *repetition is confined to the B channel;*
- (4) *the minimal representative $q = 2\ell \pm 1$ forces simple support;*
- (5) *coordinate depth two always promotes to signed-trace depth five.*

Proof. The two rows of Table 1 meet every premise of the relevant claims. The first supplies the B -channel counterexample, the second the A -channel counterexample, and both use the minimal residue representative. Exact trace depth four refutes the last assertion. $\square$

Remark 57.11 (logical boundary). The two rows are local collisions. Neither row says that both coordinates at its index are squarefull. Consequently neither refutes the signed-packet exclusion in Corollary 57.5, and neither is an *abc* counterexample. Bounded absence of a simultaneous row is recorded only as a finite observation.

57.5. Formal and computational record. The Lean module `PellSignedTraceProjector 20260903.lean` formalizes the addition and doubling laws, Theorems 57.3–57.8, the radical consequences, and the complete index-seven refutation. It compiles with `-DwarningAsError=true`. Its companion audit prints the axioms of every declaration. The union consists only of the standard imported library axioms

`propext,      Classical.choice,      Quot.sound.`

The reproducibility bundle contains the producer, matrix verifier, complete-premise collision certifier, JSON outputs, exit codes, compiler logs, environment metadata, and SHA-256 manifests. No finite no-hit is promoted to a global theorem.

57.6. Remaining problem. The prime-index squarefull gate is now the existence problem for two fourth-power support packets in the adjacent factors $A_{2\ell} - 1$ and $A_{2\ell} + 1$, together with their exact product and one explicit Newton correction step. The depth-two collisions prove that neither a single channel nor one additional trace digit can close the problem. A successful continuation must couple all support primes across both adjacent factors, or construct a genuine simultaneous packet. Since no complete-premise counterexample to that global exclusion is presently known, the route remains open.

58. PRIME-LAYER RADICAL BOUNDS AND THE CORRECTED MERSENNE ORDER LEDGER

Put $M_m = 2^m - 1$ and

$$W_m = \frac{M_m}{\text{rad}(M_m)}.$$

For every odd prime q , let $d_q = \text{ord}_q(2)$ and $w_q = v_q(2^{d_q} - 1)$. Define the base excess of the exact-order block d by

$$E_d = \prod_{\substack{q \text{ prime} \\ d_q = d}} q^{w_q - 1}.$$

The index-lifting contribution must be kept separate:

$$B_m = \prod_{d|m} E_d, \quad L_m = \prod_{\substack{q \text{ prime} \\ q|M_m}} q^{v_q(m/d_q)}.$$

Theorem 58.1 (Correct exact-order decomposition). *For every $m \geq 1$,*

$$(397) \quad W_m = L_m B_m = L_m \prod_{d|m} E_d.$$

Moreover $L_m \mid m$, so $\log L_m \leq \log m = o(m)$.

Proof. If $q \mid M_m$, then $d_q \mid m$, and odd-prime LTE gives

$$v_q(M_m) = w_q + v_q(m/d_q).$$

After one copy of q is removed into the radical, the first summand supplies q^{w_q-1} in the unique block E_{d_q} and the second supplies its factor in L_m . Multiplying over the prime support proves (397). Since $d_q \mid q-1$, one has $q \nmid d_q$ and therefore $v_q(m/d_q) = v_q(m)$. Thus L_m takes from m only the full powers of a subset of its prime divisors, proving $L_m \mid m$. $\square$

The factor L_m cannot be deleted. At $m = 6$,

$$M_6 = 63 = 3^2 \cdot 7, \quad \text{rad}(M_6) = 21, \quad W_6 = 3.$$

Here $d_3 = 2$, $w_3 = 1$, $d_7 = 3$, and $w_7 = 1$, so the base product is one, whereas $L_6 = 3$. This is a complete counterexample to the formula $W_m = \prod_{d \mid m} E_d$ without an index-lifting term. It supports the corrected decomposition rather than refuting the order-block route.

At a prime index the lifting factor disappears, and the radical admits an elementary uniform lower bound.

Theorem 58.2 (Composite prime-layer square carrier). *Let ℓ be an odd prime. If M_ℓ is composite, then it has at least two distinct prime divisors, every prime divisor q satisfies*

$$\text{ord}_q(2) = \ell, \quad 2\ell \mid q-1, \quad q \geq 2\ell+1,$$

and consequently

$$(398) \quad (2\ell+1)^2 \leq \text{rad}(M_\ell), \quad (2\ell+1)^2 E_\ell \leq \Phi_\ell(2).$$

If one prime factor is at least H , the square in (398) may be replaced by $H(2\ell+1)$.

Proof. First, $2^\ell - 1$ is not a nontrivial prime power. Indeed, if $2^\ell - 1 = q^v$, then an even v contradicts reduction modulo eight. If $v \geq 3$ is odd, the factorization

$$q^v + 1 = (q+1)(q^{v-1} - q^{v-2} + \cdots - q + 1)$$

has an odd second factor strictly greater than one, impossible for a power of two. Hence a composite value has two distinct prime factors.

For a prime divisor q , its order divides the prime ℓ and is not one, so it equals ℓ . Lagrange's theorem and oddness of q give $2\ell \mid q-1$. Two distinct factors therefore contribute at least $(2\ell+1)^2$ to the radical. Since $L_\ell = 1$ and $\text{rad}(M_\ell)E_\ell = M_\ell = \Phi_\ell(2)$, the excess form follows. If one factor is at least H , combine it with the distinct factor at least $2\ell+1$. $\square$

Writing $P(N)$ for the greatest prime factor of N , Erdős and Shorey prove $P(2^\ell - 1) \gg \ell \log \ell$ for prime ℓ [15]; the later account of Ford, Luca, and Shparlinski records the same uniform input [16]. The asymmetric part of Theorem 58.2 therefore yields

$$(399) \quad \frac{E_\ell}{\Phi_\ell(2)} \ll \frac{1}{\ell^2 \log \ell}$$

for every sufficiently large prime ℓ , including the prime M_ℓ case trivially. This is only a polynomial saving. It leaves

$$\log E_\ell \leq \ell \log 2 - 2 \log \ell - \log \log \ell + O(1),$$

whereas the Mersenne endpoint needs $\log E_\ell = o(\ell)$.

The corrected global ledger identifies a clean sufficient target.

Proposition 58.3 (Base-block totient criterion). *If*

$$\log E_d = o(\varphi(d)),$$

then $\log W_m = o(m)$ as $m \rightarrow \infty$.

Proof. Put $e_d = \log E_d \geq 0$. Given $\varepsilon > 0$, choose D so that $e_d \leq \varepsilon \varphi(d)$ for $d \geq D$, and put $C_D = \sum_{d < D} e_d$. The divisor identity $\sum_{d \mid m} \varphi(d) = m$ gives

$$\log B_m = \sum_{d \mid m} e_d \leq C_D + \varepsilon m.$$

Theorem 58.1 now gives $\log W_m \leq \log m + C_D + \varepsilon m$. Divide by m , let m tend to infinity, and then let ε tend to zero. $\square$

Proposition 58.4 (Cyclotomic reformulation). *For every $d > 1$,*

$$(400) \quad E_d = \frac{\Phi_d(2)}{\text{rad}(\Phi_d(2))}, \quad \log \Phi_d(2) = \varphi(d) \log 2 + O(1),$$

where the absolute value of the $O(1)$ term is at most two. If V_d is the powerful part of $\Phi_d(2)$, then

$$E_d \leq V_d \leq E_d^2.$$

Hence the hypothesis of Proposition 58.3 is equivalent both to

$$\log \text{rad}(\Phi_d(2)) = \varphi(d) \log 2 + o(\varphi(d))$$

and to $\log V_d = o(\varphi(d))$.

Proof. Murty and Wong record the unconditional cyclotomic classification that a prime divisor of $\Phi_d(a, b)$ is 1 (mod d), except possibly for the largest prime factor of d , and that the exceptional divisor occurs to at most the first power [17]. Pomerance gives the same classification directly for $\Phi_d(2)$ and records the sharper bounds $2^{\varphi(d)-1} \leq \Phi_d(2) < 2^{\varphi(d)+1}$ [19]. For $(a, b) = (2, 1)$ the first class consists of exact-order- d primes; the exceptional class disappears on division by the radical. This proves the first identity in (400). The cyclotomic product formula gives

$$\log \Phi_d(2) = \varphi(d) \log 2 + \sum_{e|d} \mu(d/e) \log(1 - 2^{-e}).$$

Since $-\log(1 - x) \leq 2x$ for $0 \leq x \leq 1/2$, the last sum has absolute value at most $\sum_{e \geq 1} 2^{1-e} = 2$. A prime power q^a , $a \geq 2$, contributes q^{a-1} to E_d and q^a to V_d ; now $a - 1 \leq a \leq 2(a - 1)$. Taking logarithms proves the equivalences. $\square$

The reformulation permits an unconditional removal of all sufficiently small repeated-prime support. Put

$$\begin{aligned} \mathcal{W}_d &= \{q : \text{ord}_q(2) = d, v_q(2^d - 1) \geq 2\}, \\ T_d &= \prod_{q \in \mathcal{W}_d} q, \quad D_d = \prod_{\substack{\text{ord}_q(2)=d \\ v_q(2^d-1) \geq 3}} q^{v_q(2^d-1)-2}. \end{aligned}$$

Thus $E_d = T_d D_d$.

Proposition 58.5 (Near-quadratic tail reduction). *Set*

$$Y_d = \frac{\varphi(d)^2}{\log \log(3d)}, \quad T_d^{\leq Y} = \prod_{\substack{q \in \mathcal{W}_d \\ q \leq Y_d}} q.$$

Then

$$(401) \quad \log T_d^{\leq Y} = o(\varphi(d)).$$

If $\log E_d = o(\varphi(d))$ fails, then there are some $\epsilon > 0$ and a sequence $d_j \rightarrow \infty$ such that, for each fixed $\delta > 0$, there is a subsequence on which at least one of the following persists:

- (1) $\log D_{d_j} \geq \epsilon \varphi(d_j)/4$;
- (2) the interval $Y_{d_j} < q \leq d_j^{2+\delta}$ contains at least

$$\frac{\epsilon \varphi(d_j)}{4(2 + \delta) \log d_j}$$

distinct base-two Wieferich primes, all of exact order d_j ;

(3)

$$\sum_{\substack{\text{ord}_q(2)=d_j, v_q(2^{d_j}-1) \geq 2 \\ q > d_j^{2+\delta}}} \log q \geq \epsilon \varphi(d_j)/4.$$

Every prime in the third arm has $\text{ord}_q(2) < q^{1/(2+\delta)}$ and hence belongs to the zero-density exceptional set in the Erdős–Murty almost-all order theorem [18]. That density theorem does not control the weighted mass on a sparse sequence.

Proof. Exact order d implies $q \equiv 1 \pmod{d}$. Fix $\theta \in (1/2, 1)$. The uniform bound $\varphi(d) \gg d/\log \log(3d)$ gives $d < Y_d^\theta$ for all sufficiently large d . The weighted Brun–Titchmarsh inequality of Murty–Séguin [20, Theorem 2.4] then gives

$$\log T_d^{\leq Y} \leq \sum_{\substack{q \leq Y_d \\ q \equiv 1 \pmod{d}}} \log q \leq \frac{2Y_d \log Y_d}{\varphi(d) \log(Y_d/d)} = o(\varphi(d)),$$

which proves (401). If the target fails, choose d_j with $\log E_{d_j} \geq \epsilon \varphi(d_j)$. Split $E_d = T_d D_d$ into the small factor, the two displayed support ranges, and D_d . For large j the small logarithm is at most $\epsilon \varphi(d_j)/4$; one of the other three nonnegative logarithms is therefore at least $\epsilon \varphi(d_j)/4$. Infinite pigeonhole fixes an arm. In the transition range every summand is at most $(2 + \delta) \log d_j$, giving the asserted cardinality. In the extreme range $q > d_j^{2+\delta}$ and exact order give $d_j < q^{1/(2+\delta)}$. The final literature comparison follows from the almost-all lower order bound of Erdős and Murty. $\square$

Corollary 58.6 (A fixed fraction of the square budget). *Let $s_d = |\mathcal{W}_d|$ and put*

$$B_d = \frac{(\varphi(d) + 1) \log 2}{2 \log(d + 1)}.$$

Then $s_d \leq B_d$. On the transition alternative of Proposition 58.5,

$$\frac{|\{q \in \mathcal{W}_d : Y_d < q \leq d^{2+\delta}\}|}{B_d} \geq \frac{\epsilon}{2(2 + \delta) \log 2} \frac{\varphi(d)}{\varphi(d) + 1} \frac{\log(d + 1)}{\log d}.$$

Thus that alternative consumes a fixed positive proportion of the maximum support allowed by square divisibility alone.

Proof. Every $q \in \mathcal{W}_d$ is at least $d + 1$, while $T_d^2 \mid \Phi_d(2)$. Pomerance’s bound $\Phi_d(2) < 2^{\varphi(d)+1}$ [19] therefore gives

$$s_d \log(d + 1) \leq \log T_d \leq \frac{1}{2} \log \Phi_d(2) < \frac{1}{2} (\varphi(d) + 1) \log 2.$$

Divide the transition-cardinality bound in the preceding proposition by B_d . $\square$

This is a necessary structure, not a contradiction. One very deep lift can still carry the first arm, and no unconditional theorem excludes the fixed-proportion cluster in the second arm.

Because $q \nmid (q - 1)/d_q$, LTE also gives

$$v_q(2^{q-1} - 1) = v_q(2^{d_q} - 1) = w_q.$$

Thus T_d is exactly the same-order Wieferich support and D_d is its super-Wieferich excess. A prime enters only one canonical block. Finiteness of base-two super-Wieferich primes would remove the first arm eventually but would leave the two support tails. Current fixed-base results, including Fellini–Murty’s finite-super implication, do not provide a weighted estimate inside one exact-order block [39]. Under uniformly bounded canonical valuations—in particular, under finiteness of base-two super-Wieferich primes—Murty–Séguin’s bound for the largest prime is only at a polynomial scale [20]. Pomerance’s unconditional results establish many composite primitive parts; his stronger distinct-factor theorem is *abc*-conditional [19]. Neither conclusion yields exponential radical saturation on the $\varphi(d)$ scale.

Proposition 58.7 (A nontrivial canonical block). *The canonical exact-order block at $d = 364$ is nontrivial; more precisely,*

$$1093 \mid E_{364}.$$

Proof. Direct integer arithmetic gives

$$1093^2 \mid 2^{364} - 1, \quad 1093^3 \nmid 2^{364} - 1,$$

and

$$2^{182} \equiv 1092, \quad 2^{52} \equiv 27, \quad 2^{28} \equiv 121 \pmod{1093}.$$

The number 1093 is prime and $364 = 2^2 \cdot 7 \cdot 13$. The prime-divisor criterion for multiplicative order therefore gives $\text{ord}_{1093}(2) = 364$: exponent 364 kills 2, while division by any prime divisor of 364 does not. The first two divisibilities show that the valuation at 1093 is exactly two. Thus the defining product for E_{364} contains $1093^{2^{-1}}$. $\square$

Proposition 58.7 is a complete counterexample to the strengthening $E_d = 1$ for every d . It does not contradict $\log E_d = o(\varphi(d))$, because a single fixed block has no effect on the limit at infinity.

Two further exact witnesses delimit tempting universal strengthenings:

$$2^{37} - 1 = 223 \cdot 616318177$$

has exactly two prime factors, refuting the claim that every composite prime layer with $\ell \geq 37$ has at least three; and

$$2^{11} - 1 = 23 \cdot 89, \quad \text{rad}(M_{11}) = 2047 < (2 \cdot 11 + 1)^3$$

refutes a cubic radical replacement valid at every composite odd-prime layer. These witnesses do not refute an eventual three-factor estimate, a weighted base-block theorem, or Proposition 58.3. A deterministic scan of all prime indices $3 \leq \ell \leq 61$ found no repeated factor; that is recorded only as a finite no-hit.

Lean verifies the elementary prime-layer theorems, the three prime-layer witnesses, and the finite totient-tail inequality. Companion modules additionally verify exact-order LTE, $L_m \mid m$, the finite product $W_m = L_m \prod_{d \mid m} E_d$, equality of relative and canonical blocks on divisors, the exact logarithmic divisor-sum identity, and the conditional asymptotic conclusion of Proposition 58.3, as well as the nontrivial canonical-block witness $1093 \mid E_{364}$. The finite core of the powerful-part comparison, the deep/transition/extreme mass trichotomy, and both the realized-support and exact ambient square-budget ratio implications is checked in a fifth module. That module takes the cyclotomic identification, the Brun–Titchmarsh small-arm estimate, and the prime-order distribution theorem as explicit paper inputs; it does not assert them as Lean theorems. The cited largest-prime-factor theorem remains an external published input, and $\log E_d = o(\varphi(d))$ remains an open hypothesis. The lifting remainder and the passage from that hypothesis are now kernel-checked.

59. THE WEIGHTED SMALL-ORDER TAIL

This section sharpens the third alternative in Proposition 58.5. It also records a logical limitation of the almost-all order theorem: density in the prime-size variable does not control logarithmic mass in a moving exact-order fibre. None of the results in this section closes the Mersenne route unconditionally.

For an odd prime p , write

$$f(p) = \text{ord}_p(2), \quad \alpha_p = v_p(2^{f(p)} - 1).$$

The valuation formula of Murty–Séguin and the intrinsic-prime classification give, for $d > 1$,

$$(402) \quad \log E_d = \sum_{f(p)=d} (\alpha_p - 1) \log p = \log T_d + \log D_d.$$

Here T_d contains one copy of every p with $f(p) = d$ and $\alpha_p \geq 2$, while D_d contains the remaining depth. In particular,

$$(403) \quad T_d^2 \mid \Phi_d(2), \quad \log T_d < \frac{\varphi(d) + 1}{2} \log 2.$$

Proposition 59.1 (Global order-index budget). *For $Y \geq 2$,*

$$\sum_{\substack{p \text{ odd prime} \\ f(p) \leq Y}} \alpha_p \log p \leq \sum_{d \leq Y} \log \Phi_d(2) = O(Y^2).$$

Consequently

$$\#\{p \text{ odd prime} : f(p) \leq Y\} = O(Y^2 / \log Y).$$

Proof. If $f(p) = d$, the full power p^{α_p} belongs to the exact-order part of $\Phi_d(2)$. The order is unique, so contributions belonging to different d do not overlap. Sum the logarithms and use $\Phi_d(2) < 2^{\varphi(d)+1}$ from [19]. For the counting assertion, split the primes at Y^2 ; every prime above the split contributes at least $2 \log Y$. $\square$

This is a quadratic cumulative estimate. Restricting it to a single order cannot produce the required $o(\varphi(d))$ bound.

Proposition 59.2 (Weighted consequence of the almost-all order theorem). *Fix $\delta > 0$ and set*

$$\mathcal{B}_\delta = \{p : f(p) \leq p^{1/(2+\delta)}\}.$$

Then

$$\#\{p \leq x : p \in \mathcal{B}_\delta\} = o(x / \log x), \quad \sum_{\substack{p \leq x \\ p \in \mathcal{B}_\delta}} \log p = o(x).$$

Proof. Erdős–Murty prove that, for every positive $\eta(p) \rightarrow 0$, outside $o(\pi(x))$ primes one has $f(p) > p^{1/(2+\eta(p))}$ [18]. Take $\eta(p) < \delta$ eventually. If $A_\delta(x)$ denotes the exceptional count, Abel summation gives

$$\sum_{\substack{p \leq x \\ p \in \mathcal{B}_\delta}} \log p = A_\delta(x) \log x - \int_2^x \frac{A_\delta(t)}{t} dt = o(x).$$

$\square$

At the beginning of the extreme arm, $x = d^{2+\delta}$, the last estimate need not be $o(\varphi(d))$. The weighted Brun–Titchmarsh theorem [20, Theorem 2.4] gives a little-oh precisely in the range

$$x = o(\varphi(d)^2), \quad d < x^\theta, \quad \frac{\log x}{\log(x/d)} = O(1)$$

for a fixed $\theta < 1$. It gives only the natural $O(\varphi(d))$ scale when $x \asymp \varphi(d)^2$. Thus neither result covers the extreme range.

In a dyadic shell $(X, 2X]$, the available inputs reduce to

$$(404) \quad \sum_{\substack{p \in \mathcal{W}_d \\ X < p \leq 2X}} \log p \leq \min \left\{ o(X), O\left(\frac{X \log X}{\varphi(d) \log(X/d)}\right), O(\varphi(d)) \right\}.$$

The final term is (403); at every relevant scale it is the only generally effective one, and it has no strict saving. Across larger X , the available theorems furnish no uniform $o(\varphi(d))$ saving, so the known shell estimates do not create the missing little-oh when summed.

Proposition 59.3 (Coarse information does not imply a fibrewise bound). *There exist primes p_j and integers $d_j \rightarrow \infty$ such that*

$$p_j \equiv 1 \pmod{d_j}, \quad d_j < p_j^{1/(2+\delta)}, \quad \#\{j : p_j \leq x\} = O(\log \log x),$$

for every fixed $\delta > 0$, while positive numbers ϕ_j satisfy

$$\log p_j \asymp \phi_j, \quad 2 \log p_j < (\phi_j + 1) \log 2.$$

Hence global zero density, the congruence forced by exact order, the small-order scale, and the numerical square budget do not by themselves imply $o(\phi_j)$ mass in a moving labelled fibre.

Proof. Take $d_j = 2^j$, $\phi_j = \varphi(d_j) = 2^{j-1}$, fix $0 < c < \log 2/2$, and put $X_j = \exp(c\phi_j)$. Since $d_j = O(\log X_j)$, the classical Siegel–Walfisz theorem supplies, for large j , a prime

$$p_j \in [X_j, 2X_j], \quad p_j \equiv 1 \pmod{d_j}.$$

The logarithms grow geometrically, proving the count. Moreover,

$$\sum_{p_j \leq x} \log p_j = O(\log x \log \log x) = o(x),$$

so even the global logarithmic mass is sparse. Exponential growth gives the small-order inequality, and $\log p_j = c\phi_j + O(1)$ gives the stated linear fibre mass. Finally, $\Phi_{2^j}(2) = 2^{\phi_j} + 1$, and eventually

$$2 \log p_j < \phi_j \log 2 < \log \Phi_{2^j}(2),$$

which proves the asserted numerical square budget. $\square$

In Proposition 59.3, assigning abstract labels $\tilde{f}(p_j) = d_j$ and $\tilde{\alpha}_{p_j} = 2$ makes every fibre a singleton of linear logarithmic mass. These labels are not asserted to equal the actual multiplicative order and valuation. The proposition therefore refutes only an inference from the four coarse estimates; it is not an arithmetic counterexample to the tail and does not close the route.

Proposition 59.4 (Summable-profile shell localization). *Suppose a finite nonnegative mass satisfies $H \leq M \leq \sum_{k \in I} M_k$, where $H > 0$. If $a_k \geq 0$ and $\sum_{k \in I} a_k \leq 1$, then some $k \in I$ satisfies*

$$M_k \geq H a_k.$$

Proof. Otherwise $\sum_k M_k < H \sum_k a_k \leq H$, a contradiction. $\square$

For

$$L_d = (2 + \delta) \log d, \quad \mathcal{S}_{d,k} = \{p \in \mathcal{W}_d : 2^k L_d < \log p \leq 2^{k+1} L_d\},$$

take $a_k = 6/(\pi^2(k+1)^2)$. An extreme mass at least $\epsilon\varphi(d)/4$ forces, for some k ,

$$\sum_{p \in \mathcal{S}_{d,k}} \log p \geq \frac{3\epsilon}{2\pi^2(k+1)^2} \varphi(d)$$

and consequently

$$(405) \quad |\mathcal{S}_{d,k}| \geq \frac{3\epsilon\varphi(d)}{2\pi^2(k+1)^2 2^{k+1} (2 + \delta) \log d}.$$

A summable-envelope upper bound of the reverse form, with a common coefficient tending to zero, would close the extreme arm.

There is also a useful concentration split. At any logarithmic height $h > 0$, a mass at least H either has mass at least $H/2$ above h , or contains at least $H/(2h)$ points below h . The square budget bounds the number of points above h by $O(\varphi(d)/h)$. Thus a failed tail has either a macroscopic component of rank $O(\varphi(d)/h)$ or an increasingly diffuse submacroscopic component. The first rank is bounded when h is chosen comparable to $\varphi(d)$.

The pointwise block estimate is stronger than the endpoint actually requires. Put

$$B_m = \prod_{d|m} E_d, \quad A(m) = \log B_m = \sum_{d|m} \log E_d.$$

The exact lifting identity and $1 \leq L_m \leq m$ give

$$W_m = L_m B_m, \quad \log W_m = A(m) + \log L_m, \quad \log L_m = o(m).$$

Theorem 59.5 (Exact divisor-average endpoint). *One has*

$$\log W_m = o(m) \iff \sum_{d|m} \log E_d = o(m).$$

Proof. From left to right, subtract the actual LTE term $\log L_m = o(m)$ from $\log W_m = \log B_m + \log L_m$. From right to left, add that same term. Both steps use the exact product identity. $\square$

The quantifier in both little-oh statements is $\forall \epsilon > 0 \exists M \forall m \geq M$. The identity holds for every positive m , and $L_m \mid m$ is the LTE lifting input that proves $0 \leq \log L_m \leq \log m = o(m)$. Thus both directions include the complete lifting remainder. In particular, the reverse implication is not inferred from a product inequality: it uses $W_m = L_m B_m$ and the negligible lifting factor.

Let s_d be the support mass below $Y_d = \varphi(d)^2 / \log \log(3d)$, and write

$$\log E_d = s_d + r_d,$$

where

$$(406) \quad r_d = \sum_{\substack{f(p)=d, \alpha_p \geq 2 \\ p > Y_d}} \log p + \sum_{f(p)=d} (\alpha_p - 2)_+ \log p.$$

The proved small-arm estimate is $s_d = o(\varphi(d))$.

Corollary 59.6 (Divisor-average remainder criterion). *The Mersenne endpoint follows from*

$$\forall \epsilon > 0 \exists M \forall m \geq M, \quad \sum_{d \mid m} r_d \leq \epsilon m.$$

Proof. The identity

$$\sum_{d \mid m} \varphi(d) = m$$

converts the small-arm little-oh into

$$\sum_{d \mid m} s_d = o(m).$$

Adding the displayed hypothesis and applying Theorem 59.5 completes the proof. $\square$

This criterion is strictly weaker than $r_d = o(\varphi(d))$. To see strictness, select a rapidly increasing divisibility chain of squarefree primorials $n_1 \mid n_2 \mid \dots$ such that

$$\sum_{i < k} n_i = o(n_k), \quad \varphi(n_k)/n_k \longrightarrow 0,$$

and define $r_{n_k} = \varphi(n_k)$ and $r_d = 0$ otherwise. Pointwise $o(\varphi(d))$ fails along n_k . If n_k is the largest selected primorial dividing m , then

$$\frac{1}{m} \sum_{d \mid m} r_d \leq \frac{\varphi(n_k) + \sum_{i < k} n_i}{n_k} = o(1).$$

When the largest such k remains bounded, the numerator is fixed and the ratio again tends to zero. The example is abstract and asserts nothing about the actual Mersenne remainders. The divisor-average criterion opens a second analytic attack on the tail, through divisor lattices rather than uniform control of every fibre.

For a fixed $\delta > 0$, a pointwise proof still reduces to the three nonnegative gates

$$\sum_{f(p)=d} (\alpha_p - 2)_+ \log p = o(\varphi(d)),$$

$$\#\{p \in \mathcal{W}_d : Y_d < p \leq d^{2+\delta}\} = o(\varphi(d)/\log d),$$

and

$$\sum_{\substack{p \in \mathcal{W}_d \\ p > d^{2+\delta}}} \log p = o(\varphi(d)).$$

The current literature supplies none of these three estimates in the required uniform form. The exact divisor-average criterion and the summable-profile localization are formalized in `MersenneWeightedOrderTail20260901`; all analytic inputs remain external to the Lean kernel.

60. EXACT-ORDER BLOCKS

Let

$$M_m = 2^m - 1, \quad W_m = \frac{M_m}{\text{rad}(M_m)}.$$

For an odd prime q , put

$$(407) \quad d_q = \text{ord}_q(2), \quad w_q = v_q(2^{d_q} - 1).$$

Since $d_q \mid q-1$ and $q \nmid (q-1)/d_q$, the odd-prime lifting-the-exponent formula gives

$$(408) \quad v_q(2^{q-1} - 1) = w_q.$$

Thus $w_q \geq 2$ is precisely the base-two Wieferich condition, while $w_q \geq 3$ is the base-two super-Wieferich condition.

For each $d \geq 1$, define the canonical exact-order block and its two depth components by

$$(409) \quad E_d = \prod_{d_q=d} q^{w_q-1}, \quad T_d = \prod_{\substack{d_q=d \\ w_q \geq 2}} q,$$

$$(410) \quad S_d^{(3)} = \prod_{\substack{d_q=d \\ w_q \geq 3}} q, \quad D_d = \prod_{\substack{d_q=d \\ w_q \geq 3}} q^{w_q-2}.$$

All these products are finite: every prime in the order- d fibre divides $2^d - 1$. Comparing the exponent of each prime gives the exact identity

$$(411) \quad E_d = T_d D_d.$$

The factor T_d records one copy of every repeated exact-order prime; D_d records only the multiplicity beyond the second copy.

For completeness, the order blocks recover the whole Mersenne power loss after an elementary lifting factor is retained. Set

$$(412) \quad L_m = \prod_{q \mid M_m} q^{v_q(m/d_q)}.$$

Here $d_q \mid m$ whenever $q \mid M_m$.

Proposition 60.1 (Exact Mersenne ledger). *For every $m \geq 1$,*

$$(413) \quad W_m = L_m \prod_{d \mid m} E_d, \quad L_m \mid m.$$

Consequently $0 \leq \log L_m \leq \log m = o(m)$.

Proof. If $q \mid M_m$, then $d_q \mid m$, and odd-prime LTE gives

$$v_q(M_m) = w_q + v_q(m/d_q).$$

After the radical removes one copy of q , the first term contributes q^{w_q-1} to the unique block E_{d_q} , and the second contributes its factor to L_m . Multiplication over the prime support proves the first identity in (413). Moreover, $d_q \mid q-1$ implies $q \nmid d_q$, hence $v_q(m/d_q) = v_q(m)$. The product L_m therefore takes the full q -primary component of m for only a subset of the prime divisors of m , and so $L_m \mid m$. $\square$

The lemmas of Murty and Séguin classify prime powers in cyclotomic values [20, Lemmas 4.2–4.3]. At the unique exact-order index they correspond to Wieferich and super-Wieferich primes. Their results do not give a weighted upper bound within a moving exact-order fibre. Fellini and Murty obtain quantitative lower bounds for non-Wieferich prime ideals under a number-field *abc* conjecture or under finiteness of the relevant super-Wieferich primes [39, Theorems 1.2–1.4]. Neither hypothesis supplies an unconditional estimate for the fixed-base depth masses studied below.

61. THE EXACT DEPTH LAYER CAKE

For $j \geq 3$, define the logarithmic depth layer

$$(414) \quad \Theta_{d,j} = \sum_{\substack{d_q=d \\ w_q \geq j}} \log q.$$

Proposition 61.1 (Finite layer-cake identity). *For every d ,*

$$(415) \quad \log D_d = \sum_{j \geq 3} \Theta_{d,j}.$$

The sum is finite. More precisely, if $M \geq w_q$ for every prime in the order- d super-Wieferich support, then the right-hand side may be truncated at $j = M$.

Proof. A fixed prime q contributes $(w_q - 2) \log q$ to $\log D_d$. It occurs once in every layer with $3 \leq j \leq w_q$, and therefore contributes

$$\sum_{j=3}^{w_q} \log q = (w_q - 2) \log q.$$

The prime support is finite, and every valuation is finite. Interchanging the resulting two finite sums and then summing the displayed identity over q proves (415). The same calculation proves the stated finite truncation. $\square$

The identity distinguishes the first super-Wieferich layer $\Theta_{d,3} = \log S_d^{(3)}$ from the accumulated mass at higher depths. A small first layer alone does not control the full sum unless the high-depth layers are controlled as well.

62. MOVING-DEPTH TRUNCATION

For an integer $K \geq 3$, put

$$(416) \quad R_d(K) = \sum_{\substack{d_q=d \\ w_q \geq 3}} (w_q - K)_+ \log q, \quad (x)_+ = \max\{x, 0\}.$$

Proposition 62.1 (Exact truncation and support bound). *For every $K \geq 3$,*

$$(417) \quad \log D_d = \sum_{\substack{d_q=d \\ w_q \geq 3}} \min\{w_q - 2, K - 2\} \log q + R_d(K),$$

$$(418) \quad \log D_d \leq (K - 2) \log S_d^{(3)} + R_d(K).$$

Moreover,

$$(419) \quad R_d(K) = \sum_{j \geq K+1} \Theta_{d,j}.$$

Proof. For integers $w \geq 0$ and $K \geq 2$, truncated subtraction gives

$$(420) \quad w - 2 = \min\{w - 2, K - 2\} + (w - K)_+$$

whenever $w \geq 2$. Multiplying (420) by the nonnegative number $\log q$ and summing over the super-Wieferich support proves (417). The first coefficient is at most $K - 2$; summing that pointwise inequality yields (418). Finally, $w - K$ is the number of integers j with $K + 1 \leq j \leq w$. The finite sum interchange used in Proposition 61.1 therefore proves (419). $\square$

Corollary 62.2 (Frequency–depth dichotomy). *If $K > 2$ and $\log D_d \geq A$, then*

$$(421) \quad \frac{A}{2(K - 2)} \leq \log S_d^{(3)} \quad \text{or} \quad \frac{A}{2} \leq R_d(K).$$

Proof. If the first alternative fails, then $(K - 2) \log S_d^{(3)} < A/2$. Combining this strict inequality with (418) and $\log D_d \geq A$ forces $R_d(K) > A/2$. $\square$

Thus a large deep factor cannot be explained merely by the existence of some super-Wieferich primes. At every chosen threshold it forces either large one-copy support mass or large mass above that threshold.

63. SUFFICIENT ASYMPTOTIC GATES

All little-oh statements in this section are taken along positive integers tending to infinity.

Theorem 63.1 (Moving-threshold gate). *Let $K_d \geq 3$ be an integer-valued threshold. If*

$$(422) \quad (K_d - 2) \log S_d^{(3)} = o(\varphi(d)),$$

$$(423) \quad R_d(K_d) = o(\varphi(d)),$$

then

$$(424) \quad \log D_d = o(\varphi(d)).$$

If, in addition, $\log T_d = o(\varphi(d))$, then

$$(425) \quad \log \frac{2^m - 1}{\text{rad}(2^m - 1)} = o(m).$$

Proof. Apply (418) with $K = K_d$. Its two nonnegative right-hand terms are little-oh of $\varphi(d)$ by (422)–(423), proving (424). The exact identity (411) then gives $\log E_d = o(\varphi(d))$ under the additional hypothesis.

For clarity, the last implication uses only the standard divisor identity

$$\sum_{d|m} \varphi(d) = m.$$

Given $\varepsilon > 0$, choose D so that $\log E_d \leq \varepsilon \varphi(d)$ for $d \geq D$, and put $C_D = \sum_{d < D} \log E_d$. Then

$$\sum_{d|m} \log E_d \leq C_D + \varepsilon m.$$

Proposition 60.1 and $\log L_m = o(m)$ now imply (425). □

The two hypotheses in Theorem 63.1 are sufficient, not necessary. In particular, a uniform depth bound makes (423) vanish for a fixed sufficiently large threshold but does not by itself prove the support estimate (422). Conversely, sparsity of $S_d^{(3)}$ does not control an unbounded high-depth tail.

The actual Mersenne endpoint admits a weaker divisor-average interface.

Proposition 63.2 (Exact divisor-average endpoint). *One has*

$$(426) \quad \log W_m = o(m) \iff \sum_{d|m} \log E_d = o(m).$$

Proof. Taking logarithms in (413) gives

$$\log W_m = \log L_m + \sum_{d|m} \log E_d.$$

All terms are nonnegative and $\log L_m = o(m)$. Adding or subtracting this negligible lifting term proves the two implications in (426). □

Let s_d be a controlled part of the one-copy repeated support and let U_d be the remaining part, so that

$$(427) \quad \log T_d = s_d + U_d, \quad s_d, U_d \geq 0.$$

In the Mersenne order-block route, s_d is the Brun–Titchmarsh small arm and U_d collects the surviving one-copy support tails.

Theorem 63.3 (Divisor-average threshold gate). *Assume*

$$(428) \quad \sum_{d|m} s_d = o(m).$$

Let $K_d \geq 3$. If

$$(429) \quad \sum_{d|m} \left(U_d + (K_d - 2) \log S_d^{(3)} + R_d(K_d) \right) = o(m),$$

then (425) holds.

Proof. Equations (411) and (427), together with (418), give the termwise nonnegative bound

$$\log E_d \leq s_d + U_d + (K_d - 2) \log S_d^{(3)} + R_d(K_d).$$

Sum over $d \mid m$ and apply (428)–(429). The right-hand side is $o(m)$, so Proposition 63.2 yields (425). $\square$

If $s_d = o(\varphi(d))$, then (428) follows from $\sum_{d|m} \varphi(d) = m$. Likewise, separate pointwise $o(\varphi(d))$ estimates for the three terms in (429) imply that divisor-average condition. No converse is asserted: a divisor average can tolerate large values at sparse orders. Theorem 63.3 is therefore the weaker interface and is often the more appropriate target.

64. AN EXACT ODD-ORDER COUNTEREXAMPLE

Factorizations involving $2^{d/2} \pm 1$ may suggest restricting repeated exact-order primes to even d . The second classical base-two Wieferich prime rules out that restriction.

Proposition 64.1 (The prime 3511). *The integer 3511 is prime and*

$$(430) \quad \text{ord}_{3511}(2) = 1755, \quad v_{3511}(2^{1755} - 1) = 2.$$

In particular, the exact order is odd.

Proof. First, $3511 - 1 = 3510 = 2 \cdot 3^3 \cdot 5 \cdot 13$. The congruences

$$7^{3510} \equiv 1 \pmod{3511}$$

and

r	2	3	5	13
$7^{3510/r} \bmod 3511$	3510	2754	1800	494
$\gcd(7^{3510/r} - 1, 3511)$	1	1	1	1

give a Pocklington certificate for the primality of 3511: the known factor is $F = 3510 > \sqrt{3511}$.

Direct modular exponentiation gives

$$(431) \quad 2^{1755} \equiv 1 \pmod{3511^2},$$

$$(432) \quad 2^{1755} \equiv 21954602502 \not\equiv 1 \pmod{3511^3},$$

$$(433) \quad 2^{585} \equiv 756 \pmod{3511},$$

$$(434) \quad 2^{351} \equiv 1578 \pmod{3511},$$

$$(435) \quad 2^{135} \equiv 88 \pmod{3511}.$$

The first congruence shows that the order divides $1755 = 3^3 \cdot 5 \cdot 13$. The last three congruences exclude division of the order by each prime divisor 3, 5, and 13; hence the order is exactly 1755. Finally, (431)–(432) show that its canonical valuation is exactly two. $\square$

Proposition 64.1 refutes the precise universal statement

$$(436) \quad q^2 \mid 2^{\text{ord}_q(2)} - 1 \implies 2 \mid \text{ord}_q(2).$$

The scope is narrow. Since the valuation at 3511 is two, this prime is absent from $S_{1755}^{(3)}$ and contributes nothing to D_{1755} . It is not a super-Wieferich counterexample and does not contradict any condition in Theorem 63.1 or Theorem 63.3.

65. FINITE COMPUTATIONAL AUDIT

A finite scan enumerated every prime at most 10^7 , tested $2^{q-1} \equiv 1 \pmod{q^2}$, computed the exact order for every hit, and then increased the prime-power modulus until the congruence first failed. The generator used a full Eratosthenes sieve. An independent verifier used a segmented sieve and recomputed every Fermat congruence, order certificate, and depth certificate.

The two enumerations agree on all 664,579 primes in the interval. Their complete list of base-two Wieferich hits is

q	$\text{ord}_q(2)$	w_q	order parity
1093	364	2	even
3511	1755	2	odd

Thus the interval contains no base-two super-Wieferich hit.

This is strictly a finite audit. The absence of a depth-three hit below 10^7 proves neither finiteness nor density zero and supplies no asymptotic estimate for $S_d^{(3)}$, $R_d(K)$, or D_d . It is used only to check the explicit certificate in Section 64 and the implementation of the depth definitions.

66. REMAINING ANALYTIC GATE

The exact identities reduce the deep obstruction without estimating it. A weaker decomposed sufficient target is to choose moving thresholds $K_d \geq 3$ and prove

$$(437) \quad \sum_{d|m} \left(U_d + (K_d - 2) \log S_d^{(3)} + R_d(K_d) \right) = o(m),$$

while retaining the already controlled divisor-average small arm. A stronger pointwise route would instead establish

$$\log T_d = o(\varphi(d)), \quad (K_d - 2) \log S_d^{(3)} = o(\varphi(d)), \quad R_d(K_d) = o(\varphi(d)).$$

Neither Murty–Séguin nor Fellini–Murty supplies these fixed-base, weighted exact-order estimates unconditionally. The truncation bound is a one-way sufficient mechanism, and no necessity claim is made. The super-Wieferich depth route therefore remains open.

67. TOTIENT-WEIGHTED CONCENTRATION AT THE MERSENNE ENDPOINT

The exact order-block identity of Theorem 59.5 changes the remaining Mersenne problem from a pointwise estimate into a divisor-average problem. This section identifies the natural probability measure on that divisor lattice and localizes the unresolved arithmetic mass to divisors extremely close to the ambient index.

For an integer $m \geq 1$ and a nonnegative sequence a_d put

$$A(m) = \sum_{d|m} a_d, \quad \mu_m(d) = \frac{\varphi(d)}{m} \quad (d | m).$$

The identity $\sum_{d|m} \varphi(d) = m$ makes μ_m a probability measure. Assume throughout the boundedness condition

$$(438) \quad 0 \leq a_d \leq C\varphi(d)$$

with one constant $C \geq 0$, and define

$$B_\epsilon(m) = \sum_{\substack{d|m \\ a_d > \epsilon\varphi(d)}} \varphi(d).$$

Theorem 67.1 (Exact totient exceptional-mass criterion). *Under (438),*

$$A(m) = o(m) \iff B_\epsilon(m) = o(m) \quad \text{for every fixed } \epsilon > 0.$$

Proof. For every positive ϵ one has the finite inequalities

$$\epsilon B_\epsilon(m) \leq A(m) \leq \epsilon m + C B_\epsilon(m).$$

The first implication follows from the lower inequality. For the reverse implication, given $\eta > 0$, apply the hypothesis with $\epsilon = \eta/2$ and require $B_\epsilon(m) \leq \eta m/(2(C+1))$. $\square$

The divisor probability has an exact logarithmic moment. If $m = \prod_{p^a \parallel m} p^a$ and D is distributed according to μ_m , then

$$(439) \quad \mathbb{E}_{\mu_m} \log \frac{m}{D} = \sum_{p^a \parallel m} \frac{1 - p^{-a}}{p - 1} \log p.$$

Indeed, the exponent $J_p = v_p(D)$ has probability $\varphi(p^j)/p^a$, and the tail-sum identity gives $\mathbb{E}(a - J_p) = \sum_{r=1}^a p^{-r}$.

Theorem 67.2 (Moving near-diagonal concentration). *There is an absolute C_0 such that*

$$\mathbb{E}_{\mu_m} \log \frac{m}{D} \leq C_0 \log \log(3m).$$

Consequently, if $H(m) > 0$ eventually and $H(m)/\log \log(3m) \rightarrow \infty$, then

$$\frac{1}{m} \sum_{\substack{d \mid m \\ d \leq m \exp(-H(m))}} \varphi(d) \rightarrow 0.$$

Proof. Bound (439) by $\sum_{p \mid m} \log p/(p-1)$. Enlarge the absolute constant to absorb the finitely many small m , and assume $z = \log m > 1$. Split at z . Chebyshev's estimate $\vartheta(x) = O(x)$ bounds every dyadic shell below z by an absolute constant, giving $O(\log z)$ in total. Above z the sum is at most $\log \text{rad}(m)/(z-1) = O(1)$. Markov's inequality at height $H(m)$ proves the assertion. $\square$

The growth condition on H cannot uniformly be replaced by $H = C \log \log m$ with fixed C . For the primorial $m_y = \prod_{p \leq y} p$, the complementary divisor m_y/D contains each $p \leq y$ independently with probability $1/p$. Choose an integer $r > C+1$ and disjoint intervals $0 < u_1 < v_1 < \dots < u_r < v_r < 1$ with $\sum_i u_i > C+1$. For each i , let $A_i(y)$ be the event that m_y/D contains a prime in $(y^{u_i}, y^{v_i}]$. Mertens's prime-product theorem [21] gives

$$\Pr(A_i(y)^c) = \prod_{y^{u_i} < p \leq y^{v_i}} (1 - 1/p) \rightarrow u_i/v_i.$$

The events are independent because their intervals are disjoint, hence $\liminf_y \Pr(\bigcap_i A_i(y)) = \prod_i (1 - u_i/v_i) > 0$. On this intersection $m_y/D > y^{C+1}$; Chebyshev's two-sided bounds for $\vartheta(y) = \log m_y$ imply $y^{C+1} > (\log m_y)^C$ eventually. Therefore, for every fixed $C > 0$,

$$\liminf_{y \rightarrow \infty} \mu_{m_y} \left\{ D \leq \frac{m_y}{(\log m_y)^C} \right\} > 0.$$

This is a full-premise counterexample to the proposed uniform strengthening for arbitrary bounded masses (take $a_d = \varphi(d)$). It is not a counterexample to the actual Mersenne blocks.

For those blocks let $a_d = \log E_d$. If a prime p occurs in E_d , then $\text{ord}_p(2) = d$, so $d \mid p-1$ and $p \nmid d$. In the identity $\prod_{e \mid d} \Phi_e(2) = 2^d - 1$, a divisibility $p \mid \Phi_e(2)$ with $e \mid d$ would, since $p \nmid e$, make the reduction of 2 a primitive e th root; hence $e = d$. Thus the full p -adic multiplicity of $2^d - 1$ is carried by $\Phi_d(2)$. The exponent used in E_d is one smaller, and the distinct prime-power factors are pairwise coprime, proving $E_d \mid \Phi_d(2)$. The root product then gives

$$(440) \quad 0 \leq \log E_d \leq \varphi(d) \log 3.$$

The relevant valuation identities and current distribution results are consistent with [20, 19, 39]; none supplies the following exceptional-weight estimate. The primitive-divisor refinement [22] and the recent generalized-Wieferich congruence criteria [23] likewise control local phenomena rather than the required moving totient mass.

Corollary 67.3 (Localized exact Mersenne gate). *Fix $\delta > 0$. Then*

$$\log W_m = o(m) \iff \sum_{\substack{d|m \\ d > m^{1-\delta} \\ \log E_d > \epsilon \varphi(d)}} \varphi(d) = o(m) \quad \text{for every } \epsilon > 0.$$

The same statement remains valid when the near-top range is narrowed to

$$d > m \exp(-\{\log \log(3m)\}^2) = m^{1-o(1)}.$$

Proof. Apply Theorem 67.1 using (440). Theorem 67.2 makes the complementary divisor range negligible. Finally use the exact equivalence $\log W_m = o(m) \iff \sum_{d|m} \log E_d = o(m)$. $\square$

The endpoint still forces $a_m = o(m)$, so pointwise control remains necessary on primes and fixed prime powers. What is gained is permission for large normalized blocks on orders with small $\varphi(d)/d$, provided their near-diagonal totient mass is negligible. No checked source, including the recent powerful-number count [8], proves this final gate, and no full-premise counterexample to it is known.

The Lean companion checks the exact moment, its dyadic $O(\log \log(3m))$ estimate, the general moving-threshold Markov theorem and its squared-loglog specialization. It also proves the exact-order prime-power divisibility in $\Phi_d(2)$ and hence (440). Combining these results gives a kernel-checked equivalence between the Mersenne endpoint and the remaining near-diagonal exceptional-totient estimate; that estimate itself is neither assumed nor proved.

68. FIXED-POLYLOGARITHMIC CO-DIVISORS AT THE MERSENNE ENDPOINT

The squared-loglog localization in Corollary 67.3 can be sharpened without assuming a stronger concentration theorem. The sharpening proved here is a family of fixed windows with a quantified outer limit. The present moment argument does not show that the complement of one fixed window is negligible.

Retain $a_d = \log E_d$ and put

$$B_\epsilon(m) = \sum_{\substack{d|m \\ a_d > \epsilon \varphi(d)}} \varphi(d), \quad \Lambda_m = \log \log(3m).$$

For a fixed positive integer k , define

$$B_{\epsilon,k}^{\text{poly}}(m) = \sum_{\substack{d|m \\ \log(m/d) < k\Lambda_m \\ a_d > \epsilon \varphi(d)}} \varphi(d).$$

Writing $q = m/d$, its support is the genuinely polylogarithmic range

$$q < \exp(k\Lambda_m) = \{\log(3m)\}^k.$$

Let

$$\Delta(m) = \frac{1}{m} \sum_{d|m} \varphi(d) \log \frac{m}{d}.$$

The exact moment formula in Section 67, together with the elementary Chebyshev bound, supplies constants $C_0 > 0$ and m_0 such that $\Delta(m) \leq C_0 \Lambda_m$ for $m \geq m_0$. Finite weighted Markov therefore gives, for every fixed $k \geq 1$,

$$(441) \quad \sum_{\substack{d|m \\ k\Lambda_m \leq \log(m/d)}} \varphi(d) \leq \frac{C_0}{k} m.$$

Consequently

$$(442) \quad 0 \leq B_\epsilon(m) - B_{\epsilon,k}^{\text{poly}}(m) \leq \frac{C_0}{k} m.$$

The right side is uniform in m , but it is not $o(m)$ for one fixed k .

Theorem 68.1 (Fixed-polylogarithmic co-divisor gate). *For the actual canonical Mersenne blocks,*

$$\log W_m = o(m) \iff \text{for every } \epsilon > 0 \text{ and every fixed } k \geq 1, \quad B_{\epsilon,k}^{\text{poly}}(m) = o(m).$$

The same equivalence holds when k ranges over all fixed positive real numbers.

Proof. The cyclotomic bound $0 \leq a_d \leq (\log 3)\varphi(d)$ and the exceptional-mass criterion give

$$\log W_m = o(m) \iff B_\epsilon(m) = o(m) \quad \text{for every } \epsilon > 0.$$

The forward implication follows by restricting a nonnegative sum. For the reverse implication, fix $\epsilon > 0$ and a requested normalized error $\eta > 0$. First choose one integer $k > 2C_0/\eta$. This k is now fixed, so the hypothesis eventually gives $B_{\epsilon,k}^{\text{poly}}(m) < \eta m/2$. Equation (442) then gives $B_\epsilon(m) < \eta m$. Letting η vary proves the required little-oh statement. Integer windows imply the real-window formulation, while the converse follows by specialization. $\square$

The order of quantifiers in the proof is essential:

$$\forall \eta > 0 \exists k = k(\eta) \forall m \geq m_0(\eta, k).$$

An abstract primordial divisor model shows that, for general totient-bounded masses, the far tail for a single fixed k need not be $o(m)$. Thus the Markov and moment estimates alone cannot replace the theorem by a one-window assertion. This model is not the actual Mersenne mass $\log E_d$ and does not refute a future Mersenne-specific one-window strengthening.

There is a useful dichotomy in the negation. If the endpoint fails, then for some fixed $\epsilon, \eta, k > 0$ and an unbounded sequence m_j ,

$$(443) \quad \sum_{\substack{q|m_j \\ q < \{\log(3m_j)\}^k \\ \log E_{m_j/q} > \epsilon \varphi(m_j/q)}} \frac{\varphi(m_j/q)}{m_j} \geq \eta.$$

Let $\mathcal{Q}_j$ be the set of co-divisors counted in (443), and let M_j be its largest summand. After passage to a subsequence, exactly one of the following pressure regimes remains.

- (1) If $M_j \geq \tau > 0$, then $\varphi(m_j/q)/m_j \leq 1/q$ forces $q \leq 1/\tau$; a further subsequence fixes $q = q_0$. With $d_j = m_j/q_0$ one obtains

$$\frac{\varphi(d_j)}{d_j} \geq q_0 \tau, \quad \frac{\log E_{d_j}}{d_j} > \epsilon q_0 \tau.$$

This is the atomic, fixed-co-divisor obstruction and still contains the prime-index problem.

- (2) If $M_j \rightarrow 0$, the packet contains at least $\eta/M_j \rightarrow \infty$ exceptional co-divisors, and

$$\sum_{q \in \mathcal{Q}_j} \log E_{m_j/q} > \epsilon \eta m_j.$$

This is the diffuse multi-fibre obstruction.

The transition, extreme-order, and deep-lift decomposition of Section 69 applies inside the same fixed-polylogarithmic packet.

The prime-index test confirms that no core difficulty has been hidden: for $m = \ell$ prime, $q = 1$ lies in every window with weight $(\ell - 1)/\ell$, so the theorem still requires $\log E_\ell = o(\ell)$. The certified occurrences $1093 \mid E_{364}$ and $3511 \mid E_{1755}$ refute small universal pointwise constants but are finite and do not refute the eventual gate. Conversely, the sparse primordial model from the earlier totient audit satisfies every fixed- k gate while its pointwise normalized blocks need not converge to zero. These tests verify both directions of the quantifier boundary and produce no full-premise counterexample to the actual Mersenne endpoint.

The Lean companion proves the exact exceptional-set partition, the finite C_0/k estimate, a general uniform-in-fixed-index transfer theorem, and the displayed equivalence for the repository's actual `mersennePowerLoss`. All moment and cyclotomic estimates are reused from previously checked theorems; no distribution hypothesis enters the kernel.

69. GLOBAL NEAR-DIAGONAL TRIAGE OF THE MERSENNE EXCESS

The localized gate of Corollary 67.3 identifies the moving divisor range on which an obstruction can survive. We now combine that localization with the exact-order Wieferich decomposition. The result does not estimate the remaining mass, but it proves that a failure cannot be distributed among unspecified cyclotomic effects: one of three explicit mechanisms must carry a positive linear logarithmic mass.

For an odd prime q , put

$$d_q = \text{ord}_q(2), \quad w_q = v_q(2^{d_q} - 1),$$

and, for $d \geq 1$, let

$$\mathcal{W}_d = \{q : d_q = d, w_q \geq 2\}, \quad D_d = \prod_{\substack{d_q=d \\ w_q \geq 3}} q^{w_q-2}.$$

Fix $\delta > 0$ and set

$$Y_d = \frac{\varphi(d)^2}{\log \log(3d)}.$$

The logarithm of the canonical excess block has the exact four-layer decomposition

$$(444) \quad \log E_d = S_d + T_{d,\delta} + X_{d,\delta} + G_d,$$

where

$$\begin{aligned} S_d &= \sum_{\substack{q \in \mathcal{W}_d \\ q \leq Y_d}} \log q, & T_{d,\delta} &= \sum_{\substack{q \in \mathcal{W}_d \\ Y_d < q \leq d^{2+\delta}}} \log q, \\ X_{d,\delta} &= \sum_{\substack{q \in \mathcal{W}_d \\ q > d^{2+\delta}}} \log q, & G_d &= \log D_d. \end{aligned}$$

Indeed, the exponent of q in E_d is $w_q - 1 = 1 + (w_q - 2)$. Moreover, LTE gives

$$v_q(2^{q-1} - 1) = v_q(2^{d_q} - 1) = w_q,$$

so $\mathcal{W}_d$ consists precisely of base-two Wieferich primes of exact order d .

Put

$$H(m) = \{\log \log(3m)\}^2, \quad \mathcal{N}_m = \{d \mid m : d > me^{-H(m)}\}.$$

The cyclotomic cap $E_d \leq 3^{\varphi(d)}$ and the exact logarithmic-deficit moment prove

$$(445) \quad \sum_{\substack{d \mid m \\ d \notin \mathcal{N}_m}} \log E_d = o(m).$$

The uniform Brun–Titchmarsh argument for primes $q \equiv 1 \pmod{d}$ below Y_d gives $S_d = o(\varphi(d))$. Since the least member of $\mathcal{N}_m$ tends to infinity and $\sum_{d \mid m} \varphi(d) = m$, this pointwise uniform estimate sums to

$$(446) \quad \sum_{d \in \mathcal{N}_m} S_d = o(m).$$

Theorem 69.1 (Global three-arm obstruction). *Suppose that, for some $\epsilon > 0$ and an unbounded sequence m_j ,*

$$\sum_{d \mid m_j} \log E_d \geq \epsilon m_j.$$

After passage to a subsequence, one fixed alternative holds for every sufficiently large j :

- (1) $\sum_{d \in \mathcal{N}_{m_j}} G_d \geq \epsilon m_j / 6$;
- (2) *there are at least*

$$\frac{\epsilon m_j}{6(2 + \delta) \log m_j}$$

distinct base-two Wieferich primes q such that

$$d_q \mid m_j, \quad d_q > m_j e^{-H(m_j)}, \quad Y_{d_q} < q \leq d_q^{2+\delta};$$

- (3) $\sum_{d \in \mathcal{N}_{m_j}} X_{d,\delta} \geq \epsilon m_j / 6$. Every prime counted in the last alternative satisfies $\text{ord}_q(2) < q^{1/(2+\delta)}$.

Proof. By (445) and (446), the far mass and the near-small mass are each at most $\epsilon m_j / 4$ after deleting finitely many indices. Summing (444) therefore gives

$$\sum_{d \in \mathcal{N}_{m_j}} (G_d + T_{d,\delta} + X_{d,\delta}) \geq \frac{\epsilon m_j}{2}.$$

One of the three sums is at least $\epsilon m_j / 6$; infinite pigeonhole fixes the same arm on a subsequence. In the transition case each summand is at most $(2+\delta) \log m_j$. Exact-order prime sets for different d are disjoint, so division by this cap proves the asserted cardinality. The extreme inequality $q > d_q^{2+\delta}$ gives the final order bound. $\square$

Corollary 69.2 (Three sufficient aggregate gates). *For one fixed $\delta > 0$, assume*

$$\sum_{d \in \mathcal{N}_m} G_d = o(m), \quad \#\{\text{transition primes at } m\} = o(m / \log m), \quad \sum_{d \in \mathcal{N}_m} X_{d,\delta} = o(m).$$

Then $\log W_m = o(m)$.

Proof. The transition sum is at most $(2+\delta) \log m$ times its support size. Adding the three hypotheses to (445) and (446) yields $\sum_{d|m} \log E_d = o(m)$. The exact factorization $W_m = L_m \prod_{d|m} E_d$, with $L_m \mid m$, finishes the proof. $\square$

None of the three disjuncts can be deleted by positivity alone. On a one-point near set, take target mass one, far and small masses zero, and respectively

$$(G, T, X) = (1, 0, 0), \quad (0, 1, 0), \quad (0, 0, 1).$$

These are complete counterexamples to each corresponding two-disjunct ordered-ring strengthening. They are abstract mass ledgers rather than Mersenne blocks, so they do not eliminate an arithmetic arm.

The companion Lean module checks the finite near/far subtraction, the 1/6 trichotomy, the transition-cardinality lower bound, the five-arm closure inequality, and all three full-premise counterexamples. Its analytic inputs remain the proved and cited statements above; no distribution assertion is inserted as a formal axiom. Present results on primitive cyclotomic factors and generalized Wieferich congruences [22, 39, 23] do not bound any one of the three weighted moving arms. All three routes therefore remain active.

70. A THRESHOLD-FREE FIXED-POLYLOGARITHMIC MERSENNE GATE

Let E_d be the canonical exact-order excess block and put $a_d = \log E_d$. Previous sections prove

$$(447) \quad 0 \leq a_d \leq (\log 3) \varphi(d), \quad \log W_m = o(m) \iff A(m) := \sum_{d|m} a_d = o(m).$$

For $L_m = \log \log(3m)$ and a fixed positive integer k , define

$$P_k(m) = \sum_{\substack{d|m \\ \log(m/d) < kL_m}} a_d, \quad F_k(m) = \sum_{\substack{d|m \\ kL_m \leq \log(m/d)}} a_d.$$

The strict and closed inequalities give the exact partition $A(m) = P_k(m) + F_k(m)$.

Theorem 70.1 (Actual fixed-window equivalence). *One has*

$$\log W_m = o(m) \iff \text{for every fixed } k \geq 1, \quad P_k(m) = o(m).$$

Proof. The forward implication follows from $0 \leq P_k \leq A$. Conversely, the exact logarithmic-deficit moment and weighted Markov inequality provide an absolute C_0 such that, eventually and uniformly in $k \geq 1$,

$$\sum_{\substack{d|m \\ kLm \leq \log(m/d)}} \varphi(d) \leq \frac{C_0}{k} m.$$

By (447), $F_k(m) \leq (\log 3)C_0 m/k$. Given $\eta > 0$, first fix k so large that this upper bound is below $\eta m/2$, and only then use the fixed- k hypothesis to make $P_k(m) < \eta m/2$. Hence $A(m) = o(m)$, and (447) finishes the proof. $\square$

Inside this window, write the exact-order Wieferich decomposition

$$a_d = S_d + T_{d,\delta} + X_{d,\delta} + G_d,$$

where S_d contains one copy of primes $p \leq Y_d = \varphi(d)^2 / \log \log(3d)$, $T_{d,\delta}$ contains one copy for $Y_d < p \leq d^{2+\delta}$, $X_{d,\delta}$ contains one copy for $p > d^{2+\delta}$, and

$$G_d = \sum_{\substack{\text{ord}_p(2)=d \\ v_p(2^d-1) \geq 3}} (v_p(2^d - 1) - 2) \log p.$$

The proved bound $S_d = o(\varphi(d))$ and the fact that all window indices tend to infinity combine with

$$\sum_{d|m} \varphi(d) = m$$

to show that the localized S -mass is $o(m)$.

Theorem 70.2 (Sharp fixed-window three-arm obstruction). *Fix $\delta > 0$. If $\log W_m \neq o(m)$, then there are fixed $k \geq 1$ and $\epsilon > 0$, and an unbounded sequence m_j , for which $P_k(m_j) \geq \epsilon m_j$. For every fixed $0 < \gamma < 1/3$, after passage to a subsequence one same alternative holds eventually:*

- (1) *the localized sum of G_d is at least $\gamma \epsilon m_j$;*
- (2) *at least*

$$\frac{\gamma \epsilon m_j}{(2 + \delta) \log m_j}$$

distinct base-two Wieferich primes lie in the localized transition arm;

- (3) *the localized sum of $X_{d,\delta}$ is at least $\gamma \epsilon m_j$; each such prime satisfies $\text{ord}_p(2) < p^{1/(2+\delta)}$.*

In particular, one may take $\gamma = 1/4$.

Proof. The contrapositive of Theorem 70.1 fixes k and a linear lower bound along a subsequence. After subtracting the localized $o(m_j)$ small arm, the three remaining nonnegative masses total $(\epsilon - o(1))m_j$. Because $3\gamma < 1$, one carries at least $\gamma \epsilon m_j$ for all large j ; infinite pigeonhole fixes the arm. In the transition range, $p \leq d^{2+\delta} \leq m_j^{2+\delta}$, so each summand is at most $(2 + \delta) \log m_j$. Exact-order prime sets are disjoint, yielding the cardinality bound. The extreme-order inequality is the rearrangement of $p > d^{2+\delta}$. $\square$

The constants are sharp at the stated algebraic level. The full quarter-budget mass ledger

$$(R, S, G, T, X) = (4, 1, 1, 1, 1)$$

refutes every uniform coefficient greater than $1/4$. The ledger $(3, 0, 1, 1, 1)$ refutes every coefficient greater than $1/3$ when the small arm is zero. These ledgers are not asserted to arise from Mersenne blocks.

Actual finite pressure comes from the sealed exhaustive scan through 10^7 , independently rechecked by a segmented sieve. Its only base-two Wieferich hits are

$$(p, \text{ord}_p(2), v_p(2^{\text{ord}_p(2)} - 1)) = (1093, 364, 2), \quad (3511, 1755, 2).$$

They refute the finite strengthenings that every localized block vanishes, or that every localized repeated prime must already be deep, transition, or extreme. Two isolated indices do not refute any little-oh assertion, and the absence of depth at most 10^7 has no asymptotic consequence.

For any fixed $\delta > 0$, closure now reduces to proving, for every fixed k , that the localized deep mass and extreme mass are $o(m)$ and that the transition support is $o(m/\log m)$. No full-premise counterexample to these three arithmetic gates is known, so all three remain active.

71. MULTIPLIER-INDEX COMPRESSION OF THE LOCALIZED MERSENNE LOSS

We retain the notation of Section 70. For an odd prime p put

$$d_p = \text{ord}_p(2), \quad w_p = v_p(2^{d_p} - 1), \quad r_p = \frac{p-1}{d_p}.$$

Lagrange's theorem gives $d_p \mid p-1$, so r_p is a positive integer and $p = 1 + d_p r_p$. In particular, for fixed d , the multiplier r_p determines p injectively inside the exact-order fibre $d_p = d$. This elementary observation removes one entire localized mass arm.

Set

$$L_m = \log \log(3m), \quad \Lambda_m = \log(3m)L_m^2.$$

For $d \mid m$, split the one-copy repeated-prime support as

$$\begin{aligned} U_d(m) &= \sum_{\substack{d_p=d, w_p \geq 2 \\ p \leq d^2/\Lambda_m}} \log p, \\ B_d(m) &= \sum_{\substack{d_p=d, w_p \geq 2 \\ p > d^2/\Lambda_m}} \log p, \end{aligned}$$

and retain the deep excess

$$G_d = \sum_{\substack{d_p=d \\ w_p \geq 3}} (w_p - 2) \log p.$$

Prime by prime, the canonical loss is exactly

$$(448) \quad a_d = U_d(m) + B_d(m) + G_d.$$

Proposition 71.1 (Uniform low-multiplier estimate). *For every fixed positive integer k , and all sufficiently large m ,*

$$\sum_{\substack{d \mid m \\ \log(m/d) < kL_m}} U_d(m) \leq \frac{2m \log m}{\Lambda_m} (1 + kL_m) = o(m).$$

Proof. If p contributes to $U_d(m)$, then

$$dr_p = p - 1 < p \leq d^2/\Lambda_m, \quad r_p < d/\Lambda_m.$$

The positive integral multipliers are distinct, so this fibre has at most d/Λ_m elements. Since $p \leq d^2 \leq m^2$ eventually, $U_d(m) \leq 2d \log m/\Lambda_m$. Writing $q = m/d$ and enlarging the divisor sum to all positive integers $q < (\log(3m))^k$ gives

$$\sum U_d(m) \leq \frac{2m \log m}{\Lambda_m} \sum_{q < (\log(3m))^k} \frac{1}{q} \leq \frac{2m \log m}{\Lambda_m} (1 + kL_m).$$

After division by m , the final expression is at most $2(1 + kL_m)/L_m^2$, which tends to zero. $\square$

Theorem 71.2 (Two-arm near-square-root obstruction). *Suppose that the Mersenne endpoint $\log W_m = o(m)$ fails. Then there are a fixed $k \geq 1$, an $\epsilon > 0$, and an unbounded sequence m_j such that, for every fixed $0 < \gamma < 1/2$, after passage to a subsequence one same alternative holds for all large j :*

(1) *the localized deep mass satisfies*

$$\sum_{\substack{d \mid m_j \\ \log(m_j/d) < kL_{m_j}}} G_d \geq \gamma \epsilon m_j;$$

(2) the localized B -mass satisfies

$$\sum_{\substack{d|m_j \\ \log(m_j/d) < kL_{m_j}}} B_d(m_j) \geq \gamma \epsilon m_j,$$

and every prime in this sum obeys

$$\text{ord}_p(2) < \sqrt{p\Lambda_{m_j}}.$$

Proof. The fixed-window equivalence supplies k, ϵ and m_j for which the localized sum of a_d is at least ϵm_j . Sum (448) and apply Proposition 71.1. The two unresolved arms then total at least $(\epsilon - o(1))m_j$. If both were smaller than $\gamma \epsilon m_j$, their sum would be smaller than $2\gamma \epsilon m_j$, contradicting $2\gamma < 1$ for large j . Infinite pigeonhole fixes an arm on a subsequence. Finally, $p > d^2/\Lambda_{m_j}$ implies $d < \sqrt{p\Lambda_{m_j}}$. $\square$

Thus this route closes if, for every fixed k , both localized sums of G_d and $B_d(m)$ are $o(m)$. The coefficient $1/2$ is sharp at the exact nonnegative-ledger level: $(R, U, G, B) = (2, 0, 1, 1)$ satisfies every displayed mass premise and defeats each coefficient greater than $1/2$. This tuple is an abstract full-premise counterexample to the stronger ledger conclusion; it is not a Mersenne block.

The exact-order lower bounds of Erdős–Murty [18] are global and unweighted, and leave a genuine $p^{1/2-o(1)}$ to $p^{1/2+o(1)}$ band. The valuation dictionary of Murty–Séguin [20] and the cyclotomic description in Pomerance [19] justify the block language, but do not give the localized Wieferich-weighted estimate needed here. The conditional alternatives in Fellini–Murty [39] likewise do not supply either remaining unconditional little-oh statement. Hence both arms remain active.

The formal companion proves the integral multiplier representation and injectivity, the finite cardinality and logarithmic-mass bounds, the exact two-arm subtraction, its fixed-window finite-sum form, and the sharp ledger counterexample. The harmonic asymptotic and the realization of the three mass functions remain explicit paper mathematics; no distribution or squarefreeness hypothesis is installed as a Lean axiom.

72. BALANCED MERSENNE MULTIPLIER LOCALIZATION

We refine Section 71 in two directions. The one-copy multiplier cutoff can be moved closer to its critical harmonic scale, and a published pointwise p -adic logarithm bound removes a growing low-multiplier interval from the deep-valuation arm. No density conjecture is used.

For an odd prime p , retain

$$d_p = \text{ord}_p(2), \quad w_p = v_p(2^{d_p} - 1), \quad r_p = \frac{p-1}{d_p},$$

and let $a_d = \sum_{d_p=d} (w_p - 1) \log p$. Fix a positive integer k and a real number $\eta > 0$. Put

$$(449) \quad L_m = \log \log(3m), \quad F_{m,\eta} = \log(3m) L_m^{1+\eta}, \quad H_{m,\eta} = \left\lfloor \sqrt{\frac{\log(3m)}{L_m^{1+\eta}}} \right\rfloor.$$

Split the one-copy repeated support into

$$U_d^{(\eta)}(m) = \sum_{\substack{d_p=d, w_p \geq 2 \\ p \leq d^2/F_{m,\eta}}} \log p,$$

$$B_d^{(\eta)}(m) = \sum_{\substack{d_p=d, w_p \geq 2 \\ p > d^2/F_{m,\eta}}} \log p,$$

and split the deep excess by multiplier:

$$V_d^{(\eta)}(m) = \sum_{\substack{d_p=d, w_p \geq 3 \\ r_p < H_{m,\eta}}} (w_p - 2) \log p,$$

$$G_d^{(\eta)}(m) = \sum_{\substack{d_p=d, w_p \geq 3 \\ r_p \geq H_{m,\eta}}} (w_p - 2) \log p.$$

The four terms give the exact identity

$$(450) \quad a_d = U_d^{(\eta)}(m) + B_d^{(\eta)}(m) + V_d^{(\eta)}(m) + G_d^{(\eta)}(m).$$

Proposition 72.1 (Adaptive one-copy cutoff). *For fixed $k \geq 1$ and $\eta > 0$, eventually in m ,*

$$\sum_{\substack{d|m \\ \log(m/d) < kL_m}} U_d^{(\eta)}(m) \leq \frac{2m \log m}{F_{m,\eta}} (1 + kL_m) = O_{k,\eta} \left(\frac{m}{L_m^\eta} \right) = o(m).$$

Proof. Write $q = m/d$. If p occurs in $U_d^{(\eta)}(m)$, multiplier injectivity and $p = 1 + dr_p$ give

$$r_p < \frac{p}{d} \leq \frac{d}{F_{m,\eta}}.$$

There are therefore at most $d/F_{m,\eta}$ such primes. Eventually $F_{m,\eta} \geq 1$, so $p \leq d^2 \leq m^2$ and

$$U_d^{(\eta)}(m) \leq \frac{2d \log m}{F_{m,\eta}}.$$

The window is $q < (\log(3m))^k$. Hence

$$\sum U_d^{(\eta)}(m) \leq \frac{2m \log m}{F_{m,\eta}} \sum_{\substack{q|m \\ q < (\log(3m))^k}} \frac{1}{q} \leq \frac{2m \log m}{F_{m,\eta}} (1 + kL_m).$$

Dividing by m gives $O_k((1 + L_m)/L_m^{1+\eta}) = O_k(L_m^{-\eta})$. $\square$

Thus the exponent two on L_m in the previous cutoff can be replaced by $1 + \eta$ for every fixed positive η . The endpoint $\eta = 0$ is not claimed: the same calculation then gives only a bounded normalized error. Relative to the old cutoff this is strictly sharper for $0 < \eta < 1$, equal at $\eta = 1$, and a valid but weaker localization for $\eta > 1$. That limitation is not an arithmetic counterexample, so the critical cutoff remains an active route.

The deep estimate uses Yamada's unconditional theorem [85]. With

$$C_Y = 283 \log 3 \log 6,$$

his Theorem 1.2, equation (7), states for every prime p that

$$(451) \quad v_p(2^{p-1} - 1) \leq \left\lfloor C_Y \frac{p-1}{(\log p)^2} \right\rfloor + 4.$$

Lemma 72.2 (Exact-order Yamada envelope). *For an odd prime p , with $d = d_p$, $r = r_p$, and $w = w_p \geq 2$,*

$$(w - 2) \log p \leq C_Y \frac{dr}{\log p} + 2 \log p.$$

If $d > 1$ and $r < H$, then

$$(452) \quad (w - 2) \log p \leq C_Y \frac{dr}{\log d} + 2 \log(1 + dH).$$

Proof. Since $p - 1 = dr$ and $0 < r < p$, one has $p \nmid r$. LTE gives

$$v_p(2^{p-1} - 1) = v_p((2^d)^r - 1) = v_p(2^d - 1) + v_p(r) = w.$$

Substitute this equality into (451), discard the floor, subtract two, and multiply by $\log p$. The final assertion uses $p > d$ and $p = 1 + dr < 1 + dH$. $\square$

Lemma 72.3 (Triangular multiplier energy). *If positive integral labels $r(x) < H$ are injective on a finite set S , then*

$$|S| \leq H - 1, \quad \sum_{x \in S} r(x) \leq \frac{H(H-1)}{2}.$$

Consequently, for $d > 1$ and $H \geq 2$,

$$(453) \quad \sum_{\substack{d_p=d, w_p \geq 3 \\ r_p < H}} (w_p - 2) \log p \leq \frac{C_Y d}{\log d} \frac{H(H-1)}{2} + 2(H-1) \log(1 + dH).$$

Proof. The image of the labels lies in $\{1, \dots, H-1\}$, proving both finite bounds. Apply (452) prime by prime and use injectivity of $p \mapsto r_p$ in the exact-order fibre. $\square$

Theorem 72.4 (Low-multiplier deep mass is negligible). *For every fixed $k \geq 1$ and $\eta > 0$,*

$$\sum_{\substack{d|m \\ \log(m/d) < kL_m}} V_d^{(\eta)}(m) = o(m).$$

More precisely, if $Q_m = (\log(3m))^k$ and $H = H_{m,\eta}$, then eventually

$$(454) \quad \begin{aligned} \sum V_d^{(\eta)}(m) &\leq \frac{C_Y m H(H-1)}{2 \log(m/Q_m)} (1 + kL_m) \\ &\quad + 2H Q_m \log(1 + mH). \end{aligned}$$

Proof. In the window $q = m/d < Q_m$, hence $d > m/Q_m$. Sum (453). The coefficient of the triangular term is bounded using

$$\sum_{\substack{q|m \\ q < Q_m}} d = m \sum_{\substack{q|m \\ q < Q_m}} \frac{1}{q} \leq m(1 + kL_m),$$

and $1/\log d \leq 1/\log(m/Q_m)$. There are fewer than Q_m integral co-divisors, giving the second line of (454). Finally

$$H^2 \leq \frac{\log(3m)}{L_m^{1+\eta}}, \quad \log(m/Q_m) = \log m - kL_m \sim \log m.$$

The first line divided by m is $O_k(L_m^{-\eta})$, and the second line is polylogarithmic in m , hence $o(m)$. $\square$

The same slack L_m^η therefore balances the normalized losses in Proposition 72.1 and Theorem 72.4.

Theorem 72.5 (Balanced two-arm obstruction). *Suppose the Mersenne endpoint $\log W_m = o(m)$ fails, and fix $\eta > 0$. Then there are fixed $k \geq 1$, $\epsilon > 0$, and an unbounded sequence m_j such that for every fixed $0 < \gamma < 1/2$, after passage to a subsequence one same alternative holds for all large j :*

(1)

$$\sum_{\substack{d|m_j \\ \log(m_j/d) < kL_{m_j}}} G_d^{(\eta)}(m_j) \geq \gamma \epsilon m_j,$$

and every counted prime has $r_p \geq H_{m_j,\eta}$ and $p \geq 1 + d_p H_{m_j,\eta}$;

(2)

$$\sum_{\substack{d|m_j \\ \log(m_j/d) < kL_{m_j}}} B_d^{(\eta)}(m_j) \geq \gamma \epsilon m_j,$$

and every counted prime has

$$d_p < \sqrt{p \log(3m_j) L_{m_j}^{1+\eta}}.$$

Proof. The fixed-window equivalence supplies a localized a_d -mass at least ϵm_j . Sum (450) and use the two little-oh estimates above. The $G^{(\eta)}$ and $B^{(\eta)}$ sums total $(\epsilon - o(1))m_j$, so one carries every fixed share below one half on a subsequence. The displayed prime restrictions are respectively the definition of $G^{(\eta)}$ and the rearrangement $p > d_p^2/F_{m_j,\eta}$. $\square$

Thus it now suffices to prove that the two surviving localized sums are $o(m)$ for one fixed $\eta > 0$ and every fixed k . The first concerns deep base-two lifts only after their order multiplier tends to infinity; the second lies within a factor $\sqrt{\log(3m)L_m^{1+\eta}}$ of the square-root order threshold. Neither estimate follows from the global Erdős–Murty theorem [18] or the Murty–Séguin valuation dictionary [20].

The exact scope of newer literature is also important. Li–Zhao [86] fix one prime ideal and prove eventual stabilization of successive prime-power order growth; their threshold depends on that prime ideal and therefore does not bound initial depths uniformly as p varies. Large-sieve results for Fermat quotients such as Shparlinski [89] average over intervals of the base variable and do not specialize to fixed base two with the simultaneous exact-order, divisor-window, and square-divisibility restrictions. The conditional non-Wieferich conclusions of Fellini–Murty [39] likewise do not close either surviving arm.

There are two exact sharpness boundaries. For every integer $H \geq 1$, the complete positive packet $S_H = \{1, \dots, H-1\}$ satisfies

$$\sum_{r \in S_H} r = \frac{H(H-1)}{2}.$$

For any fixed C , choosing an integer $H > 2C + 1$ gives $\sum_{r \in S_H} r > CH$; hence injectivity plus a pointwise linear envelope does not imply a uniform $O(H)$ energy bound. The member $H = 4$ already refutes the coefficient-one inequality. At the two-arm level, $(R, U, V, G, B) = (2, 0, 0, 1, 1)$ defeats every coefficient greater than one half. Both are complete abstract counterexamples to the respective stronger finite inferences. In actual arithmetic,

$$\text{ord}_{3511}(2) = 1755, \quad v_{3511}(2^{1755} - 1) = 2, \quad \frac{3511 - 1}{1755} = 2,$$

so the universal assertion that every repeated exact-order prime has multiplier at least three is also false with all premises present. This finite depth-two row does not refute either asymptotic surviving arm.

The companion Lean module formalizes triangular multiplier energy, the positive-label capacity, the exact LTE transport from $p-1$ to the canonical exact-order valuation, the affine weighted envelope and its realization on the actual exact-order fibre, the two-controlled/two-surviving finite ledger, the complete triangular sharpness family and the two-arm sharpness example, and the full 3511 order, square-divisibility, and multiplier certificate. Yamada’s theorem and the real asymptotic summation above remain cited paper mathematics; no valuation, density, or *abc* assertion is introduced as a Lean axiom.

73. CRITICAL SLOW-SLACK COMPRESSION IN THE MERSENNE ROUTE

The balanced multiplier argument of Section 72 can be pushed below every fixed positive power of $\log \log m$. For an odd prime p , retain

$$d_p = \text{ord}_p(2), \quad w_p = v_p(2^{d_p} - 1), \quad r_p = \frac{p-1}{d_p}, \quad L_m = \log \log(3m).$$

Call a positive function σ *slow-admissible* when

$$(455) \quad \sigma(m) \longrightarrow \infty, \quad \frac{L_m \sigma(m)}{\log(3m)} \longrightarrow 0.$$

Set

$$(456) \quad F_{m,\sigma} = \log(3m)L_m\sigma(m), \quad H_{m,\sigma} = \left\lfloor \sqrt{\frac{\log(3m)}{L_m\sigma(m)}} \right\rfloor.$$

The latter tends to infinity. For $d \mid m$, split the repeated one-copy mass and the deep excess as

$$\begin{aligned} U_d^{(\sigma)}(m) &= \sum_{\substack{d_p=d, w_p \geq 2 \\ p \leq d^2/F_{m,\sigma}}} \log p, & B_d^{(\sigma)}(m) &= \sum_{\substack{d_p=d, w_p \geq 2 \\ p > d^2/F_{m,\sigma}}} \log p, \\ V_d^{(\sigma)}(m) &= \sum_{\substack{d_p=d, w_p \geq 3 \\ r_p < H_{m,\sigma}}} (w_p - 2) \log p, & G_d^{(\sigma)}(m) &= \sum_{\substack{d_p=d, w_p \geq 3 \\ r_p \geq H_{m,\sigma}}} (w_p - 2) \log p. \end{aligned}$$

These four nonnegative terms still sum exactly to the block mass a_d .

Theorem 73.1 (Critical slow-slack removal). *Let σ satisfy (455). For every fixed positive integer k ,*

$$(457) \quad \sum_{\substack{d \mid m \\ \log(m/d) < kL_m}} U_d^{(\sigma)}(m) = o(m), \quad \sum_{\substack{d \mid m \\ \log(m/d) < kL_m}} V_d^{(\sigma)}(m) = o(m).$$

If also $\sigma(m) = o(L_m^\eta)$ for every fixed $\eta > 0$, both surviving cutoff regions are eventually contained in the corresponding fixed- η regions of Section 72.

Proof. Write $q = m/d$ and $Q_m = (\log(3m))^k$. In the displayed divisor window, $q < Q_m$. If p occurs in $U_d^{(\sigma)}$, then

$$0 < r_p = \frac{p-1}{d} < \frac{p}{d} \leq \frac{d}{F_{m,\sigma}}.$$

The integral multipliers are distinct on a fixed exact-order fibre. Hence there are at most $d/F_{m,\sigma}$ such primes, each at most $d^2 \leq m^2$, and

$$(458) \quad \sum U_d^{(\sigma)}(m) \leq \frac{2m \log m}{F_{m,\sigma}} \sum_{q < Q_m} \frac{1}{q} \leq \frac{2m \log m}{F_{m,\sigma}} (1 + kL_m) = o(m).$$

The last step uses only $\sigma(m) \rightarrow \infty$.

For the deep term, Yamada's pointwise theorem [85] and exact-order LTE give, for $d > 1$ and $H \geq 2$,

$$\sum_{\substack{d_p=d, w_p \geq 3 \\ r_p < H}} (w_p - 2) \log p \leq \frac{C_Y d H(H-1)}{\log d} + 2(H-1) \log(1 + dH), \quad C_Y = 283 \log 3 \log 6.$$

Summing with $H = H_{m,\sigma}$ and $d > m/Q_m$ gives

$$\begin{aligned} \sum V_d^{(\sigma)}(m) &\leq \frac{C_Y m H(H-1)}{2 \log(m/Q_m)} (1 + kL_m) \\ &\quad + 2H Q_m \log(1 + mH). \end{aligned}$$

Since $H^2 \leq \log(3m)/(L_m \sigma(m))$, the first line divided by m is $O_k(\sigma(m)^{-1})$; the second is polylogarithmic and hence $o(m)$.

Finally,

$$\frac{F_{m,\sigma}}{F_{m,\eta}} = \frac{\sigma(m)}{L_m^\eta} \rightarrow 0, \quad \frac{\sqrt{\log(3m)/(L_m \sigma(m))}}{\sqrt{\log(3m)/L_m^{1+\eta}}} = \sqrt{\frac{L_m^\eta}{\sigma(m)}} \rightarrow \infty.$$

Thus the near-square survivor threshold rises and the high-multiplier threshold rises in the required directions. Taking floors preserves the second strict inequality eventually because both raw scales diverge. $\square$

The explicit choice

$$(459) \quad \sigma_*(m) = \log(3 + L_m)$$

satisfies every hypothesis. It replaces every fixed L_m^η loss by one further iterated logarithm. The cutoff regions are strictly nested as real intervals; for a particular finite m the actual prime sets may coincide if the removed band happens to contain no prime.

Theorem 73.2 (Critical two-arm alternative). *If the Mersenne endpoint*

$$\log \frac{2^m - 1}{\text{rad}(2^m - 1)} = o(m)$$

fails, then for any slow-admissible σ there are fixed $k \geq 1$, $\epsilon > 0$, and an unbounded sequence m_j such that, after passage to a subsequence, one of the following carries at least $\gamma \epsilon m_j$ for every fixed $0 < \gamma < 1/2$:

- (1) *the deep excess with $r_p \geq H_{m_j, \sigma}$;*
- (2) *the repeated one-copy mass with*

$$d_p < \sqrt{p \log(3m_j) L_{m_j} \sigma(m_j)}.$$

The same alternative occurs for all sufficiently large indices of the chosen subsequence.

Proof. The fixed-window equivalence supplies k, ϵ and a localized total block mass at least ϵm_j . Subtract the two $o(m_j)$ terms in (457). The G and B masses then total $(\epsilon - o(1))m_j$. If both were below $\gamma \epsilon m_j$, their sum would be below $2\gamma \epsilon m_j$, a contradiction for large j . Infinite pigeonhole fixes one arm on a subsequence. The order bound is the rearrangement $p > d_p^2 / F_{m_j, \sigma}$. $\square$

Exact order also gives a useful filter that is independent of the asymptotic argument.

Proposition 73.3 (Euler character of the exact-order multiplier). *For an odd prime p , if $d = \text{ord}_p(2)$ and $p - 1 = dr$, then*

$$\left(\frac{2}{p}\right) = (-1)^r.$$

The allowed residue classes modulo eight are

$d \bmod 8$	$r \bmod 8$
0	0, 2, 4, 6
1	0, 6
2	0, 1, 4, 5
3	0, 2
4	0, 1, 2, 3, 4, 5, 6, 7
5	0, 6
6	0, 3, 4, 7
7	0, 2

Proof. If d is odd, the even number $p - 1 = dr$ forces r even, and Euler's half-power is one. If d is even, exactness implies $2^{d/2} \not\equiv 1 \pmod{p}$, while its square is one; hence it is -1 and $2^{(p-1)/2} \equiv (-1)^r$. Euler's criterion gives the identity. Combining it with the supplementary law for $(2/p)$ and $p \equiv 1 + dr \pmod{8}$ yields the table by the 64 residue checks. $\square$

The exact arithmetic row

$$p = 1093, \quad d_p = 364, \quad w_p = 2, \quad r_p = 3$$

has $1093^2 \mid 2^{364} - 1$ and $1093^3 \nmid 2^{364} - 1$. It is therefore a full-premise counterexample to the proposed assertion that every repeated base-two exact-order prime has even multiplier. That assertion is retired. It lies on the unrestricted $d \equiv 4 \pmod{8}$ row and does not refute the character theorem or either slow-slack estimate. A deterministic scan of all primes through 10^5 recovers only the known depth-two rows 1093 and 3511; this finite no-hit statement has no asymptotic consequence.

There is a separate abstract endpoint boundary. Labels $r = 1, \dots, H - 1$ with weights $x_r = r$ satisfy positivity, injectivity, the cutoff $r < H$, and the pointwise affine envelope, but have total $H(H - 1)/2$. With co-divisors $q \leq Q$, weights $x_{q,r} = r/q$ attain one half of $H(H - 1) \sum_{q \leq Q} 1/q$. This refutes the inference of a subquadratic energy bound from those abstract premises alone. It

is not an arithmetic prime packet, so the critical arithmetic target $\sigma \equiv 1$ remains active. Current order and Wieferich literature, including the exact valuation dictionary of Murty–Séguin [20], supplies no localized weighted estimate that closes it.

The Lean companion checks support monotonicity, nonnegative survivor-mass monotonicity, the exact cutoff balance, the finite two-arm ledger, the Euler half-power identity on actual exact-order fibres, and the full 1093 certificate. The real asymptotic estimates and Yamada’s theorem remain paper mathematics with explicit citations; no density statement or *abc* consequence is introduced as a Lean axiom.

74. EXACT-ORDER COUPLING AT THE MERSENNE ENDPOINT

We return to the fixed polylogarithmic co-divisor window. Put

$$A_m = \log(3m), \quad L_m = \log A_m, \quad Q_m = A_m^k,$$

where $k > 0$ is fixed, and let

$$F_m = A_m L_m, \quad H_m = \left\lfloor \sqrt{A_m / L_m} \right\rfloor.$$

For a prime p of exact base-two order d_p , write

$$w_p = v_p(2^{d_p} - 1), \quad r_p = \frac{p-1}{d_p}.$$

For $d \mid m$ with $q = m/d < Q_m$, split the repeated-prime mass as

$$(74.1) \quad \begin{aligned} U_d &= \sum_{\substack{d_p=d, w_p \geq 2 \\ p \leq d^2/F_m}} \log p, & B_d &= \sum_{\substack{d_p=d, w_p \geq 2 \\ p > d^2/F_m}} \log p, \\ V_d &= \sum_{\substack{d_p=d, w_p \geq 3 \\ r_p < H_m}} (w_p - 2) \log p, & G_d &= \sum_{\substack{d_p=d, w_p \geq 3 \\ r_p \geq H_m}} (w_p - 2) \log p. \end{aligned}$$

Their sum is exactly the logarithmic repeated-prime mass of the canonical d th exact-order block

$$E_d = \prod_{\text{ord}_p(2)=d} p^{v_p(2^d-1)-1}.$$

It need not equal the full radical loss of $2^d - 1$, because that integer can also contain primes whose exact order properly divides d . The earlier independent multiplier estimate reached only $O(m)$ when the slack parameter was set to one. Summing primes with their congruence weights removes that endpoint loss.

Theorem 74.1 (The low one-copy arm at $\sigma = 1$). *For every fixed $k > 0$,*

$$\sum_{\substack{d \mid m \\ m/d < Q_m}} U_d = o(m).$$

Proof. For one divisor d , put $X = d^2/F_m$. Every prime in U_d is $1 \pmod{d}$, hence

$$U_d \leq \vartheta(X; d, 1) = \sum_{\substack{p \leq X \\ p \equiv 1 \pmod{d}}} \log p.$$

The window gives $d > m/Q_m$. Uniformly there, $d/F_m \rightarrow \infty$ and $F_m^2 < d$ for sufficiently large m ; the latter is equivalent to $d < X^{2/3}$. The weighted Brun–Titchmarsh estimate [20, Theorem 2.4], with the fixed exponent $2/3$, gives

$$\vartheta(X; d, 1) \leq \frac{2X \log X}{\varphi(d) \log(X/d)}.$$

Moreover $\log X \leq 2 \log d$ and, uniformly in the window, $\log(X/d) = \log(d/F_m) \geq \frac{1}{2} \log d$. Consequently

$$U_d \ll \frac{d^2}{F_m \varphi(d)}.$$

The uniform estimate $d/\varphi(d) \ll \log \log(3d) \ll L_m$ yields $U_d \ll d/A_m$. On writing $d = m/q$ and enlarging the divisor sum to all positive integers $q < Q_m$, we obtain

$$\sum_{\substack{d|m \\ m/d < Q_m}} U_d \ll \frac{m}{A_m} \sum_{q < Q_m} \frac{1}{q} \ll_k m \frac{L_m}{A_m} = o(m).$$

□

The remaining low-multiplier deep arm must retain the common-index correlation. Define $\mathcal{S}_k(m)$ to consist of triples (q, r, p) such that

$$q \mid m, \quad q < Q_m, \quad d = m/q = \text{ord}_p(2), \quad w_p \geq 3, \quad p = 1 + dr, \quad 1 \leq r < H_m,$$

and put

$$E_k(m) = \sum_{(q,r,p) \in \mathcal{S}_k(m)} \frac{r}{q}, \quad N_k(m) = |\mathcal{S}_k(m)|.$$

Proposition 74.2 (Cross-fibre injection and first moment). *Distinct rows of $\mathcal{S}_k(m)$ have distinct rational values r/q , and*

$$(74.2.1) \quad E_k(m) = \frac{1}{m} \sum_{(q,r,p) \in \mathcal{S}_k(m)} (p-1).$$

Proof. If $r_1/q_1 = r_2/q_2$, then $d_1 r_1 = (m/q_1) r_1 = (m/q_2) r_2 = d_2 r_2$. Thus $p_1 = p_2$. The order of two modulo a fixed prime is unique, so $d_1 = d_2$, whence $q_1 = q_2$ and $r_1 = r_2$. For each row, $r/q = dr/(dq) = (p-1)/m$; summation proves the identity. □

Using Yamada's published pointwise valuation estimate in the form

$$(74.2.2) \quad (w_p - 2) \log p \leq C_Y \frac{d_p r_p}{\log p} + 2 \log p,$$

one obtains the following correlation-preserving transfer.

Theorem 74.3 (Farey-energy reduction). *For fixed k and all sufficiently large m ,*

$$(74.3.1) \quad \sum_{\substack{d|m \\ m/d < Q_m}} V_d \leq \frac{C_Y m}{\log(m/Q_m)} E_k(m) + 2N_k(m) \log(1 + mH_m),$$

where $N_k(m) \leq H_m Q_m$ and the second term is $o(m)$. Hence

$$(74.3.2) \quad E_k(m) = o(\log m)$$

is sufficient for the entire low-multiplier deep arm to be $o(m)$. Equivalently, the missing condition is

$$\sum_{(q,r,p) \in \mathcal{S}_k(m)} (p-1) = o(m \log m).$$

Proof. On a row, $d_p r_p = m(r_p/q)$, $\log p \geq \log d_p \geq \log(m/Q_m)$, and $p = 1 + d_p r_p \leq 1 + mH_m$. Substitution into the pointwise estimate and summation gives the displayed bound. There are fewer than Q_m possible q and fewer than H_m possible r , and the preceding injection forbids duplicate rows. Therefore

$$N_k(m) \leq H_m Q_m, \quad H_m Q_m \log(1 + mH_m) = O_k(A_m^{k+3/2} L_m^{-1/2}) = o(m).$$

Finally $\log(m/Q_m) \sim \log m$, which proves the implication. □

The condition in Theorem 74.3 is not inferred from distinct slopes alone. Indeed the reduced Farey box $\{(q, r) : q, r \leq N, (q, r) = 1\}$ has distinct slopes but total energy asymptotic to $(3/\pi^2)N^2 \log N$. This pressure example omits the common divisibility, primality, exact-order, and depth-three premises, so it refutes only that abstract inference and leaves the arithmetic route active.

There is also an exact prime-power interpretation. If $d = \text{ord}_p(2)$ and $w = v_p(2^d - 1)$, then for every $j \geq 1$,

$$(74.3.3) \quad \text{ord}_{p^j}(2) = d \iff j \leq w.$$

One direction follows from $p^j \mid 2^d - 1$; in the other, the order modulo p^j divides d , while reduction modulo p makes d divide it. Thus $B + G$ is exactly the logarithmic mass of the selected stable exact-order prime-power layers: layer two above the size cutoff, together with layers $j \geq 3$ above the multiplier cutoff. The elementary implication

$$dH_m \leq d^2/F_m, \quad p = 1 + dr > d^2/F_m \implies r \geq H_m$$

holds for every sufficiently large m in the fixed window. Hence both surviving arms eventually lie on a common high-multiplier support.

Two complete arithmetic witnesses delimit stronger pointwise claims. For $p = 1093$, $d = 364$, $q = 10^{12}$, $m = dq$, and $k = 8$, one has

$$\Phi_{364}(2) = 1093^2 \cdot 4733 \cdot 8861085190774909 \cdot 556338525912325157.$$

All four factors have exact order 364, only 1093 has depth two, and the endpoint inequalities put its mass in B_{364} with $U_{364} = V_{364} = G_{364} = 0$. This refutes $B = 0$, every universal fibrewise inequality $B \leq CG$, and the assertion that every repeated high-multiplier row is deep. For $p = 3511$, $d = 1755$, $q = 10^{71}$, and $k = 32$, the row lies in B but has $r = 2 < H_m = 5$. It refutes a finite pointwise implication $B \Rightarrow r \geq H_m$, while the sufficiently-large window theorem remains intact.

An exhaustive sieve of all 50,847,534 primes through 10^9 finds only the two repeated-order carriers 1093 and 3511, both of depth two. This is a finite certificate and has no asymptotic consequence. The exact active targets are still

$$E_k(m) = o(\log m), \quad \sum_{(d,p,j,q) \in \mathcal{C}_k(m)} \log p = o(m),$$

where $\mathcal{C}_k(m)$ is the stable high-multiplier layer packet described above. Neither target has a counterexample or a proof at this checkpoint.

75. DENOMINATOR ENTROPY AT THE MERSENNE FAREY ENDPOINT

We continue with the exact-order endpoint support

$$\mathcal{S}_k(m) = \left\{ \begin{array}{l} q \mid m, \quad q < A_m^k, \quad d = m/q, \\ (q, r, p) : \text{ord}_p(2) = d, \quad p = 1 + dr, \\ 1 \leq r < H_m, \quad v_p(2^d - 1) \geq 3 \end{array} \right\},$$

where

$$A_m = \log(3m), \quad L_m = \log A_m, \quad H_m = \left\lfloor \sqrt{A_m/L_m} \right\rfloor.$$

The surviving low-multiplier energy is

$$E_k(m) = \sum_{(q,r,p) \in \mathcal{S}_k(m)} \frac{r}{q}.$$

The preceding reduction needs $E_k(m) = o(\log m)$. We do not prove that estimate here. We instead show that any failure forces a polynomial swarm of distinct base-two super-Wieferich primes, improving the direct cardinality reduction by a fixed power of $\log m$.

Define

$$W_3(x) = \#\{p \leq x : p \text{ prime and } p^3 \mid 2^{p-1} - 1\}.$$

For $1 \leq T < A_m^k$, let

$$\Delta_3(m; T) = \# \left\{ p : p \text{ prime, } \frac{m}{A_m^k} < p < 1 + \frac{mH_m}{T}, \quad p^3 \mid 2^{p-1} - 1 \right\}.$$

Proposition 75.1 (Depth transport). *Every row of $\mathcal{S}_k(m)$ is a base-two super-Wieferich prime, and the map from rows to their primes is injective. Moreover $p \leq 1 + mH_m$.*

Proof. For a row, $p - 1 = dr$ and $p^3 \mid 2^d - 1$. Hence

$$2^d - 1 \mid 2^{dr} - 1 = 2^{p-1} - 1.$$

If two rows carry the same prime, uniqueness of the exact order gives the same d , after which $q = m/d$ and $r = (p - 1)/d$ agree. Finally $p = 1 + (m/q)r \leq 1 + mH_m$. $\square$

For an integer $1 \leq T < A_m^k$, let

$$N_{>T}(m) = \#\{(q, r, p) \in \mathcal{S}_k(m) : q > T\}, \quad \mathcal{H}_T = \sum_{q \leq T} \frac{1}{q}.$$

Theorem 75.2 (Denominator-entropy inequality). *One has*

$$E_k(m) \leq \frac{H_m(H_m - 1)}{2} \mathcal{H}_T + \frac{H_m}{T} N_{>T}(m).$$

Proof. For fixed $q \leq T$, the pair (q, r) determines $p = 1 + (m/q)r$, so the fibre contains each $1 \leq r < H_m$ at most once. Its energy is at most $H_m(H_m - 1)/(2q)$. If $q > T$, each row has $r/q < H_m/T$. Summing the two estimates proves the assertion. $\square$

Theorem 75.3 (Super-Wieferich swarm forced by endpoint failure). *Fix $k > 0$. If, for some $\varepsilon > 0$ and infinitely many m ,*

$$E_k(m) \geq \varepsilon A_m,$$

then, with

$$\eta = \min\{\varepsilon/4, k/2\} > 0,$$

one has along that sequence

$$\Delta_3(m; \lfloor A_m^\eta \rfloor) \gg_\varepsilon A_m^{1/2+\eta} L_m^{1/2},$$

and therefore

$$W_3(1 + mH_m) \gg_\varepsilon A_m^{1/2+\eta} L_m^{1/2}.$$

Proof. Set $T = \lfloor A_m^\eta \rfloor$. The prefix in Theorem 75.2 is at most

$$\frac{A_m}{2L_m} (1 + \log T) = \frac{\eta}{2} A_m + o(A_m) \leq \frac{\varepsilon}{8} A_m + o(A_m).$$

It is therefore at most $\varepsilon A_m/4$ for all sufficiently large m . The tail must contribute at least $3\varepsilon A_m/4$, whence

$$N_{>T}(m) \geq \frac{3\varepsilon}{4} \frac{A_m T}{H_m} \gg_\varepsilon A_m^{1/2+\eta} L_m^{1/2}.$$

For every tail row, $q > A_m^\eta$ and $q < A_m^k$ give

$$\frac{m}{A_m^k} < p < 1 + \frac{mH_m}{A_m^\eta};$$

depth transport and injectivity therefore give the localized assertion and then the global one. $\square$

Corollary 75.4 (Critical counting exponent one half). *If*

$$\limsup_{x \rightarrow \infty} \frac{\log \max\{1, W_3(x)\}}{\log \log x} \leq \frac{1}{2},$$

then $E_k(m) = o(\log m)$ for every fixed $k > 0$.

Proof. A failure would give Theorem 75.3 for a fixed $\eta > 0$. Since

$$\log \log(1 + mH_m) = L_m + o(1),$$

the resulting lower bound makes the displayed limsup at least $1/2 + \eta$, a contradiction. $\square$

The hypothesis of Corollary 75.4 is open. It is weaker than finiteness of super-Wieferich primes and also weaker than the direct sufficient bound $W_3(x) = o(\sqrt{\log x \log \log x})$. Current multiplicative-order, Brun–Titchmarsh, and Fermat-quotient large-sieve results [18, 20, 85, 89] do not provide it for the fixed base 2. The recent conditional results [39, 86] likewise do not upper-bound this packet unconditionally.

There is a sharp combinatorial warning. Let $m_n = \text{lcm}(1, \dots, n)$ and take all reduced pairs

$$q \leq n/2, \quad 1 \leq r < H_{m_n}, \quad (q, r) = 1.$$

Then $q \mid m_n$, the integers $1 + (m_n/q)r$ exist, and the slopes are distinct. Standard totient summation gives

$$\sum \frac{r}{q} = \left(\frac{3}{\pi^2} + o(1) \right) A_{m_n}.$$

Thus common divisibility and Farey injectivity alone do not give a little-oh. These abstract labels need not be prime, have the prescribed exact order, or have depth three, so this is not a counterexample to the Mersenne route.

The exact computation bundle constructs these common-index models through $n = 10000$ and obtains energy ratio $0.2961553070\dots$, while checking the denominator inequality with rational arithmetic. It also replays the exhaustive scan of all 50,847,534 primes through 10^9 : the only base-two Wieferich primes are 1093 and 3511, and both have depth exactly two. The published Dorais–Klyve search gives the corresponding ordinary Wieferich exclusion through 6.7×10^{15} . These are finite boundaries, not asymptotic evidence inserted into the proof.

Lean formalizes the finite prefix and tail inequalities, the resulting lower bound for the number of tail rows, depth transport, and the two exact depth-two witnesses. Its theorem-level dependency union is only `Classical.choice`, `Quot.sound`, and `propext`. The critical counting estimate and every analytic limit remain outside the kernel.

76. A QUANTITATIVE KERNEL FOR THE MERSENNE FAREY SWARM

This section turns the finite denominator split of Section 75 into a kernel-checked quantitative and subsequential statement. It also records a sharp counterexample boundary. No estimate for the distribution of base-two super-Wieferich primes is inserted.

Retain the notation

$$\mathcal{H}_T = \sum_{q=1}^T \frac{1}{q}, \quad C_H = \sum_{r=1}^{H-1} r,$$

and write $E_{\leq T}$ and $E_{>T}$ for the two pieces of the Farey energy.

Proposition 76.1 (Harmonic and square prefix bounds). *For every $T \geq 1$,*

$$\mathcal{H}_T \leq 1 + \log T.$$

If every denominator fibre is contained in $\{1, \dots, H-1\}$, then

$$E_{\leq T} \leq H^2(1 + \log T).$$

Proof. The first sum is the classical harmonic number. Isolating its first term and comparing the remaining decreasing summand with the integral of $1/x$ gives the first inequality. The exact denominator split gives $E_{\leq T} \leq C_H \mathcal{H}_T$, while $C_H = H(H-1)/2 \leq H^2$. $\square$

The common-index scale itself also admits an unconditional formal bracket. For $M_n = \text{lcm}(1, \dots, n)$,

$$n \log 2 - \log(n+1) \leq \log M_n \leq (\log 4 + 4)n.$$

Indeed, Mathlib identifies $\log M_n$ with Chebyshev’s function $\psi(n)$. The lower bound follows from $2^n \leq (n+1)M_n$, and the upper bound is the explicit elementary Chebyshev estimate for ψ . This proves the correct linear scale without the prime number theorem; the sharper ratio $\log M_n/n \rightarrow 1$ is not claimed.

Theorem 76.2 (Cleared quantitative swarm). *Let $T > 0$. Suppose the prefix fibres lie in $\{1, \dots, H-1\}$, all tail numerators are at most H , and*

$$\varepsilon A \leq E \leq E_{\leq T} + E_{> T}, \quad H^2(1 + \log T) \leq \kappa A.$$

Then

$$T(\varepsilon - \kappa)A \leq N_{> T}H.$$

Proof. Proposition 76.1 bounds the prefix by κA . The tail must therefore carry at least $(\varepsilon - \kappa)A$. Since every tail row contributes at most H/T , multiplication by T proves the assertion. $\square$

For $X \geq 0$, put

$$\mathcal{W}_3(X) = \{p \leq X : p \text{ prime and } p^3 \mid 2^{p-1} - 1\}.$$

Proposition 76.3 (Finite exact-order row transport). *Fix a common index m and a finite set S of actual exact-order endpoint rows. If every $x \in S$ satisfies*

$$p_x \text{ prime}, \quad p_x^3 \mid 2^{d_x} - 1, \quad p_x \leq X,$$

then

$$|S| \leq |\mathcal{W}_3(X)|.$$

Proof. Exact order gives $d_x = \text{ord}_{p_x}(2)$, so the depth transport theorem yields $p_x^3 \mid 2^{p_x-1} - 1$. Hence the prime-coordinate map lands in $\mathcal{W}_3(X)$. The same exact-order identity makes that coordinate map injective on endpoint rows. Restricting the injection to S proves the finite cardinality inequality. $\square$

This statement closes the finite row-to-prime bookkeeping only for rows whose primality, exact-order and depth premises have been proved. It neither turns arbitrary Farey rows into actual endpoint rows nor estimates the size of $\mathcal{W}_3(X)$.

For a predicate P_m , write $\text{Freq}(P_m)$ when $\forall N \exists m \geq N : P_m$.

Proposition 76.4 (Exact failure witness for little-oh). *If $f, g : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$, then*

$$f \neq o(g) \iff \exists \varepsilon > 0 : \text{Freq}(\varepsilon g(m) < f(m)).$$

Proof. The definition of $f = o(g)$ says that for every positive ε the inequality $|f(m)| \leq \varepsilon|g(m)|$ holds eventually. Nonnegativity removes the absolute values. Negating the two quantifiers gives precisely the displayed frequent strict reverse inequality. $\square$

Corollary 76.5 (Frequent quantitative swarm). *Suppose $E_m, A_m \geq 0$, $E \neq o(A)$, and the finite packet conditions hold eventually. If for every $\delta > 0$ one eventually has*

$$H_m^2(1 + \log T_m) \leq \frac{\delta}{2} A_m,$$

then for some fixed $\varepsilon > 0$,

$$\text{Freq} \left(T_m \frac{\varepsilon}{2} A_m \leq N_{> T_m}(m) H_m \right).$$

Proof. Proposition 76.4 supplies one fixed $\varepsilon > 0$ and a frequent set on which $\varepsilon A_m < E_m$. Intersect it with the eventual structural set and apply Theorem 76.2 with $\kappa = \varepsilon/2$. $\square$

The coefficient of the exact prefix estimate cannot be uniformly reduced. Indeed, take the full fibres $R_q = \{1, \dots, H-1\}$. Then

$$E_{\leq T} = C_H \mathcal{H}_T.$$

For $T = 1$ and $H = 2$, all three quantities equal one. Hence any proposed constant $c < 1$ satisfying $E_{\leq T} \leq c C_H \mathcal{H}_T$ for every admissible packet would force $1 \leq c$, a contradiction. This full-premise example deletes only that strict universal improvement. It leaves the arithmetic route intact: future gain must use primality, exact order, depth, or cross-fibre correlation.

The Lean module `MersenneFareyQuantitativeSwarm20260903` proves all statements in this section at their displayed quantifiers. Its theorem-level dependency union is only `propext`, `Classical.choice`, and `Quot.sound`. The critical upper bound

$$\limsup_{x \rightarrow \infty} \frac{\log \max\{1, W_3(x)\}}{\log \log x} \leq \frac{1}{2}$$

remains an open arithmetic input, so neither the Mersenne endpoint nor the abc conjecture is proved here.

77. ALTERNATIVE QUALITY METRICS AND THE PACKING BOUNDARY

Sankaran's 2026 preprint introduces doubly geometric mean (DGM) and power-mean qualities and proves unconditional divergence for an alternative quality using Chen's theorem [26]. For the standard and DGM qualities it identifies the exact packing factor

$$(AQ.1) \quad \eta = \frac{G}{A}, \quad q_{\text{std}} = \eta q_{\text{DGM}},$$

where G and A are the geometric and arithmetic means of the logarithms of the distinct primes dividing abc .

Proposition 77.1 (Exact packing-coordinate transfer). *If $q_s = \eta q_D$ and $\eta > 0$, then, for every real B ,*

$$q_s \leq B \iff q_D \leq B/\eta.$$

If also $0 \leq \eta \leq 1$ and $q_D \geq 0$, then $0 \leq q_s \leq q_D$.

Proof. The equivalence follows by dividing $\eta q_D \leq B$ by the positive factor η . Nonnegativity gives the second lower bound, and multiplication of $\eta \leq 1$ by $q_D \geq 0$ gives the upper bound. $\square$

Consequently, the inequality $q_{\text{DGM}} \leq (1 + \varepsilon)/\eta$ is an exact rewriting of the standard abc quality bound. It becomes a proof tool only after an independent estimate for the reciprocal packing efficiency is supplied. Alternative-quality divergence alone has no such force.

Proposition 77.2 (Clustered-logarithm lower bound). *If $L > 0$, $C \geq 0$, and $L \leq G \leq A \leq L + C$, then*

$$\frac{G}{A} \geq \frac{L}{L + C}.$$

Proof. The factor $L/(L + C)$ is nonnegative, so $A \leq L + C$ gives $LA/(L + C) \leq L \leq G$. Divide by $A > 0$. $\square$

For distinct primes in $[X, e^C X]$ with fixed $C > 0$, this lower bound tends to one as $X \rightarrow \infty$, regardless of the number of selected primes. This identifies a scope mismatch in the extension following Theorem 4.13 of the source: the decay part of its Lemma 4.12 fixes both the number of primes and the complementary product while one prime grows, whereas the later application allows the prime count and complementary product to vary. The claimed decay does not follow from the displayed largest-prime condition alone. Moreover, $\log P_n = O(\log c_n)$ is only a power upper bound and is not equivalent to $P_n \sim c_n^\kappa$. The critical-boundary proof of Theorem 4.10 also derives an upper estimate before calling its limiting constant exact; the limsup use in Theorem 4.13 needs only the upper estimate. These issues do not affect the exact packing identity or the Chen-based alternative-quality divergence result, and a cluster of prime coordinates is not asserted to be realized by additive abc triples.

Proposition 77.3 (Exact abstract no-go for divergence transfer). *There are positive sequences $\eta_n, q_{D,n}, q_{s,n}$ with $0 < \eta_n \leq 1$ and $q_{s,n} = \eta_n q_{D,n}$ such that q_D is unbounded above while $q_{s,n} = 1$ for every n .*

Proof. Take $q_{D,n} = n + 1$, $\eta_n = (n + 1)^{-1}$ and $q_{s,n} = 1$. The product identity and all positivity conditions are exact. The Archimedean property shows that q_D is unbounded, whereas q_s is constant. $\square$

This witness satisfies every premise of the metric-only implication it refutes. It is not a sequence of integer abc triples, so it neither refutes the standard conjecture nor invalidates the arithmetic search. It locates the surviving gate: on actual triples one must control the correlation between growth of q_{DGM} and decay of η .

The source PDF and verification capsule are archived at

research/sources/alternative_quality_metrics_2026_09_03/.

The Lean companion module

`AlternativeQualityPackingBridge20260903`

checks the displayed algebra and the abstract countermodel.

78. SYNCHRONIZED DIVISOR PACKETS AND ORIENTATION RIGIDITY

Author: ChatGPT. This section extracts a finite incidence structure directly from the equation $a + b = c$. Its points are triples of divisors whose squares collide modulo the opposite arms. The construction records exponent caps, the allocation of primes among the three arms, and local square-root signs. It is therefore different from a scalar reweighting of the usual radical. We prove the finite algebraic statements first and state the Lean boundary only after the traditional proofs. No form of the *abc* conjecture is assumed or proved.

78.1. The packet spectrum. Let

$$(460) \quad a, b, c \in \mathbf{Z}_{>0}, \quad a + b = c, \quad (a, b) = 1, \quad a > 1, \quad b > 1.$$

For positive integers r, s , write

$$\Delta(r, s) = |r^2 - s^2|.$$

Definition 78.1 (Synchronized divisor packet). A synchronized divisor packet for (a, b, c) is a triple $Q = (x, y, z)$ satisfying

$$(461) \quad \begin{aligned} x, y, z &> 1, & x &| a, & y &| b, & z &| c, \\ a &| \Delta(y, z), & b &| \Delta(x, z), & c &| \Delta(x, y). \end{aligned}$$

The finite set of all packets is denoted by $\mathcal{S}(a, b, c)$. A packet is proper if it differs from (a, b, c) . We attach to a packet the quantities

$$(462) \quad D(Q) = \Delta(y, z)\Delta(x, z)\Delta(x, y),$$

$$(463) \quad B(Q) = \max(y, z)^2 \max(x, z)^2 \max(x, y)^2,$$

$$(464) \quad T(Q) = \max(x, y, z).$$

The restriction $a, b > 1$ removes unit-arm degeneracies from this section. It is a stated scope restriction and does not remove unit-arm triples from the *abc* conjecture.

Proposition 78.2 (Nonempty finite spectrum). *The coordinates of every packet are pairwise coprime and pairwise distinct. Moreover, $(a, b, c) \in \mathcal{S}(a, b, c)$, so the spectrum is finite and nonempty.*

Proof. The equation and $(a, b) = 1$ give

$$(a, c) = (a, a + b) = 1, \quad (b, c) = (b, a + b) = 1.$$

Common divisors of packet coordinates divide the corresponding pairs of arms, so $(x, y) = (x, z) = (y, z) = 1$. The coordinates exceed one and are therefore distinct.

For the full packet, the congruences

$$c \equiv b \pmod{a}, \quad c \equiv a \pmod{b}, \quad a \equiv -b \pmod{c}$$

give the three square congruences in (461). Finally, each arm has finitely many divisors. $\square$

Theorem 78.3 (Product divisibility and height envelope). *Every $Q \in \mathcal{S}(a, b, c)$ satisfies*

$$(465) \quad abc \mid D(Q), \quad abc \leq D(Q) \leq B(Q) \leq T(Q)^6.$$

Proof. The packet congruences give positive integers α, β, γ with

$$\Delta(y, z) = a\alpha, \quad \Delta(x, z) = b\beta, \quad \Delta(x, y) = c\gamma.$$

Their product proves $abc \mid D(Q)$. The coordinate gaps are nonzero by Proposition 78.2, so a positive multiple of abc is at least abc . For every pair r, s ,

$$\Delta(r, s) = \max(r, s)^2 - \min(r, s)^2 \leq \max(r, s)^2.$$

Multiplication yields $D(Q) \leq B(Q)$, and each pair maximum is at most $T(Q)$. $\square$

The pair-max term in (465) is the exact output of the argument. Passing immediately to T^6 loses anisotropic information.

78.2. Index one and an infinite exact-gap stratum. The divisibility in Theorem 78.3 defines the positive integer

$$(466) \quad \kappa(Q) = \frac{D(Q)}{abc}.$$

Proposition 78.4 (Index-one gap rigidity). *One has $\kappa(Q) = 1$ if and only if*

$$(467) \quad \Delta(y, z) = a, \quad \Delta(x, z) = b, \quad \Delta(x, y) = c.$$

In that case z lies strictly between x and y .

Proof. Using the positive quotients α, β, γ from the preceding proof, we have $\kappa(Q) = \alpha\beta\gamma$. A product of positive integers is one exactly when every factor is one. Under (467), the equation $c = a + b$ becomes

$$|x^2 - y^2| = |y^2 - z^2| + |x^2 - z^2|.$$

Equality in the triangle inequality on the real line places z^2 between x^2 and y^2 . Positivity then places z between x and y . $\square$

Proposition 78.5 (Parametric exact-gap packets). *For every integer $t \geq 2$, put*

$$(468) \quad \begin{aligned} a_t &= 2t + 1, & b_t &= t(3t + 2), & c_t &= (t + 1)(3t + 1), \\ x_t &= 2t + 1, & y_t &= t, & z_t &= t + 1. \end{aligned}$$

Then (a_t, b_t, c_t) satisfies (460), and (x_t, y_t, z_t) is a proper packet of index one.

Proof. Expansion gives

$$a_t + b_t = 3t^2 + 4t + 1 = (t + 1)(3t + 1) = c_t.$$

We have $(2t + 1, t) = 1$, while

$$2(3t + 2) - 3(2t + 1) = 1;$$

hence $(a_t, b_t) = 1$. The three coordinate divisibilities in (461) are visible in (468). Finally,

$$\begin{aligned} (t + 1)^2 - t^2 &= 2t + 1 = a_t, \\ (2t + 1)^2 - (t + 1)^2 &= t(3t + 2) = b_t, \\ (2t + 1)^2 - t^2 &= (t + 1)(3t + 1) = c_t. \end{aligned}$$

Thus the gaps are exact and Proposition 78.4 applies. The packet is proper because $b_t > y_t$ and $c_t > z_t$. $\square$

In this family $a_t b_t c_t \asymp 18t^5$ and $T(Q_t) = 2t + 1 \asymp 2t$. Consequently no exponent strictly smaller than five can replace six in a universal estimate with constant one. The proposition is a square-difference identity and overlaps familiar quadratic constructions. Its role here is to exhibit an exact stratum of the finite packet spectrum; it is not presented as a renamed independent discovery.

78.3. Prime-power orientations and a rigid global sign. For an odd prime power $p^e \parallel a$, the first square congruence factors as

$$p^e \mid (y - z)(y + z).$$

The prime p cannot divide both factors: otherwise it divides $2y$ and $2z$, and oddness together with $(y, z) = 1$ gives a contradiction. The whole prime power therefore occupies exactly one of the channels

$$(469) \quad p^e \mid y - z \quad \text{or} \quad p^e \mid y + z.$$

The same statement holds cyclically on the other arms. Different prime powers on one composite arm may choose different signs. At the unique possible even arm the factors need not be coprime, so the 2-primary branch must be retained separately.

Definition 78.6 (Canonical orientation). A packet is canonically oriented when

$$(470) \quad y \equiv z \pmod{a}, \quad x \equiv z \pmod{b}, \quad x \equiv -y \pmod{c}.$$

These are the three whole-arm signs used by the full packet.

Theorem 78.7 (Canonical-orientation rigidity). *A canonically oriented packet is the full packet (a, b, c) .*

Proof. The divisor conditions give $x \leq a$, $y \leq b$, and $z \leq c$. The final congruence in (470) says $c \mid x + y$. Because $x + y > 0$,

$$c \leq x + y \leq a + b = c.$$

It follows that $x + y = a + b$; the two coordinate inequalities then force $x = a$ and $y = b$.

We claim that $c < ab$. Since $a, b > 1$, the product $(a - 1)(b - 1)$ is positive. If it were one, then $a = b = 2$, contradicting $(a, b) = 1$. Hence $(a - 1)(b - 1) \geq 2$, which is equivalent to $a + b < ab$.

After substituting $x = a$ and $y = b$, the first two congruences say that z and $c = a + b$ have the same residue modulo both a and b . The coprime Chinese remainder theorem yields $z \equiv c \pmod{ab}$. But $0 < z \leq c < ab$, so $z = c$. $\square$

Every proper packet therefore has an actual orientation defect. On odd prime powers it changes or fragments at least one of the canonical signs; the possible even arm carries the separate 2-adic ambiguity.

78.4. Connection with radical, height, and quality. Let

$$\mathcal{R} = \text{rad}(abc), \quad q_s(a, b, c) = \frac{\log c}{\log \mathcal{R}}.$$

The spectrum is finite and nonempty, so one may define

$$(471) \quad \mathcal{E}_{\text{sync}}(a, b, c) = \min_{Q \in \mathcal{S}(a, b, c)} \frac{\log B(Q)}{\log \mathcal{R}}.$$

Corollary 78.8 (Packet-energy majorant). *For every triple in the present scope,*

$$q_s(a, b, c) \leq \mathcal{E}_{\text{sync}}(a, b, c).$$

Proof. Theorem 78.3 gives $c \leq abc \leq B(Q)$ for every packet. Take logarithms, divide by the positive number $\log \mathcal{R}$, and then take the minimum. $\square$

The original candidate global statement is now visible. If, for each $\varepsilon > 0$ and all but finitely many primitive nonunit triples, one could produce a packet with

$$(472) \quad B(Q) \leq \mathcal{R}^{1+\varepsilon},$$

then Corollary 78.8 would imply the usual abc inequality. The infinite primitive family established in the companion radical-excess note refutes (472) at $\varepsilon = 1/3$. Only this uncompensated gate is retired. The corrected live gate inserts the factor ab :

$$(473) \quad B(Q) \leq ab \mathcal{R}^{1+\varepsilon}.$$

Together with $abc \leq B(Q)$, this would imply $c \leq \mathcal{R}^{1+\varepsilon}$. No eventual estimate of the compensated form is established or refuted here.

For comparison, Sankaran's alternative-quality construction [26] assigns to the distinct primes $p_1, \dots, p_\omega$ dividing abc the arithmetic and geometric means

$$A = \frac{1}{\omega} \sum_i \log p_i, \quad G = \left(\prod_i \log p_i \right)^{1/\omega}, \quad \eta = G/A,$$

and proves the exact factorization

$$q_s = \eta q_{\text{DGM}}, \quad q_{\text{DGM}} = \frac{\log c}{\omega G}.$$

The packet spectrum is not this scalar factorization under another name. It depends on exponent caps, arm assignment, simultaneous modular incidence, and orientation channels. For example,

$(2, 3, 5)$ and $(3, 5, 8)$ have the same radical 30 and hence the same η , but the first has one packet and the second has the additional proper packet $(3, 5, 2)$.

78.5. Counterexample and finite-search audit. The following complete-premise examples eliminate tempting stronger claims:

(a, b, c)	packet (x, y, z)	eliminated claim
$(3, 5, 8)$	$(3, 5, 2)$	uniqueness of the full packet
$(5, 7, 12)$	$(5, 7, 2)$	$abc \leq T^3$
$(5, 16, 21)$	$(5, 2, 3)$	$abc \leq T^4$ and $abc \leq (xyz)^2$
$(385, 527, 912)$	$(7, 31, 24)$	$abc \leq T^5$

In the last two rows the three gaps equal (a, b, c) exactly. In particular,

$$385 \cdot 527 \cdot 912 = 185040240 > 31^5 = 28629151.$$

These counterexamples do not touch the pair-max/sixth-power theorem.

A deterministic integer search scanned all 3,795,230 normalized primitive triples with $c \leq 5000$. It enumerated every packet for $c \leq 1000$, for the top 200 standard-quality triples through $c = 5000$, and for every triple in that range with $q_s \geq 1$. Across 151,244 enumerated triples it found 151,711 packets, including 467 proper packets and 105 index-one packets. Exact assertions checked the three packet congruences, the product divisibility, and both bounds in (465). The family in Proposition 78.5 was separately checked for $2 \leq t \leq 100000$. The selection $q_s \geq 1$ used the equivalent exact integer condition $c \geq \text{rad}(abc)$; floating-point logarithms were used for ranking and reported ratios.

The twenty highest-quality triples in the finite range had only their full packet. This is adverse finite evidence, but a finite range neither proves eventual packet rigidity nor decides the compensated all-but-finitely-many assertion.

78.6. Lean boundary and audit. The independent Lean companion formalizes the exact finite core: the primitive data, synchronized divisor packets, inherited coprimality, product divisibility, pair-max and sixth-power bounds, canonical-orientation rigidity, and generic channel lemmas. It also defines the actual squarefree radical, standard quality, packet energy, and the minimum over the finite nonempty packet spectrum, and proves the quality majorant in Corollary 78.8. Each counterexample in the preceding table is an explicit Lean datum and packet satisfying all packet premises. The deterministic computation and both global radical estimates are absent from this original formalization; in particular no upper bound on the minimum packet energy is assumed. The companion radical-excess module formalizes the counterfamily and the rational-exponent compensated implication. A separate axiom audit lists all dependencies.

Thus the surviving output of this route is a finite structural invariant and the compensated radical-control problem. The original uncompensated problem has a full-premise counterfamily. Neither result proves or disproves the abc conjecture.

79. RADICAL-EXCESS OBSTRUCTIONS TO PACKET COMPRESSION

Author: ChatGPT. We continue with the synchronized divisor packets of Section 78. The purpose of this section is twofold. First, we identify the exact order-statistic geometry hidden in the pair-max envelope. Second, we correct the proposed radical-scale packet gate by giving an infinite complete-premise counterfamily. The counterfamily disproves the packet gate, not the abc conjecture.

Let

$$\mathcal{R} = \text{rad}(abc), \quad E = \frac{abc}{\mathcal{R}}.$$

The integer E will be called the radical excess. Recall that every synchronized packet $Q = (x, y, z)$ satisfies $abc \leq B(Q)$.

79.1. The shape identity. Put

$$m(Q) = \min(x, y, z), \quad T(Q) = \max(x, y, z),$$

and

$$L(Q) = \max(y, z) \max(x, z) \max(x, y).$$

Thus $B(Q) = L(Q)^2$.

Theorem 79.1 (Symmetric packet-shape identity). *For every synchronized packet,*

$$m(Q)L(Q) = xyzT(Q), \quad m(Q)^2B(Q) = (xyzT(Q))^2.$$

Proof. Sort the coordinates as $r \leq s \leq t$. Their pair maxima are s, t, t , so $m = r$, $T = t$, and $L = st$. Consequently $mL = rst^2 = (rst)t = xyzT$. Squaring proves the second identity. Since both sides are symmetric, the computation is independent of the sorting permutation. $\square$

Proposition 79.2 (Support and full-envelope monotonicity). *Every synchronized packet satisfies*

$$xyz \mid abc, \quad \text{rad}(xyz) \mid \mathcal{R}, \quad B(Q) \leq B(a, b, c).$$

Proof. Multiply the three coordinate divisibilities. Radical divisibility is monotone under divisibility for nonzero natural numbers. Finally, $x \leq a$, $y \leq b$, and $z \leq c$; hence every pair maximum for Q is no larger than its full-packet counterpart. Square and multiply. $\square$

Equality of coordinate support with full radical support is false. The complete-premise packet $(5, 7, 2)$ over $(5, 7, 12)$ has coordinate radical 70, whereas $\text{rad}(5 \cdot 7 \cdot 12) = 210$.

79.2. The radical-excess obstruction.

Theorem 79.3 (Necessary condition for rational packet compression). *Let $m, n \geq 0$. If*

$$B(Q)^m \leq \mathcal{R}^{m+n},$$

then

$$E^m \leq \mathcal{R}^n.$$

Equivalently, $\mathcal{R}^n < E^m$ implies $\mathcal{R}^{m+n} < B(Q)^m$ for every synchronized packet Q over the same primitive triple.

Proof. Since $\mathcal{R} \mid abc$, one has $\mathcal{R}E = abc$. Therefore

$$\mathcal{R}^m E^m = (abc)^m \leq B(Q)^m \leq \mathcal{R}^{m+n} = \mathcal{R}^m \mathcal{R}^n.$$

The radical is positive, so cancellation of $\mathcal{R}^m$ gives the assertion. $\square$

This condition depends only on the underlying triple. In particular, no choice inside the packet spectrum can repair its failure.

Theorem 79.4 (Infinite four-thirds obstruction). *For $k \geq 0$, set*

$$A_k = 2^{k+4}, \quad P_k = (A_k, 3, A_k + 3).$$

These are pairwise distinct primitive nonunit triples. If $\mathcal{R}_k = \text{rad}(A_k \cdot 3 \cdot (A_k + 3))$, then every synchronized packet over P_k satisfies

$$\mathcal{R}_k^4 < B(Q)^3.$$

Consequently the locus on which every packet violates $B^3 \leq \mathcal{R}^4$ is infinite.

Proof. Put $h = 2^{k+3}$, so $A_k = 2h$ and $h \geq 8$. The triple is primitive because A_k is a power of 2 and is coprime to 3. Radical submultiplicativity gives

$$\mathcal{R}_k \leq \text{rad}(A_k) \text{rad}(3) \text{rad}(A_k + 3) \leq 2 \cdot 3 \cdot (A_k + 3) = 6(2h + 3).$$

Writing $E_k = A_k 3 (A_k + 3) / \mathcal{R}_k$, the equality $\mathcal{R}_k E_k = 6h(A_k + 3)$ and the preceding upper bound imply $h \leq E_k$. Moreover,

$$\mathcal{R}_k \leq 6(2h + 3) = 12h + 18 < 64h \leq h^3 \leq E_k^3.$$

Theorem 79.3, with $(m, n) = (3, 1)$, now gives $\mathcal{R}_k^4 < B(Q)^3$ for every packet. The map $k \mapsto P_k$ is injective because its first coordinate is strictly increasing. $\square$

For positive real $\mathcal{R}$ and B , an estimate $B \leq \mathcal{R}^{4/3}$ implies $B^3 \leq \mathcal{R}^4$. Theorem 79.4 therefore refutes the earlier all-but-finitely-many packet gate $B(Q) \leq \mathcal{R}^{1+\varepsilon}$ at the exact value $\varepsilon = 1/3$. It does not produce triples violating $c \leq \mathcal{R}^{1+\varepsilon}$, and hence gives no counterexample to abc .

79.3. A correctly compensated forward gate.

Theorem 79.5 (Compensated packet implication). *Let $m, n \geq 0$. If*

$$B(Q)^m \leq (ab)^m \mathcal{R}^{m+n},$$

then

$$c^m \leq \mathcal{R}^{m+n}.$$

Proof. The synchronized product bound gives

$$(ab)^m c^m = (abc)^m \leq B(Q)^m \leq (ab)^m \mathcal{R}^{m+n}.$$

Cancel the positive factor $(ab)^m$. □

Thus the factor ab is the exact elementary compensator that changes a packet bound for the full product into the desired bound for the sum arm. No eventual compensated estimate is proved here. Its pointwise four-thirds form already fails at $(2, 3, 5)$, whose packet spectrum consists only of the full packet. The all-but-finitely-many compensated direction remains open because this section supplies no infinite counterfamily to it.

79.4. Formal and computational audit. The companion Lean module formalizes every identity and implication above, the infinite dyadic obstruction, and the finite counterexamples. Its axiom audit contains only `propext`, `Classical.choice`, and `Quot.sound`.

An exact integer search exhausts every normalized primitive nonunit triple with $c \leq 3000$ and every divisor triple satisfying all packet premises. It checks 1,365,095 primitive triples, 1,366,531 packets, and 1,436 proper packets. The same program separately factors and exhausts the first twenty-one dyadic rows. Null finite searches are recorded only as evidence and are not promoted to universal statements.

80. CANONICAL GAIN SURFACES AND DEFECT FLAGS

Müller and Taktikos propose separating the quality of certain abc equations into an approximation gain and a power gain [27]. The factorization itself is exact and useful, but the proposed fixed bounds on the two factors do not supply the coefficient-one quantifier in the standard abc conjecture. We record the precise correlation that is needed and then introduce a finite-dimensional defect flag on which different arithmetic mechanisms can act.

The source requires two cautions. Definition 2.6 prints $\log(q_n^s k)$ in the numerator when $d_n > 0$, whereas the proof of Theorem 2.7 uses $\log(p_n^3)$ for the same sign. These are distinct when $p_n^3 = kq_n^3 + d_n$ and $d_n > 0$. Moreover, the printed proof of Theorem 2.8 obtains a limiting equality from a one-sided asymptotic estimate and refers to a finite verification below an unspecified cutoff. We therefore use no continued-fraction assertion from that theorem. The frozen source and its audit extraction are recorded under `research/sources/abc_gain_surface_2026_09_03/`.

80.1. The gain surface. Let $H, N, \mathcal{R} > 1$ and write

$$h = \log H, \quad n = \log N, \quad \rho = \log \mathcal{R}.$$

Define

$$q(H, \mathcal{R}) = \frac{h}{\rho}, \quad A(H, N) = \frac{h}{n}, \quad P(N, \mathcal{R}) = \frac{n}{\rho}.$$

Proposition 80.1 (Exact gain surface). *For every intermediate scale $N > 1$,*

$$q(H, \mathcal{R}) = A(H, N)P(N, \mathcal{R}).$$

Proof. Both n and ρ are positive, and $(h/n)(n/\rho) = h/\rho$. □

The level set $q = 1 + \varepsilon$ is the hyperbola $AP = 1 + \varepsilon$. Separate bounds $A \leq A_0$ and $P \leq P_0$ give only $q \leq A_0 P_0$. Thus the proposed constants $3/2$ and 3 give a weak exponent $9/2$, rather than the standard $1 + \varepsilon$ for every positive ε .

This failure is not an issue of endpoints. In logarithmic coordinates take $(\rho, h, n) = (2, 4, 5)$ and exponentiate. Then $1 < \mathcal{R} < H < N$ while

$$A = \frac{4}{5} < \frac{3}{2}, \quad P = \frac{5}{2} < 3, \quad q = 2 > \frac{3}{2}.$$

This is a full-premise countermodel to the abstract implication from the two independent bounds. It is not an abc triple.

80.2. The canonical arithmetic corridor. Let $a + b = c$ be a primitive positive triple with $a, b > 1$, and set

$$M = abc, \quad \mathcal{R} = \text{rad}(abc), \quad A_{\text{can}} = \frac{\log c}{\log M}, \quad P_{\text{can}} = \frac{\log M}{\log \mathcal{R}}.$$

Theorem 80.2 (Canonical gain corridor). *Every such triple satisfies*

$$c^2 < M < c^3, \quad \frac{1}{3} < A_{\text{can}} < \frac{1}{2}, \quad q(a, b, c) = A_{\text{can}} P_{\text{can}}.$$

Consequently

$$\frac{1}{3} P_{\text{can}} < q(a, b, c) < \frac{1}{2} P_{\text{can}}.$$

Proof. Because $a, b > 1$ and $(a, b) = 1$, the pair cannot be $(2, 2)$, so $(a - 1)(b - 1) > 1$. Expansion and $a + b = c$ give $ab > c$, whence $abc > c^2$. Also $a < c$ and $b < c$, so $abc < c^3$. Strict monotonicity of the logarithm gives

$$2 \log c < \log M < 3 \log c,$$

which is equivalent to the claimed corridor. The factorization is Proposition 80.1; multiplying the corridor by the positive number P_{can} proves the last display. $\square$

Corollary 80.3 (Necessary canonical power concentration). *If $q(a, b, c) \geq 1 + \varepsilon$, then*

$$P_{\text{can}} > 2(1 + \varepsilon).$$

Conversely, $P_{\text{can}} \leq 2(1 + \varepsilon)$ implies $q(a, b, c) < 1 + \varepsilon$.

Proof. Use $q < P_{\text{can}}/2$ from Theorem 80.2. $\square$

The coefficient two is the same sharp threshold found by the symmetric product transfer, but the factorized form exposes the cancellation that the coefficient statement hides.

80.3. The affine defect plane. Put $h = \log c$, $m = \log M$, and $\rho = \log \mathcal{R}$, and define

$$X = \frac{m - \rho}{\rho}, \quad Y = \frac{m - h}{\rho}.$$

Here X is the normalized multiplicity discarded by the radical and Y is the normalized product-to-height slack.

Theorem 80.4 (Exact cancellation). *One has*

$$\boxed{q(a, b, c) = 1 + X - Y.}$$

Consequently the effective logarithmic inequality

$$h \leq (1 + \varepsilon)\rho + C$$

is equivalent to

$$X - Y \leq \varepsilon + \frac{C}{\rho}.$$

Proof. Cancellation gives

$$1 + \frac{m - \rho}{\rho} - \frac{m - h}{\rho} = \frac{h}{\rho}.$$

The second assertion follows by subtracting one and dividing by $\rho > 0$. $\square$

Thus a large power excess can be harmless when the approximation slack is comparably large. Independent constant bounds erase this relation.

80.4. A scale groupoid and higher-dimensional flags. On real scale coordinates define

$$\delta(x, y) = y - x.$$

This is an additive one-cocycle on the pair groupoid:

$$\delta(x, x) = 0, \quad \delta(x, z) = \delta(x, y) + \delta(y, z).$$

On nonzero coordinates, $g(x, y) = y/x$ is the corresponding multiplicative cocycle. Both identities are immediate by cancellation.

A *defect flag* is a finite path

$$\rho = s_0, s_1, \dots, s_k = m, \quad d_i = s_i - s_{i-1}.$$

Its total is path-independent:

$$\sum_{i=1}^k d_i = m - \rho.$$

In particular every closed scale flag has trivial additive and multiplicative holonomy. Refining a factorization cannot manufacture a saving; arithmetic must restrict the admissible paths or bound their costs.

Theorem 80.5 (Defect-budget half-space). *Suppose*

$$\sum_i d_i \leq \sum_i B_i$$

and

$$\sum_i B_i \leq (m - h) + \varepsilon\rho + C.$$

Then

$$h \leq (1 + \varepsilon)\rho + C.$$

Proof. The cocycle law telescopes, so $m - \rho = \sum_i d_i \leq \sum_i B_i$. Combining this with the second budget and cancelling m gives the result. $\square$

For a fixed total allowance and no sign constraints, the vectors $(B_1, \dots, B_k)$ lie in an affine half-space. Its intersection with the nonnegative orthant is a simplex when the allowance is nonnegative. Packet compression, Pell transversality, IUT log-volume estimates, and analytic radical bounds may therefore be assigned to different coordinates, but their total must remain below the approximation slack plus the ε -allowance.

80.5. An exact counterexample boundary. The tempting universal strengthening $P_{\text{can}} \leq 3$ is false. For the primitive hit

$$3 + 125 = 128$$

one has

$$M = 48000 = 2^7 \cdot 3 \cdot 5^3, \quad \mathcal{R} = 30,$$

and $48000 > 30^3 = 27000$. Hence

$$P_{\text{can}} = \frac{\log 48000}{\log 30} > 3.$$

This retires only the universal power-three cap. It does not refute the correlated defect barrier or the *abc* conjecture.

An exhaustive computation of the 5,465,583 unordered primitive nonunit triples with $c \leq 6000$ found no violation of the canonical approximation corridor, 65 triples of quality above one, and 14 triples with canonical power gain above three. These data are finite evidence only.

The exact open gate is, for every $\varepsilon > 0$, to prove

$$\sum_i d_i \leq (m - h) + \varepsilon \rho + C_\varepsilon.$$

This is equivalent to standard *abc*, so the coordinate construction alone does not close the conjecture. It supplies a common, noncircular accounting surface on which the different routes can be combined.

81. A LABELLED VALUATION-INCIDENCE COMPLEX

This section develops a higher-dimensional object directly from the prime factorisation of an *abc* triple. It is deliberately restricted to *primitive nonunit triples*

$$a, b, c \in \mathbb{N}_{>0}, \quad a > 1, \quad b > 1, \quad a + b = c, \quad \gcd(a, b) = 1.$$

The restriction is part of every statement below. Unit-arm triples require a separate extension. The construction is finite and exact. The next two sections refute its first one-face quantitative selectors and preserve the remaining weighted, multi-face, three-arm, and homological design problem.

Put $x_A = a$, $x_B = b$, and $x_C = c$. For $i \in \{A, B, C\}$ let

$$S_i = \{p : p \text{ prime}, p \mid x_i\}, \quad e_i(p) = v_p(x_i).$$

Primitivity and $a + b = c$ make the three coordinates pairwise coprime. Thus their prime supports are disjoint, although we retain the arm label because it also records a different local congruence.

Definition 81.1 (Valuation-incidence complex). A vertex is a labelled pair (i, p) with $p \in S_i$, carrying the datum $(i, p, e_i(p))$. A face is a triple

$$F = (F_A, F_B, F_C), \quad F_i \subseteq S_i.$$

We write $F \preceq G$ if $F_i \subseteq G_i$ on every arm. Union and intersection are taken armwise. The empty face is $\emptyset_P = (\emptyset, \emptyset, \emptyset)$ and the top face is $\top_P = (S_A, S_B, S_C)$.

The face relation is the Boolean face lattice on $S_A \sqcup S_B \sqcup S_C$. In particular, it is reflexive, transitive and antisymmetric; the empty and top faces are respectively least and greatest; and armwise union is the least common upper bound. These assertions follow coordinatewise from the corresponding elementary facts for finite subsets. If

$$\nu(F) = |F_A| + |F_B| + |F_C|, \quad \dim_{\mathbb{N}}(F) = \max\{\nu(F) - 1, 0\},$$

then a nonempty top face with N vertices has dimension $N - 1$. There is no a priori bound on N .

For a face F and an arm i , define

$$\begin{aligned} R_i(F) &= \prod_{p \in F_i} p, & D_i(F) &= \prod_{p \in F_i} p^{e_i(p)-1}, \\ M_i(F) &= \prod_{p \in F_i} p^{e_i(p)}, & d_i(F) &= \sum_{p \in F_i} (e_i(p) - 1). \end{aligned}$$

Thus R_i records selected radical mass, D_i selected multiplicity beyond the first layer, M_i the corresponding prime-power modulus, and d_i its additive defect degree. The exact six-coordinate realisation is

$$T(F) = ((R_A(F), D_A(F)), (R_B(F), D_B(F)), (R_C(F), D_C(F))).$$

Applying logarithms produces an additive display in $\mathbb{R}^6$, but logarithms play no role in face membership or in the filtration below.

Proposition 81.2 (Exact face factorisation). *For every face and every arm,*

$$R_i(F)D_i(F) = M_i(F), \quad M_i(F) \mid x_i.$$

At the top face,

$$M_i(\top_P) = x_i, \quad R_i(\top_P) = \text{rad}(x_i), \quad \text{rad}(x_i)D_i(\top_P) = x_i.$$

If F and G are armwise disjoint, each of R_i, D_i, M_i is multiplicative under $F \cup G$, while $d_i(F \cup G) = d_i(F) + d_i(G)$.

Proof. For every selected prime, positivity of $e_i(p)$ gives $pp^{e_i(p)-1} = p^{e_i(p)}$. Multiplication over F_i proves the first identity. The selected product is a subproduct of the canonical factorisation of x_i , so it divides x_i ; on the top face it is the full factorisation. Retaining one copy of each support prime gives the usual squarefree radical. The disjoint-union formulas are the standard product and sum formulas over disjoint finite sets. $\square$

For a budget $u = (u_A, u_B, u_C) \in \mathbb{N}^3$, define the filtered subcomplex

$$\mathcal{K}_u(P) = \{F : d_i(F) \leq u_i \text{ for } i = A, B, C\}.$$

Proposition 81.3 (Defect filtration). *If $G \preceq F$ and $F \in \mathcal{K}_u(P)$, then $G \in \mathcal{K}_u(P)$. The empty face lies in every filtration level. If F, G are armwise disjoint, $F \in \mathcal{K}_u(P)$ and $G \in \mathcal{K}_v(P)$, then $F \cup G \in \mathcal{K}_{u+v}(P)$. Moreover,*

$$F \in \mathcal{K}_{(0,0,0)}(P) \iff e_i(p) = 1 \text{ for every selected } (i, p).$$

Proof. Every summand $e_i(p) - 1$ is nonnegative. Restricting to a subset can only decrease its sum, and the empty sums vanish. The graded-union assertion uses the additive formula in Proposition 81.2. At budget zero, a finite sum of nonnegative integers is zero precisely when each summand is zero; positivity of $e_i(p)$ then gives the final equivalence. $\square$

The arm colours also retain the additive equation at selected prime-power resolution.

Proposition 81.4 (Local incidence and CRT rigidity). *Every face satisfies*

$$c \equiv b \pmod{M_A(F)}, \quad c \equiv a \pmod{M_B(F)}, \quad a + b \equiv 0 \pmod{M_C(F)}.$$

The three moduli are pairwise coprime. Call F AB-reconstructing when $c < M_A(F)M_B(F)$. Every top face is AB-reconstructing. More generally, if F is AB-reconstructing and

$$0 \leq x < M_A(F)M_B(F), \quad x \equiv b \pmod{M_A(F)}, \quad x \equiv a \pmod{M_B(F)},$$

then $x = c$.

Proof. Proposition 81.2 gives $M_A(F) \mid a$, $M_B(F) \mid b$, and $M_C(F) \mid c$; subtracting within $a + b = c$ proves the three congruences. Since each selected modulus divides one member of a pairwise coprime triple, the moduli are pairwise coprime. At the top face, $M_A = a$ and $M_B = b$. For nonunits one has $a + b \leq ab$, with equality only at $a = b = 2$; primitivity excludes that case, so $a + b < ab$. For a general reconstructing face, the displayed residues show that x and c are congruent modulo both coprime moduli, hence modulo their product. Both lie in the same half-open interval of that length, so they are equal. $\square$

The top point recovers the ordinary conductor exactly:

$$R_A(\top_P)R_B(\top_P)R_C(\top_P) = \text{rad}(abc).$$

It also gives a precise half-space form of a rational-power abc bound.

Theorem 81.5 (Exact top-face half-space). *Write $R_i = R_i(\top_P)$ and $D_C = D_C(\top_P)$. For all $m, n \in \mathbb{N}$,*

$$c^m \leq \text{rad}(abc)^{m+n} \iff D_C^m \leq (R_A R_B)^{m+n} R_C^n.$$

Furthermore, for any face F , the stronger selected-face inequality

$$D_C^m \leq (R_A(F)R_B(F))^{m+n} R_C^n$$

implies $c^m \leq \text{rad}(abc)^{m+n}$.

Proof. The top-face identities give $c = R_C D_C$ and $\text{rad}(abc) = R_A R_B R_C$. Substitution turns the left inequality into

$$R_C^m D_C^m \leq R_C^m ((R_A R_B)^{m+n} R_C^n).$$

Since $R_C > 0$, cancellation proves the equivalence. For the second claim, $F_i \subseteq S_i$ gives $R_A(F) \leq R_A$ and $R_B(F) \leq R_B$; enlarge the right side of the selected-face inequality and apply the reverse implication of the equivalence. $\square$

Theorem 81.5 is an exact change of coordinates. It does not prove that a useful selected face exists. The following fully realised example also prevents two tempting degeneracy assumptions.

Proposition 81.6 (A five-dimensional complete-premise boundary). *The primitive nonunit triple*

$$12 + 833 = 845, \quad 12 = 2^2 \cdot 3, \quad 833 = 7^2 \cdot 17, \quad 845 = 5 \cdot 13^2$$

has a six-vertex top face, hence natural dimension five, and

$$T(\top_P) = ((6, 2), (119, 7), (65, 13)), \quad (d_A, d_B, d_C) = (1, 1, 1).$$

Every arm has nontrivial defect, and on every arm the two valuation exponents are different. The face selecting 3, 17, and 5 lies at zero budget.

Proof. The displayed factorizations list the six vertices and give all coordinates by their definitions. They also show directly that one exponent on each arm is 2 and the other is 1. Thus the defects are 2, 7, 13, all nontrivial, and every defect degree is one. Selecting the three exponent-one vertices gives defect degree zero on every arm. $\square$

This example refutes exactly the pointwise universal statements that some arm must have defect one and that the valuations must be constant on each arm. It does not refute eventual or density variants and does not refute any estimate that permits all three arms to be defective.

The first falsifiable gate proposed for this higher-dimensional route used an absolute budget. For fixed $m, n \geq 1$, it asked whether a number $t = t(m, n)$ exists such that all but finitely many primitive nonunit triples possess a face F satisfying

$$(\text{VIC-ABS-1}) \quad d_A(F), d_B(F) \leq t, \quad c < M_A(F) M_B(F), \quad D_C^m \leq (R_A(F) R_B(F))^{m+n} R_C^n.$$

Theorem 81.5 proves that VIC-ABS-1 would imply the corresponding rational-exponent height bound on this domain. The next section constructs an infinite complete-premise family that refutes this absolute budget formulation. That obstruction does not refute the complex or its weighted, multi-face, three-arm, and homological replacements.

The companion Lean module `ABCValuationIncidenceComplex20260903` constructs the actual faces with `Nat.primeFactors` and the labels with `Nat.factorization`. It kernel-checks the finite-poset, factorisation, filtration, congruence, CRT, half-space, selected-face, and witness statements above. No selector is inserted as an axiom or an inhabited interface.

82. THE FIXED-BUDGET INCIDENCE OBSTRUCTION

We now test VIC-ABS-1 against every one of its necessary reconstruction premises. The resulting obstruction is asymptotic and uses actual primitive nonunit triples; it therefore retires the exact all-but-finitely-many absolute-budget statement.

For $k \geq 0$, define

$$P_k = (2^{k+4}, 3, 2^{k+4} + 3).$$

Every P_k is primitive and nonunit, and the family is injective through its first coordinate.

Theorem 82.1 (No fixed-budget reconstruction on the dyadic tail). *Let $t, k \in \mathbb{N}$ with $t < k + 3$. No face of P_k satisfies both*

$$d_A(F) \leq t \quad \text{and} \quad c < M_A(F) M_B(F).$$

Consequently, for every fixed t , infinitely many primitive nonunit triples have no face that both reconstructs through the two summand arms and has A -defect at most t . In particular, VIC-ABS-1 is false for every pair of rational-power parameters (m, n) .

Proof. The A -coordinate of P_k is 2^{k+4} . Its support consists only of 2, with valuation $k+4$. If a face selected this vertex, then its contribution to $d_A(F)$ would be

$$v_2(2^{k+4}) - 1 = k + 3 > t,$$

contrary to the budget. Hence F_A is empty and $M_A(F) = 1$. The selected B -modulus divides the coordinate 3, so $M_B(F) \leq 3$. It follows that

$$M_A(F)M_B(F) \leq 3 < 2^{k+4} + 3 = c,$$

which rules out reconstruction. For a fixed t , every $k \geq t$ satisfies $t < k + 3$; the injective tail therefore supplies infinitely many failures. Every full VIC-ABS-1 face would satisfy the two conditions just excluded, so its additional quantitative inequality cannot repair the selector. $\square$

The theorem retires the fixed absolute defect budget and nothing more. In particular, it leaves intact all face-poset, local-incidence, CRT, and half-space results of Section 81. It also suggests where the missing scale entered: selecting one high-multiplicity arm can cost logarithmically in the height.

The first attempted correction was therefore scale-sensitive. Put

$$\ell_2(c) = \min\{s \in \mathbb{N} : c \leq 2^s\}.$$

For fixed $m, n \geq 1$, VIC-1R asks whether a slack $t = t(m, n)$ exists such that all but finitely many primitive nonunit triples possess a face satisfying

$$d_A(F) + d_B(F) \leq \ell_2(c) + t,$$

(VIC-1R)

$$c < M_A(F)M_B(F),$$

$$D_C^m \leq (R_A(F)R_B(F))^{m+n} R_C^n.$$

The dyadic obstruction does not violate the scale budget merely by selecting the prime 2. Nevertheless, the next companion section gives a different balanced two-prime family that refutes VIC-1R itself. The surviving parent route must therefore use weighted budgets, several faces with combined congruence data, all three labelled arms, or another invariant absent from this one-face selector. Each replacement remains subject to simultaneous positive proof construction and complete-premise counterexample search.

The Lean module `ABCValuationIncidenceFixedBudgetObstruction20260903` defines the full pointwise selector and its all-but-finitely-many form. It kernel-checks the dyadic valuation, support exclusion, modulus bounds, infinite failure set, and the negation of VIC-ABS-1. It does not formalise or assume VIC-1R; that separate statement is handled by the scale-budget obstruction module.

83. THE COEFFICIENT-ONE SCALE-BUDGET OBSTRUCTION

We now test the first scale-sensitive repair of the fixed-budget incidence selector. Write

$$\ell_2(c) = \min\{s \in \mathbb{N} : c \leq 2^s\}.$$

For fixed $m, n \geq 1$, VIC-1R asks for a fixed additive slack $t = t(m, n)$ such that all but finitely many primitive nonunit triples admit a face F with

$$d_A(F) + d_B(F) \leq \ell_2(c) + t,$$

(VIC-1R)

$$c < M_A(F)M_B(F),$$

$$D_C^m \leq (R_A(F)R_B(F))^{m+n} R_C^n.$$

The following obstruction satisfies all ambient arithmetic premises and fails the first two necessary conditions together on an infinite tail.

For $r \geq 1$, put

$$Q_r = (2^{2r}, 3^r, 2^{2r} + 3^r).$$

The two summands are coprime, so Q_r is primitive; all three coordinates exceed one. The family is injective through its first coordinate.

Theorem 83.1 (No coefficient-one scale-budget reconstruction). *Let $r \geq 1$ and let F be an AB -reconstructing face of Q_r . Then*

$$d_A(F) + d_B(F) = 3r - 2 \quad \text{and} \quad \ell_2(2^{2r} + 3^r) \leq 2r + 1.$$

Consequently, for every fixed $t \in \mathbb{N}$, every $r > t + 3$ fails

$$d_A(F) + d_B(F) \leq \ell_2(c) + t.$$

In particular, VIC-1R is false for every pair $m, n \geq 1$.

Proof. The A -support is the singleton $\{2\}$ and the B -support is the singleton $\{3\}$. If 2 were omitted, then $M_A(F) = 1$ and $M_B(F) \mid 3^r$, whence

$$M_A(F)M_B(F) \leq 3^r < 2^{2r} + 3^r = c,$$

contrary to reconstruction. If 3 were omitted, the symmetric estimate $M_A(F)M_B(F) \leq 2^{2r} < c$ gives the same contradiction. Thus both vertices are selected, and their excess valuations give

$$d_A(F) + d_B(F) = (2r - 1) + (r - 1) = 3r - 2.$$

Since $3^r < 4^r = 2^{2r}$ for $r \geq 1$,

$$c = 2^{2r} + 3^r < 2^{2r+1},$$

so the defining minimum satisfies $\ell_2(c) \leq 2r + 1$. If $r > t + 3$, then

$$3r - 2 > 2r + 1 + t \geq \ell_2(c) + t.$$

Hence no reconstructing face meets the budget. Infinitely many distinct Q_r occur for every fixed t . The third line of VIC-1R cannot create a face after the first two necessary conditions have become incompatible, so the all-but-finitely-many selector fails for every m, n . $\square$

This theorem retires the coefficient-one binary-scale one-face selector and nothing broader. It does not refute the valuation-incidence complex, its exact half-space theorem, or standard *abc*. Surviving constructions may use logarithmically weighted defect, a larger scale coefficient coupled to a new quantitative restriction, several faces with joint CRT data, three-arm reconstruction, or a homological invariant of the filtered complex. None of these replacements is assumed here.

The ordinary proof above was fixed before formalization. The Lean module `ABCValuationIncidenceScaleBudgetObstruction20260903` reindexes by $r = k + 1$, defines the binary scale by a least-witness construction, proves the exact defect $3k + 1$ and scale bound $2k + 3$, constructs an infinite failure set for every slack, and negates the full pointwise and all-but-finitely-many VIC-1R predicates. Its one-for-one audit covers all 34 public declarations. The dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`; there are no custom axioms, placeholders, or native decision oracles.

84. THREE-ARM INCIDENCE COVERS AND COMPLEMENT TRANSPORT

The preceding obstructions rule out two one-face selectors, but they do not rule out the labelled incidence complex itself. We now allow a face to use prime-power vertices on all three arms and charge every selected excess layer by its actual logarithmic weight.

Let $P = (a, b, c)$ be a positive primitive triple with $a + b = c$. For $i \in \{A, B, C\}$ put $x_A = a$, $x_B = b$, and $x_C = c$. A three-arm face F selects a subset F_i of the prime support of each x_i . Define

$$\begin{aligned} R_i(F) &= \prod_{p \in F_i} p, & \overline{R}_i(F) &= \prod_{\substack{p \mid x_i \\ p \notin F_i}} p, \\ D_i(F) &= \prod_{p \in F_i} p^{v_p(x_i)-1}, & M_i(F) &= \prod_{p \in F_i} p^{v_p(x_i)}. \end{aligned}$$

Write $R(F)$, $\overline{R}(F)$, $D(F)$, and $M(F)$ for the products over the three arms, and write $R(P) = \text{rad}(abc)$. We say that F covers the endpoint when $c \leq M(F)$.

Proposition 84.1 (Exact three-arm products and CRT rigidity). *Every three-arm face satisfies*

$$R(F)D(F) = M(F), \quad R(F)\bar{R}(F) = R(P) \quad (3C-1).$$

If $c < M(F)$, $0 \leq x < M(F)$, and

$$x \equiv b \pmod{M_A(F)}, \quad x \equiv a \pmod{M_B(F)}, \quad x \equiv 0 \pmod{M_C(F)},$$

then $x = c$.

Proof. For a selected prime, the identity $pp^{v_p(x_i)-1} = p^{v_p(x_i)}$ gives the first product formula. Selected and unselected primes partition the support on each arm, so their radicals multiply to $\text{rad}(x_i)$. Pairwise coprimality of the coordinates then gives the second formula.

Each selected arm modulus divides its coordinate. The equation $a + b = c$ therefore gives the three displayed congruences for c . The three moduli are pairwise coprime, so the Chinese remainder theorem gives $x \equiv c \pmod{M(F)}$. Both numbers lie in $[0, M(F))$, hence $x = c$. $\square$

The first weighted proposal asks, for every $\varepsilon > 0$, for a constant C_ε such that every P has a covering face satisfying

$$\log D(F) \leq \varepsilon \log R(P) + C_\varepsilon \quad (\text{RAW-3C}).$$

Coverage and (3C-1) would then give

$$\log c \leq \log R(P) + \log D(F) \leq (1 + \varepsilon) \log R(P) + C_\varepsilon,$$

so RAW-3C would imply standard abc . The following complete-premise family shows that this proposal is still too strong.

For $t \geq 1$, put

$$x_t = 2t + 1, \quad y_t = 2t(t + 1), \quad z_t = 2t^2 + 2t + 1,$$

and set $P_t = (x_t^2, y_t^2, z_t^2)$. Euclid's identity gives $x_t^2 + y_t^2 = z_t^2$, while the consecutive-parameter gcd calculation gives $\gcd(x_t, y_t) = 1$. Thus the P_t are distinct positive primitive triples.

Theorem 84.2 (Infinite obstruction to RAW-3C). *RAW-3C is false, already at $\varepsilon = 1/4$.*

Proof. Every selected valuation on P_t is at least two. Consequently $R(F) \leq D(F)$ and

$$M(F) = R(F)D(F) \leq D(F)^2.$$

If F covers, then $z_t^2 \leq M(F)$ and hence $z_t \leq D(F)$. On the other hand, $\text{rad}(u^2) = \text{rad}(u) \leq u$ for every positive integer u , and $x_t, y_t \leq z_t$. Therefore

$$R(P_t) \leq x_t y_t z_t \leq z_t^3.$$

If RAW-3C held at $1/4$, every t would admit a covering face with

$$\log z_t \leq \log D(F) \leq \frac{1}{4} \log R(P_t) + C \leq \frac{3}{4} \log z_t + C.$$

This would bound $\frac{1}{4} \log z_t$ by the fixed constant C , whereas z_t is unbounded. $\square$

The failure identifies the missing resource: the radical vertices omitted from a covering face. For every selected prime p and every layer above its first occurrence, create a source token of weight $\log p$ and key p . For every unselected prime q , create a sink token of capacity $\log q$ and key q . A complement transport is a nonnegative fractional flow from sources to sinks, subject to row and column capacities, in which positive flow from p to q is allowed only when $p \leq q$. Its unmatched source mass is

$$U(F, f) = \sum_s \left(w(s) - \sum_q f(s, q) \right).$$

Proposition 84.3 (Transport accounting). *For every complement transport,*

$$\sum_s w(s) = \log D(F), \quad \sum_q w(q) = \log \bar{R}(F),$$

and

$$\log D(F) \leq \log \bar{R}(F) + U(F, f) \quad (3C-2).$$

For every prime threshold T one also has the necessary Hall inequality

$$\sum_{p(s) > T} w(s) - \sum_{q > T} w(q) \leq U(F, f) \quad (3C-3).$$

Proof. A selected vertex p contributes $v_p(x_i) - 1$ source tokens, so summing their weights gives $\log D(F)$. The sink identity is the logarithm of the complementary squarefree product. If W is the carried mass, then $U = \log D(F) - W$, while the column capacities give $W \leq \log \bar{R}(F)$; this proves (3C-2). A source with key above T can send mass only to a sink whose key is also above T . Comparing that source-tail mass with the available sink-tail capacity proves (3C-3). $\square$

The ordered gate CT-3C asks, for every $\varepsilon > 0$, for a constant C_ε such that every positive primitive P has a covering face and a complement transport satisfying

$$U(F, f) \leq \varepsilon \log R(P) + C_\varepsilon \quad (\text{CT-3C}).$$

Theorem 84.4 (Complement transport implies standard *abc*). *If CT-3C holds, then the standard *abc* conjecture holds.*

Proof. Coverage, (3C-1), and (3C-2) give

$$\begin{aligned} \log c &\leq \log R(F) + \log D(F) \\ &\leq \log R(F) + \log \bar{R}(F) + U(F, f) \\ &= \log R(P) + U(F, f). \end{aligned}$$

Substitution of CT-3C yields $\log c \leq (1 + \varepsilon) \log R(P) + C_\varepsilon$ with the standard quantifier order. $\square$

Section 86 gives an infinite complete-premise refutation of CT-3C. Before that obstruction was isolated, the exact-support computation enumerated all 218,893 unordered primitive triples with $a \leq b$ and $a + b = c \leq 1200$, all covering faces of each triple, the balanced family $(2^{2r}, 3^r, 2^{2r} + 3^r)$ for $1 \leq r \leq 12$, and the first eighty members of the Pythagorean-square family. It finds 1,669 triples without a zero-defect covering face. The largest displayed value of $\min U / \log R(P)$ in the exhaustive range is approximately 0.613147, at $1 + 8 = 9$. These finite ratios can be absorbed by an additive constant and by themselves neither prove nor refute an asymptotic gate. Integer factorisation, support, coverage, radical, modulus, and defect decisions are exact. Floating-point logarithms enter only the finite flow values and their ranking.

For completeness, the descending greedy calculation is optimal for this finite nested graph. Group equal keys and process groups from largest to smallest, adding the current sink capacity before matching the current source mass. If Δ_r is cumulative source mass minus cumulative sink capacity after r groups and U_r is the greedy unmatched mass, conservation gives

$$U_r = \max(U_{r-1}, \Delta_r), \quad U_0 = 0.$$

Thus $U_r = \max(0, \Delta_1, \dots, \Delta_r)$. These cumulative differences are precisely the Hall upper-tail excesses, every one of which is a lower bound for any transport. The greedy value therefore equals the finite min-cut formula. The implementation cross-checks the two calculations and includes the regression instance with sources $(5, 1), (2, 1)$ and sink $(3, 1)$, whose unmatched optimum is 1.

The ordinary arguments above were fixed before their Lean translation. The companion module is

`ABCThreeArmIncidenceSuccessor20260903.`

It formalises the exact products, local signatures, three-modulus CRT rigidity, both implications to **ABCConjecture**, the infinite RAW-3C refutation, the source and sink mass identities, the Hall obstruction, and the exact bound $\text{height}(P) \leq \text{conductor}(P) + U(F, f)$. Its one-for-one audit covers all 65 public declarations. The dependency union is exactly **propext**, **Classical.choice**, and **Quot.sound**; no custom axiom or placeholder is used. The converse finite Hall theorem remains a useful exact description of this ordered graph, but the prime-square obstruction shows that no bound on its optimized tail excess can establish CT-3C. Subsequent work must change the transport geometry while retaining the parent incidence data.

85. SIGNED ENDPOINT PRIME-TOKEN TRANSPORT

We next attach a finite transport problem to every positive primitive abc triple; unit arms are allowed in this section. Set

$$K_c = \frac{c}{\text{rad}(c)}, \quad E_{ab} = \text{rad}(a) \text{rad}(b), \quad R = \text{rad}(abc).$$

Proposition 85.1 (Endpoint-core balance). *Every positive primitive triple satisfies*

$$E_{ab} = \text{rad}(ab), \quad R = E_{ab} \text{rad}(c), \quad E_{ab}c = RK_c.$$

Consequently

$$\delta_c(a, b, c) := \log K_c - \log E_{ab} = \log c - \log R.$$

Proof. The three coordinates are pairwise coprime, so the squarefree radical is multiplicative across their supports. Since $c = \text{rad}(c)K_c$, multiplication gives $E_{ab}c = RK_c$. Every factor is positive; taking logarithms and rearranging proves the last identity. $\square$

For $p \mid c$, write $e_p = v_p(c)$. The source tokens and their masses are

$$\mathcal{S}_c = \{(p, j) : p \mid c, 1 \leq j < e_p\}, \quad w(p, j) = \log p.$$

Thus one source token is retained for each prime layer above the first copy. The sink tokens and capacities are

$$\mathcal{T}_{ab} = \{q : q \mid ab\}, \quad v(q) = \log q,$$

with one sink for every distinct external prime. Canonical factorisation immediately gives

$$(EPT-1) \quad \sum_{s \in \mathcal{S}_c} w(s) = \log K_c, \quad \sum_{q \in \mathcal{T}_{ab}} v(q) = \log E_{ab}.$$

Definition 85.2 (Fractional monotone endpoint transport). A transport is a family $F(s, q) \in \mathbb{R}_{\geq 0}$ such that

$$\sum_q F(s, q) \leq w(s), \quad \sum_s F(s, q) \leq v(q),$$

and $F((p, j), q) > 0$ only when $p \leq q$. Write

$$U(F) = \sum_s \left(w(s) - \sum_q F(s, q) \right), \quad V(F) = \sum_q \left(v(q) - \sum_s F(s, q) \right)$$

for unmatched source mass and unused sink capacity.

The zero transport always exists. The content of the construction is the size of $U(F)$, together with the order restriction on its edges.

Theorem 85.3 (Exact accounting and height control). *Every fractional monotone endpoint transport satisfies*

$$U(F) = \delta_c(a, b, c) + V(F).$$

In particular,

$$\delta_c(a, b, c) \leq U(F), \quad \log c \leq \log R + U(F).$$

Proof. Let $C(F) = \sum_{s,q} F(s, q)$. Finite interchange of summation gives

$$U(F) = \sum_s w(s) - C(F), \quad V(F) = \sum_q v(q) - C(F).$$

Subtracting and using the two token-mass identities gives $U(F) - V(F) = \delta_c(a, b, c)$. The column-capacity inequalities make $V(F) \geq 0$, and Proposition 85.1 converts the first resulting inequality into the height bound. $\square$

The monotone edge condition carries information that the scalar identity does not see. For $t \in \mathbb{R}$, put

$$S_{>t} = \sum_{(p,j) \in \mathcal{S}_c, p>t} \log p, \quad T_{>t} = \sum_{q \in \mathcal{T}_{ab}, q>t} \log q.$$

Theorem 85.4 (Weighted Hall tail obstruction). *Every fractional monotone endpoint transport satisfies*

$$U(F) \geq S_{>t} - T_{>t} \quad (t \in \mathbb{R}).$$

Proof. Mass emitted by a source over $p > t$ can enter only a sink $q \geq p > t$. Consequently the total carried mass from that source tail is at most the total capacity $T_{>t}$ of the sink tail. Its unmatched mass is therefore at least $S_{>t} - T_{>t}$, and it is at most the total unmatched mass because every row deficit is nonnegative. $\square$

We do not use a converse max-flow formula. The proved tail inequalities are necessary tests for any proposed transport construction.

Definition 85.5 (Uniform endpoint-flow gate). For every $\varepsilon > 0$, require a constant C_ε such that every positive primitive triple admits a monotone fractional transport satisfying

$$(EPF) \quad U(F) \leq \varepsilon \log R + C_\varepsilon.$$

Theorem 85.6 (EPF implies standard *abc*). *If the uniform endpoint-flow gate holds, then the standard *abc* conjecture holds.*

Proof. For the flow supplied by EPF, Theorem 85.3 gives

$$\log c \leq \log R + U(F) \leq (1 + \varepsilon) \log R + C_\varepsilon.$$

Exponentiating yields the usual constant-times $\text{rad}(abc)^{1+\varepsilon}$ inequality, with the same quantifier order. $\square$

The gate EPF is refuted by the infinite prime-square family in Section 86. Two smaller complete-premise examples separately show why an integral replacement is too rigid. A *full integral matching* assigns every source token injectively to a sink token whose prime is at least the source prime.

Proposition 85.7 (Integral matching boundaries). *For $3 + 13 = 16$, the source multiset is $\{2, 2, 2\}$ and the sink set is $\{3, 13\}$, so no full integral matching exists. For $9 + 16 = 25$, the source has one token over 5 and the sinks are $\{2, 3\}$, so no monotone edge exists, although*

$$K_c = 5 \leq 6 = E_{ab}.$$

Proof. The first claim is the pigeonhole principle applied to three distinct layer tokens and two sinks. In the second example every available sink prime is strictly below 5. The displayed aggregate inequality follows from $\text{rad}(25) = 5$, $\text{rad}(9) = 3$, and $\text{rad}(16) = 2$. $\square$

These witnesses refute universal full integral matching and the inference from $K_c \leq E_{ab}$ to such a matching. On their own they do not settle the fractional epsilon gate; the later infinite family does.

There is also an infinite obstruction to removing the epsilon allowance from the scalar endpoint-core estimate.

Proposition 85.8 (No fixed zero-epsilon multiplier). *Let $X_k = 4^{3^k}$. Then*

$$3^{k+1} \mid X_k - 1, \quad 3^k \operatorname{rad}(X_k - 1) \leq X_k - 1,$$

and the primitive triple $(1, X_k - 1, X_k)$ satisfies

$$\frac{K_{X_k}}{\operatorname{rad}(1) \operatorname{rad}(X_k - 1)} > \frac{3^k}{2}.$$

Consequently, for every $K \in \mathbb{N}$ there is a positive primitive triple with $K_c > K \operatorname{rad}(a) \operatorname{rad}(b)$.

Proof. The divisibility is by induction. It is true at $k = 0$ because $X_0 - 1 = 3$. If $3^{k+1} \mid X_k - 1$, then $X_k \equiv 1 \pmod{3}$ and hence $3 \mid X_k^2 + X_k + 1$. The factorisation

$$X_{k+1} - 1 = X_k^3 - 1 = (X_k - 1)(X_k^2 + X_k + 1)$$

supplies the next power of 3. Write $X_k - 1 = 3^{k+1}u$. Radical submultiplicativity and $\operatorname{rad}(u) \leq u$ give

$$\operatorname{rad}(X_k - 1) \leq 3 \operatorname{rad}(u) \leq 3u,$$

which is the second inequality. Since X_k is a power of 2, $K_{X_k} = X_k/2$; therefore

$$\frac{K_{X_k}}{\operatorname{rad}(X_k - 1)} \geq \frac{(X_k/2)3^k}{X_k - 1} > \frac{3^k}{2}.$$

Choosing, for example, $k = 2K + 1$ makes $3^k > 2K$ and proves the final claim. $\square$

This is an infinite, full-quantifier counterexample to a fixed natural multiplier at epsilon zero. This particular family does not refute *abc* or the epsilon gate: the displayed certified logarithmic lower bound grows only linearly in k , while $\log X_k = 3^k \log 4$. The logically separate prime-square family in the next section does refute the ordered epsilon gate.

The companion module `ABCSignedEndpointPrimeTokenTransport20260903` formalises the exact core identity, finite weighted-flow accounting, Hall tail obstruction, the quantified implication $\text{EPF} \Rightarrow \text{ABCConjecture}$, both finite integral counterexamples, and the elementary radical-budget mechanism used by the zero-epsilon family. The module does not postulate EPF. Its quantified negation and the common CT-3C obstruction are checked in the companion module of Section 86.

86. A COMMON OBSTRUCTION TO ORDERED PRIME TRANSPORT

The three-arm and endpoint constructions use the same hard orientation: positive mass may move from a source prime r to a sink prime q only when $r \leq q$. The next family tests that orientation with every quantifier and every arithmetic premise present.

For an odd prime p , put

$$P_p = (1, p^2 - 1, p^2).$$

These are positive primitive *abc* triples. Indeed, the additive relation is immediate, the unit arm is coprime to both other arms, and $\gcd(p^2 - 1, p^2) = 1$.

Lemma 86.1 (Prime separation). *If p is an odd prime and q is a prime divisor of $p^2 - 1$, then $q < p$.*

Proof. Since $p^2 - 1 = (p - 1)(p + 1)$, Euclid's lemma gives $q \mid p - 1$ or $q \mid p + 1$. The first alternative gives $q < p$. In the second, $p + 1$ is an even composite integer greater than 2, so its prime divisor q is a proper divisor and hence $q \leq (p + 1)/2 < p$. $\square$

Theorem 86.2 (The ordered endpoint gate is false). *The uniform endpoint-flow gate (EPF) is false. More precisely, every monotone endpoint transport on P_p has*

$$U(F) = \log p.$$

Proof. The endpoint p^2 has one excess layer, at the prime p , so the source consists of one token of mass $\log p$. Every sink prime divides $p^2 - 1$ and is therefore smaller than p by Lemma 86.1. The order condition forbids every edge from the source token. Thus no source mass is carried and $U(F) = \log p$.

Moreover, radical multiplicativity and the elementary bound $\text{rad}(n) \leq n$ give

$$(OPS-1) \quad R(P_p) = p \text{ rad}(p^2 - 1) \leq p(p^2 - 1) < p^3.$$

If EPF held at $\varepsilon = 1/4$ with constant C , then every odd prime p would satisfy

$$\log p = U(F) \leq \frac{1}{4} \log R(P_p) + C < \frac{3}{4} \log p + C.$$

This bounds $\frac{1}{4} \log p$ by C , contrary to the existence of arbitrarily large odd primes. $\square$

Theorem 86.3 (The ordered three-arm gate is false). *The uniform three-arm complement-transport gate (CT-3C) is false. For every covering face of P_p and every one of its ordered complement transports,*

$$U(F, f) \geq \log p.$$

Proof. Suppose first that a face does not select the prime p on the C -arm. The A -arm has no prime vertex, the selected C -arm modulus is then 1, and the selected B -arm modulus divides $p^2 - 1$. Hence $M(F) \leq p^2 - 1 < p^2$, contradicting coverage. Every covering face must therefore select p on the C -arm. Its valuation is two, so this vertex creates a source token of mass $\log p$.

The A -arm has no complementary prime. The only C -arm prime has already been selected. Every remaining complementary prime is a divisor of $p^2 - 1$ and is smaller than p by Lemma 86.1. The distinguished source token can therefore send no mass, regardless of the other choices in the face. This proves $U(F, f) \geq \log p$. Combining the claimed CT-3C estimate at $\varepsilon = 1/4$ with (OPS-1) gives the same impossible bound $\frac{1}{4} \log p < C$ for all odd primes. $\square$

The conditional implications

$$\text{EPF} \implies abc, \quad \text{CT-3C} \implies abc$$

remain valid: the counterexample proves that their antecedents are false. It does not disprove standard *abc*, the labelled incidence complex, or a transport with a different geometry. Taking finitely many independent covering faces under the same hard order creates no admissible sink for the distinguished p -source; a repair must change either the edge geometry or the way different faces interact.

There is a useful exact description of the Hall obstruction. For a three-arm face and a threshold T , let $H_{>T}(F)$ be the logarithm of the selected full prime-power modulus contributed by selected primes above T , and let $R_{>T}(P)$ be the logarithm of the full radical contributed by all arm-labelled primes above T . Separating selected from unselected radical vertices gives

$$(OPS-2) \quad S_{>T}(F) - T_{>T}(F) = H_{>T}(F) - R_{>T}(P).$$

Indeed, a selected prime of valuation e contributes $(e - 1) \log p$ to the source side, an unselected prime contributes $-\log p$, and adding and subtracting the selected $\log p$ gives the right-hand side. At $T = p - 1$ for P_p , the right-hand side is $2 \log p - \log p = \log p$. Thus the obstruction is an exact high-prime valuation surplus, not a failure of the max-flow algorithm.

Simply deleting the order condition is not by itself a new arithmetic advance. For a complete bipartite fractional flow with nonnegative source weights of total mass A and sink capacities of total mass B , the minimum unmatched mass is

$$(OPS-3) \quad \min U = (A - B)_+.$$

The capacity inequalities give the lower bound. If $0 < A \leq B$, the product coupling $f_{ij} = a_i b_j / B$ fills every source; if $A > B > 0$, the coupling $f_{ij} = a_i b_j / A$ fills every sink and leaves $A - B$ unmatched. The zero cases are immediate. At the endpoint, (OPS-3) becomes

$$\min U = (\log c - \log R(P))_+.$$

After harmless enlargement of the additive constant to be nonnegative, a uniform epsilon-bound for this quantity is equivalent to the usual scalar *abc* inequality. Unordered transport therefore supplies no independent mechanism.

A nontrivial successor must allow the downward edges exposed by P_p while retaining information beyond total mass. One candidate assigns a nonnegative relative displacement cost

$$\kappa(r, q) = \frac{(\log r - \log q)_+}{\log r}$$

to mass sent from r to q , and studies $U + \sum f(r, q)\kappa(r, q)$ on one or several faces. A uniform epsilon-small bound for that sum still implies *abc*, because the added cost is nonnegative. The family P_p no longer causes the formal “no admissible edge” failure. No uniform cost bound is asserted here. The next section performs the required adversarial test and refutes this exact relative-drop calibration on a different infinite family. Weighted multi-face and homological incidence constructions remain active only as precisely stated future interfaces.

The ordinary arguments in this section were fixed before formalization. The companion obstruction module is

ABCThreeArmComplementTransportObstruction20260903.

It checks the complete premises, both quantified negations, the exact tail identity (OPS-2), and their shared prime-square mechanism without assuming either transport gate or **ABCConjecture**. The costed bidirectional successor and its conditional implication are isolated in

ABCBidirectionalPrimeTransportSuccessor20260903.

The two modules expose respectively 37 and 44 public declarations, with one axiom query for each declaration. Their dependency union is exactly **propext**, **Classical.choice**, and **Quot.sound**; no custom axiom or placeholder is used. No uniform instance of the successor is postulated; its exact BEP instance is refuted in Section 87.

87. A SQUARE-ROOT OBSTRUCTION TO RELATIVE-DROP ENERGY

The bidirectional endpoint model removes the false hard order and assigns an edge from p to q the relative drop

$$d(p, q) = \max \left\{ 0, \frac{\log p - \log q}{\log p} \right\}.$$

For a feasible flow f , its proposed energy is $E(f) = U(f) + \sum_{s,q} f(s, q)d(p_s, q)$. The prime-square family of Section 86 no longer fails because its downward edges are absent: all capacity-compatible edges now exist. A different infinite family shows that charging every downward edge at this scale is still too rigid.

First observe the exact finite optimization problem. Put

$$\rho(p, q) = 1 - d(p, q) = \min \left\{ 1, \frac{\log q}{\log p} \right\}.$$

If S is total source mass, then

$$(BRE-1) \quad E(f) = S - \sum_{s,q} \rho(p_s, q)f(s, q).$$

Thus minimizing energy is a maximum-reward transportation problem. Its dual minimizes

$$\sum_s w(s)u_s + \sum_q v(q)v_q \quad \text{subject to} \quad u_s + v_q \geq \rho(p_s, q), \quad u_s, v_q \geq 0.$$

For one source, the optimum fills sink capacities in nonincreasing prime order until either the source or all sinks are exhausted. This follows by moving any mass on a lower-reward sink to an unused higher-reward sink. The description is exact and supplies finite primal–dual certificates; it does not imply an asymptotic estimate.

Choose an arbitrarily large prime $p \equiv 1 \pmod{4}$. Fermat’s two-square theorem gives

$$p = m^2 + n^2, \quad m > n > 0.$$

Primality makes m and n coprime, and their parities are opposite. Put

$$X = m^2 - n^2, \quad Y = 2mn.$$

Then $X^2 + Y^2 = p^2$ and $\gcd(X, Y) = 1$, so

$$(BRE-2) \quad Q_p = (X^2, Y^2, p^2)$$

is a positive primitive *abc* triple.

Lemma 87.1 (Square-root support). *Every prime q dividing XY satisfies*

$$q \leq m + n, \quad q^2 \leq 2p.$$

Proof. The factorizations $X = (m - n)(m + n)$ and $Y = 2mn$ show that q divides one of $m - n, m + n, 2, m, n$. Each is at most $m + n$. Finally,

$$(m + n)^2 = m^2 + n^2 + 2mn \leq 2(m^2 + n^2) = 2p.$$

□

Theorem 87.2 (Bidirectional relative-drop obstruction). *The gate (BEP) is false. It is refuted at $\varepsilon = 1/12$ by the family (BRE-2).*

Proof. The endpoint p^2 has exactly one excess token, of capacity $L = \log p$. Every sink prime divides XY , and the lemma gives

$$\log q \leq \frac{L + \log 2}{2}, \quad d(p, q) \geq \delta_p := \frac{1}{2} - \frac{\log 2}{2L}.$$

For $p \geq 5$, one has $0 < \delta_p < 1$. If an arbitrary flow carries total mass W , its single row gives $0 \leq W \leq L$, while

$$U = L - W, \quad D \geq \delta_p W.$$

Consequently

$$(BRE-3) \quad E \geq L - (1 - \delta_p)W \geq \delta_p L = \frac{L - \log 2}{2}.$$

The radical of (BRE-2) satisfies

$$\text{rad}(X^2 Y^2 p^2) \leq XYp \leq p^3,$$

because $X, Y \leq p$. Hence its conductor is at most $3L$. If BEP held at $\varepsilon = 1/12$ with constant C , (BRE-3) would imply

$$\frac{L - \log 2}{2} \leq E \leq \frac{1}{12}(3L) + C,$$

or $L/4 \leq C + (\log 2)/2$. Primes congruent to one modulo four are unbounded, so this is impossible. □

The proof bounds every admissible flow and uses an infinite family with all positivity, addition, and coprimality premises present. It therefore retires the exact relative-drop energy gate. It does not refute standard *abc*, bidirectional transport as a parent construction, or a differently calibrated packet or homological cost.

The mechanism suggests the correction. Prime-hypotenuse Pythagorean squares naturally place leg factors at square-root scale. A candidate that assigns a fixed positive proportional charge to every edge down to that scale will fail for the same reason. The next successor must aggregate several sinks into a packet, record an algebraic degree, or permit cancellation across an incidence boundary. A quadratic calibration such as

$$\max \left\{ 0, \frac{\log p - 2 \log q}{\log p} \right\}$$

passes this particular scale test, but its uniform estimate is not asserted and must be subjected to the same complete-premise audit.

The ordinary proof above precedes its Lean realization in

ABCBidirectionalEnergyPythagoreanObstruction20260903.

The companion audit queries every public declaration. The module proves the unbounded prime-hypotenuse construction, the support and energy bounds, and the quantified negation of BEP without assuming **ABCConjecture**. The ordinary estimate above is the sharper square-root bound and gives $\varepsilon = 1/12$; the Lean proof uses the simpler integral consequence $q^4 \leq p^3$, obtains energy at least $(\log p)/4$, and closes the same negation at $\varepsilon = 1/24$. The module exposes 40 public declarations and its audit has exactly 40 corresponding queries; the dependency union is only **propext**, **Classical.choice**, and **Quot.sound**.

88. PRIME-PACKET BOUNDARY TRANSPORT

The two transport obstructions distinguish individual edge geometry from aggregate radical capacity. We next aggregate all excess layers at one endpoint prime into one source and assign each external radical prime, as an indivisible sink, to at most one source packet. Several small primes may therefore act together, and no cost is attached to their individual distance from the source.

Let finite sources have weights $x_i \geq 0$ and finite sinks have weights $y_j \geq 0$. A packet assignment is a map

$$o : J \longrightarrow I \sqcup \{\emptyset\}.$$

Put

$$M_i(o) = \sum_{o(j)=i} y_j, \quad B_i(o) = (x_i - M_i(o))_+, \quad B(o) = \sum_i B_i(o).$$

Proposition 88.1 (Packet boundary inequality). *Every packet assignment satisfies*

$$(PBT-1) \quad \sum_i x_i - \sum_j y_j \leq B(o).$$

Proof. The packets are disjoint, so $\sum_i M_i(o)$ is at most the total sink weight. For each source, $x_i - M_i(o) \leq (x_i - M_i(o))_+$. Summing these inequalities gives

$$\sum_i x_i - \sum_j y_j \leq \sum_i (x_i - M_i(o)) \leq \sum_i B_i(o).$$

□

For an *abc* point $P = (a, b, c)$, use one source for each prime $p \mid c$, with aggregate weight

$$x_p = (v_p(c) - 1) \log p,$$

and one indivisible sink for each prime $q \mid ab$, with weight $\log q$. Canonical factorization gives

$$\sum_{p \mid c} x_p = \log \frac{c}{\text{rad}(c)}, \quad \sum_{q \mid ab} \log q = \log \text{rad}(ab).$$

Together with (PBT-1), the exact endpoint balance yields

$$(PBT-2) \quad \log c \leq \log \text{rad}(abc) + B(o).$$

The packet-boundary gate PBT asks that, for every $\varepsilon > 0$, one constant C_ε work for all positive primitive P , with a packet assignment such that

$$(PBT) \quad B(o) \leq \varepsilon \log \text{rad}(abc) + C_\varepsilon.$$

Theorem 88.2 (Packet boundary implies standard *abc*). *PBT implies the standard abc conjecture.*

Proof. Substitute (PBT) into (PBT-2). Since $a + b = c$ and all coordinates are positive, c is their maximum. The resulting inequality has exactly the standard epsilon-constant quantifier order. □

PBT is not the divisible unordered relaxation pointwise. Two sources of weight one and a single indivisible sink of weight two have equal total source and sink mass. Divisible flow has zero residual, whereas a packet assignment can cover at most one source; its optimal packet boundary is one. Thus the packet objective cannot be recovered from the two total masses alone. We do not claim that this proves logical independence of its uniform arithmetic gate from *abc*.

The two preceding counterexample mechanisms do not automatically refute PBT. Whenever $c = p^2$, there is one aggregate source of weight $\log p$. Assigning all external primes to its packet gives exactly

$$B = (\log p - \log \text{rad}(ab))_+.$$

For both $(1, p^2 - 1, p^2)$ and the prime-hypotenuse Pythagorean-square family, the individual inequalities $q < p$ or $q \leq \sqrt{2p}$ create neither a forbidden edge nor a per-edge charge. This observation does not prove PBT: an infinite family with small aggregate external radical, or competition among several powerful endpoint primes, may still refute it.

The next positive problem is therefore an arithmetic logarithmic bin-covering theorem: partition the distinct primes of ab among the powerful primes of c while leaving only epsilon-small uncovered aggregate weight. The next counterexample search must optimize the full indivisible assignment on an unbounded family; a finite bad value is absorbed by C_ϵ and retires nothing. If the bare packet gate fails, congruence labels or a genuine signed multi-face boundary must be added without discarding the parent construction.

The ordinary proof was completed before the Lean module

`ABCPrimePacketBoundaryTransportSuccessor20260903`.

That module checks the finite boundary inequality, endpoint specialization, conditional implication, a two-source noncollapse certificate, and the one-source formula. PBT itself is a definition and has no asserted inhabitant in this construction. The next section gives a complete counterexample to that inhabitation claim. The module exposes 39 public declarations and its audit has exactly 39 corresponding queries; the dependency union is only `propext`, `Classical.choice`, and `Quot.sound`.

89. A LINNIK OBSTRUCTION TO EXCLUSIVE PRIME PACKETS

The packet-boundary gate of the preceding section has a second, genuinely different obstruction. Its exclusive ownership rule prevents one external prime from recording simultaneous congruences at several powerful endpoint primes. Linnik's least-prime theorem turns this defect into an unconditional infinite counterfamily.

For a finite packet problem, let the source and sink weights be $x_i \geq 0$ and $y_j \geq 0$, and put $X = \sum_i x_i$, $Y = \sum_j y_j$. If M_i is the mass assigned to source i , then

$$(LPO-1) \quad \sum_i (x_i - M_i)_+ = X - \sum_i \min\{x_i, M_i\}.$$

This is the clipped-reward identity. In particular, if there are k selected sources with weights $z_1 \geq \dots \geq z_k$ and only t positive sinks, disjoint ownership lets at most t of those sources receive positive mass. Every assignment therefore has residual at least

$$(LPO-2) \quad \sum_{i=t+1}^k z_i.$$

For one sink of weight y , the exact optimum is

$$(LPO-3) \quad R_* = X - \max_i \min\{x_i, y\}.$$

These formulas isolate a cardinality loss which is absent from divisible transport.

We use Linnik's unconditional theorem in the following uniform form: there are absolute constants $C \geq 1$ and $L \geq 1$ such that the least prime in any reduced residue class $a \pmod{Q}$ is at most CQ^L . This is precisely the uniform polynomial estimate proved by Linnik and sharpened, with an explicit admissible exponent, by Xylouris [83, Theorem 1.1]. No optimal value of L is needed.

Let

$$p_1 < p_2 < \cdots < p_k, \quad M_k = \prod_{i=1}^k p_i, \quad Q_k = M_k^2,$$

where the p_i are the first k primes. Apply Linnik's theorem to the reduced class $-1 \pmod{Q_k}$ and choose a prime ℓ_k with

$$(LPO-4) \quad \ell_k \equiv -1 \pmod{M_k^2}, \quad \ell_k \leq CM_k^{2L}.$$

Then

$$(LPO-5) \quad P_k = (1, \ell_k, \ell_k + 1)$$

is a positive primitive abc point. The family is infinite because $\ell_k + 1 \geq M_k^2$.

Lemma 89.1 (One sink and many compulsory sources). *Every packet assignment at P_k has residual*

$$(LPO-6) \quad R(P_k) \geq \log M_k - \log p_k.$$

Proof. The external product is the prime ℓ_k , so there is exactly one sink. For every $i \leq k$, (LPO-4) gives $p_i^2 \mid \ell_k + 1$; hence the aggregate source at p_i has weight

$$(v_{p_i}(\ell_k + 1) - 1) \log p_i \geq \log p_i.$$

The unique sink can belong to at most one source packet. Apply (LPO-2) and discard at most the largest of the k guaranteed weights. $\square$

The conductor remains comparable with $\log M_k$. Indeed,

$$(LPO-7) \quad \begin{aligned} r(P_k) &\leq \log(\ell_k(\ell_k + 1)) \\ &\leq 2 \log \ell_k + \log 2 \leq 4L \log M_k + B, \end{aligned}$$

where $B = 2 \log C + \log 2$. At the same time,

$$(LPO-8) \quad \frac{\log p_k}{\log M_k} \longrightarrow 0.$$

For completeness, Bertrand's postulate gives $p_k < 2^k$ for large k , while $p_i \geq i + 1$ gives $M_k \geq (k + 1)!$; the last half of the factorial factors already proves (LPO-8).

Theorem 89.2 (Exclusive packet obstruction). *The gate PBT is false.*

Proof. Set $\varepsilon_0 = 1/(8L)$. Combining (LPO-6) and (LPO-7) gives

$$(LPO-9) \quad R_*(P_k) - \varepsilon_0 r(P_k) \geq \frac{1}{2} \log M_k - \log p_k - \frac{B}{8L}.$$

By (LPO-8), the right side tends to infinity. Thus every proposed additive constant fails at some P_k , even after minimizing over all assignments. This has exactly the negated epsilon-constant-point-assignment quantifier order of PBT. $\square$

This is not an abc counterexample. Since M_k is even, $\text{rad}(\ell_k + 1) \geq 2$, and therefore

$$(LPO-10) \quad \text{rad}(\ell_k(\ell_k + 1)) = \ell_k \text{rad}(\ell_k + 1) \geq 2\ell_k \geq \ell_k + 1.$$

Hence every point in the family satisfies $h(P_k) \leq r(P_k)$ and has zero positive scalar defect. PBT fails solely through its indivisibility gap. Indeed, if $\Delta(P) = (h(P) - r(P))_+$ and $G(P) = R_*(P) - \Delta(P)$, then the exact identity $R_* = \Delta + G$ shows that PBT is equivalent to standard logarithmic abc together with a uniform sublinear bound on G . The Linnik family refutes the second conjunct while satisfying the first with constant zero.

The correct structural lesson is that one external prime may encode many CRT incidences at once:

$$\ell_k \equiv -1 \pmod{p_i^2} \quad (1 \leq i \leq k).$$

A successor must permit this shared incidence while charging its logarithmic capacity only once. Allowing arbitrary divisible sharing without incidence data returns to the scalar defect, so neither

repair alone is sufficient. The next positive target is a shared CRT-incidence boundary with a pointwise height bridge and an audit showing zero artificial loss on (LPO-5).

The ordinary proof above uses the published Linnik theorem. The current Mathlib tree does not contain that analytic theorem. The companion Lean module therefore checks the finite packet identities, the prime-neighbour arithmetic core, and the implication from a transparent Linnik-style escape hypothesis to the negation of PBT; it does not insert Linnik's theorem as a project axiom or mislabel the resulting source boundary as kernel-closed.

The accompanying exact computation independently enumerates all 1,368,094 normalized primitive triples with $c \leq 3000$. It finds 572 strict packet fragmentation gaps, 567 of them with zero scalar defect, and replays all optima with a second state-set algorithm. Five finite square-primorial prime-predecessor rows exhibit the same one-sink mechanism. These data are regression and discovery evidence only; Theorem 89.2 uses the infinite Linnik family and does not extrapolate from the scan.

90. SHARED CRT INCIDENCE AFTER EXCLUSIVE PACKETS

The Linnik counterfamily to exclusive prime packets identifies an arithmetic loss rather than a shortage of total radical mass. One prime $\ell \equiv -1 \pmod{M^2}$ certifies all congruences modulo the prime-power factors of M^2 simultaneously. Its capacity $\log \ell$ must therefore be shareable, but it may be charged only once. This section gives a first capacity-correct shared-incidence successor, proves its height bridge, and separates it from the scalar relaxation. Its uniform estimate remains open.

Let $P = (a, b, c)$ be a positive primitive point:

$$a, b, c \in \mathbb{Z}_{>0}, \quad a + b = c, \text{quad} \gcd(a, b) = \gcd(b, c) = \gcd(c, a) = 1.$$

Put

$$\text{core}(n) = \frac{n}{\text{rad}(n)}, \quad h(P) = \log c, \quad r(P) = \log \text{rad}(abc),$$

and define the endpoint sources and external sinks by

$$I = I(P) = \{p : p \mid c, v_p(c) \geq 2\}, \quad J = J(P) = \{q : q \mid ab\}.$$

For $e_p = v_p(c)$, set

$$x_p = (e_p - 1) \log p, \quad y_q = \log q,$$

and write

$$(SCRT-1) \quad X = \sum_{p \in I} x_p = \log \text{core}(c), \quad Y = \sum_{q \in J} y_q = \log \text{rad}(ab).$$

For $S \subseteq I$ and $T \subseteq J$, let

$$(SCRT-2) \quad \begin{aligned} m(S) &= \prod_{p \in S} p^{e_p}, \\ a_T &= \prod_{\substack{q \in T \\ q \mid a}} q^{v_q(a)}, & b_T &= \prod_{\substack{q \in T \\ q \mid b}} q^{v_q(b)}, \\ x(S) &= \sum_{p \in S} x_p, & y(T) &= \sum_{q \in T} y_q. \end{aligned}$$

The empty products are one. Thus a_T, b_T are unitary divisors carrying the complete valuation of every selected external prime.

Definition 90.1 (Shared CRT block). A pair (S, T) with $\emptyset \neq S \subseteq I$ and $\emptyset \neq T \subseteq J$ is *CRT compatible* when

$$(SCRT-3) \quad m(S) \mid a_T + b_T.$$

It is *saturated* when also

$$(SCRT-4) \quad x(S) \leq y(T).$$

A saturated compatible pair is a *shared CRT block*.

Compatibility is arithmetic, rather than merely an incidence convention: the truncated relation $a_T + b_T$ simultaneously certifies every prime-power modulus indexed by S . When I and J are nonempty, the full pair is always compatible because $a_J = a$, $b_J = b$, and $m(I) \mid c$.

Definition 90.2 (SCRT-0 configuration). An SCRT-0 configuration consists of a finite family $\mathcal{H}$ of saturated blocks (S_ν, T_ν) whose source sets are pairwise disjoint and whose sink sets are pairwise disjoint. Put

$$I_0 = I \setminus \bigcup_{\nu} S_\nu, \quad J_0 = J \setminus \bigcup_{\nu} T_\nu.$$

The remaining data are an exclusive assignment

$$o : J_0 \longrightarrow I_0 \sqcup \{\emptyset\}.$$

Thus a residual sink is unused or belongs to one residual source, while several sinks may belong to the same source. For $p \in I_0$, put

$$M_p(o) = \sum_{\substack{q \in J_0 \\ o(q)=p}} y_q, \quad R_p(o) = \max\{x_p - M_p(o), 0\},$$

and define

$$(SCRT-5) \quad B(\mathcal{H}, o) = \sum_{p \in I_0} R_p(o), \quad B_{SCRT}(P) = \min_{(\mathcal{H}, o)} B(\mathcal{H}, o).$$

The minimum exists because the underlying sets are finite; the empty block family with every sink unused is an admissible configuration. Sources in a saturated block have zero boundary. Choosing a block consumes all of $y(T_\nu)$, including any surplus over $x(S_\nu)$.

The sink union T_ν occurs in exactly one block, and its total credit is at most $y(T_\nu)$. Every residual sink occurs in at most one exclusive packet. Thus no logarithmic capacity is counted twice.

Proposition 90.3 (Once-charged height bridge). *Every SCRT-0 configuration satisfies*

$$(SCRT-6) \quad X \leq Y + B(\mathcal{H}, o)$$

and consequently

$$(SCRT-7) \quad h(P) \leq r(P) + B(\mathcal{H}, o).$$

Proof. Saturation gives $x(S_\nu) \leq y(T_\nu)$ block by block. At every remaining source, $x_p \leq M_p(o) + R_p(o)$. Disjoint blocks and exclusive ownership give

$$X \leq \sum_{\nu} y(T_\nu) + \sum_{p \in I_0} M_p(o) + B(\mathcal{H}, o) \leq Y + B(\mathcal{H}, o),$$

which is (SCRT-6). Prime factorisation gives

$$h(P) - r(P) = \log \text{core}(c) - \log \text{rad}(ab) = X - Y,$$

which proves (SCRT-7). $\square$

Let $B_{PBT}(P)$ denote the optimum of the earlier exclusive packet model and put $\Delta(P) = \max\{X - Y, 0\}$.

Proposition 90.4 (Sandwich and easy strata). *For every positive primitive point,*

$$(SCRT-8) \quad \Delta(P) \leq B_{SCRT}(P) \leq B_{PBT}(P).$$

Moreover, $X \leq Y$ implies $B_{SCRT}(P) = 0 = \Delta(P)$, and $|I| \leq 1$ implies $B_{SCRT}(P) = \Delta(P)$.

Proof. Nonnegativity and (SCRT-6) give the lower bound. Taking no shared blocks recovers exactly the PBT assignment problem, giving the upper bound. If $I = \emptyset$, then $X = 0$ and the empty configuration has zero boundary. If $I \neq \emptyset$ and $X \leq Y$, then $J \neq \emptyset$ and the full pair (I, J) is a valid saturated block, again giving zero boundary. Finally, if I has one member, assign every sink to that source; the resulting residual is $\max\{X - Y, 0\}$ and attains the lower bound. $\square$

Thus an additional shared-incidence gap can occur only when $X > Y$ and at least two endpoint primes carry positive excess weight.

Definition 90.5 (Uniform shared-CRT gate). SCRT-0 is the statement

$$(SCRT-0) \quad \forall \varepsilon > 0 \exists C_\varepsilon \forall P \exists (\mathcal{H}, o) : \quad B(\mathcal{H}, o) \leq \varepsilon r(P) + C_\varepsilon.$$

Here P ranges over all positive primitive *abc* points.

Theorem 90.6 (Conditional reduction to *abc*). *If SCRT-0 holds, then the standard logarithmic abc conjecture holds.*

Proof. For the configuration supplied by SCRT-0, Proposition 90.3 gives

$$h(P) \leq r(P) + B(\mathcal{H}, o) \leq (1 + \varepsilon)r(P) + C_\varepsilon.$$

This has the standard quantifier order. □

No instance of SCRT-0 is assumed or proved here.

Both clauses in the definition are necessary. If the capacity of one shared sink union were repeated once per incident source, the point $(1, 224, 225)$ would be assigned zero boundary: its two source weights are $\log 3, \log 5$ and its external mass is $\log 14$, available through the full relation. Yet

$$h - r = \log(15/14) > 0,$$

so the height bridge would fail. At the other extreme, if the full block may distribute arbitrary partial credits z_p with $\sum z_p \leq Y$, its optimal boundary is

$$(SCRT-9) \quad X - \min(X, Y) = (X - Y)_+.$$

The unrestricted model is therefore exactly the scalar defect. In the reverse logical implication from *abc*, the additive constant may first be replaced by its maximum with zero before taking positive parts.

SCRT-0 is strictly different pointwise. At

$$(343, 625, 968) = (7^3, 5^4, 2^3 11^2),$$

the source weights are $\log 4, \log 11$ and the sink weights are $\log 5, \log 7$. The joint block is unsaturated because $\log 44 > \log 35$. The saturated blocks are

$$(\{2\}, \{7\}), \quad (\{2\}, \{5, 7\}), \quad (\{11\}, \{5, 7\}),$$

with resulting best residuals $\log(11/5), \log 11, \log 4$, respectively. The empty block family and split assignment $5 \mapsto 2, 7 \mapsto 11$ attain $\log(11/7)$, and the finite list of other assignments is worse. Thus

$$(SCRT-10) \quad B_{\text{SCRT}}(343, 625, 968) = \log(11/7) > \log(44/35) = \Delta.$$

The recorded infinite counterfamilies do not reproduce their earlier artificial losses. Both $(1, p^2 - 1, p^2)$ and the prime-hypotenuse Pythagorean-square family have one endpoint source, so $B_{\text{SCRT}} = \Delta$. The same holds for the fixed-prime family $(1, p^N - 1, p^N)$. In the Linnik family $(1, \ell, \ell + 1)$, the single sink ℓ gives a full shared block: since ℓ is odd,

$$\text{core}(\ell + 1) \leq (\ell + 1)/2 \leq \ell,$$

so that block is saturated and has zero boundary. For the shifted family $(2, \ell, \ell + 2)$ with $\ell \equiv -2 \pmod{M^2}$ and M odd, the full sink union $\{2, \ell\}$ likewise closes every forced source once; indeed $\text{core}(\ell + 2) \leq (\ell + 2)/3 \leq \ell$. For $(4^r, 3^r, 4^r + 3^r)$, the full block works whenever $\text{core}(4^r + 3^r) \leq 6$, and a single repeated endpoint prime again gives $B_{\text{SCRT}} = \Delta$. The earlier balanced-family obstruction does not prove an infinite subsequence with several powerful endpoint primes and a bad SCRT partition, so it does not refute this gate.

A harder child model, SCRT-SAT, deletes the exclusive residual stage and charges every source outside saturated blocks in full. It is false. For

$$n = 284 + 310t, \quad P_n = (2, 15^n - 2, 15^n),$$

direct modular exponentiation gives $15^{284} \equiv 2 \pmod{31^2}$ and $15^{310} \equiv 1 \pmod{31^2}$. Hence $31^2 \mid 15^n - 2$ and

$$(\text{SCRT-11}) \quad Y = \log(2 \operatorname{rad}(15^n - 2)) < \log(2 \cdot 15^n / 31) < (n - 1) \log 15 = X.$$

Disjoint saturated blocks cannot close both endpoint sources, so SCRT-SAT leaves at least $(n - 1) \log 3$. Since its conductor is less than $n \log 15$, every fixed $0 < \varepsilon < \log 3 / \log 15$ refutes its uniform gate. SCRT-0 retains the external prime-log partition and therefore is not refuted by this argument; (SCRT-11) alone supplies only the constant total-capacity gap $X - Y > \log(31/30)$.

The saturated rule currently consumes the whole capacity $y(T)$, including surplus beyond $x(S)$. This is a design choice. A surplus-reuse successor would need a proved fractional-cover or signed-chain identity which keeps each sink charged once without admitting the unrestricted pooling in (SCRT-9).

The Lean module `ABCSharedCRTIncidenceSuccessor20260903` formalises the abstract once-charged certificate and the mass inequality (SCRT-6). It proves the endpoint height bridge and the conditional implication to `ABCConjecture`. It also checks the duplicated-capacity failure and the scalar optimum (SCRT-9). The module has 30 public declarations and a one-for-one 30-query axiom audit whose union is only `propext`, `Classical.choice`, and `Quot.sound`. The arithmetic CRT-block configuration, the optimum (SCRT-10), and the SCRT-SAT counterfamily are ordinary proofs at this checkpoint and are not represented by axioms or opaque gate inhabitants. Proving or refuting the uniform SCRT-0 estimate is the next unresolved step.

91. PROPER-SUBFACE FLAGGED CRT SURPLUS AND ENDPOINT RESIDUE CUBES

This section refines the shared-CRT boundary of Section 90. The refinement has two purposes. First, it identifies and removes a false source of apparent progress: if the surplus of a saturated block may be sent to an arbitrary residual source, the resulting optimization is exactly the scalar endpoint defect. Second, it retains surplus only when a proper subpacket supplies an independent prime-power congruence for the target source. The resulting finite boundary, denoted B_{FCRT} , obeys the same once-charge height inequality, can be strictly smaller than B_{SCRT} , and need not equal the scalar defect. The associated uniform estimate is left as an explicit open premise throughout.

91.1. Endpoint data. Let $P = (a, b, c)$ satisfy

$$a, b, c \in \mathbb{Z}_{>0}, \quad a + b = c, \quad \gcd(a, b) = 1.$$

Pairwise coprimality follows. Put

$$I = \{p : p \mid c, e_p := v_p(c) \geq 2\}, \quad J = \{q : q \mid ab\},$$

and assign the logarithmic source and sink weights

$$(\text{FCRT-1}) \quad x_p = (e_p - 1) \log p, \quad y_q = \log q, \quad X = \sum_{p \in I} x_p, \quad Y = \sum_{q \in J} y_q.$$

For $S \subseteq I$ and $T \subseteq J$, define

$$(\text{FCRT-2}) \quad \begin{aligned} m(S) &= \prod_{p \in S} p^{e_p}, & x(S) &= \sum_{p \in S} x_p, \\ a_T &= \prod_{\substack{q \in T \\ q \mid a}} q^{v_q(a)}, & b_T &= \prod_{\substack{q \in T \\ q \mid b}} q^{v_q(b)}, \\ y(T) &= \sum_{q \in T} y_q. \end{aligned}$$

Empty products are one. A pair (S, T) of nonempty subsets is a *shared CRT block* if

$$(\text{FCRT-3}) \quad m(S) \mid a_T + b_T, \quad x(S) \leq y(T).$$

The two clauses will be called compatibility and saturation, respectively. Let $\mathcal{B}(P)$ be the set of all shared CRT blocks. It is a finite subset of

$$(\mathcal{P}(I) \setminus \{\emptyset\}) \times (\mathcal{P}(J) \setminus \{\emptyset\}).$$

Prime factorization gives the exact identity

$$(FCRT-4) \quad h(P) - r(P) = \log c - \log \text{rad}(abc) = X - Y,$$

and hence the scalar lower bound is

$$\Delta(P) = (X - Y)_+.$$

91.2. The exact collapse of free-target reuse. It is useful to dispose of an unrestricted surplus rule before imposing the arithmetic flag. Let $K \subseteq \mathcal{B}(P)$ be a selected block family such that the sets S_ν are pairwise disjoint and the sets T_ν are pairwise disjoint, where $\nu = (S_\nu, T_\nu) \in K$. In particular, the blocks are distinct and K is finite. Write

$$(FCRT-5) \quad I_0 = I \setminus \bigcup_{\nu \in K} S_\nu, \quad J_0 = J \setminus \bigcup_{\nu \in K} T_\nu, \quad \sigma_\nu = y(T_\nu) - x(S_\nu) \geq 0.$$

A free-target configuration consists further of two total maps into pointed finite sets,

$$(FCRT-6) \quad o : J_0 \longrightarrow I_0 \sqcup \{\perp\}, \quad t : K \longrightarrow I_0 \sqcup \{\perp\}.$$

Here $o(q) = p$ assigns the residual sink q exclusively to p , while $t(\nu) = p$ sends the whole unsplit surplus σ_ν to p ; the value $\perp$ means unused. Several independently charged objects may have the same target. Define

$$(FCRT-7) \quad C_p = \sum_{\substack{q \in J_0 \\ o(q)=p}} y_q + \sum_{\substack{\nu \in K \\ t(\nu)=p}} \sigma_\nu, \quad B_{\text{free}}(\mathcal{C}) = \sum_{p \in I_0} (x_p - C_p)_+.$$

Proposition 91.1 (Free-target collapse). *For every positive primitive P ,*

$$(FCRT-8) \quad \min_{\mathcal{C}} B_{\text{free}}(\mathcal{C}) = (X - Y)_+.$$

Proof. Every sink weight is charged at most once. A selected block consumes $x(S_\nu)$ and leaves at most σ_ν for reuse, whereas an unselected sink contributes at most its own weight. Therefore

$$X \leq Y + B_{\text{free}}(\mathcal{C})$$

for every configuration, proving the lower bound.

If $I = \emptyset$, the empty configuration realizes zero. Suppose $I \neq \emptyset$ and $X \leq Y$. Then $J \neq \emptyset$: otherwise $a = b = 1$, $c = 2$, and I would be empty. The full pair (I, J) is compatible because $m(I) \mid c = a + b$, and it is saturated because $X \leq Y$. Selecting it realizes zero.

It remains to consider $X > Y$. Choose an inclusion-maximal subset $S \subset I$ with $x(S) \leq Y$. It exists and is proper. If $S \neq \emptyset$, select the full-sink block (S, J) , which is compatible and saturated. For every $p \in I \setminus S$, maximality gives

$$(FCRT-9) \quad x_p > Y - x(S) =: \sigma.$$

Send the unique surplus token σ to one such p . Since no residual sinks remain, all residual terms are positive and their sum is

$$x_p - \sigma + \sum_{u \in I \setminus (S \cup \{p\})} x_u = X - Y.$$

If $S = \emptyset$, maximality says $x_p > Y$ for every $p \in I$. Select no block and assign every residual sink to any one source. The boundary is again $X - Y$. This proves equality in all cases. $\square$

Proposition 91.1 rules out only this exact free-target child model. It supplies neither a counterexample to the shared-CRT route nor a reason to remove arithmetic restrictions from a surplus token.

91.3. A fully typed proper-subface configuration. Fix a finite block family as in (FCRT-5). For each $\nu \in K$, define its finite flag set

$$(FCRT-10) \quad \mathcal{F}_\nu = \left\{ (p, U) \mid \begin{array}{l} p \in I_0, \quad U \subseteq T_\nu, \quad \emptyset \neq U \neq T_\nu, \\ p^{e_p} \mid a_U + b_U \end{array} \right\}.$$

Thus U is a nonempty proper face of the block's sink packet, and the target p is residual, hence lies outside every selected block source set. An *FCRT configuration* consists of the block family, a residual-sink owner map

$$o : J_0 \longrightarrow I_0 \sqcup \{\perp\},$$

and, for every $\nu \in K$, a choice

$$(FCRT-11) \quad f_\nu \in \mathcal{F}_\nu \sqcup \{\perp\}.$$

If $f_\nu = \perp$, put $\rho_\nu = 0$. If $f_\nu = (p_\nu, U_\nu)$, put

$$(FCRT-12) \quad \rho_\nu = \min\{\sigma_\nu, y(U_\nu)\}, \quad \sigma_\nu = y(T_\nu) - x(S_\nu).$$

The token of mass ρ_ν is indivisible, has the single target p_ν , and travels one hop. The amount $\sigma_\nu - \rho_\nu$ is discarded. Credit left after a target is filled is also discarded and cannot emit a new token. Different residual sinks and different block tokens may meet at the same source; this does not repeat the charge of any originating sink.

For $p \in I_0$, set

$$(FCRT-13) \quad \begin{aligned} M_p &= \sum_{\substack{q \in J_0 \\ o(q)=p}} y_q, \\ Z_p &= \sum_{\nu \in K} z_{\nu,p}, \quad z_{\nu,p} = \begin{cases} \rho_\nu, & f_\nu = (p, U) \text{ for some } U, \\ 0, & \text{otherwise,} \end{cases} \quad B_{FCRT}(\mathcal{C}) = \sum_{p \in I_0} R_p. \\ R_p &= (x_p - M_p - Z_p)_+, \end{aligned}$$

Every set and every choice space in (FCRT-10)–(FCRT-11) is finite. The empty block family with all residual sinks unused is admissible, so the pointwise minimum

$$(FCRT-14) \quad B_{FCRT}(P) = \min_{\mathcal{C} \in \text{Conf}_{FCRT}(P)} B_{FCRT}(\mathcal{C})$$

exists.

Proposition 91.2 (Once-charge mass bridge). *Every FCRT configuration satisfies*

$$(FCRT-15) \quad X \leq Y + B_{FCRT}(\mathcal{C})$$

and consequently

$$(FCRT-16) \quad h(P) \leq r(P) + B_{FCRT}(\mathcal{C}).$$

Proof. The definition of the positive part gives $x_p \leq M_p + Z_p + R_p$. Exclusive ownership of residual sinks gives

$$\sum_{p \in I_0} M_p \leq y(J_0).$$

Each block emits at most one token, and $\rho_\nu \leq \sigma_\nu$, so

$$\sum_{p \in I_0} Z_p \leq \sum_{\nu \in K} \sigma_\nu.$$

Pairwise source disjointness, pairwise sink disjointness, and (FCRT-5) now yield

$$\begin{aligned} X &= \sum_{\nu \in K} x(S_\nu) + \sum_{p \in I_0} x_p \\ &\leq \sum_{\nu \in K} x(S_\nu) + y(J_0) + \sum_{\nu \in K} (y(T_\nu) - x(S_\nu)) + B_{FCRT}(\mathcal{C}) \\ &= Y + B_{FCRT}(\mathcal{C}). \end{aligned}$$

Combining this with (FCRT-4) proves the height inequality. $\square$

The congruence and proper-face clauses do not enter the last accounting calculation. Their role is to restrict which once-charged configurations exist; deleting them recovers the scalar collapse above.

Corollary 91.3 (Pointwise sandwich). *For every positive primitive point,*

$$(FCRT-17) \quad \Delta(P) \leq B_{FCRT}(P) \leq B_{SCRT}(P) \leq B_{PBT}(P).$$

Proof. Every boundary is nonnegative, and Proposition 91.2 gives $B_{FCRT}(\mathcal{C}) \geq X - Y$ configuration by configuration. Taking the finite minimum therefore gives $B_{FCRT}(P) \geq \max\{X - Y, 0\} = \Delta(P)$. Giving every selected block the choice $\perp$ embeds the SCRT configurations into the FCRT configurations. Taking no shared block recovers the exclusive PBT problem. $\square$

Definition 91.4 (Uniform proper-subface gate). The statement FCRT-1 is

$$(FCRT-18) \quad \forall \varepsilon > 0 \exists C_\varepsilon \in \mathbb{R} \forall P \exists \mathcal{C} \in \text{Conf}_{FCRT}(P) : \quad B_{FCRT}(\mathcal{C}) \leq \varepsilon r(P) + C_\varepsilon,$$

where P ranges over all positive primitive triples.

Theorem 91.5 (Conditional implication to *abc*). *If FCRT-1 holds, then the standard logarithmic *abc* conjecture holds.*

Proof. Insert the configuration supplied by (FCRT-18) into (FCRT-16):

$$h(P) \leq r(P) + B_{FCRT}(\mathcal{C}) \leq (1 + \varepsilon)r(P) + C_\varepsilon.$$

The quantifier order is unchanged. $\square$

No instance, inhabitant, or proof of the uniform gate (FCRT-18) is supplied here. The theorem is a conditional reduction, not an unconditional proof of *abc*.

91.4. Three exact separating points. The following proposition records complete finite optimizations, rather than merely exhibiting upper bounds.

Proposition 91.6 (Strict improvement, noncollapse, and an active cap). *The three positive primitive points below have the indicated exact boundaries:*

	P	$e^{\Delta(P)}$	$e^{B_{FCRT}(P)}$	$e^{B_{SCRT}(P)}$	$e^{B_{PBT}(P)}$
(FCRT-19)	(1, 675, 676)	26/15	26/15	2	2
	(1, 224, 225)	15/14	3/2	3/2	3/2
	(1, 65024, 65025)	255/254	3/2	3	15/2.

In particular, the last row makes every inequality in (FCRT-17) strict.

Proof. For $P = (1, 675, 676)$,

$$675 = 3^3 5^2, \quad 676 = 2^2 13^2, \quad e^X = 26, \quad e^Y = 15.$$

The proper packet $\{3\}$ satisfies $1 + 3^3 = 28 \equiv 0 \pmod{4}$, while $1 + 5^2 = 26$ is divisible by neither 4 nor 13^2 . The full packet certifies both source moduli. The complete list of saturated compatible blocks is

$$(\{2\}, \{3\}), \quad (\{2\}, \{3, 5\}), \quad (\{13\}, \{3, 5\}).$$

They share sinks, so at most one is selectable. The four complete assignments of the two residual sinks in the no-block case have boundary factors

$$13, \quad \frac{13}{5}, \quad \frac{13}{3}, \quad 2,$$

while the three one-block cases have best factors $13/5$, 13, and 2, in the order displayed above. Thus the exact SCRT minimum is $\log 2$; the no-block list also gives the PBT minimum. In FCRT, choose $(S, T) = (\{13\}, \{3, 5\})$. Its surplus is $\log(15/13)$, and $U = \{3\}$ is a legal face for

residual source 2. Because $15/13 < 3$, the surplus, rather than the face weight, is limiting: the whole surplus is reused and the residual factor is

$$\frac{2}{15/13} = \frac{26}{15}.$$

The scalar lower bound in (FCRT-17) proves optimality.

For $P = (1, 224, 225)$,

$$224 = 2^5 \cdot 7, \quad 225 = 3^2 5^2, \quad e^X = 15, \quad e^Y = 14.$$

The two nonempty proper packets give $1 + 2^5 = 33$ and $1 + 7 = 8$; neither is divisible by 9 or 25. Hence the only nontrivial saturated blocks use the full sink set with one singleton source, and neither block can emit a flagged token. The four complete PBT assignments give residual factors

$$5, \quad 3, \quad \frac{3}{2}, \quad \frac{5}{2}.$$

The full-sink singleton blocks leave factors 5 and 3. Thus all three finite models have exact factor $3/2$, strictly above the scalar factor $15/14$.

Finally,

$$65024 = 2^9 \cdot 127, \quad 65025 = 3^2 \cdot 5^2 \cdot 17^2, \quad e^X = 255, \quad e^Y = 254.$$

The face $\{2\}$ gives $1 + 2^9 = 513 = 3^3 \cdot 19$, so it certifies source 3 and neither source 5 nor source 17. The face $\{127\}$ gives 128 and certifies none of the three source moduli. Every nonempty proper source subset paired with the full sink set is compatible and saturated; the full source set is not saturated. The only compatible proper-packet block is $(\{3\}, \{2\})$, and it is not saturated. Consequently every selectable block uses the full sink set and at most one can be selected.

In PBT, the nine complete assignments of the two sinks to the three sources have residual factors among

$$\frac{15}{2}, \quad 15, \quad \frac{51}{2}, \quad 51, \quad \frac{85}{2}, \quad 85;$$

the minimum is $15/2$. In SCRT, the block $(\{5, 17\}, \{2, 127\})$ closes source factor 85 and leaves factor 3. Every other legal full-sink block leaves at least 5, 15, 17, 51, or 85, proving exactness. The same block has surplus factor $254/85$, while the legal face $U = \{2\}$ has factor 2 and targets the remaining source 3. Here the proper-face cap is active:

$$e^\rho = \min \left\{ \frac{254}{85}, 2 \right\} = 2, \quad e^{B_{\text{FCRT}}} = \frac{3}{2}.$$

If a block contains source 3, it cannot target 3; if it omits source 3, the token just described is the only possible flagged credit and is at most a factor of 2. The chosen block closes the largest possible mass in the other sources. This proves the matching lower bound. Hence

$$\frac{255}{254} < \frac{3}{2} < 3 < \frac{15}{2}.$$

□

The middle row disproves the pointwise identity $B_{\text{FCRT}} = \Delta$. It does not disprove the quantified uniform gate, which permits both an $\varepsilon r(P)$ term and an additive constant.

91.5. The endpoint residue cube. Fix a source face $S \subseteq I$ and define the finite abelian group

$$(FCRT-20) \quad G_S = \prod_{p \in S} (\mathbb{Z}/p^{e_p} \mathbb{Z})^\times.$$

For every integer n coprime to $m(S)$, let $\bar{n}_S \in G_S$ denote its diagonal residue tuple. Since $\gcd(ab, c) = 1$, the following elements are well-defined:

$$(FCRT-21) \quad g_{S,q} = \begin{cases} \overline{q^{v_q(a)}}_S, & q \mid a, \\ \overline{q^{v_q(b)}}_S^{-1}, & q \mid b. \end{cases}$$

Define the packet-label map on the Boolean cube by

$$(FCRT-22) \quad \Phi_S : \mathcal{P}(J) \longrightarrow G_S, \quad \Phi_S(T) = \prod_{q \in T} g_{S,q}.$$

Proposition 91.7 (Fixed-fibre and complement identities). *For every $T \subseteq J$,*

$$(FCRT-23) \quad \Phi_S(T) = \overline{a_{TS}} \overline{b_{TS}}^{-1}, \quad \Phi_S(J) = \overline{1}_S,$$

and

$$(FCRT-24) \quad \begin{aligned} m(S) \mid a_T + b_T &\iff \Phi_S(T) = \overline{1}_S \\ &\iff \Phi_S(J \setminus T) = 1. \end{aligned}$$

If $S' \subseteq S$ and $\pi_{S,S'} : G_S \rightarrow G_{S'}$ is coordinate projection, then

$$(FCRT-25) \quad \pi_{S,S'} \circ \Phi_S = \Phi_{S'}.$$

Proof. Separating the primes of T between the two coprime arms gives the first identity in (FCRT-23). At every coordinate p^{e_p} , $a + b \equiv 0$ and b is a unit, so $ab^{-1} \equiv -1$; this gives the second identity. Since b_T is a unit modulo every p^{e_p} , $a_T + b_T \equiv 0$ is equivalent to $\overline{a_{TS}} \overline{b_{TS}}^{-1} = \overline{1}_S$. Finally,

$$\Phi_S(T) \Phi_S(J \setminus T) = \Phi_S(J) = \overline{1}_S,$$

which proves the complement statement by cancellation. The projection identity follows coordinatewise from the definitions. $\square$

Thus compatible packets form the $\overline{1}_S$ fibre, and complementation identifies this fibre with the identity fibre. The groups and maps form an inverse system over the source-face poset. An FCRT flag is nevertheless a cross-fibre incidence. For a block source S_ν and residual target $p \notin S_\nu$, it requires

$$(FCRT-26) \quad T_\nu \in \Phi_{S_\nu}^{-1}(\overline{1}_{S_\nu}), \quad U_\nu \in \Phi_{\{p\}}^{-1}(\overline{1}_{\{p\}}), \quad U_\nu \subsetneq T_\nu.$$

Both labels can be compared by projection from $G_{S_\nu \cup \{p\}}$, but neither packet is required to lie in the $\overline{1}$ fibre for the union source. Consequently (FCRT-25) does not convert the flag into a fibre inclusion.

91.6. Signed exchange and exact Fourier enumerators. Define the residue-relation lattice

$$(FCRT-27) \quad \Lambda_S = \left\{ z \in \mathbb{Z}^J : \prod_{q \in J} g_{S,q}^{z_q} = 1 \right\}.$$

Proposition 91.8 (Same-fibre signed exchange). *If $T, T' \subseteq J$ are compatible with the same source face S , then*

$$(FCRT-28) \quad \prod_{q \in T' \setminus T} g_{S,q} = \prod_{q \in T \setminus T'} g_{S,q},$$

and

$$(FCRT-29) \quad y(T') - y(T) = y(T' \setminus T) - y(T \setminus T').$$

If $T \neq T'$, then $\mathbf{1}_{T'} - \mathbf{1}_T$ is a nonzero member of $\Lambda_S \cap \{-1, 0, 1\}^J$.

Proof. Both packet products are $\overline{1}_S$. Factor each over $T \cap T'$ and its disjoint remainder, then cancel the common factor. This proves (FCRT-28); the analogous disjoint decompositions of the weight sums prove (FCRT-29). The final statement is exactly the multiplicative identity written in exponent-vector form. $\square$

In particular, if $2^{|J|} > |G_S|$, the pigeonhole principle gives distinct packets with the same label, and hence a nonzero $\{-1, 0, 1\}$ relation in Λ_S . This conclusion alone controls neither the sign or size of (FCRT-29) nor the location of the collision in the compatible fibre.

Character orthogonality makes the missing analytic input precise. Let $\widehat{G}_S = \text{Hom}(G_S, S^1)$. For $g \in G_S$ and $t \in \mathbb{R}$,

$$(FCRT-30) \quad \sum_{\Phi_S(T)=g} e^{ty(T)} = \frac{1}{|G_S|} \sum_{\chi \in \widehat{G}_S} \overline{\chi(g)} \prod_{q \in J} (1 + e^{ty_q} \chi(g_{S,q})).$$

Formula (FCRT-30) only enumerates one prescribed source fibre. The FCRT flag instead requires a joint, nested selection of T in the S -fibre and a nonempty proper $U \subsetneq T$ in the $\{p\}$ -fibre, with $p \notin S$. A future Fourier argument would therefore need a rigorously defined two-fibre enumerator, control of its boundary terms, and a uniform nontrivial-character estimate strong enough to retain pairs with the required logarithmic weights. It would then still have to select several pairs subject to source and sink disjointness. No such joint estimate is asserted here. Neither character orthogonality, the Boolean collision count, nor the same-fibre exchange proposition supplies it.

91.7. Scope of the exact finite computation. The frozen computation uses integer factorizations and rational multiplicative residuals; floating logarithms occur only as display fields. It exhausts the normalized domain

$$(FCRT-31) \quad 1 \leq a \leq b, \quad a + b = c, \quad \gcd(a, b) = 1, \quad c \leq 3000,$$

which contains 1,368,094 triples. The easy strata $I = \emptyset$, $X \leq Y$, and $|I| = 1$ reduce the block search to proved closed formulas: all boundaries vanish when $I = \emptyset$; SCRT and FCRT vanish when $X \leq Y$ because the full block is saturated, although PBT need not vanish; and when $|I| = 1$ with $X > Y$, all three boundaries equal $X - Y$. Only six points require a direct block search. The producer enumerates every compatible saturated block, represents SCRT states by all attainable disjoint source–sink unions, retains every disjoint block family for FCRT, and solves the residual ownership problems by an exact capped-state dynamic program. The independent validator instead recursively enumerates every disjoint block family and then every residual sink and reuse-token owner word. It does not import the producer and performs a deterministic byte-for-byte replay. The six directly enumerated points are

$$(1, 224, 225), (1, 675, 676), (343, 625, 968), \\ (1, 2303, 2304), (125, 2187, 2312), (99, 2401, 2500).$$

The same validation also covers 154 fully factored structured-family rows, including the out-of-range regression point (1, 65024, 65025), and 17 detailed certificates.

Besides the three rows in (FCRT-19), the forced comparison

$$(FCRT-32) \quad \begin{array}{c|c|c|c|c} P & e^\Delta & e^{B_{FCRT}} & e^{B_{SCRT}} & e^{B_{PBT}} \\ \hline (343, 625, 968) & 44/35 & 44/35 & 11/7 & 11/7 \end{array}$$

is reproduced exactly. Five points in the bounded domain have $B_{SCRT} > \Delta$; FCRT reaches the scalar lower bound at (1, 675, 676), (343, 625, 968), and (99, 2401, 2500). The two remaining FCRT fragmentation points are (1, 224, 225) and (1, 2303, 2304). The frozen validation status is PASS; sealed hashes for the exact output, producer, validator, and validation log are stored with the computation artifacts.

These statements have finite scope. They prove no uniform upper bound and produce no complete-premise unbounded counterfamily. In particular, the $n = 284$ point in $(2, 15^n - 2, 15^n)$ is recorded only with the proved fact $v_{31}(15^{284} - 2) = 2$; its 335-digit external arm is not completely factored, so no exact FCRT optimum is asserted there.

91.8. Lean formalization and the remaining arithmetic boundary. The ordinary proofs above precede the formalization. The primary Lean module has 70 public declarations and a matching 70-query axiom audit. An independently written bridge module has 18 public declarations and a matching 18-query audit. All four files compile with warnings treated as

errors. The union reported by `#print axioms` is exactly `propext`, `Classical.choice`, and `Quot.sound`; no `sorry`, `admit`, `unsafe`, or `native_decide` escape occurs.

The formalized unconditional content includes the finite maximal-feasible subset and cancellation lemmas used in the nontrivial part of Proposition 91.1; an explicit owner kernel in which each residual sink and reuse token has at most one owner; the clipped residual and once-charge mass inequality; the cap of reuse credit by both block surplus and witness weight; the endpoint height bridge; and the conditional implication of the separately supplied uniform gate to the repository’s unchanged **ABCConjecture**. The independent bridge caps raw incoming credit by source demand,

$$\widehat{c}_i = \min\{x_i, c_i\}, \quad x_i - \widehat{c}_i = \max\{x_i - c_i, 0\},$$

constructs the aggregate certificate from the owner kernel, and proves that the clipped boundary is preserved exactly. Its endpoint conversion assumes the decomposition

$$(FCRT-33) \quad \text{signedEndpointCoreDefect}(P) = \text{sourceMass} - \text{sinkMass}$$

as an explicit parameter.

Several formal limits are mathematically substantive. The Lean selection and cancellation lemmas do not encode the full typed free-target configuration or all cases of Proposition 91.1. The formal **ProperSubfaceFlag** is not yet constructed from concrete endpoint unitary divisors, and the owner kernel itself stores neither block source sets nor the residual-target and block-disjointness proofs; it also does not identify its abstract witness weight with $y(U)$. A bare endpoint certificate is tautologically inhabited by the boundary $\max\{h - r, 0\}$, so its mere existence carries no flagged-CRT content. Concrete arithmetic admissibility, the endpoint identity (FCRT-33), and uniform smallness carry that content.

The additive commutative-group versions of the complement identity and same-fibre signed exchange are formalized. The multiplicative residue-unit construction (FCRT-20)–(FCRT-22), the inverse system, the Boolean collision criterion, and the Fourier identity (FCRT-30) are not yet formalized. The required joint cross-fibre enumerator has not yet been promoted to a theorem. The witness files check the displayed factorizations, congruences, cap inequalities, and logarithmic strict inequalities, but do not encode the exhaustive optimizer that proves the three exact minima in Proposition 91.6. Most significantly, no Lean term inhabits the uniform gate (FCRT-18). Therefore this checkpoint proves neither the standard *abc* conjecture nor its negation; the cross-fibre weighted selection estimate remains an open obligation.

92. ANCHORED PREFIX FIBRES FOR PROPER-FACE CRT FLAGS

The residue-cube formulation in Section 91 turns an FCRT flag into a nested cross-fibre problem. This section extracts a finite positive result from Bae’s anchored-fibre selection lemma [87]. An ordered reservoir with sufficiently many prefix samples, relative to its residue subgroup and weight-cell entropy, contains a light zero-sum interval. Its complement is the required proper face. The entropy hypothesis is explicit; no uniform endpoint theorem supplying it is assumed.

Let $p^{e_p} \parallel c$, with $e_p \geq 2$, be a target source and let S be a source face disjoint from p . For a sink packet $A \subseteq J$, retain the notation

$$(APF-1) \quad G_p = (\mathbb{Z}/p^{e_p}\mathbb{Z})^\times, \quad \Phi_p(A) = \prod_{q \in A} g_{p,q}, \quad y(A) = \sum_{q \in A} \log q.$$

Here $g_{p,q}$ contains the complete q -power on the a -arm and its inverse on the b -arm. Thus

$$(APF-2) \quad \Phi_p(J) = -1, \quad p^{e_p} \mid a_A + b_A \iff \Phi_p(A) = -1.$$

Fix a packet $T \subseteq J$ satisfying the joint conditions

$$(APF-3) \quad \Phi_S(T) = -1, \quad \Phi_p(T) = -1, \quad x(S) \leq y(T).$$

The second condition is stronger than ordinary FCRT block compatibility, but it is automatic when $T = J$.

92.1. The complement dictionary. Call a nonempty $D \subseteq T$ a p -zero-sum deletion if $\Phi_p(D) = 1$.

Proposition 92.1 (Zero-sum deletion and compatible proper face). *The assignments*

$$(APF-4) \quad D \mapsto U = T \setminus D, \quad U \mapsto D = T \setminus U$$

give inverse correspondences between nonempty p -zero-sum deletions of T and nonempty proper subfaces $U \subsetneq T$ satisfying $\Phi_p(U) = -1$.

Proof. If D and $U = T \setminus D$ partition T , then

$$\Phi_p(T) = \Phi_p(U)\Phi_p(D).$$

Thus $\Phi_p(T) = -1$ and $\Phi_p(D) = 1$ imply $\Phi_p(U) = -1$. Nonemptiness of D makes U proper. If U were empty, then $D = T$ and $\Phi_p(T) = 1$, contradicting $\Phi_p(T) = -1$. The two group elements are distinct because $p^{e_p} \geq 4$.

Conversely, if $U \subsetneq T$ and both U and T have target label -1 , then cancellation in G_p gives $\Phi_p(T \setminus U) = 1$. Properness makes $T \setminus U$ nonempty, and the two complement operations in (APF-4) are inverse. $\square$

The block surplus is $\sigma = y(T) - x(S)$. If $U = T \setminus D$, its capped FCRT credit is

$$(APF-5) \quad \rho = \min\{\sigma, y(U)\} = \min\{y(T) - x(S), y(T) - y(D)\}.$$

It follows directly that the full surplus is reusable precisely when

$$(APF-6) \quad y(D) \leq x(S).$$

92.2. Anchored weighted selection. Choose a nonempty reservoir $K \subseteq T$, order it as $K = (q_1, \dots, q_m)$, and put

$$(APF-7) \quad H = \langle g_{p,q} : q \in K \rangle \leq G_p, \quad W = y(K).$$

Then $m \geq 1$ and $W > 0$. Fix $L > 0$, set

$$(APF-8) \quad N = \left\lceil \frac{W}{L} \right\rceil,$$

and partition $[0, W]$ into N consecutive cells of diameter at most L . Let $\mathcal{F} \subseteq \{1, \dots, m\}$ be a set of forbidden prefix gaps.

Theorem 92.2 (Anchored prefix flag). *Under (APF-3), assume the exact entropy inequality*

$$(APF-9) \quad m + 1 > |H|N(|\mathcal{F}| + 1).$$

There are $0 \leq i < j \leq m$, with $j - i \notin \mathcal{F}$, for which

$$(APF-10) \quad D = \{q_{i+1}, \dots, q_j\} \neq \emptyset, \quad \Phi_p(D) = 1, \quad y(D) \leq L.$$

Consequently $U = T \setminus D$ is a legal nonempty proper target face and

$$(APF-11) \quad \rho \geq \min\{y(T) - x(S), y(T) - L\}.$$

If $L \leq x(S)$, then $\rho = y(T) - x(S)$, so the complete block surplus is reused.

Proof. For $0 \leq k \leq m$, define the prefix packet, residue, and weight by

$$P_k = \{q_1, \dots, q_k\}, \quad R_k = \Phi_p(P_k), \quad Z_k = y(P_k).$$

Code k by the pair consisting of $R_k \in H$ and the weight cell containing $Z_k \in [0, W]$. There are at most $|H|N$ codes. Bae's anchored-fibre lemma states that a map on $\{0, \dots, m\}$ with at most C codes has equal-code indices $i < j$ whose positive difference avoids $\mathcal{F}$, provided $m + 1 > C(|\mathcal{F}| + 1)$ [87, Lemma 5.1]. Apply it with $C = |H|N$.

Equality of residue coordinates gives

$$1 = R_j R_i^{-1} = \prod_{r=i+1}^j g_{p,q_r} = \Phi_p(D).$$

Equality of weight cells and monotonicity of the prefix weights give $y(D) = Z_j - Z_i \leq L$. Since $i < j$, the deletion is nonempty. Proposition 92.1 turns it into a legal face. Moreover, $y(U) = y(T) - y(D) \geq y(T) - L$, and (APF-5) proves (APF-11). When $L \leq x(S)$, $y(T) - L \geq y(T) - x(S)$, so the first entry is the minimum and the full surplus is reused. $\square$

The forbidden-gap clause is retained exactly. It may encode a bad tail, unwanted interval size, or another obstruction depending only on $j - i$. The theorem does not assert that endpoint arithmetic supplies a uniformly small forbidden set.

Corollary 92.3 (Light-reservoir full-surplus criterion). *If*

$$(APF-12) \quad |K| \geq |H|, \quad y(K) \leq x(S),$$

then the complete surplus of (S, T) can be sent to p .

Proof. Take $L = x(S)$ and $\mathcal{F} = \emptyset$. The hypotheses imply $0 < W \leq L$, so $N = 1$, and $|K| + 1 > |H|$. Theorem 92.2 applies. $\square$

Corollary 92.4 (Full packet and the scalar boundary). *Suppose $I \setminus \{p\} = S \neq \emptyset$, $x(S) \leq Y = y(J)$, and the hypotheses of Theorem 92.2 hold with $T = J$ and $L \leq x(S)$. The resulting one-block configuration has*

$$(APF-13) \quad B_{\text{FCRT}} = (x_p - (Y - x(S)))_+ = (X - Y)_+.$$

Proof. The full packet has label -1 in every source coordinate. The selected block covers S , consumes all sinks, and sends its full surplus $Y - x(S)$ to the sole residual source p . Its clipped residual is the first quantity in (APF-13), while

$$x_p - (Y - x(S)) = x_p + x(S) - Y = X - Y.$$

The universal once-charge lower bound proves optimality. $\square$

This corollary removes a finite fragmentation loss when its entropy premise holds. It does not estimate $(X - Y)_+$ uniformly and therefore is not an *abc* theorem.

92.3. Recovery of the two motivating flags. For $(a, b, c) = (1, 675, 676)$, choose

$$S = \{13\}, \quad p = 2, \quad T = J = \{3, 5\}, \quad K = \{5\}.$$

Modulo 4, the b -arm generator satisfies $g_{2,5} = 5^{-2} = 1$. Consequently H is trivial, $|K| = |H| = 1$, and $y(K) = \log 5 < \log 13 = x(S)$. Corollary 92.3 produces $D = \{5\}$ and $U = \{3\}$, recovering the full-surplus flag and the boundary factor $26/15$.

For $(1, 65024, 65025)$, choose

$$S = \{5, 17\}, \quad p = 3, \quad T = J = \{2, 127\}, \quad K = \{127\}.$$

Modulo 9, $g_{3,127} = 127^{-1} = 1$. With $L = \log 127$ and $\mathcal{F} = \emptyset$, one has $m = N = |H| = 1$, so the strict entropy condition is $2 > 1$. Theorem 92.2 gives $D = \{127\}$ and $U = \{2\}$. Therefore

$$\rho \geq \min \left\{ \log \frac{254}{85}, \log \frac{254}{127} \right\} = \log 2.$$

The face itself has weight $\log 2$, so equality holds. This is the cap-active flag giving FCRT boundary factor $3/2$.

92.4. A complete-premise failure of the naive Boolean shortcut. Consider the assertion

$$(NBF) \quad 2^{|J|} > |G_p| \implies \text{there is a nonempty proper } U \subsetneq J \text{ with } \Phi_p(U) = -1.$$

Its proposed FCRT consequence would give a target- p flag for a saturated full block whose source face is disjoint from p .

Proposition 92.5 (Complete-premise NBF counterexample). *The primitive endpoint*

$$(a, b, c) = (1, 4715, 4716)$$

satisfies the Boolean inequality and has a compatible saturated full block disjoint from the target $p = 3$, but it has no nonempty proper 3-compatible packet. Hence NBF is false.

Proof. The exact factorizations are

$$(APF-14) \quad 4715 = 5 \cdot 23 \cdot 41, \quad 4716 = 2^2 \cdot 3^2 \cdot 131.$$

Thus $p^2 = 9 \mid c$, $J = \{5, 23, 41\}$, and

$$2^{|J|} = 8 > 6 = |(\mathbb{Z}/9\mathbb{Z})^\times| = |G_p|.$$

The full block $S = \{2\}$, $T = J$ is disjoint from p , compatible because $4 \mid 4716$, and saturated because $x(S) = \log 2 < \log 4715 = y(J)$.

Each sink prime is congruent to 5 modulo 9 and occurs to the first power on the b -arm. Therefore

$$g_{3,q} = q^{-1} = 2 \pmod{9} \quad (q \in J),$$

and every packet has label

$$\Phi_3(U) = 2^{|U|} \pmod{9}.$$

A nonempty proper packet has size one or two, hence label 2 or 4, never $-1 = 8$. Equivalently, its arm sum is congruent to 6 or 8, rather than zero, modulo 9. Only the full packet, of size three, is compatible. Every possible FCRT face is a nonempty proper subset of J , so no flag can target 3. $\square$

The Boolean collision predicted by pigeonhole still occurs: all singleton packets have label 2, and all doubletons have label 4. Distinct packets of the same size are incomparable, so cancellation produces a signed relation rather than a positive zero-sum deletion. The counterexample does not meet Theorem 92.2's entropy premise. Every nonempty reservoir generates $H = \langle 2 \rangle = G_p$, so $|H| = 6$ while $|K| \leq 3$ and $N \geq 1$.

Accordingly, this example retires only the implication NBF. It does not refute the anchored theorem, FCRT-1, SCRT-0, the residue-cube route, or the standard *abc* conjecture.

92.5. The remaining uniform bottleneck. For a jointly compatible packet T , define the weighted zero-sum radius

$$(APF-15) \quad \zeta_p(T) = \min\{y(D) : \emptyset \neq D \subseteq T, \Phi_p(D) = 1\},$$

with value $+\infty$ when no such deletion exists. The complement dictionary identifies proper target faces with these deletions, and full surplus reuse is equivalent to $\zeta_p(T) \leq x(S)$.

The anchored theorem controls this radius only after an ordered reservoir satisfies

$$(APF-16) \quad |K| + 1 > |\langle g_{p,q} : q \in K \rangle| \left\lceil \frac{y(K)}{L} \right\rceil (|\mathcal{F}| + 1),$$

with L small enough to yield useful credit. No uniform endpoint theorem forcing (APF-16) is known. The generated subgroup may have prime-power scale, whereas $|K|$ counts only distinct sink primes. Bae's application has an exponentially longer sample interval and a separate exponential tail estimate [87]; endpoint analogues of those inputs have not been proved. The NBF counterexample shows why replacing the ordered prefixes by all Boolean packets does not by itself solve the mismatch.

The remaining positive problem is a weighted zero-sum or Davenport-type estimate using the special endpoint relation $\Phi_p(J) = -1$, or an arithmetic theorem forcing many light sink generators into a small subgroup. Absence of such an estimate records the present bottleneck and does not retire the parent route.

92.6. Lean boundary. The companion module `ABCAnchoredPrefixFlaggedCRT20260904` has 28 public declarations, and its audit file contains exactly 28 corresponding `#print axioms` queries. Both files compile with warnings treated as errors. The audit union is exactly `propext`, `Classical.choice`, and `Quot.sound`; no `sorry`, `admit`, `unsafe`, or `native_decide` escape occurs. The formalization proves the finite anchored code-fibre avoidance lemma, additive prefix cancellation, the abstract weighted selection theorem with a supplied finite cell code, both directions of the complement dictionary, the two capped-credit inequalities, the scalar residual identity, and the complete finite arithmetic assertions for $(1, 4715, 4716)$.

The formal weighted theorem assumes a cell code together with the proved implication that equal cells have interval-weight difference at most L . It does not formalize the elementary real-ceiling construction of the cells in (APF-8). It also does not construct the concrete unit group G_p uniformly from an arbitrary endpoint, prove the entropy estimate (APF-16), or inhabit the uniform FCRT-1 gate. The two motivating examples are ordinary exact recoveries; the Lean module’s new endpoint arithmetic is the complete-premise 4715 counterexample. Thus the formal development contains neither an unconditional proof nor a disproof of *abc*.

93. PRIMARY-LITERATURE GATE AUDIT THROUGH SEPTEMBER 2026

We re-audited primary sources available through September 3, 2026 against the exact gates used in this paper. This is a theorem-by-theorem compatibility check, not a survey by title or terminology. No checked source supplies a generally accepted unconditional proof or disproof of standard *abc*, and no source closes one of the remaining repository gates without an additional uniformity statement.

Laniewski’s parity coordinates give the clearest direct match to the canonical defect route [25, Proposition 3.7 and Corollary 3.9]. For $m = \log(abc)$, $r = \log \text{rad}(abc)$, and $h = \log c$, the cited identity is exactly

$$q = 1 + \frac{(m - r) - (m - h)}{r} = 1 + X - Y.$$

The equality follows by expanding the numerator: $(m - r) - (m - h) = h - r$. It therefore confirms the coordinate choice used in the canonical gain section, but it gives no upper bound for $X - Y$. Laniewski’s Theorem 4.3 similarly gives an exact parity-class transgression threshold in terms of linear radical defect and the smaller summand. It is a certificate for transgression, not an estimate excluding it. The conditional dichotomy in Theorem 10.5 starts from a hypothetical fixed-epsilon transgressive sequence and hence cannot supply that missing estimate.

The exceptional-set theorems of Bernert–Browning–Lichtman–Teräväinen give, in their notation,

$$N_\lambda(X) \ll_{\lambda, \varepsilon} X^{(23\lambda+3)/40+\varepsilon}$$

and a stronger fixed- $\lambda < 1$ exponent $3/5 + \varepsilon$ [7, Theorems 1.2–1.3]. These are unconditional aggregate counts. They do not imply a pointwise bound on $X - Y$, an eventual compensated packet estimate, or a fixed-index Pell valuation theorem. The distinction between aggregate scarcity and an all-triples uniform inequality is retained throughout our formal interfaces.

Baek and Lee prove Mason–Stothers over arbitrary fields with the necessary alternative that all three derivatives vanish [9, Theorem 1]. Their formal development is the source of the Mathlib theorem `Polynomial.abc` already used by our `MasonSpecializationBarrier` module. In positive characteristic the triple

$$-1, \quad -x^p, \quad (1+x)^p$$

has vanishing derivatives, so deleting that branch would be false. The polynomial theorem has no specialization mechanism that turns it into the needed integer radical inequality; the repository’s specialization barrier therefore remains.

For the Pell route, Laniewski’s Theorem 4.31 provides a positive-density composite-index family in which quality exceeding one is detected by non-squarefreeness of a Pell-coordinate product. Non-squarefree is weaker than squarefull. At its only odd-prime seed $\ell = 7$, the repository’s exact factorization is

$$A_7 B_7 = 239 \cdot 13^2, \quad v_{239}(A_7 B_7) = 1.$$

Thus this row satisfies the non-squarefree condition but not the simultaneous squarefull gate. Sanna’s valuation formulas determine later valuations once the rank-of-aparition valuation is known [36]; Carmichael’s primitive-divisor theorem guarantees a primitive divisor [40], but being primitive does not imply exponent one. These results leave exactly the first-rank exponent problem isolated by the signed trace module.

Browning and Verzbio obtain power-saving counts for sums of three powerful numbers in a broad log-general-type range [8, Theorem 1.2]. The squarefull exponent triple $(2, 2, 2)$ relevant

to the elementary Pell surface is explicitly outside that range. Their theorem is therefore a valid aggregate input in its stated domain, but not a squarefull-Pell exclusion.

The alternative-quality identity $q_{\text{std}} = \eta q_{\text{DGM}}$ remains usable [26]. Divergence of q_{DGM} alone does not control the product. The claimed later phase-boundary transfer uses a one-sided estimate as an equality and applies a fixed-support lemma in a growing-support regime; we do not import those conclusions. Likewise, the fixed-epsilon exceptional-set claim in Carella depends on replacing an $O(h)$ error by $O(h\rho(u))$ while $\rho(u) \rightarrow 0$ [14, Lemma 4.2]. That step does not follow. These findings reject only the dependent inferences, not the broader smooth-neighbour or alternative-metric routes. The exact gain factorization in Müller–Taktikos remains motivational, but the preprint changes the approximation-gain numerator between its definition and subsequent theorem and supplies no standard-*abc* coefficient-one estimate [27].

Finally, IUT III Corollary 3.12 is stated inside the situation of Theorem 3.11 and the preceding IUT constructions [62]. Zhou’s explicit rational inequalities expressly invoke its μ_6 version [31]. Published sources therefore exist, but they do not eliminate the source-faithful obligation carried by our IUT-facing route: all places, the Ind1–Ind3 action, actual hulls and integral orders, and pointed same-pilot transport must inhabit one compatible interface. That chain remains open in this repository; no source-dependent conclusion is silently promoted to an independent Lean theorem.

The detailed version-locked ledger, with proposition and page references, is `research/ABC_PRIMARY_LITERATURE_GATE_AUDIT_2026_09_03.md`. Its classifications are local: a flawed inference retires that inference only, while every unrefuted parent route continues.

94. STEINBERG VALUATION CONTACT SURFACES AND A VERONESE OBSTRUCTION

Author: ChatGPT. This section constructs a two-dimensional valuation invariant of a primitive *abc* point, proves its affine and five-term identities, and audits two proposed closure estimates against full-premise counterexamples. No form of the *abc* conjecture, Vojta’s conjecture, or an IUT comparison is assumed.

94.1. The divisor triangle. Let

$$\text{Div}_{\mathbf{Q}} = \bigoplus_{p \text{ prime}} \mathbf{Z}[p], \quad d(q) = \sum_p v_p(q)[p].$$

For a positive primitive equation $a + b = c$, put $A = d(a)$, $B = d(b)$, $C = d(c)$, and $x = a/c$. Since $1 - x = b/c$, define the oriented valuation contact surface

$$(474) \quad \Omega(A, B, C) = (A - C) \wedge (B - C) = d(x) \wedge d(1 - x).$$

Proposition 94.1 (Three-leg expansion and affine covariance). *One has*

$$\Omega(A, B, C) = A \wedge B + B \wedge C + C \wedge A.$$

Moreover, for every divisor vector T , integer m , and permutation $\sigma \in S_3$,

$$\Omega(A + T, B + T, C + T) = \Omega(A, B, C),$$

$$\Omega(mA, mB, mC) = m^2 \Omega(A, B, C),$$

$$\Omega(A_{\sigma(1)}, A_{\sigma(2)}, A_{\sigma(3)}) = \text{sgn}(\sigma) \Omega(A, B, C).$$

Proof. Bilinearity and alternation give

$$(A - C) \wedge (B - C) = A \wedge B - A \wedge C - C \wedge B.$$

This is the displayed three-leg sum. Translation cancels in both differences, scaling factors from both exterior arguments, cyclic permutations preserve the three-leg sum, and a transposition reverses it. $\square$

Primitivity makes the prime supports S_a, S_b, S_c pairwise disjoint. If $w_p \geq 0$ and $L_w(A) = \sum_{p \in S_a} v_p(a)w_p$, define

$$\left\| \sum_{p < q} z_{pq}[p] \wedge [q] \right\|_{1,w} = \sum_{p < q} |z_{pq}| w_p w_q.$$

Proposition 94.2 (Exact mixed contact area). *For a primitive abc point,*

$$\|\Omega(A, B, C)\|_{1,w} = L_w(A)L_w(B) + L_w(B)L_w(C) + L_w(C)L_w(A).$$

Proof. The three terms of Proposition 94.1 lie on the disjoint rectangles $S_a \times S_b$, $S_b \times S_c$, and $S_c \times S_a$. On each rectangle the absolute coefficient is a product of its two positive valuation coefficients. The double sum therefore factors, and the three rectangle norms add. $\square$

With $w_p = \log p$, write

$$(475) \quad \Phi = \log a \log b + \log b \log c + \log c \log a.$$

If $r_a = \log \text{rad}(a)$ and similarly for the other legs, the radical skeleton area is

$$\Psi = r_a r_b + r_b r_c + r_c r_a, \quad 3\Psi \leq (r_a + r_b + r_c)^2.$$

Writing $\log a = r_a + \delta_a$ and cyclically gives the exact positive excess

$$\begin{aligned} \Phi - \Psi &= r_a \delta_b + \delta_a r_b + \delta_a \delta_b + r_b \delta_c + \delta_b r_c + \delta_b \delta_c \\ &\quad + r_c \delta_a + \delta_c r_a + \delta_c \delta_a. \end{aligned}$$

94.2. Five-term gluing before the Milnor quotient. For rational x, y in the domain of the following expressions, set

$$z_1 = x, \quad z_2 = y, \quad z_3 = y/x, \quad z_4 = \frac{y(1-x)}{x(1-y)}, \quad z_5 = \frac{1-x}{1-y}.$$

Theorem 94.3 (Five-term contact relation).

$$\Omega(z_1) - \Omega(z_2) + \Omega(z_3) - \Omega(z_4) + \Omega(z_5) = 0.$$

Proof. Put $X = d(x)$, $U = d(1-x)$, $Y = d(y)$, $V = d(1-y)$, and $Z = d(x-y)$. Multiplicativity of divisors and $d(-1) = 0$ give the five pairs

$$(X, U), (Y, V), (Y - X, Z - X), (Y + U - X - V, Z - X - V), (U - V, Z - V).$$

After inserting the alternating signs, bilinear expansion cancels every basis wedge. $\square$

The terminology is deliberate. Equation (474) is the image under $\bigwedge^2 d$ of the standard Bloch boundary $\delta([x]) = x \wedge (1-x)$. It is not a new nonzero Milnor K_2 invariant: the Milnor quotient imposes $\{x, 1-x\} = 0$. We work before that quotient. Theorem 94.3 allows one contact cell to be replaced by four auxiliary cells and exposes the new divisor $d(x-y)$.

Proposition 94.4 (Quadratic Veronese peeling). *For $x \in \mathbf{Q} \setminus \{0, 1, -1\}$,*

$$\Omega(x^2) = 2\Omega(x) - 2\Omega\left(\frac{x}{1+x}\right).$$

If $x = X/Z$ with $0 < X < Z$ and $\gcd(X, Z) = 1$, this becomes

$$\Omega(X^2, Z^2 - X^2, Z^2) = 2\Omega(X, Z - X, Z) - 2\Omega(X, Z, X + Z).$$

Proof. Set $y = x^2$ in Theorem 94.3. The five arguments are $x, x^2, x, x/(1+x), 1/(1+x)$. The last is the complement of the fourth, and $\Omega(1-z) = -\Omega(z)$. Substitution and rearrangement give the first identity. The second follows by putting $x = X/Z$ in lowest terms. $\square$

This exact oriented identity replaces a coherent square cell by two cells whose entries are linear in X, Z . It exposes primes of $Z - X$ and $Z + X$ to the filling cost, and explains why a single-cell square obstruction does not rule out every five-term filling policy.

94.3. A non-circular gate, and why it is false. Put $h_a = \log a$, $h_b = \log b$, $H = \log c$, and $\rho = r_a + r_b + r_c = \log \text{rad}(abc)$. Define the mixed polarization

$$\mathcal{M} = h_a r_b + r_a h_b + h_b r_c + r_b H + H r_a + r_c h_a.$$

Consider the uniform single-cell assertion

$$(476) \quad \Phi \leq \frac{1+\epsilon}{2} \mathcal{M} + C_\epsilon H.$$

Proposition 94.5 (The gate would imply abc). *If (476) holds for every $\epsilon > 0$ with one constant valid for all primitive points, then the logarithmic abc conjecture holds.*

Proof. Every leg height is at most H , so $\mathcal{M} \leq 2H\rho$. Also $c \leq 2ab$, whence $H - \log 2 \leq h_a + h_b$ and $\Phi \geq H(H - \log 2)$. Substitute these two estimates in (476) and cancel $H > 0$ to obtain

$$H \leq (1 + \epsilon)\rho + C_\epsilon + \log 2.$$

□

This implication is non-circular: the premise compares a quadratic two-vector norm with a one-sided radical polarization. It is nevertheless false.

Theorem 94.6 (Primitive Pythagorean-square obstruction). *Gate (476) fails with all of its quantifiers and all primitive abc premises satisfied.*

Proof. For $t \geq 1$, let

$$x = 2t + 1, \quad y = 2t(t + 1), \quad z = 2t^2 + 2t + 1.$$

Expansion gives $x^2 + y^2 = z^2$. The odd number x is coprime to 2, t , and $t + 1$, so $\gcd(x, y) = 1$; the equation gives pairwise coprimality of x, y, z . Thus (x^2, y^2, z^2) is a primitive abc point.

For every $u > 0$,

$$\log \text{rad}(u^2) \leq \log u = \frac{1}{2} \log(u^2).$$

Consequently $\mathcal{M} \leq \Phi$. But

$$\frac{\Phi}{H} \geq \log(y^2) = 2 \log(2t(t + 1)) \longrightarrow \infty.$$

Taking $\epsilon = 1/2$ in (476) would imply $\Phi \leq (3/4)\Phi + C_{1/2}H$, or $\Phi \leq 4C_{1/2}H$, a contradiction. □

The same family refutes the direct skeleton gate $\Phi \leq 3(1 + \epsilon)^2 \Psi + K_\epsilon^2$ for sufficiently small positive ϵ , since $\Psi \leq \Phi/4$ and Φ is unbounded. The finite primitive point $9 + 16 = 25$ separately refutes coefficient-one thinness: its $[3] \wedge [2]$ coefficient has absolute value 8, while the radical coefficient is 1. None of these is an abc counterexample.

94.4. Veronese correction and open route. For $n > 1$, put $g(n) = \gcd_{p|n} v_p(n)$ and write $n = u^{g(n)}$. If $h = \log n$, $r = \log \text{rad}(n)$, and $u_h = h/g$, then

$$(477) \quad h - r = (g - 1)u_h + (u_h - r).$$

For a unit leg we set $g(1) = 1$ and $h = r = u_h = 0$, so both defect components vanish. Thus the definitions remain total on auxiliary primitive cells. The first summand is coherent Veronese-ray thickness; the second is the residual thickness of a base whose prime exponents have gcd one. Polarizing (477) on all three legs gives

$$2\Phi - \mathcal{M} = \mathcal{V} + \mathcal{R},$$

where, with $u_i = h_i/g_i$,

$$\begin{aligned} \nu_i &= (g_i - 1)u_i, & \sigma_i &= u_i - r_i, \\ \mathcal{V} &= \sum_{i < j} (\nu_i h_j + \nu_j h_i), & \mathcal{R} &= \sum_{i < j} (\sigma_i h_j + \sigma_j h_i). \end{aligned}$$

These coherent and residual contact losses are nonnegative and total on unit legs. The coherent Veronese term $\mathcal{V}$ alone is already large enough to supply the square-family obstruction. This does not assert that $\mathcal{R} = 0$ and does not refute separate control of $\mathcal{R}$.

For a rational cell, write $z = m/n$ in lowest terms with $n > 0$ and $m(n - m) \neq 0$. We use the signed relation $m + (n - m) = n$ for orientation and the pairwise-coprime absolute legs $|m|, |n - m|, n$ for valuations, heights, radicals, and exponent gcds. Thus every scalar cell has a canonical normalization, even when an auxiliary argument lies outside $(0, 1)$. One may instead work in the conservative positive subcomplex $0 < y < x < 1$: all five arguments then lie in $(0, 1)$ and have unique reduced positive primitive triples. The quadratic move $y = x^2$ is in this domain for $0 < x < 1$.

For a five-term-equivalent chain $\Gamma = \sum_j n_j [z_j]$, define

$$\mathcal{Q}(\Gamma) = \sum_j |n_j|(\mathcal{M}_j + \mathcal{V}_j), \quad \mathcal{R}(\Gamma) = \sum_j |n_j| \mathcal{R}_j.$$

Proposition 94.7 (Automatic filling-boundary inequality). *Every finite five-term-equivalent chain satisfies*

$$\Phi(a, b, c) \leq \frac{1}{2}(\mathcal{Q}(\Gamma) + \mathcal{R}(\Gamma)).$$

Proof. Five-term equivalence gives the exact oriented relation $\Omega_0 = \sum_j n_j \Omega_j$. The weighted ℓ^1 triangle inequality and the cellwise identity $2\Phi_j = \mathcal{M}_j + \mathcal{V}_j + \mathcal{R}_j$ give

$$\Phi_0 = \|\Omega_0\|_{1,\log} \leq \sum_j |n_j| \|\Omega_j\|_{1,\log} = \frac{1}{2} \sum_j |n_j|(\mathcal{M}_j + \mathcal{V}_j + \mathcal{R}_j).$$

Only finitely many prime coordinates occur. □

The surviving open target is to construct, uniformly, a chain satisfying

$$(478) \quad \begin{aligned} \mathcal{R}(\Gamma) &\leq \epsilon \mathcal{Q}(\Gamma) + K_\epsilon H, \\ \mathcal{Q}(\Gamma) &\leq 2H\rho + L_\epsilon H. \end{aligned}$$

Together with Proposition 94.7, these estimates give

$$\Phi \leq (1 + \epsilon)H\rho + \frac{1}{2}((1 + \epsilon)L_\epsilon + K_\epsilon)H.$$

The lower bound $\Phi \geq H(H - \log 2)$ therefore implies *abc*. Gate (478) is not a renamed height inequality: it requires an actual five-term filling and separate control of its coherent and primitive-residual costs. It remains open. The Pythagorean-square family eliminates the uncalibrated one-cell policy, but it does not eliminate every auxiliary-cell filling.

The exponent triple also defines a generalized Fermat signature with $\chi_{\text{orb}} = 1/g_a + 1/g_b + 1/g_c - 1$. Results for a fixed hyperbolic signature do not provide the uniform estimate over varying signatures that (478) requires.

Primary-source boundary. The relevant primary sources have sharply different arithmetic settings:

- Ru–Wang, *Vojta’s abc conjecture for entire curves in toric varieties highly ramified over the boundary*, <https://arxiv.org/abs/2410.19395v2>, is a complex entire-curve theorem.
- Pasten, *On the arithmetic case of Vojta’s conjecture with truncated counting functions*, <https://arxiv.org/abs/2205.07841v3>, proves arithmetic approximation inequalities and bounds toward *abc*.
- Gasbarri–Guo–Wang, *Campana conjecture for coverings of toric varieties over function fields*, <https://arxiv.org/abs/2401.13186v2>, works over function fields.

None is an unconditional integer *abc* theorem over $\mathbf{Q}$. Cached primary PDFs and SHA-256 metadata are stored under `research/sources/steinberg_contact_surface_2026_09_02/`.

Formal status. The Lean modules `SteinbergValuationContactSurface20260902.lean` and its axiom audit formalize the exterior coefficient identities, five-term relation, scalar area identities, quadratic Veronese peeling, $\text{MC} \Rightarrow \text{abc}$, the coefficient-one audit, the complete Gate-MC Pythagorean-square obstruction, the abstract scalar Veronese/residual split for supplied data, and the finite-coordinate filling-boundary inequality. The integer exponent-gcd construction, actual finite-support cell norm, and conditional exact-boundary chain instantiation are supplied by the supplemental Section 95. The algebraic and positive five-term-generated relations and

their exact boundary bridge are formalized in Section 96. What remains open is a positive filling-existence theorem for an arbitrary target and the two analytic Gate VF cost estimates. The quantified Gate-SC obstruction is formalized at the fixed admissible value $\epsilon = 1/10$. These modules do not inhabit Gate VF or close **ABCConjecture**.

95. INTEGER EXPONENT GCDS AND FINITE-SUPPORT STEINBERG CHAINS

Author: ChatGPT. This section supplies two algebraic bridges needed by the valuation-contact route. First, it constructs the canonical common-exponent decomposition of every positive integer and separates its coherent and primitive residual defects. Second, it realizes contact surfaces in an actual finite-support coordinate module and proves the exact weighted cell norm and a conditional finite-chain estimate. No form of the *abc* conjecture and no analytic filling estimate is assumed.

95.1. The canonical common-exponent decomposition. For $n > 1$, write

$$n = \prod_{p \in S(n)} p^{e_p}, \quad e_p > 0,$$

where $S(n)$ is nonempty, and put

$$(479) \quad g(n) = \gcd_{p \in S(n)} e_p, \quad u(n) = \prod_{p \in S(n)} p^{e_p/g(n)}.$$

For the unit we adopt the convention

$$(480) \quad g(1) = 1, \quad u(1) = 1.$$

The second convention does not turn the gcd of an empty exponent family into one; it only defines the decomposition at the unit.

Proposition 95.1 (Integer exponent-gcd decomposition). *For every positive integer n ,*

$$g(n) > 0, \quad n = u(n)^{g(n)}, \quad u(n) > 0, \quad \text{rad}(u(n)) = \text{rad}(n).$$

If $n > 1$, then $u(n)$ has the same nonempty prime support as n and

$$\gcd_{p|u(n)} v_p(u(n)) = 1.$$

The last assertion is deliberately restricted to $n > 1$.

Proof. The assertions at $n = 1$ follow from (480). Suppose $n > 1$. Its finite prime support is nonempty, so the gcd $g(n)$ of its positive exponents is positive and divides every e_p . Consequently

$$g(n) \frac{e_p}{g(n)} = e_p$$

for every supported prime, and unique factorization gives

$$u(n)^{g(n)} = \prod_{p \in S(n)} p^{g(n)(e_p/g(n))} = \prod_{p \in S(n)} p^{e_p} = n.$$

Every quotient $e_p/g(n)$ is positive. Thus no supported prime disappears and no new prime appears, proving support and radical equality. Dividing a finite nonzero family by its gcd makes the gcd of the quotient family one, which proves the final assertion. $\square$

Set

$$h(n) = \log n, \quad h_u(n) = \log u(n), \quad r(n) = \log \text{rad}(n),$$

and define

$$(481) \quad \nu(n) = (g(n) - 1)h_u(n), \quad \sigma(n) = h_u(n) - r(n).$$

Proposition 95.2 (Coherent–residual defect split). *For every positive integer n ,*

$$(482) \quad h(n) = g(n)h_u(n), \quad h(n) - r(n) = \nu(n) + \sigma(n), \quad \nu(n), \sigma(n) \geq 0.$$

At $n = 1$, the three heights h, h_u, r and both thicknesses ν, σ vanish (while the convention gives $g(1) = 1$).

Proof. Proposition 95.1 and the logarithm of a positive integral power give $h(n) = g(n)h_u(n)$. Since $g(n) \geq 1$ and $h_u(n) \geq 0$, one has $\nu(n) \geq 0$. The equality of radicals and the elementary inequality $\text{rad}(u(n)) \leq u(n)$ give $\sigma(n) \geq 0$. Finally,

$$g(n)h_u(n) - r(n) = (g(n) - 1)h_u(n) + (h_u(n) - r(n)).$$

The unit statement follows directly from the convention. $\square$

95.2. An actual finite-support contact module. Let

$$\mathcal{D} = \mathbb{N} \rightarrow_0 \mathbb{Z}, \quad \mathcal{E} = (\mathbb{N} \times \mathbb{N}) \rightarrow_0 \mathbb{Z},$$

where $\rightarrow_0$ denotes finitely supported functions. For $X, Y \in \mathcal{D}$ define

$$(X \otimes Y)_{p,q} = X_p Y_q, \quad X \wedge Y = X \otimes Y - Y \otimes X,$$

and set

$$\Omega(A, B, C) = (A - C) \wedge (B - C) \in \mathcal{E}.$$

This ordered-pair model stores both orientations of each alternating coordinate. For effective divisors $A, B, C : \mathbb{N} \rightarrow_0 \mathbb{N}$, a finite set S , and nonnegative weights w_p , put

$$L_{S,w}(A) = \sum_{p \in S} A_p w_p, \quad \|\Omega\|_{S,w} = \frac{1}{2} \sum_{p,q \in S} |\Omega_{p,q}| w_p w_q.$$

Proposition 95.3 (Exact finite-support mixed area). *If the supports of A, B, C are pairwise disjoint, then*

$$(483) \quad \|\Omega(A, B, C)\|_{S,w} = L_{S,w}(A)L_{S,w}(B) + L_{S,w}(B)L_{S,w}(C) + L_{S,w}(C)L_{S,w}(A).$$

Proof. The three-leg expansion gives

$$\Omega_{p,q} = A_p B_q - A_q B_p + B_p C_q - B_q C_p + C_p A_q - C_q A_p.$$

At any coordinate at most one of A_p, B_p, C_p is nonzero. Applying this observation at p and at q gives the pointwise identity

$$\begin{aligned} |\Omega_{p,q}| &= A_p B_q + B_p A_q + B_p C_q + C_p B_q \\ &\quad + C_p A_q + A_p C_q. \end{aligned}$$

After multiplication by $w_p w_q$ and finite summation, each directed rectangle factors. For example,

$$\sum_{p,q \in S} A_p B_q w_p w_q = L_{S,w}(A)L_{S,w}(B).$$

Every cross-leg rectangle occurs in both orientations; the prefactor $1/2$ removes this duplication and yields (483). $\square$

A canonical positive rational cell is a triple $P = (a, b, c)$ of positive integers such that

$$a + b = c, \quad \gcd(a, b) = 1.$$

It represents $a/c \in (0, 1)$. The sum relation also gives $\gcd(a, c) = \gcd(b, c) = 1$, so the three valuation divisors have pairwise disjoint prime supports. Let $S(P)$ be their finite union and take $w_p = \log p$. Unique factorization and Proposition 95.3 give

$$(484) \quad \Phi(P) := \|\Omega(P)\|_{S(P), \log} = \log a \log b + \log b \log c + \log c \log a.$$

Let $M(P)$ be the one-sided radical polarization of the three leg heights. Let $V(P)$ be the bilinear contact loss formed from the three coherent thicknesses ν , and let $R(P)$ be the

corresponding loss formed from the three residual thicknesses σ . Polarizing Proposition 95.2 in (484) proves the exact identity

$$(485) \quad 2\Phi(P) = M(P) + V(P) + R(P).$$

No estimate is used in (485).

95.3. Finite chains with an exact boundary. Let I be finite, let P_0 and P_j be canonical positive rational cells, and let $n_j \in \mathbb{Z}$. Assume the exact equality in the integral finite-support module

$$(486) \quad \Omega(P_0) = \sum_{j \in I} n_j \Omega(P_j).$$

Theorem 95.4 (Concrete exact-boundary filling estimate). *Under (486),*

$$(487) \quad \Phi(P_0) \leq \frac{1}{2} \left(\sum_{j \in I} |n_j| (M(P_j) + V(P_j)) + \sum_{j \in I} |n_j| R(P_j) \right).$$

Proof. Take the finite union S of every prime support occurring in the target and the chain. Evaluate (486) at each $(p, q) \in S \times S$, cast to the reals, and multiply by $\frac{1}{2} \log p \log q$. The triangle inequality gives

$$\Phi(P_0) \leq \sum_{j \in I} |n_j| \Phi(P_j).$$

Substitute (485) for each cell and separate the two sums. This is exactly (487). $\square$

Theorem 95.4 is conditional only on the displayed finite-support equality. The inductively generated equivalence relation from the permitted rational five-term moves remains *open and unformalized*: this section neither defines all generators and closure operations nor proves that every generated chain satisfies (486). It also proves neither of the two analytic bounds required by Gate VF.

95.4. Lean realization and formal boundary. The companion Lean module listed in Section 171 formalizes Propositions 95.1 and 95.2, the actual finite-support types `FiniteDivisor` and `FiniteExteriorSurface`, Proposition 95.3, the canonical rational-cell specialization (484), the split (485), and Theorem 95.4. Its final namespace theorem instantiates the earlier abstract finite-coordinate triangle inequality. The companion axiom audit uses no project-specific axiom, and direct compilation with warnings treated as errors succeeds.

96. GENERATED FIVE-TERM RELATIONS AND THE FINITE-BOUNDARY BRIDGE

Author: ChatGPT. The valuation-contact construction of Section 95 provides a concrete finite-support exterior surface for every positive reduced rational cell and an analytic estimate for every finite chain satisfying an exact surface boundary. What remained was to derive that boundary from the relation generated by five-term moves. We supply that derivation here. We first work in a free divisor-symbol module, then pass to the smaller relation generated only by positively realized rational moves. The argument is the divisor shadow of the standard pre-Bloch complex [28, 29, 30]; no K -theoretic or *abc* estimate is imported from that theory.

96.1. The free symbol chain and its boundary. Let

$$\mathcal{D} = \mathbb{N} \rightarrow_0 \mathbb{Z}, \quad \mathcal{E} = (\mathbb{N} \times \mathbb{N}) \rightarrow_0 \mathbb{Z},$$

and retain the ordered-coordinate exterior product

$$(A \wedge B)_{p,q} = A_p B_q - A_q B_p.$$

Write $\mathcal{S} = \mathcal{D} \times \mathcal{D}$ for the set of divisor symbols and let

$$\mathcal{C}_0 = \mathbb{Z}^{(\mathcal{S})}$$

be the free integral module on those symbols. Its basis element associated with (A, B) is denoted by $[A, B]$. Define a linear map

$$(488) \quad \partial : \mathcal{C}_0 \longrightarrow \mathcal{E}, \quad \partial[A, B] = A \wedge B.$$

For $X, U, Y, V, Z \in \mathcal{D}$, define the formal five-term chain

$$(489) \quad \begin{aligned} \mathfrak{t}(X, U, Y, V, Z) &= [X, U] - [Y, V] + [Y - X, Z - X] \\ &\quad - [Y + U - X - V, Z - X - V] + [U - V, Z - V]. \end{aligned}$$

Theorem 96.1 (Five-term generators are boundary cycles). *For all $X, U, Y, V, Z \in \mathcal{D}$,*

$$\partial \mathfrak{t}(X, U, Y, V, Z) = 0.$$

Proof. By linearity and (488), the claimed boundary is

$$\begin{aligned} &X \wedge U - Y \wedge V \\ &\quad + (Y - X) \wedge (Z - X) \\ &\quad - (Y + U - X - V) \wedge (Z - X - V) \\ &\quad + (U - V) \wedge (Z - V). \end{aligned}$$

Expand each wedge bilinearly. Every elementary wedge between two members of $\{X, U, Y, V, Z\}$ occurs with opposite coefficients and cancels. This is also the finite-support lift of the earlier coefficientwise five-term identity. $\square$

Let

$$(490) \quad \mathcal{R}_5 = \text{span}_{\mathbb{Z}}\{\mathfrak{t}(X, U, Y, V, Z) : X, U, Y, V, Z \in \mathcal{D}\} \subseteq \mathcal{C}_0.$$

Theorem 96.2 (Generated relation-to-boundary bridge). *One has*

$$\mathcal{R}_5 \subseteq \ker \partial.$$

Consequently, if

$$c \sim_5 d \iff c - d \in \mathcal{R}_5,$$

then $\partial c = \partial d$.

Proof. The kernel of ∂ is an integral submodule. By Theorem 96.1, it contains every generator of the span in (490); hence it contains the span. If $c - d \in \mathcal{R}_5$, then linearity gives $\partial c - \partial d = \partial(c - d) = 0$. $\square$

This proof includes arbitrary finite sums, negatives, and integer multiples. In particular, the exact boundary is a consequence of a generation certificate and is not included among its premises.

96.2. The positive rational subrelation. For a positive reduced cell

$$P = (a, b, c), \quad a + b = c, \quad \gcd(a, b) = 1,$$

define its divisor symbol by

$$(491) \quad \sigma(P) = (d(a) - d(c), d(b) - d(c)).$$

Then

$$(492) \quad \partial[\sigma(P)] = (d(a) - d(c)) \wedge (d(b) - d(c)) = \Omega(P).$$

A positive realization of a five-term generator consists of positive reduced cells $P_1, \dots, P_5$ and divisor data X, U, Y, V, Z for which

$$(493) \quad \begin{array}{c|l} P_1 & (X, U) \\ P_2 & (Y, V) \\ P_3 & (Y - X, Z - X) \\ P_4 & (Y + U - X - V, Z - X - V) \\ P_5 & (U - V, Z - V) \end{array}$$

are the five individual identities for $\sigma(P_i)$. These local symbol identities contain no signed boundary assertion. Let $\mathcal{R}_5^+$ be the integral span of the signed chains

$$(494) \quad [\sigma(P_1)] - [\sigma(P_2)] + [\sigma(P_3)] - [\sigma(P_4)] + [\sigma(P_5)]$$

over all positive realizations.

Theorem 96.3 (Positive generated boundary). *There are inclusions*

$$\mathcal{R}_5^+ \subseteq \mathcal{R}_5 \subseteq \ker \partial.$$

Thus every equivalence generated by actual positive five-term moves preserves the finite-support surface.

Proof. The five identities in (493) identify each chain (494) with the algebraic generator (489). Therefore each generator of $\mathcal{R}_5^+$ belongs to $\mathcal{R}_5$, and the first inclusion follows by taking spans. The second inclusion is Theorem 96.2. $\square$

Suppose now that a target cell P_0 and cells P_j with coefficients $n_j \in \mathbb{Z}$ satisfy the generation statement

$$(495) \quad [\sigma(P_0)] - \sum_j n_j [\sigma(P_j)] \in \mathcal{R}_5^+.$$

Theorem 96.3 and (492) derive

$$(496) \quad \Omega(P_0) = \sum_j n_j \Omega(P_j).$$

Applying the exact finite-chain theorem from the preceding section gives

$$(497) \quad \Phi(P_0) \leq \frac{1}{2} \left(\sum_j |n_j| (M(P_j) + V(P_j)) + \sum_j |n_j| R(P_j) \right).$$

Unlike the earlier conditional structure, (495) stores only membership in the relation generated by actual moves. Equation (496) is a theorem.

96.3. An infinite arithmetic family of positive moves. Let $0 < b < a < c$ and impose the five reducedness conditions

$$(498) \quad \begin{aligned} \gcd(a, c-a) &= \gcd(b, c-b) = \gcd(b, a-b) = 1, \\ \gcd(b(c-a), c(a-b)) &= 1, \\ \gcd(c-a, a-b) &= 1. \end{aligned}$$

Consider the five cells

	left	middle	total
P_1	a	$c-a$	c
P_2	b	$c-b$	c
P_3	b	$a-b$	a
P_4	$b(c-a)$	$c(a-b)$	$a(c-b)$
P_5	$c-a$	$a-b$	$c-b$

Every entry is positive and the hypotheses (498) make every row reduced. The fourth row is a cell because

$$b(c-a) + c(a-b) = a(c-b).$$

Proposition 96.4 (Common-denominator realization). *The cells (499) positively realize the five-term move with*

$$x = \frac{a}{c}, \quad y = \frac{b}{c},$$

and arguments

$$(500) \quad \frac{a}{c}, \quad \frac{b}{c}, \quad \frac{b}{a}, \quad \frac{b(c-a)}{a(c-b)}, \quad \frac{c-a}{c-b}.$$

Proof. Put

$$\begin{aligned} X &= d(a) - d(c), & U &= d(c - a) - d(c), \\ Y &= d(b) - d(c), & V &= d(c - b) - d(c), \\ Z &= d(a - b) - d(c). \end{aligned}$$

The symbols of the first two cells are (X, U) and (Y, V) by definition. Direct subtraction gives the third and fifth rows of (493). For the fourth row, multiplicativity of divisors gives

$$\begin{aligned} d(b(c - a)) - d(a(c - b)) &= Y + U - X - V, \\ d(c(a - b)) - d(a(c - b)) &= Z - X - V. \end{aligned}$$

Thus all five symbol identities hold, and Theorem 96.3 supplies the surface relation. $\square$

For every odd $c \geq 3$, the choice $a = 2$, $b = 1$ satisfies (498). Indeed, the only nontrivial checks reduce to

$$\gcd(2, c - 2) = 1, \quad \gcd(c - 2, c) = \gcd(c - 2, 2) = 1.$$

This gives an infinite parameterized family of actual positive generators.

96.4. A full-premise counterexample to literal chain cancellation. The generated quotient is essential. Define the augmentation

$$\varepsilon : \mathcal{C}_0 \longrightarrow \mathbb{Z}, \quad \varepsilon \left(\sum_s n_s [s] \right) = \sum_s n_s.$$

For every input one has

$$(501) \quad \varepsilon(\mathfrak{r}(X, U, Y, V, Z)) = 1 - 1 + 1 - 1 + 1 = 1.$$

Proposition 96.5 (A generator is never the zero free chain). *For every $X, U, Y, V, Z \in \mathcal{D}$,*

$$\mathfrak{r}(X, U, Y, V, Z) \neq 0.$$

In particular, the boundary map ∂ is not injective.

Proof. If the free chain were zero, its augmentation would be zero, contradicting (501). Its boundary is nevertheless zero by Theorem 96.1, so ∂ has a nonzero kernel. $\square$

This refutes the exact shortcut asserting that the five signed basis chains cancel before quotienting. It does not refute the five-term route. The correct statement is the kernel inclusion of Theorem 96.2.

The failure persists with all positivity and reducedness premises. Taking

$$x = \frac{1}{2}, \quad y = \frac{1}{6}$$

gives the five distinct arguments

$$\frac{1}{2}, \quad \frac{1}{6}, \quad \frac{1}{3}, \quad \frac{1}{5}, \quad \frac{3}{5}$$

and reduced cell triples

$$(1, 1, 2), \quad (1, 5, 6), \quad (1, 2, 3), \quad (1, 4, 5), \quad (3, 2, 5).$$

Their signed surface boundary is zero, whereas their signed free chain is nonzero; its coefficient at the $1/2$ symbol is one.

96.5. Formal boundary and remaining analytic gate. The Lean companion module is `SteinbergFiveTermBoundaryBridge20260903`.

It formalizes Theorems 96.1, 96.2, and 96.3, the generated filling implication (497), Proposition 96.4, the odd-denominator family, and Proposition 96.5. It also formalizes the five-distinct positive example. Its axiom audit uses only `propxt`, `Classical.choice`, and `Quot.sound`.

The algebraic part of S-I4C is therefore closed: a chain generated by actual positive five-term moves has the required finite exterior boundary. The route remains open at the harder existence and analytic stages. For an arbitrary target cell, one must still construct a finite positive generated filling satisfying useful bounds on chain length, coefficients, prime support, and the two costs on the right of (497). More precisely, put

$$\mathcal{Q}(\Gamma) = \sum_j |n_j|(M(P_j) + V(P_j)), \quad \mathcal{R}(\Gamma) = \sum_j |n_j|R(P_j).$$

For every $\epsilon > 0$, Gate VF still requires constants K_ϵ, L_ϵ and a generated filling for each primitive target of height $H = \log c$ and radical logarithm $\rho = \log \text{rad}(abc)$ such that

$$\mathcal{R}(\Gamma) \leq \epsilon \mathcal{Q}(\Gamma) + K_\epsilon H, \quad \mathcal{Q}(\Gamma) \leq 2H\rho + L_\epsilon H.$$

Neither estimate nor such a uniform construction is proved here, and this section proves neither the abc conjecture nor its negation.

97. QUADRATIC VERONESE PEELING AND VALUATION-LAYER FLAGS

Author: ChatGPT. This section isolates the specialization $y = x^2$ of the five-term contact relation, transports it to primitive integer triples, and audits the resulting one-move policy on a full-premise Pythagorean-square family. The fixed policy fails three necessary uniform estimates. The unrestricted multi-move contact-complex route remains open.

For $z \in \mathbf{Q} \setminus \{0, 1\}$, let

$$\Omega(z) = d(z) \wedge d(1 - z), \quad d : \mathbf{Q}^\times \longrightarrow \bigoplus_p \mathbf{Z}[p].$$

Recall the five-term identity

$$(502) \quad \Omega(x) - \Omega(y) + \Omega(y/x) - \Omega\left(\frac{y(1-x)}{x(1-y)}\right) + \Omega\left(\frac{1-x}{1-y}\right) = 0.$$

Theorem 97.1 (Quadratic Veronese peeling). *The specialization $y = x^2$ is admissible in (502) exactly when $x \notin \{0, 1, -1\}$. On this domain,*

$$(503) \quad \Omega(x^2) = 2\Omega(x) - 2\Omega\left(\frac{x}{1+x}\right).$$

Proof. The five-term domain says that x, x^2 are nonzero, different from one, and different from each other. Since $x^2 - 1 = (x - 1)(x + 1)$ and $x^2 - x = x(x - 1)$, this is equivalent to the three stated exclusions. Direct field algebra then gives

$$\frac{x^2}{x} = x, \quad \frac{x^2(1-x)}{x(1-x^2)} = \frac{x}{1+x}, \quad \frac{1-x}{1-x^2} = \frac{1}{1+x}.$$

The final two arguments are complementary. Alternation gives $\Omega(1-z) = -\Omega(z)$, so substitution in (502) proves (503). No sign condition on x is required, since $d(-1) = 0$ for finite-prime divisors. $\square$

Writing $X = d(x)$, $U = d(1-x)$, and $W = d(1+x)$ exposes the coefficientwise identity behind the proof:

$$(504) \quad (2X) \wedge (U + W) = 2X \wedge U - 2(X - W) \wedge (-W).$$

97.1. Primitive integer transport. Let (a, b, c) be a positive pairwise-coprime triple with $a + b = c$ and put $x = a/c$.

Proposition 97.2 (Quadratic transform). *The triple*

$$T_2(a, b, c) = (a^2, b(a + c), c^2)$$

is positive, pairwise coprime, and satisfies $a^2 + b(a + c) = c^2$. Moreover,

$$(505) \quad \Omega(a^2, b(a + c), c^2) = 2\Omega(a, b, c) - 2\Omega(a, c, a + c).$$

Proof. The sum identity follows from $b = c - a$: $a^2 + (c - a)(a + c) = c^2$. A common prime of a^2 and $b(a + c)$ would divide a and either b or $a + c$, contradicting the original pairwise coprimality. The same argument applies to $b(a + c)$ and c^2 , while a^2 and c^2 are coprime directly. Equation (505) is Theorem 97.1 with $x = a/c$. $\square$

For $t \geq 1$, set

$$X = 2t + 1, \quad Y = 2t(t + 1), \quad Z = 2t^2 + 2t + 1.$$

Then

$$X^2 + Y^2 = Z^2, \quad Z - Y = 1, \quad Y + Z = X^2,$$

and X, Y, Z are pairwise coprime. Taking $x = Y/Z$ in (503) yields the full-premise identity

$$(506) \quad \boxed{\Omega(Y^2, X^2, Z^2) = 2\Omega(Y, 1, Z) - 2\Omega(Y, Z, X^2).}$$

Thus the first auxiliary cell has no nontrivial middle divisor, and only the third leg of the second auxiliary cell retains a visible square.

97.2. Exact positive-cost redistribution. Put

$$u = \log X, \quad v = \log Y, \quad w = \log Z,$$

and

$$\beta = \log \text{rad}(X), \quad \alpha = \log \text{rad}(Y), \quad \gamma = \log \text{rad}(Z).$$

Write $\delta_X = u - \beta$, $\delta_Y = v - \alpha$, and $\delta_Z = w - \gamma$. For the declared outer-square decomposition, the full, coherent, and residual costs of the squared cell are

$$(507) \quad \begin{aligned} \Phi_{\text{sq}} &= 4(uv + uw + vw), \\ V_{\text{sq}} &= \Phi_{\text{sq}}, \\ R_{\text{sq}} &= 2\{\delta_Y(u + w) + \delta_X(v + w) + \delta_Z(u + v)\}. \end{aligned}$$

After weighting both right-hand cells in (506) by the absolute coefficient two, their costs are

$$(508) \quad \begin{aligned} \Phi_{\text{peel}} &= 2vw + 2\{vw + 2u(v + w)\}, \\ V_{\text{peel}} &= 2u(v + w), \\ R_{\text{peel}} &= 4\delta_Y(u + w) + 2\delta_X(v + w) + 4\delta_Z(u + v). \end{aligned}$$

Proposition 97.3 (Conservation and redistribution). *One has*

$$(509) \quad \Phi_{\text{peel}} = \Phi_{\text{sq}},$$

$$(510) \quad V_{\text{peel}} < \frac{1}{2}V_{\text{sq}},$$

$$(511) \quad R_{\text{peel}} - R_{\text{sq}} = 2\delta_Y(u + w) + 2\delta_Z(u + v) \geq 0.$$

In addition, $V_{\text{peel}}/V_{\text{sq}} \rightarrow 1/4$ as $t \rightarrow \infty$.

Proof. Expansion of (508) proves (509). The difference between half the old coherent cost and the new coherent cost is $2vw > 0$. Direct subtraction proves (511). Finally $u \sim \log t$ and $v, w \sim 2 \log t$, which gives the stated limit. $\square$

For calibrated cost $Q = M + V = 2\Phi - R$, the same computation gives

$$(512) \quad Q_{\text{peel}} - Q_{\text{sq}} = -2\delta_Y(u + w) - 2\delta_Z(u + v).$$

The coherent improvement has therefore been transferred to residual cost; the total $Q + R = 2\Phi$ has not decreased.

97.3. The primewise valuation-layer flag. For $k \geq 1$, define

$$R_k(n) = \prod_{p: v_p(n) \geq k} p, \quad \lambda_k(n) = \log R_k(n).$$

Then

$$(513) \quad \log n = \sum_{k \geq 1} \lambda_k(n), \quad \log \text{rad}(n) = \lambda_1(n), \quad \log \frac{n}{\text{rad}(n)} = \sum_{k \geq 2} \lambda_k(n).$$

This filtration sees repeated prime mass even when the gcd of all exponents is one; $2s^2$ with s odd is the basic example. Bilinearity gives the exact layer-pair expansion

$$(514) \quad \Phi(a, b, c) = \sum_{k, l \geq 1} (\lambda_k(a)\lambda_l(b) + \lambda_k(b)\lambda_l(c) + \lambda_k(c)\lambda_l(a)).$$

Squaring duplicates every layer:

$$(515) \quad R_{2j-1}(n^2) = R_{2j}(n^2) = R_j(n).$$

Consequently the square-cell layer-pair area is replicated four times, and the layer ledger recovers the conservation law (509).

The flag also rules out a naive uniform depth cutoff. With $t = 2^m$, $v_2(Y^2) = 2m + 2$, so the prime 2 occupies arbitrarily high layers inside the genuine primitive cells (Y^2, X^2, Z^2) . Weighted layer decay and primewise matching across several five-term moves are not refuted by this example.

97.4. Three fixed-policy no-go theorems. The first no-go is immediate from (509). If some $\eta > 0$ and K satisfied

$$\Phi_{\text{peel}} \leq (1 - \eta)\Phi_{\text{sq}} + K(2w)$$

for all t , then $\eta\Phi_{\text{sq}} \leq 2Kw$. But $Z = Y + 1 \leq 2Y$ and $Y \geq 4$, so $w \leq 3v/2$ and

$$\frac{\Phi_{\text{sq}}}{2w} \geq \frac{4}{3}u \rightarrow \infty.$$

Thus no positive one-step area contraction exists.

For the calibrated-boundary coefficient, expansion gives

$$(516) \quad Q_{\text{peel}} = 2u(v + w) + 4\alpha(u + w) + 4\gamma(u + v) + 2\beta(v + w),$$

$$(517) \quad Q_{\text{peel}} - 2H\rho = 2u(v + w) + 4\alpha u + 4\gamma(u + v - w) + 2\beta(v - w),$$

where $H = 2w$ and $\rho = \alpha + \beta + \gamma$. Since $Y < Z \leq XY$, $0 \leq \beta \leq u$, and $\alpha, \gamma \geq 0$, the last display implies

$$(518) \quad Q_{\text{peel}} - 2H\rho \geq 4uv.$$

Dividing by H and using $w \leq 3v/2$ shows that the excess is an unbounded multiple of height. Hence the fixed chain cannot satisfy $Q_{\text{peel}} \leq 2H\rho + LH$ for any constant L .

Finally, the policy requiring the following bound for every $\varepsilon > 0$ is false:

$$R_{\text{peel}} \leq \varepsilon Q_{\text{peel}} + K_\varepsilon H$$

with a uniform K_ε . Set $t = 2^m$ and $L_m = \log t$. For $m \geq 2$,

$$L_m \leq u \leq 2L_m, \quad 2L_m \leq v, w \leq 3L_m, \quad \delta_Y \geq L_m.$$

Thus

$$R_{\text{peel}} \geq 12L_m^2, \quad \Phi_{\text{sq}} \leq 84L_m^2, \quad H \leq 6L_m.$$

As $Q_{\text{peel}} \leq 2\Phi_{\text{sq}}$, at $\varepsilon = 1/100$ one obtains

$$R_{\text{peel}} - \frac{1}{100}Q_{\text{peel}} \geq \frac{258}{25}L_m^2,$$

which cannot be absorbed into any uniform $K_{1/100}H$. Thus the single admissible value $\varepsilon = 1/100$ already refutes the universal-in- ε policy; no assertion is made here that every positive ε fails. This theorem concerns the declared outer-square split. It does not claim that the same subsequence refutes the maximal common-exponent residual.

All triples used above satisfy positivity, the sum equation, and pairwise coprimality. These counterexamples retire the exact one-move policies only. Gate VF permits arbitrary additional five-term moves and remains open. The *abc* conjecture also remains open.

The Lean companion `QuadraticVeronesePeeling20260902.lean` formalizes the exact domain, integer transport, valuation-layer identities, and all three quantified fixed-policy no-go theorems. Its separate axiom audit covers all 53 public theorem declarations. A deterministic computation checks 20,000 consecutive parameters and a power-of-two subsequence through 2^{18} .

98. INTEGRATED RESEARCH EDITION: PROVENANCE AND PROOF STATUS

This edition integrates six historical full mathematical proof bodies, two explicitly labelled proof consolidations (normal forms and rank/depth), and one new moving-support paper. It does not concatenate unrelated PDFs. The two consolidations retain the principal mathematical results with full proofs, but do not repeat unconfirmed historical runtime narratives or claim to be byte-identical reprints. Existing main-text mathematics is preserved; removing the labelled integration blocks recovers the hash-pinned master. References and local notation in the supplements have isolated namespaces.

The original sealed PDF at the baseline commit has 270 pages. Rebuilding the current baseline source with pdfLaTeX gives 281 pages before these additions: the source already contains material not present in that sealed artifact. Thus the difference from 270 is not a count of newly proved pages. The old PDF remains recoverable from the baseline Git commit.

Global status. Standard `abc` is neither proved nor disproved. The new theorem is uniform only on its explicitly stated growing-support class. None of the earlier sufficient gates is silently supplied. The complex logarithmic-form theorem is an established cited input; it is not proved or formalized by the present Lean modules.

Checkpoint	Starts on page
Flagged CRT obstructions and exact divisor gaps	259
Signed endpoint transport and exact losses	267
Unitary faces, safe multiple outputs, and valuation lifting	275
Signed normal forms and arithmetic Wronskians	283
Quantitative transverse integer lifting	286
Power-radical descent and first-depth obstructions	293
Eisenstein descent and strong boundary divisibility	302
First-appearance depth and the norm-seven orbit	311
Moving norm support and compensated cubic boundaries	315

The integration workflow separately compiles the exact currently included sources with warnings as errors and an axiom whitelist restricted to `propext`, `Classical.choice`, and `Quot.sound`. It checks the two Eisenstein modules (40 named queries), the rank/depth module (18), and the new moving-support module (12). The full norm-boundary gcd and general finite-list compensated identity are among the new checks. These are scoped algebraic certificates, not formalizations of the complete analytic theorem. Earlier CRT and normal-form drafts are not thereby certified, and no full-repository Lean build is claimed. Current compiler logs, source hashes and exact replay results are recorded separately.

The original IUT all-place comparison, compensated packet gate, catalogue sparsity and Pell/Mersenne distribution obligations remain open. Algebraic coverage does not resolve unit terminals or prime-norm endpoints. The new result controls the varying-generator angular constant in a specified regime, while retaining the unrestricted multiplicity budget as an open problem. Hosted Ubuntu execution is not access to the user's personal WSL; no external peer review or independent autonomous-agent audit is claimed.

INTEGRATED CHECKPOINT: FLAGGED CRT OBSTRUCTIONS AND EXACT DIVISOR GAPS

Full mathematical proof body of the historical checkpoint. Historical execution statements retain their original scope.

Research manuscript and reproducible supplement.

Ordinary proofs and exact computations; the Lean companion has not been compiled.

No unconditional proof or disproof of the *abc* conjecture is claimed.

99. SCOPE, PROVENANCE, AND ENDPOINT NOTATION

The source snapshot is

6118955d20b4edd32e577e06d1060f3945358dd9.

The September 4 endpoint, anchored-prefix, and independent-audit notes [92, 93, 94] are the immediate antecedents. We retain their mathematical FCRT-1 definition, rather than replacing its admissibility condition with a desired height inequality. In particular, a flag must have an actual nonempty proper prime-power face. Results below are proved here; priority or novelty beyond comparison with those notes is not asserted. Computational evidence is separated from universal statements.

Let a, b, c be positive integers, $a + b = c$, and $\gcd(a, b) = 1$. Put $R = \text{rad}(abc)$. For each $p \mid c$, let $e_p = v_p(c)$, and set

$$I = \{p : e_p \geq 2\}, \quad d_p = p^{e_p-1}, \quad x_p = \log d_p, \quad J = \{q : q \mid ab\}.$$

For $S \subseteq I$, $T \subseteq J$, write

$$D_S = \prod_{p \in S} d_p, \quad m_S = \prod_{p \in S} p^{e_p}, \quad Q_T = \prod_{q \in T} q,$$

$$a_T = \prod_{q \in T, q \mid a} q^{v_q(a)}, \quad b_T = \prod_{q \in T, q \mid b} q^{v_q(b)}.$$

Empty products are one. These are *whole-prime-power packets*: an exponent may not be split between faces. A block (S, T) has nonempty source and sink sets and is admissible and saturated when

$$(519) \quad m_S \mid a_T + b_T, \quad D_S \leq Q_T.$$

The selected blocks have pairwise disjoint source sets and pairwise disjoint sink sets. Let I_0, J_0 be the respective residual sets. A residual sink has zero or one owner in I_0 . A selected block may emit zero or one token to one $p \in I_0$, witnessed by

$$(520) \quad \emptyset \neq U \subsetneq T, \quad p^{e_p} \mid a_U + b_U.$$

The logarithmic credit is

$$\rho = \min\{\log Q_T - \log D_S, \log Q_U\}.$$

No credit is re-emitted. Several independent tokens may have the same target. These rules are precisely the one-hop, capped FCRT-1 rules of the snapshot.

Let $X = \sum_{p \in I} x_p$, $Y = \log Q_J$, $h = \log c$, $r = \log R$. Unique factorization and pairwise coprimality give

$$(521) \quad \prod_{p \in I} d_p = \frac{c}{\text{rad}(c)}, \quad X - Y = h - r, \quad \Delta = \max\{h - r, 0\}.$$

We reserve $B_{\text{FCRT}}, B_{\text{SCRT}}, B_{\text{PBT}}$ for *logarithmic* optimized boundaries. Their exponentials are called boundary factors.

100. A CONCRETE MULTIPLICATIVE CERTIFICATE

This section supplies a mathematical arithmetic-to-accounting construction. It is not a claim that the repository's Lean owner types have now been inhabited or connected by compiled code.

For a used block ν , define the rational number

$$(522) \quad f_\nu = \min\{Q_{T_\nu}/D_{S_\nu}, Q_{U_\nu}\};$$

for an unused block put $f_\nu = 1$. Saturation implies $f_\nu \geq 1$. For a residual source p , let

$$K_p = \prod_{q \in J_0: o(q)=p} q \prod_{\nu: t(\nu)=p} f_\nu, \quad F = \prod_{p \in I_0} \max\{d_p/K_p, 1\}.$$

Everything in this definition uses positive integers, rational arithmetic, finite sets, and divisibility. Its logarithm is exactly the original clipped FCRT boundary.

Theorem 100.1 (Arithmetic once-charge certificate). *Every configuration satisfying the actual disjointness, saturation, owner, and face conditions above satisfies*

$$(523) \quad c \leq RF.$$

Consequently $\Delta \leq \log F$. The construction does not assume this inequality as a field of its input.

Proof. For every residual source, $d_p \leq K_p \max\{d_p/K_p, 1\}$. Multiplication, disjointness, and at-most-once ownership imply

$$\begin{aligned} \prod_{p \in I} d_p &\leq \left(\prod_{\nu} D_{S_\nu} \right) \left(\prod_{p \in I_0} K_p \right) F \\ &\leq \left(\prod_{\nu} D_{S_\nu} f_\nu \right) Q_{J_0} F \leq \left(\prod_{\nu} Q_{T_\nu} \right) Q_{J_0} F = Q_J F. \end{aligned}$$

Unused sinks only increase the middle upper bound because their primes are at least two. Substitute (521) and multiply by $\text{rad}(c)$. Since $F \geq 1$, taking logarithms also bounds the positive part. $\square$

Remark 100.2 (What the certificate does and does not use). Congruence and witness-face conditions make the configuration arithmetically admissible. The inequality itself uses disjointness and the surplus cap. This distinction is essential: dropping arithmetic conditions preserves a valid inequality but may destroy all additional arithmetic information. The construction maps source weights to $\log d_p$, block weights to $\log D_S, \log Q_T$, witness weights to the *actual* $\log Q_U$, and owners to the given partial maps. The finite partitions explicitly establish the source and sink mass decompositions required by the earlier independent audit [94].

101. OPTIMAL FACES AS MINIMUM-PRODUCT DELETIONS

Fix a source p , and put $M = p^{e_p} \geq 4$. All primes in J are units modulo M . Define

$$g_q = \begin{cases} q^{v_q(a)} & q \mid a, \\ q^{-v_q(b)} & q \mid b \end{cases} \quad \text{in } (\mathbb{Z}/M\mathbb{Z})^\times, \quad \Phi(T) = \prod_{q \in T} g_q.$$

Then $\Phi(U) = -1$ is equivalent to $M \mid a_U + b_U$, and $\Phi(J) = -1$. For any nonempty $T \subseteq J$, define

$$(524) \quad z_p(T) = \min\{Q_D : \emptyset \neq D \subseteq T, \Phi(D) = -\Phi(T)\},$$

with value $+\infty$ when the set is empty.

Theorem 101.1 (Exact optimal-face normal form). *For a fixed saturated block (S, T) and fixed residual target p , a proper flag exists if and only if $z_p(T) < \infty$. If one exists, the maximum credit factor over its legal witnesses is*

$$(525) \quad f_{p,S,T}^* = \frac{Q_T}{\max\{D_S, z_p(T)\}}.$$

In particular its discarded logarithmic surplus is

$$(526) \quad \log \frac{Q_T}{D_S} - \log f_{p,S,T}^* = \max\{\log z_p(T) - \log D_S, 0\}.$$

Proof. For complementary subsets U, D in T , multiplication in the unit group gives $\Phi(U) = -1$ if and only if $\Phi(D) = -\Phi(T)$. The nonempty condition on D says $U \neq T$. Also $D = T$ cannot satisfy its displayed label: it would imply $\Phi(T) = -\Phi(T)$, hence $M \mid 2$, contrary to $M \geq 4$. Thus U is automatically nonempty. Since $Q_U = Q_T/Q_D$, maximizing the face weight is equivalent to minimizing Q_D . Monotonicity of the minimum in (522) gives (525); cancellation gives (526). $\square$

When $\Phi(T) = -1$, the deletion label in (524) is one. For a general block it is *not* necessarily one. This distinction prevents an incorrect application of an anchored identity-deletion result to an arbitrary cross-source block. Optimizing each fixed target independently is valid: increasing its incoming credit never increases the clipped boundary.

101.1. Exact finite computation. Process the distinct primes $q_1, \dots, q_n$ of T . A table indexed by reachable residues stores the least product of a *nonempty* subset and a witness. Initially it is empty. At stage k , retain all previous entries, insert the singleton (g_{q_k}, q_k) , and, using only the previous table, insert

$$(rg_{q_k}, wq_k) \quad \text{from every previous entry } (r, w).$$

At each residue retain the smaller product. The empty subset is not allowed to overwrite a nonempty identity entry.

Proposition 101.2 (Dynamic-program invariant). *After stage k , each present table entry is the minimum product among nonempty subsets of $\{q_1, \dots, q_k\}$ with that residue; an absent entry means no such subset exists. Thus the entry at $-\Phi(T)$ computes (524) exactly.*

Proof. A nonempty subset either omits q_k , is the singleton $\{q_k\}$, or is $\{q_k\}$ together with a nonempty earlier subset. These are exactly the three transitions. Multiplication by the positive integer q_k preserves order, so discarding a larger product with the same earlier residue cannot remove a later optimum. Induction proves the assertion. $\square$

The sparse table has at most the size of the subgroup generated by the labels. Arithmetic-operation complexity is $O(n|H|)$ in a dense implementation; this excludes bit-complexity and makes no claim of polynomial time in the bit length of M . No numerical logarithm is used.

101.2. An arithmetic cost obstruction.

Proposition 101.3. *Suppose $\Phi(T) = -1$, and let D be a nonempty identity deletion. Write $E = \max_{q \in T} v_q(ab)$ and $V_T = a_T b_T / Q_T$. Then*

$$Q_D^E \geq M + 1, \quad Q_D V_T \geq M + 1.$$

Consequently, full-surplus reuse at a block requires both $D_S^E \geq M + 1$ and $D_S V_T \geq M + 1$.

Proof. The positive coprime integers a_D, b_D are congruent modulo M . They are unequal: equality and coprimality would make both one, contrary to a nonempty packet. Hence $|a_D - b_D| \geq M$ and $a_D b_D \geq \max\{a_D, b_D\} \geq M + 1$. Furthermore $a_D b_D \leq Q_D^E$, and $a_D b_D = Q_D V_D \leq Q_D V_T$. Full reuse means $Q_D \leq D_S$ for some deletion, by Theorem 101.1. Substitute this inequality. $\square$

102. A SHARP OBSTRUCTION FOR UNIT-GAP ENDPOINTS

The next result is stronger than a weight obstruction: in its range the required face does not exist at any weight.

Lemma 102.1 (Bilinear factor gap). *Let $M \geq 2$, $s, t \geq 1$ be integers, and suppose*

$$C + t = s(Mt + 1), \quad C \neq M.$$

Then $C \geq 2M - 1$.

Proof. If $s = t = 1$, the equation gives $C = M$, excluded by hypothesis. If $t \geq 2$, then

$$C = s(Mt + 1) - t \geq (M - 1)t + 1 \geq 2M - 1.$$

If $t = 1$, necessarily $s \geq 2$, and $C = s(M + 1) - 1 \geq 2M + 1$. $\square$

Theorem 102.2 (Prime-power no-face threshold). *Let c be a positive integer and $p^e \parallel c$, with $e \geq 2$. Put $M = p^e$. If a nonempty proper prime-power packet U of $c - 1$ satisfies $M \mid 1 + b_U$, then*

$$(527) \quad c \geq M(2M - 1).$$

In fact the conclusion holds for any factorization $c - 1 = du$, $d, u > 1$, with $u \equiv -1 \pmod{M}$.

Proof. Write $c = MC$; exactness of $p^e \parallel c$ implies $p \nmid C$, so $C \neq M$. The complementary packet gives $c - 1 = du$ with $d, u > 1$. Since $du \equiv -1 \pmod{M}$ and $u \equiv -1 \pmod{M}$, we have $d \equiv 1 \pmod{M}$. Thus

$$d = Mt + 1, \quad u = Ms - 1 \quad (s, t \geq 1).$$

Expanding $(Mt + 1)(Ms - 1) = MC - 1$ and dividing by M gives $C + t = s(Mt + 1)$. Lemma 102.1 yields (527). $\square$

The exact prime-power assumption is significant. Without it, the case $C = M$ is possible; the weaker threshold $c \geq M^2$ is all the same argument yields. The proof does not rely on estimating the number of prime factors or on random residue heuristics.

Proposition 102.3 (Sharpness on actual prime-power packets). *For $M = 2^{2k+1}$, $k \geq 1$, put $c = M(2M - 1)$. Then $v_2(c) = 2k + 1$ and a proper packet modulo the full 2-power exists.*

Proof. We have $c - 1 = (M - 1)(2M + 1)$. Since $M \equiv 2 \pmod{3}$, $\gcd(M - 1, 2M + 1) = \gcd(M - 1, 3) = 1$. Therefore the two factors are unions of disjoint whole-prime-power packets. Both exceed one, and $M - 1 \equiv -1$, $2M + 1 \equiv 1 \pmod{M}$. The factor $2M - 1$ is odd, proving the valuation assertion. $\square$

Corollary 102.4 (Balanced two-source endpoints). *Suppose $c = 2^e 3^f$, $e, f \geq 2$, and*

$$(528) \quad \max\{2^e, 3^f\} < 2 \min\{2^e, 3^f\} - 1.$$

For $(1, c - 1, c)$, no nonempty proper packet of the full sink set is compatible with either source. In particular, no FCRT-1 configuration at this endpoint contains a used flag, regardless of its selected blocks.

Proof. For either source modulus M , the complementary factor $C = c/M$ is less than $2M - 1$, by (528). Apply Theorem 102.2. A proper face of any block is also a proper face of the full sink set, so choosing a smaller block cannot evade the result. $\square$

103. INFINITELY MANY POSITIVE-DEFECT ENDPOINTS WITHOUT FLAGS

Theorem 103.1. *There are infinitely many distinct primitive triples $(1, c - 1, c)$ such that $c = 2^e 3^f$, (528) holds, $49 \mid c - 1$, and*

$$(529) \quad \text{rad}(c(c - 1)) \leq \frac{6(c - 1)}{7} < c.$$

All their proper FCRT flags are absent.

Proof. The number $\alpha = 2 \log 3 / \log 2$ is irrational. Otherwise an equality between a positive power of 2 and a positive power of 3 would contradict unique factorization. The elementary pigeonhole approximation argument produces positive integers u_k, v_k , with $v_k \rightarrow \infty$, such that $u_k - \alpha v_k \rightarrow 0$. For completeness, place the $Q + 1$ fractional parts of $0, \alpha, \dots, Q\alpha$ into Q equal intervals. A collision gives $1 \leq v \leq Q$ and an integer u with $|v\alpha - u| \leq 1/Q$. These denominators cannot remain bounded as $Q \rightarrow \infty$, since irrationality gives a positive minimum error on any fixed finite set. Pass to a subsequence with unbounded denominators; its numerators are eventually positive.

Set $e_k = 21u_k$, $f_k = 42v_k$. Then

$$\log \frac{2^{e_k}}{3^{f_k}} = 21 \log 2 (u_k - \alpha v_k) \longrightarrow 0.$$

Both factors tend to infinity. Their larger-to-smaller ratio tends to one, so (528) holds eventually. The elementary congruences $2^{21} \equiv 1 \pmod{49}$ and $3^{42} \equiv 1 \pmod{49}$ imply $49 \mid 2^{e_k} 3^{f_k} - 1$.

A number divisible by 7^2 has radical at most one seventh of itself. Also $\text{rad}(c) = 6$ and $\gcd(c, c-1) = 1$. This proves (529). Select an increasing subsequence of c to ensure distinctness, and invoke Corollary 102.4. $\square$

Corollary 103.2 (Unbounded absolute defect without flags). *For every integer $k \geq 2$, there are infinitely many balanced two-source endpoints without flags such that $7^k \mid c-1$ and*

$$\frac{c}{\text{rad}(c(c-1))} > \frac{7^{k-1}}{6}.$$

Consequently the scalar defect Δ is unbounded on the no-flag stratum. For every fixed threshold there are infinitely many no-flag endpoints with defect above that threshold.

Proof. The congruences

$$2^{3 \cdot 7^{k-1}} \equiv 1 \pmod{7^k}, \quad 3^{6 \cdot 7^{k-1}} \equiv 1 \pmod{7^k}$$

follow by induction from $2^3 \equiv 3^6 \equiv 1 \pmod{7}$: raising a number congruent to one modulo 7^j to the seventh power makes it congruent to one modulo 7^{j+1} , as the binomial expansion shows. In the proof of Theorem 103.1, replace the constants 21, 42 by $3 \cdot 7^{k-1}$, $6 \cdot 7^{k-1}$, keeping k fixed during the approximation argument. The same ratio tends to one, and $\text{rad}(c-1) \leq (c-1)/7^{k-1}$. Now let k increase to exceed any prescribed absolute defect threshold. There is still no estimate of this defect as a positive proportion of $\log R$. $\square$

Remark 103.3 (Exact statement refuted). Theorem 103.1 refutes the following proposed shortcut, including its “all sufficiently large” variant: every positive-defect two-source unit-gap endpoint has at least one proper compatible face. Corollary 103.2 also rules out rescue by imposing only a sufficiently large absolute defect. It also prevents a global application of any sufficient condition that would force such a face at *every* endpoint in this stratum. It does not contradict the anchored-prefix theorem [93]; that theorem has additional hypotheses, which cannot all hold here. No original FCRT uniform gate is refuted. Nor does a lower bound $c/R > 7/6$ refute *abc*: it does not establish that $c/R^{1+\varepsilon}$ is unbounded for any fixed $\varepsilon > 0$.

104. THE REMAINING FINITE PROBLEM IS EXACTLY A DIVISOR GAP

Let the hypotheses of Corollary 102.4 hold. Write

$$Q = \text{rad}(c-1), \quad A = 2^{e-1}, \quad B = 3^{f-1}, \quad D = \log(AB/Q).$$

In this section assume $AB > Q$, so $D = \Delta > 0$.

Theorem 104.1 (Exact reduction on the no-flag stratum). *At these endpoints,*

$$(530) \quad B_{\text{FCRT}} = B_{\text{SCRT}} = B_{\text{PBT}} = \log \min_{H \mid Q} (\max\{A/H, 1\} \max\{BH/Q, 1\}).$$

Equivalently, for the nonempty real interval $\mathcal{I} = [\log(Q/B), \log A]$,

$$(531) \quad B_{\text{FCRT}} = D + \min_{H \mid Q} \text{dist}(\log H, \mathcal{I}).$$

In particular, the scalar lower bound is attained if and only if there is a divisor $H \mid Q$ with $Q/B \leq H \leq A$.

Proof. Every compatible nonempty block has the full sink set: a proper one would contradict Corollary 102.4. Disjointness permits at most one such block. The block with both sources is not saturated, because $AB > Q$. A saturated singleton-source block with all sinks has exactly the same boundary as assigning every sink exclusively to that source. Thus it cannot improve upon the exclusive partition optimum. Flags do not exist, so all three optima coincide.

An unused residual sink can be assigned without increasing the boundary. Since Q is squarefree, assigning all sinks between the two sources is the same as choosing a divisor $H \mid Q$; the capacities are $H, Q/H$. This proves (530). To obtain (531), put $t = \log H$, $L = \log(Q/B)$, $U = \log A$. Since $L < U$, direct consideration of $t < L$, $L \leq t \leq U$, and $t > U$ gives

$$(\log A - t)_+ + (\log B - \log Q + t)_+ = D + \text{dist}(t, [L, U]).$$

Minimize over the finite divisor set. □

104.1. A dense-divisor upper bound, not an abc estimate. The maximum ratio of consecutive divisors is a classical object in the study of dense divisors [95]. We include the elementary squarefree calculation to make the precise inequality self-contained. Write $Q = q_1 \cdots q_n$, with distinct increasing primes, and define

$$G(Q) = \max_{1 \leq j \leq n} \frac{q_j}{q_1 \cdots q_{j-1}}.$$

The empty denominator is one, so $G(Q) \geq 2$.

Lemma 104.2. *The largest ratio of consecutive positive divisors of squarefree Q is exactly $G(Q)$.*

Proof. Adjoin the primes in increasing order. In logarithmic coordinates, the new divisor set is the union of the old set and its translate by $\log q_j$. If their convex hulls are disjoint, the new bridge gap is $\log(q_j/(q_1 \cdots q_{j-1}))$; all other gaps are old ones. If the hulls overlap, a gap in the union cannot cross either hull endpoint in its interior, since those endpoints are themselves divisor logarithms. It therefore lies in one hull and between two consecutive points of that translated or untranslated set. Thus no gap exceeds the previous maximum. This gives the upper bound by induction. For the reverse bound, whenever $q_j > q_1 \cdots q_{j-1}$, a divisor smaller than q_j uses only earlier primes and is at most their full product. Hence that product and q_j are consecutive divisors of the full Q . The case $j = 1$ supplies the initial gap from one. □

Corollary 104.3. *On the positive-defect no-flag stratum,*

$$D \leq B_{\text{FCRT}} \leq D + \frac{1}{2} \max\{\log G(Q) - D, 0\}.$$

If $\log G(Q) \leq D$, the scalar optimum is attained.

Proof. The interval $\mathcal{I}$ has length D and intersects $[0, \log Q]$, since $A, B \geq 1$. If it contains a divisor logarithm, the penalty is zero. Otherwise it lies strictly inside one consecutive-divisor gap, of length $g \leq \log G(Q)$. The nearer endpoint is at distance at most $(g - D)/2$ from the interval. Apply Theorem 104.1. □

There is no proved uniform bound here for either D or the endpoint's divisor-gap penalty. A theorem about the density of integers with dense divisors does not automatically apply pointwise to the special sequence $\text{rad}(2^e 3^f - 1)$. The exact reduction changes the finite optimization problem; it does not eliminate the Diophantine difficulty.

105. AN EXACT 25-DIGIT ENDPOINT AND COMPUTATIONAL REPLAY

Take

$$\begin{aligned} e &= 41, & f &= 26, & M &= 2199023255552, & N &= 2541865828329, \\ & & & & c &= 5589622068988418728132608. \end{aligned}$$

Direct integer arithmetic gives

$$\begin{aligned} c - 1 &= 7^2 \cdot 439 \cdot 857 \cdot 2729 \cdot 292183 \cdot 380261663, \\ Q &= 7 \cdot 439 \cdot 857 \cdot 2729 \cdot 292183 \cdot 380261663. \end{aligned}$$

All six displayed factors are prime; the companion program checks this by deterministic trial division. The two no-face inequalities are

$$N < 2M - 1, \quad M < 2N - 1.$$

Consequently neither source has a proper flag, while $R = 6Q = 6(c - 1)/7 < c$. Nevertheless the divisor

$$H = 2729 \cdot 380261663 = 1037734078327$$

attains the scalar optimum. Indeed

$$H \leq A = 1099511627776, \quad Q/H = 769481753663 \leq B = 847288609443.$$

These are exact integer comparisons. Thus

$$\exp B_{\text{FCRT}} = \frac{AB}{Q} = \frac{931603678164736454688768}{798517438426916961161801} = \frac{c}{R}.$$

The absence of flags therefore does *not* imply failure of the full uniform-boundary strategy. It identifies a stratum requiring a different mechanism, here an ordinary sink partition.

105.1. Reproducibility protocol. The standard-library Python companion performs: independent dynamic-program versus exhaustive-subset comparisons; bounded direct checks of the no-face threshold; exact rational FCRT optimization for the repository’s finite witnesses; deterministic primality and factorization checks for the endpoint above; and arithmetic checks on a large symbolic member with exponents (399, 252). In this replay there were 12,231 ordered primitive triples with $c \leq 200$, 26,832 dynamic-program/exhaustive-table comparisons, 5,082 arithmetic-cost checks, and 3,700 concrete owner-bridge checks. The separate unit-gap scan through $c = 200000$ checked 6,741 eligible endpoint–modulus pairs and 35,756 proper unitary packets, with no counterexample. The dense-divisor identity was checked on 12,159 squarefree values through 20,000; the bilinear factor lemma was checked on 297,381 nonexceptional bounded parameter triples. Its JSON output records exact bounds, factors, and selected configurations. All assertions are executable checks rather than floating-point estimates. The results and invocation are in the accompanying verification report. Bounded tests do not prove a uniform theorem, and a Python replay is not a Lean kernel proof.

106. INDEPENDENT ROUTE AND SOURCE AUDIT

106.1. IUT: retain the route without assuming its comparison. The original IUT III discusses the three indeterminacies, log-Kummer compatibility, and log-volume comparison in the proof of Corollary 3.12 [96]. Inspection of those passages does not supply the missing source-faithful global construction in the repository. Its own audit still distinguishes local accounting interfaces from actual arithmetic data. Joshi’s report [97] takes a distinct position: it criticizes both sides of the controversy and claims that his arithmetic Teichmüller spaces provide additional structures. These are attributed claims, not results validated by this manuscript. We neither import the disputed comparison as an axiom nor declare the entire IUT route false on the basis of a simplified model. No IUT bridge is closed here.

106.2. Anchored selection and recent literature. The anchored code-fibre lemma in Bae’s revised manuscript [98] is an elementary sufficient selection principle. Its proof selects two indices in the same fibre with a permitted positive difference. It does not establish the arithmetic reservoir hypotheses at every *abc* endpoint. Theorem 103.1 exhibits infinitely many endpoints where any sufficient condition implying a proper flag must fail.

Lichtman’s discussion of the almost-all *abc* result [99] concerns quantitative exceptional-set bounds; such a bound does not itself exclude infinitely many exceptions. Dujella’s prime-power tuple theorem [100] controls a different Diophantine configuration. No uniform map from the endpoint packets above into that theorem’s hypotheses has been constructed here. These sources are relevant background, not an imported solution to the uniform gate.

106.3. Other maintained routes. The repository’s Pell, Mersenne/Wieferich, synchronized-packet, Steinberg, and Frey/Szpiro ledgers identify separate unresolved valuation or uniformity obligations. This manuscript does not settle those obligations and does not retire those routes. In particular, fixed-parameter finiteness, polynomial function-field analogies, or identities for radical defects cannot be promoted to a constant uniform in all integer triples without another

proof. The present ordinary results use neither GRH nor abc , IUT, a conjectural prime-producing polynomial, or a conjecture about Wieferich primes.

107. FORMALIZATION BOUNDARY AND REMAINING UNIFORM TARGET

The mathematical proofs above precede the Lean companion. That companion contains candidate proof scripts for the bilinear arithmetic obstruction, the proper-divisor implication, the real interval identity, and concrete integer certificates. It has *not* been compiled in this execution environment. Accordingly there is no claimed Lean declaration count, no claimed successful axiom audit, and no claimed unconditional closed term of type **ABCConjecture**. The infinite-family approximation argument, the full arithmetic owner construction, the dynamic-program invariant, and the complete optimization theorems are not fully formalized by that file.

The pinned repository toolchain is `leanprover/lean4:v4.32.0`. Official release notes for 4.32.1 and 4.32.2 describe kernel soundness fixes [101, 102]; the latter also warns that the affected checking path included comparator. This is not evidence that the repository exploited a bug or that any particular theorem is false. It does mean that the eventual trusted replay should rebuild compatible dependencies on a patched toolchain, run a fresh-process check, and, where appropriate, an updated independent checker. Merely printing a familiar list of axioms is not a substitute for that check.

For the original target, a sufficient FCRT statement remains

$$(532) \quad \forall \varepsilon > 0 \exists C_\varepsilon \in \mathbb{R} \forall (a, b, c), \quad B_{\text{FCRT}}(a, b, c) \leq \varepsilon \log \text{rad}(abc) + C_\varepsilon.$$

Together with Theorem 100.1, it gives the usual $c \leq K_\varepsilon \text{rad}(abc)^{1+\varepsilon}$. Neither (532) nor its negation is established. On the positive-defect no-flag stratum, Theorem 104.1 says precisely that the boundary is the sum of two nonnegative terms:

$$\log \frac{c}{\text{rad}(c(c-1))} \quad \text{and} \quad \min_{H|\text{rad}(c-1)} \text{dist}(\log H, [\log(Q/B), \log A]).$$

A uniform sublinear bound for their sum is equivalent to such bounds for both terms separately: necessity follows from nonnegativity, and sufficiency uses $\varepsilon/2$ in each bound. Thus further work must control arithmetic radical defect as well as any transport fragmentation. Creating a formal hypothesis with those bounds would not constitute proving them.

Repository integration status. This supplement is based on the pinned commit and is provided as an additive patch. No remote branch, commit, pull request, or merge was created in this session. Existing build imports and verified-status ledgers are deliberately unchanged. A local research result, a locally replayed computation, a compiled Lean proof, and a merged remote theorem are four different deliverables.

INTEGRATED CHECKPOINT: SIGNED ENDPOINT TRANSPORT AND EXACT LOSSES

Full mathematical proof body of the historical checkpoint. Historical execution statements retain their original scope.

Ordinary proofs and exact reproducible computations.

The partial Lean companion is uncompiled.

No unconditional proof or disproof of the standard *abc* conjecture is claimed.

108. THE TARGET AND THE PRECISE CONTINUATION

For positive coprime integers a, b , write $c = a + b$, normalize $a \leq b$, and put

$$R = \text{rad}(abc), \quad \Delta = \max\{\log c - \log R, 0\}.$$

The target is

$$(533) \quad \forall \varepsilon > 0 \exists K_\varepsilon > 0 \forall (a, b, c), \quad c \leq K_\varepsilon R^{1+\varepsilon}.$$

The repository's FCRT-1 boundary is a sufficient auxiliary target, not an already proved estimate. The first September 5 supplement proved sharp obstructions to proper flags and reduced balanced two-source sum endpoints to a divisor-partition problem [103]. The present paper changes the endpoint rather than silently relaxing the face condition. It does not modify the meaning of the original boundary.

The inspected remote branch remains at

$$6118955d20b4edd32e577e06d1060f3945358dd9.$$

The route ledger [104] retains the IUT comparison, canonical defect, compensated packet, first-aparition valuation, and geometric uniformity obligations. Our arguments neither assume these obligations nor close them. There was no web literature search in this round, and no claim of priority beyond the inspected project sources is made.

109. SIGNED CONFIGURATIONS AND AN EXACT LOSS IDENTITY

Fix $\sigma \in \{+1, -1\}$, positive coprime x, y , and a positive target

$$n = x + \sigma y.$$

For $\sigma = -1$ we require $x > y$. These three integers are pairwise coprime. Set

$$I = \{p : v_p(n) \geq 2\}, \quad d_p = p^{v_p(n)-1}, \quad J = \{q : q \mid xy\}, \quad R = \text{rad}(xyn).$$

For source and sink subsets define

$$D_S = \prod_{p \in S} d_p, \quad M_S = \prod_{p \in S} p^{v_p(n)}, \quad Q_T = \prod_{q \in T} q, \\ x_T = \prod_{q \in T} q^{v_q(x)}, \quad y_T = \prod_{q \in T} q^{v_q(y)}.$$

Empty products are one. The exponent of a selected prime is never split.

Definition 109.1 (Signed FCRT configuration). A block (S, T) has nonempty source and sink sets and satisfies

$$(534) \quad M_S \mid x_T + \sigma y_T, \quad D_S \leq Q_T.$$

Blocks are source-disjoint and sink-disjoint. Each residual sink has at most one residual source owner. A block may emit at most one one-hop token to a residual source p , witnessed by

$$(535) \quad \emptyset \neq U \subsetneq T, \quad p^{v_p(n)} \mid x_U + \sigma y_U.$$

The token factor is $f = \min\{Q_T/D_S, Q_U\}$. An unused token has factor one. Tokens are not re-emitted.

For a residual source p , let K_p be the product of its owned sink primes and incoming token factors. Define the configuration's boundary factor

$$F = \prod_{p \text{ residual}} \max\{d_p/K_p, 1\}.$$

This is a positive rational number. The finite minimum over legal configurations is denoted $\mathcal{F}_\sigma(x, y; n)$; its logarithm is $\mathcal{B}_\sigma$. Finiteness follows from the finite sets of blocks, owners, targets, and faces.

Theorem 109.2 (Exact arithmetic loss identity). *For any signed configuration put*

$$U_0 = \prod_{q \text{ residual and unowned}} q, \quad S_0 = \prod_{\nu} \frac{Q_{T_\nu}}{D_{S_\nu} f_\nu}, \quad O_0 = \prod_{p \text{ residual}} \max\{K_p/d_p, 1\}.$$

Then $U_0, S_0, O_0 \geq 1$, and

$$(536) \quad \boxed{F = \frac{n}{R} U_0 S_0 O_0.}$$

Consequently $n \leq RF$, and $\log F \geq \max\{\log(n/R), 0\}$.

Proof. For every positive d, K ,

$$\max\{d/K, 1\} = (d/K) \max\{K/d, 1\}.$$

The product of all demands is $n/\text{rad}(n)$. Disjointness gives

$$\prod_{p \text{ residual}} d_p = \frac{n/\text{rad}(n)}{\prod_{\nu} D_{S_\nu}}.$$

The product of all residual credits is the product of the owned residual sink primes times $\prod_{\nu} f_\nu$. Thus

$$\frac{\prod_{p \text{ residual}} d_p}{\prod_{p \text{ residual}} K_p} = \frac{n}{\text{rad}(n) Q_J} U_0 \prod_{\nu} \frac{Q_{T_\nu}}{D_{S_\nu} f_\nu}.$$

Since $\text{rad}(n) Q_J = R$, multiplication by O_0 proves the identity. Saturation and the surplus cap give $f_\nu \leq Q_{T_\nu}/D_{S_\nu}$; all three loss factors are therefore at least one. Finally $F \geq 1$ by definition. $\square$

The proof uses actual prime factorizations and actual owners. The congruences restrict which configurations may enter the minimum, although the accounting equality itself uses the partition and cap conditions. This is not a wrapper whose input already assumes an abc -strength bound.

Let $L = \log U_0 + \log S_0 + \log O_0$ and let L_σ^* be its minimum. The theorem gives, for either sign,

$$(537) \quad \mathcal{B}_\sigma = \log(n/R) + L_\sigma^*.$$

This separates the scalar arithmetic defect from configuration loss exactly, not merely by an upper estimate.

110. CHANGING THE ENDPOINT WITHOUT CHANGING THE RADICAL

For a normalized primitive triple define

$$\mathcal{F}_c = \mathcal{F}_+(a, b; c), \quad \mathcal{F}_b = \mathcal{F}_-(c, a; b).$$

Both use precisely the radical $R = \text{rad}(abc)$. The second orientation is new relative to simply optimizing the first one.

Definition 110.1 (Two-endpoint selector).

$$\mathcal{F}_{\text{sel}} = \min \left\{ \mathcal{F}_c, \frac{c}{b} \mathcal{F}_b \right\}, \quad E = \log \mathcal{F}_{\text{sel}}.$$

Theorem 110.2 (Adaptive height bridge and exact loss). *The selector satisfies*

$$(538) \quad c \leq R\mathcal{F}_{\text{sel}}, \quad \Delta \leq E \leq \mathcal{B}_+(a, b; c).$$

If L_c^*, L_b^* denote the respective minimum logarithmic losses, then

$$(539) \quad E = \log(c/R) + \min\{L_c^*, L_b^*\}.$$

In particular, when $c > R$, this is $\Delta + \min\{L_c^*, L_b^*\}$.

Proof. Theorem 109.2 gives $c \leq R\mathcal{F}_c$ and $b \leq R\mathcal{F}_b$. Multiplying the second inequality by c/b proves the first statement. Both candidate factors are at least one because $c/b \geq 1$. This proves the lower bound; choosing the first candidate proves the upper bound. For the exact formula, substitute (537) into

$$E = \min\{\mathcal{B}_+, \log(c/b) + \mathcal{B}_-\}.$$

The scalar terms are equal because $\log(c/b) + \log(b/R) = \log(c/R)$. $\square$

Thus a uniform estimate

$$(540) \quad \forall \varepsilon > 0 \exists C_\varepsilon \in \mathbb{R} \forall (a, b, c), \quad E \leq \varepsilon \log R + C_\varepsilon$$

implies (533), with $K_\varepsilon = \exp C_\varepsilon$. This implication does not supply (540). In the positive-defect region, the outstanding upper bound has two nonnegative components in (539).

Proposition 110.3 (Easy signed strata). *For either sign, if $n \leq R$, then $\mathcal{F}_\sigma = 1$. If the target has at most one prime of exponent at least two, then*

$$\mathcal{F}_\sigma = \max\{n/R, 1\}.$$

Proof. If there are sources and $n \leq R$, the full block has $M_I \mid n = x + \sigma y$ and $D_I = n/\text{rad}(n) \leq Q_J$. It is saturated and leaves no residual source. If there are no sources, the empty configuration already has boundary one. With exactly one source, assigning every sink to it gives $\max\{d_p/Q_J, 1\} = \max\{n/R, 1\}$. The lower bound follows from Theorem 109.2. $\square$

Corollary 110.4. *If c has at most one powerful prime, then $E = \Delta$. If b has at most one powerful prime and $R \leq b$, then again $E = \Delta$. Without $R \leq b$, the latter condition gives*

$$E \leq \Delta + \log(c/b) \leq \Delta + \log 2.$$

Proof. The first assertion uses the sum candidate. For the second, the difference candidate is $(c/b)(b/R) = c/R$, and the lower bound is exact. If $R > b$, the difference candidate is c/b , so $E \leq \log(c/b) \leq \Delta + \log(c/b)$. Finally $a \leq b$ implies $c \leq 2b$. $\square$

The corollary removes allocation loss on these strata; it does not independently bound their scalar defect in all parameter ranges.

111. PROPER DELETIONS: A NECESSARY SIGN CORRECTION

Fix a target source modulus $M = p^{v_p(n)} \geq 4$. In the unit group modulo M , set

$$\Phi(T) = x_T y_T^{-1}.$$

Compatibility of a face is $\Phi(U) = -\sigma$. For complementary subsets U, D of T , this is equivalent to

$$\Phi(D) = -\sigma\Phi(T).$$

Define

$$z_\sigma(T) = \min\{Q_D : \emptyset \neq D \subsetneq T, \Phi(D) = -\sigma\Phi(T)\},$$

with value $+\infty$ when the set is empty.

Proposition 111.1 (Signed optimal-face formula). *A proper compatible face exists exactly when $z_\sigma(T) < \infty$. For a fixed saturated block (S, T) and residual target, the best credit factor is*

$$f^* = \frac{Q_T}{\max\{D_S, z_\sigma(T)\}}.$$

Proof. Complementation bijects the permitted faces and deletions. Since $Q_U = Q_T/Q_D$, a smallest deletion maximizes face weight. Substitution into $\min\{Q_T/D_S, Q_U\}$ yields the formula. $\square$

For the plus sign, the equation automatically excludes $D = T$, because otherwise $\Phi(T) = -\Phi(T)$, forcing $M \mid 2$. For the minus sign it does not: $D = T$ always meets its own residue target. One must explicitly forbid this empty complementary face. For example, $17 - 1 = 16$ has one sink prime 17; its full deletion has residue one modulo 16, but there is no nonempty proper face. Treating this full deletion as a flag is invalid.

112. SHARP ABSENCE THRESHOLDS FOR THE DIFFERENCE ENDPOINT

An integer factor u formed by a whole-prime-power packet is a unitary divisor: $u \mid N$ and $\gcd(u, N/u) = 1$.

Theorem 112.1 (Sharp signed unitary-gap threshold). *Let p be prime, $e \geq 2$, and $M = p^e \parallel n$. Suppose*

$$n + 1 = du, \quad d, u > 1, \quad \gcd(d, u) = 1, \quad u \equiv 1 \pmod{M}.$$

Then

$$(541) \quad n \geq \begin{cases} M(2M + 3), & p \neq 3, \\ M(4M + 5), & p = 3. \end{cases}$$

Each bound is attained for every permitted prime power M . Consequently, below these thresholds the difference endpoint $(n + 1) - 1 = n$ has no nonempty proper compatible whole-prime-power face at that source.

Proof. Since $du \equiv 1 \pmod{M}$ and $u \equiv 1 \pmod{M}$, also $d \equiv 1 \pmod{M}$. Write

$$d = Ms + 1, \quad u = Mt + 1, \quad s, t \geq 1.$$

Coprimality excludes $s = t$, since then $d = u > 1$. Interchange the factors to assume $s < t$. Expansion gives

$$(542) \quad n/M = Mst + s + t.$$

Here $st \geq 2$ and $s + t \geq 3$, proving the first lower bound.

For $p = 3$, the ordered pair $(s, t) = (1, 2)$ would give $n/M = 2M + 3$, divisible by 3, contradicting exactness. The pair $(1, 3)$ would give two even factors $M + 1$ and $3M + 1$, since M is odd, contradicting coprimality. If $s = 1$, then $t \geq 4$, giving $Mst + s + t \geq 4M + 5$. If $s \geq 2$, then $t \geq 3$, giving $Mst + s + t \geq 6M + 5 \geq 4M + 5$.

For sharpness at $p \neq 3$, take

$$n = M(2M + 3), \quad n + 1 = (M + 1)(2M + 1).$$

The factors have gcd one, and $p \nmid 2M + 3$ because its reduction modulo p is 3. Thus $M \parallel n$. Both factors are one modulo M , and their coprimality makes them actual whole-prime-power packets.

For $p = 3$, take

$$n = M(4M + 5), \quad n + 1 = (M + 1)(4M + 1).$$

Their gcd is $\gcd(M + 1, 3) = 1$; also $3 \nmid 4M + 5$. This attains the exceptional bound with exact source valuation and legal packets. $\square$

The extra factor at the prime three uses both exact valuation and unitary divisibility. Omitting either premise would invalidate that strengthened conclusion.

113. A CERTIFIED LARGE REPAIR AND A SURVIVING OBSTRUCTION

113.1. **A 39-digit no-face sum endpoint.** Set

$$c = 2^{64}3^{41} = 672808029771005150108072916419239477248$$

and

$$q = 3981112602195296746201614890054671463.$$

The exact factorization is

$$(543) \quad c - 1 = 13^2 q.$$

The integer q is prime; Section 114 supplies a deterministic certificate protocol and its complete recursive data are in the supplement.

Write $A = 2^{63}$, $B = 3^{40}$, so $c = 6AB$. Direct integer comparisons give

$$\max(2^{64}, 3^{41}) < 2 \min(2^{64}, 3^{41}) - 1, \quad 13 < A < B < q.$$

The preceding supplement's plus-sign threshold therefore excludes every proper compatible face at both sources. To recall its proof: if $c - 1 = du$ with $u \equiv -1 \pmod{M}$, then $d = Mt + 1$, $u = Ms - 1$, and $c/M = s(Mt + 1) - t$. The pair $s = t = 1$ contradicts exactness $M \parallel c$; every other positive pair gives $c/M \geq 2M - 1$.

Here $Q = 13q$ and $AB > Q$. Every compatible block uses the full sink set, so at most one block can be selected. The two-source block is unsaturated, while a singleton block is no better than assigning all sinks to that source. There are no flags. The complete optimum is consequently

$$\mathcal{F}_c = \min_{H|Q} \max\{A/H, 1\} \max\{BH/Q, 1\}.$$

The four divisors are $1, 13, q, 13q$. Their boundary factors are, respectively,

$$A, \quad A/13, \quad B/13, \quad B.$$

Thus

$$(544) \quad \boxed{\mathcal{F}_c = 2^{63}/13.}$$

At the difference endpoint $c - 1 = b = 13^2 q$, the sole powerful prime is 13. Its demand is 13, and the sink primes are 2, 3, with product 6. Proposition 110.3 gives

$$\mathcal{F}_b = 13/6.$$

Since $R = 78q < b$, the selected factor is exactly

$$(545) \quad \boxed{\mathcal{F}_{\text{sel}} = \frac{c}{78q} = \frac{c}{R}.}$$

The original excess factor removed by selection is

$$(546) \quad \frac{\mathcal{F}_c}{c/R} = \frac{q}{3^{40}} = \frac{3981112602195296746201614890054671463}{12157665459056928801}.$$

It exceeds $3 \cdot 10^{17}$, an integer comparison verified by the replay. This is a large repair of an auxiliary bound, not a large *abc* defect: the selected scalar factor is only slightly greater than $13/6$.

More generally, every endpoint $c = 2^e 3^f$, $e, f \geq 1$, with $c - 1 = 13^2 s$, s squarefree and $13 \nmid s$, has one powerful prime on the difference side. Its radical is $78s$, so the same selector attains c/R . This is a statement about every member satisfying the hypotheses, not an assertion that infinitely many such members have been produced or that these hypotheses hold for arbitrary balanced endpoints.

113.2. The selector does not always attain the scalar defect.

Proposition 113.1 (Complete-premise noncollapse). *For $1 + 3024 = 3025$,*

$$R = 2310, \quad \mathcal{F}_c = 11/7, \quad \mathcal{F}_b = 8/5, \quad \mathcal{F}_{\text{sel}} = 11/7 > 55/42 = c/R.$$

The adaptive excess factor is $6/5$.

Proof. The factorizations are

$$3024 = 2^4 3^3 7, \quad 3025 = 5^2 11^2.$$

At the sum endpoint the source demands are 5, 11 and the sink weights are 2, 3, 7. The proper whole-power packets of 3024 are

$$16, 27, 7, 432, 112, 189.$$

None is minus one modulo either 25 or 121. Hence every compatible block uses all sinks; the two-source block is unsaturated because $55 > 42$, and no flag exists. The complete partition optimum is obtained by assigning $2 \cdot 3$ to demand 5, and 7 to demand 11, giving $11/7$. To check minimality explicitly, the possible divisor products are 1, 2, 3, 6, 7, 14, 21, 42; substituting them in $\max\{5/H, 1\} \max\{11H/42, 1\}$ gives no smaller value. Singleton full blocks give at least 5, so do not improve it.

At the difference endpoint, the demands are 8, 9, and the sink weights are 5, 11. Neither proper packet 25 nor 121 is one modulo 16 or 27, because their differences from one are 24, 120. Thus again only full blocks are compatible. The two-source block is unsaturated since $72 > 55$. Assigning 5 to demand 8 and 11 to demand 9 gives $8/5$. The other split gives $9/5$; assigning all sinks to one source or using a singleton full block leaves boundary at least 8. Therefore the optimum is $8/5$.

Finally

$$\min \left\{ \frac{11}{7}, \frac{3025}{3024} \frac{8}{5} \right\} = \frac{11}{7}, \quad \frac{11/7}{55/42} = \frac{6}{5}.$$

All positivity, coprimality, packet, saturation, and face premises have been checked. $\square$

This refutes only universal pointwise scalar collapse. A finite positive loss does not refute the quantified estimate (540); the adaptive route remains open.

114. DETERMINISTIC COMPUTATIONAL EVIDENCE

114.1. An elementary recursive primality certificate. The certificate for q begins with the full factorization

$$q - 1 = 2 \cdot 61 \cdot 563 \cdot 921889 \cdot 2532974063 \cdot 24821433536218331.$$

For a candidate $N > 2$, recursively certify all primes r in a complete factorization of $N - 1$. For each such r , supply an integer a_r with

$$(547) \quad a_r^{N-1} \equiv 1 \pmod{N}, \quad \gcd(a_r^{(N-1)/r} - 1, N) = 1.$$

The root uses $a_2 = 5$ and $a_r = 2$ for each other displayed factor. The recursive file contains 25 certified primes and 70 order-witness checks, including the separate prime 13.

For completeness, this criterion is sufficient without any probable-prime assumption. Let ℓ be any prime divisor of N . If $r^v \parallel N - 1$, the order of a_r modulo ℓ divides $N - 1$, but does not divide $(N - 1)/r$ by the gcd condition. Therefore r^v divides that order and hence $\ell - 1$. Doing this for every r gives $N - 1 \mid \ell - 1$, so $\ell \geq N$. Since $\ell \mid N$, necessarily $N = \ell$ is prime. Recursion terminates at 2.

The discovery of factors used a separate symbolic program. The delivered verifier does not depend on that program: it checks every product, recursive dependency, modular power, and gcd using only Python integers. This distinction is important because finding a factorization and verifying its prime factors are different tasks.

114.2. **Finite scans and their exact scope.**

Check	Completed count
Normalized primitive triples filtered, $c \leq 3000$	1,368,094
Rows in that range with $R < b$, both optima computed	49
Rows among those 49 with nonzero adaptive loss	0
Consecutive endpoints with $c \leq 10^6$, $R < c - 1$	135
Rows among those 135 with nonzero adaptive loss	3
Eligible difference-target/modulus pairs, $n \leq 200000$	7,167
Proper packets checked below the new thresholds	36,668
Signed face/deletion comparisons, larger input ≤ 100	5,736
Sharp threshold examples checked	30

The three consecutive noncollapse endpoints are $c = 3025, 50176, 998001$, with excess factors $6/5, 669/512, 37/25$. The scan through 3000 alone would have missed every one of them. This demonstrates concretely why finite scalar attainment must not be promoted to a theorem.

The optimizer retains the first supplement's exhaustive block-family, residual-owner, target, and proper-face enumeration. Only packet compatibility receives the sign parameter. An independent multiplicative evaluator reconstructs unused capacity, discarded surplus, and overflow from each returned configuration and checks (536) exactly. The signed deletion test separately enumerates both complementary subsets and explicitly forbids full deletions. The threshold scan enumerates every proper unitary packet in its stated range.

These are finite implementation and certificate checks. The proofs of Theorems 109.2, 110.2, and 112.1 do not depend on extrapolating these computations. Repeating a deterministic run verifies reproducibility, not independent mathematical peer review.

115. THE REMAINING MATHEMATICAL AND FORMAL OBLIGATIONS

The completed step is not merely another name for Δ . The selector uses concrete signed blocks and genuine nonempty proper faces, repairs a large loss in Section 6, and still has the complete-premise obstruction of Proposition 113.1. Its live uniform target is (540). In the region $R < b$, it requires control of both

$$\log(c/R) \quad \text{and} \quad \min\{L_c^*, L_b^*\}.$$

The first term is still the *abc* defect. Setting it to $o(\log R)$ as an assumption would restate the missing conclusion rather than prove it. The second term is now weaker than requiring either orientation's loss to be small separately, but no uniform estimate for it is established here.

The original FCRT-1 route is not retired: the 39-digit example is finite. The adaptive scalar-collapse child is retired by $1 + 3024 = 3025$; the adaptive uniform route is not. The IUT all-place/Ind3/pointed comparison, compensated divisor-packet, Pell first-apparition valuation, Mersenne depth-three counting, and geometric uniformity routes remain as recorded in the inspected ledger. No new unconditional theorem from those separate routes is claimed in this supplement.

115.1. **Formal status.** The mathematical proofs preceded the Lean scripts. The partial file

`Lean/drafts/ABCSignedEndpointSelection20260905.lean`

contains algebraic loss identities, the abstract endpoint selection inequality, two elementary bilinear cores, sharpness identities, and rational arithmetic for the small witness. It includes one axiom query for each of its 12 named theorems. These queries have *not* been executed. The file is explicitly uncompiled and is not imported into the repository's verified build.

No local `lean` or `lake` executable was present, and the available installation path did not yield one. The full signed arithmetic adapter, prime-certificate verification in Lean, optimizer completeness, and the universal estimate are not formalized by this partial companion. A Python pass is not a Lean kernel pass. There is no claimed term of type `ABCConjecture`.

115.2. Integration status. The GitHub connector supplied repository reads but no write action. The local Git remote test failed with DNS resolution failure. No remote branch, commit, pull request, workflow dispatch, or merge was created. The deliverable is an additive source patch and a research archive, not an assertion that remote `main` was changed. Patch-application tests are distinct from a full repository build and from remote integration.

INTEGRATED CHECKPOINT: UNITARY FACES, SAFE MULTIPLE OUTPUTS, AND VALUATION
LIFTING

Full mathematical proof body of the historical checkpoint. Historical execution statements retain their original scope.

116. TARGET, PROVENANCE, AND EXACT SCOPE

For a positive integer m , let $\text{rad}(m) = \prod_{p|m} p$. Our target is

$$(548) \quad \forall \epsilon > 0 \exists K_\epsilon > 0 \forall a, b, c \in \mathbb{Z}_{>0}, \quad a + b = c, (a, b) = 1 \implies c \leq K_\epsilon \text{rad}(abc)^{1+\epsilon}.$$

The constant must be independent of the triple, prime support, base and exponent. A conditional implication to (548) is not a proof of its premise.

The starting repository snapshot is `d05220b6ddd59e40da976ba4238121f3b3b0b012`, which contains the two preceding supplements [106, 107]. We preserve their whole-prime-power faces, exact congruences, and no-double-counting conventions. The new proofs below are developed here, rather than attributed to those sources. No general web literature search was performed in this round; external novelty and peer review are not asserted. The valuation argument in Section 120 is an elementary lifting calculation, not a claim to have discovered the classical lifting phenomenon.

Write $R = \text{rad}(abc)$. The preceding signed selector has

$$F_{\text{sel}} = \min\{F_c, (c/b)F_b\}, \quad c \leq RF_{\text{sel}},$$

for $a \leq b$, and its logarithm decomposes exactly into the scalar defect $\log(c/R)$ and the smaller of two allocation losses [107]. Reducing an allocation loss alone does not bound the scalar defect. This distinction is maintained throughout.

117. A SHARP THRESHOLD FOR GENERAL UNITARY FACES

Definition 117.1. A divisor d of $m > 0$ is *unitary* if $(d, m/d) = 1$. A whole-prime-power face of a primitive triple $a + b = c$ is a choice

$$a = AX, \quad b = BY,$$

where $A \mid a$ and $B \mid b$ are unitary. The face is nonempty and proper when $(A, B) \neq (1, 1)$ and $(X, Y) \neq (1, 1)$. It is compatible with M when $M \mid A + B$.

All four integers A, B, X, Y are then pairwise coprime. This includes the within-summand conditions $(A, X) = (B, Y) = 1$, not only the cross-summand conditions.

Theorem 117.2 (Sharp general threshold). *Let $a + b = c$ be a primitive positive triple, and let $M \geq 4$ be a unitary divisor of c . If there is a compatible nonempty proper whole-prime-power face, then*

$$(549) \quad ab \geq T(M) := \begin{cases} (2M - 1)(M + 1), & M \equiv 1 \pmod{3}, \\ (M - 1)(2M + 1), & M \not\equiv 1 \pmod{3}. \end{cases}$$

For every integer $M \geq 4$, equality is attained by an actual primitive triple and an actual unitary proper face.

Lemma 117.3 (Normal form and excluded unit cell). *Under the hypotheses of Theorem 117.2, after interchanging the summands when necessary, there are positive integers k, ℓ, Y such that*

$$(550) \quad A + B = kM, \quad X = Y + \ell M, \quad c/M = kY + \ell A.$$

The case $k = \ell = Y = 1$ is impossible.

Proof. Since $M \mid c$ and $(a, c) = 1$, the integer A is invertible modulo M . From $A + B \equiv 0$ and $AX + BY \equiv 0 \pmod{M}$, we get $X \equiv Y \pmod{M}$. If $X = Y$, pairwise coprimality gives $X = Y = 1$, contrary to properness. Interchange the summands so that $X > Y$; this gives (550). Unitarity of M says $(M, kY + \ell A) = 1$.

In the unit cell, $X = M + 1$, $A + B = M$, and $c/M = A + 1$. If M is odd, X is even, so coprimality with X forces both A and B odd, contradicting their odd sum. If M is even, coprimality of A, B and their even sum forces both odd; then $c/M = A + 1$ is even, contradicting $(M, c/M) = 1$. $\square$

Proof of Theorem 117.2. For positive A, B ,

$$AB - (A + B - 1) = (A - 1)(B - 1) \geq 0.$$

Hence $AB \geq kM - 1$, and $XY = Y(Y + \ell M)$. Outside the excluded unit cell, at least one of the following applies:

$$(551) \quad Y \geq 2 : \quad ab \geq 2(M - 1)(M + 2),$$

$$(552) \quad k \geq 2 : \quad ab \geq (2M - 1)(M + 1),$$

$$(553) \quad \ell \geq 2 : \quad ab \geq (M - 1)(2M + 1).$$

Each is at least $(M - 1)(2M + 1)$. This proves the baseline threshold.

Suppose now $M \equiv 1 \pmod{3}$. The difference between the right sides of (551) and (552) is $M - 3 \geq 1$. Thus only $Y = k = 1$ requires further treatment. If $\ell \geq 3$,

$$(M - 1)(3M + 1) - (2M - 1)(M + 1) = M(M - 3) \geq 0.$$

If $\ell = 2$, then $X = 2M + 1$ is divisible by 3. Neither A nor B can equal 1, since the other would equal $M - 1$, also divisible by 3, contradicting its coprimality with X . Therefore $A, B \geq 2$, and

$$(A - 2)(B - 2) \geq 0 \implies AB \geq 2(M - 2).$$

Consequently

$$ab \geq 2(M - 2)(2M + 1) \geq (2M - 1)(M + 1),$$

since the difference is

$$2M^2 - 7M - 3 = 2(M - 4)^2 + 9(M - 4) + 1 > 0.$$

This proves the strengthened threshold.

For sharpness take $a = 1, b = uv, c = uv + 1$. If $M \not\equiv 1 \pmod{3}$, set

$$u = M - 1, \quad v = 2M + 1, \quad c = M(2M - 1).$$

Here $(u, v) = (M - 1, 3) = 1$, $(M, 2M - 1) = 1$, and $1 + u = M$. If $M \equiv 1 \pmod{3}$, set

$$u = 2M - 1, \quad v = M + 1, \quad c = M(2M + 1).$$

Then $(u, v) = (M + 1, 3) = 1$, $(M, 2M + 1) = 1$, and $1 + u = 2M$. In each case $u, v > 1$, the face $(1, u)$ is nonempty and proper, and its product is exactly $T(M)$. $\square$

Corollary 117.4 (A source exclusion rule). *For every complete prime power $p^e \parallel c$ with $p^e \geq 4$, the inequality $ab < T(p^e)$ rules out every compatible nonempty proper whole-prime-power face at that source. In particular it rules out all proper-face transfers to that source from any smaller packet.*

Proof. A complete prime power is a unitary divisor of c . A nonempty proper subface of any packet is also a nonempty proper face of the full endpoint. Apply the theorem. $\square$

Remark 117.5. For $a = 1$, the baseline recovers the earlier $c \geq M(2M - 1)$ obstruction, and the residue-one case improves it to $c \geq M(2M + 1)$. The theorem does not claim that a face exists whenever the threshold is exceeded. It only proves necessity and sharpness. Omitting within-arm unitarity invalidates the parity argument; arbitrary divisors must not be substituted for these faces.

118. MULTI-OUTPUT TRANSPORT WITH A SINGLE TOTAL CHARGE

We give a complete definition because relaxing the old rules silently would change the problem. Consider a signed endpoint

$$n = x + \sigma y > 0, \quad \sigma \in \{1, -1\}, \quad (x, y) = 1.$$

Then n, x, y are pairwise coprime. The sources are $p_i^{e_i} \parallel n$, $e_i \geq 2$, with demands $d_i = p_i^{e_i-1}$. The sinks are the distinct primes of xy . For a set T of sinks, let x_T, y_T select their full powers from x, y , and set $Q_T = \text{rad}(x_T y_T)$.

A block (S, T) has nonempty source and sink sets, satisfies

$$\prod_{i \in S} p_i^{e_i} \mid x_T + \sigma y_T, \quad D_S := \prod_{i \in S} d_i \leq Q_T,$$

and is disjoint in both kinds of indices from every other selected block. A residual sink is assigned to at most one residual source. Write K_i for the product assigned to source i , with the empty product equal to 1.

Definition 118.1 (Multi-output extension). A selected block $k = (S_k, T_k)$ may emit factors $f_{ki} \geq 1$ to several residual sources. Whenever $f_{ki} > 1$, there must exist a nonempty proper subset $U_{ki} \subsetneq T_k$ with

$$p_i^{e_i} \mid x_{U_{ki}} + \sigma y_{U_{ki}}, \quad f_{ki} \leq Q_{U_{ki}}.$$

The *joint* block budget is

$$(554) \quad \prod_i f_{ki} \leq Q_{T_k} / D_{S_k}.$$

The witnesses need not be disjoint from one another: they provide congruence and individual-capacity restrictions. The available block surplus, not a sum of independent face budgets, is the resource charged by (554). The residual factor is

$$F = \prod_{i \text{ residual}} \max \left\{ \frac{d_i}{K_i \prod_k f_{ki}}, 1 \right\}.$$

This is an explicitly defined successor of FCRT, not an assertion that the old single-output optimizer already permitted such splitting.

Theorem 118.2 (Exact accounting and height certificate). *Let $R_n = \text{rad}(nxy)$. For every multi-output configuration,*

$$(555) \quad F = \frac{n}{R_n} U_0 S_0 O_0 \geq \frac{n}{R_n},$$

where

$$U_0 = \prod_{\text{unowned residual sinks}} q, \quad S_0 = \prod_k \frac{Q_{T_k}}{D_{S_k} \prod_i f_{ki}}, \quad O_0 = \prod_i \max \left\{ \frac{K_i \prod_k f_{ki}}{d_i}, 1 \right\}.$$

All three factors are at least one. The minimum new boundary is no greater than the preceding single-output FCRT boundary.

Proof. Use $\max(d/z, 1) = (d/z) \max(z/d, 1)$ at each residual source. The product of all source demands is $n/\text{rad}(n)$. Multiply the covered and residual factors, and partition the sink primes into owned residual sinks, unowned sinks, and disjoint blocks. Cancellation gives (555). The joint budget gives $S_0 \geq 1$; the other inequalities are immediate. Each old configuration is a new one with at most one nontrivial factor per block, proving the optimizer comparison. $\square$

For the two orientations of $a + b = c$, the same argument gives a selector

$$E_{\text{multi}} = \log \min \{ F_{\text{multi},c}, (c/b) F_{\text{multi},b} \}, \quad c \leq R \exp(E_{\text{multi}}), \quad E_{\text{multi}} \leq E.$$

The pointwise lower bound is still $\max\{\log(c/R), 0\}$.

119. THE FINITE CUT FORMULA AND A GENUINELY INDEPENDENT VERIFIER

Fix the disjoint blocks and residual ownership. Define

$$r_i = \log \max\{d_i/K_i, 1\}, \quad s_k = \log(Q_{T_k}/D_{S_k}),$$

and let W_{ki} be the maximum Q_U over legitimate nonempty proper compatible faces of block k for source i , with $W_{ki} = 1$ if none exist. Set $u_{ki} = \log W_{ki}$.

Theorem 119.1 (Exact residual cut formula). *The optimal logarithmic residual for these fixed blocks and ownership is*

$$(556) \quad B_* = \max_{A \subseteq I} \left\{ \sum_{i \in A} r_i - \sum_k \min \left(s_k, \sum_{i \in A} u_{ki} \right) \right\}.$$

Equivalently, with $\rho_i = \max(d_i/K_i, 1)$ and $\beta_k = Q_{T_k}/D_{S_k}$,

$$(557) \quad e^{B_*} = \max_{A \subseteq I} \frac{\prod_{i \in A} \rho_i}{\prod_k \min(\beta_k, \prod_{i \in A} W_{ki})}.$$

In particular, its exact value is rational; logarithms need not be numerically evaluated.

Proof. Put edges of capacities s_k from a new source to block vertices, edges of capacities u_{ki} from block k to demand vertex i , and edges of capacities r_i from demand vertices to a new sink. Removing useless overfill cannot increase the residual, so usable transfers are exactly feasible flows in this finite real-capacity network. The total unmatched mass is $\sum_i r_i$ minus maximum flow.

For completeness, maximum flow equals minimum cut here without requiring a termination assumption for arbitrary real augmentations. The feasible flow polytope is nonempty and compact, so a maximizing flow exists. Its residual graph cannot contain a source-to-sink path, since augmenting by the minimum positive residual along that finite path would increase its value. Let the source side be the vertices reachable in the residual graph. Every forward edge crossing the resulting cut is saturated, and every backward crossing edge carries zero flow. Summing conservation shows equality of flow value and cut capacity. Every flow is bounded by every cut, establishing the equality.

For a cut with demand vertices A on the sink side, putting a block on the sink side costs s_k , while putting it on the source side costs $\sum_{i \in A} u_{ki}$. Choose the smaller independently for each block. Demand vertices outside A contribute $\sum_{i \notin A} r_i$. Subtract the minimum cut from total demand to obtain (556); exponentiate to get (557). $\square$

Proposition 119.2 (Two outputs and an obstruction that persists). *For one block and two residual demands, put $a = \min(r_1, u_1)$, $b = \min(r_2, u_2)$. The usable transfer is $\min(s, a + b)$, versus $\min(s, \max(a, b))$ for a single output. A feasible maximizing choice is*

$$t_1 = \min(s, a), \quad t_2 = \min(s - t_1, b).$$

If every proper-face cap vanishes, multi-output transport emits nothing and cannot repair the no-face obstruction.

Proof. Any feasible sum is at most both s and $a + b$. The displayed greedy pair attains their minimum by separating the cases $s \leq a$, $a < s \leq a + b$, and $a + b < s$. A single output is bounded by $\min(s, a)$ or $\min(s, b)$. If all $u_{ki} = 0$, every usable transfer is zero. $\square$

The model $s = 2$, $u_1 = u_2 = r_1 = r_2 = 1$ gives a strict abstract improvement. No arithmetic realization of this model is asserted. In particular the preceding complete counterexample $1 + 3024 = 3025$ has no proper faces in either orientation, so its excess factor $6/5$ persists. More flexibility is not itself a uniform estimate.

The supplied verifier has two separate algorithms. One enumerates (557); the other implements breadth-first augmenting paths with rational multiplicative residuals. It starts each reverse residual at 1, augments a path by its smallest residual factor $\alpha > 1$, divides forward residuals by α , and multiplies reverse residuals by α . These are exact exponentiated flow operations. It

also checks edge capacities and multiplicative conservation. The two algorithms agree on 5,000 seeded rational networks and all 1,273 fixed arithmetic configurations arising in the 38 selected triples described in Section 121.

120. A UNIFORM EXPONENT-LIFTING BUDGET

Let $x \geq 2$, $n \geq 1$, and $N = x^n - 1$. For an odd prime $p \mid N$, put

$$r_p = \text{ord}_p(x), \quad t_p = v_p(x^{r_p} - 1).$$

The order divides n and $p - 1$, and hence $p \nmid r_p$. The divisibility by $p - 1$ follows, for example, by partitioning the finite group of nonzero residues into cosets of the cyclic subgroup generated by x .

Lemma 120.1 (Valuation lifting, with the prime two separated). *For odd $p \mid x^n - 1$,*

$$v_p(x^n - 1) = t_p + v_p(n).$$

If x is odd, then

$$v_2(x^n - 1) = \begin{cases} v_2(x - 1), & n \text{ odd}, \\ v_2(x^2 - 1) + v_2(n) - 1, & n \text{ even}. \end{cases}$$

Proof. For odd p , write $z = 1 + p^t u$, where $p \nmid u$, $t \geq 1$. If m is coprime to p , the first term of the binomial expansion of $z^m - 1$ has valuation t , while all later terms have larger valuation. Thus raising to such an exponent preserves valuation. Raising to p increases it by exactly one: the first term has valuation $t + 1$; for $2 \leq j < p$, the j -th term has valuation at least $jt + 1 \geq t + 2$; the last term has valuation $pt \geq t + 2$. Apply this repeatedly to $z = x^{r_p}$ and exponent n/r_p .

For odd x , an odd exponent contributes an odd geometric sum and does not change $v_2(x - 1)$. For $n = 2^h m$, $h \geq 1$, m odd, remove the odd exponent in the same way and use

$$x^{2^h} - 1 = (x - 1)(x + 1) \prod_{j=1}^{h-1} (x^{2^j} + 1).$$

Each factor in the product has valuation exactly one, since an odd square is 1 (mod 8). This proves the formula, including $h = 1$. $\square$

Define

$$T_{\text{odd}} = \prod_{\substack{p \mid N \\ p \text{ odd}}} p^{t_p - 1}, \quad L_{\text{odd}} = \prod_{\substack{p \mid N \\ p \text{ odd}}} p^{v_p(n)}.$$

For the two-part set

$$(T_2, L_2) = \begin{cases} (1, 1), & x \text{ even}, \\ (2^{v_2(x-1)-1}, 1), & x, n \text{ odd}, \\ (2^{v_2(x^2-1)-2}, 2^{v_2(n)}), & x \text{ odd}, n \text{ even}. \end{cases}$$

All exponents are nonnegative. Let $T = T_2 T_{\text{odd}}$ and $L = L_2 L_{\text{odd}}$.

Theorem 120.2 (Uniform budget for repeated lifting). *For every $x \geq 2, n \geq 1$,*

$$(558) \quad \frac{x^n - 1}{\text{rad}(x^n - 1)} = TL, \quad L \mid n, \quad 1 \leq L \leq n.$$

Consequently

$$0 \leq \frac{\log L}{\log(x^n)} \leq \frac{\log n}{n \log 2},$$

which tends to zero uniformly over all bases $x \geq 2$ as $n \rightarrow \infty$.

Proof. For each odd prime of N , Lemma 120.1 splits its excess exponent as $v_p(N) - 1 = (t_p - 1) + v_p(n)$. The two-part does exactly the same splitting using its separate formula. Multiplication proves the equality in (558). Every prime exponent in L is either zero or its full exponent in n , so $L \mid n$. The logarithmic bound follows from $x \geq 2$. $\square$

Corollary 120.3 (A uniform *abc* bound on an explicit stratum). *Fix $A \geq 0$. Among all triples*

$$(1, x^n - 1, x^n), \quad x \geq 2, \quad n \geq 1, \quad T(x, n) \leq n^A,$$

*the standard *abc* inequality holds with a constant depending only on A, ϵ , not on x, n .*

Proof. Let $c = x^n$. Since $(x, x^n - 1) = 1$, $\text{rad}(x) \geq 2$, and $c - 1 \geq c/2$, Theorem 120.2 gives

$$R = \text{rad}(x) \frac{c-1}{TL} \geq \frac{c}{n^{A+1}}.$$

Thus, with $B = (A+1)(1+\epsilon)$,

$$\frac{c}{R^{1+\epsilon}} \leq \frac{n^B}{c^\epsilon} \leq n^B 2^{-n\epsilon} \leq \max \left\{ 1, \left(\frac{B}{\epsilon \log 2} \right)^B \right\}.$$

The last bound follows by maximizing $t^B \exp(-\epsilon(\log 2)t)$ for real $t > 0$. $\square$

The condition $T \leq n^A$ is an explicit restriction, not a newly proved bound on arbitrary power neighbours. The uncontrolled first-appearance valuations remain in T . In particular $n = 1$ already leaves unrestricted squarefullness in $x - 1$, so a claim that this corollary settles all consecutive triples would be false.

Lemma 120.4 (Geometric-sum gcd core). *For $x \geq 1, n \geq 0$, let $G(x, 0) = 0$ and $G(x, n+1) = xG(x, n) + 1$. Then*

$$G(x, n) \equiv n \pmod{x-1}, \quad \gcd(x-1, G(x, n)) = \gcd(x-1, n).$$

For $x = 1$, interpret the congruence as equality.

Proof. Induction gives the congruence since $x \equiv 1 \pmod{x-1}$. Applying the Euclidean gcd recurrence gives the equality. At $x = 1$, $G(1, n) = n$ directly. $\square$

This is the lifting route's formalized gcd core, not a replacement for the full valuation proof.

121. EXACT COMPUTATIONS AND ADVERSARIAL AUDIT

All reported comparisons use integers or rational numbers. Numerical logarithms and probable-prime declarations are not accepted as certificates.

Check	Count
Normalized primitive triples with $c \leq 5000$	3,800,229
Nonempty proper whole-prime-power faces examined	70,740,182
Compatible unitary-modulus cases $M \geq 4$	8,435,196
Cases in the residue-one refinement	3,866,583
Equality cases in that scan	1,156
Sharpness constructions, $4 \leq M \leq 10000$	9,997
Independent cut/augmenting checks on seeded networks	5,000
Arithmetic fixed-configuration cross-checks	1,273
Selected positive-defect triples with three or more powerful sources	38
Certified supplied power-neighbour factorizations	498
Distinct factor primes checked by deterministic trial division	502

The first scan found no threshold counterexample. Its range is finite; the theorem is proved in Section 117 independently of this scan.

For transport, an exact radical sieve considered $c \leq 10^6$, at least three source primes of exponent at least two, and $R < c$. It tested 3,278 eligible targets and 1,554,459 candidate summand choices, obtaining 38 normalized primitive triples. For each, all legal block selections and residual ownership assignments were enumerated, and both network solvers were run. The complete new optimum equals the old FCRT optimum in all 38 cases. This is *no demonstrated strict arithmetic improvement*, not evidence of a global equality theorem.

The lifting certificate contains 498 supplied rows from the discovery rectangle $2 \leq x \leq 20, 1 \leq n \leq 30$; it is not a complete scan of that rectangle. Factor discovery was separated from

verification. The standard-library verifier trial-checks every included prime, checks each full factorization product, derives each order by exact modular exponentiation, and checks (558). No probabilistic discovery result is relied upon without this deterministic replay.

Three adversarial checks matter structurally. First, within-arm coprimality cannot be dropped from the face definition. Second, an abstract network's strict gain is not an arithmetic example. Third, $L \leq n$ says nothing by itself about the first-apparition factor T . These restrictions prevent three different ways of turning a local result into an unsupported global assertion.

122. LEAN VERIFICATION AND DEPENDENCY AUDIT

The standalone module `CapacityAndLifting.lean` imports only `Std`. At commit

539db3ad12faeb5a78ebc18b2a44895218744a6a

it passed

```
lean -DwarningAsError=true \
```

```
research/checkpoints/2026_09_05_multiflag/Lean/CapacityAndLifting.lean
```

on Lean 4.32.2, compiler commit f3b06c705e6c85f5314019d5d3baab0fec5b580c. GitHub Actions run 33968189781, job 101311920225, returned success. The compiler archive checksum was checked before execution. The source SHA-256 is

1bb648834b0570daac70f8a4dbe021f35ddd35ba0adceb69667345376bd929c0.

All 14 named theorems have executed axiom queries. The union is `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx` occurs. The strict finite abstract example uses no axioms. Two earlier compilation attempts failed and were repaired; only the stated successful source is described as checked.

The formalized scope comprises the natural-capacity two-output optimizer, no-face zero-transfer and monotonicity, elementary positive-product estimates, parity exclusion of the unit cell, the baseline threshold in the explicit normal form (550), and geometric-sum congruence/gcd identities. The normal-form theorem assumes the four relevant gcd conditions and positivity hypotheses explicitly. It does not assume its conclusion.

The complete translation from actual unitary faces into that normal form, the $M \equiv 1 \pmod{3}$ refinement, sharpness for all M , the real-capacity max-flow theorem, the exact signed arithmetic accounting map, and full valuation lifting have ordinary proofs here but are *not fully formalized in this module*. The discrete capacity theorem is not silently presented as a theorem on arbitrary real logarithms. The existing project toolchain, verified imports, and earlier drafts were not changed by this standalone check.

Dependency	Status
Unitary factorization $\Rightarrow$ normal form	Ordinary proof, Lemma 117.3
Normal form $\Rightarrow$ baseline threshold	Ordinary proof and checked Lean core
Mod-3 refinement and sharpness	Ordinary proof, Theorem 117.2
Joint budget $\Rightarrow$ exact accounting	Ordinary proof, Theorem 118.2
Fixed network $\Rightarrow$ cut formula	Ordinary proof, Theorem 119.1; two finite implementations
Valuation identities $\Rightarrow L \mid n$	Ordinary proof, Theorem 120.2
Geometric gcd identity	Ordinary proof and checked Lean theorem
Uniform first-apparition control	Unresolved
Uniform global multi-output bound	Unresolved
Unconditional <code>ABCConjecture</code> closed term	Not obtained

No independent autonomous research agents or outside peer review are claimed. Independent *algorithms*, not independent mathematicians, supplied the numerical cross-check.

123. ROUTE REGISTER AND THE REMAINING DECISIVE STATEMENT

The new exclusion theorem supplies a sharp arithmetic test for which transport edges cannot exist. The multi-output construction supplies an exact optimizer for the edges that do exist.

The lifting budget controls one source of high valuations uniformly. None bounds the total scalar defect on arbitrary primitive triples.

The actual sufficient target for the successor remains

$$(559) \quad \forall \epsilon > 0 \exists C_\epsilon \forall (a, b, c), \quad E_{\text{multi}}(a, b, c) \leq \epsilon \log R + C_\epsilon.$$

Its implication to (548) follows from the certificate, but (559) has not been proved. In a no-face region all new edge caps vanish; therefore splitting cannot by itself supply the missing global mechanism. Replacing E_{multi} by its scalar lower bound would reverse the direction needed for a proof.

For power neighbours, the independently proved $L \mid n$ removes repeated exponent lifting as a potential large uniform error. The remaining task is arithmetic control of T , including first-rank super-Wieferich behaviour, rather than a further estimate on L . The restricted Corollary 120.3 does not cover all T .

The IUT source-faithful all-place tensor, integral-order and Ind1–Ind3/same-pilot bridges remain unresolved. The canonical correlated defect $X - Y$, compensated divisor-packet construction, Pell first-appearance valuations, Mersenne depth-three prime count, positive five-term filling, and geometric uniformity gates retain their prior open status [108]. They were reviewed for their exact dependencies, not declared solved by the local results of this supplement. No entire parent route is retired on the basis of a child obstruction or a null finite search.

Conclusion. The completed outputs are the sharp general unitary-face theorem, a fully specified once-charged multi-output successor and its cut formula, the uniform lifting decomposition, a restricted uniform abc stratum, deterministic finite checks, and actually compiled elementary Lean proofs. No unconditional proof or disproof of (548) and no Lean closed term of its full statement has been obtained in this round.

INTEGRATED CHECKPOINT: SIGNED NORMAL FORMS AND ARITHMETIC WRONSKIANs

Edited proof consolidation; provenance and historical verification boundaries are stated below.

124. PROVENANCE AND SCOPE

This integrated edition restates with complete proofs the principal mathematical results of the September 5 normal-form supplement. It is not a byte-identical reproduction of that historical standalone paper: its historical runtime narrative and finite-count table are not repeated as current verification claims. The original source is identified by Git blob hash `ce5ae32428f1e5c710386acfff4b8e5f1cfd9e02` in the conversation research bundle. Neither this edition nor its predecessor proves standard *abc*. The earlier partial Lean draft is not certified by integrating these ordinary proofs. Later supplements supersede the historical assessment of the uniform derivative gate by proving its equivalence to *abc*.

125. EXACT SIGNED PACKET CONSTRUCTION

Fix $M \geq 4$, $0 < a < M$, $b > 0$, $\gcd(a, b) = 1$, and $\eta \in \{-1, 1\}$, with $n = b + \eta a > 0$ divisible by M . Put $C = n/M$. A unitary packet consists of

$$a = xX, \quad b = yY, \quad \gcd(x, X) = \gcd(y, Y) = 1.$$

It is nonempty if $(x, y) \neq (1, 1)$, proper if $(X, Y) \neq (1, 1)$, and compatible if $M \mid y + \eta x$.

Theorem 125.1 (Complete normal form). *All nonempty proper compatible packets are in bijection with unitary divisors x of a and positive integers r, s satisfying, with $X = a/x$,*

$$C = Mrs + sX - \eta rx, \quad y = Ms - \eta x, \quad Y = Mr + X, \quad \gcd(y, Y) = 1.$$

Proof. Primitivity implies $\gcd(x, M) = 1$. Write $y + \eta x = Ms$. For the minus sign this quantity exceeds $-M$; if it were zero, $x = y = 1$, contradicting nonemptiness. The plus sign is positive. Thus $s \geq 1$. Reducing $n = yY + \eta xX$ modulo M gives $X \equiv Y \pmod{M}$. Since $X < M$, $Y - X > -M$. Equality $Y = X$ would imply $X = Y = 1$, contradicting properness. Hence $Y - X = Mr$, $r \geq 1$. Expansion gives the stated formula for C . All recovered parameters are unique. Conversely the displayed data have $y > 0$, $Y > 1$, product $yY = n - \eta a = b$, and the required unitary gcds. Their selected signed sum is Ms . An empty packet would give that sum zero or two, impossible for $Ms \geq M$. The packet is proper because $Y > X \geq 1$. $\square$

For each unitary divisor x , all candidates can be constructed without factoring b by testing

$$1 \leq r \leq \max\left(0, \left\lfloor \frac{C - X}{M - \eta x} \right\rfloor\right), \quad s = \frac{C + \eta rx}{Mr + X}.$$

Indeed $s \geq 1$ is exactly the stated upper restriction on r . Testing integrality and the gcd completes the enumeration. This removes factorization of b for packet existence, not necessarily for radical weights or the full FCRT optimization; nor is it uniformly faster than subset enumeration.

126. SHARP DYADIC THRESHOLDS

Assume now $M = 2^e \parallel n$, $e \geq 2$. Then a, x, X, C are odd, so the normal form gives $r + s$ odd and excludes $(r, s) = (1, 1)$.

Theorem 126.1 (Plus threshold). *For $n = a + b$, the existence of a compatible proper packet requires*

$$n \geq M(2M + 1 - 2a).$$

For each fixed odd $a \geq 1$ infinitely many complete dyadic sources attain equality.

Proof. For $r \geq 2$,

$$Mrs + sX - rx - (2M + X - 2x) = (r - 2)(M - x) + (s - 1)(Mr + X) \geq 0.$$

For $r = 1$, $s \geq 2$, and the same difference equals $(s - 2)(M + X) + X + x > 0$. Hence $C \geq 2M + X - 2x \geq 2M + 1 - 2a$. Choose $L \geq 1$ with $2^L \equiv 1 \pmod{a(2a + 1)}$ and let $M = 2^{kL-1} > a$

with exponent at least two. Put $b = (M - a)(2M + 1)$. Then $a + b = M(2M + 1 - 2a)$ and the quotient by M is odd. The congruence $2M \equiv 1 \pmod{a(2a + 1)}$ gives $\gcd(a, b) = 1$. Also a common divisor of $M - a$ and $2M + 1$ divides $2a + 1$, while $2M + 1 \equiv 2 \pmod{2a + 1}$; that gcd is one. The packet $(x, X, y, Y) = (a, 1, M - a, 2M + 1)$ is genuine and has $s = 1, r = 2$. $\square$

Define

$$\mu(a) = \min_{x|a, \gcd(x, a/x)=1} (2x + a/x).$$

Theorem 126.2 (Difference threshold). *For $n = b - a$, a compatible proper packet requires $n \geq M(2M + \mu(a))$. When $a = 1$ or a is an odd prime power, $\mu(a) = a + 2$, and equality is attained infinitely often by actual complete dyadic sources.*

Proof. For fixed x, X , $C = Mrs + sX + rx$. If $r \geq 2$, then $C \geq 2M + X + 2x$; if $r = 1, s \geq 2$, then $C \geq 2M + 2X + x$. Both candidate minima are covered by minimizing $2x + X$ over unitary divisors, since exchanging x, X preserves that set. If a is a prime power, its unitary divisors are $1, a$, giving $\mu(a) = a + 2$; the formula also holds at $a = 1$. Choose $L \geq 1$ such that $2^L \equiv 1 \pmod{a|a-2|}$, omitting modulus-one factors, and let $M = 2^{kL} > a$ have exponent at least two. Use $(x, X, y, Y) = (1, a, M + 1, 2M + a)$, so $s = 1, r = 2$ and $n = M(2M + a + 2)$. Both factors of $b = (M + 1)(2M + a)$ are congruent to two modulo the odd integer a , so $\gcd(a, b) = 1$. Their gcd divides $a - 2$ and also $M + 1 \equiv 2 \pmod{a-2}$; hence it is one. The quotient $2M + a + 2$ is odd, proving completeness of the dyadic source. These infinitely many choices attain equality. $\square$

127. ARITHMETIC DERIVATIVE IMAGE AND TRANSVERSE SIZE

For a fixed primitive triple $a + b = c$ choose integer weights λ_p and define

$$\partial_\lambda(n) = \sum_{p|n} \frac{n}{p} v_p(n) \lambda_p, \quad H(\lambda) = \max_{p|abc} |\lambda_p|/p.$$

Only require $\partial(a) + \partial(b) = \partial(c)$ at this triple. Put

$$W = a\partial(b) - b\partial(a), \quad R = \text{rad}(abc), \quad D = abc/R.$$

For $n > 1$ let G_n be the gcd of the coefficients $(n/p)v_p(n)$, and set $G_1 = 0$.

Theorem 127.1 (Exact image). *The set of all such W is $I\mathbb{Z}$, with*

$$I = \frac{\gcd(cG_aG_b, bG_aG_c, aG_bG_c)}{\gcd(G_a, G_b, G_c)}.$$

Also $D \mid I$. Except for $(1, 1, 2)$ this image is nonzero.

Proof. The values of the derivative on each arm are exactly $G_n\mathbb{Z}$, by Bezout applied to its coefficients. Distinct arms have disjoint prime support. Thus admissible triples of derivative values are parametrized by the integer kernel of the row $A = (G_a, G_b, -G_c)$, and W is the row $B = (-bG_a, aG_b, 0)$ on that kernel. Let g be the gcd of A and $A' = A/g$. Choose an integer column u with $A'u = 1$. For any integer column v , $v - u(A'v)$ is in the kernel. Conversely the kernel is generated by the columns $A'_je_i - A'_ie_j$: if $A'v = 0$, summing $u_jv_i(A'_je_i - A'_ie_j)$ over i, j gives v . Their B -images are exactly the signed two-by-two minors of A', B . Their gcd is the displayed value of I . Each $n/\text{rad}(n)$ divides $\partial(n)$. The identities $W = a\partial(b) - b\partial(a) = a\partial(c) - c\partial(a)$ show that $a/\text{rad}(a)$, $b/\text{rad}(b)$ and $c/\text{rad}(c)$ all divide W . They are pairwise coprime, so $D \mid W$, whence $D \mid I$. For nontrivial triples at least two of a, b, c exceed one, making one of the displayed minors nonzero. At $(1, 1, 2)$ the additivity condition forces the remaining derivative to be zero. $\square$

Let $J = I/D > 0$ in the nontrivial case and $\Omega(n) = \sum_{p|n} v_p(n)$, with $\Omega(1) = 0$.

Theorem 127.2 (Three operator bounds). *Every admissible λ with $W \neq 0$ satisfies*

$$\begin{aligned} cJ &\leq RH(\lambda)(\Omega(a) + \Omega(b)), \\ bJ &\leq RH(\lambda)(\Omega(a) + \Omega(c)), \quad aJ \leq RH(\lambda)(\Omega(b) + \Omega(c)). \end{aligned}$$

Proof. The definition gives $|\partial(n)| \leq n\Omega(n)H$. Therefore $|W| \leq abH(\Omega(a) + \Omega(b))$. The other two representations using additivity give analogous bounds with ac and bc . Since a nonzero image value has absolute value at least $I = abcJ/R$, divide to get the assertions. $\square$

128. A FULLY SPECIFIED TRANSVERSE OPTIMIZATION

Consider $a = 2$, $b = 3^{10}109 = 6436341$, $c = 23^5 = 6436343$. The support is $(2, 3, 109, 23)$ and the exact additivity row is

$$\lambda_2 + 21454470\lambda_3 + 59049\lambda_{109} - 1399205\lambda_{23} = 0.$$

Here $R = 15042$ and $I = D = 5508110403$. There is a nonzero tangent vector $(0, 3, -1090, 0)$ of norm ten, whereas the second operator bound gives every transverse vector norm at least $6561/92 > 71$. Thus every shortest nonzero vector of this actual integer kernel is tangent.

Theorem 128.1 (Exact optimum). *The minimum norm among admissible vectors with $W \neq 0$ is $1644/23$.*

Proof. The vector $(-141, 107, 79, 1644)$ satisfies the displayed row, has $W = D$, and has norm $1644/23$. Suppose a smaller transverse vector existed. The image theorem and the second operator bound force $W = \pm D$, since $6ac(1644/23) < 2D$. Change signs to take $W = D$. Additivity then gives

$$W = 23^4(10\lambda_{23} - 23\lambda_2), \quad 10\lambda_{23} - 23\lambda_2 = 19683.$$

The strict norm bound gives $|\lambda_2| \leq 142$ and $|\lambda_{23}| \leq 1643$. Modulo ten, $\lambda_2 \equiv 9$, so $\lambda_2 \geq -141$. The left side is at most $16430 + 3243 = 19673 < 19683$, a contradiction. $\square$

129. THE EXCESS-CONTACT MODULE AND THE ACTUAL REMAINING GATE

Since $I = DJ$, reduction modulo D gives the exact sequence

$$0 \longrightarrow D\mathbb{Z}/I\mathbb{Z} \longrightarrow \mathbb{Z}/I\mathbb{Z} \longrightarrow \mathbb{Z}/D\mathbb{Z} \longrightarrow 0, \quad D\mathbb{Z}/I\mathbb{Z} \simeq \mathbb{Z}/J\mathbb{Z}.$$

Consequently $\log I = \log D + \log J$. This adds a nonnegative index; it does not bound $\log D$. In the preceding triple $J = 1$ and $c/R > 427$, refuting the shortcut $c/R \leq J$.

If for every $\eta > 0$ one could independently find a uniformly bounded transverse certificate $H(\lambda) \leq C_\eta JR^\eta$, the first operator bound would imply $c \ll_\eta R^{1+\eta} \log c$. For a prescribed $\varepsilon > 0$, choose $\eta = \varepsilon/2$ and absorb $\log c \leq A_\delta c^\delta$ with $\delta = \varepsilon/(2(1+\varepsilon))$; then $(1+\eta)/(1-\delta) = 1+\varepsilon$. This is only a conditional implication. The later transverse-lifting supplement proves that its hypothesis is in fact at full abc strength. The packet normal form, exact image and the finite optimum do not prove that global hypothesis. They neither close IUT's all-place comparison nor any unrestricted radical estimate.

INTEGRATED CHECKPOINT: QUANTITATIVE TRANSVERSE INTEGER LIFTING

Full mathematical proof body of the historical checkpoint. Historical execution statements retain their original scope.

Keywords. Arithmetic derivations; radical; Chinese remainder theorem; integer lattices; Wronskian; formal verification.

130. PROBLEM, SCOPE, AND PRINCIPAL RESULT

A primitive triple consists of positive integers a, b, c with $a + b = c$ and $\gcd(a, b) = 1$. Its entries are pairwise coprime. Put

$$R = \text{rad}(abc), \quad D = \frac{abc}{R}, \quad \Omega(n) = \sum_{p|n} v_p(n), \quad \Omega(1) = 0.$$

Standard abc asserts that for each $\varepsilon > 0$ there is $K_\varepsilon > 0$, independent of the triple, such that

$$(560) \quad c \leq K_\varepsilon R^{1+\varepsilon}.$$

The triple $(1, 1, 2)$ satisfies this with $K_\varepsilon = 1$ and will be excluded from transverse statements.

For integer weights $\lambda = (\lambda_p)_{p|abc}$, define

$$(561) \quad \partial_\lambda(n) = \sum_{p|n} \frac{n}{p} v_p(n) \lambda_p, \quad H(\lambda) = \max_{p|abc} \frac{|\lambda_p|}{p}.$$

Weights outside the displayed support can be set to zero. This operator satisfies the Leibniz identity by additivity of valuations, but is not assumed additive on arbitrary pairs. We impose only

$$\mathcal{L} = \{\lambda \in \mathbb{Z}^{\{p:p|abc\}} : \partial_\lambda(a) + \partial_\lambda(b) = \partial_\lambda(c)\}.$$

The arithmetic Wronskian is

$$W(\lambda) = a\partial_\lambda(b) - b\partial_\lambda(a).$$

Write $G_1 = 0$ and, for $n > 1$,

$$G_n = \gcd \left(\frac{n}{p} v_p(n) \right)_{p|n}.$$

The exact image calculation from the preceding checkpoint [110] is re-proved in Section 132:

$$(562) \quad W(\mathcal{L}) = I\mathbb{Z}, \quad I = \frac{\gcd(cG_aG_b, bG_aG_c, aG_bG_c)}{\gcd(G_a, G_b, G_c)}, \quad J = I/D \in \mathbb{Z}_{>0}.$$

For $m \neq 0$ let H_m be the least norm with $W = mI$, and set

$$H_\perp = \min_{\lambda \in \mathcal{L}, W(\lambda) \neq 0} H(\lambda).$$

These minima exist: the image is nonzero, every bounded weight box contains finitely many integer vectors, and a nonempty set of such vectors has a least norm.

Theorem 130.1 (Uniform additive lifting bound). *For every nontrivial primitive triple, define*

$$(563) \quad h_0 = \frac{J}{R} \max \left\{ \frac{c}{\Omega(a) + \Omega(b)}, \frac{b}{\Omega(a) + \Omega(c)}, \frac{a}{\Omega(b) + \Omega(c)} \right\} > 0.$$

There is an explicit nonnegative $E = E(a, b, c)$, specified in Theorem 132.4, such that for every nonzero integer m ,

$$(564) \quad |m|h_0 \leq H_m \leq |m|h_0 + E, \quad E \leq \frac{1}{2}(\log_2 c)^2 + \log_2 c.$$

In particular

$$(565) \quad h_0 \leq H_\perp \leq H_1 \leq h_0 + \frac{1}{2}(\log_2 c)^2 + \log_2 c.$$

The upper bound is realized by a constructive procedure given the three prime factorizations. It is not a claim about the complexity of obtaining those factorizations.

The theorem isolates the rounding cost from the scalar term h_0 , which itself contains c/R . Section 133 proves that the previously proposed subpower bound for $H_\perp/J$ is equivalent to abc . That equivalence is an obstruction audit, not a proof of either assertion.

131. LOCAL CONTENT AND A BOUNDED LIFTING CONSTRUCTION

For $n > 1$, write $n = \prod_{p|n} p^{e_p}$ and define

$$(566) \quad \begin{aligned} r_n &= \text{rad}(n), & D_n &= n/r_n, & k_n &= \gcd_{p|n} e_p, & f_p &= e_p/k_n, \\ \rho_n &= \prod_{\substack{p|n \\ p \nmid f_p}} p, & r'_n &= r_n/\rho_n, & S_n &= \sum_{p|n} f_p = \Omega(n)/k_n. \end{aligned}$$

Thus $\gcd_p f_p = 1$. The factor ρ_n records which normalized exponents are divisible by their own primes; this is its definition, not an assumed distributional property.

Lemma 131.1 (Exact local content). *The attainable values of $\partial_\lambda(n)$ are exactly $G_n\mathbb{Z}$, where*

$$(567) \quad G_n = D_n k_n \rho_n.$$

Moreover

$$(568) \quad h_n := \frac{G_n}{\gcd(G_n, n)} = \frac{k_n}{\gcd(k_n, r'_n)} \mid k_n.$$

Proof. The image of an integer linear form is the ideal generated by its coefficients, by Bézout's identity. After dividing the coefficients by $D_n k_n$, their gcd is the gcd of $f_p r_n/p$. A prime outside r_n does not divide that gcd because $\gcd_p f_p = 1$. For $p \mid r_n$, the p -valuation of the gcd is zero when $p \nmid f_p$. When $p \mid f_p$, all the coefficients are divisible by p , and some f_q with $q \neq p$ is not divisible by p ; the coefficient at q then has p -valuation exactly one. Thus the gcd is ρ_n , proving (567). Since r_n is squarefree,

$$\gcd(k_n \rho_n, r_n) = \rho_n \gcd(k_n, r'_n),$$

which gives (568). $\square$

For a prime power $n = p^e$, one has $k_n = e$, $f_p = 1$, $\rho_n = 1$, $G_n = ep^{e-1}$ and $h_n = e/\gcd(e, p)$. At $n = 12 = 2^2 \cdot 3$, one instead has $k_n = 1$, $\rho_n = 2$ and $G_n = 4$. At $n = 108 = 2^2 \cdot 3^3$, every support prime contributes to ρ_n , and $G_n = 108$. These examples include the one-coordinate and all-resonant extremes of the construction below.

Lemma 131.2 (Bounded correction by a residue graph). *Let $m_1, \dots, m_t$ be positive integers of gcd one, with $M = \max m_i$ and $m = \min m_i$. For any integer u , choose an index i_0 with $m_{i_0} = m$. There are integers z_i with $\sum m_i z_i = u$, all $z_i \geq 0$ for $i \neq i_0$, such that*

$$\sum_{i \neq i_0} z_i \leq m - 1, \quad |z_{i_0}| \leq \frac{|u| + M(m - 1)}{m}.$$

Proof. On $\mathbb{Z}/m\mathbb{Z}$ put directed edges $x \mapsto x + m_i$. Every residue is reachable from zero: the steps generate the group since their gcd together with m is one, and in a finite group the inverse of a step is a nonnegative multiple of that step. A shortest path to $u \bmod m$ has no repeated vertex and has at most $m - 1$ edges. It uses no self-loop, in particular no step m_{i_0} . Let its step counts be z_i and its integer sum be u_0 . Then $0 \leq u_0 \leq M(m - 1)$ and $u_0 \equiv u \pmod{m}$. Set $z_{i_0} = (u - u_0)/m$. This proves all assertions, including $m = 1$, when the path is empty. $\square$

Theorem 131.3 (Local lifting near the real minimizer). *For $n > 1$ and every $d \in G_n\mathbb{Z}$, integer weights realizing $\partial_\lambda(n) = d$ can be chosen so that*

$$(569) \quad \max_{p|n} \left| \frac{\lambda_p}{p} - \frac{d}{n\Omega(n)} \right| \leq S_n.$$

When n is a prime power, the left side can be zero.

Proof. Suppress the subscript n , set $q = d/(n\Omega(n))$, and write $u = d/G \in \mathbb{Z}$. Equation (567) gives

$$\frac{d}{nk} = \frac{u}{r'}.$$

For each regular prime $p \mid r'$ choose λ_p in the residue class

$$(570) \quad \lambda_p \equiv u(f_p(r'/p))^{-1} \pmod{p}$$

such that $|\lambda_p/p - q| \leq 1/2$. This is possible by choosing the nearest point in an arithmetic progression of spacing one in the normalized coordinate. For each remaining prime, choose an integer nearest to pq , giving error at most $1/(2p) \leq 1/2$.

The residual

$$v = \frac{d}{nk} - \sum_{p|n} \frac{f_p \lambda_p}{p}$$

is an integer. Indeed, the terms at the primes of ρ are integers; multiplication by r' and reduction at each regular prime gives zero by (570). The pairwise coprimeness of these primes gives the assertion, also when $r' = 1$. Moreover

$$(571) \quad |v| = \left| \sum_{p|n} f_p \left(q - \frac{\lambda_p}{p} \right) \right| \leq S/2.$$

Put

$$m_p = \begin{cases} f_p, & p \mid r', \\ f_p/p, & p \mid \rho. \end{cases}$$

These are positive integers of gcd one, since every m_p divides f_p and $\gcd_p f_p = 1$. Apply Lemma 131.2 to v . Correct a regular coordinate by $\lambda_p \mapsto \lambda_p + pz_p$, and a remaining coordinate by $\lambda_p \mapsto \lambda_p + z_p$. The change in the normalized derivative is exactly $\sum m_p z_p = v$.

For the distinguished coordinate, using $M \leq S$, (571), and $m \leq S$ gives

$$|z_{i_0}| \leq \frac{S}{2m} + \frac{S(m-1)}{m} = S - \frac{S}{2m} \leq S - \frac{1}{2}.$$

Every other z_p is at most $m-1 \leq S-1$. The change in each normalized coordinate is at most $|z_p|$. Adding its initial error at most $1/2$ proves (569).

For $n = p^e$, the only allowed value is $d = ep^{e-1}u$. Taking $\lambda_p = u$ gives $\lambda_p/p = d/(ne)$ exactly. $\square$

The construction uses a graph with $m \leq S_n$ vertices after the prime factorization is known. It provides a norm guarantee, not necessarily the local norm optimum. Define the available error bound

$$(572) \quad \kappa_n = \begin{cases} 0, & n = 1 \text{ or } n \text{ is a prime power,} \\ \Omega(n)/k_n, & \text{otherwise.} \end{cases}$$

132. WRONSKIAN IMAGE, REAL RELAXATION, AND INTEGER FIBRES

Lemma 132.1 (Image of a linear form on an integer kernel). *If an integer row $A = (A_1, \dots, A_t)$ is primitive and B is an integer row of the same size, then*

$$B(\ker_{\mathbb{Z}} A) = \gcd_{i < j} (A_i B_j - A_j B_i) \mathbb{Z}.$$

Proof. Choose $v \in \mathbb{Z}^t$ with $A \cdot v = 1$. For every $x \in \ker_{\mathbb{Z}} A$,

$$x = \sum_{i < j} (v_i x_j - v_j x_i) (A_i e_j - A_j e_i).$$

This follows by comparing each coordinate, or by expanding $(A \cdot v)x - (A \cdot x)v$. Thus the displayed pair vectors generate the integer kernel, not just its rational span. Applying B proves the image formula by Bézout's identity. $\square$

Proposition 132.2 (Exact arithmetic image and divisibility). *Equation (562) holds, and $I > 0$ except at $(1, 1, 2)$.*

Proof. The disjoint prime supports let the derivative values range independently over $G_a\mathbb{Z}, G_b\mathbb{Z}, G_c\mathbb{Z}$. Divide the row $(G_a, G_b, -G_c)$ by $g = \gcd(G_a, G_b, G_c)$ and apply Lemma 132.1 with $B = (-bG_a, aG_b, 0)$. Its three minors give (562). At least one of a, b is greater than one, so at least one of aG_bG_c, bG_aG_c is positive.

For every n , $D_n \mid \partial_\lambda(n)$. The identities

$$W = a\partial_\lambda(b) - b\partial_\lambda(a) = a\partial_\lambda(c) - c\partial_\lambda(a)$$

show that each of D_a, D_b, D_c divides W . These integers are pairwise coprime, so their product D divides every W , hence divides the positive generator I . $\square$

Proposition 132.3 (Exact real minimum). *Allow real weights, retain triple additivity, and fix $W = I$. Then the least norm is exactly h_0 in (563). For $W = mI$ the real minimum is $|m|h_0$.*

Proof. For a fixed aggregate derivative value d_n , the condition $H \leq h$ implies $|d_n/n| \leq \Omega(n)h$. Conversely this inequality is sufficient: take every coordinate $\lambda_p/p = d_n/(n\Omega(n))$ when $n > 1$; at $n = 1$ require $d_n = 0$.

Write $x = d_a/a$. Additivity and $W = I$ force

$$d_b/b = x + I/(ab), \quad d_c/c = x + I/(ac).$$

With offsets $o = (0, I/(ab), I/(ac))$, the three admissible intervals for x are

$$[-\Omega(n)h - o_n, \Omega(n)h - o_n], \quad n = a, b, c.$$

Intervals on a line have a common point precisely when they overlap pairwise. Their least admissible radius is

$$\max_{i < j} \frac{|o_i - o_j|}{\Omega(n_i) + \Omega(n_j)}.$$

All denominators are positive in the nontrivial cases; a zero $\Omega(n)$ simply produces a point interval. Since $I = abcJ/R$, this is (563). Scaling all weights proves the last assertion. $\square$

Theorem 132.4 (Discrete fibre and global lifting). *For $a, b > 1$, put*

$$\ell = \text{lcm}(h_a, h_b, h_c), \quad E = \frac{\ell}{2 \min\{\Omega(a), \Omega(b), \Omega(c)\}} + \max\{\kappa_a, \kappa_b, \kappa_c\}.$$

If $a = 1$ or $b = 1$, instead put $E = \max\{\kappa_a, \kappa_b, \kappa_c\}$. Then (564) holds for every $m \neq 0$.

Proof. There is at least one integral aggregate triple $d = (d_a, d_b, d_c)$ with $W = mI$ by Proposition 132.2. Two such aggregate triples differ by

$$(d'_a - d_a, d'_b - d_b, d'_c - d_c) = t(a, b, c), \quad t \in \mathbb{Z}.$$

Indeed $a(d'_b - d_b) = b(d'_a - d_a)$ and $\gcd(a, b) = 1$. For each n , the new derivative remains in $G_n\mathbb{Z}$ exactly when $h_n \mid t$. Thus the allowed values of $x = d_a/a$ form one coset of $\ell\mathbb{Z}$.

Choose a real minimizer from Proposition 132.3. A point in the allowed coset lies at distance at most $\ell/2$ from it. The resulting aggregate values satisfy

$$\max_{n=a,b,c} \frac{|d_n|}{n\Omega(n)} \leq |m|h_0 + \frac{\ell}{2 \min_n \Omega(n)}.$$

Apply Theorem 131.3 independently on the disjoint prime supports. Its error at each n is at most κ_n , so the combined vector proves the upper bound.

When $a = 1$ or $b = 1$, the vanishing derivative at one fixes the real aggregate triple uniquely. Proposition 132.2 already makes those values attainable integrally. There is no period-rounding error in this case. The lower bound in all cases is Proposition 132.3. $\square$

Proof of Theorem 130.1. For $n \leq c$, $k_n \leq \Omega(n) \leq L := \log_2 c$. If $a, b > 1$, then $\ell \leq k_a k_b k_c$ by Lemma 131.1. Dividing this product by the smallest $\Omega(n)$ leaves a bound at most the product of the other two exponent sums, hence at most L^2 . Also $\kappa_n \leq L$. Thus $E \leq L^2/2 + L$; the one-entry-equals-one case is stronger. Taking the minimum over levels gives (565). $\square$

Corollary 132.5 (Non-power entries and level truncation). *If $a, b, c > 1$ and $k_a = k_b = k_c = 1$, then $E \leq \log_2 c + 1/2$. In general an optimal nonzero level must satisfy*

$$1 \leq |m| \leq \left\lfloor \frac{H_1}{h_0} \right\rfloor \leq \left\lfloor 1 + \frac{E}{h_0} \right\rfloor.$$

In particular $h_0 > E$ implies that every transverse minimizer has $|m| = 1$.

Proof. The first assertion follows from $\ell = 1$. For the second, a level with $|m|h_0 > H_1$ cannot minimize the norm. The final assertion follows from $2h_0 > h_0 + E \geq H_1$. $\square$

133. THE REMAINING UNIFORM CONDITION IS EQUIVALENT TO ABC

Definition 133.1. The small transverse-derivative assertion, denoted SD, is

$$\forall \tau > 0 \exists C_\tau > 0 \forall \text{ nontrivial primitive triples}, \quad H_\perp \leq C_\tau J R^\tau.$$

No constant here may depend on the prime support, its size, the triple, or its exponent pattern.

Theorem 133.2 (Equivalence, not an unconditional height theorem). *Standard abc is equivalent to SD. It is also equivalent to the same assertion with H_1 in place of $H_\perp$.*

Proof. Assume SD. The first term of (563) gives

$$c \leq C_\tau R^{1+\tau} (\Omega(a) + \Omega(b)) \leq \frac{2C_\tau}{\log 2} R^{1+\tau} \log c.$$

Given $\varepsilon > 0$, set $\tau = \varepsilon/2$ and $\delta = \varepsilon/(2(1 + \varepsilon))$. Elementary maximization gives $\log x \leq x^\delta/(e\delta)$ for $x \geq 1$. Therefore

$$c^{1-\delta} \leq \frac{2C_\tau}{e\delta \log 2} R^{1+\tau}.$$

Since $(1 + \tau)/(1 - \delta) = 1 + \varepsilon$, raising to the positive power $1/(1 - \delta)$ proves (560). Enlarge the constant to one to include $(1, 1, 2)$.

Conversely, assume abc and fix $\tau > 0$. Choose $K \geq 1$, depending only on τ , such that $c \leq KR^{1+\tau/2}$. Each denominator in (563) is at least one, so $h_0/J \leq c/R \leq KR^{\tau/2}$. Moreover

$$\log_2 c \leq \frac{\log K + (1 + \tau/2) \log R}{\log 2}.$$

For $\sigma > 0$, $(\log R)^2 \leq 4R^\sigma/(e^2\sigma^2)$ for $R \geq 1$. Applying Theorem 130.1 with $\sigma = \tau/2$ and $J \geq 1$ yields

$$H_1/J \leq KR^{\tau/2} + E/J \leq C_\tau R^\tau$$

with a constant depending only on τ . This proves the strongest of the two transverse assertions. Finally $H_\perp \leq H_1$ gives their equivalence with abc . $\square$

The preceding sufficient condition in [110] therefore contains the full uniform height problem, even though its local integer-realization cost has now been controlled. The new object J cannot be regarded as a replacement for D : it is normalized out on both sides of the height inequality. No estimate for $c/R^{1+\varepsilon}$ has been obtained here independently of abc .

A counterexample to standard abc would require a fixed $\varepsilon > 0$ for which $c/R^{1+\varepsilon}$ is unbounded over primitive triples. A single high-quality triple, or either counterexample to an auxiliary optimization assertion below, does not meet that requirement.

134. EXACT OPTIMIZATION AND FAILED SHORTCUTS

134.1. A short transverse vector need not lie over a primitive image value.

Proposition 134.1. *For $5 + 7 = 12$, $I = 4$, $D = 2$, $J = 2$, and*

$$H_\perp = \frac{1}{5}, \quad H_1 = H_{-1} = \frac{2}{5}.$$

Every transverse minimizer has $W = 12$ or $W = -12$, hence level 3 or -3 .

Proof. In coordinates $(\lambda_2, \lambda_3, \lambda_5, \lambda_7)$, the actual derivative constraint and Wronskian are

$$\lambda_5 + \lambda_7 = 12\lambda_2 + 4\lambda_3, \quad W = 5\lambda_7 - 7\lambda_5.$$

The coefficient 12 includes the valuation $v_2(12) = 2$. Substitution gives

$$W = 4(15\lambda_2 + 5\lambda_3 - 3\lambda_5).$$

The vector $(0, -1, -2, -2)$ has $W = 4$ and norm $2/5$, so the image is exactly $4\mathbb{Z}$. The vector $(0, 0, -1, 1)$ has $W = 12$ and norm $1/5$.

Below radius $1/5$, the first three coordinates vanish and additivity forces the fourth to vanish. At radius at most $1/5$, the first two vanish, the remaining ones are opposites, and each is in $\{-1, 0, 1\}$. This proves the transverse minimum and the level assertion.

Below radius $2/5$, $\lambda_5 \in \{-1, 0, 1\}$. If $W = 4$, reduction modulo five gives $\lambda_5 \equiv -2 \pmod{5}$, impossible in that range. If $W = -4$, it gives $\lambda_5 \equiv 2 \pmod{5}$, likewise impossible. The primitive witness and its negative prove the claimed primitive-class minimum. $\square$

This refutes pointwise primitive-class optimality, but not the primitive-class upper bound in Theorem 130.1. Its proof and all-radius exclusion statements are implemented in `PrimitiveClass.lean`.

134.2. Reconstructing the large benchmark without a lattice search. The preceding checkpoint considered

$$2 + 3^{10} \cdot 109 = 23^5, \quad b = 6436341, \quad c = 6436343, \quad R = 15042, \quad I = D = 5508110403.$$

Here $J = 1$, $(\Omega(a), \Omega(b), \Omega(c)) = (1, 11, 5)$, $\ell = 5$ and $h_0 = 6561/92$. The new construction, using nearest CRT representatives and the residue graph, returns

$$(573) \quad (\lambda_2, \lambda_3, \lambda_{109}, \lambda_{23}) = (-141, 98, 3349, 1644), \quad H = \frac{1644}{23}.$$

Its aggregate values are $(-141, 2300293161, 2300293020)$ and its Wronskian is D . The earlier optimal witness was $(-141, 107, 79, 1644)$; the two differ by three times the tangent vector $(0, 3, -1090, 0)$. Thus the construction reaches the already proved optimum, with actual excess

$$\frac{1644}{23} - \frac{6561}{92} = \frac{15}{92}.$$

This instance does not assert that the construction is always pointwise optimal.

For completeness, the all-vector exclusion underlying the optimum is as follows. Additivity implies

$$10\lambda_{23} - 23\lambda_2 = 19683m, \quad W = 5508110403m, \quad m \in \mathbb{Z}.$$

At radius strictly below $1644/23$, $|\lambda_2| \leq 142$ and $|\lambda_{23}| \leq 1643$. Thus $m \in \{-1, 0, 1\}$. For $m = 1$, the congruence implies $\lambda_2 \equiv 9 \pmod{10}$, hence $\lambda_2 \geq -141$. Then the left side is at most $19673 < 19683$, a contradiction. The case $m = -1$ follows by changing signs. Hence every smaller vector has $W = 0$. The tangent vector has norm ten, so every shortest nonzero lattice vector is tangent. This disproves the shortcut that a shortest nonzero lattice vector is automatically transverse. These assertions, including the factorization and coefficient checks, are implemented in `TransverseBenchmark.lean`.

135. VERIFICATION, DEPENDENCIES, AND THE NEXT UNRESOLVED NODE

135.1. Exact computation. The programs use Python integer arithmetic and `Fraction`; no floating-point solver decides a congruence, a norm comparison, or an optimum. The local content formula is compared with a direct coefficient gcd. The aggregate constructor is checked by separately evaluating the derivative definition. A second real-minimum implementation enumerates the breakpoints of the piecewise-linear objective, rather than using the pairwise interval formula. Both methods use the same elementary arithmetic library; they are distinct implementations, not independent autonomous researchers.

Check	Completed cases
Local content profiles, $1 \leq n \leq 30000$	30,000
Local signed/large-value lifts, $n \leq 5000$	29,995
Generated large factorizations / additional lifts	1,200 / 2,400
Normalized primitive triples, $3 \leq c \leq 600$	54,749
Extra negative/positive levels with swapped addends	2,256
Nonzero points in the small example's unit box	86

The generated profiles include 695 with smallest correction step at least two, and four with all support primes resonant. Two full default replays produced byte-identical JSON. The file's SHA-256 is

9b1a73607615b4bf732e8c651c4b381e4c156fbd258d597ce239a0c8e94d196.

These are finite implementation checks. The universal statements in Sections 2–4 rest on the proofs, not on extrapolation from this table.

135.2. Formal scope and reproducibility. Two standalone modules contain 24 named theorems with 24 axiom queries, using Std. Both passed Lean 4.32.2 with warnings treated as errors in Actions run 33970484302, at source commit 3bf8bd92cef6bbd74f571eeb50035cc01271b1e4. The axiom union is `propext`, `Classical.choice`, `Quot.sound`; no `sorryAx` occurs. Positive rational norm balls are encoded by exact integer inequalities. The scope is Section 134 and a centered-residue lemma, not the general lifting theorem, real relaxation, or equivalence theorem.

The script `wsl/verify_lean.sh` installs an isolated, checksum-pinned Lean 4.32.2 and runs both modules without changing existing toolchains. A companion parser checks each reported axiom dependency against an explicit whitelist. The supplied verification record distinguishes the source check from the final integration check. Hosted Ubuntu execution is not execution on the user's own WSL; that machine was not accessible.

135.3. Proof dependency register.

Node	Dependencies	Status here
Local content $G_n = D_n k_n \rho_n$	Factorization, Bézout	Ordinary proof
Local bounded lifting	Content, CRT, residue graph	Ordinary proof
Exact Wronskian image	Primitive integer kernel	Ordinary proof; two instances Lean-checked
Global additive error bound	Local lifting, fibre period, intervals	Ordinary proof
Smallest vectors need not be transverse	Large exact benchmark	Exact Lean certificate
Primitive image values need not minimize	$5 + 7 = 12$	Exact Lean certificate
$SD \iff abc$	Global bound, elementary log estimates	Ordinary equivalence proof
Uniform SD or uniform abc	Independent height input still required	Unresolved

The baseline was 1e791475504efdcfc951b85ed6ccc14f9532d603. The existing FCRT/signed-endpoint and unitary-face/multi-output routes [111, 112] remain active. Their first-appearance problem and the IUT/geometric gates are neither assumed nor closed here; new proofs in this article concern the derivative route.

The uniform assertion in Section 133 remains unresolved. By Theorem 133.2, proving it without an equivalent conjectural input is exactly an *abc* proof. Neither such a proof, external peer review, nor a full formalization is asserted; priority outside these project sources is unverified.

INTEGRATED CHECKPOINT: POWER-RADICAL DESCENT AND FIRST-DEPTH OBSTRUCTIONS

Full mathematical proof body of the historical checkpoint. Historical execution statements retain their original scope.

Keywords: *abc* conjecture; radicals; multiplicative order; valuation lifting; arithmetic derivatives; Lean.

136. PROBLEM, STARTING POINT, AND SCOPE

For a positive integer N , put $\text{rad}(N) = \prod_{p|N} p$. A primitive positive triple satisfies $a + b = c$ and $\gcd(a, b) = 1$, hence is pairwise coprime. The target is

$$(574) \quad \forall \varepsilon > 0 \exists K_\varepsilon > 0 \forall (a, b, c), \quad c \leq K_\varepsilon \text{rad}(abc)^{1+\varepsilon}.$$

The constant must be independent of the triple, its prime support, and every auxiliary base or exponent. None of the unrestricted estimates equivalent to (574) is used as a premise in our unconditional results.

The starting repository snapshot is commit `d3f7b33b1115538920cb5fcd851a97f0c3a26a3d` of `wyx66624/ABCConjecture`. Its previous transverse-lifting checkpoint [113] controls integer realization of arithmetic derivatives, but identifies its proposed uniform small-transverse-derivative condition as equivalent to *abc*. Its earlier exponent-lifting checkpoint [114] separates repeated lifting from first appearance for $x^n - 1$. We extend the latter to relative radical budgets, descent between actual triples, and base changes. An independent energy argument sharpens local realization without claiming to bound the original scalar defect.

For $m \geq 1$ define the cleared defect

$$(575) \quad \mathcal{Q}_m(a, b, c) = \frac{c^m}{\text{rad}(abc)^{m+1}}.$$

Uniform boundedness of (575) for every positive integer m is equivalent to (574). Indeed, *abc* with $\varepsilon = 1/m$ gives the bound after taking the m th power. Conversely, given $\varepsilon > 0$, choose m with $1/m \leq \varepsilon$, take the m th root, and use $\text{rad}(abc) \geq 1$. The exponent choice depends only on ε .

137. VALUATION LIFTING AND THE EXACT RELATIVE RADICAL

Throughout this section, $x > y \geq 1$ and $\gcd(x, y) = 1$.

Lemma 137.1 (Elementary valuation lifting). *If an odd prime p divides $U - V$ and $p \nmid UV$, then*

$$v_p(U^t - V^t) = v_p(U - V) + v_p(t) \quad (t \geq 1).$$

The formula also holds with $U + V$ and $U^t + V^t$ when t is odd. If U, V are odd and t is even, then

$$v_2(U^t - V^t) = v_2(U - V) + v_2(U + V) + v_2(t) - 1.$$

For odd t , $v_2(U^t - V^t) = v_2(U - V)$, and $v_2(U^t + V^t) = v_2(U + V)$.

Proof. Write $U = V + p^h w$ with $p \nmid Vw$. Raising to a positive exponent prime to p and expanding shows a unique lowest-valuation term $tV^{t-1}p^h w$, of valuation h . Raising to p , the first term has valuation $h + 1$; the middle binomial terms have valuation at least $1 + 2h > h + 1$, and the last has valuation $ph > h + 1$ for odd p , except $p = 3, h = 1$ where it is still $3 > 2$. Iteration and the decomposition $t = p^s u$ prove the odd-prime formula. Using $-V$ proves the odd sum case. For two, odd geometric or alternating geometric sums have odd length and are odd modulo two. The first squaring contributes $v_2(U - V) + v_2(U + V)$. Every later factor $U^{2^j} + V^{2^j}$, $j \geq 1$, is 2 modulo 8, and contributes exactly one. An odd remaining exponent contributes zero additional valuation. $\square$

137.1. **Difference powers.** Let

$$b_n = x^n - y^n, \quad g_n = \frac{x^n - y^n}{x - y}, \quad R_1 = \text{rad}(xy(x - y)), \quad R_n = \text{rad}(xyb_n).$$

The actual lifted triple is (y^n, b_n, x^n) and has radical R_n . A prime divisor of b_n is called *new relative to this base pair* when it does not divide $x - y$. Every such prime is odd. For a new prime set

$$d_p = \text{ord}_{\mathbb{F}_p^\times}(xy^{-1}), \quad \alpha_p = v_p(x^{d_p} - y^{d_p}).$$

The inverse exists because $p \nmid xy$. The order is greater than one, divides both n and $p - 1$, and is prime to p . Define

$$(576) \quad T_n(x, y) = \prod_{\substack{p|b_n \\ p \nmid x-y}} p^{\alpha_p-1}, \quad L_n(x, y) = \prod_{p|b_n} p^{v_p(n)},$$

$$(577) \quad E_n(x, y) = \begin{cases} 2^{v_2(x+y)-1}, & x, y \text{ odd and } n \text{ even,} \\ 1, & \text{otherwise,} \end{cases} \quad B_n(x, y) = E_n L_n T_n.$$

Empty products are one. Notice that $L_n \mid n$.

Theorem 137.2 (Exact relative radical identity). *For every $n \geq 1$,*

$$(578) \quad R_n B_n = R_1 g_n.$$

In particular, $R_1 \mid R_n$ and R_n/R_1 is the product of the new primes, each counted once.

Proof. All primes in $x - y$ persist in b_n , and neither integer has a prime factor in common with xy . For an odd old prime, Lemma 137.1 gives $v_p(g_n) = v_p(n)$. For a new prime it gives

$$v_p(g_n) = v_p(b_n) = \alpha_p + v_p(n).$$

Dividing g_n by the product of new primes thus leaves $p^{\alpha_p-1+v_p(n)}$ at a new prime and $p^{v_p(n)}$ at an odd old prime. If two occurs, it is old; its exponent in this quotient is $v_2(n)$ for odd n (both are zero) and $v_2(n) + v_2(x + y) - 1$ for even n . This is exactly (577). These identities exhaust the prime divisors of g_n . $\square$

Remark 137.3 (A necessary two-adic correction). At $(x, y, n) = (7, 1, 2)$ one has $g_n = 8$, $R_1 = R_n = 42$, $L_n = 2$, $T_n = 1$, and $E_n = 4$. Omitting E_n gives a false radical identity. This correction cannot be silently included in the factor claimed to divide n .

137.2. **Odd sum powers.** For odd n put

$$b_n^+ = x^n + y^n, \quad g_n^+ = \frac{x^n + y^n}{x + y}, \quad R_1^+ = \text{rad}(xy(x + y)), \quad R_n^+ = \text{rad}(xyb_n^+).$$

For $p \mid b_n^+$, $p \nmid x + y$, let $d_p = \text{ord}(-xy^{-1})$. This is an odd divisor of n , greater than one. Use $\alpha_p = v_p(x^{d_p} + y^{d_p})$ and define T_n^+, L_n^+ as in (576) with the new seed and sum. Set $B_n^+ = L_n^+ T_n^+$.

Proposition 137.4. *For odd n , $L_n^+ \mid n$ and*

$$(579) \quad R_n^+ B_n^+ = R_1^+ g_n^+.$$

Proof. For a new prime, apply the odd-sum part of Lemma 137.1 at d_p and n/d_p , both odd. Its valuation is $\alpha_p + v_p(n)$ because $d_p \mid p - 1$. For an old odd prime, the valuation increment is $v_p(n)$. Two, if present, is old and its valuation does not change. Counting the new primes once proves the identity exactly as before. $\square$

138. SAME-CONSTANT DESCENT AND COMPOSITION

Write $\mathcal{Q}_m^-(x, y; n) = \mathcal{Q}_m(y^n, x^n - y^n, x^n)$ and $\mathcal{Q}_m^+(x, y; n) = \mathcal{Q}_m(x^n, y^n, x^n + y^n)$. In this section $m \geq 1$.

Theorem 138.1 (Exact defect transfer). *For difference powers,*

$$(580) \quad \frac{\mathcal{Q}_m^-(x, y; n)}{\mathcal{Q}_m^-(x, y; 1)} = \frac{x^{m(n-1)} B_n^{m+1}}{g_n^{m+1}} \leq \frac{B_n^{m+1}}{x^{n-1}}.$$

For odd sum powers,

$$(581) \quad \frac{\mathcal{Q}_m^+(x, y; n)}{\mathcal{Q}_m^+(x, y; 1)} = \frac{(B_n^+)^{m+1}}{g_n^+}.$$

Consequently a bound $\mathcal{Q}_m \leq K$ at the seed transfers with the same constant if $B_n^{m+1} \leq x^{n-1}$ in the difference case, or $(B_n^+)^{m+1} \leq g_n^+$ in the sum case.

Proof. Substitute $R_n/R_1 = g_n/B_n$ in the quotient of the two cleared defects. In the difference case the height ratio is x^{n-1} ; the geometric sum $g_n = \sum_{j=0}^{n-1} x^{n-1-j} y^j$ is at least x^{n-1} . In the sum case the height ratio equals g_n^+ , so its m th power cancels all but one of the $m+1$ radical-ratio powers. Every denominator used here is positive. $\square$

Corollary 138.2 (An obstruction in a minimal counterexample). *Fix $K > 0$. If a triple with minimal height among those satisfying $\mathcal{Q}_m > K$ has the difference-power form with $n \geq 2$, then*

$$B_n^{m+1} > x^{n-1}.$$

For odd n this implies

$$(582) \quad T_n > \frac{x^{(n-1)/(m+1)}}{n}.$$

In the odd sum-power form with $n \geq 3$, it must satisfy $(B_n^+)^{m+1} > g_n^+ \geq x^{n-1}/2$.

Proof. The seed is a primitive triple of smaller height, so its cleared defect is at most K . Apply Theorem 138.1 contrapositively. For odd n , $B_n = L_n T_n \leq n T_n$. Finally $g_n^+ = (x^n + y^n)/(x + y) \geq x^{n-1}/2$. $\square$

Remark 138.3 (What descent does not give). A smallest counterexample to a fixed inequality is a legitimate conditional object. Its nonexistence is not proved here. Most triples do not have two entries with a common exponent greater than one, so these power maps do not supply a predecessor for every triple. Furthermore, (582) is not contradicted by our bounds. The next section constructs genuinely large first-appearance factors after common-power normalization.

Proposition 138.4 (Complete-budget composition). *For difference powers and positive integers u, v ,*

$$B_{uv}(x, y) = B_u(x, y) B_v(x^u, y^u).$$

For odd u, v the corresponding sum identity holds.

Proof. In the difference case, $g_{uv}(x, y) = g_u(x, y) g_v(x^u, y^u)$, and the intermediate radical is both $R_u(x, y)$ and $R_1(x^u, y^u)$. Multiply the two exact radical identities and cancel the positive intermediate radical. For sums, both quotients are integral when u, v are odd, and the same argument applies. This proves composition without assigning a prime's contribution twice. $\square$

First-appearance depth alone does not obey this composition law. For an odd prime with order d at (x, y) , the order at (x^u, y^u) is $d/\gcd(d, u)$. If $p \mid x^{uv} - y^{uv}$ but $p \nmid x^u - y^u$, its new first valuation is

$$\alpha_p + v_p(u),$$

because $p \nmid d$. Dividing the two first-appearance products gives the more precise identity

$$(583) \quad T_v(x^u, y^u) = \frac{T_{uv}(x, y)}{T_u(x, y)} \prod_{\substack{p \mid x^{uv} - y^{uv} \\ p \nmid x^u - y^u}} p^{v_p(u)}.$$

The final product divides u . All primes in it are odd. For example, $\text{ord}_7(2) = 3$ and $v_7(2^3 - 1) = 1$, whereas for the base 2^{7^j} and outer exponent three the first valuation at seven is $j + 1$.

A canonical way to remove this ambiguity is to write $x = X^s, y = Y^s$, where s is the greatest common divisor of all the positive prime exponents in x and y (ignore the empty profile for $y = 1$). Unique factorization proves uniqueness and maximality of s . The pair (X, Y) is not a simultaneous nontrivial power. Formula (583) keeps track of what this normalization reallocates. The obstruction below shows that normalization is insufficient to make T sublinear.

138.1. A uniform estimate on a restricted large-exponent stratum.

Theorem 138.5. *Fix $A \geq 0$ and $\varepsilon > 0$. For odd $n \geq 2(1 + \varepsilon)/\varepsilon$, suppose that the relevant first-appearance product satisfies $T_n \leq n^A$ (or $T_n^+ \leq n^A$). Then the corresponding primitive power triple has height C and radical R satisfying*

$$C \leq K_{\varepsilon, A} R^{1+\varepsilon},$$

where, with $b = (A + 1)(1 + \varepsilon)$, one may take

$$K_{\varepsilon, A} = 2^{2+\varepsilon} \max \left\{ 1, \left(\frac{2b}{e\varepsilon \log 2} \right)^b \right\}.$$

This constant is independent of x, y, n within the stated stratum.

Proof. The exact balance and $L_n \mid n$ give $R \geq x^{n-1}/n^{A+1}$ for differences and $R \geq x^{n-1}/(2n^{A+1})$ for sums. In both cases $C \leq 2x^n$. Thus

$$\frac{C}{R^{1+\varepsilon}} \leq 2^{2+\varepsilon} n^b x^{1+\varepsilon-\varepsilon n} \leq 2^{2+\varepsilon} n^b 2^{-\varepsilon n/2}.$$

The maximum of $t^b e^{-ct}$ over $t > 0$ is $(b/(ec))^b$; use $c = \varepsilon \log 2/2$. $\square$

The hypothesis on T is not proved for all bases. Even among powers this theorem leaves all bounded outer exponents with unbounded bases untreated. Those families cannot be absorbed into a finite exceptional list.

139. A NORMALIZED FIRST-APPEARANCE OBSTRUCTION OF LINEAR SIZE

Let $\phi(X) = X^2 + X + 1$. We now give a fully explicit infinite construction, without factoring its growing values.

Lemma 139.1 (A polynomial lifting orbit). *Set*

$$r_0 = 226, \quad r_{k+1} = r_k + 32\phi(r_k) \quad (k \geq 0).$$

Then $r_k \equiv 30 \pmod{49}$, $r_k \equiv 2 \pmod{4}$, and

$$v_7(\phi(r_k)) = v_7(r_k^3 - 1) = k + 2.$$

The sequence is strictly increasing, $\text{ord}_7(r_k) = 3$, and no r_k is a perfect power of exponent at least two.

Proof. The exact polynomial identity is

$$(584) \quad \phi(X + 32\phi(X)) = \phi(X)(1024X^2 + 1088X + 1057).$$

At $X \equiv 30 \pmod{49}$, $\phi(X) \equiv 0 \pmod{49}$, so the first residue is preserved. The multiplier in (584) is 42 modulo 49, so it has seven-adic valuation one. Since $\phi(226) = 49 \cdot 1047$ and $1047 \equiv 4 \pmod{7}$, induction proves the exact valuation claim for ϕ . The residue modulo four is preserved because the increment is a multiple of 32. Hence $v_2(r_k) = 1$, which excludes $r_k = z^e$ for $e \geq 2$. Modulo seven the residues of r_k, r_k^2, r_k^3 are 2, 4, 1, respectively, proving the order statement. Finally $r_k^3 - 1 = (r_k - 1)\phi(r_k)$ and $r_k - 1 \equiv 1 \pmod{7}$. The increment is strictly positive. $\square$

The orbit grows unnecessarily quickly. A bounded representative retains its useful divisibility and the normalization condition.

Theorem 139.2 (Small normalized representatives). *For every $k \geq 0$, define*

$$x_k = r_k \bmod (4 \cdot 7^{k+2}),$$

where the remainder lies between zero and its modulus. Then

$$(585) \quad 2 \leq x_k < 28 \cdot 7^{k+1}, \quad x_k \equiv 2 \pmod{4}, \quad x_k \equiv 2 \pmod{7},$$

$$(586) \quad 7^{k+2} \mid \phi(x_k), \quad 7^{k+2} \leq \phi(x_k).$$

No x_k is a nontrivial perfect power, and $x_k \rightarrow \infty$.

Proof. Reduction modulo $4 \cdot 7^{k+2}$ preserves the residues modulo four and seven and the polynomial value modulo 7^{k+2} . The residue modulo four implies $x_k \geq 2$ and excludes nontrivial perfect powers. The positive integer $\phi(x_k)$ is a multiple of 7^{k+2} and is therefore at least that large. Since $X^2 + X + 1$ is bounded on every bounded interval of nonnegative integers, (586) implies $x_k \rightarrow \infty$. The sequence of small representatives need not be strictly increasing; no monotonicity of these remainders is used. $\square$

Corollary 139.3 (No independent sublinear first-depth bound). *There are infinitely many integers x that are not nontrivial perfect powers and satisfy*

$$(587) \quad T_3(x, 1) > x/28.$$

Consequently, for each $0 \leq \theta < 1$ and each $C > 0$, some such x satisfies $T_3(x, 1) > Cx^\theta$. In particular, no bound depending only on the outer exponent can bound T_n uniformly over normalized bases.

Proof. Seven is new relative to $(x_k, 1)$ and has order three. Its first valuation is at least $k + 2$, so $T_3(x_k, 1) \geq 7^{k+1} > x_k/28$. Since $x_k \rightarrow \infty$, $x_k^{1-\theta}/28 \rightarrow \infty$, proving all the assertions. $\square$

For example the first eight small representatives are

$$30, 1010, 3754, 51774, 387914, 3211490, 19682350, 65800758.$$

A separate implementation obtains the same representatives by lifting the simple root 2 of ϕ modulo seven one digit at a time, and imposing $x \equiv 2 \pmod{4}$. At a root r modulo $M = 7^e$, the next digit is $t \equiv -(\phi(r)/M)(2r+1)^{-1} \pmod{7}$, and r is replaced by $r + tM$. This implementation does not use the recurrence (584).

Remark 139.4 (Why this is not an *abc* counterexample). The true *abc* ratio for $(1, x^3 - 1, x^3)$ also contains $\text{rad}(x(x-1))$ and all new radical primes in $\phi(x)$. An unbounded first-appearance factor, even one linear in x , does not bound that entire radical from above strongly enough to contradict (574). The refuted claim is independent control of T , not a coupled estimate involving the seed's radical or defect. At fixed $\varepsilon > 0$, an *abc* disproof would require the entire ratio $c/\text{rad}(abc)^{1+\varepsilon}$ to be unbounded. No such family is constructed here.

140. AN INDEPENDENT PAIRWISE IMPROVEMENT FOR ARITHMETIC DERIVATIVES

This section sharpens local realization instead of imposing a new unproved global estimate. Let

$$N = \prod_{i=1}^t p_i^{e_i} > 1, \quad k = \gcd(e_1, \dots, e_t), \quad f_i = e_i/k, \quad S = \sum_i f_i.$$

For $\lambda \in \mathbb{Z}^t$, define

$$\partial_\lambda(N) = \sum_i \frac{N}{p_i} e_i \lambda_i.$$

A target d is *attainable* when it is one of these integer values.

Lemma 140.1 (The attainable-value ideal). *Put $r = \prod_i p_i$ and $\rho = \prod_{p_i \mid f_i} p_i$. The attainable values are exactly $G_N \mathbb{Z}$, where*

$$G_N = \frac{N}{r} k \rho.$$

Proof. The image is the ideal generated by the integer coefficients $(N/p_i)e_i$. Remove the common factor $(N/r)k$. At a prime outside r , the remaining coefficients have minimum valuation zero since $\gcd_i f_i = 1$. At a prime $p_j \mid r$, the j th coefficient is $f_j(r/p_j)$. If $p_j \nmid f_j$, the minimum is zero. Otherwise all coefficients are divisible by p_j , while some f_i with $i \neq j$ is prime to p_j and its coefficient has valuation exactly one. Thus their gcd is exactly ρ . The single-prime case has $f_1 = 1$ and fits the same formula. Bezout's identity also proves existence of an integer vector for every multiple of G_N . $\square$

Set

$$s_i = \begin{cases} 1, & p_i \nmid f_i, \\ 1/p_i, & p_i \mid f_i, \end{cases} \quad m_i = f_i s_i \in \mathbb{Z}_{>0}, \quad q = \frac{d}{N\Omega(N)}, \quad \Omega(N) = kS.$$

The s_i are the natural spacings of λ_i/p_i after the forced regular-prime residues have been imposed. Only their displayed values, not this interpretation, are needed for the proof.

Theorem 140.2 (Pair-energy proximity). *For every attainable integer d there is an integer vector with $\partial_\lambda(N) = d$ and*

$$(588) \quad \max_i \left| \frac{\lambda_i}{p_i} - q \right| \leq \kappa_N, \quad \kappa_N = \max_i \frac{1}{2S} \sum_{j \neq i} \frac{f_j(f_i + f_j)s_i s_j}{\gcd(m_i, m_j)}.$$

In particular,

$$\kappa_N \leq \max_i f_i.$$

For $t = 1$ it is zero. If every normalized exponent f_i equals one, $\kappa_N = (t - 1)/t < 1$.

Proof. Write $\delta_i = \lambda_i/p_i - q$. The target equation is equivalent to $\sum_i f_i \delta_i = 0$. Among feasible integer vectors, minimize

$$\mathcal{E}(\lambda) = \sum_i f_i \delta_i^2.$$

A feasible vector exists by Lemma 140.1; each bounded sublevel contains only finitely many integer vectors, so the minimum exists. For $i \neq j$, let $g = \gcd(m_i, m_j)$ and $v = s_i s_j / g$. The two-coordinate move

$$(\delta_i, \delta_j) \mapsto (\delta_i + f_j v, \delta_j - f_i v)$$

is a legal integer-weight move: the corresponding increments are $p_i s_i m_j / g$ and $-p_j s_j m_i / g$, both integers. It preserves the target. Its energy change, with either sign, is

$$\pm 2f_i f_j v (\delta_i - \delta_j) + f_i f_j v^2 (f_i + f_j).$$

Minimality for both signs gives $|\delta_i - \delta_j| \leq v(f_i + f_j)/2$. Since the weighted mean of the δ_i is zero,

$$|\delta_i| = \left| \frac{1}{S} \sum_{j \neq i} f_j (\delta_i - \delta_j) \right| \leq \frac{1}{2S} \sum_{j \neq i} f_j (f_i + f_j) v,$$

proving (588). Finally $v \leq 1$ and $(f_i + f_j)/2 \leq \max_i f_i$; sum the weights and divide by S . The two special cases follow by direct substitution. $\square$

Corollary 140.3 (A terminating pair-move algorithm). *Starting from any feasible vector, repeatedly perform a strictly energy-decreasing integral multiple of a two-coordinate move whenever one exists. This process terminates and its output satisfies (588). No polynomial bound on the number of moves is asserted.*

Proof. All visited vectors lie in the finite sublevel set determined by the initial energy, and strict decrease prevents revisiting a vector. At termination, the positive and negative unit moves are nondecreasing for every pair. The proof of Theorem 140.2 only needs these inequalities, not global optimality. $\square$

The previous local bound was $S = \Omega(N)/k$ [113]. The new bound is at most $\max_i f_i$, and is less than one when all $f_i = 1$, regardless of the number or size of the supporting primes. For actual finite target choices in the replay we obtain:

Profile N	Previous bound S	New envelope κ_N	Realized error
$2 \cdot 3 \cdot 5 \cdot 7$	4	$3/4$	419/840
$2 \cdot 3^{100}$	101	50	10150/303
$2^2 3^3$	5	$1/4$	$1/5$
$2^{64} 3^{41}$	105	16	736/315

The “realized error” column is one constructed vector per displayed profile, not an assertion of the optimal error for every target.

140.1. Why the global scalar obstruction remains. The prior global construction first chooses admissible values of $\partial(a), \partial(b), \partial(c)$ in a one-dimensional aggregate fibre. When all entries exceed one its period is

$$\ell = \text{lcm}(h_a, h_b, h_c), \quad h_N = \frac{G_N}{\gcd(G_N, N)} \mid k_N.$$

Replacing its local lifting by Theorem 140.2 changes the overhead to at most

$$\frac{\ell}{2 \min\{\Omega(a), \Omega(b), \Omega(c)\}} + \max\{\kappa_a, \kappa_b, \kappa_c\}.$$

Entries equal to one have zero derivative and are treated without this three-variable period term. This sharpens realization, but the real minimum is still

$$h_0 = \frac{J}{R} \max \left\{ \frac{c}{\Omega(a) + \Omega(b)}, \frac{b}{\Omega(a) + \Omega(c)}, \frac{a}{\Omega(b) + \Omega(c)} \right\},$$

where $I\mathbb{Z}$ is the exact Wronskian image, $D = abc/R$, and $J = I/D$. In particular, the scalar factor c/R has not disappeared. The uniform small-transverse condition remains at *abc* strength as established in the earlier checkpoint; the new proximity estimate is not an independent proof of that condition.

141. PROOF DEPENDENCIES, VALIDATION, AND FORMAL SCOPE

141.1. Dependency audit. The main unconditional chains are

$$\begin{aligned} \text{binomial valuation lemma} &\implies \text{exact radical balance} \implies \text{defect transfer and cocycle;} \\ &\text{polynomial orbit identity} \implies \text{all-depth valuation induction} \\ &\implies \text{small representatives} \implies T_3 > x/28; \\ \text{coefficient gcd and Bezout} &\implies \text{feasible energy minimization} \\ &\implies \text{pairwise proximity envelope.} \end{aligned}$$

None uses *abc*, Szpiro, Vojta, or a uniform assertion about first appearance. The large-exponent stratum theorem has the explicit input $T \leq n^A$; it must not be labeled an unrestricted result. The transfer theorem has an explicit seed bound and budget condition, not an unconditional conclusion for every triple.

The examples $x = 7, y = 1, n = 2$ and x_k refute, respectively, omission of the two-adic correction and independent sublinear first-depth control. The rebasing example explains how ordinary repeated lifting can be misclassified as first appearance. The derivative proof attacks integer realization, not the already identified scalar height obstruction.

141.2. Exact finite checks. The standard-library program `computation/replay.py` uses integers and rational numbers throughout. It rejects optimized Python execution because assertions are part of the checks. The default run records:

Check	Count
Signed radical identities	3,553
Cleared-defect transfer checks	14,212
Complete-budget composition checks	405
Base-change first-depth checks	180
Exact polynomial-orbit valuations	13
Small representatives, two independent algorithms	129
Local energy lifts on actual factorizations	7,497

The power grid covers coprime $1 \leq y < x \leq 32$, difference exponents one through seven, and sum exponents one, three, five, seven. Local profiles include every $2 \leq N \leq 2000$ and 500 generated factorizations, each with three signed or positive attainable targets. Finite checks validate implementations; all infinite assertions above have proofs independent of these counts. No scan is used to prove uniformity.

141.3. What Lean actually verifies. Lean/PowerDescent.lean imports only Std. Its 29 named theorems and 29 axiom queries passed Lean 4.32.2 with this recorded identity:

```
Source commit    67af725edf41a7b4103eae055e9ee19c6be716c6
Source blob      eb665612004cc072255419b6fc5af464f4dc1ffe
Actions run / job 33976085784 / 101332907717
```

Warnings are treated as errors. The axiom union is `propext`, `Classical.choice`, and `Quot.sound`. There is no `sorryAx`. Two failed development versions were repaired and are not counted as successful checks.

The formalized infinite statements are genuine universal theorems about the recurrence, exact divisibility by 7^{k+2} , residues, exclusion of all nontrivial perfect powers, and existence of the bounded representatives for every k . In particular the module proves

$$\forall k \exists x, \quad 2 \leq x < 28 \cdot 7^{k+1}, \quad x \equiv 2 \pmod{4}, \quad x \equiv 2 \pmod{7}, \\ 7^{k+2} \mid x^2 + x + 1, \quad 7^{k+2} \leq x^2 + x + 1, \quad \forall z \in \mathbb{N} \forall e \geq 2, \quad x \neq z^e,$$

where in the final clause z is any natural number and $e \geq 2$. The Lean source states the quantifiers without the displayed shorthand. It also checks abstract integer transfer and composition given an explicit radical-balance equality, and the polynomial energy-exchange identity.

Not fully formalized here: multiplicative-order theory, the general valuation-to-radical map, all real-power statements, the asymptotic interpretation of $T_3 > x/28$, the real-valued energy-minimum theorem, and the full abc statement. The root-family's arithmetic certificates are stronger than finite samples, but they do not make the whole paper kernel-checked.

A checksum-pinned isolated installer and verifier is provided for x86_64 Ubuntu/WSL. Hosted Ubuntu execution is not access to the user's personal WSL. Existing project toolchains, dependencies, and verified imports are not changed. There was no independently running research subagent, external peer review, or full-repository build in this work.

142. REMAINING MATHEMATICAL OBLIGATIONS

Power descent would need a new mechanism jointly controlling first appearance and the seed's radical or defect. Corollary 139.3 prevents treating the first factor as uniformly sublinear on normalized bases independently of that seed information. Descent additionally needs coverage of triples without two entries sharing a nontrivial exponent; writing a canonical root decomposition does not create such an exponent. Neither obligation is proved here.

The derivative route needs an independent height estimate, not merely a smaller integer-realization error. The FCRT and multi-output route still needs its uniform endpoint estimate; no new edge exists in a no-face region solely because the scalar bookkeeping has been sharpened. The IUT all-place compatibility, geometric uniformity, Pell first-appearance, and other recorded routes retain their earlier unresolved status. They have not been refuted as entire approaches, nor newly closed in this supplement.

Thus the strongest completed results of this work are the exact descent mechanism, the normalized infinite obstruction to independent first-depth bounds, and the local proximity improvement. The final node (574) is still unresolved. No completed *abc* proof, disproof, or Lean closed term for it is asserted.

INTEGRATED CHECKPOINT: EISENSTEIN DESCENT AND STRONG BOUNDARY DIVISIBILITY

Full mathematical proof body of the historical checkpoint. Historical execution statements retain their original scope.

Keywords. abc conjecture; Eisenstein integers; radical; descent; Lucas sequence; first apparition; Lean formalization.

143. THE TARGET AND THE INHERITED BREAKPOINT

For a positive integer n , put $\text{rad}(n) = \prod_{p|n} p$. The standard target is

$$(589) \quad \forall \varepsilon > 0 \exists K_\varepsilon > 0 \forall a, b, c \in \mathbb{Z}_{>0}, \quad a + b = c, \gcd(a, b) = 1 \implies c \leq K_\varepsilon \text{rad}(abc)^{1+\varepsilon}.$$

All constants in this assertion are independent of the triple. Its negation requires one fixed positive ε and a family for which the displayed ratio is unbounded. Neither a large finite example nor a counterexample to an auxiliary estimate proves that negation.

The inspected project checkpoint [116] contains several distinct lines: proper-face transport, compensated divisor packets, first-apparition estimates, arithmetic derivatives, and IUT/geometric interfaces. Earlier power descent compares a triple with a common-power lift and keeps the entire first-appearance factor separate from the controlled repeated-lifting factor [117]. Earlier transverse lifting controls integer-realization overhead, but its proposed uniform small-norm condition was proved equivalent to abc [118]. We do not assume either missing estimate.

The present mechanism uses a factorization of $a + b\rho$, rather than a common integer power in two arms. Every primitive positive triple is covered by the resulting algebraic descent. We then study the actual change in its boundary prime support. Our results distinguish three separate questions:

existence of a smaller predecessor; exact common prime support; uniform radical growth.

The first two are resolved below in the stated settings. The third is not.

The source snapshot is commit `9edd965d6df645dbebc0869530f3f5a40fb8cddc`. Its endpoint-residue obstruction [119] is retained: no positive-face existence assumption is used here. This manuscript develops its mathematical arguments directly. It does not assert priority relative to the external literature or external peer review.

144. THE QUADRATIC MODEL AND HEIGHT NORMALIZATION

Let $\rho = (1 + \sqrt{-3})/2$ and $\mathcal{E} = \mathbb{Z}[\rho]$. An element $x + y\rho$ is represented by the integer pair (x, y) . Its operations are

$$(590) \quad (u, v)(x, y) = (ux - vy, vx + (u + v)y),$$

$$(591) \quad \overline{(x, y)} = (x + y, -y),$$

$$(592) \quad N(x, y) = x^2 + xy + y^2.$$

Direct multiplication proves $z\bar{z} = (N(z), 0)$ and $N(zw) = N(z)N(w)$. The norm is positive away from zero, since

$$4N(x, y) = (2x + y)^2 + 3y^2.$$

An element is a unit exactly when its norm is one: the conjugate supplies an integral inverse in that case, and multiplicativity proves the converse.

Definition 144.1. A pair (x, y) is primitive when $\gcd(|x|, |y|) = 1$. Define

$$F(x, y) = xy(x + y), \quad H(x, y) = \max\{|x|, |y|, |x + y|\}.$$

For a primitive nonunit pair, define $R(x, y) = \text{rad}(|F(x, y)|)$.

The boundary is nonzero at every primitive nonunit. Indeed, if one of $x, y, x + y$ vanishes, primitivity forces the remaining nonzero coordinate to have absolute value one, and the norm is one.

Lemma 144.2 (Normalization and height corridor). *Every primitive nonunit is unit-equivalent to a pair (a, b) with $a, b > 0$. The three positive integers obtained by sorting $|x|, |y|, |x + y|$ form a primitive *abc* triple with largest entry $H(x, y)$ and radical $R(x, y)$. Moreover*

$$(593) \quad \frac{3}{4}H(z)^2 \leq N(z) \leq H(z)^2.$$

Proof. Multiplication by ρ sends (x, y) to $(-y, x + y)$. Its six iterates permute $x, y, x + y$ up to sign. They preserve H and N , and change F only by sign. Their six closed cones cover the plane, so one representative has both coordinates nonnegative. A zero coordinate would make a primitive representative a unit. For the positive representative, $H = a + b$ and

$$N = (a + b)^2 - ab, \quad 4N - 3(a + b)^2 = (a - b)^2.$$

These prove the corridor. Pairwise coprimality of $a, b, a + b$ follows from $\gcd(a, b) = 1$. $\square$

Lemma 144.3 (Coprime norm and boundary). *For every primitive pair z , $\gcd(N(z), |F(z)|) = 1$.*

Proof. Modulo x, y , and $x + y$, the norm is respectively y^2, x^2 , and y^2 . Each of these squares is coprime to its modulus. Since the three boundary factors are pairwise coprime, the conclusion follows. This also covers units, where $N = 1$. $\square$

Thus factoring the norm does not directly factor the *abc* radical. This distinction is important when interpreting the descent.

145. PRIME-NORM DESCENT WITH COMPLETE ALGEBRAIC COVERAGE

Lemma 145.1 (Euclidean division). *For $z, w \in \mathcal{E}$, $w \neq 0$, there are $q, r \in \mathcal{E}$ such that*

$$z = qw + r, \quad N(r) \leq \frac{3}{4}N(w) < N(w).$$

Proof. Write $z/w = s + t\rho$ with $s, t \in \mathbb{Q}$. Choose integers s_0, t_0 within $1/2$ of the respective coordinates, and put $q = s_0 + t_0\rho$. If $A = s - s_0, B = t - t_0$, then

$$N(A + B\rho) = A^2 + AB + B^2 \leq \frac{3}{4}.$$

Multiplication by w proves the bound. The remainder is integral by construction. Consequently the Euclidean algorithm terminates and gives a gcd together with a Bezout identity; back substitution proves the identity at every step. $\square$

Theorem 145.2 (Every primitive pair has a prime-norm predecessor). *Let $z \in \mathcal{E}$ be primitive and nonunit. For every rational prime $p \mid N(z)$, there are integral π, w satisfying*

$$(594) \quad z = \pi w, \quad N(\pi) = p, \quad N(w) = N(z)/p,$$

where w is primitive, $p \geq 3$, and

$$(595) \quad H(w) \leq \frac{2}{3}H(z).$$

*Iterating produces a finite path from every primitive *abc* triple to a unit.*

Proof. Write $z = x + y\rho$. If $p \mid y$, then $N(z) \equiv x^2 \pmod{p}$ would force $p \mid x$, contradicting primitivity. Thus y is invertible modulo p , and $r = -x/y$ in $\mathbb{F}_p$ satisfies $r^2 - r + 1 = 0$. Evaluation $\rho \mapsto r$ defines a ring homomorphism $\mathcal{E} \rightarrow \mathbb{F}_p$ which kills both p and z .

Let π be a Euclidean gcd of p and z . It is not a unit: a Bezout identity for a unit gcd would, after multiplying by its inverse, express 1 as a combination of p and z , contradicting evaluation in $\mathbb{F}_p$. Because $\pi \mid p$, its positive integer norm divides p^2 . The only possible nonunit norms are p and p^2 . If $N(\pi) = p^2$, the integral quotient p/π has norm one; hence π is associated to p . Since $\pi \mid z$, this would make both coordinates of z divisible by p . Therefore $N(\pi) = p$.

The quotient $w = z/\pi$ is primitive: any rational integer dividing both its coordinates divides both coordinates of z . A primitive norm is odd, as inspection modulo two shows, so $p \neq 2$. Applying Lemma 144.2 to w and z yields

$$H(w)^2 \leq \frac{4}{3}N(w) = \frac{4}{3p}N(z) \leq \frac{4}{9}H(z)^2.$$

The positive integral norm strictly decreases at each step, and norm one is precisely the terminal unit case. $\square$

This proof does not assume that either arm is an integer power. It also does not assert that a radical inequality improves along the path. The latter statement is false, even when all prime-norm predecessor choices are allowed; see Section 149.

146. THE EXACT BOUNDARY GCD IDENTITY

Theorem 146.1 (Exact boundary overlap). *For arbitrary $u, v, x, y \in \mathbb{Z}$ with $\gcd(|x|, |y|) = 1$,*

$$(596) \quad \gcd(|F(x, y)|, |F((u, v)(x, y))|) = \gcd(|F(x, y)|, |F(u, v)|).$$

This identity has been fully formalized in the supplied standalone Lean modules.

Proof. Put $A = F(u, v)$ and $T = F((u, v)(x, y))$. Formula (590) gives the exact congruences

$$(597) \quad T \equiv -Ay^3 \pmod{x},$$

$$(598) \quad T \equiv Ax^3 \pmod{y},$$

$$(599) \quad T \equiv Ay^3 \pmod{x+y}.$$

For example, modulo x the transformed coordinates are $-vy$ and $(u+v)y$, whose sum is uy ; their product is $-uv(u+v)y^3$. The other two substitutions give the remaining congruences.

The integers $x, y, x+y$ are pairwise coprime in absolute value. Thus y^3 can be cancelled in the gcd modulo x and $x+y$, and x^3 can be cancelled modulo y . We obtain

$$\begin{aligned} \gcd(|x|, |T|) &= \gcd(|x|, |A|), & \gcd(|y|, |T|) &= \gcd(|y|, |A|), \\ \gcd(|x+y|, |T|) &= \gcd(|x+y|, |A|). \end{aligned}$$

For pairwise coprime integers r_i , prime valuations show $\gcd(|\prod_i r_i|, |t|) = \prod_i \gcd(|r_i|, |t|)$. Multiplying the three equalities proves the theorem. If a boundary factor is zero, the other factors have absolute value one; the same gcd calculation remains valid. $\square$

The assertion is stronger than an equality of prime supports: it preserves the exact minimum valuation at every common prime. It requires no coprimality condition on the multiplier (u, v) .

Corollary 146.2 (Exact overlap on a descent edge). *For a nonterminal edge $z = \pi w$ from Theorem 145.2, with w nonunit, the common radical factor is*

$$\gcd(R(w), R(z)) = \text{rad}(\gcd(|F(w)|, |F(\pi)|)).$$

In particular, if $\pi = 2 + \rho$, all common primes lie in $\{2, 3\}$.

The formula controls overlap, not the magnitudes of the new primes or their multiplicities. Replacing those missing magnitude bounds by this overlap formula would be an invalid inference.

147. A NORM-SEVEN ORBIT AND STRONG DIVISIBILITY

Set $\alpha = 2 + \rho$ and write

$$(600) \quad \alpha^n = x_n + y_n \rho, \quad (x_0, y_0) = (1, 0).$$

Then

$$(601) \quad x_{n+1} = 2x_n - y_n, \quad y_{n+1} = x_n + 3y_n,$$

and $N(x_n, y_n) = 7^n$.

Lemma 147.1 (Primitivity of the entire orbit). *Every pair (x_n, y_n) is primitive. For $n \geq 1$,*

$$x_n \equiv 2y_n \pmod{7}, \quad y_n \not\equiv 0 \pmod{7}.$$

Proof. The congruences hold at $n = 1$. Under (601), their difference becomes a multiple of seven, and $y_{n+1} \equiv 5y_n \pmod{7}$, so they persist. A common divisor of x_n, y_n divides their norm 7^n , but is coprime to seven by the second congruence. It is therefore one. The case $n = 0$ is immediate. $\square$

Definition 147.2. Let $F_n = F(x_n, y_n)$ and $B_n = |F_n|$, including $B_0 = 0$. Define the integer sequence

$$U_0 = 0, \quad U_1 = 1, \quad U_{n+2} = 20U_{n+1} - 343U_n.$$

Theorem 147.3 (A whole-boundary strong divisibility sequence). *For all $m, n \in \mathbb{N}$,*

$$(602) \quad \gcd(B_m, B_n) = B_{\gcd(m, n)},$$

and

$$(603) \quad F_n = 6U_n.$$

Both statements have been proved in Lean for arbitrary indices.

Proof. The orbit multiplication law and Theorem 146.1 give

$$\gcd(B_m, B_{m+n}) = \gcd(B_m, B_n).$$

Repeated subtraction, or Euclidean induction on (m, n) , proves (602); at an index zero it reduces to $\gcd(0, B_n) = B_n$.

For the sequence identity, direct polynomial expansion for the map $S(x, y) = (2x - y, x + 3y)$ gives

$$F(S^2 z) = 20F(Sz) - 343F(z).$$

Since $F_0 = 0$ and $F_1 = 6$, induction proves (603). As an independent explanation, the coefficient of ρ in z^3 is $3F(z)$, whereas α^3 has trace 20 and norm 343. $\square$

Corollary 147.4. *For $n \geq 1$, $B_n > 0$. If $\gcd(m, n) = 1$, then $\gcd(B_m, B_n) = 6$; hence their prime supports outside two and three are disjoint.*

Proof. A zero boundary at a primitive pair would force norm one, whereas the norm here is $7^n > 1$. The gcd assertion follows from $B_1 = 6$. $\square$

This gives an exact connection with first-appearance questions in recurrence sequences. It does not bound the first-appearance valuation. Distinct prime support at different indices alone supplies no pointwise exponential radical lower bound.

148. A FULL QUADRATIC QUOTIENT WITH ONE PRIME SUPPORT

The same orbit constructs a stronger obstruction to an independently small first-appearance factor when homogeneous bases are allowed. Signs must be retained: the boundary in this section is $xy(x - y)$, not $F(x, y) = xy(x + y)$.

Lemma 148.1 (A non-power subsequence). *For $n = 6j + 1$, $j \geq 0$,*

$$x_n \equiv 2 \pmod{4}, \quad y_n \equiv 1 \pmod{4}.$$

Consequently $|x_n|$ is not a nontrivial perfect power, so the two absolute bases cannot have a common nontrivial power exponent.

Proof. An exact computation gives $\alpha^6 = (-323, 360)$. Multiplication by this pair has matrix

$$\begin{pmatrix} -323 & -360 \\ 360 & 37 \end{pmatrix} \equiv I \pmod{4}.$$

The residues at $n = 1$ therefore persist. The two-adic valuation of $|x_n|$ is exactly one. If $|x_n| = z^e$ with $e \geq 2$, an odd z gives an odd power and an even z gives a multiple of four, both contradictions. $\square$

Theorem 148.2 (Quadratic first-appearance excess). *For every $n = 6j + 1$, let $x = x_n$, $y = y_n$, and $h = \max(|x|, |y|)$. Then $x, y, x - y$ are nonzero and pairwise coprime, and*

$$(604) \quad x^3 - y^3 = (x - y)7^n, \quad 7 \nmid xy(x - y).$$

The ratio x/y modulo seven has exact order three, and

$$v_7(x^3 - y^3) = n.$$

Thus the entire excess of the new-prime quadratic quotient is

$$(605) \quad T = 7^{n-1}, \quad \frac{3}{28}h^2 \leq T \leq \frac{3}{7}h^2.$$

The resulting signed cubic relation gives a positive primitive abc triple, and the bases are normalized as in Lemma 148.1.

Proof. The polynomial identity $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$ proves the first equation. The residues from Lemma 147.1 give $x/y = 2$ modulo seven. This residue has order three, and neither x , y , nor $x - y$ is zero modulo seven. The exact valuation follows. In particular $x - y \neq 0$; the other coordinates are also nonzero. Their pairwise coprimality follows from that of x, y .

For arbitrary real x, y with $h = \max(|x|, |y|)$,

$$\frac{3}{4}h^2 \leq x^2 + xy + y^2 \leq 3h^2.$$

For the lower bound, fix the coordinate of absolute value h and complete the square in the other coordinate. Divide these inequalities by seven and use $N = 7^n$ to obtain (605).

Sorting the positive quantities $|x|^3, |y|^3, |x - y|7^n$ gives an abc triple. If x, y have the same sign, it is a difference of positive cubes. If they have opposite signs, it is a sum of positive cubes, with quadratic quotient $|x|^2 - |x||y| + |y|^2$. Coprimality was already proved. Finally $h \geq 7^{n/2}/\sqrt{3} \rightarrow \infty$, so infinitely many distinct base pairs occur. $\square$

Corollary 148.3 (An independent subquadratic bound fails). *For every fixed $\theta < 2$ and $C > 0$, the normalized signed homogeneous family in Theorem 148.2 contains infinitely many members with $T > Ch^\theta$.*

This extends the previous one-variable obstruction to a homogeneous signed setting; it is not a claim that the stronger exponent holds with $y = 1$. Nor is it a counterexample to abc.

Proposition 148.4 (The exact coupled radical still required). *For the positive cubic triple just constructed, write C_n for its largest entry and $S_n = \text{rad}(|x_n y_n (x_n - y_n)|)$. Its radical is exactly $7S_n$, and*

$$(606) \quad \frac{1}{27}7^{3n} \leq C_n^2 \leq 4 \cdot 7^{3n}.$$

Consequently abc restricted to this family is equivalent to the collection of lower bounds

$$(607) \quad \forall \delta > 0 \exists c_\delta > 0 \forall n = 6j + 1, \quad S_n \geq c_\delta 7^{(3/2-\delta)n}.$$

No proof of (607) is supplied here.

Proof. Equation (604) and $7 \nmid xy(x - y)$ give the radical formula. If x, y have the same sign, then $C_n = h^3$ and $h^2 \leq N \leq 3h^2$. If they have opposite signs, write $u = |x|$, $v = |y|$; then $C_n = (u + v)N$, with $N = u^2 - uv + v^2$ and $N \leq (u + v)^2 \leq 4N$. These prove (606).

If the restricted abc statement holds for every $\varepsilon > 0$, the lower height bound gives $S_n \geq c_\varepsilon 7^{3n/(2(1+\varepsilon))}$. For any $0 < \delta < 3/2$, choose ε small enough that $3/(2(1+\varepsilon)) \geq 3/2 - \delta$. Larger δ are immediate. Conversely, for a given $\varepsilon > 0$, choose $0 < \delta < 3\varepsilon/(2(1+\varepsilon))$ and use (607) together with the upper height bound. The constants depend only on the chosen exponent, not on n . $\square$

The sequence S_n in this proposition is not $\text{rad}(B_n)$. Substituting the strong-divisibility conclusion for the latter into (607) would confuse two different boundary forms.

149. A COMPLETE OBSTRUCTION TO POINTWISE NO-LOSS DESCENT

Theorem 149.1 (Every prime-norm predecessor can have smaller normalized defect). *For $z = \alpha^6 = (-323, 360)$, every prime-norm factorization $z = \pi w$ has a quotient unit-equivalent to $\alpha^5 = (-87, 149)$. Their associated triples and radicals are*

Pair	Positive triple	Height	Radical
$(-87, 149)$	$(62, 87, 149)$	149	803706
$(-323, 360)$	$(37, 323, 360)$	360	358530

For every integer $m \geq 1$,

$$(608) \quad \frac{149^m}{803706^{m+1}} < \frac{360^m}{358530^{m+1}}.$$

Hence a policy requiring the normalized *abc* ratio never to increase on a forward prime-norm edge cannot cover every triple.

Proof. The complete factorizations are

$$|F_5| = 2 \cdot 3 \cdot 29 \cdot 31 \cdot 149, \quad |F_6| = 2^3 \cdot 3^2 \cdot 5 \cdot 17 \cdot 19 \cdot 37.$$

The displayed factors are prime by trial division up to their square roots, and multiplication verifies the products. Thus $R_5 = 803706$ and $R_6 = 358530$. Inequality (608) follows from $149 < 360$ and $358530 < 803706$. Its denominator-cleared form is formally proved for every $m \in \mathbb{N}$.

Every prime norm dividing $N(z) = 7^6$ is seven. A pair of norm seven has coordinates between -3 and 3 , by $N \geq 3u^2/4$ and $N \geq 3v^2/4$. The twelve possibilities are exhausted by $(2u + v)^2 + 3v^2 = 28$. To test integrality of $z/(u + v\rho)$, the two numerators are

$$(u + v)(-323) + 360v = -323u + 37v, \quad 323v + 360u.$$

They are divisible by seven exactly when $v \equiv 4u \pmod{7}$. The six surviving pairs, with their quotients, are

u	v	w_1	w_2
-3	2	149	-62
-2	-1	87	-149
-1	3	62	87
1	-3	-62	-87
2	1	-87	149
3	-2	-149	62

These quotients are precisely unit rotations of α^5 . This checks every prime-norm predecessor, not just one chosen algorithmic predecessor. $\square$

Both displayed triples have height strictly below their radical. The theorem refutes only the pointwise monotonicity policy, not any asymptotic *abc* estimate. It also does not refute a scheme allowing controlled losses on several edges.

Proposition 149.2 (Adding both factors does not restore the same constant). *The triple $(1, 48, 49)$ has $q_{25} > 1$, but every one of its prime-norm factorizations $z = \pi w$ has $q_{25}(\pi) < 1$ and $q_{25}(w) < 1$. Thus an induction rule that transfers the same constant from both factors is false, even on a positive-defect child.*

Proof. In the pair ring,

$$(1, 48) = (1, -4)(-15, 4), \quad N(1, -4) = 13, \quad N(-15, 4) = 181.$$

Both 13 and 181 are prime. Every prime norm in a factorization is one of these. A gcd of an element of norm 13 and one of norm 181 has norm dividing both primes, so is a unit. Its Bezout identity shows that any norm-13 divisor of the product divides the first factor, and hence is associated to it; the analogous assertion holds for 181. Thus, up to units and interchanging the factors, the displayed factorization is the only prime-norm factorization.

The associated smaller triples are $(1, 3, 4)$ and $(4, 11, 15)$, with radicals 6 and 330. The original radical is 42. The exact integer certificates are

$$4^{25} < 6^{26}, \quad 15^{25} < 330^{26},$$

$$49^{25} - 42^{26} = 197132423399647862770020139227092550013105 > 0.$$

These prove the claims at the single fixed exponent $\varepsilon = 1/25$ and constant $K = 1$. Allowing a larger constant removes this finite exception; it is not a counterexample to standard *abc*. $\square$

150. WHAT REMAINS AFTER ALGEBRAIC COVERAGE

For a primitive nonunit z , put

$$q_m(z) = \frac{H(z)^m}{R(z)^{m+1}}, \quad m \geq 1.$$

For terminal units only, define $q_m = 1$ as a bookkeeping convention; this is not a definition of $\text{rad}(0)$. A forward edge $z = \pi w$ has logarithmic loss

$$\ell_m(w, z) = \log q_m(z) - \log q_m(w).$$

Along any complete path from a unit to z , the sum of these losses is exactly $\log q_m(z)$. Consequently an unspecified uniform bound for the complete path loss is merely a reformulation of abc. It is not a new proved bridge.

Theorem 145.2 supplies actual predecessors for all triples; Theorem 146.1 identifies their shared boundary divisors exactly; Theorem 149.1 prevents free pointwise monotonicity, and Proposition 149.2 prevents a same-constant transfer from both factors. The unresolved task is to prove a quantitative estimate on new boundary primes and their multiplicities that, when accumulated, bounds q_m independently of the final triple. No such estimate follows from the gcd identity alone. If the norm itself is prime, the predecessor is a unit and the multiplier is the original element; thus coverage alone has not reduced the height problem on that stratum.

There is also a useful separation within the fixed orbit. Lemma 144.2 gives

$$7^{n/2} \leq H(x_n, y_n) \leq \frac{2}{\sqrt{3}} 7^{n/2}.$$

The original boundary orbit would require $\text{rad}(6U_n) \geq c_\delta 7^{(1/2-\delta)n}$ for every $\delta > 0$; the signed cubic family instead requires (607). Each is a restricted-family condition, not a resolution for arbitrary triples. Strong divisibility resolves support overlap across indices but not either radical growth condition.

The existing IUT same-pilot and all-place comparisons, compensated packet bounds, Mersenne depth-three counting, Pell first-appearance valuations, and geometric uniformity gates in [116] remain unresolved in this work. Their parent routes are retained. The fixed nonreal Lucas sequence above is not identified with the earlier real Pell sequence. What is shared is the precise first-valuation bottleneck, not an unproved transfer of estimates.

151. PROOF AUDIT, FINITE REPLAY, AND FORMAL VERIFICATION

151.1. Dependency scope. The ordinary proof dependencies are

- quadratic algebra and coordinate rounding $\implies$ Euclidean gcd and prime-norm descent;
- polynomial remainders and coprimality $\implies$ boundary gcd $\implies$ orbit strong divisibility;
- norm recurrence and modular invariants $\implies$ single-prime quadratic quotient;
- all norm-seven divisors $\implies$ failure of pointwise no-loss descent.

None of these arrows uses abc, a Szpiro conjecture, a Vojta conjecture, or an unproved height bound. The final global radical estimate has no completed incoming proof arrow.

The signed case, unit case, index zero, repeated prime powers, and all prime-norm parents of the finite obstruction have been treated explicitly. The claimed quadratic first-appearance size uses a homogeneous signed family, not an unstated one-variable specialization. The two boundary forms $xy(x+y)$ and $xy(x-y)$ are deliberately separated.

151.2. Actual formal scope. The two supplied files import only Lean's standard library and the first local module:

File	Scope
<code>EisensteinDescent.lean</code>	29 named theorems: pair algebra, norm and polynomial remainders, orbit identities and residues, Lucas relation, and the denominator-cleared finite counterexample for all exponents.
<code>BoundaryGcd.lean</code>	11 named theorems: gcd algebra, the full general boundary gcd theorem with its actual coprimality premise, primitive orbit, and full strong divisibility for all indices.

At commit `b228479c215d45d6555e56a1f6b42c08b665a8fb`, workflow run `34005200961` compiled both files using checksum-pinned Lean 4.32.2, with warnings treated as errors. All 40 executed axiom queries passed the whitelist

`{propxt, Classical.choice, Quot.sound}`.

One intervening development run for the second module failed on an API name, a cast, and a linter warning, then was repaired. The failed version is not certified.

The Euclidean algorithm’s complete abstract existence proof, height contraction for every primitive input, asymptotic quadratic-excess interpretation, enumeration of all norm-seven parents, and real-power *abc* equivalence are ordinary proofs in this paper, not fully formalized here. The arithmetic product certificates in Lean do not silently stand in for a general radical definition. No theorem named or equivalent to unconditional **ABCConjecture** has been proved.

151.3. Finite replay. The dependency-free Python replay uses exact integer arithmetic, coordinate-rounded Euclidean division, Bezout-checked gcds, and an independently generated Lucas recurrence. Default checks are:

Check	Count
Full boundary gcd cases on signed coordinate boxes	230,000
Normalized primitive triples, height at most 400	24,339
Prime-norm descent steps on those triples	51,381
Orbit norm and Lucas indices, including zero	476
Strong-divisibility index pairs	58,081
Normalized single-prime quadratic quotients	80
Small cubic triples with complete radical checks	10
Prime-norm factorizations of the positive-defect obstruction	12
Cleared exponents in the finite obstruction replay	101

All six prime-norm parents of the orbit obstruction and all twelve prime-norm factorizations of the positive-defect obstruction are checked separately. Two complete local replays produced byte-identical JSON, with SHA-256

`a8c86d05cc111bd41d2909580338bf0bc79a43db6ab6c6cbd1ff64e7706f2d3a`.

These are finite implementation checks, not proofs of the universal statements. In particular, full factorization of the large orbit terms is not claimed.

151.4. Reproduction and environment boundary. The checkpoint supplies `verify.py` for source integrity and exact replay, and `wsl/verify_lean.sh` for actual compilation and axiom checking. The latter installs a separate checksum-pinned compiler on x86_64 Linux, including an existing WSL Ubuntu distribution. It does not change the original repository toolchain or elan defaults. Hosted Ubuntu execution is not access to the user’s personal WSL. No independent autonomous subagent audit, external peer review, or complete build of the existing repository is claimed.

152. CONCLUSION

The new descent covers all primitive abc triples algebraically, and the exact boundary gcd theorem supplies a multiplicity-sensitive coupling law. Its fixed norm-seven orbit has a fully verified strong-divisibility boundary and yields a full single-prime quadratic quotient with normalized bases. The same orbit gives a complete-premise counterexample to pointwise no-loss factor descent. These are concrete structural and formal results. They do not imply the uniform radical estimate. Any continuation must control that estimate independently, rather than reinterpret algebraic coverage, support disjointness, or successful finite and formal tests as a completed proof of abc.

INTEGRATED CHECKPOINT: FIRST-APPEARANCE DEPTH AND THE NORM-SEVEN ORBIT

Edited proof consolidation; provenance and historical verification boundaries are stated below.

153. PROVENANCE AND THE FIXED ARITHMETIC OBJECT

This is a proof consolidation of the preceding rank/depth supplement, not a byte-identical reprint of its historical runtime narrative. The original conversation source is identified by Git blob 69dc72153b9dc58e7b512b20a57cb46e03a733. The local laws, truncation arguments, modular certificate and fixed-angle application are restated explicitly below. The complex logarithmic-form input is cited in a directly checked form. Neither the original nor this edition proves standard *abc*.

Let $\zeta = (1 + \sqrt{-3})/2$, let $z_n = (2 + \zeta)^n = (x_n, y_n)$, and define

$$N(x, y) = x^2 + xy + y^2, \quad P(x, y) = xy(x + y), \quad H(x, y) = \max(|x|, |y|, |x + y|).$$

Then

$$z_{n+1} = (2x_n - y_n, x_n + 3y_n), \quad N(z_n) = 7^n.$$

These pairs are primitive: a common prime after one step must divide both $7x_n$ and $7y_n$ and hence can only be seven; but for $n \geq 1$, $z_n \equiv 5^{n-1}(2, 1) \pmod{7}$, excluding it. Unit rotation supplies a positive triple for every $n \geq 1$. A primitive zero-boundary pair is a unit, incompatible with norm $7^n > 1$.

Put $H_n = H(z_n)$, $\mathcal{R}_n = \text{rad}(|P(z_n)|)$. Cubic expansion yields

$$P(z_n) = 6U_n, \quad U_0 = 0, \quad U_1 = 1, \quad U_{n+2} = 20U_{n+1} - 343U_n.$$

Equivalently, with $\alpha = 10 + 9\sqrt{-3}$ and $\beta = 10 - 9\sqrt{-3}$,

$$U_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}.$$

The earlier full boundary gcd theorem gives $\gcd(|U_m|, |U_n|) = |U_{\gcd(m, n)}|$. In particular a prime with least positive appearance index d_p divides U_n exactly when $d_p \mid n$.

154. THE RANK AND VALUATION LAWS, INCLUDING EXCEPTIONAL PRIMES

For every prime that appears, define $s_p = v_p(U_{d_p})$.

Theorem 154.1 (Complete local laws). *Seven never divides U_n for $n \geq 1$. Every other prime has a first index, and*

$$v_p(U_n) = 0 \quad (d_p \nmid n), \quad v_p(U_n) = s_p + v_p(n/d_p) \quad (d_p \mid n).$$

At the exceptional primes,

$$d_2 = 2, \quad s_2 = 2, \quad v_2(U_n) = \begin{cases} 0 & n \text{ odd}, \\ v_2(n) + 1 & n \text{ even}, \end{cases} \quad d_3 = 3, \quad s_3 = 1, \quad v_3(U_n) = v_3(n).$$

For $p \geq 5$, $p \neq 7$, set $\chi_p = 1$ when $p \equiv 1 \pmod{3}$ and $\chi_p = -1$ when $p \equiv 2 \pmod{3}$. Then

$$d_p \mid (p - \chi_p)/3, \quad p \geq 3d_p - 1.$$

Proof. Modulo seven, the recurrence gives $U_n \equiv 20^{n-1} \neq 0$. For $p \neq 2, 3, 7$ work in the unramified quadratic algebra with $s^2 = -3$, which is either a field or a product of two fields. Let $\gamma = (5 + s)/2$, of norm seven. Since $\gamma^3 = \alpha$, the quotient $\alpha/\beta = (\gamma/\bar{\gamma})^3$ is a cube in the norm-one group. Frobenius is identity or conjugation according as χ_p is one or minus one. Thus $(\gamma/\bar{\gamma})^{p-\chi_p} = 1$. As three divides $p - \chi_p$, the order of α/β divides $(p - \chi_p)/3$. The denominator $\alpha - \beta$ and β are units at p , so this order is precisely the first index of U_n . In each unramified component, write $(\alpha/\beta)^{d_p} = 1 + p^{s_p}u$ with u a unit. For a positive multiplier coprime to p , binomial expansion leaves the first term at valuation s_p . Raising to the p -th power increases it by exactly one, since the first binomial term has uniquely least valuation. Iteration proves $s_p + v_p(n/d_p)$. The component valuations agree with the valuation of the rational integer U_n , since the Binet denominator and β are units.

At three use the ramified field $\mathbb{Q}_3(\sqrt{-3})$, with valuation extending v_3 on $\mathbb{Q}$. Here $v_3((\alpha/\beta) - 1) = 5/2$. The same binomial argument is valid at this depth: the first term remains uniquely of least valuation, giving $v_3((\alpha/\beta)^n - 1) = 5/2 + v_3(n)$. Subtracting the denominator valuation gives $v_3(U_n) = v_3(n)$. At two, the recurrence shows U_n is odd exactly when n is odd. Put $V_n = \alpha^n + \beta^n$. Then $U_{2n} = U_n V_n$, and $V_0 = 2, V_1 = 20$ satisfy the same recurrence. Modulo eight it has period two, with $V_n \equiv 4$ for odd n and 2 for even n . Thus $v_2(V_n) = 2$ in the first case and one in the second. Starting at odd n and doubling repeatedly proves the displayed two-adic law. The first indices and exponents follow, including $U_2 = 20$ and $U_3 = 57$. $\square$

155. EXACT BOOKKEEPING AND ABSORPTION THROUGH DEPTH TWO

Define

$$r_n = \text{rad}(|U_n|), \quad \mathcal{L}_n = \prod_{d_p|n} p^{v_p(n/d_p)}, \quad \mathcal{D}_n = \prod_{d_p|n} p^{(s_p-2)_+}, \quad F_n = \prod_{d_p|n} p^{(s_p-3)_+}.$$

Every product is over the actual finite prime support of U_n ; $u_+ = \max(u, 0)$. Theorem 154.1 implies $\mathcal{L}_n \mid n$ and

$$|U_n| = \mathcal{L}_n \prod_{d_p|n} p^{s_p}.$$

In particular

$$|U_n| \leq nr_n^2 \mathcal{D}_n, \quad |U_n| \leq nr_n^3 F_n.$$

The index-lifting factor is not discarded: it is bounded by n because $v_p(n/d_p) \leq v_p(n)$ for every supported prime.

For positive rotated coordinates x, y , let $H = x + y$. Since $xy \geq x + y - 1$ and $xy \geq 1$, $H \leq 2xy$, and therefore $H^2 \leq 2xyH = 2P$. Also

$$P + 1 - N = (x + y + 1)(x - 1)(y - 1) \geq 0.$$

Consequently the following is unconditional:

Theorem 155.1 (Elementary depth-two absorption). *For every $n \geq 1$,*

$$H_n^2 \leq 12n\mathcal{R}_n^2 \mathcal{D}_n.$$

Proof. Use $H_n^2 \leq 2|P(z_n)| = 12|U_n|$ and $r_n \leq \mathcal{R}_n$ in the preceding inequality. $\square$

Thus first squares do not occur in the remaining excess $\mathcal{D}_n$, and ordinary lifting costs only a factor n . This is not a proof that the remaining product is small.

156. EXACT CYCLOTOMIC BLOCKS

For $m \geq 1$ define the positive rational product

$$A_m = \prod_{d|m} |U_d|^{\mu(m/d)}.$$

All U_d for $d \geq 1$ are nonzero. For a prime p with first index d_p , its valuation in this product is zero when $d_p \nmid m$. If $m = d_p k$, it is

$$\sum_{e|k} \mu(k/e)(s_p + v_p(e)) = \begin{cases} s_p & k = 1, \\ 1 & k = p^j, j \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Indeed the constant term has divisor sum zero except at $k = 1$; write $v_p(e) = \sum_{j \geq 1} \mathbf{1}_{p^j|e}$ and apply the same divisor cancellation. All valuations are nonnegative, so A_m is an integer. This proves both its first-appearance factorization and the fact that recycled exceptional factors have only exponent one. In particular repeated-prime excess in A_m comes only from first-appearance primes. A bound on how many such primes appear does not by itself bound the first exponents s_p .

157. COMPLEX LOGARITHMS ABSORB FIRST CUBES ON THE FIXED ORBIT

We use the classical complex logarithmic-form estimate as recorded in Bugeaud, Theorem 1.1, equation (1.2) [121]: a nonzero integer linear combination of fixed logarithms of fixed algebraic numbers has absolute value bounded below by a negative power of the maximum coefficient. It is an established external theorem, not a Lean axiom and not a real-positive-logarithm theorem applied outside its hypotheses.

Lemma 157.1 (Fixed-angle lower bound). *For $\lambda = \alpha/\beta = (-143 + 180\sqrt{-3})/343$, there are effective fixed constants $c_0 > 0$, $\kappa \geq 0$ such that*

$$|\lambda^n - 1| \geq c_0 n^{-\kappa} \quad (n \geq 1).$$

Proof. The number λ has absolute value one, is nonrational, and has quadratic trace $-286/343$, not an integer. Hence it is not an algebraic integer and not a root of unity. Fix $\log \lambda = i\theta$ with $|\theta| \leq \pi$, and choose $j \in \mathbb{Z}$ so that $|n\theta - 2\pi j| \leq \pi$. Then $2|j| \leq n + 1$, and the form

$$\Lambda_n = n \log \lambda - 2j \log(-1)$$

is nonzero. Applying the cited theorem to these two fixed logarithms gives $|\Lambda_n| \geq c_1 n^{-\kappa}$ after absorbing finitely many small coefficients. For $|t| \leq \pi$, $|e^{it} - 1| \geq 2|t|/\pi$. This proves the assertion. $\square$

Theorem 157.2 (Depth-three absorption). *There is a fixed effective $C_0 > 0$ such that for all $n \geq 1$,*

$$H_n^3 \leq C_0 n^{\kappa+1} \mathcal{R}_n^3 F_n.$$

Proof. Since $|\alpha| = |\beta| = 7^{3/2}$ and $|\alpha - \beta| = 18\sqrt{3}$, the Binet formula and Lemma 157.1 give

$$|U_n| \geq \frac{c_0}{18\sqrt{3}} 7^{3n/2} n^{-\kappa}.$$

Norm-height comparison gives $H_n \leq (2/\sqrt{3})7^{n/2}$. Consequently $H_n^3 \leq (48/c_0)n^\kappa |U_n|$. Now use $|U_n| \leq nr_n^3 F_n$ and $r_n \leq \mathcal{R}_n$. $\square$

Corollary 157.3 (Uniform restricted class). *For every fixed $A \geq 0$ and $\varepsilon > 0$ there is $K_{A,\varepsilon} > 0$ such that*

$$H_n \leq K_{A,\varepsilon} \mathcal{R}_n^{1+\varepsilon}$$

whenever $F_n \leq n^A$.

Proof. The theorem gives $H_n \leq C_0^{1/3} \mathcal{R}_n n^B$, where $B = (\kappa + 1 + A)/3$. Since $H_n \geq 7^{n/2}$, we have $n \leq 2 \log H_n / \log 7$. For $\delta = \varepsilon/(1 + \varepsilon)$,

$$(\log t)^B \leq (B/(e\delta))^B t^\delta \quad (t \geq 1).$$

Substituting, rearranging, and raising to $1/(1 - \delta)$ proves the result, with constants depending only on the fixed displayed parameters. $\square$

No proof that this class contains all indices, or any specified infinite family, is asserted. In particular $\log F_n = o(n)$ remains an unproved sufficient estimate for the entire fixed orbit. If $H_n > K \mathcal{R}_n^{1+\varepsilon}$, the theorem instead forces

$$F_n > \frac{K^{3/(1+\varepsilon)}}{C_0 n^{\kappa+1}} H_n^{3\varepsilon/(1+\varepsilon)}.$$

Thus any fixed-exponent *abc* counterfamily within this orbit requires exponentially large first-depth-four excess, after the index contribution is removed. This is a necessary condition, not a construction or an exclusion of such a family.

158. A GENUINE FIRST SQUARE, CHECKED BY EXACT MODULAR ARITHMETIC

The prime $p = 38629$ has $d_p = 12876 = 2^2 \cdot 3 \cdot 29 \cdot 37$ and $s_p = 2$. Trial division by integers through 196 proves primality. In the pair algebra modulo p , the element $\alpha = 1 + 18\zeta$ to the full exponent 12876 is $(1, 0)$. At exponents $12876/q$ for $q = 2, 3, 29, 37$, the results are respectively

$$(25926, 25406), \quad (38628, 1), \quad (10564, 19690), \quad (23253, 26998).$$

None is scalar, so none has U -value zero modulo p . The rank divides the full index by Theorem 154.1; any proper divisor divides one of the four tested exponents, so its exact value is 12876. At modulus $p^2 = 1492199641$ the full result is $(89232991, 0)$; at modulus $p^3 = 57642179932189$ it is

$$(43039603478354, 40547540844893), \quad 40547540844893 = 27173p^2, \quad 0 < 27173 < p.$$

The second coordinate is $18U_{12876}$, proving exact first valuation two. These data refute the first-squarefree shortcut, but are absorbed by both of the depth estimates above. They are not an abc counterexample. The current exact replay verifies the second coordinates independently by the linear recurrence, not only by pair repeated squaring.

159. FORMAL AND GLOBAL BOUNDARY

The existing `RankDepth.lean` verifies its explicit finite-list accounting, first-index implication under a stated strong-divisibility hypothesis, elementary boundary bounds and modular certificates. Its successful source run is recorded separately from this ordinary proof edition. The full finite-field order law, ramified and unramified valuation arguments, Binet interpretation, complex logarithmic-form theorem and real-exponent absorption are not all formalized by that module.

The new moving-support supplement studies how to replace the fixed constants in Lemma 157.1 when several norm generators vary. This does not remove the unresolved first-depth product on the fixed orbit, nor the prime-norm terminal and general mixed-factor support problems. No global abc theorem follows from the displayed sufficient classes alone.

INTEGRATED CHECKPOINT: MOVING NORM SUPPORT AND COMPENSATED CUBIC BOUNDARIES

New research supplement, with external analytic input and formal scope explicitly separated.

160. OBJECTS AND THE EXACT SCOPE OF THE CONTINUATION

Throughout a, b, c are positive integers with $a + b = c$ and $\gcd(a, b) = 1$. Set

$$T = abc, \quad R = \text{rad}(T), \quad M = a^2 + ab + b^2 = c^2 - ab.$$

The target remains

$$\forall \varepsilon > 0 \exists K_\varepsilon > 0 \forall (a, b, c), \quad c \leq K_\varepsilon R^{1+\varepsilon}.$$

A theorem on a restricted class is not this target. We explicitly retain the distinction between the radical of the boundary T and the distinct primes in the auxiliary norm M .

The project previously gave exact boundary overlap, norm descent, and a fixed norm-seven orbit, but no radical induction [124]. Its next supplement bounded the integer multiplicity contribution using complex logarithmic forms only for that fixed orbit [125]. The present argument joins that mechanism to varying prime-norm factorizations. The uniform constants derived below are not inherited from an unspecified fixed-base theorem.

For an integer $k \geq 1$ and $t \geq 1$ define

$$E_k(t) = \prod_{p|t} p^{(v_p(t)-k)_+}, \quad C_k(t) = \prod_{p|t} p^{(k-v_p(t))_+}, \quad (u)_+ = \max(u, 0).$$

Empty products equal one. We call E_k the excess beyond depth k and C_k the complementary truncated credit. These are concrete functions of the full prime factorization, not asserted small quantities.

161. ARITHMETIC FACTORIZATION WITH SUPPORT KEPT VISIBLE

Let $\zeta = (1 + \sqrt{-3})/2$ and $\mathcal{E} = \mathbb{Z}[\zeta]$. For $z = x + y\zeta$ put

$$N(z) = x^2 + xy + y^2, \quad P(z) = xy(x + y), \quad H(z) = \max(|x|, |y|, |x + y|).$$

Multiplication in coordinates is

$$(x, y)(u, v) = (xu - yv, xv + yu + yv),$$

and conjugation is $(x, y) \mapsto (x + y, -y)$. A pair is primitive when $\gcd(x, y) = 1$.

Lemma 161.1 (Norm and boundary support are disjoint). *For every primitive pair z , including the zero-boundary units,*

$$\gcd(N(z), |P(z)|) = 1.$$

Moreover, after positive unit rotation of a nonzero-boundary pair,

$$(609) \quad \frac{3}{4}H(z)^2 \leq N(z) \leq H(z)^2, \quad 4|P(z)| \leq H(z)^3.$$

Proof. The three integers $x, y, x + y$ are pairwise coprime. Modulo these integers the norm is respectively y^2, x^2, y^2 . Thus each individual gcd is one, proving the first assertion by multiplication. If a factor is zero, primitivity forces the pair to be a unit, with norm one. Multiplication by ζ sends (x, y) to $(-y, x + y)$ and permits a rotation to $x, y > 0$. Here $H = x + y$ and $N = H^2 - xy$; use $0 \leq xy \leq H^2/4$. The final inequality is the identity

$$H^3 - 4P = H(x - y)^2 \geq 0.$$

□

Lemma 161.2 (The norm-prime representation). *For a primitive nonunit z there are distinct rational primes $q_i \equiv 1 \pmod{3}$, elements π_i of norm q_i , integers $e_i \geq 1$, a unit u , and $e_0 \in \{0, 1\}$ such that*

$$(610) \quad z = u(1 + \zeta)^{e_0} \prod_{i=1}^r \pi_i^{e_i}, \quad N(z) = 3^{e_0} \prod_{i=1}^r q_i^{e_i}.$$

One cannot use both orientations π_i and $\bar{\pi}_i$ for the same q_i . If $z = (a, b)$ with $a, b > 0$ and $r = 0$, then $(a, b, c) = (1, 1, 2)$.

Proof. In the complex quotient of two nonzero elements, round the two real coordinates in the basis $1, \zeta$. Both errors have absolute value at most $1/2$, and their norm is at most $3/4 < 1$. This is norm-Euclidean division; hence $\mathcal{E}$ is a unique factorization domain. For a rational prime $q \neq 3$, the polynomial $X^2 - X + 1$ has a root modulo q exactly when $q \equiv 1 \pmod{3}$: its roots correspond to nontrivial cube roots of unity after changing sign, with the prime two checked directly. Thus a prime congruent to two modulo three is inert, whereas a split prime has two conjugate prime ideals of index q . In this principal ideal domain they have generators of norm q . An inert prime dividing z would divide both coordinates. Likewise, both conjugate split factors dividing z imply that the rational integer q divides both coordinates. Neither can occur. Finally $(1 + \zeta)^2 = 3\zeta$, so two ramified factors would again violate primitivity. Unique factorization proves (610). With $r = 0$, a nonunit primitive element is associated to $1 + \zeta$; its only positive sector pair is $(1, 1)$. $\square$

This representation supplies an explicit variable support for logarithmic forms, not a radical descent. Lemma 161.1 prevents charging these norm primes directly to R . At a prime-norm endpoint there is still no smaller positive abc triple supplied by this factorization.

162. AN EXTERNAL LOGARITHMIC-FORM INPUT WITH ITS DEPENDENCE RETAINED

We use only inequality (1.2) of Theorem 1.1 in Bugeaud [123], stated there as a consequence of the classical complex logarithmic-form estimates of Waldschmidt and Matveev. In degree D , for fixed choices of logarithms and a nonzero form

$$\Lambda = b_1 \log \alpha_1 + \cdots + b_m \log \alpha_m, \quad B = \max(3, |b_1|, \dots, |b_m|),$$

there is an effective $c_D > 0$, depending only on D , for which

$$(611) \quad \log |\Lambda| \geq -c_D^m \prod_{i=1}^m h^*(\alpha_i) \log B, \quad h^*(\alpha) = \max(1, h(\alpha)).$$

The statement requires a nonzero coefficient in the last position; we may reorder or delete zero coefficients. It does not require multiplicative independence. This is an external established theorem, not proved in this paper and not encoded as a Lean axiom. We do not use the refinements involving B' or B'' , or a real-only version of Matveev's theorem.

For an algebraic number θ of degree d with primitive minimal polynomial $a_d \prod_j (X - \theta_j)$, our logarithmic height convention is

$$h(\theta) = \frac{1}{d} \left(\log |a_d| + \sum_j \log \max(1, |\theta_j|) \right).$$

The standard height laws $h(uv) \leq h(u) + h(v)$, $h(u^{-1}) = h(u)$ and $h(u^m) = |m|h(u)$ follow equivalently by summing the corresponding local inequalities at all places.

Theorem 162.1 (A uniform varying-support angular penalty). *There is an effective absolute constant $C_* > 2$ with the following property. For any primitive positive triple other than $(1, 1, 2)$, use the primes q_i of Lemma 161.2, set $t = \log c$, and define*

$$(612) \quad \mathcal{A}(z) = C_*^{r+1} \log(3 + t) \prod_{i=1}^r \log q_i.$$

Then, for $\lambda_z = (z/\bar{z})^3$,

$$(613) \quad |\lambda_z - 1| \geq \exp(-\mathcal{A}(z)).$$

In particular C_ does not depend on the primes, their exponents, or their number.*

Proof. Put $\eta_i = (\pi_i/\bar{\pi}_i)^3$. These elements lie in the same degree-two field and have complex absolute value one. The two conjugates of π_i have absolute value $\sqrt{q_i}$, so $h(\pi_i) = \frac{1}{2} \log q_i$ and

$$h^*(\eta_i) \leq 3 \log q_i$$

by the height laws and $q_i \geq 7$. Also $(1 + \zeta)/(1 + \bar{\zeta}) = \zeta$, and a unit contributes its sixth power, which is one. Thus

$$\lambda_z = (-1)^{e_0} \prod_i \eta_i^{e_i}.$$

Each $\eta_i \neq 1$: equality would make π_i associated to its conjugate, impossible for a split prime. Cubic expansion gives

$$(614) \quad z^3 - \bar{z}^3 = 3\sqrt{-3}P(z).$$

Hence $\lambda_z \neq 1$ because $P(z) \neq 0$.

Choose principal arguments $\log \eta_i = i\theta_i$, with $|\theta_i| \leq \pi$, and choose an integer j so that

$$\Lambda = \sum_i e_i \log \eta_i + (e_0 - 2j) \log(-1)$$

has imaginary part in $[-\pi, \pi]$. It is nonzero, and

$$|e_0 - 2j| \leq 1 + \sum_i e_i, \quad \sum_i e_i \leq \frac{\log N(z)}{\log 7} \leq \frac{2t}{\log 7}.$$

The coefficient size is therefore at most $3 + 2t/\log 7$. Apply (611) with at most $r + 1$ logarithms in degree two and $h^*(-1) = 1$. Since $|e^{iv} - 1| \geq 2|v|/\pi$ for $|v| \leq \pi$, we get

$$-\log |\lambda_z - 1| \leq c_2^{r+1} 3^r \prod_i \log q_i \log(3 + 2t/\log 7) + \log(\pi/2),$$

where we enlarge c_2 to at least one. Both the fixed factors and the last additive constant are absorbed by a fixed C_* in (612), for example by taking $C_* \geq 12 \max(1, c_2)$. No generator is kept fixed in this argument. $\square$

163. CUBIC RADICAL CONTROL ON A GROWING-SUPPORT CLASS

Theorem 163.1 (A general estimate with a visible penalty). *For every primitive positive triple, after increasing the same fixed constant if necessary,*

$$(615) \quad c^3 \leq 8e^{A(z)} R^3 E_3(abc).$$

For $(1, 1, 2)$ one may simply set $A = 0$.

Proof. Taking absolute values in (614) gives

$$abc = \frac{N(z)^{3/2}}{3\sqrt{3}} |\lambda_z - 1|.$$

Since $N(z) \geq 3c^2/4$, Theorem 162.1 yields

$$abc \geq c^3 e^{-A(z)}/8.$$

Prime by prime $v_p(abc) \leq 3 + (v_p(abc) - 3)_+$, so $abc \leq R^3 E_3(abc)$. Combining proves (615). The omitted small triple satisfies it directly. $\square$

The penalty in (615) is not asserted sublinear in $\log c$ for all triples. In particular, a large prime norm has $r = 1$ and $\log q \asymp \log c$, producing a penalty of order $\log c \log \log c$, not the required $o(\log c)$.

Corollary 163.2 (Fixed rank, varying primes). *Fix an integer $r_0 \geq 0$ and real numbers $A > 0$, $B \geq 0$. For every $\varepsilon > 0$ there is $K_{r_0, A, B, \varepsilon} > 0$, independent of the triple and its prime support, such that*

$$c \leq K_{r_0, A, B, \varepsilon} R^{1+\varepsilon}$$

for all triples with $r \leq r_0$, every $q_i \leq (\log c)^A$, and $E_3(abc) \leq (\log c)^B$.

Proof. With $t = \log c$, (612) gives $\mathcal{A} = O_{r_0, A}((\log t)^{r_0+1}) = o(t)$ uniformly on this class. Also $\log E_3 \leq B \log t = o(t)$. Taking logarithms in (615) and using these uniform bounds gives, for sufficiently large c ,

$$3 \log c \leq 3 \log R + \frac{3\varepsilon}{1+\varepsilon} \log c.$$

Thus $\log c \leq (1+\varepsilon) \log R$. The finitely many smaller triples are absorbed in K , with a threshold depending only on the displayed parameters. $\square$

Theorem 163.3 (The number of norm primes may grow). *Fix $A > 0$, $B \geq 0$ and $0 < \delta < 1$. For every $\varepsilon > 0$ there exists $K_{A,B,\delta,\varepsilon} > 0$ such that the standard $1+\varepsilon$ inequality holds uniformly on the class defined, for $c > \exp(\exp(e))$, by*

$$(616) \quad q_i \leq (\log c)^A, \quad r \leq (1-\delta) \frac{\log \log c}{\log \log \log c}, \quad E_3(abc) \leq (\log c)^B.$$

Any smaller triples can be included by enlarging the same constant. No assertion that this class exhausts the triples, or contains an infinite sequence, is needed or made.

Proof. Put $t = \log c$, $s = \log t$, $u = \log s$. As c grows, all three tend to infinity. The first two conditions imply

$$\log \mathcal{A} \leq r(u + \log(AC_*)) + O(u) \leq (1-\delta)s \left(1 + \frac{\log(AC_*)}{u}\right) + O(u).$$

All implied constants are independent of the primes and their exponents. Since $u = o(s)$, the right side is at most $(1-\delta/2)s$ past a threshold depending only on A, δ . Consequently

$$\mathcal{A} \leq t^{1-\delta/2}, \quad \log E_3 \leq B \log t.$$

These are uniformly $o(t)$. Apply the explicit absorption step in the preceding proof. Equivalently choose a threshold where $\log 8 + t^{1-\delta/2} + B \log t \leq 3\varepsilon t/(1+\varepsilon)$. For the smaller range $c \leq c_0$ the choice $K = \max(1, c_0)$ already works, because $R \geq 1$. This proves the required constant order without any dependence on individual prime supports. $\square$

164. COMPENSATION, AND A PRECISE LIMIT OF THE ABSORPTION MECHANISM

Proposition 164.1 (Exact truncation, rather than discarding credit). *For every $t \geq 1$ and $k \geq 1$,*

$$(617) \quad tC_k(t) = \text{rad}(t)^k E_k(t).$$

More generally, for any finite list (p_i, s_i, ℓ_i) of positive integer bases and nonnegative exponents, put $V = \prod_i p_i^{s_i+\ell_i}$, $R_ = \prod_i p_i$, $L = \prod_i p_i^{\ell_i}$, $F = \prod_i p_i^{(s_i-k)_+}$ and $C = \prod_i p_i^{(k-s_i)_+}$. Then*

$$VC = R_*^k LF.$$

No primality or distinctness is needed for this algebraic identity. Those hypotheses are necessary when interpreting R_ as an actual radical.*

Proof. The exponent identity $s + \ell + (k-s)_+ = k + \ell + (s-k)_+$ holds in the cases $s \leq k$ and $s \geq k$. Multiply the corresponding prime-power equalities. For (617), use the distinct actual prime factors, $s_i = v_{p_i}(t)$ and $\ell_i = 0$. $\square$

For example $t = 2^6 \cdot 3$, $k = 3$, gives $E_3 = 8$, $C_3 = 9$, and $192 \cdot 9 = 6^3 \cdot 8$. A large uncompensated excess is not on its own a statement that a radical bound fails.

For the preceding norm-seven sequence, Proposition 164.1 gives

$$|U_n| \prod_{d_p|n} p^{(3-s_p)_+} = \text{rad}(|U_n|)^3 \mathcal{L}_n \prod_{d_p|n} p^{(s_p-3)_+}.$$

It therefore retains the low-first-depth credits discarded in the one-sided bound. A uniform estimate on the resulting correlated quotient is still required; this exact identity is not advertised as an independent solution.

Proposition 164.2 (Cubic degree is a real barrier for this mechanism). *On the primitive norm-seven orbit $z_n = (2 + \zeta)^n$, no constants $C > 0$, $B \geq 0$ can satisfy*

$$H(z_n)^k \leq Cn^B |P(z_n)| \quad \text{for all } n \geq 1$$

when the fixed real exponent k is greater than three.

Proof. By (609) and $N(z_n) = 7^n$,

$$\frac{H(z_n)^k}{|P(z_n)|} \geq 4H(z_n)^{k-3} \geq 4 \cdot 7^{n(k-3)/2}.$$

An exponential with positive rate exceeds every fixed polynomial, proving the assertion. $\square$

Thus first fourth powers cannot be absorbed merely by replacing a cubic size estimate by a quartic one with polynomial index loss. This refutes that precise modification, not all higher-order or multi-object methods.

165. NON-ARCHIMEDEAN LOCALIZATION OF THE REMAINING MULTIPLICITY

The angular estimate controls the infinite place, but does not bound the excess product. A second, independent input gives a uniform elimination of small boundary primes within the same norm-support regime.

We use the first inequality of Bugeaud's Theorem 1.3 [123], a stated consequence of Yu's logarithmic-form estimates. Let $m \geq 2$ and let α_i lie in a number field of degree D . For integers b_i with $\prod_i \alpha_i^{b_i} \neq 1$, put $B_0 = \max(3, |b_1|, \dots, |b_m|)$. At a place above the rational prime p , normalize the valuation by $v_p(p) = 1$. There is an effective $c_D > 0$, depending only on D , such that

$$(618) \quad v_p \left(\prod_i \alpha_i^{b_i} - 1 \right) < c_D^m p^D \prod_i h^*(\alpha_i) \log B_0.$$

Only this first inequality is used, not the conditional refinements or the open improvement discussed later in that source. This theorem is an established external input and is not formalized by the accompanying Lean module.

Theorem 165.1 (Uniform local multiplicity bound). *Enlarge the absolute constant C_* in Theorem 162.1 once so it also dominates the degree-two constant in (618). For every nontrivial primitive positive triple and every prime $p \mid abc$,*

$$v_p(abc) \leq p^2 \mathcal{A}(z).$$

Consequently, for $Y \geq 2$ and the full small-prime part

$$S_Y(abc) = \prod_{p \mid abc, p \leq Y} p^{v_p(abc)},$$

one has

$$(619) \quad \log S_Y(abc) \leq \mathcal{A}(z) Y^3 \log Y.$$

Proof. Use the actual factorization of Lemma 161.2. Since $\gcd(N(z), abc) = 1$, both z and $\bar{z}$ are units at every place above $p \mid abc$. The exact cubic identity implies

$$v_p(\lambda_z - 1) = v_p(abc) + v_p(3\sqrt{-3}) \geq v_p(abc).$$

This also treats $p = 2$ and $p = 3$: the added valuation is zero except at three, where it is $3/2$ under the stated normalization.

Write $\lambda_z = (-1)^{e_0} \prod_i \eta_i^{e_i}$ as in the angular proof. There are $r + 1 \geq 2$ algebraic factors, all in the fixed quadratic field. Zero coefficients are permitted in (618); hence the factor -1 can be retained also when $e_0 = 0$. The product is not one, by $P(z) \neq 0$. The same coefficient and height estimates give

$$\log B_0 \leq \log(3 + 2 \log c / \log 7), \quad h^*(\eta_i) \leq 3 \log q_i, \quad h^*(-1) = 1.$$

Apply (618) with $D = 2$ and absorb fixed factors into the common C_* . This proves the local bound uniformly in the rational prime p . Summing $v_p(abc) \log p$ for $p \leq Y$ and using

$$\sum_{p \leq Y} p^2 \log p \leq \lfloor Y \rfloor Y^2 \log Y \leq Y^3 \log Y$$

proves (619), without a prime-number asymptotic. $\square$

Corollary 165.2 (Small primes require no separate multiplicity hypothesis). *Fix $A > 0$ and $0 < \delta < 1$. Suppose only the first two norm-support conditions in (616) hold. With $t = \log c$ and $Y = t^{\delta/12}$, for all sufficiently large triples in this class,*

$$\log S_Y(abc) \leq \frac{\delta}{12} t^{1-\delta/4} \log t = o(t)$$

uniformly in the triple and its prime support. In particular the same bound holds for the contribution to $\log E_3(abc)$ from $p \leq Y$.

Proof. The proof of Theorem 163.3, applied with the enlarged but still absolute constant, gives $\mathcal{A}(z) \leq t^{1-\delta/2}$ past a uniform threshold. Substitute $Y = t^{\delta/12}$ into (619). Since $Y^3 = t^{\delta/4}$ and $\log Y = (\delta/12) \log t$, the bound follows. The ratio to t tends to zero because $t^{-\delta/4} \log t \rightarrow 0$. No hypothesis on the small-prime multiplicities was used. $\square$

Theorem 165.3 (A larger, explicitly localized uniformity class). *Fix $A > 0$, $B \geq 0$, $0 < \delta < 1$ and $\varepsilon > 0$. There exists a constant $K_{A,B,\delta,\varepsilon} > 0$ such that $c \leq K_{A,B,\delta,\varepsilon} R^{1+\varepsilon}$ for every primitive positive triple satisfying the first two conditions of (616) and only the large-prime excess condition*

$$\prod_{\substack{p|abc \\ p > (\log c)^{\delta/12}}} p^{(v_p(abc)-3)_+} \leq (\log c)^B.$$

All sufficiently small triples may again be included by enlarging the same constant. In particular, final exponents at most three are required only above the moving cutoff, not at the small primes.

Proof. The low-prime contribution to $\log E_3$ is uniformly $o(\log c)$ by Corollary 165.2. The remaining contribution is at most $B \log \log c$, also uniformly $o(\log c)$. Together with $\mathcal{A}(z) = o(\log c)$, insert these estimates into (615) and use the same explicit $3\varepsilon/(1+\varepsilon)$ absorption as before. All thresholds depend only on the displayed parameters and the fixed external theorem constants. $\square$

This is a genuine removal of a separate hypothesis on small-prime multiplicities, but only inside the stated norm-support class. The high-prime excess, large norm primes and unrestricted growing factor support are still not controlled. The theorem does not assert that its restricted class is infinite or that it covers all triples. The non-Archimedean estimates and their analytic use are ordinary/source-dependent proofs, not additional kernel-checked declarations.

Proposition 165.4 (From first-depth excess to the present budget). *On the norm-seven orbit, with $P(z_n) = 6U_n$, the exact local valuation law of the preceding supplement implies*

$$E_3(|P(z_n)|) \leq 6\mathcal{L}_n F_n \leq 6n F_n.$$

Proof. For each supported prime write $v_p(U_n) = s_p + \ell_p$. The elementary inequality

$$(s_p + \ell_p + v_p(6) - 3)_+ \leq (s_p - 3)_+ + \ell_p + v_p(6)$$

gives the claim on multiplying; any prime supplied only by six contributes at most its factor in six. Finally $\mathcal{L}_n \mid n$. Thus a fixed polynomial bound on F_n implies a polynomial logarithmic bound on the actual-boundary excess because $n \ll \log H_n$. This connects the preceding sufficient class to the present one without confusing first exponents with final exponents. $\square$

166. DEPENDENCY LEDGER AND FORMAL SCOPE

The proof chain for Theorem 163.3 is

$$\begin{aligned}
 & \text{Euclidean factorization and heights} \\
 & + \text{established complex logarithmic forms} \\
 & \implies \text{uniform angular penalty} \\
 & \implies \text{general cubic estimate} \\
 & \implies \text{the explicitly restricted uniform theorem.}
 \end{aligned}$$

The first arrow uses the external theorem (611); it is not a new proof of that theorem. None of the arrows asserts the unrestricted conditions (616).

The companion `MovingSupport.lean` formalizes the full norm-boundary gcd theorem, the cubic cap, cubic conjugate coordinates, and the general finite-list compensated identity and its elementary consequences. It imports the earlier checked boundary module. The external logarithmic estimate, algebraic height bounds, Euclidean classification and analytic asymptotic absorption remain ordinary/source-dependent mathematics. Compiler output and axiom dependencies are recorded only after actual execution, separately from this mathematical text.

The test program uses exact integer pairs and trial division for bounded factorization ranges; list identities use generated integer data. It does not numerically test the unknown effective constant in Theorem 162.1, and it does not certify the unrestricted asymptotic conjecture. There is no claim of independent autonomous-agent review or external peer review.

Direction	Reliable input	Remaining exact obligation
Eisenstein / angular	Varying-support penalty, this paper	Large norm primes and general mixed support
First-depth / packets	Exact compensated accounting	Uniform high-depth mass after low-depth credit
Arithmetic derivatives	Constructive integer lifting	Global real transverse term, still at <i>abc</i> strength
FCRT / multi-output	Once-charged accounting	Actual arithmetic uniform residual bound
Pell / Mersenne	Rank and valuation ledgers	First-appearance weighted distribution at the required scale
IUT / geometry	Source-labelled local interfaces	Full same-pilot all-place comparison and uniform height bound

The new restricted theorem handles moving primes and a growing number of them, improving the fixed-generator scope of the preceding argument. It does not settle prime-norm terminal triples, prove a bound for $E_3(abc)$ on all triples, or show that the exact low-depth credits suffice globally. These are the places where a complete *abc* proof would still require genuinely new estimates. No unconditional `ABCConjecture` term is asserted.

167. EXPONENT BLOCKS INSTEAD OF PRIME-BY-PRIME LOGARITHMS

Continuation status. Author: ChatGPT. The preceding moving-support supplement proves a restricted theorem, not standard *abc*. We retain its objects and replace only the representation used in its logarithmic forms. The new results below are ordinary mathematical proofs using two explicitly stated established inputs. The companion Lean file covers elementary regrouping and obstruction certificates, not these external estimates or a closed term of type `ABCConjecture`. No claim of priority or independent peer review is made.

Write $T = abc$, $R = \text{rad}(T)$, $t = \log c$, and

$$z = a + b\zeta, \quad \zeta^2 - \zeta + 1 = 0, \quad M = N(z) = a^2 + ab + b^2.$$

As before, a, b are positive coprime integers and $c = a + b$. For $N \geq 1$ define

$$E_3(N) = \prod_{p|N} p^{(v_p(N)-3)_+}, \quad C_3(N) = \prod_{p|N} p^{(3-v_p(N))_+}.$$

For a real $Y \geq 1$, a superscript $> Y$ restricts either product to primes $p > Y$. Define also the positive rational, compensated tail

$$W_Y(T) = E_3^{>Y}(T)/C_3^{>Y}(T).$$

It can be less than one. These are functions of actual factorizations, not abstract capacities.

We use the prime-norm factorization already established in the moving-support supplement:

$$(620) \quad z = u(1 + \zeta)^{e_0} \prod_{i=1}^r \pi_i^{e_i}, \quad M = 3^{e_0} \prod_{i=1}^r q_i^{e_i},$$

where $e_0 \in \{0, 1\}$, $q_i = N(\pi_i) \equiv 1 \pmod{3}$ are distinct primes, $e_i \geq 1$, and a conjugate pair of factors is never used together. The only positive triple with $r = 0$ is $(1, 1, 2)$; it is handled directly. The norm-boundary identity $\gcd(M, T) = 1$ and

$$(621) \quad z^3 - \bar{z}^3 = 3\sqrt{-3}T, \quad \frac{3}{4}c^2 \leq M \leq c^2$$

will be used with their full arithmetic hypotheses.

Definition 167.1 (A partition into exponent blocks). Let $\mathcal{P} = \{I_1, \dots, I_m\}$ be any partition of $\{1, \dots, r\}$ into nonempty subsets. Put

$$k_j = \gcd\{e_i : i \in I_j\}, \quad w_j = \prod_{i \in I_j} \pi_i^{e_i/k_j}, \quad Q_j = N(w_j) = \prod_{i \in I_j} q_i^{e_i/k_j}, \quad S = \sum_{j=1}^m k_j.$$

Thus $k_j \geq 1$, $Q_j \geq 7$, and

$$(622) \quad z = u(1 + \zeta)^{e_0} \prod_{j=1}^m w_j^{k_j}, \quad \sum_j k_j \log Q_j = \log M - e_0 \log 3 \leq 2t.$$

The one-block partition has $k_1 = g := \gcd(e_1, \dots, e_r)$. Grouping equal values of e_i is another allowed partition; its number of blocks is the number of distinct positive exponents, not the number of primes.

All assertions in (622) follow from integer divisibility and associative commutative multiplication. In particular grouping does not create an algebraic root: the displayed w_j are elements of the original quadratic ring. Allowing a partition with a few blocks also handles $g = 1$, for example when two blocks have coprime contents k and $k + 1$.

167.1. The two established inputs and their exact use. We use inequalities (1.2) and the first inequality of Theorem 1.3 of Bugeaud [127]. In a number field of fixed degree D , the first asserts, for a nonzero integer linear form in s fixed determinations of logarithms,

$$\log |b_1 \log \alpha_1 + \dots + b_s \log \alpha_s| \geq -c_D^s \prod_i h^*(\alpha_i) \log \max(3, |b_1|, \dots, |b_s|).$$

Zero coefficients can be deleted and a nonzero last coefficient chosen. The second, for $s \geq 2$ and a product different from one, asserts

$$v_p \left(\prod_i \alpha_i^{b_i} - 1 \right) < d_D^s p^D \prod_i h^*(\alpha_i) \log \max(3, |b_1|, \dots, |b_s|).$$

Here $v_p(p) = 1$ and $h^*(\alpha) = \max(1, h(\alpha))$. The constants depend only on D . We use neither a conjectural refinement nor one branch of an unproved alternative. Our applications have $D = 2$ throughout. These results are external established inputs, not new theorems proved or axiomatized in the companion Lean development.

Theorem 167.2 (Blockwise archimedean and non-archimedean bounds). *There is an effective absolute constant $C > 2$ such that, for every partition in Definition 167.1, the quantity*

$$(623) \quad \mathcal{B}_{\mathcal{P}}(z) = C^{m+1} \log(3 + S) \prod_{j=1}^m \log Q_j$$

satisfies, with $\lambda = (z/\bar{z})^3$,

$$(624) \quad |\lambda - 1| \geq \exp(-\mathcal{B}_{\mathcal{P}}(z)),$$

$$(625) \quad v_p(T) \leq p^2 \mathcal{B}_{\mathcal{P}}(z) \quad (p \mid T),$$

$$(626) \quad c^3 \leq 8 \exp(\mathcal{B}_{\mathcal{P}}(z)) R^3 \frac{E_3(T)}{C_3(T)},$$

$$(627) \quad \log \prod_{\substack{p \mid T \\ p \leq Y}} p^{v_p(T)} \leq \mathcal{B}_{\mathcal{P}}(z) Y^3 \log Y \quad (Y \geq 2).$$

The same conclusions hold with the minimum of $\mathcal{B}_{\mathcal{P}}(z)$ over all partitions. No sublinear bound for this minimum on all triples is asserted.

Proof. Put $\eta_j = (w_j/\bar{w}_j)^3$. Both conjugates of w_j have absolute value $\sqrt{Q_j}$, so $h(w_j) = \frac{1}{2} \log Q_j$ and the height laws give $h^*(\eta_j) \leq 3 \log Q_j$. None is one: otherwise $w_j^3 = \bar{w}_j^3$, so each split prime factor of w_j would also divide its conjugate, contrary to (620). The unit and ramified factors contribute respectively one and $(-1)^{e_0}$. Consequently

$$\lambda = (-1)^{e_0} \prod_j \eta_j^{k_j} \neq 1,$$

where nonvanishing also follows directly from (621) and $T > 0$. Choose principal arguments $\log \eta_j = i\theta_j$, $|\theta_j| \leq \pi$, and an integer v for which

$$\Lambda = \sum_j k_j \log \eta_j + (e_0 - 2v) \log(-1)$$

has imaginary part in $[-\pi, \pi]$. It is nonzero, and $|e_0 - 2v| \leq 1 + S$. Apply the complex bound with at most $m + 1$ logarithms, then use $|e^{ix} - 1| \geq 2|x|/\pi$ on that interval. All factors 3^m , the constants for degree two, and the additive $\log(\pi/2)$ are absorbed in one absolute C , proving (624).

For $p \mid T$, the identity $\gcd(M, T) = 1$ implies that $z, \bar{z}$ and every $w_j, \bar{w}_j$ are units at places over p . Equation (621) gives

$$v_p(\lambda - 1) = v_p(T) + v_p(3\sqrt{-3}) \geq v_p(T).$$

Apply the second external bound to the $m + 1 \geq 2$ numbers $\eta_1, \dots, \eta_m, -1$ with coefficients $k_1, \dots, k_m, e_0$. Retaining a zero last coefficient is permitted here. This proves (625), including the primes two and three. Summing and using $\sum_{p \leq Y} p^2 \log p \leq Y^3 \log Y$ proves (627).

Finally (621) and $M \geq 3c^2/4$ give $T \geq c^3 \exp(-\mathcal{B}_{\mathcal{P}})/8$. Prime by prime $TC_3(T) = R^3 E_3(T)$, proving (626). Dropping $C_3(T) \geq 1$ recovers the one-sided bound. The compensated version retains all low-depth credit. There are finitely many partitions and every one gives all the stated inequalities. Choosing a minimizing partition is therefore legitimate. $\square$

168. REMOVING BOTH PRIME-SIZE AND PRIME-COUNT RESTRICTIONS ON EXPLICIT PROFILES

Corollary 168.1 (One block, without fixing the root). *Let $g = \gcd(e_1, \dots, e_r)$. Then an admissible penalty is*

$$C^2 \frac{\log M - e_0 \log 3}{g} \log(3 + g) \leq 2C^2 t \frac{\log(3 + g)}{g}.$$

Neither the size nor the number of norm primes is restricted. For $g \geq 64$, put $Y = g^{1/6}$. The full small-prime part satisfies

$$(628) \quad \log \prod_{\substack{p \mid T \\ p \leq g^{1/6}}} p^{v_p(T)} \leq \frac{C^2}{3} t \frac{\log(3 + g) \log g}{\sqrt{g}}.$$

Both ratios to t tend to zero as $g \rightarrow \infty$, uniformly in all other arithmetic data.

Proof. Take the one-block partition in Theorem 167.2; $\log Q_1 = (\log M - e_0 \log 3)/g$. Substitute $Y = g^{1/6}$ into (627). The displayed limits are elementary. $\square$

Corollary 168.2 (A uniform restriction on unbalanced norm powers). *For each $0 < \eta < 1$ there is an effective G_η such that every primitive positive triple with $g \geq G_\eta$ has*

$$\min(a, b) > c^{1-\eta}.$$

In particular this is uniform even when $M = 3^{e_0} Q^g$ has arbitrarily large or numerous split prime factors in Q .

Proof. If $\min(a, b) \leq c^{1-\eta}$ then $T \leq c^{3-\eta}$. The lower bound for T in the proof of Theorem 167.2 yields

$$\eta t \leq \log 8 + 2C^2 t \frac{\log(3+g)}{g}.$$

But $t \geq g \log 7/2$. Choose G_η so large that, for every $g \geq G_\eta$, each of $2 \log 8/(g \log 7)$ and $2C^2 \log(3+g)/g$ is at most $\eta/3$. Dividing by t gives a contradiction. Such a threshold is effective and depends only on η . $\square$

Theorem 168.3 (A high-content conditional class with no norm-support bound). *Fix $B \geq 0$ and $\varepsilon > 0$. There is an effective integer $G_{B,\varepsilon}$ such that every primitive positive triple with*

$$g \geq G_{B,\varepsilon}, \quad W_{g^{1/6}}(T) \leq g^B$$

satisfies $c \leq R^{1+\varepsilon}$. In particular the stronger condition $E_3^{>g^{1/6}}(T) \leq g^B$ suffices. The large-prime hypothesis is not asserted for all triples. In particular this is not a proof of standard abc.

Proof. The small-prime contribution to $\log(E_3/C_3)$ is at most the left side of (628); the large-prime contribution is at most $B \log g$. Since $t \geq g \log 7/2$, the sum of $\log 8$, the angular penalty, and $\log(E_3/C_3)$ divided by t is at most

$$\frac{2 \log 8}{g \log 7} + 2C^2 \frac{\log(3+g)}{g} + \frac{C^2 \log(3+g) \log g}{3 \sqrt{g}} + \frac{2B \log g}{g \log 7}.$$

This tends to zero independently of the norm primes. Choose the threshold so that it is at most $3\varepsilon/(1+\varepsilon)$ and apply (626) after taking logarithms. $\square$

There is also a usual uniform-constant formulation on any explicitly fixed class for which $g \geq f(\log c)$ with a fixed function $f(t) \rightarrow \infty$ and the same excess hypothesis: beyond a threshold Theorem 168.3 applies, and below it finitely many triples are absorbed into K_ε . No assertion that g diverges on all prospective counterexamples is made.

Theorem 168.4 (Several blocks, including global content one). *Fix an integer $d \geq 1$ and reals $0 < \delta < 1$, $B \geq 0$. For every $\varepsilon > 0$ there is a constant $K_{d,\delta,B,\varepsilon} > 0$ such that the standard $1 + \varepsilon$ inequality holds for every primitive positive triple admitting a partition with*

$$(629) \quad m \leq d, \quad \prod_{j=1}^m k_j \geq t^{m-1+\delta}, \quad W_{t^{\delta/12}}(T) \leq t^B.$$

These conditions define the class for $t \geq 2$; the finitely many smaller triples may be included by enlarging the same constant. No restrictions on individual q_i or their total count are imposed.

Proof. Weighted AM–GM in (622) gives

$$\prod_j \log Q_j \leq \frac{(2t/m)^m}{\prod_j k_j}, \quad S \leq \frac{2t}{\log 7}.$$

For $m \leq d$, Theorem 167.2 therefore gives

$$\mathcal{B}_P \leq C_d t^{1-\delta} \log(3+t) \leq t^{1-\delta/2}$$

past a threshold depending only on d, δ . With $Y = t^{\delta/12}$, (627) is at most $(\delta/12)t^{1-\delta/4} \log t = o(t)$ uniformly. The small-prime contribution to $\log(E_3/C_3)$ is no greater than this full small-prime

bound; the high-prime signed cost is at most $B \log t = o(t)$. In (626) absorb the total cost below $3\epsilon t/(1 + \epsilon)$; this proves the result above a uniform threshold. Below it $K = \max(1, c_0)$ suffices. $\square$

168.1. Actual profiles not covered by a polylogarithmic norm-prime cutoff. There are infinitely many primes $q \equiv 1 \pmod{3}$. An elementary proof is to suppose a finite list, take x equal to three times their product, and take a prime divisor of $x^2 + x + 1$. It is neither three nor any listed prime, and the order of x modulo it is three, so it is another prime congruent to one. Choose π_q of norm q in the same ring, with one fixed orientation.

First take $k = \lceil (\log q)^2 \rceil$ and $z = \pi_q^k$. Its coordinates are primitive; a common rational prime divisor would require a conjugate factor. Its boundary is nonzero because its norm is greater than one and a primitive zero-boundary pair is a unit. After rotation it gives an actual triple with

$$g = k, \quad \log c = \frac{k}{2} \log q + O(1) \asymp (\log q)^3.$$

Thus $q > (\log c)^A$ eventually for every fixed A while the one-block angular and small-prime bounds apply uniformly. This demonstrates an infinite extension of the *profile* domain, not an infinite family satisfying the additional unproved high-prime excess hypothesis.

For a content-one example, let $q \neq 7$, put $k = \lceil (\log q)^3 \rceil$, and use $z = \pi_q^k(2 + \zeta)^{k+1}$. It is again primitive and nondegenerate, its global content is $\gcd(k, k+1) = 1$, and the two indicated blocks satisfy

$$\log c \asymp (\log q)^4, \quad k(k+1) \asymp (\log q)^6.$$

Hence their product exceeds $t^{1+\delta}$ for any fixed $\delta < 1/2$ and large q . Again q exceeds every fixed power of t . The profile hypothesis in Theorem 168.4 is consequently not just the earlier bounded-prime condition or a disguised common-power condition.

Proposition 168.5 (Exact limit on squarefree split-norm profiles). *Suppose all split exponents e_i equal one. Then the minimum of (623) over all partitions is exactly*

$$\min_{\mathcal{P}} \mathcal{B}_{\mathcal{P}}(z) = C^2 \log 4 (\log M - e_0 \log 3).$$

It is attained by the single block. In particular regrouping removes the previous artificial factor $\log \log c$ at a prime-norm endpoint, but cannot make this particular bound $o(\log c)$ there.

Proof. Every block has $k_j = 1$, hence $S = m$ and $\sum_j \log Q_j = \log M - e_0 \log 3 =: V$. Set $l = \log 7 > 1$. Since every $\log Q_j/l \geq 1$, their product is at least their average; therefore $\prod_j \log Q_j \geq l^{m-1}V/m$. Also $C^{m-1}l^{m-1} \geq 2^{m-1} \geq m$ for $m \geq 1$. Thus every partition has penalty at least $C^2 \log 4 V$, with equality for the single block. Since $V = 2 \log c + O(1)$, the last assertion follows. This concerns the proved bound, not a lower bound on the true angular defect. $\square$

169. A LARGE-PRIME EXCESS BOUND THAT CANNOT HOLD GLOBALLY

An attempted global proof must not simply assert that every large-prime excess factor is small. That assertion is false even on triples that already satisfy the stronger bound $c < R$.

We use only the classical Linnik theorem: there are absolute $A_0, L > 0$ such that each reduced progression $a \bmod m$ contains a prime at most $A_0 m^L$. Its quantitative form is recorded, for example, in Xylouris [128]. We do not require the best numerical exponent.

Theorem 169.1 (Deep large-prime excess with negative *abc* defect). *Fix an integer $h \geq 4$. For each odd prime p let ℓ_p be the least prime in the progression $p^h - 1 \pmod{p^{h+1}}$, and put*

$$(a_p, b_p, c_p) = (1, \ell_p, \ell_p + 1).$$

Then $v_p(c_p) = h$ and

$$p^h \leq c_p \leq (A_0 + 1)p^{(h+1)L'}$$

for an absolute $L' \geq \max(1, L)$, after harmless enlargement of A_0 . Moreover

$$R_p = \text{rad}(a_p b_p c_p) \geq 2\ell_p > c_p.$$

For every fixed $A > 0$, $p > (\log c_p)^A$ eventually and

$$(630) \quad \liminf_{p \rightarrow \infty} \frac{\log E_3^{>(\log c_p)^A}(a_p b_p c_p)}{\log c_p} \geq \frac{h-3}{(h+1)L'} > 0.$$

Thus neither a uniform polylogarithmic bound on this raw excess nor its uniform $o(\log c)$ logarithmic bound is true on all primitive triples. The family is not a counterexample to *abc* and does not refute the restricted norm-profile theorems above or in the preceding manuscript.

Proof. The residue $p^h - 1$ is coprime to p^{h+1} . Linnik supplies the upper bound. Write $\ell_p = p^h - 1 + jp^{h+1}$ with $j \geq 0$. Then $c_p = p^h(1 + jp)$, which proves exact valuation h and the lower bound. The prime ℓ_p is larger than two, so it is odd and $2 \mid c_p$; also $\gcd(\ell_p, c_p) = 1$. These distinct supported primes imply $R_p \geq 2\ell_p > c_p$.

The height upper bound gives $\log c_p \leq \log(A_0 + 1) + (h+1)L' \log p$. Consequently $p > (\log c_p)^A$ for large p . Its contribution to the restricted excess is p^{h-3} . Divide the logarithm of this contribution by the height bound and take the lower limit, proving (630). Finally $c_p \geq p^h$ makes the family height-unbounded. $\square$

Proposition 169.2 (Explicit compensation and an angular obstruction). *On the family in Theorem 169.1,*

$$E_3(T) = E_3(c_p), \quad C_3(T) = \ell_p^2 C_3(c_p), \quad \frac{E_3(T)}{C_3(T)} \leq \frac{c_p}{\ell_p^2}.$$

For every triple $1 + b = c$, including this family,

$$-\log |(z/\bar{z})^3 - 1| = \log c + O(1) \quad (c \rightarrow \infty),$$

with an absolute implied constant. Therefore a uniform sublinear angular penalty on all triples is also impossible.

Proof. The prime ℓ_p occurs with exponent one in $T = \ell_p c_p$ and is absent from c_p , giving the first two equalities. Use $E_3(c_p) \leq c_p$ and $C_3(c_p) \geq 1$ for the inequality. For $a = 1, b = c - 1$, $M = c^2 - c + 1$ and (621) gives exactly

$$|(z/\bar{z})^3 - 1| = \frac{3\sqrt{3}c(c-1)}{(c^2 - c + 1)^{3/2}}.$$

After multiplication by c , this tends to $3\sqrt{3}$, proving the claim. $\square$

This identifies an obstruction to bounding the archimedean and positive multiplicity losses independently, not to correlated methods. Indeed the exact identity with $\theta = 4ab/c^2$ is

$$\left(\frac{c}{R}\right)^3 = \frac{4E_3(T)}{\theta C_3(T)}.$$

Proving the corresponding global correlated bound without new arithmetic input would be another expression of *abc*, not a completed bridge theorem.

170. DEPENDENCY AUDIT, INDEPENDENT COMPUTATIONS, AND PRECISE REMAINING GATES

The positive proof chain is

$$\begin{aligned} &\text{actual prime-norm factors} \longrightarrow \text{integer exponent blocks} \\ &\Downarrow \quad \text{two established logarithmic-form bounds} \\ &\text{uniform block penalty and small-prime bound.} \end{aligned}$$

The uniform $1 + \varepsilon$ consequences additionally require their explicitly stated high-prime excess conditions. The Linnik construction independently proves that an unrestricted positive-excess smallness claim is false. The prime-norm terminal case has just one split exponent equal to one: there is no nontrivial exponent block to extract. It remains unresolved; the present method does not supply a hidden small predecessor for it.

The exact verifier factors bounded norms, reconstructs their oriented ring factors, compares exponent-block products to the original coordinates, and independently checks norm products and primitive coordinates. It also certifies finite prime-neighbour examples by trial division and checks their exact valuation, radical and credit. Finite computations neither prove Linnik nor estimate unknown logarithmic-form constants. Machine-readable bounds and counts are recorded only after execution.

The Lean companion proves the finite commutative multiplication and norm regrouping cores, universal arithmetic progression divisibility and nondivisibility, and explicit integer inequalities. These statements do not formalize the external logarithmic-form estimates, the general splitting classification, the Linnik existence theorem, or all asymptotic absorption. A successful scoped compile must not be reported as a full-repository build, independent peer review, autonomous-subagent audit, or a proof of standard *abc*.

The retained gates are: control the *correlated* large-prime multiplicity and low-depth credit; handle triples with bounded block contents or many arithmetically incompatible contents; and supply a new input for prime-norm terminal cases. The original IUT all-place comparison, FCRT arithmetic residual, Pell/Mersenne first-depth estimates, and geometric uniformity routes remain open at their previously recorded statements. The universal upper bounds on raw high-prime excess and on the angular defect alone are now specifically refuted, while their parent correlated routes remain active.

170.1. Even full low-depth credit cannot close this particular envelope. The squarefree-profile calculation permits a stronger audit of the proposed uniform gate. Let $\mathcal{B}_*(z)$ be the minimum partition penalty and set

$$\Psi(z) = \mathcal{B}_*(z) + \log(E_3(T)/C_3(T)).$$

This is the remainder in the proved cubic upper bound, not the true defect.

Proposition 170.1 (An unbounded obstruction at prime-norm endpoints). *Along the infinite family of primitive positive triples with $M = q$ prime,*

$$\Psi(z) \geq (2C^2 \log 4 - 6) \log c + O_C(1).$$

*In particular $\Psi = o(\log c)$ fails, and the uniform sufficient gate $\Psi \leq 3\varepsilon \log R + K_\varepsilon$ fails already at $\varepsilon = 1/4$. Neither statement is a failure of standard *abc*.*

Proof. Every prime $q \equiv 1 \pmod{3}$ is the norm of a primitive element of the Eisenstein ring; rotation gives a positive nonzero-boundary pair. Infinitely many such primes were constructed above, and $c \geq \sqrt{q}$ is unbounded. There is only one split prime exponent and it equals one, so $\mathcal{B}_* = C^2 \log 4 \log q$. Exact truncation gives $E_3(T)/C_3(T) = T/R^3$. Since $R \leq T \leq c^3/4$,

$$\log(T/R^3) \geq -2 \log T \geq -6 \log c + 2 \log 4.$$

Also $q \geq 3c^2/4$. Combining proves the stated lower bound. For $C > 2$, $2C^2 \log 4 - 6 > 5$, whereas $(3/4) \log R \leq (9/4) \log c$. Their difference is unbounded, so no additive constant works at the displayed epsilon. The argument retires only the explicit logarithmic-form envelope as a universal sufficient gate. A sharper input for prime-norm endpoints, or a method using additional additive information, remains a different route. $\square$

171. FORMAL VERIFICATION AND REMAINING OBLIGATIONS

The companion development uses Lean 4.32.0 and the repository's pinned dependencies. The principal companion modules are listed below. Audit modules only display their types and dependency reports.

Module (within IUTThreeClosures)	Checked mathematical scope
SmoothLowOmegaCounting	Actual prime-power encoding, finite cardinality bounds, and the finite first moment.

Module (within IUTThreeClosures)	Checked mathematical scope
CarellaGlobalOmegaHypothesis20260901	Positive-multiple injection and the exact N/Q high- ω lower bound; the $\sigma < 2/5 - \kappa$ disproof gate, the density-compatible exponent-window threshold $k > 5$, and a uniform one-fifth feasible exponent for $k \geq 6$.
SignedPrimeSupport	Actual prime-exponent sums and the coupled equivalence with the original target.
SignedEndpointDyadicObstruction	The genuine dyadic triples and failure of a separate-endpoint slope below one.
ReachableTensorLatticeCriterion	Unit-determinant span certificate, symmetric span and its missing off-diagonal tensor.
FreyPelleEffectiveResidualCorridor20260830	The integral elliptic identity, integer square-factor descent, explicit scalar implications, and the Frey denominator bound.
AnalyticAmplificationContinuation20260830	CRT uniqueness and finite counts; actual radical inequalities; conic and height identities.
DVRReachableHaar20260830	A minimum determinant from actual columns, the exact quotient cardinality, closed span, and full-rank Haar and log-volume.
GeometryUniformityContinuation20260830	Norm and ratio identities, logarithmic absorption, explicit corridor implications, and the primitive Frey–Mordell point.
IUTReachabilityContinuation20260830	Tensor units, independent scaled vectors, and integral tensor-span containment in a coefficient-module hull.
MatveevPellFinitePacket20260830	The strict Pell approximation, normalized-coefficient algebra, absolute-index absorption, and the final height implication with external bounds stated as hypotheses.
IUTTargetReset20260830	Covariance of an all-isomorphism collation with repeated labels, its span, and source reindexing under an explicit compatibility identity.
AnalyticActualRadicalUniformGate20260830	Actual new-prime radical and integer excess identities, the conic budget, exponent-window algebra, and exact numerical counterexamples.
GeometryGlobalUniformGate20260830	Canonical integer cube decomposition, auxiliary point and cubic-root identities, finite matrix determinants, and explicit algebraic height bridges.
IUTAdmissibleGaloisUniformGate20260830	Kernel and valuation consequences of an explicit trace-transport hypothesis, affine-depth separation, and invariance of module span under unit scalars.

Module (within IUTThreeClosures)	Checked mathematical scope
IUTTracePreserving Transvection20260830	Actual linear automorphisms achieving common avoidance while preserving a functional.
IUTAllOpenLatticeRigidity20260830	Scalar rigidity from all shrinking line neighborhoods; an actual $\mathbb{Z}_p$ specialization, with surjectivity forcing a unit scalar.
IUTFullGaloisWordLift20260830	An actual free-group automorphism, its inverse words, fixed boundary, and abelian images.
IUTThreeLabelMinimumLayer20260830	Actual linear automorphisms, finite-kernel avoidance, and three-label specialization.
Frey139Tate210Realization20260830	Actual Frey Weierstrass invariants and point types over finite fields, exact integer certificates and real logarithmic intervals for two curves.
SL2TransvectionGeneration20260830	Elementary matrices in the actual special linear group and the normal-subgroup coprime-index argument.
Frey43BalancedRealization20260830	The genuine primordial and all-prime exponent bound, actual curve invariants and finite-field point cardinality, and the real logarithmic window.
IUTGeneralTameSquareLabels20260830	Uniqueness of the actual exceptional quadratic congruence, floor and ceiling bounds, finite avoidance budget, and the hull-exponent scale identities.
TraceDualPreidealHull20260831	The actual algebra trace and integral dual, exact two-factor product duality, a scaled order-ideal inclusion, and its transported B -module span bound.
ABCTwoPrimeSupport20260831	Actual prime-support classification, the sharp radical bound on the full two-prime subclass, and its equality cases.
TraceCovariantRational Return20260831	Actual algebra traces and dimensions, rational scalar return, the entire scalar-line action forced by one nonzero return, and valuation consequences.
ABCOddPartFibre20260831	Actual odd parts of primitive triples, the finite fibre and bound two, fixed-parity uniqueness, and the exact two-point fibre over $(1, 3, 7)$.
FreyEntireIsogenyArithmetic20260831	Actual primitive family, four Weierstrass models, finite-field point count, and displayed discriminant and rational absolute j -bounds.
FreyIsogenyWeilHeight20260831	Actual reduced numerators and denominators, library Weil heights of the four curves, the exact least height, its unique minimizer, and explicit bounds.

Module (within IUTThreeClosures)	Checked mathematical scope
FreyIsogenyConductor Sharpness20260831	Actual radical of the reduced j denominator, the power subfamily, conductor-proxy bound, and subcritical gap for all four actual models.
ABCMixedFullCampana20260831	Denominator-free mixed-full radical compression, exact reciprocal slope, abc-conditional height bound, and the standard-target disproof implication from an unbounded family.
ABCThreePrimeSignatures20260831	Elementary factor rigidity and all output placements of the prime-base signatures $(2, 3, 3)$ and $(2, 3, 6)$, without importing external Diophantine classifications.
IUTFiniteProductProjection Span20260831	Central-idempotent reconstruction of a finite product-ring span from coordinate spans and finite nonnegative weighted-sum monotonicity.
ABCSubcriticalLocus Uniformity20260831	Exact equivalence between standard abc and all fixed subcritical-locus height bounds, integer and pointwise amplification thresholds, and an infinite square-gap counting countermodel.
PellCampanaCounterexample20260831	Actual Pell recurrence and unboundedness, fullness transfer to the strict $(7, 3, 2)$ signature, and the conditional implication from an unbounded squarefull-root family to the unchanged negation of standard abc .
PellAdjacentFactor Counterexample20260831	Exact half-factorization, the primitive adjacent point, one-half conductor slope, fourth-full transfer, and the standard-target conditional disproof gate.
PellSquareRootDescent20260831	The $(1 + \sqrt{2})^n$ orbit, norm and coprimality, exact adjacent square shapes, and the squarefull product equivalence.
PellPrimeIndexDichotomy20260831	Odd squarefull valuation depth, the exact index-seven simple divisor, and the two-channel numerical product bound.
PellPrimeRank Counterexamples20260901	The exact index-seven repeated divisor and nonsingular-derivative counterexample, together with arithmetic counterexamples to two defective cyclotomic intermediate claims exposed by the source audit.
HallSquarefull Counterexample20260831	Primitive Hall points, the twelfth-power radical inequality, the strict $11/12$ conductor slope, and the unbounded-family disproof gate.
AffineShearAmplification20260831	The complete affine equation, support avoidance, cofactor and endpoint coprimality, actual ABCPoint construction, and parameter injectivity.

Module (within IUTThreeClosures)	Checked mathematical scope
AffineExcessUpperBound20260831	All three pair projections, same-prime exclusion, and the integer geometric-mean step.
AffineRadicalStep20260901	Support-killing moduli, primitive endpoints, all three pair injections, nesting of multiple-step fibres, the exact matching-lower quantifier gate, and the two square-row radical nonexamples.
DanilovGlobalIndexSieve20260831	Uniform modular linearization, ten prime-square packets, exact CRT, and the explicit global index progression.
DanilovRecursiveLift20260901	The recursive state invariant, finite fixed-index support obstruction, and the complete abstract one-step continuation countermodel.
DanilovSimplePrimitiveNoGo20260901	A full standard real-Lucas counterexample to sequence-uniform simplicity, powerful primitive parts, finite correction support, and the elementary $41n + 1$ tail.
IUTLanaSpecificationNoGo20260901	Uninhabitability of the pinned upstream RHSData and assembled variant records, the total set-volume no-go, normalized-weight nonemptiness, and the pointed same-pilot sufficiency lemma.
AffineDensityAttack20260901	Exact gap recovery, strict-positive reverse boundary, the complete $U = 1$ counterexample to its weaker form, finite-field fibre cardinality, squarefree excess identities, and the two-long-arm cancellation kernel.
AffineTwoArmCRTPacket20260901	Coprime joint excess, the unique diagonal CRT class and squarefree-carrier transfer, the exact 318322715-point canonical image, and a complete counterexample to two-marginal-gate sufficiency.
MersennePrimeLayerRadical20260901	Non-prime-power composite Mersenne values, exact prime-index order and radical carriers, finite totient-tail and lifting-mass bridges, and complete $m = 6, 11$, and 37 counterexamples to three stronger formulas.
MersenneOrderBlock Decomposition20260901	Actual exact-order divisibility and LTE, $L_m = \gcd(m, 2^m - 1)$, the finite fibre grouping $B_m = \prod_{d m} E_d$, the corrected product for W_m , and the complete index-six ledger.
MersenneOrderBlock Asymptotic20260901	Canonical index-independent blocks, relative/canonical agreement, exact logarithmic divisor sums, and the implication $\log E_d = o(\varphi(d)) \Rightarrow \log W_m = o(m)$ with its open antecedent explicit.

Module (within IUTThreeClosures)	Checked mathematical scope
MersenneCanonicalBlock Witness20260901	Exact order $\text{ord}_{1093}(2) = 364$, exact valuation two, the divisibility $1093 \mid E_{364}$, and the negation of the strengthening that every canonical block equals one.
MersenneWieferichTail Reduction20260901	Finite powerful-part comparison, the deep/transition/extreme mass trichotomy, and the transition-support and square-budget cardinality bounds, with all analytic inputs kept as explicit premises.
AffineTemplateEntropy20260901	Three-form cubic-product separation including every zero branch, affine membership cancellation, the canonical separated-set bound $ S R^2 < 12c^4$ under explicit box and separation hypotheses, a generic finite-union inequality, and the complete counterexample to deleting an individual modulus cap.
MersenneWeightedOrderTail20260901	Summable-profile localization, heavy-or-diffuse mass alternatives, divisor-mass little-oh transfer, and the exact equivalence between Mersenne power loss and the $o(m)$ divisor-average block endpoint.
MersenneSuperWieferichDepth20260901	Exact finite depth layer cake, moving-threshold support/tail bounds, the pointwise sufficient implication, and the certified odd-order depth-two witness at 3511.
PellFourPrimeCoupling20260901	Arbitrary finite-product quadratic truncation, quotient cancellation, the exact Pell coordinate identity, second-order channel transfer modulo $4\ell^2$, and the index-seven certificate.
PellResidueParity Localization20260901	The exact eight-step Pell recurrence and modulo-eight coordinate table, involution product extraction of odd depth at least three under squarefullness, the finite sign-product step, and the complete index-eleven Legendre counterexample to an all-pairs strengthening.
DanilovWSSEscape20260901	Primitive-rank injectivity, multiplicity transfer, the exact 2^{637} and $2^{638} - 622$ witness-count kernels, the $41n + 1$ arithmetic implication, and the derivative no-go counterexample.
IUTCorrectedVolumeHolonomy20260901	Actual valuation balls, normalized packet shifts, zero transport holonomy, rational-prime log independence, prime-local cancellation, and the positive normalized scalar collision.

Module (within IUTThreeClosures)	Checked mathematical scope
IUTPrimeUnitLabelVector Bridge20260901	Rational, actual p -adic, and arbitrary-field scale reconstruction; faithful labelled signature-image transfer, vector holonomy, and three exact counterexamples to weakened coordinate interfaces.
IUTRefinedFactorZeroAware Signature20260901	Zero-aware reconstruction on each primitive field factor, faithful dependent products, generic fixed-stage finite-etale packet composition once its instances are supplied, and covariance under explicit refined field transport.
IUTLanaCurrentConcrete ImplicationAudit20260901	The universal empty-set obstruction to total real log-volume translation, the pinned public specialization, and the explicit missing bridge from abstract StatementI to integer ABCConjecture .
IUTAdmissibleVolumeInteger Bridge20260901	An extensional nonempty admissible-region interface, the actual $\mathbb{Q}_p$ valuation-ball model, fixed/periodic/finite-region no-go theorems, and the explicit uniform-comparison transfer from StatementI to the exact integer ABCConjecture .
AffineDeterminantLayer Entropy20260901	Product-modulus determinant divisibility, forced collinearity, one-dimensional capacity interfaces, canonical scale constants, fractional covers, and two exact arithmetic counterexamples plus the zero-density logical counterexample.
AffineAdaptiveCommonKernel20260901	Sharp square triangle-determinant bounds, the derivation from actual pointwise arm divisors and triplewise gcds to six difference congruences, factor-three arm caps, the long-arm square-excess alternative, factor-27 support bounds, and two exact full-premise local counterexamples.
AffineCommonKernelTriple Selection20260901	The lattice line cap, fixed-label packing, large-label fibre and incidence-multiplicity bounds, catalogue selection, the exact third-gcd incidence-energy expansion, and the actual canonical-box collinear counterexample to inferring noncollinearity from common-product size.
MersenneTotientDivisor Concentration20260901	Exact exceptional-totient criteria, logarithmic-deficit factorization, $O(\log \log(3m))$ moment and moving-window concentration, unconditional cyclotomic cap for the actual blocks, the resulting near-diagonal Mersenne equivalence, and the finite power-of-two weight witness.

Module (within IUTThreeClosures)	Checked mathematical scope
MersennePolylogCodivisor Gate20260901	Exact partition into fixed-polylogarithmic co-divisor windows, the uniform C/k tail, uniform-in-fixed-index transfer, and the actual Mersenne endpoint equivalence.
MersenneNearDiagonalGlobal Triage20260901	Aggregate near/far subtraction, the deep/transition/extreme $1/6$ trichotomy, transition-cardinality control, five-arm closure, and three full-premise one-point counterexamples to deleting a disjunct.
MersenneFixedPolylogBlockMass Triage20260901	The exact actual-mass window partition, the threshold-free little-oh equivalence, localized deep/transition/extreme selection for every coefficient below $1/3$, transition cardinality, and sharp full-premise mass-ledger counterexamples.
AffineCollinearPeriodEnergy20260901	Exact reduced direction periods and capture factorization, actual ray divisibility, residue-class capacity, nonconstant-ray cube-root and weighted cubic-energy bounds, the vertical square-root bound under an arm cap, and two full-premise period counterexamples plus the ambient-positivity boundary model.
IUTRationalTripodShadow Comparison20260901	Actual open rational points, exact Weil-height and truncated-count conversions, the zero-error shadow comparison, the equivalence of its Statement I with integer abc , and three full-premise semantic pressure models.
MersenneMultiplierIndex TwoArm20260901	Actual exact-order multiplier representation and injectivity, finite low-multiplier cardinality and log-mass bounds, two-arm finite-sum subtraction, and the sharp one-half mass-ledger counterexample.
PellOddKernelThirdOrder Packet20260901	Cubic-square odd-kernel factorization, congruence transport, Jacobi reciprocity rows, arbitrary finite third-order products, quotient cancellation, and the modulo- $8\ell^3$ two-channel ledger.
AffineCatalogueWeight Overlap20260901	Exact three-coordinate Euler-totient catalogue mass, divisor-count tail and top-label baseline, monotone cubic aggregation under coincident sublabels, weighted shifted-incidence bounds, and the complete $(9, 25, 1)$ counterexample to subtracting only the threshold.

Module (within IUTThreeClosures)	Checked mathematical scope
IUTRationalDegreeOneSource Realization20260901	Exact degree-one recurrence, source-realization transport and equivalence with integer <i>abc</i> , the rational-shadow witness, and full-premise pressure models for missing height, different, conductor, and degree semantics.
MersenneBalancedMultiplierDepth Localization20260901	Triangular multiplier energy, exact-order LTE transport, finite weighted envelopes, the two-controlled/two-surviving mass ledger, its sharpness family, and the complete multiplier-two certificate at 3511.
PellLucasAllOrderStaircase20260901	Even-power tail coprimality, coefficient-support exclusion, paired companion congruences, channel-square reconstruction, local-surjectivity logic, and the kernel recurrence certificate refuting fixed-zero rigidity.
IUTAdmissibleScalingOrder Index20260901	Empty- and whole-set obstructions to overbroad volume shifts, exact congruence-order kernel and quotient, additive index, properness, and surjective coordinate projections.
AffineSignedRayCanonical Caps20260901	Signed direction period and capture, strict large-label squeeze, all three exact arm captures, optimal shifted cubic bridge, weighted owner-catalogue bounds, and exact countermodels to stronger threshold, period, and premise deletions.
MersenneCriticalSlowSlack Gate20260901	Nested survivor supports and masses, exact cutoff balance, two-arm subtraction, Euler half-power character on actual exact-order fibres, and the complete odd-multiplier certificate at 1093.
PellLucasCorrelatedAllOrder Exclusion20260901	Rational coefficient correlation, exact weighted tail identity, coherent every-order splitters, cross-order determinant, sixth-power channel signs, negative-row extraction, and quartic-two power algebra.
IUTActualHaarAdmissible Orbit20260902	Actual normalized additive Haar preimages on finite-positive and compact-open regions, uniformizer orbit and no-envelope theorem, local-degree packet normalization, the raw-weight coefficient counterexample, and the $\mathbb{Q}_p$ residue cardinality.
AffineInversePeriodCatalogue Novelty20260902	Optimal incidence-plus-shift inequalities, exact finite class-incidence exchange, multiplicity/novelty identities, owner-tail and pair-catalogue double counts, inverse-period capture algebra, and complete finite period-one certificates.

Module (within IUTThreeClosures)	Checked mathematical scope
PellLucasFactorQuotientProjective Coupling20260902	Depth-three quotient fingerprints, two companion jets, exact endpoint curvature and U^2 sharpness, together with the local-ledger and genuine index-seven counterexamples.
MersenneSigmaOneExactOrder Coupling20260902	Common-index slope injection, exact Farey first-moment identities, stable prime-power order layers, cutoff and two-arm ledgers, and the finite 1093 and 3511 certificates; no analytic limit is asserted in Lean.
IUTRefinedTensorHaarTheta SamePilot20260902	Primitive local-factor identities, the ef coefficient, total-degree Haar normalization, relative factor weights, compact-open orbit control, and a conditional pointed same-pilot sandwich; no source-level IUT comparison is asserted.
MersenneFareyDenominator Entropy20260902	Exact finite Farey prefix–tail denominator splitting, a generic finite tail row-count lower bound, exact-order depth-three implications, injectivity, and finite short-window certificates; the complete swarm theorem and critical asymptotic count are not asserted.
AffineOwnershipMaximalIntersection Aggregation20260902	Abstract finite-poset maximal-support, owner, and Cauchy skeleton; algebraic strict-ray consequences; and numerical cores of the boundary examples. The concrete maximal cover, large-label migration, and catalogue-sparsity estimate remain outside Lean.
PellPolynomialHensel Specialization20260902	Exact index-three polynomial identities, one-digit Hensel lifting, Taylor/CRT data, and the displayed $T = 282$ numerical packet; no bundled full-premise witness or fixed- $D = 2$ exclusion is claimed in this module.
PellPolynomialAllIndex Formalization20260902	All-index Pell/Lucas polynomial recurrences, norm and derivative identities, arbitrary polynomial Hensel lifting, conditional all-support displacement equivalence, finite CRT steering, and the bundled moving witness.
SteinbergValuationContact Surface20260902	Divisor exterior contact, standard five-term boundary cancellation, abstract finite-coordinate ℓ^1 boundary control, and complete Pythagorean counterexamples to two single-cell gates.

Module (within IUTThreeClosures)	Checked mathematical scope
SteinbergIntegerFiniteChain20260902	Positive-integer exponent-gcd normalization, radical/coherent/residual splitting, actual finite-support divisor cells, the exact mixed-area norm, and the filling inequality conditional on an explicit integral surface-boundary equality.
QuadraticVeronesePeeling20260902	Exact quadratic five-term peeling, primitive Pythagorean specialization, layer expansion, and full-premise failures of three fixed one-move policies; arbitrary multi-move fillings remain open.
MersenneFareyQuantitativeSwarm20260903	Classical harmonic identification and logarithmic prefix control, cleared and divided finite swarm inequalities, exact frequent-subsequence quantifiers, and the finite injection of actual exact-order depth-three rows into the bounded super-Wieferich set.
AlternativeQualityPackingBridge20260903	Exact packing coordinate equivalence, the AM–GM comparison, a clustered-coordinate lower bound, and an abstract full-premise example in which DGM quality is unbounded while standard quality is constant.
PellFixedTwoTransversality20260903	Matrix Frobenius congruences at prime indices, unconditional derivative transversality at every actual fixed- $T = 2$ support prime, the exact squarefull/simultaneous-displacement equivalence, and the individual (7, 13) collision boundary.
PellSignedTraceProjector20260903	Exact adjacent signed trace-square identities, support-depth doubling, the all-support fourth-power packet equivalence, radical consequences, a sharp raw channel projector and its explicit Newton lift, and the exact index-seven fifth-power boundary.
SteinbergFiveTermBoundaryBridge20260903	Algebraic and positive-realization-generated five-term submodules, their inclusion in the boundary kernel, conversion to calibrated finite chains, realization families, and nonzero-cycle counterexamples to literal cancellation and boundary injectivity.

Module (within IUTThreeClosures)	Checked mathematical scope
ABCSynchronizedDivisor Packets20260903	Actual primitive nonunit abc packet data, finite nonemptiness, product and sixth-power bounds, positive synchronization index, canonical-orientation rigidity, the real-log minimum-energy majorant, an infinite proper exact-gap family, and actual finite packets refuting the recorded stronger height candidates.
SynchronizedPacketRadicalExcess Obstruction20260903	Exact packet shape and radical-excess identities, an infinite primitive counterfamily to uncompensated four-thirds compression, and the surviving <i>ab</i> -compensated implication from packet size to the abc height bound.
ABCCanonicalGainSurface20260903	Strict canonical gain corridor, exact gain-product and excess/slack identities, additive and multiplicative scale cocycles, equivalence of the uniform defect-flag budget with ABCConjecture , and complete countermodels to two stronger gain bounds.
ABCValuationIncidenceComplex 20260903	Labelled three-arm prime-support faces, exact radical/defect/modulus coordinates, defect filtration, local incidence signatures, CRT reconstruction, the rational-power top-face half-space, a filtered-face implication, and the complete five-dimensional $12 + 833 = 845$ boundary. No eventual selector is asserted.
ABCValuationIncidenceFixedBudget Obstruction20260903	The dyadic family $P_k = (2^{k+4}, 3, 2^{k+4} + 3)$, exact support and valuation certificates, forced omission of the <i>A</i> -vertex below a fixed budget, failure of <i>AB</i> reconstruction on an infinite tail, and the negation of the original all-but-finitely-many VIC-ABS-1 selector.
ABCValuationIncidenceScaleBudget Obstruction20260903	The balanced family $Q_r = (2^{2r}, 3^r, 2^{2r} + 3^r)$, forced selection of both singleton summand supports, exact defect $3r - 2$, binary-scale upper bound $2r + 1$, an infinite tail of reconstruction-budget failures, and the negation of the full coefficient-one VIC-1R selector.

Module (within IUTThreeClosures)	Checked mathematical scope
ABCThreeArmIncidence Successor20260903	Exact selected/complement radical partitions, three-arm local signatures and CRT rigidity, the implication of RAW-3C to ABCConjecture , its infinite primitive Pythagorean-square refutation, complement-flow mass identities and Hall tails, and the implication of the ordered CT-3C gate to ABCConjecture . CT-3C is not asserted and is refuted by the companion module below.
ABCSignedEndpointPrimeToken Transport20260903	Exact endpoint core/radical balance, finite fractional-flow accounting, weighted Hall tail obstruction, implication of the uniform small-unmatched-mass gate to ABCConjecture , integral matching counterexamples, and the zero-epsilon prime-power radical-budget obstruction. The ordered uniform fractional gate is not asserted.
ABCThreeArmComplementTransport Obstruction20260903	The odd-prime family $(1, p^2 - 1, p^2)$, the strict separation of its endpoint source from every external prime, forced selection of that source in every covering three-arm face, the exact full-modulus-minus-radical tail identity, conductor growth at most $3 \log p$, and the quantified negations of both CT-3C and EPF at $\varepsilon = 1/4$.
ABCBidirectionalPrimeTransport Successor20260903	An all-edge endpoint flow with a nonnegative relative downward-displacement cost, its exact accounting and conditional implication to ABCConjecture , and the distinction between that candidate and the scalar completely unordered optimum. The companion obstruction below refutes its exact uniform BEP gate.
ABCBidirectionalEnergyPythagorean Obstruction20260903	The unbounded prime-hypotenuse Pythagorean-square family, positivity and primitivity of every point, the square-root bound for every external prime, energy at least $(\log p)/4$ for every flow, conductor at most $3 \log p$, and the quantified negation of BEP at $\varepsilon = 1/24$.
ABCPrimePacketBoundaryTransport Successor20260903	Aggregate endpoint source packets and indivisible external-prime sinks, the finite packet-boundary inequality, the endpoint height bridge, the conditional implication $\text{PBT} \Rightarrow \text{ABCConjecture}$, an exact two-source/one-sink noncollapse certificate, and the one-source formula for $c = p^2$. No uniform PBT instance is asserted.

Module (within IUTThreeClosures)	Checked mathematical scope
ABCPrimePacketBoundaryLinnik Obstruction20260903	The clipped-reward identity, unique-sink residual lower bound, actual prime-neighbour point, square-product endpoint sources, conductor and zero-scalar-defect bounds, and the implication from the transparent arithmetic proposition LinnikPrimeNeighborEscape to the negation of PBT. The published Linnik theorem supplies that proposition in the ordinary proof; it is not inserted as a Lean axiom.
ABCSharedCRTIncidence Successor20260903	The abstract once-charged capacity certificate, its source-minus-sink and endpoint height bridges, the conditional implication for an explicit arithmetic admissibility predicate, the exact scalar optimum of unrestricted pooling, and the failure of duplicated capacity. Arithmetic CRT blocks, the strict noncollapse optimum, and the SCRT-SAT counterfamily remain ordinary proofs rather than axioms.
ABCFlaggedCRTSurplusResidue Cube20260904	The maximal feasible-subset selection, once-owned residual and capped-reuse accounting kernels, source-minus-sink and endpoint height bridges, conditional FCRT implication, additive commutative-group packet identities, and exact arithmetic/logarithmic checks for three finite witnesses. The uniform gate has no inhabitant.
ABCFlaggedCRTSurplusResidue Cube20260904IndependentBridge	The exact owner-to-aggregate conversion after capping overfill, preservation of the clipped boundary, an endpoint constructor under the displayed defect decomposition, and the observation that a bare endpoint certificate is tautological. Concrete proper-face admissibility remains open.
ABCAncoredPrefixFlaggedCRT20260904	The finite anchored code-fibre lemma, additive prefix cancellation, zero-deletion/proper-face dictionary, capped-credit consequences, scalar full-surplus identity, and a complete Boolean-packet verification of the (1, 4715, 4716) counterexample to the raw cardinality shortcut. No uniform entropy premise is asserted.

The flagged-CRT primary module has 70 public declarations and a one-for-one 70-query axiom audit. Its independent bridge has 18 declarations and 18 queries. The anchored-prefix module adds 28 declarations and 28 queries. All six files compile directly with warnings treated as errors,

and the combined dependency union is only `propext`, `Classical.choice`, and `Quot.sound`. This audit proves the logical accounting bridges; it does not formalize the multiplicative residue groups, the arithmetic map from proper faces to owners, the finite optimum enumerations, the real-ceiling construction behind the weight cells, or the uniform FCRT entropy and boundary estimates.

The six modules added in the dual-route continuation contain 94 public theorems and 15 definitions or structures. A declaration-level audit checks all 109 items: five have no kernel axioms, and every other dependency is a subset of `propext`, `Classical.choice`, and `Quot.sound`. There is no `sorryAx`, declaration-style axiom, or unsafe proof. The full 9142-job build succeeds; its 265 warning lines are exactly the frozen previous multiset, so these modules add no warning. The validation record lists the exact commands, results, hashes, and kernel dependency reports. The pre-existing signed-layer module required an explicit import, closure of an elementary algebra branch, and the correct order of unfolding before rewriting; these were repaired without changing its statements.

The four balanced-persistence modules add 62 public theorems and 15 definitions or structures, for 77 audited declarations. Their dedicated audit reports no `sorryAx`; every dependency is a subset of `propext`, `Classical.choice`, and `Quot.sound`, and the axiom-free declarations are recorded separately in the validation artifact. Direct compilation of the four modules and their audit succeeds without warnings. The audit target builds 8767 jobs with 19 warnings from historical dependencies and none from these files. Both the library and repository-default builds complete 9147 jobs with 265 warnings; the warning multisets agree exactly with the frozen dual-route baseline.

The September 1 continuation adds three more modules with 57 theorems and 30 definitions or structures, for 87 further declarations. Direct compilation succeeds, as does the aggregate 9151-job build. In particular, the Danilov finite certificates use kernel reduction rather than native evaluation; declaration-level `#print axioms` reports only `propext`, `Classical.choice`, and `Quot.sound`, with no native-evaluation axiom and no `sorryAx`.

The global-packet and same-pilot continuation adds five modules with 122 theorem or lemma declarations and 42 definitions, structures, or abbreviations, for 164 declarations in their source. Each module compiles directly, and the aggregate target completes 9194 jobs. A token-boundary source scan finds no axiom declaration or proof placeholder and no use of native evaluation, opaque declarations, or unsafe definitions in these files. The embedded kernel-dependency reports contain only `propext`, `Classical.choice`, and `Quot.sound` where dependencies occur. The permanent validation bundle records the commands, complete logs, source hashes, and computation-manifest replays.

The present holonomy-depth continuation adds four modules with 65 theorem or lemma declarations, 18 definitions, four abbreviations, and five structures, for 92 declarations. The files contain 58 embedded `#print axioms` queries; the union of their reported kernel dependencies is exactly `propext`, `Classical.choice`, and `Quot.sound`. A code-only source scan finds no declaration-style axiom, proof placeholder, native evaluation, opaque declaration, or unsafe definition. All four direct compilations are warning-free, and the aggregate target completes 9198 jobs. The analytic squarefree-bulk estimate, the published Pell and Fibonacci inputs, and the large finite certificates retain the explicit paper-and-computation boundary stated in Section 28.

The literal Carella audit adds one module with eleven theorem declarations and two definitions. Direct compilation is warning-free. Its eleven embedded `#print axioms` reports are all either axiom-free or have dependency set contained in `propext`, `Classical.choice`, and `Quot.sound`; there is no proof placeholder or custom analytic axiom. Lean checks the finite positive-multiple count and exact exponent arithmetic. Bertrand's postulate, the Rankin radical count, the prime number theorem, and the passage to the moving asymptotic threshold remain paper inputs rather than new Lean axioms.

The prime-unit-label continuation adds one module with 42 theorem declarations, 19 definitions, four structures, and three abbreviations. Its 43 embedded axiom reports, one of which also audits a definition, have union exactly `propext`, `Classical.choice`, and `Quot.sound`; direct compilation is warning-free. Lean checks reconstruction on both rational multiplicative inputs

and Mathlib’s actual complete p -adic field, the arbitrary-field scale template, labelled region transfer, vector transport, and the three finite counterexamples. It does not identify those models with the complete IUT packet carrier or assert the missing global coordinate-preservation theorem.

The affine two-arm continuation adds 26 theorem declarations and six definitions. Every theorem has an embedded kernel report. Their dependency union consists of `propext`, `Classical.choice`, and `Quot.sound`. Lean checks the general CRT equivalence, squarefree-carrier transfer, both directions of canonical interval membership, injectivity and exact image cardinality, and every hypothesis of the insufficiency counterexample. The independent replay recomputes the complete factorizations and strict integer inequalities; it is not used to infer an asymptotic statement.

The five Mersenne modules add 97 theorem declarations, ten definitions, and 97 dependency reports, again with union exactly the same three standard kernel principles. They contain no native evaluation or proof placeholder. Lean checks the elementary prime-layer carrier, the finite totient-tail and abstract total-mass inequalities, the exact $m = 6, 11$, and 37 witnesses, the actual exact-order LTE grouping, $L_m \mid m$, the corrected finite product, the limiting implication from the canonical block hypothesis, the nontrivial canonical block $1093 \mid E_{364}$, the finite powerful-part comparison, the three-arm mass dichotomy, and the transition-support cardinality, realized-support, and ambient square-budget ratio implications. The stronger complete identity $E_d = \Phi_d(2)/\text{rad}(\Phi_d(2))$ and the analytic cyclotomic distribution estimates, Brun–Titchmarsh and totient estimates, the Erdős–Murty order theorem, the Erdős–Shorey theorem, and the hypothesis $\log E_d = o(\varphi(d))$ retain the external/open boundary stated in Section 58.

Together, the eight modules in this continuation contain 176 theorem declarations, 37 definitions, three abbreviations, and four structures, for 220 top-level declarations. Their 177 embedded `#print axioms` queries have dependency union exactly `propext`, `Classical.choice`, and `Quot.sound`. All eight direct compilations and the aggregate 9206-job target succeed.

The preceding affine-entropy and Mersenne-depth checkpoint adds three modules with 74 theorem declarations and 21 definitions, for 95 further top-level declarations. Every theorem has an embedded `#print axioms` query. Lean checks the complete finite separation algebra and canonical packing constant, the exact divisor-average endpoint, the finite depth layer cake and threshold bounds, and all premises of the two narrow auxiliary counterexamples. The analytic fixed-base distribution estimates and the final affine exceptional-template lower bound remain paper obligations rather than Lean axioms. Together with the preceding eight-module checkpoint, these eleven modules contain 250 theorems, 58 definitions, three abbreviations, and four structures, for 315 counted top-level declarations and 251 embedded dependency reports. The direct module checks and the aggregate 9209-job target succeed; the dependency union remains exactly `propext`, `Classical.choice`, and `Quot.sound`.

The current determinant, totient-concentration, and refined-factor checkpoint adds three modules with 105 theorem declarations, 20 lemma declarations, and 26 definitions, for 151 counted declarations. A generated same-scope audit covers all 125 proofs one for one, including the 13 private affine proofs that cannot be named after import. Its dependency union is again exactly `propext`, `Classical.choice`, and `Quot.sound`. All three direct compilations and the generated audit are warning-free, and the aggregate 9212-job target succeeds. The sealed replay freezes 469 local Lean and Lake inputs and explicitly records that these checks do not close the standard *abc* conjecture.

The present adaptive-kernel, fixed-polylogarithmic, near-diagonal, and current-IUT audit adds four modules with 52 theorem declarations, four lemma declarations, and sixteen definitions, for 72 declarations in all; three of the affine theorems are private implementation lemmas. A generated same-scope audit issues one `#print axioms` query for each of the 56 proof declarations, including the private proofs; their union is exactly `propext`, `Classical.choice`, and `Quot.sound`. All four direct compilations are warning-free. The aggregate target completes 9216 jobs with no errors; its warning diagnostics are inherited from historical imports and none arises in these modules. A code-only scan finds no proof placeholder, declaration-style axiom, native evaluation, unsafe definition, or opaque declaration, and `git diff --check` succeeds. This aggregate

build retains the pinned LANA commit `ddaddc2`. Separately, against a detached checkout of commit `6e963070`, the public `Iut` target builds and an additional audit theorem proves $\text{LocalTheory}(K) \rightarrow \text{False}$. Successful construction of the library therefore does not provide an inhabitant of that structure.

The selection, admissible-volume, Pell-residue, and fixed-block-mass checkpoint adds four modules with 92 theorem declarations, two lemma declarations, 22 definitions, and eight structures, for 124 counted declarations. The generated same-scope audit issues exactly 94 `#print axioms` queries, including all eight private affine proofs. Its dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`. The four direct compilations and the audit are warning-free, and the aggregate target completes 9220 jobs. The associated finite Mersenne replay independently checks all 664579 primes through 10^7 , recovering precisely the depth-two witnesses 1093 and 3511 and no deeper hit. These computations refute only the explicitly stated finite strengthenings; they do not convert any surviving asymptotic gate into an assumption or close the standard *abc* statement.

The current direction-period, rational-shadow, multiplier-index, and odd-kernel checkpoint adds four modules with 83 theorem declarations, 40 definitions, two structures, and one instance, for 126 counted declarations. A generated same-scope audit covers all 83 proofs, including three private affine helpers. Its dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`; all four direct compilations and the audit are warning-free. The aggregate target completes 9224 jobs. The independent Pell replay checks 648189 bounded repeated-factor candidates, 42 exact simple-divisor certificates covering every odd prime index through 191, and one complete Pocklington certificate. It records 187 unresolved bounded-search indices as unresolved and makes no asymptotic inference. These checks certify the stated formal reductions and narrow counterexamples, not the standard *abc* conjecture or its negation.

The source-realization, balanced-multiplier, catalogue-overlap, and all-order Lucas continuation adds four modules with 67 theorem declarations, 18 definitions, and one structure, for 86 counted declarations. A generated same-scope audit covers all 67 proofs one for one. Its dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`; all four direct compilations and the audit are warning-free, and no target source contains a proof placeholder, declaration-style axiom, native evaluation, unsafe definition, or opaque declaration. The aggregate target completes 9229 jobs. The independent Lucas replay checks all five local residues modulo five, including the fixed-zero counterexample, and checks the full coefficient staircase and companion splitter at $\ell = 3, 5, 7, 11$. Its finite no-hit results are pressure tests, not an asymptotic exclusion or a proof of *abc*.

The signed-ray, critical-slow-slack, correlated-Lucas, and admissible-index checkpoint adds four modules with 90 theorem declarations and 16 definitions, for 106 counted declarations. A generated same-scope audit issues one `#print axioms` query for every theorem. The dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`; all four direct warning-as-error compilations and the generated audit pass. The aggregate target completes 9233 jobs. Exact replays cover 1,776,807 affine cubic ledgers, 2,390,018 quadratic ledgers, 15,840 signed directions, 43,403 arm captures, the complete Mersenne character packet, all 57 Pell prime indices through 271, 138,675 product-derived coefficient pairs, and 527,352 bounded depth candidates. The current-source IUT patch replays byte for byte and the patched upstream `Iut` and `Iut4Sec1` targets build all 8767 jobs. The owner-energy theorem is audited only on the selected powerful-kernel catalogue union; the recorded $972,496 > 1,072$ witness prevents an invalid all-divisor enlargement. No bounded no-hit, source-interface repair, or local formal theorem is promoted to an unconditional proof or disproof of *abc*.

The actual-Haar, sigma-one, catalogue-novelty, and endpoint-curvature checkpoint adds four modules with 102 theorem declarations, 18 definitions, two structures, and one abbreviation, for 123 counted declarations. Same-scope audits issue exactly 102 `#print axioms` queries, one for every theorem; their dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`. All four direct warning-as-error compilations and both separate audit modules pass, and the aggregate target completes 9239 jobs. The finite replays check the actual local-degree normalization witness, all 50,847,534 primes through 10^9 , six affine catalogue boxes

and every recorded full-premise witness, and exponent-one certificates at all 57 odd prime Pell indices through 271. The weighted Brun–Titchmarsh and Yamada-type analytic inputs, the tensor/same-pilot realization, the final affine intersection aggregation, and the global squarefull Pell exclusion remain outside Lean and are not introduced as axioms.

The refined-Haar, Farey-entropy, ownership, polynomial-Hensel, contact, and quadratic-peeling checkpoint adds eight modules with 297 theorem declarations, 113 definitions, five abbreviations, four structures, and two named instances, for 421 counted declarations. Its generated cross-audit issues one `#print axioms` query for every counted declaration, not only for proof declarations. All 421 reports are present, and their dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`. Each module and the generated audit passes direct warning-as-error compilation, and the aggregate target completes 9255 jobs. Deterministic replays cover the refined tensor-weight identities, the finite Mersenne denominator certificates, two independent affine checks, the Pell producer/verifier and bounded moving-parameter scan, and the quadratic peeling producer/verifier. At the date of that checkpoint, the fixed- $D = 2$ Pell transversality and rank gate, critical Mersenne count, affine catalogue sparsity, source-level IUT transport, five-term generated-move boundary theorem, and two Gate VF cost estimates remain open and are not inserted as axioms. Thus this checkpoint supplies neither a term of `ABCConjecture` nor a term of its negation.

The September 3 quantitative-transversality and generated-packet checkpoint adds five modules with 169 theorem declarations, 79 definitions, one abbreviation, seven structures, and two named instances, for 258 counted declarations. The generated audit issues one `#print axioms` query for every declaration. Its dependency union is contained in `propext`, `Classical.choice`, and `Quot.sound`; every module, hand-written audit, and generated audit passes direct warning-as-error compilation, and the umbrella target completes 9265 jobs. The fixed- $T = 2$ Pell support-transversality premise and the algebraic five-term boundary bridge are now closed. The Mersenne harmonic and quantifier seams, the synchronized-packet finite energy majorant, and all recorded full-premise local counterexamples are also kernel checked. The global Pell squarefull exclusion, actual Mersenne row supply and critical count, and arbitrary-target positive five-term filling with its two costs remain open. The synchronized radical-scale compression proposed at that checkpoint is refuted below at exponent $4/3$ and replaced by the compensated form of Theorem 79.5. None of the open gates is inserted as an axiom, and the abstract alternative-quality countermodel is not an integer abc family. Thus this repository contains no unconditional proof or disproof of the standard conjecture.

The gain/packet/trace follow-up adds three principal modules with 114 one-for-one axiom queries. Direct warning-as-error compilation succeeds for each module and audit, and the dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`. Exact finite replays cover 5,465,583 primitive triples for the canonical gain surface, 1,365,095 primitive triples and 1,366,531 synchronized packets for the radical-excess audit, and 63,950 Pell prime indices with 764,366 candidate-prime tests. The two Pell collision rows are also checked by complete trial division and three independent orbit algorithms. These finite records do not decide any unbounded gate.

The incidence/endpoint continuation adds eleven principal modules with 496 public declarations and 496 one-for-one axiom queries. The dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`. The first two one-face selectors, the raw weighted three-arm gate, CT-3C, the ordered fractional endpoint gate, the first relative-drop bidirectional gate, the exclusive-ownership PBT gate, and the saturated-only SCRT-SAT child are refuted by infinite complete-premise families. The once-charged SCRT-0 successor has a proved pointwise bridge and conditional implication but no asserted uniform estimate. The parent incidence complex remains active through SCRT-0, surplus-reuse, weighted multi-face, and homological successors. The endpoint replay enumerates 3,795,230 primitive nonunit triples, and the three-arm replay enumerates 218,893 unordered primitive positive triples. The exact packet optimizer adds 1,368,094 normalized primitive triples and 1,038 structured rows; an independent implementation replays the full scope. All three searches keep their finite scope explicit and do not infer an asymptotic claim from a bounded scan.

The formal scope is deliberately narrower than the full paper. Theorem 2.3 and Corollary 2.4 use Younis and classical smooth-number asymptotics in their mathematical proofs; those external analytic theorems are not formalized here. The full-rank case of Theorem 6.1, including its actual index and Haar measure, is now checked in its normalized-probability formulation on the compact integral lattice. Identifying this with the restriction of ambient local-field Haar measure and the rank-deficient measure-zero case remain paper proofs. The ellipse count and asymptotic amplification estimates, source-specific Kummer and local-field identifications, general independent-action theorem, Matveev’s theorem, (35), the small-generator approximation inputs, and the endomorphism-algebra argument retain their stated paper-proof boundary. In particular, the complete effective finiteness theorem is mathematically unconditional by the cited published theorems, but it has not been formalized as an unconditional Lean theorem. A Lean implication with an explicit approximation or height hypothesis is not a Lean proof of that hypothesis. The new radical count uses the cited S -unit theorem and an analytic divisor sum outside Lean. The actual Ism classification, local class field theory, Galois cohomology, root-logarithm calculation, and categorical representative reconstruction are also paper proofs. The shrinking-line rigidity theorem proves their algebraic core without assuming LCFT as an axiom. The global cubic construction does not formalize Pasten’s theorem, number-field orders, Siegel’s local approximation, or Nagell–Lutz merely by checking their coordinate identities. These distinctions apply equally to the new module-hull and fixed-support balance statements. Likewise, the free-group word proof does not formalize the Jannsen–Wingberg presentation of a local Galois group, and the elementary matrix development does not formalize an elliptic Tate representation. The integral local logarithm, trace-dual lattices, finite-*etale* covers, the initial-data construction, Linnik’s theorem and the analytic power-free count retain their mathematical proof boundary. The algebraic trace-dual core is checked with the actual library trace and dual submodules, without formalizing the local inverse different or the Galois witness by that fact alone. Exact finite arithmetic is checked against actual library objects, without inserting these external theorems as axioms. The complete two-prime inequality and arithmetic odd-part fibre bound are unconditional Lean theorems about the unchanged canonical objects. The scalar-return module uses the actual algebra trace, with trace covariance as an explicit hypothesis; it does not formalize the Galois, Kummer, or principal-unit logarithm inputs that establish the hypothesis in the mathematical application. The entire rational isogeny-class argument uses the cited cyclic-degree classification and Frobenius and minimal-model theory outside Lean. The companion module checks its arithmetic models and bounds without inserting those inputs as axioms. The height companion additionally checks the reduced fractions and the least absolute logarithmic Weil height on those four actual curves, using the library height and its normalized-height interface over $\mathbb{Q}$. It does not formalize the classification of the entire isogeny class merely by proving the four-model minimum. The new conductor-obstruction companion deliberately calls $16 \operatorname{rad}(c(c-1))$ a *conductor proxy*: the identification with the Néron conductor, the 2-adic reduction type, and isogeny invariance are the paper inputs of Proposition 24.1. The mixed-full module consumes the unchanged standard **ABCConjecture**; it neither assumes the Browning–Verzobio count nor asserts the existence of an unbounded Campana family. The finite-product module formalizes the heterogeneous finite product of its coordinate rings. Source compactness, finite-*etale* field decomposition, Haar normalization, and the identification with the IUT hull remain the explicitly cited paper layer. The support-three classifications that use Fermat’s Last Theorem and the Arif–Abu Muriefah theorem are not inserted into Lean as axioms. The arithmetic bundles, their metrics and descent also remain mathematical proofs; equality of their degrees alone is not a formal or mathematical proof of source-family membership. The finite theta point-hull theorem uses the previously proved full Galois constructions and original theta evaluation formulas outside Lean. Its common arrow is allowed to depend on the tuple of torsion choices. The per-arrow logarithmic bridge equates principal ideals in the declared carrier, not the original multiplicative Kummer inputs. The source-defined canonical local membership in Section 21 uses the cited reconstruction theorems and local reciprocity outside Lean. The single base branch and its lower ideal are not represented as a formal proof of the complete global comparison.

For the global-packet continuation, Lean checks the exact finite Pell counterexamples, the affine support-modulus algebra and matching-lower quantifier logic, the Danilov recursion invariant and abstract continuation countermodel, the sequence-uniform and parity-free Lucas counterexamples, the elementary powerful-packet consequences, and the literal pinned LANA record contradiction. The repaired Fellini–Murty prime-order/Siegel argument and its rank-tower descent remain paper proofs. The Fibonacci cyclotomic correction, rank-of-apparition valuation transfer, Carmichael–Yabuta primitive-divisor theorem, and Hong’s large-prime theorem are cited mathematical inputs rather than Lean axioms. The pinned LANA no-go proves only that the pinned low-resolution record is uninhabited; no corrected Haar-volume interface or object-level pointed same-pilot diagram has been constructed. Thus none of these formal modules turns the surviving Pell, Fibonacci–Danilov, affine, or IUT obligations into an unconditional abc conclusion.

The surviving proof target is a uniform estimate for the coupled expression (25), or a source-derived theorem strong enough to imply it. The IUT route still needs the comparison of the actual output family with the region in the global upper estimate, under compatible normalizations. The analytic disproof route still needs a height-unbounded family satisfying one fixed radical exponent gap; the finite-support bound neither constructs nor forbids such a sparse family. The specified Pell–Chebyshev residual is now effectively finite; the fundamental-unit cases and the absence of a global abc reduction remain outside that conclusion. The actual-radical amplifier still requires a lower count in the surviving window; the cubic route must control the growing omitted endpoint or denominators uniformly. On the local IUT route, explicit full-Galois arrows and exact pre-ideal hulls have now been obtained. The original word weights are explicit on the rational branch, and the computed local hulls admit global determinant objects over the field of moduli. The remaining task is to establish membership and comparison for the complete globally synchronized family with its markings, including the Ind3 and cross-Frobenius comparison for the same measured set and reference. The canonical local membership of one fixed base branch is now supplied by Theorem 21.5; it does not settle these remaining comparisons. The newest public conditional implication does not bypass this gate: its current local-volume interface is empty by Theorem 32.1. The valuation-ball model of Theorem 33.1 proves that an explicit extensional admissible-region replacement is consistent, and Theorem 33.2 removes the logical ambiguity from the final integer bridge. The rational shadow now realizes that bridge with zero error, but Theorem 34.1 shows exactly why this is not a proof: Statement I for the shadow is equivalent to *abc*. The actual degree-one source computation now identifies the rational points, canonical height, different, and conductor with their standard arithmetic counterparts, and Theorem 35.1 proves the exact two-way transport for any source-faithful realization. What remains is to construct the intended arithmetic **HeightTheory** instance in Lean and prove its upstream Statement I from the actual IUT data, including the disputed Corollary 3.12 comparison. The full-premise pressure models show that the current abstract record and proof package cannot manufacture those semantic identifications.

The public source at commit `c65b28c9f9631635e742294c3a5df15759e7c74c` now constructs the old concrete theta-data existence interface from curve data and explicit providers. This closes one assembly seam but retains the anabelian, local-theory, tower, analytic, height-package, and Corollary 3.12 premises. The all-set log-volume law makes a literal local-theory provider impossible; Theorem 36.1 gives the exact counterexample, and the patched admissible interface compiles throughout the current upstream tree. Theorem 36.3 also shows that componentwise surjectivity cannot erase an integral-order index. The active IUT gate is therefore an admissible, index-aware construction of the actual local volumes followed by the full horizontal and Ind3 covariance and the concrete comparison theorem.

On the affine route, determinant divisibility now survives point-adaptive certificates. Theorem 43.2 converts the selection problem into a catalogue-multiplicity inequality; alternatively, one may lower-bound the large-product part of (345) beyond every collinear contribution. The exact ray-period theorem now controls the previously unbounded nonconstant collinear occupancy by a weighted cubic-energy ceiling. The actual downward catalogue now has total totient weight exactly equal to the canonical kernel product, asymptotically all above the determinant threshold,

and coincident sublabeled only increase the cubic energy. The signed-direction capture squeeze now handles all primitive rays and gives exact linear caps for the three constant-arm directions. The global ceiling (359) leaves one non-arm inverse-square-period sum, supported only on large directions, and the comparison of the singleton baseline with catalogue multiplicities. These are the active gates. The canonical vertical example proves that the label product itself cannot be used as a period, while the $(9, 25, 1)$ catalogue and the new exact pressure tuples disprove only their stated stronger inferences.

On the Mersenne route, multiplier injectivity removes the low one-copy arm in every fixed-polylogarithmic co-divisor packet. The remaining gates are the high-multiplier deep-lift estimate and the localized Wieferich-weighted near-square-root order estimate. The balanced cutoff works with every fixed $\eta > 0$, and the critical slow-slack theorem now works for any divergent σ with $L_m\sigma = o(\log m)$; the explicit $\sigma = \log(3 + L_m)$ narrows both surviving regions beyond every fixed positive log-log power. The Euler-character identity filters the exact multiplier residues, but its $d \equiv 4 \pmod{8}$ row is unrestricted. No finite computation or current global density theorem proves either surviving little-oh statement at the arithmetic endpoint $\sigma = 1$. The exact primes 3511 and 1093 refute only the stated multiplier-at-least-three and even-multiplier strengthenings.

The Pell packet now carries nontrivial odd kernels, a complete aggregate Jacobi triangle, the third-order quotient ledger, and an all-order norm-one Lucas staircase. The paired companion correction recovers opposite signs on the two channel squares. The companion staircase is now explicit and coefficientwise correlated with the first; every order recovers the same splitter, and all cross-order determinants vanish modulo U^2 . For a hypothetical squarefull packet, vertexwise character laws force a nonresidue opposite pair with depth at least three when $\ell \equiv 3, 5 \pmod{8}$. Certified simple divisors now exclude squarefull packets at every odd prime index through 271; the finite boundary is kept literal. The $k = 2451$ recurrence witness and the mixed-sign row at index 11 refute only fixed-zero one-channel rigidity and the assertion that all edges of a negative row are negative. No known theorem excludes or constructs an unbounded packet whose two channels satisfy all depth, residue, character, staircase, cross-order, and third-order quotient constraints.

Constructing initial data, descending a determinant, localizing a divisor average, or eliminating Ind2 from a fixed B -module hull does not by itself supply the missing global comparison or distribution estimate. No unconditional closed term of type **ABCConjecture**, or of its negation, is produced by this work.

172. CURRENT BOUNDARY AND CONTINUING PROGRAMME

The developments above do not prove or disprove the standard *abc* conjecture. They replace several informal bottlenecks by exact interfaces: an index-aware pointed Haar comparison, a critical exact-order counting problem, an ownership-preserving affine energy inequality, a fixed-parameter Pell displacement problem, a calibrated multi-cell contact filling, and the SCRT-0 shared-incidence uniform estimate or a weighted multi-face or homological successor in the labelled valuation-incidence complex. The two formerly listed ordered transport estimates, the first relative-drop bidirectional estimate, and the exclusive prime-packet estimate are now refuted. In each retained route the local algebra or finite combinatorics is proved before a remaining uniform statement is named.

The counterexamples have a narrower and useful role. They exclude raw or factor-count Haar normalization, reduced Farey reasoning without arithmetic premises, owner-free affine estimates, moving-parameter Hensel exclusions, single-cell contact truncations, fixed one-step quadratic fillings, the proposed $A < 3/2$, $P < 3$ gain inference, uncorrected packet compression, and automatic Pell depth-five promotion. They also exclude universal trivial arm-defect and constant-per-arm valuation claims, both the fixed and coefficient-one binary-scale one-face incidence selectors, the raw weighted three-arm defect gate, both ordered CT-3C and EPF, the relative-drop BEP gate, the exclusive-ownership PBT gate, the saturated-only SCRT-SAT child, full integral endpoint matching, and a fixed natural multiplier at epsilon zero. Each counterexample retires only the exact statement whose complete premises it satisfies. No bounded

search is used to infer an asymptotic assertion. We therefore retain every surviving parent route and continue proof construction and counterexample search in parallel.

The next decisive advances would be any one of the following: construct the source-faithful pointed IUT comparison with its integral-order correction; prove the critical one-half counting bound for the relevant depth-three exact-order primes; establish the canonical affine catalogue-sparsity bound; control the canonical power-excess/approximation-slack difference; prove an eventual ab -compensated packet bound; exclude or construct simultaneous fixed- $T = 2$ signed fourth-trace Pell packets; or construct a five-term filling satisfying both calibrated cost estimates; construct a weighted, multi-face, or homological incidence successor with a non-tautological sufficient inequality; or prove or refute the exact SCRT-0 uniform estimate, including the composite-base prime-log partition and a capacity-safe surplus-reuse rule. A full-premise counterexample to any one of these exact statements would retire that statement alone and trigger a corrected formulation, while a proof of a sufficient gate would be transported through the already checked reduction.

173. A PARALLEL RESEARCH CONTINUATION AND ITS VERIFICATION BOUNDARIES

This continuation separates three research directions and subjects their ordinary proofs to review by a different research agent before integration. The positive direction enlarges disjoint exponent blocks to generators that may share the same oriented prime factors. It provides a uniform angle and small-prime estimate and an unconditional balance conclusion on the resulting class. The signed large-prime estimate required to obtain standard *abc* on that class remains an explicit hypothesis.

The two independent directions establish precise boundaries. A normalized infinite family makes the actual relative *abc* defect under the cubic power map arbitrarily large; this does not imply that its absolute defect is unbounded. A geometric argument classifies globally regular maps from a finite product of three-punctured projective lines to one such line, and identifies a new boundary divisor when genuinely mixed maps are allowed. Neither result retires its parent arithmetic or geometric program.

The proofs in the next three sections are ordinary mathematical arguments. Their companion Lean files cover actual integer-pair exponent reconstruction, normalized polynomial-root escape and explicitly conditional numerical amplification, and the integer rectangle identities underlying the geometric argument. They do not formalize the cited analytic inputs, the complete geometric classification, the general radical compression lemma, or standard *abc*. Fresh local compiler and axiom records specify their exact scope in the checkpoint verification directory. Independent agent review here is distinct from external peer review.

Artifact provenance. The user-designated August 31 PDF originally had 93 pages. The main source used as this round's baseline already contained the subsequent integrated research; its last tracked canonical PDF had 353 pages. This edition continues that source and updates the user-designated artifact. Page-count differences are not counts of newly proved mathematics. The earlier files remain recoverable from Git.

The next obligations. The shared-generator route must still control the signed large-prime tail on its new class and address profiles outside that class. The geometric successor must estimate the arithmetic cost of an unavoidable new boundary divisor. The power route must retain the seed's actual arithmetic credit. Existing IUT, Pell, Mersenne, packet, and valuation-incidence gates remain open unless a complete-premise theorem explicitly decides them. Standard *abc* is neither proved nor disproved by this continuation.

174. SHARED EXPONENT GENERATORS AND A STRICT IMPROVEMENT OF THE PROFILE ENVELOPE

Status. This continuation proves new ordinary mathematical results using established logarithmic-form estimates. It gives unconditional balance and small-prime estimates on actual content-one families, and a conditional *abc* class with a visibly retained large-prime hypothesis. It does not prove or disprove standard *abc*. The elementary two-column reconstruction has a separate Lean verification; the analytic theorems below are not thereby kernel-checked.

Write $T = abc$, $R = \text{rad}(T)$, $t = \log c$, $c = a + b$, where a, b are positive coprime integers. Retain the actual factorization from the preceding exponent-profile supplement:

$$z = a + b\zeta = u(1 + \zeta)^{e_0} \prod_{i=1}^r \pi_i^{e_i}, \quad M = N(z) = 3^{e_0} \prod_{i=1}^r q_i^{e_i}, \quad q_i = N(\pi_i) \geq 7.$$

The q_i are distinct split primes, $e_0 \in \{0, 1\}$, and each split prime is used with only one conjugate orientation. The prior identities are

$$(631) \quad \gcd(M, T) = 1, \quad \frac{3}{4}c^2 \leq M \leq c^2, \quad z^3 - \bar{z}^3 = 3\sqrt{-3}T.$$

The case with no split factors is $(1, 1, 2)$ and satisfies $c \leq R$ directly. It is excluded from all logarithmic-form applications below.

174.1. Allowing the same oriented prime in several generators.

Definition 174.1. A shared exponent representation has $m \geq 1$ columns, integers $k_j \geq 1$, and nonnegative integers f_{ij} such that

$$e_i = \sum_{j=1}^m k_j f_{ij}, \quad w_j = \prod_i \pi_i^{f_{ij}}, \quad Q_j = N(w_j), \quad S = \sum_j k_j.$$

Every column has a positive entry. Different columns may contain the same oriented prime. The ramified factor stays outside the matrix.

Lemma 174.2 (Exact shared reconstruction). *Every such representation satisfies*

$$z = u(1 + \zeta)^{e_0} \prod_j w_j^{k_j}, \quad \sum_j k_j \log Q_j = \log M - e_0 \log 3 \leq 2t, \quad Q_j \geq 7.$$

Proof. Expand the powers and commute the finite products. The total exponent of π_i is $\sum_j k_j f_{ij} = e_i$. Apply the multiplicative norm and take logarithms. A nonempty column contains a prime of norm at least seven. $\square$

Theorem 174.3 (Shared-generator bounds at both kinds of places). *There is an effective absolute $C > 2$ such that, for every representation,*

$$\mathcal{B} = C^{m+1} \log(3 + S) \prod_j \log Q_j$$

satisfies, with $\lambda = (z/\bar{z})^3$,

$$(632) \quad |\lambda - 1| \geq e^{-\mathcal{B}}, \quad v_p(T) \leq p^2 \mathcal{B} \quad (p \mid T),$$

$$(633) \quad \log \prod_{\substack{p \mid T \\ p \leq Y}} p^{v_p(T)} \leq \mathcal{B} Y^3 \log Y \quad (Y \geq 2),$$

$$(634) \quad c^3 \leq 8e^{\mathcal{B}} R^3 \frac{E_3(T)}{C_3(T)}.$$

Here $E_3(T) = \prod_{p \mid T} p^{(v_p(T)-3)_+}$ and $C_3(T) = \prod_{p \mid T} p^{(3-v_p(T))_+}$.

Proof. Put $\eta_j = (w_j/\bar{w}_j)^3$. Both complex conjugates of w_j have absolute value $\sqrt{Q_j}$, so $h(w_j) = \frac{1}{2} \log Q_j$ and the Weil height inequalities imply $h^*(\eta_j) \leq 3 \log Q_j$. No η_j equals one: otherwise the prime-ideal factorizations of w_j^3 and $\bar{w}_j^3$ would agree, although a selected split prime occurs only on the first side. This argument needs no coprimality between different columns.

The unit contribution disappears and the ramified factor contributes $(-1)^{e_0}$; hence $\lambda = (-1)^{e_0} \prod_j \eta_j^{k_j} \neq 1$ by (631). Choose principal logarithms and an integer v so that

$$\Lambda = \sum_j k_j \log \eta_j + (e_0 - 2v) \log(-1)$$

has imaginary part in $[-\pi, \pi]$. It is nonzero and $|e_0 - 2v| \leq S + 1$. Apply Bugeaud's Theorem 1.1, equation (1.2) [127], with at most $m + 1$ logarithms and field degree at most two. Delete zero coefficients if needed and select a nonzero last coefficient. Use $|e^{ix} - 1| \geq 2|x|/\pi$ on the displayed interval. Absorb the height factors and fixed constants into one absolute C .

For $p \mid T$, the norm coprimality in (631) shows that every $w_j, \bar{w}_j, z, \bar{z}$ is a unit at each place above p , including when the w_j overlap. With valuation normalized by $v_p(p) = 1$,

$$v_p(\lambda - 1) = v_p(T) + v_p(3\sqrt{-3}) \geq v_p(T).$$

Apply the first inequality of Bugeaud's Theorem 1.3 to $\eta_1, \dots, \eta_m, -1$ and coefficients $k_1, \dots, k_m, e_0$. It permits zero coefficients and does not require multiplicative independence. Its p^D factor is at most p^2 . This proves (632) after enlarging C once; summation with $\sum_{p \leq Y} p^2 \log p \leq Y^3 \log Y$ gives (633). Finally (631) gives $T \geq c^3 e^{-\mathcal{B}}/8$. Combine this with the exact identity $TC_3(T) = R^3 E_3(T)$ to prove (634). $\square$

Only the two explicitly identified established inequalities are used. In particular the conditional refinements and open improvements later in the non-Archimedean source are not substituted for its first inequality.

174.2. Quotients, remainders, and a new uniform class. Choose $g \geq 1$ and nonnegative profiles f_i, r_i with $e_i = gf_i + r_i$ and some $f_i > 0$. Set

$$w = \prod_i \pi_i^{f_i}, \quad v = \prod_i \pi_i^{r_i}, \quad Q = N(w) \geq 7, \quad V = N(v),$$

$$h = \max(1, \log V), \quad \rho = \frac{h \log(4+g)}{g}.$$

Then $z = u(1 + \zeta)^{e_0}vw^g$. If v is a unit, its column is omitted. The parameter g need not divide the original exponents.

Corollary 174.4. *There is an effective absolute $A \geq 1$ for which an admissible penalty satisfies $\mathcal{B} \leq Apt$. If $0 < \rho \leq 1/64$ and $Y = \rho^{-1/6}$, then*

$$\log \prod_{\substack{p|T \\ p \leq Y}} p^{v_p(T)} \leq \frac{A}{6} \sqrt{\rho} \log(1/\rho)t.$$

Proof. When $V > 1$, use $\mathcal{B} = C^3 \log(4+g) \log V \log Q$ and $g \log Q \leq 2t$. When $V = 1$, the one-column bound is smaller than the same estimate with $h = 1$. Thus $A = 2C^3$ suffices. Substitute $Y^3 = \rho^{-1/2}$ and $\log Y = \log(1/\rho)/6$ in (633); the threshold ensures $Y \geq 2$. $\square$

Theorem 174.5 (Unconditional balance). *For every $0 < \eta < 1$ there is an effective $\rho_\eta > 0$ such that every triple with this representation and $\rho \leq \rho_\eta$ satisfies $\min(a, b) > c^{1-\eta}$. No hypothesis on its radical or large-prime tail is required.*

Proof. If $\min(a, b) \leq c^{1-\eta}$, then $T \leq c^{3-\eta}$ and $\eta t \leq \log 8 + \mathcal{B}$. Also $t \geq g \log 7/2$ and $\rho \geq 1/g$. Consequently $\eta \leq (A + 2 \log 8 / \log 7) \rho$, a contradiction after choosing a smaller positive threshold. $\square$

Theorem 174.6 (An explicit conditional *abc* class). *For every $B_0 \geq 0$ and $\varepsilon > 0$ there is an effective $\rho_{B_0, \varepsilon} > 0$ such that*

$$0 < \rho \leq \rho_{B_0, \varepsilon}, \quad W_{\rho^{-1/6}}(T) := \frac{E_3^{>\rho^{-1/6}}(T)}{C_3^{>\rho^{-1/6}}(T)} \leq g^{B_0} \implies c \leq R^{1+\varepsilon}.$$

The threshold is independent of the norm primes and their number.

Proof. Make the threshold at most $1/64$. The small-prime part of $\log(E_3/C_3)$ is at most the full mass in Corollary 174.4, and the remaining part is at most $B_0 \log g$. Since $t \geq g \log 7/2$ and $\log g \leq \log(4+g)$, the remainder in (634), divided by t , is at most

$$A\rho + \frac{A}{6} \sqrt{\rho} \log(1/\rho) + \frac{2(\log 8 + B_0)}{\log 7} \rho.$$

Choose the threshold to make this at most $3\varepsilon/(1+\varepsilon)$. Taking logarithms in (634) and rearranging gives $t \leq (1+\varepsilon) \log R$. $\square$

The signed tail is an explicit unproved condition. The theorem asserts no global bound on it and no infinitude of the class satisfying that condition.

174.3. A strict separation from all disjoint partition envelopes.

Theorem 174.7. *There are height-unbounded primitive positive triples with split-exponent gcd one such that every old disjoint partition penalty satisfies*

$$\mathcal{B}_{\mathcal{P}} \geq \frac{\log 4}{8} \log M,$$

whereas a shared representation with two columns has $\mathcal{B}_{\text{shared}} = o(\log c)$. These triples satisfy the unconditional balance and small-prime estimates above. Their large-prime tail remains open.

Proof. Take r distinct split primes $q_i \in [X, 2X]$, put $L = \log X \geq \log 7$, $g = \lceil r^4 L^2 \rceil$, and $e_i = g + i$. Choose one orientation π_i of each and form $z = \prod_i \pi_i^{e_i}$. A rational prime dividing both coordinates would force a conjugate factor to occur, so z is primitive. A primitive zero-boundary pair is a unit, whereas $N(z) > 1$. Unit rotation therefore gives an actual positive triple. For $r \geq 2$ the exponent gcd is one. Writing $V_{\text{tot}} = \log M$, we have

$$(635) \quad grL \leq V_{\text{tot}} \leq 4grL \leq 8r^5 L^3.$$

For $w = \prod_i \pi_i$ and $v = \prod_i \pi_i^{e_i}$, $\log N(v) \leq 2r^2 L$, and hence

$$\rho \leq \frac{2 \log(4 + g)}{r^2 L} \rightarrow 0$$

uniformly in $L \geq \log 7$. Indeed $\log(4 + g) \leq \log 6 + 4 \log r + 2 \log L$; use the lower bound on L and boundedness of $(\log L)/L$. Corollary 174.4 proves the new sublinear estimate.

Consider any old partition into m blocks. For a block of size h_j , write k_j for its exponent gcd and $L_j = \log Q_j$. Always $L_j \geq L$. If $h_j \geq 2$, its indices are distinct and congruent modulo k_j ; therefore

$$k_j \leq \frac{r-1}{h_j-1}, \quad L_j \geq \frac{gh_j(h_j-1)L}{r}.$$

If $m \leq r/2$, a largest block has $h \geq r/m \geq 2$, giving $L_j \geq grL/(2m^2)$. Bound the other columns by L and use (635) to obtain

$$\frac{\mathcal{B}_{\mathcal{P}}}{V_{\text{tot}}} \geq \frac{C^{m+1} \log 4 L^{m-1}}{8m^2} \geq \frac{\log 4}{8}.$$

The last inequality uses $C > 2$, $L > 1$, and $2^{m+1} \geq m^2$. This elementary inequality holds directly for $m = 1, 2, 3$ and propagates for $m \geq 3$ because $(m+1)^2 \leq 2m^2$.

If $m > r/2$, use $L_j \geq L$ in every column:

$$\frac{\mathcal{B}_{\mathcal{P}}}{V_{\text{tot}}} \geq \frac{C^{m+1} \log 4 L^{m-3}}{8r^5}.$$

For $r \geq 8$, this is at least

$$\frac{\log 4}{4(\log 7)^3} \frac{(2 \log 7)^{r/2}}{r^5},$$

which tends to infinity independently of X . Thus all partitions, including those whose number of columns grows with r , satisfy the required lower bound once r is large.

Finally such prime lists exist with $r \rightarrow \infty$. As a separately identified established input, the explicit fixed-modulus estimate of [Bennett, Martin, O'Bryant, and Rechnitzer](#) gives $|\theta(x; 3, 1) - x/2| < x/(160 \log x)$ for $x \geq 8 \cdot 10^9$. Subtract the estimates at $2X$ and X to get $\theta(2X; 3, 1) - \theta(X; 3, 1) \geq X/3$ eventually. The interval consequently contains at least $X/(3 \log(2X))$ such primes. This supplies $r \rightarrow \infty$; (635) supplies unbounded height. This completes the proof. $\square$

The lower bound is for the previous explicit envelope, not for the true angle. The new representation therefore improves the available estimate on genuine arithmetic inputs, without contradicting the earlier theorems. Its conditional tail conclusion is not silently promoted to an unconditional *abc* result.

174.4. Dependencies, review, and precise formal scope. The ordinary proof chain is actual oriented factorization, nonnegative matrix reconstruction, the two stated established logarithmic-form bounds, and then uniform balance and small-prime control. The conditional *abc* arrow requires the independent signed-tail hypothesis in Theorem 174.6. Prime availability in Theorem 174.7 is an additional established input used only in that family's infinitude argument.

The companion `OverlapColumns.lean` proves reconstruction and the norm identity for a finite list of actual integer pairs with two shared exponent columns, as well as Euclidean quotient-remainder splitting of exponents. Its signature imposes no disjoint-support assumption. It does not formalize the external analytic estimates, the splitting classification, all asymptotic absorption, or standard *abc*.

The remaining arithmetic node for this new class is $W_{\rho^{-1/6}}(T) \leq g^{B_0}$ or a comparably sufficient correlated bound. Prime-norm terminal profiles remain outside $\rho \rightarrow 0$. Existing parent routes remain open, and no claim of a closed Lean term proving **ABCConjecture** is made.

175. AN UNBOUNDED OBSTRUCTION TO UNIFORM CUBIC DEFECT DESCENT

This section gives an unconditional obstruction to one precise child of the power-descent route. It concerns actual radicals of actual primitive triples, and is stronger than an obstruction to an independently estimated lifting factor. The parent route using the defect of the seed remains open.

For $x \geq 2$ put

$$S_x = (1, x-1, x), \quad L_x = (1, x^3-1, x^3), \quad R_1 = \text{rad}(x(x-1)), \quad R_3 = \text{rad}(x(x^3-1)).$$

For an integer $m \geq 1$ write

$$\mathcal{Q}_m(a, b, c) = c^m / \text{rad}(abc)^{m+1}.$$

Both displayed triples are positive and primitive.

Lemma 175.1 (A direct radical compression bound). *Suppose that $h \geq 1$ and $7^h \mid x^2 + x + 1$. Then*

$$7^{h-1} R_3 \leq R_1(x^2 + x + 1).$$

If also $x < 4 \cdot 7^h$, then $R_3 < 84xR_1$.

Proof. Set $G = x^2 + x + 1$. The factorization $x^3 - 1 = (x-1)G$ gives $R_3 \leq R_1 \text{rad}(G)$. Write $G = 7^v u$ with $v \geq h$ and $7 \nmid u$. Then

$$7^{h-1} \text{rad}(G) = 7^h \text{rad}(u) \leq 7^h u \leq G.$$

This proves the first inequality without requiring any unproved information about the other prime factors of G . Since $x \geq 2$, $G \leq 3x^2$. Also $x < 4 \cdot 7^h = 28 \cdot 7^{h-1}$, so $R_3 \leq 3x^2 R_1 / 7^{h-1} < 84xR_1$. $\square$

Lemma 175.2 (Infinitely many normalized compressed bases). *For every integer $h \geq 2$ there exists an integer x_h such that*

$$2 \leq x_h < 4 \cdot 7^h, \quad x_h \equiv 2 \pmod{4}, \quad x_h \equiv 2 \pmod{7}, \quad 7^h \mid x_h^2 + x_h + 1.$$

Every x_h is not a perfect power of exponent at least two, and $x_h \rightarrow \infty$ as $h \rightarrow \infty$.

Proof. The polynomial $F(X) = X^2 + X + 1$ satisfies $F(2) = 7$ and $F'(2) = 5 \not\equiv 0 \pmod{7}$. A completely elementary digit lift suffices: if r is a root modulo 7^j , choose the unique $d \in \{0, \dots, 6\}$ such that

$$F(r)/7^j + d(2r+1) \equiv 0 \pmod{7}.$$

Then $r + d7^j$ is a root modulo 7^{j+1} and is still 2 modulo seven. Induction gives a root r_h modulo 7^h . The Chinese remainder theorem gives the unique $x_h \in [0, 4 \cdot 7^h)$ satisfying $x_h \equiv r_h \pmod{7^h}$ and $x_h \equiv 2 \pmod{4}$. Its latter residue implies $x_h \geq 2$ and $v_2(x_h) = 1$, excluding every nontrivial perfect power. Finally $7^h \leq F(x_h)$; since F is bounded on every bounded interval of nonnegative integers, this proves $x_h \rightarrow \infty$. The sequence need not be monotone. $\square$

Theorem 175.3 (Unbounded amplification of the actual cleared defect). *For every integer $m \geq 2$, the normalized bases in Lemma 175.2 satisfy*

$$\frac{\mathcal{Q}_m(L_{x_h})}{\mathcal{Q}_m(S_{x_h})} > \frac{x_h^{m-1}}{84^{m+1}} \rightarrow \infty.$$

Consequently there is no finite constant A_m such that $\mathcal{Q}_m(L_x) \leq A_m \mathcal{Q}_m(S_x)$ for every integer $x \geq 2$ which is not a nontrivial perfect power.

Proof. The exact quotient of the two positive defects is

$$\frac{\mathcal{Q}_m(L_x)}{\mathcal{Q}_m(S_x)} = x^{2m}(R_1/R_3)^{m+1}.$$

Lemma 175.1 bounds this from below by $x^{2m}/(84x)^{m+1} = x^{m-1}/84^{m+1}$. For $m \geq 2$ this tends to infinity along Lemma 175.2. All arithmetic premises of the refuted estimate, including normalization, hold throughout this family. $\square$

Corollary 175.4 (The entire relevant real-exponent interval). *For $\varepsilon > 0$ define $D_\varepsilon(a, b, c) = c/\text{rad}(abc)^{1+\varepsilon}$. For every fixed $0 < \varepsilon < 1$,*

$$\frac{D_\varepsilon(L_{x_h})}{D_\varepsilon(S_{x_h})} > \frac{x_h^{1-\varepsilon}}{84^{1+\varepsilon}} \longrightarrow \infty.$$

Proof. The exact quotient is $x_h^2(R_1/R_3)^{1+\varepsilon}$. Apply Lemma 175.1. The exponent $1 - \varepsilon$ is positive. $\square$

Scope and the surviving bottleneck. The theorem refutes even a bounded-loss universal cubic defect comparison, not only a universal nonincrease comparison. It does not prove the target defects themselves unbounded: their seed defects may tend to zero. In particular the precise remaining obligation in a seed-sensitive approach is to control the product

$$\mathcal{Q}_m(S_x) x^{2m}(R_1/R_3)^{m+1} = \mathcal{Q}_m(L_x).$$

Treating this identity as a new independent estimate would be circular. The counterfamily excludes only the universal relative-comparison child; power descent with arithmetic seed credit and all other parent routes remain active. The endpoint $\varepsilon = 1$ is not decided by this lower bound. Neither this theorem nor finite computations below prove or disprove standard *abc*.

Formal and computational scope. The companion `CubicAmplification.lean` was checked using Lean 4.32.0 with warnings as errors. Its nine named declarations prove the cofactor cap and monotonicity, an explicit normalized root above any prescribed bound, the compression-to-amplification implications, and the impossibility of a uniform relative bound conditional on the displayed compression inequalities. The imported 29-declaration power-descent module was freshly compiled with the same compiler. The reported axiom dependencies are limited to `propxt`, `Classical.choice` and `Quot.sound`. This is a scoped check; the actual radical function and its prime-factorization inequality in Lemma 175.1 remain ordinary proofs, and are not claimed as formalized by this module. An independent exact-integer replay checks six fully factorized rows and 100 lifting levels. These finite checks neither certify an asymptotic estimate nor replace the infinite argument.

176. PRODUCT-TRIPOD RIGIDITY AND THE UNAVOIDABLE MIXING DIVISOR

Status. The geometric statements in this section are ordinary mathematical theorems, checked by a separate adversarial research agent. The companion Lean file verifies ten integer algebra statements, including a two-coordinate matrix separation theorem. It does not encode the geometric classification or prove the standard *abc* conjecture.

Let k be a field of characteristic zero, $n \geq 1$, and

$$U = \mathbb{P}_k^1 \setminus \{0, 1, \infty\}, \quad R_n = k[x_1, \dots, x_n, x_1^{-1}, (1 - x_1)^{-1}, \dots, x_n^{-1}, (1 - x_n)^{-1}].$$

The one-dimensional Riemann–Hurwitz obstruction does not by itself classify maps from a fixed product of tripods. The following theorem supplies that separate higher-dimensional obstruction.

Lemma 176.1 (A rectangle and its complement). *In an integral domain, suppose*

$$AD = BC, \quad (1 - A)(1 - D) = (1 - B)(1 - C).$$

Then $A + D = B + C$ and $(A - B)(A - C) = 0$. Consequently either $A = B$, $C = D$, or $A = C$, $B = D$.

Proof. Expand the second equality and cancel the first to obtain $A + D = B + C$. Substitute $D = B + C - A$ in $AD = BC$ and factor. The zero-product law and the additive equality give the two alternatives. $\square$

Theorem 176.2 (Product-tripod rigidity). *Every nonconstant regular k -morphism $U^n \rightarrow U$ is one of*

$$x_i, \quad 1 - x_i, \quad x_i^{-1}, \quad (1 - x_i)^{-1}, \quad \frac{x_i}{x_i - 1}, \quad \frac{x_i - 1}{x_i}$$

for some i . Every dominant regular endomorphism of U^n is therefore a coordinate permutation followed by these transformations coordinate by coordinate, and is an automorphism.

Proof. The polynomial ring is a unique factorization domain. After inverting the irreducibles $x_i, 1 - x_i$, every unit is $c \prod_i x_i^{a_i} (1 - x_i)^{b_i}$ with $c \in k^*$ and $a_i, b_i \in \mathbb{Z}$. Indeed a different irreducible cannot divide a unit or its inverse. A map to U is exactly a function $f \in R_n$ such that f and $1 - f$ are units. Thus both have separated product expressions

$$f = c \prod_i u_i(x_i), \quad 1 - f = d \prod_i v_i(x_i).$$

Pass to an algebraic closure K . If u_i is nonconstant, choose $s_i, t_i \in U(K)$ with $u_i(s_i) \neq u_i(t_i)$. Fix $j \neq i$, arbitrary $s_j, t_j \in U(K)$, and all other coordinates. The four values of f at the rectangle, with rows indexed by s_i, t_i , satisfy both determinant identities of Lemma 176.1. Its first-column entries differ because all factors are nonzero. The lemma therefore forces each row to be constant. Cancelling the other nonzero factors gives $u_j(s_j) = u_j(t_j)$. Infinitude of K implies that u_j is constant as a rational function. Thus a nonconstant f depends on one coordinate only.

Set every other coordinate equal to 2; this is a valid localization retraction in characteristic zero. It shows explicitly that f and $1 - f$ are units in the corresponding one-variable localization. Write $f = P/Q$ in lowest terms. The nonzero polynomials $P, Q, Q - P$ are pairwise coprime, and every root of their product lies in $\{0, 1\}$. Its reduced radical has degree at most two. Mason–Stothers yields $\max(\deg P, \deg Q, \deg(Q - P)) < 2$; its derivative-zero alternative would make all three polynomials constant in characteristic zero. The nonconstant map therefore has degree one and permutes the three boundary points. Their six permutations give the displayed list.

Finally, dominance of an endomorphism requires its output functions to be algebraically independent. None can be constant and no two can read the same input coordinate. The resulting coordinate assignment is a permutation; the inverse coordinate assignment and inverse boundary permutations are regular on U^n . $\square$

The one-active-coordinate argument is characteristic-free after passage to an algebraic closure. Degree-one rigidity is not: in characteristic p , $f = x_i^p$ is a regular map to U , since $1 - f = (1 - x_i)^p$, and has degree p in that coordinate. This is a complete-premise counterexample only to the characteristic-free strengthening.

Corollary 176.3 (Codimension-two deletion does not create an amplifier). *If $Z \subset U^n$ is closed of codimension at least two, every regular map $U^n \setminus Z \rightarrow U$ extends uniquely to U^n and satisfies Theorem 176.2.*

Proof. The rational functions $f, 1/f, 1 - f, 1/(1 - f)$ are regular at the generic point of every irreducible divisor of $\text{Spec } R_n$, since that point is outside Z . In a reduced fraction in the factorial ring R_n , this excludes every nonunit irreducible denominator factor. All four functions therefore belong to R_n . Hence f and $1 - f$ are units and define the extension; uniqueness follows from equality in the function field. $\square$

This closes the specific gate of globally regular, genuinely mixed or degree-amplifying maps of a fixed finite product of tripods with no new boundary. It does not close the routes through correspondences, proper subvarieties, non-product moduli spaces, moving maps, or explicitly charged new divisors.

176.1. The exact arithmetic cost of a minimal mixed map. The function $f(x, y) = xy$ defines a mixed map $U^2 \setminus \{xy = 1\} \rightarrow U$. At the primitive positive point $(x, y) = (a/c, b/c)$ with $a + b = c$, put

$$N = c^2 - ab = a^2 + ab + b^2.$$

The output is the primitive triple (ab, N, c^2) . A prime dividing N and one of a, b, c would divide a second entry of the original triple, by reducing the displayed formula modulo the first. Thus

$$\gcd(N, abc) = 1, \quad \text{rad}(abNc^2) = \text{rad}(abc) \text{rad}(N), \quad \frac{3}{4}c^2 \leq N < c^2.$$

The size inequalities follow from $(a - b)^2 \geq 0$ and $ab > 0$. Writing $h = \log c$, $r = \log \text{rad}(abc)$, and $s = \log \text{rad}(N)$, the output quality and excess are exactly

$$q' = \frac{2h}{r + s}, \quad E'_\varepsilon = 2E_\varepsilon + (1 + \varepsilon)(r - s), \quad E_\varepsilon = h - (1 + \varepsilon)r.$$

The new factor has quadratic size and no overlap with the original radical. Its radical need not have quadratic size. Any argument using this mixed map must supply independent arithmetic control of s or another way to pay for the new divisor. No such uniform control is proved or assumed here.

The formal file `ProductTripodRectangle.lean` verifies the integer rectangle identities, the row-or-column dichotomy, a global one-coordinate dependence theorem for integer-valued matrices with those identities, and the four displayed mixed-map algebra and size statements. Its ten axiom queries report only `propext`, `Classical.choice`, and `Quot.sound`. Unit classification, scheme extension, and the full geometric theorem remain outside that formal scope.

177. SHIFTED RANKS AND SUCCESSIVE NORM TRANSPORT: A SECOND PARALLEL CONTINUATION

This continuation follows different unresolved gates from the preceding round. On the forward arithmetic route, exact shifted residue classes give an elementary average bound for the full multiplicity of primes in a growing range. Retaining the residual norm in the height denominator makes the bound uniform in the residual and in the root. Finite unions then permit adaptive selection from specified growing norm boxes. Exceptional indices and primes outside the controlled range remain explicit.

A further two-place refinement depends on the remainder proportion $\lambda = \max(1, \log N(v))/g$ alone. It treats the ramified sign, unit residuals and multiplicatively dependent factors explicitly. Actual representations with $\lambda \rightarrow 0$ and displayed $\rho \rightarrow \infty$ demonstrate the gain over the earlier sufficient test. The refinement also proves a positive second- λ floor when the first root is fixed and its first λ is bounded. It does not compare that floor with the finite two-step threshold or bound it uniformly for moving first roots.

On the reverse route, actual radical transport through two norm transforms gives a sufficient criterion for a disproof. A fixed quartic field and unit describe the second norm's prime depths. Local lifting supplies simultaneous arbitrarily deep divisibility at two distinct primes on actual primitive families. Such local compatibility does not give compression of the entire second norm, and the required unbounded family is still unconstructed.

The homogeneous signed-rank decomposition remains a separate route. Its rankwise polynomial hypothesis implies that the positive part of the normalized signed tail tends to zero. An upper bound tending to zero is not asserted to be a two-sided limit for the signed quantity itself. The residual counterfamily refutes a precise homogeneous lifting rule transferred to shifted orbits; its fixed prime eventually lies below the moving tail cutoff.

Verification boundary. The ordinary proofs in the next sections were checked by other research agents before integration. The new formal modules cover actual residue-class parameterizations and exact counting errors, an infinite finite-depth orbit, integer radical identities and compression, and an implication from an explicit unbounded-family premise to the negation of the repository's standard abc definition. The family premise is not supplied. The analytic estimates, complete

local group classification, and full ordinary research program are not silently included in that formal scope. Independent agent review does not constitute external peer review.

Continuing obligations. The forward route still needs a signed large-prime bound and control of exceptional indices. The reverse route still needs either an actual unbounded family meeting all compression conditions or a theorem excluding a precisely stated family. Moving roots, shared remainders, and the broader geometric, IUT, Pell, Mersenne and incidence routes stay open unless their full premises are decided by proof or counterexample. This continuation does not prove or disprove standard *abc*.

178. SIGNED RANK PACKETS AND THE RESIDUAL-SHIFT BOUNDARY

The shared-generator theorem leaves an arithmetic signed tail. The present section gives an exact decomposition on homogeneous orbits, a sufficient rankwise budget with its full quantifiers, and an actual content-one family which explains why that decomposition cannot be transferred unchanged to a residual orbit. The rankwise arithmetic budget remains unproved. No result in this section proves or disproves standard *abc*.

For a positive integer T and a real $Y \geq 1$, write

$$W_Y(T) = \prod_{\substack{p|T \\ p > Y}} p^{v_p(T)-3}.$$

This is a positive rational number. In particular, primes of exponent one or two contribute negative logarithmic credit; they are not silently removed. For an actual representation $z = u(1 + \zeta)^{e_0}vw^g$, the shared-generator parameter is

$$\rho = \frac{\max(1, \log N(v)) \log(4 + g)}{g}.$$

The previously proved sufficient class assumes small ρ and a bound $W_{\rho^{-1/6}}(abc) \leq g^B$. Its exact height absorption also works under the weaker conclusion $(\log W_{\rho^{-1/6}}(abc))_+ = o(\log c)$, provided that the little-oh estimate is uniform on the stated class. This observation is not a proof of either arithmetic estimate.

178.1. An exact homogeneous decomposition. Let w be a primitive nonunit Eisenstein integer with no ramified factor and no conjugate pair of split factors. Set

$$Q = N(w) \geq 7, \quad T_n = |P(w^n)| \quad (n \geq 1), \quad P(x + y\zeta) = xy(x + y).$$

Every w^n remains primitive. Its boundary is nonzero, since a primitive zero-boundary element is a unit. Rotate w^n into the positive sector and denote the largest entry of the resulting primitive triple by c_n . Then

$$(636) \quad \log c_n \geq \tfrac{1}{2}n \log Q.$$

For $p > 3$ with $p \nmid Q$, let d_p be the residue order of

$$\alpha = (w/\bar{w})^3$$

at a place above p . At a split prime the two components give inverse residues, so this order does not depend on the component. At an inert prime the residue lies in the norm-one group of order $p + 1$. In particular $d_p \mid p - 1$ or $d_p \mid p + 1$, and $p \nmid d_p$. Put

$$s_p = v_p(T_{d_p}).$$

Lemma 178.1 (Exact homogeneous first-depth law). *For every such prime and every $n \geq 1$,*

$$p \mid T_n \iff d_p \mid n, \quad v_p(T_n) = s_p + v_p(n) \quad \text{if } d_p \mid n.$$

Proof. The cubic identity

$$w^{3n} - \bar{w}^{3n} = 3\sqrt{-3}P(w^n)$$

identifies the valuation of $\alpha^n - 1$ with that of T_n : both Q and $3\sqrt{-3}$ are units at p . Reduction modulo p proves the support criterion. If $n = d_p m$, the usual odd-prime binomial argument in the unramified local quadratic ring gives

$$v_p(\alpha^n - 1) = v_p(\alpha^{d_p} - 1) + v_p(m).$$

Raising a principal unit to a p -coprime power preserves its valuation; raising it to p increases it by one, since the linear binomial term is uniquely least. There is no ramification correction at $p > 3$. Finally $p \nmid d_p$, so $v_p(m) = v_p(n)$. At split primes, norm-one conjugacy gives the same normalized valuation in both components. $\square$

For $Y \geq 3$ define the finite signed rank packet

$$A_d(Y; w) = \prod_{\substack{p > Y \\ d_p = d}} p^{s_p - 3}, \quad L_{n,Y} = \prod_{\substack{p > Y \\ d_p | n}} p^{v_p(n)}.$$

Every prime in A_d divides the fixed nonzero integer T_d , so the product is finite. Primes dividing Q never divide T_n and are not included.

Proposition 178.2 (The signed packet identity). *For every $n \geq 1$ and $Y \geq 3$,*

$$(637) \quad \boxed{W_Y(T_n) = L_{n,Y} \prod_{d|n} A_d(Y; w), \quad L_{n,Y} \mid n.}$$

Proof. For each supported prime above Y , Lemma 178.1 separates its exponent $v_p(T_n) - 3$ into $(s_p - 3) + v_p(n)$. Group the first terms by their exact rank $d_p \mid n$. Each exponent in the second product is at most the corresponding exponent in n , proving the divisibility assertion. No positivity assumption on the first product is used. $\square$

178.2. A rankwise polynomial budget is sufficient. For the homogeneous representation take

$$\rho_n = \frac{\log(4+n)}{n}, \quad Y_n = \rho_n^{-1/6}.$$

The next theorem restricts to $Y_n \geq 3$, the domain of the rank packets; this includes all sufficiently large n .

Theorem 178.3 (Uniform absorption from rankwise budgets). *Fix $\kappa \geq 0$. Consider actual pairs (w, n) as above, with $n \geq 1$ and $Y_n \geq 3$, satisfying*

$$(638) \quad A_d(Y_n; w) \leq (Qd)^\kappa \quad \text{for every } d \mid n.$$

Then

$$(639) \quad \log W_{Y_n}(T_n) \leq \log n + \kappa \tau(n) \log Q + \frac{\kappa}{2} \tau(n) \log n.$$

Consequently $(\log W_{Y_n}(T_n))_+ / \log c_n \rightarrow 0$ uniformly on this class as $n \rightarrow \infty$. For every $\varepsilon > 0$ there is a threshold depending only on κ, ε and the absolute constants in the established shared-generator theorem such that every pair in this class above that threshold satisfies

$$c_n \leq \text{rad}(T_n)^{1+\varepsilon}.$$

The arithmetic hypothesis (638) is not proved for all roots or all ranks.

Proof. Take logarithms in Proposition 178.2, use $L_{n,Y_n} \leq n$, and sum (638). The involution $d \mapsto n/d$ of the divisors gives

$$\sum_{d|n} \log d = \frac{1}{2} \tau(n) \log n,$$

including a possible fixed divisor $\sqrt{n}$. This proves (639). Pairing small and large divisors also gives the elementary bound $\tau(n) \leq 2\sqrt{n}$. Divide by (636), use $Q \geq 7$, and obtain the explicit uniform estimate

$$(640) \quad \frac{\log W_{Y_n}(T_n)}{\log c_n} \leq \frac{2 \log n}{n \log 7} + \frac{4\kappa}{\sqrt{n}} + \frac{2\kappa \log n}{\sqrt{n} \log 7} \rightarrow 0.$$

The one-column shared-generator theorem supplies a uniformly sublinear angular penalty and full small-prime mass at Y_n . In the compensated cubic height inequality, the remaining high-prime signed mass is bounded by (640). Its nonnegative right side also bounds the positive part of the left side; no two-sided sublinear claim is made about a signed quantity which may retain substantial negative credit. Choose the threshold at which their combined ratio, including the constant $\log 8$, is at most $3\varepsilon/(1+\varepsilon)$. The standard absorption yields the asserted inequality. No threshold depends on w or its prime support. $\square$

This theorem asks for polynomial dependence on both Q and the individual rank; it does not demand that their complete divisor product be polynomial in n . That latter inference would be false without an additional arithmetic restriction. For example, the abstract equality model $A_d = d^\kappa$ gives

$$\prod_{d|n} A_d = n^{\kappa\tau(n)/2},$$

which exceeds any fixed power of n for $\kappa > 0$ along $n = 2^j$. This is a counterexample to that abstract aggregation inference, not an arithmetically realized counterexample to a rank-packet theorem. Estimate (640) shows why the false inference is unnecessary. Also, a bound on a full signed packet at cutoff zero does not automatically survive the removal of small primes carrying negative credit. The actual cutoff Y_n in (638) is essential.

178.3. A small-rho content-one residual counterfamily. The divisibility $L_{n,Y} | n$ uses the homogeneous first-depth law. The next construction refutes one precise attempted transplantation of that law to vw^g while preserving every structural hypothesis of the new small-rho class.

Fix $w = 2 + \zeta$, with $N(w) = 7$ and $w^2 = 3 + 5\zeta$. For each $k \geq 1$ put

$$g_k = 2^k, \quad w^{g_k} = A_k + B_k\zeta.$$

Modulo five, this is the nonzero scalar $3^{g_k/2}$. Hence $A_k + B_k$ is invertible modulo 5^k . By the modular inverse and the Chinese remainder theorem, there is a unique $y_k \in [0, 84 \cdot 5^k)$ satisfying

$$(641) \quad B_k + y_k(A_k + B_k) \equiv 0 \pmod{5^k}, \quad y_k \equiv 42 \pmod{84}.$$

Set

$$v_k = 1 + y_k\zeta, \quad V_k = 1 + y_k + y_k^2, \quad z_k = v_k w^{g_k}, \quad T_k = |P(z_k)|.$$

Theorem 178.4 (Deep residual hits with total content one). *Every z_k is primitive with nonzero boundary and gives a primitive positive triple after unit rotation. Its complete split-norm exponent gcd equals one. Its displayed shared representation satisfies*

$$\rho_k = \frac{\log V_k \log(4 + 2^k)}{2^k} \rightarrow 0, \quad v_5(T_k) \geq k, \quad v_5(g_k) = 0.$$

Proof. The pair $(1, y_k)$ is primitive. Since y_k is divisible by three and seven, neither prime divides V_k . Thus the factorization of v_k has no ramified factor, no conjugate pair, and no factor above seven. Multiplication by w^{g_k} preserves primitivity and the oriented factorization. Its norm is greater than one, so its primitive boundary cannot vanish.

Since $y_k \equiv 2 \pmod{4}$, one has $V_k \equiv 3 \pmod{4}$. At least one of its prime exponents is odd, since a factorization with all exponents even would make V_k an odd square. This prime is not seven. Its odd exponent has gcd one with the norm exponent 2^k at seven, proving that the complete split-exponent gcd is one.

The second coordinate of z_k is $B_k + y_k(A_k + B_k)$, divisible by 5^k . Modulo five, $B_k = 0$, $A_k \neq 0$, and (641) gives $y_k = 0$. The first coordinate $A_k - y_k B_k$ and the sum of the coordinates are both units modulo five. Hence $v_5(T_k)$ is exactly the valuation of the second coordinate and is at least k . Clearly $5 \nmid 2^k$.

Finally $y_k < 84 \cdot 5^k$ and $y_k \geq 2$ give

$$\log V_k < \log(3 \cdot 84^2) + 2k \log 5, \quad \log(4 + 2^k) \leq (k + 2) \log 2.$$

Thus $\rho_k = O(k^2/2^k) \rightarrow 0$. This argument does not factor the growing integer V_k . $\square$

For the fixed step root $w = 2 + \zeta$, the homogeneous data at five are $d_5 = 2$ and $s_5 = v_5(P(w^2)) = v_5(120) = 1$. The displayed family therefore refutes the exact proposed extension

$$v_5(P(vw^g)) \leq s_5 + v_5(g)$$

when s_5 is retained from the homogeneous step root: its proposed bound is one while the actual depth is arbitrarily large. This failure persists under small ρ and complete exponent gcd one. A theorem defining new residual-dependent first depths or shifted residue classes is a different claim and is not refuted.

This counterfamily does not refute the signed-tail gate. Five eventually lies below $\rho_k^{-1/6}$; the construction also provides no upper bound on the entire boundary radical or on all its signed credits. In particular it supplies no ABC counterexample. The proper successor is the shifted residue lattice at each prime and depth, treated separately.

178.4. Exact local replay and formal scope. Independent modular exponentiation and CRT gave, for example,

k	g_k	y_k	$v_5(T_k)$
16	65536	7181029215570	16
18	262144	144280296684930	20
20	1048576	5276983743390870	21

Each exact depth was checked by vanishing modulo 5^e and nonvanishing modulo 5^{e+1} , with the other boundary coordinates verified to be units. The same procedure reaches $k = 60$ without constructing its enormous full boundary. These are local certificates; no complete factorization of those boundary integers is claimed.

The homogeneous packet and residual-family ordinary proofs were independently reviewed by the critical-bottleneck research agent. The arithmetic budget (638) remains open after that review.

The separate second-round module `ShiftedResidueCounts.lean` proves eight scoped declarations using only `Std`. It proves the actual residue-hit parameterization

$$n < N, \ n \bmod d = a \iff \exists j < C(d, a, N), \ n = a + dj,$$

its uniqueness, and the strict one-period counting error via explicit Euclidean division. It also checks the actual infinite family $(1 + 5\zeta)(1 + 50\zeta)^n$ whose boundary is 5 (mod 25) at every index, while the step boundary has exact five-adic depth two. All eight declarations were checked by Lean 4.32.0 with warnings as errors and axiom dependencies limited to `propext`, `Classical.choice`, and `Quot.sound`. This formal scope does not include the general rank law, the rankwise arithmetic budget, the complete norm-one group classification, or the real weighted global tail limit. No such stronger formal claim is made here.

179. SHIFTED RANK TOWERS AND UNIFORM MOVING-PRIME WINDOWS

Status. This second continuation proves exact local residue structures and uniform average estimates over moving prime ranges. It also controls a common exceptional set for bounded families of actual residuals and roots. These are ordinary mathematical results with the dependencies stated below. The full signed tail, individual exceptional indices, and standard *abc* remain unresolved. The companion Lean module verifies actual residue counts and an actual stopping tower, with its narrower scope stated at the end.

179.1. Actual arithmetic inputs and shifted residue classes. Let $\mathcal{O} = \mathbb{Z}[\zeta]$, where $\zeta^2 - \zeta + 1 = 0$, and write $P(x + y\zeta) = xy(x + y)$. Let $w, v \in \mathcal{O}$ be primitive with compatible oriented split factors, with w unramified and the ramified exponent of v at most one. Assume $Q = N(w) \geq 7$, put $V = N(v) \geq 1$, and set

$$A_n = |P(vw^n)|, \quad B_n = |P(w^n)| \quad (n \geq 1).$$

All these elements are primitive. Their norms exceed one, so their boundaries are nonzero. Let c_n be the height after positive unit rotation. Then

$$(642) \quad \log c_n \geq \frac{1}{2}(\log V + n \log Q), \quad \log A_n \leq \frac{3}{2}(\log V + n \log Q).$$

The second inequality follows directly from $|P(z)| \leq 2N(z)^{3/2}/(3\sqrt{3}) \leq N(z)^{3/2}$, which is a consequence of the cubic identity. The already proved full boundary overlap identity gives

$$\gcd(A_m, A_n) = \gcd(A_m, B_{n-m}) \quad (n > m \geq 1).$$

Fix a prime $p > 3$ with $p \nmid VQ$, and put

$$\alpha = (w/\bar{w})^3, \quad \beta = (v/\bar{v})^3, \quad d_p = \text{ord}(\alpha \bmod p), \quad s_p = v_p(B_{d_p}).$$

At split primes one can use the product residue algebra; conjugation makes the component orders and valuations agree. At inert primes use the quadratic residue field. With $\chi_p = 1$ for $p \equiv 1 \pmod{3}$ and -1 otherwise, the norm-one group has order $p - \chi_p$. Since α is a cube,

$$(643) \quad d_p \mid (p - \chi_p)/3, \quad v_p(B_n) = \begin{cases} 0, & d_p \nmid n, \\ s_p + v_p(n/d_p), & d_p \mid n. \end{cases}$$

For the valuation formula, write $\alpha^{d_p} = 1 + p^{s_p}u$ in the unramified local algebra. Raising to a power prime to p preserves the valuation, and raising to p increases it by one, since the linear binomial term has uniquely least valuation. The cubic identity has unit prefactor at $p > 3$, so it identifies this with the displayed rational integer valuation.

Theorem 179.1 (Exact shifted rank lattice). *For every $e \geq 1$, the positive integers n with $p^e \mid A_n$ form either an empty set or one residue class modulo*

$$D_{p,e} = d_p p^{(e-s_p)+}.$$

Nonempty successive classes are coherent under reduction. In every interval of N consecutive positive integers, the number of hits differs from $N/D_{p,e}$ by less than one for a nonempty class, and is zero for an empty class.

Proof. The cubic identity and local units give $p^e \mid A_n$ if and only if $\beta\alpha^n = 1 \pmod{p^e}$. Once a solution exists, all solutions differ by multiples of the multiplicative order of α modulo p^e . Formula (643) says that this order is exactly $D_{p,e}$. Reduction proves coherence. A single arithmetic progression has the stated interval count. In particular its residue need not be zero; homogeneous divisibility of the exponent cannot replace this class. $\square$

Theorem 179.2 (The three possible tower types). *At a fixed good prime, either all levels are empty, the nonempty levels are exactly $1, \dots, t_p$ for some $1 \leq t_p < s_p$, or every level is nonempty.*

Proof. Emptiness persists under projection. Let G_e be the norm-one group of $(\mathcal{O}/p^e\mathcal{O})^\times$. For $j \geq 1$, the kernel of $G_{j+1} \rightarrow G_j$ consists of $1 + p^j u$ with $\text{Tr}(u) = 0 \pmod{p}$. Indeed its norm is $1 + p^j \text{Tr}(u) \pmod{p^{j+1}}$. Trace on the two-dimensional residue algebra is surjective because $\text{Tr}(1) = 2 \neq 0$. The kernel consequently has p elements. Reduction is surjective: correct an arbitrary lift of norm $1 + p^j a$ by a factor with trace $-a$.

Thus $\ker(G_e \rightarrow G_{s_p})$ has size p^{e-s_p} . By (643), α^{d_p} has exactly this order and generates the whole kernel. A solution at depth s_p therefore extends to every depth $e \geq s_p$ by adding a multiple of d_p to its exponent. This proves the classification. $\square$

179.2. Signed local density and its finite error. For $H \geq 1$ define

$$D_{p,H}(n) = \sum_{e=1}^H \mathbf{1}_{p^e \mid A_n} - 3\mathbf{1}_{p \mid A_n}.$$

When $H \geq v_p(A_n)$ this is zero on unsupported primes and is $v_p(A_n) - 3$ otherwise, retaining first- and second-power credit. Let $\delta_{p,e} = 1/D_{p,e}$ for nonempty levels and zero for empty levels. On any interval I of N consecutive positive integers,

$$(644) \quad \left| \sum_{n \in I} D_{p,H}(n) - N \left(\sum_{e=1}^H \delta_{p,e} - 3\delta_{p,1} \right) \right| \leq H + 3.$$

This follows by summing the actual progression errors from Theorem 179.1; an empty level has exactly zero error.

The corresponding infinite density is

$$(645) \quad \mu_p = \begin{cases} 0, & \text{empty tower,} \\ (t_p - 3)/d_p, & \text{finite tower,} \\ (s_p - 3 + 1/(p-1))/d_p, & \text{full tower.} \end{cases}$$

For the last case sum the s_p initial levels of density $1/d_p$ and the subsequent geometric series of total $1/(d_p(p-1))$. The finite case has exactly t_p initial levels, all with period d_p . In particular,

$$\mu_p \leq \frac{\max(0, s_p - 3 + 1/(p-1))}{d_p}.$$

Every nonempty tower with $s_p \leq 2$ has negative mean. At $s_p = 3$, the only possible positive mean is $1/(d_p(p-1))$.

Proposition 179.3 (A fixed-finite-prime mean law). *For fixed v, w and a fixed finite set S of good primes,*

$$\frac{1}{N} \sum_{n=1}^N \log \prod_{\substack{p \in S \\ p|A_n}} p^{v_p(A_n)-3} \longrightarrow \sum_{p \in S} \mu_p \log p.$$

Proof. The first p-adic inequality of Bugeaud's Theorem 1.3 [127], applied to the fixed β, α with exponents $1, n$, gives $v_p(A_n) \leq K_p \log(4+n)$. All numbers are local units and the product is nontrivial because $A_n \neq 0$. Choose $H_N = O(\log N)$ covering the valuations for $p \in S$, $n \leq N$, and tending to infinity. Then the capped quantities equal the actual signed valuations. Divide (644) by N : the finite- S error is $O(\log N/N)$. The geometric sums tend to (645), proving the claim. This is the only external analytic input to this proposition; it does not justify replacing the fixed set S by all primes. $\square$

179.3. An elementary estimate for a moving prime range. For real $Z \geq 5$ set

$$F_Z(n) = \sum_{3 < p \leq Z} v_p(A_n) \log p.$$

This is the full nonnegative prime-power mass in the specified range. The following estimate uses elementary height caps and actual residue counts.

Lemma 179.4 (First-rank density bound). *For every good prime,*

$$\frac{s_p + 1/(p-1)}{d_p} \log p \leq 2 \log Q.$$

Proof. The norm bound gives $s_p \log p \leq \log B_{d_p} \leq (3d_p/2) \log Q$. Also $\log p \leq (p-1)/2$ for $p \geq 5$: the difference $(x-1)/2 - \log x$ is positive at five and increasing thereafter. The lifting contribution is therefore at most $1/2 \leq (\log Q)/2$. $\square$

Theorem 179.5 (Uniform moving-prime block bound). *For every $N \geq 2$ and $Z \geq 5$,*

$$(646) \quad \frac{1}{N} \sum_{n=N}^{2N-1} \frac{F_Z(n)}{\log c_n} \leq \frac{10\pi(Z)}{N}.$$

The constant is uniform over all compatible v, w , with no upper bound on V or Q and no small- ρ assumption. Within each block the factors v, w are fixed.

Proof. Write $q = \log Q$, $v_0 = \log V$, and $H_{\max} = \frac{3}{2}v_0 + 3Nq$. By (642), every valuation in the block is at most $H_p = \lfloor H_{\max}/\log p \rfloor$; this cap may be zero. Counting each layer with Theorem 179.1 gives

$$\sum_{n=N}^{2N-1} v_p(A_n) \leq N \frac{s_p + 1/(p-1)}{d_p} + H_p.$$

Use Lemma 179.4 and $H_p \log p \leq H_{\max}$ to obtain

$$\sum_{n=N}^{2N-1} v_p(A_n) \log p \leq 5Nq + \frac{3}{2}v_0.$$

Primes dividing VQ have zero boundary support by the norm-boundary gcd. Sum the remaining primes and retain the full denominator $\log c_n \geq (v_0 + Nq)/2$. The resulting bound is

$$\frac{\pi(Z)}{N} \frac{10Nq + 3v_0}{Nq + v_0} \leq \frac{10\pi(Z)}{N}.$$

Every endpoint error was included; no p-adic logarithmic-form cap is needed for this average estimate. $\square$

Lemma 179.6 (An elementary prime-counting bound). *For real $x \geq 2$, one has $\pi(x) \leq 8x/\log x$.*

Proof. Every prime in $(m, 2m]$ divides $\binom{2m}{m}$, so $\theta(2m) - \theta(m) \leq 2m \log 2$. Sum at powers of two to obtain $\theta(2^k) < 2^{k+1} \log 2$, hence $\theta(x) \leq 4x \log 2$. Separate primes at $\sqrt{x}$:

$$\pi(x) \leq \sqrt{x} + \frac{2\theta(x)}{\log x} \leq \sqrt{x} + \frac{8(\log 2)x}{\log x}.$$

The function $\sqrt{x} - \log x$ has a positive minimum at $x = 4$, so $\sqrt{x} \leq x/\log x$. Finally $1 + 8 \log 2 < 8$; for example the trapezoid bound for the integral of $1/t$ on $[1, 2]$ gives $\log 2 \leq 3/4$. $\square$

Corollary 179.7 (A linear prime range for almost all block indices). *For all sufficiently large N , apart from at most $80N/\sqrt{\log N}$ indices in the block,*

$$F_N(n) \leq \frac{\log c_n}{\sqrt{\log N}}.$$

Proof. At $Z = N$, Theorem 179.5 and Lemma 179.6 give mean at most $80/\log N$. Apply Markov's inequality to the nonnegative normalized masses with threshold $1/\sqrt{\log N}$. More generally, (646) applies to any moving range with $\pi(Z) = o(N)$. $\square$

179.4. A common exceptional set for bounded residuals and roots. There are at most $25B$ integer pairs with $0 < N(v) \leq B$, for $B \geq 1$. Indeed the norm bounds both coordinates in absolute value by $2\sqrt{B}$, giving at most $(4\sqrt{B} + 1)^2 \leq 25B$ pairs. Primitivity or compatibility restrictions only reduce this count.

Theorem 179.8 (Simultaneous residual control). *Fix w , $N \geq 2$, $B \geq 1$, $Z \geq 5$, and $\eta > 0$. The proportion of indices in $[N, 2N)$ for which some compatible actual residual v of norm at most B satisfies $F_Z(n) > \eta \log c_n$ is at most*

$$\frac{250B\pi(Z)}{N\eta}.$$

Thus, for fixed $0 \leq \beta < 1$, $B = N^\beta$, $Z = N^{1-\beta}$, and $\eta = 1/\sqrt{\log N}$, the exceptional proportion is at most

$$\frac{2000}{(1-\beta)\sqrt{\log N}}$$

eventually. At every remaining index the bound holds simultaneously for all compatible residuals in the displayed norm range.

Proof. Markov and (646) bound each residual's bad indices by $10\pi(Z)/\eta$. Take the union over at most $25B$ actual residuals and divide by N . The specialized bound follows from Lemma 179.6. Because the good set is common to the entire finite family, the residual may be chosen separately at each good index within that family. $\square$

Corollary 179.9 (Simultaneous roots and residuals). *If the roots are also restricted to $7 \leq N(w) \leq C$, the exceptional proportion for some compatible pair is at most*

$$\frac{6250BC\pi(Z)}{N\eta}.$$

For fixed $\beta \geq 0$, $\gamma > 0$, $\beta + \gamma < 1$, take $B = N^\beta$, $C = N^\gamma$, $Z = N^{1-\beta-\gamma}$ and $\eta = 1/\sqrt{\log N}$. The bound becomes

$$\frac{50000}{(1 - \beta - \gamma)\sqrt{\log N}}$$

eventually.

Proof. There are at most $625BC$ ordered pairs of small-norm factors. Sum the same uniform Markov bound over compatible pairs and use the elementary prime count. $\square$

179.5. Actual stopping examples and formal scope. For $p = 5$, $w = (1, 50)$ and $v = (1, 5)$ have coprime norms 2551 and 31, neither divisible by three. Their products are primitive. Here $d_5 = 1$ and $s_5 = v_5(2550) = 2$. Since $w \equiv (1, 0) \pmod{25}$, $vw^n \equiv (1, 5) \pmod{25}$ and $P(vw^n) \equiv 5 \pmod{25}$ for every n . The tower stops at $t_5 = 1$. With the same w and $v = (2, 1)$ the boundary is constantly $6 \pmod{5}$, giving an empty tower. A full-tower example is $w = (2, 1)$, $v = (3, 1)$: $d_5 = 2$, $s_5 = 1$, and a first hit exists. Exact replay gives depth- e residues

$$(e, D_{5,e}, a_{5,e}) = (1, 2, 1), (2, 10, 7), (3, 50, 17), (4, 250, 67), (5, 1250, 567).$$

The tower theorem supplies all higher depths; the finite replay is not used to prove its infinitude.

The companion `ShiftedResidueCounts.lean` proves a bijective parameterization of actual residue hits before a bound, uniqueness, and the denominator-cleared count discrepancy. It also proves the actual recursive integer-pair stopping orbit's coordinates modulo 25, its boundary's exact depth one, and the step's exact depth two. The finite norm-one group classification, real weighted means, moving-window theorems, factor compatibility, and external analytic estimate used in the fixed-finite-prime proposition remain ordinary mathematics in this section.

179.6. What remains in the signed tail. The new estimates control every prime power in the specified moving range for all but a quantified exceptional set of indices. In the simultaneous residual range $V \leq N^\beta$, the shared parameter is uniformly $\rho = O(\log^2 N/N)$, so the established angular bound and pointwise treatment of primes two and three also apply. The unresolved signed product can therefore be placed beyond the new cutoff Z on these good indices. A uniform $o(\log c_n)$ upper bound for the positive part of its logarithm would complete the usual conditional $1 + \varepsilon$ absorption on those indices.

No such large-prime bound is proved here. A chosen exponent could remain in the exceptional set, and unbounded roots or residuals are not covered by the finite-family union argument. The actual progression count, every endpoint error, and the precise bounded-family quantifiers are retained. These results provide a larger proved range and a smaller residual arithmetic task; they do not establish standard *abc* or a closed Lean term of its type.

180. PAID NORM DIVISORS AND TWO-STEP COMPATIBILITY

Status. This section contains ordinary mathematical proofs, checked by separate research agents, and separately identified exact finite computations. It proves sufficient criteria for an *abc* counterexample family, but does not prove the existence premises of those criteria. Arbitrary depth at one or two primes is not identified with compression of an entire norm. No full Lean verification of this section is asserted.

180.1. From a shared representation to the complete norm radical. Let a, b be coprime positive integers, $c = a + b$, and

$$z = a + b\zeta, \quad \zeta^2 - \zeta + 1 = 0, \quad M = N(z) = a^2 + ab + b^2, \quad t = \log c, \quad r = \log \text{rad}(abc), \quad q = t/r.$$

Suppose there is an actual oriented representation

$$(647) \quad z = u(1 + \zeta)^{e_0}vw^g, \quad e_0 \in \{0, 1\}, \quad g \geq 1, \quad V = N(v) \geq 1, \quad Q = N(w) \geq 7.$$

Set $h_v = \max(1, \log V)$, $\lambda = h_v/g$, $\rho = \lambda \log(4 + g)$, and $s = \log \text{rad}(M)$. Here λ is a positive real compression ratio, not the complex quotient used earlier in the logarithmic-form argument. The norm identity and height comparison give

$$(648) \quad M = 3^{e_0}VQ^g, \quad M \leq c^2, \quad t \geq g \log 7/2.$$

Theorem 180.1 (Complete norm radical bound). *Every representation (647) satisfies*

$$\text{rad}(M) \leq 3^{e_0} VQ, \quad s \leq \frac{\log M}{g} + \left(1 - \frac{1}{g}\right) (\log V + e_0 \log 3) \leq \frac{2t}{g} + \log V + \log 3.$$

Consequently, with

$$K_* = 2 + \frac{2(1 + \log 3)}{\log 7},$$

one has

$$(649) \quad 0 < \frac{s}{t} \leq K_* \lambda = \frac{K_* \rho}{\log(4 + g)}.$$

If $e_0 = 0$, the constant can be replaced by $2 + 2/\log 7$.

Proof. Radicals are invariant under positive powers and submultiplicative, so

$$\text{rad}(3^{e_0} VQ^g) \leq 3^{e_0} \text{rad}(V) \text{rad}(Q) \leq 3^{e_0} VQ.$$

Overlapping support of V, Q only improves this bound. Write $x = \log V + e_0 \log 3$. The identity $\log M = x + g \log Q$ rewrites $s \leq x + \log Q$ as the first logarithmic inequality. Then use $g \geq 1$ and (648). Finally

$$\frac{s}{t} \leq \frac{2}{g} + \frac{2(h_v + \log 3)}{g \log 7} \leq K_* \frac{h_v}{g},$$

because $h_v \geq 1$. Omitting the ramified factor proves the last assertion. $\square$

This theorem controls the norm radical, not $\text{rad}(abc)$. Neither it nor the established balance theorem implies that the input quality exceeds one half.

180.2. Exact quality transport and its quantified conditional criterion. The primitive mixed transform is

$$\mathcal{T}(a, b, c) = (ab, M, c^2).$$

The norm-boundary gcd identity gives $\gcd(M, abc) = 1$, hence its logarithmic height and radical are $t' = 2t$ and $r' = r + s$. Therefore

$$(650) \quad q' = \frac{2q}{1 + q(s/t)}, \quad \frac{1}{q'} = \frac{1}{2q} + \frac{s}{2t}, \quad \frac{1}{2q} \leq \frac{1}{q'} \leq \frac{1}{2q} + \frac{K_*}{2} \lambda.$$

These identities follow by division of $t' = 2t$ by $r' = r + s$. In particular, along a sequence with $\rho \rightarrow 0$, $1/q' - 1/(2q) \rightarrow 0$. Claiming $q'/q \rightarrow 2$ additionally requires $q(s/t) \rightarrow 0$, for example bounded input qualities; it is not asserted without that extra condition.

Theorem 180.2 (Half-quality sufficient criterion). *Suppose a sequence of actual primitive triples admits (647), has $\rho_n \rightarrow 0$, and satisfies $q_n \geq q_0 > 1/2$ eventually for one fixed q_0 . Then, for one fixed $\varepsilon > 0$,*

$$\frac{c_n^2}{(\text{rad}(a_n b_n c_n) \text{rad}(M_n))^{1+\varepsilon}} \rightarrow \infty.$$

Thus such a sequence would disprove standard abc .

Proof. Choose once and for all $0 < \varepsilon < 2q_0 - 1$, and put $d = 2 - (1 + \varepsilon)/q_0 > 0$. The output excess is exactly

$$E'_\varepsilon = \left(2 - \frac{1 + \varepsilon}{q} - (1 + \varepsilon) \frac{s}{t}\right) t.$$

Equation (649) implies $s_n/t_n \rightarrow 0$, so eventually $E'_{\varepsilon,n} \geq dt_n/2$. Also $\rho_n \geq 1/g_n$, hence $g_n \rightarrow \infty$, and (648) gives $t_n \rightarrow \infty$. The fixed-epsilon excess therefore tends to infinity. Exponentiating defeats every uniform constant. The outputs remain primitive by the gcd identity. $\square$

The same statement follows from $\limsup q_n > 1/2$ by passage to a subsequence. No such sequence is constructed here. The previously constructed small- ρ families are proved to be balanced; that does not prove this quality premise. The elementary bound $\text{rad}(abc) \leq abc \leq c^3/4$ supplies only an asymptotic one-third lower bound for quality.

180.3. Fresh divisors over several steps. Iterate the actual transform by

$$(a_{j+1}, b_{j+1}, c_{j+1}) = (a_j b_j, M_j, c_j^2), \quad M_j = a_j^2 + a_j b_j + b_j^2.$$

Every new M_j is coprime to the entire accumulated support. Thus the M_j are pairwise coprime and coprime to the initial boundary, and induction gives

$$(651) \quad t_n = 2^n t_0, \quad R_n = R_0 \prod_{j < n} \text{rad}(M_j), \quad r_n = r_0 + \sum_{j < n} s_j.$$

Writing $d_j = r_j/t_j$ and $\theta_j = s_j/t_j$, the exact recurrence and its iterated form are

$$d_{j+1} = (d_j + \theta_j)/2, \quad d_n = 2^{-n} d_0 + \sum_{j < n} 2^{j-n} \theta_j.$$

Every θ_j is charged from its actual new divisor; none is discarded.

Theorem 180.3 (Two-step sufficient criterion). *If a seed and its first transform both admit (647), with ratios λ_0, λ_1 , then*

$$\frac{r_2}{t_2} \leq \frac{3}{4} + \frac{K_*}{4}(\lambda_0 + 2\lambda_1).$$

If $\lambda_0 + 2\lambda_1 \leq 1/(5K_)$, the second transform has quality at least $5/4$. Unbounded seed heights satisfying this condition would disprove abc at the fixed value $\varepsilon = 1/8$.*

Proof. The elementary estimate $r_0 \leq 3t_0$, the bounds $s_0 \leq K_* \lambda_0 t_0$, $s_1 \leq 2K_* \lambda_1 t_0$, and (651) give $r_2 \leq (3 + K_* \lambda_0 + 2K_* \lambda_1)t_0$. Divide by $t_2 = 4t_0$. Under the stated condition this is $r_2/t_2 \leq 4/5$. Therefore

$$E_{2,1/8} = 4t_0 - \frac{9}{8}r_2 \geq 4t_0 - \frac{9}{8} \frac{16}{5}t_0 = \frac{2}{5}t_0.$$

An unbounded sequence of seed heights has a subsequence on which this fixed-epsilon excess tends to infinity. $\square$

The arithmetic existence premise is open. A sequence with both $\rho_0 \rightarrow 0$ and $\rho_1 \rightarrow 0$ would suffice, but is not supplied by the theorem. The following analysis gives new local compatibility results and precise restrictions on attempts to construct that sequence.

180.4. The second norm and a fixed quartic unit. Put $U = ab$ and retain $M_0 = a^2 + ab + b^2$. The second norm is

$$(652) \quad M_1 = U^2 + UM_0 + M_0^2 = c^4 - Uc^2 + U^2 = F(a, b),$$

where

$$F(X, Y) = X^4 + 3X^3Y + 5X^2Y^2 + 3XY^3 + Y^4.$$

Since $0 < U/M_0 \leq 1/3$, $M_0^2 < M_1 \leq 13M_0^2/9$. Moreover $\gcd(M_1, abcM_0) = 1$, and

$$(653) \quad M_1 \equiv 1 \pmod{3}.$$

For the last assertion, reduce F to $(a^2 + b^2)^2$ modulo three; primitivity excludes the simultaneous zero residue and every remaining squared sum has square one.

Let $K_0 = \mathbb{Q}(\zeta)$ and let α satisfy

$$P(X) = X^2 + (2 - \zeta)X + 1, \quad L = K_0(\alpha).$$

The discriminant $-1 - 3\zeta$ has norm thirteen, so it is not a square in K_0 and $[L : \mathbb{Q}] = 4$. The equality $\zeta = (\alpha + 1)^2/\alpha$ shows $L = \mathbb{Q}(\alpha)$. Multiplying P by its coefficient conjugate gives the irreducible polynomial $F(X, 1)$, with polynomial discriminant 117: the two discriminants have product thirteen and the squared resultant is nine. No maximal-order identification is needed for the following argument.

Lemma 180.4 (Actual unit coefficients). *One has*

$$M_1 = N_{L/\mathbb{Q}}(a - \alpha b), \quad (\alpha + \zeta)(\alpha + \bar{\zeta}) = -\bar{\zeta}\alpha.$$

The element $\kappa = (\alpha + \zeta)/(\alpha + \bar{\zeta})$ is an algebraic unit outside K_0 , is not a root of unity, and satisfies, for $\nu = \kappa/\zeta$,

$$\nu + \nu^{-1} = 1 - 3\zeta, \quad \nu^4 + \nu^3 + 9\nu^2 + \nu + 1 = 0.$$

Finally

$$(654) \quad a - \alpha b = \frac{(\alpha + \zeta)\bar{z} - (\alpha + \bar{\zeta})z}{\zeta - \bar{\zeta}}.$$

Proof. The relative norm is $a^2 + (2 - \zeta)ab + b^2 = \zeta^{-1}(ab + \zeta M_0)$, whose norm to $\mathbb{Q}$ is M_1 . The quadratic relation gives the product identity. As α is an algebraic unit, that identity makes both coefficient factors units. If $\kappa \in K_0$, it cannot equal one and $\alpha = (\zeta - \kappa\bar{\zeta})/(\kappa - 1)$ would belong to K_0 , a contradiction. Directly taking the relative norm and trace of κ gives ζ^2 and $3 - 2\zeta$. Division by ζ and multiplication by the coefficient conjugate give the two assertions about ν . The nonreal value $1 - 3\zeta$ prevents ν from having complex absolute value one, so neither ν nor κ is a root of unity. Solving the two linear equations $z = a + b\zeta$, $\bar{z} = a + b\bar{\zeta}$ proves (654). $\square$

Theorem 180.5 (Second-norm small-prime mass). *Suppose the first profile is the actual pure-power representation*

$$z = u(1 + \zeta)^{e_0}w^g, \quad e_0 \in \{0, 1\}, \quad Q = N(w) \geq 7.$$

There is an effective absolute $D > 0$, independent of w and its support, such that

$$v_p(M_1) \leq Dp^4 \log Q \log(3 + g) \quad (p \mid M_1),$$

and, for $Y \geq 2$,

$$\log \prod_{\substack{p \mid M_1 \\ p \leq Y}} p^{v_p(M_1)} \leq D \log Q \log(3 + g) Y^5 \log Y.$$

For $g \geq 1024$, take $Y = g^{1/10}$. The ratio of this mass to $\log M_1$ is at most $\frac{D}{20} \log(3 + g) \log g / \sqrt{g}$, which tends to zero uniformly in the size and number of norm primes of w .

Proof. Put $\eta = w/\bar{w}$ and $\kappa_{u,e_0} = \kappa u^{-2}\zeta^{-e_0}$. Since $z/\bar{z} = u^2\zeta^{e_0}\eta^g$, equation (654) gives, at every place above $p \mid M_1$,

$$v_P(a - \alpha b) = v_P(\eta^g \kappa_{u,e_0}^{-1} - 1).$$

Every omitted factor is a unit there: $p \nmid 3Q$ by (653) and the norm-boundary gcd identity, the two coefficient factors are global units, and $\zeta - \bar{\zeta}$ is supported only above three. This includes the ramified prime thirteen.

Apply the first inequality of Theorem 1.3 of Bugeaud, *B prime*, with coefficients $g, -1$ in a field of degree at most four. The product is not one since $\eta^g \in K_0$ and $\kappa_{u,e_0} \notin K_0$. One has $h^*(\eta) \leq \log Q$, and the finite root-of-unity twists of κ have fixed height. With normalized valuations $v_P(p) = 1$, the source theorem gives the stated bound for each place, up to an absolute constant. For the rational norm one must sum the places:

$$v_p(M_1) = \sum_{P \mid p} f_P \operatorname{ord}_P(a - \alpha b) = \sum_{P \mid p} e_P f_P v_P(a - \alpha b), \quad \sum_{P \mid p} e_P f_P = 4.$$

Enlarge one absolute constant to obtain the assertion, including ramification. Summation uses $\sum_{p \leq Y} p^4 \log p \leq Y^5 \log Y$. Finally $\log M_1 \geq 2g \log Q$ and substitution of $Y = g^{1/10}$ give the displayed ratio. $\square$

This positive distribution theorem does not bound the second radical from above. It forces any successful second compression to account for depth at moving large primes.

180.5. What fixed first roots force on second compression.

Theorem 180.6 (A fixed-root bound on second pure content). *Fix w in the preceding theorem. Let h be the gcd of the positive prime exponents of M_1 , and choose any $p \mid Q$ with $e = v_p(Q) > 0$. If $C > 2$ is the absolute constant of the established one-block two-place estimate, then*

$$\frac{h}{\log(3+h)} \leq \frac{2C^2 p^2}{e} (\log Q + \log 4).$$

In particular h has an effective upper bound independent of g and of the unit rotation choosing positive coordinates.

Proof. By (653), the oriented factorization of z_1 has no ramified factor. Dividing all its exponents by h gives an actual representation $z_1 = u_1 W^h$. The boundary of the first transformed triple contains M_0 , hence has p -valuation at least ge . The previously established one-block estimate applied to this triple yields, with $t_1 = \log(c^2)$,

$$ge \leq p^2 C^2 \log(3+h) \log N(W) \leq 2C^2 p^2 t_1 \frac{\log(3+h)}{h}.$$

Since $M_0 \geq 3c^2/4$, $t_1 \leq g \log Q + \log 4$. Divide by g and use $g \geq 1$. Effectivity follows because $h/\log(3+h) \rightarrow \infty$. $\square$

The following earlier ρ estimate alone leaves a remainder window. More generally, if $z = u(1+\zeta)^{e_0} v_0 w^{g_0}$ with fixed w , the established shared bound $v_p(T_1) \leq Ap^2 \rho_1 t_1$ for a representation of the first transform implies

$$\rho_1 \geq \frac{e}{Ap^2(\log Q + \lambda_0 + (\log 4)/g_0)}.$$

This follows from $v_p(T_1) \geq g_0 e$ and $t_1 \leq g_0 \log Q + \log N(v_0) + \log 4$. If $\rho_0 \rightarrow 0$, then $\lambda_0 \rightarrow 0$, $g_0 \rightarrow \infty$, and $\liminf \rho_1 \geq e/(Ap^2 \log Q) > 0$. Taken alone, this estimate does not exclude $\lambda_1 = \rho_1/\log(4+g_1) \rightarrow 0$ with a growing remainder. The subsequent refinement in Corollary 181.4 proves a positive λ_1 floor when w is fixed and λ_0 is bounded, closing that asymptotic window. Neither bound has been compared with Theorem 180.3's finite lambda threshold, and roots with unbounded norm remain subject to the separate necessary conditions in Corollary 181.5.

180.6. An exact local obstruction and positive arbitrary-depth lifting.

Theorem 180.7 (The thirteen-adic boundary). *For every primitive integer pair (a, b) , $13 \mid F(a, b)$ if and only if $a \equiv b \pmod{13}$, and in that case $v_{13}(F(a, b)) = 1$.*

Proof. Modulo thirteen, $F(a, b) = (a-b)^2(a^2+5ab+b^2)$. The second quadratic has discriminant eight, outside the square residues 0, 1, 3, 4, 9, 10, 12. If $b = 0$ modulo thirteen, primitivity gives $F = a^4 \neq 0$; otherwise divide by b^4 . This proves the equivalence. For the exact depth, write $a = b + d$:

$$F(b+d, b) = d^4 + 7bd^3 + 20b^2d^2 + 26b^3d + 13b^4.$$

At $d = 13k$, this is $13b^4 \not\equiv 0 \pmod{169}$. $\square$

For the fixed root $w = 2 + \zeta$, exact arithmetic gives $w^{12} = 1 \pmod{13}$ and $w^{10} = 6 + 6\zeta \pmod{13}$. Below, $F(w^g)$ denotes F applied to the two integer coordinates of w^g . Infinitely many exponents $g = 10 + 12n$ have positive raw coordinates: if $\theta = \arg w$, then θ/π is irrational, since a root-of-unity quotient $w/\bar{w}$ would equate its distinct oriented prime ideals above seven. The irrational rotation by 12θ visits the open sector $(0, \pi/3)$ infinitely often. Density follows from the arbitrarily small angles in the closure of an infinite cyclic subgroup of the circle; positive orbits and their tails have that same closure. These actual positive pairs are primitive, because a common rational prime divisor would force a conjugate factor into w^g . Their first norm is w^g , while their second content is exactly one by Theorem 180.7. This refutes automatic pure-content transfer, not a transfer with remainders.

Theorem 180.8 (Arbitrarily deep compatible contact at sixty-seven). *For each $k \geq 1$ there is a unique class $g = g_k \pmod{66 \cdot 67^{k-1}}$ within the initial class $g = 1 \pmod{66}$ for which $67^k \mid F(w^g)$. The classes lift compatibly, with $g_1 = 1$, $g_2 = 199$, $g_3 = 9043$, and $g_4 = 7415893$. For each fixed depth, infinitely many positive primitive pairs of first norm 7^g have this second-norm divisibility.*

Proof. Exact calculations give

$$F(2, 1) = 67, \quad w^{66} \equiv 3954 + 1541\zeta = 1 + 67(59 + 23\zeta) \pmod{67^2}.$$

Put $\Delta = (w^{66} - 1)/67$. Modulo 67, its coordinates are $(59, 23)$ and those of $w\Delta$ are $(28, 61)$. Since $\partial_a F(2, 1) = 91$, $\partial_b F(2, 1) = 86$, the exponent derivative is $91 \cdot 28 + 86 \cdot 61 = 22 \pmod{67}$. The binomial theorem gives

$$(w^{66})^{67^{k-1}} \equiv 1 + 67^k \Delta \pmod{67^{k+1}}.$$

Inductively, all terms after the first order vanish at the required modulus on raising to the 67th power. A Taylor expansion of the quartic therefore gives

$$F(w^{g+66 \cdot 67^{k-1}j}) \equiv F(w^g) + 22j67^k \pmod{67^{k+1}},$$

since $w^g = w \pmod{67}$ in the initial class. Exactly one $j \pmod{67}$ kills the next digit. The first is $j = 3$, yielding $g_2 = 199$. Induction proves existence and uniqueness of the compatible classes. The irrational-rotation argument above gives positive raw coordinates infinitely often in each fixed class, and oriented factorization gives primitivity and first norm 7^g . $\square$

At $g = 199$, multiplication of the negative raw coordinates by minus one gives a positive pair without changing the quartic. Exact division verifies that its 337-digit second norm has valuation exactly two at 67. Its full factorization is not asserted, and this local witness is not described as a compressed second norm.

Theorem 180.9 (Two distinct prime targets are simultaneously compatible). *For any $k, \ell \geq 1$, infinitely many actual positive primitive pairs of first norm 7^g satisfy $67^k 967^\ell \mid M_1$.*

Proof. Use the following exact local data; the derivative is along multiplication by $(w^d - 1)/p$ in the exponent direction:

p	d	g_1	$(w^d - 1)/p \pmod{p}$	$w^{g_1} \pmod{p}$	D_p
67	66	21	(59, 23)	(33, 66)	7
967	966	651	(759, 279)	(53, 172)	39

Both quartic residues vanish, and both derivatives are units. The preceding binomial and Taylor argument gives a class modulo $66 \cdot 67^{k-1}$ and a class modulo $966 \cdot 967^{\ell-1}$. Their representatives remain three modulo six. Since $66 = 2 \cdot 3 \cdot 11$ and $966 = 2 \cdot 3 \cdot 7 \cdot 23$, the two moduli have gcd exactly six for every depth pair. The generalized Chinese remainder theorem supplies a common class, beginning with $g = 7413 \pmod{10626}$ at depth one. Positive raw pairs occur infinitely often in each common class by irrational rotation, and they remain primitive. $\square$

These simultaneous local towers do not satisfy the whole-norm premise merely by construction. Indeed Theorem 180.5 shows that any fixed finite set of primes carries only $o(\log M_1)$ mass as $g \rightarrow \infty$. The unresolved requirement is high depth across enough of the full moving support, with the exact remainder norm below the two-step threshold.

180.7. Exact computation and remaining scope. The standalone integer replay `two_step_compatibility.py` verifies twelve complete second-norm factorizations for the first twelve positive rotations of $(2 + \zeta)^g$. Primality is checked by trial division and each product is reconstructed exactly. All twelve are squarefree, which proves only that those twelve inputs do not meet the second compression threshold. The $g = 199$ witness already refutes extrapolating universal squarefreeness. The same replay checks all relevant thirteen-adic residue cases, eight sixty-seven-adic lifting levels, and thirty-six simultaneous depth pairs between one and six at 67 and 967. Local checks never stand in for a missing complete factorization.

The remaining branches are both active: fixed roots with growing second remainders, and moving roots of unbounded norm. The new quartic unit gives a concrete shifted residue target;

the local towers prove nontrivial compatibility; and the small-prime theorem and fixed-root content bound give precise limitations. None establishes the unbounded family required by Theorem 180.2 or Theorem 180.3.

181. A REMAINDER-PROPORTION REFINEMENT OF THE TWO-PLACE BOUNDS

The bounds in this section depend on $\lambda = \max\{1, \log V\}/g$, instead of requiring $\rho = \lambda \log(4+g)$ to be small. They give unconditional balance and small-prime estimates, a stronger obstruction to fixed-root two-step compression, and two actual separation families. The complementary large-prime tail and the existence of a family meeting the two-step radical gate remain unproved. The transcendence and height arguments here are ordinary proofs; they are not claimed as completed Lean formalizations.

181.1. Actual representations and exact heights. Let a, b be coprime positive integers, $c = a + b$, $T = abc$, $t = \log c$, and $M = a^2 + ab + b^2$. In $\mathbb{Z}[\zeta]$, where $\zeta^2 - \zeta + 1 = 0$, set $\gamma = 1 + \zeta$ and suppose there is an actual factorization

$$(655) \quad z = a + b\zeta = u\gamma^e v w^g, \quad e \in \{0, 1\}, \quad g \geq 1, \quad Q = N(w) \geq 7, \quad V = N(v) \geq 1.$$

Here u is a unit and v, w contain only split prime elements, with at most one of each conjugate pair in their combined support. Shared prime factors are allowed. The ramified factor is kept outside v and w . Write

$$h = \max\{1, \log V\}, \quad \lambda = h/g, \quad \rho = \lambda \log(4 + g).$$

The elementary norm and boundary relations give

$$(656) \quad \gcd(M, T) = 1, \quad \frac{3}{4}c^2 \leq M \leq c^2, \quad M = 3^e V Q^g, \quad 2t \geq g \log Q.$$

Lemma 181.1 (Exact oriented heights). *For an oriented split nonunit x of norm q ,*

$$h_W(x/\bar{x}) = \frac{1}{2} \log q, \quad h^*((x/\bar{x})^3) = \frac{3}{2} \log q,$$

where h_W is the absolute logarithmic Weil height and $h^* = \max\{1, h_W\}$. In particular, for

$$\eta = (w/\bar{w})^3, \quad \xi = (-1)^e (v/\bar{v})^3, \quad R = \xi \eta^g = (z/\bar{z})^3,$$

one has

$$(657) \quad h^*(\eta) = \frac{3}{2} \log Q, \quad h \leq h^*(\xi) \leq \frac{3}{2} h.$$

Moreover $R \neq 1$ and

$$(658) \quad |R - 1| = \frac{3\sqrt{3}T}{M^{3/2}}, \quad v_p(R - 1) = v_p(T) + \frac{3}{2}v_p(3) \geq v_p(T) \quad (p \mid T).$$

The valuations in the second identity are normalized by $v_p(p) = 1$ at any place of the quadratic field above p .

Proof. The ideals (x) and $(\bar{x})$ are coprime, the two complex absolute values of $x/\bar{x}$ are one, and its denominator ideal has norm q . The degree-two normalized height formula therefore gives $(\log q)/2$. Cubing multiplies the height by three, and $q \geq 7$ makes the result larger than one. Multiplication by a root of unity does not change the height. If v is a unit, both h and $h^*(\xi)$ equal one; otherwise they are $\log V$ and $(3/2)\log V$ respectively.

The equalities $u^6 = 1$, $\gamma/\bar{\gamma} = \zeta$, and $\zeta^3 = -1$ prove the formula for R . Direct expansion gives

$$z^3 - \bar{z}^3 = 3(\zeta - \bar{\zeta})T.$$

Its absolute value gives the first formula in (658); it is nonzero since $T > 0$. If $p \mid T$, then $\bar{z}$ is a unit at every place above p , by (656). Taking normalized valuations gives the second formula. This includes $p = 3$ when $e = 0$; when $e = 1$, the condition $\gcd(M, T) = 1$ excludes $3 \mid T$. $\square$

181.2. A uniform theorem covering dependence and units. We use two precise established estimates from Bugeaud, B' , arXiv:2209.00275v1, Theorem 1.1, equation (1.3), and Theorem 1.4. The former bounds a nonzero complex linear form in logarithms from below using the height product times $\log B'$, with

$$B' = \max \left\{ 3, \max_{j < n} \left(\frac{|b_n|}{h^*(\alpha_j)} + \frac{|b_j|}{h^*(\alpha_n)} \right) \right\}, \quad b_n \neq 0.$$

For fixed number of logarithms and field degree its constant is effective and absolute; it requires no multiplicative independence. Theorem 1.4 states that for multiplicatively independent p -units α, β and positive integers k, l ,

$$(659) \quad v_p(\alpha^k - \beta^l) \leq Cp^D h^*(\alpha) h^*(\beta) (\log B')^2, \quad B' = \max \left\{ 3, \frac{k}{h^*(\beta)} + \frac{l}{h^*(\alpha)} \right\},$$

where $D = [\mathbb{Q}(\alpha, \beta) : \mathbb{Q}]$ and C is absolute and effective. We do not discard the alternative in the distinct disjunctive refinement of that source's Theorem 1.3.

Theorem 181.2 (Remainder-proportion two-place estimates). *There are absolute effective constants A_1, A_2 such that every actual representation (655) with $0 < \lambda \leq 1$ satisfies*

$$(660) \quad 3t - \log T \leq A_1 \lambda \log(4/\lambda)t,$$

$$(661) \quad v_p(T) \leq A_2 p^2 \lambda \log^2(4/\lambda)t \quad (p \mid T).$$

The constants are independent of all prime supports, norms, units, ramified exponents, and multiplicative dependence between v and w .

Proof. Take principal logarithms of η and ξ . Choose $k \in \mathbb{Z}$ so

$$\Lambda = g \log \eta - 2k \log(-1) + \log \xi$$

lies in $i[-\pi, \pi]$. Then $|2k| \leq g + 2$. Since $\exp \Lambda = R \neq 1$, the form is nonzero. Apply the complex estimate with three logarithms and put ξ last with coefficient one. The degree is at most two, even when $\xi = \pm 1$. By (657),

$$B' \leq \max\{3, 1 + (g + 2)/h\} \leq 4/\lambda, \quad h^*(\eta)h^*(-1)h^*(\xi) \leq \frac{9}{4}h \log Q.$$

Here $h \geq 1$ and $g \geq h$ give $h + g + 2 \leq 4g$. Hence $\log |\Lambda| \geq -Ch \log Q \log(4/\lambda)$. The elementary chord bound $|\exp \Lambda - 1| \geq 2|\Lambda|/\pi$ now combines with (658) and $M \geq 3c^2/4$ to give

$$3t - \log T \leq Ch \log Q \log(4/\lambda) + C'.$$

Since $2t \geq g \log Q$ and $\lambda t \geq (\log 7)/2$, the absolute constant and the height term are absorbed into (660). This argument includes unit v and needs no independence.

For the non-Archimedean estimate put $\alpha = \eta$ and $\beta = \xi^{-1} = (-1)^e(\bar{v}/v)^3$. First suppose that they are multiplicatively independent. Then v is a nonunit. They are p -units at every $p \mid T$, and $\alpha^g - \beta$ differs from $R - 1$ by a p -unit. Equation (659) applies with positive coefficients $g, 1$ and degree at most two. Moreover

$$B' \leq \max\{3, g/h + 1\} \leq 3/\lambda, \quad \frac{h^*(\alpha)h^*(\beta)}{t} \leq \frac{9}{2}\lambda.$$

Together with (658), this proves (661) in this branch.

It remains to cover multiplicative dependence, including unit v . Choose oriented prime elements on the union of the supports and let f_i, r_i be their exponents in w, v . A relation $\alpha^m \beta^n = 1$ implies $mf_i = nr_i$ at each corresponding prime ideal. If r is the zero vector, write $f = sd$ with d primitive and set $\ell = 0$. Otherwise a nontrivial relation forces m, n to be nonzero with the same sign and the vectors to be proportional. Since $\gcd(d_i) = 1$, the proportionality factor in $r = \ell d$ is an integer. Thus in all cases $s \geq 1, \ell \geq 0$, and actual unique factorization gives, for $x = \prod_i \pi_i^{d_i}$,

$$w = \text{unit} \cdot x^s, \quad v = \text{unit} \cdot x^\ell, \quad z = \text{unit} \cdot \gamma^e x^H, \quad H = sg + \ell \geq g.$$

The exponent H concerns split primes; the ramified factor is still γ^e . To bound this branch directly, apply the first inequality of Bugeaud Theorem 1.3 to the two algebraic numbers $((x/\bar{x})^3, -1)$ with integer coefficients (H, e) . Their product is $R \neq 1$. That inequality allows $e = 0$ and needs no independence. The degree is at most two, the height product is $(3/2) \log N(x)$, and its exponent parameter is at most $3 + H$. Consequently

$$v_p(T) \leq C_0 p^2 \log N(x) \log(3 + H), \quad \frac{v_p(T)}{t} \leq 2C_0 p^2 \frac{\log(3 + H)}{H}.$$

The latter uses $2t \geq H \log N(x)$. The function $x \mapsto \log(3 + x)/x$ is decreasing for $x > 0$. Since $H \geq g \geq 1/\lambda$,

$$\frac{\log(3 + H)}{H} \leq \lambda \log(3 + 1/\lambda) \leq \lambda \log(4/\lambda).$$

Finally $\log(4/\lambda) > 1$ absorbs this into (661). The two branches exhaust all cases. $\square$

Corollary 181.3 (Balance and full small-prime mass). *Consider an actual representation (655) with $0 < \lambda \leq 1$. For every $\delta > 0$, sufficiently small effective λ in (655) implies $\min(a, b) > c^{1-\delta}$. For every $Y \geq 2$,*

$$(662) \quad \sum_{\substack{p \leq Y \\ p|T}} v_p(T) \log p \leq A_2 \lambda \log^2(4/\lambda) Y^3 \log Y t.$$

For $\lambda \leq 1/64$ and $Y = \lambda^{-1/6}$, the ratio of this mass to t is at most

$$\frac{A_2}{6} \sqrt{\lambda} \log^2(4/\lambda) \log(1/\lambda) \rightarrow 0.$$

Proof. If $\min(a, b) \leq c^{1-\delta}$, then $T \leq c^{3-\delta}$, which contradicts (660) once $A_1 \lambda \log(4/\lambda) < \delta$. Summing (661) and bounding $\sum_{p \leq Y} p^2 \log p$ by $Y^3 \log Y$ proves the remaining assertions. $\square$

These are unconditional balance and small-prime conclusions, not an ABC quality bound above one half. The complementary large-prime multiplicity excess is still unbounded by these arguments.

181.3. Second remainder floors and necessary escape of prime support.

Corollary 181.4 (Fixed first roots force a second remainder floor). *Fix a nonunit oriented w_0 , $Q_0 = N(w_0)$, and a finite constant L_0 . For every first representation (655) with this w_0 and $\lambda_0 \leq L_0$, every actual two-column representation of (ab, M_0, c^2) has λ_1 bounded below by a positive effective constant depending only on w_0, L_0 .*

Proof. Choose $p \mid Q_0$ and put $d = v_p(Q_0) > 0$. Writing $T_1 = abM_0c^2$ and $t_1 = \log(c^2)$, the inherited prime support and (656) give

$$v_p(T_1) \geq g_0 d, \quad t_1 \leq g_0 \log Q_0 + \log V_0 + \log 4.$$

Hence

$$\frac{v_p(T_1)}{t_1} \geq \frac{d}{\log Q_0 + L_0 + \log 4} =: \delta_0 > 0.$$

If $\lambda_1 \leq 1$, Theorem 181.2 gives $\delta_0 \leq A_2 p^2 \lambda_1 \log^2(4/\lambda_1)$. The right side tends to zero with λ_1 , yielding an effective positive floor; $\lambda_1 > 1$ is harmless. All estimates hold for every choice of the second representation. $\square$

This closes the asymptotic fixed-root remainder window left by the earlier ρ_1 estimate. Its constant has not been compared with the finite threshold in Theorem 180.3. Thus that finite gate is not excluded, and moving roots need a separate analysis.

Corollary 181.5 (Necessary escape of moving first-root support). *For any individual first representation, set*

$$J_0 = \max_{p|Q_0} \frac{v_p(Q_0)}{p^2(\log Q_0 + \lambda_0 + (\log 4)/g_0)}.$$

Every second representation with $\lambda_1 \leq 1$ satisfies

$$(663) \quad \lambda_1 \log^2(4/\lambda_1) \geq J_0/A_2.$$

Under the same assumption $\lambda_1 \leq 1$, if $S_Y(Q_0) = \sum_{p \leq Y, p|Q_0} v_p(Q_0) \log p$, then

$$(664) \quad \frac{S_Y(Q_0)}{\log Q_0} \leq A_2 \lambda_1 \log^2(4/\lambda_1) Y^3 \log Y \left(1 + \frac{\lambda_0 + (\log 4)/g_0}{\log Q_0} \right) \quad (Y \geq 2).$$

Consequently, along any family with bounded λ_0 and $\lambda_1 \rightarrow 0$, one necessarily has

$$\frac{S_{\lambda_1^{-1/6}}(Q_0)}{\log Q_0} \rightarrow 0.$$

Proof. Retain the individual λ_0, g_0 in the preceding proof and divide (661) by g_0 . This proves (663); multiplying the resulting primewise inequalities by $\log p$ and summing proves (664). If $\lambda_0 \leq L_0$, its parenthesis is at most $1 + (L_0 + \log 4)/\log 7$. Substitute $Y = \lambda_1^{-1/6}$ as in Corollary 181.3. $\square$

These necessary conditions are computed from the first representation. They still allow sufficiently large moving primes; they do not presuppose any depth or radical bound on the second norm.

181.4. A strict separation from all old disjoint penalties.

Theorem 181.6 (A family with small λ and divergent ρ). *There are actual primitive positive triples and specified two-column representations for which $\lambda \rightarrow 0$, $\rho \rightarrow \infty$, while every old disjoint-partition penalty is at least $\kappa \log M$ for an absolute $\kappa > 0$. Nevertheless both normalized bounds of Theorem 181.2 tend to zero.*

Proof. Choose $r \geq 3$ distinct primes $q_i \equiv 1 \pmod{3}$ in $[X, 2X]$, $X \geq 7$, and oriented elements $N(\pi_i) = q_i$. Put

$$L = \log X, \quad j = \lceil \sqrt{\log r} \rceil, \quad g = r^2 \lceil L \rceil j, \quad e_i = g + i, \quad z = \prod_{i=1}^r \pi_i^{e_i}.$$

Its support is oriented, so its coordinates are primitive. A unit rotation makes both coordinates strictly positive: a sector boundary would make z a unit times a rational integer, contradicting its nonempty oriented support. The consecutive exponents have gcd one. Use the actual shared representation $w = \prod_i \pi_i$, $v = \prod_i \pi_i^i$. Since $L \leq \log q_i \leq 2L$, $\lceil L \rceil \leq 2L$, and $\sqrt{\log r} \leq j \leq 2\sqrt{\log r}$,

$$\frac{r(r+1)L}{2} \leq h = \log V \leq 2r^2L, \quad r^2L\sqrt{\log r} \leq g \leq 4r^2L\sqrt{\log r}.$$

It follows that, uniformly in $L \geq \log 7$,

$$(665) \quad \frac{1}{8\sqrt{\log r}} \leq \lambda \leq \frac{2}{\sqrt{\log r}}, \quad \rho \geq \frac{1}{4}\sqrt{\log r} \rightarrow \infty.$$

The last inequality uses $\log(4+g) \geq 2 \log r$. Thus $\lambda \log(4/\lambda)$ and $\lambda \log^2(4/\lambda)$ tend to zero, and the new bounds apply.

Write $S = \log M$. An old partition into m blocks has penalty

$$B_{\mathcal{P}} = C^{m+1} \log(3 + \sum_j k_j) \prod_j L_j,$$

where $C > 2$, k_j is the gcd of its block's exponents, and L_j is its reduced logarithmic norm. Here

$$grL \leq S \leq 4grL \leq 16r^3L^2\sqrt{\log r}, \quad L_j \geq L.$$

For a block of size $h_j \geq 2$, the index differences imply $k_j \leq (r-1)/(h_j-1)$, hence $L_j \geq gh_j(h_j-1)L/r$. If $m \leq r/2$, a largest block has size at least r/m , so its reduced logarithmic norm is at least $grL/(2m^2)$. Therefore

$$\frac{B_{\mathcal{P}}}{S} \geq \frac{C^{m+1} \log 4 L^{m-1}}{8m^2} \geq \frac{\log 4}{8},$$

using $L > 1$ and $2^{m+1} \geq m^2$ for every integer $m \geq 1$. If $m > r/2$, use $L_j \geq L$ for every block to obtain

$$\frac{B_{\mathcal{P}}}{S} \geq \frac{C^{m+1} \log 4 L^{m-2}}{16r^3 \sqrt{\log r}} \geq \frac{\log 4}{8(\log 7)^2} \frac{(2 \log 7)^{r/2}}{r^3 \sqrt{\log r}}$$

for sufficiently large r . The last expression tends to infinity uniformly in X . Thus $\kappa = \log 4/8$ works eventually for all partitions, including those with growing block count.

The established prime number theorem in the fixed progression $1 \pmod{3}$ supplies dyadic lists with $r \rightarrow \infty$. As in the preceding shared-generator separation theorem, a quantitative primary source is Bennett–Martin–O’Bryant–Rechnitzer, arXiv:1802.00085v3, Theorem 1.2. This input supplies prime availability only; the representation and penalty comparisons above are explicit. $\square$

This compares the specified old ρ sufficient test and all old disjoint penalties with the new estimates. It does not assert that every alternative shared representation has large ρ , nor that these triples have high ABC quality.

181.5. An explicit fixed-root family without factorization.

Theorem 181.7 (An explicit parameter separation). *For $k \geq 1$, define*

$$g_k = 2^{k^2}, \quad m_k = \lfloor g_k/k \rfloor, \quad y_k = 42 + 84 \cdot 2^{m_k}, \quad v_k = 1 + y_k \zeta, \quad w = 2 + \zeta, \quad z_k = v_k w^{g_k}.$$

These are primitive elements which rotate to positive triples. Their split exponent content is one, and the specified representation satisfies

$$\lambda_k \sim \frac{2 \log 2}{k} \longrightarrow 0, \quad \rho_k \sim 2(\log 2)^2 k \longrightarrow \infty.$$

No prime availability or complete factorization is required for these assertions.

Proof. The coordinates $1, y_k$ are primitive, and y_k is divisible by $3, 7$. Thus $V_k = 1 + y_k + y_k^2$ is coprime to 21 , and v_k has oriented split support disjoint from both factors above seven. The product with w^{g_k} remains primitive and oriented and rotates to strictly positive coordinates. Since $y_k \equiv 2 \pmod{4}$, one has $V_k \equiv 3 \pmod{4}$, so some prime factor of V_k has odd valuation. The valuation at seven of $N(z_k)$ is exactly g_k , a power of two. Hence the gcd of all split exponents is one.

Finally $y_k/2^{m_k} = 84 + 42/2^{m_k}$ stays in a fixed positive interval, as does V_k/y_k^2 . It follows that

$$\log V_k = 2m_k \log 2 + O(1), \quad m_k = g_k/k + O(1),$$

with absolute constants. Dividing by g_k and multiplying in turn by $\log(4 + g_k) = k^2 \log 2 + o(1)$ gives the stated asymptotics. $\square$

The preceding explicit family makes parameter separation independent of prime selection, while Theorem 181.6 gives the stronger all-disjoint-penalties comparison. Because the first root w here is fixed and λ_k is bounded, Corollary 181.4 also gives a positive floor for every second representation’s λ_1 on this family. This is compatible with its unconditional balance and small-prime conclusions; neither is a claim of two-step whole-norm compression.

182. ACTUAL RADICALS AND A CONDITIONAL ABC OBSTRUCTION

The preceding sufficient conditions involve actual radical arithmetic. The following elementary bridge is also checked against the standard logarithmic *abc* definition in the repository. Its family-existence premise remains open.

For coprime positive integers a, b , put $c = a + b$ and

$$M = a^2 + ab + b^2, \quad N = (ab)^2 + abM + M^2.$$

The two transformed triples are (ab, M, c^2) and (abM, N, c^4) . The next identities charge the new prime supports exactly; no upper bound for them is assumed in the identities.

Proposition 182.1 (Exact two-step radical transport). *Both displayed triples are primitive, and*

$$\begin{aligned}\operatorname{rad}(abMc^2) &= \operatorname{rad}(abc) \operatorname{rad}(M), \\ \operatorname{rad}(abMNC^4) &= \operatorname{rad}(abc) \operatorname{rad}(M) \operatorname{rad}(N).\end{aligned}$$

For any positive integers d, V, Q, g , with no disjoint-support requirement,

$$\operatorname{rad}(dVQ^g) \leq dVQ.$$

Proof. Modulo a and c , the norm M reduces to b^2 , and modulo b it reduces to a^2 . Hence $\gcd(M, abc) = 1$. The identity $ab + M = c^2$ shows that the first transform is primitive. Apply the same argument to the coprime pair ab, M ; its norm N is coprime to $abMc^2$ and therefore to $abcM$. Radical multiplicativity for coprime factors and invariance under positive powers give both equalities. For the last inequality use $\operatorname{rad}(xy) \mid \operatorname{rad}(x) \operatorname{rad}(y)$, $\operatorname{rad}(Q^g) = \operatorname{rad}(Q)$, and $\operatorname{rad}(x) \leq x$ for positive x . $\square$

Theorem 182.2 (An integer sufficient condition for a disproof). *If coprime positive seeds (a, b) satisfying*

$$(666) \quad (\operatorname{rad}(M) \operatorname{rad}(N))^5 \leq c$$

have unbounded $\log c$, then standard abc is false.

Proof. Write $R_2 = \operatorname{rad}(abMNC^4)$. Since $\operatorname{rad}(abc) \leq abc \leq c^3$, the preceding proposition gives

$$R_2^5 \leq c^{15} (\operatorname{rad}(M) \operatorname{rad}(N))^5 \leq c^{16}.$$

Standard abc with $\varepsilon = 1/8$, applied to the actual primitive triple (abM, N, c^4) , would provide one constant C such that

$$4 \log c \leq \frac{9}{8} \log R_2 + C \leq \frac{18}{5} \log c + C.$$

Thus $(2/5) \log c \leq C$, contradicting the stated unboundedness. $\square$

Formal contract and open premise. The companion modules `RadicalTransport` and `TwoStepABCObstruction` use Mathlib's radical and identify it with the repository's product of distinct prime divisors. The last formal statement has conclusion `not ABCConjecture`, conditional on the explicit unbounded family in Theorem 182.2. The source definition of `ABCConjecture` is imported without alteration. Neither the ordinary proof nor the formal implication constructs that family. In particular, the implication is not a formal disproof of abc . The new verification record states the checked source hashes, compiler, Mathlib revision, and axiom audit separately from the earlier algebra-only checkpoint.

183. PRIVATE DEPTH AND INDEPENDENT ELLIPTIC GEOMETRY

This continuation retains two distinct approaches to the remaining arithmetic gates. Exact residual collisions eliminate the common large factor from a boundary gcd and yield a pointwise bound for all depth that is repeated at another residual. A joint finite residue object then gives an exact large-prime excess law and a first-minimum distribution estimate. Its full-state entropy transfer has a proved cardinality obstruction. Passing to the exact excess observable removes that irrelevant loss, but the resulting entropy is comparable to the positive excess itself; an independent arithmetic estimate is still required.

The geometric direction gives an infinite primitive positive family whose second norm is a square, together with an explicit elliptic parametrization. A different two-isogeny descent proves that the product of the first two norms is neither a rational square nor three times a rational square. Thus simultaneous even extraction with square residual norms is impossible, including moving roots. Odd exponents and nonsquare residuals remain available. A dynamic family of square-root covers has explicitly computed genus and gives a further fixed-square-class finiteness theorem; it supplies no uniform height estimate over moving twists.

The ordinary arguments were developed and independently reviewed before formalization. Twenty-seven new Lean declarations verify exact residual cancellation, annotated determinant inequalities, the elementary local obstructions used in the descent, and actual polynomial identities. They do not constitute formal verification of the full elliptic descent or the probability arguments. Neither the forward signed-tail gate nor an unbounded family satisfying the earlier radical disproof condition is proved in this continuation. Standard abc remains unproved and undisproved in this manuscript.

184. RESIDUAL DIVISOR COLLISIONS AND PRIVATE DEPTH

The results in this section use exact integer identities and finite lattice geometry. They control prime-power depth repeated across small residuals; the maximal depth belonging to one residual remains a separate problem. No logarithmic-form estimate or hypothesis that large primes are simple is used. Write $\mathcal{O} = \mathbb{Z}[\zeta]$, $\zeta^2 - \zeta + 1 = 0$, $P(a + b\zeta) = ab(a + b)$ and $N(a + b\zeta) = a^2 + ab + b^2$.

184.1. Exact cancellation of a common large factor. Recall the established primitive boundary identities

$$(667) \quad \gcd(N(x), |P(x)|) = 1, \quad \gcd(|P(x)|, |P(yx)|) = \gcd(|P(x)|, |P(y)|),$$

where x is primitive and y is arbitrary. For completeness, if $x = (a, b)$ and $y = (r, s)$, the transformed boundary is congruent to $-P(y)b^3$, $P(y)a^3$, and $P(y)b^3$ modulo a , b , and $a + b$, respectively. These three boundary factors are pairwise coprime, so the unit factors can be cancelled in their gcds. Multiplication of the three gcd equalities proves the second identity. Reducing the norm modulo each boundary factor proves the first.

Theorem 184.1 (Exact cross-residual overlap). *Let $H, v_1, v_2 \in \mathcal{O}$, suppose $z_i = v_i H$ are primitive, and put $A_i = |P(z_i)| > 0$, $V_i = N(v_i)$. Then*

$$\gcd(A_1, A_2) = \gcd(A_1, |P(v_2 \bar{v}_1)|).$$

This equality retains all prime-power multiplicities, including those at two and three.

Proof. In (667) take $x = z_1$, $y = v_2 \bar{v}_1$. Then $yx = V_1 z_2$ and $|P(yx)| = V_1^3 A_2$. Since $N(z_1) = V_1 N(H)$, the first identity in (667) gives $\gcd(A_1, V_1) = 1$. Cancel V_1^3 in the left gcd. $\square$

Lemma 184.2 (Small residual overlap). *If the residuals in Theorem 184.1 are not associates under the six units, then $P(v_2 \bar{v}_1) \neq 0$ and*

$$\gcd(A_1, A_2) \leq (V_1 V_2)^{3/2}.$$

Proof. Each v_i is primitive because $v_i H$ is primitive. A nonzero Eisenstein integer has zero boundary exactly when it is an integer multiple of a unit. Thus zero cross-boundary would imply $v_2 = (m/V_1)uv_1$ for an integer m and a unit u . A rational scalar between primitive integer coordinate pairs is 1 or -1 , by cancelling its denominator and then its numerator. This would make the residuals associates. Finally the exact cubic identity gives

$$3\sqrt{3} |P(D)| = |D^3 - \bar{D}^3| \leq 2N(D)^{3/2}.$$

Apply this to $D = v_2 \bar{v}_1$, of norm $V_1 V_2$, and use Theorem 184.1. $\square$

184.2. A total budget and exact signed compensation. Let $v_1, \dots, v_K$ be pairwise nonassociate residuals, $K \geq 1$, with all $z_i = v_i H$ primitive and $A_i = |P(z_i)| > 0$. Set $V_i = N(v_i)$ and $L = \text{lcm}(A_1, \dots, A_K)$.

Theorem 184.3 (Repeated-multiplicity budget). *The integer $D = \prod_i A_i / L$ satisfies*

$$D \mid \prod_{i < j} \gcd(A_i, A_j), \quad \log D \leq \frac{3}{2}(K-1) \sum_i \log V_i.$$

In particular $V_i \leq B$ implies $\log D \leq \frac{3}{2}K(K-1) \log B$.

Proof. At a fixed prime, arrange its valuations increasingly as $e_1 \leq \dots \leq e_K$. The valuation in D is $\sum_{i < K} e_i$, while that in the product of pairwise gcds is $\sum_{i < K} (K - i)e_i$. This proves the divisibility. Bound each gcd by Lemma 184.2; each $\log V_i$ occurs in $K - 1$ pairs. $\square$

For $Y \geq 1$ put $W_Y(A) = \prod_{p|A, p > Y} p^{v_p(A)-3}$. Assign each prime dividing L to an index of maximal valuation, breaking ties by the smallest index. This index is its *owner*; ownership is defined from actual integer valuations.

Theorem 184.4 (Signed compensation). *Define*

$$D_Y = \prod_{p > Y} p^{\sum_i v_p(A_i) - v_p(L)}, \quad E_Y = \prod_{p|L, p > Y} p^{\#\{i: p|A_i\} - 1}.$$

Then $D_Y \mid D$ and

$$\prod_i W_Y(A_i) = W_Y(L) \frac{D_Y}{E_Y^3}.$$

For each index define

$$U_i(Y) = \prod_{\substack{p > Y \\ i \text{ owns } p}} p^{v_p(A_i) - 3}, \quad R_i(Y) = \sum_{\substack{p > Y, p|A_i \\ i \text{ does not own } p}} v_p(A_i) \log p.$$

Then

$$\log W_Y(A_i) \leq \log U_i(Y) + R_i(Y), \quad \sum_i R_i(Y) = \log D_Y.$$

Proof. At a prime occurring in s of the A_i , the signed exponent identity is

$$\sum_i e_i - 3s = (\max_i e_i - 3) + \left(\sum_i e_i - \max_i e_i\right) - 3(s - 1).$$

This proves the product formula. The individual inequality drops only the negative -3 credits at nonowned primes, and their full multiplicities sum to the exponent of D_Y . $\square$

Suppose additionally $H = w^g$, $Q = N(w) \geq 7$, and $V_i \leq B$. After rotating each product to the positive sector, let $t_i = \log c_i$; then $t_i \geq g \log Q/2$. Consequently

$$(668) \quad \frac{1}{K} \sum_i \frac{R_i(Y)}{t_i} \leq \delta := \frac{3(K-1) \log B}{g \log Q}.$$

Apply Markov to the nonnegative quantities $R_i(1)/t_i$. If $\delta > 0$, outside at most a $\sqrt{\delta}$ proportion of indices, every $Y \geq 1$ satisfies $R_i(Y)/t_i \leq \sqrt{\delta}$. If $\delta = 0$, every contribution is zero. Thus $K \log B = o(g \log Q)$ gives a simultaneous sublinear repeated-depth cost for most residuals. No Markov inequality is applied to a signed aggregate: negative credit could conceal positive individual signed costs.

Corollary 184.5 (A specified packet). *If $V_i \leq B$ and an integer $q > B^3$ divides two of the A_i , those residuals are associates. In a pairwise nonassociate family, at most one member can therefore contain a specified such packet.*

Proof. Otherwise $q \leq \gcd(A_i, A_j) \leq (V_i V_j)^{3/2} \leq B^3$. $\square$

184.3. Annotated congruences and a lattice of exact index. Let $q = \prod p^{e_p}$ be coprime to $N(H)$. At each prime choose one of the three boundary linear forms $\ell_{\sigma(p)} \in \{a, b, a + b\}$ and define

$$\mathcal{L}(q, \sigma; H) = \{v \in \mathbb{Z}^2 : \ell_{\sigma(p)}(vH) \equiv 0 \pmod{p^{e_p}} \text{ for all } p \mid q\}.$$

Theorem 184.6 (Annotated lattice). *This lattice has index q . Any two rationally independent v, v' in it satisfy*

$$N(v)N(v') \geq \frac{3}{4}q^2.$$

If $q > 2B/\sqrt{3}$, all primitive vectors in it of norm at most B belong to at most one opposite pair.

Proof. For $H = r + s\zeta$, multiplication has matrix

$$\begin{pmatrix} r & -s \\ s & r + s \end{pmatrix}$$

of determinant $N(H)$, invertible modulo each retained prime power. Each chosen row is unimodular modulo that prime, so its map onto $\mathbb{Z}/p^e\mathbb{Z}$ is surjective. The Chinese remainder theorem on the two coordinates makes the combined map surjective, proving the index. The same row vanishes on both vectors modulo each prime power, so $q \mid \det(v, v')$. Write $v = (a, b)$, $v' = (a', b')$. The exact identity

$$4N(v)N(v') - (2aa' + ab' + ba' + 2bb')^2 = 3(ab' - ba')^2$$

gives the norm-product bound. Primitive integer vectors on one rational line differ only by sign. $\square$

Corollary 184.7 (Fixed-packet capacity). *If $\gcd(q, N(H)) = 1$ and $q > 2B/\sqrt{3}$, at most $2 \cdot 3^{\omega(q)}$ primitive integer pairs of norm at most B satisfy $q \mid P(vH)$. For $B \geq 1$ this is at most $3^{\omega(q)-1}$ associate classes.*

Proof. At each prime, primitive coordinates and invertible multiplication imply that exactly one boundary arm contains the full prime power. There are at most $3^{\omega(q)}$ annotations, each with at most one opposite pair by Theorem 184.6. The raw solution set is freely closed under the six units, which preserve norm and boundary divisibility. Divide its bound by six. The threshold implies $q > 1$ when $B \geq 1$. $\square$

The packet in this corollary is specified. A sum over packets chosen after seeing the residual needs an additional bound on that collection. Likewise, the lattice theorem constrains the product of two independent solution heights; it does not lower-bound the first minimum. A lattice may have one very short primitive vector and a very long independent vector.

184.4. A pointwise repeated-depth bound.

Theorem 184.8 (Linear prime-power collision). *For nonassociate residuals in Theorem 184.1, any common prime satisfies*

$$p^{\min(v_p(A_1), v_p(A_2))} \leq \frac{2}{\sqrt{3}} \sqrt{V_1 V_2}.$$

Proof. A common boundary prime divides neither residual norm. Thus the cross element $D = v_2 \bar{v}_1$ has unit norm modulo that prime, and at most one of its three boundary factors is divisible by it. The whole common prime power in Theorem 184.1 lies in that one nonzero factor. Each factor has absolute value at most $2\sqrt{N(D)}/\sqrt{3}$, by

$$4N(a, b) = 3a^2 + (a + 2b)^2 = 3b^2 + (2a + b)^2 = 3(a + b)^2 + (a - b)^2.$$

Use $N(D) = V_1 V_2$. $\square$

Theorem 184.9 (Pointwise nonowner budget). *When $V_i \leq B$, every index and every $Y \geq 1$ satisfy*

$$R_i(Y) \leq \log \text{lcm}(1, \dots, \lfloor 2B/\sqrt{3} \rfloor) < 4B.$$

For $H = w^g$ as above, $R_i(Y)/t_i \leq 8B/(g \log Q)$. Hence $B = o(g \log Q)$ gives sublinear nonowner cost at every residual, with no exceptional indices.

Proof. For a nonowned prime the owner has at least the same depth, so Theorem 184.8 gives $p^{v_p(A_i)} \leq X = 2B/\sqrt{3}$. Their product divides $\text{lcm}(1, \dots, \lfloor X \rfloor)$, whose logarithm is $\psi(X) = \sum_{p^e \leq X} \log p$. Every prime power in $(m, 2m]$ contributes to the corresponding valuation of $\binom{2m}{m}$, with nonnegative contributions at other levels. Thus $\psi(2m) - \psi(m) \leq 2m \log 2$. Dyadic summation yields $\psi(X) < 4X \log 2 \leq 3X$, where $\log 2 \leq 3/4$ follows from the trapezoidal upper bound for $\int_1^2 dx/x$. Therefore $\psi(X) \leq 2\sqrt{3}B < 4B$. The height lower bound proves the final assertion. $\square$

184.5. An exact limit of the collision argument. Take $w = 2 + \zeta$, $v_1 = 1$, $v_2 = 1 + 5\zeta$ and $g_k = 2 \cdot 5^k$ for $k \geq 1$. The products are primitive since the residual norm 31 is disjoint from $N(w) = 7$. Their displayed lambda parameters tend to zero. The elementary homogeneous rank law at five has rank two and first depth one: $w^2 = 3 + 5\zeta$ has boundary 120. Hence

$$v_5(|P(w^{g_k})|) = k + 1.$$

But $P(v_2) = 30$, so Theorem 184.1 forces $v_5(|P(v_2 w^{g_k})|) = 1$. The maximal depth is unbounded while its repeated depth is exactly one. This disproves only the attempted inference that a collision budget bounds maximal depth. The fixed prime five is eventually below the moving lambda cutoff, so the family does not refute that tail gate.

The exact signed formulas, capacity bound and pointwise nonowner theorem leave the owner packets $U_i(Y)$ unbounded. For primes greater than $2B/\sqrt{3}$, all supported depth is already private to one associate class. The new arithmetic task is to constrain that private first short direction using the special common factor w^g , without replacing a specified packet by an uncontrolled union or assuming prime simplicity. The companion finite residue measure section develops one such reference object and states its transfer gap explicitly. The results here do not establish standard *abc*.

185. A JOINT FINITE RESIDUE OBJECT FOR PRIVATE DEPTH

This section proves a large-prime positive-excess law and a first-minimum distribution theorem for explicitly defined finite reference measures. These measures are not asserted to describe small-height integer residuals or a specified power orbit. The entropy transfer cost, and precise obstacles to two proposed shortcuts, are retained in the statements.

185.1. Exact local counts and a common multiplication action. Let S be a finite set of primes greater than a real $Y \geq 5$, choose integers $E_p \geq 1$, and set $m = \prod_{p \in S} p^{E_p}$. Put uniform product measure μ on

$$\Omega = \prod_{p \in S} (\mathcal{O}/p^{E_p} \mathcal{O})^\times, \quad \mathcal{O} = \mathbb{Z}[\zeta], \quad \zeta^2 - \zeta + 1 = 0.$$

Fix $H \in \mathcal{O}$ with $\gcd(N(H), m) = 1$. For a finite residual state define $\nu_p(vH)$ as the largest $e \in \{0, \dots, E_p\}$ for which $P(vH) \equiv 0 \pmod{p^e}$. This is a *capped* valuation, determined by the finite state; it is not the uncapped valuation of a chosen integer lift. Multiplication by the same H permutes the product space. In particular, all actions $v \mapsto vw^g$ preserve this measure when w is a unit at the retained primes.

Theorem 185.1 (Exact joint local law). *Write $\chi_p = 1$ if $p \equiv 1 \pmod{3}$ and $\chi_p = -1$ otherwise. For $1 \leq e \leq E_p$,*

$$\Pr_\mu(\nu_p(vH) \geq e) = \frac{3}{p^{e-1}(p - \chi_p)}.$$

The depth variables are independent across primes and their joint law is independent of H .

Proof. The unit count modulo p^e is $p^{2e-2}(p-1)(p-\chi_p)$: modulo p the algebra is split or quadratic inert, and each further pair of digits multiplies the count by p^2 . Each boundary arm $a = 0$, $b = 0$, or $a + b = 0$ has exactly $\varphi(p^e)$ unit solutions. The three sets are disjoint on units. If the boundary product vanishes modulo p^e , only one factor can be divisible by p , so that factor contains the entire depth. Divide the count $3\varphi(p^e)$ by the unit count. Projection from E_p to e is uniform, multiplication by H is a permutation, and product measure gives independence. $\square$

Define the retained nonnegative large-prime excess

$$F_Y(v; H) = \sum_{p \in S} (\nu_p(vH) - 3)_+ \log p.$$

Theorem 185.2 (Whole retained positive tail). *Uniformly in the precision, number of primes and common factor,*

$$\mathbb{E}_\mu F_Y \leq \sum_{p \in S} \frac{3 \log p}{p^2(p-1)(p-\chi_p)} \leq \frac{10 \log Y}{Y^3}.$$

Proof. Use $(n-3)_+ = \sum_{e \geq 4} \mathbf{1}_{n \geq e}$, then Theorem 185.1 and the geometric series. For $p \geq 5$ the summand is at most $5 \log p/p^4$. The decreasing function $\log x/x^4$ has integral $\log a/(3a^3)+1/(9a^3)$ on $[a, \infty)$. Compare the integer sum with the integral from $\lfloor Y \rfloor \geq 4Y/5$ and enlarge the constant to ten. $\square$

This controls primes *above* the cutoff in the stated reference measure. It follows from exact counts, without assuming that large primes usually have valuation one.

185.2. A reference first-minimum theorem. Now let $q = \prod p^{e_p}$ have primes at least five, and sample H uniformly among the units of $\mathcal{O}/q\mathcal{O}$. At each prime specify one arm $\sigma(p)$, and form the actual lattice $\mathcal{L}(q, \sigma; H)$ from Theorem 184.6.

Theorem 185.3 (First-minimum distribution). *For $B \geq 1$,*

$$\begin{aligned} \Pr_H(\exists \text{ primitive } v : N(v) \leq B, v \in \mathcal{L}(q, \sigma; H)) &\leq \frac{25B(5/4)^{\omega(q)}}{q}, \\ \Pr_H(\exists \text{ primitive } v : N(v) \leq B, q \mid P(vH)) &\leq \frac{25B(15/4)^{\omega(q)}}{q}. \end{aligned}$$

Proof. The norm ball contains at most $25B$ integer pairs, since each coordinate has absolute value at most $2\sqrt{B}$. Fix a primitive v . If a retained prime divides $N(v)$, its arm event is impossible: vH has zero norm but a nonzero coordinate vector modulo that prime, whereas every nonzero boundary-arm vector has nonzero norm. Otherwise multiplication by v permutes the local units. One specified arm then has probability $1/[p^{e_p-1}(p-\chi_p)]$. The product probability is

$$\frac{1}{q} \prod_{p \mid q} \frac{p}{p-\chi_p} \leq \frac{(5/4)^{\omega(q)}}{q}.$$

Sum over residuals, then over at most $3^{\omega(q)}$ annotations. $\square$

The packet is specified, and the random common factor is sampled from the stated unit space. A special sequence $H = w^g$ is not thereby uniform. The common multiplication action supplies exact covariance, but controlling the visits of that arithmetic orbit to a rare first-minimum event remains open.

185.3. An exponential moment and its finite entropy cost.

Theorem 185.4 (Finite entropy transfer). *One has*

$$\log \mathbb{E}_\mu e^{F_Y/2} \leq 4Y^{-5/2}.$$

For every probability measure ν on the same finite space,

$$\mathbb{E}_\nu F_Y \leq 2D(\nu \parallel \mu) + 8Y^{-5/2}, \quad D(\nu \parallel \mu) = \sum_x \nu(x) \log \frac{\nu(x)}{\mu(x)}.$$

Zero-probability summands are set to zero.

Proof. For $K_p = (\nu_p - 3)_+$ and $0 < \theta < 1$, summation by tails yields

$$\mathbb{E}_\mu p^{\theta K_p} \leq 1 + \frac{3(p^\theta - 1)}{p^3(p - \chi_p)(1 - p^{\theta-1})}.$$

At $\theta = 1/2$, $(\sqrt{p} - 1)/(1 - 1/\sqrt{p}) = \sqrt{p}$. Independence and $\log(1+x) \leq x$ therefore give

$$\log \mathbb{E}_\mu e^{F_Y/2} \leq \sum_{p > Y} \frac{3}{p^{5/2}(p - \chi_p)} \leq \frac{15}{4} \sum_{n > Y} n^{-7/2} \leq 4Y^{-5/2}.$$

The integral from $\lfloor Y \rfloor \geq 4Y/5$ gives the last constant $(3/2)(5/4)^{5/2} < 4$. Tilt μ by $e^{F_Y/2}$ and normalize by $Z = \mathbb{E}_\mu e^{F_Y/2}$. Nonnegativity of relative entropy against this tilted measure gives $\frac{1}{2}\mathbb{E}_\nu F_Y \leq D(\nu\|\mu) + \log Z$. The nonnegativity itself follows from $\log x \leq x - 1$. $\square$

Theorem 185.5 (A full-state entropy obstruction). *If ν is supported on at most K states, then*

$$D(\nu\|\mu) \geq \left(2 - \frac{2}{(Y-1)\log Y}\right) \log m - \log K.$$

Proof. Uniformity gives $D = \log |\Omega| - \mathcal{H}(\nu)$, and the Shannon entropy is at most $\log K$. The exact count is

$$|\Omega| = m^2 \prod_{p \in S} (1 - 1/p)(1 - \chi_p/p).$$

Each logarithmic factor is at least $-2/(p-1)$, using $-\log(1 - 1/p) \leq 1/(p-1)$ and noting that the inert second term is positive. Now $\omega(m) \leq \log m / \log Y$ proves the bound. $\square$

If $\log m \geq \delta t$, $\log K = o(t)$ and $Y \rightarrow \infty$, this implies $\liminf D/t \geq 2\delta$. Residuals of norm at most B provide at most $25B$ states. Thus $\log B = o(t)$ cannot give negligible full-state entropy loss when the retained modulus has logarithmic size comparable with t . This rules out that concrete shortcut; it does not exclude probability or entropy methods using more appropriate observables or conditioned measures.

185.4. A primitive-state reference containing every integer residual. Let Ω_p^{prim} be the pairs modulo p^{E_p} not simultaneously zero modulo p , and put uniform product measure μ_{prim} on their product. Every primitive integer residual reduces to this space. Multiplication by H with norm coprime to m still permutes it.

Theorem 185.6 (Primitive reference law). *Its independent capped depths satisfy*

$$\Pr_{\mu_{\text{prim}}} (\nu_p(vH) \geq e) = \frac{3}{p^{e-1}(p+1)}, \quad 1 \leq e \leq E_p.$$

Theorems 185.2 and 185.4 hold with the same numerical bounds and this measure. If a distribution has at most K states, then

$$D(\nu\|\mu_{\text{prim}}) \geq \left(2 - \frac{1}{(Y^2-1)\log Y}\right) \log m - \log K.$$

Proof. There are $p^{2e-2}(p^2-1)$ primitive pairs modulo p^e . Each arm has $\varphi(p^e)$ such points, and the arms are mutually exclusive; this proves the law. Replacing $p - \chi_p$ by the larger $p + 1$ in the previous moment proofs preserves their bounds. Finally

$$|\Omega^{\text{prim}}| = m^2 \prod_{p \in S} (1 - p^{-2}),$$

and $-\log(1 - p^{-2}) \leq 1/(p^2 - 1)$ proves the entropy bound. $\square$

This variant fixes the norm-unit conditioning mismatch for actual residuals, but retains the full-state cardinality obstruction.

185.5. The excess observable: lossless quotient and an exact boundary. Push the primitive reference measure to the vector $K = (K_p)$, where $K_p = (\nu_p - 3)_+$, and call its law π . Then $F_Y = \sum_p K_p \log p$ descends exactly to this observable.

Theorem 185.7 (Observable transfer and comparability). *For any residual-state distribution and its observable pushforward ν_K ,*

$$\mathbb{E}F_Y \leq 2D(\nu_K\|\pi) + 8Y^{-5/2}, \quad D(\nu_K\|\pi) \leq D(\nu\|\mu_{\text{prim}}).$$

Moreover

$$\frac{1}{2}\mathbb{E}F_Y - 4Y^{-5/2} \leq D(\nu_K\|\pi) \leq 4\mathbb{E}F_Y + 10Y^{-3}.$$

If the observable distribution has at most K states, the additional bound is

$$D(\nu_K \parallel \pi) \geq \mathbb{E} F_Y + \mathbb{E} \sum_{p: K_p > 0} (3 \log p - \log 3) - \log K.$$

Proof. Pushforward preserves the exponential moment of F_Y , so the same tilted measure proof applies. The finite entropy chain rule gives the second inequality by dropping the nonnegative conditional entropies in the fibers. For $k \geq 1$, the nonterminal atom and terminal cap atom are respectively

$$\pi_p(k) = \frac{3(p-1)}{(p+1)p^{k+3}}, \quad \pi_p(k) = \frac{3p}{(p+1)p^{k+3}}.$$

Both lie between $p^{-(k+3)}$ and $3p^{-(k+3)}$. For caps at least four, $\pi_p(0) = 1 - 3/[p^3(p+1)]$; smaller caps give a certain zero. The total zero-coordinate surprisal is at most $10Y^{-3}$, by $-\log(1-x) \leq 2x$ and integral comparison of $\sum_{n>Y} n^{-4}$. The positive atoms have surprisal at most $(k+3) \log p \leq 4k \log p$. Subtracting nonnegative Shannon entropy proves the upper bound. Their lower surprisal is $(k+3) \log p - \log 3$, while zero-coordinate cost is nonnegative; subtract at most $\log K$ to get the final inequality. The first transfer inequality gives the remaining lower bound. $\square$

The full-state obstruction need not pass to this quotient. For the actual primitive pair $v = (1, 1)$ and $H = 1$, the boundary is two, so every retained excess is zero and

$$D(\delta_0 \parallel \pi) = - \sum_{p: E_p \geq 4} \log \left(1 - \frac{3}{p^3(p+1)} \right) \leq 10Y^{-3}.$$

Increasing the caps at a fixed retained prime makes the full-state Dirac entropy unbounded while this observable entropy remains bounded. This example only compares entropy losses; it is not a theorem about nontrivial small-lambda power profiles.

The comparability theorem also gives a precise remaining boundary: at growing height, $D(\nu_K \parallel \pi) = o(t)$ is equivalent to $\mathbb{E} F_Y = o(t)$ for this retained target. The quotient removes an irrelevant state-space cost, but it is not a newly proved weaker arithmetic premise. An upper bound must come from independent geometry or orbit structure; computing it from the same unbounded depths would be circular.

Finally F_Y is the sum of positive parts *at individual primes*. It upper-bounds the positive part of the signed logarithm, but negative prime credits can make these quantities different. The signed compensation in Theorem 184.4 remains available. This probability direction pursues a stronger sufficient target and does not discard methods using negative credits. Neither the reference law, its first-minimum estimate, nor the entropy reformulation proves the required special-orbit transfer or establishes standard *abc*.

186. SQUARE NORM FAMILIES, EXACT PARITY OBSTRUCTIONS, AND A DYNAMIC COVER TOWER

This section gives a positive infinite family whose second norm is an actual square, and a precise obstruction to making both norms squares up to the ramified factor three. The proofs use explicit elliptic curves and descent. A separate multiquadratic tower gives fixed-square-class finiteness for successive mixed transforms. None of these conclusions supplies the unbounded family required by the two-step radical criterion. The geometric arguments are ordinary mathematics; any companion Lean arithmetic identities have their narrower scope stated separately.

Put

$$N(x) = x^2 + x + 1, \quad F(x) = x^4 + 3x^3 + 5x^2 + 3x + 1.$$

For coprime positive integers a, b , write $c = a + b$, $U = ab$ and

$$M_0 = a^2 + ab + b^2, \quad M_1 = U^2 + UM_0 + M_0^2 = a^4 + 3a^3b + 5a^2b^2 + 3ab^3 + b^4.$$

186.1. **An actual infinite family with square second norm.**

Lemma 186.1 (Explicit elliptic coordinates). *The smooth projective curve $y^2 = F(x)$ is birational over $\mathbb{Q}$ to*

$$(669) \quad E: \quad Y^2 = X^3 - X^2 - 3X.$$

On the open sets where the displayed denominators are nonzero, the maps are

$$(670) \quad u = \frac{y - 1 - 3x/2}{x^2}, \quad X = 2 - 2u, \quad Y = \frac{X}{2}((X - 4)x - 3),$$

$$(671) \quad x = \frac{2Y + 3X}{X(X - 4)}, \quad y = 1 + \frac{3}{2}x + \left(1 - \frac{X}{2}\right)x^2.$$

Proof. Substitute $y = 1 + 3x/2 + ux^2$ and divide the quartic equation by x^2 :

$$(u^2 - 1)x^2 + 3(u - 1)x + 2u - \frac{11}{4} = 0.$$

The discriminant of this quadratic in x is $-8u^3 + 20u^2 - 10u - 2$. Substitution $X = 2 - 2u$ changes this to $X^3 - X^2 - 3X$, and completing the square gives (670). Equivalently, after $u = 1 - X/2$ the quartic difference is

$$\frac{x^2}{4}(X(X - 4)x^2 - 6Xx - 4X - 3).$$

Solving for x gives (671); the two maps are inverse on their open domains. The discriminant of F is 117, and the cubic in (669) also has distinct roots, so their projective normalizations are smooth genus-one curves. $\square$

Theorem 186.2 (Infinitely many positive square second norms). *There are infinitely many coprime positive pairs (a, b) , with unbounded $a + b$, for which M_1 is an integer square. Each such pair gives an actual second Eisenstein extraction with exponent two and unit residual, hence $\lambda_1 = 1/2$ for that extraction.*

Proof. On E take $P = (3, 3)$. Exact group operations give

$$2P = (4, -6), \quad 3P = (75, 645), \quad 4P = \left(\frac{361}{144}, \frac{2413}{1728}\right).$$

The doubling formula is

$$(672) \quad X(2Q) = \frac{(X(Q)^2 + 3)^2}{4X(Q)(X(Q)^2 - X(Q) - 3)}.$$

If $s = v_2(X(Q)) < 0$, the numerator and denominator have valuations $4s$ and $2 + 3s$. Thus $v_2(X(2Q)) = s - 2$. Starting at $4P$, whose X coordinate has valuation -4 , gives infinitely many distinct doubles. Therefore P has infinite order without a database rank computation.

The real identity component contains infinity and the branch $X \geq (1 + \sqrt{13})/2$, hence contains P . It is a compact connected one-dimensional Lie group, isomorphic to a circle by integration of an invariant differential. The infinite-order point P corresponds to an irrational angle, so its integer multiples are dense. At $3P$, the inverse map gives

$$x = \frac{101}{355} > 0, \quad y = -\frac{192529}{126025}.$$

A neighborhood avoids the exceptional denominators and maps continuously to $x > 0$. Infinitely many distinct multiples lie there. Birationality and the fact that each x supports at most two y give infinitely many distinct positive rational ratios $x = a/b$ in lowest terms. Now $(yb^2)^2 = F(a, b) \in \mathbb{Z}$, and a rational number whose square is integral is integral. Thus M_1 is an integer square. Infinitely many reduced positive ratios have unbounded numerator plus denominator.

The first mixed output (ab, M_0, c^2) is primitive and its norm M_1 has no factor three. In its oriented Eisenstein factorization all split exponents are therefore even. Unique factorization gives $z_1 = \text{unit} \cdot W^2$, so the specified unit-residual extraction has $\lambda_1 = 1/2$. $\square$

For example, $F(101, 355) = 192529^2$. Exponent two alone is much weaker than the NT4 sufficient threshold; no stronger extraction or high ABC quality is inferred. Exact searches of the existing repository found no prior use of this pair or elliptic model in this connection. The elliptic curve itself is familiar, and no assertion of literature novelty is made.

186.2. Two explicit descent calculations. We use Mordell–Weil finite generation and prime-to- p torsion injectivity at good reduction, as stated in the author-hosted J. S. Milne, *Elliptic Curves*, second edition, Chapter IV, finite-basis theorem, and Chapter II, Corollary 4.2: <https://www.jmilne.org/math/Books/EC2.pdf>. The following descent maps and local exclusions are explicit, so no rank or torsion assertion is taken from a database.

For integers a, b , let $E(a, b) : y^2 = x^3 + ax^2 + bx$, where $b(a^2 - 4b) \neq 0$, and let $E' = E(-2a, a^2 - 4b)$. The two-isogeny is

$$(673) \quad \phi(x, y) = \left(x + a + \frac{b}{x}, y \left(1 - \frac{b}{x^2} \right) \right).$$

Applying the same formula to E' and scaling by $1/4, 1/8$ gives the dual ψ , with $\psi\phi = [2]$. The square-class map $\alpha : E(\mathbb{Q}) \rightarrow \mathbb{Q}^\times / \mathbb{Q}^{\times 2}$ takes the class of x away from $O, (0, 0)$ and satisfies $\alpha(O) = 1, \alpha((0, 0)) = [b]$. Its kernel is $\psi(E'(\mathbb{Q}))$.

Here the image criterion can be checked directly. The first coordinate in (673) is $(y/x)^2$. Conversely, for a target $X = t^2 \neq 0$, the equation $x^2 + (a - X)x + b = 0$ has discriminant $(X - a)^2 - 4b = (Y/t)^2$. Choosing its root and the sign in $y = tx$ gives a rational preimage. A target $(0, 0)$ has a preimage exactly when the target's b coefficient is square, in agreement with its square-class value. The homomorphism property follows from the product of the three x coordinates cut out by a line $y = mx + n$ being n^2 . For $n = 0$, the product of the other two roots is b , which is consistent with the assigned value at $(0, 0)$; vertical lines and tangencies give the remaining cases. More explicitly, $T = (0, 0)$ on E lies in $\psi(E')$ exactly when b is square, and O always lies in the image. The formula $x(P + T) = b/x(P)$ verifies the class law through T , while vertical lines give $\alpha(P)\alpha(-P) = 1$.

Every image class is a signed squarefree divisor d of b . A prime outside b cannot have odd positive valuation in x , and a negative valuation in x is even by comparing y^2 with the monic cubic. Writing $x = dU^2/V^2$ and clearing denominators gives, for any such nontrivial class, primitive integers satisfying

$$(674) \quad W^2 = dU^4 + aU^2V^2 + (b/d)V^4, \quad \gcd(U, V) = 1.$$

The covering coordinate W is initially rational with integral square, so it is an integer.

Lemma 186.3 (Complete points on two small twists). *The rational points of $E_1 : y^2 = x(x-1)(x+3)$ are exactly*

$$O, (0, 0), (1, 0), (-3, 0), (-1, \pm 2), (3, \pm 6).$$

The rational points of $E_3 : y^2 = x(x-3)(x+9)$ are exactly $O, (0, 0), (3, 0), (-9, 0)$.

Proof. For E_1 one has $a = 2, b = -3$, and its dual has $a' = -4, b' = 16$. The image of α consists of the four classes $1, -1, 3, -3$, all realized by the listed points. On the dual, $x \geq 0$ because $x^2 - 4x + 16 = (x - 2)^2 + 12 > 0$. Thus the possible image classes are $1, 2$. Class two would force

$$W^2 = 2U^4 - 4U^2V^2 + 8V^4.$$

In the primitive parity cases (odd, even), (odd, odd), (even, odd), the right side is respectively $2, 6, 8$ modulo sixteen, all nonsquares. The dual image is therefore trivial and $E'_1(\mathbb{Q}) = \phi(E_1(\mathbb{Q}))$. Hence $\ker \alpha = \psi(E'_1) = 2E_1$ and $|E_1(\mathbb{Q})/2E_1(\mathbb{Q})| = 4$. There are four rational two-torsion points; finite generation then gives rank zero. Both good reductions at five and seven have eight points. The two prime-to- p torsion injections eliminate odd torsion and bound the two-primary torsion by eight. All eight listed points occur, so the list is complete.

For E_3 , $a = 6, b = -27$, and its dual has $a' = -12, b' = 144$. The four rational two-torsion points already realize the four classes $1, -1, 3, -3$ of α . The dual has $x \geq 0$, since

$x^2 - 12x + 144 = (x - 6)^2 + 108 > 0$, and its possible classes are 1, 2, 3, 6. For classes two and six the quartics in (674) are

$$2U^4 - 12U^2V^2 + 72V^4, \quad 6U^4 - 12U^2V^2 + 24V^4.$$

Their residues for the same three parity cases are respectively (2, 14, 8) and (6, 2, 8) modulo sixteen, excluding both. For class three one would have

$$W^2 = 3U^4 - 12U^2V^2 + 48V^4.$$

Writing $W = 3W_0$ and dividing by three gives, modulo three, $0 = U^4 - U^2V^2 + V^4$. Its right side is one unless both U, V are divisible by three, contradicting primitivity. Thus the dual image is again trivial, the quotient by doubling has order four, and the rank is zero. The good reductions at five and seven have four and eight points, so the torsion order divides four. Its four rational two-torsion points exhaust the group. $\square$

Theorem 186.4 (The combined norm is neither a square nor three times one). *For every positive rational x , neither $N(x)F(x)$ nor $N(x)F(x)/3$ is a rational square. Consequently, for every primitive positive integer seed (a, b) , neither M_0M_1 nor $M_0M_1/3$ is a rational square.*

Proof. Suppose $N(x)F(x) = dh^2$ with $d \in \{1, 3\}$. Set

$$u = x + x^{-1}, \quad v = 2u + 3, \quad t = 4h(x + 1)/x^2.$$

Since $N(x)/x = u + 1$ and $F(x)/x^2 = u^2 + 3u + 3$,

$$(675) \quad dt^2 = 16(u + 2)(u + 1)(u^2 + 3u + 3) = v^4 + 2v^2 - 3 = (v^2 - 1)(v^2 + 3).$$

Thus $(X, Y) = (v^2, vt)$ lies on $dY^2 = X(X - 1)(X + 3)$. For $d = 1$ this is E_1 ; for $d = 3$ the coordinates $(3X, 9Y)$ lie on E_3 . Positivity gives $u \geq 2, v \geq 7$. The respective x coordinates are at least 49 and 147, contradicting Lemma 186.3. Finally $M_0M_1 = b^6N(a/b)F(a/b)$, and b^6 is a square. $\square$

Corollary 186.5 (A precise even-exponent obstruction). *In actual successive norm representations*

$$M_0 = 3^e V_0 Q_0^{g_0}, \quad M_1 = V_1 Q_1^{g_1}, \quad e \in \{0, 1\},$$

the exponents g_0, g_1 cannot both be even when both residual norms V_0, V_1 are squares. In particular, two unit-residual even extractions are impossible, even with arbitrarily moving roots.

Proof. Those assumptions would make M_0M_1 equal to 3^e times a square, contrary to Theorem 186.4. $\square$

This excludes a specific exact route, not arbitrary residuals, odd extraction exponents, or the general finite lambda threshold.

186.3. Fixed residual square classes and a genus-three curve.

Theorem 186.6 (Fixed-class finiteness). *For any fixed positive rational square classes d_0, d_1 , only finitely many coprime positive pairs (a, b) satisfy $M_0 = d_0A^2, M_1 = d_1B^2$ with rational A, B . No effective height bound is asserted.*

Proof. Take the smooth projective normalization with function field

$$\mathbb{Q}(x) \left(\sqrt{N(x)/d_0}, \sqrt{F(x)/d_1} \right).$$

The discriminants are -3 and 117 . The polynomials have no common root, since $F \bmod N = x^2$ and $N(0) = 1$; indeed their resultant is one. Disjoint simple zeros prove that their geometric square classes are independent. The cover is therefore geometrically connected of degree four. There are six finite branch points. Each has two points of ramification index two, contributing two to Riemann–Hurwitz. Infinity is unramified because both degrees are even. Thus $2g - 2 = -8 + 12 = 4$, so $g = 3$. Faltings’s theorem gives finitely many rational points on this fixed curve. Every seed gives $x = a/b$ with square coordinates $A/b, B/b^2$, and a reduced positive ratio determines its primitive pair uniquely. $\square$

The external finiteness input is G. Faltings, *Endlichkeitssätze für abelsche Varietäten über Zahlkörpern*, *Inventiones Mathematicae* **73** (1983), 349–366, with the 1984 erratum; the original article is indexed at <https://eudml.org/doc/143051>.

For two actual even extractions the square classes are $([3^e V_0], [V_1])$. Along a family whose seed height tends to infinity, this *pair* must eventually leave each fixed finite product of square-class sets. The quantifier is joint: it need not be one fixed coordinate that escapes along the whole sequence. Moving square classes and odd exponents remain possible.

186.4. A dynamic tower with explicit genus. Define

$$(676) \quad A_0 = x, \quad B_0 = 1, \quad A_{j+1} = A_j B_j, \quad B_{j+1} = A_j^2 + A_j B_j + B_j^2, \quad f_j = B_{j+1}.$$

Then A_j/B_j is the j th iterate of $r(x) = x/(x^2 + x + 1)$, and $f_0 = N, f_1 = F$.

Theorem 186.7 (Dynamic multiquadratic covers). *Every f_j is squarefree of degree 2^{j+1} , and distinct f_j have disjoint root sets. For fixed nonzero rational $d_0, \dots, d_{k-1}$, the smooth projective cover with function field*

$$\mathbb{Q}(x) \left(\sqrt{f_0/d_0}, \dots, \sqrt{f_{k-1}/d_{k-1}} \right)$$

is geometrically connected of degree 2^k and has genus

$$(677) \quad G_k = 1 + 2^{k-2}(2^{k+1} - 6), \quad k \geq 1.$$

In particular $G_1 = 0, G_2 = 3, G_3 = 21$. For $k \geq 2$, fixing this vector of square classes allows only finitely many primitive positive seeds for the successive mixed transforms.

Proof. The degree-two map r has exactly the critical points $1, -1$, since $r'(x) = (1 - x^2)/(x^2 + x + 1)^2$. The point -1 is fixed, and the forward orbit of 1 remains positive and finite. Neither orbit reaches infinity; infinity maps to the fixed point zero. Thus infinity is not postcritical. All points in $(r^{j+1})^{-1}(\infty)$ are unramified, giving exactly 2^{j+1} distinct finite points.

The numerator and denominator in (676) are coprime by induction: a common zero after one step would be a common zero before it. For $j \geq 1$ their degrees are $2^j - 1, 2^j$, with monic leading coefficients. Hence these preimages are exactly the roots of f_j and $\deg f_j = 2^{j+1}$. If two different levels shared a root, a positive iterate of infinity would return to infinity, contrary to its orbit.

Any nonempty product of distinct f_j has an odd valuation at a root of one selected factor. Thus their geometric square classes are independent, proving degree 2^k and geometric connectedness. There are $2^{k+1} - 2$ finite branch points, each with inertia group of order two and total ramification contribution 2^{k-1} . All degrees are even, so infinity is unramified. Riemann–Hurwitz gives

$$2G_k - 2 = -2 \cdot 2^k + (2^{k+1} - 2)2^{k-1},$$

which is (677). Faltings gives finiteness for $k \geq 2$. Homogenizing $f_j(a/b)$ multiplies it by $b^{2^{j+1}}$, a square. Consequently the classes are those of the actual successive norm integers. $\square$

The tower gives explicit higher-genus objects and a fixed-class restriction. It does not control the rational points uniformly over twists or settle the moving-class two-step compression problem.

187. FORMAL CROSS-RESIDUAL CANCELLATION AND ANNOTATED DETERMINANTS

The preceding integer collision mechanism admits a direct formalization using the repository's existing Eisenstein multiplication, norm, boundary, and exact boundary gcd theorem. The new module `ResidualCollisions.lean` contains fifteen declarations. Its central statement is the actual integer equality

$$\gcd(P(v_1 H), P(v_2 H)) = \gcd(P(v_1 H), P(v_2 \bar{v}_1)), \quad v_1 H \text{ primitive.}$$

Only the first product needs to be primitive in this formal statement. Integer gcd retains signs by taking the usual nonnegative gcd; zero second boundary and the primes two and three are included. The proof first establishes coprimality of $N(v_1)$ with $P(v_1 H)$, and then cancels the actual scalar cube in $(v_2 \bar{v}_1)(v_1 H) = N(v_1)v_2 H$.

The module also verifies the exact Gram identity and its integer norm consequence. For a specified boundary arm $\ell \in \{a, b, a + b\}$, it proves

$$\begin{aligned} q \mid \ell(vH), \quad q \mid \ell(wH), \quad \gcd(q, N(H)) = 1, \quad \det(v, w) \neq 0 \\ \implies 3q^2 \leq 4N(v)N(w). \end{aligned}$$

Here q is an arbitrary integer. The chosen arm is the same for the entire modulus. The proof obtains divisibility of the determinant after multiplication, cancels $N(H)$, and uses the nonzero integer quotient to deduce $q^2 \leq \det(v, w)^2$. Different arm choices at different primes, the full CRT lattice index, and the first-minimum question are not silently inserted into this formal signature.

The separate module `QuarticTwistArithmetic.lean` adds twelve declarations supporting the independent elliptic descent. Its actual primitive integer statements exclude squares for the three dual quartics with coefficients $(2, -4, 8)$, $(2, -12, 72)$, and $(6, -12, 24)$ modulo sixteen, and for the quartic with coefficients $(3, -12, 48)$ modulo three. Finite residue propositions are proved by kernel reduction, and a general polynomial congruence lemma transfers them to arbitrary coprime integer coordinates. The module also proves the cleared-denominator identity linking the two norms to the elliptic twist, and the exact example $M_1(101, 355) = 192529^2$. It does not formalize the elliptic group law, isogeny descent, Mordell–Weil theorem, torsion classification, or the complete combined-norm obstruction.

A fresh standalone Lake build with Lean 4.32.0 and warnings treated as errors accepted these twenty-seven declarations and recompiled all fifty-two declarations in the three unchanged local dependency modules. Every declaration was subjected to a printed-axiom audit; the union consists of `propext`, `Classical.choice`, and `Quot.sound`. The saved inventory identifies the exact source bytes by SHA-256. The full lcm envelope and finite-reference probability theorems remain ordinary proofs. No private-depth estimate, height-to-measure transfer, or complete ABC proof is asserted by this scoped formal verification.

188. SIGNED SUPPORT AND HIGHER-POWER ARITHMETIC CLASSES

The fourth continuation keeps the forward signed-tail problem and the independent higher-power geometry separate while identifying a precise connection between their arithmetic supports. On the forward side, conditioning on the actual supported primes retains the negative contributions of depths one and two. An exact relative-entropy identity, with an elementary uniform height error, replaces the unnecessarily strong positive-excess target by a support-dependent allowance. The required arithmetic upper bound for that allowance remains open.

On the geometric side, a finite Kummer descent proves finiteness of primitive positive seeds for every fixed higher exponent and residual power class. It removes the denominator-class restriction from a direct affine lift by using integral primitivity. Its actual-profile consequences force residual support or exponent escape, without supplying a bound uniform in the exponent. A separate construction realizes arbitrary specified simultaneous extraction exponents with nonsquare residuals. A separate consecutive-seed family shows why a fixed finite list of direct norm-residue tests cannot exclude the whole branch.

The ordinary proofs were independently reviewed before their arithmetic pieces were formalized. The new Lean module verifies an unbounded actual primitive family passing all direct tests whose moduli divide a specified integer, together with polynomial, primitive-coordinate, and denominator arithmetic. It does not formalize Hensel lifting, the finite Kummer descent, Faltings’s theorem, or the entropy estimates. Neither the actual signed allowance nor the uniform moving-exponent radical gate has been proved. Standard *abc* remains unproved and undisproved here.

189. SIGNED DEPTH ENTROPY WITH THE ACTUAL SUPPORT RETAINED

The positive-excess observable of Section 185 discards the negative contributions of primes of depths one and two. We now retain these contributions and condition on the actual supported primes. An elementary height count bounds the remaining conditional Shannon entropy. The

result is an exact signed allowance criterion, not an estimate of that allowance on a specified arithmetic orbit.

For a positive integer A and a real $Y \geq 5$, put

$$S_Y(A) = \log W_Y(A) = \sum_{\substack{p > Y \\ p|A}} (v_p(A) - 3) \log p, \quad R_Y(A) = \prod_{\substack{p > Y \\ p|A}} p.$$

The sum is signed. For the finite primitive-pair reference measure modulo p^E , $p > 3$, write ν for the capped boundary valuation. The local count proved in Section 185 gives

$$\begin{aligned} \Pr(\nu = 0) &= \frac{p-2}{p+1}, \\ \Pr(\nu = e) &= \frac{3(p-1)}{(p+1)p^e} \quad (1 \leq e < E), \quad \Pr(\nu = E) = \frac{3}{(p+1)p^{E-1}}. \end{aligned}$$

Indeed the primitive state count modulo p^e is $p^{2e-2}(p^2 - 1)$, and each of the three disjoint boundary arms has $\varphi(p^e)$ such states. Multiplication by a fixed H with $p \nmid N(H)$ permutes the primitive states. Distinct primes have independent reference laws by CRT. These are reference statements; no uniformity of actual small-height residuals or power orbits is assumed.

189.1. Signed moments before and after conditioning on support. Define $j(0) = 0$ and $j(e) = e - 3$ for $e \geq 1$.

Theorem 189.1 (Signed local moments). *For $0 < \theta < 1$ the infinite-precision signed moment is*

$$Z_p(\theta) = 1 - \frac{3}{p+1} + \frac{3(p-1)}{(p+1)p^{1+2\theta}(1-p^{\theta-1})}.$$

The finite capped moment is at most $Z_p(\theta)$. For $p \geq 5$,

$$Z_p(1/2) = 1 - \frac{3}{p+1}(1 - p^{-1} - p^{-3/2}) \leq 1 - \frac{2}{p+1}.$$

Thus for a finite set J of retained primes and the signed reference variable $S = \sum_{p \in J} j(\nu_p) \log p$,

$$\log \mathbb{E}_\mu e^{S/2} \leq -2 \sum_{p \in J} \frac{1}{p+1}.$$

For every probability law ν on the same finite space,

$$\mathbb{E}_\nu S \leq 2D(\nu \| \mu) - 4 \sum_{p \in J} \frac{1}{p+1}.$$

Proof. Sum the geometric series over the positive depths. Capping at a positive integer E decreases j on the positive branch, and hence decreases its exponential moment. At $\theta = 1/2$, cancellation uses $(1 - p^{-1})/(1 - p^{-1/2}) = 1 + p^{-1/2}$. The inequality $1 - p^{-1} - p^{-3/2} \geq 2/3$ holds at $p = 5$ and its left side increases. Multiply the local moments and use $\log(1 - x) \leq -x$. Finally, the tilted measure $\mu_\theta(x) = e^{\theta S(x)} \mu(x) / \mathbb{E}_\mu e^{\theta S}$ and $D(\nu \| \mu_\theta) \geq 0$ give the stated entropy bound. $\square$

The unconditioned negative term above is a reciprocal-prime sum. Conditioning on the supported primes gives a radical-sized allowance.

Theorem 189.2 (Support-conditioned signed moments). *Given a hit $\nu_p \geq 1$, the infinite-precision reference law is*

$$q_p(e) = (1 - p^{-1})p^{-(e-1)} \quad (e \geq 1).$$

At finite precision E , this formula holds for $e < E$, with $q_p(E) = p^{-(E-1)}$. In either case

$$\mathbb{E}[p^{\theta(\nu_p-3)} \mid \nu_p \geq 1] \leq p^{-2\theta} \frac{1 - p^{-1}}{1 - p^{\theta-1}}.$$

In infinite precision equality holds, and at $\theta = 1/2$ the value is $p^{-1}(1 + p^{-1/2})$. Fix a supported set J and write $R_J = \prod_{p \in J} p$. If D_J is the relative entropy of an actual depth law on this support against the conditional product reference, then

$$\mathbb{E}S \leq 2D_J - 2\log R_J + 2 \sum_{p \in J} \log(1 + p^{-1/2}).$$

More generally, for $\theta = 1 - \delta \in (0, 1)$,

$$\mathbb{E}S \leq \frac{D_J}{\theta} - 2\log R_J + \frac{1}{\theta} \sum_{p \in J} (\log(1 - p^{-1}) - \log(1 - p^{-\delta})).$$

Proof. Divide the positive-depth probabilities by $3/(p+1)$, sum the resulting geometric series, and apply the same tilted-measure inequality. Capping only decreases the signed moment. For primes $p > Y$, the last sum is at most $|J|Y^{-\delta}/(1 - Y^{-\delta})$. For example, $\delta = 1/\sqrt{\log Y}$ makes it $o(\log R_J)$, uniformly over finite nonempty supports as $Y \rightarrow \infty$; the empty support has zero error. $\square$

189.2. An exact identity with a uniform height error. Fix $t > 0$ and any probability distribution on finitely many positive integers A satisfying $\log A \leq 3t$. This is a common upper height bound for the ensemble. It applies in particular to an ensemble of actual primitive boundaries $A = abc$ with $\log c \leq t$. Let

$$J = \{p > Y : p \mid A\}, \quad e = (v_p(A))_{p \in J}, \quad L = \sum_{p \in J} e_p \log p, \quad r = \log R_J(A).$$

Keep the support law exactly as in the actual ensemble, and put $H_{\text{depth}} = H(e \mid J)$, with natural logarithms. For each prime in the finite union of the actual supports choose an integer $E_p > 3t/\log p$. Given J , let q_J be the product of the finite capped geometric laws in Theorem 189.2, and set

$$D_{\text{depth}} = \mathbb{E}_J D(\text{Law}(e \mid J) \parallel q_J), \quad c_J = \sum_{p \in J} -\log(1 - p^{-1}).$$

Every actual depth is strictly below its reference precision cap. The terminal reference atoms are needed for normalization but do not enter the cross entropy at actual states.

Theorem 189.3 (Height-sensitive signed entropy identity). *With the preceding common-height assumptions,*

$$\mathbb{E}S_Y(A) = D_{\text{depth}} - 2\mathbb{E}r + H_{\text{depth}} - \mathbb{E}c_J.$$

Moreover,

$$0 \leq H_{\text{depth}} \leq \frac{3(\log 2)t}{\log Y}, \quad 0 \leq \mathbb{E}c_J \leq \frac{3t}{(Y-1)\log Y}.$$

Consequently

$$-\frac{3t}{(Y-1)\log Y} \leq \mathbb{E}S_Y(A) - (D_{\text{depth}} - 2\mathbb{E}r) \leq \frac{3(\log 2)t}{\log Y}.$$

These estimates are uniform in the ensemble size, actual prime supports, common factor, and the chosen precisions satisfying the strict bounds.

Proof. On every actual depth vector, $-\log q_J(e) = \sum_{p \in J} (e_p - 1) \log p + c_J = L - r + c_J$. Thus $D_{\text{depth}} = \mathbb{E}(L - r + c_J) - H_{\text{depth}}$; subtracting $2\mathbb{E}r$ proves the identity since $S_Y = L - 3r$.

Put $M = \lfloor 3t/\log Y \rfloor$. Since $p > Y$ and $L \leq 3t$, every possible depth vector satisfies $\sum e_p \leq M$. For a fixed support of size r_0 , stars and bars gives exactly $\binom{M}{r_0}$ positive integer vectors with sum at most M . The empty support contributes one vector. As $\binom{M}{r_0} \leq 2^M$, its conditional entropy is at most $M \log 2$. Also $|J| \leq 3t/\log Y$ and $-\log(1 - p^{-1}) \leq 1/(p-1) \leq 1/(Y-1)$. These prove both nonnegative error bounds and the final estimate. $\square$

For a sequence of such ensembles with $Y \rightarrow \infty$, the uniform one-sided upper bound $\mathbb{E}S_Y(A) \leq o(t)$ is therefore equivalent to

$$D_{\text{depth}} \leq 2\mathbb{E} \log R_Y(A) + o(t).$$

This permits negative linear signed credit; it does not assert that the signed cost converges to zero after division by height. For a point mass the identity still holds, with $H_{\text{depth}} = 0$. No distributional assumption has been introduced into that single seed.

189.3. Arithmetic scope and the remaining allowance bound. This conditional allowance retains credits that the positive-excess observable loses. For a concrete retained-prime comparison, take primes $p < q < 2p$ and the actual primitive seed $(a, b) = (p^4, q)$ with common factor $H = 1$. At the specified primes $\{p, q\}$ its boundary has depths four and one. Its retained positive excess is $\log p$, whereas its retained signed cost is $\log p - 2 \log q < 0$ and $\log(a + b) \sim 4 \log p$. This comparison does not control the other prime factors of the boundary and does not supply a nontrivial power orbit with small λ .

The new allowance has not been bounded arithmetically. Neither data processing nor a sparse set of actual states proves it. On a homogeneous orbit, the exact rank law gives $e_p = s_p + v_p(g)$ at the supported ranks $d_p \mid g$, with total lifting cost at most $\log g$. The unresolved part is the actual joint first-depth and supported-prime budget. Shifted residuals require their own shifted ranks and stopping towers; the homogeneous divisibility law cannot be transferred unchanged. The dependency chain here consists of the proved local count, finite geometric series, the relative-entropy identity, and stars and bars. These ordinary probability statements are not claimed as new Lean formalizations or as a resolution of the signed-tail gate or ABC.

190. HIGHER-POWER NORM CLASSES AND FINITE ARITHMETIC DESCENT

The second actual mixed norm is the binary quartic

$$F(a, b) = a^4 + 3a^3b + 5a^2b^2 + 3ab^3 + b^4, \quad F(x) = F(x, 1).$$

Its discriminant is 117. We first give the exact denominator correction for a direct affine lift. We then use integral primitivity and finite Kummer descent to remove that denominator restriction from the fixed-exponent finiteness theorem. This is an explicit application of the classical Darmon–Granville mechanism [32, Theorem 1 and its consequence on p. 514], not a new general finiteness theorem. Throughout, a seed means positive coprime integers a, b ; its height is $a + b$.

190.1. The exact direct lift. For $h \geq 2$ put

$$\delta = \gcd(h, 4), \quad m = h/\delta, \quad k = 4/\delta.$$

A positive integer is h -free when every prime valuation belongs to $\{0, \dots, h-1\}$. Every positive rational class modulo h -th powers has a unique positive h -free integer representative. For $m = 1$, the only m -free representative is 1.

Lemma 190.1 (Integral lifting and the denominator class). *Fix an h -free positive integer D and an m -free positive integer β . Primitive positive solutions*

$$(678) \quad F(a, b) = DW^h, \quad b = \beta B^m, \quad W, B \in \mathbb{Z}_{>0}$$

correspond bijectively to positive rational points (x, y) satisfying

$$(679) \quad y^h = \frac{\beta^4 F(x)}{D}$$

whose reduced denominator b has the indicated m -th power class. The maps are $x = a/b$, $y = W/B^k$, and conversely $W = yB^k$. The denominator-class map

$$\mathbb{Q}_{>0}^\times / \mathbb{Q}_{>0}^{\times m} \longrightarrow \mathbb{Q}_{>0}^\times / \mathbb{Q}_{>0}^{\times h}, \quad [b] \longmapsto [b^4]$$

is injective.

Proof. The identity $4m = hk$ gives $b^4 = \beta^4(B^k)^h$; homogeneity proves the displayed maps. To check integral lifting, suppose $q > 0$ is rational, Dq^h is an integer, and D is h -free. For every prime p ,

$$hv_p(q) = v_p(Dq^h) - v_p(D) \geq -(h-1).$$

The left side is a multiple of h , hence is nonnegative. Thus q is an integer. Apply this to B and then to W ; when $m = 1$ the first step is immediate. Positivity and the unique reduced fraction a/b give the bijection. Finally, $h \mid 4e$ if and only if $m \mid e$ for each integer e , proving the class-map assertion prime by prime. $\square$

The correction is necessary. The primitive seed $(a, b) = (1, 2)$ satisfies $F(a, b) = 67$. With $D = 67$ and $W = 1$,

$$\frac{F(1/2)}{67} = \frac{1}{16},$$

which is not a rational h -th power when $h \nmid 4$. Its 2-adic valuation is -4 . Thus $W/b^{4/h}$ cannot be used as a rational coordinate in general. This is a counterexample to that direct map, not to the integer finiteness theorem below.

Theorem 190.2 (Direct-cover genus and coupled class). *For $d > 0$ rational, the smooth projective curve with function field $\mathbb{Q}(x)(y)$, $y^h = F(x)/d$, is geometrically connected and has genus*

$$(680) \quad g_h = \frac{3h - 2 - \gcd(h, 4)}{2}.$$

For fixed $h \geq 3$, only finitely many primitive positive seeds $F(a, b) = VW^h$ can have the coupled class $[Vb^{-4}]_h$ in any fixed finite set of h -th power classes. In particular fixing $[V]_h$ and $[b]_m$ suffices.

Proof. Over the algebraic closure, valuation one at a root gives Kummer degree h . The four simple roots have inertia h , contributing $4(h-1)$. At infinity the pole order is four: there are δ points of ramification index h/δ , contributing $h - \delta$. There is no other ramification. Thus

$$2g_h - 2 = -2h + 4(h-1) + (h - \delta) = 3h - 4 - \delta.$$

For $h \geq 3$, this genus is at least three, so Faltings's theorem gives finitely many rational points on each fixed twist. If $Vb^{-4} = dq^h$, then $F(a/b) = d(qW)^h$, which supplies such a rational point. Each positive reduced rational a/b determines its seed uniquely. A finite union over twists proves the result. $\square$

For $h = 4$, $m = 1$, so the denominator correction is automatic and each fixed fourth-power class of M_1 gives only finitely many seeds. For $h = 2$, the genus is one, in agreement with the infinite square family of Section 186. The next theorem strengthens the fixed-class conclusion for every $h \geq 3$ without fixing a denominator class.

190.2. Finite descent using the four linear factors.

Lemma 190.3 (Finite Kummer classes). *Let K be a number field, S a finite set of its nonarchimedean primes, and $h \geq 2$. Then*

$$K(S, h) = \{z \in K^\times : v_{\mathfrak{p}}(z) \in h\mathbb{Z} \text{ for } \mathfrak{p} \notin S\} / K^{\times h}$$

is finite.

Proof. In the ring $\mathcal{O}_{K,S}$, write $(z) = \mathfrak{a}^h$. The map to $[\mathfrak{a}]$ has image in the finite group $\text{Cl}(\mathcal{O}_{K,S})[h]$. Its kernel is the group of S -units modulo h -th powers: if $\mathfrak{a} = (t)$, then z/t^h is an S -unit, and conversely an S -unit has trivial ideal. The S -unit group is finitely generated. Thus both kernel and image are finite. This uses finiteness of the ideal class group, with no principal-ideal or element-factorization assumption. $\square$

Theorem 190.4 (Finite seeds for a fixed higher-power residual class). *Fix $h \geq 3$ and a positive rational h -th power class $[D]_h$. There are only finitely many primitive positive seeds satisfying*

$$(681) \quad F(a, b) = DW^h, \quad W \in \mathbb{Q}_{>0}.$$

The conclusion holds for a fixed finite set of residual classes, and also when the positive integer residual has prime support in one fixed finite set. No denominator-class hypothesis is required.

Proof. Choose the positive h -free integer representative D ; the integral lifting argument above makes W an integer. Let K contain the four roots $\alpha_1, \dots, \alpha_4$ of F and the h -th roots of unity. These roots are integral because F is monic. Put

$$L_i = a - \alpha_i b, \quad \prod_{i=1}^4 L_i = DW^h.$$

Let S consist of the primes of K over rational primes dividing $117D$. Outside S all differences $\alpha_i - \alpha_j$ are units, since their squared product is 117. If a prime $\mathfrak{p} \notin S$ divided both L_i and L_j , it would divide b and then a , contradicting an integer Bezout identity for the primitive seed. Hence at most one L_i has positive valuation at $\mathfrak{p}$. Taking valuations of the product shows $h \mid v_{\mathfrak{p}}(L_i)$ for every i outside S .

Lemma 190.3 gives only finitely many possible classes of the L_i , and hence only finitely many triples of classes of L_i/L_4 for $i = 1, 2, 3$. Choose their representatives ξ_i . Every seed gives a K -point, with $x = a/b$, on one of the finitely many curves

$$(682) \quad y_i^h = \frac{x - \alpha_i}{\xi_i(x - \alpha_4)}, \quad i = 1, 2, 3.$$

These points avoid the branch values since $F(x) > 0$ for $x > 0$.

Over an algebraic closure the three Kummer classes are independent modulo h : evaluating the valuation of a relation at α_i forces its i -th exponent to be divisible by h . This holds also for composite h . The smooth projective normalization of (682) is therefore geometrically connected of degree h^3 , with group $(\mathbb{Z}/h\mathbb{Z})^3$.

At each α_i for $i \leq 3$ the inertia order is h . At α_4 the three functions all have a simple pole. Geometric local units have h -th roots in characteristic zero, so the three extensions there adjoin the same h -th root of a uniformizer. The inertia group is diagonal cyclic of order h , not of order h^3 . At infinity all three ratios are units, and there is no ramification. Consequently the four branch points each contribute $h^2(h-1)$, and Riemann–Hurwitz gives

$$(683) \quad 2G_h - 2 = -2h^3 + 4h^2(h-1), \quad G_h = 1 + h^2(h-2) > 1.$$

Faltings's theorem over the fixed number field K gives finitely many K -points on each curve. There are finitely many curves, and each positive reduced rational x determines a, b uniquely. This proves the fixed-class statement; a finite union proves its finite-class version. If the residual support is in S_0 , its h -free representative has at most $h^{|S_0|}$ possibilities, proving the last assertion. $\square$

The arithmetic covers capture every actual primitive seed. They do not assert that the direct coordinate $W/b^{4/h}$ is rational. Primitivity outside a finite set supplies the extra information that the direct affine normalization lacked.

190.3. An exponent restriction for possible unbounded families.

Corollary 190.5 (Residual support or exponent escape). *Consider actual second profiles $M_1 = V_1 Q_1^{g_1}$ along a sequence of primitive positive seeds with $a + b \rightarrow \infty$. For every fixed $h \geq 3$, among terms with $h \mid g_1$, the class $[V_1]_h$ eventually leaves every fixed finite set of classes. If all V_1 have prime support in one fixed finite set, then each fixed $h \geq 3$ divides g_1 only finitely often. In particular, eventually $v_2(g_1) \leq 1$, and for each fixed B no odd prime $p \leq B$ divides g_1 at sufficiently large indices. If also $g_1 \rightarrow \infty$, its least odd prime factor tends to infinity.*

Proof. For $h \mid g_1$, substitute $W = Q_1^{g_1/h}$ in Theorem 190.4. Each indicated finite class set gives only finitely many seeds, which a height-tending-to-infinity sequence eventually leaves. With fixed residual support, apply this to $h = 4$ and each of the finitely many odd primes at most B . If g_1 tends to infinity, after excluding multiples of four the values with no odd prime divisor, namely 1 and 2, no longer occur. $\square$

Corollary 190.6 (Finite total support). *For each fixed finite prime set S , there are only finitely many primitive positive seeds with $\text{supp}(M_1) \subseteq S$. Thus, along a sequence with $a + b \rightarrow \infty$, if $\text{supp}(Q_1) \subseteq S_Q$ for one fixed finite set, then for every fixed finite S_V one eventually has $\text{supp}(V_1) \not\subseteq S_V$. No condition on divisibility of g_1 is needed.*

Proof. Write $M_1 = DW^3$ with D cube-free. There are at most $3^{|S|}$ possible D , so Theorem 190.4 applies. For the profile consequence use $S = S_Q \cup S_V$; a height-tending-to-infinity sequence eventually leaves the resulting finite set of seeds. $\square$

This is a concrete Thue–Mahler consequence, with no quantitative prime-growth estimate. The unit residual case $V_1 = 1$ is included in Corollary 190.5, even with arbitrary moving roots. Each fixed exponent divisor $h \geq 3$ supports only finitely many seeds. This does not assert that such seeds do not exist, and it is not uniform in h . Large odd prime exponents, twice such exponents, and moving residual support remain outside this obstruction. A small $\lambda_1 = \max\{1, \log V_1\}/g_1$ need not bound the residual support or its power class, so the two-step finite threshold remains unresolved.

The number-field and geometric dependencies here are the finite ideal class group, finite generation of S -units, Kummer theory, Riemann–Hurwitz, and Faltings’s theorem. This section supplies ordinary proofs; it does not claim Lean verification of those dependencies or of the resulting finiteness theorem. The companion exact arithmetic replay checks the denominator identities and examples only.

191. ACTUAL EXPONENT REALIZATIONS BEYOND THE PARITY OBSTRUCTION

For a primitive positive seed (a, b) put

$$M_0 = a^2 + ab + b^2, \quad U = ab, \quad M_1 = U^2 + UM_0 + M_0^2 = F(a, b).$$

The actual Eisenstein integers are $z_0 = a + b\zeta$ and $z_1 = U + M_0\zeta$, where $\zeta^2 - \zeta + 1 = 0$. Both are primitive, since $\gcd(ab, M_0) = 1$. We distinguish direct norm-residue tests, local power solutions, and actual simultaneous extractions. None of the constructions below supplies the small-lambda two-step gate.

191.1. A precise boundary of direct fixed-modulus tests.

Theorem 191.1 (Simultaneous direct power residues). *Fix $m, n \geq 2$ and finitely many positive integer moduli. There are infinitely many primitive positive seeds, with $a/(a+b) \rightarrow 1/2$ and $a+b \rightarrow \infty$, for which M_0 and M_1 are respectively m -th and n -th power residues modulo every specified modulus.*

Proof. Let L be the least common multiple of the moduli and take $(a, b) = (Lk, Lk+1)$, $k \geq 1$. These integers are coprime. Their residues are $(0, 1)$ modulo L , so $M_0 \equiv M_1 \equiv 1 \pmod{L}$. The integer one is both specified power residues. The limiting height and balance assertions follow directly. $\square$

This excludes a universal obstruction consisting solely of finitely many fixed power-residue tests on the original norm integers. It does not exclude every argument using local information: the successful QG3 descent uses auxiliary covering classes and additional global input. The seeds here are not claimed to have power norms. Indeed the theorem also applies to the globally excluded square/square positive branch.

Theorem 191.2 (Primitive local power solutions). *For every $m, n \geq 2$ and every prime p , there are $a, b \in \mathbb{Z}_p$ with $ab(a+b) \neq 0$, at least one coordinate a unit, such that M_0 is an m -th power and M_1 an n -th power in $\mathbb{Z}_p$. The same system has a positive real solution.*

Proof. Put $s = 2 \max\{v_p(m), v_p(n)\} + 1$, $a = p^s$, and $b = a + 1$. For consecutive seeds, direct expansion gives

$$(684) \quad N_c(a) = M_0 = 1 + 3a + 3a^2,$$

$$(685) \quad F_c(a) = M_1 = 1 + 7a + 20a^2 + 26a^3 + 13a^4.$$

Both differences from one have valuation at least s . Applying strong Hensel lifting at one to $T^m - M_0$ and $T^n - M_1$ is valid because the error valuation exceeds twice the derivative valuation. It gives unit roots in $\mathbb{Z}_p$. The chosen integers have the stated nonzero and primitive properties. Over the reals, take the positive roots of the norms of any positive seed. $\square$

Different primes may use different seeds and roots. Local solubility therefore does not supply a common rational or integer power solution.

191.2. Prescribed actual depths with nonsquare residuals.

Theorem 191.3 (Arbitrary two-step extraction exponents). *For every $g_0, g_1 \geq 2$, there are infinitely many primitive positive consecutive seeds, balanced in the limit, with actual representations*

$$z_0 = v_0(1 + 2\zeta)^{g_0}, \quad z_1 = v_1(2 + 7\zeta)^{g_1}.$$

Writing $V_i = N(v_i)$, the residual norms are positive nonsquares and

$$\begin{aligned} M_0 &= V_0 7^{g_0}, & M_1 &= V_1 67^{g_1}, \\ v_7(V_0) &= v_{67}(V_1) = 0, & v_{13}(V_0) &= v_{31}(V_1) = 1. \end{aligned}$$

In particular both exponents may be arbitrary odd numbers, or arbitrary numbers congruent to two modulo four.

Proof. At $a = 1$ the consecutive polynomials satisfy

$$N_c(1) = 7, \quad N'_c(1) = 9, \quad F_c(1) = 67, \quad F'_c(1) = 177 \equiv 43 \pmod{67}.$$

Each is a simple root at its indicated prime. A simple root modulo p lifts uniquely to each modulus p^e . Of the p extensions of a root modulo p^e to modulus p^{e+1} , exactly one remains a root; choosing any other extension gives exact valuation e . Thus choose a class modulo 7^{g_0+1} with $v_7(N_c(a)) = g_0$, and a class modulo 67^{g_1+1} with $v_{67}(F_c(a)) = g_1$. Both reduce to $a = 1$ at their base prime.

Also impose $a \equiv 5 \pmod{13^2}$ and $a \equiv 14 \pmod{31^2}$. The exact identities

$$N_c(5) = 91 = 7 \cdot 13, \quad F_c(14) = 574771 = 31 \cdot 18541, \quad 18541 \equiv 3 \pmod{31}$$

give residual-prime depth one. The four moduli are pairwise coprime, so CRT provides an infinite progression of positive a satisfying all four exact valuation requirements. Set $b = a + 1$.

The two indicated Eisenstein primes have norms seven and sixty-seven. Modulo seven, z_0 reduces to $1 + 2\zeta$; modulo sixty-seven, z_1 reduces to $2 + 7\zeta$. Thus each prime occurs with the stated orientation. Primitivity prevents its conjugate from also dividing the same integer. Unique factorization and the exact norm valuations give the displayed powers of these fixed prime elements. Dividing produces actual primitive Eisenstein quotients v_i , without any assumption about the factorization of their remaining norms.

For consecutive seeds, $M_0 \equiv 1 \pmod{3}$, so its ramified exponent is zero; M_1 is also prime to three. Division by 7^{g_0} and 67^{g_1} leaves the exact depths at thirteen and thirty-one, proving that the residual norms are nonsquares. The exact root valuations give their asserted coprimality with the displayed root norms. Height and balance follow along the positive CRT progression. $\square$

This is a complete integer realization beyond the square-residual parity obstruction. It does not show that the residuals are analytically small. At fixed g_0, g_1 their logarithms tend to infinity along the progression, so both displayed $\lambda_i = \max\{1, \log V_i\}/g_i$ diverge. For varying exponents, the earlier fixed-first-root obstruction still applies whenever its bounded-first-lambda premise holds.

Corollary 191.4 (Residual support escape in the realized family). *Along each progression in Theorem 191.3, at fixed g_0, g_1 , the support of V_1 eventually fails to lie in each fixed finite prime set. No effective rate of prime growth is asserted.*

Proof. If $\text{supp}(V_1) \subseteq S$, then $\text{supp}(M_1) \subseteq S \cup \{67\}$. The finite-total- support theorem, Corollary 190.6, permits only finitely many such primitive seeds. Their heights are eventually left by the infinite positive progression. $\square$

This last corollary uses the separate global finite-descent and Faltings input. The residue and realization constructions themselves do not.

An exact integer replay verifies actual division and reconstruction in the Eisenstein ring for twenty-one members with varied exponent pairs. A separate scan of 76,116 primitive pairs with $1 \leq a \leq b \leq 500$ finds ten first-norm cubes and no second-norm cubes. The null scan is only finite diagnostic evidence and proves no global nonexistence. No statement in this section establishes high ABC quality, the required two-step radical compression, or a proof or disproof of ABC.

192. FORMAL ARITHMETIC FOR DIRECT POWER TESTS AND DENOMINATOR CLASSES

The module `LocalPowerArithmetic.lean` contains fifteen declarations supporting the preceding ordinary proofs. It imports the existing Eisenstein pair norm and defines the second norm by that actual norm,

$$M_1(a, b) = N(ab, N(a, b)),$$

then proves its quartic expansion. Its principal quantified statement says that for every positive natural number L and every natural height bound T , there exist positive integers a, b with $a + b > T$ such that

$$\gcd(a, b) = \gcd(ab, N(a, b)) = 1$$

and, simultaneously for every integer $m \mid L$ and natural exponents e, f ,

$$m \mid N(a, b) - 1^e, \quad m \mid M_1(a, b) - 1^f.$$

The witness is $a = L(T + 1)$, $b = a + 1$. Thus this is an actual unbounded primitive family, with the same seed working for every stated test. Taking L to be a common multiple gives the direct finite-modulus obstruction. The formal theorem does not assert that either norm is a perfect power as an integer or that the seeds have high ABC quality.

Further declarations prove the two consecutive-coordinate polynomial identities, their exact linearizations at the simple roots modulo seven and sixty-seven, the coprime derivatives, and the exact arithmetic supporting the residual depth-one conditions at thirteen and thirty-one. These are the arithmetic inputs to the ordinary Hensel and CRT argument; the infinite family of actual Eisenstein power extractions is not itself claimed as formalized by these declarations.

For the direct-lift denominator, the module checks the integer valuation implication

$$h > 0, \quad 0 \leq d < h, \quad d + hv \geq 0 \quad \implies \quad v \geq 0.$$

It also proves, for natural h, z ,

$$h \mid 4z \iff \frac{h}{\gcd(h, 4)} \mid z, \quad 4 \frac{h}{\gcd(h, 4)} = h \frac{4}{\gcd(h, 4)}.$$

The first equivalence is stated at the nonnegative valuation level in Lean. The passage through rational valuations, rational power-class groups, the full lifting bijection, and the higher-genus finiteness argument remain ordinary proofs with their stated external dependencies.

A fresh standalone Lake build with Lean 4.32.0, warnings treated as errors, verifies the fifteen new declarations and recompiles all twenty-nine declarations in the unchanged norm dependency. Every declaration has a printed-axiom inventory; only `propext`, `Classical.choice`, and `Quot.sound` occur. The checkpoint records the exact source hashes and reproducible finite arithmetic replays. This is scoped formal verification, with no claim to a full repository build, formalized geometric or probability theorems, or a solution of ABC.

193. ACTUAL RANK PACKETS AND VARYING-EXPONENT COVERS

The fifth continuation attacks the arithmetic content left open by the support-entropy and fixed-class reductions. On a homogeneous actual Eisenstein orbit, integer cyclotomic factors isolate each prime's first rank and its later lifting tower. The resulting packet bound has totient weight, whose sum over the divisors is the exponent itself. It proves a specified sparse-rank subclass without a divisor-count loss. A separate actual construction places a deep prime at the maximal rank beyond the moving cutoff, refuting automatic sparsity of ranks merely containing a deep prime. It does not determine the sign of the complete rank packet.

The independent geometric direction works in a fixed splitting field while the extraction exponent varies. Ideal allocations and the unit lattice give an explicit bound for the number of covers, including the cost of choosing a moving small residual, and a bound for their coefficient heights. Some unit power classes necessarily retain a linear height cost. The number of possible covers supplies no height bound for their individual points. The actual-point and moving-exponent issues therefore remain open.

The ordinary proofs were independently reviewed before formalization. Twenty-one new Lean declarations verify a general actual Lucas recurrence and its full solution, together with exact bad-prime support statements for every primitive integer second norm. The full cyclotomic, totient, number-field and geometric arguments remain ordinary proofs with their stated dependencies. Neither the unrestricted signed tail nor the two-step radical threshold is resolved; standard abc remains unproved and undisproved.

194. ACTUAL CYCLOTOMIC RANK PACKETS AND THE TOTIENT BUDGET

The support-conditioned entropy criterion retains signed credit but does not itself estimate the arithmetic allowance. We now use an integer factorization of the actual homogeneous orbit. Its first-depth packets have a $\varphi(d)$ budget, whose sum over the divisors of the exponent is exactly the exponent. This proves a sparse-rank subclass while retaining the primitive top-layer saving as an open step.

Let $\mathcal{O} = \mathbb{Z}[\zeta]$, $\zeta^2 - \zeta + 1 = 0$, and let w be primitive and unramified, with $Q = N(w) \geq 7$ and $3 \nmid Q$. Put $\eta = w^3$, $\bar{\eta} = \bar{w}^3$, and, for $n \geq 1$, let $T_n = |P(w^n)|$, where $P(a + b\zeta) = ab(a + b)$. Oriented factorization shows that all w^n remain primitive.

194.1. An integer factorization with exact rank valuations.

Theorem 194.1 (Actual cyclotomic integer factors). *For the homogeneous cyclotomic polynomials, define*

$$C_1 = P(w), \quad C_d = \Phi_d(\eta, \bar{\eta}) \quad (d \geq 2).$$

Every C_d is a nonzero rational integer, and

$$P(w^n) = \prod_{d|n} C_d.$$

For $d \geq 2$,

$$\log |C_d| \leq \varphi(d) \left(\frac{3}{2} \log Q + \log 2 \right),$$

while $\log |C_1| \leq \frac{3}{2} \log Q$. In particular, $\log |C_d| \leq 2\varphi(d) \log Q$ for every $d \geq 1$.

Proof. The ideals (w) and $(\bar{w})$ are coprime. A common prime away from three would divide $b\sqrt{-3}$ and then both integer coordinates; a prime above three is excluded by $3 \nmid Q$. If $\eta/\bar{\eta}$ were a root of unity, unique ideal factorization would give $(w)^3 = (\bar{w})^3$ and hence force w to be a unit, contrary to $Q \geq 7$.

For $d \geq 2$, reciprocity gives $\Phi_d(X, Y) = \Phi_d(Y, X)$, including $\Phi_2(X, Y) = X + Y$. Its value is an algebraic integer fixed by conjugation in $\mathbb{Q}(\sqrt{-3})$, so it lies in $\mathbb{Z}$; nonvanishing follows from the non-root-of-unity ratio. Direct expansion gives

$$(a + b\zeta)^3 - (a + b\bar{\zeta})^3 = 3\sqrt{-3}ab(a + b).$$

Apply the homogeneous cyclotomic factorization to $\eta^n - \bar{\eta}^n$ and cancel its first factor $3\sqrt{-3}P(w)$. This proves the exact signed integer product. Each complex linear factor of $\Phi_d(\eta, \bar{\eta})$ has

modulus at most $2Q^{3/2}$. The same cubic identity gives $|P(w)| \leq 2Q^{3/2}/(3\sqrt{3}) \leq Q^{3/2}$. Finally $\log 2 \leq \frac{1}{2} \log Q$ for $Q \geq 7$. $\square$

For a prime $p > 3$, $p \nmid Q$, let

$$d_p = \text{ord}((w/\bar{w})^3 \bmod p), \quad s_p = v_p(T_{d_p}).$$

The norm-one group has order $p - \chi_p$, where $\chi_p = 1$ in the split case and -1 in the inert case. Thus $p \nmid d_p$. The previously proved homogeneous rank and LTE law is

$$v_p(T_n) = \begin{cases} 0, & d_p \nmid n, \\ s_p + v_p(n), & d_p \mid n. \end{cases}$$

Theorem 194.2 (The only cyclotomic occurrences of a prime). *For every $n \geq 1$,*

$$v_p(C_n) = \begin{cases} s_p, & n = d_p, \\ 1, & n = d_p p^j \text{ for some } j \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. The integer factorization says that $v_p(T_n)$ is the divisor sum of $v_p(C_d)$. The proposed right side has divisor sum zero if $d_p \nmid n$. Otherwise its divisor sum is s_p plus the number of positive j for which $d_p p^j \mid n$, which is $s_p + v_p(n)$ because $p \nmid d_p$. Uniqueness of divisor inversion proves the claim. In particular, at the first rank the Möbius product has only one term of positive valuation, and all later occurrences lie on that rank's p -power tower with depth one per factor. $\square$

194.2. Totient-sized packets and exact signed assembly. For $Y \geq 5$ put

$$B_d = \prod_{\substack{p > Y \\ d_p = d}} p^{s_p}, \quad R_d = \prod_{\substack{p > Y \\ d_p = d}} p, \quad A_d = B_d / R_d^3.$$

These finite products depend on Y, w . Every prime in them divides T_d . A prime dividing Q cannot divide a primitive boundary: one zero boundary arm together with zero norm would force both coordinates to vanish. Thus no supported prime above the cutoff is missing from this rank notation.

Theorem 194.3 (Actual packet budget and signed assembly). *For every $d \geq 1$,*

$$B_d \mid |C_d|, \quad \log B_d \leq 2\varphi(d) \log Q.$$

For every $n \geq 1$,

$$\log W_Y(T_n) = \sum_{d \mid n} \log A_d + \ell_{n,Y}, \quad \ell_{n,Y} = \sum_{\substack{p > Y \\ p \mid T_n}} v_p(n) \log p, \quad 0 \leq \ell_{n,Y} \leq \log n.$$

Proof. The primes of rank d occur to exact depth s_p in C_d , so their full product divides that integer. Apply Theorem 194.1. For every supported prime the rank law gives $v_p(T_n) - 3 = (s_p - 3) + v_p(n)$. Partition the first terms by their unique rank. The product represented by the remaining sum divides n . $\square$

The identity $\sum_{d \mid n} \varphi(d) = n$ bounds the total full packet mass by $2n \log Q$. There is no $\sigma(n)$ or divisor-count loss. This is still a linear budget and supplies no sublinear radical saving by itself.

Let $E(n, w, Y)$ be the divisors $d \mid n$ for which the *net* signed packet $\log A_d$ is positive, and write

$$m_\varphi(n, w, Y) = \frac{1}{n} \sum_{d \in E(n, w, Y)} \varphi(d).$$

Theorem 194.4 (A quantitative sparse-rank subclass). *Let c_n be the largest side of the primitive positive triple from the three boundary arms of w^n , and $t_n = \log c_n$. Then*

$$\frac{\log W_Y(T_n)}{t_n} \leq 4m_\varphi(n, w, Y) + \frac{2 \log n}{n \log Q}.$$

If every positive net packet has rank at most an integer $D \geq 0$, the right side may instead be bounded by

$$\frac{2D(D+1)}{n} + \frac{2 \log n}{n \log Q}.$$

Thus $m_\varphi \rightarrow 0$, or the sufficient condition $D = o(\sqrt{n})$, gives a vanishing positive part of the signed ratio as $n \rightarrow \infty$, uniformly in moving $Q \geq 7$ and the cutoff. At most K exceptional ranks, all at most D , also suffice when $KD = o(n)$.

Proof. Discard only the nonpositive whole rank packets and apply their full-mass bounds. The norm satisfies $Q^n \leq c_n^2$, so $t_n \geq (n/2) \log Q$. This proves the first inequality. Use $\varphi(d) \leq d$ and $\sum_{d \leq D} d = D(D+1)/2$, or bound the sum for K ranks by KD . The resulting nonnegative upper bounds give the positive-part conclusion, not two-sided convergence of a signed quantity. $\square$

At the established small- λ homogeneous cutoff $Y = n^{1/6}$, $\lambda = 1/n$, this supplies the required one-sided signed tail on the stated subclass. It combines with the earlier low-prime and archimedean estimates along sequences with $n \rightarrow \infty$. It does not control bounded exponents with arbitrarily moving roots, and membership of a general actual orbit in this subclass is unproved.

194.3. A saving condition with no divisor-count inflation.

Theorem 194.5 (Totient-normalized rank saving). *Consider any sequence of actual data (n, w, Y) with $n \rightarrow \infty$ and $Q \geq 7$. Suppose a function $\epsilon(d) \geq 0$, tending to zero, satisfies*

$$\log A_d(Y; w) \leq \epsilon(d) \varphi(d) \log Q$$

at every divisor d of each actual n in the sequence. Then $\limsup \log W_Y(T_n)/t_n \leq 0$.

Proof. Given $\delta > 0$, choose D so that $\epsilon(d) \leq \delta$ for $d > D$. Bound the finitely many smaller ranks by the unconditional packet bound. The exact totient sum gives

$$\log W_Y(T_n) \leq 2 \log Q \sum_{d \leq D} \varphi(d) + \delta n \log Q + \log n.$$

Divide by $t_n \geq (n/2) \log Q$, first let $n \rightarrow \infty$ and then $\delta \downarrow 0$. The small-rank bound is uniform in Q and Y . $\square$

The older hypothesis $A_d \leq (Qd)^\kappa$ implies this saving condition. Indeed $\varphi(d) \geq \sqrt{d/2}$: in $\varphi(d)^2/d$, all odd prime-power factors are at least one, while the only factor below one is $1/2$ from 2^1 . Thus $\kappa(1 + \log d / \log 7) / \varphi(d) \rightarrow 0$. The new hypothesis permits larger upper budgets, such as $\log Q \sqrt{\varphi(d)}$. This compares hypotheses; it does not exhibit an arithmetic family violating the earlier hypothesis.

It is essential to distinguish E from the larger set of ranks containing one prime with $s_p \geq 4$. Other primes at the same rank may cancel that prime's positive contribution. Also one rank near n can carry totient mass comparable to n ; rank cardinality alone is insufficient. For prime n , the factorization has only ranks one and n . The first rank costs $O(\log Q)$ and the lifting term costs $O(\log n)$. The unproved issue is precisely the net primitive top-layer packet on the scale $n \log Q$.

These ordinary statements use actual Eisenstein ideals, cyclotomic factorization, the previously established homogeneous LTE law, and elementary divisor identities. No assertion that actual large primes usually have simple depth enters them. Neither these packet budgets nor their finite arithmetic checks prove the unrestricted signed-tail gate or ABC, and no new Lean coverage is asserted for this section.

195. A HIGH-DEPTH PRIME AT THE MAXIMAL ACTUAL RANK

The net sign of a rank packet differs from the occurrence of a single high-depth prime in that packet. We give actual homogeneous profiles with small λ for which the latter event occupies the maximal rank. The construction does not determine the net packet sign.

Write $\mathcal{O} = \mathbb{Z}[\zeta]$, $\zeta^2 - \zeta + 1 = 0$, $P(A + B\zeta) = AB(A + B)$ and $N(A + B\zeta) = A^2 + AB + B^2$. For an unramified primitive root w with $N(w) > 1$, and $p > 3$ prime to $N(w)$, let $d_p = \text{ord}((w/\bar{w})^3 \bmod p)$ and $s_p = v_p(P(w^{d_p}))$. At a cutoff Y , distinguish

$$E_{\text{net}}(g, Y) = \left\{ d \mid g : \sum_{\substack{p > Y \\ d_p = d}} (s_p - 3) \log p > 0 \right\},$$

$$E_{\text{hit}}(g, Y) = \{d \mid g : \text{some } p > Y \text{ has } d_p = d, s_p \geq 4\}.$$

These sums and sets range only over $p > 3$ with $p \nmid N(w)$, as in the rank definition, even if an initial cutoff is below three. One always has $E_{\text{net}} \subseteq E_{\text{hit}}$, but the reverse inclusion need not follow. Our counterfamily addresses only automatic totient sparsity of the larger set.

Lemma 195.1 (An elementary prime with the required order). *For each prime $\ell \geq 5$ there is a prime $p \equiv 1 \pmod{6\ell}$.*

Proof. Put $n = 6\ell$ and take a prime divisor p of $\Phi_n(n)$. Its complex product has absolute value at least $(n-1)^{\varphi(n)} > 1$. The constant term is one, so no prime dividing n divides this value. In particular $p \nmid n$. Modulo p , the factorization $X^n - 1 = \prod_{d|n} \Phi_d(X)$ is squarefree. The residue n is a root of Φ_n and of no factor at a proper divisor of n . It therefore has multiplicative order n , giving $n \mid p-1$. $\square$

Only this elementary existence is needed, with no least-prime estimate.

Theorem 195.2 (Prescribed first depth at a maximal rank). *Fix a prime $\ell \geq 5$, a prime $p \equiv 1 \pmod{6\ell}$ and $s \geq 4$. There is a positive integer $a \equiv 0 \pmod{3}$ such that $w = a + \zeta$ satisfies*

$$\gcd(N(w), 3p) = 1, \quad d_p = \ell, \quad s_p = s, \quad p^s \leq a < 3p^{s+1}.$$

Proof. In $\mathbb{F}_p$ choose z of order six and R of order ℓ , and put $\bar{z} = 1 - z$. Set

$$a_0 = \frac{R\bar{z} - z}{1 - R}.$$

Then $a_0 + \bar{z} = (\bar{z} - z)/(1 - R) \neq 0$ and $a_0 + z = R(a_0 + \bar{z}) \neq 0$. Their cubed ratio has exact order ℓ , and their product is nonzero. For the integer polynomial $G_\ell(T) = P((T + \zeta)^\ell)$ the cubic identity gives

$$3(z - \bar{z})G_\ell(T) = (T + z)^{3\ell} - (T + \bar{z})^{3\ell} \quad \text{over } \mathbb{F}_p.$$

At a_0 the right side vanishes. If $R(T) = (T + z)/(T + \bar{z})$, its derivative there is the derivative of $(T + \bar{z})^{3\ell}(R(T)^{3\ell} - 1)$, which is nonzero since $p \nmid 3\ell$ and $R'(a_0) = (\bar{z} - z)/(a_0 + \bar{z})^2 \neq 0$. Thus a_0 is a simple root of G_ℓ .

Lift it to the unique root b modulo p^s , with $0 \leq b < p^s$. Of the p residues $b + jp^s$ modulo p^{s+1} , exactly one lifts to a root. Choose a nonlifting $j \in \{1, 2\}$. Then $A = b + jp^s$ has exact depth s and $p^s \leq A < 3p^s$. Choose $k \in \{0, 1, 2\}$ by CRT so that $a = A + kp^{s+1} \equiv 0 \pmod{3}$. Since $p > 3$, this preserves the depth and gives the stated upper bound. The norm is a unit modulo p by the first calculation, and it is one modulo three. $\square$

Theorem 195.3 (Failure of automatic coarse rank sparsity). *There is a sequence of actual primitive homogeneous profiles w^g , with unit residual and $g \rightarrow \infty$, for which $\lambda = 1/g \rightarrow 0$ and, at the actual cutoff $Y = g^{1/6}$,*

$$\frac{1}{g} \sum_{d \in E_{\text{hit}}(g, Y)} \varphi(d) \geq \frac{g-1}{g} \rightarrow 1.$$

Thus this coarse exceptional-rank mass is not automatically $o(g)$.

Proof. Take $g = \ell$ through the primes at least five and apply the previous two results, for example with $s = 4$. The root $w = a + \zeta$ is primitive and has norm prime to three. Oriented factorization has no inert prime and no pair of conjugate split factors, so every w^g is primitive. A root-of-unity quotient $w/\bar{w}$ would make their coprime ideals equal and force w to be a unit. Its norm exceeds one, so no power has zero boundary. A unit rotation gives an actual positive primitive triple and preserves the absolute boundary and the norm. The profile has unit residual.

The selected prime satisfies $p \geq 6g + 1 > g^{1/6}$, $d_p = g$ and $s_p = 4$. Therefore $g \in E_{\text{hit}}$, while $\varphi(g) = g - 1$. $\square$

This construction quantifies the loss in marking a whole rank after seeing one deep prime. For its positive rotation with $t = \log c$, the size choice in Theorem 195.2 gives

$$N(w) \geq p^{2s}, \quad t \geq \frac{g}{2} \log N(w) \geq gs \log p, \quad \frac{(s-3) \log p}{t} \leq \frac{s-3}{gs} \leq \frac{1}{g}.$$

An arbitrarily large prescribed first depth can therefore occur at the maximal totient rank while its own signed cost is sublinear in height. Other primes in the same rank may supply negative credits; their full packet sign and size have not been determined here.

In particular this is not a counterexample to a condition on E_{net} , a rankwise saving hypothesis, or the global signed tail. The selected prime moves beyond the actual cutoff, but every other retained contribution must still be accounted for. Exact integer replay checks fourteen profiles with varied ranks and depths, including primitivity, positive unit rotations, exact first depth and all displayed size premises. It does not factor the whole boundary or claim a net packet sign. No ABC proof or disproof follows.

196. UNIFORM COVER COUNTS FOR SMALL SECOND-NORM RESIDUALS

We quantify the arithmetic covers of Section 190 in a fixed number field as the exponent varies. Write

$$F(a, b) = a^4 + 3a^3b + 5a^2b^2 + 3ab^3 + b^4 = VQ^g, \quad \gcd(a, b) = 1, \quad a, b, V > 0, \quad Q \geq 7,$$

with all displayed parameters integers. Put $\lambda = \max\{1, \log V\}/g$. The results control the number of covers and the heights of their coefficients. They supply no uniform bound on the heights of points on those covers.

196.1. The fixed field and the bad primes.

Lemma 196.1 (A fixed splitting field). *Let $\zeta^2 - \zeta + 1 = 0$, $K_0 = \mathbb{Q}(\zeta)$, and $d_0 = -1 - 3\zeta$. The splitting field of $F(x) = F(x, 1)$ is*

$$K = K_0(\sqrt{d_0}, \sqrt{13}).$$

It has degree eight, signature $(0, 4)$, and unit rank three, and is unramified outside 3 and 13.

Proof. With $t = x + x^{-1}$, one has $F(x)/x^2 = t^2 + 3t + 3$. The two values of t are $\zeta - 2$ and $-1 - \zeta$; the corresponding quadratic discriminants are d_0 and $\bar{d}_0$, with product 13. This gives the stated splitting field. Neither d_0 nor $\bar{d}_0$ is a square in K_0 , since each has norm 13. Also $\sqrt{13} \notin K_0$, whose real elements are rational. For a nonsquare d in a field of characteristic zero, an element t becomes a square in the quadratic extension by $\sqrt{d}$ only if t or t/d was already a square: expand $(u + v\sqrt{d})^2 = t$ and use $uv = 0$. Applying this to $t = 13$, $d = d_0$, gives degree $2 \cdot 2 \cdot 2 = 8$. The field contains the imaginary quadratic field K_0 , so its signature is $(0, 4)$; the unit theorem gives rank three.

The discriminant of F is 117. At a prime outside 117 its residual polynomial is separable. Its roots lift in an unramified extension after extending the residue field to split it, by simple-root Hensel lifting. Hence its splitting field is unramified there. The root differences are also units at all such primes. $\square$

Fix roots $\alpha_1, \dots, \alpha_4$ in K . Let H be the class number of K , and w the number of its roots of unity. Their values are not assumed. The standard ideal and unit inputs, including the full logarithmic unit lattice, are supplied by [129, Chapters 3–5, Theorems 5.1 and 5.9]. There is no

need to adjoin g -th roots of unity to the arithmetic field: the geometric Kummer calculations take place over its algebraic closure.

Lemma 196.2 (Bad-prime depth). *For every primitive positive seed, $F(a, b) \equiv 1 \pmod{3}$, and $13 \mid F(a, b)$ implies $v_{13}(F(a, b)) = 1$.*

Proof. Modulo three, $F(a, b) = (a^2 + b^2)^2 = 1$. Modulo thirteen,

$$F(a, b) \equiv (a - b)^2(a^2 + 5ab + b^2).$$

The quadratic factor has nonsquare discriminant 8 modulo thirteen. Thus $13 \mid F$ forces $a \equiv b \not\equiv 0 \pmod{13}$. Writing $a = b + 13t$ and using $F(1, 1) = 13$, $F_x(1, 1) = 26$ gives $F(b + 13t, b) \equiv 13b^4 \pmod{169}$, proving exact depth one. $\square$

196.2. A count uniform in the exponent.

Theorem 196.3 (Fixed-residual cover count). *Let $g > 8$ and let V be a positive g -free integer. For fixed g, V , all primitive positive seeds $F(a, b) = VQ^g$, with arbitrary Q , lift to one of at most*

$$(686) \quad C_K 4^{8\omega(V)} g^9, \quad C_K = 9^{32} H^4 w^3,$$

curves over K , of the form

$$(687) \quad y_i^g = \frac{x - \alpha_i}{\xi_i(x - \alpha_4)}, \quad i = 1, 2, 3, \quad \xi_i \in K^\times.$$

Each smooth projective normalization is geometrically connected, with degree g^3 over $\mathbb{P}^1$ and genus $1 + g^2(g - 2)$. The constant is independent of g, V, Q ; repetitions in this list are allowed.

Proof. Set $L_i = a - \alpha_i b$ and factor ideals as

$$(L_i) = A_i B_i^g, \quad 0 \leq v_p(A_i) < g.$$

All these ideals are integral. Outside 3, 13, at most one L_i is divisible by a given prime $\mathfrak{p}$; otherwise the unit root difference forces divisibility of both a and b . Unramifiedness and $\prod_i L_i = VQ^g$ show that the nonzero remainder valuation is exactly $v_p(V)$. At each prime above $p \mid V$ it can be assigned to one of four ideals A_i . There are at most eight primes above p , giving at most 4^8 choices per rational prime.

There is no contribution above three. At a prime above thirteen the total valuation of $\prod_i L_i$ is at most its ramification index, which is at most eight. As $g > 8$, this entire contribution remains in the A_i . Bound its four allocations by 9^4 for each of at most eight primes above thirteen. The possible ordered quadruples (A_i) therefore number at most $9^{32} 4^{8\omega(V)}$. These are deliberately coarse bounds. The valuation argument also gives

$$(688) \quad \prod_i A_i = (V), \quad N A_i \leq V^8.$$

For each A_i , the class $[B_i]$ solves $[B_i]^g = [A_i]^{-1}$ and has at most H possibilities. Choose an integral ideal C_i in each possible class and a generator δ_i of $A_i C_i^g$. Write $B_i = (t_i) C_i$; then

$$L_i = \delta_i \varepsilon_i t_i^g, \quad \varepsilon_i \in \mathcal{O}_K^\times.$$

The four ideal classes have at most H^4 choices. For the three ratios L_i/L_4 only the three unit classes of $\varepsilon_i/\varepsilon_4$ matter. The unit theorem gives

$$|\mathcal{O}_K^\times / (\mathcal{O}_K^\times)^g| = \gcd(g, w) g^3 \leq w g^3.$$

Thus there are at most $w^3 g^9$ ratio-class choices, proving (686). These give actual K -points of (687), with $x = a/b$; positive x avoids the roots.

Over the algebraic closure, valuation at each α_i for $i \leq 3$ proves the three Kummer classes independent modulo g , including composite g . The degree is g^3 . The inertia at each of the four roots is g ; at α_4 it is the diagonal subgroup because the three simple poles adjoin the same root of a uniformizer. Infinity is unramified. Thus

$$2G_g - 2 = -2g^3 + 4g^2(g - 1), \quad G_g = 1 + g^2(g - 2).$$

$\square$

Corollary 196.4 (Small residuals, fixed and varying). *If $g > 8$ and $\lambda = \max\{1, \log V\}/g < \log 2$, then V is g -free, and a fixed (g, V) list has size $N_{g,V}$ satisfying*

$$(689) \quad \frac{\log \max\{1, N_{g,V}\}}{g} \leq 16\lambda + \frac{9 \log g + \log C_K}{g}.$$

For $0 < L < \log 2$, all seeds with exponent $g > 8$ and $1 \leq V \leq \exp(Lg)$ are covered by a combined list of size at most

$$(690) \quad C_K g^9 \exp(17Lg).$$

In particular, when $L = L_g \rightarrow 0$ and $g \rightarrow \infty$, the logarithm of this combined list size divided by g tends to zero.

Proof. For each p , $v_p(V) \leq \log V / \log 2 < g$. Also $\omega(V) \leq \log V / \log 2$, so $8 \log 4 \omega(V) \leq 16 \log V$. This proves (689). For the union, there are at most $\exp(Lg)$ integer residuals, each with at most $C_K g^9 \exp(16Lg)$ covers. Sum these bounds. $\square$

The factor for the choice of V is retained in (690); a bound for one fixed residual is not silently used as a bound for all moving residuals. The construction applies to prime exponents, twice a prime, and other $g > 8$ alike.

196.3. Coefficient heights and a strict unit-class boundary. Write h_K for the absolute logarithmic Weil height. For a reduced positive rational a/b , it is $\log \max(a, b)$.

Theorem 196.5 (Coefficient normalization). *The lists above may be chosen so that*

$$(691) \quad h_K(\xi_i) \leq 2 \log V + C'_K g.$$

Consequently the coefficient heights of the three equations are $O_K(\log V + g)$, uniformly in g, V, Q .

Proof. Choose a fixed integral representative of every ideal class, of norm at most M_K . By (688),

$$\log N(A_i C_i^g) \leq 8 \log V + g \log M_K.$$

A principal integral ideal J has a generator δ satisfying

$$(692) \quad h_K(\delta) \leq \frac{1}{8} \log N J + B_K.$$

Indeed, subtract the average $\log N J / 8$ from the four complex logarithmic absolute values of an integral generator. Reduce this sum-zero vector modulo the full logarithmic unit lattice into a fixed bounded fundamental domain. Multiplication by the corresponding unit keeps the generator integral and bounds every logarithmic absolute value by that average plus a fixed constant. The average is nonnegative; summing positive parts proves (692).

Choose the δ_i in this way. Then

$$h_K(\delta_i) \leq \log V + \frac{g}{8} \log M_K + B_K.$$

For fixed fundamental units $\epsilon_1, \epsilon_2, \epsilon_3$, every unit class modulo g -th powers has a representative $\rho = \omega \prod_j \epsilon_j^{e_j}$, with $0 \leq e_j < g$ and ω a root of unity. Its height is at most $g \sum_j h_K(\epsilon_j)$. Taking $\xi_i = (\delta_i / \delta_4) \rho$ for each required ratio class and using the multiplication and inversion height inequalities gives (691). These choices preserve every actual lift. $\square$

Lemma 196.6 (Linear height is necessary for some unit classes). *For a fixed fundamental unit ϵ_1 there is $c_K > 0$ such that, for every $g \geq 3$, every representative in $K^\times$ of the class $[\epsilon_1^{\lfloor g/2 \rfloor}]$ modulo $K^{\times g}$ has height at least $c_K g$.*

Proof. Write $k = \lfloor g/2 \rfloor$ and $\xi = \epsilon_1^k t^g$. If t is not a unit, some finite valuation of ξ has absolute value at least g . Applying the denominator contribution to ξ or to ξ^{-1} , whose heights agree, gives $h_K(\xi) \geq g \log 2 / 8$.

If t is a unit, its coordinates in the fundamental-unit basis give the exponent vector $(k + gn_1, gn_2, gn_3)$ for ξ , up to torsion. Its first coordinate has absolute value at least $\lfloor g/2 \rfloor \geq g/3$. Weil height on the real logarithmic unit space is a norm: the weighted sum is zero, so the sum of positive parts is half the weighted absolute sum. The fundamental-unit logarithms are linearly

independent. Equivalence with the maximum norm in these fixed coordinates therefore gives $h_K(\xi) \geq c'_K g/3$. $\square$

Thus arbitrary unit classes cannot all be normalized to height $o(g)$. They are already present in the unrestricted class list with trivial residual ideals. This does not show that these particular classes have actual positive seeds. To remove their possible contribution one must use the compatibility of actual points, rather than merely choosing different class representatives. The order- g term in (691) is not absorbed into $O(\log V)$.

196.4. The remaining point-height problem. Put $u = ab/(a+b)^2 \in (0, 1/4]$. Then

$$\frac{F(a,b)}{(a+b)^4} = 1 - u + u^2, \quad \frac{13}{16}(a+b)^4 \leq F(a,b) < (a+b)^4.$$

Every actual seed therefore satisfies

$$(693) \quad h_K(a/b) \geq \frac{g \log Q + \log V}{4} - \log 2 \geq \frac{g \log 7}{4} - \log 2.$$

A family with $\lambda \rightarrow 0$ would require points of height at least linear in g , on a list whose logarithmic size is $o(g)$ and whose coefficient heights are $O(g)$. These facts are compatible: the number of possible curves gives no height bound for a point on one curve. No estimate uniform in g for these point heights has been proved here. Moving residual support, large-prime exponents, and the two-step small-residual threshold remain unresolved.

This section proves ordinary arithmetic-geometric reductions. It does not assert a uniform Faltings theorem or Lean verification of the number-field, unit-lattice, or geometric arguments.

197. FORMAL GENERAL LUCAS IDENTITIES AND ACTUAL BAD-PRIME SUPPORT

The module `GeneralLucasBoundary.lean` contains ten declarations using the existing actual Eisenstein pair multiplication, norm and boundary. For arbitrary $w = a + b\zeta$ it defines

$$\tau(w) = 2a^3 + 3a^2b - 3ab^2 - 2b^3, \quad R(w) = N(w)^3.$$

The kernel checks the cubic trace and discriminant identities and the general shifted recurrence

$$\tau(w)^2 + 27P(w)^2 = 4N(w)^3, \quad P(w^2z) + R(w)P(z) = \tau(w)P(wz).$$

These are actual polynomial identities for all integer pairs w, z , with no primitivity or nonzero assumption. Ordinarily, the recurrence follows by multiplying the characteristic equation of w^3 by z^3 and taking its difference with its conjugate. The formal proof directly checks the integer polynomial identity.

For arbitrary integer parameters t, r , put $U_0 = 0$, $U_1 = 1$ and $U_{n+2} = tU_{n+1} - rU_n$. A general formal recurrence theorem proves

$$f(n+1) = U_{n+1}f(1) - rU_nf(0)$$

for every integer sequence with that recurrence. Applied to the actual Eisenstein powers, it gives

$$P(w^{n+1}z) = U_{n+1}P(wz) - N(w)^3U_nP(z), \quad (t, r) = (\tau(w), N(w)^3). \\ P(w^n) = P(w)U_n, \quad P(w) \mid P(w^n),$$

The zero exponent and zero-boundary cases are included. Norm multiplicativity is also lifted to every actual power. This is a general actual Lucas bridge; the whole cyclotomic factorization and its rank valuation classification are not silently inserted into its signature.

The separate module `QuarticSupportArithmetic.lean` contains eleven declarations. It imports the actual second norm $F(a, b) = N(ab, N(a, b))$ from the preceding continuation, proves polynomial congruence transfer for arbitrary integer coordinates, and checks

$$\gcd(a, b) = 1 \implies F(a, b) \equiv 1 \pmod{3}, \\ \gcd(a, b) = 1, \quad 13 \mid F(a, b) \implies 169 \nmid F(a, b).$$

The proof uses finite residue tables reduced in the kernel, transfers them to all coprime integers, and applies the exact identity

$$F(b + 13t, b) = 13b^4 + 169(2b^3t + 20b^2t^2 + 91bt^3 + 169t^4).$$

It then proves that, whenever $g \geq 2$ and $F(a, b) = VQ^g$ is an actual primitive integer profile, one has $13 \nmid Q$. Thus thirteen cannot be absorbed into a high-exponent extraction root. No positivity restriction is needed in these formal integer conclusions.

A fresh standalone Lake build with Lean 4.32.0 and warnings treated as errors accepted all twenty-one new declarations and recompiled forty-four unchanged dependency declarations. Each declaration has a printed-axiom inventory; only `propext`, `Classical.choice`, and `Quot.sound` occur. The exact source hashes and independent signature review are recorded in the checkpoint. This verifies neither the complete cyclotomic and geometric sections nor standard ABC, and is not a full repository build.

198. ACTUAL ROOT INTERVALS, COMPATIBLE COVERS AND MODULAR CONSTRAINTS

The sixth continuation advances three distinct arithmetic interfaces. Counting actual integer roots in an interval, with their exact rank stratification, removes an exponent factor from the prime-window error. This proves a larger finite-window result for most actual roots while retaining both the exceptional roots and the primes outside that window. A separate descent through a class-number-one quartic field preserves the conjugacy of the four factors. It reduces the actual coefficient choices and gives explicit smooth complete intersections over $\mathbb{Q}$. The uniform height of their individual points remains uncontrolled.

The independent modular direction attaches an actual degree-two Frey $\mathbb{Q}$ -curve to the second norm. Complete ordinary local calculations and explicitly named representation-theoretic inputs restrict the level and weight of a necessary residual newform. The candidate dimension budget includes both finite-flat and non-finite-flat branches at the exponent prime. These spaces need not be empty. An exactly studied non-CM boundary curve explains why one specified finite residue-product elimination can fail; its positive primitive local shadows are not global power solutions.

Ordinary proofs and independent agent review preceded eighteen new scoped Lean declarations verifying the actual Frey polynomial arithmetic. The full counting, geometric, Tate-algorithm and modularity arguments retain their ordinary status and external dependencies. No unrestricted signed tail estimate, uniform point-height theorem, or unbounded two-step radical family is supplied. Standard *abc* remains unproved and undisproved.

199. ACTUAL MOVING ROOTS THROUGH A GROWING PRIME WINDOW

We average over an actual interval of integer roots and retain every endpoint error in the prime-power congruence counts. Exact rank stratification and the spacing of its possible primes improve the window substantially. The conclusion concerns most roots in that interval; the primes beyond the window and its exceptional roots remain open.

Write $\mathcal{O} = \mathbb{Z}[\zeta]$, $\zeta^2 - \zeta + 1 = 0$, $N(a + b\zeta) = a^2 + ab + b^2$ and $P(a + b\zeta) = ab(a + b)$. For integers $n, B \geq 1$ and $B \leq k < 2B$, put

$$w_k = 3k + \zeta, \quad Q_k = N(w_k), \quad T_{k,n} = |P(w_k^n)|, \quad t_{k,n} = \log c_{k,n},$$

where $c_{k,n}$ is the largest side of the positive unit rotation of w_k^n . For real $5 \leq Y \leq Z$, define

$$E_{k,n}(Y, Z) = \sum_{Y < p \leq Z} (v_p(T_{k,n}) - 3)_+ \log p.$$

This is the primewise positive excess of the actual integer boundary in the specified finite window.

Lemma 199.1 (Actual heights and nonzero boundaries). *All these roots and their powers are primitive and unramified, their boundaries are nonzero, and*

$$t_{k,n} \geq n \log(3B), \quad \log |P(w_k^m)| \leq 3m \log(6B + 1) \quad (m \geq 1).$$

Proof. The root is primitive and $Q_k = 9k^2 + 3k + 1 \equiv 1 \pmod{3}$. Oriented Eisenstein factorization preserves primitivity under all powers. A zero boundary would make $w_k/\bar{w}_k$ a root of unity; primitivity and the absence of a ramified factor would then force w_k to be a unit, contrary to $Q_k \geq 13$. Unit rotation preserves norm and absolute boundary. Its positive coordinates give $Q_k^n \leq c_{k,n}^2$, while $Q_k \geq (3B)^2$. Also $Q_k \leq (6B+1)^2$ and the cubic difference identity yields $|P(w_k^m)| \leq Q_k^{3m/2}$. $\square$

Lemma 199.2 (Simple roots after removing the exponent's prime part). *For $p > 3$ and $p \nmid m$, every root modulo p of $G_m(X) = P((3X + \zeta)^m) \in \mathbb{Z}[X]$ is simple. There are at most $3m$ such roots, each lifting uniquely at every precision. If $a_p = v_p(n)$ and $m_p = n/p^{a_p}$, then at each actual k*

$$v_p(T_{k,n}) = \begin{cases} v_p(|G_{m_p}(k)|) + a_p, & v_p(|G_{m_p}(k)|) > 0, \\ 0, & v_p(|G_{m_p}(k)|) = 0. \end{cases}$$

Proof. Over the split algebra or the quadratic extension of $\mathbb{F}_p$, let $z, \bar{z}$ be the distinct roots of $X^2 - X + 1$. The identity

$$3(z - \bar{z})G_m(X) = (3X + z)^{3m} - (3X + \bar{z})^{3m}$$

shows that neither factor $3X + z$, $3X + \bar{z}$ can vanish at a root. Their ratio has derivative $3(\bar{z} - z)/(3X + \bar{z})^2 \neq 0$. Since $p \nmid 3m$, differentiating its $3m$ -th power at the root gives a nonzero result. Thus the root is simple, and the polynomial degree bound and simple-root lifting prove the first assertions.

If $p \nmid Q_k$, its homogeneous rank is prime to p , so it divides n if and only if it divides m_p . The established rank and lifting law gives the displayed identity. If $p \mid Q_k$, primitivity implies that p divides neither boundary; both valuations are zero. Consequently, at every actual root,

$$(v_p(T_{k,n}) - 3)_+ \leq (v_p(|G_{m_p}(k)|) - 3)_+ + v_p(n).$$

The total additional lifting cost in any window is at most $\log n$. $\square$

Proposition 199.3 (An unstratified actual interval estimate). *One has*

$$\frac{1}{B} \sum_{B \leq k < 2B} \frac{E_{k,n}(Y, Z)}{t_{k,n}} \leq \frac{10 \log Y}{Y^3 \log(3B)} + \frac{18n\pi(Z)}{B} + \frac{\log n}{n \log(3B)}.$$

Proof. For $m = m_p$, each of the at most $r_{p,m} \leq 3m$ lifted residue classes meets the interval in at most $B/p^e + 1$ integers. The cap $H = \lfloor 3m \log(6B+1)/\log p \rfloor$ covers every actual depth. Summing the positive-excess layers $4 \leq e \leq H$ gives

$$\sum_k (v_p(|G_m(k)|) - 3)_+ \log p \leq \frac{3mB \log p}{p^3(p-1)} + 9m^2 \log(6B+1).$$

Empty layer ranges contribute zero. Use $m \leq n$, divide by $Bn \log(3B)$, and use $\log(6B+1) \leq 2 \log(3B)$. The elementary estimate

$$(694) \quad 3 \sum_{p > Y} \frac{\log p}{p^3(p-1)} \leq \frac{10 \log Y}{Y^3} \quad (Y \geq 5)$$

follows by $p/(p-1) \leq 5/4$ and comparison of the decreasing function $\log x/x^4$ with its integral from $\lfloor Y \rfloor$. Its integral is $(\log \lfloor Y \rfloor / 3 + 1/9) / \lfloor Y \rfloor^3$; $\lfloor Y \rfloor \geq 4Y/5$ gives the stated loose constant. Add the lifting cost from Lemma 199.2. Every denominator comparison has a nonnegative numerator. $\square$

Lemma 199.4 (The exact rank count). *For $p > 3$, put $\chi_p = 1$ at split primes and -1 at inert primes, and $N_p = p - \chi_p$. For each integer $d \geq 1$, the number of residues $k \bmod p$ for which $((3k + \zeta)/(3k + \bar{\zeta}))^3$ has exact order d is*

$$r_p(d) = \begin{cases} 3\varphi(d) - \mathbf{1}_{d=1}, & d \mid N_p/3, \\ 0, & \text{otherwise.} \end{cases}$$

Only nonzero-norm residues enter this definition. Each counted root of G_d lifts uniquely to every p^e , retaining its exact rank modulo p .

Proof. The fractional-linear map $a \mapsto (a + \zeta)/(a + \bar{\zeta})$ is a bijection onto the norm-one group minus its identity. Its inverse is $(R\bar{\zeta} - \zeta)/(1 - R)$; conjugation fixes this inverse in the inert case, and the split case follows directly in the split algebra. The norm-one group is cyclic of order N_p , divisible by three. Cubing has kernel three and image of order $N_p/3$. Each order- d element has three preimages, and only at $d = 1$ must the omitted identity be subtracted. Multiplication by three permutes the parameter residues. Since $p \nmid d$, Lemma 199.2 applies. Retaining the initial residue retains its rank. $\square$

Theorem 199.5 (A rank-stratified actual prime window). *For all the above data,*

$$(695) \quad \frac{1}{B} \sum_{B \leq k < 2B} \frac{E_{k,n}(Y, Z)}{t_{k,n}} \leq \frac{10 \log Y}{Y^3 \log(3B)} + \frac{12(Z+1)}{B} + \frac{\log n}{n \log(3B)}.$$

Proof. At each supported prime use its actual rank d and first depth $s = v_p(T_{k,d})$. Then $d \mid n$, $d \mid N_p/3$, and $v_p(T_{k,n}) = s + v_p(n)$. It suffices to count $(s-3)_+$, with at most $\log n$ additional lifting cost at each root.

For each fixed rank-prime pair, Lemma 199.4 gives at most $3\varphi(d)(B/p^e + 1)$ parameters at depth at least e . The cap $\log |G_d(k)| \leq 3d \log(6B+1)$ yields the bound

$$\frac{3\varphi(d)B \log p}{p^3(p-1)} + 9\varphi(d)d \log(6B+1)$$

for their first-depth positive cost. At fixed p , summing the first term over $d \mid \gcd(n, N_p/3)$ uses $\sum \varphi(d) = \gcd(n, N_p/3) \leq n$ and (694).

For the endpoint errors, reverse the order of summation. A prime of rank d lies in $p = 3dj + 1$ or $p = 3dj - 1$, with $j \geq 1$. Thus at most $2(Z+1)/(3d)$ candidates occur up to Z . The excluded integer $p = 1$ is not counted, including when $d = 1$. Therefore

$$\sum_{d \mid n} 9\varphi(d)d \log(6B+1) \frac{2(Z+1)}{3d} = 6n(Z+1) \log(6B+1).$$

Here the depth-cap factor d cancels the progression spacing. Divide by $Bn \log(3B)$, use $\log(6B+1) \leq 2 \log(3B)$, and add the lifting cost. Norm-dividing primes have zero boundary valuation, so the rank partition omits no supported prime. $\square$

Corollary 199.6 (A window entering the maximal-rank range). *Take $B = n^4$, $Y = n^{1/6}$ and $Z = n^3$. For sufficiently large n ,*

$$\frac{1}{B} \sum_k \frac{E_{k,n}(Y, Z)}{t_{k,n}} \leq \frac{5}{12\sqrt{n}} + \frac{12}{n} + \frac{12}{n^4} + \frac{1}{4n}.$$

The proportion for which $E_{k,n}/t_{k,n} > n^{-1/4}$ is at most

$$\frac{5}{12n^{1/4}} + \frac{12+1/4}{n^{3/4}} + \frac{12}{n^{15/4}}.$$

Proof. Substitute into (695), use $\log(3B) \geq 4 \log n$, and apply Markov's inequality to the nonnegative actual cost. $\square$

All these profiles are actual homogeneous profiles with unit residual and $\lambda = 1/n$; their root norms are comparable to n^8 . The window includes possible maximal-rank primes even when n is prime. More generally, $Z = o(B)$ makes the discrepancy vanish whenever the main and lifting terms vanish. For $B = n^4$ one may take $Z = \lfloor B/(\log n)^2 \rfloor$, obtaining average $O((\log n)^{-2} + n^{-1/2})$ and exceptional proportion $O(1/\log n)$ at threshold $1/\log n$.

These are finite-window statements about most actual roots. They do not bound a specified exceptional root, primes above Z , or the complete signed tail of any prescribed orbit. The original unstratified bound also gives, for every fixed integer $A \geq 2$, vanishing average in $B = n^{A+2}$ through $Z = n^A$. The rank stratification is stronger and uses no prime-distribution estimate.

The full ordinary proof was independently reviewed, including the split and inert rank counts, the prime-part removal, endpoint caps, and the exceptional-proportion constants. Separate exact replays check actual integer blocks and complete finite residue spaces at specified precisions. They are diagnostics of the ordinary proof, not a complete-tail certificate or a Lean formalization.

200. ACTUAL CONJUGATE COMPATIBILITY IN ONE QUARTIC FIELD

The splitting-field bound can be sharpened by retaining the fact that the four linear factors come from one actual algebraic integer. Put

$$f(X) = X^4 + 3X^3 + 5X^2 + 3X + 1, \quad L = \mathbb{Q}(\alpha), \quad f(\alpha) = 0,$$

and let K be its degree-eight splitting field. In this section a, b are positive coprime integers and

$$F(a, b) = VQ^g, \quad g \geq 3, \quad V, Q \geq 1, \quad V \text{ is } g\text{-free}.$$

Shared support of V and Q is allowed. All heights are absolute logarithmic Weil heights.

Theorem 200.1 (A principal quartic root field). *The field L has degree four, signature $(0, 2)$, class number one, and unit rank one. Its discriminant has absolute value at most 117.*

Proof. Over $\mathbb{Q}(\zeta)$ the two factors of f are $X^2 + (2 - \zeta)X + 1$ and $X^2 + (1 + \zeta)X + 1$. Choose α in the first. Then $\zeta = \alpha + \alpha^{-1} + 2$ and $L = \mathbb{Q}(\zeta, \sqrt{-1 - 3\zeta})$. The norm of $-1 - 3\zeta$ is 13, so it is not a square in $\mathbb{Q}(\zeta)$. This proves the degree and signature. The exact resultant calculation gives order discriminant 117, so the field discriminant is no larger and the order index is supported only at three. Minkowski's ideal-class bound [129, Theorem 4.3] gives every class an integral representative of norm at most

$$\left(\frac{4}{\pi}\right)^2 \frac{4!}{4^4} \sqrt{|\text{Disc } L|} \leq \frac{3\sqrt{117}}{2\pi^2} < \frac{11}{6} < 2.$$

The integral norm must be one, so every class is trivial. Dirichlet's unit theorem gives rank one. Fix $\mathcal{O}_L^* = \mu_L \langle \epsilon \rangle$ and $w_L = |\mu_L|$. These classical number-field theorems are external inputs. $\square$

Theorem 200.2 (Extraction in the actual root field). *There are integral ideals $\mathfrak{A}, \mathfrak{B}$ with*

$$(a - \alpha b) = \mathfrak{A}\mathfrak{B}^g, \quad N\mathfrak{A} = V, \quad N\mathfrak{B} = Q.$$

For fixed V , at most $4^{\omega(V)}$ choices for $\mathfrak{A}$ occur. Choosing one integral generator δ of each such ideal, every actual seed has an expression

$$a - \alpha b = \delta u \beta^g, \quad \beta \in \mathcal{O}_L, \quad u \in \{\xi \epsilon^k : \xi \in \mu_L, 0 \leq k < g\}.$$

Proof. The norm of $a - \alpha b$ is $F(a, b)$. If $p \nmid 117$ divides this norm, then b is nonzero modulo p . Every prime ideal dividing the element has $\alpha \equiv a/b$ in its residue field. Factorization away from the order index and the distinct roots of f modulo p show that exactly one prime ideal above p divides the element. Its residue degree is one and its valuation is $v_p(F)$; there are at most four possible such ideals as the seed varies.

At three no factor occurs. At thirteen the primitive exact-depth theorem gives $v_{13}(F) = 1$ whenever thirteen occurs. The norm formula $\sum_{\mathfrak{p}|13} f_{\mathfrak{p}} v_{\mathfrak{p}}(a - \alpha b) = 1$ again forces one degree-one prime of depth one. This argument makes no unramifiedness assumption at thirteen. Write $v_p(F) = r_p + gq_p$ with $r_p = v_p(V) < g$ and $q_p = v_p(Q)$, and allocate these exponents at the unique prime above p . This proves the ideal identities and the count. Principality supplies integral generators; their quotient is a unit, which can be reduced modulo unit g th powers by changing β . $\square$

Theorem 200.3 (A smaller compatible cover budget). *For fixed V, g , every actual seed lifts to one of at most*

$$4w_L 4^{\omega(V)} g$$

covers over the fixed field K . Each is geometrically connected of degree g^3 over $\mathbb{P}^1$ and genus $1 + g^2(g - 2)$. Its three coefficients ξ_i can be chosen with

$$h(\xi_i) \leq \frac{1}{2} \log V + C_L g.$$

For $\lambda = \max\{1, \log V\}/g$ the logarithm of the cover count divided by g is at most

$$2\lambda + \frac{\log g + \log(4w_L)}{g}.$$

Allowing all integral $V \leq e^{Ag}$ costs at most $4w_L g e^{3Ag}$ covers.

Proof. Put $\kappa = \delta u$ and apply the four embeddings $\sigma_i : L \rightarrow K$. All four equations are conjugates of the same equation. Their ratios give

$$y_i^g = \frac{x - \alpha_i}{\xi_i(x - \alpha_4)}, \quad \xi_i = \frac{\sigma_i(\kappa)}{\sigma_4(\kappa)}, \quad x = a/b, \quad 1 \leq i \leq 3.$$

The ideal choice and the single unit class determine all three ratios; there are no three independent splitting-field unit choices. The looser factor four in the stated count is harmless. Divisors at the four distinct roots give Kummer independence also for composite g . Each of the four inertia groups has order g , so Riemann–Hurwitz gives the stated genus.

Balance the two complex logarithmic absolute values of δ in a fixed interval for the unit lattice. Their mean is $(\log V)/4$, and integrality gives $h(\delta) \leq (\log V)/4 + C$. Since $h(u) \leq gh(\epsilon)$, the height inequalities for a ratio yield $h(\xi_i) \leq 2h(\delta u) \leq (\log V)/2 + C_L g$. Finally $4^{\omega(V)} \leq V^2$. Summing the resulting bound over at most e^{Ag} residual integers proves the union estimate, including the cost of choosing V . $\square$

Theorem 200.4 (A rational complete-intersection model). *Each compatible class above has a smooth geometrically connected complete-intersection model of degrees (g, g) in $\mathbb{P}_{\mathbb{Q}}^3$, capturing all its actual seeds. Its defining forms have logarithmic projective coefficient heights $O(\log V + g)$, with fixed-field constants.*

Proof. Write $\beta = X_0 + X_1\alpha + X_2\alpha^2 + X_3\alpha^3$. Expand $\kappa\beta^g$ in this rational basis and set the coefficients of α^2, α^3 equal to zero. These are two homogeneous degree- g forms over $\mathbb{Q}$. Every actual seed supplies a rational point; the reverse implication needs separate integral and primitive conditions.

Over an algebraic closure use the invertible embedding coordinates $Z_i = \sigma_i(\beta)$. The image $(\sigma_i(\kappa)Z_i^g)_i$ must lie on the projective line $(a - \alpha_i b)_i$. The line meets the four coordinate hyperplanes at distinct points, so at most one Z_i vanishes. The intersection is nonempty by taking g th roots of any line point. A dependence of the two gradients would give a nonzero vector in the annihilator of the line supported solely on a vanishing coordinate. No coordinate vanishes identically on the line, so this is impossible. The intersection is therefore smooth of pure dimension one. Kummer independence makes the chart $Z_4 \neq 0$ connected; its complement is finite and contains no hidden curve component. The genus is $1 + g^2(g - 2)$.

In the fixed basis, the rational coordinate height of κ is $O(\log V + g)$. Clear its common denominator first. The monic relation for α bounds all multiplication coefficients in its powers exponentially in g , while the multinomial coefficients have total sum 4^g . Thus both defining forms have the asserted coefficient height. $\square$

These statements keep actual conjugate compatibility and improve the preceding cover budget. They do not identify generic hard unit classes with actual seeds. A small number of covers and controlled coefficient heights still supply no uniform height bound for their individual points. The actual lower bound $h(a/b) \geq (g \log Q + \log V)/4 - \log 2$ is consistent with all these results. The uniform actual-point and simultaneous-compression questions remain open.

201. A FREY Q-CURVE AND A RESIDUAL MODULAR-LEVEL BUDGET

This section gives a modular entry for the second mixed norm. Let

$$F = F(a, b) = a^4 + 3a^3b + 5a^2b^2 + 3ab^3 + b^4, \quad a, b \in \mathbb{Z}_{>0}, \quad \gcd(a, b) = 1,$$

and put $x = a^2 + b^2$, $y = a + b$. We derive an exact conductor over $\mathbb{Q}(\sqrt{-3})$ and a finite set of necessary modular levels for the pure-power branch. The remaining newforms are not eliminated. All representation-theoretic inputs are ordinary mathematical inputs; this is not a full Lean formalization of Tate’s algorithm or modularity.

201.1. The actual generalized-Fermat coordinates.

Lemma 201.1 (The square gate and two parity charts). *One has*

$$(696) \quad x^2 + 3y^4 = 4F, \quad 2x - y^2 = (a - b)^2, \quad \gcd(x, y) \mid 2.$$

If a, b have opposite parity, then $x, y, d = a - b$ are odd, $\gcd(x, y) = 1$, $d^2 = 2x - y^2$, and $|d| < y$. Conversely, these conditions with $y > 0$ recover a primitive positive seed by $a = (y + d)/2$, $b = (y - d)/2$.

If a, b are both odd, then $X = x/2$, $Y = y/2$, $D = (a - b)/2$ satisfy

$$(697) \quad X^2 + 12Y^4 = F, \quad X - Y^2 = D^2, \quad |D| < Y, \quad X \text{ odd}, \quad \gcd(X, Y) = 1.$$

Conversely these conditions with $Y > 0$ recover the primitive positive odd seed $a = Y + D$, $b = Y - D$.

Proof. The identities follow by expansion. A common divisor of x, y divides $2ab$, while $\gcd(ab, a + b) = 1$. In the first inverse, an odd common prime of a, b would divide x, y , and both seeds cannot be even since x is odd. In the second inverse, $X = Y^2 + D^2$ odd makes Y, D opposite in parity, and $\gcd(X, Y) = 1$ implies $\gcd(Y, D) = 1$. The strict inequalities give positivity in both cases. $\square$

An arbitrary solution of $x^2 + 3y^4 = 4VQ^g$ is therefore not yet an actual seed: it must satisfy the additional square and positivity gate. Also, the known equation $A^4 + 3B^2 = C^p$ has different coefficients. The substitution $A = 3y, B = 3x$ gives $A^4 + 3B^2 = 108VQ^g$ with non-coprime parameters, so its equation-specific non-existence results do not apply.

The quartic F is odd and $F \equiv 1 \pmod{3}$. The latter follows from $F \equiv (a^2 + b^2)^2 \pmod{3}$; in particular $3 \nmid x$. Lemma 196.2 gives the exact additional fact $13 \mid F \Rightarrow v_{13}(F) = 1$. Thus

$$(698) \quad F = Q^g, \quad g \geq 2 \quad \Rightarrow \quad \gcd(Q, 78) = 1.$$

201.2. An explicit degree-two Q -curve. Put $r = \sqrt{-3}$, $K = \mathbb{Q}(r)$ and $\alpha_{\pm} = 3y^2 \pm xr$. Attach

$$(699) \quad E_{a,b}: \quad \mathcal{Y}^2 = \mathcal{X}^3 + 12y\mathcal{X}^2 + 6\alpha_+\mathcal{X}.$$

Lemma 201.2 (Isogeny and invariant identities). *The curve (699) is a degree-two Q -curve, with the displayed isogenies defined over $K(\sqrt{-2})$. Its invariants are*

$$(700) \quad \begin{aligned} \alpha_+\alpha_- &= 12F, & c_4 &= 2^5 3^2 (5y^2 - xr), \\ c_6 &= 2^8 3^3 y(3xr - 7y^2), & \Delta &= 2^{11} 3^4 F \alpha_+. \end{aligned}$$

At every rational prime $q > 3$ not dividing F it has good reduction. If $q \mid F$, then q splits in K , the displayed equation is minimal and multiplicative at both primes, and, for $m = v_q(F)$,

$$(701) \quad \{v_q(\Delta_{\min}), v_{\bar{q}}(\Delta_{\min})\} = \{m, 2m\}.$$

Proof. Write $A = 12y$ and $B = 6\alpha_+$. The quotient by the order-two point $(0, 0)$ has coefficients $-2A = -24y$ and $A^2 - 4B = 24\alpha_-$. Hence it is the quadratic twist by -2 of the conjugate curve. This is the explicit isogeny construction also used in [130, Proposition 2.2]; it does not require a primitive solution of the equation studied there. The formulas $c_4 = 16(A^2 - 3B)$, $c_6 = -64A^3 + 288AB$ and $\Delta = 16B^2(A^2 - 4B)$ prove (700).

If $q \nmid F$, the discriminant is a unit. If $q \mid F$, neither x nor y is divisible by q : otherwise (696) would force both, contradicting $\gcd(x, y) \mid 2$. The same equation makes -3 a nonzero square modulo q , so q splits. The elements α_+, α_- cannot both vanish at a prime above q , since their sum and difference would force x, y to vanish. Their local valuations are $m, 0$ in one order. The factor $5y^2 - xr$ in c_4 reduces to $8y^2$ or $2y^2$, a unit. Thus the equation is minimal and multiplicative, with the two valuations in (701).

Since $F \geq 13$ and $\gcd(F, 6) = 1$, such a multiplicative prime always exists. Its negative valuation of j excludes complex multiplication, because CM j -invariants are algebraic integers. In the non-CM case the isogenies to the conjugate form a rank-one module, so their degrees are the minimum degree times integer squares. The existing degree-two isogeny forces the minimum degree to be exactly two. $\square$

201.3. Uniform Tate calculations at two and three. Let $\mathfrak{p}_2 = (2)$, an inert prime with residue field $\mathbb{F}_4$, and $\mathfrak{p}_3 = (r)$, with $v_{\mathfrak{p}_3}(3) = 2$. We use valuations normalized by $v_{\mathfrak{p}_2}(2) = 1$ and $v_{\mathfrak{p}_3}(r) = 1$.

Theorem 201.3 (The exact conductor over the quadratic field). *The curve has minimal discriminant valuation 12, Kodaira type I_2^* and conductor exponent 6 at $\mathfrak{p}_2$. At $\mathfrak{p}_3$ the corresponding data are 9, III^* and 2. Consequently*

$$(702) \quad \mathcal{N}(E) = \mathfrak{p}_2^6 \mathfrak{p}_3^2 \prod_{q|F, q>3} q\bar{q}.$$

Proof. At $\mathfrak{p}_3$, $3 \nmid x$ gives $v(\alpha_+) = 1$, hence $v(\Delta) = 9$. The coefficients satisfy

$$a_1 = a_3 = a_6 = 0, \quad v(a_2) \geq 2, \quad v(a_4) = 3.$$

The auxiliary cubic in Tate's additive normalization is T^3 . After its zero triple root, the quadratic for the IV^* test is T^2 . The stronger divisibility of a_3, a_6 after its zero repeated root already holds. The next test is nonzero since a_4/r^3 is a unit; thus the type is III^* . Its eight geometric components give conductor exponent $9 + 1 - 8 = 2$. These branch tests also establish minimality.

At $\mathfrak{p}_2$, put $\zeta = (1 + r)/2$. For integers U, V and $n \geq 1$, the integral-basis identity $U + Vr = (U - V) + 2V\zeta$ gives

$$(703) \quad U + Vr \in 2^n \mathcal{O}_{K, \mathfrak{p}_2} \iff 2^n \mid U - V, \quad 2^{n-1} \mid V.$$

Conjugation preserves the valuation and $\alpha_+ \alpha_- = 12F$, so $v(\alpha_+) = 1$ and $v(\Delta) = 12$. Make the integral substitution $\mathcal{X} = X' + R$, $\mathcal{Y} = Y' + T$ specified by the following table.

condition	R	T
$y \equiv 0 \pmod{4}$	2	$2(r + 1)$
$y \equiv 2 \pmod{4}$	2	$2(r - 1)$
y odd, even seed 0 $\pmod{4}$	$1 + r$	4
y odd, even seed 2 $\pmod{4}$	$1 + r$	$2(r + 1)$

The transformed coefficients are

$$(704) \quad \begin{aligned} a'_1 &= 0, & a'_2 &= A + 3R, & a'_3 &= 2T, \\ a'_4 &= B + 2AR + 3R^2, & a'_6 &= R^3 + AR^2 + BR - T^2. \end{aligned}$$

We verify the uniform pattern

$$(705) \quad v(a'_2) = 1, \quad v(a'_3) \geq 3, \quad v(a'_4) = 3, \quad v(a'_6) \geq 5.$$

First suppose y is even. Then $x \equiv 2 \pmod{8}$ and $a'_2 = 12y + 6$ has valuation one. With $T = 2(r + s)$, where $s = 1$ or -1 according to the table, write

$$a'_4 = 6(H + xr), \quad H = 3y^2 + 8y + 2, \quad a'_6 = C + Dr,$$

where $C = 16 + 48y + 36y^2$ and $D = 12x - 8s$. Both $H/2, x/2$ are odd. For odd integers u, v , the norm $u^2 + 3v^2 \equiv 4 \pmod{8}$ makes $v_{\mathfrak{p}_2}(u + vr) = 1$. This proves $v(a'_4) = 3$. Also $16 \mid D$ and

$$(C - D)/4 = 4 + 12y + 9y^2 - 3x + 2s \equiv 0 \pmod{8},$$

using $x \equiv 2 \pmod{8}$ and the two even values of $y \pmod{4}$. Equation (703) gives $v(a'_6) \geq 5$. Finally $v(T) = 2$, giving $v(a'_3) = 3$.

If y is odd, then

$$a'_2 = 3(4y + 1 + r), \quad a'_4 = 6(H + Jr), \quad H = 3y^2 + 4y - 1, \quad J = x + 4y + 1.$$

The first bracket has valuation one; $H/2, J/2$ are odd, so again $v(a'_4) = 3$. For $T = 4$, the coefficients of $a'_6 = C + Dr$ obey

$$D = 24y + 18y^2 + 6x, \quad C - D = -24(1 + 2y + x).$$

Here $x \equiv 1 \pmod{8}$ makes $16 \mid D$, and $x \equiv 1 \pmod{4}$ gives $32 \mid C - D$. For $T = 2(r + 1)$ the corresponding formulas are

$$D = 24y + 18y^2 + 6x - 8, \quad C - D = 8(1 - 6y - 3x).$$

Now $x \equiv 5 \pmod{8}$ proves $16 \mid D$, and again $32 \mid C - D$. In both cases $v(a'_3) = 3$, completing (705).

Tate's auxiliary cubic is $Z^2(Z + b)$, where $b = a'_2/2$ is nonzero in $\mathbb{F}_4$. Its double root is zero. The first quadratic in the I_n^* branch is Z^2 , since $a'_3/4, a'_6/16$ vanish in the residue field. No further translation is needed: $8 \mid a'_3$ and $32 \mid a'_6$ already hold. The second quadratic is

$$(a'_2/2)Z^2 + (a'_4/8)Z + a'_6/32,$$

whose derivative $a'_4/8$ is a unit. The algorithm stops at I_2^* ; its seven geometric components give conductor exponent $12 + 1 - 7 = 6$. All changes have scaling one. These are residue-field polynomial tests, valid over $\mathbb{F}_4$; no shortcut peculiar to $\mathbb{F}_2$ is used. For the explicit double-root and triple-root branches see [132, Chapter 3, pp. 66–68]. $\square$

201.4. Five necessary levels in the pure-power branch.

Theorem 201.4 (Residual modular levels). *If $F(a, b) = Q^p$ for $Q \in \mathbb{Z}_{>0}$ and a prime $p > 7$, then there is a weight-two newform with character $\epsilon = \psi_3$ of conductor 12, and level*

$$(706) \quad N = 2^e 3^2, \quad 2 \leq e \leq 6, \quad N \in \{36, 72, 144, 288, 576\},$$

whose residual representation restricts to the specified twist of $E_{a,b}[p]$ over K .

Proof. We spell out the external representation inputs because the level formula for the different Fermat equation cannot be copied. By [130, Theorem 3.2 and Table 3.3], with $d = 3$, there is a fixed order-four character χ of G_K satisfying

$$\text{cond}(\chi) \mid \mathfrak{p}_2^3, \quad \chi^2 = \epsilon|_{G_K}, \quad {}^\sigma\chi = \chi\psi_{-2}|_{G_K}.$$

In particular it is unramified at three. This character construction does not require a primitive Fermat solution. The isogeny of Lemma 201.2 makes $\rho_E \otimes \chi$ invariant under quadratic conjugation. The extension proof in [130, Theorem 4.2] applies to this non-CM Q-curve. Its oddness invokes the general Q-curve result of Ribet cited there; it does not follow from Schur's lemma alone. The determinant is $\epsilon\chi_{\text{cyc}}$.

At every $q > 3$, Lemma 201.2 makes the minimal discriminant valuations divisible by p . The Tate-curve criterion gives unramified p -torsion away from $6p$ and finite flat p -torsion at p . Indeed, at p the Tate parameter has valuation divisible by p , so its Kummer class has a unit representative; good reduction gives finite flatness directly. The extension $K/\mathbb{Q}$ and the character χ are unramified at p , preserving these properties under twist and descent. The same local mechanism is explained in [131, Propositions 3.6–3.9].

At $\mathfrak{p}_2$, the curve is potentially good: the invariant formulas give $v(c_4) = 6$, $v(\Delta) = 12$, hence $v(j) = 6 > 0$. On every finite higher inertia group its characteristic-zero representation has determinant one. Its fixed subspace cannot have dimension one: semisimplicity would make the complementary character equal to the determinant, hence trivial as well. Thus its ramification has a single break. The conductor exponent 6 gives break 2. The character χ has break at most 2, so the twist has conductor exponent at most 6. Since $K_{\mathfrak{p}_2}/\mathbb{Q}_2$ is unramified, restriction preserves that exponent. Its determinant has conductor exponent 2, furnishing the lower bound $e \geq 2$.

At three, the curve is potentially good and has conductor exponent two, so it has no wild ramification. Twisting by χ preserves this. The quadratic extension $K_{\mathfrak{p}_3}/\mathbb{Q}_3$ is tame, so both extensions to $G_{\mathbb{Q}_3}$ have trivial wild inertia and conductor exponent at most two. They differ by ψ_{-3} . The conductor of their direct sum, the induced representation, has exponent $2 \cdot 1 + 1 \cdot 2 = 4$ by [130, Theorem 4.1]. Hence each exponent equals two.

The big-image input is the general Q-curve statement [130, Theorem 5.2]: squarefree isogeny degree and a multiplicative prime greater than three suffice. Lemma 201.2 checks both conditions, independently of the nonprimitive substitution $(3y, 3x)$. For $K = \mathbb{Q}(\sqrt{-3})$ the source gives $N_3 = 7$; the improvement uses [131, Proposition 5.4]. This is a published external input, not a new modular-symbol certificate in this repository.

Large image, oddness and finite flatness give Serre weight two and the listed prime-to- p levels, by the modularity and lowering arguments in the cited sources. The finite inertial images at two

and three have order supported at two and three; adjoining the order-four character and the quadratic descent does not change that support. Reduction modulo $p > 7$ therefore preserves their conductors and the character ϵ . This proves the necessary level list. $\square$

201.5. A direct budget for moving residual support.

Theorem 201.5 (Residual level and dimension budgets). *Suppose $F = VQ^p$, where $V, Q \in \mathbb{Z}_{>0}$, $p > 7$ is prime, V is p -power-free, and $p \nmid V$. Then the necessary weight-two newform has character $\epsilon = \psi_3$ and exact prime-to- p level*

$$(707) \quad N = 2^e 3^2 \text{rad}(V), \quad 2 \leq e \leq 6.$$

For $\lambda = \max\{1, \log V\}/p$ this implies

$$(708) \quad \frac{\log N}{p} \leq \lambda + \frac{\log 576}{p}.$$

For all $1 \leq V \leq \exp(Lp)$ together, with $L \geq 0$, the sum of the dimensions of the five possible newspaces is at most $965 \exp(3Lp)$. In particular this budget is subexponential when $L \rightarrow 0$ and $p \rightarrow \infty$.

Proof. At $q > 3$, write $m = v_q(V) + pv_q(Q)$. The two local minimal discriminant valuations are $m, 2m$. The Tate residual inertia is nontrivial precisely when p does not divide its parameter's valuation. Since p is odd and V is p -power-free, this occurs precisely at $q \mid V$. The twist is unramified at these primes. At p , the condition $p \nmid V$ ensures finite flatness. All other steps in Theorem 201.4 are unchanged, proving (707). Then $N \leq 576V$ gives (708).

Let $D = \text{rad}(V)$; it is prime to six. The index satisfies

$$I = [\text{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = 2N \prod_{q \mid D} (1 + q^{-1}) \leq 2ND \leq 1152V^2.$$

The weight-two Sturm bound gives dimension at most $\lfloor I/6 \rfloor + 1 \leq 193V^2$ for each space. The five dimensions sum to at most $965V^2$. Put $H = \lfloor \exp(Lp) \rfloor$. Even allowing inadmissible V , there are at most H terms of size at most $965H^2$, giving the asserted bound. Dimensions count a Galois orbit with its coefficient-field degree, not merely as one orbit. The index and Sturm-bound conventions agree with [133, mfsturm]. $\square$

The condition $p \nmid V$ cannot be inferred from a small λ ; Section 201.6 includes the non-finite-flat branch $p \mid V$ at weight $p + 1$, with uniform candidate elimination still open. The level budget does not make the newspaces empty and gives no seed height bound. Even in the pure branch the five spaces must be examined against the actual seed family and its inverse square gate. If a prime p divides a larger pure exponent g , write $Q^g = (Q^{g/p})^p$. An eventual contradiction for all large p , together with the earlier fixed-exponent finiteness, would give finiteness for pure-power seeds admitting an exponent $g \geq 3$. No such contradiction is proved here. The known infinite square family is outside that proposed implication. The moving Kummer-cover and point-height routes remain open alongside this modular entry.

201.6. The non-finite-flat branch and a uniform modular budget. The condition $p \nmid V$ in Theorem 201.5 can be removed by retaining a second weight. This is a necessary modular condition and a dimension estimate; candidate elimination remains an additional problem.

Theorem 201.6 (Both residual-weight branches). *Let a, b be coprime positive integers, let $p > 7$ be prime, and suppose*

$$F(a, b) = VQ^p, \quad V, Q \in \mathbb{Z}_{>0},$$

where V is p -power-free. Put $r = v_p(V)$, $R = V/p^r$, and

$$(709) \quad k(V, p) = \begin{cases} 2, & r = 0, \\ p + 1, & r > 0. \end{cases}$$

The descended residual representation constructed in Theorem 201.4 arises from a normalized characteristic-zero newform of weight $k(V, p)$, nebentypus ψ_3 of conductor 12, and exact prime-to- p level

$$(710) \quad N = 2^e 3^2 \text{rad}(R), \quad 2 \leq e \leq 6.$$

If $r > 0$, then $p \equiv 1 \pmod{3}$. In both branches, for $\lambda = \max\{1, \log V\}/p$,

$$(711) \quad \frac{\log N}{p} \leq \lambda + \frac{\log 576}{p}.$$

Proof. The actual Frey curve is the non-CM Q -curve of minimum degree 2 already studied above. Its multiplicative prime greater than 3, the fixed order-four splitting character χ , the odd descent with determinant ψ_3 times the cyclotomic character, and the general large-image theorem giving residual absolute irreducibility for $p > 7$ do not require $p \nmid V$.

Suppose $r > 0$. Since $p \mid F(a, b)$, the local calculation of the Frey curve shows that p splits in $K = \mathbb{Q}(\sqrt{-3})$, and either completion is $\mathbb{Q}_p$. The two minimal discriminant valuations are $m, 2m$, where

$$(712) \quad m = r + p v_p(Q), \quad 1 \leq r \leq p - 1.$$

The corresponding c_4 valuations are zero, so the j valuations are $-m, -2m$, neither divisible by p . The character χ is unramified at p , so it leaves the inertia representation and Serre weight unchanged. Serre's semistable weight formula [134, §2.9, Proposition 5] therefore gives exactly $k = p + 1$. That formula includes nonsplit multiplicative reduction by an unramified quadratic extension. If $r = 0$, the good-reduction or finite-flat Tate argument of Theorem 201.5 instead gives weight 2, including an inert prime p in the good-reduction case.

For $q > 3$, $q \neq p$, the Tate monodromy modulo p is nonzero precisely when $p \nmid v_q(F)$. Since V is p -power-free, this is equivalent to $q \mid V$. Thus the prime-to- p conductor factor away from 6 is $\text{rad}(R)$. The conductor arguments at 2 and 3 are unchanged: their exponents are $2 \leq e \leq 6$ and exactly 2. Reduction modulo $p > 7$ preserves these finite inertia conductors. This proves (710) for the residual conductor.

The completed strong form of Serre modularity, due to Khare–Wintenberger with the Kisin input described in their work [135], supplies a residual eigensystem at this Serre weight and conductor. Serre's 1987 paper supplies the local weight formula and characteristic-zero lifting, not a proof of the then-conjectural full modularity theorem. Both weights satisfy $k - 1 \equiv 1 \pmod{p - 1}$, so the determinant identifies the residual nebentypus as ψ_3 . Its Teichmüller lift is the same quadratic character. The characteristic-zero eigensystem lift preserves weight, level and this character [134, §3.1.6]. It can be chosen new at exact level N : an oldform from a proper divisor of N would give residual conductor dividing that smaller level, contradicting the exact conductor just computed. Finally $N \leq 576V$ gives (711). $\square$

Corollary 201.7 (Uniform dimension budget). *For each $p > 7$ and each positive p -power-free V coprime to 6, the sum of the dimensions of the five newspaces in (709)–(710) is at most*

$$(713) \quad 485(p + 1)V^2.$$

For every $L \geq 0$, their dimension sum over all $1 \leq V \leq e^{Lp}$ satisfying these hypotheses is at most

$$(714) \quad 485(p + 1)e^{3Lp}.$$

Each V contributes its one prescribed weight $k(V, p)$. This counts Galois orbits with their coefficient-field degrees, and includes residual integers for which no actual seed exists.

Proof. Put $D = \text{rad}(R)$. Since $\gcd(V, 6) = 1$, the modular-group index satisfies

$$I = [\text{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = 2N \prod_{q \mid D} (1 + q^{-1}) \leq 2ND \leq 1152V^2.$$

The factors at 2 and 3 have product 2, and $1 + q^{-1} \leq q$ bounds the other factors. The Sturm coefficient bound [133] bounds the whole cusp space, hence its newspace, by $1 + \lfloor kI/12 \rfloor \leq$

$1 + 96(p+1)V^2$. Summing over five exponents e gives (713). For $H = \lfloor e^{Lp} \rfloor$, use $\sum_{V=1}^H V^2 \leq H^3$ to obtain (714). In particular,

$$\frac{\log(485(p+1)e^{3Lp})}{p} = 3L + \frac{\log 485 + \log(p+1)}{p} \rightarrow 0$$

when $L \rightarrow 0$ and $p \rightarrow \infty$. $\square$

The remaining condition. Every actual power profile in this budget gives a congruence, at a place above p , between its descended Frey representation and a member of the indicated finite newform collection. Its seed also satisfies the exact inverse-square and positivity conditions above. The missing input is an exclusion of these congruences for actual integer seeds, uniformly in moving residual integers and primes. A dimension bound does not bound seed height or exclude a single congruence. The nontrivial compression target has $Q > 1$. The necessary theorem also includes $Q = 1$; fixed seeds can have such trivial representations for arbitrarily large p , so no blanket exclusion of every profile is intended.

For example, a theorem forcing every such congruence into a finite explicitly classified seed list, together with verification that the list contains no required profiles, would close that branch. No such theorem is asserted here. A Frey–Mazur type claim turning a congruence with the boundary curve into an isogeny is likewise an additional conditional input; finite matching traces do not prove it. The local boundary shadows are not asserted to be global power profiles.

For $V = 1$, exclusion at all sufficiently large prime exponents, combined with the earlier fixed-exponent finiteness, would settle the pure second-norm family with exponent at least 3. It would not settle arbitrary residual profiles or ABC. Moreover, if $p \mid g$, first write $V = V_p C^p$ with V_p p -power-free. The new profile is $F = V_p (CQ^{g/p})^p$, and its parameter satisfies $\lambda_p = \max\{1, \log V_p\}/p \leq \max\{1, \log V\}/p$. It is at most $1/\theta$ times $\max\{1, \log V\}/g$ when $p \geq \theta g$ for a fixed $\theta > 0$; without this condition its smallness does not follow from the original parameter's smallness. The independent Kummer point-height route and these remaining modular branches stay open.

202. A NON-CM BOUNDARY CURVE AND ITS ACTUAL LOCAL SHADOWS

This section separates an exact boundary curve, its unbounded actual local shadows, and a conditional obstruction to one specified modular elimination procedure. It gives no global pure-power family and does not identify compatible Galois representations from finitely many matching traces.

Put $r^2 = -3$, $K = \mathbb{Q}(r)$, $\zeta = (1+r)/2$, and

$$F(a, b) = a^4 + 3a^3b + 5a^2b^2 + 3ab^3 + b^4, \quad x = a^2 + b^2, \quad y = a + b.$$

The actual Frey equation is

$$E_{a,b}: \quad Y^2 = X^3 + 12yX^2 + 6(3y^2 + xr)X.$$

At the excluded boundary seed $(a, b) = (0, 1)$ it specializes to

$$E_0: \quad Y^2 = X^3 + 12X^2 + 6(3+r)X, \quad \Delta(E_0) = 2^{11}3^4(3+r).$$

Since $N_{K/\mathbb{Q}}(3+r) = 12$, this equation has good reduction at every prime above a rational prime greater than three.

Theorem 202.1 (The boundary curve has no geometric CM). *The elliptic curve E_0 has no complex multiplication over $\overline{\mathbb{Q}}$.*

Proof. Full finite-field enumerations give the following exact data, including the point at infinity:

residue field	r	$\#E_0$	t	$t^2 - 4q$
$\mathbb{F}_{25}$	$r^2 = -3$	34	-8	-36
$\mathbb{F}_7$	2	12	-4	-12
$\mathbb{F}_7$	5	4	4	-12

Here $X^2 + 3$ is irreducible modulo five. The certificate enumerates all 625 affine pairs over $\mathbb{F}_{25}$ and all 49 pairs in each of the two prime-field reductions.

Use the auxiliary Tate-module prime 17. If E_0 had geometric CM, its geometric endomorphism algebra would be an imaginary quadratic field M . The algebra $M \otimes \mathbb{Q}_{17}$ embeds into $\text{End}(V_{17}(E_0))$. All these endomorphisms are defined over some finite number-field extension. Consequently their action commutes with some positive power of Frobenius at each good place. These are the usual injectivity and good-reduction properties of Tate modules; see Milne, *Elliptic Curves*, second edition, Proposition II.6.7 and Proposition V.8.3, in the [author's text](#). The latter Frobenius argument applies over the completion of K just as it does over a completion of $\mathbb{Q}$.

At norm 25 the eigenvalues are $-4 \pm 3i$, and their ratio is $(7 - 24i)/25$. Its real part $7/25$ excludes the roots of unity in $\mathbb{Q}(i)$. At either norm-seven place the eigenvalues are $2 \pm r$ or $-2 \pm r$, and their ratio has real part $1/7$. This excludes the sixth roots of unity in $\mathbb{Q}(r)$. Thus every positive power of either Frobenius still has distinct eigenvalues. Its centralizer has dimension two and is the same as the centralizer of the original Frobenius.

The two-dimensional algebra $M \otimes \mathbb{Q}_{17}$ must therefore equal each centralizer. At norm 25 the latter is

$$\mathbb{Q}_{17}[T]/(T^2 + 8T + 25) \simeq \mathbb{Q}_{17} \times \mathbb{Q}_{17},$$

since -1 is a square modulo 17. At norm seven it is $\mathbb{Q}_{17}[T]/(T^2 \pm 4T + 7)$, a quadratic field: its discriminant has nonsquare unit part -3 modulo 17. A product algebra with nontrivial idempotents is not a field. This contradiction proves the theorem. $\square$

Theorem 202.2 (Simultaneous actual positive local shadows). *For every pair of integers $L \geq 1$ and $H \geq 0$, there are positive coprime integers a, b with $a + b > H$ such that*

$$(a, b) \equiv (0, 1) \pmod{L}, \quad (x, y, F(a, b)) \equiv (1, 1, 1) \pmod{L}, \quad 2x - y^2 = (a - b)^2 = 1.$$

The same seed passes every direct congruence test that $F(a, b)$ be a g -th power modulo every divisor of L , simultaneously for all integers $g \geq 1$.

Proof. Choose a sufficiently large integer $k \geq 1$ and put $a = Lk$, $b = Lk + 1$. Consecutiveness proves primitivity, positivity and the last equality, and their sum tends to infinity with k . The congruences follow by substitution. The residue 1 is the g -th power of 1 for every positive g . $\square$

The Frey coefficients are integer polynomials in x, y . Thus these actual curves reduce to E_0 at every prime above any $q > 3$ dividing L . Arbitrary finitely many precisions can be prescribed by including the required q^e in L . This statement retains the exact inverse square condition and positive primitive integer seeds. It does not assert the integer equation $F(a, b) = Q^g$.

Proposition 202.3 (A specified Mazur product has a zero factor). *Suppose a descended eigenform f_0 has independently been identified with the compatible representation of E_0 twisted by the fixed splitting character. At an auxiliary $q > 3$, consider an elimination product over all original seed residue states admitted by positivity, primitivity, the inverse-square condition and the direct F power residue condition. Suppose its factors are the differences of the compatible candidate and curve Frobenius traces. This product is zero at every such q .*

Proof. The admitted state $(0, 1)$ yields the twisted trace of E_0 , equal by hypothesis to the candidate trace. Its factor is zero. Theorem 202.2 additionally gives actual positive primitive seeds shadowing these states simultaneously at any finite set of auxiliary primes. $\square$

The proposition applies to the specified product with its stated state space. It does not restrict auxiliary global covers, additional local conditions proved from those covers, deformation arguments, or other modular information. An identification of f_0 is an explicit premise, not a conclusion drawn from a few computed coefficients.

Lemma 202.4 (The fixed character gives trace minus four at thirteen). *For the $d = 3$ splitting character in [130, Table 3.3 and the following definition], both twisted Frobenius traces of E_0 at primes above thirteen are -4 .*

Proof. The local character θ on $(\mathcal{O}_K/8)^\times$ has values

$$\theta(\zeta - 1) = 1, \quad \theta(r) = i, \quad \theta(3 + 2r) = 1, \quad \theta(-1) = 1.$$

The four generators have orders 3, 4, 2, 2. Their 48 products are distinct. Indeed, reduction modulo two removes the order-three factor. Modulo four the remaining exponent of r must be even. Now $r^2 \equiv 5 \pmod{8}$, while $3 + 2r = 1 + 4\zeta$; the ζ coefficient removes its exponent, and the remaining scalar factors 5 and -1 are independent modulo eight. This also verifies directly that the prescribed values define a character on all units. It is trivial on global units. Since $r^2 \equiv 1 \pmod{4}$ has character -1 , its conductor is exactly eight.

For $\pi = 3 + \zeta$ and $\bar{\pi} = 4 - \zeta$, both of norm thirteen,

$$\pi \equiv (\zeta - 1)(3 + 2r), \quad \bar{\pi} \equiv (\zeta - 1)^2 r^2 (3 + 2r) \pmod{8}.$$

Their local character values are therefore 1, -1 . At the associated primes r has images 6, 7 respectively. Full point counts give 18, 10 points, hence untwisted traces $-4, 4$ respectively. Global principality and the absence of other ramified finite places give $\chi((\pi)) = \theta(\pi)^{-1}$ and the analogous equality for $\bar{\pi}$. Both twisted traces are consequently -4 . $\square$

An independent exact PARI/GP 2.15.4 newspace replay, with character $\psi_3 = (12/\cdot)$, gives

N	$\dim S_2^{\text{new}}(N, \psi_3)$	eigenorbit fields	CM discriminants
36	2	$\mathbb{Q}(\sqrt{-2})$	-4
72	0	none	none
144	2	$\mathbb{Q}(\sqrt{-2})$	-4
288	4	$\mathbb{Q}(\zeta_8)$	0
576	8	$\mathbb{Q}(\sqrt{-2}), \mathbb{Q}(\sqrt{-2}), \mathbb{Q}(\zeta_8)$	$-4, -4, 0$

Here zero is the software's non-CM return value. The field-degree sums equal the dimensions, and the exact coefficients through 25 are retained. The non-CM orbits at 288 and 576 have rational $a_{13} = 4$ and -4 , respectively. Thus an independently justified membership of the E_0 descent in this list would select the level 576 orbit. The positive-seed large-image argument cannot simply be applied to $F(0, 1) = 1$, which has no multiplicative prime above three. A separate general modularity or sufficiently-large-prime argument for this fixed non-CM curve remains necessary for that unconditional identification.

Verification scope. The ordinary non-CM and local-shadow proofs were independently reviewed, and the point counts and character sign were separately replayed. The certificate enumerates all 48 local units, checks all 2304 multiplication pairs and both norm-thirteen reductions. The newspace computation is an independently repeated exact software calculation; it is not a Lean proof of modular-form algorithms or an exclusion of the remaining eigenforms. The corresponding scripts, canonical JSON files and exact hashes are recorded in the sixth-round adversarial-audit checkpoint.

203. FORMAL ARITHMETIC OF THE ACTUAL FREY SEED

The module `FreySeedArithmetic.lean` contains eighteen new declarations. It imports the actual second norm and the previously checked primitive residue statements. For all integers a, b , it defines $x = a^2 + b^2$, $y = a + b$ and proves

$$x^2 + 3y^4 = 4F(a, b), \quad 2x - y^2 = (a - b)^2.$$

It also checks the two polynomial identities for the odd-seed chart $a = Y + D$, $b = Y - D$, including $F(Y + D, Y - D) = (Y^2 + D^2)^2 + 12Y^4$. These chart identities do not silently assert positivity, primitivity or the full inverse parametrization.

All algebraic coefficients are actual Eisenstein integer pairs, with $r = 2\zeta - 1$ and $r^2 = -3$. The module defines

$$\alpha_+ = 3y^2 + xr, \quad \alpha_- = 3y^2 - xr, \quad A = 12y, \quad B = 6\alpha_+,$$

and checks conjugation, $N(\alpha_+) = 12F$, and $\alpha_+\alpha_- = 12F$. It defines the usual polynomial invariants of $Y^2 = X^3 + AX^2 + BX$ by

$$c_4 = 16(A^2 - 3B), \quad c_6 = -64A^3 + 288AB, \quad \Delta = 16B^2(A^2 - 4B).$$

The kernel verifies their actual values

$$\begin{aligned} c_4 &= 288(5y^2 - xr), \\ c_6 &= 6912y(3xr - 7y^2), \\ \Delta &= 165888F\alpha_+, \quad A^2 - 4B = 24\alpha_- = 4\overline{B}. \end{aligned}$$

Thus the quotient-coefficient and twisted-conjugate identities are checked as actual polynomial equalities. The complete elliptic isogeny construction is not itself encoded by those coefficient identities.

Finite kernel-reduced residue tables, transferred to all integers under the actual hypothesis $\gcd(a, b) = 1$, prove that F is odd and $3 \nmid x$. These and the norm identity give $\Delta \neq 0$ for every primitive integer pair, including negative and boundary inputs. Finally the module proves, for every pair of integers L, k ,

$$L \mid x(Lk, Lk + 1) - 1, \quad L \mid y(Lk, Lk + 1) - 1, \quad L \mid F(Lk, Lk + 1) - 1.$$

The preceding module supplies the corresponding positive unbounded primitive family; the new statements check its actual Frey residues.

A fresh standalone Lake build with Lean 4.32.0 and warnings treated as errors accepts all eighteen new declarations and recompiles sixty-five unchanged dependency declarations. Every declaration has an axiom inventory, using only `propext`, `Classical.choice` and `Quot.sound`. Exact source hashes, logs and an independent signature review are recorded. These are scoped formal arithmetic results, not a full repository build or a formal proof of Tate reduction, class number, modularity, the prime-window theorem, or standard ABC.

204. SEVENTH CONTINUATION: ACTUAL BOUNDARY CLASSES AND GLOBAL ESCAPE

Standard ABC remains unproved and undisproved. This continuation combines several distinct advances while retaining their exact dependencies and unresolved gates. An actual interval sieve extends the controlled prime window beyond the length of the root-parameter block on specified exponent subclasses. It still leaves exceptional roots and primes above that window. Effective three-chart Diophantine bounds force hypothetical power seeds to escape every fixed affine slice. Elementary square gaps give explicit bounds in a narrower class.

The modular route now identifies the specified non-CM boundary descent using characteristic-zero modularity and its actual conductor before applying the complete newspace enumeration. Congruence to any coefficient embedding of this boundary orbit forces a pure power and restricts its moving prime support; it does not construct such a congruence. An independent rational two-isogeny descent completely classifies the actual residual square class thirteen. The integral map and actual square-axis inequalities have scoped Lean verification; the elliptic group and modular theorems remain ordinary proofs with explicit external inputs.

The class-thirteen result closes precisely that even-power residual class. Other residual classes, odd prime exponents, moving affine slices and the full signed prime tail remain active. Finite local shadows, generic cover counts and sparse allowed prime support are not promoted to a global nonexistence argument.

205. A SIEVED ENDPOINT FOR ACTUAL ROOT INTERVALS

For positive integers n, B and $B \leq k < 2B$, put $w_k = 3k + \zeta$, $T_n(k) = |P(w_k^n)|$, and let t_k be the logarithm of the largest side of its primitive positive sector normalization. The preceding actual root-interval argument proves

$$t_k \geq n \log(3B), \quad \log |P(w_k^d)| \leq 3d \log(6B + 1).$$

For real $5 \leq Y \leq Z$, write

$$E_{Y,Z}(k) = \sum_{Y < p \leq Z} (v_p(T_n(k)) - 3)_+ \log p.$$

These are actual integer roots and valuations, not a uniform reference distribution substituted for a power orbit.

Theorem 205.1 (A sieved actual interval estimate). *If $Z > 3n$, and $\sigma(n) = \sum_{d|n} d$, then*

$$\frac{1}{B} \sum_{B \leq k < 2B} \frac{E_{Y,Z}(k)}{t_k} \leq \frac{10 \log Y}{Y^3 \log(3B)} + \frac{36Z\sigma(n)}{Bn \log(Z/(3n))} + \frac{\log n}{n \log(3B)}.$$

The middle term may also be replaced by its minimum with $12(Z+1)/B$.

Proof. At an eligible prime of exact homogeneous rank $d \mid n$, the actual root count is $3\varphi(d) - \mathbf{1}_{d=1}$, and each root lifts uniquely. The first-depth positive cost summed over the interval is therefore at most

$$\frac{3\varphi(d)B \log p}{p^3(p-1)} + 9\varphi(d)d \log(6B+1).$$

The first term follows by summing depth layers at least four; the second uses their actual polynomial-height cap. At a fixed prime, the sum of $\varphi(d)$ over eligible divisors of n is at most n . The already proved elementary prime sum gives the first term of the asserted normalized estimate.

For the endpoint term, a rank- d prime lies in a reduced class 1 or -1 modulo $3d$. The established Brun–Titchmarsh inequality gives, for $X > q$ and $(a, q) = 1$,

$$\pi(X; q, a) \leq \frac{2X}{\varphi(q) \log(X/q)}.$$

The precise source is Montgomery–Vaughan, *The large sieve*, Theorem 2, printed page 121, equation (1.10), in the [author-hosted primary paper](#). Its interval form gives the displayed version by taking the left endpoint down to zero. The bound is uniform for $X > q$. Thus the two classes through Z contain at most $4Z/(\varphi(3d) \log(Z/(3d)))$ primes. Since $Z > 3n$ and $\varphi(3d) \geq 2\varphi(d)$, their total endpoint cost is at most

$$36Z \log(6B+1) \sum_{d|n} \frac{d\varphi(d)}{\varphi(3d) \log(Z/(3d))} \leq \frac{18Z\sigma(n) \log(6B+1)}{\log(Z/(3n))}.$$

Divide by $Bn \log(3B)$ and use $6B+1 \leq (3B)^2$. The actual additional lifting cost is at most $\log n$ at each root, by $(s+u-3)_+ \leq (s-3)_+ + u$ and $\sum_p v_p(n) \log p = \log n$. This proves the estimate. Minimizing with the previously proved unsieved endpoint bound gives the last assertion. $\square$

Corollary 205.2 (A window beyond the root-interval length). *Let $n \rightarrow \infty$ with $\sigma(n)/n \leq C$ for a fixed constant. Set $B = n^4$, $Y = n^{1/6}$ and $Z = \lfloor B\sqrt{\log n} \rfloor$. Then the mean in Theorem 205.1 is $O_C((\log n)^{-1/2})$, and the proportion exceeding $(\log n)^{-1/4}$ is $O_C((\log n)^{-1/4})$.*

Proof. The first and last terms are at most $5/(12\sqrt{n})$ and $1/(4n)$. Furthermore $\log(Z/(3n)) = 3 \log n + \frac{1}{2} \log \log n - \log 3 + o(1)$. This bounds the middle term as asserted. Apply Markov's inequality to the nonnegative actual cost. $\square$

Prime-power indices satisfy the divisor-sum hypothesis with $C = 2$. Here $Z/B \rightarrow \infty$; for prime indices the window extends beyond the first possible top-rank progression, without asserting that it contains a prime. Every prime above Z and every exceptional root remain uncontrolled. No estimate for a prescribed root's full signed packet has been deduced from this average.

206. EFFECTIVE ESCAPE FROM FIXED AFFINE SLICES

Let a, b be positive coprime integers, let V, Q, g be integers, and suppose

$$F(a, b) = VQ^g, \quad V \geq 1, \quad Q > 1, \quad g \geq 2.$$

No power-free or disjoint-support premise on V is needed here.

Theorem 206.1 (Three-chart effective exponent bound). *There is an effectively computable constant $C \geq 1$, depending only on the fixed quartic, such that*

$$\log g \leq C(1 + \log V + \log(1 + M)), \quad M = \min\{a, b, |a - b|\}.$$

Consequently $1 + M \geq e^{-1}g^{1/C}/V$.

Proof. For a nonzero integer c , the three seed charts use the two polynomials

$$P_c(T) = F(T + c, T) = 13T^4 + 26cT^3 + 20c^2T^2 + 7c^3T + c^4,$$

$$R_c(T) = F(T, c) = T^4 + 3cT^3 + 5c^2T^2 + 3c^3T + c^4.$$

Their discriminants are $117c^{12}$ and their logarithmic coefficient heights are at most $\log 26 + 4\log(1 + |c|)$. Equivalently their roots are $c/(\alpha_i - 1)$ and $c\alpha_i$, respectively, where the α_i are the four distinct roots of $F(X, 1)$. The denominators do not vanish since $F(1, 1) = 13$.

Apply Bérczes–Evertse–Györy, *Effective results for Diophantine equations over finitely generated domains*, Theorem 2.3, in the [primary paper](#). Use the fixed presentation $\mathbb{Z} = \mathbb{Z}[U]/(U)$, with $r = d = 1$ in their notation and polynomial degree four. Their equation $H(t) = \delta_0 y^m$ then has the effective bound $\log m \leq C_0(h + 1)$, where h bounds the logarithmic coefficient and δ_0 heights. Take $\delta_0 = V$, $y = Q$ and $m = g$. The root-of-unity exception is absent because $Q > 1$.

The coefficient bounds above give the asserted inequality with $|c|$ in place of M . Apply this with $c = a, b$, and $a - b$, using symmetry for the first choice. All are nonzero: $a = b$ would force $(a, b) = (1, 1)$ and $F = 13$, which has no such nontrivial power divisor. Taking the minimum proves the claim and its rearrangement. $\square$

Theorem 206.2 (Effective finiteness on bounded slices). *For each fixed V and fixed H , only finitely many such primitive positive seeds have $\min\{a, b, |a - b|\} \leq H$. They are effectively determinable in principle.*

Proof. For each nonzero fixed c and fixed V , the preceding argument bounds all possible exponents in $P_c(t) = VQ^g$ or $R_c(t) = VQ^g$ effectively, even without primitivity. For each of the finitely many exponents, Theorem 2.2 of the same primary paper supplies an effective finite list. Its hypotheses hold for these degree-four polynomials with nonzero discriminant, including exponent two. There are only finitely many charts and nonzero parameters with $|c| \leq H$. $\square$

Thus pure profiles with $g \rightarrow \infty$ must escape all three boundaries at least polynomially in g . If $V = g^{o(1)}$ the same holds with exponent $1/C - o(1)$. This does not exclude general small $\lambda = \max\{1, \log V\}/g$, which permits much larger residuals. The consecutive local shadows $(Lk, Lk + 1)$ all have fixed difference -1 , even when L varies, so only finitely many can be global pure powers across all $g \geq 2$. This is a global restriction beyond their finite local compatibility; it does not settle moving affine slices. The two cited effective Diophantine theorems remain external inputs, not Lean theorems proved in this checkpoint.

207. ELEMENTARY GAPS BETWEEN ACTUAL NORM SQUARES

The following bounds are elementary restrictions on particular residual square classes. They use actual polynomial identities and integer square gaps, with no probabilistic transfer.

Theorem 207.1 (A residue-sensitive axis gap). *Let a, b be positive coprime integers and $F(a, b) = q^2$, $q > 0$. Put $S = 8a^2 + 12ab + 11b^2$ and let $r \in \{1, \dots, 8\}$ be its least positive residue modulo eight. Then*

$$2ra < 9b^3, \quad \max\{a, b\} < \frac{3}{2} \min\{a, b\}^3.$$

In particular no such seed has $\min\{a, b\} = 1$.

Proof. Expansion gives

$$D := S^2 - 64F = 72ab^3 + 57b^4 > 0.$$

Thus $8q < S$, and congruence modulo eight gives $8q \leq S - r$. Both sides are nonnegative, so $D \geq r(2S - r)$. If $2ra \geq 9b^3$, then $72ab^3 \leq 16ra^2$ and $24rab \geq 108b^4$, giving

$$r(2S - r) - D \geq 51b^4 + 22rb^2 - r^2 > 0$$

because $b \geq 1$ and $r \leq 8$. This is a contradiction. For odd b , $S \equiv 4a + 3 \pmod{8}$, so $r = 3$ or 7 . For even b , primitivity forces a odd, giving $r = 4$ or 8 . Hence $r \geq 3$ and $2a < 3b^3$; interchange a, b . If the minimum is one, the bound forces both to be one, whereas $F(1, 1) = 13$ is not a square. $\square$

This applies to every actual even exponent with a square residual, and in particular to pure even powers. It gives an explicit finite axis search bound, while remaining compatible with the earlier infinite family of positive square second norms.

Proposition 207.2 (A complementary diagonal identity). *Let a, b be positive coprime integers and $F(a, b) = 13q^2$ with $q > 0$. If $c = a - b \neq 0$, then*

$$52ab + 14c^2 + 1 \leq 3c^4.$$

For each fixed nonzero c , all positive solutions in this square class admit a finite exact divisor parametrization.

Proof. Set $A = 7a^2 + 12ab + 7b^2 = 26ab + 7c^2$. The exact identity

$$52F = A^2 + 3c^4$$

gives $(26q - A)(26q + A) = 3c^4$. The smaller factor is positive, so the difference of squares is at least $2A + 1$, proving the inequality.

For completeness, choose positive divisor pairs $d_1 < d_2$ of $3c^4$. Retain only integral positive

$$q = (d_1 + d_2)/52, \quad A = (d_2 - d_1)/2, \quad u = (A - 7c^2)/26.$$

Require $c^2 + 4u = h^2$ for a positive integer $h > |c|$ with $h - c$ even. Then $a = (h + c)/2$, $b = (h - c)/2$ give exactly all positive solutions; impose their gcd condition if desired. Indeed these tests are precisely $a - b = c$, $ab = u$ and the displayed difference of squares. Conversely each solution supplies $d_1 = 26q - A$, $d_2 = 26q + A$. $\square$

The diagonal identity and its divisor classification remain useful explicit arithmetic even if a separate global argument further restricts that square class. These gap arguments alone assert neither such a global exclusion nor control of odd exponents, arbitrary residual classes, or a full signed top-rank packet.

208. IDENTIFICATION OF THE FIXED BOUNDARY DESCENT

We discharge the explicit modular-membership premise for the boundary curve of Section 202. The argument uses one fixed Tate-module prime and a finite character comparison, followed by the compatibility of an already constructed modular form. It does not assume that unrelated choices of descent at different primes are compatible. The resulting orbit identification combines ordinary modularity theorems with the independently repeated exact newspace enumeration; it is not a Lean formalization of these inputs.

Write $K = \mathbb{Q}(r)$, $r^2 = -3$, and

$$E_0 : Y^2 = X^3 + 12X^2 + 6(3 + r)X.$$

Theorem 202.1 proves geometric non-CM. The same explicit two-isogeny as in the actual family identifies its quotient with the (-2) -twist of its conjugate, so E_0 is a Q-curve. Fix the order-four character χ used in the earlier descent. Its square is $\psi_3|_{G_K}$, and its conjugation relation cancels the character of that isogeny.

Lemma 208.1 (Finite-character comparison). *Let H be a finite-index normal subgroup of G . Let ρ_1, ρ_2 be representations over an algebraically closed characteristic-zero field whose restrictions to H are isomorphic and absolutely irreducible. Then $\rho_2 \simeq \rho_1 \otimes \nu$ for a character ν of the finite group G/H .*

Proof. Identify the two restrictions. For $g \in G$, the matrix $\rho_2(g)\rho_1(g)^{-1}$ commutes with their common action of H , by normality. Schur's lemma makes it scalar, say $\nu(g)$. The two representation laws give $\nu(gg') = \nu(g)\nu(g')$, and $\nu|_H = 1$. For continuous Galois representations this is a finite continuous character; over $G_{\mathbb{Q}}$ it is a Dirichlet character. $\square$

Theorem 208.2 (Modularity of the specified boundary descent). *Either extension of $V_{17}(E_0) \otimes \chi$ to $G_{\mathbb{Q}}$ is the representation of a weight-two newform with nebentypus ψ_3 .*

Proof. The conjugation relation and Schur normalization in the proof of [130, Theorem 4.2] give the extension. Its two choices differ by ψ_{-3} . Here only the general representation argument is used, not that source's equation-specific conductor formula.

Ribet's Theorem 6.1 and its construction give a primitive abelian variety A of GL_2 -type over $\mathbb{Q}$ and a finite Galois extension $L/\mathbb{Q}$ such that A_L is isogenous to a power of E_0 ; see the author's *Abelian varieties over $\mathbb{Q}$ and modular forms*, Section 6. Enlarge L to contain K and kill χ . The completed modularity theorem for GL_2 -type abelian varieties, [135, Corollary 10.2(i)], makes A modular. Choose a two-dimensional coefficient-field component ρ_A of its Tate module at 17. It comes from a weight-two newform f_A .

On G_L , this component is isomorphic to $V_{17}(E_0)$ after scalar extension: the whole Tate module is a sum of copies and the chosen component has dimension two. Faltings' semisimplicity and endomorphism theorem, together with $\mathrm{End}_L(E_0) \otimes \mathbb{Q} = \mathbb{Q}$, make this restriction absolutely irreducible. The specified extension $\tilde{\rho}$ has the same restriction, since χ is killed there. Lemma 208.1 gives

$$\tilde{\rho} \simeq \rho_A \otimes \nu$$

for a finite Dirichlet character ν . The primitive newform associated to $f_A \otimes \nu$ still has weight two, and realizes $\tilde{\rho}$. In particular it is odd: a finite character has square 1 on complex conjugation. On G_K the determinant is ψ_3 times the cyclotomic character; oddness gives equality on complex conjugation and therefore on $G_{\mathbb{Q}}$. Its nebentypus is exactly ψ_3 . $\square$

The multiplicative-prime large-image result for positive seeds has not been used; it would not apply to $F(0, 1) = 1$. Nor has any nonzero Tate twist been introduced.

Lemma 208.3 (Compatibility supplied by the identified modular form). *For the form f in Theorem 208.2 and every auxiliary prime ℓ , after fixing compatible coefficient embeddings,*

$$\rho_{f,\ell}|_{G_K} \simeq V_{\ell}(E_0) \otimes \chi_{\ell}.$$

Proof. At every prime of K outside a finite set, the restricted Frobenius polynomials agree by the isomorphism at 17. They are polynomials in algebraic numbers, and the embedding into $\overline{\mathbb{Q}}_{17}$ is injective. Compatibility for the elliptic curve, finite character and already constructed modular form gives equality at any other auxiliary prime. Chebotarev and semisimplicity then give the displayed isomorphism. $\square$

Proposition 208.4 (The actual characteristic-zero level). *The newform in Theorem 208.2 has exact level*

$$N = 2^e 3^2, \quad 2 \leq e \leq 6.$$

Proof. The local Tate branches hold at $(a, b) = (0, 1)$, and can be checked directly. At the inert prime over 2, translate X by $1 + r$ and Y by 4. The new coefficients are

$$a'_1 = 0, \quad a'_2 = 15 + 3r, \quad a'_3 = 8, \quad a'_4 = 36 + 36r, \quad a'_6 = -48 + 48r.$$

Their relevant valuations are 1, 3, 3, 5. For example $a'_6 = 96(\zeta - 1)$ with $\zeta = (1 + r)/2$ and $\zeta - 1$ a unit at 2. The discriminant valuation is 12. The first Tate cubic has the form $Z^2(Z + b)$ with b a unit; the first starred quadratic has double root zero, and the next has unit linear coefficient

$a'_4/8$. Thus the minimal type is I_2^* and its conductor exponent is 6, by the earlier reviewed Tate algorithm.

At r over 3, with $v_r(3) = 2$, the original coefficients have $v_r(a_2) = 2$, $v_r(a_4) = 3$, $a_1 = a_3 = a_6 = 0$ and $v_r(\Delta) = 9$. The successive multiple roots are zero and a_4/r^3 is a unit. The minimal type is III^* , of conductor exponent 2. At all primes above $q > 3$ the original curve has good reduction.

The character χ is unramified at 3 and has conductor exponent 3 at 2. At 2 the elliptic representation has finite inertia and unramified determinant. Every inertia-subgroup fixed space has dimension zero or two: a finite determinant-one matrix fixing a nonzero vector must be the identity. The conductor 6 therefore consists of tame codimension 2 and Swan conductor 4, so its final upper break is 2. The character also has upper break 2. Their tensor product is trivial on all upper groups with index greater than 2, giving conductor at most 6. Unramified restriction from $\mathbb{Q}_2$ to K_2 preserves the exponent. The determinant ψ_3 has conductor exponent 2 there, giving the lower bound 2.

At 3 the restricted representation is tame with no inertia invariants. The extension $K_3/\mathbb{Q}_3$ is tame, so its wild inertia groups agree. The descent is tame, and its inertia invariants are contained in those of the restriction. Its conductor exponent is consequently exactly 2 for either descent.

For $q > 3$ choose $\ell \neq q$ and apply Lemma 208.3. The elliptic curve and character are unramified at q , and $K/\mathbb{Q}$ is unramified there, so the descent is unramified. In particular the rational prime 17 can be checked using $\ell = 19$. Equality of characteristic-zero newform conductor and the conductor of its representation away from the auxiliary prime now gives the asserted level. $\square$

Theorem 208.5 (The non-CM level-576 orbit). *Together with the complete exact newspace enumeration recorded in Section 202, the preceding propositions identify either fixed boundary descent with the unique non-CM Galois orbit of weight two, nebentypus ψ_3 , and level 576.*

Proof. The form cannot have CM. Over a finite extension a CM form's two-dimensional representation decomposes into characters, whereas the Tate representation of E_0 remains absolutely irreducible. Lemma 208.3 gives a contradiction after enlarging the finite extension to kill χ .

Among the five levels of Proposition 208.4, the complete enumeration has exactly two non-CM orbits, at 288 and 576. Their rational coefficients a_{13} are 4 and -4 respectively. Lemma 202.4 gives $a_{13} = -4$. Since thirteen splits in K , the ambiguity ψ_{-3} between the two descents does not change that trace. The level-576 orbit is therefore the unique possibility. $\square$

This uses a proved membership theorem followed by a complete finite enumeration. It is not a conclusion of equality of representations from a few matching coefficients. The finite computation and its independent exact replay remain a separate, explicit dependency.

For the specified descent, this discharges the premise in Proposition 202.3: those particular finite-state products have a zero factor at every auxiliary prime. It supplies no global power equation for their positive local shadows. Additional global covers or other restrictions can still eliminate actual integer seeds. In particular the independent effective affine-slice argument limits the consecutive shadow family globally.

The complete ordinary comparison, local-conductor calculation and orbit-identification argument passed two independent reviews. Both reviewers opened Ribet's construction and Khare–Wintenberger's GL_2 -type modularity corollary. No new Lean theorem is asserted.

209. BOUNDARY CONGRUENCES FORCE PURE NORMS AND MOVING FROBENIUS SUPPORT

This section assumes an actual isomorphism of residual Galois modules. Its arithmetic and density conclusions do not require an eigenform identification. The final corollary separately uses the boundary orbit identified in Theorem 208.5. No existence of such a congruence, uniform point-height theorem, or Frey–Mazur statement is assumed.

Put $K = \mathbb{Q}(r)$, $r^2 = -3$, and write

$$F = a^4 + 3a^3b + 5a^2b^2 + 3ab^3 + b^4, \quad x = a^2 + b^2, \quad y = a + b,$$

$$E_{a,b} : Y^2 = X^3 + 12yX^2 + 6(3y^2 + xr)X, \quad E_0 = E_{0,1}.$$

The local results of Section 201 give the following actual seed facts. For coprime positive a, b , F is prime to 6, each rational prime $q \mid F$ splits in K , and both places are multiplicative with minimal discriminant valuations $m, 2m$, where $m = v_q(F)$. The fixed curve E_0 has good reduction above every $q > 3$ and no geometric CM, as proved in the boundary section.

Theorem 209.1 (A boundary congruence forces a pure norm). *Let $p > 7$ be prime, let a, b be coprime positive integers, and let ν be a quadratic character of G_K unramified outside the primes over 6, including the trivial character. Suppose*

$$(715) \quad E_{a,b}[p] \otimes \overline{\mathbb{F}}_p \simeq (E_0[p] \otimes \overline{\mathbb{F}}_p) \otimes \nu \quad \text{as } G_K\text{-modules.}$$

Then $F(a, b) = Q^p$ for an integer $Q > 1$. Every prime $q \mid Q$ satisfies

$$(716) \quad q \geq (\sqrt{p} - 1)^2.$$

For $q \neq p$, at either place of K above q , let t_q be the trace of E_0 over $\mathbb{F}_q$. Then

$$(717) \quad p \mid (q + 1)^2 - t_q^2.$$

In addition, neither 7 nor 13 divides Q .

Proof. For $q \mid F$ with $q \neq p$, the boundary representation and its allowed quadratic twist are unramified. At a multiplicative place the Tate representation modulo p is unramified exactly when the valuation of its Tate parameter is divisible by p . Its valuations here are $m, 2m$, and p is odd. Thus (715) forces $p \mid m$.

If $p \mid F$, the prime p splits in K and both completions are $\mathbb{Q}_p$. The good boundary representation has Serre weight 2; the unramified character ν does not change this inertia condition. For a multiplicative elliptic curve over $\mathbb{Q}_p$, the semistable weight formula of [134, §2.9, Proposition 5] gives weight 2 exactly when p divides the negative valuation of j . These negative valuations are $m, 2m$, so again $p \mid m$. Equivalently, the full local representation retains its finite-flat property under the isomorphism. This argument concerns the full representations, rather than only local semisimplified traces, and is unchanged after extending the coefficient field.

Every prime valuation of F is now a multiple of p . Since $F \geq 13$, unique factorization gives $F = Q^p$ with $Q > 1$. For a divisor $q \neq p$, the unramified residual Tate representation has trace $\pm(q + 1)$. The additional sign of ν is absorbed in this sign, so $t_q \equiv \pm(q + 1) \pmod{p}$, proving (717). The Hasse bound makes the corresponding integer difference nonzero and gives

$$p \leq |t_q \mp (q + 1)| \leq q + 1 + 2\sqrt{q} = (\sqrt{q} + 1)^2.$$

This is (716). If $q = p$, that cutoff holds directly; no unramified Frobenius assertion at p is used. The actual depth-one theorem at 13 excludes $13 \mid Q$. Complete boundary point counts at 7 give $t_7 = \pm 4$, so $p \mid 8^2 - 4^2 = 48$ would be necessary if $7 \mid Q$. This is impossible for $p > 7$. $\square$

Lemma 209.2 (An exact Frobenius count). *For an odd prime p , put*

$$C_p = \{A \in \mathrm{GL}_2(\mathbb{F}_p) : (\mathrm{tr} A)^2 = (\det A + 1)^2\}.$$

Then

$$(718) \quad |C_p| = p(2p^2 - p - 5), \quad \frac{|C_p|}{|\mathrm{GL}_2(\mathbb{F}_p)|} = \frac{2p^2 - p - 5}{(p - 1)^2(p + 1)}.$$

Proof. The equation $\mathrm{tr} A = \det A + 1$ means that 1 is an eigenvalue. For each other eigenvalue $u \in \mathbb{F}_p^* \setminus \{1\}$, the class of $\mathrm{diag}(1, u)$ has size $p(p + 1)$; there are $p - 2$ choices. With repeated eigenvalue 1, the identity and the nontrivial unipotent class together have size p^2 . Thus the number of matrices with eigenvalue 1 is

$$(p - 2)p(p + 1) + p^2 = p(p^2 - 2).$$

Negation gives the same count for eigenvalue -1 . Their intersection is exactly the class of $\mathrm{diag}(1, -1)$, of size $p(p + 1)$. Inclusion-exclusion gives the first formula, and the second follows from $|\mathrm{GL}_2(\mathbb{F}_p)| = p(p - 1)^2(p + 1)$. $\square$

Theorem 209.3 (Density of the permitted support). *Fix an odd prime p for which the residual image of E_0 over K is $\mathrm{GL}_2(\mathbb{F}_p)$. Among the good prime ideals away from p , let S_p consist of those whose residue-field order n and trace t satisfy $t^2 \equiv (n+1)^2 \pmod{p}$. Then S_p has natural density*

$$(719) \quad \delta_p = \frac{2p^2 - p - 5}{(p-1)^2(p+1)}$$

among the prime ideals of K , ordered by norm. The corresponding permitted split rational primes have natural density $\delta_p/2$ among all rational primes. The full-image hypothesis holds for every sufficiently large p , with no effective threshold asserted here; along those primes, $\delta_p \sim 2/p$.

Proof. In $K(E_0[p])/K$, Frobenius has trace t and determinant n modulo p . The condition defining C_p is conjugacy invariant. Chebotarev natural density, in the form of [129, Definition 8.30 and Theorem 8.31], and Lemma 209.2 give (719).

At a split rational prime, the traces at the two places have the same square. Indeed, the explicit two-isogeny relates E_0 to the (-2) -twist of its conjugate, so these traces differ by the quadratic sign $(-2/q)$. Hence both places are permitted or neither is. Prime ideals above inert rational primes counted up to norm X make only an $O(\sqrt{X})$ contribution. Each permitted split rational prime contributes two prime ideals, giving density $\delta_p/2$ among rational primes. The split primes themselves have density $1/2$.

Finally E_0 has no geometric CM. Serre's open-image theorem gives full mod- p image over the fixed number field for all but finitely many p ; see [136, pp. 259–260, assertions (3), (6), (7)]. The displayed expression for δ_p has the stated asymptotic. $\square$

These density assertions concern each fixed p . They provide no Chebotarev error bound uniform in p and the norm cutoff. For every fixed sufficiently large p , the permitted support still has positive density and is infinite. The allowed twists ν only change traces by sign away from $6p$, so this support set is unchanged; no image threshold uniform over varying twists is required. Under (715), every prime divisor other than possibly p belongs to this permitted split support and also satisfies (716). Neither restriction alone excludes all prime factors of a polynomial value from lying in that support.

Corollary 209.4 (All embeddings of the boundary orbit). *Suppose the descended residual representation of a coprime positive seed, at a prime $p > 7$, is congruent to any coefficient embedding of the level-576 boundary orbit identified in Theorem 208.5. Then $F(a, b)$ is a pure p -th power and satisfies the support conditions of Theorem 209.1. In particular, in the canonical decomposition $F = VQ^p$ with V p -power-free, that orbit cannot occur when $V > 1$.*

Proof. The specified order-four character in the descent is χ . Compatibility and the algebraic Frobenius identities of the boundary identification show that restriction to G_K compares $E_{a,b}[p] \otimes \chi$ with $E_0[p] \otimes \chi^\sigma$. This comparison uses the compatible representations, not finitely many coefficient values. The checked large-image input gives absolute irreducibility for the actual representation, so the resulting semisimplified isomorphism is the full module isomorphism needed above. An embedding either fixes or inverts the values of χ , and hence

$$\chi^\sigma / \chi \in \{1, \chi^{-2}\} = \{1, \psi_3|_{G_K}\}.$$

Both possibilities are quadratic and unramified outside 6. Cancelling the specified χ therefore gives (715), possibly with its nontrivial quadratic twist. Theorem 209.1 applies. This does not exclude the pure branch. $\square$

Verification and remaining scope. The ordinary proofs have independent reviews. Exact standard-library replay exhausts all matrices for $p = 3, 5, 7, 11, 13, 17, 19$ and all points of the eight boundary reductions over $q = 7, 13, 19, 31$. The general matrix formula is proved above; the finite checks supplement it. The local trace divisibility test always retains the condition $q \neq p$. The orbit corollary has the additional modular-identification dependency stated in its hypothesis. No Lean formalization of the Tate, Serre, Chebotarev or modular comparison theorems is asserted by these checks.

210. COMPLETE CLASSIFICATION OF THE RESIDUAL SQUARE CLASS THIRTEEN

The following closes one actual nonunit residual class. It is independent of the modular candidate enumeration and does not assert uniformity over moving residual classes or exponents. The rank conclusion below is proved by explicit descent; a preliminary software rank calculation is not an input to the proof.

Lemma 210.1 (The required rational elliptic group). *For*

$$E : Y^2 = X^3 - 13X^2 - 507X$$

one has $E(\mathbb{Q}) = \{O, (0, 0)\}$.

Proof. Its rational two-isogenous curve is

$$E' : Y^2 = X^3 + 26X^2 + 2197X.$$

Use the explicit descent of (673)–(674). For $Y^2 = X^3 + AX^2 + BX$, the homomorphism to rational square classes has values 1 at O , B at $(0, 0)$, and X elsewhere. Its kernel is the image of the dual isogeny. Every nonexceptional image class is represented by a signed square-free divisor d of B with a cover

$$(720) \quad N^2 = dU^4 + AU^2V^2 + (B/d)V^4, \quad \gcd(U, V) = 1.$$

These statements include the exceptional points: the dual preimage of $(0, 0)$ exists exactly when B is a square. They follow from the explicit isogeny and substitution $X = d(U/V)^2$, as in the earlier ordinary descent proof.

For E , the candidate classes are $\pm 1, \pm 3, \pm 13, \pm 39$. The classes $1, -3$ occur at $O, (0, 0)$. Multiplication by -3 pairs the other classes, so it suffices to exclude $-1, 13, -13$. For $d = -1$, the cover is

$$N^2 = -U^4 - 13U^2V^2 + 507V^4.$$

Coprimality leaves the following exhaustive possibilities modulo 16:

parity	right side modulo 16
U odd, V even	15, 11
U even, V odd	11, 7
U, V odd	13, 5

None is a square residue.

For $d = 13$, the cover gives $N^2 = 13(U^4 - U^2V^2 - 3V^4)$, so $13 \mid N$ and $U^4 - U^2V^2 - 3V^4 \equiv 0 \pmod{13}$. If $13 \mid V$, then also $13 \mid U$, a contradiction. Otherwise $z = (U/V)^2$ modulo 13 satisfies

$$z^2 - z - 3 = (z - 7)^2 \equiv 0 \pmod{13},$$

but 7 is not a square modulo 13. The class $d = -13$ similarly forces $z^2 + z - 3 = (z - 6)^2 \equiv 0$, and 6 is not a square either. Thus the image is exactly $\alpha(E) = \{1, -3\}$.

On E' , the quadratic factor $X^2 + 26X + 2197 = (X + 13)^2 + 2028$ is positive, so every affine point with $X \neq 0$ has positive X . The signed-divisor test for $2197 = 13^3$ leaves only classes $1, 13$, both attained by O and $T' = (0, 0)$. Hence $\alpha'(E') = \{1, 13\}$. Writing the isogenies as $\phi : E \rightarrow E'$ and $\psi : E' \rightarrow E$, every $P' \in E'(\mathbb{Q})$ has either P' or $P' + T'$ in $\ker \alpha' = \phi(E(\mathbb{Q}))$. Since $\psi(T') = O$ and $\psi\phi = [2]$, this proves

$$\psi(E'(\mathbb{Q})) = 2E(\mathbb{Q}), \quad \ker \alpha = 2E(\mathbb{Q}), \quad [E(\mathbb{Q}) : 2E(\mathbb{Q})] = 2.$$

There is exactly one nonzero rational point of order two: the other quadratic factor has discriminant 2197, a nonsquare. Mordell–Weil finite generation therefore identifies this index as $2^{\text{rank } E+1}$, giving rank zero.

The discriminant $16 \cdot 507^2 \cdot 2197$ has support in $\{2, 3, 13\}$. At the good primes 5, 7, complete counts of affine ordinates at successive abscissas are

$$(1, 2, 0, 2, 0), \quad (1, 0, 0, 0, 0, 2, 0).$$

Including infinity gives orders 6 and 4. Good-reduction injection on prime-to-residue-characteristic torsion shows that the rational torsion order divides 2: its 5-primary part is

excluded at 7, its 7-primary part at 5, and every other primary part divides both orders. The known point $(0, 0)$ gives equality. This proves the lemma. The finite-generation and torsion-injection inputs are the standard ones already specified in the preceding descent section. $\square$

Lemma 210.2 (An integral map to the thirteen twist). *For arbitrary integers a, b, s , put*

$$c = a - b, \quad y = a + b, \quad A = 7a^2 + 12ab + 7b^2, \quad U = 13(A - 26s), \quad V = 13Uy.$$

Then

$$(721) \quad A^2 + 3c^4 = 52F(a, b), \quad c^2 + 13y^2 = 2A,$$

$$(722) \quad V^2 - U^3 + 13U^2c^2 + 507Uc^4 = 8788U(F(a, b) - 13s^2).$$

If $F(a, b) = 13s^2$ and $c \neq 0$, then $U \neq 0$, and

$$(X, Y) = (U/c^2, V/c^3)$$

is a finite rational point of E with $X \neq 0$.

Proof. The first two identities follow by expansion. Factoring the left side of (722) by U , its bracket becomes

$$169Uy^2 - U^2 + 13Uc^2 + 507c^4 = 26AU - U^2 + 507c^4 = 169(A^2 - 676s^2 + 3c^4),$$

which is $8788(F - 13s^2)$ by (721). If $U = 0$, then $A = 26s$, and the first identity with $F = 13s^2$ forces $3c^4 = 0$, a contradiction. Dividing the resulting homogeneous cubic identity by c^6 proves the curve equation. The only denominator condition is $c \neq 0$; there is no division by a, b, s, A or V . $\square$

Theorem 210.3 (The actual thirteen-square class). *Let a, b be coprime positive integers and let $s \in \mathbb{Z}$. Then*

$$F(a, b) = 13s^2 \iff a = b = 1 \text{ and } s \in \{-1, 1\}.$$

Consequently no such seed has $F = 13Q^g$ with $Q > 1$ and a positive even exponent g .

Proof. If $a \neq b$, Lemma 210.2 gives a finite point of E with nonzero X , contradicting Lemma 210.1. Thus $a = b$, and positivity and coprimality give $a = b = 1$. Substitution gives $s^2 = 1$. The converse is direct. For the final assertion put $s = Q^{g/2}$; the classification gives $Q^{g/2} = 1$, contrary to $Q > 1$. $\square$

Corollary 210.4 (The homogeneous integer classification). *For arbitrary integers a, b, s ,*

$$F(a, b) = 13s^2 \iff a = b \text{ and } s \in \{-a^2, a^2\}.$$

Proof. The integral map and rational group classification impose $a = b$ without positivity or coprimality. Substitution gives $s^2 = a^4$, hence $(s - a^2)(s + a^2) = 0$. The converse follows by the same substitution, including $a = b = s = 0$. $\square$

Verification scope. The three local descent exclusions, both good-prime point counts and the explicit rational map were independently reviewed. The integral map is a separately reviewed simplification whose sole denominator is $a - b$. Exact replay checks the polynomial identities and all admissible residue pairs and the original rational-map identities; it does not supply the elliptic group theorem by a rank database call. Scoped Lean verification of the integer identities and nonvanishing verifies that bridge, not Mordell–Weil, the two-isogeny descent, or the complete rational-point classification. The final Lean diagonal statement has the integral-point obstruction as an explicit hypothesis.

211. THE HOMOGENEOUS INTEGER CONCLUSION

The integral map in Lemma 210.2 has only the nonzero difference $c = a - b$ as a denominator condition. It therefore gives the following stronger domain statement, beyond positive primitive seeds.

Corollary 211.1 (All integer points of the thirteen square class). *For arbitrary integers a, b, s ,*

$$F(a, b) = 13s^2 \iff a = b \text{ and } (s = a^2 \text{ or } s = -a^2).$$

Proof. If $a \neq b$, Lemma 210.2 contradicts Lemma 210.1. Otherwise $F(a, a) = 13a^4$, and $(s - a^2)(s + a^2) = 0$ gives the two signs. The converse follows by substitution, including $a = b = s = 0$. $\square$

212. FORMAL SCOPE OF THE SQUARE GAP AND INTEGRAL BRIDGE

Two new modules contain 23 declarations accepted by a fresh scoped Lean 4.32.0 build. The same build recompiles 65 unchanged dependency declarations, with warnings treated as errors and an axiom inventory for every declaration. Only `propext`, `Classical.choice` and `Quot.sound` occur. This is a scoped build, not a full repository build.

`SquareGapArithmetic` contains 16 declarations. It proves the actual polynomial gap, transfers the modulo-eight residue table to all primitive integer seeds, establishes positivity and the required integer square separation, and concludes, for arbitrary signed integer q ,

$$a, b > 0, \quad \gcd(a, b) = 1, \quad F(a, b) = q^2 \implies 2a < 3b^3, \quad 2b < 3a^3, \quad a, b > 1.$$

The gap is derived from the actual power equation; it is not an extra hypothesis. The sharper residue-dependent coefficient $2r$ in the ordinary argument is not claimed as the conclusion of this module.

`ThirteenMapArithmetic` contains seven declarations: the three integral identities above, the actual weighted curve equation, nonvanishing of U , construction of a nontrivial integral point, and an explicitly conditional diagonal consequence. The final theorem takes the following antecedent as an ordinary theorem parameter:

$$\forall U, V, c \in \mathbb{Z}, \quad c \neq 0, \quad V^2 = U^3 - 13U^2c^2 - 507Uc^4 \implies U = 0.$$

It does not declare that antecedent as an axiom or prove it. The passage to rational coordinates and the elliptic group, isogeny descent, Mordell–Weil and torsion arguments are not formalized by these seven declarations. Thus the complete thirteen-square-class classification has an independently reviewed ordinary proof, while its integral bridge has the stated narrower formal verification.

The sieved interval estimate, effective affine-slice theorems, boundary modularity, residual support and density arguments retain their named external inputs. None of the new modules proves the large-prime tail, a uniform actual-point height bound, candidate elimination in all residual classes, or standard ABC.

213. EIGHTH CONTINUATION: GLOBAL IMAGE AND ACTUAL PURE-POWER SUPPORT

Standard ABC remains unproved and undisproved. A global projective-image argument now excludes every CM residual candidate for the actual positive Frey seeds at prime exponents greater than seven. Actual unbounded local shadows of a CM boundary remain valid; they pass the specified finite tests but do not satisfy the asserted global representation comparison.

The two remaining fixed-level non-CM orbits are exact quadratic twists. Two independent computations compare every coefficient through the proved common-level Sturm bound, including bad-prime indices. This establishes a common boundary module for every actual pure prime power, retaining all coefficient embeddings and character twists. The boundary's rational two-torsion then excludes the exponent prime from the norm support and gives the stronger necessary cutoff $q \geq (\sqrt{2p} - 1)^2 > p$. It yields a height lower bound and an effective exponent budget, without a uniform upper bound for actual points.

The analytic route separately extends its root-window average to every exponent index. Primes above the growing endpoint and prescribed exceptional roots remain open. Ten new Lean declarations verify the elementary trace and actual-height consequences with their modular antecedents explicit. The global modular, local representation and analytic proofs retain their stated external inputs and ordinary independent review status.

214. AN ACTUAL SIEVED WINDOW AT EVERY EXPONENT INDEX

The sieved estimate also yields a window beyond the integer-root interval length without restricting the exponent to a bounded-divisor-sum subsequence. The endpoint is slightly smaller; the unbounded upper tail and exceptional actual roots remain outside this conclusion.

Lemma 214.1 (An elementary uniform divisor-sum bound). *For every integer $n \geq 8$,*

$$\frac{\sigma(n)}{n} \leq e^{18}(1 + \log \log n).$$

Proof. For $x \geq 2$ put $s = 1 + 1/\log x$. The nonnegative logarithmic series and $1 - e^{-u} \leq u$ give

$$\log \prod_{p \leq x} \frac{1 - p^{-s}}{1 - p^{-1}} = \sum_{p \leq x} \sum_{j \geq 1} \frac{p^{-j} - p^{-sj}}{j} \leq (s - 1) \sum_{p \leq x} \frac{\log p}{p - 1}.$$

The already proved elementary estimate $\theta(t) \leq 3t$ and partial summation imply

$$\sum_{p \leq x} \frac{\log p}{p - 1} \leq 2 \left(\frac{\theta(x)}{x} + \int_2^x \frac{\theta(t)}{t^2} dt \right) \leq 6 + 6 \log x.$$

Hence the logarithmic ratio is at most $6 + 6/\log 2 < 15$. The finite Euler product with exponent s is at most $\zeta(s)$, while the elementary integral comparison gives $\zeta(s) \leq 1 + 1/(s - 1)$. It follows that

$$\prod_{p \leq x} (1 - p^{-1})^{-1} \leq e^{15}(1 + \log x).$$

Set $x = \log n \geq 2$. The number of distinct prime factors of n exceeding x is at most $x/\log x$. Their additional logarithmic product is at most

$$\sum_{\substack{p|n \\ p > x}} -\log(1 - p^{-1}) \leq \sum_{\substack{p|n \\ p > x}} \frac{1}{p - 1} \leq \frac{x}{(x - 1) \log x} \leq \frac{2}{\log 2} < 3.$$

Combine the two bounds with $\sigma(n)/n \leq \prod_{p|n} (1 - p^{-1})^{-1}$. This is the standard elementary Euler-product argument, with a loose constant sufficient for the application. $\square$

For the actual roots $w_k = 3k + \zeta$, $B \leq k < 2B$, retain $T_n(k)$, t_k and $E_{Y,Z}(k)$ from Section 205.

Theorem 214.2 (An all-index window beyond the root interval). *Let n tend to infinity through all positive integers. Put*

$$B = n^4, \quad Y = n^{1/6}, \quad Z = \left\lfloor \frac{n^4 \sqrt{\log n}}{1 + \log \log n} \right\rfloor.$$

Then $Z/B \rightarrow \infty$ and

$$\frac{1}{B} \sum_{B \leq k < 2B} \frac{E_{Y,Z}(k)}{t_k} = O((\log n)^{-1/2}),$$

with an absolute constant. The proportion of actual roots for which $E_{Y,Z}(k)/t_k > (\log n)^{-1/4}$ is $O((\log n)^{-1/4})$.

Proof. Eventually $Z > 3n$ and $5 \leq Y \leq Z$, so Theorem 205.1 applies. Its first and last terms are at most $5/(12\sqrt{n})$ and $1/(4n)$. Lemma 214.1 cancels the factor $1 + \log \log n$ in the chosen endpoint, and

$$\log(Z/(3n)) = 3 \log n + \frac{1}{2} \log \log n - \log(1 + \log \log n) - \log 3 + o(1) \sim 3 \log n.$$

Thus its middle term is $O((\log n)^{-1/2})$ for every index. Markov's inequality applied to the nonnegative actual cost gives the exceptional proportion. No distributional assumption on the power-map output is used. $\square$

The larger endpoint $B\sqrt{\log n}$ remains available on the previously specified bounded- $\sigma(n)/n$ subclasses. Both conclusions leave all primes above their endpoints and the exceptional roots uncontrolled. Neither bounds the complete signed packet of a prescribed root.

215. A CM LOCAL BOUNDARY AND ITS GLOBAL PROJECTIVE-IMAGE OBSTRUCTION

The boundary considered here differs from the non-CM seed $(0, 1)$. It supplies actual positive seeds satisfying a specified finite collection of local tests, while a separate global theorem excludes congruence to CM newforms. Both assertions concern their exact stated tests and representations.

Write $N(a, b) = a^2 + ab + b^2$ and retain the actual quartic $F(a, b)$ and Frey curve $E_{a,b}$ defined in the preceding modular sections.

Lemma 215.1 (An explicit CM boundary). *At $(a, b) = (-1, 1)$ one has $N(a, b) = F(a, b) = 1$, and the actual Frey curve specializes over $K = \mathbb{Q}(r)$, $r^2 = -3$, to*

$$E_* : Y^2 = X^3 + 12rX.$$

It has geometric CM by $\mathbb{Z}[i]$, good reduction away from the primes over 6, and trace zero at both places over every rational prime $q \equiv 7 \pmod{12}$.

Proof. The specialization follows by substitution. Its discriminant is $-64(12r)^3$, and its j -invariant is 1728. Over $K(i)$ the origin-preserving automorphism $(X, Y) \mapsto (-X, iY)$ squares to elliptic negation, giving the stated endomorphisms. For $q \equiv 7 \pmod{12}$, the prime splits in K and -1 is a quadratic nonsquare modulo q . At either place choose $r_q^2 = -3$. The polynomial $f(X) = X^3 + 12r_qX$ is odd, so the quadratic-character sum of $f(X)$ cancels in pairs $X, -X$, with zero contribution at $X = 0$. Thus the full point count is $q + 1$ and its trace is zero. $\square$

Theorem 215.2 (Actual positive finite local shadows). *For every positive integer L and every height target, there exist positive coprime a, b of greater height satisfying*

$$(a, b) \equiv (-1, 1) \pmod{L}, \quad N(a, b) \equiv F(a, b) \equiv 1 \pmod{L}.$$

Their Frey coefficients agree with those of E_ modulo L . Consequently they satisfy all direct norm power-residue tests of any positive exponent at divisors of L , and all the trace-zero tests at primes $q \mid L$ with $q \equiv 7 \pmod{12}$.*

Proof. Take $b = 1$ and $a = Lk - 1$ for arbitrarily large positive integers k . Primitivity is immediate and $a + b = Lk$ tends to infinity. Evaluate the two integer polynomials at $(-1, 1)$ to get the norm congruences; the root 1 certifies every stated power-residue test. The polynomial Frey coefficients also specialize modulo L . Their reductions at the stated primes are good and identical to E_* , so their traces are zero by Lemma 215.1. Taking L divisible by specified prime powers preserves any specified finite coefficient precision. The mixed seed remains primitive because a prime dividing both ab and N would divide both a, b ; its square identities hold exactly for these actual integer seeds. $\square$

These shadows assert no global power equation. In fact Theorem 207.1 excludes a square F throughout this family, since $b = 1$. Theorem 206.2 makes all global pure powers $F(a, 1) = Q^g$, $Q > 1$, $g \geq 2$, effectively finite across all exponents. It does not classify the remaining odd-exponent possibilities. Nor does the local shadow identify a particular CM newform or supply an isomorphism of residual Galois modules.

Theorem 215.3 (Global exclusion of CM residual candidates). *Let a, b be positive coprime integers and $p > 7$ a prime. The specified residual descent of $E_{a,b}[p] \otimes \chi$ used in the Frey modular reduction is not isomorphic, over an algebraic closure of its residue coefficient field, to the semisimplified residual representation of any characteristic-zero CM newform.*

Proof. The actual quartic satisfies $F \geq 13$ and $\gcd(F, 6) = 1$, so some prime $q > 3$ divides it. The proved local calculation makes $E_{a,b}$ multiplicative at the places over q , and its explicit cyclic isogeny to its conjugate has square-free degree two. The general Q-curve large-image theorem, with the same published $N_3 = 7$ input already checked in the Frey modular section, therefore applies.

For precision about its target, Ellenberg's [author-hosted original paper](#), Theorem 3.14, displays a surjective projective extension

$$\mathbb{P}\bar{\rho}_{E,p} : G_{\mathbb{Q}} \longrightarrow \mathrm{PGL}_2(\mathbb{F}_p),$$

unless the curve has potentially good reduction at all primes outside 6. The multiplicative place excludes that alternative. The explicit $N_3 = 7$ specialization is the general Theorem 5.2 in [Pacetti–Villagra Torcomian](#); it is an external published input, not a new computation or an inference from a nonprimitive substitution in another Diophantine equation.

On G_K , the image H of this projective representation is a normal subgroup of index at most two in $\mathrm{PGL}_2(\mathbb{F}_p)$. It contains $\mathrm{PSL}_2(\mathbb{F}_p)$. Indeed the latter is perfect for $p > 3$: $\mathrm{SL}_2(\mathbb{F}_p)$ is generated by upper and lower elementary unipotents, and, for $t^2 \neq 1$,

$$[\mathrm{diag}(t, t^{-1}), U(x/(t^2 - 1))] = U(x).$$

The analogous lower formula proves perfectness before and after projectivization. Every abelian quotient of the projective group kills that subgroup. Its projective image is nontrivial, so it is not solvable; neither is H . We have not asserted that the image on G_K itself must be the full projective linear group.

A CM newform's characteristic-zero representation is induced from a one-dimensional character of the Galois group of its imaginary quadratic CM field. This CM representation description is also used explicitly in Section 6 of the same primary Q-curve paper. In an induced stable lattice its two cosets give diagonal matrices on the index-two subgroup and antidiagonal matrices on the other coset. After reduction, if the two inducing characters differ, the resulting module is irreducible of this monomial form; if they coincide, it splits over $\mathbb{F}_p$, since p is odd and the coset matrix has two distinct square-root eigenvalues. Thus the semisimplified projective image is abelian or cyclic-by-a-group-of-order-at-most-two, and is solvable. The diagonal projective image is finite cyclic as a finite subgroup of $\mathbb{F}_p^*$.

Any proposed residual isomorphism can now be restricted to G_K . Projectivization removes the scalar character χ . The actual side has the nonsolvable image H , whereas the CM side has a subgroup of a solvable group. This is a contradiction. Coefficient embeddings and scalar twists do not affect it. Large image also ensures that semisimplifying the actual module loses no extension in this comparison. $\square$

This uses uniform large projective image. The fact that an elliptic curve individually has no geometric CM would not suffice to exclude all its CM residual congruences at the stated primes.

Corollary 215.4 (Two remaining pure-branch orbits). *If an actual seed satisfies $F = Q^p$ with $Q > 1$ and prime $p > 7$, its fixed-level modular candidate belongs to one of the two non-CM Galois orbits at levels 288 and 576 in the complete exact newspace enumeration. All coefficient embeddings remain candidates.*

Proof. The pure-branch modular theorem supplies weight two, character ψ_3 and one of the five levels 36, 72, 144, 288, 576. The independently repeated complete enumeration in the preceding manuscript lists precisely two non-CM orbits. Apply Theorem 215.3 to all the CM candidates. $\square$

The same projective-image argument removes CM candidates at the varying levels and non-flat weights of the other modular reductions. Those spaces are not reduced to the same two-orbit list. The two non-CM pure-branch orbits are not excluded here, and the level-576 boundary comparison continues to retain the distinction between χ and χ^{-1} .

Verification and remaining scope. All five ordinary claims passed complete independent reviews by the other two research agents; both reopened the original projective-image source and the published small-prime input. A separate exact replay enumerates all 81,376 affine states in the sixteen listed finite reductions and checks thirty-two actual positive congruence shadows. These finite checks do not establish the global representation theorem. No new Lean proof of projective-image surjectivity, CM induction, or modular elimination is asserted. The finite local shadows and the global CM obstruction coexist: the latter closes that specific candidate route while leaving the non-CM modular and analytic tail routes open.

216. EXACT TWISTS OF THE TWO NON-CM BOUNDARY ORBITS

The two surviving pure-branch orbits are related by quadratic twists. This section proves the relation using a complete exact coefficient comparison and a proved common-level determination bound. It then derives the boundary module premise for every actual pure prime power. The modular-form coefficients are exact software inputs, independently repeated; they are not asserted to have Lean proofs.

Retain $K = \mathbb{Q}(\sqrt{-3})$, the actual quartic F , the Frey curves $E_{a,b}$, the boundary curve E_0 , and the order-four character χ from the preceding sections. Write $\epsilon = \psi_3$ for the quadratic character of conductor 12. Let f be the normalized representative of the non-CM orbit at level 576 identified in Theorem 208.5, and let g be the representative of the unique non-CM orbit at level 288 in the exact enumeration. For both, fix the recorded field model $L = \mathbb{Q}[z]/(z^4 + 1)$.

Lemma 216.1 (Common level and coefficient determination). *Both a coefficient embedding of g and either quadratic coefficient twist $f \otimes \psi_{-2}$ or $f \otimes \psi_2$ belong to*

$$S_2(\Gamma_0(M), \epsilon), \quad M = 36864.$$

Equality of their Fourier coefficients in degrees 0 through 12288 implies equality of the modular forms.

Proof. A quadratic twist of conductor $m = 8$ has a valid level bound Nm^2 and the same character as a form of level N and character ϵ . Here is a direct check. For $T_u = \begin{pmatrix} 1 & u/m \\ 0 & 1 \end{pmatrix}$, the nonzero Gauss sum gives the twist as

$$\tau(\psi)^{-1} \sum_{u \bmod m} \psi(u) f|_2 T_u.$$

Its coefficients are $\psi(n)a_n(f)$. Given $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(Nm^2)$, choose $v \equiv ud^2 \pmod{m}$. Then

$$T_u \gamma T_{-v} = \begin{pmatrix} a + uc/m & b + (ud - av)/m - ucv/m^2 \\ c & d - cv/m \end{pmatrix} \in \Gamma_0(N).$$

Its lower-right entry equals d modulo N . The permutation $u \mapsto ud^2$ preserves the quadratic character, giving exactly the transformation character $\epsilon(d)$. Rational translates preserve holomorphy and vanishing at cusps. This proves the level bound, independently of a software metadata level.

Take $N = 576$. Then $M = 576 \cdot 8^2 = 36864$, and $288 \mid M$. The index and safe cutoff are

$$(723) \quad I = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(M)] = M(1 + 1/2)(1 + 1/3) = 73728, \quad B = \left\lfloor \frac{2I}{12} \right\rfloor = 12288.$$

To prove coefficient determination, suppose a nonzero difference H vanishes through degree B . Form the product of its weight-two translates over the I right cosets of $\Gamma_0(M)$ in $\mathrm{SL}_2(\mathbb{Z})$. Its square is a nonzero holomorphic level-one form of weight $4I$: squaring removes the possible transformation signs of the quadratic character. Its order at infinity is at least $2(B + 1)$, as every other cusp contribution is nonnegative. This exceeds $4I/12$, contradicting the level-one valence formula; see [137, Proposition 4.12 and Example 4.13]. $\square$

Theorem 216.2 (The two exact boundary twists). *In the recorded common field model, let $\sigma_j(z) = z^j$ for odd j modulo 8. The normalized forms satisfy*

$$(724) \quad g^{\sigma_7} = f \otimes \psi_{-2}, \quad g^{\sigma_5} = f \otimes \psi_2.$$

Proof. The polynomial $z^4 + 1$ is the eighth cyclotomic polynomial, so these are field automorphisms. In the basis $(1, z, z^2, z^3)$ they act by

$$\begin{aligned}\sigma_7(c_0, c_1, c_2, c_3) &= (c_0, -c_3, -c_2, -c_1), \\ \sigma_5(c_0, c_1, c_2, c_3) &= (c_0, -c_1, c_2, -c_3).\end{aligned}$$

The retained PARI/GP 2.15.4 computation initializes both complete newspaces, selects the unique non-CM form in each by its CM test, and checks their field and character models. Their dimensions are 4, 8, and the automatically selected orbit indices are 1, 3. It outputs every coefficient from degree 0 to degree 12288 in four rational coordinates. The independent Python comparison applies the two displayed coordinate transformations and the quadratic Kronecker values of discriminants $-8, 8$. All coefficients agree, including every degree divisible by 2 or 3.

The canonical evidence retains both complete arrays, each with 12289 rows. A second independent implementation selects the forms afresh, uses PARI's actual twist operator, and compares the four coordinates; all 98312 coordinate equalities agree. It does not reuse the first program's coefficient cache or polynomial substitution. These are complete exact modular-form computations, rather than floating point approximations. Lemma 216.1 upgrades these full-cutoff equalities to (724). $\square$

Theorem 216.3 (The common boundary module for every actual pure power). *Let a, b be positive coprime integers, let $p > 7$ be prime, and suppose $F(a, b) = Q^p$ for an integer $Q > 1$. Then for some character*

$$\nu \in \{1, \psi_3|_{G_K}, \psi_{-2}|_{G_K}, \psi_{-6}|_{G_K}\}$$

one has the full residual module isomorphism

$$(725) \quad E_{a,b}[p] \otimes \overline{\mathbb{F}}_p \simeq (E_0[p] \otimes \overline{\mathbb{F}}_p) \otimes \nu.$$

Proof. The actual pure-branch modular theorem, the CM exclusion and the complete enumeration give exactly the two possible orbits in Corollary 215.4. By (724), every embedding of the g orbit is the quadratic twist by ψ_{-2} of some embedding of the f orbit; rational character values commute with embeddings.

For the f orbit, the compatible boundary identification restricts to the system of E_0 times χ^σ , whereas the actual system restricts to $E_{a,b}$ times χ . The ratio is either 1 or $\chi^{-2} = \psi_3|_{G_K}$. The g orbit adds $\psi_{-2}|_{G_K}$, producing exactly the four displayed characters. An equality of twisted modular forms gives the corresponding twist of compatible representations: their Frobenius polynomials agree at every prime outside the finite ramification set, so Chebotarev and semisimplicity identify the representations. The checked large projective-image input gives absolute irreducibility of the actual residual representation. Hence the semisimplified congruence becomes the full module isomorphism (725). All four characters are quadratic and unramified outside 6. $\square$

Thus the boundary premise of Theorem 209.1 is now derived for every actual pure p -th power. In particular every prime $q \mid Q$ satisfies $q \geq (\sqrt{p} - 1)^2$, neither 7 nor 13 divides Q , and for $q \neq p$ one has $p \mid (q + 1)^2 - t_q(E_0)^2$. For every sufficiently large fixed p , its permitted split rational prime support has density

$$\frac{2p^2 - p - 5}{2(p - 1)^2(p + 1)} \sim \frac{1}{p}.$$

This remains a fixed- p statement with no uniform Chebotarev error term. The next section uses two-torsion to remove the exception $q = p$ and improve the cutoff.

Already the displayed cutoff and $F \leq (a + b)^4$ give

$$\log(a + b) \geq \frac{p}{2} \log(\sqrt{p} - 1).$$

Alternatively $F \leq 13 \max(a, b)^4$ gives the same lower bound for $\log \max(a, b)$ minus $(\log 13)/4$. These are point-height lower bounds, compatible with an $O(p)$ coefficient-height bound for a defining Kummer cover. Neither is an upper bound for its points.

Evidence and scope. The exact primary coefficient output has SHA256 c8181912aa36023ff86ac94c72bc59cf0279a67a42849035450de866ea115549; the independent twist comparison has SHA256 9864149fe7dc844d34f71306192e49b05c90d40bb22728fe71940b173a9807e0. The scripts, full arrays, failure checks and source references are retained in this checkpoint. Their modular-form algorithms are explicit exact-software dependencies; no Lean proof of those algorithms or the compatible-system comparison is claimed. Large moving prime factors, uniform point-height bounds, and compatibility with the first compressed norm remain open. Varying nonunit residual levels are not reduced to these two fixed orbits by this argument.

217. TWO-TORSION EXCLUDES THE EXPONENT PRIME FROM THE SUPPORT

The fixed boundary curve has the nonzero rational two-torsion point $(0, 0)$. At every split rational prime $q > 3$, good reduction preserves it, so

$$(726) \quad t_q(E_0) = q + 1 - \#E_0(\mathbb{F}_q) \quad \text{is even.}$$

This elementary parity condition gives a stronger consequence of the boundary module comparison, including at the exponent prime itself.

Theorem 217.1 (Boundary support with no exponent-prime exception). *Let a, b be positive coprime integers and $p > 7$ prime. Let ν be a quadratic character of G_K unramified outside the primes over 6. Suppose*

$$(727) \quad E_{a,b}[p] \otimes \overline{\mathbb{F}}_p \simeq (E_0[p] \otimes \overline{\mathbb{F}}_p) \otimes \nu.$$

Then $p \nmid F(a, b)$, and every prime $q \mid F(a, b)$ satisfies

$$(728) \quad q \geq (\sqrt{2p} - 1)^2 > p, \quad p \mid (q + 1)^2 - t_q(E_0)^2.$$

Proof. We first specify the local inputs at p . For multiplicative reduction over $\mathbb{Q}_p$, the full Tate representation is an extension

$$0 \longrightarrow \mathbb{F}_p(\omega\eta) \longrightarrow E[p] \longrightarrow \mathbb{F}_p(\eta) \longrightarrow 0,$$

where ω is the mod- p cyclotomic character and η is the unramified quadratic splitting character, possibly trivial. For good supersingular reduction over $\mathbb{Q}_p$, the residual representation is irreducible over $\mathbb{F}_p$. These local facts are the Tate and good-reduction statements of [138, Propositions 2.12(b) and 2.11(c)]. For good ordinary reduction, the connected-étale filtration has characters $\omega\lambda^{-1}$ and λ , where λ is unramified. Its arithmetic Frobenius value modulo p is the unit root of $T^2 - t_p T + p$, and hence is t_p modulo p ; see [139, §1.3.1 and Exercise 1.30]. These are local statements over $\mathbb{Q}_p$, as in their proofs by the Tate curve and finite-flat or formal groups over $\mathbb{Z}_p$.

Suppose $p \mid F$. The actual local calculation makes p split in K , identifies both completions with $\mathbb{Q}_p$, and gives multiplicative reduction of $E_{a,b}$. Work at either completion. The character ν is unramified and quadratic. From the Tate subcharacter and (727), the boundary representation has a subcharacter $\omega\eta\nu$ after coefficient extension. This character has values in $\mathbb{F}_p$. Existence of an intertwiner after extending coefficients implies existence over $\mathbb{F}_p$: the intertwining maps are the kernel of linear equations over $\mathbb{F}_p$, and scalar extension preserves this kernel. The representations have finite image, so finitely many equations suffice. Thus the supersingular irreducibility statement just cited rules out supersingular reduction. No absolute-irreducibility upgrade of that local statement has been assumed. The good boundary curve is ordinary.

Its Jordan–Holder characters after twisting by ν are $\lambda\nu$ and $\omega\lambda^{-1}\nu$. The actual Tate characters are η and $\omega\eta$. Since ω is nontrivial on inertia and all other characters here are unramified, each pair has exactly one inertia-trivial character. The isomorphism therefore gives

$$(729) \quad \lambda\nu = \eta, \quad t_p(E_0) \equiv \pm 1 \pmod{p}.$$

This compares the unique inertia-trivial Jordan–Holder character, even if an extension splits. It does not take an unramified trace of the full two-dimensional Tate representation at p .

By (726), t_p is even. But the Hasse bound and $p > 7$ give $|t_p| \leq 2\sqrt{p} < p - 1$. The only integers in this interval congruent to 1 or -1 modulo p are 1 and -1 , which are odd. This contradiction proves $p \nmid F$.

Now every prime divisor q of F differs from p . The boundary comparison of Theorem 209.1 gives $t_q \equiv \pm(q+1) \pmod{p}$. Both t_q and $q+1$ are even, so the corresponding nonzero integer difference is divisible by $2p$. It is nonzero by the strict Hasse inequality $|t_q| \leq 2\sqrt{q} < q+1$, and thus

$$2p \leq |t_q \mp (q+1)| \leq q+1 + 2\sqrt{q} = (\sqrt{q}+1)^2.$$

This gives the first cutoff in (728). The strict inequality with p follows from $p+1 > 2\sqrt{2p}$ for $p > 7$, equivalently $p^2 - 6p + 1 > 0$. The remaining divisibility is exactly the boundary trace condition, now valid at every prime divisor. $\square$

Corollary 217.2 (Unconditional support of the actual pure branch). *Let a, b be positive coprime integers, let $p > 7$ be prime, and suppose $F(a, b) = Q^p$ for an integer $Q > 1$. Then $p \nmid Q$ and every prime $q \mid Q$ satisfies (728). Neither 7 nor 13 divides Q , and*

$$(730) \quad Q \geq (\sqrt{2p} - 1)^2 > p.$$

Proof. Theorem 216.3, using the exact twist identity and its stated modular dependencies, gives (727) with a quadratic character unramified outside 6. Apply Theorem 217.1 and the previously proved boundary exclusions at 7 and 13. Since $Q > 1$ has a prime divisor, the prime cutoff gives (730). $\square$

For every sufficiently large fixed p , every prime factor now lies in the permitted split rational-prime set of Theorem 209.3. Its density is

$$\frac{2p^2 - p - 5}{2(p-1)^2(p+1)}.$$

There is no exponent-prime exception. Removing the finite interval below the cutoff does not change this fixed- p density or supply an error term uniform in p .

Corollary 217.3 (Actual height lower bounds). *Under the hypotheses of Corollary 217.2, put $c = a + b$ and $H = \max(a, b)$. Then*

$$(731) \quad \log c \geq \frac{p}{2} \log(\sqrt{2p} - 1),$$

$$(732) \quad \log H \geq \frac{p}{2} \log(\sqrt{2p} - 1) - \frac{\log 13}{4}.$$

In particular $p^p < F(a, b) \leq 13H^4$.

Proof. The identity $4F = (a^2 + b^2)^2 + 3(a+b)^4$ gives $F \leq c^4$; termwise comparison in the quartic gives $F \leq 13H^4$. Combine these with $F = Q^p$ and (730), then take logarithms. Its strict inequality $Q > p$ gives the last assertion. $\square$

Verification and remaining scope. The full local module proof, the distinction between ordinary and supersingular reduction, and the scalar-extension step have independent ordinary reviews with the cited primary passages actually opened. The elliptic representation inputs are not claimed to be formally proved in Lean by arithmetic checks of the final inequalities. These necessary height lower bounds remain compatible with arbitrarily large seeds and moving large root primes. They do not give a point-height upper bound, a contradiction to a cover's coefficient budget, or an ABC proof. The finite local residue shadows remain valid because they are not asserted to be global pure powers.

218. AN EFFECTIVE EXPONENT BUDGET AND ITS FORMAL ARITHMETIC CORE

Corollary 218.1 (An explicit pure-power exponent budget). *Let a, b be positive coprime integers, $p > 7$ prime, and $F(a, b) = Q^p$ for an integer $Q > 1$. Put $H = \max(a, b)$ and $T = \log 13 + 4 \log H$. Then*

$$p^p < F(a, b) \leq 13H^4, \quad p \log p < T.$$

If $T \geq e^2$, then

$$p < \frac{2T}{\log T}.$$

In particular $p = O(\log H / \log \log H)$ as $H \rightarrow \infty$, with an effective absolute constant.

Proof. Corollary 217.2 gives $Q > p$, and Corollary 217.3 gives the polynomial height bound. Taking logarithms proves the first two assertions. For the explicit rearrangement, suppose $p \geq 2T / \log T$. The function $x \log(x/2)$ increases for $x \geq 2$, and $2T / \log T \geq e^2$ when $T \geq e^2$. For $z = \log T \geq 2$, one has $2 \log z < z$: the difference $z - 2 \log z$ is positive at 2 and has nonnegative derivative thereafter. Therefore

$$p \log(p/2) \geq \frac{2T}{\log T} \log \frac{T}{\log T} = 2T \left(1 - \frac{\log \log T}{\log T}\right) > T,$$

contrary to $p \log p < T$. The asymptotic form follows since $T = 4 \log H + O(1)$. No rearranged bound is needed when $T < e^2$. $\square$

This budget bounds the exponent in terms of the actual point height. It supplies no height upper bound. In particular, it does not rule out arbitrarily high seeds with moving prime factors, or treat nonunit residuals $F = VQ^p$ by the same fixed-level argument.

Ordinary proofs before kernel verification. The reviewed local proof uses the following elementary facts. If $q \geq 7$ and $t^2 \leq 4q$, then

$$1 - q < t < q - 1, \quad (q - 1)^2 - 4q = q(q - 6) + 1 > 0.$$

Hence an even such trace cannot be congruent to 1 or -1 modulo q . For odd p, q and even t , if p divides either $q + 1 - t$ or $q + 1 + t$, then $2p$ divides that positive integer. The displayed interval places it below $2q$, giving $p < q$. The decomposition into one divisible factor uses primality in the separate ordinary argument; it is not hidden in the formal signature.

The polynomial height estimates also have direct certificates:

$$\begin{aligned} (a + b)^4 - F(a, b) &= ab(a^2 + ab + b^2), \\ 13a^4 - F(a, b) &= (a - b)(12a^3 + 9a^2b + 4ab^2 + b^3). \end{aligned}$$

For $a, b \geq 0$ the first gives $F \leq (a + b)^4$; the second gives $F \leq 13a^4$ when $b \leq a$. Symmetry gives $F \leq 13H^4$.

Exact scope of the new Lean module. The module `BoundarySupportArithmetic.lean` checks ten declarations: the strict Hasse interval, a small-multiple lemma, the exclusion of unit residues for even traces, doubling an odd divisor of an even number, the trace-factor prime cutoff, the sum-height identity, the sum-height inequality, the left-coordinate height inequality, its maximum-coordinate version, and the pure-power height implication.

Its last signature explicitly assumes $p < Q$ and $F(a, b) = Q^p$. Its trace cutoff explicitly assumes the disjunction $p \mid q + 1 - t$ or $p \mid q + 1 + t$. These are ordinary modular/local conclusions supplied outside the module, not additional axioms or claims of their complete formalization. The new proofs use the existing actual quartic definition. A fresh scoped Lean 4.32.0 build accepts all ten declarations and recompiles fifty-five unchanged dependency declarations. The complete axiom inventory is contained in `propext`, `Classical.choice`, and `Quot.sound`.

No Lean proof of modularity, the full-Sturm modular-form algorithms, the local elliptic representations, the analytic logarithm estimate, or ABC is claimed by that arithmetic build. The exact software replays, ordinary independent reviews, and kernel verification remain separate evidence with separately recorded source hashes.

219. NINTH CONTINUATION: SIGNED COMPENSATION AND SIMULTANEOUS NORM SUPPORT

Standard ABC remains unproved and undisproved. This continuation develops two distinct connections. First, the actual three boundary-arm quotients give a signed compensation inequality: sufficient radical credit in two arms can absorb arbitrary depth in the third. The resulting asymptotic subclass is proved uniformly at large extraction exponents, while membership of arbitrary actual roots remains open. Second, simultaneous pure powers of the two actual norms

force much of the original sum coordinate into specific prime progressions. The support budget retains the complete exceptional-prime valuation and extends from primes to prime powers and then to general common exponents with an explicit totient cost.

Fifty-four new Lean declarations verify actual arm arithmetic, finite integer-weight compensation, the common-exponent root gap, and elementary support assembly. Their complete scoped axiom audit also recompiles fifty-four unchanged dependency declarations. The ordinary prime allocation, real logarithmic estimates and arbitrary-root membership obligations remain distinct from these kernel-checked statements. In particular, a lower bound on full valuation mass is not a bound on the radical.

220. ACTUAL ARM QUOTIENTS AND SIGNED COMPENSATION OF A PRIVATE DEEP TAIL

This section keeps the negative radical contributions in the signed large-prime cost. It proves a sufficient subclass in which two actual quotient arms compensate for unrestricted depths in the third. General membership in that subclass remains unproved.

Let $\mathcal{O} = \mathbb{Z}[\zeta]$, where $\zeta^2 - \zeta + 1 = 0$, and let $w = a + b\zeta$ be primitive and unramified, with

$$Q = N(w) \geq 7, \quad \gcd(a, b) = 1, \quad 3 \nmid Q.$$

The three input arms $a, b, a + b$ are nonzero. For an integer $n \geq 5$ with $\gcd(n, 6) = 1$, define

$$(733) \quad z_n = A + B\zeta = \begin{cases} w^n, & n \equiv 1 \pmod{6}, \\ \bar{w}^n, & n \equiv 5 \pmod{6}, \end{cases} \quad D_1 = \frac{A}{a}, \quad D_2 = \frac{B}{b}, \quad D_3 = \frac{A+B}{a+b}.$$

Write $P(x + y\zeta) = xy(x + y)$ and $T_j = |P(w^j)|$.

Lemma 220.1 (Three actual integer quotients). *The D_i are evaluations of integer homogeneous polynomials of degree $n - 1$. They are nonzero and pairwise coprime, and satisfy*

$$(734) \quad aD_1 + bD_2 = (a + b)D_3, \quad |D_1D_2D_3| = \frac{T_n}{T_1}.$$

For prime n , their product is the actual top cyclotomic factor C_n up to sign. For general n in this domain it is, up to sign, $\prod_{d|n, d>1} C_d$.

Proof. The normalized power map in (733) fixes each boundary line individually. At $a = 0$ its value is $b^n\zeta$, at $b = 0$ it is a^n , and at $a = -b$ it is $b^n\zeta^2$. This uses $n \equiv 1 \pmod{6}$ in the first branch and $-n \equiv 1 \pmod{6}$ in the second. Consequently its first coordinate is divisible by a in $\mathbb{Z}[a, b]$, its second by b , and the sum by $a + b$. For the last assertion, substitute $a = -b$ and divide by the monic linear polynomial $a + b$. Homogeneity gives the degrees.

The established oriented factorization shows that the unramified primitive power w^n is primitive and has no zero boundary arm. Conjugation preserves these properties. Thus $A, B, A + B$ are pairwise coprime nonzero integers, and their respective integer divisors D_i are pairwise coprime. Substitution and multiplication give (734). Finally $P(\bar{z}) = -P(z)$ and the actual cyclotomic identity of Theorem 194.1 give the last statements. $\square$

The unramified hypothesis matters: for $w = 1 + \zeta$ and $n = 5$ the normalized output is $-9 - 9\zeta$, and all three quotients are -9 . It is excluded by $3 \nmid Q$.

Set

$$c = \max\{|A|, |B|, |A + B|\}, \quad t = \log c, \quad c_1 = \max\{|a|, |b|, |a + b|\}, \quad \Delta = 3t - \log T_n.$$

For a real cutoff $Y \geq 5$, all sums below are over primes and valuations of signed integers mean valuations of their absolute values. Define

$$\begin{aligned} L(Y) &= \sum_{q \leq Y} v_q(T_n) \log q, & S_i(Y) &= \sum_{q > Y} v_q(D_i) \log q, \\ R_i(Y) &= \sum_{\substack{q > Y \\ q|D_i}} \log q, & E_i(Y) &= \sum_{q > Y} (v_q(D_i) - 2)_+ \log q. \end{aligned}$$

As before,

$$\log W_Y(H) = \sum_{\substack{q > Y \\ q|H}} (v_q(H) - 3) \log q \quad (H \neq 0).$$

Theorem 220.2 (Finite signed-credit compensation). *For every choice of distinct $i, j, k \in \{1, 2, 3\}$,*

$$(735) \quad \log W_Y(T_n) \leq \Delta + \log c_1 + \log T_1 + \frac{L(Y)}{2} + \frac{3}{2}(E_i(Y) + E_j(Y)) - 3R_k(Y).$$

The third quotient's entire multiplicity is included; no individual depth bound is imposed on it.

Proof. Each output arm has absolute value at most c . Since the sum of their logarithms is $3t - \Delta$, each has logarithm at least $t - \Delta$. The nonzero integer input arms have absolute values between one and c_1 . Therefore, for $M_i = \log |D_i|$,

$$(736) \quad t - \Delta - \log c_1 \leq M_i \leq t.$$

Let $L_i = M_i - S_i$ be the small-prime mass of D_i . Equation (734) gives $L_i + L_j \leq L(Y)$. At every supported prime, $e \leq 2 + (e - 2)_+$, so $2R_i \geq S_i - E_i$ and $2R_j \geq S_j - E_j$. Pairwise coprimality makes the signed cost additive on the quotient product. It follows that

$$\log W_Y(D_1 D_2 D_3) \leq S_k - \frac{S_i + S_j}{2} + \frac{3}{2}(E_i + E_j) - 3R_k.$$

By (736), the first two terms on the right are at most $\Delta + \log c_1 + (L_i + L_j)/2$.

For arbitrary nonzero integers U, V , adjoining the valuations of U can increase $\log W_Y(V)$ by at most $\log |U|$. At an old prime the increase is exactly $v_q(U) \log q$; at a new prime it is $(v_q(U) - 3) \log q \leq v_q(U) \log q$. Applying this to $T_n = T_1 |D_1 D_2 D_3|$ proves (735). This step makes no coprimality assumption between T_1 and the quotient product and does not discard a negative contribution at their overlapping primes. $\square$

Theorem 220.3 (Uniform explicit error and a signed-tail subclass). *There are absolute effective constants A_1, A_2 such that, on the whole actual domain above,*

$$(737) \quad \frac{\Delta}{t} \leq A_1 \frac{\log(4n)}{n}, \quad \frac{v_q(T_n)}{t} \leq A_2 q^2 \frac{\log^2(4n)}{n} \quad (q \text{ prime}),$$

$$(738) \quad \frac{\log c_1 + \log T_1}{t} \leq \frac{5}{n}, \quad \frac{L(Y)}{t} \leq A_2 \frac{Y^3 \log Y \log^2(4n)}{n}.$$

Consequently, writing

$$\epsilon(n, Y) = A_1 \frac{\log(4n)}{n} + \frac{5}{n} + \frac{A_2 Y^3 \log Y \log^2(4n)}{2n},$$

one has

$$(739) \quad \frac{\log W_Y(T_n)}{t} \leq \epsilon(n, Y) + \frac{3}{2t}(E_i + E_j - 2R_k).$$

For $Y = n^{1/6}$, $\epsilon(n, Y) \rightarrow 0$ uniformly in the moving root. In particular, if two actual quotient arms have $v_q(D_i), v_q(D_j) \leq 2$ at every prime $q > Y$, then

$$(740) \quad \frac{(\log W_Y(T_n))_+}{t} \rightarrow 0 \quad (n \rightarrow \infty)$$

uniformly in that subclass. More generally it suffices that $(E_i + E_j - 2R_k)_+/t \rightarrow 0$.

Proof. To apply Theorem 181.2, multiply z_n by a unit to place it in the open positive sector. No boundary arm is zero, so such a unit exists. Units and conjugation only permute the three absolute arm values, preserving c, T_n, Δ and all valuations. The resulting actual representation has ramified exponent zero, unit residual, root w or $\bar{w}$ of norm Q , and exponent $g = n$. Thus $\lambda = 1/n$, proving (737). This includes $q = 2, 3$; primes dividing Q have no boundary contribution.

The quadratic and cubic height inequalities give $Q^n \leq c^2$, $Q \geq 3c_1^2/4$ and $T_1 \leq Q^{3/2}$. The numerator $\log c_1 + \log T_1$ is at most $2 \log Q + \log(2/\sqrt{3})$. Divide by $t \geq n \log Q/2$ and use

$$4 + \frac{2 \log(2/\sqrt{3})}{\log 7} < 5.$$

There are at most Y primes below Y , and each satisfies $q^2 \log q \leq Y^2 \log Y$. Summing (737) proves the remaining inequality in (738). Substitution in Theorem 220.2 gives (739). For $Y = n^{1/6}$ its error tends to zero and eventually $Y \geq 5$. If the two selected arms have depth at most two, then $E_i = E_j = 0$; the term $-3R_k/t$ is nonpositive. Taking positive parts gives (740). The more general condition follows directly from the same upper bound. $\square$

Together with the previously proved archimedean and small-prime absorption, this gives the restricted ABC estimate for this explicitly specified homogeneous subclass, uniformly for sufficiently large n for each fixed target epsilon. It does not settle bounded n with arbitrarily moving roots. Finitely many indices are not finitely many seeds. The signed upper bound is one-sided; a persistent negative contribution is compatible with it.

Exact finite evidence and formal scope. The exact replay checks six polynomial quotient identities, 113 actual unramified primitive examples and 678 rational inequalities obtained by exponentiating twice (735). It uses integer arithmetic, rational fractions and complete trial division, not floating-point logarithms. For example, $w = 54345 + \zeta$, $n = 5$ gives

$$\begin{aligned} D_1 &= 5 \cdot 31^4 \cdot 168541 \cdot 11208719, \\ D_2 &= -571 \cdot 640049 \cdot 119336975731, \\ D_3 &= 181 \cdot 659 \cdot 2131 \cdot 263129 \cdot 130411. \end{aligned}$$

Every displayed factor is prime and every product is checked exactly. Thus two actual quotients are squarefree while the third has private depth four at 31. This is a finite illustration, not an asymptotic membership theorem.

The separate Lean module `SignedArmArithmetic` proves 32 integer declarations using the actual pair arithmetic: normalized unit powers and congruences, the three actual divisibilities, quotient reconstruction, the additive and boundary-product identities, the exact norm, and quotient coprimality under an explicit output primitivity hypothesis. It also proves the signed linear budget for integer weights, its specialization to finite lists of natural depths with nonnegative integer weights, and the local weighted overlap bound. Its fresh Lean 4.32.0 build checks all 32 declarations and 39 dependency declarations with warnings treated as errors and only `propext`, `Classical.choice`, `Quot.sound`. It does not formalize preservation of primitivity, real logarithmic weights, the full prime-valuation identities, Theorem 181.2, or general membership in the sufficient subclass.

The unresolved arithmetic problem is to constrain the actual coupled quantity $E_i + E_j - 2R_k$ using (734) and the common norm. No claim that two arms are usually cubefree is made. The criterion discards some shallow contributions in the two selected arms and is a sufficient condition stronger than the general signed-tail target. The exceptional roots of earlier averaging windows and their complete large-prime tails remain uncontrolled.

221. COMMON PURE EXPONENTS FORCE PROGRESSION SUPPORT IN THE ACTUAL SEED

Let a, b be positive coprime integers, put $c = a + b$, and retain the actual norms

$$M = a^2 + ab + b^2 = c^2 - ab, \quad F = a^4 + 3a^3b + 5a^2b^2 + 3ab^3 + b^4.$$

Their elementary identity is

$$(741) \quad F - M^2 = abc^2.$$

This section studies the simultaneous pure-power hypotheses

$$(742) \quad M = R^p, \quad F = Q^p, \quad p \geq 3 \text{ an odd prime}, \quad R, Q \in \mathbb{Z}_{>0}.$$

It proves necessary conditions on such actual seeds. No existence, nonexistence, modularity, or uniform point-height theorem is assumed. The first norm $3R^p$ is a different branch.

Lemma 221.1 (Exact cyclotomic allocation and the exponent-prime exception). *Under (742), set*

$$D = Q - R^2, \quad S = \Phi_p(Q, R^2) = \sum_{j=0}^{p-1} Q^{p-1-j} R^{2j}.$$

Then D is a positive integer, $DS = abc^2$, and $\gcd(Q, R) = 1$. For every prime $q \mid S$ with $q \neq p$,

$$(743) \quad \text{ord}_{\mathbb{F}_q^\times}(Q/R^2) = p, \quad q \equiv 1 \pmod{p}.$$

Moreover $p \mid S$ if and only if $p \mid D$, and in that case $v_p(S) = 1$.

Proof. Reducing M modulo a, b, c gives respectively b^2, a^2, a^2 . Primitivity therefore implies $\gcd(M, abc) = 1$, and (741) gives $\gcd(M, F) = 1$ and hence $\gcd(Q, R) = 1$. Also $F > M^2$, so $Q > R^2$. Factoring $Q^p - R^{2p}$ proves $DS = abc^2$.

If a prime q divides Q or R , exactly one extreme term in S is nonzero modulo q . Thus every $q \mid S$ makes Q, R units. For $t = Q/R^2$ modulo q , the identity $(t-1)(1+t+\dots+t^{p-1}) = t^p - 1$ gives $t^p = 1$. If $q \neq p$ and $t = 1$, the sum would instead be the nonzero element p . Consequently the order is exactly p , giving (743).

In characteristic p , the polynomial S equals $(Q - R^2)^{p-1}$, so $p \mid S$ exactly when $p \mid D$. In that case Q, R are p -units. The expansion

$$S = pR^{2p-2} + \binom{p}{2} R^{2p-4} D + \dots + D^{p-1}$$

has first term of valuation one. Every subsequent intermediate term has a factor pD , and the final term has valuation at least $p-1 \geq 2$. Hence $v_p(S) = 1$, including the complete exceptional prime contribution. $\square$

For a positive integer n , define its full prime-power parts by

$$n_{\text{bad},p} = \prod_{\substack{q \text{ prime} \\ q \not\equiv 1 \pmod{p}}} q^{v_q(n)}, \quad n_{\text{good},p} = n/n_{\text{bad},p}.$$

Theorem 221.2 (Full bad mass and the actual root gap). *Under (742), put $U = (abc^2)_{\text{bad},p}$. Then*

$$(744) \quad U \mid pD, \quad U = a_{\text{bad},p} b_{\text{bad},p} c_{\text{bad},p}^2,$$

and

$$(745) \quad 1 \leq D \leq \frac{4R^2}{9p}, \quad U \leq \frac{4}{9} R^2.$$

If the exponent prime p is omitted as well, the remaining part of abc^2 outside $1 \pmod{p}$ divides D without the extra factor p .

Proof. At every prime outside $1 \pmod{p}$ other than p , the preceding lemma gives $v_q(S) = 0$. At p it gives $v_p(S) \leq 1$. The identity $DS = abc^2$ therefore proves $U \mid pD$ prime by prime, with the stated stronger conclusion when p is omitted. Additivity of valuations gives the equality in (744). No premise about the support of D is required.

The following exact integral certificate supplies the required archimedean inequality:

$$(746) \quad 4M^2 - 9ab(a+b)^2 = (a-b)^2(4a^2 + 7ab + 4b^2) \geq 0.$$

Since $Q > R^2$, every term of S is at least R^{2p-2} . As $M^2 = R^{2p}$, we obtain

$$D = \frac{abc^2}{S} \leq \frac{abc^2}{pR^{2p-2}} \leq \frac{4R^2}{9p}.$$

The integrality and positivity of D , followed by $U \mid pD$, prove (745). $\square$

Corollary 221.3 (Most denominator mass is in the progression). *Every representation (742) satisfies*

$$(747) \quad c_{\text{bad},p} \leq \frac{2}{3}R < \frac{2}{3}c^{2/p}, \quad c_{\text{good},p} > \frac{3}{2}c^{1-2/p}.$$

In particular, c has an actual prime divisor $q \equiv 1 \pmod{p}$; every such prime satisfies $q \geq 2p+1$. More precisely,

$$(748) \quad \sum_{\substack{q|c \\ q \equiv 1 \pmod{p}}} v_q(c) \log q > \left(1 - \frac{2}{p}\right) \log c + \log \frac{3}{2},$$

where all sums are over primes. The complementary weighted mass obeys

$$(749) \quad \sum_{q \not\equiv 1 \pmod{p}} (v_q(a) + v_q(b) + 2v_q(c)) \log q \leq \frac{2}{p} \log M + \log \frac{4}{9} < \frac{4}{p} \log c + \log \frac{4}{9}.$$

There is also an elementary height restriction

$$(750) \quad R^2 \geq \frac{9p}{4}, \quad \log c > \frac{p}{4} \log \frac{9p}{4}.$$

Proof. Equation (744) gives $c_{\text{bad},p}^2 \leq U \leq 4R^2/9$. Also $M < c^2$, so $R < c^{2/p}$. Division and logarithms prove (747)–(749). The lower bound for $c_{\text{good},p}$ is strictly greater than one. It therefore has a prime divisor. If $q = kp+1$ is prime, then k is positive and even because p is odd; hence $q \geq 2p+1$. Finally $D \geq 1$ in (745) gives $R^2 \geq 9p/4$, and $R^p = M < c^2$ gives (750). $\square$

Corollary 221.4 (Shared prime divisors of unequal pure exponents). *Suppose $M = A^h$ and $F = B^g$ for positive integer bases and $h, g \geq 2$. Every odd prime $p \mid \gcd(h, g)$ gives all the preceding conclusions on taking $R = A^{h/p}$ and $Q = B^{g/p}$. For a sequence with chosen common prime divisors $p_j \rightarrow \infty$,*

$$\frac{\sum_{\substack{q|c_j \\ q \equiv 1 \pmod{p_j}}} v_q(c_j) \log q}{\log c_j} \rightarrow 1.$$

Proof. The substituted bases satisfy (742). The last ratio is at most one and greater than $1 - 2/p_j$ by (748). The height bound also gives $c_j \rightarrow \infty$. $\square$

Verification and remaining scope. Both research peers and the root researcher completed independent ordinary reviews. The exact finite supplement checks 116 cyclotomic factor and valuation instances and 3931 actual primitive seed identities. It does not assert that any of those seeds has two pure-power norms. Two separate Lean modules prove fifteen integer algebra and budget declarations using the actual repository norms. They prove the general geometric-sum factorization and lower bound, then derive the integer root-gap budget directly from actual common power representations, for every positive integer exponent. The bad-part interface still exposes its antecedent. Prime-order and full valuation allocation, and the real logarithmic conclusions, are not claimed as formalized by these modules.

The mass in (748) consists of full valuations, not the radical. The theorem leaves large primes and large valuations in the surviving progression uncontrolled. It does not give a point-height upper bound or an ABC contradiction. Relatively prime exponents, nonunit residuals and the first norm $3A^h$ remain open: the latter changes the difference to $Q^p - 9R^{2p}$, which is not the homogeneous cyclotomic identity above. Fixed-exponent finiteness is not promoted to a uniform theorem as the exponent varies.

222. THE ACTUAL FIRST POWER SHARPENS COMMON-EXPONENT SUPPORT TO A SPLIT PROGRESSION

Let a, b be positive coprime integers, $c = a + b$, and suppose the actual norms satisfy

$$(751) \quad M = a^2 + ab + b^2 = R^p, \quad F = a^4 + 3a^3b + 5a^2b^2 + 3ab^3 + b^4 = Q^p,$$

where $p \geq 5$ is prime and R, Q are positive integers. For a positive integer modulus s , write

$$c_{\text{bad}}(s) = \prod_{q \not\equiv 1 \pmod s} q^{v_q(c)}, \quad c_{\text{good}}(s) = c/c_{\text{bad}}(s),$$

with products over primes. These are full prime-power parts, not radicals. This section combines the common-exponent factorization with one specific input arm of the actual Eisenstein power.

Lemma 222.1 (One input arm pays for the extra excluded primes). *There is a primitive unramified $w = x + y\zeta$ of norm R with $w^p = a + b\zeta$. Set*

$$\ell = \begin{cases} x + y, & p \equiv 1 \pmod{6}, \\ x, & p \equiv 5 \pmod{6}. \end{cases}$$

Then $\ell \neq 0$ and

$$(752) \quad J = \prod_{\substack{q|c \\ q \equiv 1 \pmod p \\ q \not\equiv 1 \pmod{6p}}} q^{v_q(c)} \mid |\ell|, \quad 3J^2 \leq 3\ell^2 \leq 4R.$$

Proof. A primitive Eisenstein integer has no inert rational factor and no pair of conjugate split prime factors. If $3 \mid M$, primitiveness gives $a \equiv b \not\equiv 0 \pmod{3}$, and direct expansion gives $v_3(M) = 1$. This is incompatible with $M = R^p$. Each remaining oriented split exponent is divisible by p , because its norm exponent is. Unique factorization gives $a + b\zeta = uw_0^p$ with $N(w_0) = R$ and $u \in \mu_6$. The p th-power map on μ_6 is bijective, so a unit p th root of u can be absorbed in w_0 . The resulting w is primitive, unramified and nonunit; hence $R \geq 7$ and its boundary arms are nonzero.

Apply Lemma 220.1 at exponent p . If $p \equiv 1 \pmod{6}$, the sum of the normalized output coordinates is c , with input arm $x + y$. If $p \equiv 5 \pmod{6}$, the normalized output is $\overline{a} + \overline{b}\zeta$, whose first coordinate is c , with input arm x . Thus for one actual quotient D_j ,

$$(753) \quad c = \ell D_j, \quad T_p/T_1 = |D_1 D_2 D_3|.$$

Let q occur in J . It is at least $2p + 1 > 3$, so $q \neq p$. If it split in the Eisenstein field, then $q \equiv 1 \pmod{3}$; together with its oddness and $q \equiv 1 \pmod{p}$, this would give $q \equiv 1 \pmod{6p}$. Hence it is inert. Also $\gcd(M, c) = 1$, so $q \nmid R$.

The actual homogeneous rank $d_q = \text{ord}((w/\bar{w})^3 \pmod q)$ divides p , since $q \mid T_p$. In the inert case the norm-one group has order $q + 1$, and its cube subgroup has order $(q + 1)/3$. Rank p would force $p \mid (q + 1)/3$, impossible because $q + 1 \equiv 2 \pmod{p}$. Therefore $d_q = 1$. The homogeneous valuation law, as used in Theorem 194.2, gives

$$v_q(T_p) = v_q(T_1) + v_q(p) = v_q(T_1).$$

By (753), all three nonnegative quotient valuations are zero. Hence $v_q(c) = v_q(\ell)$, with its full depth, and $J \mid |\ell|$. Finally

$$4R - 3(x + y)^2 = (x - y)^2, \quad 4R - 3x^2 = (x + 2y)^2$$

proves the size estimate. $\square$

Theorem 222.2 (Actual mass in the split progression). *Every actual representation (751) satisfies*

$$(754) \quad 27c_{\text{bad}}(6p)^2 \leq 16R^3,$$

$$(755) \quad c_{\text{good}}(6p) > \frac{3\sqrt{3}}{4} c^{1-3/p}.$$

In particular c has a prime divisor $q \equiv 1 \pmod{6p}$, necessarily $q \geq 6p + 1$, and

$$(756) \quad \sum_{\substack{q|c \\ q \equiv 1 \pmod{6p}}} v_q(c) \log q > \left(1 - \frac{3}{p}\right) \log c + \log \frac{3\sqrt{3}}{4}.$$

Proof. The common-exponent allocation gives $c_{\text{bad}}(p) \leq 2R/3$. For completeness, put $D = Q - R^2 > 0$ and $S = \sum_{j=0}^{p-1} Q^{p-1-j} R^{2j}$. Then $DS = abc^2$. Every prime factor of S other than p has exact multiplicative order p for Q/R^2 ; its only possible prime contribution outside $1 \pmod{p}$ is one factor of p . Thus the full bad part of abc^2 divides pD . The inequalities $S \geq pR^{2p-2}$ and $9abc^2 \leq 4M^2$ give $pD \leq 4R^2/9$, proving the asserted bound on $c_{\text{bad}}(p)$, with the exponent-prime exception included.

The supports of $c_{\text{bad}}(p)$ and J are disjoint, and their product is exactly $c_{\text{bad}}(6p)$. Multiplication of $9c_{\text{bad}}(p)^2 \leq 4R^2$ and $3J^2 \leq 4R$ proves (754). Dividing, taking square roots, and using $R < c^{2/p}$ from $M < c^2$ proves (755). Its right side exceeds one for $p \geq 5$, so the good part has a prime divisor. Logarithms give (756). $\square$

If $M = A^h$ and $F = B^g$ for positive integer bases and $h, g \geq 2$, every prime $p \geq 5$ dividing $\gcd(h, g)$ gives the same conclusions by setting $R = A^{h/p}$ and $Q = B^{g/p}$. Along a sequence with chosen common primes $p \rightarrow \infty$, the left side of (756) divided by $\log c$ tends to one. This is a statement about the actual seed's valuations, with no independence or prime-density hypothesis.

Primes already present in the input arm are fully included in its budget. Large new primes in the surviving progression and their high depths remain possible. The argument gives no sufficient radical bound for ABC. Nonunit residuals, the first norm $3A^h$, and pairs of exponents without a common prime at least five remain separate open branches. The rank and valuation part of this section is an ordinary proof, not a claim that the integer arm module formalizes the entire progression argument.

223. A TOTIENT BUDGET AT A GENERAL COMMON PURE EXPONENT

Retain positive coprime a, b , $c = a + b$, $M = a^2 + ab + b^2$ and $F = M^2 + abc^2$. Suppose

$$(757) \quad M = R^n, \quad F = Q^n, \quad R, Q \in \mathbb{Z}_{>0}, \quad n > 1, \quad \gcd(n, 6) = 1.$$

Let $\varphi = \varphi(n)$, let P be the largest prime divisor of n , and put

$$(758) \quad L = \Phi_n(1) = \Phi_n(1, 1), \quad \kappa = P/L.$$

Here $\Phi_n(X, Y)$ is the homogeneous cyclotomic polynomial. We verify below that $L = P$ for a prime power n and $L = 1$ otherwise. Thus κ is a positive integer with $\kappa \leq n$. All conclusions are necessary conditions on actual simultaneous pure powers, without an existence or nonexistence claim.

Lemma 223.1 (The complete exceptional-prime contribution). *Let $X > Y$ be coprime positive integers and let $n > 1$ be odd. Every prime divisor of $\Phi_n(X, Y)$ outside $1 \pmod{n}$ is the largest prime divisor P of n , and its valuation is one. Consequently the full part outside that progression divides P .*

Proof. Monicity and the constant term $Y^{\varphi(n)}$ exclude every prime dividing XY . For an odd prime q which is a unit in both coordinates, put $d = \text{ord}_{\mathbb{F}_q^\times}(X/Y)$ and $u = v_q(X^d - Y^d)$. Then $d \mid q - 1$ and

$$v_q(X^m - Y^m) = \begin{cases} 0, & d \nmid m, \\ u + v_q(m/d), & d \mid m. \end{cases}$$

To verify the second case, divide by unit powers of Y and write $B = 1 \pmod{q}$. A power prime to q preserves $v_q(B - 1)$, since its geometric sum is that exponent modulo q . Taking a q -th power adds one: the binomial leading term $q(B - 1)$ has valuation $u + 1$, and all subsequent terms have valuation at least $u + 2$. This proves the formula by iteration.

The factorization $X^m - Y^m = \prod_{e \mid m} \Phi_e(X, Y)$ determines the cyclotomic valuations uniquely by divisor induction. The values

$$v_q(\Phi_d(X, Y)) = u, \quad v_q(\Phi_{dq^j}(X, Y)) = 1 \quad (j \geq 1),$$

with all other values zero, have precisely the required divisor sums: when $d \mid m$ their sum is $u + v_q(m/d)$, and otherwise it is zero. This includes $d = 1$ and $\Phi_1 = X - Y$.

If $n = d$, then $n \mid q - 1$, so q belongs to the progression. Every other supported index has $n = dq^j$, $j \geq 1$, and valuation one. Since $d \mid q - 1$, all prime divisors of d are smaller than q , which must therefore equal P .

At $q = 2$, different coordinate parities make $X^n - Y^n$ odd. If both coordinates are odd, every odd m instead satisfies $v_2(X^m - Y^m) = v_2(X - Y)$, by the odd geometric sum. Divisor induction gives $v_2(\Phi_n(X, Y)) = 0$ for every odd $n > 1$. $\square$

Lemma 223.2 (A lower bound from conjugate primitive roots). *For positive real $X > Y$ and $n > 2$,*

$$\Phi_n(X, Y) \geq \Phi_n(1)Y^{\varphi(n)}.$$

Moreover $\Phi_n(1)$ equals P when n is a prime power and equals one otherwise.

Proof. Pair each primitive n -th root ζ with its distinct conjugate. For each pair,

$$|X - Y\zeta|^2 - Y^2|1 - \zeta|^2 = (X - Y)^2 + 2Y(X - Y)(1 - \operatorname{Re} \zeta) \geq 0.$$

Multiplying proves the bound, because the product of the factors $|1 - \zeta|^2$ is $\Phi_n(1) > 0$.

Evaluation at one of the polynomial $(T^n - 1)/(T - 1) = \prod_{d|n, d>1} \Phi_d(T)$ gives $\prod_{d|n, d>1} \Phi_d(1) = n$. Induct on n . For a non-prime-power n , its proper prime-power divisors already contribute the entire product n ; all other proper divisors contribute one. For $n = P^j$, the proper prime powers contribute P^{j-1} . These give the two asserted values. $\square$

Write $z_{\text{bad}}(s)$ for the full prime-power part of a positive integer z outside $1 \bmod s$, and $z_{\text{good}}(s) = z/z_{\text{bad}}(s)$.

Theorem 223.3 (Actual support with its totient and exceptional-prime costs). *Under (757),*

$$(759) \quad 9c_{\text{bad}}(n)^2 \leq 4\kappa R^{2(n-\varphi)}, \quad 27c_{\text{bad}}(6n)^2 \leq 16\kappa R^{2(n-\varphi)+1}.$$

Consequently

$$(760) \quad c_{\text{good}}(6n) > \frac{3\sqrt{3}}{4\sqrt{\kappa}} c^{2\varphi/n-1-1/n},$$

and, summing over primes,

$$(761) \quad \sum_{\substack{q|c \\ q \equiv 1 \bmod 6n}} v_q(c) \log q > \left(\frac{2\varphi}{n} - 1 - \frac{1}{n} \right) \log c + \log \frac{3\sqrt{3}}{4} - \frac{1}{2} \log \kappa.$$

Proof. Actual primitivity gives $\gcd(Q, R) = 1$, and $F > M^2$ gives $Q > R^2$. Put

$$T = \Phi_n(Q, R^2), \quad C = (Q^n - R^{2n})/T.$$

These are positive integers and $TC = abc^2$. By Lemma 223.1, the full bad part $U = (abc^2)_{\text{bad}}(n)$ divides PC . Lemma 223.2 and the actual four-ninths certificate give

$$C \leq \frac{4R^{2n/9}}{LR^{2\varphi}}, \quad U \leq \frac{4\kappa}{9} R^{2(n-\varphi)}.$$

As $c_{\text{bad}}(n)^2 \mid U$, this proves the first budget.

The actual oriented factorization gives an Eisenstein root $w = x + y\zeta$ with $N(w) = R$ and $w^n = a + b\zeta$. Indeed, the ramified norm valuation is at most one and is excluded by $M = R^n$; every oriented split exponent is divisible by n . The unit factor can be absorbed because $\gcd(n, 6) = 1$. The normalized three-arm construction gives $c = \ell D_j$, where $\ell = x + y$ for $n \equiv 1 \bmod 6$, and $\ell = x$ for $n \equiv 5 \bmod 6$. The root is primitive, unramified and a nonunit; hence $\ell \neq 0$ and $3\ell^2 \leq 4R$.

Let J be the full part of c at primes $q \equiv 1 \bmod n$ but $q \not\equiv 1 \bmod 6n$. Such a prime is inert, exceeds n , and does not divide R . Its homogeneous boundary rank d divides both n and $(q+1)/3$. But $q+1 \equiv 2 \bmod n$ and n is odd, so $\gcd(n, q+1) = 1$ and $d = 1$. The exact homogeneous valuation law gives $v_q(T_n) = v_q(T_1) + v_q(n) = v_q(T_1)$. Thus the entire actual three-arm quotient product has valuation zero at q . From $c = \ell D_j$ it follows that $v_q(c) = v_q(\ell)$. Therefore $J \mid |\ell|$ with full depths, and $3J^2 \leq 4R$.

The disjoint support identity $c_{\text{bad}}(6n) = c_{\text{bad}}(n)J$ now gives the second budget in (759). Divide c^2 by that bound, take square roots, and use $R < c^{2/n}$, since $M < c^2$. This proves (760) and its logarithm. The strict inequality remains valid even if the final exponent of c is nonpositive: the substituted power of R in the denominator has positive exponent $2(n - \varphi) + 1$. $\square$

Corollary 223.4 (A totient gate guarantees a large progression prime). *If (757) holds and $2\varphi(n) > n + 1$, then c has an actual prime divisor $q \equiv 1 \pmod{6n}$, and $q \geq 6n + 1$. The condition includes every prime power $n = P^k$ with $P \geq 5$.*

Proof. Put $\rho = 2\varphi(n) - n - 1$. Since n is odd, this is an even integer, and the hypothesis gives $\rho \geq 2$. The actual common-exponent size budget gives $R^2 \geq 9n/4$. To recall its direct proof, $D_0 = Q - R^2 \geq 1$ and the full geometric sum $S_n \geq nR^{2n-2}$ satisfy $D_0 S_n = abc^2 \leq 4R^{2n}/9$. Cancellation gives $9nD_0 \leq 4R^2$. Using $c^2 > R^n$ in (759), we obtain

$$c_{\text{good}}(6n)^2 > \frac{27}{16\kappa} R^\rho \geq \frac{27}{16\kappa} R^2 \geq \frac{243n}{64\kappa} \geq \frac{243}{64} > 1.$$

The good part therefore has a prime divisor. For $n = P^k$, $2\varphi(n) - n - 1 = n(1 - 2/P) - 1 \geq 2$, proving the last assertion. $\square$

Any selected common divisor n of unequal pure exponents gives these same conclusions when $\gcd(n, 6) = 1$ and the stated totient condition holds. If the totient condition fails, the support budgets remain valid; this particular prime-existence argument is not claimed to succeed.

Verification and remaining scope. Both research peers completed independent full ordinary review. The exact finite supplement checks fourteen fully factored homogeneous cyclotomic values and four additional exceptional-prime valuations. For example, the valuations at 11 of $\Phi_{55}(3, 1)$ and $\Phi_{605}(3, 1)$ are one, but that of $\Phi_{275}(3, 1)$ is zero. These are generic homogeneous examples, not actual simultaneous pure-power seeds. The classical cyclotomic and root-pair arguments were proved explicitly to separate their role from the actual norm and arm hypotheses.

The exceptional prime is retained in κ . No coefficient-height estimate is treated as a point-height bound. Large surviving primes and their valuations remain uncontrolled, and the root-size estimate is a lower bound. Nonunit residuals, the ramified first norm and exponents without a suitable common divisor remain separate live routes. No ABC proof or disproof follows.

224. PRIME-POWER REFINEMENT, FULL-EXPONENT GROWTH AND FORMAL SCOPE

The general totient budget has a particularly sharp form at a common prime-power divisor of the two actual pure exponents. Let $a, b > 0$ be coprime, $c = a + b$, $M = a^2 + ab + b^2$ and $F = M^2 + abc^2$. For the full prime-power part outside $1 \pmod{s}$, use the notation $c_{\text{bad}}(s)$ of the preceding section.

Corollary 224.1 (The full prime-power modulus). *Suppose $p \geq 5$ is prime, $k \geq 1$, $n = p^k$, $m = p^{k-1}$, and $M = R^n$, $F = Q^n$ with positive integer bases. Then*

$$(762) \quad c_{\text{bad}}(n) \leq \frac{2}{3} R^m < \frac{2}{3} c^{2/p},$$

$$(763) \quad 27c_{\text{bad}}(6n)^2 \leq 16R^{2m+1},$$

$$(764) \quad \sum_{\substack{q|c \\ q \equiv 1 \pmod{6p^k}}} v_q(c) \log q > \left(1 - \frac{2}{p} - \frac{1}{p^k}\right) \log c + \log \frac{3\sqrt{3}}{4}.$$

In particular, an actual prime divisor of c is $1 \pmod{6p^k}$, and is at least $6p^k + 1$.

Proof. Here $\Phi_n(1) = p$, $\kappa = 1$ and $n - \varphi(n) = m$. Substitution in the general support budgets proves all three estimates. The exponent in (764) is positive, and $3\sqrt{3}/4 > 1$, so the good part exceeds one.

There is also a direct allocation proof that makes the p -exception transparent. Set

$$D = Q^m - R^{2m}, \quad S = \Phi_n(Q, R^2) = \sum_{j=0}^{p-1} (Q^m)^{p-1-j} (R^{2m})^j.$$

Then $DS = abc^2$. At $q \neq p$ dividing S , the order of Q/R^2 is exactly n : its n -th power is one, its m -th power is not one, and every proper divisor of n divides m . At p , the binomial argument gives $p \mid S$ exactly when $p \mid D$, and then $v_p(S) = 1$. Consequently $(abc^2)_{\text{bad}}(n) \mid pD$. Since $S \geq pR^{2m(p-1)}$ and $abc^2 \leq 4R^{2n}/9$,

$$1 \leq D \leq \frac{4R^{2m}}{9p}, \quad (abc^2)_{\text{bad}}(n) \leq \frac{4}{9}R^{2m}.$$

This proves (762) with full valuations. The actual input-arm allocation for the remaining inert primes gives $3J^2 \leq 4R$ and $c_{\text{bad}}(6n) = c_{\text{bad}}(n)J$, proving (763) again. $\square$

At $k = 1$ this recovers the split progression bound at $6p$. For fixed p and growing k , the guaranteed fraction tends to $1 - 2/p$; it does not tend to one merely because the modulus grows. Any common prime-power divisor $p^k \mid \gcd(h, g)$ of unequal pure exponents $M = A^h$, $F = B^g$ gives the same result by replacing their bases.

Theorem 224.2 (The entire common exponent controls the extraction root). *For any positive integer n , simultaneous representations $M = R^n$, $F = Q^n$ with positive integer bases satisfy*

$$9n(Q - R^2) \leq 4R^2, \quad R^2 \geq \frac{9n}{4}, \quad \log c > \frac{n}{4} \log \frac{9n}{4}.$$

Proof. The inequality $F > M^2$ gives the positive integer $D_0 = Q - R^2$. The full geometric sum has n terms, each at least R^{2n-2} , and its product with D_0 is $abc^2 \leq 4R^{2n}/9$. Cancellation gives the first inequality. Use $D_0 \geq 1$ for the second, and $R^n = M < c^2$ for the logarithmic conclusion. This includes $n = 1$ and needs no prime-order assertion. $\square$

This is a lower bound on the height of the actual seed. It is not an upper bound, and does not contradict the existence of a moving family.

Exact scope of the fifty-four new Lean declarations. The signed-arm module proves thirty-two statements. It defines the actual normalized Eisenstein power, proves boundary-arm divisibility, reconstructs the three integer quotients, proves their additive relation and norm identity, and derives pairwise coprimality with output primitivity explicitly supplied. It also proves the signed linear budget, its two-arm specialization, finite nonnegative integer-weight depth bounds, and the exact overlap-cost inequality. The finite weights are not silently identified with real logarithms.

Two common-exponent modules prove fifteen statements. Besides the actual polynomial certificates, they prove the geometric-sum identity and lower bound for all relevant integer bases. In particular, the integer conclusions of Theorem 224.2 follow directly from actual norm representations; the sum lower bound is proved rather than postulated. The separate bad-part interface retains its explicit antecedent.

The support-assembly module proves seven statements. For every odd integer n , the rank premises $n \mid q - 1$, $d \mid n$, $d \mid q + 1$ and $d > 0$ give $d = 1$: they force $d \mid 2$, and $d = 2$ contradicts oddness. The module proves the two actual input-arm height inequalities using

$$4N(x, y) - 3x^2 = (x + 2y)^2, \quad 4N(x, y) - 3(x + y)^2 = (x - y)^2.$$

It combines explicit budgets $9B^2 \leq 4U$ and $3J^2 \leq 4R$ into $27(BJ)^2 \leq 16UR$, including its power form. Finally it verifies the integer core of the totient support-existence step. For $R \geq 1$, $0 < K \leq N$, $9N \leq 4R^2$ and integer $\rho \geq 2$, the inequality $16KG^2 > 27R^\rho$ forces $G^2 > 1$. The correspondence of these parameters with actual prime parts remains the independently reviewed ordinary input.

All fifty-four new declarations and fifty-four unchanged dependency declarations pass a fresh scoped Lake build under Lean 4.32.0 with warnings treated as errors. Every theorem's axiom list is checked; only `propext`, `Classical.choice` and `Quot.sound` occur. Three exact finite supplements are independently replayed: 113 actual signed-arm rows and 678 rational inequalities;

116 cyclotomic instances and 3931 primitive seed identities; and 14 fully factored general-index cyclotomic values with four additional exceptional-prime valuations. These are finite checks, not an infinite family or a substitute for the ordinary universal proofs.

The full prime-order and valuation allocation, unique factorization, real logarithmic estimates, and arbitrary-root signed-tail membership are outside this continuation's Lean scope. The complementary high multiplicities and the absence of a suitable common exponent remain live bottlenecks. Neither standard ABC nor its negation is asserted by any of these results.

225. RECIPROCAL DEPTH BUDGETS FOR THE THREE ACTUAL ARMS

Retain the actual primitive unramified root and quotient domain of Lemma 220.1: $Q \geq 7$, $3 \nmid Q$, $n \geq 5$ and $\gcd(n, 6) = 1$. The three nonzero integer quotients D_i are pairwise coprime. Use its notation $T_n, T_1, c, t, c_1, \Delta$ and, for $Y \geq 5$, put

$$S_i = \sum_{q>Y} v_q(D_i) \log q, \quad R_i = \sum_{\substack{q>Y \\ q|D_i}} \log q, \quad L_i = \sum_{q \leq Y} v_q(D_i) \log q, \quad L = \sum_{q \leq Y} v_q(T_n) \log q.$$

All sums are over primes. The actual height relations give

$$(765) \quad t - \Delta - \log c_1 - L_i \leq S_i \leq t, \quad L_i \geq 0, \quad \sum_i L_i \leq L.$$

Choose caps $h_i \in \mathbb{Z}_{>0} \cup \{\infty\}$. An infinite cap places no depth restriction on that quotient. Let I consist of the finite-cap indices and define

$$\begin{aligned} \beta_i &= 3/h_i \quad (i \in I), & \beta_i &= 0 \quad (i \notin I), \\ E_i^{(h_i)} &= \sum_{q>Y} (v_q(D_i) - h_i)_+ \log q \quad (i \in I), \\ C_h &= \sum_i \beta_i, & A_h &= \sum_i (\beta_i - 1)_+, & B_h &= \max_i (\beta_i - 1)_+. \end{aligned}$$

Thus $0 \leq A_h \leq 6$ and $0 \leq B_h \leq 2$, even for caps that vary with the root and exponent.

Theorem 225.1 (Exact finite reciprocal-cap compensation). *On this actual domain,*

$$(766) \quad \begin{aligned} \log W_Y(T_n) &\leq (3 - C_h)t + A_h(\Delta + \log c_1) + B_h L + \log T_1 \\ &\quad + \sum_{i \in I} \beta_i E_i^{(h_i)} - 3 \sum_{i \notin I} R_i. \end{aligned}$$

The complete depths of every infinite-cap arm are included and its negative radical contribution remains on the right.

Proof. At a supported prime, $e \leq h_i + (e - h_i)_+$, so $3R_i \geq \beta_i(S_i - E_i^{(h_i)})$ for $i \in I$. Pairwise coprimality gives exact signed additivity on the quotient product. Hence

$$\log W_Y(D_1 D_2 D_3) \leq \sum_i (1 - \beta_i) S_i + \sum_{i \in I} \beta_i E_i^{(h_i)} - 3 \sum_{i \notin I} R_i.$$

When $\beta_i \leq 1$, use the upper bound $S_i \leq t$. When $\beta_i > 1$, its coefficient is negative and (765) bounds its contribution by

$$(1 - \beta_i)t + (\beta_i - 1)(\Delta + \log c_1 + L_i).$$

Summation yields $(3 - C_h)t + A_h(\Delta + \log c_1) + B_h L$. Finally use $T_n = T_1 |D_1 D_2 D_3|$ and the general overlap inequality $\log W_Y(UV) \leq \log W_Y(V) + \log |U|$ proved in Theorem 220.2. No coprimality of T_1 and the quotient product is assumed. $\square$

Write

$$\mathcal{E}_h = \sum_{i \in I} \beta_i E_i^{(h_i)} - 3 \sum_{i \notin I} R_i.$$

Theorem 225.2 (Uniform signed-tail subclasses). *With the absolute effective constants of Theorem 220.3,*

$$(767) \quad \frac{\log W_Y(T_n)}{t} \leq 3 - C_h + \epsilon_h(n, Y) + \frac{\mathcal{E}_h}{t},$$

where

$$\epsilon_h(n, Y) = A_h A_1 \frac{\log(4n)}{n} + \frac{2A_h + 3}{n} + B_h A_2 \frac{Y^3 \log Y \log^2(4n)}{n}.$$

For $Y = n^{1/6}$, this error tends to zero uniformly in all actual moving roots and caps. In particular, if

$$(768) \quad \sum_{i \in I} \frac{1}{h_i} \geq 1$$

and the actual finite-cap arms satisfy $v_q(D_i) \leq h_i$ at every $q > Y$, then $(\log W_Y(T_n))_+/t \rightarrow 0$ uniformly as $n \rightarrow \infty$. The same conclusion follows under the more general condition $(\mathcal{E}_h)_+/t \rightarrow 0$. If $C_h \geq 3 + \delta$ for a fixed $\delta > 0$ and the caps are met, then

$$\limsup \frac{\log W_Y(T_n)}{t} \leq -\delta.$$

Proof. Sector normalization gives the same unit-residual representation with $\lambda = 1/n$ as in Theorem 220.3. Thus its all-prime two-place bounds control Δ/t and L/t as stated there. The remaining numerator obeys

$$A_h \log c_1 + \log T_1 \leq \left(\frac{A_h}{2} + \frac{3}{2} \right) \log Q + A_h \log(2/\sqrt{3}).$$

Divide by $t \geq n \log Q/2$, use $Q \geq 7$ and $2 \log(2/\sqrt{3}) < \log 7$, and obtain the bound $(2A_h + 3)/n$. Substitution proves (767). The bounds $A_h \leq 6$, $B_h \leq 2$ give uniform convergence of the error. Under the depth caps all finite excesses vanish, so $\mathcal{E}_h \leq 0$. Condition (768) is exactly $C_h \geq 3$. The positive-part and fixed negative-margin conclusions follow. $\square$

The earlier absorption therefore gives the restricted ABC estimate for these specified subclasses, uniformly for sufficiently large n for each fixed target epsilon. Bounded indices with arbitrarily moving roots remain outside this argument. The signed estimate is one-sided, rather than a claim that negative contributions disappear.

Proposition 225.3 (The five maximal depth-cap patterns). *Up to permutation, the maximal integer-cap patterns satisfying (768), under componentwise enlargement, are*

$$(769) \quad (1, \infty, \infty), \quad (2, 2, \infty), \quad (2, 3, 6), \quad (2, 4, 4), \quad (3, 3, 3).$$

Proof. Sort $h_1 \leq h_2 \leq h_3$, with infinity largest. If $h_1 = 1$ the triple is below the first pattern. If $h_1 = h_2 = 2$, it is below the second. If $h_1 = 2, h_2 = 3$, the reciprocal condition forces $h_3 \leq 6$. If $h_1 = 2, h_2 \geq 4$, it forces $h_2 = h_3 = 4$. If $h_1 = 3$, the other two must also be three; $h_1 \geq 4$ is impossible. Each displayed pattern has reciprocal sum exactly one. $\square$

The first pattern is a new useful form of the compensation mechanism: one quotient that is squarefree above Y pays for the full depths in both other arms. More precisely, at that pattern

$$(770) \quad \log W_Y(T_n) \leq 2\Delta + 2 \log c_1 + 2L + \log T_1 + 3(E_i^{(1)} - R_j - R_k).$$

Thus squarefreeness can be weakened to $(E_i^{(1)} - R_j - R_k)_+/t \rightarrow 0$. The pattern $(2, 2, \infty)$ recovers the preceding SA criterion. Patterns $(2, 3, 6)$ and $(2, 4, 4)$ permit positive signed contributions at depth above three, while another actual arm compensates them; both have $A_h = B_h = 1/2$. No automatic arithmetic membership is asserted for these conditions.

Proposition 225.4 (Sharpness only for the abstract finite ledger). *When fixed caps have $C_h < 3$, their depth restrictions and equal total arm masses alone do not force a sublinear signed integer-weight ledger. This statement does not give an actual Eisenstein counterexample.*

Proof. Let H be the least common multiple of the finite caps, or one if there are none, and set $t_m = mH$. For a finite cap h_i , take a one-entry list of depth h_i and integer weight t_m/h_i . For an infinite cap take depth t_m and weight one. All finite caps hold and all three masses S_i equal t_m . If r caps are infinite, the exact signed sum is

$$\sum_i (S_i - 3R_i) = (3 - C_h)t_m - 3r.$$

Its ratio to t_m tends to $3 - C_h > 0$. These finite lists are not asserted to be prime factorizations or to satisfy the actual additive arm relation and common norm. They show that extra arithmetic input is needed beyond this ledger; they exclude no actual norm route. $\square$

Formal verification and unresolved scope. The separate `ReciprocalDepthArithmetic` module proves seventeen declarations using finite natural depths and nonnegative integer weights. It proves arbitrary-cap depth/radical inequalities, their common-denominator interface $b_i h_i = 3d$, the negative-coefficient height costs, the canonical three-arm budget and its uniform bounds, and exact one-, two- and three-finite-cap classifications after clearing positive denominators. The fresh Lean 4.32.0 build checks all seventeen new declarations and seventy-one dependency declarations with warnings treated as errors and only the standard three axioms. This module does not define actual prime factorizations or real logarithms, formalize the analytic estimates, or prove cap membership.

The unresolved arithmetic object is the actual excess-minus-radical term $\mathcal{E}_h$, constrained by the common power, additive arm relation and norm. Large private depths can invalidate every simple cap pattern without refuting its net-credit sufficient condition. Neither the abstract sharpness family nor the proved sufficient classes settle the general signed tail or ABC.

226. ACTUAL PRIME LOGARITHMS AND A FINITE COMPENSATION THEOREM IN LEAN

The earlier integer-weight ledger has a direct arithmetic realization. This section uses the prime factorization of positive natural numbers and real logarithms. Its finite height hypotheses are discharged from integer input and output bounds. It supplies no asymptotic membership theorem for an arbitrary orbit.

For a natural cutoff Y and a positive integer N , define

$$\begin{aligned} S_Y(N) &= \sum_{\substack{p|N \\ p > Y}} v_p(N) \log p, & R_Y(N) &= \sum_{\substack{p|N \\ p > Y}} \log p, \\ L_Y(N) &= \sum_{\substack{p|N \\ p \leq Y}} v_p(N) \log p, & E_{h,Y}(N) &= \sum_{\substack{p|N \\ p > Y}} (v_p(N) - h)_+ \log p. \end{aligned}$$

Here p always denotes a rational prime, h is a natural number, and the positive part agrees with natural-number subtraction. Write $\mathcal{S}_Y(N) = S_Y(N) - 3R_Y(N)$ for the signed logarithmic cost. It is the logarithm of the corresponding multiplicative cost in the preceding sections, not that multiplicative cost itself.

Lemma 226.1 (The actual factorization ledger). *For positive integers U, V, N ,*

$$\begin{aligned} S_Y(N), L_Y(N) &\geq 0, & S_Y(N) + L_Y(N) &= \log N, \\ S_Y(N) &\leq hR_Y(N) + E_{h,Y}(N), \\ S_Y(UV) &= S_Y(U) + S_Y(V), & L_Y(UV) &= L_Y(U) + L_Y(V), \\ S_Y(UV) &\leq \mathcal{S}_Y(V) + \log U. \end{aligned}$$

If U, V are coprime, then $R_Y(UV) = R_Y(U) + R_Y(V)$ and their signed costs are additive. If all prime depths of N above Y are at most h , then $E_{h,Y}(N) = 0$.

Proof. The prime factorization of N and the logarithm of a finite product give the partition identity. Every prime logarithm is nonnegative. At a supported prime of depth e , the inequality $e \leq h + (e - h)_+$ gives the depth-cap estimate after multiplication by $\log p$ and summation. Valuations add under multiplication, which proves additivity of both full masses. Every prime divisor of V divides UV , so $R_Y(UV) \geq R_Y(V)$. Therefore

$$S_Y(UV) \leq S_Y(U) + \mathcal{S}_Y(V) \leq \log U + \mathcal{S}_Y(V).$$

For coprime factors their prime supports are disjoint; radical mass, and hence signed cost, then adds exactly. Under the stated cap every summand in the excess is zero. $\square$

Theorem 226.2 (Finite compensation from integer arm data). *Let a, b, d, A, B, C, H, H_1 be positive integers. Suppose A, B, C are pairwise coprime and*

$$aA \leq H, \quad bB \leq H, \quad dC \leq H, \quad a, b \leq H_1.$$

Put $U = abd$, $T = (aA)(bB)(dC)$ and $\Delta = 3 \log H - \log T$. Then

$$(771) \quad \begin{aligned} S_Y(T) &\leq \Delta + \log H_1 + \log U + \frac{1}{2}L_Y(T) \\ &\quad + \frac{3}{2}(E_{2,Y}(A) + E_{2,Y}(B)) - 3R_Y(C). \end{aligned}$$

No coprimality between U and ABC is assumed.

Proof. Abbreviate the masses of A, B, C by S_A, S_B, S_C , and similarly for R, L, E . The depth-two estimates give $R_A \geq (S_A - E_A)/2$ and $R_B \geq (S_B - E_B)/2$. Additivity of signed cost on the three coprime quotients yields

$$(772) \quad S_Y(ABC) \leq S_C - \frac{1}{2}(S_A + S_B) + \frac{3}{2}(E_A + E_B) - 3R_C.$$

Since the other two output logarithms are at most $\log H$,

$$\log(aA) \geq \log T - 2 \log H = \log H - \Delta.$$

Subtracting $\log a \leq \log H_1$ and using the partition identity gives $S_A \geq \log H - \Delta - \log H_1 - L_A$; the same argument applies to B . Also $C \leq dC \leq H$, so $S_C \leq \log H$. Substitution in (772) gives

$$S_Y(ABC) \leq \Delta + \log H_1 + \frac{1}{2}(L_A + L_B) + \frac{3}{2}(E_A + E_B) - 3R_C.$$

The identity $T = UABC$ and nonnegativity and additivity of L_Y imply $L_A + L_B \leq L_Y(T)$. Apply the overlap estimate of Lemma 226.1 to adjoin U , proving the theorem. $\square$

Proposition 226.3 (A squarefree-arm finite interface). *Suppose A, B, C are positive pairwise coprime integers, and every prime depth of A above Y is at most one. For real t, δ, l , assume $S_Y(A) \geq t - \delta - l$ and $S_Y(B), S_Y(C) \leq t$. Then*

$$S_Y(ABC) \leq 2\delta + 2l - 3(R_Y(B) + R_Y(C)).$$

Proof. The depth-one estimate has zero excess, so $R_Y(A) \geq S_Y(A)$. Additivity gives $S_Y(ABC) \leq -2S_Y(A) + S_Y(B) + S_Y(C) - 3R_Y(B) - 3R_Y(C)$. Insert the assumed mass bounds. $\square$

Formal scope. The modules `ActualPrimeLogCompensation` and `ActualPrimeLogHeight` contain respectively 17 and 7 proved declarations, using `Nat.factorization`, `Nat.primeFactors` and `Real.log`. They include the finite inequalities above, the actual mass partition, product and overlap laws, logarithmic monotonicity, and the derivation of the height hypotheses from the displayed natural-number data. The second module's final theorem is `finite_actual_arm_compensation`. The natural cutoff is part of the formal statement; no real-cutoff asymptotic limit is inferred from it without an additional argument.

Both modules were freshly compiled with Lean 4.32.0 and warnings as errors against the pinned Mathlib revision 81a5d257c8e410db227a6665ed08f64fea08e997. The pinned dependency cache was reused; this is not a fresh build of all Mathlib. Every new declaration has an explicit axiom query, with only `propext`, `Classical.choice` and `Quot.sound` in the union. No ordinary number-theoretic claim is added as an axiom. The sources and their mathematical scope were independently reviewed.

In the orbit application, a, b, d and A, B, C are the absolute input arms and actual quotients of Section 220. The ordinary primitivity and quotient theorems supply their reconstruction and pairwise coprimality. Those orbit facts, the logarithmic-form input, and arbitrary-root membership in a sufficient signed-compensation class are still outside these two formal modules. The finite factorization bridge therefore narrows the formal gap without closing the remaining number-theoretic one.

227. SIMULTANEOUS PRIVATE TOP-RANK DEPTHS IN ALL THREE ACTUAL ARMS

The sufficient signed-compensation classes do not have automatic membership. This section constructs actual primitive unramified roots whose three quotient arms simultaneously have prescribed private deep primes above the moving cutoff. It also accounts for the height paid by the construction. Neither the net-credit criterion nor the full signed tail is refuted.

Theorem 227.1 (Three prescribed actual first depths). *Let $n \geq 5$ be an integer coprime to six, and let $s_1, s_2, s_3 \geq 4$ be integers. There are distinct primes $q_i \equiv 1 \pmod{6n}$ and infinitely many positive integers $a \equiv 0 \pmod{3}$ such that $w = a + \zeta$ is primitive and unramified and has the following properties. Form the normalized power*

$$z_n = A + B\zeta = \begin{cases} w^n, & n \equiv 1 \pmod{6}, \\ \bar{w}^n, & n \equiv 5 \pmod{6}, \end{cases} \quad D_1 = A/a, \quad D_2 = B, \quad D_3 = (A + B)/(a + 1).$$

These are the actual integer quotients of Lemma 220.1. For $i \neq j$,

$$(773) \quad v_{q_i}(D_i) = s_i, \quad v_{q_i}(D_j) = 0, \quad q_i \nmid a(a + 1)N(w), \quad \text{ord}((w/\bar{w})^3 \bmod q_i) = n.$$

In particular $v_{q_i}(T_n) = s_i$ for $T_n = |P(w^n)|$, and these are genuinely new first depths at rank n .

Proof. We first give an elementary supply of distinct primes. For any integer $N \geq 3$ and finite set of previously chosen primes, put X equal to N times their product. The integer $\Phi_N(X)$ is greater than one: pairing primitive roots with their conjugates gives positive factors at least $(X - 1)^2 > 1$. Its constant term is one, so any prime divisor q is coprime to X , and hence to N and all the previous primes. The polynomial $Z^N - 1$ is separable over $\mathbb{F}_q$. Its cyclotomic factors are therefore pairwise coprime after reduction. A root of Φ_N cannot also be a root of $Z^d - 1$ at a proper divisor d of N . Thus $X \bmod q$ has order exactly N , and $q \equiv 1 \pmod{N}$. Repeat this three times with $N = 6n$ to obtain the required distinct q_i and elements t_i of order $6n$. No prime-density theorem is used.

Fix such a pair q, t and put $\theta = t^n$ in $\mathbb{F}_q$. It is a primitive sixth root, with $\theta^2 - \theta + 1 = 0$ and $\bar{\theta} = 1 - \theta = \theta^{-1}$. On the norm-unit locus define

$$\alpha(X) = \frac{X + \theta}{X + \bar{\theta}}.$$

For $\alpha_0 \neq 1$ the inverse coordinate is

$$(774) \quad X_0 = \frac{\alpha_0 \bar{\theta} - \theta}{1 - \alpha_0}.$$

In particular

$$X_0 + \bar{\theta} = \frac{\bar{\theta} - \theta}{1 - \alpha_0}, \quad X_0 + \theta = \alpha_0 \frac{\bar{\theta} - \theta}{1 - \alpha_0}.$$

Choose α_0 successively as t^2, t^6, t^4 . Their n th powers are $\theta^2, 1, \theta^4$, respectively, and their cubes each have order exactly n . Their orders exceed three, so X_0 is neither zero nor minus one. The two displayed expressions are nonzero, so the norm and all input arms are units at X_0 .

Write the raw power as $(X + \zeta)^n = A_n(X) + B_n(X)\zeta$. Its ratio to its conjugate is $\alpha(X)^n$. Cross multiplication on the norm-unit locus gives

$\alpha(X)^n$	vanishing raw arm
θ^2	$A_n(X)$
1	$B_n(X)$
θ^4	$A_n(X) + B_n(X)$

Two arms cannot vanish together there. Each difference of ratios is a nonzero constant times the corresponding arm divided by the unit $(X + \bar{\theta})^n$. Moreover

$$\alpha'(X) = \frac{\bar{\theta} - \theta}{(X + \bar{\theta})^2} \neq 0.$$

Since $q > n$, the indicated raw arm has a simple zero at X_0 . Dividing by the relevant input $X, 1$ or $X + 1$, all units there, preserves this simple zero of the actual quotient polynomial.

For $n \equiv 1 \pmod{6}$, the choices t^2, t^6, t^4 target D_1, D_2, D_3 in that order. For $n \equiv 5 \pmod{6}$, conjugation sends the raw coordinates to $(A_n + B_n, -B_n)$. The correct choices are then t^4, t^6, t^2 . This exchange is what identifies the labeled actual quotients in (773).

An integer polynomial with a simple root modulo q has exactly one lift to a root at the next precision: for $e \geq 1$,

$$D_i(r + jq^e) \equiv D_i(r) + jq^e D'_i(r) \pmod{q^{e+1}}.$$

Lift the chosen root uniquely through precision q^{s_i} . At the next step choose any of the $q - 1$ nonroot digits. This gives a residue $r_i \bmod q_i^{s_i+1}$ of exact depth s_i . The other quotient arms, input arms and norm remain units modulo q_i , and the cube ratio keeps exact order n .

The Chinese remainder theorem combines the three residues with $a \equiv 0 \pmod{3}$. Write

$$(775) \quad M_0 = 3 \prod_{i=1}^3 q_i^{s_i+1}, \quad 0 \leq r < M_0$$

for the resulting modulus and representative. Every $a = r + jM_0$, $j \geq 1$, is positive and preserves all those exact depths. The root has coordinate gcd one, and $N(w) = a^2 + a + 1 \equiv 1 \pmod{3}$. It is an actual primitive unramified nonunit root. Its powers have nonzero primitive boundary by the established oriented factorization. Since $T_1 = a(a + 1)$ is coprime to each chosen prime and $T_n = T_1 |D_1 D_2 D_3|$, the stated full depths and first ranks follow. $\square$

Corollary 227.2 (Failure of automatic depth-cap membership). *The constructed roots give a family with $n \rightarrow \infty$ and unit residual, hence $\lambda = 1/n$ and $\rho = \log(4 + n)/n$. Every $q_i > n^{1/6}$. No pair of actual quotient arms has all its prime depths above that cutoff at most two.*

More generally, if caps $h_i \in \mathbb{Z}_{>0} \cup \{\infty\}$ bound all those actual depths, with $1/\infty = 0$, then

$$\sum_{i=1}^3 \frac{1}{h_i} \leq \frac{3}{4} < 1.$$

Thus the union of reciprocal-cap sufficient classes does not contain every actual root, even when a cap may be selected separately for each root. This does not refute the quantitative net-credit criteria.

Proof. The representation uses exponent n and unit residual. The selected primes are at least $6n + 1$, and every selected depth is at least four. Every finite valid cap is consequently at least four; an infinite cap contributes zero to the reciprocal sum. The assertions follow directly from Theorem 227.1 for each allowed n . $\square$

Proposition 227.3 (The selected depths have a visible height cost). *For the representatives $a \geq M_0$ used above, put $c = \max(|A|, |B|, |A + B|)$ and $t_{\text{ht}} = \log c$. The contribution of the selected primes alone to the excess above depth two satisfies*

$$(776) \quad E_{\text{selected}} = \sum_{i=1}^3 (s_i - 2) \log q_i < \frac{t_{\text{ht}}}{n}.$$

The selected positive signed depths $(s_i - 3) \log q_i$ have a smaller total. The estimate is uniform even when all prescribed depths vary with n .

Proof. The actual quadratic height gives $Q^n \leq c^2$, while $Q = a^2 + a + 1 > a^2$. Hence

$$t_{\text{ht}} > n \log a \geq n \log M_0.$$

From (775), $E_{\text{selected}} < \log M_0$, which proves (776). $\square$

Evidence and remaining scope. The root researcher and both research peers independently reviewed the complete ordinary proof. The exact finite replay checks $n = 5, 7, 11, 25, 35$, with depths $(4, 5, 6)$ and two actual CRT roots per exponent: ten actual roots and thirty selected full-depth checks. It proves the selected primes prime by trial division, checks every required order against all proper divisors, and exhaustively checks each selected lifting level. All actual coordinates, norm and quotient identities, coprimality, selected depths and the integer height-cost

certificate are recomputed exactly. One other researcher independently reread the replay and reran its canonical check successfully.

The quotient arms are not completely factored in that replay. No simultaneous second pure norm, full signed-tail failure or Lean proof of this construction is asserted. The unselected prime factors may supply substantial negative radical credit. Equation (776) also shows that these selected deep primes alone pay only a vanishing normalized cost. General control of the actual excess-minus-radical quantity, including unusually short congruence directions and all unselected factors, remains open.

228. THE REMAINING TAIL CARRIES ALMOST ALL ACTUAL BOUNDARY MASS

Keep the actual roots $w_k = 3k + \zeta$ for $B \leq k < 2B$, where B, n are positive integers. Put

$$Q_k = 9k^2 + 3k + 1, \quad T_n(k) = |P(w_k^n)|, \quad t_k = \log c_n(k).$$

Their primitive unramified powers have nonzero boundary, and

$$(777) \quad t_k \geq n \log(3B), \quad \log |P(w_k^d)| \leq 3d \log(6B + 1).$$

Here the observable is full valuation mass, including shallow primes. It is different from both positive excess and signed cost.

Theorem 228.1 (Full-mass mean with rank spacing). *For real $Z \geq 5$ with $Z > 3n$, define $H_Z(k) = \sum_{5 \leq q \leq Z} v_q(T_n(k)) \log q$, over primes. Then*

$$(778) \quad \frac{1}{B} \sum_{B \leq k < 2B} \frac{H_Z(k)}{t_k} \leq \frac{6\tau(n) \log Z(1 + \log Z)}{n \log(3B)} + \frac{36Z\sigma(n)}{Bn \log(Z/(3n))} + \frac{\log n}{n \log(3B)}.$$

The first $\tau(n)$ can be replaced by $\sum_{d|n} \varphi(d)/d$.

Proof. Partition supported actual roots at $q \geq 5$ by their rank $d = \text{ord}((w_k/\bar{w}_k)^3 \bmod q)$. It divides n , and its first depth $s = v_q(T_d(k))$ satisfies $v_q(T_n(k)) = s + v_q(n)$. Primes dividing Q_k have no boundary contribution. The total additional lifting is at most $\log n$ at each root.

For fixed d, q , the established exact-rank count has at most $3\varphi(d)$ simple roots, each lifting uniquely. Thus the actual interval count at depth at least e is at most $3\varphi(d)(B/q^e + 1)$. Sum from $e = 1$, with the cap $\lfloor 3d \log(6B + 1)/\log q \rfloor$ from (777). The full first-depth contribution, summed over the actual block, is at most

$$(779) \quad \frac{3\varphi(d)B \log q}{q - 1} + 9\varphi(d)d \log(6B + 1).$$

This remains valid if the cap is zero; no actual simplicity of depth is assumed.

Eligible primes have $q = 3dj + 1$ or $q = 3dj - 1$, $j \geq 1$. In the first progression $1/(q-1) = 1/(3dj)$; in the second it is at most $1/(dj)$, since $3dj - 2 \geq dj$. All relevant j are at most Z , and the harmonic sum gives

$$\sum_{\substack{q \leq Z \\ q \text{ eligible at } d}} \frac{\log q}{q - 1} \leq \frac{2 \log Z(1 + \log Z)}{d}.$$

The first part of (779), summed over $d | n$ and divided by $Bn \log(3B)$, gives the first term of (778). We used $\varphi(d)/d \leq 1$.

For the endpoint part, the previously stated Brun–Titchmarsh bound counts at most $4Z/[\varphi(3d) \log(Z/(3d))]$ eligible primes. Its premise holds for every divisor since $Z > 3n$. Using $\varphi(3d) \geq 2\varphi(d)$ gives a total endpoint cost at most $18Z\sigma(n) \log(6B + 1)/\log(Z/(3n))$. Divide by $Bn \log(3B)$ and use $\log(6B + 1) \leq 2 \log(3B)$. Finally append the lifting contribution. All normalized costs are nonnegative, so the lower height in (777) is sufficient. This proves the theorem using actual interval counts. $\square$

Theorem 228.2 (Actual full mass beyond the all-index window). *Let $n \rightarrow \infty$ through all positive integers, and put*

$$B = n^4, \quad Z = \left\lfloor \frac{B\sqrt{\log n}}{1 + \log \log n} \right\rfloor, \quad \eta_n = (\log n)^{-1/4}.$$

There are absolute constants C, N such that for every $n \geq N$,

$$(780) \quad \frac{1}{B} \sum_{B \leq k < 2B} \frac{\sum_{q \leq Z} v_q(T_n(k)) \log q}{t_k} \leq C(\log n)^{-1/2}.$$

This sum includes $q = 2, 3$. Outside a proportion at most $C\eta_n$ of the actual roots,

$$(781) \quad 3 - A_1 \frac{\log(4n)}{n} - \eta_n \leq \frac{\sum_{q > Z} v_q(T_n(k)) \log q}{t_k} \leq 3.$$

If n tends to infinity through primes at least five, the mass restricted further to exact rank n obeys, on that same set,

$$(782) \quad \frac{\sum_{\substack{q > Z \\ d_q(k)=n}} v_q(T_n(k)) \log q}{t_k} \rightarrow 3.$$

At these primes the valuation is exactly the first depth.

Proof. The elementary inequality $\tau(n) \leq 2\sqrt{n}$ makes the first term in (778) $O(\log n/\sqrt{n})$. Lemma 214.1 bounds $\sigma(n)/n$ by $e^{18}(1 + \log \log n)$; its extra factor is canceled by the same factor in the denominator of Z/B . Since $\log(Z/(3n)) \sim 3 \log n$, the endpoint term is $O((\log n)^{-1/2})$, and the lifting term is $O(1/n)$.

The all-prime theorem with unit residual and $\lambda = 1/n$ gives

$$\frac{v_2(T_n) \log 2 + v_3(T_n) \log 3}{t_k} \leq A_2(4 \log 2 + 9 \log 3) \frac{\log^2(4n)}{n}.$$

Sector normalization meets its positive-coordinate domain and preserves these valuations and the height. This proves (780). Markov's inequality at η_n leaves at most the stated exceptional proportion. For every remaining root, subtract its complete small-prime mass from

$$\frac{\log T_n}{t_k} = 3 - \frac{\Delta_n}{t_k}, \quad 0 \leq \frac{\Delta_n}{t_k} \leq A_1 \frac{\log(4n)}{n}.$$

This proves (781).

When n is prime, only ranks one and n occur. Above $Z > 3n$ there is no exponent lifting. The rank-one full mass is bounded by $\log T_1(k)$, and

$$\frac{\log T_1(k)}{t_k} \leq \frac{(3/2) \log Q_k}{(n/2) \log Q_k} = \frac{3}{n}.$$

Removing it from (781) proves (782), uniformly on the nonexceptional set. $\square$

This locates the main amount of integer boundary size in the actual remaining tail. At prime index, it is the first-depth top-rank packet, not an old lower-rank contribution. It does not supply a lower bound on the signed cost: a prime of depth one contributes positive full mass but signed credit $-2 \log q$. Almost all the mass above the window could have that useful shallow form. Neither its radical nor its deep private owners are controlled by the theorem.

The proof is an ordinary analytic argument using the established actual rank/lifting counts, the Brun–Titchmarsh estimate already used in Theorem 205.1, and the two-place theorem only for the fixed primes two and three and the archimedean place. It uses no modularity or probabilistic independence and is not a claim of a Lean formalization. Exceptional roots and the net signed tail remain open.

229. ELEMENTARY INPUTS FOR THE ACTUAL ROOT-BLOCK MASS THEOREM

For the actual blocks in Section 228, the archimedean and two fixed-prime inputs have an elementary replacement when $B \geq n$. This gives a proof of their full-mass conclusion without logarithmic forms. It does not estimate the signed cost of the remaining primes. Throughout this section $B \geq n \geq 1$ and $B \leq k < 2B$ are integers,

$$a = 3k, \quad w = a + \zeta, \quad Q = a^2 + a + 1, \quad T_d = |P(w^d)|, \quad t = \log c_n, \quad \Delta = 3t - \log T_n.$$

The auxiliary index d is a positive integer. Here c_n is the largest absolute boundary arm. The elementary norm bounds give

$$(783) \quad Q^{n/2} \leq c_n \leq \frac{2}{\sqrt{3}} Q^{n/2}, \quad t \geq n \log(3B), \quad \log T_d \leq 3d \log(6B + 1).$$

These primitive unramified nonunit roots have nonzero boundaries; for the index n this also follows directly from the angle below.

Lemma 229.1 (A direct angular estimate). *On every such actual block,*

$$(784) \quad T_n \geq \frac{2n}{\pi(12B + 1)} Q^{3n/2}, \quad 0 \leq \frac{\Delta}{t} \leq \frac{4}{n}.$$

Proof. Write $w = \sqrt{Q} e^{i\theta}$, where $\theta = \arctan(\sqrt{3}/(6k + 1))$. The elementary bounds $x/2 \leq \arctan x \leq x$ for $0 \leq x \leq 1$ imply

$$\frac{\sqrt{3}}{2(12B + 1)} \leq \theta, \quad 0 < 3n\theta < \frac{\sqrt{3}}{2} < \frac{\pi}{2}.$$

Thus $\sin(3n\theta) \geq 3n\sqrt{3}/(\pi(12B + 1))$. The exact identity $z^3 - \bar{z}^3 = 3\sqrt{-3}P(z)$ gives

$$T_n = \frac{2}{3\sqrt{3}} Q^{3n/2} \sin(3n\theta),$$

proving the first bound. Every arm has absolute value at most c_n , so $\Delta \geq 0$. Combining the first bound with (783) gives

$$\Delta \leq \log \frac{4\pi(12B + 1)}{3\sqrt{3}n} \leq 4 \log(3B).$$

For the last inequality, $12B + 1 \leq 13B$, $\pi < 4$ and $\sqrt{3} > 1$ bound the logarithm's argument by $81B/n \leq (3B)^4$. Division by $t \geq n \log(3B)$ completes the proof. $\square$

Lemma 229.2 (Exact fixed-prime depths). *For all the above roots and indices,*

$$(785) \quad v_3(T_n) = v_3(T_1) + v_3(n),$$

$$(786) \quad v_2(T_n) = \begin{cases} v_2(T_1), & n \text{ odd}, \\ v_2(T_2) + v_2(n/2), & n \text{ even}. \end{cases}$$

Consequently

$$(787) \quad \frac{v_2(T_n) \log 2 + v_3(T_n) \log 3}{t} \leq \frac{20}{n}.$$

Proof. We first supply the lifting step. Let v be a nonarchimedean valuation above p , normalized by $v(p) = 1$, and write $u = 1 + \delta$. If $e = v(\delta) > 1/(p - 1)$, the linear term in $(1 + \delta)^p - 1$ has valuation $1 + e$. Every intermediate term has valuation $1 + je > 1 + e$, and the last has valuation $pe > 1 + e$. Thus $v(u^p - 1) = e + 1$. For an exponent r prime to p , the weaker condition $e > 0$ suffices: the linear term has valuation e and all higher terms have larger valuation. Hence $v(u^r - 1) = e$. Iteration proves $v(u^m - 1) = v(u - 1) + v_p(m)$ under the first threshold. This binomial argument applies also at the ramified quadratic place.

Put $\eta = (w/\bar{w})^3$. Since Q is prime to both two and three, $w, \bar{w}$ are units there, and

$$(788) \quad \eta^d - 1 = \frac{3\sqrt{-3}P(w^d)}{\bar{w}^{3d}}.$$

At three its valuation is $v_3(T_d) + 3/2$. The integer $T_1 = a(a+1)$ is divisible by three, so $v_3(\eta - 1) \geq 5/2 > 1/2$. Apply the lifting step and subtract $3/2$ to obtain (785).

At two the constant $3\sqrt{-3}$ is a unit and T_1 is even, so the odd-exponent assertion follows from the weaker lifting step. For an even exponent, direct multiplication gives

$$(789) \quad T_2 = a(a-1)(a+1)(a+2)(2a+1).$$

The four consecutive factors $a-1, a, a+1, a+2$ have product divisible by eight. Thus $v_2(\eta^2 - 1) = v_2(T_2) \geq 3 > 1$. Apply the stronger lifting step to η^2 at exponent $n/2$ to obtain (786).

Write $L = \log(6B+1)$. Equations (783)–(786) bound the two-prime cost by $9L + 2 \log n$; the odd case even has a smaller bound. Since $L \leq 2 \log(3B)$ and $\log n \leq \log(3B)$, this is at most $20 \log(3B)$. Normalize by $t \geq n \log(3B)$. $\square$

Theorem 229.3 (An elementary replacement in the mass-location chain). *Set $B = n^4$,*

$$Z = \left\lfloor \frac{B\sqrt{\log n}}{1 + \log \log n} \right\rfloor, \quad \eta_n = (\log n)^{-1/4}.$$

For all sufficiently large integers n , outside at most a proportion $C\eta_n$ of the actual roots in the block,

$$(790) \quad 3 - \frac{4}{n} - \eta_n \leq \frac{1}{t_k} \sum_{p > Z} v_p(T_n(k)) \log p \leq 3.$$

If n is prime, the same roots also satisfy

$$(791) \quad 3 - \frac{7}{n} - \eta_n \leq \frac{1}{t_k} \sum_{\substack{p > Z \\ d_p(k)=n}} v_p(T_n(k)) \log p \leq 3.$$

Every included valuation in (791) is its top-rank first depth.

Proof. The full-mass interval estimate of Section 228 for primes at least five uses actual rank counts, a harmonic sum and Brun–Titchmarsh. Its all-index specialization has mean $O((\log n)^{-1/2})$, by the proved elementary divisor estimates. Add (787). Markov at η_n bounds the full small-prime mass by $\eta_n t_k$ outside the stated proportion. Subtract this mass from $\log T_n = 3t_k - \Delta$ and apply (784), obtaining (790). At prime index the only ranks are one and n . Above $Z > n$ there is no exponent lifting, and the entire rank-one mass is at most $\log T_1 \leq 3t_k/n$. Subtract it to obtain (791). $\square$

This proof retains the established Brun–Titchmarsh input but uses no logarithmic-form estimate. Its exceptional roots and the signed prime costs remain uncontrolled. Full mass at an exponent-one prime is positive while its signed cost is $-2 \log p$; the displayed concentration therefore asserts neither a radical lower bound nor a failure of the signed-tail gate. The ordinary proofs were independently reviewed. A separate exact replay checks 571 actual roots and 1142 valuation equalities, without using floating-point angles as proof. The present estimates are not claimed as Lean formalizations.

230. MIXED EXPONENT COVERS FROM AN ACTUAL FIRST-PROFILE COEFFICIENT

The two norm exponents in this section may be relatively prime, and either residual may be a nonunit. The construction controls a finite list of covers and their coefficient heights. It gives no upper bound on the heights of their points.

Let $\zeta^2 - \zeta + 1 = 0$, $K_0 = \mathbb{Q}(\zeta)$, and $\gamma = 1 + \zeta$. Retain the degree-eight splitting field K of Lemma 196.1, which contains K_0 and the four roots $\alpha_1, \dots, \alpha_4$ of

$$f(X) = X^4 + 3X^3 + 5X^2 + 3X + 1.$$

Put $\beta_+ = -\zeta$ and $\beta_- = -\bar{\zeta}$, the roots of $N(X) = X^2 + X + 1$. These six branch values are distinct: both polynomials are separable, and $f \bmod N = X^2$, so $\text{Res}(N, f) = 1$.

For positive coprime integers a, b , write $c = a + b$, $x = a/b$, $M = a^2 + ab + b^2$, and $F = b^4 f(x)$. Consider genuine oriented profiles

$$(792) \quad a + b\zeta = u\gamma^e v w^h, \quad M = 3^e V_0 R^h, \quad V_0 = N v, \quad R = N w \geq 7,$$

where u is an Eisenstein unit, $e \in \{0, 1\}$, $h \geq 2$, and v, w are primitive oriented unramified Eisenstein integers as in (655). Their oriented supports may overlap. Also suppose

$$(793) \quad F = V_1 Q^g, \quad g \geq 9, \quad V_1, Q \in \mathbb{Z}_{>0}, \quad Q > 1, \quad 1 \leq V_1 < 2^g.$$

All exponents are integers. The last condition ensures that V_1 is g -free. No condition relates h and g .

The domain $Q \geq 7$ used in Section 196 follows from these hypotheses. Indeed, F is odd and $F \equiv 1 \pmod{3}$. The values of $f(r)$ for $r = 0, 1, 2, 3, 4$ modulo five are 1, 3, 2, 2, 1. If $5 \mid b$, primitivity gives $F \equiv a^4 \neq 0$ instead. Thus 2, 3, 5 do not divide F , and $Q > 1$ implies $Q \geq 7$.

Lemma 230.1 (An actual first-profile coefficient). *Define*

$$\xi_0 = (u^2 \zeta^e v / \bar{v})^{-1}.$$

Then

$$(794) \quad (w/\bar{w})^h = \xi_0 \frac{x - \beta_+}{x - \beta_-}, \quad h_K(\xi_0) = \frac{1}{2} \log V_0,$$

where h_K denotes absolute logarithmic Weil height. Over all actual residuals with $V_0 \leq X$, $X \geq 1$, there are at most CX possible ξ_0 , for an absolute constant C . If v is a unit, the coefficients lie in a fixed finite group of roots of unity, including when h is divisible by two or three.

Proof. Conjugating (792), using $u\bar{u} = 1$ and $\gamma/\bar{\gamma} = \zeta$, gives

$$\frac{a + b\zeta}{a + b\bar{\zeta}} = u^2 \zeta^e \frac{v}{\bar{v}} \left(\frac{w}{\bar{w}} \right)^h.$$

Divide by b and rearrange. The exact oriented-height calculation of Lemma 181.1 gives $h_K(v/\bar{v}) = \frac{1}{2} \log N v$: all archimedean absolute values are one, and the numerator and denominator ideals are disjoint conjugates. Overlap between the supports of v and w does not create a conjugate pair within v . Height is unchanged by extending the number field, multiplication by a root of unity, or inversion. No h -th root of u is taken.

Write $v = r + s\zeta$. Since $N v = r^2 + rs + s^2 \geq 3r^2/4, 3s^2/4$, the number of integer coordinate pairs with norm at most X is at most

$$\left(\frac{4\sqrt{X}}{\sqrt{3}} + 1 \right)^2 \leq 16X.$$

The finitely many choices of u, e change only the constant. This already counts the entire range $V_0 \leq X$; it is not a count to be summed again over possible norm values. $\square$

Theorems 196.3 and 196.5 provide a finite list of $\xi_1, \xi_2, \xi_3 \in K^\times$ and actual coordinates $Z_i \in K^\times$ satisfying

$$(795) \quad Z_i^g = \xi_i \frac{x - \alpha_i}{x - \alpha_4}, \quad i = 1, 2, 3.$$

Here we replace the coefficients of (687) by their inverses; the list sizes and heights are unchanged.

Theorem 230.2 (The mixed cover at arbitrary exponents). *Combine (795) with*

$$(796) \quad Y^h = \xi_0 \frac{x - \beta_+}{x - \beta_-}.$$

The smooth projective normalization $C_{h,g,\xi}$ is geometrically connected. Its degree over $\mathbb{P}_x^1$ and its genus are

$$(797) \quad D = hg^3, \quad G = 1 + D \left(2 - \frac{1}{h} - \frac{2}{g} \right) = 1 + 2hg^3 - g^3 - 2hg^2.$$

Every actual profile (792)–(793) gives a K -point of one such curve above $x = a/b$.

Proof. Work over $\bar{K}$, which contains all required roots of unity and roots of the nonzero twist coefficients. The three quartic ratios are independent modulo g -th powers even when g is composite: valuation at α_i is one for its own ratio and zero for the other two. Their Kummer cover is Galois of degree g^3 . Its four branch points are the α_i , each of inertia g . At α_4 the three simple poles adjoin the same root of a local uniformizer, so the inertia is diagonal cyclic of order g . Infinity is unramified, as in the proof of Theorem 196.3.

The first cover is cyclic of degree h , by its simple zero at β_+ , and is branched exactly at β_+, β_- , each of inertia h . Its branch set is disjoint from that of the quartic cover. The intersection of their function fields over $\bar{K}(x)$ is therefore unramified at every point of the projective line: an intermediate extension can ramify only where its containing extension ramifies. Riemann–Hurwitz excludes a nontrivial connected unramified cover of $\mathbb{P}^1$ in characteristic zero. Thus this intersection is trivial. Both extensions are Galois, so their compositum is a field of degree hg^3 . This proves geometric connectedness without any coprimality condition on the exponents.

In the compositum the six ramification contributions sum to $2D(1 - 1/h) + 4D(1 - 1/g)$. Infinity remains unramified, and hence

$$2G - 2 = -2D + 2D(1 - 1/h) + 4D(1 - 1/g),$$

which gives (797). The actual positive rational x lies outside the six branch values. All four Kummer coordinates are nonzero, so the affine cover is smooth there in characteristic zero. The actual point therefore lifts to the normalization over K . This does not assert that every K -point arises from a primitive positive integer seed. $\square$

Theorem 230.3 (A mixed residual budget). *For fixed h, g, V_1 as above and all first residuals $V_0 \leq X$, $X \geq 1$, the actual seeds lift to a list of at most*

$$(798) \quad C_K X g^9 4^{8\omega(V_1)}$$

mixed covers. For $L_0 \geq 0$ and $0 \leq L_1 < \log 2$, the union over $V_0 \leq \exp(L_0 h)$ and $V_1 \leq \exp(L_1 g)$ is covered by at most

$$(799) \quad C_K g^9 \exp(L_0 h + 17L_1 g)$$

models. The maximum absolute Weil height of an individual coefficient of the defining equations can be bounded by

$$(800) \quad C_K (g + \log V_0 + \log V_1 + 1).$$

The same bound, with a different constant, holds for their projective coefficient height. The degrees in Y and Z_i remain h and g , respectively.

Proof. The QC count gives $C_K g^9 4^{8\omega(V_1)}$ second-norm models for fixed g, V_1 , with $h_K(\xi_i) \leq 2 \log V_1 + C_K g$. Its proof includes the fixed field's unit rank three, class-group choices and the actual depth-one contribution above thirteen. Lemma 230.1 adds at most CX first coefficients for the entire residual range. There is no additional independent h^3 unit-class factor: the actual first profile has already supplied its precise coefficient. This proves (798).

Since $V_1 \geq 2^{\omega(V_1)}$, one has $4^{8\omega(V_1)} \leq V_1^{16}$. Summing over the integers $V_1 \leq Z$, $Z \geq 1$, costs at most Z^{17} . Substituting $X = \exp(L_0 h)$ and $Z = \exp(L_1 g)$ proves (799). No second summation over first residual norms is required.

Clear the linear denominators to obtain

$$(x - \beta_-)Y^h - \xi_0(x - \beta_+) = 0, \quad (x - \alpha_4)Z_i^g - \xi_i(x - \alpha_i) = 0.$$

The branch values are fixed, and product heights are bounded by the sums of factor heights. Equation (794) and the QC coefficient bound prove (800). There are a fixed number of nonzero coefficients, so their projective coefficient height has the same bound up to a constant. This reasoning supplies no point-height estimate. $\square$

Corollary 230.4 (Height separation when the first exponent dominates). *Consider any sequence of actual profiles as above with*

$$(801) \quad h \longrightarrow \infty, \quad g/h \longrightarrow 0, \quad \frac{\log V_0}{h} \longrightarrow 0, \quad \frac{\log V_1}{g} \text{ bounded}.$$

For the chosen models, let H_{coef} be the coefficient-height bound in (800), and put $H_{\text{point}} = h_K(a/b) = \log \max(a, b)$. Then

$$(802) \quad \frac{H_{\text{coef}} + \log h + \log g}{H_{\text{point}}} \rightarrow 0.$$

Proof. The hypotheses give $H_{\text{coef}} = o(h)$. In the other direction, (792) implies $M \geq R^h \geq 7^h$, while $M < c^2$. Consequently

$$H_{\text{point}} \geq \log c - \log 2 \geq \frac{h}{2} \log 7 - \log 2.$$

Also $\log h = o(h)$ and $\log g = o(h)$ because $9 \leq g = o(h)$. Division proves (802). $\square$

This separation allows two nonunit residuals and relatively prime exponents. It asserts neither the existence of the displayed sequence nor a contradiction. A sufficient further input would be an upper bound

$$H_{\text{point}} \leq A(H_{\text{coef}} + \log h + \log g + 1)$$

for the actual lifted points in this family, with A independent of the exponents and residuals. Such a point-height theorem is not proved here. Comparable exponents, larger moving residuals, and points whose heights exceed the coefficient budget remain unresolved. The present ordinary arithmetic-geometric proof is independently reviewed; the Kummer, ramification and height arguments in this section are not Lean formalizations.

231. TWO ACTUAL ORIENTED PROFILES IN A QUADRATIC-FIELD COVER

The actual quartic norm admits a more direct lifting than the generic splitting-field construction of Sections 196 and 230. Its defining integer is itself a primitive Eisenstein norm. This lets us use two actual residual coefficients, with no coefficient-height term proportional to either exponent. The resulting point-height problem remains open.

Retain $\mathcal{O} = \mathbb{Z}[\zeta]$, $\zeta^2 - \zeta + 1 = 0$, $K_0 = \mathbb{Q}(\zeta)$ and $\gamma = 1 + \zeta$. For positive coprime integers a, b , put

$$c = a + b, \quad x = a/b, \quad M = a^2 + ab + b^2, \quad F = a^4 + 3a^3b + 5a^2b^2 + 3ab^3 + b^4.$$

Assume an actual first oriented profile

$$(803) \quad a + b\zeta = u_0\gamma^e v_0 w_0^h, \quad M = 3^e V_0 R^h, \quad V_0 = N v_0, \quad R = N w_0 \geq 7,$$

where u_0 is a unit, $e \in \{0, 1\}$, $h \geq 2$, and v_0, w_0 are primitive oriented unramified elements, as in (655). Their oriented supports may overlap. For the second norm assume only

$$(804) \quad F = V_1 Q^g, \quad g \geq 2, \quad V_1, Q \in \mathbb{Z}_{>0}, \quad Q > 1.$$

All exponents are integers. The residual V_1 need not be g -free; that is an optional canonical convention. The exponents need not have a common divisor, or be coprime to six. As checked in Section 230, 2, 3, 5 do not divide F , so $Q \geq 7$.

Lemma 231.1 (An actual second oriented factorization). *There are a unit u_1 and primitive oriented unramified elements $v_1, w_1 \in \mathcal{O}$ such that*

$$(805) \quad ab + M\zeta = u_1 v_1 w_1^g, \quad N v_1 = V_1, \quad N w_1 = Q.$$

Proof. Direct expansion gives

$$(806) \quad N(ab + M\zeta) = (ab)^2 + abM + M^2 = F.$$

Moreover $\gcd(ab, M) = 1$: a prime dividing a or b and M would divide both a and b . Thus $ab + M\zeta$ is primitive, and it is unramified because $3 \nmid F$.

Use unique factorization in the Euclidean Eisenstein ring. An inert rational prime in the norm of a primitive element would divide both coordinates, as would the simultaneous occurrence of conjugate split factors. There is therefore a factorization

$$ab + M\zeta = u_1 \prod_q \pi_q^{E_q}, \quad N \pi_q = q,$$

with precisely one orientation chosen at each rational prime in the support. Equation (804) gives $E_q = v_q(V_1) + gv_q(Q)$, with both summands nonnegative. Set

$$v_1 = \prod_q \pi_q^{v_q(V_1)}, \quad w_1 = \prod_q \pi_q^{v_q(Q)}.$$

Their product and norms give (805) exactly. They inherit primitivity, unramifiedness and the absence of conjugate pairs within either factor. They may share oriented factors. No bound $v_q(V_1) < g$ is used. $\square$

This is an existence proof using unique factorization, not an assertion that a practical factorization of a large F has already been computed.

Put $N(X) = X^2 + X + 1$, $\beta_+ = -\zeta$, $\beta_- = -\bar{\zeta}$, and

$$(807) \quad G(X) = X + \zeta N(X), \quad \bar{G}(X) = X + \bar{\zeta} N(X).$$

Define, in the fixed quadratic field K_0 ,

$$(808) \quad \eta_0 = (u_0^2 \zeta^e v_0 / \bar{v}_0)^{-1}, \quad \eta_1 = (u_1^2 v_1 / \bar{v}_1)^{-1}.$$

Theorem 231.2 (A degree- hg cover from two actual profiles). *The actual profiles yield nonzero K_0 -valued coordinates satisfying*

$$(809) \quad Y^h = \eta_0 \frac{x - \beta_+}{x - \beta_-}, \quad Z^g = \eta_1 \frac{G(x)}{\bar{G}(x)}, \quad Y = w_0 / \bar{w}_0, \quad Z = w_1 / \bar{w}_1.$$

Their coefficients have exact absolute logarithmic Weil heights

$$(810) \quad h_{K_0}(\eta_0) = \frac{1}{2} \log V_0, \quad h_{K_0}(\eta_1) = \frac{1}{2} \log V_1.$$

The smooth projective normalization is geometrically connected over K_0 , with degree $D = hg$ over $\mathbb{P}_x^1$ and genus

$$(811) \quad G_C = 1 + hg \left(2 - \frac{1}{h} - \frac{2}{g} \right) = 1 + 2hg - g - 2h.$$

Every actual profile gives a K_0 -point of this curve above a/b .

Proof. The first equation follows by conjugating the first profile, using $\gamma/\bar{\gamma} = \zeta$ and $u_0/\bar{u}_0 = u_0^2$. For the second, $(ab + M\zeta)/b^2 = G(x)$; divide (805) by its conjugate and rearrange. Neither step takes an exponent-th root of a unit. Lemma 181.1 gives $h_{K_0}(v_i/\bar{v}_i) = \frac{1}{2} \log N v_i$. Multiplication by roots of unity and inversion preserve height. This proves (810), including unit residuals.

For the branch calculation, $G(X)/\zeta = X^2 + (2 - \zeta)X + 1$ has discriminant $-1 - 3\zeta$ of norm thirteen. Thus G has two distinct roots, as does $\bar{G}$. They have no common root: their difference is $(\zeta - \bar{\zeta})N(X)$, whereas at either root of N their value is $X \neq 0$. Their four roots are also disjoint from β_+, β_- . At infinity $G/\bar{G}$ tends to the nonzero constant $\zeta/\bar{\zeta}$.

Over $\bar{K}_0$ the second equation therefore defines a cyclic cover of degree g , with exactly four branch points, each of inertia g . Simple zeros and poles give this conclusion also for composite g . The first equation gives a cyclic cover of degree h , with exactly the two branch points β_+, β_- , each of inertia h . Both are unramified at infinity.

The two geometric Galois extensions have disjoint branch sets. Their intersection is everywhere unramified over $\mathbb{P}^1$, and is trivial by Riemann–Hurwitz. Their compositum is thus a field of degree hg , proving geometric connectedness for arbitrary $\gcd(h, g)$. Roots of unity are only adjoined for this geometric calculation, not to the arithmetic field K_0 . Summing the six ramification contributions gives

$$2G_C - 2 = -2D + 2D(1 - 1/h) + 4D(1 - 1/g),$$

and hence (811).

The actual positive x avoids the roots of N . Also $G(x)\bar{G}(x) = F/b^4 > 0$, so it avoids the other four branch values. Its nonzero coordinates give a smooth affine point, since the derivatives in Y, Z are nonzero in characteristic zero. It lifts to the normalization over K_0 . No converse asserting that every point is a primitive integer seed is used. $\square$

At $h = g = 2$ the genus is three, consistently with the earlier two-square-class geometry. This does not assert a seed in the already excluded subbranch with two unit residuals and even exponents.

Theorem 231.3 (Two cumulative residual counts). *Fix h, g . All actual profiles with $V_0 \leq X_0$, $V_1 \leq X_1$, where $X_0, X_1 \geq 1$, lift to a list of at most*

$$(812) \quad CX_0X_1$$

models, for an absolute constant C . The models can be written as

$$(813) \quad (x - \beta_-)Y^h - \eta_0(x - \beta_+) = 0, \quad \bar{G}(x)Z^g - \eta_1G(x) = 0.$$

Their maximum individual coefficient height, and their projective coefficient height up to an absolute factor, are bounded by

$$(814) \quad H_{\text{coef}} = C_0(1 + \log V_0 + \log V_1),$$

where C_0 is fixed. The degrees h, g remain explicit and are not included in this coefficient-height assertion.

Proof. The lattice count of Lemma 230.1 bounds the number of Eisenstein integers of norm at most X_i by $16X_i$. There are only six choices for each unit and two choices for e . Multiplying these two cumulative counts proves (812). Restricting to compatible actual coefficients can only reduce the list. No additional sum over residual norms or independent unit-class factor is added.

The coefficients of $G, \bar{G}$ and the branch values are fixed. The product-height inequality and (810) give (814). There are a fixed number of nonzero coefficients, independent of h, g , so their projective height is bounded by a constant multiple of the same expression. $\square$

For any $L_0, L_1 \geq 0$, the union with $V_0 \leq \exp(L_0h)$ and $V_1 \leq \exp(L_1g)$ consequently has size at most

$$(815) \quad C \exp(L_0h + L_1g).$$

This uses the actual quadratic-norm structure of F . It does not provide small representatives for every class in the generic QC list. The difficult unit classes in Lemma 196.6 were not proved to have actual seeds, so their lower bound is fully consistent with the present coefficient construction.

Corollary 231.4 (Height separation with unrestricted exponent ratios). *Write $H = \max(a, b)$ and $H_{\text{point}} = \log H$. Every actual profile satisfies*

$$(816) \quad H_{\text{point}} \geq \max \left\{ \frac{h}{2} \log 7 - \log 2, \frac{g}{4} \log 7 - \frac{1}{4} \log 13 \right\}.$$

For any sequence of actual profiles such that

$$(817) \quad T = \max(h, g) \longrightarrow \infty, \quad \log V_0 + \log V_1 = o(T),$$

the selected models obey

$$(818) \quad \frac{H_{\text{coef}} + \log h + \log g}{H_{\text{point}}} \longrightarrow 0.$$

In particular this holds whenever

$$(819) \quad \lambda_0 = \frac{\max\{1, \log V_0\}}{h} \longrightarrow 0, \quad \lambda_1 = \frac{\max\{1, \log V_1\}}{g} \longrightarrow 0,$$

without a condition on h/g or on the common divisors of h, g . For the combined list in (815) with $L_i = \lambda_i$, its logarithmic size divided by T also tends to zero.

Proof. We have $M \geq 7^h$, $M < c^2$ and $H \geq c/2$. Also $F \geq 7^g$ and $F \leq 13H^4$. These imply (816), whose right side is at least $\frac{1}{4}T \log 7 - \log 2$. Under (817), (814) is $o(T)$ and $\log h + \log g \leq 2 \log T = o(T)$. Division proves (818).

Under (819), both exponents tend to infinity and

$$\frac{\log V_0 + \log V_1}{T} \leq \lambda_0 + \lambda_1 \longrightarrow 0.$$

Finally the logarithm of (815), divided by T , is at most $\log C/T + \lambda_0 + \lambda_1$. $\square$

These conclusions are necessary properties of an actual sequence, not a construction of one. High points on varying high-genus curves are compatible with all the displayed estimates. A uniform bound

$$H_{\text{point}} \leq A(H_{\text{coef}} + \log h + \log g + 1)$$

for actual lifted points, with A independent of the profiles, would exclude (817). No such upper bound is proved here; coefficient height, genus and the number of twists alone do not give it. Moving roots, nonunit residuals, and the fixed positive two-step threshold remain unresolved. The present result is an independently reviewed ordinary proof; its number-field, geometric and height arguments are not claimed as Lean formalizations.

232. A RATIONAL-MAP DESCENT OF THE ACTUAL DOUBLE COVER

The quadratic-field model of Section 231 has an explicit descent to a rational fiber product. Its expanded rational equation has a different coefficient-height budget; the exponent-free sparse bound is not transferred to this expanded model.

Retain all actual DC hypotheses, and put

$$\tau_0 = u_0 \gamma^e v_0, \quad \tau_1 = u_1 v_1, \quad \rho(T) = \frac{T + \zeta}{T + \bar{\zeta}}, \quad r(X) = \frac{X}{X^2 + X + 1}.$$

Lemma 232.1 (Projective norm-power maps). *For $0 \neq \tau \in \mathbb{Z}[\zeta]$ and $n \in \mathbb{Z}_{\geq 1}$, define homogeneous integer polynomials by*

$$(820) \quad \tau(T + S\zeta)^n = A_{\tau,n}(T, S) + B_{\tau,n}(T, S)\zeta.$$

Then $L_{\tau,n} = [A_{\tau,n} : B_{\tau,n}]$ is a morphism $\mathbb{P}_{\mathbb{Q}}^1 \rightarrow \mathbb{P}_{\mathbb{Q}}^1$ of degree n . Over K_0 it satisfies the identity of projective maps

$$(821) \quad \rho(L_{\tau,n}(t)) = (\tau/\bar{\tau})\rho(t)^n.$$

For $n \geq 2$, its critical points and branch values are exactly $\beta_+ = -\zeta$, $\beta_- = -\bar{\zeta}$, with index n at each.

Proof. A common projective zero of A, B would make both $\tau(T + S\zeta)^n$ and $\bar{\tau}(T + S\bar{\zeta})^n$ vanish. Both coefficients are nonzero, so $T + S\zeta = T + S\bar{\zeta} = 0$, forcing $T = S = 0$. Thus there is no common projective zero. Conjugating (820) and dividing gives (821) first as a rational identity and hence as an identity of projective maps. The map ρ is a projective linear automorphism; the other side is a power map followed by nonzero scaling. Its degree is n , and its two critical points and values correspond to zero and infinity under ρ . This also accounts for all poles of the affine expression A/B . $\square$

The map r is the degree-two morphism

$$[X : Y] \longmapsto [XY : X^2 + XY + Y^2].$$

Its only critical points are $1, -1$, both of index two, with values $1/3, -1$. Indeed, $r'(x) = (1 - x^2)/(x^2 + x + 1)^2$. Its poles β_+, β_- are simple, and infinity is unramified, since $r(1/t) = t/(1+t+t^2)$. Consequently

$$(822) \quad F_h = r \circ L_{\tau_0,h}$$

has degree $2h$ and branch values exactly $\infty, 1/3, -1$. Above infinity its two ramified points are β_+, β_- , of index h . Above each other branch value are the h distinct inverse images under $L_{\tau_0,h}$ of 1 or -1 , each of index two. The total ramification is $2(h-1) + 2h = 4h - 2$, as required.

Theorem 232.2 (A smooth rational fiber product). *The full projective fiber product*

$$(823) \quad L_{\tau_1, g}(t_1) = r(L_{\tau_0, h}(t_0))$$

is a smooth geometrically connected curve D over $\mathbb{Q}$. Its equation in $\mathbb{P}_{t_0}^1 \times \mathbb{P}_{t_1}^1$ has bidegree $(2h, g)$, and

$$(824) \quad g(D) = (2h - 1)(g - 1) = 1 + 2hg - g - 2h.$$

After extension to K_0 , it is isomorphic to the smooth projective curve of Theorem 231.2. Every actual profile gives a $\mathbb{Q}$ -point of D .

Proof. The branch values of $L_{\tau_1, g}$ are β_+, β_- ; they are disjoint from $\infty, 1/3, -1$, the branch values of F_h . Thus at every geometric point in the fiber product at least one of the two maps is unramified. In local parameters, the defining relation has a nonzero derivative in at least one variable. This proves smoothness at every projective point, including those above infinity.

Abbreviate the two pairs of homogeneous polynomials by A_i, B_i . The homogeneous equation is

$$(825) \quad A_1(A_0^2 + A_0B_0 + B_0^2) - B_1A_0B_0 = 0.$$

Each pair has no common zero, and the maps have degrees $g, 2h$, so this equation has bidegree $(2h, g)$.

On a dense open set put

$$x = L_{\tau_0, h}(t_0), \quad Y = \rho(t_0), \quad Z = \rho(t_1).$$

The conjugacy identity and

$$\tau_0/\bar{\tau}_0 = \eta_0^{-1}, \quad \tau_1/\bar{\tau}_1 = \eta_1^{-1}, \quad \rho(r(x)) = G(x)/\bar{G}(x)$$

give precisely (809). Conversely, invert ρ to recover t_0, t_1 from Y, Z . Thus the generic function field is the geometrically connected field of Theorem 231.2. No further component can be confined to the removed finite set of base values: those fibers are finite, whereas every smooth component here is a curve. This proves geometric connectedness. The birational identification between smooth projective curves extends to an isomorphism over K_0 , by the standard extension and function-field equivalence [140, Lemma 53.2.2 and Theorem 53.2.6]. This proves the claimed rational descent.

Projection to t_0 has degree g . The $2h$ inverse images of each β value under F_h are distinct, since F_h is unramified there. The resulting $4h$ branch points each have inertia g , and the characteristic-zero Riemann–Hurwitz formula [140, Section 53.12] gives

$$2g(D) - 2 = -2g + 4h(g - 1).$$

This is (824).

For the actual point, write $w_i = r_i + s_i\zeta$. Primitivity and $N w_i > 1$ force $s_i \neq 0$: otherwise $\gcd(r_i, 0) = 1$ would make w_i a unit. Thus $t_i = r_i/s_i$ is rational, and the exact power identities give

$$L_{\tau_0, h}(t_0) = a/b, \quad L_{\tau_1, g}(t_1) = ab/M = r(a/b).$$

The actual displayed denominators are nonzero because $b, M > 0$. These are therefore actual rational points on the full projective curve. No converse from an arbitrary rational point to a primitive positive integer seed is asserted. $\square$

Proposition 232.3 (Height after rational expansion). *The integral equation (825) has individual and projective coefficient height bounded by*

$$(826) \quad C(h + g + \log V_0 + \log V_1 + 1).$$

Proof. The coordinates of every power of ζ are in $\{0, 1, -1\}$. The binomial expansion therefore bounds the coordinate coefficients of $(T + S\zeta)^n$ by 2^n . The coordinates of τ_0 are at most $2\sqrt{V_0}$ in absolute value, since $N \tau_0 = 3^e V_0$ and $e \leq 1$. Those of τ_1 are at most $2\sqrt{V_1}$. Multiplication by these elements changes the bounds by an absolute factor. The coefficients in the first pair are thus bounded by $C\sqrt{V_0}2^h$, and those in the second by $C\sqrt{V_1}2^g$.

Multiplying two binary forms of degree h adds at most $h + 1$ terms per coefficient. Hence (825) has coefficients bounded by

$$C(h + 1)V_0\sqrt{V_1}2^{2h+g}.$$

Take logarithms and use $\log(h + 1) \leq h$. The projective height of an integral vector is at most the logarithm of its largest absolute entry: removing its common divisor only lowers the bound. This proves (826). $\square$

The sparse quadratic-field model has coefficient height $O(1 + \log V_0 + \log V_1)$; the expanded rational model has the valid larger bound (826). The transformations include exponent-dependent expansions, and no exponent-free bound is claimed for the latter model. Neither coefficient budget bounds an actual point. The explicit rational maps instead identify a concrete compatibility problem for the two norm profiles, with all projective poles and branch fibers accounted for. These are ordinary geometric proofs, not claimed Lean formalizations.

233. THE TENTH CONTINUATION: DISTINCT REMAINING PROOF OBLIGATIONS

The present continuation gives reviewed results on three different obstructions. The reciprocal-depth theorem permits unequal caps and retains radical credit from every unrestricted arm. Its five maximal cap patterns are sufficient but do not exhaust actual roots: the three-arm private-depth family disproves that precise automatic membership assertion. Its selected deep primes have a vanishing normalized height cost, so the same family does not disprove the net compensation criterion. Controlling the full excess-minus-credit quantity at each actual moving root remains the nearest analytic gate; an average full-mass result does not supply that control.

The independent block theorem locates almost all full multiplicity mass above a growing prime threshold for most roots of a specified block. Its elementary replacement removes the archimedean and fixed-prime logarithmic-form inputs on those blocks. It retains the established Brun–Titchmarsh estimate and an exceptional set. This chain remains useful for identifying where a missing signed estimate must act; it proves neither the sign of the remaining cost nor a pointwise result on the exceptional roots.

The geometric route retains both the generic mixed Kummer construction and the stronger construction using the two actual Eisenstein norms. The latter has degree hg , genus $1 + 2hg - g - 2h$, and a sparse coefficient budget independent of the exponents. Its smooth rational descent exhibits the same compatibility through two explicit rational maps. The expanded rational coefficient budget is different. Neither a small coefficient height nor a large genus bounds the actual point height uniformly in this family. A suitable height upper bound, or an actual sequence violating it while meeting the norm requirements, remains to be established.

The new formal inventory has 41 proved declarations: 17 in the integer reciprocal-depth module and 24 in the actual prime-logarithm modules. The first fresh scoped build also recompiled 71 dependency declarations. The prime-logarithm build reused pinned Mathlib compiled dependencies. The geometric constructions, infinite private-depth family and analytic block estimates remain ordinary reviewed proofs, not Lean theorems. The two exact finite replays provide separately identified arithmetic evidence. None of these distinctions is removed by publishing the files together.

Thus no complete proof or disproof of abc is claimed. The earlier unrefuted routes remain available. Only the particular automatic-cap claim has been excluded by the actual counterfamily; difficulty, lack of a ready formal library, and disagreement with another route are not reasons to discard a direction.

234. A COMMON ACTUAL SET FOR LARGE-CUTOFF SIGNED COMPENSATION

The full-mass theorem and the reciprocal-depth ledger can be combined without discarding the signed radical credits. This replaces a small-cutoff sufficient test by a test only above the root interval, on an explicitly specified majority of actual roots. Membership in that new sufficient class remains a separate arithmetic problem.

Let $n \geq 5$ be an integer prime to six, and set

$$B = n^4, \quad Z = \left\lfloor \frac{B\sqrt{\log n}}{1 + \log \log n} \right\rfloor, \quad \eta_n = (\log n)^{-1/4}.$$

For $B \leq k < 2B$ use $w = 3k + \zeta$ and the actual normalized quotient arms D_1, D_2, D_3 of Section 220. Write

$$T = T_n(k), \quad t = \log c_n(k), \quad T_1 = |P(w)|, \quad \Delta = 3t - \log T,$$

and let c_1 be the largest input arm. The quotients are nonzero and pairwise coprime, and $T = T_1|D_1D_2D_3|$. Define

$$L_Z(T) = \sum_{p \leq Z} v_p(T) \log p, \quad \mathcal{J}(T) = \log T - 3 \log \text{rad}(T).$$

The global cost $\mathcal{J}$ is signed; it is not replaced by its positive part in the finite argument.

Theorem 234.1 (One root set for all cap choices). *There are absolute constants C, N_0 such that for every eligible $n \geq N_0$ there is a set G_n of actual root indices with*

$$(827) \quad |G_n| \geq (1 - C\eta_n)B, \quad L_Z(T_n(k))/t_k \leq \eta_n \quad (k \in G_n).$$

The set is independent of every cap choice below. For any $k \in G_n$ and any

$$h_i \in \mathbb{Z}_{\geq 1} \cup \{\infty\}, \quad \sum_i 1/h_i \geq 1, \quad 1/\infty = 0,$$

put $\beta_i = 3/h_i$ for finite h_i , and zero otherwise, and define

$$(828) \quad \begin{aligned} \mathcal{E}_Z(h; k) = & \sum_{h_i < \infty} \beta_i \sum_{p > Z} (v_p(|D_i|) - h_i)_+ \log p \\ & - 3 \sum_{h_i = \infty} \sum_{\substack{p > Z \\ p | D_i}} \log p. \end{aligned}$$

Then, simultaneously for all such moving choices,

$$(829) \quad \frac{\mathcal{J}(T)}{t} \leq \frac{39}{n} + 3\eta_n + \frac{\mathcal{E}_Z(h; k)}{t},$$

$$(830) \quad \frac{\log \text{rad}(T)}{t} \geq 1 - \frac{43}{3n} - \eta_n - \frac{(\mathcal{E}_Z(h; k))_+}{3t}.$$

Proof. The reviewed full small-prime mean in Sections 228 and 229 is $O((\log n)^{-1/2})$. Markov at η_n gives (827). This construction only tests L_Z/t , so the same set works for every cap triple.

The direct angle bound gives $\Delta/t \leq 4/n$. Also

$$(831) \quad \frac{\log c_1}{t} \leq \frac{2}{n}, \quad \frac{\log T_1}{t} \leq \frac{3}{n}.$$

Indeed $c_1 \leq (2/\sqrt{3})\sqrt{Q} \leq Q$ for $Q \geq 7$, while $t \geq (n/2) \log Q$. The cubic boundary identity gives $T_1 \leq 2Q^{3/2}/(3\sqrt{3}) \leq Q^{3/2}$. Thus $\delta = \Delta + \log c_1$ satisfies $\delta/t \leq 6/n$.

In the reciprocal-depth ledger write

$$C_h = \sum_i \beta_i, \quad A_h = \sum_i (\beta_i - 1)_+, \quad B_h = \max_i (\beta_i - 1)_+.$$

Then $C_h \geq 3$, $A_h \leq 6$ and $B_h \leq 2$. Its exact inequality at cutoff Z is

$$\log W_Z(T) \leq (3 - C_h)t + A_h\delta + B_hL_Z(T) + \log T_1 + \mathcal{E}_Z(h; k).$$

The first term is nonpositive. Insert the good-set and input bounds to obtain

$$\log W_Z(T)/t \leq 39/n + 2\eta_n + \mathcal{E}_Z(h; k)/t.$$

The omitted signed cost is at most its full mass:

$$\mathcal{J}(T) - \log W_Z(T) = \sum_{\substack{p \leq Z \\ p|T}} (v_p(T) - 3) \log p \leq L_Z(T).$$

This proves (829). Finally $3 \log \text{rad}(T) = 3t - \Delta - \mathcal{J}(T)$. Use $\Delta/t \leq 4/n$ and replace the net excess by its positive part to obtain (830). No union bound over cap choices appears, and the constants are uniform even for infinite caps. $\square$

Corollary 234.2 (A sufficient large-prime subclass). *If $k \in G_n$ and every finite-cap arm has all actual prime depths above Z at most its chosen cap, then*

$$(832) \quad \log \text{rad}(T)/t \geq 1 - 43/(3n) - \eta_n.$$

Any of the five maximal reciprocal patterns may be used, now only at primes above Z . In particular one quotient arm squarefree above Z suffices, with the other two arms unrestricted. More generally, along any sequence of roots in G_n with admissible moving caps and $(\mathcal{E}_Z)_+/t \rightarrow 0$, for each $\epsilon > 0$ all sufficiently large members satisfy

$$c_n \leq \text{rad}(T_n)^{1+\epsilon}.$$

Proof. Under the actual finite caps every finite excess is zero and the unrestricted-arm terms are nonpositive. Apply (830). For the general sequence its error tends to zero, so the radical logarithm is at least $t/(1+\epsilon)$ eventually. $\square$

The cap-only estimate is uniform at sufficiently large n for each fixed ϵ . No positive proportion of G_n is proved to meet the cap or net-budget condition. The excluded roots remain excluded, and bounded exponents with moving roots are not handled by a finite-seed argument. The gain is the location of the test: $Z/B \rightarrow \infty$, and arbitrary smaller-prime depths are allowed on G_n because their entire mass is already negligible there.

The scope of the known private-depth family. The TD construction selects $a \geq M_0 = 3 \prod_i q_i^{s_i+1}$, with $s_i \geq 4$. Thus $q_i \leq (a/3)^{1/(s_i+1)} \leq a^{1/5}$. If a chosen representative belongs to a comparable block $a = 3k$, $B \leq k < 2B$, its selected primes satisfy $q_i < (2B)^{1/5}$, well below Z for sufficiently large n , when $Z > B$. In fact the published representatives obey $a \geq 3(6n+1)^{15}$ and therefore do not belong to $B = n^4$ for $n \geq 5$. The conditional cutoff comparison does not assert otherwise.

Those selected depths alone cannot refute the new large-cutoff test. Their other prime factors were not controlled, so neither can that construction prove membership in it. The old automatic lower-cutoff cap assertion remains refuted exactly as stated. The unresolved problem is actual excess-minus-credit control above Z , including a treatment of the exceptional roots. This section is an independently reviewed ordinary proof, not a claim that the sufficient condition holds on every root.

235. A FORMAL PASSAGE FROM THE SIGNED TAIL TO THE GLOBAL RADICAL

Retain the actual prime-factorization observables of Section 226: S_Y, L_Y, R_Y and the signed logarithmic cost $\mathcal{S}_Y = S_Y - 3R_Y$, with a natural cutoff Y . For positive N , $R_0(N) = \log \text{rad}(N)$. The following finite step is independent of how a tail estimate or a small-prime estimate is obtained.

Lemma 235.1 (Global signed comparison). *For every positive integer N and natural Y ,*

$$(833) \quad \mathcal{S}_0(N) \leq \mathcal{S}_Y(N) + L_Y(N).$$

If real numbers t, δ, b, l satisfy

$$3t - \delta \leq \log N, \quad \mathcal{S}_Y(N) \leq b, \quad L_Y(N) \leq l,$$

then

$$(834) \quad t - (\delta + b + l)/3 \leq R_0(N).$$

Proof. A prime's logarithmic weight can only decrease when the cutoff increases. Summing over the actual prime support gives $R_Y(N) \leq R_0(N)$. There is no prime at most zero, so $L_0(N) = 0$ and $S_0(N) = \log N = S_Y(N) + L_Y(N)$. Substitute these identities into the signed costs to obtain (833). Since $S_0(N) = \log N - 3R_0(N)$, the three assumed inequalities then give (834). $\square$

In particular, for a positive integer n and $t > 0$, the finite hypotheses

$$3t - 4t/n \leq \log N, \quad \mathcal{S}_Y(N) \leq 39t/n + 2\eta t + e, \quad L_Y(N) \leq \eta t$$

imply exactly

$$(835) \quad 1 - 43/(3n) - \eta - e/(3t) \leq R_0(N)/t.$$

This is the numerical step used by the large-cutoff bridge; the analytic availability of its hypotheses is not assumed universally.

Under the positive integer arm hypotheses of Theorem 226.2, with $T = (aA)(bB)(dC)$, $U = abd$ and $\Delta = 3 \log H - \log T$, the same comparison proves

$$(836) \quad \begin{aligned} \mathcal{S}_0(T) &\leq \Delta + \log H_1 + \log U + \tfrac{3}{2}L_Y(T) \\ &\quad + \tfrac{3}{2}(E_{2,Y}(A) + E_{2,Y}(B)) - 3R_Y(C). \end{aligned}$$

The coefficient of the small-prime mass is three halves because (833) adds its full mass to the half-mass term already present in the actual finite tail theorem. This argument preserves arbitrary overlap with the old factor U .

Formal scope and verification. The module `ActualGlobalSignedBridge` contains nine proved declarations. They establish antitonicity of actual prime weights and radical sums, the zero-cutoff identities, the global comparison, the height transfer, the numerical constants in (835), and the actual finite global arm theorem (836). Its definitions are the actual `Nat.factorization`, `Nat.primeFactors` and `Real.log` observables imported from the preceding two modules. The formal radical is the actual finite sum of prime logarithms.

A fresh scoped run compiled these nine declarations and all 24 local prime-logarithm dependency declarations with Lean 4.32.0 and warnings as errors. The pinned Mathlib cache was reused at revision 81a5d257c8e410db227a6665ed08f64fea08e997; no complete Mathlib rebuild is claimed. All 33 declarations were queried for axioms, with only `propext`, `Classical.choice` and `Quot.sound` in the union. Independent source review checked all nine statements and proofs and matched the final source hashes to the manifest.

The numerical theorem explicitly assumes its height, tail and small-mass bounds, as well as $t > 0$ and a nonzero natural n . It does not prove the common-set analytic estimate, actual cap membership, or exceptional-root control. Thus the finite global prime-factorization step is formalized, while the large-cutoff analytic and pointwise number-theoretic obligations remain distinct.

236. JOINT SQUAREFREEDOM BEYOND THE ACTUAL ROOT INTERVAL

The following result proves simultaneous first-depth membership on a growing interval of primes for a majority of actual roots. It does not assume that supported primes are usually simple. The larger-prime tail and the exceptional roots remain separate problems.

Let $n \geq 5$ be prime, put $B = n^4$, and for integers $B \leq k < 2B$ write

$$a = 3k, \quad w_k = a + \zeta, \quad Q_k = a^2 + a + 1, \quad T_n(k) = |P(w_k^n)|, \quad t_k = \log c_n(k).$$

These are primitive unramified roots, and $T_1(k) = a(a+1)$. At a supported prime $q \geq 5$, the rank is $d_q = \text{ord}((w_k/\bar{w}_k)^3 \bmod q)$.

Theorem 236.1 (An actual joint repeated-prime bound). *For a real number $U > B$, let $\mathcal{B}(n, U)$ be the set of indices in the block for which some prime $B < q \leq U$ satisfies $q^2 \mid T_n(k)$. Then*

$$(837) \quad \frac{|\mathcal{B}(n, U)|}{B} \leq \frac{3n}{B} + \frac{6U}{B \log(U/(3n))}.$$

In particular, with

$$U_n = \left\lfloor \frac{B \log n}{\log \log n} \right\rfloor,$$

one has $U_n > B$ for all sufficiently large prime n , and $|\mathcal{B}(n, U_n)|/B = O(1/\log \log n)$. Outside this single exceptional set, every supported prime in $(B, U_n]$ has exact depth one simultaneously.

Proof. A supported prime does not divide Q_k . Since $q > B > n$, the homogeneous valuation law has no exponent-lifting term and its rank divides n . Thus its rank is one or n .

Rank one cannot produce a repeated prime in this interval. Its depth equals its depth in $T_1 = a(a+1)$. These two factors are coprime, so $q^2 \mid T_1$ would force q^2 to divide one individual factor. Each is less than $6B+1$, whereas $q^2 > B^2 > 6B+1$ for $B \geq 625$. The comparison is with the individual factors, not their product.

Every bad prime therefore has rank n . The reviewed finite-field rank calculation gives at most $3\varphi(n)$ simple roots in the k -variable modulo q . Each has a unique lift modulo q^2 , with rank unchanged. Counting actual integers in the block gives

$$\#\{k : d_q = n, q^2 \mid T_n(k)\} \leq 3\varphi(n)(B/q^2 + 1).$$

Every eligible prime lies in one of the two classes $\pm 1 \pmod{3n}$. The first term in the union bound, after division by B , is at most

$$3\varphi(n) \sum_{m>B} m^{-2} \leq 3n/B.$$

The two-progression Brun–Titchmarsh bound used in the earlier rank-window proof gives

$$\#\{q \leq U : q \equiv \pm 1 \pmod{3n}\} \leq \frac{4U}{\varphi(3n) \log(U/(3n))}.$$

Here $U > B > 3n$ and $\varphi(3n) = 2\varphi(n)$, yielding the second term of (837). No independence of primes or uniform distribution of actual roots has been assumed. Finally $U_n/B \sim \log n / \log \log n$ and $\log(U_n/(3n)) \sim 3 \log n$, giving the stated asymptotic bound. The exceptional set is defined by the existence of any repeated prime, so its complement gives the simultaneous assertion. $\square$

The exact rank and simple-lifting inputs are the ordinary results underlying the sixth-round actual-root window. The sieve input is the same real-endpoint specialization of Montgomery–Vaughan’s Theorem 2, equation (1.10), already verified in the seventh-round sieved-window proof. No stronger version of that input is invoked.

Lemma 236.2 (Actual negative credit in the window). *For $k \notin \mathcal{B}(n, U_n)$, define, over supported primes,*

$$M_{\text{win}} = \sum_{B < q \leq U_n} v_q(T_n) \log q, \quad W_{\text{far}} = \sum_{q > U_n} (v_q(T_n) - 3) \log q, \quad H_{\text{low}} = \sum_{q \leq B} (v_q(T_n) - 3) \log q.$$

Then

$$(838) \quad M_{\text{win}} = \sum_{\substack{B < q \leq U_n \\ q \mid T_n}} \log q,$$

$$(839) \quad \mathcal{J}(T_n) = W_{\text{far}} - 2M_{\text{win}} + H_{\text{low}} \leq W_{\text{far}} - 2M_{\text{win}} + L_B(T_n).$$

Proof. The preceding theorem makes each supported depth in the window equal to one. Its signed contribution is therefore $-2 \log q$. Partition the global supported-prime sum at B and U_n , and bound the remaining small-prime signed contribution by its full mass. $\square$

This is proved credit for actual roots, but there is no positive lower bound for M_{win} . The window may contain no supported primes. The next result also shows that its typical mass is sublinear.

Theorem 236.3 (The remaining packet still carries the full mass). Put $\epsilon_n = (\log \log n)^{-1/2}$. There is an absolute constant C and, for every sufficiently large prime n , a set $\mathcal{H}_n$ of actual indices of size at least $(1 - C\epsilon_n)B$, such that every member has the joint squarefree property above and

$$(840) \quad L_{U_n}(T_n(k))/t_k \leq \epsilon_n, \quad 0 \leq M_{\text{win}}(k)/t_k \leq \epsilon_n,$$

$$(841) \quad 3 - \frac{4}{n} - \epsilon_n \leq \frac{1}{t_k} \sum_{q > U_n} v_q(T_n(k)) \log q \leq 3.$$

Every supported prime in (841) has exact rank n , and its valuation is its first depth.

Proof. Apply the full-mass mean in Section 228 at U_n . For prime n , $\tau(n) = 2$ and $\sigma(n)/n = 1 + 1/n$. The three terms of that bound are respectively $O(\log n/n)$, $O(1/\log \log n)$ and $O(1/n)$. The elementary estimate of Section 229 adds at most $20/n$ for primes two and three. Hence

$$\frac{1}{B} \sum_{B \leq k < 2B} \frac{L_{U_n}(T_n(k))}{t_k} \leq \frac{C}{\log \log n}.$$

Markov at ϵ_n excludes at most $C\epsilon_n$ of the relative mass of indices. Intersect with the complement of $\mathcal{B}(n, U_n)$, whose exceptional fraction is $O(\epsilon_n^2)$. This proves (840). The elementary angular estimate $3 - 4/n \leq \log T_n/t_k \leq 3$ gives (841) by subtraction.

Eventually $U_n > 6B + 1$, so no prime above U_n divides either old input factor $a, a + 1$. Rank one is therefore absent altogether. Also $q > n$, so there is no exponent lifting. The homogeneous valuation law identifies its depth with its rank- n first depth. $\square$

For an actual sequence in $\mathcal{H}_n$, the condition

$$(W_{\text{far}} - 2M_{\text{win}})_+/t_k \rightarrow 0$$

is sufficient for $\limsup \mathcal{J}(T_n)/t_k \leq 0$ and hence for the restricted ABC estimate, for each fixed positive tolerance at sufficiently large indices. This is a conditional conclusion. On the proved set, the available window credit is itself $o(t_k)$, whereas the farther packet carries $(3 - o(1))t_k$ of full valuation mass. The squarefree window does not pay for a positive linear signed cost in that packet. Neither that signed cost nor the excluded roots have been controlled. The analytic membership theorem here is an independently reviewed ordinary proof; no completed Lean proof of its sieve or exceptional-set statements is asserted.

237. PROJECTIVE CONTENT AND A BOUNDARY TO THE SINGLE-MAP INVERSE

The rational maps of Section 232 send every actual profile to a rational point. A converse must account for integral content. That content admits a bound independent of the exponent, but the bound alone does not preserve the prescribed exponent after primitive reduction. The following counterfamily concerns a single power map; it does not assert a rational point on the full double fiber product.

Retain $\mathcal{O} = \mathbb{Z}[\zeta]$, $\zeta^2 - \zeta + 1 = 0$. For $z = A + B\zeta \neq 0$, write $\text{cont}(z) = \gcd(|A|, |B|)$.

Lemma 237.1 (A content divisor). If $0 \neq \tau \in \mathcal{O}$ and $W \in \mathcal{O}$ is primitive, then

$$\text{cont}(\tau W) \mid N\tau.$$

Proof. Write $\tau = r + s\zeta$, $W = u + v\zeta$ and $\tau W = A + B\zeta$. Multiplication and its adjugate give

$$A = ru - sv, \quad B = su + (r + s)v,$$

$$(r + s)A + sB = (N\tau)u, \quad -sA + rB = (N\tau)v.$$

The content divides both right sides. Bezout for $\gcd(|u|, |v|) = 1$ proves the claim. No coprimality between τ and W is needed. $\square$

For an unramified primitive root w , its powers are primitive by the reviewed power theorem. Thus the content C of τw^n divides $N\tau$, independently of n . If (a, b) is the primitive output, however, its norm is

$$(842) \quad a^2 + ab + b^2 = \frac{N\tau}{C^2} (Nw)^n.$$

The residual displayed here is rational and need not be integral.

Theorem 237.2 (Bounded content does not preserve the specified exponent). *Put $\pi = 2 + \zeta$ and $w = \bar{\pi} = 3 - \zeta$. For every integer $n \geq 2$ there is a unit u_n such that, with $\tau_n = u_n\pi$, the primitive output of $\tau_n w^n$ has positive coordinates a_n, b_n and*

$$(843) \quad \begin{aligned} \gcd(a_n, b_n) &= 1, & a_n^2 + a_n b_n + b_n^2 &= 7^{n-1}, \\ \text{cont}(\tau_n w^n) &= 7, & N\tau_n &= Nw = 7. \end{aligned}$$

There are no positive integers V, R with $R > 1$ satisfying

$$a_n^2 + a_n b_n + b_n^2 = VR^n.$$

The multipliers τ_n belong to a fixed set of six elements, and the input parameter in the rational power map is always $t = -3$.

Proof. The identity $\pi\bar{\pi} = 7$ gives

$$\pi w^n = 7w^{n-1}.$$

Every power w^m , $m \geq 1$, is primitive. Indeed, its norm is 7^m , so any common rational prime divisor of its coordinates would be seven. Evaluation at $\zeta = 5$ defines a homomorphism $\mathcal{O} \rightarrow \mathbb{F}_7$, since $5^2 - 5 + 1 = 0$ modulo seven. It sends w to $-2 \neq 0$, so w^m cannot have both coordinates divisible by seven.

No boundary arm of w^m vanishes: for a primitive element this would force norm one. A unit sends it into the open positive sector. To see all sign cases explicitly, multiplication by ζ and ζ^2 sends (A, B) to $(-B, A + B)$ and $(-A - B, A)$. If the signs are opposite, negate first if needed to have $A > 0 > B$; the two transformations work respectively when $A + B > 0$ and when $A + B < 0$. Equal signs require only the identity or negation. Choose u_n for w^{n-1} in this way. Units preserve content and norm, proving (843).

If $VR^n = 7^{n-1}$ with $R > 1$, every prime divisor of R is seven. Thus $R = 7^j$ for some $j \geq 1$, and the valuation of VR^n at seven is at least n , a contradiction. Finally the root $w = 3 - \zeta$ has rational parameter $3/(-1) = -3$. Replacing it by $t + \zeta = -w$ only scales the output by $(-1)^n$, so the projective output is unchanged. $\square$

An explicit positive instance is $n = 5$, $u_5 = \zeta^2$. Then

$$\begin{aligned} \tau_5 &= -3 + 2\zeta, & w^4 &= 39 - 55\zeta, \\ \tau_5 w^5 &= 112 + 273\zeta = 7(16 + 39\zeta), & 16^2 + 16 \cdot 39 + 39^2 &= 2401 = 7^4. \end{aligned}$$

In this family the canonical n -free residual of the reduced norm is 7^{n-1} and its canonical n -th-power root is one. Thus its normalized logarithmic residual tends to $\log 7$, although the given multiplier norm has normalized logarithm $\log 7/n \rightarrow 0$. The bounded cancellation leaves the rational residual $1/7$ in (842); integral normalization then changes the residual exponent by almost n .

This excludes only an automatic single-map inverse with the same prescribed exponent and an integral nonunit root. The evident alternative exponent $n - 1$ remains available. The family is not asserted to meet the second map of the RD fiber product, and does not refute the forward DC lift, simultaneous compatibility, signed compensation, or *abc*. The statements here are ordinary proofs; no Lean verification is claimed.

238. AN INTEGRAL INVERSE GATE AND THE RESIDUAL COST OF CANCELLATION

The rational models of Section 232 require an integral inverse condition. We give both a sufficient local domain and a sharp cost of cancellation for any residual at the prescribed exponent. Neither supplies a point-height upper bound.

Let $\mathcal{O} = \mathbb{Z}[\zeta]$, $\zeta^2 - \zeta + 1 = 0$ and $\gamma = 1 + \zeta$. For an integer $n \geq 2$, suppose

$$(844) \quad \tau = u\gamma^e v, \quad e \in \{0, 1\}, \quad V = Nv, \quad w \text{ primitive unramified}, \quad R = Nw > 1,$$

where u is a unit and v is primitive oriented unramified. Then τ is primitive, $N\tau = 3^e V$ and $3 \nmid VR$. Put

$$(845) \quad z = \tau w^n = A + B\zeta, \quad C = \gcd(|A|, |B|), \quad M_{\text{red}} = N(z/C) = \frac{3^e V R^n}{C^2}.$$

Here $C > 0$. The reduced pair is primitive, without a positivity assumption in the following single-profile statements.

Lemma 238.1 (The exact opposite-orientation content). *The integer C is prime to three and divides V . Its prime divisors are exactly the split rational primes where τ and w have opposite orientations. At such a prime p , put $a_p = v_p(V) > 0$ and $f_p = v_p(R) > 0$. Then*

$$(846) \quad v_p(C) = \min(a_p, n f_p), \quad v_p(M_{\text{red}}) = |a_p - n f_p|.$$

A shared prime with the same orientation contributes no content.

Proof. Primitivity in the Eisenstein UFD means that at a split rational prime each input contains at most one orientation. No inert prime occurs in either primitive norm. At the ramified prime γ , the root is a unit and the multiplier has depth $e \leq 1$. Rational divisibility by three would require depth at least two, because $\gamma^2 = 3\zeta$. Thus $3 \nmid C$.

At a split p , divisibility by the rational prime is divisibility by both conjugate Eisenstein primes. Equal orientations retain only one factor. Opposite orientations give depths a_p and $n f_p$, so the rational content has their minimum as depth. After division, the norm exponent is their absolute difference. This proves (846) and the support assertion. Finally $\min(a_p, n f_p) \leq a_p$, so $C \mid V$. $\square$

The more general content bound of Lemma 237.1 follows from an adjugate and Bezout argument. The exact minimum above uses primitive oriented factors; primitivity of τ is material to the next result.

Theorem 238.2 (A bill for any output residual). *Suppose the reduced norm has any representation*

$$(847) \quad M_{\text{red}} = 3^e V'(R')^n, \quad V', R' \in \mathbb{Z}_{>0}.$$

The root R' may differ from R and may be one. Then

$$(848) \quad \text{rad}(C)^n \mid VV',$$

where $\text{rad}(1) = 1$. In particular $C > 1$ implies

$$(849) \quad \frac{\log V + \log V'}{n} \geq \log 7.$$

Proof. Every prime $p \mid C$ is split, so $p \geq 7$. If $a_p \geq n$, V alone contains p^n . If $1 \leq a_p < n$, then $n f_p \geq n > a_p$, and (846) gives $v_p(M_{\text{red}}) = n f_p - a_p$. The nonnegative valuation $v_p(V')$ in (847) is congruent to $-a_p$ modulo n , so it is at least $n - a_p$. Thus in both cases

$$v_p(V) + v_p(V') \geq n.$$

Multiplication over the distinct primes dividing C proves (848). Its logarithm gives $(\log V + \log V')/n \geq \log \text{rad}(C)$, which is at least $\log 7$ when $C > 1$. $\square$

A canonical n -free representation (847) always exists by taking the norm exponents modulo n away from three. The exponent at three remains exactly e by (845). Thus the theorem bounds the smallest possible output residual and continues to hold when the numerical root is changed. Its threshold is sharp: the family of Theorem 237.2 has

$$C = 7, \quad V = 7, \quad V' = 7^{n-1},$$

so equality holds in (848) and (849). This is a single-map example, not a rational point asserted to lie on the full double fiber product.

Theorem 238.3 (An explicit integral inverse domain). *Fix integers $h, g \geq 2$ and $\tau_0 = u_0 \gamma^e v_0$, $\tau_1 = u_1 v_1$ as in Section 232, with primitive oriented unramified v_i and $V_i = N v_i$. Let (t_0, t_1) be a rational point of the full fiber product, with*

$$(850) \quad x = L_{\tau_0, h}(t_0) > 0 \text{ finite}, \quad L_{\tau_1, g}(t_1) = \frac{x}{x^2 + x + 1}.$$

Choose primitive integral homogeneous coordinates for t_i and write $w_i = r_i + s_i \zeta$. Assume these roots are unramified nonunits. Put

$$R = N w_0, \quad Q = N w_1, \quad C_0 = \text{cont}(\tau_0 w_0^h), \quad C_1 = \text{cont}(\tau_1 w_1^g).$$

If $x = a/b$ in lowest positive terms, then

$$(851) \quad a^2 + ab + b^2 = \frac{3^e V_0 R^h}{C_0^2}, \quad F(a, b) = \frac{V_1 Q^g}{C_1^2}.$$

In particular $C_0 = C_1 = 1$ gives actual oriented profiles with the original residuals and exponents, up to unit signs. More generally, numerical profiles with the same R, Q, h, g and positive integral residuals at most V_0, V_1 exist exactly when

$$(852) \quad C_0^2 \mid V_0, \quad C_1^2 \mid V_1.$$

Proof. The first output has ratio a/b . Dividing its coordinates by their content and changing the common sign gives exactly (a, b) . Taking norms proves the first equation of (851). Put $M = a^2 + ab + b^2$. Since $\gcd(ab, M) = 1$, the second output, whose ratio is $ab/M > 0$, reduces in the same way to (ab, M) . Its norm is $F(a, b)$, proving the other equation. All actual denominators are nonzero since $a, b, M > 0$. A coordinate $s_i = 0$ would make the primitive root a unit, already excluded here.

When both contents are one these are the actual oriented factorizations, with at most sign changes to their units. For general content, (851) determines the residuals with the same numerical root norms uniquely as V_i/C_i^2 . They are positive integers precisely under (852), and then are at most V_i . The numerical profiles have actual oriented liftings by the same UFD argument as Lemma 231.1. The original root orientations need not be preserved after division. Nor does failure of (852) exclude a profile with a different numerical root norm. $\square$

The finite local condition in a low-residual domain. On either side, write n for its exponent, V for the initial residual, and V' for any desired output residual at that exponent. If $(\log V + \log V')/n < \log 7$, Theorem 238.2 forces content one, even if the numerical root is changed.

Content one has a finite local description. For each $p \mid V$, the primitive multiplier is divisible by one split prime factor. Its product with w^n has no content at p exactly when w is not divisible by the opposite factor. In primitive coordinates $[r : s]$, this excludes the single point $r/s = -\theta$ of $\mathbb{P}^1(\mathbb{F}_p)$, where θ is the image of ζ for the opposite factor. A norm-zero primitive residue has $s \neq 0$, so no point at infinity is omitted. No other prime gives a content condition; unramifiedness of the root at three is the separate hypothesis already stated.

These exclusions on both sides suffice to reconstruct the original integral residuals V_0, V_1 . They do not guarantee an arbitrarily specified smaller output-residual budget. For such a smaller target, the assertion is the necessary exclusion of nonzero content under the joint input/output bound. The statement concerns the particular rational point and its root coordinates, not uniqueness of every profile of the resulting seed.

The bill (848) and the integral inverse domain do not prove the existence of suitable points or bound their heights. They identify the arithmetic domain on which that remaining question

must be addressed. These are ordinary proofs; no full Lean formalization of the oriented or geometric assertions is claimed.

239. EXACT CANONICAL RESIDUAL REFLECTION AFTER PROJECTIVE CANCELLATION

The general residual bill admits an exact formula when the initial residual is power-free. Work in $\mathcal{O} = \mathbb{Z}[\zeta]$, where $\zeta^2 - \zeta + 1 = 0$, and put $\gamma = 1 + \zeta$. Fix an integer $n \geq 2$ and

$$\tau = u\gamma^e v, \quad e \in \{0, 1\}, \quad V = N(v),$$

where u is a unit and v is primitive and unramified. Let w be primitive and unramified with $R = N(w) > 1$. In this section assume that V is n -free: $0 \leq v_p(V) < n$ for every rational prime p . The case $V = 1$ is included, and V and R need not be coprime. Write

$$C = \gcd(|\text{coord}_1(\tau w^n)|, |\text{coord}_2(\tau w^n)|) > 0, \quad D = \text{rad}(C).$$

Primitive reduction gives the actual integer norm

$$(853) \quad M = N(\tau w^n / C) = 3^e V R^n / C^2.$$

Theorem 239.1 (Exact canonical pair). *There are unique positive integers $V_{\text{can}}, R_{\text{can}}$ with V_{can} n -free, $3 \nmid V_{\text{can}} R_{\text{can}}$, and $M = 3^e V_{\text{can}} R_{\text{can}}^n$. They satisfy*

$$(854) \quad C \mid V, \quad \gcd(C, V/C) = 1, \quad D \mid R,$$

$$(855) \quad V_{\text{can}} = \frac{V D^n}{C^2}, \quad R_{\text{can}} = \frac{R}{D}.$$

In particular, each canceled prime removes exactly one layer from the canonical root norm, independently of its content depth.

Proof. Primitive unramified norm supports contain only split rational primes $p \equiv 1 \pmod{3}$, hence $p \geq 7$. The exponent of γ in τw^n is exactly $e \leq 1$, so no rational factor 3 is removed. At a prime common to V and R , set $a = v_p(V)$ and $f = v_p(R)$. Each primitive factor chooses only one of the two prime elements over p . If those choices agree, there is no rational p -content and the reduced norm has depth $a + nf$.

If the choices are opposite, the rational content depth is $\min(a, nf)$. Here $1 \leq a < n \leq nf$, so this depth is exactly a , and the reduced output depth is

$$(856) \quad nf - a = (n - a) + n(f - 1), \quad 1 \leq n - a < n.$$

Thus a canceled prime contributes its complete factor p^a from V to C , contributes p to D , changes the canonical residual depth from a to $n - a$, and changes the root depth from f to $f - 1$. Every other prime keeps its residual and root depths, allowing zero depths when it appears in only one input norm. This proves (854) and the primewise formulas (855). In particular, V/C and D^n/C are coprime integers whose positive prime exponents are less than n ; their product $V D^n / C^2$ is therefore an n -free integer. Together with (853) this proves existence and uniqueness of the canonical pair, with the first ramified contribution 3^e unchanged. $\square$

The canonical root may be one. It is a nonunit precisely when $R > D$; if $R = D$, every root prime has depth one and every root orientation is canceled against its opposite residual orientation. The theorem is a statement about the norm and the actual reduced output. It does not assert that an oriented algebraic root is divided by the rational integer D inside $\mathcal{O}$.

Corollary 239.2 (Exact residual bill). *Under the same hypotheses,*

$$(857) \quad V V_{\text{can}} = (V/C)^2 D^n, \quad \frac{\log V + \log V_{\text{can}}}{n} = \log D + \frac{2}{n} \log(V/C).$$

If $C > 1$, this logarithmic bill is at least $\log D \geq \log 7$. Equality with $\log D$ holds exactly when $V = C$, and equality with $\log 7$ holds exactly when $D = 7$ and $V = C$. Every other positive integer decomposition $M = 3^e V'(R')^n$ has

$$(858) \quad V' = V_{\text{can}} S^n, \quad R_{\text{can}} = S R'$$

for a positive integer S .

Proof. The identity and its logarithm follow by substituting (855). The equality criteria use $V/C \geq 1$ and, when $C > 1$, the fact that all its primes are at least seven. For the final assertion, each prime exponent in V' has remainder equal to its exponent in V_{can} modulo n . The quotient is a nonnegative integer. Multiplying the corresponding prime powers gives S ; the equality of norms then gives the root identity in (858). Thus a noncanonical residual can only increase the canonical bill. $\square$

The actual norm-seven family gives equality for every $n \geq 2$. With $\pi = 2 + \zeta$ and $w = \bar{\pi} = 3 - \zeta$, choose a unit u_n so that $u_n w^{n-1}$ has positive coordinates. The identity

$$(u_n \pi) w^n = 7 u_n w^{n-1}$$

has $V = R = C = D = 7$, hence $R_{\text{can}} = 1$ and $V_{\text{can}} = 7^{n-1}$. The previously proved primitivity of every w^m makes this the actual content and the actual reduced norm. Its bill is exactly $\log 7$. It remains a single-map inverse example and does not assert a point on a prescribed simultaneous double fiber product.

Proposition 239.3 (The power-free premise cannot be omitted). *The exact canonical-pair formula need not hold for a primitive unramified multiplier whose norm is not n -free.*

Proof. Keep $\pi = 2 + \zeta$ and let $n \geq 3$. First take $\tau = \pi^{n+1}$ and $w = \bar{\pi}$. Then

$$\tau w^n = 7^n \pi, \quad V = 7^{n+1}, \quad R = 7, \quad C = 7^n.$$

Here C no longer contains the complete prime-power factor of V , so $\gcd(C, V/C) = 1$ already fails, although the displayed canonical pair formulas happen to give the correct pair $(7, 1)$.

For an actual failure of the formulas themselves, take instead $\tau = \pi^{n+1}$ and $w = \bar{\pi}^2$. Primitivity and the oriented prime factorization give

$$V = 7^{n+1}, \quad R = 7^2, \quad C = 7^{n+1}, \quad N(\tau w^n / C) = 7^{n-1}.$$

The canonical pair is $(7^{n-1}, 1)$, whereas $R/D = 7$ and $VD^n/C^2 = 1/7$. All ring inputs remain primitive and unramified; the failed hypothesis is precisely the n -freeness of V . $\square$

This exact reflection result determines cancellation for supplied actual inputs and their orientations. It does not construct rational points on the simultaneous inverse curves, provide upper bounds for their heights, or settle arbitrary-root signed-tail membership. The general residual-divisibility bill continues to apply outside this power-free subdomain. No Lean formalization of the exact canonical pair formula is claimed here.

240. AN INTEGRAL CONVERSE FOR THE PURE-COEFFICIENT CURVES

For unit residuals, every rational point above a positive finite base coordinate has a genuine integral interpretation, including inputs at the ramified prime or at infinity. Retain $\gamma = 1 + \zeta$, $\gamma^2 = 3\zeta$, and integers $h, g \geq 2$. Fix units u_0, u_1 and $e_0 \in \{0, 1\}$, and set

$$\tau_0 = u_0 \gamma^{e_0}, \quad \tau_1 = u_1.$$

Consider the full rational fiber product of Section 232,

$$(859) \quad L_{\tau_1, g}(t_1) = r(L_{\tau_0, h}(t_0)), \quad r(x) = \frac{x}{x^2 + x + 1}.$$

Theorem 240.1 (Every positive point gives actual pure profiles). *For every rational point (t_0, t_1) of (859) with $x = L_{\tau_0, h}(t_0) > 0$ finite, write $x = a/b$ in lowest positive terms. There are $e \in \{0, 1\}$ and integers $R, Q \geq 7$ such that*

$$(860) \quad a^2 + ab + b^2 = 3^e R^h, \quad F(a, b) = Q^g.$$

Both are actual oriented profiles with unit residuals. No unramifiedness or nonunit assumption is imposed on the input coordinates. If h is even, $e = e_0$; for odd h the first ramified parity may change.

Proof. Represent each rational projective input by a primitive integer pair (r_i, s_i) and put $w_i = r_i + s_i\zeta$. This includes $s_i = 0$ and the point at infinity. The element w_i is nonzero. Its γ -depth ϵ_i is zero or one: depth at least two would imply that the rational prime three divides both coordinates. Write

$$w_i = \gamma^{\epsilon_i} v_i, \quad \epsilon_i \in \{0, 1\}, \quad v_i \text{ primitive unramified.}$$

Dividing by γ cannot introduce a common rational factor into the coordinates, since that factor would then divide w_i . The oriented UFD factorization shows that every power of v_i is primitive and unramified.

For either output set $E = e_{\text{initial}} + \epsilon_i n$, where the initial exponent is e_0 on the first side and zero on the second. The exact identity

$$(861) \quad u\gamma^E v_i^n = 3^{\lfloor E/2 \rfloor} u\zeta^{\lfloor E/2 \rfloor} \gamma^{E \bmod 2} v_i^n$$

has a primitive final factor: it has only one split orientation at every prime and γ -depth at most one. Thus the full content is exactly $3^{\lfloor E/2 \rfloor}$, and the reduced norm is

$$(862) \quad 3^{E \bmod 2} (N v_i)^n.$$

Put $M = a^2 + ab + b^2$. The first primitive output, after changing the common sign if necessary, is exactly (a, b) . Therefore

$$M = 3^e R^h, \quad R = N v_0, \quad e = (e_0 + \epsilon_0 h) \bmod 2.$$

The second primitive output has ratio ab/M , with $\gcd(ab, M) = 1$, so its positive primitive coordinates are (ab, M) . Consequently

$$F(a, b) = 3^{\epsilon_1 g \bmod 2} Q^g, \quad Q = N v_1.$$

Since $3 \nmid F$, the displayed parity is zero. Also $F(a, b) \geq 13$, so $Q > 1$. A nonunit primitive unramified norm is an integer at least seven.

If $R = 1$, then $M = 3^e \leq 3$. Positive a, b give $M \geq 3$, with equality only at $a = b = 1$. This would imply $13 = Q^g$, impossible for $g \geq 2$ and $Q > 1$ because thirteen is prime. Thus $R > 1$ and $R \geq 7$ as well. The primitive factorizations remaining in (861) give the actual oriented identities after the unit sign adjustments. All projective input cases have been included. The formula for e proves the stated parity distinction. $\square$

Conversely, every actual positive pure-profile seed gives a rational point of one of these curves by Sections 231 and 232. Hence, after allowing the finite unit choices and both initial ramified parities, their positive rational points correspond exactly to actual pure-profile seeds. This asserts neither uniqueness of the point nor uniqueness of factorization.

Corollary 240.2 (The even/even positive rational locus is empty). *If both h and g are even, no curve (859) has a rational point whose base coordinate x is positive and finite.*

Proof. Such a point would give (860). Thus $MF/3^e$ would be a rational square. The exact combined-norm obstruction of Theorem 186.4 says that neither MF nor $MF/3$ is a rational square for a positive primitive seed. This is a contradiction. $\square$

Only the positive rational base locus is excluded. Points over other fields, negative coordinates and boundary base values are not ruled out by this argument. The remaining exponent pairs stay open unless a separate proved obstruction applies. The nonunit-multiplier counterfamily of Section 237 is consistent with this converse: here the only cancellation is the ramified power of three accounted for in (861). No general point-height upper bound follows. These statements are ordinary results; the complete geometric and oriented converse is not claimed as a Lean proof.

241. AN ACTUAL GAUSSIAN PROFILE AT AN EVEN FIRST EXPONENT

Let a, b be positive coprime integers, and write

$$c = a + b, \quad A = ab, \quad M = a^2 + ab + b^2 = c^2 - A, \quad F = a^4 + 3a^3b + 5a^2b^2 + 3ab^3 + b^4.$$

The elementary primitive identities give

(863)

$$F = M^2 + Ac^2 = A^2 + Mc^2, \quad \gcd(A, c) = \gcd(A, M) = \gcd(M, c) = 1, \quad \gcd(F, AMc) = 1.$$

Also F is odd, prime to 3, and at least 13. Every prime divisor q of F is 1 modulo 3: the nonzero residue M/A satisfies $X^2 + X + 1 = 0$ and has exact order 3, since $q \neq 3$.

Theorem 241.1 (Quadratic support condition). *Suppose $M = DU^2$ with positive integers D, U , without requiring D to be squarefree. Every prime $q \mid F$ satisfies*

$$(864) \quad q \nmid DUC, \quad \left(\frac{-D}{q} \right) = 1.$$

Thus an actual first profile $M = 3^e V_0 R^h$, where $e \in \{0, 1\}$ and $h \geq 2$ is even, imposes on every prime divisor q of the whole second norm, and hence on every prime divisor of any second root Q , the conditions

$$(865) \quad e = 0 : \quad \left(\frac{-V_0}{q} \right) = 1; \quad e = 1 : \quad \left(\frac{V_0}{q} \right) = 1.$$

No equality of the two exponents or power-free residual is assumed.

Proof. Equation (863) becomes $F = A^2 + D(Uc)^2$, with $\gcd(A, Uc) = 1$. At a prime divisor q of F , the quotient $A/(Uc)$ is nonzero and its square is $-D$. Set $D = 3^e V_0$ and use $(-3/q) = 1$, which follows from $q \equiv 1 \pmod{3}$, to obtain (865). $\square$

The corresponding quadratic field has discriminant dividing $4D$; the logarithm of its absolute value is at most $\log 12 + \log V_0$. This gives no uniform prime distribution estimate while the residual, and therefore the field, moves.

Theorem 241.2 (An actual Gaussian decomposition). *Assume $M = U^2$ for a positive integer U . Then*

$$(866) \quad z_G = A + iUc \in \mathbb{Z}[i], \quad N(z_G) = F, \quad \gcd(A, Uc) = 1.$$

The Gaussian integer is unramified at 2, and every prime $q \mid F$ satisfies $q \equiv 1 \pmod{12}$. For any integers $g \geq 2$, $V_1 > 0$, $Q > 1$ with $F = V_1 Q^g$, there are primitive Gaussian integers v_G, w_G and a unit $u_G \in \{1, -1, i, -i\}$ such that

$$(867) \quad z_G = u_G v_G w_G^g, \quad N(v_G) = V_1, \quad N(w_G) = Q.$$

Neither factor contains a pair of conjugate primes or a prime over 2. Their supports may overlap; V_1 need not be g -free.

Proof. Primitivity and the norm identity follow from (863); F odd excludes the prime over 2. The Gaussian Euclidean algorithm follows by choosing a nearest Gaussian integer to a complex quotient, leaving a remainder of norm at most half the divisor norm. Thus $\mathbb{Z}[i]$ is a UFD. An inert rational prime cannot divide a primitive Gaussian integer, and the two conjugate primes over a split rational prime cannot both divide it. Hence every $q \mid F$ is 1 modulo 4, and also 1 modulo 3 as above.

For its uniquely occurring orientation π_q , the exponent in z_G is exactly $v_q(F) = v_q(V_1) + g v_q(Q)$. Assign these two nonnegative parts to v_G and w_G . Equality is then up to a Gaussian unit, proving the decomposition and its stated primitive properties. $\square$

If $V_1 = 1$, this is the pure power $z_G = u_G w_G^g$. For odd g , the unit can be absorbed into w_G , since the g -power map permutes the four Gaussian units. The unit is retained for a general exponent.

At the same time the actual second Eisenstein element has the reviewed oriented decomposition

$$(868) \quad z_E = A + U^2\zeta = u_E v_E w_E^g, \quad N(v_E) = V_1, \quad N(w_E) = Q.$$

Thus the two powers in different quadratic UFDs share the rational coordinate A , with the further constraint

$$(869) \quad (Uc)^2 = U^2(A + U^2).$$

Independently chosen powers need not satisfy either this constraint or equality of their rational coordinates.

Corollary 241.3 (Height of the actual Gaussian coefficient). *With the data of (867), put $\eta_G = (u_G^2 v_G / \bar{v}_G)^{-1}$. Then*

$$\left(\frac{w_G}{\bar{w}_G}\right)^g = \eta_G \frac{z_G}{\bar{z}_G}, \quad h(\eta_G) = \frac{1}{2} \log V_1,$$

where h is absolute logarithmic Weil height.

Proof. Conjugating the decomposition proves the identity. There is no conjugate-prime cancellation in $v_G/\bar{v}_G$, whose denominator ideal has norm V_1 and whose complex absolute value is one. The normalized height is therefore $\frac{1}{2} \log V_1$; roots of unity and inversion leave it unchanged. $\square$

This is a coefficient-height statement for actual points, with no point-height upper bound or independence assertion for the two covers.

Theorem 241.4 (Cyclotomic equation and exact integer inverse). *If $M = U^2$ for a positive integer U , then, with $d = a - b$,*

$$(870) \quad F = U^4 - U^2 c^2 + c^4 = \Phi_{12}(U, c), \quad d^2 = 4U^2 - 3c^2, \quad |d| < c,$$

and

$$\gcd(U, c) = 1, \quad \frac{\sqrt{3}}{2} < \frac{U}{c} < 1.$$

Conversely, let U, c be positive coprime integers with $U < c$ and let $d \in \mathbb{Z}$ satisfy $d^2 = 4U^2 - 3c^2$ and $|d| < c$. Then

$$(871) \quad a = (c + d)/2, \quad b = (c - d)/2$$

are positive coprime integers with $a^2 + ab + b^2 = U^2$, and their second norm is $\Phi_{12}(U, c)$.

Proof. Use $A = c^2 - U^2$ and $d^2 = c^2 - 4A$ in (863). The upper interval bound follows from $A > 0$. Equality in the lower bound would give $a = b$, hence the primitive seed $(1, 1)$ with nonsquare first norm 3; it cannot occur here.

For the inverse, $d^2 \equiv c^2 \pmod{4}$ gives matching parity, and $|d| < c$ gives positivity. Substitution proves $a^2 + ab + b^2 = U^2$ and $ab = c^2 - U^2$. An odd prime dividing both a, b would divide c, d and therefore $4U^2$, contrary to $\gcd(U, c) = 1$. If both a, b are even, their norm identity forces U even, while c is even, the same contradiction. This proves coprimality and all the claimed identities. $\square$

The extra square condition in (870) is part of the inverse. No identification with arbitrary rational points of $y^g = \Phi_{12}(x, 1)$ is made. To retain a specified larger even first exponent h , one must additionally impose $U = R^{h/2}$ with an integer $R > 1$. The ramified branch $M = 3U^2$ instead gives $F = 9U^4 - 3U^2 c^2 + c^4$ and has not been replaced by the same integral cyclotomic equation.

Remaining scope. These statements apply through the pure-coefficient inverse theorem to every finite positive rational point whose reconstructed first norm is an unramified even power. The even/even exclusion is the earlier combined-square result. The odd second exponent remains unresolved. For a pure second prime exponent $p > 7$, the earlier modular support theorem gives, for every prime $q \mid Q$, $q \geq (\sqrt{2p} - 1)^2 > p$, now intersected with $q \equiv 1 \pmod{12}$. This is a necessary support condition, not control of the large-prime tail. The ramified first branch and moving nonunit residuals remain open. The proofs here are ordinary arithmetic proofs, independently reviewed; no complete geometric or modular Lean formalization is claimed.

242. A SIMULTANEOUS COVER FOR TWO ACTUAL QUADRATIC-DOMAIN POWERS

Work in the fixed degree-four field $K = \mathbb{Q}(i, \sqrt{-3})$. Write

$$\zeta = (1 + \sqrt{-3})/2, \quad \alpha = (\sqrt{3} - i)/2, \quad \bar{\alpha} = (\sqrt{3} + i)/2, \quad \alpha^2 = \bar{\zeta}.$$

Complex conjugation acts on coefficients and leaves the variable t fixed. Put

$$(872) \quad a_0 = t^2 - 1, \quad b_0 = 2t + 1, \quad c_0 = t^2 + 2t, \quad U_0 = t^2 + t + 1, \quad A_0 = a_0 b_0,$$

$$(873) \quad E = A_0 + \zeta U_0^2, \quad G = A_0 + i U_0 c_0.$$

Then $c_0^2 - A_0 = U_0^2$ and $E\bar{E} = G\bar{G} = F(a_0, b_0)$.

Lemma 242.1 (The full branch inventory). *Set*

$$B_1 = c_0 - \alpha U_0, \quad B_2 = c_0 + \alpha U_0, \quad B_3 = c_0 - \bar{\alpha} U_0, \quad B_4 = c_0 + \bar{\alpha} U_0.$$

These four quadratics have eight distinct simple roots, none at infinity, and

$$(874) \quad E = B_1 B_2, \quad \bar{E} = B_3 B_4, \quad G = B_1 B_4, \quad \bar{G} = B_2 B_3.$$

Proof. The factorizations use $A_0 = c_0^2 - U_0^2$, $1 - \zeta = \bar{\zeta}$, $\bar{\alpha} - \alpha = i$ and $\alpha\bar{\alpha} = 1$. Each quadratic is $c_0 - sU_0$, with $s \in \{\alpha, -\alpha, \bar{\alpha}, -\bar{\alpha}\}$. These four constants are distinct and different from one. Its leading coefficient is $1 - s$, and its discriminant is

$$(2 - s)^2 + 4s(1 - s) = 4 - 3s^2 \neq 0.$$

Here s^2 is ζ or $\bar{\zeta}$, and the displayed discriminant has norm 13 from $\mathbb{Q}(\sqrt{-3})$. A root shared by two quadratics would force $c_0 = U_0 = 0$. But $U_0 - c_0 = 1 - t$ and $U_0(1) = 3$. There is no common finite root, and the nonzero leading coefficients exclude infinity. $\square$

Theorem 242.2 (Exact component count and genus). *For an integer $g \geq 2$ and constants $\eta_E, \eta_G \in K^*$, consider*

$$(875) \quad Y^g = \eta_E \frac{E}{\bar{E}}, \quad Z^g = \eta_G \frac{G}{\bar{G}}.$$

Take the smooth projective normalizations of these equations. Put $\delta = \gcd(g, 2)$. There are exactly δ geometric components. Each has degree g^2/δ over $\mathbb{P}_t^1$ and genus

$$(876) \quad 1 + \frac{3g^2 - 4g}{\delta}.$$

Thus the odd-exponent curve is geometrically connected. At an even exponent there are two geometric components; their degrees are $g^2/2$, not g^2 .

Proof. Over an algebraic closure, the two functions in (875) have, on the two roots of each successive B_j , the valuation vectors

$$(877) \quad (1, 1), \quad (1, -1), \quad (-1, -1), \quad (-1, 1).$$

A relation with exponents u, v modulo g that is a g th power must satisfy $u + v \equiv u - v \equiv 0 \pmod{g}$. This gives one relation when g is odd and two when g is even. They are sufficient: the nontrivial even relation is $(u, v) = (g/2, g/2)$ and uses

$$\frac{E}{\bar{E}} \frac{G}{\bar{G}} = \left(\frac{B_1}{B_3} \right)^2.$$

Constants have g th roots over the algebraic closure. Kummer theory therefore gives subgroup order, and function-field degree, g^2/δ . The two equations have total generic degree g^2 and split into exactly δ geometric components of that degree.

At each of the eight points of Lemma 242.1, both functions have valuation 1 or -1 . After taking g th roots of local units, the local extension is generated by one g th root of a uniformizer. Its inertia order on each component is exactly g . At infinity both functions have order zero and are local units, and there is no ramification there or at any other point. Set $d = g^2/\delta$. Characteristic-zero Riemann–Hurwitz gives

$$2g - 2 = -2d + 8d(1 - 1/g),$$

which proves (876). $\square$

For the pure unit constants below, the two even components are already defined over K . Indeed $\eta_E \eta_G \in \mu_6$, and its square roots lie in $\mu_{12} \subset K$. The components are distinguished by

$$(878) \quad \frac{(YZ)^{g/2}}{B_1/B_3} = \sqrt{\eta_E \eta_G} \quad \text{or} \quad -\sqrt{\eta_E \eta_G}.$$

This arithmetic component assertion is restricted to those constants; the preceding theorem asserts the geometric count for arbitrary $\eta_E, \eta_G \in K^*$.

Theorem 242.3 (Actual seeds lift to finitely many unit twists). *Let a, b be positive coprime integers with $M = U^2$ for a positive integer U and $F = Q^g$, where $Q > 1$ and $g \geq 2$ are integers. Then this seed gives a nonbranch K -rational point on one of the at most six curves (875) with $\eta_E \in \mu_3$, $\eta_G \in \mu_2$, at base coordinate*

$$(879) \quad t = (a + U)/b > 1.$$

For odd g only the three η_E twists are required. If $\gcd(g, 6) = 1$, both constants can be absorbed: every such seed lifts to the single curve with $\eta_E = \eta_G = 1$, of degree g^2 and genus $1 + 3g^2 - 4g$.

Proof. The equality $U^2 - (a - b)^2 = 3ab > 0$ gives $t > 1$. For $x = a/b$ one has $t = x + \sqrt{x^2 + x + 1}$ and $t^2 - 2xt - x - 1 = 0$. Thus $a_0/b_0 = x$ and $b_0 = 2t + 1 > 0$. With $k = b_0/b > 0$,

$$(880) \quad (a_0, b_0, c_0) = k(a, b, c), \quad U_0 = kU, \quad E = k^2(ab + U^2\zeta), \quad G = k^2(ab + iUc).$$

The sign of $U_0 = kU$ is fixed by $U_0 = t^2 + t + 1 > 0$.

The two actual primitive UFD decompositions give

$$ab + U^2\zeta = u_E w_E^g, \quad ab + iUc = u_G w_G^g, \quad N(w_E) = N(w_G) = Q.$$

Consequently $Y = w_E/\bar{w}_E$ and $Z = w_G/\bar{w}_G$ satisfy (875) with $\eta_E = u_E^{-2} \in \mu_3$ and $\eta_G = u_G^{-2} \in \mu_2$. The positive rational factor k^2 cancels from both ratios. None of $E, \bar{E}, G, \bar{G}$ vanishes at this point, since their norms are $k^4 F > 0$. Hence the point is smooth and nonbranch and lifts to the normalization.

For odd g , raising to g permutes μ_2 , so η_G can be absorbed into Z . If $3 \nmid g$, it also permutes μ_3 , so η_E can be absorbed into Y . These changes take place in the fixed field K and require no varying cyclotomic field. $\square$

Verification and remaining scope. The ordinary proof computes the exact relation module, including the even-exponent defect, before asserting a degree. Its geometric inputs are the same characteristic-zero Kummer and Riemann–Hurwitz facts used in the preceding reviewed cover constructions. The supplemental exact replay checks the four polynomial factorizations, their norm identity, four discriminants and six nonzero pairwise resultants in $\mathbb{Q}(\alpha)$, and the relation kernels at $2 \leq g \leq 120$. It also checks 52 actual square-first seeds with $1 \leq a \leq b \leq 300$, their complete trial factorizations, and 74 integer inverse instances with $c \leq 600$. These finite checks do not establish rational-point completeness.

This is a further cover to which actual points lift, not a birational replacement of the earlier double-oriented curve and not an integer inverse for arbitrary K -points. The larger genus may be useful for fixed-exponent rational-point calculations; it supplies no calculation or height bound uniform in g . The even-exponent geometric components exist despite the already proved emptiness of the positive seed locus. The odd second exponent and the ramified first branch remain open. No complete Lean formalization of these geometric assertions is claimed.

243. ACTUAL CONTENT AND THE LOGARITHMIC CANCELLATION BILL IN LEAN

The ordinary content and residual theorems have finite arithmetic cores that can be checked directly on integer coordinates and actual prime factorizations. Their formalization retains the boundary between the adjugate argument and the oriented factorization argument.

For an integer pair $z = (x, y)$ put

$$C(z) = \gcd(|x|, |y|), \quad z_{\text{red}} = (x/C(z), y/C(z)), \quad N(z) = x^2 + xy + y^2.$$

In the Lean definition, $C(z)$ is `Int.gcd` and the divisions are integer division. All reconstruction identities below also hold at the zero pair, under the library's total convention for division by zero. Primitivity and positivity assertions explicitly exclude the zero pair.

Proposition 243.1 (The actual coordinate core). *For every integer pair z ,*

$$C(z)z_{\text{red}} = z, \quad C(z)^2N(z_{\text{red}}) = N(z).$$

If $z \neq 0$, then $C(z) > 0$, $N(z) > 0$ and $C(z_{\text{red}}) = 1$. With the multiplication of $\mathbb{Z}[\zeta]$, for arbitrary τ, W ,

$$C(\tau W)^2N((\tau W)_{\text{red}}) = N(\tau)N(W).$$

If $C(W) = 1$, then $C(\tau W)$, cast to an integer, divides $N(\tau)$. If also $\tau \neq 0$, then $C(\tau W) > 0$.

Proof. The actual gcd divides both coordinates, so multiplying their integer quotients by the gcd reconstructs them, including at zero. Substitute these two identities into the quadratic norm to obtain its scaling law. Dividing a nonzero pair by its gcd makes it primitive. Positivity of the norm follows from $4N(x, y) = (2x + y)^2 + 3y^2$. Its multiplicativity gives the product norm identity. Finally the adjugate identities in Lemma 237.1, combined with the integer gcd's own Bezout coefficients for the primitive pair W , prove the stated divisibility. The product is nonzero by positivity and norm multiplicativity. $\square$

The module `ActualProjectiveContent` checks these statements in nine declarations, including the actual norm-seven example

$$(-3 + 2\zeta)(62 - 149\zeta) = 112 + 273\zeta = 7(16 + 39\zeta).$$

It checks both gcds and all displayed norms in this finite instance. The infinite family of Section 237 is an ordinary proof; it is not inferred from this closed instance.

Lemma 243.2 (A local reflected depth). *Let n, a, f be positive integers and b, r nonnegative integers. If $|a - nf| = b + nr$, then $n \leq a + b$.*

Proof. If $a \geq n$, use $b \geq 0$. Otherwise $nf \geq n > a$, so the equality becomes $a + b = n(f - r) > 0$. A positive integer multiple of the positive integer n is at least n . $\square$

For a natural number C , define the actual prime product

$$D(C) = \prod_{p \in \text{primeFactors}(C)} p.$$

It is positive, with empty product one. For positive C it is the ordinary radical; the formal total convention also assigns one at zero.

Proposition 243.3 (An actual product bill and its strict log threshold). *Let n, C, V, V', R, R' be positive integers. Suppose that at every prime $p \mid C$ one has $v_p(V) > 0$, $v_p(R) > 0$, and*

$$|v_p(V) - nv_p(R)| = v_p(V') + nv_p(R').$$

Then $D(C)^n \mid VV'$. If also every prime divisor of C is at least seven and

$$(881) \quad \log V + \log V' < n \log 7,$$

then $C = 1$.

Proof. Lemma 243.2 gives $n \leq v_p(V) + v_p(V') = v_p(VV')$. Thus $p^n \mid VV'$ at every prime in the actual support of C . Distinct prime powers are coprime; their product is $D(C)^n$, proving the bill.

Strict monotonicity of the logarithm on positive reals and the finite product and power laws give the exact equivalence

$$\log V + \log V' < n \log 7 \iff VV' < 7^n.$$

If $C > 1$, choose an actual prime divisor p of C . Its n th power divides VV' , so $7^n \leq p^n \leq VV'$, a contradiction. $\square$

The local depth premise is supplied ordinarily by the oriented calculation of Lemma 238.1. It has not been silently discharged from the integer gcd theorem in Lean. The bill is for the product of distinct prime divisors, not for C itself. The strict inequality cannot be weakened to a non-strict one: the norm-seven family has $C = 7$, $V = 7$, $V' = 7^{n-1}$ and equality. Connecting the global radical to a positive integer. With the actual prime-logarithm definitions of Section 226, the finite product formula proves, for every natural C ,

$$R_0(C) = \log D(C).$$

Consequently the identity and height transfer of Section 235 become

$$S_0(N) = \log N - 3 \log D(N), \quad t - (\delta + b + l)/3 \leq \log D(N),$$

with exactly the same explicit height, signed-tail and small-mass hypotheses. This verifies the passage from finite prime sums to the usual integer radical without adding any analytic premise. Exact formal scope. The modules `ResidualReflectionArithmetic` and `ActualResidualLogThreshold` contain seven and six proved declarations respectively. They use integer absolute values, `Nat.factorization`, actual finite prime products and `Real.log`. Their source-level audit checks the explicit local reflection premise, positivity, the prime lower bound and strictness of the threshold. The total natural-number statements occasionally allow zero where the mathematical application uses positive inputs; this does not omit any required positive-domain premise.

Together with the nine content declarations and the nine declarations in `ActualGlobalSignedBridge`, the joint fresh run checks 31 new declarations and 53 unchanged local dependency declarations. It uses Lean 4.32.0, warnings as errors, and the pinned Mathlib revision 81a5d257c8e410db227a6665ed08f64fea08e997. The compiled Mathlib cache is reused. All 84 declarations have explicit axiom queries; the union is only `propext`, `Classical.choice` and `Quot.sound`. The ordinary proofs precede these implementations.

The Eisenstein UFD, general integral inverse, rational-point existence, analytic mean estimates and arbitrary-root signed bounds are not formalized by these modules. Independent review and exact finite replay complement the scoped compilation; none constitutes a proof or disproof of *abc*.

244. THE ELEVENTH CONTINUATION: INTEGRAL GATES AND DISTINCT LIVE TARGETS

The inverse problem now has an exact arithmetic boundary. An actual integer gcd divides the multiplier norm, but the norm-seven family shows why this alone does not preserve a prescribed nonunit power after primitive reduction. Opposite orientations give an exact local reflection and a residual bill. On the power-free subdomain the canonical output residual and root norm are determined explicitly. The strict joint logarithmic threshold excludes cancellation, including when the output root norm changes. These results identify a usable integral domain; they do not produce compatible points in it.

For pure coefficients the positive rational locus has a complete ordinary integral converse, with ramified and infinite inputs retained. Together with the previous combined-square obstruction, this excludes the even/even positive locus. The second quadratic-domain construction provides simultaneous actual Gaussian and Eisenstein powers when the first norm is a square. Its further biquadratic cover has eight branch points. The even-exponent relation causes two geometric components; the odd-exponent cover is geometrically connected. Neither its larger genus nor its finite unit-twist list bounds the actual point heights uniformly in the exponent. The odd branch remains a distinct live arithmetic-geometric problem, with ramified and moving residual branches also retained.

On the analytic route, the common large-cutoff set is independent of the moving reciprocal caps. The finite signed estimate now transfers to the global integer radical with explicit constants. A separate prime-index theorem proves joint squarefreeness throughout a growing prime interval beyond the root block. However, that interval's typical full mass is sublinear; almost all full mass still lies in the farther top-rank packet. Thus actual depth-one membership on the proved window does not supply the missing signed bound on that packet. Controlling deep excess

together with its shallow-prime credit there, and treating all exceptional roots, remains the primary analytic target.

The new joint formal inventory is 31 proved declarations: nine actual content statements, seven residual-reflection arithmetic statements, six integer-radical and logarithmic-threshold statements, and nine global signed-bridge statements. The fresh run also compiles 53 unchanged local dependency declarations and audits all 84 axiom queries against the pinned Lean and Mathlib environment. It reuses the Mathlib compiled cache. One exact replay checks 24,192 coordinate identities and 128 members of the norm-seven family. The other checks the biquadratic polynomial identities, 119 finite relation kernels, 52 actual square-first seeds with complete elementary factorizations, and 74 integer inverse cases. Those checks do not prove an infinite or analytic theorem, or determine the full rational-point locus.

The complete oriented factorization, integral inverses, canonical reflection, sieve estimates and geometric constructions have ordinary proofs with independent internal review. They have not all been formalized. In particular the formal local reflection premise is explicit, and no uniform signed estimate or point-height bound is introduced as an axiom. There is still no complete proof or disproof of abc . Only the specific inverse assertions contradicted by the actual examples are excluded; all unrefuted parent routes remain available for further research.

245. ACTUAL DEEP ROOTS AND MULTIPLICATIVE PHASE COHERENCE

Let $n \geq 7$ be prime, $B = n^4$, and let k range over the integers $B \leq k < 2B$. Put $a_k = 3k$, $w_k = a_k + \zeta$, $Q_k = a_k^2 + a_k + 1$, $T_n(k) = |P(w_k^n)|$, $t_k = \log c_n(k)$, $Y = 6n^3$. The elements and their powers are primitive and unramified. Here $P(A + B\zeta) = AB(A + B)$, and $c_n(k)$ is the largest side after positive-sector normalization. Thus $n < Y < B$. In the pair arguments, a, b, c, d denote root parameters a_k , not the three entries of the resulting ABC triple.

Lemma 245.1 (A finite group at actual fourth depth). *For a prime $q > Y$, put*

$$\mathcal{A}_q = \{k : B \leq k < 2B, q^4 \mid T_n(k)\}.$$

For each such root, Q_k is a unit modulo q , and

$$\alpha_k = \frac{a_k + \zeta}{a_k + \bar{\zeta}} \text{ belongs to } H_q = \{x \in ((\mathbb{Z}/q^4\mathbb{Z})[\zeta])^\times : x\bar{x} = 1, x^{3n} = 1\}.$$

One has $|H_q| \leq 3n$.

Proof. The exact identity

$$z^3 - \bar{z}^3 = 3\sqrt{-3}P(z)$$

has a unit scalar at $q > 3$. If $q \mid Q_k$, in the inert case the reduction of w_k would be zero, contrary to its second coordinate being one. In the split case exactly one component is zero and the other is not, so the cubes of w_k^n and $\bar{w}_k^n$ could not coincide. Thus a boundary hit implies $q \nmid Q_k$, and the identity proves the stated group membership.

In the split case the quadratic ring is a product of two copies of $\mathbb{Z}/q^e\mathbb{Z}$, and a norm-one element is (u, u^{-1}) . Modulo q , at most $3n$ elements are killed by $3n$. Each lifts uniquely at every depth because the derivative of $X^{3n} - 1$ is a unit at a root. In the inert case the residue norm-one group is cyclic of order $q + 1$. It too has at most $3n$ such elements. Substituting $x + q^e y$ gives a unique linear lifting equation over $\mathbb{F}_{q^2}$, with invertible derivative. Reduction therefore injects H_q into a set of size at most $3n$. This counts a finite group; no distribution assumption is made for the actual roots. $\square$

Lemma 245.2 (Integer phase rigidity and positive divisor fibers). *Two ordered pairs of roots in $\mathcal{A}_q$ having the same product in H_q have the same actual rational phase:*

$$\frac{ab - 1}{a + b + 1} = \frac{cd - 1}{c + d + 1}.$$

For a fixed nonempty phase u/v , in lowest positive terms, its ordered-pair fiber in the whole block has at most $\tau(u^2 + uv + v^2)$ members, and

$$1 \leq u^2 + uv + v^2 \leq 2500B^4.$$

Proof. The actual product coordinates are

$$(a + \zeta)(b + \zeta) = (ab - 1) + (a + b + 1)\zeta.$$

Their positive coordinates satisfy $ab - 1 \leq 36B^2$ and $a + b + 1 \leq 12B$. Equality of the two ratio products modulo q^4 , after multiplication by unit denominators, implies

$$q^4 \mid D, \quad D = (ab - 1)(c + d + 1) - (cd - 1)(a + b + 1).$$

The conjugate cross difference is a unit multiple of this integer. But

$$|D| \leq 864B^3 < 1296B^3 < q^4$$

because $q > 6B^{3/4}$, so $D = 0$.

Reducing the positive phase gives $u \leq 36B^2$, $v \leq 12B$. For any pair with that phase,

$$(882) \quad (va - u)(vb - u) = u^2 + uv + v^2 =: K.$$

Both factors are positive, since $(ab - 1)/(a + b + 1) < \min(a, b)$. The first factor is a positive divisor of K , determines a , and then determines b . This is an injection of the actual ordered fiber into the positive divisors, including diagonal and exchanged pairs. Finally

$$K \leq 1296B^4 + 432B^3 + 144B^2 \leq 2500B^4.$$

□

Theorem 245.3 (A uniform count of actual deep roots). *For every $\varepsilon > 0$ there is a constant C_ε , independent of n, q , such that*

$$|\mathcal{A}_q| \leq C_\varepsilon \sqrt{n} B^{2\varepsilon}.$$

If $\mathcal{A}_q \neq \emptyset$, its prime q has exact homogeneous rank n and $q \equiv 1$ or $-1 \pmod{3n}$.

Proof. For every $m \geq 1$, the elementary divisor estimate is $\tau(m) \leq C_\varepsilon m^\varepsilon$. Indeed for $p \geq 2^{1/\varepsilon}$ and $j \geq 0$, $j + 1 \leq 2^j \leq p^{\varepsilon j}$. For each of the finitely many smaller primes, $\sup_{j \geq 0} (j + 1)p^{-\varepsilon j}$ is finite. Multiplication over prime powers proves the estimate.

All $|\mathcal{A}_q|^2$ ordered pairs map into H_q . Each occupied image has one actual phase by Lemma 245.2, and at most $C_\varepsilon(2500B^4)^\varepsilon$ pairs. There are at most $3n$ images. Taking square roots proves the bound, including the empty-set case.

The rank of $((w_k/\bar{w}_k)^3 \bmod q)$ divides the prime n . Rank one would give $v_q(T_n(k)) = v_q(T_1(k))$, since $q > Y > n$. Here $T_1(k) = a_k(a_k + 1)$ has two coprime factors, each below $6B + 1$, whereas $q^2 > 36n^6 > 6B + 1$. Rank one cannot even have depth two. The rank is therefore n . The norm-one group gives $n \mid q - \chi_q$, where $\chi_q = 1$ in the split case and -1 in the inert case. Also $3 \mid q - \chi_q$, and $\gcd(3, n) = 1$; hence the stated two progressions. □

Theorem 245.4 (A simultaneous interval of depth at most three). *For real $Z > Y$, let $\mathcal{E}(n, Z)$ be the actual set of indices with some prime $Y < q \leq Z$ of boundary depth at least four. Then*

$$\frac{|\mathcal{E}(n, Z)|}{B} \leq \frac{C_\varepsilon Z B^{2\varepsilon}}{B \sqrt{n} \log(Z/(3n))}.$$

Fix $0 < \delta < 1/2$. For all sufficiently large prime n , put $Z_\delta = \lfloor Bn^{1/2-\delta} \rfloor$. Then

$$(883) \quad \frac{|\mathcal{E}(n, Z_\delta)|}{B} = O_\delta \left(\frac{n^{-\delta/2}}{\log n} \right).$$

Outside this one set, every supported prime in the entire interval $(Y, Z_\delta]$ has depth at most three.

Proof. The two-progression Brun–Titchmarsh estimate used in Section 205 gives at most

$$\frac{4Z}{\varphi(3n) \log(Z/(3n))} = \frac{2Z}{(n-1) \log(Z/(3n))}$$

eligible primes. Apply Theorem 245.3 at every such prime and take a union of actual index sets. This proves the first bound, with $n/(n-1)$ absorbed into the constant. For the second take $\varepsilon = \delta/16$. Since $B^{2\varepsilon} = n^{8\varepsilon}$, the remaining power is $-\delta + 8\varepsilon = -\delta/2$; the denominator is asymptotic to $(7/2 - \delta) \log n$. The floor has no effect on this limit. □

Write

$$J(T) = \sum_{q|T} (v_q(T) - 3) \log q, \quad L_Y(T) = \sum_{\substack{q \leq Y \\ q|T}} v_q(T) \log q,$$

$$W_\delta(k) = \sum_{\substack{q > Z_\delta \\ q|T_n(k)}} (v_q(T_n(k)) - 3) \log q.$$

For every root outside $\mathcal{E}(n, Z_\delta)$, the intermediate signed cost is exactly

$$-2 \sum_{\substack{Y < q \leq Z_\delta \\ v_q(T_n)=1}} \log q - \sum_{\substack{Y < q \leq Z_\delta \\ v_q(T_n)=2}} \log q \leq 0.$$

Thus $J(T_n) \leq W_\delta + L_Y(T_n)$, without assuming that the full mass in this interval is small.

Theorem 228.1 and the elementary fixed-prime estimates in Section 229, evaluated at $Y = 6n^3$, give

$$\frac{1}{B} \sum_k \frac{L_Y(T_n(k))}{t_k} = O\left(\frac{\log n}{n}\right).$$

The harmonic term is $O(\log n/n)$, the endpoint is $O(1/(n \log n))$, and lifting with primes two and three costs $O(1/n)$. Put $\eta_n = \sqrt{\log n/n}$. Markov's inequality and (883) give one actual set of relative size at least

$$1 - C_\delta \left(\eta_n + \frac{n^{-\delta/2}}{\log n} \right)$$

on which

$$(884) \quad \frac{J(T_n)}{t_k} \leq \frac{W_\delta}{t_k} + \eta_n,$$

$$(885) \quad \frac{\log \text{rad}(T_n)}{t_k} \geq 1 - \frac{4}{3n} - \frac{\eta_n}{3} - \frac{(W_\delta)_+}{3t_k}.$$

The second exceptional term is not absorbed into η_n . The last inequality uses $\log T_n/t_k \geq 3 - 4/n$ from Lemma 229.1. A sequence in these sets with $(W_\delta)_+/t_k \rightarrow 0$ satisfies the restricted ABC estimate at every fixed tolerance eventually. The far condition has not been proved.

The new mechanism is that congruences at an actual fourth depth become integer phase identities with finite divisor fibers. The fifteen declarations of `DeepRootPhaseArithmetic.lean` formalize the actual product identities, positivity, injection into `Nat.divisors`, the fiber cardinality bound, the determinant height bound, and large-modulus rigidity. Its independent source reviews and fresh Lean 4.32.0 verification are recorded separately. Finite-ring torsion, lifting, the asymptotic divisor estimate, Brun–Titchmarsh, and the final analytic consequences remain ordinary proofs. No assertion that all those steps are formalized is made.

The interval has nonpositive signed cost on the stated majority, but its negative credit has no proved positive proportion of the height. Its full mass has not been shown negligible at this larger cutoff, so the earlier almost-full far-mass statement is not transferred here. The farther private top-rank primes and the exceptional roots remain unresolved.

246. PRIVATE NORM PRIMES AND HIGHER MOMENTS ON A POSITIVE-DENSITY CLASS

For a positive integer B , retain the actual roots

$$a_k = 3k, \quad w_k = a_k + \zeta, \quad Q_k = a_k^2 + a_k + 1, \quad \alpha_k = w_k/\bar{w}_k, \quad B \leq k < 2B.$$

Define the actual subset

$$\mathcal{S}_B = \{k : B \leq k < 2B, \ Q_k \text{ has a prime divisor greater than } 12B\}.$$

This section proves membership in a useful subclass; it does not assume that the actual roots have independent random prime factors.

Theorem 246.1 (A positive-density class of independent ratios). *As $B \rightarrow \infty$,*

$$|\mathcal{S}_B| \geq \frac{B}{2} - O\left(\frac{B \log \log B}{\log B}\right).$$

For every $k \in \mathcal{S}_B$ there is exactly one prime $r_k > 12B$ in Q_k , its depth is one, and it divides no other norm in the entire block. The algebraic numbers $\{\alpha_k : k \in \mathcal{S}_B\}$ are multiplicatively independent.

Proof. Uniformly in the block,

$$9B^2 \leq Q_k < 49B^2, \quad \sum_k \log Q_k = 2B \log B + O(B).$$

The polynomial $9X^2 + 3X + 1$ has no factor two or three. Every supported prime is $1 \pmod{3}$, and at such a prime r it has exactly two simple roots, each with one lift at every depth. Its

discriminant is -27 . Put $H_r = \lfloor \log(49B^2)/\log r \rfloor$. Counting the actual interval at each depth gives

$$\sum_k v_r(Q_k) \log r \leq \frac{2B \log r}{r-1} + 2H_r \log r.$$

We use only the fixed-modulus consequence

$$\vartheta(x; 3, 1) = \frac{x}{2} + O(x/\log x)$$

of Theorem 1.2, formula (1.12), of [Bennett, Martin, O'Bryant and Rechnitzer](#). Partial summation and the convergent change from $1/r$ to $1/(r-1)$ give

$$\sum_{\substack{r \leq x \\ r \equiv 1 \pmod{3}}} \frac{\log r}{r-1} = \frac{1}{2} \log x + O(\log \log x).$$

Indeed the error integral is $O(\int^x dt/(t \log t))$. The sum of the endpoint terms at $r \leq 12B$ is $O(B)$: each is at most $2 \log(49B^2)$, and $\pi(12B) = O(B/\log B)$. Thus all small norm primes together contribute at most

$$B \log B + O(B \log \log B).$$

The mass above $12B$ is at least $B \log B - O(B \log \log B)$. Each root contributes at most $2 \log B + O(1)$, proving the count. This is a fixed progression estimate, not a moving-field input.

Since $(12B)^2 > 49B^2$, a norm has at most one such prime, and its valuation is one. If $r > 12B$ divided norms at two distinct parameters a, b , it would divide

$$Q(a) - Q(b) = (a-b)(a+b+1).$$

Both nonzero factors have absolute value less than r , which is impossible. Finally, the primitive unramified w_k has one orientation above r_k . At that prime ideal α_k has valuation 1 or -1 , while every other block ratio has valuation zero. Applying these valuations to a multiplicative relation forces every exponent to vanish. $\square$

Theorem 246.2 (Higher moments at deep boundary primes). *Fix an integer $\nu \geq 2$. For every sufficiently large prime n , put $B = n^4$. For every prime $q > B$,*

$$\#\{k \in \mathcal{S}_B : q^{2\nu} \mid T_n(k)\} \leq (3\nu!n)^{1/\nu}.$$

The sufficiently-large condition depends on the fixed ν .

Proof. At every hit, the root ratio lies in the norm-one group killed by $3n$ modulo $q^{2\nu}$. The split and inert lifting proof of Lemma 245.1 works at this precision too and bounds that group by $3n$.

For two ordered ν -tuples write

$$z = \prod_{j=1}^{\nu} (a_j + \zeta) = R + S\zeta, \quad z' = R' + S'\zeta.$$

The norm coordinate bound and a telescoping expansion give

$$|R| \leq 2(7B)^\nu, \quad \left| z - \prod_j a_j \right| \leq \nu(7B)^{\nu-1}, \quad |S| \leq 2\nu(7B)^{\nu-1}.$$

The middle comparison is with a real integer. Hence

$$|RS' - R'S| \leq C_\nu B^{2\nu-1}, \quad C_\nu = 8\nu 7^{2\nu-1}.$$

When $B > C_\nu$, this is smaller than $q^{2\nu}$. Equal tuple images imply divisibility of this integer by $q^{2\nu}$, so their actual algebraic ratio products are equal. The independence in Theorem 246.1 implies equal multiplicity of every root index. Thus each image has at most $\nu!$ ordered tuples, also when entries repeat. If the set of hits has size M , its M^ν tuples have at most $3n$ images. Therefore $M^\nu \leq 3n\nu!$. No assertion uniform in a varying ν is used. $\square$

Theorem 246.3 (Full positive excess in a farther interval). *Fix an integer $\nu \geq 3$ and $0 < \delta < 1 - 1/\nu$, and for sufficiently large prime n put*

$$B = n^4, \quad Z = \lfloor Bn^{1-1/\nu-\delta} \rfloor > B, \quad F_{B,Z}(k) = \sum_{B < q \leq Z} (v_q(T_n(k)) - 3)_+ \log q.$$

Then

$$\frac{1}{B} \sum_{k \in \mathcal{S}_B} \frac{F_{B,Z}(k)}{t_k} = O_{\nu,\delta} \left(\frac{n^{-\delta}}{\log n} \right).$$

Proof. Use the exact identity $(e - 3)_+ = \sum_{j \geq 4} \mathbf{1}_{e \geq j}$. For each prime, every layer $4 \leq j < 2\nu$ has at most $\sqrt{6n}$ roots in $\mathcal{S}_B$, by Theorem 246.2 with parameter two. Every higher layer has at most $(3\nu!n)^{1/\nu}$ roots. The elementary actual height bound caps the depth by

$$H_q \leq \frac{3nL}{\log q}, \quad L = \log(6B + 1).$$

Consequently the complete positive excess at a single prime obeys

$$(886) \quad \sum_{k \in \mathcal{S}_B} (v_q(T_n(k)) - 3)_+ \log q \leq (2\nu - 4)\sqrt{6n} \log q + 3nL(3\nu!n)^{1/\nu}.$$

This remains an upper bound if a range of layers is empty. It explicitly pays for the intermediate depths as well as the deepest ones.

Since $t_k \geq n \log(3B)$, $q \leq Z \leq Bn$, and $B = n^4$, division by $Bn \log(3B)$ bounds (886) by $C_\nu n^{1/\nu}/B$. A prime contributing positive excess has rank n : rank one would put q^2 in one of $a_k, a_k + 1$, each below $6B + 1 < q^2$. Thus $q \equiv \pm 1 \pmod{3n}$. Brun–Titchmarsh bounds the number of eligible primes by

$$\frac{2Z}{(n-1) \log(Z/(3n))}.$$

The total is therefore

$$O_\nu \left(\frac{Zn^{1/\nu}}{Bn \log n} \right) = O_{\nu,\delta} \left(\frac{n^{-\delta}}{\log n} \right).$$

The normalization is by the entire block size B , not an unstated replacement by $|\mathcal{S}_B|$. $\square$

Corollary 246.4 (An actual positive-density domain). *For fixed ν, δ as above, there is a set $\mathcal{P}_{n,\nu,\delta}$ of actual roots of relative size at least $1/2 - o(1)$, lying in $\mathcal{S}_B$, such that*

$$\frac{F_{B,Z}(k)}{t_k} \leq n^{-\delta/2}, \quad \frac{L_B(T_n(k))}{t_k} \leq (\log n)^{-1/2}.$$

Writing

$$W_Z(k) = \sum_{\substack{q > Z \\ q | T_n(k)}} (v_q(T_n(k)) - 3) \log q,$$

one has on this set

$$(887) \quad \frac{J(T_n(k))}{t_k} \leq \frac{W_Z(k)}{t_k} + (\log n)^{-1/2} + n^{-\delta/2},$$

$$(888) \quad \frac{\log \text{rad}(T_n(k))}{t_k} \geq 1 - \frac{4}{3n} - \frac{(\log n)^{-1/2} + n^{-\delta/2}}{3} - \frac{(W_Z(k))_+}{3t_k}.$$

Proof. Markov's inequality in Theorem 246.3 removes $O_{\nu,\delta}(n^{-\delta/2}/\log n)$ of the whole block. The full-mass mean of Theorem 228.1, with the elementary inputs of Section 229, at B is $O(1/\log n)$. Markov at $(\log n)^{-1/2}$ removes $O((\log n)^{-1/2})$. Intersect with Theorem 246.1. The signed cost up to B is at most L_B , and that between B and Z is at most $F_{B,Z}$. This proves (887); the angular estimate $\log T_n/t_k \geq 3 - 4/n$ proves (888). $\square$

The intermediate interval need not have negligible full mass. Its negative credits can be retained in a sharper application. For arbitrarily large *fixed* ν , its upper exponent $1 - 1/\nu - \delta$ can approach one; this does not interchange the ν and n limits. The private-norm condition is proved for at least half the roots asymptotically, while the remaining roots and the unbounded farther signed cost are not controlled. The independent ordinary reviews cover these conclusions. No formalization of the progression estimate, the private-norm density theorem, or the higher-moment analytic consequences is claimed. A vanishing positive far cost is still required for the restricted ABC consequence along a sequence of these roots.

247. EXACT POSITIVE PHASE FIBERS AND THEIR DENOMINATOR BOUNDARY

For a positive integer B , put

$$\mathcal{A}_B = \{3k : k \in \mathbb{Z}, B \leq k < 2B\}.$$

Since $(a + \zeta)(b + \zeta) = (ab - 1) + (a + b + 1)\zeta$, the phase of an ordered pair is $(ab - 1)/(a + b + 1)$. An unordered pair below permits $a = b$, counted once. These are actual root parameters; no common deep-prime condition is implicit in the definition.

Theorem 247.1 (Exact divisor parameterization). *Let u, v be coprime positive integers and $K = u^2 + uv + v^2$. Positive integer pairs of phase u/v are in bijection with positive divisors $P \mid K$ satisfying $P \equiv -u \pmod{v}$, by*

$$(889) \quad P = va - u, \quad Q = vb - u, \quad a = \frac{P + u}{v}, \quad b = \frac{K/P + u}{v}.$$

For pairs with both coordinates divisible by three, the phase necessarily satisfies

$$(890) \quad u + v \equiv 0 \pmod{3}, \quad 3 \nmid uv.$$

Under (890), the exact divisor congruence is $P \equiv -u \pmod{3v}$. The additional conditions for the actual block are

$$(891) \quad 3vB - u \leq P < 6vB - u, \quad 3vB - u \leq K/P < 6vB - u.$$

Proof. The phase equation gives

$$(va - u)(vb - u) = K, \quad b(va - u) = u(a + 1) + v > 0, \quad a(vb - u) = u(b + 1) + v > 0.$$

Thus both factors are positive. Conversely the congruence for P implies $\gcd(P, v) = 1$. Since $K \equiv u^2 \pmod{v}$, its complementary divisor Q automatically satisfies $Q \equiv -u \pmod{v}$. Both reconstructed coordinates are positive integers, and expansion of $PQ = K$ proves their phase equation. These constructions are inverse.

For coordinates divisible by three the unreduced numerator is -1 modulo three and its denominator is 1 modulo three. Their gcd is prime to three, so reduction proves (890). Under that condition $\gcd(u, 3v) = 1$ and $K - u^2 = v(u + v)$ is divisible by $3v$. The same complementary-divisor argument now works modulo $3v$. The coordinate interval is exactly (891). Before knowing that a candidate phase has a nonempty block fiber, positivity of P, Q must still be imposed. The next theorem proves that its lower endpoint is positive whenever the fiber is nonempty. $\square$

Theorem 247.2 (A denominator-sensitive fiber bound). *A nonempty block fiber of reduced phase u/v satisfies*

$$(892) \quad \frac{9B^2 - 1}{6B + 1} \leq \frac{u}{v} \leq \frac{(6B - 3)^2 - 1}{12B - 5}, \quad B < \frac{u}{v} < 3B.$$

Its unordered and ordered cardinalities satisfy respectively

$$(893) \quad N_{\text{unordered}} \leq 1 + \left\lfloor \frac{2B - 2}{v} \right\rfloor,$$

$$(894) \quad N_{\text{ordered}} \leq \min \left\{ \tau(K), 2 \left(1 + \left\lfloor \frac{2B - 2}{v} \right\rfloor \right) \right\}.$$

In particular $v > 2B - 2$ implies at most one unordered pair.

Proof. At fixed b , the difference of phases at a and a' is

$$\frac{(a - a')(b^2 + b + 1)}{(a + b + 1)(a' + b + 1)}.$$

Monotonicity in both coordinates proves the first two bounds of (892). Its lower endpoint exceeds B because $3B^2 - B - 1 > 0$ for $B \geq 1$. Also $(ab - 1)/(a + b + 1) < ab/(a + b) < 3B$ when $a, b < 6B$. Consequently $L = 3vB - u > 0$. With $U = 6vB - u$, the interval condition can equivalently be written $L \leq P < U$ and $K/U < P \leq K/L$, with the displayed strict endpoints.

Put $s = a + b + 1$. The reduced phase forces $v \mid s$, while the block gives

$$6B + 1 \leq s \leq 12B - 5, \quad s \equiv 1 \pmod{3}.$$

Since $3 \nmid v$, the possible sums form one progression of step $3v$. The interval length is $6B - 6$, so there are at most $1 + \lfloor (2B - 2)/v \rfloor$ such sums. For each s , the product is $ab = us/v + 1$. The quadratic with this product and sum $s - 1$ determines at most one unordered pair. This proves (893); ordering at most doubles the number, and Theorem 247.1 supplies the bound $\tau(K)$. Diagonal pairs contribute only one ordered pair. The proof includes $B = 1$ and both endpoints of the sum interval. $\square$

This proves an actual subclass with an explicit denominator condition. It does not establish that small-denominator fibers are rare in a specified deep-root set.

Proposition 247.3 (Two actual failures of stronger injectivity). *For $n = 7$, $B = n^4 = 2401$, phase 4526 has the distinct unordered pairs*

$$(7809, 10767), \quad (8397, 9819).$$

Phase 4799 has at least three distinct unordered pairs,

$$(7668, 12828), \quad (8922, 10386), \quad (9480, 9720).$$

All coordinates belong to $\mathcal{A}_B$.

Proof. For the first phase,

$$K = 20489203 = 7^2 \cdot 67 \cdot 79^2 = 3283 \cdot 6241 = 3871 \cdot 5293.$$

The factors are $(a - 4526, b - 4526)$ in the two rows. For the second,

$$K = 23035201 = 7 \cdot 19 \cdot 31 \cdot 37 \cdot 151,$$

$$K = 2869 \cdot 8029 = 4123 \cdot 5587 = 4681 \cdot 4921.$$

These factors are $(a - 4799, b - 4799)$. The inverse formulas in Theorem 247.1 prove all phase equalities. Each coordinate is a multiple of three in $[7203, 14406]$. $\square$

The first example rules out unconditional unordered injectivity; the second rules out a universal bound of two unordered pairs. Neither meets the large-denominator hypothesis of Theorem 247.2. They do not assert common fourth-depth membership at a prime, and they do not refute the signed-tail gate.

Theorem 247.4 (A finite multiple-mark interface). *Let $S \subseteq \mathcal{A}_B$, and let H be a finite set with a binary operation. Assign images $\phi(a) \in H$ for $a \in S$. Suppose an integer $m > 864B^3$ has the property*

$$(895) \quad \phi(a)\phi(b) = \phi(c)\phi(d) \implies m \mid (ab - 1)(c + d + 1) - (cd - 1)(a + b + 1)$$

for $a, b, c, d \in S$. For a positive integer v_0 , let $E_{<v_0}$ count the ordered pairs in S^2 whose reduced phase denominator is less than v_0 . Then

$$(896) \quad |S|^2 \leq E_{<v_0} + 2|H| \left(1 + \left\lfloor \frac{2B - 2}{v_0} \right\rfloor \right).$$

Proof. The actual degree-three determinant has absolute value at most $864B^3$: each pair has coordinates between zero and $36B^2$, and between zero and $12B$, respectively. Its divisibility by m therefore forces it to vanish. Every nonempty product-image fiber has a single phase. Remove the low-denominator ordered pairs and partition the rest by their image in H . Each occupied fiber has denominator at least v_0 , so Theorem 247.2 gives its stated bound. Summing over H proves the result. No bound on $E_{<v_0}$ is presumed. $\square$

For an application with several marked primes, let $q_i > 3$ be distinct, $q_i \nmid n$, and impose actual conditions $q_i^{e_i} \mid T_n(a)$, with $e_i \geq 1$, where $T_n(a)$ is the absolute cubic boundary of $(a + \zeta)^n$. A hit forces $N(a + \zeta)$ to be a unit modulo q_i : the cubic difference would otherwise force both conjugates to vanish, while their difference is a unit at $q_i > 3$. Thus $\alpha_a = (a + \zeta)/(a + \bar{\zeta})$ is defined and belongs to the $3n$ -torsion subgroup H_i modulo $q_i^{e_i}$. The split or inert norm-one residue group is cyclic of order $q_i - \chi_i$, where $\chi_i = 1$ or -1 , respectively. Simple lifting of $X^{3n} - 1$, using $q_i \nmid 3n$, gives

$$|H_i| = \gcd(3n, q_i - \chi_i).$$

Take $H = \prod_i H_i$ and $m = \prod_i q_i^{e_i}$. Equal pair images give the conjugate-cross-product divisibility at every prime power, and coprimality gives (895). If $m > 864B^3$, Theorem 247.4 applies.

This uses combined precision, but its target size is $\prod_i |H_i|$, not one copy of $3n$. It yields no pointwise far-tail bound by itself. Actual incidence of small phase denominators, or additional dependence among product images, remains an explicit question.

The accompanying exact integer replay checks 2870 ordered pairs from blocks $1 \leq B \leq 20$, their 1540 distinct block fibers, and 1599 positive input pairs from $1 \leq a, b \leq 40$. It fully factors the two displayed constants and enumerates their divisors. A bounded search of 1201 integer phases $u \equiv 2 \pmod{3}$ in $[[3B/2], 3B]$ at $B = 2401$ finds the displayed three-pair phase. This is not a search over all rational phases and proves no uniform bound by testing. The ordinary proofs above and the finite evidence are separate; no Lean proof of the new denominator refinement is claimed.

248. THE EXACT RATIONAL DOMAIN OF THE SIMULTANEOUS QUADRATIC COVER

Put $K_E = \mathbb{Q}(\zeta)$, $K_G = \mathbb{Q}(i)$ and $K = K_E K_G$, where $\zeta^2 - \zeta + 1 = 0$. Throughout this section $g \geq 3$ is an odd integer. Define homogeneous degree- g integer forms by

$$(S + T\zeta)^g = A_E(S, T) + B_E(S, T)\zeta, \quad (S + Ti)^g = A_G(S, T) + B_G(S, T)i.$$

For $u \in \{1, \zeta, \zeta^2\}$, let $A_{E,u}, B_{E,u}$ be the coordinates of $u(S + T\zeta)^g$. These three choices suffice projectively, since changing the sign of u changes neither coordinate ratio. Retain the functions

$$(897) \quad a_0 = t^2 - 1, \quad b_0 = 2t + 1, \quad c_0 = t^2 + 2t, \quad U_0 = t^2 + t + 1, \quad A_0 = a_0 b_0,$$

$$(898) \quad E = A_0 + \zeta U_0^2, \quad G = A_0 + i U_0 c_0.$$

In homogeneous coordinates $[T_0 : W_0]$, the form b_0 means $W_0(2T_0 + W_0)$; all four forms a_0, b_0, c_0, U_0 have degree two.

Lemma 248.1 (Rational norm-one coordinates). *The maps*

$$\rho_E(s) = \frac{s + \zeta}{s + \bar{\zeta}}, \quad \rho_G(v) = \frac{v + i}{v - i}, \quad \rho_E(\infty) = \rho_G(\infty) = 1$$

are bijections from $\mathbb{P}^1(\mathbb{Q})$ to the norm-one elements of K_E and K_G , respectively. The g -power map on the latter group is injective. It is also injective on the former when $\gcd(g, 6) = 1$.

Proof. The finite denominators do not vanish at rational parameters, and conjugation proves norm one. For a norm-one element different from one the unique inverse formulas are

$$s = \frac{\zeta - Y\bar{\zeta}}{Y - 1}, \quad v = i \frac{Z + 1}{Z - 1}.$$

Their conjugates equal themselves because $\bar{Y} = Y^{-1}$ and $\bar{Z} = Z^{-1}$; hence they are rational. The value one corresponds exactly to infinity.

The quotient of two g -th roots is a g -th root of unity. The roots of unity in K_G and K_E are μ_4 and μ_6 . Indeed the trace of a root of unity in an imaginary quadratic field is an integer in $\{-2, -1, 0, 1, 2\}$ and its norm is one. The resulting quadratic polynomials give only orders 1, 2, 3, 4, 6; the two distinct fields exclude respectively orders 3, 6 or order 4. The asserted injectivity follows. $\square$

Theorem 248.2 (A complete model over $\mathbb{Q}$). *Let $C_{g,u}$ be the smooth projective normalization of the fiber product in $\mathbb{P}_t^1 \times \mathbb{P}_s^1 \times \mathbb{P}_v^1$ defined by*

$$(899) \quad U_0^2 A_{E,u}(s) - A_0 B_{E,u}(s) = 0, \quad U_0 c_0 A_G(v) - A_0 B_G(v) = 0.$$

It is a curve defined over $\mathbb{Q}$. After base change to K it is isomorphic to the smooth projective model of

$$(900) \quad Y^g = u^{-2} \frac{E}{\bar{E}}, \quad Z^g = \frac{G}{\bar{G}}.$$

It is geometrically connected, has degree g^2 over $\mathbb{P}_t^1$, and has genus $1 + 3g^2 - 4g$.

Proof. The coordinate pairs $(A_{E,u}, B_{E,u})$ and (A_G, B_G) have no common projective zero: such a zero would make powers of both distinct conjugate linear forms vanish. They therefore define finite degree- g maps of projective lines. The target pairs (A_0, U_0^2) and $(A_0, U_0 c_0)$ are coprime too. The finite numerator zeros are 1, -1 , $-1/2$; none is a zero of U_0 , and the zeros of c_0 are 0, -2 . At infinity the numerator vanishes but both denominators are nonzero. Both target maps have degree four. The fiber product is finite over the base, has total generic degree g^2 , and has no vertical curve component.

On a dense open set, use $Y = \rho_E(s)$ and $Z = \rho_G(v)$. The power identities give

$$\rho_E(A_{E,u}/B_{E,u}) = u^2 \rho_E(s)^g, \quad \rho_G(A_G/B_G) = \rho_G(v)^g.$$

They turn (899) into (900); the fractional linear inverses supply the inverse function-field map. The odd-exponent calculation in Theorem 242.2 gives geometric connectedness, degree and genus. Uniqueness of the smooth projective model extends the birational map to an isomorphism [140]. For real $t > 1$, both norms of E and G are $F(a_0, b_0) > 0$, so this chart avoids every branch point. The raw fiber product is already smooth there. The even-exponent component defect is not omitted: g is odd throughout this theorem. $\square$

Theorem 248.3 (An exact rational descent test). *On (900), let $t \in \mathbb{Q}$ and $t > 1$. The conditions*

$$(901) \quad Y \in K_E, \quad Y\bar{Y} = 1, \quad Z \in K_G, \quad Z\bar{Z} = 1$$

are exactly the conditions for a K -point on this chart to come from a rational point of $C_{g,u}$. They imply positive coprime integers a, b and integers $U, Q \geq 7$ such that

$$(902) \quad a^2 + ab + b^2 = U^2, \quad F(a, b) = Q^g.$$

Equivalently, if σ_E conjugates K_E and fixes i , and σ_G conjugates i and fixes K_E , the descent test is

$$\sigma_E(Y) = Y^{-1}, \quad \sigma_G(Y) = Y, \quad \sigma_E(Z) = Z, \quad \sigma_G(Z) = Z^{-1}.$$

Proof. Lemma 248.1 recovers unique rational projective parameters s, v , proving both directions of the rationality test. Reduce the positive rational number $x = a_0/b_0 = (t^2 - 1)/(2t + 1)$ to a/b . Then

$$a^2 + ab + b^2 = (bU_0/b_0)^2.$$

A rational square root of an integer is an integer, by valuations in a reduced fraction, so $U = bU_0/b_0$ is a positive integer. The first map is $x = L_{1,2}(t)$. The Eisenstein equation gives a rational power-map input with output $A_0/U_0^2 = ab/(a^2 + ab + b^2)$. These are precisely the two maps in the pure-coefficient inverse at exponents $(2, g)$, with coefficients 1, u . Theorem 240.1 supplies (902) and $U, Q \geq 7$. Its first ramified parity is zero because the first exponent is two and its coefficient is one. Its proof includes ramified and infinite projective inputs; none is silently excluded here. In particular $3 \nmid F$ and oddness force the second primitive input to be unramified. The Gaussian equation provides a coordinate on the cover but is not needed for this integer norm conclusion once the positive rational Eisenstein input is given. $\square$

Theorem 248.4 (Unique Gaussian lifting on the positive rational locus). *Let $D_{g,u}$ be the smooth projective normalization of the first equation in (899). Each point of $D_{g,u}(\mathbb{Q})$ with finite $t > 1$ has exactly one rational lift to $C_{g,u}$. If $\gcd(g, 6) = 1$, the points of the single curve $C_{g,1}(\mathbb{Q})$ with $t > 1$ are in bijection with the ordered positive primitive seeds satisfying (902).*

Proof. The same pure inverse just used gives an actual seed from the point of $D_{g,u}$. Theorem 241.2 writes $ab + iUc = u_G w_G^g$. Since g is odd, raising to g permutes μ_4 and the unit is absorbed into w_G . The ratio $Z = w_G/\bar{w}_G$ is a Gaussian norm-one g -th root of the actual ratio; the positive rational scale from t cancels. Lemma 248.1 yields $v \in \mathbb{P}^1(\mathbb{Q})$. Two lifts would have ratios differing by an element of $\mu_g(K_G) = \{1\}$, so v is unique. The chart is smooth and nonbranch, so normalization introduces no further lifts there.

When $\gcd(g, 6) = 1$, an actual seed gives $t = (a + U)/b > 1$. In addition to the Gaussian unit, the unit in $ab + U^2\zeta = u_E w_E^g$ can now be absorbed because raising to g permutes μ_6 . The two norm-one ratios give rational s, v , and hence a point of $C_{g,1}$. Conversely the earlier theorem gives the seed. The parameter t is the unique root greater than one of $t^2 - 2(a/b)t - a/b - 1 = 0$, the other root being negative. The Eisenstein root is unique by $\gcd(g, 6) = 1$, and the Gaussian root by oddness. This proves the bijection. $\square$

For odd g divisible by three the three unit choices must be retained; single-curve uniqueness is not asserted. The Gaussian lift remains unique over each positive rational Eisenstein point. Thus the higher cover removes none of those points. An actual point computation or a further theorem is still needed to use it. For a prescribed first exponent $h > 2$ even, additionally impose $U = R^{h/2}$; the first-square inverse does not preserve that larger exponent automatically. No uniform height bound, odd-exponent emptiness or ABC theorem follows here. All curves called rational models in this section are curves defined over $\mathbb{Q}$, not elliptic $\mathbb{Q}$ -curves in the modularity terminology.

249. AN UNRAMIFIED GAUSSIAN COVER AND AN EXACT JACOBIAN TORSION CLASS

Let $g \geq 3$ be an odd integer, and write $D = D_{g,1}$ and $C = C_{g,1}$ for the curves defined over $\mathbb{Q}$ in Section 248. Forgetting the Gaussian coordinate gives $\pi : C \rightarrow D$. Work also over the fixed field $K = \mathbb{Q}(i, \sqrt{-3})$. With the four quadratics of Lemma 242.1, the function fields are

$$(903) \quad K(D) = K(t, Y), \quad Y^g = \frac{B_1 B_2}{B_3 B_4}, \quad K(C) = K(t, Y, Z), \quad Z^g = \frac{B_1 B_4}{B_2 B_3}.$$

The eight simple roots of the four B_j are distinct and finite. The preceding geometric degree computations give degrees g and g^2 over $\mathbb{P}_t^1$.

Theorem 249.1 (An everywhere unramified relative cover). *The map π is finite étale of degree g over $\mathbb{Q}$, and*

$$(904) \quad g(D) = 3g - 3, \quad g(C) = 1 + 3g^2 - 4g = 1 + g(g(D) - 1).$$

Proof. The function-field inclusion extends to a finite nonconstant map of smooth projective curves, of degree g . On the base $\mathbb{P}_t^1$, the Kummer function defining D has a simple zero at roots of B_1, B_2 and a simple pole at roots of B_3, B_4 . Over an algebraic closure, each has a unique point above it, of ramification index g . At other points, including infinity, its valuation is zero and the local unit Kummer equation is unramified. Riemann–Hurwitz gives $2g(D) - 2 = -2g + 8(g - 1) = 6g - 8$.

On C the local valuation pairs at those roots are $(1, 1), (1, -1), (-1, -1), (-1, 1)$. Extracting roots of local units over an algebraically closed field shows that both Kummer extensions lie in one extension generated by a g -th root of a uniformizer. Its index is exactly g . Elsewhere the index is one. Indices multiply in the tower $C \rightarrow D \rightarrow \mathbb{P}_t^1$, so the relative index is one everywhere. In characteristic zero this finite map of smooth curves is étale; this property descends to $\mathbb{Q}$. The genus of C is also Theorem 242.2, and agrees with the unramified Riemann–Hurwitz formula [140]. $\square$

Theorem 249.2 (One Kummer function and an exact torsion divisor). *Put $f = B_4/B_2$ and $m = (g + 1)/2$. Then*

$$(905) \quad W = \frac{(Z/Y)^m}{f}, \quad W^g = f, \quad W^2 = Z/Y, \quad K(C) = K(D)(W), \quad Z = YW^2.$$

For $j = 2, 4$, let R_j be the sum of the unique points of D above the two roots of B_j . These are degree-two divisors defined over K , and

$$\Theta = [R_4 - R_2] \in \text{Pic}^0(D_K)$$

has exact order g , also geometrically. Its natural image in $\text{Jac}(D)(K)$ has exact order g .

Proof. The factorizations give

$$\frac{G/\bar{G}}{E/\bar{E}} = (B_4/B_2)^2 = f^2.$$

Thus $(Z/Y)^g = f^2$, and $2m = g + 1$ proves both identities for W in (905). Conversely $Z = YW^2$ gives the original second equation, proving equality of the fields over K itself.

The roots of B_j form a Galois-stable pair and each has a unique point above it, so R_j is defined over K . Pullback multiplies the zero and pole valuations of f by g , and infinity has valuation zero. Therefore

$$(906) \quad \text{div}_D(f) = g(R_4 - R_2).$$

Suppose over an algebraic closure k the class had order $d < g$. Then $d \mid g$ and some $h \in k(D)^*$ would have $\text{div}(h) = d(R_4 - R_2)$. Since D is projective and geometrically integral, $f/h^{g/d}$ is a nonzero constant. Taking a constant root in k , write $f = (ch)^{g/d}$. If $r = g/d > 1$, then

$$T^g - f = \prod_{\xi \in \mu_r(k)} (T^d - \xi ch).$$

Every root has degree at most d over $k(D)$, contradicting the geometric degree g of $k(C)/k(D)$ from Theorem 249.1. This proves exact geometric order, and (906) gives exact order over K too.

The divisor gives an actual degree-zero line bundle over K , hence a K -rational point under the natural Picard-to-Jacobian map [141, Theorem 1.1 and Remark 1.5]. After base change to k this map is the usual identification with the Jacobian. Exact geometric order therefore proves the claimed order of the K -rational image. No rational base point on D , or equality of all of $\text{Pic}^0(D_K)$ with $\text{Jac}(D)(K)$, is assumed. $\square$

There is a precise arithmetic interpretation. Locally, (906) writes $f = s^g u$ with s a rational local equation for $R_4 - R_2$ and u a regular unit. The normalization adjoins W/s satisfying $(W/s)^g = u$. This is finite étale, and scaling W gives a torsor under the group scheme μ_g over D_K . Only geometrically do all g scalar deck transformations occur. We have not assumed $\mu_g \subset K$, nor asserted a constant cyclic arithmetic deck group of order g .

Corollary 249.3 (The actual positive rational fibers). *Write $D(\mathbb{Q})_{>1}$ and $C(\mathbb{Q})_{>1}$ for points with finite rational base coordinate $t > 1$. The map*

$$\pi : C(\mathbb{Q})_{>1} \longrightarrow D(\mathbb{Q})_{>1}$$

is a bijection. If $\gcd(g, 6) = 1$, both sets are in bijection with the ordered positive primitive seeds $a^2 + ab + b^2 = U^2$, $F(a, b) = Q^g$, with $U, Q \geq 7$.

Proof. This is exactly Theorem 248.4. Existence comes from the actual Gaussian decomposition, while uniqueness uses $\mu_g(K_G) = \{1\}$ for odd g . The geometric degree is not the number of rational lifts. The additional coprimality condition is the one needed to absorb the Eisenstein unit and obtain the single-curve ordered-seed bijection. $\square$

Thus the nontrivial étale cover and its exact-order Jacobian class remove no point of this positive rational locus. Every such fiber has a rational lift, and after base change to K its μ_g -torsor has a K -point. The class provides an explicit invariant for a fixed-exponent descent or Jacobian computation. Its ambient dimension is $3g - 3$ and grows with g ; no fixed-dimension

uniform torsion conclusion applies. No point-height upper bound, full rational-point computation, odd-exponent emptiness or ABC theorem is obtained. Ramified first norms and moving nonunit residuals remain separate branches.

250. THREE SMALLER JACOBIANS FOR THE ACTUAL FIRST-SQUARE LOCUS

Fix an odd integer $g \geq 3$ and $u \in \{1, \zeta, \zeta^2\}$. Write $A = A_{E,u}(S, T)$ and $B = B_{E,u}(S, T)$ for the homogeneous degree- g integer forms in Section 248, so that $A + B\zeta = u(S + T\zeta)^g$, and put $r = A/B$. The curve $D = D_{g,u}$ is the smooth projective normalization of

$$(907) \quad r(s) = \frac{a_0(t)b_0(t)}{U_0(t)^2}, \quad a_0 = t^2 - 1, \quad b_0 = 2t + 1, \quad U_0 = t^2 + t + 1.$$

Theorem 250.1 (Three hyperelliptic quotients). *There is an equality of function fields over $\mathbb{Q}$*

$$(908) \quad \mathbb{Q}(D) = \mathbb{Q}(s)(v, w), \quad v^2 = 1 + r, \quad w^2 = 1 - 3r.$$

This is a geometrically connected V_4 cover of $\mathbb{P}_s^1$, of degree four and genus $3g - 3$. Its three quadratic quotient curves have homogeneous hyperelliptic equations

$$(909) \quad H_1 : y_1^2 = B(A + B), \quad H_2 : y_2^2 = B(B - 3A), \quad H_3 : y_3^2 = (A + B)(B - 3A).$$

Each right side is a squarefree binary form of degree $2g$; the y_j have weight g , and the equations denote their smooth projective models. Each H_j has genus $g - 1$. On a dense affine chart the quotient maps are

$$y_1 = Bv, \quad y_2 = Bw, \quad y_3 = Bvw.$$

Proof. Put $c_0 = t^2 + 2t$ and $d_0 = t^2 - 2t - 2 = a_0 - b_0$. The identities

$$c_0^2 = U_0^2 + a_0b_0, \quad d_0^2 = U_0^2 - 3a_0b_0$$

give $v = c_0/U_0$ and $w = d_0/U_0$. Conversely the inverse on a dense open is

$$(910) \quad t = \frac{v + w + 2}{v - w}.$$

Indeed $w^2 + 3v^2 = 4$ and $a_* = (v + w)/2$, $b_* = (v - w)/2$ satisfy $a_*^2 + a_*b_* + b_*^2 = 1$. Substitution gives $a_0/b_0 = a_*/b_*$ and $U_0/b_0 = 1/b_*$, recovering both v, w and $r = a_*b_*$. This proves the function-field identity, which extends to the smooth projective models [140]. The dense-open formulas do not discard exceptional points from that projective assertion.

Over K_E , the map $r(s)$ is conjugate by ρ_E to a nonzero scalar times a g -th power. Its only branch values are $-\zeta, -\bar{\zeta}$. The rational target values $\infty, -1, 1/3$ are distinct and avoid those values. Their inverse images are therefore three disjoint sets of g simple points. They are exactly the zeros of $B, A + B, B - 3A$, including any zero at infinity in homogeneous coordinates.

At a zero of $A + B$ the valuation of $1 + r$ detects its nontrivial square class, and at a zero of $B - 3A$ the valuation of $1 - 3r$ detects the other class. They are independent, so the extension in (908) has geometric group V_4 . At zeros of $A + B$ only v ramifies, at zeros of $B - 3A$ only w ramifies, and at zeros of B their common simple poles give the diagonal inertia subgroup of order two. There are no other branch points. Thus

$$2g(D) - 2 = -8 + (3g)2 = 6g - 8.$$

The three nontrivial square classes give the three quadratic subfields. Clearing by the square B^2 gives (909). Each has $2g$ simple branch points, so $2g(H_j) - 2 = -4 + 2g$. This argument includes infinity and the normalization at common branch points of the raw fiber product of H_1 and H_2 . $\square$

Theorem 250.2 (An isogeny over $\mathbb{Q}$ and transport of odd torsion). *Let $p_j : D \rightarrow H_j$ be the quotient maps, with Jacobians J_D, J_j . Define homomorphisms over $\mathbb{Q}$ by*

$$(911) \quad \Phi : J_D \longrightarrow J_1 \times J_2 \times J_3, \quad P \longmapsto ((p_1)_*P, (p_2)_*P, (p_3)_*P),$$

$$(912) \quad \Psi : J_1 \times J_2 \times J_3 \longrightarrow J_D, \quad (P_1, P_2, P_3) \longmapsto p_1^*P_1 + p_2^*P_2 + p_3^*P_3.$$

Then $\Psi\Phi = [2]$, and Ψ is an isogeny over $\mathbb{Q}$. For $u = 1$, the class Θ of Theorem 249.2 maps under Φ to a K -rational point of exact order g on the product. For prime g , at least one coordinate has exact order g .

Proof. Finite maps of smooth curves are finite locally free. Pullback and norm of line bundles therefore induce the displayed homomorphisms on Jacobians. On divisors these are the usual pullback and pushforward, with norms of rational functions ensuring that principal divisors map to principal divisors. The constructions and all p_j are defined over $\mathbb{Q}$; this is the standard Picard/Jacobian functoriality [141].

If σ_j generates the group of $D \rightarrow H_j$, the geometric divisor identity is $p_j^*(p_j)_* = 1 + \sigma_j$. At ramified points it still holds: the pullback multiplicity is two and the two right-hand terms coincide. For $q : D \rightarrow \mathbb{P}_s^1$, the analogous identity is

$$q^*q_* = 1 + \sigma_1 + \sigma_2 + \sigma_3.$$

It vanishes on the Jacobian because $\text{Pic}^0(\mathbb{P}^1) = 0$. Hence

$$(913) \quad \Psi\Phi = 3 + \sigma_1 + \sigma_2 + \sigma_3 = [2].$$

The equality of homomorphisms holds over $\mathbb{Q}$ since it holds after algebraic closure. Multiplication by two is surjective, so Ψ is surjective. The source and target dimensions are both $3(g-1)$, and its kernel is therefore finite. This proves the isogeny.

The point $\Phi(\Theta)$ is killed by g . If it is killed by an integer n , (913) shows that $2n$ kills Θ . Since Θ has exact odd order g , this forces $g \mid n$. Thus the tuple has exact order g . For prime g at least one nonzero coordinate has that order; for composite g the tuple's exact order need not occur in one coordinate, and no such stronger statement is made. $\square$

Theorem 250.3 (The positive rational chart). *On the nonbranch rational chart, the points of D with $t > 1$ are exactly the rational triples (s, v, w) in (908) satisfying*

$$(914) \quad 0 < r(s) < 1/3, \quad v > 0.$$

Proof. These conditions imply $1 < v < 2/\sqrt{3}$ and $|w| < 1$. The denominator $v - w$ in (910) is positive, and the numerator exceeds it by $2(w+1) > 0$, so $t > 1$. Conversely rational $t > 1$ gives $v = c_0/U_0 > 0$ and $r = a_0b_0/U_0^2 > 0$. The identity for d_0^2 gives $r \leq 1/3$. Equality would require $t^2 - 2t - 2 = 0$, whose roots $1 \pm \sqrt{3}$ are irrational. Thus the inequality is strict. The dense-open inverse formulas are defined throughout this chart, and keep the ordered sign choice of w . $\square$

By Section 248, these points give actual primitive seeds with square first norm and g -th-power second norm. When $\gcd(g, 6) = 1$, the single $u = 1$ curve gives the ordered-seed bijection. For $3 \mid g$, retain all three unit choices. A point on one quotient alone is merely necessary: the points on H_1, H_2 must have the same source coordinate s and satisfy (914).

For the explicit case $g = 3, u = 1$, the affine forms are

$$(915) \quad \begin{aligned} A &= s^3 - 3s - 1, & B &= 3s(s+1), \\ A + B &= s^3 + 3s^2 - 1, & B - 3A &= -3s^3 + 3s^2 + 12s + 3. \end{aligned}$$

They define three genus-two curves in (909), whose Jacobian product is isogenous to the genus-six Jacobian of $D_{3,1}$. The homogeneous degree-three B includes a simple zero at infinity. The other unit choices follow by integer coordinate multiplication and have not been discarded. This explicit reduction supplies a smaller computation problem. It is not a rank computation or a determination of all rational points on any quotient. No fixed-exponent solution, uniform height bound, or control of varying residuals follows without additional work.

251. ACTUAL PAIR PHASES AND A FINITE ROOT-COUNT BRIDGE IN LEAN

The integer rigidity step in multiplicative coherence admits a finite formalization that retains the actual pairs. For integers a, b put

$$r(a, b) = ab - 1, \quad s(a, b) = a + b + 1, \quad K(u, v) = u^2 + uv + v^2.$$

Then $(a + \zeta)(b + \zeta) = r(a, b) + s(a, b)\zeta$. For two such pairs, the conjugate cross difference has coordinates $(D, -2D)$, where

$$D = r(a, b)s(c, d) - r(c, d)s(a, b).$$

These are identities in the actual integer-coordinate multiplication of $\mathbb{Z}[\zeta]$.

Lemma 251.1 (An actual divisor fiber). *Let u, v be positive integers and let T be a finite set of ordered positive integer pairs (a, b) satisfying $v(ab - 1) = u(a + b + 1)$. Then*

$$|T| \leq \tau(K(u, v)).$$

There is no coprimality requirement on u, v .

Proof. Expansion gives $(va - u)(vb - u) = K(u, v)$. Also $b(va - u) = u(a + 1) + v > 0$, and similarly $a(vb - u) > 0$. Thus both factors are positive. The map $(a, b) \mapsto va - u$ lands in the actual positive divisors of $K(u, v)$. Its value determines a since $v > 0$, then determines $vb - u = K(u, v)/(va - u)$ and hence b . It is injective, including on diagonal and exchanged pairs. $\square$

For a natural number N , define the finite envelope

$$\mathcal{D}(N) = \max_{0 \leq k \leq N} |\text{divisors}(k)|.$$

The formal definition is a finite supremum over `Finset.range(N + 1)`, with the library's empty divisor set at zero. Only positive K occurs below. No asymptotic divisor estimate is part of this definition.

Proposition 251.2 (Finite counting with an explicit image interface). *Let n be a natural number, and let B, m be integers with $B > 0$ and $m > 864B^3$. Let S be a finite set of integers in $[3B, 6B)$, and let H be a finite set with a binary operation. Suppose $\phi : S \rightarrow H$, $|H| \leq 3n$, and for every $a, b, c, d \in S$,*

$$\phi(a)\phi(b) = \phi(c)\phi(d) \implies m \mid D.$$

Then

$$(916) \quad |S|^2 \leq 3n \mathcal{D}(2500B^4).$$

Proof. On this block the positive coordinates satisfy $r \leq 36B^2$ and $s \leq 12B$. Thus $|D| \leq 864B^3 < m$, and the assumed divisibility forces $D = 0$. Partition S^2 by the product image in H . In every nonempty fiber choose an actual representative (a, b) and take $u = r(a, b)$, $v = s(a, b)$, without reducing their fraction. All pairs in the fiber have this phase. Lemma 251.1 applies, and

$$0 < K(u, v) \leq 2500B^4.$$

The last bound follows from the coordinate bounds and $B \geq 1$: $u^2 + uv + v^2 \leq 1296B^4 + 432B^3 + 144B^2 \leq 2500B^4$. Every fiber therefore has size at most the displayed envelope, including empty fibers. Summing over all H proves (916). $\square$

For natural numbers n, q with $n > 0$ and $q > 6n^3$ one has

$$q^4 > 1296n^{12} > 864(n^4)^3.$$

Consequently, when $B = n^4$ and the explicit divisibility modulus is q^4 , the same finite theorem gives

$$|S|^2 \leq 3n \mathcal{D}(2500n^{16}), \quad |S| \leq \sqrt{3n} \sqrt{\mathcal{D}(2500n^{16})}.$$

The square-root conclusion is a statement in the real numbers, obtained by casting the natural-number inequality and using monotonicity of the real square root. The finite theorem itself does not require primality or any group law beyond the specified binary operation. The ordinary number-theoretic application supplies a group and the two explicit interface hypotheses; it is not built into the formal statement by an extra axiom.

Formal inventory and limits. `DeepRootPhaseArithmetic` contains 15 declarations: actual product and conjugate-difference identities, positive divisor factorization and injectivity, actual finite fiber counting, block height and determinant bounds, large-modulus rigidity, a bound for K , and a closed nontrivial pair collision. `ActualDeepRootCount` adds six declarations for the finite envelope, image fiber, total square count, fourth-power threshold, specialized count and real square-root bound. All 21 new declarations and 29 unchanged Eisenstein dependency declarations were freshly compiled with Lean 4.32.0 and pinned Mathlib revision 81a5d257c8e410db227a6665ed08f64fea08e997. All 50 explicit axiom queries passed; their union is only `propext`, `Classical.choice` and `Quot.sound`. The existing compiled Mathlib cache was reused. Ordinary proofs and independent internal review preceded the formal implementation.

This formal core does not construct the finite norm-one torsion group, prove its lifting theorem, formalize divisor asymptotics or the sieve, or control the unbounded signed packet. The ordinary private-support and geometric results in this continuation are separate from these 21 formal declarations. There is still no formal proof or disproof of *abc*.

252. THE TWELFTH CONTINUATION: COHERENCE, PRIVATE SUPPORT AND QUOTIENT CURVES

The analytic route now uses multiplication of actual root ratios. At sufficiently deep prime powers, the height of an integer determinant is smaller than the modulus. Congruent pair products then have exactly equal rational phase, and their multiplicity is bounded by actual divisors of a quadratic integer. This gives a simultaneous cap-three window beyond the root block on a set of relative size tending to one. It controls the signed contribution of the window without asserting that its full prime mass is small.

A distinct refinement uses primes of the root norms. At least half of the block asymptotically has a simple norm prime larger than $12B$; that prime occurs in no other root norm in the block. The corresponding oriented valuations prove multiplicative independence on this actual subset. Higher ordered moments then have only permutation fibers. For every fixed moment order this controls positive excess in a still larger bounded interval, on a set of relative size at least $1/2 - o(1)$. The constants depend on the fixed moment order. The unbounded farther signed contribution and the complementary roots remain open.

The exact phase analysis also records the limits of stronger counting claims. Divisors obey a precise congruence modulo three times the reduced denominator. Large denominators give an injective unordered subclass; small denominators admit actual repeated fibers, including three distinct unordered pairs in one fiber. These are counterexamples to the stated uniform fiber bounds, not counterexamples to deep-prime membership or *abc*. Multiple prime marks retain the product target size and an explicit low-denominator remainder.

The geometric branch remains separate. The positive rational locus on the odd biquadratic cover has an exact integral inverse. Every positive rational point on the first Eisenstein curve has a unique Gaussian lift. Thus that additional equation does not exclude such points. Nevertheless the resulting unramified cover exhibits a divisor class of exact order g on a varying Jacobian. The first curve is also a rational biquadratic cover with three hyperelliptic quotients of genus $g - 1$. Their Jacobians are isogenous in product to its Jacobian, and the odd torsion class survives in the product. At $g = 3$ these are three explicit genus-two curves. Their full rational-point sets and ranks have not been computed here. Uniform exponent and height control, ramified first norms and moving residual coefficients remain live targets.

The formal contribution is the 21-declaration actual phase and finite counting core, with 29 freshly checked unchanged dependency declarations. The finite-group size and modular divisibility are explicit inputs. The asymptotic density and sieve estimates, ideal-factorization argument and geometric constructions have ordinary proofs with independent internal review; they have not all been formalized. Finite replays check only their stated bounded domains. None of these results supplies a complete proof or disproof of *abc*, and no unrefuted parent route is removed because its remaining tools or bounds are not yet available.

253. UNIFORM MOMENTS AND A POSITIVE-EXCESS WINDOW NEAR n^5

Retain the actual roots $a_k = 3k$, $w_k = a_k + \zeta$, $Q_k = a_k^2 + a_k + 1$, $\alpha_k = w_k/\bar{w}_k$, for integers $B \leq k < 2B$. The subset $\mathcal{S}_B$ consists of roots whose norm has a prime factor above $12B$. Theorem 246.1 proves, for the actual roots,

$$|\mathcal{S}_B| \geq B/2 - O(B \log \log B / \log B),$$

and proves multiplicative independence of all their ratios. The following uniform estimate removes the former fixed-moment restriction while retaining this proved domain.

Lemma 253.1 (Tuple rigidity uniform in its length). *Let $B \geq 5$ be an integer and $q > 8B$ a prime. For every integer $\nu \geq 1$, two ordered ν -tuples of block roots, with unit conjugate denominators modulo q , whose ratio products agree modulo $q^{2\nu}$, have exactly equal algebraic ratio products in $\mathbb{Q}(\zeta)$.*

Proof. Write their actual element products as $z = R + S\zeta$ and $z' = R' + S'\zeta$. The norm coordinate bound and the finite telescoping expansion, valid at every finite ν , give

$$|R| \leq 2(7B)^\nu, \quad |S| \leq 2\nu(7B)^{\nu-1}.$$

Thus for $D = RS' - R'S$,

$$|D| \leq 8\nu 7^{2\nu-1} B^{2\nu-1}.$$

The conjugate cross difference implies $q^{2\nu} \mid D$. Put $x = 49/64$. Since $q > 8B$, the height bound divided by $q^{2\nu}$ is less than

$$\frac{8}{7B} \nu x^\nu < \frac{8}{7B} \sum_{j=0}^{\nu-1} x^j < \frac{8}{7B} \frac{1}{1-x} = \frac{512}{105B} < \frac{5}{B} \leq 1.$$

Therefore $D = 0$, and $zz' = z'\bar{z}$ over the number field. Their ratios agree. The products are nonzero; no sign condition on R, S or bound on ν is needed. The hypothesis $q > 8B$ is used explicitly and is not replaced by the earlier smaller cutoff. $\square$

Theorem 253.2 (A multiset bound at every precision). *Let $n > 324$ be prime and $B = n^4$. For a prime $q > 8B$ and a positive integer ν , put*

$$M_\nu(q) = \#\{k \in \mathcal{S}_B : q^{2\nu} \mid T_n(k)\}.$$

If $M_\nu(q) > 0$, then

$$(917) \quad \binom{M_\nu(q) + \nu - 1}{\nu} \leq 3n.$$

For an empty set the number of positive-length multisets is zero. In particular, with $r = \lceil \sqrt{n} \rceil$,

$$M_2(q) \leq \sqrt{6n}, \quad M_r(q) \leq 3.$$

Proof. Every hit gives a norm-one element killed by $3n$ modulo $q^{2\nu}$. The split and inert simple-lifting proof of Lemma 245.1 bounds the group by $3n$, since $q > n$. Every multiset of ν roots gives one group product. Equal images imply equality of actual ratio products by Lemma 253.1. Independence from Theorem 246.1 then recovers all index multiplicities. The multiset map is injective. Stars and bars gives (917), keeping repeated entries.

At $\nu = 2$, the inequality $M(M+1)/2 \leq 3n$ proves the first bound. If $M_r \geq 4$, the multisets on just four roots would instead give

$$3n \geq \binom{r+3}{3} = \frac{(r+1)(r+2)(r+3)}{6} > \frac{r^3}{6} \geq \frac{n^{3/2}}{6} > 3n.$$

The last inequality uses $\sqrt{n} > 18$, a contradiction. Here r grows with n ; the determinant estimate was first proved uniformly for all finite lengths. $\square$

Theorem 253.3 (An explicit complete excess mean). *For every prime $n > 324$, $B = n^4$, and real $8B < Z \leq B^2$, put*

$$F_{8B,Z}(k) = \sum_{8B < q \leq Z} (v_q(T_n(k)) - 3)_+ \log q.$$

Then

$$(918) \quad \frac{1}{B} \sum_{k \in \mathcal{S}_B} \frac{F_{8B,Z}(k)}{t_k} \leq \frac{80Z}{Bn \log(Z/(3n))}.$$

Proof. The complete positive excess satisfies $(e - 3)_+ = \sum_{j \geq 4} \mathbf{1}_{e \geq j}$. Fix $q > 8B$, and put $r = \lceil \sqrt{n} \rceil$. Every layer $4 \leq j < 2r$ has at most $\sqrt{6n}$ roots in $\mathcal{S}_B$. There are at most $2r$ layers, with $r \leq 2\sqrt{n}$, so their contribution is at most

$$4\sqrt{6}n \log q < 10n \log q.$$

Every higher layer has at most three roots. Writing $L = \log(6B + 1)$, the actual height bound $\log T_n(k) \leq 3nL$ caps its depth by $3nL/\log q$. All higher layers together contribute at most $9nL$. Consequently

$$(919) \quad \sum_{k \in \mathcal{S}_B} (v_q(T_n(k)) - 3)_+ \log q \leq 10n \log q + 9nL.$$

This includes empty layer ranges and pays every positive depth.

For $q \leq Z \leq B^2$ and $B \geq 7$, $\log q \leq 2 \log B$ and $L \leq 2 \log B$. Also $t_k \geq n \log(3B) \geq n \log B$. Thus the normalized contribution at one prime is at most $38/B$. Positive excess excludes rank one: $q > n$ gives no exponent lifting, and q^2 cannot divide either coprime old factor $a_k, a_k + 1 < 6B + 1$. Hence the exact rank is n , and $q \equiv \pm 1 \pmod{3n}$. The two-progression Brun–Titchmarsh estimate bounds the number of eligible primes up to Z by

$$\frac{2Z}{(n-1) \log(Z/(3n))}.$$

Multiplying by $38/B$ and using $76n/(n-1) < 80$ proves (918). The normalization is by the whole block B . $\square$

Corollary 253.4 (A logarithmic expansion beyond n^5). *Fix $\delta > 0$. For all sufficiently large prime n , put*

$$Z_n = \left\lfloor Bn(\log n)^{1-\delta} \right\rfloor, \quad B = n^4.$$

There is an actual subset $\mathcal{P}_{n,\delta}$ of relative size at least

$$\frac{1}{2} - O_\delta \left(\frac{\log \log n}{\log n} + (\log n)^{-\delta/2} + (\log n)^{-1/2} \right)$$

on which

$$\frac{F_{8B,Z_n}(k)}{t_k} \leq (\log n)^{-\delta/2}, \quad \frac{L_{8B}(T_n(k))}{t_k} \leq (\log n)^{-1/2}.$$

Proof. For fixed δ , $8B < Z_n \leq B^2$ eventually, and

$$\log(Z_n/(3n)) = 4 \log n + (1 - \delta) \log \log n - \log 3 + o(1) \sim 4 \log n.$$

Theorem 253.3 gives mean $O_\delta((\log n)^{-\delta})$. Markov at $(\log n)^{-\delta/2}$ gives its displayed error. The full-mass estimate in Theorem 228.1, with Section 229’s elementary inputs, at cutoff $8B$ has mean $O(1/\log n)$. Markov at $(\log n)^{-1/2}$ gives the other error. Intersect these sets with Theorem 246.1. In particular the interval $(B, 8B]$ is paid within the full mass up to $8B$, without imposing a new cap on its depths. $\square$

Let $W_{Z_n}(k)$ denote the signed contribution above Z_n . The exact prime decomposition gives on this set

$$(920) \quad \frac{J(T_n(k))}{t_k} \leq \frac{W_{Z_n}(k)}{t_k} + (\log n)^{-\delta/2} + (\log n)^{-1/2},$$

$$(921) \quad \frac{\log \text{rad}(T_n(k))}{t_k} \geq 1 - \frac{4}{3n} - \frac{(\log n)^{-\delta/2} + (\log n)^{-1/2}}{3} - \frac{(W_{Z_n}(k))_+}{3t_k}.$$

The second inequality uses the actual elementary angular estimate. Taking $\delta = 1$ reaches $Z_n = n^5$; any fixed $0 < \delta < 1$ gives the stated logarithmic expansion. The complete positive excess in this interval is controlled on the proved domain. Its full mass need not be small, and no size of compensating negative credit is presumed.

The fifteen declarations of `UniformMomentArithmetic.lean` formalize the numeric binomial gate, an explicit integer square-root threshold, the full finite-depth list ledger and its real weighted versions, and the constants 38 and 80 under explicit interfaces. The list count equals the actual filtered-list length. The threshold $\lfloor \sqrt{n} \rfloor + 1$ used in the code has the same required bounds and also permits square n in the numeric statements. Finite-ring construction, ideal valuations, the uniform tuple height estimate and the analytic estimates are not included in those fifteen formal claims. The unbounded farther signed cost, the complementary roots whose norms are smooth at this threshold, and the remaining exceptional subsets are still unresolved.

254. COLORED NORM SUPPORT AND A DENSITY-WINDOW TRADEOFF

For an integer $B \geq 1$, retain the actual block

$$I_B = \{k \in \mathbb{Z} : B \leq k < 2B\}, \quad a_k = 3k, \quad w_k = a_k + \zeta, \quad Q_k = a_k^2 + a_k + 1, \quad \alpha_k = w_k / \bar{w}_k.$$

Here $\zeta^2 = \zeta - 1$. The roots are primitive and unramified, and $9B^2 \leq Q_k < 49B^2$. For real $x \geq 3$, define

$$\mathcal{S}(B, x) = \{k \in I_B : \text{some prime } p > x \text{ divides } Q_k\}.$$

This is a condition on the actual norms, fixed before a boundary prime, depth or tuple length is chosen. All prime sums below range over rational primes.

Theorem 254.1 (An explicit finite independent coloring). *For every integer $B \geq 1$ and real $x \geq 3$, put*

$$m = \left\lfloor \frac{\log(49B^2)}{\log x} \right\rfloor, \quad D = 1 + m \left(2 \left\lfloor \frac{B}{x} \right\rfloor + 1 \right).$$

Then $\mathcal{S}(B, x)$ is a disjoint union of at most D classes on each of which the actual ratios α_k are multiplicatively independent. The empty case is included.

Proof. A prime dividing Q_k is neither 2 nor 3. Since $a_k^3 \equiv 1 \pmod{p}$ and $a_k \not\equiv 1 \pmod{p}$, it is 1 modulo 3. The polynomial $9X^2 + 3X + 1$ has exactly two simple roots modulo such a prime: its discriminant is -27 .

Join distinct indices in $\mathcal{S}(B, x)$ if their norms share some prime above x . A norm has at most m distinct such primes, since their product is at most $Q_k < 49B^2$. For each one of them, the two residue classes meet the block in at most $2\lfloor (B-1)/p \rfloor + 2$ indices. Removing the given vertex leaves at most

$$2 \left\lfloor \frac{B-1}{p} \right\rfloor + 1 \leq 2 \left\lfloor \frac{B}{x} \right\rfloor + 1$$

possible neighbors. Thus every vertex has degree at most $D-1$. Color vertices in increasing order, always taking the first of D colors not used by earlier neighbors. At most $D-1$ colors are forbidden, so this finite algorithm succeeds. It does not invoke a matroid-partition theorem.

Within a color, choose for each root one prime $p > x$ dividing its norm. It divides no other norm of that color. At one oriented prime ideal above p , the ratio α_k has nonzero integer valuation d_k . Both conjugate factors cannot occur, since a common ideal divisor would divide $w_k - \bar{w}_k = \sqrt{-3}$. Every other ratio in the same color has valuation zero at this ideal. Applying the valuation to a multiplicative relation gives $c_k d_k = 0$, hence $c_k = 0$, for each index. The

witness need not have depth one; nonzero depth suffices. Small norm primes may be shared. If $m = 0$, the norm inequality forces the original vertex set to be empty. $\square$

The colors depend on the actual norm factorizations, B, x , and the chosen finite ordering, but not on any later boundary prime. No independent coloring of the norm-smooth complement has been proved by this argument.

Theorem 254.2 (Full valuation mass with its cutoff endpoint). *There is an absolute constant C such that, for all integers $B \geq 3$ and reals $3 \leq x \leq B$, with $\Lambda_B = \log(49B^2)$,*

$$(922) \quad \sum_{k \in I_B} \sum_{p \leq x} v_p(Q_k) \log p \leq B \log x + CB \log \log(3x) + \frac{Cx\Lambda_B}{\log x}.$$

Consequently

$$(923) \quad |\mathcal{S}(B, x)| \geq \max \left\{ 0, \frac{2B \log B - B \log x - CB \log \log(3x) - Cx\Lambda_B / \log x}{\Lambda_B} \right\}.$$

Proof. For a supported prime p , each of the two simple residue roots lifts uniquely at every depth. Put $H_p = \lfloor \Lambda_B / \log p \rfloor$. Counting the two residue classes at every level gives

$$\sum_{k \in I_B} v_p(Q_k) \log p \leq \frac{2B \log p}{p-1} + 2H_p \log p.$$

Only $p \equiv 1 \pmod{3}$ occur. The fixed-modulus estimate $\vartheta(x; 3, 1) = x/2 + O(x/\log x)$, from Theorem 1.2, printed page 5, formula (1.12), of [Bennett, Martin, O'Bryant and Rechnitzer](#), implies by partial summation

$$\sum_{\substack{p \leq x \\ p \equiv 1 \pmod{3}}} \frac{\log p}{p-1} = \frac{1}{2} \log x + O(\log \log(3x)).$$

The error integral is $O(\int^x dt/(t \log t))$, and replacing $1/p$ by $1/(p-1)$ costs a convergent series. Increasing an absolute constant covers all $x \geq 3$. For the complete endpoint,

$$\sum_{\substack{p \leq x \\ p \equiv 1 \pmod{3}}} 2H_p \log p \leq 2\Lambda_B \pi(x) \ll \frac{x\Lambda_B}{\log x}.$$

This proves (922). The total actual norm mass is at least $2B \log B$; the mass above x is supported on $\mathcal{S}(B, x)$ and is at most Λ_B per member. Subtraction and division prove (923). $\square$

Corollary 254.3 (Uniformity on a fixed range of support exponents). *Fix $0 < \theta_0 \leq 1$. Uniformly for $\theta \in [\theta_0, 1]$, as integer B tends to infinity,*

$$(924) \quad |\mathcal{S}(B, B^\theta)| \geq (1 - \theta/2)B - O_{\theta_0}(B \log \log B / \log B),$$

$$(925) \quad D \leq (1 + 9/\theta_0)B^{1-\theta}.$$

The second bound holds already for $B \geq 49$ and $B^\theta \geq 3$.

Proof. Here $\Lambda_B \leq 3 \log B$, so the endpoint in (922) is at most $3CB^\theta/\theta_0$. The main term is $\theta B \log B$; division by $\Lambda_B = 2 \log B + O(1)$ gives (924). Also $m \leq 3/\theta_0$ and $2\lfloor B^{1-\theta} \rfloor + 1 \leq 3B^{1-\theta}$, proving (925). No uniform limit as θ_0 tends to zero is asserted. This uses the whole block, not an iteration of its mass estimate on an arbitrary discarded subset. $\square$

For the boundary estimates, let $n > 324$ be prime and $B = n^4$. Writing $w_k^n = A + C\zeta$, put

$$T_n(k) = |AC(A + C)|, \quad c_n(k) = \max\{|A|, |C|, |A + C|\}, \quad t_k = \log c_n(k).$$

A positive-sector unit rotation preserves c_n, T_n . The elementary actual bounds are

$$t_k \geq n \log(3B), \quad \log T_n(k) \leq 3n \log(6B + 1).$$

There are no zero boundary coordinates for these primitive unramified powers.

Theorem 254.4 (Uniform moments summed over the actual colors). *For every prime $n > 324$, $B = n^4$, real $x \geq 3$ and $8B < Z \leq B^2$, define*

$$F_{8B,Z}(k) = \sum_{8B < q \leq Z} (v_q(T_n(k)) - 3)_+ \log q.$$

With the exact color budget D from Theorem 254.1,

$$(926) \quad \frac{1}{B} \sum_{k \in S(B,x)} \frac{F_{8B,Z}(k)}{t_k} \leq \frac{80DZ}{Bn \log(Z/(3n))}.$$

Proof. We spell out why the uniform-moment proof applies to each color. For every positive integer ν , two actual ν -fold products of block roots have coordinates satisfying

$$|R|, |R'| \leq 2(7B)^\nu, \quad |S|, |S'| \leq 2\nu(7B)^{\nu-1}.$$

The second estimate is the finite telescoping bound for the imaginary coordinate. For a prime $q > 8B$, the determinant height divided by $q^{2\nu}$ is strictly less than

$$\frac{8}{7B} \nu (49/64)^\nu < \frac{512}{105B} < 1.$$

The finite geometric sum proves this uniformly for every finite ν . Thus congruent ratio products with unit denominators are exactly equal over $\mathbb{Q}(\zeta)$.

Every actual $q^{2\nu}$ -hit has unit norm and gives a norm-one $3n$ -torsion element modulo $q^{2\nu}$. Since $q > n$, the split/inert simple-lifting calculation of Lemma 245.1 gives at most $3n$ such elements at every depth. Within a color, Theorem 254.1 recovers the multiplicities from equal actual products. Thus multisets of ν deep roots inject into this group, giving

$$\binom{M_\nu + \nu - 1}{\nu} \leq 3n$$

when $M_\nu > 0$, with zero positive-length multisets when it is empty. In particular $M_2 \leq \sqrt{6n}$. For $r = \lceil \sqrt{n} \rceil$, four roots would give $3n \geq \binom{r+3}{3} > n^{3/2}/6 > 3n$. Hence $M_r \leq 3$.

Every positive layer is paid by $(e-3)_+ = \sum_{j \geq 4} \mathbf{1}_{e \geq j}$. In one color the layers $4 \leq j < 2r$ contribute at most $4\sqrt{6}n \log q < 10n \log q$. The remaining layers have at most three roots and the full height cap gives at most $9nL$, where $L = \log(6B+1)$. Summing colors therefore gives

$$\sum_{k \in S(B,x)} (v_q(T_n(k)) - 3)_+ \log q \leq D(10n \log q + 9nL).$$

For $q \leq Z \leq B^2$, both $\log q$ and L are at most $2 \log B$. The full-block normalized cost is at most $38D/B$ per prime.

Positive excess has homogeneous rank n . Rank one would, since $q > n$, force q^2 into one of the coprime old factors $a_k, a_k + 1 < 6B + 1$, which is impossible. Thus n divides the norm-one group order $q - \chi(q)$, where $\chi(q) = (-3/q)$. That order is also divisible by 3, and $(n, 3) = 1$, so $3n \mid q - \chi(q)$. The order of α_k itself may be n or $3n$; the argument does not require the latter. In particular $q \equiv \pm 1 \pmod{3n}$.

Montgomery–Vaughan’s Brun–Titchmarsh inequality, Theorem 2, printed page 121, formula (1.10), in the [author-hosted primary paper](#), bounds these two classes through Z by

$$\frac{2Z}{(n-1) \log(Z/(3n))}.$$

Its interval form gives the through- Z version by decreasing the left endpoint to zero. Multiply by $38D/B$, and use $76n/(n-1) < 80$, to obtain (926). $\square$

Theorem 254.5 (A density–window tradeoff). *Fix $3/4 < \theta \leq 1$ and $\delta > 0$. For sufficiently large prime n , put $B = n^4$, $\kappa = 4\theta - 3$, and*

$$Z_n = \left\lfloor Bn^\kappa (\log n)^{1-\delta} \right\rfloor.$$

Then $8B < Z_n \leq B^2$, the complete positive-excess mean on $\mathcal{S}(B, B^\theta)$ is $O_{\theta,\delta}((\log n)^{-\delta})$, and there is an actual subset $\mathcal{P}_{n,\theta,\delta}$ of I_B with relative size at least

$$1 - \frac{\theta}{2} - O_{\theta,\delta}\left(\frac{\log \log n}{\log n} + (\log n)^{-\delta/2} + (\log n)^{-1/2}\right)$$

on which

$$k \in \mathcal{S}(B, B^\theta), \quad \frac{F_{8B,Z_n}(k)}{t_k} \leq (\log n)^{-\delta/2}, \quad \frac{L_{8B}(T_n(k))}{t_k} \leq (\log n)^{-1/2}.$$

Here $L_X(T) = \sum_{q \leq X} v_q(T) \log q$ is the full mass.

Proof. For fixed $\kappa > 0$, the factor $n^\kappa (\log n)^{1-\delta}$ tends to infinity and is $o(n^4)$. Thus the cutoff satisfies the asserted domain, and

$$\log(Z_n/(3n)) = (3 + \kappa) \log n + (1 - \delta) \log \log n - \log 3 + o(1).$$

By (925), $D = O_\theta(n^{4(1-\theta)})$, and the power of n in (926) cancels exactly: $4(1 - \theta) + \kappa - 1 = 0$. This proves the mean bound. Markov at $(\log n)^{-\delta/2}$ removes the corresponding fraction of the entire block. Theorem 228.1, with Section 229's elementary inputs, gives at cutoff $8B$ a full-block mean $O(1/\log n)$; Markov removes $O((\log n)^{-1/2})$ more. Intersect these two actual good sets with (924). All means remain normalized by B . In particular $(B, 8B]$ is paid as full mass, without a cap imposed there. For fixed $\theta_0 > 3/4$ the constants and sufficiently-large threshold are uniform in $\theta \in [\theta_0, 1]$, with fixed δ . $\square$

Define $J(T) = \log T - 3 \log \text{rad}(T)$ and let $W_Z(T)$ be the signed contribution of supported primes above Z . On the preceding set, with $\varepsilon_n = (\log n)^{-\delta/2} + (\log n)^{-1/2}$, exact disjoint prime decomposition gives

$$(927) \quad \frac{J(T_n(k))}{t_k} \leq \frac{W_{Z_n}(T_n(k))}{t_k} + \varepsilon_n,$$

$$(928) \quad \frac{\log \text{rad}(T_n(k))}{t_k} \geq 1 - \frac{4}{3n} - \frac{\varepsilon_n}{3} - \frac{(W_{Z_n}(T_n(k)))_+}{3t_k}.$$

The first inequality uses the full low mass and the middle positive excess. The second uses the elementary block bound $\log T_n(k)/t_k \geq 3 - 4/n$. The far signed term keeps its negative credit; its positive part remains an unproved membership gate, not a consequence of this finite-window average.

The exponent $\kappa \in (0, 1]$ has proved domain fraction $(5 - \kappa)/8$. For example $\theta = 15/16$ gives $\kappa = 3/4$, fraction $17/32$, and cutoff $\lfloor Bn^{3/4}(\log n)^{1-\delta} \rfloor$. For $\kappa < 1/2$, the earlier MC result already gives an intermediate-window domain of fraction tending to one, so this coloring is additional structure rather than improved coverage there. The useful coverage tradeoff is $1/2 < \kappa < 1$, beyond the MC power range and with a fraction larger than one half. No unproved endpoint comparison at $\kappa = 1/2$ is asserted.

The norm-smooth complement, its possible rank deficiency, and the unbounded farther signed tail remain open. Exact pair-product collisions disprove a universal one-color assertion, but do not refute a stronger multi-color theorem for all actual roots. The full-block norm-mass proof cannot simply be iterated on an arbitrary discarded subset. The coloring and ideal witnesses above are ordinary proofs; no new Lean formalization of them or of the analytic domain theorem is claimed by this section.

255. A COMPLETE LOCAL OBSTRUCTION FOR THE IDENTITY CUBIC UNIT

The identity unit $u = 1$ must be distinguished from $u = \zeta$. The explicit cubic quotient in Section 250 gives a complete local exclusion for the former, while leaving the two nonidentity unit classes open.

Theorem 255.1 (The projective cubic quotient has no 3-adic point). *Put*

$$C(S, T) = S^3 + 3S^2T - T^3,$$

$$E_3(S, T) = S^3 - S^2T - 4ST^2 - T^3, \quad D(S, T) = -3E_3(S, T).$$

Let H be the smooth projective double cover

$$(929) \quad Y^2 = C(S, T)D(S, T),$$

where Y has weight 3. Then $H(\mathbb{Q}_3) = \emptyset$. This is the third quotient in Theorem 250.1 at $g = 3$, $u = 1$.

Proof. The binary sextic has six simple branch points by the proved HG calculation. A rational point maps to $\mathbb{P}^1(\mathbb{Q}_3)$. Choose $S, T \in \mathbb{Z}_3$ with $\min(v_3(S), v_3(T)) = 0$. Its fibre would supply a $Y \in \mathbb{Q}_3$ satisfying

$$Y^2 = -3C(S, T)E_3(S, T).$$

This includes infinity, represented by $(1, 0)$. Scaling the base coordinates changes the right side by a sixth power, so the square criterion does not depend on that choice.

On the four residue points represented by $(1, 0), (0, 1), (1, 1), (-1, 1)$, the values of E_3 are respectively $1, -1, 1, 1$ modulo 3. Thus $E_3(S, T)$ is always a unit. If $S \not\equiv T \pmod{3}$, then $C(S, T) \equiv (S - T)^3$ is a unit as well. The right side has valuation 1, so cannot be a square.

Otherwise T is a unit and $S = T + 3K$, with $K \in \mathbb{Z}_3$. The exact expansion is

$$(930) \quad C(T + 3K, T) = 3(T^3 + 9KT^2 + 18K^2T + 9K^3).$$

Hence $v_3(C) = 1$, $C/3 \equiv T^3 \pmod{3}$, and also $E_3 \equiv T^3 \pmod{3}$. The right side has valuation 2, but after division by 9 its unit part is $-T^6 \equiv -1 \pmod{3}$. A 3-adic unit square has residue 1. This is another contradiction, covering every projective fibre. $\square$

Corollary 255.2 (Propagation to identity-unit exponents divisible by three). *For every odd positive integer g divisible by 3, the curves of Section 248 satisfy*

$$D_{g,1}(\mathbb{Q}_3) = C_{g,1}(\mathbb{Q}_3) = \emptyset.$$

Their entire $\mathbb{Q}$ -rational loci are therefore empty.

Proof. For $g = 3$, the quotient morphism $D_{3,1} \rightarrow H$ from Theorem 250.1 and Theorem 255.1 give the claim. For $g = 3k$, successive powers give the identity of projective maps

$$L_{1,g} = L_{1,3} \circ L_{1,k}.$$

On the generic fibre, $(t, s) \mapsto (t, L_{1,k}(s))$ defines a nonconstant rational map $D_{g,1} \dashrightarrow D_{3,1}$. The curve extension theorem [140] extends it over the smooth projective source, including all infinite and zero-denominator chart points. It is defined over $\mathbb{Q}$, hence over $\mathbb{Q}_3$. A local point would map to a forbidden point of $D_{3,1}$. The forgetful morphism $C_{g,1} \rightarrow D_{g,1}$ proves the assertion for the higher cover. $\square$

For an actual primitive seed with $M = a^2 + ab + b^2 = U^2$, the elementary norm calculation already gives $3 \nmid M$: its primitive 3-adic depth is at most one and must be even. An identity-unit integral cube $(S + T\zeta)^3$, however, has ζ -coefficient $3ST(S + T)$, so cannot equal $ab + M\zeta$. Thus this integral subcase already has an elementary obstruction. The new curve statement records the full local obstruction, including nonpositive and infinite charts, and its projective propagation. It does not exclude $u = \zeta, \zeta^2$, moving residuals, or exponents not divisible by three.

The six declarations of `CubicUnitLocalArithmetic.lean` prove the actual shift identity, modular reduction of the sextic, the full kernel-checked `Fin(27)` table, its transfer to all integer coordinates, and the impossibility of an integer square at primitive integer coordinates. The finite table has no axiom dependencies; the other queries use only the standard Lean axioms. These arithmetic statements do not formalize $\mathbb{Q}_3$, the projective curves, or the propagation morphism.

256. AN EXACT ELLIPTIC GATE FOR A NONIDENTITY CUBIC UNIT

The unit $u = \zeta$ at $g = 3$ remains after the identity-unit exclusion. Its first quotient gives a concrete genus-two curve and an elliptic lifting condition. The following ordinary proofs use no software rank output.

Theorem 256.1 (Two explicit elliptic quotients). *Let H be the smooth projective genus-two curve*

$$(931) \quad y^2 = P(s), \quad P(s) = s^6 + 3s^5 - 3s^4 - 11s^3 - 3s^2 + 3s + 1.$$

This is the first HG quotient for $g = 3, u = \zeta$. It has degree-two morphisms to

$$(932) \quad E : Y^2 = X^3 - 9X - 9, \quad E' : (Y')^2 = (X')^3 - 189X' + 999,$$

given on finite charts by

$$(933) \quad X = -\frac{2s^2 + 5s + 2}{(s+1)^2}, \quad Y = \frac{y}{(s+1)^3},$$

$$(934) \quad X' = \frac{3(2s^2 - 7s + 2)}{(s-1)^2}, \quad Y' = -\frac{9y}{(s-1)^3}.$$

Moreover $\text{Jac}(H)$ is isogenous over $\mathbb{Q}$ to $E \times E'$.

Proof. The identity-unit cubic coordinates are $A_0 = s^3 - 3s - 1$, $B_0 = 3s(s+1)$. Multiplication by ζ gives $A = -B_0$, $B = A_0 + B_0$; thus $B(A+B) = A_0(A_0+B_0) = P$. On the chart $s \neq 1$, put

$$t = \frac{s+1}{s-1}, \quad v = y(t-1)^3.$$

Direct expansion yields

$$(935) \quad v^2 = -9t^6 + 99t^4 - 27t^2 + 1.$$

The inverse is $s = (t+1)/(t-1)$, $y = v/(t-1)^3$; the birational change extends to the smooth projective models. In this chart the elliptic coordinates are

$$\begin{aligned} X &= (1/t^2 - 9)/4, & Y &= v/(8t^3), \\ X' &= (33 - 9t^2)/4, & Y' &= -9v/8. \end{aligned}$$

Substitution proves the equations and (933)–(934). The two discriminants are 11664 and 944784, respectively. The second map is the quotient by $\sigma : (t, v) \mapsto (-t, v)$; the first is the quotient by $\sigma\iota : (t, v) \mapsto (-t, -v)$, where ι is the hyperelliptic involution. The indicated invariants together with t^2 generate the quotient fields. The involutions are nontrivial, so both maps have degree two, and extend over the smooth projective models.

Let Φ be the pair of quotient norms and Ψ the sum of pullbacks. The hyperelliptic quotient is $\mathbb{P}^1$; its divisor norm/pullback identity gives $\iota = -1$ on $\text{Jac}(H)$. The other two double covers therefore give

$$(936) \quad \Psi\Phi = (1 + \sigma) + (1 + \sigma\iota) = [2].$$

These divisor identities include the ramified fibres. Multiplication by two is surjective, hence so is Ψ . Both sides have dimension two, so Ψ is a $\mathbb{Q}$ -isogeny. This is the explicit arithmetic norm/pullback argument of Theorem 250.2. $\square$

Theorem 256.2 (The complete rational image of the first quotient). *Under (933), the image of $H(\mathbb{Q})$ is exactly*

$$(937) \quad \{O\} \cup \{(X, Y) \in E(\mathbb{Q}) : 4X + 9 \text{ is a square in } \mathbb{Q}\}.$$

Proof. Clearing the abscissa formula gives

$$(938) \quad (X+2)s^2 + (2X+5)s + (X+2) = 0,$$

with discriminant $4X+9$. At $s = -1$, $P(-1) = 1$, and the two rational points map to O . The two rational points at infinity, and those at $s = 0$, map to $(-2, 1)$, $(-2, -1)$. Thus every chart exception is also in (937).

Conversely, let a finite elliptic point have $4X + 9$ square. If $X = -2$, then $Y = \pm 1$, and $s = 0, y = Y$ is a preimage. Otherwise (938) has a rational root s . This root cannot be -1 , since the left side at -1 is -1 . Put $y = Y(s + 1)^3$. The polynomial identity behind (933) gives $y^2 = P(s)$. The repeated-discriminant case cannot occur on $E(\mathbb{Q})$: at $X = -9/4$ its cubic has value $-9/64$. Finally O has the two rational preimages already listed. $\square$

This is one quotient condition for the actual seed problem. A point of $D_{3,\zeta}$ must also have the second HG square root at the same source s , with $0 < A(s)/B(s) < 1/3$. Arbitrary elliptic points, or quotient points at different source parameters, do not construct an actual seed.

Theorem 256.3 (Two independent rank contributions). *Both elliptic curves in (932) have points of infinite order. Thus $\text{Jac}(H)(\mathbb{Q})$ has Mordell–Weil rank at least two.*

Proof. Take $P = (-2, 1) \in E(\mathbb{Q})$ and $P' = (6, 9) \in E'(\mathbb{Q})$. Direct doubling gives

$$2P = (25/4, -107/8), \quad 2P' = (33/4, 9/8).$$

For a short integral equation $y^2 = x^3 + ax + b$ with a, b odd,

$$x(2Q) = \frac{x^4 - 2ax^2 - 8bx + a^2}{4(x^3 + ax + b)}.$$

If $v_2(x) = s < 0$, the unique smallest numerator and denominator valuations are $4s$ and $2 + 3s$. Hence $v_2(x(2Q)) = s - 2$. Both displayed doubled points start at valuation -2 , and their repeated doublings have distinct abscissas with valuations $-2, -4, -6, \dots$. Both original points therefore have infinite order. The $\mathbb{Q}$ -isogeny of Theorem 256.1 and the Mordell–Weil theorem give the rank lower bound. No assertion that these points generate the full groups is used. $\square$

Consequently a rank-less-than-genus argument for this genus-two quotient cannot be justified. Other descents and simultaneous square conditions remain available. The gate itself is strict: P lifts but $2P$ does not, since $4(25/4) + 9 = 34$ is not a rational square. This makes no claim about all higher multiples.

The separate bounded PARI/GP search, checked by exact rational arithmetic, found 24 affine points across the nine cubic quotient models with numerator and denominator bounded by 1000. All had $s = 0$ or -1 , and none supplied the positive common-source locus. A distinct exhaustive rational-abscissa check at bound 80 checked 70767 curve-abscissa cases. The same replay checked all 648 primitive coordinate classes modulo 27 for Theorem 255.1. These finite checks do not determine all rational points. Diagnostic software rank outputs are not used in the ordinary proofs above, and no genus-two or elliptic-curve theorem here is claimed as a complete Lean formalization.

257. ACTUAL PRIVATE VALUATIONS AND COMPLETE FINITE DEPTH BUDGETS IN LEAN

The finite formalization starts with actual products and actual index sets. Let I be a finite set, G a commutative monoid, and $a_i \in G$. Suppose homomorphisms $v_i : G \rightarrow (\mathbb{Z}, +)$ satisfy

$$v_i(a_j) = 0 \quad (j \neq i), \quad d_i = v_i(a_i) \neq 0.$$

In Lean the additive target is expressed as `Multiplicative Int`, so these are ordinary multiplicative monoid homomorphisms. The integers d_i may have either sign.

Lemma 257.1 (Actual multiset injectivity). *The map taking a finite multiset of indices to the product of its actual elements a_i is injective.*

Proof. For a multiset s , write $c_i(s)$ for its multiplicity at i . Commutativity gives

$$\prod_{i \in s} a_i = \prod_{i \in I} a_i^{c_i(s)}, \quad v_i\left(\prod_{j \in s} a_j\right) = c_i(s)d_i.$$

Equal products therefore imply $c_i(s)d_i = c_i(t)d_i$ for every i . Cancel the nonzero integer d_i to obtain equality of all multiplicities, and hence of the multisets. No tuple-injectivity assumption is needed. $\square$

Give each actual index a natural depth e_i , and define $I_h = \{i \in I : e_i \geq h\}$. Private witnesses restrict to every I_h . Let H_h be a finite commutative monoid with $|H_h| \leq 3n$, and suppose $\phi_h : I_h \rightarrow H_h$. At a specified positive integer moment ν , assume that equality of the mapped products of two ν -multisets implies equality of their original products in G . Lemma 257.1 then gives

$$(939) \quad \binom{|I_h| + \nu - 1}{\nu} \leq 3n.$$

This counts actual multisets with repetitions, including the empty-index case when $\nu > 0$. In the formal library the cardinality is that of `Sym` applied to the actual depth subtype.

Proposition 257.2 (The finite private-root depth bridge). *Let $n > 324$ and C be natural numbers, put $r = \lfloor \sqrt{n} \rfloor + 1$, and assume $e_i \leq C$ for every i . Suppose the two finite targets and actual-product rigidity assumptions above are supplied at $(h, \nu) = (4, 2)$ and $(2r, r)$. Then*

$$(940) \quad \sum_{i \in I} (e_i - 3)_+ \leq 10n + 3C.$$

For $w \geq 0$ with $Cw \leq 3nL$, one also has

$$(941) \quad w \sum_{i \in I} (e_i - 3)_+ \leq 10nw + 9nL.$$

If $B, s > 0$, $w \leq 2s$ and $L \leq 2s$, division by Bns bounds the left side of (941) after normalization by $38/B$.

Proof. The two-multiset instance of (939) gives $|I_4|^2 \leq 6n$. Also $n \leq r^2 \leq 4n$. If $|I_{2r}| \geq 4$, then

$$3n \geq \binom{|I_{2r}| + r - 1}{r} \geq \binom{r + 3}{3} > \frac{r^3}{6} \geq \frac{n^{3/2}}{6} > 3n,$$

a contradiction. Hence $|I_{2r}| \leq 3$. For each index separately,

$$(e_i - 3)_+ \leq 2r \mathbf{1}_{e_i \geq 4} + C \mathbf{1}_{e_i \geq 2r}.$$

Summing counts each actual index once, even when different indices have the same depth. The first contribution is at most $2\sqrt{4n}\sqrt{6n} < 10n$ and the second at most $3C$. Multiplication by the nonnegative weight and the stated height bound prove (941). Finally its right side is at most $38ns$; all three factors in the denominator are positive. $\square$

The precise formal scope. There are 37 new declarations in four modules. `PrivateValuationMultisets` contributes eight actual product, private-valuation, symmetric-product and finite-count theorems. `UniformMomentArithmetic` contributes 15 numerical, actual-list, complete-layer and normalization theorems. `ActualUniformDepthBudget` contributes eight theorems on restricted private witnesses, actual depth subtypes, finite sums, and the complete connection in Proposition 257.2. The formal weighted sums explicitly perform natural-number truncated subtraction before casting the complete sum to the reals.

Finally, `CubicUnitLocalArithmetic` contributes six theorems on the actual integer sextic in the local obstruction. A kernel-checked table modulo 27 implies that an integer square value forces both homogeneous coordinates to be divisible by three. It therefore excludes every primitive integer coordinate pair. The finite table uses `decide`, with no native computation axiom.

All four sources were freshly compiled together with Lean 4.32.0 and Mathlib revision 81a5d257c8e410db227a6665ed08f64fea08e997. All 37 explicit axiom queries passed; their union consists only of `propext`, `Classical.choice` and `Quot.sound`. The fixed Mathlib compiled cache was reused. The ordinary proofs and independent internal review preceded the corresponding implementation.

The final private-root theorem derives both binomial count premises from the supplied private witnesses and actual product rigidity. It does not construct the arithmetic ideal valuations, finite torsion groups, their lifts, or the height cap. The formal natural depths remain a supplied

function on a finite index type. The ideal, sieve, density, $\mathbb{Q}_3$ -geometric and elliptic arguments retain their separately reviewed ordinary-proof status. None of these declarations proves *abc*.

258. THIRTEENTH CONTINUATION: DEPENDENCIES AND REMAINING OBSTACLES

The analytic and geometric branches have advanced independently. Uniform determinant control permits every finite moment at the same lower prime cutoff $8B$. Private norm support supplies an actual multiplicatively independent domain. The complete layer budget reaches $Bn(\log n)^{1-\delta}$ on a domain of asymptotic relative size at least one half. Coloring actual norm support gives the further tradeoff $(5 - \kappa)/8$ at cutoff $Bn^\kappa(\log n)^{1-\delta}$, for fixed $0 < \kappa \leq 1$. All positive layers of these intervals are paid. The argument preserves the signed cost beyond their upper endpoint.

The geometric local obstruction excludes the identity-unit curves for all odd exponents divisible by three, including their projective points. This is a proved obstruction for that specified branch. The two nonidentity units remain live. For one cubic nonidentity quotient, two explicit elliptic maps give a rational product isogeny, an exact square condition for lifting, and a proved Jacobian rank lower bound of two. Mordell–Weil is used only in its standard finite-generation form; see Milne’s Theorem 16.7, printed page 69¹. The bounded rational-point search and software rank observations are distinct from these proofs.

The main unresolved dependencies are the full unbounded signed packet, roots outside the proved norm-support domains, and the positive common-source rational locus for the remaining geometric unit classes. Neither fixed genus at one exponent nor finite searches give a uniform height bound as the exponent or residual coefficients vary. The far-tail incidence, norm-smooth complement, higher descent and moving-residual routes are retained. A local obstruction retires only its precisely stated branch, and a rank obstruction only the specific rank-less-than-genus criterion.

The 37 finite formal declarations are compiled and audited separately from the ordinary analytic and geometric arguments. The proof chain still has the substantive open dependencies just listed. There is no complete proof or disproof of *abc*, and no unconditional Lean theorem asserting it. Subsequent adaptive-precision and owner-concentration candidates are maintained separately for the next continuation.

¹J. S. Milne, *Abelian Varieties*, <https://www.jmilne.org/math/CourseNotes/AVc.pdf>.

259. ADAPTIVE PRECISION AND TWO MAXIMAL DEPTHS AT EVERY LARGE PRIME

Let $n > 324$ be prime, $B = n^4$, and $L = \log(8B)$. For integers $B \leq k < 2B$, put

$$w_k = 3k + \zeta, \quad Q_k = N(w_k), \quad \alpha_k = w_k/\bar{w}_k, \quad \zeta^2 = \zeta - 1.$$

Fix a subset I of this actual block on which the ratios α_k are multiplicatively independent. The private-norm class of Theorem 246.1, or one fixed color of Theorem 254.1, supplies such a set. For $w_k^n = A_k + C_k\zeta$, write

$$T_k = |A_k C_k (A_k + C_k)|, \quad c_k = \max(|A_k|, |C_k|, |A_k + C_k|), \quad t_k = \log c_k.$$

The actual powers are primitive and unramified. Their three nonzero boundary arms are therefore pairwise coprime.

Lemma 259.1 (The individual-arm cap). *For every $k \in I$,*

$$n \log(3B) \leq t_k < nL.$$

For every prime $q > 8B$ dividing T_k ,

$$(942) \quad v_q(T_k) \log q \leq t_k < nL.$$

Its homogeneous rank is n , and $q \equiv 1$ or $-1 \pmod{3n}$.

Proof. The norm satisfies $9B^2 \leq Q_k < 36B^2$. The quadratic-coordinate bound and the largest-arm norm inequality give

$$Q_k^{n/2} \leq c_k \leq \frac{2}{\sqrt{3}} Q_k^{n/2} < \frac{2}{\sqrt{3}} (6B)^n < (8B)^n.$$

The last strict inequality follows from $2/\sqrt{3} < 4/3 \leq (4/3)^n$. At a prime dividing T_k , exactly one arm is divisible by that prime; its entire multiplicity is bounded by the logarithm of that arm. This proves (942). Such a prime does not divide the norm, since the norm reduces to a nonzero coordinate square on each arm. The established homogeneous rank formula gives rank dividing n . Rank one would divide the old boundary $3k(3k+1)$, which is impossible for $q > 8B > 3k+1$. Thus the rank is n . The split/inert norm-one order formula gives $3n \mid q-1$ or $3n \mid q+1$, respectively. $\square$

Lemma 259.2 (Separate moment length and precision). *Let $q > 8B$ be prime, and let e, ν be positive integers with*

$$e \log q \geq 2\nu L.$$

If $M_e = \#\{k \in I : q^e \mid T_k\} > 0$, then

$$(943) \quad \binom{M_e + \nu - 1}{\nu} \leq 3n.$$

An empty hit set has zero positive-length multisets.

Proof. The proof of Lemma 253.1 bounds the integer determinant of any two actual ν -fold root products by

$$|D| \leq 8\nu 7^{2\nu-1} B^{2\nu-1} < (8B)^{2\nu}.$$

For completeness, division by $(8B)^{2\nu}$, with $x = 49/64$, bounds it by

$$\frac{8}{7B} \nu x^\nu < \frac{8}{7B} \sum_{j=0}^{\nu-1} x^j < \frac{512}{105B} < 1.$$

The hypothesis gives $q^e \geq (8B)^{2\nu}$. Equal ratio products modulo q^e imply $q^e \mid D$, hence $D = 0$ and exact ratio equality in $\mathbb{Q}(\zeta)$.

At every hit the conjugate denominator is a unit. The cubic boundary identity implies that the ratio is killed by $3n$ modulo q^e . The split/inert simple-lifting result of Lemma 245.1 bounds that norm-one torsion group by $3n$, since $q \nmid 3n$. The multiset product map lands in this group. Equality of images gives exact equality of algebraic ratio products; independence then recovers all multiplicities. The map is injective, and stars and bars proves (943). Repeated indices are included. $\square$

Theorem 259.3 (A complete bound at each prime). *For every prime $q > 8B$,*

$$(944) \quad \sum_{k \in I} (v_q(T_k) - 3)_+ \log q \leq 13nL,$$

$$(945) \quad \frac{1}{B} \sum_{k \in I} \frac{(v_q(T_k) - 3)_+ \log q}{t_k} \leq \frac{26}{B}.$$

There is no upper restriction on q .

Proof. Put $r = \lceil \sqrt{n} \rceil$ and

$$h = \max \left(4, \left\lceil \frac{2rL}{\log q} \right\rceil \right).$$

Lemma 259.2 with $(e, \nu) = (4, 2)$ gives $M_4 \leq \sqrt{6n}$. With $(e, \nu) = (h, r)$, it gives $M_h \leq 3$. Indeed, $M_h \geq 4$ would imply

$$3n \geq \binom{r+3}{3} > \frac{r^3}{6} \geq \frac{n^{3/2}}{6} > 3n,$$

using $n > 324$.

If $h = 4$, at most three indices have any positive excess, and (942) pays their entire depths by $3nL$. Otherwise h is the indicated ceiling and $(h-4) \log q \leq 2rL$. The full layers $4, \dots, h-1$ cost at most

$$(h-4)M_4 \log q \leq 2r\sqrt{6n}L \leq 4\sqrt{6}nL < 10nL.$$

Every layer from h onward is supported on at most three indices; paying their full depths adds at most $3nL$. The identity $(e-3)_+ = \sum_{j \geq 4} \mathbf{1}_{j \leq e}$ accounts for all layers. Subtracting four absorbs the ceiling error, without an extra $\log q$ term. This proves (944). Now use $t_k \geq n \log(3B)$ and $L \leq 2 \log(3B)$. $\square$

If the actual support is partitioned into D independent colors chosen before q , both bounds are multiplied by D . For a real $Z > 8B$, the same Brun–Titchmarsh input as Theorem 253.3 counts the two eligible progressions and gives

$$(946) \quad \frac{1}{B} \sum_{k \in I} \frac{\sum_{8B < q \leq Z} (v_q(T_k) - 3)_+ \log q}{t_k} \leq \frac{52Z}{B(n-1) \log(Z/(3n))} \leq \frac{55Z}{Bn \log(Z/(3n))}.$$

Here $\varphi(3n) = 2(n-1)$, and $52n/(n-1) < 55$ for $n > 324$. This remains true for $Z > B^2$, although its right side is generally too large there. On the established logarithmic window it improves a constant, rather than the asymptotic range.

For a real $Z \geq 8B$, define the actual finite set

$$\mathcal{D}_Z(I) = \{q > Z : q \text{ prime and } v_q(T_k) \geq 4 \text{ for some } k \in I\}.$$

Every label divides the nonzero integer $\prod_{k \in I} T_k$. Summing (945) over just these labels proves

$$(947) \quad \frac{1}{B} \sum_{k \in I} \frac{\sum_{q > Z} (v_q(T_k) - 3)_+ \log q}{t_k} \leq \frac{26|\mathcal{D}_Z(I)|}{B}.$$

The sufficient condition $|\mathcal{D}_Z(I)| = o(B)$ has not been proved. Counting roots at a fixed prime does not count different primes owned by different roots. This positive target is also stronger than a signed target that can use depth-one and depth-two credits; no equivalence is claimed.

Theorem 259.4 (Two maximal depths and the complete remainder). *For a prime $q > 8B$, put $e_k = v_q(T_k)$ and*

$$r = \lceil \sqrt{6n} \rceil, \quad h_0 = \max \left(4, \left\lceil \frac{6L}{\log q} \right\rceil \right), \quad h_1 = \max \left(h_0, \left\lceil \frac{2rL}{\log q} \right\rceil \right).$$

Let O_q be the $\min(2, |I|)$ indices of largest actual depth, with ties broken by index order. Then

$$(948) \quad \sum_{k \in I \setminus O_q} (e_k - 3)_+ \log q \leq 39n^{5/6}L,$$

$$(949) \quad \sum_{k \in I} (e_k - 3)_+ \log q \leq (2n + 39n^{5/6})L.$$

The normalized remainder is at most $78n^{-1/6}/B$.

Proof. At precisions 4, h_0, h_1 , respectively, Lemma 259.2 gives

$$M_4 \leq \sqrt{6n}, \quad M_{h_0} < 3n^{1/3}, \quad M_{h_1} \leq 2.$$

The middle bound follows from $M^3 \leq 6 \binom{M+2}{3} \leq 18n < 27n$. For the last bound, $M \geq 3$ would give

$$3n \geq \binom{r+2}{2} = \frac{(r+1)(r+2)}{2} > \frac{r^2}{2} \geq 3n.$$

Every h_1 -deep index belongs to O_q : an omitted such index and the two selected indices would contradict $M_{h_1} \leq 2$. Empty and singleton sets satisfy this directly.

For each remaining depth $e < h_1$, the exact finite inequality is

$$(950) \quad (e - 3)_+ \leq (h_0 - 4)\mathbf{1}_{e \geq 4} + (h_1 - h_0)\mathbf{1}_{e \geq h_0}.$$

It follows by splitting $e < 4$, $4 \leq e < h_0$, and $h_0 \leq e < h_1$. Both ceilings satisfy

$$(h_0 - 4) \log q \leq 6L, \quad (h_1 - h_0) \log q \leq 2rL,$$

including an empty range. Summing (950) bounds the complete remainder by

$$(6M_4 + 2rM_{h_0})L \leq 15\sqrt{n}L + 24n^{5/6}L \leq 39n^{5/6}L,$$

where $r \leq \sqrt{6n} + 1 \leq 4\sqrt{n}$. The at most two removed indices cost at most $2nL$ by the individual-arm cap. Normalizing as before proves the last assertion. $\square$

If $\log q \geq (r/2)L$, then $h_0 = h_1 = 4$, so every such prime with positive excess has at most two hit roots. Its actual prime-labelled incidence graph allows singleton edges and repeated edges with different prime labels. The choices O_q vary with q . Neither this graph nor the uniform remainder per prime bounds their unrestricted cumulative costs.

Finite formal scope. The separate module `AdaptiveOwnerArithmetic.lean` has twelve freshly compiled declarations. They prove the cubic and two-index binomial thresholds, actual natural ceiling bounds, selection of two maxima from a finite set, deep-set containment, (950) and its complete finite sum, the exact weighted remainder $(6M_4 + 2rM_{h_0})L$, and addition of at most two full caps. The final theorem derives a genuine selected set O and both finite budgets from explicit real-height and symmetric-count interfaces. Natural truncated subtraction is performed before the cast to real numbers. The old fifteen local declarations were also freshly compiled, with the pinned Mathlib cache reused; all axiom queries use only the standard three axioms.

The arithmetic construction of valuations, private witnesses, torsion and determinant rigidity remains ordinary mathematics here. The constants 39, 78 involving fractional real powers and the Brun–Titchmarsh consequences are not claimed as new Lean theorems. The roots outside I , pointwise exceptions, the changing maximal depths across different primes, and the full signed far tail remain unresolved.

260. A PRIVATE NORM DOMAIN STRICTLY LARGER THAN ONE HALF

For a positive integer B , put

$$I_B = \{k \in \mathbb{Z} : B \leq k < 2B\}, \quad Q(k) = 9k^2 + 3k + 1.$$

This section improves actual membership in the independent private-norm class. It does not estimate the unbounded boundary-prime tail.

Lemma 260.1 (A fixed discriminant after extracting a cofactor). *Let $\rho(m)$ count roots of Q modulo the positive integer m . Then ρ is multiplicative, $\rho(1) = 1$, and $\rho(p^j) = 2$ for $p \equiv 1 \pmod{3}$, $j \geq 1$, with zero otherwise. If $\rho(m) > 0$ and $0 \leq b < m$ is a root, the integer polynomial*

$$f_{m,b}(t) = \frac{Q(b+mt)}{m} = 9mt^2 + (18b+3)t + \frac{Q(b)}{m}$$

has discriminant -27 . Its prime root counts are

$$\rho_m(\ell) = \begin{cases} 2, & \ell \equiv 1 \pmod{3}, \ell \nmid m, \\ 1, & \ell \mid m, \\ 0, & \text{otherwise.} \end{cases}$$

They are independent of the chosen root b .

Proof. The prime-root and simple-lifting calculation for Q was used in Theorem 246.1; its discriminant is -27 and it has no root modulo two or three. Expanding the new discriminant gives

$$(18b+3)^2 - 36Q(b) = -27.$$

When $\ell \nmid m$, the affine variable change and division by m are invertible. When $\ell \mid m$, its quadratic coefficient vanishes and $18b+3 = Q'(b)$ is a unit, because the original root is simple. Thus it becomes a nonconstant linear polynomial. $\square$

For squarefree d , counting residue classes of the integer t in the real interval $[(B-b)/m, (2B-b)/m)$ gives

$$(951) \quad \#\{t : d \mid f_{m,b}(t)\} = \frac{B}{m} \frac{\rho_m(d)}{d} + R_d, \quad |R_d| \leq \rho_m(d).$$

No distribution theorem for quadratic prime values is assumed.

Lemma 260.2 (A cofactor-uniform upper sieve). *Define*

$$A(m) = \prod_{\ell \mid m} \frac{\ell}{\ell-2}.$$

Uniformly for every allowed m, b and every real $z \geq 2$, the number of integers in the preceding interval for which $f_{m,b}$ is prime and greater than z is at most

$$(952) \quad C \frac{BA(m)}{m \log z} + z^4$$

for an absolute constant C .

Proof. We use the finite upper-bound Selberg sieve of Ford, *Sieve Methods Lecture Notes, Spring 2023*, Theorem 4.1, printed page 45, with its parameter $D = z^2$. For local density $g(d) = \rho_m(d)/d$, the bound is

$$\frac{B/m}{J_m(z)} + \sum_{\substack{d_1, d_2 \leq z \\ d_1, d_2 \text{ squarefree}}} |R_{[d_1, d_2]}|, \quad J_m(z) = \sum_{\substack{d \leq z \\ d \text{ squarefree}}} h_m(d), \quad h_m(\ell) = \frac{g(\ell)}{1 - g(\ell)}.$$

All sums are restricted to the sieving support. Its local densities are less than one. Ford's equations (4.2)–(4.5) give the minimizing weights with absolute value at most one, and hence precisely this complete least-common-multiple remainder.

Put $h_0(\ell) = 2/(\ell-2)$ for $\ell \equiv 1 \pmod{3}$, zero otherwise; extend multiplicatively on squarefree integers, and put it equal to zero on nonsquarefree integers. Write $J_0(z) = \sum_{d \leq z} h_0(d)$. We first prove

$$(953) \quad J_0(z) \geq c \log z \quad (z \geq 2)$$

with an absolute positive c . The fixed-modulus prime estimate used in Theorem 246.1, from Theorem 1.2, formula (1.12), of [Bennett, Martin, O'Bryant and Rechnitzer](#), gives by partial summation

$$\sum_{\substack{\ell \leq y \\ \ell \equiv 1 \pmod{3}}} \frac{1}{\ell} = \frac{1}{2} \log \log y + O(1),$$

$$\sum_{\substack{\ell \leq y \\ \ell \equiv 1 \pmod{3}}} \frac{\log \ell}{\ell} = \frac{1}{2} \log y + O(\log \log(3y)).$$

The bounded error in the first line uses the extra logarithm in the integrated prime-counting error.

The total h_0 -weight of all squarefree divisors of the product of the supported primes through y is

$$E(y) = \prod_{\substack{\ell \leq y \\ \ell \equiv 1 \pmod{3}}} (1 + h_0(\ell)) \asymp \log y.$$

Its exact weighted mean of $\log d$ equals

$$\sum_{\substack{\ell \leq y \\ \ell \equiv 1 \pmod{3}}} \frac{h_0(\ell)}{1 + h_0(\ell)} \log \ell = \sum_{\substack{\ell \leq y \\ \ell \equiv 1 \pmod{3}}} \frac{2 \log \ell}{\ell}.$$

With $y = z^{1/4}$, this mean is at most $\frac{1}{2} \log z$ for large z . The finite weighted cutoff inequality puts at least half the total weight on $d \leq z$. Thus $J_0(z) \geq E(y)/2$, proving (953); reducing c covers bounded $z \geq 2$. This auxiliary finite measure does not assert a distribution of the actual root parameters.

Outside m , the weights h_m equal h_0 ; on m they are nonnegative. Splitting a squarefree d into its part supported on m and its coprime part proves

$$J_0(z) \leq \left(\prod_{\ell|m} (1 + h_0(\ell)) \right) \sum_{\substack{d \leq z \\ (d,m)=1}} h_0(d) \leq A(m) J_m(z).$$

Therefore $J_m(z) \geq c \log z / A(m)$, uniformly in both m, b . Finally, $\rho_m([d_1, d_2]) \leq d_1 d_2$, so the complete remainder in (951) is bounded by

$$\left(\sum_{1 \leq d \leq z} d \right)^2 \leq z^4.$$

The asserted bound follows. In particular it contains no uncontrolled factor depending on $\log \log m$. $\square$

Theorem 260.3 (An epsilon-uniform exceptional set). *There is an absolute C_0 such that, uniformly for $0 < \epsilon \leq 1/10$,*

$$(954) \quad \#\{k \in I_B : P^+(Q(k)) > B^{2-\epsilon}\} \leq C_0 \epsilon B + O\left(\frac{B}{\log B} + B^{3/5}\right).$$

The constants and sufficiently-large threshold are independent of ϵ .

Proof. For an indicated prime $r = P^+(Q(k))$, put $m = Q(k)/r$. Then $1 \leq m < X = 49B^\epsilon$. Each parameter is in one of the $\rho(m)$ classes of Lemma 260.2. Take $z = B^{1/10}$; then $r > B^{2-\epsilon} > z$. Overcounting is harmless.

The needed cofactor weight satisfies

$$\sum_{m \leq X} \frac{\rho(m) A(m)}{m} \ll \log(3X).$$

Indeed, at every supported prime and every $j \geq 1$, $\rho(p^j)A(p^j) = 2p/(p-2)$. The unrestricted Euler product on primes through X majorizes the sum, with local factor

$$1 + \frac{2p}{(p-2)(p-1)} = 1 + \frac{2}{p} + O(p^{-2}).$$

The reciprocal-prime estimate in the preceding proof gives the bound. It includes $m = 1$ and all prime powers.

The main terms of (952) now sum to

$$CB \frac{\log(3X)}{\log z} \leq C_0 \epsilon B + O(B/\log B).$$

The entire error is bounded by

$$z^4 \sum_{m \leq X} \rho(m) \leq z^4 \sum_{m \leq X} m \ll X^2 z^4 \ll B^{2\epsilon+2/5} \leq B^{3/5}.$$

The constants 49 and $\log 147$ are absolute, proving all the stated uniformities. $\square$

Theorem 260.4 (An actual domain strictly larger than one half). *There exist absolute $\delta_0 > 0$ and B_0 such that*

$$\mathcal{S}_B = \{k \in I_B : Q(k) \text{ has a prime divisor above } 12B\}$$

satisfies

$$|\mathcal{S}_B| \geq (1/2 + \delta_0)B \quad (B \geq B_0).$$

All its actual ratios are multiplicatively independent.

Proof. The full-block norm-mass calculation in Theorem 246.1 gives

$$\sum_{k \in I_B} \sum_{r > 12B} v_r(Q(k)) \log r \geq B \log B - O(B \log \log B).$$

Every such prime is globally private to its parameter and has depth one. A norm has at most one of them, since $(12B)^2 > 49B^2 > Q(k)$; denote it by r_k .

Fix $0 < \epsilon \leq 1/10$ small enough that $C_0 \epsilon \leq 1/4$. Let E_B be the exceptional count in (954). Outside that exceptional set, $\log r_k \leq (2 - \epsilon) \log B$; inside it, $\log r_k \leq 2 \log B + \log 49$. Hence the same full-block mass is at most

$$(2 - \epsilon)|\mathcal{S}_B| \log B + \epsilon E_B \log B + B \log 49.$$

Divide by $B \log B$ and use Theorem 260.3 to obtain

$$\liminf_{B \rightarrow \infty} \frac{|\mathcal{S}_B|}{B} \geq \frac{1 - C_0 \epsilon^2}{2 - \epsilon} \geq \frac{1 - \epsilon/4}{2 - \epsilon} = \frac{1}{2} + \frac{\epsilon}{4(2 - \epsilon)}.$$

Thus $\delta_0 = \epsilon/[8(2 - \epsilon)]$ works for all sufficiently large B . Private oriented prime valuations prove independence exactly as in Theorem 246.1. $\square$

The established UM logarithmic window therefore holds on an actual domain of fraction at least $1/2 + \delta_0 - o(1)$, with its same finite-window errors and unchanged unproved far signed term. No optimized numerical value of δ_0 is claimed. The argument uses an independent upper sieve for one exceptional set and the original full-block mass bound once. It does not iterate that mass bound on an arbitrary discarded subset.

These are complete ordinary proofs using the explicitly located primary sieve and fixed-modulus prime inputs. Root's separate `PrivateDensityMass.lean` formalizes selected cofactor identities, actual finite mass cuts, the exact finite remainder-to-density inequality and the finite positive-index weighted logarithmic cutoff. It does not formalize the sieve, prime asymptotics, Euler weights or the full density theorem. The complementary roots and the unbounded boundary-prime tail remain unresolved.

261. VERY LARGE PRIME INCIDENCE ON THE ACTUAL BOUNDARY ARMS

Let $n > 324$ be prime, $B = n^4$, $r = \lceil \sqrt{6n} \rceil$, and $L = \log(8B)$. Retain the actual roots

$$I_B = \{k \in \mathbb{Z} : B \leq k < 2B\}, \quad a_k = 3k, \quad w_k = a_k + \zeta,$$

$$Q_k = a_k^2 + a_k + 1, \quad \alpha_k = w_k/\bar{w}_k, \quad \zeta^2 = \zeta - 1.$$

Fix a subset $I \subseteq I_B$ whose actual ratios are multiplicatively independent. The established private-norm set or any fixed color from Theorem 254.1 supplies such a set. No independence assertion for an arbitrary subset of the whole block is made.

Write $w_k^n = A_k + C_k\zeta$, and set

$$b_{k,1} = |A_k|, \quad b_{k,2} = |C_k|, \quad b_{k,3} = |A_k + C_k|, \quad T_k = \prod_{j=1}^3 b_{k,j}, \quad c_k = \max_j b_{k,j}, \quad t_k = \log c_k.$$

These powers are primitive and unramified, so the arm sizes are positive and pairwise coprime. A zero boundary would make w_k^n a unit times a rational integer, contradicting the nonzero oriented split-prime valuations of $w_k/\bar{w}_k$. The actual bounds $9B^2 \leq Q_k < 36B^2$ give

$$(955) \quad n \log(3B) \leq t_k < nL.$$

Indeed, $c_k \geq Q_k^{n/2}$, while

$$c_k \leq \frac{2}{\sqrt{3}} Q_k^{n/2} < \frac{2}{\sqrt{3}} (6B)^n < (8B)^n.$$

The norm and each arm are coprime: reducing the norm modulo a prime dividing A_k , C_k , or $A_k + C_k$ leaves the square of a nonzero coordinate.

Theorem 261.1 (Two endpoints at very large primes). *For every rational prime q satisfying*

$$(956) \quad \log q \geq \frac{r}{2}L,$$

the set $E_q = \{k \in I : v_q(T_k) \geq 4\}$ has at most two elements.

Proof. The cutoff gives $q > 8B > n$ and $q^4 \geq (8B)^{2r}$. Every hit supplies a unit norm and unit conjugate denominator modulo q^4 . The identity

$$(A + C\zeta)^3 - (A + C\bar{\zeta})^3 = 3AC(A + C)(\zeta - \bar{\zeta})$$

therefore implies $\alpha_k^{3n} = 1 \pmod{q^4}$. Since $q \nmid 3n$, the established split/inert simple-lifting count bounds this norm-one torsion group by $3n$.

The coordinate calculation in the proof of Lemma 253.1 gives, for two actual products of r roots, a determinant bounded by

$$|D| \leq 8r 7^{2r-1} B^{2r-1} < (8B)^{2r}.$$

The strict inequality is uniform in r : after division, with $x = 49/64$, its upper bound is

$$\frac{8}{7B} r x^r < \frac{8}{7B} \frac{1}{1-x} = \frac{512}{105B} < 1.$$

Equal ratio products modulo q^4 thus give $q^4 \mid D$, hence $D = 0$ and exact ratio equality in $\mathbb{Q}(\zeta)$. Independence recovers every multiplicity. Consequently the multisets of length r from E_q inject into the torsion group, giving

$$\binom{|E_q| + r - 1}{r} \leq 3n$$

when E_q is nonempty. If $|E_q| \geq 3$, its left side is at least

$$\binom{r+2}{2} = \frac{(r+1)(r+2)}{2} > \frac{r^2}{2} \geq 3n,$$

a contradiction. Empty hit sets cause no difficulty. $\square$

Theorem 261.2 (Actual simultaneous packets and arm degrees). *Let $V = I \times \{1, 2, 3\}$. Take as labels precisely the primes satisfying (956) for which $E_q \neq \emptyset$, and put*

$$e(q) = \{(k, j) : k \in E_q, q \mid b_{k,j}\}.$$

Each label has one or two endpoints, with distinct roots in the two-endpoint case. For an incident endpoint $v = (k, j)$, write $\epsilon_v(q) = v_q(b_{k,j}) \geq 4$. For every subset $U \subseteq V$ and every subset P of the labels,

$$(957) \quad \prod_{q \in P} q^{\sum_{v \in U \cap e(q)} \epsilon_v(q)} \mid \prod_{v \in U} b_v.$$

Writing $d_U(q) = |U \cap e(q)|$, this implies

$$(958) \quad 4 \sum_q d_U(q) \log q \leq \sum_q \sum_{v \in U \cap e(q)} \epsilon_v(q) \log q \leq \sum_{v \in U} \log b_v.$$

With $d = \lfloor n/(2r) \rfloor$, every arm vertex has degree at most d , and every root has at most $3d$ incident prime labels.

Proof. The arm sizes are pairwise coprime, so each hit root belongs to exactly one endpoint at this prime. Theorem 261.1 then gives the endpoint count. The label set is finite, since its members divide the fixed nonzero integer $\prod_{k \in I} T_k$. Different prime labels may have the same endpoint set, and singleton edges are allowed.

For each prime q , the exponent on the left of (957) is at most its valuation in the integer on the right. Applying this simultaneously to the distinct primes proves the divisibility. Taking logarithms proves (958). Primes omitted from the graph, including depths below four at other vertices, only add nonnegative mass on the right.

If m labels meet a fixed arm $v = (k, j)$, then

$$2rLm \leq \sum_{q \ni v} 4 \log q \leq \log b_v \leq t_k < nL.$$

Thus $m < n/(2r)$. Since n is odd and $2r$ is even, this ratio is not an integer, giving $m \leq d$. Summing over the three different arms gives the root-degree bound. $\square$

The total number of labels is at most $3d|I|$, by summing incidences. If every label had two endpoints, it would be at most $3d|I|/2$; singleton labels have not been excluded. In particular, the available cardinality bound is $O(|I|\sqrt{n})$, and does not provide an $o(B)$ estimate.

Theorem 261.3 (A joint matching decomposition). *The entire actual prime-labelled graph on arm vertices is a disjoint union of at most $2d - 1$ matchings. Within each matching, different prime labels have disjoint endpoint sets. They may use different arms of the same root.*

Proof. Here $d \geq 1$, since $2r \leq 2\sqrt{6n} + 2 < n$ for $n > 324$. Process the finite labels in increasing order. At an edge with two endpoints, at most $2(d - 1)$ earlier labels meet either endpoint: each endpoint has total degree at most d , including the current edge. A singleton edge has at most $d - 1$ earlier neighbors. Multiple edges only cause overcounting in these upper bounds. Thus one of $2d - 1$ colors is absent from all earlier adjacent labels. Assign the first available color. Its color classes are the claimed matchings. If I is empty, there are no labels to color. This is a direct finite construction, without a multigraph-coloring or matroid-partition assumption. $\square$

This statement applies to all of the actual labels simultaneously. It does not bound the total weight of a matching or resolve the independence of roots outside the established domain.

Proposition 261.4 (The remaining weighted budget). *For every subset U of the actual arm vertices,*

$$(959) \quad \sum_q \sum_{v \in U \cap e(q)} (\epsilon_v(q) - 3) \log q \leq \sum_{v \in U} \log b_v - 3 \sum_q d_U(q) \log q.$$

For $U = V$, its left side is precisely the positive excess in the region (956).

Proof. Subtract $3 \sum_q d_U(q) \log q$ from the second inequality of (958). Each incidence has depth at least four. Pairwise coprimality allocates the entire depth of T_k to its unique arm, proving the last assertion. $\square$

This retains the graph's radical saving, but does not account for the negative contributions at depths one and two outside this positive graph. The estimates do not make its right side $o(\sum_{k \in I} t_k)$.

For a precise limitation of the finite inequalities alone, consider an abstract weighted ledger for sufficiently large n . Give each of N formal roots three arms and $m = \lfloor n/(4r) \rfloor$ private labels on each arm. Give every label depth four and logarithmic weight $w = (r/2)L$. All edges are singletons, and the full mass on an arm is

$$4mw = 2rmL \leq nL/2.$$

Its degree is at most d , and every moment support at a fixed label has cardinality one. Yet its positive excess per root is

$$3mw \sim \frac{3}{8}nL.$$

Thus the degree, moment-cardinality and strict arm-cap inequalities alone permit a linear positive cost. These formal labels and weights have not been realized by distinct rational primes or by Eisenstein powers. This is not a counterexample to any actual root-family statement, to signed compensation, or to *abc*. The stronger simultaneous integer structure in (957) remains a possible source of improvement.

The outstanding interfaces are actual private-label counts and weights, interactions between different primes, the omitted signed credits, and the roots outside the proved independent domain. None is declared automatically satisfied or impossible. All results in this section are ordinary proofs. No new formal Lean statement, prime-distribution certificate or density experiment is claimed.

262. THE SIMULTANEOUS CUBIC-UNIT GATE ON A FIXED ELLIPTIC CURVE

Fix $g = 3, u = \zeta$. Write

$$A_0 = s^3 - 3s - 1, \quad B_0 = 3s(s + 1), \quad A = -B_0, \quad B = A_0 + B_0.$$

The smooth projective normalization $D = D_{3,\zeta}$ of

$$(960) \quad v^2 = 1 + r(s), \quad w^2 = 1 - 3r(s), \quad r = A/B$$

has genus six by HG. Its positive rational locus is $0 < r < 1/3, v > 0$; its seed parameter is $t = (v + w + 2)/(v - w) > 1$. Both square roots must have the same s . Their quotient coordinates are

$$y_1 = Bv, \quad y_1^2 = A_0(A_0 + B_0), \quad y_2 = Bw, \quad y_2^2 = (A_0 + B_0)(A_0 + 4B_0).$$

Theorem 262.1 (A quartic over the fixed elliptic quotient). *Put*

$$\begin{aligned} f &= X^3 - 9X - 9, & h &= X + 2, \\ H &= f + 18h^2 = X^3 + 18X^2 + 63X + 63, & R &= f - 108h^2 = X^3 - 108X^2 - 441X - 441, \\ J &= 6Xh, & d &= 4X + 9. \end{aligned}$$

Over $E : Y^2 = f(X)$, the equation

$$(961) \quad Z^4 - 2H(X)Z^2 + f(X)R(X) = 0$$

has smooth projective normalization isomorphic over $\mathbb{Q}$ to D . It is geometrically connected of genus six and has degree four over E .

Proof. On $s \neq -1$, define

$$(962) \quad \begin{aligned} X &= -\frac{2s^2 + 5s + 2}{(s+1)^2}, & Y &= \frac{y_1}{(s+1)^3}, & Z &= \frac{y_2}{(s+1)^3}, \\ \kappa &= \frac{s-1}{s+1}, & \gamma &= \frac{A_0 + B_0}{(s+1)^3}. \end{aligned}$$

The first quotient identity gives $Y^2 = f$. Clearing denominators gives

$$(963) \quad \begin{aligned} \kappa^2 &= d, & \frac{B_0}{(s+1)^3} &= -3h, & \gamma &= \frac{-X\kappa - 3h}{2}, \\ Y^2 &= \gamma^2 + 3h\gamma, & Z^2 &= \gamma^2 - 9h\gamma = H + J\kappa. \end{aligned}$$

Indeed $A_0/(s+1)^3 = \gamma + 3h$. Multiplying the normalized cubic factors proves the square formulas; their sum is $-X\kappa$. Expansion gives

$$(964) \quad H^2 - fR = J^2d.$$

Eliminating κ proves the quartic. Conversely, on the open with nonzero denominators, put

$$(965) \quad \kappa = \frac{Z^2 - H}{J}, \quad s = \frac{1 + \kappa}{1 - \kappa}, \quad y_1 = Y(s+1)^3, \quad y_2 = Z(s+1)^3.$$

The quartic and (964) imply $\kappa^2 = d$. The identities (963) recover the original two equations at this same s . The maps are inverse on nonempty dense opens.

The maps $D \rightarrow H_1$ and $H_1 \rightarrow E$ both have degree two. Thus $\mathbb{Q}(D)/\mathbb{Q}(E)$ has degree four; the inverse formulas show that Z generates it. The monic quartic is its minimal polynomial. The same argument over the algebraic closure proves geometric irreducibility. The function-field isomorphism extends to the smooth projective models by [140], Lemma 53.2.2 and Theorem 53.2.6. $\square$

Theorem 262.2 (The exact positive rational domain). *The positive rational locus of D is in bijection with the rational triples satisfying*

$$(966) \quad Y^2 = f(X), \quad Z^4 - 2HZ^2 + fR = 0, \quad 0 < Z^2 < R(X), \quad Y(X+2) > 0.$$

This is a bijection of curve points with their power-map parameter, not a claim of a unique such point above each integer seed at $g = 3$.

Proof. For a positive point, $s = -1$ and $s = \infty$ are impossible: both have homogeneous r -value zero. Hence (962) is defined. Here $r = 3h/\gamma$, $v = Y/\gamma$, $w = Z/\gamma$, and

$$(967) \quad \Delta := Y^2 - Z^2 = 12h\gamma, \quad r = \frac{36h^2}{\Delta}.$$

The value $h = 0$ forces $s = 0$ or ∞ in the EQ projective abscissa relation, again with $r = 0$. Thus $0 < r < 1/3$ implies $\Delta > 108h^2 > 0$ and $Z^2 < R$. A rational positive point cannot have $w = 0$, since this would give $3v^2 = 4$. Therefore $Z^2 > 0$. Since γ and h have the same sign, positivity of v is exactly $Yh > 0$.

Conversely assume (966). The value $X = 0$ is impossible because $f(0) = -9$. At $X = -2$, one has $f = H = R = 1, J = 0$; the quartic becomes $(Z^2 - 1)^2 = 0$, inconsistent with $0 < Z^2 < R$. Thus $J \neq 0$. Formula (965) gives rational κ with $\kappa^2 = 4X + 9$. Neither $\kappa = 1$ nor -1 occurs, since either would force $X = -2$. The resulting s is finite and not -1 , and both hyperelliptic equations hold there. Set

$$(968) \quad \Delta = f - Z^2, \quad \gamma = \frac{\Delta}{12h}, \quad v = \frac{Y}{\gamma}, \quad w = \frac{Z}{\gamma}.$$

The inequality gives $\Delta > 108h^2 > 0$. Equations (963) recover the normalized B -coordinate γ , so $r = 36h^2/\Delta \in (0, 1/3)$; $Yh > 0$ makes $v > 0$. Equivalently the quartic rearranges to

$$(969) \quad (f - Z^2)^2 = 36h^2(3f + Z^2),$$

giving $3v^2 + w^2 = 4$. These constructions are inverse.

The point O of E corresponds to $s = -1$, outside the positive locus. The repeated-discriminant value $X = -9/4$ is not in $E(\mathbb{Q})$, since $f(-9/4) = -9/64$. Positivity avoids all HG branch values; normalization therefore introduces no extra choice of local branch in this bijection. $\square$

Together with CI and RL, this is a sufficient as well as necessary test for the actual unit- ζ first-square/second-cube branch. A triple passing the test reconstructs the seed through t . No such triple is asserted to exist. The two signs of Z retain the ordered seed directions.

Theorem 262.3 (Integral factorization and exact prime depth). *Write a point of (966) in primitive coordinates*

$$X = A/d_0^2, \quad Y = B/d_0^3, \quad d_0 > 0, \quad \gcd(A, d_0) = \gcd(B, d_0) = 1.$$

Then $C = Zd_0^3 \in \mathbb{Z}$. With $K = A + 2d_0^2 \neq 0$,

$$(970) \quad \begin{aligned} B^2 &= A^3 - 9Ad_0^4 - 9d_0^6, \\ (B^2 - C^2)^2 &= 36d_0^2K^2(3B^2 + C^2), \\ L &:= \frac{B^2 - C^2}{6d_0K} \in \mathbb{Z}, \quad L^2 = 3B^2 + C^2. \end{aligned}$$

Every prime $p > 3$ dividing d_0K satisfies

$$(971) \quad v_p(B^2 - C^2) = v_p(d_0K), \quad p \nmid BCL.$$

The entire p -primary part of d_0K divides exactly one of $B - C$ and $B + C$, with no excess depth there.

Proof. The coordinates exist: at a prime of negative X -valuation, the cubic term is uniquely smallest in the equation, so that valuation is even and the Y -denominator exponent is three times the corresponding half-exponent. Elsewhere Y is integral. Both $H_0 = d_0^6 H(A/d_0^2)$ and $R_0 = d_0^6 R(A/d_0^2)$ are integers. Clearing the quartic makes C a rational root of the monic polynomial

$$C^4 - 2H_0C^2 + B^2R_0 = 0,$$

so it is an integer. Clearing (969) gives (970). The square of the nonzero integer $6d_0K$ divides $(B^2 - C^2)^2$; prime valuations imply $6d_0K \mid B^2 - C^2$, and hence the assertions about L .

One has $\gcd(d_0, K) = 1$. If $p \mid d_0$, then $B^2 \equiv A^3 \not\equiv 0 \pmod{p}$. If $p \mid K$, then d_0 is a unit and $A \equiv -2d_0^2$, giving $B^2 \equiv d_0^6 \not\equiv 0 \pmod{p}$. In either case (970) forces $C^2 \equiv B^2 \pmod{p}$. Thus $3B^2 + C^2 \equiv 4B^2$ is a unit for $p > 3$. Taking valuations proves (971), and the equation for L gives $p \nmid L$. Such an odd prime cannot divide both $B - C$ and $B + C$; their product is $B^2 - C^2$, proving the allocation. $\square$

This necessary factorization gives no upper bound for A, d_0 or the point height, and does not force a prime violating the allocation.

Theorem 262.4 (A three-isogeny between the elliptic quotients). *For the curves in (932), put*

$$F_3(X) = X + \frac{36}{X+3} - \frac{36}{(X+3)^2}, \quad F'_3(X) = 1 - \frac{36}{(X+3)^2} + \frac{72}{(X+3)^3}.$$

The chart map

$$(972) \quad \phi : E \longrightarrow E', \quad (X, Y) \longmapsto (F_3(X), YF'_3(X))$$

extends to a degree-three isogeny over $\mathbb{Q}$, with geometric kernel $\{O, (-3, 3i), (-3, -3i)\}$. Consequently $\text{Jac}(H_1) \sim_{\mathbb{Q}} E \times E$.

Proof. Clearing $(X+3)^6$ verifies

$$(973) \quad f(X)F'_3(X)^2 = F_3(X)^3 - 189F_3(X) + 999.$$

The nonconstant rational map extends between the smooth projective curves. At O , $F_3(X)$ has leading term X , so O maps to O . The origin-preserving morphism theorem (Milne, *Elliptic Curves*, Chapter II, Proposition 1.5) makes it a homomorphism. The numerator of F_3 is cubic, its denominator is $(X+3)^2$, and the numerator at -3 is -36 . Thus its degree on $\mathbb{P}^1$ is three. Both

elliptic abscissa maps have degree two; their commutative diagram gives $\deg \phi = 3$. The poles are exactly O and the two points $X = -3, Y^2 = -9$; these are the preimages of O . Composition with the established EQ isogeny proves the final assertion. $\square$

For $P = (-2, 1)$, $P' = (6, 9)$, one has

$$\phi(P) = (-2, 37) = -3P'.$$

Indeed $2P' = (33/4, 9/8)$; its secant with P' has slope $-7/2$, giving $3P' = (-2, -37)$. Neither this identity nor the isogeny certifies a rank upper bound, a full Mordell–Weil basis, or the rational points passing (966).

Verification and remaining scope. Both peer routes and the root group reviewed the ordinary proofs in full. An exact standard-library replay checks fifteen polynomial identities and the displayed point relation. It also checks the eighty specified points nP , $1 \leq |n| \leq 40$, and their isogeny images; none passes the positive gate. This bounded check is not a complete group or rational-point computation. The twelve declarations in `SimultaneousEllipticArithmetic` also passed fresh compilation and all axiom queries with only the standard axioms. They prove the actual polynomial identities, the homogeneous elliptic/quartic-to-conic implication, integer square divisibility and its quotient, and the boundary unit from $\text{Int.gcd}(A, d_0) = 1$. These are combined to prove the full product-depth equality in (971) from the actual homogeneous equations and stated prime hypotheses, without a supplied depth or unit conclusion. The rational three-isogeny identity is formalized as well. The monic rational-to-integer bridge, primitive rational-coordinate representation, projective point bijection, geometric isogeny degree/kernel, Jacobian statement and final allocation to one of the two linear factors remain ordinary-only. The simultaneous rational-point problem remains unresolved, as do the other nonidentity unit and the general ABC statement.

263. FINITE DENSITY, ACTUAL OWNERS AND INTEGER DEPTH IDENTITIES IN LEAN

The following finite bridge records the mass inequalities behind the strict private-domain improvement. It retains actual indices and both arithmetic error terms.

Proposition 263.1 (An actual finite density gain). *Let S be a finite set, $E \subseteq S$, and w_i real weights. Let $B, \ell > 0$, $0 < \epsilon \leq 1/10$, $c \geq 0$, and suppose*

$$|S| \leq B, \quad |E| \leq C\epsilon B + R, \quad C\epsilon \leq \frac{1}{4}.$$

Assume

$$(1 - \lambda)B\ell \leq \sum_{i \in S} w_i, \quad w_i \leq 2\ell + c \quad (i \in S), \quad w_i \leq (2 - \epsilon)\ell \quad (i \in S \setminus E).$$

If

$$(974) \quad \lambda + \frac{\epsilon R}{B} + \frac{c}{\ell} \leq \frac{\epsilon}{8},$$

then

$$(975) \quad \frac{|S|}{B} \geq \frac{1}{2} + \frac{\epsilon}{8(2 - \epsilon)} > \frac{1}{2}.$$

The weights need not be nonnegative for this proposition.

Proof. At every actual index,

$$w_i \leq (2 - \epsilon)\ell + c + \epsilon\ell \mathbf{1}_{i \in E}.$$

Summing and using the two cardinality bounds gives

$$(1 - \lambda)B\ell \leq (2 - \epsilon)|S|\ell + C\epsilon^2 B\ell + \epsilon R\ell + cB.$$

Put $\lambda' = \lambda + \epsilon R/B + c/\ell$. Division by positive $B\ell(2 - \epsilon)$ yields

$$\frac{|S|}{B} \geq \frac{1 - C\epsilon^2 - \lambda'}{2 - \epsilon} \geq \frac{1 - 3\epsilon/8}{2 - \epsilon} = \frac{1}{2} + \frac{\epsilon}{8(2 - \epsilon)}.$$

The second inequality uses $C\epsilon \leq 1/4$ and (974); the final extra term is strictly positive. $\square$

For the ordinary private-density argument, take S to be the actual private norm support, E its extreme-prime subset, $w_i = \log r_i$, $\ell = \log B$, and $c = \log 49$. The full norm-mass error supplies λ and the uniform sieve supplies R . For fixed sufficiently small ϵ , their displayed bounds make (974) hold eventually. The finite formal theorem keeps these estimates as explicit inputs; it does not itself prove their asymptotic decay.

Lemma 263.2 (A finite logarithmic weight cutoff). *Let D be a finite set of positive natural numbers with nonnegative real weights u_d , and let $z > 1$. If*

$$\sum_{d \in D} u_d \log d \leq \frac{\log z}{2} \sum_{d \in D} u_d,$$

then the total weight on $d \leq z$ is at least half the total.

Proof. Terms with $d \leq z$ have nonnegative logarithmic cost. Every term with $d > z$ costs at least $u_d \log z$. Thus the weight above z is at most half the total, since $\log z > 0$. The exact finite partition gives the conclusion. No distribution of actual arithmetic roots is asserted by this deterministic weighted inequality. $\square$

There are also two direct cofactor identities. For integers b, m, c with $9b^2 + 3b + 1 = mc$, set

$$f_{m,b,c}(t) = 9mt^2 + (18b + 3)t + c.$$

Expansion proves

$$mf_{m,b,c}(t) = 9(b + mt)^2 + 3(b + mt) + 1, \quad (18b + 3)^2 - 4(9m)c = -27.$$

These identities do not assume or establish the later sieve's local root-count formula.

The 35 new formal declarations. `PrivateDensityMass` contains eleven declarations: the two actual cofactor identities, exact exceptional-indicator counting, finite mass cuts, strict numerical gains with and without error, the actual density connection including both remainders, and Lemma 263.2. The input weights and errors are explicit.

`AdaptiveOwnerArithmetic` contains twelve declarations. They prove the cubic and two-owner binomial thresholds, actual natural ceilings, existence of the two largest depth indices in a finite set, containment of every sufficiently deep index in that selected set, and complete natural truncated-excess sums on its complement. For the two actual ceiling thresholds h_0, h_1 , the final theorem produces an actual set O of size $\min(2, |I|)$, with every depth in O at least every remaining depth. Its remaining weighted cost is at most

$$(6|I_4| + 2r|I_{h_0}|)L,$$

and its total cost is at most this expression plus twice the supplied single-root cap. The high-depth multiset-count hypothesis is explicit; the arithmetic valuation and torsion-group construction is ordinary. The further fractional-power estimates giving 39 and 78 are not claimed as Lean statements here.

`SimultaneousEllipticArithmetic` contains twelve declarations for the actual quartic and conic polynomials, integral square divisibility, an integer conic lift, and exact prime depth. Writing $K = A + 2d^2$, with $dK \neq 0$ as in the positive chart, the actual primitive elliptic and quartic equations give, for every prime $p > 3$ dividing dK ,

$$v_p(B^2 - C^2) = v_p(dK).$$

The formal proof derives the needed unit from $\gcd(A, d) = 1$ and the actual equation in $\mathbb{Z}/p\mathbb{Z}$; it does not assume the unit or this depth equality. It also verifies the rational three-isogeny formula's polynomial identity. The curve maps, geometric degree, Jacobian decomposition, rational-to-integral coordinate existence and complete prime allocation retain their separately stated ordinary scope.

All 35 new declarations and 15 unchanged uniform-moment dependencies were freshly compiled together. All 50 explicit axiom queries passed with only `propext`, `Classical.choice` and `Quot.sound` in their union. The compiler is Lean 4.32.0 and Mathlib is pinned to

81a5d257c8e410db227a6665ed08f64fea08e997; its compiled cache was reused. Source and ordinary-scope reviews preceded publication. These finite cores neither control the full signed tail nor determine all points of the remaining geometric curves.

264. FOURTEENTH CONTINUATION: THE REMAINING ARITHMETIC AND GEOMETRIC GATES

Adaptive precision removes the upper-prime restriction from the individual complete-cost estimate. Two actual largest-depth roots contain the high layers, and outside them the per-prime cost is smaller by a power of n . For very large primes, simultaneous integer divisibility supplies an actual arm graph with at most two endpoints per prime, bounded degree and a finite matching decomposition. Its prime labels and total owner weight are still uncontrolled at the strength needed for the full signed gate. The abstract weighted example only limits the displayed finite inequalities; it is not an arithmetic counterfamily.

An independent Selberg cofactor argument proves that the actual private norm domain has relative size at least $1/2 + \delta_0$ for some absolute $\delta_0 > 0$. Its uniform endpoint bounds are retained. Combining this with the established moment window enlarges the proved domain while leaving the same far signed term. The remaining smooth-support and exceptional roots are separate targets, and the full-block mass argument is never iterated on an arbitrary discarded subset.

The surviving cubic geometric branch now has a reversible same-source quartic equation over one fixed elliptic curve, exact positive-domain inequalities and an integral prime-allocation constraint. The two elliptic quotients are rationally three-isogenous, so the genus-two Jacobian is isogenous to a square of one elliptic curve. These identities do not solve the simultaneous rational-point problem. Higher descent and uniform control of exponents, heights and moving residuals remain live.

At the user's request, the research group has also located the official open-source Fermat formalization². It exports the complete natural-number theorem and a guarded derivation of Mathlib's Fermat statement. Its stated toolchain is Lean 4.33.1 with Mathlib 4.33.0. In a separate directory, two unmodified official source files for the vanishing of weight-two cusp forms on $\Gamma_0(2)$ have been freshly compiled with the present Lean 4.32.0 and pinned Mathlib cache. An actual bridge calls the official solution on Mathlib's cusp-form type; both explicit axiom queries have only the standard three axioms. The original sources emit two deprecation warnings. This verifies that particular dependency's local reuse. The complete Fermat import and the project's independent full-kernel replays have not been rerun locally; the published verification records retain their official-source provenance. The 35 new local declarations above do not depend on an unverified Fermat import. Exact applicable established theorems may be reused; excluding pure equal-exponent powers alone does not estimate the near-power signed tail.

All new ordinary arguments have independent internal review, and the finite cores have the explicit formal verification just described. The open global gates remain substantive. No complete proof or disproof of abc , and no gap-free formal proof of either, is claimed.

²Anthropic, *Fermat's Last Theorem in Lean 4*, <https://github.com/anthropics/fermats-last-theorem>, source commit aa2d8b34692b16c70f699536de0d8e75b9a3e9ef.

265. UNIT-NORMALIZED SIGNED MOMENTS AND A SINGLE MAXIMAL DEPTH

Let $n > 324$ be prime, $B = n^4$, $L = \log(8B)$, and

$$w_k = 3k + \zeta, \quad \alpha_k = w_k/\bar{w}_k, \quad \zeta^2 = \zeta - 1, \quad B \leq k < 2B.$$

Fix a subset I of this actual integer block on which the α_k are multiplicatively independent. Write

$$w_k^n = A_k + C_k \zeta, \quad T_k = |A_k C_k (A_k + C_k)|, \quad t_k = \log \max(|A_k|, |C_k|, |A_k + C_k|).$$

The primitive and unramified power identities, together with the individual-arm cap of Lemma 259.1, give

$$(976) \quad N(w_k) < 36B^2, \quad n \log(3B) \leq t_k < nL, \quad v_q(T_k) \log q \leq t_k.$$

The last bound holds for primes $q > 8B$. In fact the three nonzero arms are pairwise coprime, so every such prime divides exactly one arm and its entire power divides that arm.

Lemma 265.1 (A coherent unit choice at one actual prime). *Fix a prime $q > 8B$. For every $k \in I$ with $q \mid T_k$, there is a global $\tau_{q,k} \in \{1, \zeta^2, \zeta^4\}$, independent of the precision, such that*

$$\beta_{q,k} = \tau_{q,k} \alpha_k, \quad \beta_{q,k}^n = 1 \pmod{q^e} \quad (1 \leq e \leq v_q(T_k)).$$

The $\beta_{q,k}$ remain multiplicatively independent and are ratios of unit-rotated actual roots to their conjugates. The norm-one n -torsion group modulo q^e has at most n elements in both the split and inert cases.

Proof. In the Eisenstein residue algebra the unique zero arm gives

$$\alpha_k^n = \begin{cases} 1, & C_k = 0, \\ \zeta^2, & A_k = 0, \\ \zeta^4, & A_k + C_k = 0. \end{cases}$$

For the third identity use $\zeta - 1 = \zeta^2$ and $\bar{\zeta} - 1 = -\zeta$. All denominators are units: a prime dividing a primitive boundary arm does not divide the norm. The arm is already determined modulo q , and remains the same at every available precision. Since $\gcd(n, 3) = 1$, its global cube root of unity has a unique inverse n -th root in this group. Choose that root as $\tau_{q,k}$.

The map $v \mapsto v/\bar{v}$ on the six global units has image exactly $\{1, \zeta^2, \zeta^4\}$. Thus $\beta_{q,k}$ has the claimed actual representation. Cubing any integer-exponent relation among the $\beta_{q,k}$ eliminates all unit factors; independence of the α_k then recovers every exponent.

At precision one the norm-one group is cyclic: it is $\mathbb{F}_q^\times$ in the split case, and the norm-one subgroup of $\mathbb{F}_{q^2}^\times$ in the inert case. It has at most n elements killed by n . Since $q \nmid n$, the n -th power map is invertible on the principal q -kernels, and each such element has exactly one lift through each successive precision. This proves the same cardinal bound at every level. $\square$

The choices in Lemma 265.1 are coherent for each fixed q , but can change when q changes. They do not define one common normalized map for all primes.

Theorem 265.2 (An actual signed-ball injection). *Let $q > 8B$ be prime and let e, ν be positive integers with $e \log q \geq 2\nu L$. Put*

$$I_e(q) = \{k \in I : q^e \mid T_k\}, \quad M_e = |I_e(q)|.$$

Then

$$(977) \quad C(M_e, \nu) := \sum_{j=0}^{\min(M_e, \nu)} 2^j \binom{M_e}{j} \binom{\nu}{j} \leq n.$$

Proof. For an integer vector $x = (x_k)$ with $\sum_k |x_k| \leq \nu$, represent $\prod_k \beta_{q,k}^{x_k}$ by $z_x/\bar{z}_x$, using the unit-rotated w_k for positive exponents and its conjugate for negative exponents, with their indicated multiplicities. The empty product is $z_0 = 1$. These are nonzero actual Eisenstein integers with

$$|z_x| = \sqrt{N(z_x)} \leq (6B)^\nu.$$

Units and conjugation preserve the norm. For $z = R + S\zeta$ and $z' = R' + S'\zeta$, the exact integer Gram identity is

$$(978) \quad 4N(z)N(z') - (2RR' + RS' + SR' + 2SS')^2 = 3(RS' - R'S)^2.$$

Consequently the determinant $D = RS' - R'S$ of two representatives obeys

$$(979) \quad |D| \leq \frac{2}{\sqrt{3}}|z_x||z_y| \leq \frac{2}{\sqrt{3}}(6B)^{2\nu} < (8B)^{2\nu}.$$

Indeed $(2/\sqrt{3})(3/4)^{2\nu} \leq 9/(8\sqrt{3}) < 1$ for every $\nu \geq 1$. This includes unequal lengths and empty products.

If the two residue products agree modulo q^e , their conjugate cross difference is divisible by q^e . The factor $\zeta - \bar{\zeta}$ is a unit at $q > 3$, so $q^e \mid D$. The strict bound (979) and the precision assumption force $D = 0$. The exact algebraic ratios are equal, and Lemma 265.1 recovers $x = y$. Thus the actual integer ℓ^1 ball injects into the norm-one n -torsion group.

To count this ball, select its j nonzero coordinates and their signs. The positive absolute values with sum at most ν have cardinality $\binom{\nu}{j}$, by subtracting one from each and adjoining a slack coordinate. The $j = 0$ term counts the zero vector, including $M_e = 0$. This gives (977). The slack coordinate is only part of the counting argument; it is not an extra independent root. $\square$

Expanding this finite sum gives

$$(980) \quad \begin{aligned} C(M, 2) &= 2M^2 + 2M + 1, & 3C(M, 3) &= 4M^3 + 6M^2 + 8M + 3, \\ C(2, \nu) &= 2\nu^2 + 2\nu + 1. \end{aligned}$$

The formulas include the small-dimensional and zero-radius cases where applicable.

Theorem 265.3 (The complete single-prime budget). *Put $r = \lceil \sqrt{n/2} \rceil$. For every prime $q > 8B$,*

$$(981) \quad \begin{aligned} \sum_{k \in I} (v_q(T_k) - 3)_+ \log q &\leq \frac{5}{2}nL, \\ \frac{1}{B} \sum_{k \in I} \frac{(v_q(T_k) - 3)_+ \log q}{t_k} &\leq \frac{5}{B}. \end{aligned}$$

There is no upper restriction on q .

Proof. One has $2r^2 \geq n$ and $r \leq \sqrt{n}$: use $\sqrt{n/2} \leq (3/4)\sqrt{n}$ and $1 \leq \sqrt{n}/4$. Write $w = \log q$ and $h = \max(4, \lceil 2rL/w \rceil)$. Theorem 265.2 at $(e, \nu) = (4, 2)$ gives $M_4 \leq \sqrt{n/2}$. At $(e, \nu) = (h, r)$, two indices would give $n \geq C(2, r) = 2r^2 + 2r + 1 > n$. Thus $M_h \leq 1$.

All layers from four through $h - 1$ cost at most

$$(h - 4)M_4w \leq 2r\sqrt{n/2}L \leq \sqrt{2}nL < \frac{3}{2}nL.$$

The subtraction of four absorbs the ceiling error; the range is empty when $h = 4$. The remaining layers have at most one actual index, whose entire depth costs at most nL by (976). This proves the first inequality. Since $L \leq 2\log(3B)$, the lower height in (976) proves the second. $\square$

Theorem 265.4 (One actual maximum and two support thresholds). *For a prime $q > 8B$, set*

$$h_0 = \max(4, \lceil 6L/\log q \rceil), \quad h_1 = \max(h_0, \lceil 2rL/\log q \rceil).$$

Let O_q contain one index of maximal actual depth, or be empty if I is empty. Then

$$(982) \quad \begin{aligned} \sum_{k \in I \setminus O_q} (v_q(T_k) - 3)_+ \log q &\leq (6M_4 + 2rM_{h_0})L \\ &\leq 7n^{5/6}L. \end{aligned}$$

Its normalized remainder is at most $14n^{-1/6}/B$ per prime. If $\log q \geq (r/2)L$, at most one root of I has depth at least four. If $\log q \geq 2rL$, at most one root of I has any positive depth.

Proof. The signed counts give

$$M_4 \leq \sqrt{n/2}, \quad M_{h_0} < n^{1/3}, \quad M_{h_1} \leq 1.$$

For the middle inequality use $(4/3)M^3 < C(M, 3) \leq n$. Maximality implies that O_q contains every h_1 -deep index. For each remaining depth $e < h_1$, the complete truncated inequality is

$$(e - 3)_+ \leq (h_0 - 4)\mathbf{1}_{e \geq 4} + (h_1 - h_0)\mathbf{1}_{e \geq h_0}.$$

The two ceiling inequalities $(h_0 - 4) \log q \leq 6L$ and $(h_1 - h_0) \log q \leq 2rL$ prove the first bound in (982), including empty ranges. For the second,

$$6M_4 + 2rM_{h_0} \leq 5\sqrt{n} + 2n^{5/6} \leq 7n^{5/6}.$$

The height lower bound gives the normalized statement. Adding back the entire selected index costs at most nL .

At $\log q \geq (r/2)L$, both thresholds equal four, hence the positive-excess support has at most one index. At the stronger threshold, Theorem 265.2 applies with $e = 1, \nu = r$, which bounds the full positive-depth support by one. $\square$

The first threshold allows other roots of depth one, two or three; the second excludes them. Neither bounds the number of different prime labels that can occur on one root.

Prime summation and the remaining problem. Every supported $q > 8B$ has rank n , hence $q \equiv 1$ or $-1 \pmod{3n}$. The same primary Brun–Titchmarsh bound used in the adaptive-precision section, together with (981), gives, for real $Z > 8B$,

$$(983) \quad \frac{1}{B} \sum_{k \in I} \frac{\sum_{8B < q \leq Z} (v_q(T_k) - 3)_+ \log q}{t_k} \leq \frac{10Z}{B(n-1) \log(Z/(3n))} \leq \frac{11Z}{Bn \log(Z/(3n))}.$$

For the actual finite set

$$\mathcal{D}_Z(I) = \{q > Z : q \text{ prime, } \exists k \in I, v_q(T_k) \geq 4\},$$

the far positive mean is at most $5|\mathcal{D}_Z(I)|/B$. The needed bound $|\mathcal{D}_Z(I)| = o(B)$ has not been proved. The constants and support thresholds improve; this result alone does not enlarge the known useful asymptotic prime window.

In the strongest support region, a prime cannot supply both a high-depth debit at one root of I and a depth-one or depth-two credit at another root of I . Credits at different primes and outside I remain possible. The unit choices also depend on the prime. Repeating the fixed-prime argument therefore supplies neither one common finite target nor a bound for the full signed far tail. Private-label counts and weights, pointwise exceptions, and the independent-domain complement remain separate problems.

The finite arithmetic formally checked here. The module `SignedMomentArithmetic.lean` contains 22 new theorems, freshly compiled with its 27 unchanged local dependencies under the pinned Lean 4.32.0 and Mathlib snapshot. Every declaration has a complete axiom query using only the standard three axioms. Its `ballCount` is the actual finite binomial sum: its special formulas, monotonicity and numerical consequences are proved. Interpreting that sum as an arithmetic torsion image remains an explicit input to this module.

The actual finite depth filters, a singleton of maximal depth, natural truncated subtraction, every remaining positive layer and the per-index whole-depth caps enter the final finite theorems. They prove the $5/2$ total budget, the exact two-range remainder, and the normalized $5/B$ bound using individual heights $t_i \geq t_0 > 0$ with $nL \leq 2t_0$. A separate numerical bridge accepts the concrete two-coordinate diamond lower bound. For a completely natural threshold the module uses $r = \lfloor \sqrt{\lfloor n/2 \rfloor} \rfloor + 1$; it proves $2r^2 \geq n$ and $r^2 \leq n$ for $n \geq 18$. This is not asserted to equal the real ceiling in every even perfect-square case. The polynomial inequality

$$4rM_4 \leq r^2 + 4M_4^2 \leq 3n$$

proves the finite $5/2$ budget without a real square-root input. The coefficient change from $10/(n-1)$ to $11/n$ is also formally checked. The $7n^{5/6}$ simplification, prime distribution, actual

residue torsion, and global tail membership are not claimed as formal consequences of this numerical module.

266. TWO QUADRATIC CUBE DESCENTS AND EXACT ELLIPTIC RANKS

Keep the elliptic curves and three-isogeny of Theorem 262.4:

$$E : Y^2 = f(X) = X^3 - 9X - 9, \quad E' : V^2 = f_*(T) = T^3 - 189T + 999,$$

$$\phi(X, Y) = (F(X), YF'(X)), \quad F(X) = X + \frac{36}{X+3} - \frac{36}{(X+3)^2}.$$

Put $\epsilon = 2 + \sqrt{3}$. We prove exact rank statements by explicit descent. These statements do not determine the rational points of the simultaneous quartic in Theorem 262.1.

Theorem 266.1 (The automatic Gaussian cube). *For every finite $(X, Y) \in E(\mathbb{Q})$, there is a unique $\alpha + i\beta \in \mathbb{Q}(i)$ such that*

$$(984) \quad (\alpha + i\beta)^3 = Y + 3i(X + 2).$$

Proof. Use primitive coordinates

$$X = A/d^2, \quad Y = B/d^3, \quad d > 0, \quad \gcd(A, d) = \gcd(B, d) = 1, \quad B^2 = A^3 - 9Ad^4 - 9d^6.$$

One has $3 \nmid B$. Otherwise $3 \mid A$, so $3 \mid d$, and reduction modulo 27 gives $B^2 \equiv 18 \pmod{27}$. Writing $B = 3b$ would give $b^2 \equiv 2 \pmod{3}$, a contradiction. Set $K = A + 2d^2$ and $S = B + 3idK$. Then

$$(985) \quad N(S) = B^2 + 9d^2K^2 = (A + 3d^2)^3 > 0.$$

For a prime $q > 3$ dividing dK , one has $q \nmid B$: if $q \mid d$, the curve equation gives $B^2 \equiv A^3$, a unit; if $q \mid K$, then d is a unit and $B^2 \equiv d^6$. Thus no such q divides both coefficients of S .

In $\mathbb{Z}[i]$, the unique prime above 2 has norm 2; its exponent in S is $v_2(N(S))$, hence a multiple of three. There is no prime above 3 in S . For $q > 3$, an inert prime cannot divide S , since this would divide both coefficients. At a split prime at most one conjugate orientation occurs, for the same reason; its exponent is $v_q(N(S))$, again divisible by three. These quadratic splitting and ramification facts follow from [129], Examples 2.41 and 3.44–3.45.

The ring is norm-Euclidean: rounding both coordinates leaves a difference of norm at most $1/2$. It is therefore a UFD. Its four units are permuted by cubing. Consequently $S = W^3$ for an actual Gaussian integer W , and W/d proves existence in (984). This root is nonzero. The field $\mathbb{Q}(i)$ has no nontrivial cube root of unity, so the root is unique. $\square$

The cube condition is automatic on $E(\mathbb{Q})$; it is not an additional exclusion of elliptic points.

Theorem 266.2 (The opposite isogeny is bijective on rational points). *Define*

$$(986) \quad G(T) = \frac{T + 108/(T-9) + 108/(T-9)^2}{9}, \quad c(T) = \frac{1 - 108/(T-9)^2 - 216/(T-9)^3}{27}.$$

The map $\psi(T, V) = (G(T), Vc(T))$ extends to a degree-three isogeny $E' \rightarrow E$, induces a bijection $E'(\mathbb{Q}) \rightarrow E(\mathbb{Q})$, and satisfies $\psi\phi = [3]$. For a finite point on E , the inverse is

$$(987) \quad T = 9 + \frac{3}{\beta - 1}, \quad V = \frac{9\alpha}{\beta - 1},$$

where $\alpha + i\beta$ is the root in Theorem 266.1.

Proof. Clearing denominators proves $f_*(T)c(T)^2 = f(G(T))$. The rational map extends to the smooth projective curves and maps O to O , so it is a homomorphism by [142], Chapter II, Proposition 1.5. For $u = T - 9$, one has $G = 1 + u/9 + 12/u + 12/u^2$. Its reduced numerator is cubic and nonzero at $u = 0$; the abscissa degree diagram gives degree three. Its geometric kernel consists of $O, (9, 3\sqrt{3}), (9, -3\sqrt{3})$, so its rational kernel is trivial.

Norm and imaginary-coordinate comparison in (984) give

$$\alpha^2 + \beta^2 = X + 3, \quad \beta(3\alpha^2 - \beta^2) = 3(X + 2).$$

If $\beta = 1$, these imply $3X + 5 = 3X + 6$. Thus (987) is defined. Substitution proves its curve equation and its image under ψ . For explicit checks in the reverse direction, if $u = T - 9$, set $\beta = 1 + 3/u$, $\alpha = V/(3u)$. Then

$$(988) \quad \begin{aligned} \alpha^2 &= \frac{f_*(T)}{9u^2}, & \alpha^2 + \beta^2 &= G(T) + 3, \\ \beta(3\alpha^2 - \beta^2) &= 3(G(T) + 2), & \alpha(\alpha^2 - 3\beta^2) &= Vc(T). \end{aligned}$$

Conversely the two norm and imaginary equations imply $4\beta^3 - 3(X + 3)\beta + 3(X + 2) = 0$; with $\beta \neq 1$, this solves for X as $G(T)$ in (987). The real coordinate recovers Y . Every finite rational point therefore has a rational preimage. The only rational preimage of O is O , and the trivial rational kernel proves uniqueness everywhere. The model scalings $1/9$ and $1/27$ in (986) are retained.

Finally, on a dense chart put

$$\begin{aligned} x_2 &= \frac{(3X^2 - 9)^2}{4f(X)} - 2X, & b_2 &= \frac{(3X^2 - 9)(X - x_2)}{2f(X)} - 1, \\ \ell &= \frac{b_2 - 1}{x_2 - X}, & x_3 &= f(X)\ell^2 - X - x_2, & b_3 &= \ell(X - x_3) - 1. \end{aligned}$$

The chord law gives $[3](X, Y) = (x_3, Yb_3)$. The exact rational-function identities

$$(989) \quad G(F(X)) = x_3, \quad F'(X)c(F(X)) = b_3$$

follow by clearing denominators. Equality of morphisms on this dense chart extends over every omitted point. Thus $\psi\phi = [3]$, with the asserted orientation. $\square$

Lemma 266.3 (Three real-quadratic classes). *For every finite $(T, V) \in E'(\mathbb{Q})$, the nonzero element*

$$\theta(T, V) = V + 3\sqrt{3}(T - 8)$$

has cube class in the three distinct classes $1, \epsilon, \epsilon^2$ in $\mathbb{Q}(\sqrt{3})^\times / \mathbb{Q}(\sqrt{3})^{\times 3}$.

Proof. The value $T = 9$ is impossible because $V^2 = 27$. Choose primitive coordinates $T = A/d^2$, $V = B/d^3$. Then

$$(990) \quad S = B + 3\sqrt{3}d(A - 8d^2), \quad N(S) = (A - 9d^2)^3 \neq 0.$$

The integer norm may be negative. For $q > 3$, if $q \mid d$ the curve equation gives a unit $B^2 \equiv A^3$; if $q \mid A - 8d^2$, it gives $B^2 \equiv (8^3 - 189 \cdot 8 + 999)d^6 = -d^6$, a unit. Thus q cannot divide both coefficients of S .

The ring of integers is $\mathbb{Z}[\sqrt{3}]$, of discriminant 12. Each of 2 and 3 has one ramified prime of residue degree one, so its exponent in S is a multiple of three by (990). At $q > 3$, inert divisibility is impossible and only one split orientation occurs; again the exponent is a multiple of three.

This ring is norm-Euclidean: rounding the two coordinates gives $|a|, |b| \leq 1/2$, so $|a^2 - 3b^2| \leq 3/4 < 1$. Its units are exactly $\{\pm\epsilon^j : j \in \mathbb{Z}\}$. Indeed norm -1 is impossible modulo 3. After changing sign and dividing by a power of ϵ , a unit lies in $[1, \epsilon)$ in the chosen real embedding. A unit strictly above 1 has integral coordinates $x \geq 2, y \geq 1$, since its conjugate is its positive reciprocal; it would therefore be at least ϵ . This proves the unit assertion.

Unique factorization now gives $S = \epsilon^j W^3$, absorbing the sign into W . Division by d^3 proves the three-class bound. A cube root of ϵ or ϵ^2 in the field would be an integral unit by its prime valuations, contradicting the unit exponents just proved. The three classes are distinct. $\square$

Theorem 266.4 (The exact descent quotient). *With $\delta(O) = 1$ and $\delta(T, V) = [\theta(T, V)]$, the map*

$$\delta : E'(\mathbb{Q}) \longrightarrow \mathbb{Q}(\sqrt{3})^\times / \mathbb{Q}(\sqrt{3})^{\times 3}$$

is a homomorphism with kernel $\phi(E(\mathbb{Q}))$ and image of order three. Consequently

$$(991) \quad |E'(\mathbb{Q})/\phi(E(\mathbb{Q}))| = 3.$$

Proof. Put $T_0 = (9, -3\sqrt{3})$. Its tangent is $Y = m_0X + b_0$, with $m_0 = -3\sqrt{3}$, $b_0 = 24\sqrt{3}$; the line function is $\theta = Y - m_0X - b_0$. One has the exact flex identity

$$X^3 - 189X + 999 - (m_0X + b_0)^2 = (X - 9)^3.$$

Let a nonvertical rational chord or tangent be $Y = mX + b$, with three intersections P, Q, R , counted with multiplicity, so $P + Q + R = O$. Its monic intersection polynomial is $F_{\text{line}}(X) = X^3 - 189X + 999 - (mX + b)^2$. If $m \neq m_0$, set $x_0 = -(b - b_0)/(m - m_0)$. The two lines agree at x_0 , so $F_{\text{line}}(x_0) = (x_0 - 9)^3$. Multiplication of their linear evaluations at the three roots gives

$$(992) \quad \theta(P)\theta(Q)\theta(R) = [b - b_0 + 9(m - m_0)]^3.$$

For $m = m_0$, the evaluations are all $b - b_0$, giving the same conclusion. This polynomial identity includes repeated intersections. Its right side is nonzero: a rational line cannot contain T_0 , whose abscissa is rational and ordinate irrational. Moreover

$$(993) \quad \theta(P)\theta(-P) = (9 - X(P))^3 \neq 0.$$

Equations (992)–(993) prove the homomorphism; the latter also handles vertical lines, and $\delta(O) = 1$ handles identity cases.

For the kernel, suppose $\theta(T, V) = (u + v\sqrt{3})^3$, $u, v \in \mathbb{Q}$. Norm and imaginary-coordinate comparison give

$$u^2 - 3v^2 = T - 9, \quad v(u^2 + v^2) = T - 8.$$

Equality of rational cubes justifies the first equality for either sign of the norm. The value $v = 1$ would give both $u^2 = T - 6$ and $u^2 = T - 9$. Thus

$$(994) \quad X = -3 + \frac{3}{1 - v}, \quad Y = \frac{3u}{1 - v}$$

is defined, and substitution gives a rational point of E mapping to (T, V) under ϕ . Explicitly, with $z = X + 3$, $u = Y/z$, $v = 1 - 3/z$, the identities are

$$u^2 - 3v^2 = F(X) - 9, \quad v(u^2 + v^2) = F(X) - 8, \quad u(u^2 + 9v^2) = YF'(X).$$

They also prove that every finite image has trivial cube class. There is no rational point with $X = -3$, since its ordinate would satisfy $Y^2 = -9$. The identity point and its preimage O complete the kernel statement.

Finally $P' = (6, 9)$ gives

$$\theta(P') = 9 - 6\sqrt{3} = \epsilon^2(3 - 2\sqrt{3})^3.$$

Its class has order three. Lemma 266.3 therefore gives the exact image and (991). $\square$

Corollary 266.5 (Exact ranks, without a generator claim). *The ranks of $E(\mathbb{Q})$ and $E'(\mathbb{Q})$ are both one. For the nontrivial-unit quotient H_1 , the rank of $\text{Jac}(H_1)(\mathbb{Q})$ is exactly two.*

Proof. The bijective rational homomorphism ψ , with $\psi\phi = [3]$, identifies (991) with $|E(\mathbb{Q})/3E(\mathbb{Q})| = 3$. By Mordell–Weil ([142], Chapter IV, printed p. 105), write $E(\mathbb{Q}) \simeq \mathbb{Z}^r \oplus T$ with T finite. Then $3 = 3^r|T/3T|$. The ordinary EQ doubling proof already gives the infinite-order point $(-2, 1)$, so $r \geq 1$. It follows that $r = 1$; ψ gives the same rank for E' . The established $\mathbb{Q}$ -isogeny $\text{Jac}(H_1) \sim E \times E$ proves the final assertion. $\square$

Neither this argument nor the finite arithmetic replay proves that $(-2, 1)$ or $(6, 9)$ generates its full group. The simultaneous same-source quartic and its positive inequalities remain to be solved. The Gaussian cube is a reversible descent, not a new point exclusion. The exact replay checks eleven rational-function identities, the displayed unit-class witness, and the complete primitive mod-27 model table; it uses no software rank upper bound.

The arithmetic formally checked here. The module `GaussianDescentArithmetic.lean` contains 15 new theorems. A fresh copy of the source was compiled to a fresh `olean` with warnings treated as errors, under Lean 4.32.0 and the pinned Mathlib snapshot. All 15 complete axiom queries use only the standard three axioms.

The module proves the two actual homogeneous integer norm identities, reduction modulo an arbitrary integer modulus, a complete kernel-checked `Fin(27)` table, its transfer to signed integer congruences, and the actual primitive-curve consequence $3 \nmid B$. It also proves the opposite-isogeny rational-function identity, the three Gaussian inverse and three real-quadratic inverse identities, and both coordinate expressions for $\psi\phi$ and chord-law tripling on the displayed rational charts. Every denominator assumption remains explicit in the theorem signatures; no projective exception is hidden inside a formal global curve assertion.

The quadratic rings, their units and prime splitting, existence and uniqueness of cube roots, the homomorphism δ , isogeny degree, extension of the maps, Mordell–Weil and the exact ranks remain the reviewed ordinary proof above. In particular these arithmetic theorems do not constitute a formal proof of the entire isogeny descent or of the rational points of the simultaneous curve.

267. ACTUAL SIGNED PRODUCTS AND THEIR FINITE DEPTH CONNECTION

The arithmetic argument of Section 265 uses actual integer products. The following finite results give explicit formal connections between those products, their determinant and the single maximal-depth bound. The construction of the arithmetic residue maps remains a separate part of the ordinary proof.

Lemma 267.1 (An integer Gram criterion). *For integer coordinate pairs $z = (R, S)$ and $z' = (R', S')$, put*

$$\begin{aligned} N(z) &= R^2 + RS + S^2, & D &= RS' - R'S, \\ K &= 2RR' + RS' + SR' + 2SS'. \end{aligned}$$

Then $N(z) \geq 0$ and

$$(995) \quad 4N(z)N(z') - K^2 = 3D^2.$$

If an integer m divides D and $4N(z)N(z') < 3m^2$, then $D = 0$.

Proof. The identity $4N(z) = (2R + S)^2 + 3S^2$ proves nonnegativity. Expansion proves (995), hence $3D^2 \leq 4N(z)N(z') < 3m^2$. Since $m^2 \mid D^2$, the nonnegative integer $D^2 < m^2$ must vanish. The strict premise already excludes $m = 0$, so no separate sign condition on m is needed. $\square$

Proposition 267.2 (Private homomorphisms and signed products). *Let I be a finite type and G a commutative group. Suppose $a_i \in G$ and additive integer-valued group homomorphisms v_i satisfy*

$$v_i(a_j) = 0 \quad (j \neq i), \quad v_i(a_i) \neq 0.$$

Then the actual product map

$$P_a : \mathbb{Z}^I \longrightarrow G, \quad P_a(x) = \prod_{i \in I} a_i^{x_i}$$

is injective. If $u_j \in G$ satisfy $v_i(u_j) = 0$ for all i, j , the same holds for P_{ua} .

Let $\mathcal{A} \subset \mathbb{Z}^I$ be finite, H a finite commutative group, and $\phi_i \in H$. If, for actual $x, y \in \mathcal{A}$,

$$(996) \quad P_\phi(x) = P_\phi(y) \implies P_a(x) = P_a(y),$$

then $|\mathcal{A}| \leq |H|$.

Proof. Applying v_i to $P_a(x) = P_a(y)$ gives $x_i v_i(a_i) = y_i v_i(a_i)$, and cancellation proves $x_i = y_i$. This allows negative exponents and negative nonzero diagonal values. Also $v_i(u_j a_j) = v_i(a_j)$, proving the assertion for actual unit normalization. Finally, (996) makes the actual product map from the membership set of $\mathcal{A}$ into H injective. $\square$

Lemma 267.3 (A concrete signed diamond). *For every integer $r \geq 0$, the disjoint union*

$$\mathcal{D}_r = (\{0, \dots, r\} \times \{0, \dots, r\}) \sqcup (\{0, \dots, r-1\} \times \{0, \dots, r-1\})$$

has an injective map into $\{(x, y) \in \mathbb{Z}^2 : |x| + |y| \leq r\}$. Its cardinality is $2r^2 + 2r + 1$.

Proof. On the first square use $(x, y) = (-r + i + j, i - j)$, and on the second use $(x, y) = (-r + 1 + i + j, i - j)$. Within either square the two linear equations recover i, j . Between squares, $x + y$ has different parity, so the images are disjoint. The coordinate bounds give $x + y, x - y, -x + y, -x - y \leq r$, implying $|x| + |y| \leq r$. The empty second square at $r = 0$ is included. The cardinality is $(r + 1)^2 + r^2$. Surjectivity onto the entire lattice diamond is unnecessary for this lower bound. $\square$

Proposition 267.4 (Actual pairs give the high-depth bound). *Let s be a finite index set with the private homomorphisms of Proposition 267.2, and let n, r be natural numbers. Suppose $|H| \leq n \leq 2r^2$. For each distinct $i, j \in s$, assume the rigidity implication (996) for the actual two-element arrays (a_i, a_j) and (ϕ_i, ϕ_j) on $\mathcal{D}_r$. Then $|s| \leq 1$.*

Proof. Two distinct indices restrict the private homomorphisms to two nonzero diagonal values and zero off-diagonal values. The actual diamond products therefore inject into H , yielding $2r^2 + 2r + 1 \leq |H| \leq n \leq 2r^2$, a contradiction. $\square$

For an actual finite depth function $e_i \in \mathbb{N}$ and real L, w , apply Proposition 267.4 to the filtered set

$$I_h = \{i \in I : e_i \geq h\}, \quad h = \max(4, \lceil 2rL/w \rceil).$$

If additionally $r^2 \leq n$, $L \geq 0$, $w > 0$, $e_i w \leq nL$ and $2|I_4|^2 \leq n$, the complete finite-layer argument of Section 265 gives

$$(997) \quad 2 \sum_{i \in I} (e_i - 3)_+ w \leq 5nL.$$

Here $(e_i - 3)_+$ is natural truncated subtraction. To see explicitly that all layers occur, remove one actual maximal-depth index O . Every remaining depth is less than h , so its positive excess costs at most $(h - 4)w \leq 2rL$; only indices in I_4 contribute. The removed cost is at most nL . Finally $(r - 2|I_4|)^2 \geq 0$, together with the two squared bounds, gives $4r|I_4| \leq 3n$, proving (997).

If actual heights $t_i \geq t_0 > 0$ satisfy $nL \leq 2t_0$ and $B > 0$, positivity of the excess terms also gives

$$\frac{1}{B} \sum_{i \in I} \frac{(e_i - 3)_+ w}{t_i} \leq \frac{5}{B}.$$

The low-depth count, the size of H and the precise rigidity implication remain explicit in the formal connection; the cardinality conclusion is derived from the actual two-element maps.

Formal scope. **ActualGramRigidity** contains four declarations and uses the actual previously formalized Eisenstein norm on $\mathbb{Z} \times \mathbb{Z}$. **ActualSignedProducts** contains twelve declarations for actual signed products, private homomorphisms, unit normalization, finite injection and the two explicit squares. **ActualSingleOwnerBridge** contains two declarations deriving actual cardinality at most one and then (997). These eighteen statements complement the twenty-two finite signed-moment and depth declarations in **SignedMomentArithmetic**. Identifying the finite binomial sum with the cardinality of the full integer ℓ^1 ball retains its ordinary counting proof; the special binomial-sum identities and the concrete two-coordinate lower bound are formal. Constructing the arithmetic valuations, residue torsion and unit choices, and controlling the sum over all primes, are distinct obligations.

268. FIFTEENTH CONTINUATION: SIGNED RIGIDITY AND THE REMAINING POINT PROBLEM

The fixed-prime argument now uses actual signed products after a coherent unit choice. Its target has at most n elements. The complete positive excess costs at most $(5/2)n \log(8B)$, and a single actual maximal-depth index contains every sufficiently deep layer. Outside that index the cost is at most $7n^{5/6} \log(8B)$. The stronger support threshold excludes even depth-one hits at a second root of the independent domain. These conclusions include arbitrarily large primes

individually. They still require a bound on the number and total weight of different prime labels before summation over all primes can close the signed gate. The strict private-domain density improvement from the preceding continuation remains available; its complement remains a separate target.

On the surviving cubic geometric branch, explicit descents in $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{3})$ determine an isogeny quotient of order three. They prove, by ordinary mathematics, that both displayed elliptic curves have rank one and the first genus-two quotient has Jacobian rank two. The Gaussian cube is automatic on every rational point of the first elliptic curve. It supplies a reversible map and does not by itself exclude any of those points. The simultaneous quartic, its positive-domain conditions and the other quotient curves remain live rational-point problems. Neither exact rank nor a finite list of tested multiples is a complete rational-point determination.

The finite formal package consists of 55 new declarations: 22 signed count and depth statements, 18 actual determinant/product/diamond connections, and 15 integer and rational-function descent identities. They were freshly compiled together with 56 unchanged local dependency declarations. All 111 explicit axiom queries were checked; their union is exactly `propext`, `Classical.choice` and `Quot.sound`. The compiler is Lean 4.32.0 and Mathlib is pinned to 81a5d257c8e410db227a6665ed08f64fea08e997; its compiled dependency cache was reused. The exact arithmetic replay independently checks eleven rational-function identities, the nontrivial unit-class witness and all 486 primitive residue hits of the specified model modulo 27. Sampling is not used to prove the ordinary rank upper bound.

Local reuse of the official Fermat source has also advanced to a bounded sample of eighteen unchanged dependency modules at the previously cited upstream commit. The six named solution declarations and their six exported wrappers have actual local axiom queries using only the standard three axioms. This sample was compiled with the present compiler and Mathlib cache, requesting only Lean object output. It is separate from the 55 new declarations above. The complete upstream import remains uncompiled locally; the full-source inventory contains 60,475 local modules for its final check. Neither the sample nor the inventory establishes compatibility of every untried source. The successful cusp-form bridge and the official full-Fermat verification records keep their distinct local and upstream provenance.

The next substantive obligations are joint arithmetic control across different primes, the remaining independent-domain complement, complete points on the simultaneous geometric locus, and uniform control of moving exponents, heights and residual coefficients. A failed special test does not exclude any of these parent routes. The research group retains independent analytic, arithmetic, geometric and formal audits. There is still no proof or disproof of *abc*, and no complete formal proof of either conclusion.

269. A PRIME-INDEPENDENT QUOTIENT AND COMBINED-PRECISION PACKETS

Let $n > 324$ be prime, $B = n^4$, $L = \log(8B)$, and

$$K = \mathbb{Q}(\zeta), \quad \zeta^2 - \zeta + 1 = 0, \quad w_k = 3k + \zeta, \quad \alpha_k = w_k/\bar{w}_k \quad (B \leq k < 2B).$$

Here the indices are integers. Fix a subset I on which the actual α_k are multiplicatively independent, and write

$$w_k^n = A_k + C_k \zeta, \quad T_k = |A_k C_k (A_k + C_k)|.$$

The previously proved primitive-power arithmetic makes these three arms nonzero and pairwise coprime. Put

$$G = \{z \in K^\times : z\bar{z} = 1\}, \quad \mu_3 = \{1, \zeta^2, \zeta^4\}, \quad a_k = [\alpha_k] \in G/\mu_3.$$

This section keeps a single actual quotient while using distinct finite reductions. It does not estimate the number or total weight of different primes with different owners.

Lemma 269.1 (Actual quotient and finite reductions). *The elements a_k , $k \in I$, are multiplicatively independent. For every prime $q > 8B$ and positive integer e , the quotient of the norm-one residue group modulo q^e by its embedded μ_3 has an n -torsion subgroup of cardinality at most n . If $q^e \mid T_k$, the actual element a_k reduces into this subgroup.*

Proof. A relation $\prod_k \alpha_k^{x_k} \in \mu_3$ becomes $\prod_k \alpha_k^{3x_k} = 1$ after cubing. Independence of the actual α_k gives $x_k = 0$ for every integer exponent.

Reduction is applied only to ratios whose numerator and denominator are q -units. Every marked root has this property, because a prime dividing one primitive boundary arm does not divide the norm. At precision one the norm-one group is cyclic of order $q - \chi_q$, where $\chi_q = (-3/q)$. Its quotient by the three distinct roots of unity has order $(q - \chi_q)/3$. At higher precision the reduction kernel is an abelian q -group. Since $q \nmid n$, taking n -th powers is invertible on that kernel. Thus n -torsion injects into the precision-one quotient and has cardinality at most n .

The actual arm identities give $\alpha_k^n \in \{1, \zeta^2, \zeta^4\} \pmod{q^e}$ when $q^e \mid T_k$. Consequently the quotient class has n -th power one. This argument works in both the split and inert residue algebras. $\square$

Theorem 269.2 (Combined-precision rigidity). *Let S be a nonempty finite set of distinct primes $q > 8B$. Assign positive integer precisions e_q , set*

$$m = \prod_{q \in S} q^{e_q},$$

and suppose $J \subseteq I$ satisfies

$$q^{e_q} \mid T_k \quad \text{for every } k \in J \text{ and every } q \in S.$$

For a positive integer ν , assume

$$\log m \geq 6\nu L.$$

Then the integer exponent vectors $x = (x_k)_{k \in J}$ with $\sum_k |x_k| \leq \nu$ inject into the product of the finite n -torsion groups in Lemma 269.1.

Proof. Represent the actual product $\prod_k \alpha_k^{x_k}$ as $z_x/\bar{z}_x$, using w_k for positive occurrences and $\bar{w}_k$ for negative occurrences. The zero vector uses the empty product 1. These nonzero Eisenstein integers satisfy

$$|z_x| \leq (6B)^\nu$$

and are units at every marked prime.

Suppose x, y have equal images in all the finite quotient groups. For each $q \in S$ there is a unique $\tau_q \in \mu_3$ such that their ratios differ by τ_q modulo q^{e_q} . For each of the three global τ , choose an Eisenstein unit v_τ with $v_\tau/\bar{v}_\tau = \tau$. Let D_τ be the integer coordinate determinant of

z_x and $v_\tau z_y$. The conjugate cross difference and $q > 3$ give $q^{e_q} \mid D_{\tau_q}$. Pairwise coprimality of the marked prime powers therefore gives the single integer divisibility

$$m \mid D_1 D_{\zeta^2} D_{\zeta^4}.$$

No common value of τ_q is assumed.

The actual Gram identity for Eisenstein coordinates gives

$$|D_\tau| \leq \frac{2}{\sqrt{3}} |z_x| |z_y| \leq \frac{2}{\sqrt{3}} (6B)^{2\nu} < (8B)^{2\nu}.$$

The last strict inequality holds for every positive integer ν , and multiplying by a unit does not change the norm. Hence

$$|D_1 D_{\zeta^2} D_{\zeta^4}| < (8B)^{6\nu} \leq m.$$

An integer divisible by m and of smaller absolute value must vanish. Some D_τ is zero, so the exact algebraic ratios differ by that global element of μ_3 . Independence in Lemma 269.1 forces $x = y$.

All signs, unequal product lengths and the empty vector are included. Equivalently one may partition the primes according to their three phases; some phase has product modulus at least $m^{1/3}$. The integer-product argument also includes an empty phase class. $\square$

Corollary 269.3 (Exact product-target capacity). *With the hypotheses of Theorem 269.2, put $M = |J|$ and $s = |S|$. Then*

$$\sum_{j=0}^{\min(M,\nu)} 2^j \binom{M}{j} \binom{\nu}{j} \leq n^s.$$

Proof. Choose the j nonzero coordinates and their signs. Their positive absolute values have sum at most ν , giving $\binom{\nu}{j}$ choices by adjoining a slack coordinate. The $j = 0$ term is the zero vector. This counts precisely the domain of the injection; the product of the target cardinalities is at most n^s . $\square$

The theorem combines precision from several primes, but every root of J must satisfy every marked prime-power condition. Its domain is a common-hit rectangle in the actual incidence relation. The size of the product target cannot be replaced by n merely because every factor has exponent n . For $M = 1$ the displayed bound is only $2\nu + 1 \leq n^s$; no claim of its optimality on a singleton is needed. If a separate premise supplied a common phase for every comparison at all marked primes, a single determinant and $2\nu L$ would suffice. That additional premise is not used here.

In particular the result neither bounds singleton prime labels nor controls their weighted depths. Repeated fixed-prime arguments do not produce a common finite target of cardinality n . The full positive and signed far tails, roots outside the independent domain, and global ABC coverage remain unresolved. This section records the complete ordinary JP proof; it makes no additional Lean claim about residue groups, their actual arithmetic construction, or these open interfaces.

270. RATIONAL SIMPLICITY AND POSITIVE RANK OF THE REMAINING GENUS-TWO FACTORS

This section gives reviewed ordinary proofs, supported by complete finite tables. It gives no upper rank or complete rational-point classification. For the fixed branch $g = 3, u = \zeta$, put

$$f_t(s) = s^3 + 3ts^2 + (3t - 3)s - 1, \quad H_1 : y^2 = f_0 f_1, \quad H_2 : y^2 = f_1 f_4, \quad H_3 : y^2 = f_0 f_4.$$

All curves mean their smooth projective models, and $J_j = \text{Jac}(H_j)$. These are the three actual quotients of Section 250.

Theorem 270.1 (Order-three symmetry and even ranks). *Each H_j has a rational automorphism of exact order three*

$$\sigma(s, y) = \left(-\frac{1}{s+1}, -\frac{y}{(s+1)^3} \right).$$

Its quotient has genus zero. There is an embedding $\mathbb{Q}(\sqrt{-3}) \hookrightarrow \text{End}_{\mathbb{Q}}(J_j) \otimes \mathbb{Q}$, and $\text{rank } J_j(\mathbb{Q})$ is even.

Proof. For every t , homogeneous substitution gives

$$(s+1)^3 f_t\left(-\frac{1}{s+1}\right) = -f_t(s).$$

The matrix $\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$ has cube $-I$, and the weight-three ordinate transformation has the corresponding sign, so $\sigma^3 = 1$. Its action on s is nontrivial. The map extends to an automorphism of the smooth projective curve, including $s = -1$ and infinity.

Set $A = -3s(s+1)$, $B = f_1(s)$, $r = A/B$. The two homogeneous cubic coordinates transform by the same factor. The invariant ordinate y/B satisfies, for the three curves respectively,

$$v^2 = 1 + r, \quad w^2 = 1 - 3r, \quad z^2 = (1+r)(1-3r).$$

Since $s \mapsto r$ has degree three, the extensions of these conic function fields have degree at most three; the automorphism of order three forces equality. These are therefore the fixed fields. Each conic is smooth of genus zero and has the rational point $(r, \text{ordinate}) = (0, 1)$. The norm followed by pullback factors through its zero Jacobian and is $1 + \sigma_* + \sigma_*^2$. Thus σ_* satisfies $X^2 + X + 1 = 0$. The associated unit map from the quadratic field is injective. It makes $J_j(\mathbb{Q}) \otimes \mathbb{Q}$ a vector space over that field, proving that its rational dimension is even, including rank zero. $\square$

Theorem 270.2 (Exact certificates at five). *The curves H_2, H_3 have good reduction at five, with counts*

	$\#H(\mathbb{F}_5)$	$\#H(\mathbb{F}_{25})$
H_2	6	36
H_3	6	12

Their Jacobian Frobenius characteristic polynomials are

$$(998) \quad P_2(X) = X^4 + 5X^2 + 25, \quad P_3(X) = X^4 - 7X^2 + 25.$$

Proof. The discriminant of f_t is $81(t^2 - t + 1)^2$, nonzero modulo five for $t = 0, 1, 4$. Distinct chosen factors differ by $3(t-u)s(s+1)$, with nonzero coefficient, and have values $-1, 1$ at $s = 0, -1$. Hence their products have six distinct geometric roots modulo five. Their leading coefficient is one. In the infinity chart the two points have ordinate ± 1 and are smooth. The two standard hyperelliptic charts give a smooth proper model over $\mathbb{Z}_5$, and its Jacobian has good reduction.

The sextic values for $s = 0, 1, 2, 3, 4$ are respectively

$$[1, 3, 2, 3, 1], \quad [1, 2, 3, 2, 1].$$

In both cases the character sum is $1 - 1 - 1 - 1 + 1 = -1$; adding the five affine baseline fibres and two points at infinity gives six. Here is the complete extension-field certificate. In $\mathbb{F}_{25} = \mathbb{F}_5[\omega]$, $\omega^2 = 2$, rows below index $b = 0, \dots, 4$ and columns $a = 0, \dots, 4$, for inputs $a + b\omega$. The entries are quadratic characters of the sextic values:

$$\begin{array}{c} H_2 \\ \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & 1 & 1 \\ 1 & -1 & -1 & -1 & 1 \\ 1 & -1 & -1 & -1 & 1 \\ 1 & 1 & -1 & 1 & 1 \end{pmatrix} \end{array} \quad \begin{array}{c} H_3 \\ \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 \\ -1 & -1 & -1 & -1 & -1 \\ -1 & -1 & -1 & -1 & -1 \\ -1 & -1 & -1 & -1 & -1 \end{pmatrix} \end{array}$$

Their sums are 9, -15 , giving $25 + 9 + 2 = 36$ and $25 - 15 + 2 = 12$. These entries can be checked using the Legendre symbol of $a^2 - 2b^2$: the norm-character identity is $\chi_{25}(x) = x^{12} = N_{25/5}(x)^2$, including the zero convention. There are no zero entries in these two tables.

For $S_1 = 6 - \#H(\mathbb{F}_5)$, $S_2 = 26 - \#H(\mathbb{F}_{25})$, the curve/Jacobian trace formula and Newton's identity give

$$P(X) = X^4 - S_1X^3 + \frac{S_1^2 - S_2}{2}X^2 - 5S_1X + 25.$$

Here $S_1 = 0$ and $S_2 = -10, 14$, proving (998). The standard trace and good-Jacobian reduction inputs are [141, Theorem 11.1, Corollaries 11.4 and 12.3]. $\square$

Theorem 270.3 (No rational elliptic factors). *Both J_2, J_3 are simple over $\mathbb{Q}$ and are not $\mathbb{Q}$ -isogenous to each other. Neither H_2 nor H_3 has a nonconstant morphism over $\mathbb{Q}$ to a genus-one curve.*

Proof. If one of these abelian surfaces were not simple over $\mathbb{Q}$, Poincaré reducibility over the ground field would give an isogeny to $E_a \times E_b$. Its good reduction at five, the rational Tate-module decomposition and the Neron–Ogg–Shafarevich criterion give good reduction at five for both elliptic factors; see [143, I, Proposition 10.1; IV, Theorem 3.5 and Corollary 3.6]. Writing their integer traces as a, b would give

$$P_j(X) = (X^2 - aX + 5)(X^2 - bX + 5).$$

The cubic coefficient implies $b = -a$, and the quadratic coefficient is then $10 - a^2$. For $j = 2$ one obtains $a^2 = 5$; for $j = 3$, $a^2 = 17$. Neither is possible for an integer.

A nonconstant genus-one quotient induces a surjection from J_j to a dimension-one Jacobian, contrary to simplicity. One may also use the rational infinity point on H_j to identify the target with its elliptic Jacobian. An isogeny between J_2, J_3 would identify their rational Tate modules and hence their polynomials at five, which differ. $\square$

This is simplicity over $\mathbb{Q}$, not absolute simplicity. Combining Sections 250 and 266 gives

$$(999) \quad \text{Jac}(D_{3,\zeta}) \sim_{\mathbb{Q}} E^2 \times J_2 \times J_3,$$

$$(1000) \quad \text{rank Jac}(D_{3,\zeta})(\mathbb{Q}) = 2 + \text{rank } J_2(\mathbb{Q}) + \text{rank } J_3(\mathbb{Q}).$$

There are no rational elliptic factors for the two remaining surfaces; working over an extension field or directly with those surfaces remains possible. This does not yet determine their ranks or the simultaneous positive source locus.

Theorem 270.4 (Trivial torsion and rank at least six). *Both $J_2(\mathbb{Q})$ and $J_3(\mathbb{Q})$ have trivial torsion and rank at least two. The genus-six Jacobian in (999) therefore has rank at least six.*

Proof. The discriminants and factor differences used above also prove good reduction at seven. The sextic values at $s = 0, \dots, 6$ are

$$H_2 : [1, 0, 1, 0, 3, 0, 1], \quad H_3 : [1, 0, 3, 0, 1, 0, 1].$$

Both character sums are two and both curve counts are eleven. For a complete check over $\mathbb{F}_{49} = \mathbb{F}_7[\omega]$, $\omega^2 = 3$, the character tables, with rows b and columns a in $a + b\omega$, are

$$\begin{array}{c} H_2 \\ \left(\begin{array}{cccccc} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & -1 & 1 & 1 & 1 & -1 & 1 \\ 1 & -1 & -1 & -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & -1 & 1 & 1 & -1 \\ -1 & -1 & 1 & -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 & 1 & -1 & 1 \end{array} \right) \end{array} \quad \begin{array}{c} H_3 \\ \left(\begin{array}{cccccc} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & -1 & 1 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 & 1 & -1 & -1 \\ -1 & 1 & 1 & -1 & 1 & -1 & -1 \\ 1 & 1 & 1 & -1 & -1 & -1 & 1 \\ 1 & -1 & 1 & 1 & 1 & -1 & 1 \end{array} \right) \end{array}$$

Both sums are ten, and $\#H_j(\mathbb{F}_{49}) = 49 + 10 + 2 = 61$. The norm-character rule now uses $a^2 - 3b^2$ modulo seven. Newton's identities give in both cases

$$P_7(X) = X^4 + 3X^3 + 10X^2 + 21X + 49, \quad \#J_j(\mathbb{F}_7) = 84.$$

At five, (998) gives $\#J_2(\mathbb{F}_5) = 31$, $\#J_3(\mathbb{F}_5) = 19$. Prime-to- p rational torsion injects into the special fibre at a good prime p , by the finite étale multiplication kernels and smooth proper base

change (equivalently the specialization used in [143, IV, Theorem 3.5]). For a torsion prime other than five or seven its full primary order must divide both 84 and 31, or both 84 and 19, impossible. Five-primary torsion is excluded at seven, and seven-primary torsion is excluded at five. All rational torsion therefore vanishes.

On either H_j , let $P = (0, 1)$ and let $Q = \infty_+$ have leading ordinate $+1$. The class $[P - Q]$ is nonzero. Otherwise a function with precisely the single simple pole Q would define a degree-one map to $\mathbb{P}^1$, contrary to genus two. With torsion zero this class has infinite order. The rank is positive and, by Theorem 270.1, even, hence at least two. Formula (999) completes the proof. $\square$

These are lower bounds. They prevent directly using the classical strict rank-less-than-genus condition on $D_{3,\zeta}, H_2, H_3$; they do not exclude quadratic Chabauty, further descent, or methods over extension fields. No upper rank, Mordell–Weil basis, point-height bound or positive seed follows from these tables.

Finite verification and ordinary scope. The standard-library program `replay_rational_simple_quotients.py --check` regenerates the complete four curve/prime certificate, directly counting square fibres and comparing them with norm-character tables. Its canonical SHA-256 is 6ba65a743aa190819b6d3b43c7e357d0f5aacb1e53508078cab882a46f240883. Both peers independently recomputed the finite counts. The Jacobian, Frobenius, simplicity and Mordell–Weil conclusions are the ordinary arguments above with their cited inputs; finite arithmetic is not a formal proof of the complete geometric conclusions.

271. A QUADRATIC CHABAUTY ENTRY THROUGH THE FIRST QUOTIENT

We give the now independently reviewed ordinary application of the quadratic Chabauty finiteness criterion. No Chabauty set is computed in this section. Put $D = D_{3,\zeta}$, let H_j be the three curves of Section 270, and let $J_D = \text{Jac}(D)$. The established decompositions and exact elliptic rank give

$$J_D \sim_{\mathbb{Q}} E^2 \times J_2 \times J_3, \quad J_1 \sim_{\mathbb{Q}} E^2, \quad E: Y^2 = X^3 - 9X - 9, \quad \text{rank } E(\mathbb{Q}) = 1.$$

The full projective morphism $\pi: D \rightarrow H_1$ has degree two. On its dense chart it is $(s, v, w) \mapsto (s, y = Bv)$, where $B = f_1(s)$.

Lemma 271.1 (Rational divisor classes and good reduction). *For the Neron–Severi rank over the ground field,*

$$\rho_{\mathbb{Q}}(J_1) \geq 3, \quad \rho_{\mathbb{Q}}(J_D) \geq 5.$$

Both H_1 and D have rational basepoints and good reduction at five.

Proof. On E^2 , the rational divisor classes of $E \times \{O\}$, $\{O\} \times E$ and the diagonal have intersection matrix

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad \det = 2.$$

They are independent modulo algebraic equivalence since algebraically trivial divisors are numerically trivial. Hence $\rho_{\mathbb{Q}}(E^2) \geq 3$. Pullback by an isogeny is injective on Neron–Severi groups tensored with $\mathbb{Q}$: the norm of a pulled-back line bundle is its degree-th tensor power, and norm preserves algebraic equivalence. There are isogenies in both directions, so the rational ranks agree. This proves the first inequality. On $E^2 \times J_2 \times J_3$, add a rational ample class from each J_j . Such classes exist by projectivity over $\mathbb{Q}$. Restriction to the three factors proves the independence of all five classes, giving the second inequality. No assertion of equality, absolute Picard rank, or absence of CM is needed.

The smoothness argument of Theorem 270.2 applies to $f_0 f_1$ as well: both discriminants are units at five, their difference is $3s(s+1)$, and neither vanishes at $0, -1$. The two infinity points with leading ordinate ± 1 are smooth; either is a rational basepoint.

For D , the three branch divisors on $\mathbb{P}^1$ are the zeros of f_0, f_1, f_4 . Each is finite étale over $\mathbb{Z}_5$ and they are disjoint. Normalize in the biquadratic extension

$$v^2 = f_0/f_1, \quad w^2 = f_4/f_1.$$

Away from these divisors the cover is étale. At a zero of f_0 or f_4 , only one square root ramifies. Étale-locally on the base its normalized equation is $z^2 = t$ times a unit; it is smooth over $\mathbb{Z}_5$ because the coefficient of the local parameter t is a unit. The other square root is an étale unit extension. At a zero of f_1 , use $z = 1/v$ and w/v : the first gives the same simple ramification and the latter has square a unit. Thus the normalization is smooth at every branch. It is proper and finite over $\mathbb{P}_{\mathbb{Z}_5}^1$ and has the smooth genus-six generic fibre, so it supplies good reduction. Infinity is unramified. The point $(s, v, w) = (0, 1, 1)$ is rational and nonbranch and supplies a basepoint. $\square$

Theorem 271.2 (A finite five-adic container through the first quotient). *Let $H_1(\mathbb{Q}_5)_2$ denote the quadratic Chabauty set of Balakrishnan–Dogra. Then*

$$\begin{aligned} H_1(\mathbb{Q}_5)_2 &\text{ is finite,} \\ \mathcal{C}_5 = \{P \in D(\mathbb{Q}_5) : \pi(P) \in H_1(\mathbb{Q}_5)_2\} &\text{ is finite,} \\ D(\mathbb{Q}) &\subseteq \mathcal{C}_5. \end{aligned}$$

Proof. The exact GD rank is $\text{rank } J_1(\mathbb{Q}) = 2$, while the preceding lemma gives

$$2 < 2 + \rho_{\mathbb{Q}}(J_1) - 1,$$

whose right side is at least four. All hypotheses of [144, Lemma 3.2] apply: the field is $\mathbb{Q}$, the curve is smooth geometrically integral and projective of genus greater than one, it has a rational basepoint, and five is a good reduction prime and splits completely in the base field. The Picard number in that criterion is precisely over $\mathbb{Q}$. It follows that $H_1(\mathbb{Q}_5)_2$ is finite; its defining Selmer condition includes $H_1(\mathbb{Q})$.

The projective quotient π is finite of degree two. Every fibre over a $\mathbb{Q}_5$ -point has at most two geometric points, including ramified fibres and all chart exceptions. Its inverse image of the finite set is therefore finite. Every rational point maps to a rational point of H_1 , proving the asserted inclusion. $\square$

This theorem does not compute $\mathcal{C}_5$, its cardinality or coordinates, or determine which of its points are rational. It does not identify $\mathcal{C}_5$ with the intrinsic set $D(\mathbb{Q}_5)_2$. The quotient entry avoids an upper rank bound for J_2, J_3 for this container. It does not classify the positive common-source locus. An actual calculation still needs rigorous heights and integrals, finite local-height value sets, sufficient precision, and a rational-point or Mordell–Weil sieve. A finite search for multiples of a displayed elliptic point supplies none of these completeness assertions; GD by itself does not prove that point to be a global generator.

Proposition 271.3 (A separate conditional entry on the full curve). *If the additional bound*

$$\text{rank } J_2(\mathbb{Q}) + \text{rank } J_3(\mathbb{Q}) \leq 6$$

holds, then $D(\mathbb{Q}_5)_2$ itself is finite.

Proof. The bound would give $\text{rank } J_D(\mathbb{Q}) \leq 8$, while

$$\text{genus}(D) + \rho_{\mathbb{Q}}(J_D) - 1 \geq 6 + 5 - 1 = 10.$$

The other source hypotheses were verified in Lemma 271.1; the same finiteness criterion applies. $\square$

The extra upper bound is unproved. The positive even lower bounds from Theorem 270.4 mean that it would allow total genus-six Jacobian rank six or eight. If the total rank were ten or greater, the particular estimate $\rho \geq 5$ would simply not establish this criterion; larger Picard number or other constructions remain possible.

Source and scope. The setup and the exact criterion were checked in [144, printed p. 3, Section 2, and Lemmas 3.1–3.2 on printed p. 12]. They concern general smooth projective curves, not only hyperelliptic ones, and use the ground-field Picard number. The explicit height formula in that paper’s Theorem 1.2 has additional assumptions, including rank equal to genus and a finite-index closure of the rational Jacobian group in the local group. Those algorithmic assumptions are not claimed or needed by the abstract finiteness argument of this section. The

intersection calculation and divisor norm argument above prove the stated Neron–Severi bounds. No Chabauty, Neron–Severi, rank or complete point-set theorem is claimed here as a Lean formalization.

272. AN EXACT THREE-PHASE DETERMINANT AND JOINT FINITE CAPACITY

The joint packet argument has an arithmetic refinement that keeps all three phases inside one actual additive relation. It improves a constant in the determinant comparison, without providing the missing global radical estimate.

Lemma 272.1 (The three actual phase determinants). *For integer Eisenstein coordinate pairs z, w , put*

$$N(a, b) = a^2 + ab + b^2, \quad R(a, b) = (-b, a + b),$$

and define

$$x = \det(z, w), \quad y = \det(z, R^2 w), \quad d = \det(z, R w).$$

Then

$$(1001) \quad d = x + y, \quad N(z)N(w) = x^2 + xy + y^2,$$

$$(1002) \quad 4(N(z)N(w))^3 - 27(xy d)^2 = ((x - y)(2x + y)(x + 2y))^2.$$

If an integer m divides $xy d$ and

$$(1003) \quad 4(N(z)N(w))^3 < 27m^2,$$

then at least one of x, y, d is zero.

Proof. Writing $z = (a, b)$ and $w = (c, e)$ gives $x = ae - bc$, $y = ac + bc + be$ and $d = ac + ae + be$. These expressions give (1001). Expanding $4(x^2 + xy + y^2)^3 - 27x^2y^2(x + y)^2$ proves (1002). In particular $27(xy d)^2 \leq 4(N(z)N(w))^3 < 27m^2$. But $m^2 \mid (xy d)^2$, so this nonnegative integer strictly below m^2 must vanish. The strict premise excludes $m = 0$ automatically; negative moduli and zero coordinate pairs require no extra convention. $\square$

Proposition 272.2 (Different phases, one product modulus). *Let $(m_i)_{i \in S}$ be finitely many pairwise coprime integer moduli. If each m_i divides at least one of x, y, d , then $\prod_{i \in S} m_i \mid xy d$. If (1003) holds for this product modulus, one phase determinant vanishes.*

Proof. Each modulus divides the full product $xy d$. Pairwise coprimality implies divisibility by their product, and Lemma 272.1 applies. The selected phase may differ for every i . $\square$

Proposition 272.3 (A finite joint-capacity theorem). *Let A and S be finite sets, with A represented by actual integer pairs z_u , and let $f_i : A \rightarrow H_i$ be maps to finite sets for $i \in S$. Suppose:*

- (1) *the integer moduli m_i are pairwise coprime;*
- (2) *$f_i(u) = f_i(v)$ forces m_i to divide one of the three phase determinants for z_u, z_v ;*
- (3) *for every u, v , (1003) holds with $z = z_u$, $w = z_v$ and $m = \prod_i m_i$;*
- (4) *a zero phase determinant for z_u, z_v implies $u = v$.*

Then the joint map $u \mapsto (f_i(u))_{i \in S}$ is injective, and

$$|A| \leq \prod_{i \in S} |H_i| \leq n^{|S|} \quad \text{if } n \in \mathbb{N} \text{ and } |H_i| \leq n \text{ for every } i.$$

Proof. Equality of the joint images gives a phase divisibility at every coordinate. Proposition 272.2 and the size bound force a zero determinant, so the separation premise identifies the two inputs. Counting the finite product target proves both inequalities. $\square$

Lemma 272.4 (Private independence survives a cubic kernel). *Let G be a commutative group, $\phi : G \rightarrow H$ a group homomorphism whose kernel is killed by three, and $(a_i)_{i \in I}$ a finite family with integer-valued homomorphisms v_i satisfying $v_i(a_j) = 0$ for $j \neq i$ and $v_i(a_i) \neq 0$. Then $x \mapsto \phi(\prod_i a_i^{x_i})$ is injective on $\mathbb{Z}^I$.*

Proof. Equality of two images places their quotient in the kernel. Cubing and applying v_i gives $3v_i(a_i)(x_i - y_i) = 0$. The nonzero diagonal value gives $x_i = y_i$ for every i , including negative exponents. $\square$

Exact scope. The formal module proves the displayed integer identities, product divisibility, kernel lemma and finite-capacity implication. In the JP application, actual modular reductions, the n -torsion sizes, product height bounds and separation of distinct signed vectors are supplied by the separate ordinary argument. The capacity theorem does not construct those objects by taking them as hypotheses. Moreover, $n^{|S|}$ cannot be replaced by n merely because every target has exponent dividing n . The actual relation $d = x + y$ is also a warning: postulating an unproved radical bound for these three determinants would introduce another ABC-shaped estimate.

273. SIXTEENTH CONTINUATION: VERIFIED SCOPE AND REMAINING DEPENDENCIES

Standard ABC remains unproved and undisproved. The present continuation adds the complete ordinary joint-packet argument, the rational simplicity and rank lower bounds for the two remaining genus-two factors, a finite quadratic Chabauty container through the first quotient, and the actual three-phase integer bridge. Their ordinary proofs preceded implementation and received independent internal review. This is not external peer review.

The two new Lean modules contain 27 declarations: ten for actual phase arithmetic and joint finite capacity, and seventeen for homogeneous curve identities, complete finite square-fibre counts, integer trace obstructions and the resulting numerical gcds. All 27 new declarations and 45 unchanged local dependency declarations were freshly compiled with warnings treated as errors. Every one of the 72 declarations has an explicit axiom query; only `propext`, `Classical.choice` and `Quot.sound` occur. The checked compiler is Lean 4.32.0 and the exact Mathlib source pin is 81a5d257c8e410db227a6665ed08f64fea08e997. Its compiled dependency cache was reused; the five listed local source modules were freshly built.

The formal statements retain their mathematical inputs. In particular, the joint-capacity theorem assumes collision divisibility, actual pair heights, phase separation and the finite target sizes. It does not prove the construction of the arithmetic norm-one quotient or its torsion. The finite curve counts use the displayed quadratic pair multiplication and add two as the number of infinity points. Good reduction, the interpretation of those pairs as the extension field, the curve/Jacobian trace formula, rational simplicity, Mordell–Weil rank and torsion groups are supplied by the separately reviewed ordinary proof. The Chabauty criterion and the Neron–Severi arguments are also ordinary inputs, not Lean theorems in these modules.

The mathematical dependency chain has two distinct open ends. For the analytic route, combined precision on a common-hit rectangle still gives the product target $n^{|S|}$. It does not control the nonrectangular incidence of different primes with different roots, the full signed far tail, exceptional individual roots, or global ABC coverage. For the geometric route, the existence of a finite five-adic container has not computed that container or determined its rational survivors. Even a complete classification of the fixed cubic branch would leave changing exponents and residual parameters, integral lifting and uniform point heights unresolved. None of these gaps is replaced by a formal hypothesis and then counted as a solved conjecture.

The full verification logs and exact source hashes are recorded in the accompanying checkpoint. Its finite replay independently regenerates all four small-field curve certificates. Subsequent research candidates about collective input content, actual divisor partitions and local Jacobian closure are kept in the next continuation until their own proof, formal-scope and publication reviews are complete. No unrefuted parent route is abandoned.

274. COLLECTIVE CONTENT, PRIMITIVE NORM PROFILES, AND EXACT COMPLEMENT CANCELLATION

Throughout this section let

$$\mathcal{O} = \mathbb{Z}[\zeta], \quad \zeta^2 - \zeta + 1 = 0, \quad N(a + b\zeta) = a^2 + ab + b^2.$$

For a nonzero $Z = A + C\zeta$, put

$$\text{cont}(Z) = \gcd(|A|, |C|) > 0, \quad Z_{\text{prim}} = Z / \text{cont}(Z).$$

Fix an integer $B \geq 2$, and define the actual input block

$$K_B = \{B, \dots, 2B - 1\}, \quad w_k = 3k + \zeta, \quad Q_k = N(w_k) = 9k^2 + 3k + 1.$$

The results below concern these input norms. They do not identify their prime factors with the boundary primes of $P(w_k^n)$, nor assume that the whole block of ratios $w_k/\bar{w}_k$ is multiplicatively independent.

274.1. The exact oriented content and its interval average. Each Q_k is odd, is one modulo three, and has only split prime factors $p \equiv 1 \pmod{3}$. For each such prime choose the two conjugate prime ideals (π_p) , $(\bar{\pi}_p)$, and write

$$d_{k,p} = v_{\pi_p}(w_k) - v_{\bar{\pi}_p}(w_k).$$

The two valuations cannot both be positive, since the coordinates $(3k, 1)$ are primitive. Hence $|d_{k,p}| = v_p(Q_k)$.

Lemma 274.1 (CC1: exact signed incidence and content). *For arbitrary integers x_k , $k \in K_B$, define the canonical product*

$$Z_x = \prod_{x_k > 0} w_k^{x_k} \prod_{x_k < 0} \bar{w}_k^{-x_k}, \quad U_p(x) = \sum_k |x_k| v_p(Q_k), \quad D_p(x) = \sum_k x_k d_{k,p}.$$

The empty product is one. Then

$$\begin{aligned} 2v_p(\text{cont}(Z_x)) &= U_p(x) - |D_p(x)|, \\ \log N((Z_x)_{\text{prim}}) &= \sum_p |D_p(x)| \log p. \end{aligned}$$

All sums have finite support.

Proof. Let E_p^+, E_p^- be the two ideal valuations of Z_x . Their sum is $U_p(x)$, and their difference is $D_p(x)$. A rational power p^j divides both integer coordinates exactly when it divides both conjugate ideal factors, so

$$v_p(\text{cont}(Z_x)) = \min(E_p^+, E_p^-).$$

Now $2 \min(s, t) = s + t - |s - t|$ proves the first identity. Moreover

$$N(Z_x) = \prod_k Q_k^{|x_k|}, \quad N((Z_x)_{\text{prim}}) = N(Z_x) / \text{cont}(Z_x)^2.$$

Taking logarithms and using the first identity proves the second. Neither the ramified prime three nor an inert prime occurs. The proof includes an empty product and individual zero coordinates. $\square$

Lemma 274.2 (CC2: support of collective content). *For every subset $J \subseteq K_B$, put $W_J = \prod_{k \in J} w_k$. Every prime dividing $\text{cont}(W_J)$ is at most $12B - 5$. The same bound holds for the canonical signed products in Lemma 274.1.*

Proof. A content prime occurs with both orientations. Primitivity of each root forces occurrences at two distinct indices k, l . If $p > 12B - 5$, then

$$p \mid Q_k - Q_l = 3(k - l)(3(k + l) + 1).$$

But $p > 3$, $0 < |k - l| \leq B - 1 < p$, and $0 < 3(k + l) + 1 \leq 12B - 5 < p$, a contradiction. For a canonical signed product a prime above this cutoff can occur at only one index and in only its selected orientation, giving the same conclusion. Opposite occurrences of the same index must first be cancelled: the statement is false for unrestricted products such as $w_k \bar{w}_k = Q_k$. $\square$

Theorem 274.3 (CC3: uniform interval content). *For every interval $J = \{a, \dots, a + H - 1\} \subseteq K_B$, where $0 \leq H \leq B$ is an integer, as $B \rightarrow \infty$,*

$$\begin{aligned}\log \text{cont}(W_J) &= \frac{H}{2} \log B + O(H \log \log B + B), \\ \log N((W_J)_{\text{prim}}) &= H \log B + O(H \log \log B + B).\end{aligned}$$

The constants are absolute and independent of a, H . In particular,

$$\begin{aligned}\log \text{cont}(W_{K_B}) &= (1/2 + o(1))B \log B, \\ \log N((W_{K_B})_{\text{prim}}) &= (1 + o(1))B \log B,\end{aligned}$$

whereas $\log N(W_{K_B}) = 2B \log B + O(B)$.

Proof. The polynomial $9X^2 + 3X + 1$ has discriminant -27 . At each split prime $p \geq 7$, its two roots modulo p are simple and each lifts uniquely at every prime power. Label the two towers by the two orientations. Since $9B^2 < Q_k < 36B^2$, put

$$h_p = \left\lfloor \frac{\log(36B^2)}{\log p} \right\rfloor.$$

No individual valuation exceeds h_p . For either fixed tower, the number of points in an interval of H consecutive integers belonging to its residue class modulo p^j differs from H/p^j by at most one. Therefore the total oriented valuation satisfies

$$\left| E_p^\pm(J) - \frac{H}{p-1} \right| \leq h_p + 1.$$

Indeed the first h_p errors sum to at most h_p , and the omitted geometric tail is

$$\frac{H}{p^{h_p}(p-1)} < 1,$$

using $p^{h_p+1} > 36B^2$, $H \leq B$, and $p \geq 7$. The minimum of the two valuations obeys the same error bound. Lemma 274.2 then gives

$$\log \text{cont}(W_J) = H \sum_{\substack{p \leq 12B \\ p \equiv 1 \pmod{3}}} \frac{\log p}{p-1} + O\left(\sum_{p \leq 12B} (h_p + 1) \log p \right).$$

The endpoint sum is $O(B)$: each summand is at most $\log(36B^2) + \log(12B)$, and the elementary Chebyshev bound gives $\pi(12B) = O(B/\log B)$. This estimate includes primes absent from W_J ; their zero contributions still satisfy the oriented error bound.

The only progression-distribution input is the fixed-modulus estimate

$$\vartheta(x; 3, 1) = x/2 + O(x/\log x).$$

It follows from Theorem 1.2, equation (1.12), printed page 5 of [M. A. Bennett, G. Martin, K. O'Bryant and A. Rechnitzer, *Explicit bounds for primes in arithmetic progressions*](#). That theorem gives explicit positive constants for each fixed modulus; only modulus three is used here. Partial summation yields

$$\sum_{\substack{p \leq x \\ p \equiv 1 \pmod{3}}} \frac{\log p}{p} = \frac{1}{2} \log x + O(\log \log x).$$

The difference obtained by replacing $1/p$ with $1/(p-1)$ is bounded by the convergent series $\sum_{m \geq 2} \log m/(m(m-1))$. Substitution proves the first assertion. Finally $\log Q_k = 2 \log B + O(1)$ uniformly in K_B , and

$$N(W_J) = \text{cont}(W_J)^2 N((W_J)_{\text{prim}})$$

proves the remaining formulas. The asymptotic formulation avoids any small- B convention for $\log \log B$. $\square$

Lemma 274.4 (CC4: the exact deletion cost). *If $K_B = J \sqcup F$, then*

$$\text{cont}(W_{K_B}) \mid \text{cont}(W_J) \frac{N(W_F)}{\text{cont}(W_F)}.$$

The right side is an integer. Consequently

$$\log \text{cont}(W_J) \geq \log \text{cont}(W_{K_B}) - \sum_{k \in F} \log Q_k.$$

Proof. At a split prime write (a, b) for the valuations of W_J and (c, d) for those of W_F . Then

$$\min(a + c, b + d) \leq \min(a, b) + \max(c, d), \quad \max(c, d) = c + d - \min(c, d).$$

These are exactly the primewise divisibility inequalities. They also show that $N(W_F)/\text{cont}(W_F)$ is an integer. Taking logarithms and discarding the nonnegative $\log \text{cont}(W_F)$ proves the weaker displayed lower bound. $\square$

Deleting $o(B)$ arbitrary roots preserves the full-block main term. Deleting a fixed positive fraction has a charged cost; this lemma does not transfer that main term to the known independent domains of roughly half the block. More fundamentally, the primes cancelled here divide input norms Q_k , whereas a boundary prime dividing $P(w_k^n)$ does not divide Q_k . No formula above identifies the cancelled input content with the negative boundary contributions $(3 - v_q(P(w_k^n)))_+ \log q$. A prime hitting just one boundary need not make the full-product ratio an n -torsion element, because the unhit factors remain.

274.2. Exact cancellation for a common multiplier and complement. Write $R(a, b) = (-b, a + b)$ and $\det((a, b), (c, d)) = ad - bc$. Let $W = (a, b)$ be primitive and let $V \neq 0$ be an Eisenstein integer. Put

$$c = \text{cont}(VW), \quad d = \text{cont}(V), \quad U = VW/c, \quad V_0 = V/d,$$

and define

$$x = \det(U, V_0), \quad D = \det(U, RV_0), \quad y = \det(U, R^2V_0).$$

Primitivity makes $W \neq 0$, so c, d are positive integers and U, V_0 are actual primitive pairs.

Theorem 274.5 (CD1: the precise common scalar). *There is a positive integer r such that*

$$N(V) = cdr, \quad (x, D, y) = r(-b, a, a + b), \quad \gcd(|x|, |D|, |y|) = r.$$

Proof. Multiplication by $V = v_0 + v_1\zeta$ has integer matrix

$$\begin{pmatrix} v_0 & -v_1 \\ v_1 & v_0 + v_1 \end{pmatrix}$$

of determinant $N(V)$, and commutes with R . Apply this determinant identity to W and $1, \zeta, \zeta^2$, and clear the positive denominators c, d . The result is

$$cdx = -bN(V), \quad cdD = aN(V), \quad cdy = (a + b)N(V).$$

Choose integers u, v with $ua + vb = 1$. Then

$$N(V) = cd(uD - vx).$$

Thus $cd \mid N(V)$, and $r = N(V)/(cd)$ is a positive integer. Cancelling cd proves the three coordinates. Their common gcd is r , since $\gcd(a, b, a + b) = 1$. The proof permits individual zero coordinates of W ; only its primitivity and $V \neq 0$ are needed. In particular integrality of r follows from Bézout, not merely from knowing that the three determinants are integers separately. $\square$

Corollary 274.6 (CD2: the primitive boundary ledger is unchanged). *Assume $ab(a + b) \neq 0$. Dividing by the full common content in Theorem 274.5 gives the additive relation $-b + (a + b) = a$. The absolute primitive boundary product is*

$$|(x/r)(D/r)(y/r)| = |ab(a + b)|.$$

Its radical and its complete signed depth sum are therefore exactly those of W :

$$\sum_{q \mid |(x/r)(D/r)(y/r)|} (v_q(|(x/r)(D/r)(y/r)|) - 3) \log q = \sum_{q \mid |ab(a+b)|} (v_q(ab(a+b)) - 3) \log q.$$

Proof. These identities follow from the three exact coordinates, including their signs. Before division the determinant product is $-r^3 ab(a+b)$. If $r > 1$, that triple is not primitive and cannot be inserted into a primitive radical assertion before cancelling r . There is no omitted content term in the stated equalities. $\square$

Corollary 274.7 (CD3: simultaneous audit of every complement owner). *In the actual primitive unramified power block, let n be a positive integer, and take any finite set J of indices and any $k \in J$. Set*

$$W = w_k^n, \quad V = \prod_{l \in J \setminus \{k\}} w_l^n.$$

The empty complement is one. The preceding identities hold for every k simultaneously, and the primitive three-phase output is precisely that owner's original additive triple.

Proof. The actual unramified primitive-power statement makes W primitive with nonzero boundary arms, and every complement factor is nonzero. Apply Theorem 274.5 and Corollary 274.6 separately to each k ; neither argument imposes a common boundary prime condition across the other indices. $\square$

This nonrectangular construction retains a different complement for each owner but returns exactly the original primitive ledger. It audits this particular common-multiplier map, without ruling out other mixed determinants, divided differences, or additional relations between their primitive outputs.

274.3. The powerful part of the primitive collective input norm. For an interval $J \subseteq K_B$ of length H , define

$$N_J = N((W_J)_{\text{prim}}), \quad R_J = \prod_{\substack{p > 12B \\ p \mid Q_k \text{ for some } k \in J}} p, \quad M_B = \prod_{\substack{p \leq 12B \\ p \equiv 1 \pmod{3}}} p^{h_p},$$

where $h_p = \lfloor \log(36B^2) / \log p \rfloor$. These are finite products of actual rational primes.

Theorem 274.8 (CP1: localization of the entire powerful part). *There is a positive integer D_J such that*

$$N_J = R_J D_J, \quad \gcd(R_J, D_J) = 1, \quad D_J \mid M_B.$$

Moreover

$$\frac{N_J}{\text{rad}(N_J)} \mid D_J \mid M_B, \quad 0 \leq \log N_J - \log \text{rad}(N_J) \leq \log M_B = O(B).$$

The implied constant is absolute; the divisibilities include an empty interval.

Proof. For $p \leq 12B$, let N_j^+, N_j^- count the two oriented residue classes modulo p^j in J . Each is either $\lfloor H/p^j \rfloor$ or $\lceil H/p^j \rceil$. Thus $|N_j^+ - N_j^-| \leq 1$, and Lemma 274.1 gives

$$v_p(N_J) = |E_p^+ - E_p^-| = \left| \sum_{j=1}^{h_p} (N_j^+ - N_j^-) \right| \leq h_p.$$

If $p > 12B$, the difference-of-norms argument in Lemma 274.2 shows that it divides at most one norm in the entire block. Its depth is one because $p^2 > (12B)^2 > 36B^2 > Q_k$. Its sole orientation survives primitive cancellation. These are exactly the primes of R_J . No inert or ramified primes occur. The actual prime factorization proves all the divisibilities. Finally the Chebyshev bound gives

$$\log M_B \leq \pi(12B) \log(36B^2) = O(B).$$

No prime number theorem in a progression is used for this last bound. $\square$

Corollary 274.9 (CP2: radical mass of the primitive input norm). *For each fixed $\delta > 0$, uniformly on intervals with $H \geq \delta B$, as $B \rightarrow \infty$,*

$$\begin{aligned}\log N_J &= H \log B + O(H \log \log B + B), \\ \log R_J &= H \log B + O(H \log \log B + B), \\ \frac{\log \text{rad}(N_J)}{\log N_J} &\geq 1 - O_\delta(1/\log B).\end{aligned}$$

Proof. The first formula is Theorem 274.3. Subtracting $\log D_J = O(B)$ proves the second. For all sufficiently large B depending only on δ , $\log N_J \geq (\delta/2)B \log B$. Divide the logarithmic powerful-part bound of Theorem 274.8 by this lower bound to obtain the last assertion. The constants in the first two errors are absolute. $\square$

Theorem 274.10 (CP3: obstruction to compressing this aggregate). *Suppose positive integers V, Q, g satisfy*

$$N_J = VQ^g, \quad Q > 1, \quad g \geq 2.$$

Then

$$R_J \mid V, \quad Q^g \mid D_J, \quad \frac{\log V}{g} \geq \frac{(\log 2)(\log R_J)}{\log D_J}.$$

Here $D_J > 1$. For every fixed $\delta > 0$, there are positive c_δ, B_δ such that for every integer $B \geq B_\delta$, every interval $H \geq \delta B$, and every such representation,

$$\frac{\max(1, \log V)}{g} \geq c_\delta \log B.$$

No power-free condition on V , or coprimality of V, Q , is assumed.

Proof. Every prime of R_J has total depth one. Since $g \geq 2$, its exponent in Q must be zero and its exponent in V must be one. All prime factors of Q instead belong to D_J , and their g -fold exponents are bounded by the corresponding exponents in D_J . This proves the two divisibilities. Since $Q \geq 2$,

$$g \log 2 \leq g \log Q \leq \log D_J.$$

Combine this with $\log V \geq \log R_J$. For the uniform conclusion, use $\log D_J \leq CB$ with one absolute C , and $\log R_J \geq (\delta/2)B \log B$ for sufficiently large B . If $D_J = 1$, no representation with $Q > 1$ exists. $\square$

Theorem 274.11 (CP4: a primewise exponent ceiling and linear obstruction). *Under the hypotheses of Theorem 274.10,*

$$g \leq \frac{\log(36B^2)}{\log 7}, \quad \frac{\log V}{g} \geq \frac{(\log 7)(\log R_J)}{\log(36B^2)}.$$

Consequently, for every fixed $\delta > 0$, all sufficiently large integers B , all intervals $H \geq \delta B$, and every representation in that theorem,

$$\frac{\max(1, \log V)}{g} \geq \frac{\delta \log 7}{6} B.$$

Proof. Choose an actual prime $p \mid Q$, possible because $Q > 1$. It divides D_J , is split, and is at least seven. The actual exponent comparison and Theorem 274.8 give

$$g \leq gv_p(Q) \leq v_p(D_J) \leq h_p \leq \frac{\log(36B^2)}{\log 7}.$$

Together with $\log V \geq \log R_J$, this proves both finite bounds. When $\log R_J = 0$, the second bound is still the valid zero lower bound. On the specified macroscopic intervals, $\log R_J \geq (\delta/2)B \log B$ for sufficiently large B , and $B \geq 36$ gives $\log(36B^2) \leq 3 \log B$. Substitution proves the linear lower bound. The ceiling uses a single actual prime of Q , not merely the total mass $\log D_J$. $\square$

The height decrease of the primitive collective product is genuine, and almost all of its input norm mass is radical mass. Nevertheless this specific object cannot produce small- λ compressed norm seeds on macroscopic intervals: every possible displayed representation has λ growing at least linearly in B . This excludes neither useful representations of original roots nor a different combined map.

The proofs in this section are ordinary proofs. The sole progression input has been stated with its primary theorem and exact fixed-modulus scope. Separate finite replays of the content identities, complement identities, and norm profiles test exact instances; they supply no evidence in place of the asymptotic proofs or their universal quantifiers. This section makes no additional Lean claim. Any separately formalized arithmetic core has its own explicit signatures and verification scope. In particular the unrestricted boundary signed tail, individual exceptional roots, the independent-domain complement, and general ABC coverage remain open.

275. ACTUAL AFFINE PRIME SELECTORS AND COMPLETE SIGNED COMPENSATION

This section gives an additive construction in which different primes may select different actual input roots. Its output is a new primitive Eisenstein integer. In particular, preserving the selected prime powers does not preserve the input power representation. All data in the positive-density theorem below are fixed before the parameter tends to infinity.

Write

$$\mathcal{O} = \mathbb{Z}[\zeta], \quad \zeta^2 - \zeta + 1 = 0, \quad N(a + b\zeta) = a^2 + ab + b^2,$$

and identify an element with its pair of integer coordinates. Put

$$P(a + b\zeta) = ab(a + b), \quad |Z| = \sqrt{N(Z)}.$$

For nonzero Z , its content is the positive gcd of its coordinates. When $P(Z) \neq 0$, the complete signed boundary cost is

$$(1004) \quad J(Z) = \sum_{p|P(Z)} (v_p(P(Z)) - 3) \log p = \log |P(Z)| - 3 \log \text{rad}(|P(Z)|).$$

In particular, depth one and depth two contribute respectively $-2 \log p$ and $-\log p$. These terms are retained throughout.

Proposition 275.1 (The actual additive quotient and a short kernel). *For a finite set I and a positive integer n , let*

$$S : \mathbb{Z}^I \longrightarrow \mathcal{O}, \quad S(c) = \sum_{k \in I} c_k w_k^n, \quad w_k = 3k + \zeta.$$

Its integer image has rank at most two. Every family of homogeneous congruences depending only on $S(c)$, such as $\ell_q(S(c)) \equiv 0 \pmod{q^{e_q}}$ for integer linear forms ℓ_q , contains $\ker S$. On any $n + 2$ consecutive indices, there is the actual identity

$$(1005) \quad \sum_{j=0}^{n+1} (-1)^j \binom{n+1}{j} w_{k+j}^n = 0.$$

Its coefficient vector is nonzero and every coefficient has absolute value at most 2^{n+1} .

Proof. Both coordinates of $(3X + \zeta)^n$ are integer polynomials of degree at most n . For a monomial of degree $m \leq n$, apply $(T d/dT)^m$ to $(1 - T)^{n+1}$ and evaluate at $T = 1$. Every resulting term still vanishes there. Expanding gives the alternating binomial identity, and linearity gives (1005). Its first coefficient is one, while $\sum_j \binom{n+1}{j} = 2^{n+1}$ bounds each coefficient. The rank statement follows from the literal identification $\mathcal{O} = \mathbb{Z}^2$, and every displayed congruence vanishes on the kernel. $\square$

For prime $n > 324$, the block $B = n^4$, $B \leq k < 2B$, contains such consecutive indices. Thus a nonzero coefficient vector satisfying arbitrarily many congruences need not have nonzero actual output. This is a counterexample to that inference on the original block. It does not exclude an

argument that separately proves escape from the kernel or retains a higher-dimensional actual output.

Theorem 275.2 (An exact selector for different actual owners). *Let $Z_1, Z_2 \in \mathcal{O}$ be primitive, with*

$$\Delta = \det(Z_1, Z_2) \neq 0, \quad P(Z_1)P(Z_2) \neq 0.$$

For $i = 1, 2$, let S_i be a nonempty finite set of primes greater than three, with $S_1 \cap S_2 = \emptyset$. For every $q \in S_i$ assume

$$e_q = v_q(P(Z_i)) \geq 1, \quad q \nmid N(Z_i).$$

Set

$$M_i = \prod_{q \in S_i} q^{e_q+1}, \quad M = M_1 M_2.$$

For every integer t satisfying

$$(1006) \quad t \equiv 1 \pmod{M_1}, \quad t \equiv 0 \pmod{M_2},$$

define $U_t = tZ_1 + (1-t)Z_2$, $g_t = \text{cont}(U_t)$, and $V_t = U_t/g_t$. These elements are well-defined and

$$(1007) \quad g_t \mid |\Delta|, \quad \gcd(g_t, M) = 1, \quad v_q(P(V_t)) = e_q \quad (q \in S_1 \cup S_2).$$

In particular the exact selected signed cost is

$$(1008) \quad \sum_{q \in S_1 \cup S_2} (v_q(P(V_t)) - 3) \log q = \sum_{i=1}^2 \sum_{q \in S_i} (e_q - 3) \log q.$$

There are infinitely many such integers t .

Proof. The two moduli are coprime, so the integer Chinese remainder theorem gives one residue class modulo M . Linear independence of Z_1, Z_2 over $\mathbb{Q}$ makes $U_t \neq 0$, since t and $1-t$ cannot both vanish. The actual determinant identities

$$\det(U_t, Z_2) = t\Delta, \quad \det(U_t, Z_1) = -(1-t)\Delta$$

imply $g_t \mid \Delta$ by subtraction. For $q \in S_i$, $U_t \equiv Z_i \pmod{q^{e_q+1}}$. The integer cubic therefore has exactly valuation e_q , and $N(U_t)$ is a q -unit. Consequently $q \nmid g_t$, and dividing the cubic by g_t^3 preserves the exact valuation. This proves both asserted identities. $\square$

There is no requirement that both input roots hit each marked prime. The additional digit in q^{e_q+1} fixes the depth exactly, including depths one and two. The theorem by itself makes no claim about unmarked primes.

Proposition 275.3 (The cost after actual primitive cancellation). *For every selector in Theorem 275.2,*

$$|t| \geq M_2, \quad |1-t| \geq M_1,$$

$$(1009) \quad \log |V_t| \geq \log(\sqrt{3}/2) + \max\{\log M_2 - \log |Z_2|, \log M_1 - \log |Z_1|\}.$$

If n, B are positive integers and $Z_i = w_{k_i}^n$ for integers $B \leq k_i < 2B$, then

$$(1010) \quad \log |V_t| \geq \frac{1}{2} \log M - n \log(6B) + \log(\sqrt{3}/2).$$

The representative $0 \leq t < M$ also satisfies

$$(1011) \quad |V_t| \leq |U_t| < M(|Z_1| + |Z_2|).$$

Proof. The congruences exclude $t = 0, 1$. Thus the nonzero integers t and $1-t$ are respectively multiples of M_2 and M_1 . The Gram determinant inequality in these coordinates is

$$|\det(X, Y)| \leq \frac{2}{\sqrt{3}} |X| |Y|.$$

Apply it to (V_t, Z_2) . Since $|\Delta|/g_t$ is a positive integer, it gives

$$\frac{2}{\sqrt{3}} |V_t| |Z_2| \geq |t| \frac{|\Delta|}{g_t} \geq |t|.$$

The other determinant gives the analogous inequality for $|1 - t|$ and Z_1 . Taking logarithms proves (1009). The bounds $|w_k| < 6B$ and $\max(\log M_1, \log M_2) \geq \frac{1}{2} \log M$ prove (1010). For the chosen representative, $t \neq 0, 1$, $|t| < M$, and $|1 - t| < M$; the triangle inequality and $g_t \geq 1$ give (1011). $\square$

The modulus cost is therefore present in an actual integer selector; it does not disappear just by dividing out content. This is a bound for this construction, not an optimality assertion for every map.

Theorem 275.4 (A fixed-data positive-density family with full credit). *Fix all the data of Theorem 275.2. Suppose also*

$$Z_2 = (a_2, b_2), \quad Z_1 - Z_2 = (a_0, b_0), \quad a_0, b_0 > 0.$$

Choose a positive t_ in the CRT class with U_{t_*} having positive coordinates. Let $\mathcal{E}$ be the finite set of primes dividing the nonzero integer*

$$6M\Delta a_0 b_0 (a_0 + b_0).$$

For $p \in \mathcal{E}$ put $F_p = v_p(P(U_{t_}))$, and define*

$$(1012) \quad H = \prod_{p \in \mathcal{E}} p^{F_p+1}, \quad t(j) = t_* + Hj \quad (j \geq 1), \quad g = \text{cont}(U_{t_*}),$$

$$f_p = F_p - 3v_p(g), \quad K = \prod_{p \in \mathcal{E}} p^{f_p}.$$

Then $M \mid H$, $f_p \geq 0$, and for every positive integer j , the actual primitive output

$$V_j = U_{t(j)}/g = (a_j, b_j)$$

has positive coordinates. Its three arms are affine integer polynomials in j , with positive slopes.

There is a set $\mathcal{G}$ of positive integers with natural density

$$(1013) \quad d(\mathcal{G}) = \prod_{p \notin \mathcal{E}} (1 - 3/p^2) \geq \frac{1}{4}$$

such that, for every $j \in \mathcal{G}$,

$$(1014) \quad T_j = a_j b_j (a_j + b_j) = K R_j, \quad R_j \text{ squarefree, } \gcd(K, R_j) = 1,$$

and all primes of R_j lie outside $\mathcal{E}$. The complete signed identity and its limit are

$$(1015) \quad J(V_j) = -2 \log T_j + 3 \log \frac{K}{\text{rad}(K)},$$

$$(1016) \quad \frac{J(V_j)}{\log(a_j + b_j)} \rightarrow -6 \quad (j \in \mathcal{G}, j \rightarrow \infty).$$

For a constant C depending on these fixed data, $a_j + b_j \leq C \text{rad}(T_j)^{1/3}$ for every $j \in \mathcal{G}$.

Proof. At a marked prime, the selector theorem gives $F_q = e_q$, so $M \mid H$. For any $p \in \mathcal{E}$, the congruence $U_{t(j)} \equiv U_{t_*} \pmod{p^{F_p+1}}$ preserves the exact cubic valuation. It also preserves the common coordinate valuation: if $v = \min(v_p(a_*), v_p(b_*))$, then $F_p \geq 3v$, whence $F_p + 1 > v$, and a coordinate of valuation v retains it. No prime outside $\mathcal{E}$ divides the content, by (1007). Thus $\text{cont}(U_{t(j)}) = g$ for every j . In particular $g \mid H$, so the quotient arms are affine integer polynomials, with positive slopes and positive initial coordinates. Dividing the cubic by g^3 gives the nonnegative exact depths f_p .

For $p \notin \mathcal{E}$, all three slopes are p -units. The determinant of the slope and constant coordinate pairs is $H\Delta/g^2$, up to sign, and is also a p -unit. Each of the three affine arms therefore has one simple root modulo p^2 , and their roots are distinct already modulo p . Exactly three classes modulo p^2 are forbidden by $p^2 \mid T_j$. Notice that every such prime is at least five.

For any fixed cutoff Y , finite CRT gives density $\prod_{p \leq Y, p \notin \mathcal{E}} (1 - 3/p^2)$ for avoiding these finitely many classes. The entire omitted tail is elementary. There is a fixed $C_0 > 0$ such that every individual positive affine arm at $1 \leq j \leq X$ is at most $C_0 X$, for every real $X \geq 1$. Distinct roots

modulo p ensure that $p^2 \mid T_j$ forces p^2 to divide one arm, rather than arising from two different arms. Such a prime is at most $\sqrt{C_0 X}$. Counting the three residue classes gives

$$(1017) \quad \#\{1 \leq j \leq X : \exists p > Y, p \notin \mathcal{E}, p^2 \mid T_j\} \leq 3X \sum_{p>Y} p^{-2} + 3\sqrt{C_0 X}.$$

Divide by X , let $X \rightarrow \infty$, and then let $Y \rightarrow \infty$. The finite CRT upper and lower bounds prove existence of the natural density and its exact infinite product. Its lower bound follows from

$$\sum_{p \geq 5} \frac{3}{p^2} \leq 3 \sum_{m \geq 5} \frac{1}{m^2} < 3 \int_4^\infty x^{-2} dx = \frac{3}{4}$$

and the finite inequality $\prod(1 - u_i) \geq 1 - \sum u_i$ for $0 \leq u_i \leq 1$. Passing to the convergent product proves (1013); removing additional primes in $\mathcal{E}$ only increases it. No distribution theorem for primes is needed.

At each prime of $\mathcal{E}$ the exact depth is f_p . Outside it a good parameter has depth zero or one. This proves (1014), and hence

$$\text{rad}(T_j) = \text{rad}(K)R_j = \frac{T_j}{K/\text{rad}(K)}.$$

Substitution in (1004) gives (1015). The positive affine slopes imply

$$\log T_j = 3 \log j + O(1), \quad \log(a_j + b_j) = \log j + O(1),$$

which prove (1016). They also give $T_j \geq c_0(a_j + b_j)^3$ for a fixed $c_0 > 0$, decreasing c_0 if necessary for finitely many small indices. The radical identity proves the last bound. $\square$

The data, progression and threshold in this theorem are fixed. The numbers K and C_0 can be very large when the input roots or packets change. The density lower bound is a natural-density statement in this fixed progression; it is not a density statement in a moving original root block.

Proposition 275.5 (Actual source roots and nonempty examples). *Let $n \geq 2$, $B \geq n$ be integers, and let $B \leq k_2 < k_1 < 2B$. The two actual powers $Z_i = (3k_i + \zeta)^n$ are primitive and unramified, have $P(Z_i) > 0$, are nonparallel, and have both coordinates of $Z_1 - Z_2$ positive.*

Proof. The element $3k + \zeta$ is primitive and its norm is one modulo three. Its powers remain primitive: modulo any prime other than three the Eisenstein algebra is a product of fields or a field, so an element with zero power is zero; the coordinate one prevents that. At three the norm is a unit. This proves the required property at every prime and also nonramification.

For real $x \geq B$, put

$$\theta(x) = \arg(3x + \zeta) = \arctan \frac{\sqrt{3}}{6x + 1}.$$

This function is positive, strictly decreasing, and less than $1/(3x)$. Thus $0 < n\theta(x) < 1/3 < \pi/3$, so each power lies in the open positive coordinate cone. Their two arguments are distinct and differ by less than π , proving nonparallelism. Finally

$$(3k_1 + \zeta)^n - (3k_2 + \zeta)^n = \int_{k_2}^{k_1} 3n(3x + \zeta)^{n-1} dx.$$

Since $n \geq 2$, the integrand lies strictly in the same cone; both coordinates of its integral are positive. $\square$

For $n = 2$, $B = 2$, the roots $w_2 = 6 + \zeta$, $w_3 = 9 + \zeta$ give

$$Z_2 = (35, 13), \quad P(Z_2) = 35 \cdot 13 \cdot 48, \quad Z_1 = (80, 19), \quad P(Z_1) = 80 \cdot 19 \cdot 99.$$

The prime 19 belongs only to the first boundary, and 13 only to the second. Their depths are one and their selected norms are units. The choices $S_1 = \{19\}$, $S_2 = \{13\}$ therefore give a nonempty infinite actual output family in Theorem 275.4.

Positive signed costs can also be exhibited with different owners. Still take $n = 2$, and set

$$k_1 = 21720 + 2 \cdot 19^5 = 4973918, \quad k_2 = 61882 + 13^6 = 4888691.$$

Both belong to the extended block $B = 4880000$. Their second coordinates are respectively $19^4 \cdot 229$ and $13^5 \cdot 79$. If $q > 3$ divides the second coordinate $6k + 1$, the first coordinate $(3k)^2 - 1$ is $-3/4 \pmod{q}$. Thus the actual whole boundary depths are four and five. The other owner's boundary is a unit at the marked prime, as follows from $k_1 \equiv 1 \pmod{13}$ and $k_2 \equiv 10 \pmod{19}$. The selected positive signed cost is exactly $\log 19 + 2 \log 13$. The selector preserves it, while the new squarefree support in the proved affine family pays it.

These examples have exponent two. They belong to the extended actual-power family of Proposition 275.5; they do not satisfy the original restrictions of prime $n > 324$, $B = n^4$, or large marked primes beyond that block. The selector lemma applies to roots in that original block when the required actual packets are supplied, but these examples do not establish such moving-block membership.

Evidence and the remaining arithmetic constraint. The standard-library script `replay_affine_selectors.py --check` checks ten exact CRT selectors, including the depth-four and depth-five packet, 256 actual fixed-progression content/depth rows, and the three distinct simple roots modulo p^2 for 162 eligible primes up to 997. It also checks six literal finite-difference identities at exponents 2, 5, 7, 11, 17, 331; the last is in the original block $B = n^4$. The canonical certificate SHA256 is

70945a8da8fdecee2a8857dd9ea71ab2bb12f9054a8f25b35ad87f342df6d767.

These are finite integer checks. They do not factor every unmarked arm or prove density by sampling. The complete infinite tail and density proof is (1017), with the finite prime exceptions retained in K . There is no new Lean claim here.

The positive theorem is about new primitive affine outputs and their full signed costs. It neither bounds the original inputs' tails nor proves that the outputs are powers or have small-residual power representations. Its constants cannot be applied uniformly to moving packets. The additional constraint would be to retain useful original compression while proving a sufficient supply of new negative credit. Original boundary tails, exceptional roots, the complement of the independent domain, and general ABC coverage remain open.

276. EXACT FIVE-ADIC CLOSURE FOR THE FIRST GENUS-TWO QUOTIENT

This section proves the local closure and saturation input for the fixed curve $H = H_1$. It does not determine all rational points or a global Mordell–Weil basis. Use the previously established curves

$$E : y^2 = x^3 - 9x - 9, \quad P = (-2, 1), \quad E' : y^2 = x^3 - 189x + 999, \quad P' = (6, 9),$$

and

$$H : y^2 = s^6 + 3s^5 - 3s^4 - 11s^3 - 3s^2 + 3s + 1.$$

Write $J = \text{Jac}(H)$. The two degree-two morphisms $\pi_1 : H \rightarrow E$, $\pi_2 : H \rightarrow E'$ from the earlier explicit quotient construction induce

$$\Phi = (\pi_{1*}, \pi_{2*}) : J \longrightarrow E \times E', \quad \Psi = \pi_1^* + \pi_2^* : E \times E' \longrightarrow J.$$

The reviewed norm–pullback identities give $\Psi\Phi = [2]$. Both maps are isogenies, and hence $\Phi\Psi = [2]$ as well: compose the first identity with Φ and use its geometric surjectivity. All maps and identities are defined over $\mathbb{Q}$. The ordinary descent has proved $\text{rank } E(\mathbb{Q}) = \text{rank } E'(\mathbb{Q}) = 1$, $\text{rank } J(\mathbb{Q}) = 2$, and $E(\mathbb{Q})[3] = 0$.

Lemma 276.1 (The formal logarithm at five). *For an integral Weierstrass model smooth at five, let $E_1(\mathbb{Q}_5)$ be the kernel of reduction and put $t = -x/y$. Its formal logarithm is a topological group isomorphism*

$$\log_E : E_1(\mathbb{Q}_5) \xrightarrow{\sim} 5\mathbb{Z}_5, \quad \log_E(Q) \equiv t(Q) \pmod{25}.$$

Proof. The integral formal group and the identification of its parameters with the reduction kernel are the usual Weierstrass constructions ([142], II.2.7 and II.4.4). Its invariant differential

is $(1 + \sum_{m \geq 1} b_m t^m) dt$, with $b_m \in \mathbb{Z}_5$. Termwise integration gives

$$L(t) = t + \sum_{m \geq 1} \frac{b_m}{m+1} t^{m+1}.$$

For $n \geq 2$, $n - v_5(n) \geq 2$. Thus $L(t) - t \in 25\mathbb{Z}_5$ on $5\mathbb{Z}_5$. The difference of the degree- n term at $t, u \in 5\mathbb{Z}_5$ has valuation at least

$$v_5(t - u) + (n - 1) - v_5(n) \geq v_5(t - u) + 1.$$

Consequently $L(t) - t$ is a strict contraction. Successive approximation proves that L is a bijective isometry of $5\mathbb{Z}_5$. Invariance of the differential, followed by differentiation and evaluation at zero, gives $L(F(t, u)) = L(t) + L(u)$. This proves the topology and group assertions. $\square$

Theorem 276.2 (Two dense cyclic local groups). *Both elliptic curves have good ordinary reduction at five, and*

$$\overline{\mathbb{Z}P} = E(\mathbb{Q}_5), \quad \overline{\mathbb{Z}P'} = E'(\mathbb{Q}_5).$$

Proof. Their discriminants are 11664 and 944784, both units at five. Complete finite enumeration gives nine points on each reduced curve; the reductions of P, P' each have order nine. Their exact multiples are as follows, with the two coordinates displayed separately:

$$\begin{aligned} x(9P) &= -\frac{1625193766960369990795405419794}{1135192428045421857524125001025}, \\ y(9P) &= \frac{1179179153363180072193201096933126009785708521}{1209495420803976369128841727712564695256198625}, \\ x(9P') &= \frac{31544182486}{4119714225}, \quad y(9P') = -\frac{229868810028359}{264423857531625}. \end{aligned}$$

In both cases $v_5(x) = -2$, $v_5(y) = -3$, and exact rational arithmetic gives

$$(1018) \quad \frac{t(9P)}{5} \equiv 1 \pmod{5}, \quad \frac{t(9P')}{5} \equiv 3 \pmod{5}.$$

The points lie in the respective reduction kernels. By Lemma 276.1, their logarithms are five times units. Integer multiples of either logarithm are dense in $5\mathbb{Z}_5$, so the multiples of $9P$, respectively $9P'$, are dense in those kernels. The multiples of P, P' already fill all nine reduction cosets, proving the asserted density. The common Frobenius trace is $6 - 9 = -3$, so the good reduction is ordinary. $\square$

Corollary 276.3 (Finite index and five-saturation). *Both rational elliptic groups are torsion-free. The displayed cyclic subgroups have finite index, and both indices are prime to five.*

Proof. The local reduction kernels are torsion-free by Lemma 276.1; hence any finite local subgroup injects into the reduction group of order nine. The already proved $E(\mathbb{Q})[3] = 0$ excludes rational torsion on E . The ordinary descent's bijective rational homomorphism $\psi : E'(\mathbb{Q}) \rightarrow E(\mathbb{Q})$ gives the same result for E' . The rank-one Mordell–Weil theorem gives finite index.

Neither displayed point is divisible by five even locally. If $P = 5R$, then $5 \operatorname{red}(9R) = 0$ in a group of order nine, so $9R \in E_1$. This would give $\log_E(9P) = 5 \log_E(9R) \in 25\mathbb{Z}_5$, contradicting (1018). The proof for P' is identical. In a torsion-free rank-one group, an index divisible by five would make the corresponding point divisible by five. This proves the claim. It does not determine either index at all other primes. $\square$

Theorem 276.4 (An explicit dense divisor lattice). *Let ∞_+, ∞_- be the two rational points of H with y/s^3 tending to $+1, -1$, respectively. Set*

$$A = (0, 1), \quad D_1 = [A - \infty_+], \quad D_2 = [\infty_- - \infty_+] \in J(\mathbb{Q}).$$

Then

$$(1019) \quad \Phi(D_1) = (0, 2P'), \quad \Phi(D_2) = (-2P, 2P'), \quad \overline{\mathbb{Z}D_1 + \mathbb{Z}D_2} = J(\mathbb{Q}_5).$$

The rational group $J(\mathbb{Q})$ is torsion-free, and this lattice has finite index prime to five in $J(\mathbb{Q})$.

Proof. The exact projective quotient formulas give all six images

$$\begin{array}{c|ccc} & A & \infty_+ & \infty_- \\ \hline \pi_1 & P & P & -P \\ \pi_2 & P' & -P' & P' \end{array}$$

including the leading-term limits at infinity. They prove the first two identities in (1019). In the rational bases P, P' their columns are $\begin{pmatrix} 0 & -2 \\ 2 & 2 \end{pmatrix}$, with determinant four.

The earlier smooth-model proof gives good reduction of H and J at five. Moreover $\#J(\mathbb{F}_5) = 81$, either from the isogeny and good-reduction Frobenius polynomial or from the full counts $\#H(\mathbb{F}_5) = 12$, $\#H(\mathbb{F}_{25}) = 28$. The latter give trace -6 , middle coefficient 19, and $1 + 6 + 19 + 30 + 25 = 81$. Multiplication by two is bijective on the formal kernel of J , since its linear coefficient is a unit, and on its finite reduction group of odd order. Hence it is bijective on $J(\mathbb{Q}_5)$, which in particular has no two-torsion. The same argument applies to the elliptic local groups.

It follows from $\Psi\Phi = \Phi\Psi = [2]$ that $\Phi : J(\mathbb{Q}_5) \rightarrow E(\mathbb{Q}_5) \times E'(\mathbb{Q}_5)$ is a topological isomorphism. Its kernel is killed by two and its image contains twice every point of the target. Continuity of the inverse follows from compactness and the Hausdorff property. Now $\Phi(D_1) = (0, 2P')$ and $\Phi(D_2 - D_1) = (-2P, 0)$. Their generated image is exactly $2\mathbb{Z}P \times 2\mathbb{Z}P'$, dense by Theorem 276.2 and invertibility of doubling. This uses the two coordinate axes separately; it does not apply a cross-coordinate integer matrix to two different elliptic groups.

The two classes are independent by their images and hence have finite index, since $\text{rank } J(\mathbb{Q}) = 2$. Any rational torsion maps to zero in the torsion-free elliptic groups and lies in the two-torsion kernel of Φ ; it is therefore zero. If $5D = mD_1 + nD_2$ with $D \in J(\mathbb{Q})$, apply Φ and Corollary 276.3 to obtain $5 \mid n$ and $5 \mid (m + n)$, hence $5 \mid m$. Subtract the corresponding integral combination from D . The result is five-torsion and is zero. Thus the lattice is five-saturated, equivalently its finite index is prime to five. $\square$

There is also an actual logarithm determinant certificate. Extend the elliptic logarithm by $\log_E(Q) = \log_E(9Q)/9$, and similarly for E' . Equation (1018) gives $\log_E(P)/5 \equiv 4$, $\log_{E'}(P')/5 \equiv 2 \pmod{5}$. In product differential coordinates, the logarithm columns of D_1, D_2 , divided by five, are therefore

$$\begin{pmatrix} 0 & 2 \\ 4 & 4 \end{pmatrix} \pmod{5}, \quad \det \equiv 2 \pmod{5}.$$

Their determinant has valuation exactly two in these coordinates; no conjectural regulator enters the argument.

The available height formula and what remains. The earlier QH theorem proves that $H(\mathbb{Q}_5)_2$ is finite. The additional hypotheses of [144], Theorem 1.2(ii), now hold as well: rank equals genus two, the rational Neron–Severi rank is at least three, and the rational group has finite-index closure in $J(\mathbb{Q}_5)$, in fact index one. The rational basepoint and good ordinary reduction at five have also been verified. Once the correspondence and local-height data are specified, that theorem supplies an explicit height expression whose finite level set contains the quadratic Chabauty locus on its stated open chart. The excluded finite correspondence-support set must be handled separately. No such set has yet been completely computed here.

For either rank-one elliptic group, a fixed quadratic 5-adic height h satisfies

$$h(Q) = \alpha \log_E(Q)^2 \quad (Q \in E(\mathbb{Q})), \quad \alpha = \frac{h(P)}{\log_E(P)^2}.$$

Indeed write $P = mG, Q = nG$ in the torsion-free rank-one group and use quadraticity and additivity. Thus a nonzero infinite-order point suffices to compute the ratio; proving that P is a global generator is unnecessary for this step. A later Mordell–Weil sieve still needs its own appropriate saturation information.

The unresolved computation is to certify the height normalization between minimal elliptic models and the chosen even-sextic chart, determine the bad-prime local-height value sets, construct analytic expressions in all necessary five-adic charts, and certify every zero and its multiplicity before a rational sieve. In particular neither bounded saturation nor agreement of

two precision runs gives a complete rational-point set. The separate numerical normalization probe is not part of the proof above. Finally, lifting from H to $D_{3,\zeta}$ requires the second square at the same source and the positive real conditions. Even a complete fixed-curve classification does not settle varying exponents, nonunit residuals, integral content, or the uniform point-height problem.

Exact evidence and formal scope. The standard-library replay `replay_padic_entry.py --check` recomputes both complete elliptic tables modulo five, exact ninefold coordinates, valuation and unit tests, a full $\mathbb{F}_{25}$ hyperelliptic count, and the integer matrix arithmetic. Its canonical certificate has SHA256 `2bc0673769cc10abba959725e1683676d162918afa7fbf89a463f664c9b24696`. These are finite arithmetic checks. The formal logarithm, topological closure, Jacobian functoriality and quadratic Chabauty input remain the complete ordinary proofs and explicitly cited standard results; this section makes no new Lean claim for those theories.

277. EXACT HEIGHT TRANSPORT TO THE QUADRATIC CHABAUTY MODELS

The fixed minimal elliptic curves and rational points are

$$E : y^2 = x^3 - 9x - 9, \quad P = (-2, 1), \quad E' : y^2 = x^3 - 189x + 999, \quad P' = (6, 9).$$

The monic even-sextic chart of the genus-two curve and its standard elliptic quotients are

$$(1020) \quad C : W^2 = z^6 - 27z^4 + 99z^2 - 9,$$

$$(1021) \quad F_1 : Y^2 = X^3 - 27X^2 + 99X - 9,$$

$$(1022) \quad F_2 : Y^2 = X^3 + 99X^2 + 243X + 81.$$

The isomorphisms from the minimal models are

$$(1023) \quad i_1(x, y) = (4x + 9, 8y), \quad i_2(x, y) = (4x - 33, 8y).$$

Thus both have scale $u = 2$, and translation $r = 9$ or -33 , in the notation $X = u^2x + r$, $Y = u^3y$. The two displayed short models are globally minimal: their discriminants are $2^4 3^6$ and $2^4 3^{10}$, respectively. At each prime the valuation is below twelve, whereas replacing an integral model by one with smaller integral discriminant would decrease that valuation by a positive multiple of twelve. The curve substitutions in (1023) are exact polynomial identities.

The height convention. We use the cyclotomic character

$$\chi_5 = \log_5, \quad \log_5(5) = 0, \quad \chi_\ell(a) = -v_\ell(a) \log_5(\ell) \quad (\ell \neq 5),$$

the unit-root splitting at five, and the diagonal canonical height $H_E(Q)$. This is twice Silverman's local-height normalization, as explicitly fixed in [144, §8.3, printed p. 35]. It differs from the introductory scalar h_p of [145]: that scalar uses character $\log_p/p$ and is $-1/2$ of the corresponding diagonal pairing. For an infinite-order rational point Q and a positive integer m , after this conversion formula (1.1) of that source reads

$$(1024) \quad H_E(Q) = -\frac{2}{m^2} \log_5 \left(\frac{\sigma_E(mQ)}{d(mQ)} \right),$$

where mQ lies in the formal group at five and in the identity components at the bad primes, and $d(mQ)$ is the positive square root of its minimal-model x -denominator.

Theorem 277.1 (Scalar transport and the differential basis). *For either isomorphism $i : E_i \rightarrow F_i$, let $\omega = dx/(2y)$, $\eta = x\omega$, and let s be the slope for which $\eta + s\omega$ spans the unit-root line. The exact transport laws are*

$$(1025) \quad H_{F_i}(iQ) = H_{E_i}(Q), \quad \log_{F_i}(iQ) = u^{-1} \log_{E_i}(Q),$$

$$(1026) \quad s_{F_i} = u^2 s_{E_i} - r, \quad \alpha_{F_i} = u^2 \alpha_{E_i},$$

where $\alpha_E = H_E(P)/\log_E(P)^2$ for a nonzero infinite-order rational point. In particular, the two height/logarithm-square ratios in the standard quotient models are four times those on the minimal models.

Proof. Direct pullback gives

$$(1027) \quad i^*\omega_F = u^{-1}\omega_E, \quad i^*\eta_F = u\eta_E + (r/u)\omega_E.$$

Frobenius commutes with the isomorphism, so uniqueness of the ordinary unit-root line implies

$$i^*(\eta_F + s_F\omega_F) = u(\eta_E + s_E\omega_E).$$

This is exactly the asserted formula for s_F . Integration from the origin gives the logarithm formula on the formal group; multiplication by a common integer extends it to all local points.

The scalar height is intrinsic. In this case its invariance can also be checked directly in the sigma construction. Theorem 1.3 of [145] characterizes the normalized odd integral sigma function by its differential equation. The changes of formal parameter are integral at five because $u = 2$ is a unit. Together with (1027) and the slope formula, the defining differential equation and its leading coefficient give

$$(1028) \quad \sigma_F(iQ) = u^{-1}\sigma_E(Q).$$

Indeed the parameter change is odd with leading coefficient $1/u$, and both sides have coefficient one when expressed in the parameter on F . Uniqueness proves the equality. In the four-point sigma quotient defining the local pairing, [145, §2.3], there are two factors in each of the numerator and denominator; all four factors $1/u$ cancel. At primes away from five the construction uses the intrinsic minimal Néron model, [145, §2.4], and is unchanged under this isomorphism. Bilinearity extends the equality from the four-point representations to the rational group. Equivalently all auxiliary local data may be transported through i . This proves invariance of H ; division by the squared logarithm proves the formula for α . The rank-one quadraticity argument requires no assertion that the chosen point generates the full Mordell–Weil group. $\square$

For local integration, the complementary differential scales by u and ω by $1/u$, consistently transporting their cup-dual normalization. Individual tangential-basepoint constants must still be fixed explicitly. The theorem does not identify an arbitrary local software output with an intrinsically normalized scalar.

Proposition 277.2 (The pinned raw-output transformation). *In the official PARI/GP 2.15.4 source, let $[a, b]$ denote the vector returned on the minimal model and $[A, B]$ the vector returned on F for the corresponding point. For the changes above, the source performs*

$$(1029) \quad A = \frac{a + rb}{u}, \quad B = ub.$$

Its inverse and the recovery of the intrinsically normalized scalar are

$$(1030) \quad a = uA - rB/u, \quad b = B/u,$$

$$(1031) \quad H_E = uA - \frac{r + s_E}{u}B = uA - \frac{(u^2 + 1)r + s_F}{u^3}B.$$

Consequently the uncorrected expression $A - s_FB$ cannot in general be used as the canonical scalar on this nonminimal model.

Proof. This first assertion is a version-specific source statement. In [146], `src/basemath/ellpadic.c`, lines 688–689 obtain the global minimal model and transform the point to it. Lines 741–748 perform the final transformation (1029). The direction of the change is also explicit in `src/basemath/elliptic.c`, lines 4993–5016, and in the official `ellchangecurve` documentation [147]: the coordinates satisfy $X = u^2x + r$, $Y = u^3y + su^2x + t$. Here the latter s, t are zero and are unrelated to the unit-root slope. Inverting the two-by-two transformation is ordinary rational linear algebra. On the minimal model the canonical scalar is $a - s_Eb$; substitution proves the first recovery formula. Substituting $s_E = (s_F + r)/u^2$ proves the second.

For the scalar convention, on the minimal short model the source computes

$$a = -2m^{-2}\log_5(\sigma_0(mP)/d), \quad b = -m^{-2}\log_E(mP)^2.$$

The routine `ellformallogsigma.t`, lines 427–440 of `ellpadic.c`, implements the zero-slope differential equation. Replacing σ_0 by $\sigma_s = \sigma_0 \exp(-s_E \log_E^2/2)$ gives $a - s_Eb$ and exactly (1024).

Thus the raw final vector transformation and the unit-root slope do not yield the same scalar through the uncorrected combination. The discrepancy is not a rounding effect. This conclusion is about these pinned source bytes and inputs; it is not a statement about every PARI release or every convention for a vector-valued height. $\square$

The simplest implementation computes $[a, b]$ and s_E on the minimal model, forms $H_E = a - s_E b$ there, and then uses (1026) for the standard-model logarithm and ratio. It need not compute the nonminimal raw vector at all.

Theorem 277.3 (An exact leading-digit discrepancy). *For each of the actual points P, P' , the difference $A - s_F B - H_E(P)$ has five-adic valuation exactly one. This statement has a finite integer proof independent of PARI's height algorithm.*

Proof. Write the exact ninefold point as $9P = (A_0/d^2, B_0/d^3)$, with $d > 0$, and similarly for P' . Integer elliptic addition and reduction modulo 25 give

	E	E'
$-A_0/B_0 \pmod{25}$	14	4
$\log_5(-A_0/B_0) \pmod{25}$	10	20
$H(P) \pmod{25}$	5	10

We justify the last row with an error bound for the entire omitted sigma series. The exact ninefold coordinates are nonsingular at both bad primes 2 and 3: if a prime divides the denominator the point reduces to the origin; otherwise at least one of $2y, 3x^2 + a$ is nonzero in the residue field. These alternatives are checked by integer arithmetic in the certificate below. The formal parameter t at five has valuation one, as proved in the preceding closure calculation. The normalized sigma function is odd and belongs to $t\mathbb{Z}_5[[t]]$, by [145, Theorem 1.3]. Therefore

$$\frac{\sigma(t)}{t} \in 1 + t^2\mathbb{Z}_5[[t]], \quad \log_5 \frac{\sigma(t)}{t} \in 25\mathbb{Z}_5.$$

Since $t/d = -A_0/B_0$, formula (1024) gives

$$H(P) \equiv -\frac{2}{81} \log_5(-A_0/B_0) \pmod{25}.$$

For any unit $q \in \mathbb{Z}_5^\times$, $\log_5(q) \equiv (q^4 - 1)/4 \pmod{25}$: write $q^4 = 1 + 5k$ and bound every subsequent logarithm term in $25\mathbb{Z}_5$. This evaluates all entries using modular inverses. The same argument applies to P' .

Finally $b = -\log_E(P)^2$ has valuation two, both slopes are integral, and $u = 2$ is a unit. Equations (1029)–(1031) imply

$$A - s_F B - H_E(P) \equiv (1/2 - 1)H_E(P) \pmod{25}.$$

The residues are 10 and 20, respectively. Each is a nonzero multiple of five modulo 25, proving the exact valuation. No comparison between two numerical truncations is used for this conclusion. $\square$

Reproducible evidence and remaining scope. The standard-library script `replay_height_transport.py --check` first performs an independent binary elliptic addition, the exact modulo-25 proof, and rational inverse-matrix checks. It then checks the source-predicted relations with explicit PARI five-adic balls at precisions 12 and 24. In each run every relation must vanish to the common declared absolute precision; mere agreement of the two runs is not the proof of Theorem 277.3. The canonical certificate has SHA256 70ec75594f205336757496605750b1302e5f3b271afd6acf5b12275a36c9de05. The high-precision sigma and Frobenius routines remain trusted PARI computations. The exact algebra, scalar transport and leading-digit error proof are the ordinary mathematical arguments given above. No new Lean verification of height theory is claimed.

The official source archive used for this analysis has SHA256 c3545bf0e0c6dfb40b77fb4bbabaf999d82e60069b9f6d28bcb6cf004c8c5c0f. Local copies of the relevant C files and function documentation were inspected during the audit. They can be recovered from the pinned versioned archive; those copies are not distributed with this publication. No installed library was patched. The result supplies the global height coefficients in the

prospective quadratic Chabauty expression. Local-height value sets, compatible local integrals and chart constants require their own proofs; certification of every analytic zero and the rational sieve remain further tasks. Neither a Jacobian rank calculation nor a complete rational-point classification is asserted here.

278. THE COMPLETE LOCAL HEIGHT-VALUE SET AWAY FROM FIVE

Let C be the smooth projective curve

$$W^2 = z^6 - 27z^4 + 99z^2 - 9.$$

The standard elliptic quotients in [144, Theorem 1.4] are

$$(1032) \quad F_1 : Y^2 = X^3 - 27X^2 + 99X - 9, \quad f_1(z, W) = (z^2, W),$$

$$(1033) \quad F_2 : Y^2 = X^3 + 99X^2 + 243X + 81, \quad f_2(z, W) = (-9/z^2, -9W/z^3).$$

The maps extend to the smooth projective curves. Their globally minimal models and corresponding points are

$$(1034) \quad E : y^2 = x^3 - 9x - 9, \quad R_1 = ((z^2 - 9)/4, W/8),$$

$$(1035) \quad E' : y^2 = x^3 - 189x + 999, \quad R_2 = ((33 - 9/z^2)/4, -9W/(8z^3)).$$

Both isomorphisms have scale $u = 2$, although their translations differ. Their minimal discriminants are $2^4 3^6$ and $2^4 3^{10}$.

For a finite prime $\ell \neq 5$, put $\chi_\ell(a) = -v_\ell(a) \log_5(\ell)$, where $\log_5(5) = 0$. Use local Néron heights $h_{E,\ell}, h_{E',\ell}$ in twice Silverman's normalization on these minimal models. The local-height coefficient is the ordinary rational coefficient of the real local height, with $\log(\ell)$ replaced by $\log_5(\ell)$. In particular, at a point of nonsingular reduction it is

$$h_{E,\ell}(R) = \max(0, -v_\ell(x(R))) \log_5(\ell).$$

At singular reduction we will use precisely the valuation formula stated below.

On the standard F_i models use the corresponding Weierstrass tangential normalization. The exact change-of-variables rule is

$$(1036) \quad h_{F_i,\ell}(f_i(P)) = h_{E_i,\ell}(R_i(P)) + 2\chi_\ell(2), \quad E_1 = E, \quad E_2 = E'.$$

Indeed, the normalized local Néron function for $2O$ has principal term $-2\chi_\ell(t)$, with $t = -x/y$. Under $X = u^2x + r$, $Y = u^3y$, one has $t_F = t_E/u + O(t_E^3)$. Thus its tangential normalization changes by $2\chi_\ell(u)$, and uniqueness with this normalization proves (1036). This also agrees with the local addition identity of [144, Lemma 7.4], since $x_F(P) - x_F(Q) = u^2(x_E(P) - x_E(Q))$. The global scalar is unchanged because $\sum_v \chi_v(u) = 0$; the local functions individually have the stated constants.

Those two constants are equal and cancel. Hence the exact combination used in Theorem 1.4 of that source is

$$(1037) \quad \begin{aligned} J_\ell(P) &= h_{F_1,\ell}(f_1(P)) - h_{F_2,\ell}(f_2(P)) - 2\chi_\ell(z) \\ &= h_{E,\ell}(R_1(P)) - h_{E',\ell}(R_2(P)) - 2\chi_\ell(z). \end{aligned}$$

Initially z is finite and nonzero. The locally constant extensions over zero and infinity are included in the proofs; we do not insert a separate infinite height of an elliptic origin into this expression.

We recall the two cases of the local theorem that suffice here. For a minimal Weierstrass model let

$$A = v_\ell(3x^2 + 2a_2x + a_4 - a_1y),$$

$$B = v_\ell(2y + a_1x + a_3),$$

$$C_0 = v_\ell(3x^4 + b_2x^3 + 3b_4x^2 + 3b_6x + b_8).$$

If $A \leq 0$ or $B \leq 0$, the height coefficient is $\max(0, -v_\ell(x))$. Otherwise, if $v_\ell(c_4) > 0$ and $C_0 \geq 3B$, the coefficient is $-2B/3$. These are cases (a) and (c) of [132, Proposition 3.4.1, printed pp. 72–73]. The text explicitly states its applicability to points over $\mathbb{Q}_\ell$ and the twice-Silverman

normalization. We use $v_\ell(0) = +\infty$. No inference from a software Kodaira type or a finite table is required.

Theorem 278.1 (The full three-adic value set). *For every $P \in C(\mathbb{Q}_3)$, with the extended convention above,*

$$J_3(P) = \frac{4}{3} \log_5(3).$$

Thus the full local value set is the displayed singleton.

Proof. First suppose $z \neq 0, \infty$ and let $m = v_3(z)$. If $m > 0$, the right side of the sextic has valuation exactly two and nonsquare unit part -1 modulo three: the other three valuations are $6m, 3 + 4m, 2 + 2m$, all larger than two. At $z = 0$ the same obstruction is $W^2 = -9$. Thus every finite point has $m \leq 0$. If $m = 0$, then W is a unit; if $m < 0$, the leading term is uniquely dominant, so $v_3(W) = 3m$.

For the second minimal point $(x, y) = R_2(P)$, both cases give

$$(1038) \quad v_3(x) = 1, \quad v_3(y) = 2, \quad X := x/3 = \frac{11}{4} - \frac{3}{4z^2} \equiv 2 \pmod{3}.$$

For E' the third division polynomial has the identities

$$(1039) \quad \psi_3(x) = 3x^4 - 1134x^2 + 11988x - 35721,$$

$$(1040) \quad \psi_3(3X) = 3^5(X^4 + 148X) - 3^6(14X^2 + 49).$$

Since $X \equiv 2 \pmod{3}$, one has $X^4 + 148X \equiv 0 \pmod{3}$; hence $C_0 \geq 6$. Also

$$B = 2, \quad A = v_3(27(X^2 - 7)) > 0, \quad v_3(c_4(E')) = 4 > 0.$$

The second stated case of the local theorem therefore applies and gives

$$h_{E',3}(R_2) = -\frac{4}{3} \log_5(3).$$

For E , when $m = 0$ the ordinate $W/8$ is a unit, so case (a) gives zero. When $m < 0$, the abscissa has valuation $2m$ and $v_3(3x^2 - 9) = 1 + 4m < 0$, giving $h_{E,3}(R_1) = -2m \log_5(3)$. Thus that latter formula holds for both ranges $m \leq 0$. The term $-2\chi_3(z) = 2m \log_5(3)$ cancels it, proving the result on the finite chart.

There are two rational points at infinity because the sextic is monic. Every sufficiently close punctured three-adic neighborhood has $m < 0$, where the combination is the same constant. It extends with that value at both infinity points. There is no zero-fiber point over $\mathbb{Q}_3$, as already shown. This exhausts the smooth projective local curve. Finally $(z, W) = (1, 8)$ is a rational point, proving nonemptiness and therefore equality of the value set with the singleton. $\square$

Theorem 278.2 (Two and all remaining primes away from five). *For every finite prime $\ell \neq 3, 5$ and every $P \in C(\mathbb{Q}_\ell)$, the extended combination is*

$$J_\ell(P) = 0.$$

Proof. At two, put $m = v_2(z)$ on the finite nonzero chart. If $m < 0$, the two abscissae in (1035) have valuations $2m - 2$ and -2 . If $m > 0$, their valuations are -2 and $-2m - 2$. For either short model and any negative abscissa valuation, the derivative $3x^2 + a_4$ has valuation $2v_2(x) < 0$. The local theorem gives height $-v_2(x) \log_5(2)$. Subtracting the two heights and adding $2m \log_5(2)$ gives zero in both ranges.

If $m = 0$, then $z^2 \equiv 1 \pmod{8}$. Both abscissae are integral and even, since $z^2 - 9 \equiv 0 \pmod{8}$ and $33 - 9/z^2 \equiv 0 \pmod{8}$. The two cubic equations then force the ordinates to be integral odd units. The derivative $3x^2 + a_4$ is odd, so both heights vanish; this also covers an abscissa equal to zero. The character term vanishes as well. Thus every valuation range gives $J_2 = 0$. The fiber $z = 0$ has no $\mathbb{Q}_2$ -point because -1 is not a square in $\mathbb{Q}_2$. At either infinity point the constant punctured neighborhoods with $m < 0$ give the extension.

Now let $\ell > 3$, $\ell \neq 5$. Both minimal elliptic curves have good reduction. With $m = v_\ell(z)$, if $m < 0$, R_1 has abscissa valuation $2m$ and R_2 is integral. If $m > 0$, R_1 is integral and R_2 has abscissa valuation $-2m$. For $m = 0$ both are integral. Possible divisibility of the

constant 33 causes no exception: when $-9/z^2$ is dominant its coefficient is a unit, and otherwise extra cancellation leaves an integral abscissa. The good-reduction height formula and the term $2m \log_5(\ell)$ again cancel exactly in all three cases.

At any local point above $z = 0$ or infinity, sufficiently close points outside that fiber have respectively $m > 0$ or $m < 0$, so the combination extends as zero. This covers every projective exception. The rational point $(1, 8)$ ensures nonemptiness at every prime. $\square$

Corollary 278.3 (The exact target value in quadratic Chabauty). *With the stated character and height conventions, the set Ω of [144, Theorem 1.4] for this curve and $p = 5$ is exactly*

$$\Omega = \left\{ -\frac{4}{3} \log_5(3) \right\}.$$

Proof. That theorem defines Ω as the values of $-\sum_{\ell \neq 5} J_\ell(P_\ell)$ on the product of the local curves. Theorems 278.1 and 278.2 evaluate every summand on its entire local domain, and their domains are nonempty. Only the summand at three is nonzero, with the value asserted above. $\square$

For the next calculation, the minimal-model expression on the finite nonzero five-adic chart can be written

$$(1041) \quad \begin{aligned} \mathcal{F}(P) = & h_{E,5}(R_1) - h_{E',5}(R_2) - 2 \log_5(z) \\ & - \alpha_E \log_E(R_1)^2 + \alpha_{E'} \log_{E'}(R_2)^2. \end{aligned}$$

The equal local model constants cancel. Moreover the exact scaling of Section 277 gives $\alpha_F \log_F^2 = \alpha_E \log_E^2$. Thus this is the standard expression, with its models and signs fixed. Every rational point in that chart satisfies

$$\mathcal{F}(P) = -\frac{4}{3} \log_5(3).$$

One can independently check the sign by summing the local heights and using the rank-one global quadratic identities. Since $b = (1, 8)$ is rational, the same equation is $\mathcal{F}(P) - \mathcal{F}(b) = 0$. This relative form cancels a uniform additive convention if the same convention is used at both points.

No analytic zero list or multiplicity calculation is claimed here. The remaining work is to construct compatible Coleman or sigma expressions, prove complete truncation bounds on all required five-adic charts, certify every analytic zero, and apply the rational sieve. The zero fiber is over $\mathbb{Q}(i)$; it has $\mathbb{Q}_5$ -points but no rational points. The two infinity points are rational and must be recorded separately if omitted from a chart. In particular the translated formula cannot simply assume that $Q_1 = (0, 3i) \in F_1$ is rational over $\mathbb{Q}$; the direct Theorem 1.4 expression was used above. After a classification of $C(\mathbb{Q})$, the second square and positive common-source conditions for the original curve D remain additional tests. Moving exponents, residuals and the uniform point-height problem are unchanged. These universal local-value statements are ordinary proofs with stated standard inputs, not a numerical sampling result or a new Lean formalization.

279. SCALAR-OPTIMAL ACTUAL RADICAL PARTITIONS WITHOUT REUSABLE FLAGS

This section gives an arithmetic subclass of the unchanged FCRT problem. It neither bounds the scalar ABC defect from above nor constructs an ABC counterexample. All divisors below are divisors of actual integer radicals.

For the primitive endpoint $(1, c-1, c)$, with $c = 2^e 3^f$ and integers $e, f \geq 2$, put

$$M = 2^e, \quad N = 3^f, \quad A = M/2, \quad B = N/3, \quad Q = \text{rad}(c-1).$$

The previously established exact optimization on the stratum

$$\max(M, N) < 2 \min(M, N) - 1, \quad AB > Q$$

is

$$(1042) \quad B_{\text{FCRT}} = B_{\text{SCRT}} = B_{\text{PBT}} = D + \min_{H|Q} \text{dist}(\log H, [\log(Q/B), \log A]),$$

$$(1043) \quad D = \log(AB/Q) = \log\left(\frac{c}{\text{rad}(c(c-1))}\right) > 0.$$

Here no nonempty proper compatible face exists at either source, and every sink prime is assigned with its entire power in the endpoint. We recall the two inherited facts needed for this reduction.

If $p^e \| c$, $m = p^e$, and $c - 1 = du$ with $d, u > 1$ and $u \equiv -1 \pmod{m}$, then $c \geq m(2m - 1)$. Indeed, write $c = mC$, $d = mt + 1$, $u = ms - 1$ with positive integers s, t . Exactness of $p^e \| c$ excludes $C = m$. Expansion gives $C + t = s(mt + 1)$. If $s = t = 1$ then $C = m$, excluded; otherwise $C \geq 2m - 1$. The balance condition above therefore excludes every proper compatible packet for both exact source moduli. Every compatible block must use the full sink set; a block with both sources is unsaturated, and a singleton block cannot improve on giving all sinks to that source. Thus the full optimization is

$$\log \min_{H|Q} \max(A/H, 1) \max(BH/Q, 1).$$

Splitting into the three positions of $\log H$ relative to the stated interval proves (1042). These are the ordinary no-face and optimization results used here; no uncompiled draft is reclassified as a Lean proof.

Theorem 279.1 (A complete two-factor partition). *Let $u, v \geq 1$ and $k \geq 2$ be integers, set $x = 2^u 3^v$, $e = 2u$, $f = 2v$, and suppose*

$$(1044) \quad x > 6, \quad \frac{3}{4} \leq \rho := \frac{2^u}{3^v} \leq \frac{4}{3}, \quad 7^k \mid x - 1, \quad 5^k \mid x + 1.$$

Then the hypotheses of (1042) hold, all proper flags are absent, and the actual coprime radicals

$$H = \text{rad}(x - 1), \quad J = \text{rad}(x + 1), \quad Q = HJ$$

satisfy $H \leq A$ and $J \leq B$. Consequently

$$(1045) \quad B_{\text{FCRT}} = D > (k - 1) \log 35 - \log 6.$$

Proof. Since x is even, $x - 1, x + 1$ are odd and coprime. Their radicals therefore multiply to exactly $Q = \text{rad}(x^2 - 1)$. For any positive integer m with $p^k \mid m$, $\text{rad}(m) \leq m/p^{k-1}$, regardless of all other prime powers. Hence

$$H \leq \frac{x - 1}{7^{k-1}} < \frac{x}{7}, \quad J \leq \frac{x + 1}{5^{k-1}} \leq \frac{x + 1}{5} \leq \frac{x}{4}.$$

The last inequality uses $x \geq 4$. Meanwhile

$$A = \frac{x\rho}{2} \geq \frac{3x}{8}, \quad B = \frac{x}{3\rho} \geq \frac{x}{4}.$$

Thus $Q/B \leq H \leq A$. The first packet uses all prime powers of $x - 1$, and the second all prime powers of $x + 1$. They are disjoint and exhaust every sink, with no duplicated credit or putative subface.

For the no-flag hypothesis, $M = x\rho$, $N = x/\rho$, $\max(M, N)/\min(M, N) \leq 16/9$, and $\min(M, N) \geq 3x/4$. Therefore

$$2 \min(M, N) - \max(M, N) \geq \frac{2}{9} \min(M, N) \geq \frac{x}{6} > 1.$$

The powers M, N are exact full prime powers in c , so the no-face result applies. Moreover

$$Q \leq \frac{x^2 - 1}{35^{k-1}} < \frac{x^2}{6} = AB.$$

The exhibited divisor makes the distance in (1042) zero, while

$$\frac{AB}{Q} \geq \frac{35^{k-1}x^2}{6(x^2 - 1)} > \frac{35^{k-1}}{6}.$$

Taking logarithms proves (1045). No factorization of the remaining parts of $x - 1$ or $x + 1$ was required. $\square$

Lemma 279.2 (Simultaneous actual residues). *For every integer $k \geq 2$, put $L = 12 \cdot 35^{k-1}$. For integers $r \geq 0$, $s \geq 1$, define*

$$(1046) \quad u = Lr + L/2 = 6 \cdot 35^{k-1}(2r + 1), \quad v = Ls.$$

Then $x = 2^u 3^v$ satisfies $7^k \mid x - 1$ and $5^k \mid x + 1$.

Proof. The binomial induction $z \equiv 1 \pmod{p} \Rightarrow z^{p^{k-1}} \equiv 1 \pmod{p^k}$ gives

$$2^{3 \cdot 7^{k-1}} \equiv 1 \pmod{7^k}, \quad 2^{2 \cdot 5^{k-1}} \equiv -1 \pmod{5^k}.$$

For the second congruence apply it to $-4 \equiv 1 \pmod{5}$ and use that 5^{k-1} is odd. The exponent u is a multiple of $3 \cdot 7^{k-1}$, while

$$\frac{u}{2 \cdot 5^{k-1}} = 3 \cdot 7^{k-1}(2r + 1)$$

is odd. Thus 2^u has the required signs at both moduli. The same induction applied to $3^6 \equiv 1 \pmod{7}$ and $3^4 \equiv 1 \pmod{5}$, together with the respective divisibilities $6 \cdot 7^{k-1} \mid v$ and $4 \cdot 5^{k-1} \mid v$, gives $3^v \equiv 1$ at both moduli. Multiplication finishes the proof. $\square$

Theorem 279.3 (Infinitely many scalar-optimal no-flag endpoints). *For every fixed integer $k \geq 2$, infinitely many pairs r, s in (1046) satisfy (1044); they may be chosen with $\rho \rightarrow 1$ and $x \rightarrow \infty$. Across these fixed- k families, the absolute scalar defect D is unbounded, while the radical partition penalty in (1042) is identically zero.*

Proof. Let $\alpha = \log 3 / \log 2$, irrational by unique factorization. Pigeonhole approximation gives positive integers q_j and integers a_j with $\beta_j = |q_j \alpha - a_j| > 0$ tending to zero. Choose a positive integer h_j nearest to $1/(2\beta_j)$. Then $h_j \beta_j \rightarrow 1/2$. For $s_j = h_j q_j$, the fractional part of αs_j tends to $1/2$, whether the approximation error has positive or negative sign. Also $s_j \rightarrow \infty$. Take $r_j = \lfloor \alpha s_j \rfloor$, eventually positive. For the fixed number $L = 12 \cdot 35^{k-1}$,

$$r_j + \frac{1}{2} - \alpha s_j \rightarrow 0, \quad \log \rho_j = L \log 2 (r_j + \frac{1}{2} - \alpha s_j) \rightarrow 0.$$

Thus the balance conditions and $x > 6$ hold eventually, while Lemma 279.2 supplies both divisibilities. Theorem 279.1 applies to all those actual inputs.

For any real threshold T , first choose k so that $(k - 1) \log 35 - \log 6 > T$, and then take arbitrarily large members of that fixed- k family. Their defects exceed T . A diagonal choice makes both c and D increase without bound. The radical partition penalty stays zero throughout. $\square$

This disproves the particular added claim that absence of proper flags and sufficiently large absolute defect force a positive partition penalty. It does not disprove FCRT-1 or SCRT-0. In particular, no fixed $\varepsilon > 0$ is exhibited for which the ABC excess $\log c - (1 + \varepsilon) \log \text{rad}(c(c - 1))$ is unbounded. Nor is an upper bound for that expression, for D , or for the partition penalty on all other endpoints proved here.

The global maximum consecutive divisor gap of Q can be much less relevant than the required central interval. The present proof supplies one actual divisor in that interval, without estimating the maximum gap over the entire divisor set. All unspecified prime factors remain in their full integer packets. The surviving general problem includes both the scalar defect and actual central divisor gaps when such witnesses are unavailable.

Ordinary and finite verification scope. The complete ordinary argument received independent internal review. The standard-library program `replay_divisor_partition.py --check` checks three actual balanced inputs by exact integer inequalities and the prescribed prime-power divisibilities, together with 66 residue-only instances for $k = 2, \dots, 12$. The latter are not asserted to be balanced. Its canonical certificate SHA-256 is `6b7a363b91b7fad5c570696ed8b5fcd50be50d73607a7d0c0d829bc1b75c551d`. It neither factors the remaining neighbours nor computes their radicals or logarithms, and it does not prove the infinite-family theorem. No new Lean theorem is claimed in this section.

280. AN ACTUAL LARGE-PRIME OBSTRUCTION TO CENTRAL RADICAL PARTITIONS

Let $e, f \geq 2$ be integers, and set

$$c = 2^e 3^f, \quad M = 2^e, \quad N = 3^f, \quad A = M/2, \quad B = N/3, \quad Q = \text{rad}(c - 1).$$

Throughout this section impose the complete balanced two-source hypotheses

$$(1047) \quad \max(M, N) < 2 \min(M, N) - 1, \quad AB > Q.$$

The endpoint $(1, c - 1, c)$ is primitive, and its exact source prime powers are M, N . The no-proper-face theorem and optimization proved in Section 279 give

$$(1048) \quad \mathcal{B}_{\text{FCRT}} = \mathcal{B}_{\text{SCRT}} = \mathcal{B}_{\text{PBT}} = D + \Gamma, \\ D = \log(AB/Q) > 0, \\ \Gamma = \min_{H|Q} \text{dist}(\log H, [\log(Q/B), \log A]).$$

Every divisor in this formula is a divisor of the full actual squarefree integer Q . When a prime is assigned to a packet, its full prime power on the original endpoint is assigned. No unknown cofactor is omitted.

Theorem 280.1 (The exact large-prime partition formula). *Under (1047), suppose an actual prime $q \mid c - 1$ satisfies $q > \max(A, B)$. Put $R = Q/q$. Then*

$$\Gamma = \log \frac{q}{\max(A, B)} > 0, \quad D + \Gamma = \log \frac{\min(A, B)}{R}.$$

No primality condition on R , prescription of the remaining prime depths, or square condition on c is required.

Proof. Since $AB > qR$ and $q > \max(A, B)$,

$$R < \min(A, B), \quad R < Q/B, \quad A < q.$$

Each divisor $H \mid Q$ either omits q , hence divides R and satisfies $H \leq R$, or contains q , hence satisfies $H \geq q$. Both extremal choices R, q are actual divisors. The central interval is nonempty because $AB > Q$, and lies strictly between their logarithms. The distances from its endpoints to the nearest divisor below and above are therefore

$$\log \frac{Q/B}{R} = \log \frac{q}{B}, \quad \log \frac{q}{A}.$$

Their minimum proves the formula for Γ . Adding $\log(AB/(qR))$ proves the budget formula. This optimizes over all actual divisors. There is at most one such large prime, since two would give $Q > \max(A, B)^2 \geq AB$. $\square$

Theorem 280.2 (A fully certified nonsquare endpoint). *Take $e = 64, f = 41$, and put*

$$q = 3981112602195296746201614890054671463.$$

Then q is prime, (1047) holds, and

$$\begin{aligned} c &= 672808029771005150108072916419239477248 = 13^2 q + 1, \\ M &= 18446744073709551616, & N &= 36472996377170786403, \\ A &= 9223372036854775808, & B &= 12157665459056928801, \\ Q &= 51754463828538857700620993570710729019 = 13q. \end{aligned}$$

The exact optimum is

$$D = \log \frac{c}{78q}, \quad \Gamma = \log \frac{q}{3^{40}}, \quad \mathcal{B}_{\text{FCRT}} = \log \frac{2^{63}}{13}.$$

Moreover $0 < D < 1$ and $\Gamma > 52D$.

Proof. Primality is established by the complete recursive certificate proved below. Direct integer arithmetic gives

$$2 \min(M, N) - 1 - \max(M, N) = 420491770248316828 > 0.$$

The full factorization is $c - 1 = 13^2 q$; thus the four divisors of its radical are exactly 1, 13, q , $13q$. Also

$$AB/Q = c/(78q) > 1, \quad q > B > A > 13.$$

The preceding theorem gives all three logarithmic formulas. The exact integer comparison

$$(1049) \quad q(78q)^{52} > 3^{40} c^{52}$$

is checked by the integer certificate, and implies $\Gamma > 52D$ by strict monotonicity of the logarithm. Finally

$$1 < \frac{169q + 1}{78q} < \frac{7}{3} < \exp(1)$$

proves $0 < D < 1$. The last inequality follows, for example, from $\exp(1) > 1 + 1 + 1/2 > 7/3$. No decimal approximation proves any of these strict inequalities. Since $f = 41$ is odd, c is not a square, so this example does not belong to the DP square family. $\square$

For scale only, D is approximately 0.77319, Γ approximately 40.33013, and their sum approximately 41.10332. The exact formulas and comparison (1049) are the mathematical statements.

Corollary 280.3 (A scalar-control strengthening fails). *The assertion that every actual endpoint satisfying (1047) has $\Gamma \leq D$ is false, even if one adds $0 < D < 1$. In particular, universal vanishing of Γ on that class is false.*

Proof. The fully factored endpoint of Theorem 280.2 satisfies every premise and $\Gamma > 52D > D > 0$. $\square$

This is a finite counterexample to the two displayed strengthenings. It does not disprove a bound with an unspecified additive constant C_ε , the parent FCRT uniform gate, or ABC. All reusable proper flags are absent by the same inherited theorem used in (1048); no flag or radical factor has been suppressed to manufacture the failure.

Proposition 280.4 (A conditional distinction between two uniform gates). *Fix a prime $p \geq 7$. If there are infinitely many distinct balanced endpoints $c = 2^e 3^f$, $e, f \geq 2$, such that*

$$c - 1 = p^2 q, \quad q \text{ prime},$$

then, along these endpoints as $c \rightarrow \infty$,

$$D = \log(p/6) + o(1),$$

$$\mathcal{B}_{\text{FCRT}} = \frac{1}{2} \log c + O_p(1),$$

$$\log \text{rad}(c(c-1)) = \log c + O_p(1).$$

Consequently the particular budget in (1048) would fail its ε -log-radical upper bound, with any additive constant, for each fixed $0 < \varepsilon < 1/2$. Meanwhile the scalar ABC excess

$$\log c - (1 + \varepsilon) \log \text{rad}(c(c-1))$$

would tend to minus infinity for every fixed $\varepsilon > 0$.

Proof. The distinct positive integers c tend to infinity along any increasing enumeration. The balance condition gives $1/2 < M/N < 2$; hence A, B are comparable to $\sqrt{c}$, with absolute constants. Eventually $q = (c-1)/p^2$ is distinct from p and larger than both A, B . Then $Q = pq$,

$$D = \log \frac{c}{6pq} \longrightarrow \log(p/6) > 0.$$

All hypotheses of Theorem 280.1 eventually hold, and it gives

$$\mathcal{B}_{\text{FCRT}} = \log \frac{\min(A, B)}{p} = \frac{1}{2} \log c + O_p(1).$$

Since $\text{rad}(c(c-1)) = 6pq = 6(c-1)/p$, the radical asymptotic follows. Subtracting the corresponding multiples of $\log c$ proves the two contrasting conclusions. $\square$

No infinitude of these prime values has been proved or used. This conditional proposition identifies an additional arithmetic input which would separate this specific certificate gate from scalar ABC. It must not be reported as an actual infinite counterexample. An equivalence proved for a different canonical ABC gate does not give the reverse implication for the present FCRT optimization.

280.1. A deterministic recursive primality and endpoint certificate.

Lemma 280.5 (Full predecessor-factor certificate). *Let $m > 2$ be an integer whose complete factorization of $m-1$ into proved primes is known. Suppose that for every prime $r \mid m-1$ there is an integer a_r satisfying*

$$a_r^{m-1} \equiv 1 \pmod{m}, \quad \gcd(a_r^{(m-1)/r} - 1, m) = 1.$$

Then m is prime. The witnesses for different primes need not agree.

Proof. Let ℓ be any prime divisor of m . The order of a_r modulo ℓ divides $m-1$, but does not divide $(m-1)/r$. Its r -adic valuation is therefore the entire r -adic valuation of $m-1$. This order also divides $\ell-1$. Applying this to each prime $r \mid m-1$ shows $m-1 \mid \ell-1$, whence $\ell \geq m$. Since $\ell \mid m$, we obtain $m = \ell$. Induction on m , beginning with the elementary prime two, justifies a recursive certificate using only smaller prime nodes. $\square$

For the prime in Theorem 280.2, the top decomposition is

$$q-1 = 2 \cdot 61 \cdot 563 \cdot 921889 \cdot 2532974063 \cdot 24821433536218331.$$

The witnesses for these six factors are respectively 5, 2, 2, 2, 2, 2. The complete data contains the predecessor factorizations and witnesses for all smaller factors, and a node proving thirteen. The standard-library program `replay_large_prime_partition.py` checks all twenty-five prime nodes and seventy modular/gcd witnesses. It uses no factorization oracle or probable-prime predicate. Thus generation of candidate data with a computer algebra system leaves no primality-test assumption in the verifier's proof chain.

After primality, the program checks the exact endpoint factorization, balance, positive defect, the complete four-divisor list, the rational identity

$$\exp(D) \exp(\Gamma) = \exp(\mathcal{B}_{\text{FCRT}}),$$

and (1049), using only integers and rational numbers. Default mode writes a canonical result; `--check` recomputes it and compares bytes without changing it. Both modes have actually returned PASS. The canonical result SHA256 is

$$3b4de60c0f5fb1fcb68b7def3b655db4b406047de5ea764f7b29b66f729fb51.$$

This certifies one endpoint and its deterministic primality inputs. The full-class optimization and the conditional implication retain their ordinary proofs above. Neither the certificate nor this section adds a Lean theorem, prime-value infinitude, a fixed- ε ABC counterexample, or a proof or disproof of the parent route.

281. FORMAL SCOPE OF PRIMITIVE CONTENT AND POWER EXTRACTION

The ordinary arguments in the preceding new sections were completed before their selected integer and real-arithmetic consequences were formalized. The present formal supplement contains forty new declarations: twenty-four in `ComplementContent.lean` and sixteen in `SquarefreePowerExtraction.lean`. All forty were freshly compiled together with fifty-five unchanged dependency declarations. Every one of these ninety-five declarations has an explicit axiom query. The axiom union is precisely `propext`, `Classical.choice`, and `Quot.sound`. The compiler is Lean 4.32.0 and the Mathlib revision is 81a5d257c8e410db227a6665ed08f64fea08e997. Only the Mathlib compiled cache is reused; the six listed project modules are compiled from their recorded source bytes in a fresh directory.

The first module uses the actual integer-pair Eisenstein multiplication, its positive norm, the integer gcd of the coordinates, and exact integer division to define primitive reduction. It constructs the actual normalizer cd and proves its divisibility into $N(V)$ by Bézout. It then proves the three recovered phase coordinates, their exact common gcd, and equality of the primitive absolute boundary product. Its radical is the product of the actual natural prime factors; for $T > 0$, its signed ledger is

$$\sum_{p|T} (v_p(T) - 3) \log p.$$

Both functions are therefore exactly unchanged. The cleared phase equalities and the integrality of the common multiplier are proved from the actual map, not supplied as hypotheses to its final theorem.

The second module starts with the explicit natural-number profile

$$RD = VQ^g, \quad R \text{ squarefree}, \quad (R, D) = 1, \quad g \geq 2.$$

It proves $R \mid V$ and $Q^g \mid D$ without requiring $(V, Q) = 1$. For $Q > 1$, an actual prime divisor of Q bounds g by its multiplicity in D . The module derives both logarithmic compression inequalities and, from the explicitly stated prime profile and mass hypotheses, the lower bound

$$\frac{\log V}{g} \geq \frac{\delta \log 7}{6} B.$$

The hypotheses realizing this profile in the interval product, and its large- B mass estimate, remain the preceding ordinary proofs. They have not been made formal by passing their finite consequences.

Eight finite replay records cover the collective content, complement determinants, primitive power profile, elliptic local-entry coordinates, actual divisor-partition witnesses, the recursive Lucas certificates for the large-prime partition example, the height-transport check, and the affine selectors' exact depths, content, finite local roots and kernels. The portable height check recomputes the exact rational leading-digit calculations and checks algebra on the recorded precision balls; it does not recompute those balls with PARI. The separate full PARI 2.15.4 replay was actually run locally and is distinguished from this portable check. These replays do not prove prime asymptotics, an infinite family, a formal-logarithm theorem, Jacobian closure, a height pairing, or rational-point completeness. Those results retain their ordinary proofs and source attributions. The independent reviews are internal mathematical reviews, not external peer review.

The completed statements do not bound the full boundary signed tail, cover every actual root or every ABC triple, compute a complete quadratic Chabauty set, or establish uniform point heights as exponents and residuals vary. Standard ABC remains unproved and undisproved. In particular, the input-norm height gain cannot be counted as boundary radical credit; the exact complement theorem identifies one specific construction where that proposed credit disappears under primitive normalization.

282. AFFINE OUTPUTS AND THE ORIGINAL POWER-ROOT CHART

This section gives complete ordinary intersection bounds for the one-parameter chart of actual Eisenstein powers. The bounds include integer residual coefficients. They concern a fixed exponent and this root chart; they do not describe all two-coordinate roots.

Let $\mathcal{O} = \mathbb{Z}[\zeta]$, where $\zeta^2 - \zeta + 1 = 0$ and $\text{Im } \zeta > 0$. In the basis $(1, \zeta)$, write $\det$ for the coordinate determinant and $N(a + b\zeta) = a^2 + ab + b^2$. Fix integers $n \geq 2$, $B \geq n$, and set

$$Z(x) = (3x + \zeta)^n = A(x) + C(x)\zeta, \quad x \geq B.$$

All lines below are affine lines in these two real coordinates.

Proposition 282.1 (Strict curvature of the complete chart). *Both $A'(x)$ and $C'(x)$ are positive, and*

$$\det(Z'(x), Z''(x)) = -27n^2(n-1)N(3x + \zeta)^{n-2} < 0.$$

Every real affine line meets this chart in at most two points. The same bound holds after any invertible real linear change of coordinates.

Proof. Put $\theta(x) = \arg(3x + \zeta)$. Then $0 < \theta(x) < 1/(3x)$, so $0 < (n-1)\theta(x) < \pi/3$. Consequently $Z'(x) = 3n(3x + \zeta)^{n-1}$ has both coordinates positive. Also $Z''(x) = 9n(n-1)(3x + \zeta)^{n-2}$. Multiplication by an element W scales coordinate determinants by $N(W)$, while $\det(3x + \zeta, 1) = -1$. Factoring the common $(3x + \zeta)^{n-2}$ proves the displayed determinant identity.

Since $A' > 0$, use A as a coordinate and write $C = F(A)$. Then

$$F''(A(x)) = \frac{\det(Z'(x), Z''(x))}{A'(x)^3} < 0.$$

A vertical line meets this graph at most once. Three intersections with a nonvertical line would contradict strict concavity at the middle abscissa, using the chord through the outer two intersections. An invertible linear map takes affine lines to affine lines under its inverse, so it preserves the bound. $\square$

Proposition 282.2 (Fixed-content and unfrozen affine selectors). *Let $Z_i = Z(k_i)$, where $B \leq k_2 < k_1$, put $D = Z_1 - Z_2$ and $\Delta = \det(Z_1, Z_2) > 0$. In a fixed-content selector progression*

$$V_j = \frac{Z_2 + (t_* + Hj)D}{g}, \quad j \geq 1,$$

with $H, g > 0$, the distinct outputs lie on the line

$$\det(D, V_j) = \Delta/g.$$

At most two outputs equal $Z(k)$ for an integer $k \geq B$. If $g = 1$ and the two nonempty disjoint CRT packets exclude the selector parameters $t = 0, 1$, there are no such equalities.

For all CRT selectors of these same two roots before content is frozen, there are at most $2\tau(|\Delta|)$ parameters whose primitive output equals $Z(k)$ for an integer $k \geq B$.

Proof. Both coordinates of D are positive, and $H/g > 0$, so the progression is injective. Taking its determinant with D gives the stated line because $\det(D, Z_2) = \Delta$. Proposition 282.1 bounds the intersection by two. For $g = 1$, the two known intersections are already $Z(k_2)$ and $Z(k_1)$. No third intersection is possible, and equality to these two outputs forces $t = 0$ and $t = 1$, respectively.

The actual content theorem for affine selectors, Theorem 275.2, gives $g_t \mid |\Delta|$. For each possible positive divisor g_t , the outputs lie on its one determinant line and have at most two possible values $Z(k)$. For fixed content and output, the nonzero vector D determines t uniquely. Sum over the positive divisors of $|\Delta|$. This is an upper bound only, with no assertion that any of those intersections exists. $\square$

Theorem 282.3 (Uniform counting with every small integer residual). *Let $V_j = V_0 + jV_1$ be any affine integer progression in $\mathcal{O}$, with $V_1 \neq 0$, and let $X \geq 1$ be real. The number of positive integer parameters j admitting*

$$V_j = \tau(3k + \zeta)^n, \quad \tau \in \mathcal{O} \setminus \{0\}, \quad N(\tau) \leq X, \quad k \in \mathbb{Z}, \quad k \geq B$$

is at most

$$2 \left[\left(2 \lfloor \sqrt{2X} \rfloor + 1 \right)^2 - 1 \right] < 30X.$$

If the conjugate of $(3k + \zeta)^n$ is also allowed, the bound is less than $60X$. These constants are independent of n, B , the affine line, and any packet primes or depths. Unit multipliers are already included among the possible τ .

Proof. For each fixed nonzero τ , multiplication by τ is invertible over $\mathbb{R}$, with determinant $N(\tau) > 0$. Its image of the power chart meets the progression line at most twice by Proposition 282.1. Since $V_1 \neq 0$, each output determines j uniquely. Summing these bounds may overcount multiple representations, which is harmless.

For $\tau = a + b\zeta$, one has $N(\tau) \geq (a^2 + b^2)/2$. Thus both coordinates are at most $\lfloor \sqrt{2X} \rfloor$ in absolute value, giving at most $(2\lfloor \sqrt{2X} \rfloor + 1)^2 - 1$ nonzero pairs. For $X \geq 1$,

$$(2\sqrt{2X} + 1)^2 \leq (9 + 4\sqrt{2})X < 15X.$$

This proves the first bound. Conjugation is an additional invertible real linear transformation, so summing the two orientation bounds gives $60X$. No power-free or coprimality restriction on the residual was used; such restrictions could only reduce the count. $\square$

Corollary 282.4 (Scope of transfer from the paid affine family). *Apply Theorem 282.3 to a fixed-data progression in Theorem 275.4. For every real $L \geq 0$, fewer than $60e^{nL}$ parameters in the entire progression admit either orientation with*

$$\log N(\tau) \leq nL.$$

More generally, if $X(J) \geq 1$ and $X(J) = o(J)$, then only $o(J)$ indices $1 \leq j \leq J$ admit such a representation with $N(\tau) \leq X(J)$. All but $o(J)$ of the paid good indices up to J have no such representation. In particular, for fixed $0 < \eta < 1$, the good indices with a representation satisfying $N(\tau) \leq j^\eta$ form a zero-density subset.

Proof. Use $X = e^{nL}$ for the first assertion and $X = X(J)$ for the second. Intersecting the resulting bound with the good set can only decrease it; that good set has fixed-data density at least $1/4$ by Theorem 275.4. For the last assertion, $j \leq J$ and $N(\tau) \leq j^\eta$ imply $N(\tau) \leq J^\eta$; the exceptional count is therefore less than $60J^\eta = o(J)$. $\square$

These bounds explain a specific obstruction to identifying the affine outputs with many same-index outputs in the original root chart. They do not exclude a finite useful intersection, a different two-coordinate root chart, a changing exponent, a transfer between different outputs, or a nonlinear selector. The $60X$ bound is uniform, whereas the paid affine density has fixed-data thresholds. No uniform statement for moving packets, no original far-tail budget, and no control of the dependent-root complement follows. The results in this section are ordinary proofs, not a claim of complete Lean formalization.

283. NONLINEAR SELECTORS PRESERVING AN ACTUAL PURE SQUARE

Imposing the different-owner congruences on the root parameter before squaring preserves the exact power representation. The following ordinary proof concerns exponent two and fixed marked data. It does not concern the original moving prime-index block.

Retain $\mathcal{O} = \mathbb{Z}[\zeta]$ and $\zeta^2 - \zeta + 1 = 0$. For $k \geq 1$ integral put

$$(3k + \zeta)^2 = A_k + C_k\zeta, \quad c_k = A_k + C_k, \quad T(k) = A_k C_k c_k.$$

Proposition 283.1 (The actual square and its full boundary). *The three arms are positive and pairwise coprime, with*

$$A_k = 9k^2 - 1, \quad C_k = 6k + 1, \quad c_k = 3k(3k + 2),$$

and

$$(1050) \quad T(k) = (3k - 1)(3k + 1)(6k + 1)(3k)(3k + 2).$$

The input norm $9k^2 + 3k + 1$ is 1 (mod 3). Every prime divisor of the full boundary $T(k)$ is a unit for this input norm.

Proof. Squaring in the stated ring gives the coordinate formulas and factorization. If a positive integer divides both A_k, C_k , it divides $4A_k - C_k^2 = -2C_k - 3$ and $2C_k$, hence divides 3. Since $A_k \equiv -1 \pmod{3}$, their gcd is one. The other arm gcds follow from $c_k = A_k + C_k$.

Norm multiplicativity gives

$$(9k^2 + 3k + 1)^2 = A_k^2 + A_k C_k + C_k^2.$$

At a prime dividing any one of the three pairwise coprime arms, the right side is the square of a unit, including the case that the sum arm vanishes. Thus the input norm is nonzero modulo that prime. No factorization of the unmarked primes is assumed. $\square$

Theorem 283.2 (Exact finite packets with different actual owners). *Fix a finite nonempty set $\mathcal{S}$ of primes $q > 5$. For each q , choose a positive integer $k(q)$ with actual depth $e_q = v_q(T(k(q))) \geq 1$. Choose a positive integer k_0 by CRT satisfying*

$$k_0 \equiv k(q) \pmod{q^{e_q+1}} \quad (q \in \mathcal{S}).$$

Put

$$\mathcal{E} = \mathcal{S} \cup \{2, 3, 5\}, \quad F_p = v_p(T(k_0)), \quad H = \prod_{p \in \mathcal{E}} p^{F_p+1}, \quad K = \prod_{p \in \mathcal{E}} p^{F_p}.$$

For every integer $j \geq 1$, set $k_j = k_0 + Hj$. Then $(3k_j + \zeta)^2$ is a positive primitive unramified pure square with exact depth F_p at every $p \in \mathcal{E}$. In particular its marked depths are e_q , and its entire selected signed cost is

$$\sum_{q \in \mathcal{S}} (e_q - 3) \log q.$$

Proof. The integer polynomial T preserves its value modulo each q^{e_q+1} . Fixing one digit beyond the actual valuation therefore gives $F_q = e_q$, and $\prod_{q \in \mathcal{S}} q^{e_q+1} \mid H$. For every $p \in \mathcal{E}$, $k_j \equiv k_0 \pmod{p^{F_p+1}}$, so its exact depth remains F_p . Proposition 283.1 gives positivity, primitivity and all norm-unit conclusions. The signed sum keeps the terms of depths one and two as well as the positive ones. $\square$

Theorem 283.3 (Full squarefree sieve on the unchanged pure family). *With all data in Theorem 283.2 fixed, there is a set $\mathcal{G}$ of positive integers of natural density*

$$d(\mathcal{G}) = \prod_{p \notin \mathcal{E}} (1 - 5/p^2) \geq \frac{1}{6}$$

such that every $j \in \mathcal{G}$ has

$$T(k_j) = KR_j, \quad R_j \text{ squarefree}, \quad \gcd(K, R_j) = 1,$$

and all prime divisors of R_j lie outside $\mathcal{E}$. For the complete signed sum $J(T) = \sum_{p|T} (v_p(T) - 3) \log p$,

$$(1051) \quad J(T(k_j)) = -2 \log T(k_j) + 3 \log \frac{K}{\text{rad } K}.$$

Along these good indices,

$$\frac{J(T(k_j))}{\log c_{k_j}} \rightarrow -5, \quad \frac{\log \text{rad}(T(k_j))}{\log c_{k_j}} \rightarrow \frac{5}{2}.$$

Every good index satisfies

$$(1052) \quad c_{k_j} \leq \frac{15}{324^{2/5}} \left(\frac{K}{\text{rad } K} \right)^{2/5} \text{rad}(T(k_j))^{2/5}.$$

Proof. For every $p \notin \mathcal{E}$, the five factors of (1050) become affine polynomials in j with p -unit slopes. In the coordinate $a = 3k$, their roots are

$$1, \quad -1, \quad -\frac{1}{2}, \quad 0, \quad -2.$$

They are distinct modulo $p > 3$: multiply them by two and note that every nonzero difference between $2, -2, -1, 0, -4$ has prime divisors only 2, 3. The change $a = 3k_0 + 3Hj$ is invertible modulo p^2 . Thus exactly five simple residue classes modulo p^2 are forbidden. Since the factors have distinct roots already modulo p , a square divisor p^2 of their product must divide one individual factor.

For $1 \leq j \leq X$, every positive factor is at most $C_0 X$, where $C_0 = 6(k_0 + H) + 1$. A prime square dividing a factor has $p \leq \sqrt{C_0 X}$. Counting its five residue classes gives the complete large-prime estimate

$$(1053) \quad \begin{aligned} & \#\{1 \leq j \leq X : \exists p > Y, p \notin \mathcal{E}, p^2 \mid T(k_j)\} \\ & \leq 5X \sum_{p > Y} p^{-2} + 5\sqrt{C_0 X}. \end{aligned}$$

For a fixed finite cutoff Y , CRT gives the finite product density. Divide (1053) by X , first let X tend to infinity, and then let Y tend to infinity. The upper and lower finite-CRT bounds establish the exact natural density. Since all omitted primes are at least seven,

$$\sum_{p \notin \mathcal{E}} \frac{5}{p^2} \leq 5 \sum_{m \geq 7} m^{-2} < 5 \int_6^\infty x^{-2} dx = \frac{5}{6}.$$

The finite inequality $\prod(1 - u_i) \geq 1 - \sum u_i$, followed by the convergent product limit, proves the density lower bound. This treats all prime squares; there is no unbounded-degree squarefree-value assumption or unexamined large-prime range.

The frozen depths inside $\mathcal{E}$ and squarefreeness outside it give $T = KR_j$ and

$$\text{rad } T = T/(K/\text{rad } K).$$

This proves (1051), including every negative term. The five affine factors give $\log T = 5 \log j + O(1)$, while $\log c = 2 \log j + O(1)$, proving the two limits.

For all $k \geq 1$, the five lower bounds $2k, 3k, 6k, 3k, 3k$ give $T(k) \geq 324k^5$, and $c_k \leq 15k^2$. Eliminating k and using the radical identity gives (1052). $\square$

The set of resulting root indices has density $d(\mathcal{G})/H > 0$ among all positive integers. It consists of actual pure squares satisfying the fixed marked data, not merely compatible local labels.

A positive-cost packet and the boundary of the conclusion. Take the two actual owners

$$k(19) = 4973918 = 21720 + 2 \cdot 19^5,$$

$$k(13) = 4888691 = 61882 + 13^6.$$

Their second arms are $19^4 \cdot 229$ and $13^5 \cdot 79$. At $q > 3$ dividing $6k + 1$, the other coordinate is $-3/4 \pmod{q}$, and the input norm is $3/4 \pmod{q}$. The whole depths are therefore exactly four and five. The other owner has no hit: $k(19) \equiv 1 \pmod{13}$ and $k(13) \equiv 10 \pmod{19}$, and substitution in the five factors proves all are units at that other prime. The selected cost is $\log 19 + 2 \log 13 > 0$. Theorem 283.3 then supplies infinitely many actual pure-square outputs with this fixed positive packet and the full negative credit in (1051). The earlier finite selector certificate checks the displayed owner arithmetic; it is not an infinite squarefree-density proof.

The residual coefficient remains one and the exponent is two, so the earlier convention $\lambda = 1/n$ gives the fixed value $1/2$, which does not approach zero. All packet primes, depths, k_0, H, K, C_0 are fixed in the density limit. The size of $K/\text{rad } K$, the progression step and the effective tail threshold can depend on the full packet. The construction does not give a moving-packet estimate in the original $n > 324$ prime, $B = n^4$ block.

The elementary arithmetic of these literal squares is separately amenable to the scoped selector Lean module. The CRT, complete sieve, natural density and asymptotic statements here are ordinary proofs, not claims of complete formalization. The original far signed tail, its exceptions, dependent-root complement, general residuals and full ABC coverage remain open. The linear-factor proof above cannot be applied to an arbitrary higher-degree boundary without first checking its actual factors.

284. ACTUAL PURE-CUBE SELECTORS AND A PROVED SQUAREFREE-VALUE INPUT

The following construction imposes the marked congruences on the root before cubing. A published unconditional theorem supplies the complete squarefree-value asymptotic after its hypotheses have been checked on the resulting integer polynomial. Its constants depend on the fixed packet; no assertion for changing exponents is made.

Work in $\mathcal{O} = \mathbb{Z}[\zeta]$, where $\zeta^2 - \zeta + 1 = 0$. For a positive integer k , put $a = 3k$ and

$$(a + \zeta)^3 = A + C\zeta, \quad c = A + C, \quad T_3(k) = ACc.$$

Proposition 284.1 (The actual cube and every irreducible factor). *The three arms are positive and pairwise coprime, with*

$$A = a^3 - 3a - 1, \quad C = 3a(a + 1), \quad c = a^3 + 3a^2 - 1.$$

In particular,

$$(1054) \quad T_3(k) = 3a(a+1)f(a)g(a), \quad f(a) = a^3 - 3a - 1, \quad g(a) = a^3 + 3a^2 - 1.$$

Both cubics are irreducible over $\mathbb{Q}$, each of discriminant 81, and $a(a+1)f(a)g(a)$ is squarefree over $\mathbb{Q}$. Modulo every prime $p > 3$ this product has no repeated root over an algebraic closure and has at most eight roots in $\mathbb{F}_p$. Every prime dividing the full boundary is a unit for the actual input norm $a^2 + a + 1$, which is 1 (mod 3).

Proof. Direct multiplication gives the formulas and positivity for $a \geq 3$. The values of f at zero and minus one are respectively $-1, 1$; also $f(a) \equiv -1 \pmod{3}$ for $a = 3k$. Thus A is coprime to $3a(a+1) = C$. The other two gcds follow by taking their sum c . Norm multiplicativity gives

$$(a^2 + a + 1)^3 = A^2 + AC + C^2.$$

At a prime dividing one of the three coprime arms, the right side is a nonzero square modulo that prime. This proves the norm-unit assertion for every boundary prime.

The rational-root test at $1, -1$ proves irreducibility of both monic cubics. The cubic discriminant formula gives 81 for each. Moreover $g - f = 3a(a+1)$, while $f(0) = -1$ and $f(-1) = 1$. Outside characteristic three, these identities exclude common roots between the cubics or between a cubic and a linear factor. The discriminants exclude repeated roots inside either cubic. They also prove the corresponding assertions over $\mathbb{Q}$. The substitution $a = 3k$ preserves these conclusions outside three. $\square$

Theorem 284.2 (Exact packets and the actual quotient polynomial). *Fix a finite nonempty set $\mathcal{S}$ of primes $q > 7$. For each q , fix an actual positive owner $k(q)$ with $e_q = v_q(T_3(k(q))) \geq 1$. Choose a positive integer k_0 with*

$$k_0 \equiv k(q) \pmod{q^{e_q+1}} \quad (q \in \mathcal{S}).$$

Define

$$\mathcal{E} = \mathcal{S} \cup \{2, 3, 5, 7\}, \quad F_p = v_p(T_3(k_0)), \quad K = \prod_{p \in \mathcal{E}} p^{F_p}, \quad H = \prod_{p \in \mathcal{E}} p^{F_p+1}.$$

For $j \geq 1$ integral put $k_j = k_0 + Hj$. The outputs $(3k_j + \zeta)^3$ remain positive primitive unramified pure cubes, and their exact depths at all $p \in \mathcal{E}$ are F_p . In particular the marked depths remain e_q .

The literal polynomial

$$(1055) \quad G(J) = T_3(k_0 + HJ)/K$$

belongs to $\mathbb{Z}[J]$. It is nonconstant and squarefree over $\mathbb{Q}$, each irreducible factor has degree at most three, and it has no fixed prime divisor.

Proof. CRT applies to the pairwise coprime prime powers. An integer polynomial preserves its value modulo these powers, so the extra digit gives $F_q = e_q$. Since H contains p^{F_p+1} for every $p \in \mathcal{E}$, all these exact valuations remain fixed along the progression. Proposition 284.1 proves the actual power, gcd and norm assertions.

To prove the stronger assertion $G \in \mathbb{Z}[J]$, its constant coefficient is divisible by K , and every nonconstant coefficient of $T_3(k_0 + HJ)$ is divisible by H . This follows by expanding each integer monomial. Since $K \mid H$, division by K is coefficientwise integral.

For $p \in \mathcal{E}$, the constant $T_3(k_0)/K$ is a p -unit, whereas all nonconstant coefficients after division are multiples of H/K , which is divisible by p . Thus G is a fixed nonzero constant modulo p . For $p \notin \mathcal{E}$, the quantities $3, H, K$ are units and $p \geq 11$. The affine substitution is invertible; Proposition 284.1 gives a nonzero squarefree polynomial of degree eight with at most eight roots modulo p . It cannot vanish at every residue class. This excludes a fixed divisor at every prime. A nonzero affine substitution and a nonzero scalar preserve factor degrees and squarefreeness over $\mathbb{Q}$, proving the remaining claims. $\square$

Theorem 284.3 (Positive density with the complete signed budget). *Keep all data of Theorem 284.2 fixed. There is a set $\mathcal{G}$ of positive integers of natural density*

$$d(\mathcal{G}) = \prod_p \left(1 - \frac{\rho_G(p^2)}{p^2}\right) \geq \frac{1}{5}, \quad \rho_G(m) = \#\{j \bmod m : G(j) \equiv 0 \pmod{m}\},$$

such that each $j \in \mathcal{G}$ has

$$T_3(k_j) = KR_j, \quad R_j \text{ squarefree}, \quad \gcd(K, R_j) = 1,$$

and the prime divisors of R_j all lie outside $\mathcal{E}$. For $J(T) = \sum_{p|T} (v_p(T) - 3) \log p$, the exact identity is

$$(1056) \quad J(T_3(k_j)) = -2 \log T_3(k_j) + 3 \log \frac{K}{\text{rad } K}.$$

Along these good indices tending to infinity,

$$\frac{J(T_3(k_j))}{\log c_{k_j}} \rightarrow -\frac{16}{3}, \quad \frac{\log \text{rad}(T_3(k_j))}{\log c_{k_j}} \rightarrow \frac{8}{3}.$$

Every good output satisfies the explicit bound

$$(1057) \quad c_{k_j} \leq 2 \left(\frac{K}{\text{rad } K} \right)^{3/8} \text{rad}(T_3(k_j))^{3/8}.$$

Proof. The precise external input is Booker–Browning [148, Theorem 1.2, printed p. 4]. For a nonconstant squarefree integer polynomial with no fixed prime divisor and irreducible factors of degree at most three, their theorem gives

$$(1058) \quad \sum_{1 \leq j \leq X} \mu^2(G(j)) = X \prod_p \left(1 - \frac{\rho_G(p^2)}{p^2}\right) + O_G(X/\log X).$$

The no-fixed-prime assumption is imposed in the authors' opening setup on printed p. 1; their equation (2) defines ρ . All these hypotheses were individually checked in Theorem 284.2. The theorem is unconditional; its large-square input is proved using the low-degree results described in their Section 4. We do not substitute an elementary degree-eight union estimate for that input, and no ABC conjecture is assumed.

For $p \in \mathcal{E}$, the polynomial is always a p -unit, so $\rho_G(p^2) = 0$. Outside $\mathcal{E}$, its at most eight simple roots lift uniquely to p^2 , so $\rho_G(p^2) \leq 8$. Hence

$$\sum_{p \notin \mathcal{E}} \frac{\rho_G(p^2)}{p^2} \leq 8 \sum_{m \geq 11} m^{-2} < 8 \int_{10}^{\infty} x^{-2} dx = \frac{4}{5}.$$

The product inequality $\prod (1 - u_i) \geq 1 - \sum u_i$, followed by a convergent product limit, proves the lower bound $1/5$. Take $\mathcal{G}$ to be the indices for which $G(j)$ is squarefree. All original factors are positive for $j \geq 1$; exact freezing excludes $\mathcal{E}$ from $G(j)$. Thus $R_j = G(j)$ has all the asserted properties.

Now

$$\text{rad}(T_3(k_j)) = T_3(k_j)/(K/\text{rad } K).$$

This proves (1056), retaining every frozen depth and every unmarked negative contribution. The polynomial degrees give $\log T_3 = 8 \log j + O(1)$ and $\log c = 3 \log j + O(1)$, which prove the limits. For $a \geq 3$,

$$f(a) \geq a^3/2, \quad g(a) \geq a^3, \quad 3a(a+1) \geq 3a^2.$$

The first inequality follows from $a^3 - 6a - 2 \geq 0$, increasing on $[3, \infty)$. Therefore $T_3 \geq a^8$, while $c = a^3 + 3a^2 - 1 \leq 2a^3$. Eliminating a and using the radical identity proves (1057). $\square$

Actual distinct owners and arbitrary fixed positive depths. At $k = 1, 2$, respectively, the literal triples are

$$(A, C, c) = (17, 36, 53), \quad (197, 126, 323).$$

The integers 53 and 197 are prime by trial division up to their square roots. The first triple has no factor 197, and the second no factor 53. They already give a nonempty different-owner packet.

Larger prescribed finite depths do not require prime values of a polynomial. At $k = 1$, the derivative of the c factor is 135, a unit modulo 53; the other arms are units. At $k = 2$, the derivative of the A factor is 315, a unit modulo 197, and again the other arms are units. For either simple root, the expansion

$$F(r + q^j u) \equiv F(r) + q^j u F'(r) \pmod{q^{j+1}}$$

has exactly one digit retaining the root. Lift to depth e , then choose a different next digit; the resulting residue has exact depth e . This realizes every prescribed positive integer depth. CRT can additionally require the 53-owner to be 1 (mod 197) and the 197-owner to be 2 (mod 53), which excludes the other marked prime at each owner. Positive representatives exist. For example depths four and five give the marked cost $\log 53 + 2 \log 197 > 0$. Theorem 284.3 supplies infinitely many actual pure cubes retaining that fixed cost and the full new negative support.

The constant in (1058) depends on the fixed polynomial G , whose coefficients depend on the whole packet and its CRT representative. The good root indices have density $d(\mathcal{G})/H > 0$ among positive integers, with no uniform threshold for moving packets. The residual coefficient is one and the exponent remains three, so the earlier convention $\lambda = 1/n$ gives the fixed value $1/3$.

These are fixed low-degree nonlinear selectors. The result does not control the original $n > 324$ prime, $B = n^4$ block, nor allow the squarefree theorem to be invoked at an arbitrary changing exponent without checking its irreducible factors and constants. The elementary literal-cube identities have a separate scoped Lean verification; CRT, squarefree density and its analytic input remain ordinary proofs here. Other roots, general residuals, exceptional sets, the independent-domain complement and full ABC coverage remain open.

285. ACTUAL COEFFICIENT PACKETS AND THE COPRIME RESULTANT TEST

This section returns to the original block. It disproves a proposed bound based on one short primitive coefficient vector, using actual different owners, and then proves the correct two-polynomial certificate with its full signed accounting. The second polynomial must be coprime to the boundary polynomial; constructing it with a useful degree and height remains a separate problem.

Let $n > 324$ be prime, $B = n^4$, and $\mathcal{O} = \mathbb{Z}[\zeta]$, where $\zeta^2 - \zeta + 1 = 0$ with positive imaginary part. Put

$$P(A + C\zeta) = AC(A + C), \quad w_k = 3k + \zeta, \quad z_k = w_k^n, \quad T_k = P(z_k) \quad (B \leq k < 2B).$$

Here and throughout the block indices are integers. These three raw arms are positive: if $\theta = \arg w_k$, then $0 < n\theta < n/(3k) \leq 1/3$, so the first coordinate is

$$|w_k|^n \left(\cos(n\theta) - \frac{\sin(n\theta)}{\sqrt{3}} \right) > 0,$$

and the second is positive as well.

The powers are primitive. For a prime $p \neq 3$, the algebra $\mathcal{O}/p\mathcal{O}$ is reduced because the discriminant is -3 . Thus $w_k^n = 0$ in this algebra would imply $w_k = 0$, contrary to its second coordinate being one. At $p = 3$, the norm $9k^2 + 3k + 1$ is a unit, so again the two coordinates of the power cannot vanish. Hence the three arms are pairwise coprime. A prime dividing one arm has nonzero norm at the power and at its root.

A full assigned T -packet means a finite set of distinct rational primes p , an actual block owner $k(p)$ for each, and its full positive depth $e_p = v_p(T_{k(p)})$. For a degree limit d its evaluation lattice is

$$\mathcal{L} = \{G \in \mathbb{Z}[X]_{\leq d} : G(k(p)) \equiv 0 \pmod{p^{e_p}} \text{ for every packet prime}\}.$$

The statement being tested is that one nonzero primitive vector of small coefficient height in this lattice bounds its index by a fixed power of that height, uniformly in n .

Proposition 285.1 (Literal primitive boundary polynomial). *Set*

$$P_n(a) = P((a + \zeta)^n), \quad T_n(X) = P_n(3X), \quad F_n(X) = T_n(X)/3.$$

The content of T_n is exactly three. Consequently F_n is a nonzero primitive integer polynomial of degree $d = 3n - 1$, and

$$(1059) \quad \|F_n\|_1 \leq 64^n/3 \leq 64^n.$$

Moreover there is an $R_n \in \mathbb{Z}[a]$ such that

$$(1060) \quad P_n(a) = na + a^2 R_n(a).$$

Proof. The sixth roots of unity have coordinate pairs

$$(1, 0), (0, 1), (-1, 1), (-1, 0), (0, -1), (1, -1).$$

Since $n \equiv 1$ or $5 \pmod{6}$, one has $P_n(0) = 0$ and $P'_n(0) = n$. In the first case $\zeta^n = \zeta$; the zero first arm has derivative n . In the second case $\zeta^n = 1 - \zeta$; the zero sum arm has derivative $-n$ and the product of the other two arms is -1 . This proves (1060). All coefficients of T_n are divisible by three, and its linear coefficient is $3n$.

Frobenius in $(\mathcal{O}/n\mathcal{O})[X]$ gives

$$(1061) \quad T_n(X) \equiv \begin{cases} (3X)^n((3X)^n + 1) & n \equiv 1 \pmod{6}, \\ -(3X)^n((3X)^n + 1) & n \equiv 5 \pmod{6} \end{cases} \pmod{n}.$$

This is a nonzero polynomial over $\mathbb{F}_n$. The content divides $3n$ and is not divisible by n , so it is exactly three.

Writing $w_X^n = A_n(X) + C_n(X)\zeta$, the degrees of $A_n, C_n, A_n + C_n$ are $n, n-1, n$, with leading coefficients $3^n, n3^{n-1}, 3^n$. Thus $\deg F_n = 3n - 1$. Every coordinate and the coordinate sum of ζ^j has absolute value at most one. The binomial expansion consequently bounds each of the three coefficient ℓ^1 -norms by $\sum_j \binom{n}{j} 3^{n-j} = 4^n$. Submultiplicativity, followed by division by three, proves (1059). $\square$

Proposition 285.2 (Two different actual positive-depth owners). *For $k_0 = B$,*

$$v_3(T_{k_0}) = 1, \quad v_n(T_{k_0}) = 5, \quad v_5(T_{k_0}) = 0.$$

There is an integer $k_1 \in [B, 2B)$ with

$$k_1 \equiv 5^4 \pmod{5^5}, \quad k_1 \equiv 1 \pmod{n}.$$

Every such k_1 satisfies

$$v_5(T_{k_1}) = 4, \quad v_n(T_{k_1}) = 0.$$

Thus the two actual distinct owners supply the positive signed contributions $2 \log n$ and $\log 5$, respectively.

Proof. Since $3 \nmid nB$, equation (1060) at $a = 3B$ gives $v_3(T_B) = 1$. At the prime n , use $a = 3n^4$. At $a = 0$ exactly one arm vanishes, with linear term $\pm na$ of valuation five. Every term of degrees two through $n-1$ has valuation at least $1 + 4 \cdot 2 = 9$, since the corresponding binomial coefficient is divisible by n ; the final term has valuation $4n$. The other two arms remain units modulo n . This proves the full depth five.

Modulo five, $B = n^4 = 1$, so $w_B = 3 + \zeta$ in $\mathbb{F}_{25}$. Its norm is $13 = 3 \pmod{5}$, whence $(3 + \zeta)^6 = 3$, and its fifth power is its conjugate $4 - \zeta$. As $n \equiv 1$ or $5 \pmod{6}$, the power w_B^n is a nonzero scalar multiple of one of these two elements. Their P -values are $12, -12$, both nonzero modulo five.

CRT supplies the stated congruences with modulus $3125n$. The least solution at least B is less than $B + 3125n < 2B$, because $n^3 > 3125$. Its five-adic depth as an index is four. Equation (1060) then has linear term $3nk_1$ of five-adic depth four and every later term of depth at least eight. Finally (1061) at $k_1 \equiv 1 \pmod{n}$ gives $T_{k_1} \equiv \pm 12 \pmod{n}$, a unit. The owners differ modulo n , so they are distinct. $\square$

These primes are small relative to the moving far-tail cutoffs. The construction is not a counterexample to a far-tail estimate, and makes no claim that these particular owners lie in the multiplicatively independent subdomain.

Theorem 285.3 (Exact index and the complete selected ledger). *Let $m_0 = F_n(B) = T_B/3$. Assign every prime dividing m_0 to k_0 , with its full depth in T_B , and assign the additional prime five to k_1 , with depth four. Then all labels are norm units at their actual owners, and*

$$(1062) \quad M = 625m_0, \quad [\mathbb{Z}^{d+1} : \mathcal{L}] = M, \quad F_n \in \mathcal{L}.$$

This is not a common-hit rectangle: n hits only the first of these owners, and five only the second. For the complete $J(T) = \log T - 3 \log \text{rad } T$, its selected ledger is

$$(1063) \quad \sum_{p|m_0} (v_p(m_0) - 3) \log p + \log 5 = J(T_B) + 2 \log 3 + \log 5.$$

Proof. Proposition 285.2 gives $\gcd(m_0, 15) = 1$. Primitivity in the setup proves the norm-unit assertion. The additive homomorphism

$$\mathbb{Z}[X]_{\leq d} \longrightarrow \left(\prod_{p|m_0} \mathbb{Z}/p^{v_p(m_0)}\mathbb{Z} \right) \times \mathbb{Z}/625\mathbb{Z}$$

given by evaluation at the assigned owners is surjective: constant polynomials already realize all residue tuples by CRT. Its kernel is $\mathcal{L}$, proving the index assertion. The polynomial F_n satisfies each first-owner congruence by its value m_0 , and at k_1 it has exact five-adic depth four because division by three does not affect that depth. Its coefficient vector is primitive by Proposition 285.1, and its value $F_n(B) = m_0$ is positive. Thus this is neither the zero polynomial nor a vanishing integer evaluation used in an additive selector kernel.

The only prime removed from T_B is three at depth one, whose contribution to $J(T_B)$ is $-2 \log 3$. The added five-adic depth four contributes $\log 5$. This proves (1063), with every depth-one and depth-two term retained. Neither its sign nor its asymptotic size is asserted. $\square$

Theorem 285.4 (Failure of a fixed-power short-vector bound). *For $H_n = \max(2, \|F_n\|_1)$, every fixed $C, C_0 > 0$ satisfies*

$$[\mathbb{Z}^{d+1} : \mathcal{L}] > C_0 H_n^C$$

for all sufficiently large prime n in the construction above.

Proof. Put $r = |3B + \zeta| \geq 3B$ and $u = n \arg(3B + \zeta) \in (0, 1/3)$. The inequalities $\cos u \geq 1 - u^2/2$ and $\sin u \leq u$ give

$$A_B = r^n (\cos u - \sin u / \sqrt{3}) > r^n / 2.$$

The second coordinate C_B is a positive integer, and $A_B + C_B > A_B$. Therefore

$$T_B > (3B)^{2n}/4, \quad M > (625/12)(3B)^{2n}.$$

Together with (1059), this yields

$$\log M > 8n \log n + 2n \log 3 + \log(625/12), \quad \log H_n \leq n \log 64.$$

The ratio tends to infinity along primes, proving the assertion. $\square$

Only bounds with an absolute fixed exponent based on one primitive short coefficient vector are refuted. The dimension grows with n . A dimension-dependent bound, a bound paying the evaluation height B , and Hadamard applied to a full independent basis are not refuted. In particular, one short primitive lattice vector is insufficient evidence for a small covolume.

Theorem 285.5 (A coprime second polynomial certificate). *Let $F, G \in \mathbb{Z}[X] \setminus \{0\}$, of actual degrees $d, e \geq 1$, be coprime over $\mathbb{Q}$. For a finite set of distinct primes, positive integers h_p , and arbitrary integer owners a_p , suppose*

$$p^{h_p} \mid F(a_p), \quad p^{h_p} \mid G(a_p).$$

Then

$$(1064) \quad \prod_p p^{h_p} \mid \text{Res}(F, G), \quad \sum_p h_p \log p \leq e \log \|F\|_2 + d \log \|G\|_2.$$

No monicity or invertibility of a leading coefficient is required. The empty packet is allowed, with modulus product one.

Proof. The integer Sylvester determinant is nonzero by coprimality. View its matrix as the coefficient map $(U, V) \mapsto UF + VG$, for $\deg U < e$ and $\deg V < d$. Applying its integer adjugate to the constant basis vector gives $U, V \in \mathbb{Z}[X]$ satisfying

$$UF + VG = \pm \text{Res}(F, G).$$

Evaluation at each assigned owner proves divisibility by the full prime power p^{h_p} . Multiplying these pairwise coprime divisibilities proves the first assertion. This argument never passes to a field root or divides by a leading coefficient.

The Sylvester determinant has e shifted coefficient rows of F and d of G . Their Euclidean lengths are respectively $\|F\|_2, \|G\|_2$. Hadamard's inequality gives

$$|\text{Res}(F, G)| \leq \|F\|_2^e \|G\|_2^d.$$

Taking logarithms of the nonzero integer multiple proves the second assertion, including all prime-power depths. $\square$

For a nonzero constant $G = c$, the direct identity $\text{Res}(F, c) = c^d$ covers the degree-zero case. Each packet modulus must then divide c ; it is no useful short certificate unless the modulus is already small. Theorem 285.5 genuinely allows different primes to have different owners. It requires one coprime second polynomial, rather than a complete lattice basis, but its saving depends on constructing that polynomial with suitable height.

Proposition 285.6 (Automatic owner polynomial and the exact correction at three). *First take a nonempty full assigned T -packet whose primes are different from three, and let $\mathcal{K}$ be its set of distinct owners, $s = |\mathcal{K}|$. Then*

$$G_{\mathcal{K}}(X) = \prod_{k \in \mathcal{K}} (X - k)$$

is a coprime second certificate for F_n , with

$$(1065) \quad |\text{Res}(F_n, G_{\mathcal{K}})| = \prod_{k \in \mathcal{K}} F_n(k),$$

$$(1066) \quad \|G_{\mathcal{K}}\|_2 \leq \|G_{\mathcal{K}}\|_1 = \prod_{k \in \mathcal{K}} (1 + k) \leq (2B)^s.$$

Thus its Sylvester estimate gives only

$$\log M \leq s \log \|F_n\|_2 + (3n - 1)s \log(2B).$$

For an arbitrary nonempty full T -packet including three, let k_3 be its assigned owner. Its exact depth is

$$(1067) \quad e_3 = 1 + v_3(k_3).$$

Form the adjusted F_n -packet with $h_p = e_p$ for $p \neq 3$ and $h_3 = e_3 - 1$, omitting three when $h_3 = 0$. Its product M_F and the original product M_T satisfy

$$(1068) \quad M_T = 3M_F.$$

The same owner polynomial is a certificate for the adjusted packet, even when it is empty, and the logarithmic bound gains exactly one $\log 3$ for M_T . For the respective positive-depth signed supports,

$$(1069) \quad J_T = \begin{cases} J_F - 2 \log 3, & e_3 = 1, \\ J_F + \log 3, & e_3 \geq 2. \end{cases}$$

Proof. At every owner $F_n(k) = T_k/3 > 0$; hence none is a root of F_n , so the owner polynomial is coprime to it. It vanishes as an integer at every owner, proving all adjusted congruences. The product formula for the resultant with its monic linear factors proves (1065). Since every $k > 0$, the absolute elementary symmetric coefficients sum to $\prod(1+k)$; this proves (1066). Theorem 285.5 gives the displayed bound.

Division by three is the reason the first assertion excluded that prime. For a general block owner, substitute $a = 3k_3$ into (1060). The linear term has valuation $1 + v_3(k_3)$, because $3 \nmid n$; every later term has valuation at least $2 + 2v_3(k_3)$. This proves (1067), with no assumption that k_3 is a three-adic unit. Division of the value by three reduces this depth by exactly one, proving (1068).

If $e_3 = 1$, then $\text{rad } M_T = 3 \text{rad } M_F$; if $e_3 \geq 2$ their radicals are equal. Subtracting three times their radical logs from $\log M_T = \log M_F + \log 3$ proves (1069). Thus even the depth-one and depth-two negative contributions are translated exactly. $\square$

Corollary 285.7 (Signed degree-height test with the original radical). *Let $G \in \mathbb{Z}[X] \setminus \{0\}$ be coprime to F_n over $\mathbb{Q}$ and satisfy the adjusted F_n -packet congruences. For the adjusted packet,*

$$J_F \leq \deg(G) \log \|F_n\|_2 + (3n - 1) \log \|G\|_2 - 3 \log \text{rad } M_F.$$

Equivalently, for its original full T -packet,

$$(1070) \quad J_T \leq \deg(G) \log \|F_n\|_2 + (3n - 1) \log \|G\|_2 + \delta_3 \log 3 - 3 \log \text{rad } M_T,$$

where δ_3 is one if three is present and zero otherwise.

Proof. For positive degree apply Theorem 285.5 to the adjusted product; for constant $G = c \neq 0$, use the preceding direct case $\text{Res}(F_n, c) = c^{3n-1}$. Then use $J = \log M - 3 \log \text{rad } M$. If three is present, use (1068); otherwise the products coincide. This gives (1070) without discarding a negative term. $\square$

What the automatic certificate does and does not prove. Equation (1065) recovers the bound obtained by assigning each prime power to its owner and multiplying the owner values. In the actual two-owner example it is $(X - B)(X - k_1)$, with resultant $F_n(B)F_n(k_1)$. This pays the old owner height and is consistent with Theorem 285.4. It does not itself supply the negative prime-depth compensation needed for a signed saving.

A genuine new gain requires a coprime certificate whose

$$\deg(G) \log \|F_n\|_2 + (3n - 1) \log \|G\|_2$$

improves the relevant full signed budget. A small lattice vector may simply be a multiple of F_n , while the automatic product of owner factors gives (1065). None of the results cited here constructs the required shorter coprime certificate on the original independent domain, and no dependence on the packet modulus is hidden in its claimed height.

Ordinary and finite scope. The arguments in this section are complete ordinary proofs. The separate exact coefficient-packet replay checks three prime indices 331, 337, 347, six actual powers, coefficient content and degree, independent polynomial evaluation, both exact marked depths and CRT indices, and the automatic resultant divisibility. It neither fully factors the first-owner boundary nor computes its full signed ledger, and finite cases are not the proof of the asymptotic obstruction. No Lean formalization of this resultant and lattice chain is claimed here.

The positive fixed-exponent selector families of Sections 283 and 284 are separate. These results do not exclude useful small intersections or other nonlinear maps, do not control the original far signed tail, and do not close its exceptional set, the dependent-root complement, or the reduction from this special block to every primitive ABC triple.

286. AN EXACT RATIONAL EXPANSION OF THE FIXED HEIGHT FUNCTION

Retain the character, twice-normalized local heights and minimal models of Sections 277 and 278:

$$(1071) \quad C : W^2 = z^6 - 27z^4 + 99z^2 - 9,$$

$$(1072) \quad E_1 : y^2 = x^3 - 9x - 9, \quad P_1 = (-2, 1),$$

$$(1073) \quad E_2 : y^2 = x^3 - 189x + 999, \quad P_2 = (6, 9),$$

$$(1074) \quad R_1 = \left(\frac{z^2 - 9}{4}, \frac{W}{8} \right), \quad R_2 = \left(\frac{33 - 9/z^2}{4}, -\frac{9W}{8z^3} \right).$$

The maps are interpreted projectively at their exceptional fibers. Write $\omega = dx/(2y)$, $t = -x/y$, and $L = \log_E$. The unit-root complementary line is spanned by $(x + c)\omega$, with $c \in \mathbb{Z}_5$. The line spanned by $x\omega$ is also an allowed isotropic complement. Subscripts c and 0 denote their local heights λ , global diagonal heights H , and sigma functions. The local principal term is $-2\log_5 t$.

Proposition 286.1 (Cancellation of the complementary line). *For every nonzero local point, and for the infinite-order rational points P_i above,*

$$(1075) \quad \lambda_c(Q) = \lambda_0(Q) + cL(Q)^2,$$

$$(1076) \quad H_c(P_i) = H_0(P_i) + cL(P_i)^2, \quad \alpha_c = \alpha_0 + c,$$

$$(1077) \quad \lambda_c(Q) - \alpha_c L(Q)^2 = \lambda_0(Q) - \alpha_0 L(Q)^2,$$

where $\alpha_j = H_j(P_i)/L(P_i)^2$. In particular the fixed rank-one expression does not require computing either unit-root slope.

Proof. The defining sigma differential equation gives, on the formal group,

$$\sigma_c(t) = \sigma_0(t) \exp(-cL(t)^2/2).$$

Taking $-2\log_5$ proves the first formula there. The normalized multiplication law is

$$(1078) \quad \lambda_j([n]Q) = n^2 \lambda_j(Q) - 2\log_5 \psi_n(Q), \quad j \in \{0, c\},$$

for positive integers n , away initially from the displayed poles. Here ψ_n is the usual division polynomial. With our parameter, its leading term is

$$\psi_n(t) = (-1)^{n-1} n t^{1-n^2} (1 + O(t));$$

in particular $\psi_2 = 2y \sim -2t^{-3}$. The sign has logarithm zero.

For completeness, the addition identity of [144, Lemma 7.4] yields (1078) with the standard division-polynomial recurrences. Its divisor differences must first be identified with group differences: translation preserves ω , acts trivially on H^1 , preserves the selected complementary line, and transports the invariant tangent vectors. Functoriality of the local height then identifies the height of $(Q_1) - (Q_2)$ with $\lambda_j(Q_1 - Q_2)$. The recurrence step uses

$$x([n]Q) - x(Q) = -\frac{\psi_{n+1}(Q)\psi_{n-1}(Q)}{\psi_n(Q)^2}.$$

Substitution in the addition identity gives the $n + 1$ law from the $n, n - 1$ laws. The tangent limit supplies the doubling case; the leading term at O fixes the constant. These conventions agree with [149, §5.5 and Theorem 5.21]. Continuity deals with removable exceptional charts.

Both reductions have nine points over $\mathbb{F}_5$. Hence $[9]Q$ lies in the formal group for every $Q \in E_i(\mathbb{Q}_5)$. Subtract the two versions of (1078) with $n = 9$ and divide by 81. This proves the first formula off the nonzero nine-torsion points, and continuity extends it across them. The heights at the other places do not change. Summing on a rational point gives the global formula. The already established logarithms of P_i are nonzero, so division gives the remaining identities. No assertion that P_i generates the global group is used. $\square$

Proposition 286.2 (Rational recursions and the entire omitted tail). *For either integral pair $(a, b) = (-9, -9), (-189, 999)$, let*

$$(1079) \quad w = t^3 + atw^2 + bw^3, \quad w \in t^3\mathbb{Z}[[t]],$$

$$(1080) \quad x = t/w, \quad A(t) = \frac{tw'(t)/w(t) - 1}{2},$$

$$(1081) \quad L(t) = \int A(t) dt, \quad R_0(t) = - \int \left[\frac{1}{t} + A(t) \int x(t)A(t) dt \right] dt.$$

All integrals have constant zero, including the Laurent primitive. These are the actual invariant logarithm and $R_0 = \log(\sigma_0/t)$. They satisfy

$$L \in t\mathbb{Q}[[t^2]], \quad L = t + O(t^5), \quad R_0 \in t^4\mathbb{Q}[[t^2]].$$

Writing $L = \sum l_j t^j$ and $R_0 = \sum r_j t^j$, for $j \geq 1$,

$$(1082) \quad v_5(l_j) \geq -\lfloor \log_5 j \rfloor, \quad v_5(r_j) \geq -2\lfloor \log_5 j \rfloor.$$

If L_N, R_N retain degrees at most the integer $N \geq 1$, then, uniformly for $t \in 5\mathbb{Z}_5$, both omitted tails have valuation at least $K = \lceil (N+1)/2 \rceil$.

Proof. Coefficient induction gives the unique integral solution for w : each substitution strictly raises the order of an unknown correction. It is the formal coordinate $-1/y$. Direct differentiation gives $\omega = A(t)dt$, $A = 1 + O(t^4)$, and $xA = t^{-2} + O(t^2)$. Thus the apparent $1/t$ terms cancel. Integrate the defining equation

$$-d \left(\frac{d \log \sigma_0}{\omega} \right) = x\omega.$$

Oddness fixes the inner integration constant; removing $\log t$ gives precisely (1081), including the stated parity.

The invariant differential is integral at five, so its termwise primitive gives the bound for l_j . By [145, Theorem 1.3], the canonical odd sigma has $\sigma_c/t \in 1 + t^2\mathbb{Z}_5[[t]]$, and $c \in \mathbb{Z}_5$. Proposition 286.1 gives

$$R_0 = \log(\sigma_c/t) + (c/2)L^2.$$

In degree j , the logarithm denominators are integers at most $j/2$; this part has valuation at least $-\lfloor \log_5 j \rfloor$. Every product contributing to L^2 has valuation at least $-2\lfloor \log_5 j \rfloor$. This proves (1082), using only the existence and integrality of the canonical sigma, not its computed coefficients.

For $k = \lfloor \log_5 j \rfloor \geq 1$, induction gives $j \geq 5^k \geq 4k$. Thus $j - 2\lfloor \log_5 j \rfloor \geq \lceil j/2 \rceil$; the case $k = 0$ is immediate. Every omitted evaluated term has valuation at least K , tending to infinity with j . Completeness and the ultrametric inequality bound the full tails. The bound for L is stronger still. $\square$

Proposition 286.3 (A rational-function and rational-series formula). *Set, as rational functions after cancellation,*

$$(1083) \quad \delta_i(Q) = \frac{t([9]Q)}{\psi_9(Q)}, \quad T_i = t([9]R_i), \quad \Xi = z^{81} \frac{\delta_1(R_1)}{\delta_2(R_2)}.$$

Each T_i is analytic on $C(\mathbb{Q}_5)$ with values in $5\mathbb{Z}_5$, and Ξ is finite and nonzero there. With $\alpha_i^0 = H_0(P_i)/L(P_i)^2$, the extended fixed height function is

$$(1084) \quad F_0 = -\frac{2}{81}(\log_5 \Xi + R_{01}(T_1) - R_{02}(T_2)) - \alpha_1^0 \left(\frac{L_1(T_1)}{9} \right)^2 + \alpha_2^0 \left(\frac{L_2(T_2)}{9} \right)^2.$$

It equals the canonical-splitting expression of Section 278. We shall test $f = F_0 - \Omega$, where $\Omega = -\frac{4}{3} \log_5 3$.

Proof. Near O , the leading terms give $\delta_i = t^{81}(1 + O(t))$. At a nonzero nine-torsion point, $t([9]Q)$ and $\psi_9(Q)$ both vanish simply, since $[9]$ is étale in characteristic zero; cancellation gives a nonzero value. At any other nonzero local point, $[9]Q$ is a nonzero formal point, where t is finite and nonzero. Thus δ_i has no other local zero or pole.

Away from zero and infinity, neither R_i is O . At $z = 0$, with $W_0^2 = -9$, $t(R_2)/z \rightarrow -2/W_0$, so the order 81 in δ_2 cancels z^{81} . At infinity put $q = 1/z$ and $W/z^3 \rightarrow \epsilon \in \{1, -1\}$. Then $t(R_1)/q \rightarrow -2/\epsilon$, cancelling the other pole. The second quotient is finite in each respective exceptional case. This proves all asserted local extensions, without assuming that the zero-fiber points are rational over $\mathbb{Q}$.

Off these fibers, (1078) gives

$$\lambda_0(Q) = -\frac{2}{81}(\log_5 \delta_i(Q) + R_{0i}(t([9]Q))).$$

Subtract the two heights and $2 \log_5 z$, and use $L(R_i) = L_i(T_i)/9$, to obtain (1084). The preceding cancellations extend it across the exceptional fibers; (1077) identifies the two splittings. $\square$

Exact constants and precision. Write the already established ninefold rational points as $9P_i = (A_i/d_i^2, B_i/d_i^3)$, and set $t_i = t(9P_i)$, $q_i = -A_i/B_i = t_i/d_i$. The checked identity-component hypotheses from Section 277 give

$$(1085) \quad H_0(P_i) = -\frac{2}{81}(\log_5 q_i + R_{0i}(t_i)), \quad L(P_i) = L_i(t_i)/9.$$

This follows either by changing the splitting in the canonical global formula or by its sigma definition. Each q_i is a rational five-adic unit, whose logarithm is $\frac{1}{4} \log(q_i^4)$. The exact leading-digit calculation gives $v_5 H_0(P_i) = v_5 L(P_i) = 1$, hence $v_5 \alpha_i^0 = -1$. Changing the slope adds $c_i L(P_i)^2 \in 25\mathbb{Z}_5$, so it does not alter the height's first nonzero digit.

For the integer $K = \lceil (N+1)/2 \rceil$, truncating the two R_0, L series at degree N in (1084) changes their combined contribution by an error in $5^K \mathbb{Z}_5$, if the constants are exact. Indeed each logarithm and its truncation has valuation at least one; the squaring error has valuation at least $K+1$, and multiplication by α_i^0 loses one digit. An absolute error at least $K-2$ in α_i^0 also suffices, since its multiplier is a square in $25\mathbb{Z}_5$. Evaluation errors in the rational functions and in $\log_5 \Xi$ must be bounded separately.

The rational recurrence replay through degree forty, with (1085), yields the following exact residues:

	$H_0(P_i) \bmod 5^8$	$L(P_i)/5 \bmod 5^8$	$5\alpha_i^0 \bmod 5^8$
$i = 1$	245730	36459	352766
$i = 2$	96385	378472	277078

The full-tail bounds are twenty-one absolute digits for the series and nineteen for α_i^0 ; thus the stated residues are within the proved precision. The smaller residues needed below are

$$(1086) \quad \begin{array}{c|ccc} & H_0(P_i) \bmod 125 & L(P_i)/5 \bmod 25 & 5\alpha_i^0 \bmod 25 \\ \hline i = 1 & 105 & 9 & 16 \\ i = 2 & 10 & 22 & 3 \end{array}$$

All local charts and verification scope. The sextic is 1 at $z = 0$ modulo five and 4 at $z = \pm 1, \pm 2$. Each of the five residues has two nonzero simple ordinates, and infinity has two points. Thus all local points lie in exactly twelve smooth residue disks. Their lifts may be based at

$$(0, \pm\sqrt{-9}), \quad (\pm 1, \pm 8), \quad (\pm 2, \pm\sqrt{19}), \quad \infty_{\pm}.$$

The finite charts use $z - a \in 5\mathbb{Z}_5$; infinity uses $1/z \in 5\mathbb{Z}_5$. The derivative $2W$ being a unit gives each unique analytic square-root branch. This is an inventory of local disks, not of rational points.

The exact rational recurrences and residues are replayed by `next_replay_zero_slope.py`; the canonical result and exact source hashes are listed in the accompanying geometry inventory. The entire-tail estimates, splitting comparison and projective extensions above are ordinary proofs using the cited primary inputs. They are not assertions of Lean formalization or consequences of finite coefficient inspection. The following disk certificates and the separate rational sieve use

these estimates to complete the fixed-curve calculation; none supplies a uniform height bound for varying exponents or residuals.

287. RATIONAL DESCENT AND FOUR RESIDUE-DISK ORBITS

This section concerns the single smooth projective curve

$$C : W^2 = z^6 - 27z^4 + 99z^2 - 9$$

and the two minimal elliptic models and actual quotient maps

$$\begin{aligned} E_1 : y^2 &= x^3 - 9x - 9, & R_1 &= ((z^2 - 9)/4, W/8), \\ E_2 : y^2 &= x^3 - 189x + 999, & R_2 &= ((33 - 9/z^2)/4, -9W/(8z^3)). \end{aligned}$$

The notation $F_0, T_i, \delta_i, \Xi, \alpha_i^0, R_{0i}, L_i$ is that of the preceding zero-slope expansion. Put $f = F_0 + (4/3)\log_5 3$. All statements here are ordinary proofs, with the established removable extensions on $C(\mathbb{Q}_5)$.

RD1: an actual multiplication formula. On either short model put $\psi = \psi_9$, $\phi = x\psi_9^2 - \psi_{10}\psi_8$, and $D = \phi'\psi - 2\phi\psi'$. These are integral polynomials in x : the two even-index factors contribute $(2y)^2 = 4(x^3 + ax + b)$. Then the rational-function identities are

$$x([9]Q) = \frac{\phi}{\psi^2}, \quad y([9]Q) = \frac{yD}{9\psi^3}, \quad T(Q) = -\frac{9\phi\psi}{yD}, \quad \delta(Q) = -\frac{9\phi}{yD} = -\frac{\phi}{\omega_9}.$$

Indeed the first is the division-polynomial multiplication formula [149]. If $X = \phi/\psi^2$, the identity $[9]^*(dx/(2y)) = 9 dx/(2y)$ gives $X'(x)/(2y([9]Q)) = 9/(2y)$ and $X' = D/\psi^3$. This fixes both the factor nine and the ordinate sign. The remaining identities follow from $t = -x/y$. They extend as rational identities; $D \neq 0$ because X is nonconstant in characteristic zero.

RD2: descent to $u = z^2$. Write $S(u) = u^3 - 27u^2 + 99u - 9$, $x_1 = (u - 9)/4$, $x_2 = (33 - 9/u)/4$. Define rational functions over $\mathbb{Q}$ by

$$\begin{aligned} \mathcal{X}(u) &= -9u^{39} \frac{\phi_1(x_1)D_2(x_2)}{\phi_2(x_2)D_1(x_1)}, \\ V_1(u) &= \frac{5184\phi_1(x_1)^2\psi_1(x_1)^2}{S(u)D_1(x_1)^2}, \\ V_2(u) &= \frac{64u^3\phi_2(x_2)^2\psi_2(x_2)^2}{S(u)D_2(x_2)^2}. \end{aligned}$$

For actual local points $\Xi = \mathcal{X}(z^2)$ and $T_i^2 = V_i(z^2)$. To check the constants, $y_2/y_1 = -9/z^3$ makes the delta ratio times z^{81} equal to the first expression, while $y_1^2 = S/64$, $y_2^2 = 81S/(64u^3)$ give the other two. The identities on the common open chart extend to the previously proved removable values. Thus $\mathcal{X}$ is finite and nonzero and $V_i \in 25\mathbb{Z}_5$ on the image of the actual local curve. These assertions are not made for arbitrary $u \in \mathbb{Q}_5$.

RD3: a square-root-free expression. There are unique rational series $\widehat{R}_i, \widehat{M}_i$ with $\widehat{R}_i(t^2) = R_{0i}(t)$ and $\widehat{M}_i(t^2) = L_i(t)^2$. On the actual curve image,

$$\begin{aligned} F_0 &= -\frac{2}{81} \left(\log_5 \mathcal{X}(u) + \widehat{R}_1(V_1(u)) - \widehat{R}_2(V_2(u)) \right) \\ &\quad - \frac{\alpha_1^0}{81} \widehat{M}_1(V_1(u)) + \frac{\alpha_2^0}{81} \widehat{M}_2(V_2(u)). \end{aligned}$$

This follows by regrouping the convergent even series and the convergent Cauchy product L_i^2 . The proved tail bounds apply at the original t -degrees. For finite nonzero residue classes, $2z$ is a unit and u is a local coordinate. At zero it has ramification order two; at infinity $1/u = q^2$ has order two. Algebraic cancellation alone does not supply analytic denominator or logarithm error bounds.

DS1: exact symmetry. The commuting involutions $\iota(z, W) = (z, -W)$ and $\sigma(z, W) = (-z, W)$ generate a group of order four preserving f . Indeed

$$R_1\iota = -R_1, \quad R_2\iota = -R_2, \quad R_1\sigma = R_1, \quad R_2\sigma = -R_2.$$

Since ψ_9 depends only on x , $\delta(-Q) = -\delta(Q)$. The two delta signs cancel under ι ; under σ the signs of z^{81} and δ_2 cancel. Thus Ξ is invariant. The remaining terms are unchanged because R_{0i} is even and L_i^2 is even. Analytic continuation proves invariance at the exceptional fibers too.

DS2: complete coverage. There are twelve smooth residue disks and exactly four orbits: two over $z = 0$, four over $z = \pm 1$, four over $z = \pm 2$, and two at infinity. Modulo five the sextic takes value one at zero and four at every nonzero residue, with nonzero ordinates. The monic leading coefficient supplies the two infinities. The involutions give precisely the asserted orbits. Representatives can be chosen through $(0, \sqrt{-9})$, $(1, 8)$, $(2, \sqrt{19})$, and $W/z^3 = 1$ at infinity. Their finite parameter $z - a$ or infinity parameter $q = 1/z$ ranges over $5\mathbb{Z}_5$. Transport sends the parameter to itself or its negative, so is an analytic isomorphism and preserves each zero multiplicity. A complete certificate on each representative therefore covers all disks.

DS3: even orders at the fixed fibers. At zero the unique Hensel branch satisfies $W(-z) = W(z)$, and σ preserves the representative disk. At infinity let $v = Wq^3$; its equation is $v^2 = 1 - 27q^2 + 99q^4 - 9q^6$. The composite $\sigma\iota$ preserves $v(0) = 1$ and sends q to $-q$. Hence f is exactly even in both parameters. Its central derivative vanishes. If its central value is zero and the series is nonzero, its order there is positive and even. The following disk certificates determine those values and orders.

288. COMPLETE CERTIFICATES ON THE TEN FINITE RESIDUE DISKS

Use the curves, exact constants $\alpha_i = \alpha_i^0$, and $f = F_0 + \frac{4}{3}\log_5 3$ of Section 286. For a disk coordinate s , write $\mathcal{A} = \mathbb{Z}_5\langle s \rangle$, the ring of restricted series, with Gauss valuation $v_{\mathcal{A}}(\sum c_j s^j) = \inf_j v_5(c_j)$. Its coefficients tend to zero, and its derivative belongs to the same ring. We prove congruences in this complete ring; evaluations at a few points do not replace the entire-series assertions. The involutions and four disk orbits are established in Section 287.

288.1. The representative disk above one. Take

$$(1087) \quad z = 1 + 5s, \quad W(1) = 8, \quad s \in \mathbb{Z}_5.$$

The square-root branch is the unique one with this residue sign. For either short elliptic model put

$$(1088) \quad \psi = \psi_9, \quad \phi = x\psi_9^2 - \psi_{10}\psi_8,$$

$$(1089) \quad \omega_9 = \frac{\psi_{11}\psi_8^2 - \psi_7\psi_{10}^2}{4y}, \quad \delta = -\phi/\omega_9, \quad T = -\phi\psi/\omega_9.$$

These are the normalized division polynomials: $[9](x, y) = (\phi/\psi^2, \omega_9/\psi^3)$. Thus $\delta = t([9]Q)/\psi_9(Q)$ and $T = t([9]Q)$, with no remaining division by ψ_9 at nine-torsion.

Lemma 288.1 (Integral functions and complete tails on the first disk). *On (1087), $\phi_i, \omega_{9,i}, \delta_i, \Xi$ are units of $\mathcal{A}$, $\psi_{9,i}, T_i \in 5\mathcal{A}$, and $g = f/5 \in \mathcal{A}$. If $u = z - 1$ and $\beta = (d \log \Xi/du)(0)$, then*

$$(1090) \quad \log_5 \Xi(s) - \log_5 \Xi(0) = 5\beta s \pmod{25\mathcal{A}}.$$

All series estimates remain valid after termwise differentiation in s .

Proof. The equation and $2W(0) = 16$ give $W(u) \in 8 + u\mathbb{Z}_5[[u]]$, with $W'(0) = 6$. Both quotient coordinates are integral series and both ordinates are units. The division recurrences divide only by $2y$ or $4y$, which are units. The complete calculation below gives $\psi_9 = 0, \phi = \omega_9 = 1$ modulo five at the center, for both quotients. An integral formal series with unit constant has an integral inverse. Substitution $u = 5s$ gives a restricted series whose degree- j coefficient is divisible by 5^j . It follows that ϕ, ω_9, δ are units on the entire disk and that $\psi_9, T \in 5\mathcal{A}$. Since z is also a unit, $\Xi = z^{81}\delta_1/\delta_2$ is a unit.

to infinity. The change in the squared-log terms, divided by five, is therefore

$$-(4 + 4s)^2 + 3(3 + s)^2 - (-4^2 + 3 \cdot 3^2) = s + 2s^2 \pmod{5}.$$

Finally $f(0) = 0$ exactly: $(1, 8)$ is a rational point in the finite chart, so the global-height identity and the full away-from-five set of Section 278 give $F_0(1, 8) = \Omega$. Adding the logarithmic slope proves (1091). $\square$

Theorem 288.3 (Two simple zeros on the first representative disk). *The disk (1087) has exactly two zeros of f , both simple: the point $(1, 8)$, and a unique point with $s = 4 \pmod{5}$, equivalently $z = 21, W = 3 \pmod{25}$. The four disks above $z = \pm 1 \pmod{5}$ have exactly eight simple zeros in total, four known rational points and four further local points in one symmetry orbit.*

Proof. The only roots of $2s(s + 1)$ in $\mathbb{F}_5$ are $0, 4$, with derivatives $2, 3$. For an integral restricted series the Taylor estimate $g(x + h) = g(x) + hg'(x) \pmod{h^2\mathbb{Z}_5}$ holds termwise and extends by completeness. Thus Newton iteration gives a root in each of these two residue disks. For two roots x, y in the same disk,

$$g(x) - g(y) = (x - y)(g'(y) + (x - y)H), \quad H \in \mathbb{Z}_5,$$

whose second factor is a unit; uniqueness follows. The other residues are excluded by (1091). The first root is the exact center. The derivatives at both roots are units for g , so $f' = 5g' \neq 0$, giving simplicity. The expansion $W = 8 + 30s \pmod{25}$ gives the second ordinate residue. The two involutions from Section 287 transport these roots and their multiplicities to the other three disks. This argument by itself makes no rationality assertion about the second orbit. $\square$

288.2. The representative disk above two. Take

$$(1099) \quad z = 2 + 5s, \quad W(2) = \sqrt{19} \equiv 2 \pmod{5}.$$

Here $u = z - 2$ again gives integral quotient coordinates with unit ordinates. The center reductions are $\psi_{9,i} = 0$, $(\phi_1, \phi_2) = (4, 1)$, $(\omega_{9,1}, \omega_{9,2}) = (3, 4)$ modulo five. Consequently the whole unit-series argument of Lemma 288.1 applies: $T_i \in 5\mathcal{A}$, $\Xi \in \mathcal{A}^*$, and $g = f/5 \in \mathcal{A}$, with the same derivative and tail control. An integral series with unit constant is being inverted, so these assertions cover the disk, not just its center.

Lemma 288.4 (The first reduction above two). *On (1099),*

$$(1100) \quad g(s) = 4(s + 1)^2 \pmod{5\mathcal{A}}.$$

Proof. The exact lift is $W(2) = 12 \pmod{25}$, with $W' = -138/W = 1 \pmod{25}$. The recurrences (1096) give the following constant residues and dual values; the latter use $z = 2 + e, W = 2 + e$.

mod25	E_1	E_2	(value, derivative) mod 5	E_1	E_2
ψ_9	15	15	ϕ	(4, 0)	(1, 3)
ϕ	24	1	ω_9	(3, 0)	(4, 3)
ω_9	18	24	δ	(2, 0)	(1, 1)
δ	7	1	$(\log \delta)'$	0	1
T	5	15			

Hence $\beta = 81/2 + 0 - 1 = 2 \pmod{5}$, and the logarithmic local-height slope in g is one. The center calculation also gives

$$\Xi = 14, \quad \log_5 \Xi = 10, \quad \log_5 3 = 20 \pmod{25}, \quad \ell_1/5 = 4, \quad \ell_2/5 = 2 \pmod{5}.$$

The differential values in (1097) are $2, 4$ modulo five. The full integrated-tail estimate used in (1098) therefore gives

$$\ell_1/5 = 4 + 2s, \quad \ell_2/5 = 2 + 4s \pmod{5\mathcal{A}}.$$

With $5\alpha_i = (1, 3)$ and $\Omega/5 = 3$ modulo five, substitution in (1084) yields constant 4, linear coefficient 3, and quadratic coefficient 4. This is (1100). The center has not been assumed rational or an exact zero. $\square$

Lemma 288.5 (Sharper tails at precision one hundred twenty-five). *For either curve and any $T \in 5\mathcal{A}$,*

$$R_0(T) \in 5^4\mathcal{A}, \quad L(T) - T \in 5^4\mathcal{A}.$$

Replacing $L(T)/9$ by $T/9$ in a squared-log term multiplied by α_i makes an error in $5^4\mathcal{A}$. For a five-adic unit q , with $h = q^4 - 1$,

$$(1101) \quad \log_5 q = \frac{1}{4}(h - h^2/2) \pmod{125}.$$

Proof. The even series R_0 starts at degree four. The valuation bound for degree four is four; degree five is absent; degrees six and seven have bounds four and five; all degrees at least eight have bound $\lceil j/2 \rceil \geq 4$. For $L - T$, only degrees $j \geq 5$ occur, and $j - \lfloor \log_5 j \rfloor \geq 4$. Both bounds tend to infinity, proving the claims for the entire compositions. The difference of the two squares is in $5^5\mathcal{A}$, and $v_5\alpha_i = -1$. Finally, for every $j \geq 3$, the omitted term h^j/j has valuation at least $j - v_5(j) \geq 3$, tending to infinity. Changing the unit representative modulo 125 also changes its logarithm only by a multiple of 125. This proves the exact logarithm congruence. $\square$

Theorem 288.6 (No zero above two). *There is no zero of f in (1099), or in any of the four disks above $z = \pm 2 \pmod{5}$.*

Proof. The only possible root residue in (1100) is $s = 4$. At its exact center $z = 22$, the selected branch has $W = 32 \pmod{125}$. This satisfies the sextic congruence with unit derivative. Unit denominators in (1089) therefore make evaluation at this residue exact modulo 125. The same recurrences give

$$\Xi = 24, \quad T_1 = 90, \quad T_2 = 10, \quad \log_5 \Xi = 100, \quad \log_5 3 = 95 \pmod{125}.$$

The logarithms have the full-tail justification (1101). Lemma 288.5 allows omission of the R_0 terms and replacement of ℓ_i by $T_i/9$ at this precision. Thus $\ell_1/5 = 2, \ell_2/5 = 3 \pmod{25}$. The exact constants (1086) give $5\alpha_1 = 16, 5\alpha_2 = 3 \pmod{25}$; equivalently these follow from $21/9^2 = 16, 2/22^2 = 3$ modulo 25. All used digits lie within the previously proved nineteen-digit absolute error for α_i . Substitution now gives

$$(1102) \quad g(4) = -\frac{2}{81} 20 - 16 \cdot 2^2 + 3 \cdot 3^2 + \frac{4}{3} 19 = 15 \pmod{25}.$$

Every denominator is a five-adic unit. The positive $4/3$ sign comes from $f = F_0 - \Omega$.

Since $g \in \mathcal{A}$ and $g'(4) = 0 \pmod{5}$, its entire Taylor expansion gives

$$g(4 + 5t) = g(4) + 5tg'(4) = 15 \pmod{25\mathbb{Z}_5\langle t \rangle}.$$

Every higher term is a multiple of 25. Completeness justifies this expansion for the full restricted series and every $t \in \mathbb{Z}_5$. It excludes the only possible root residue; the other residues were already excluded by (1100). The involutions transport this exclusion to the three companion disks. $\square$

288.3. The zero fiber and its removable pole. Take

$$(1103) \quad z = 5s, \quad W(0) = W_0 = \sqrt{-9} \equiv 1 \pmod{5}.$$

Its center is not being assumed rational or a zero of the height function.

Lemma 288.7 (An integral even removable factor). *On (1103), Ξ extends to an integral even unit, $T_i \in 5\mathcal{A}$, and $g = f/5 \in \mathcal{A}$ is even. Its central value is obtained from*

$$(1104) \quad \Xi(0) = \delta_1(-9/4, W_0/8)(-W_0/2)^{81}.$$

Proof. For a good integral short model at five the integral formal group law and the leading coefficient of the odd division polynomial give

$$\frac{t([9]Q)}{t} = 9 + t\mathbb{Z}_5[[t]], \quad t^{80}\psi_9(Q) = 9 + t\mathbb{Z}_5[[t]].$$

Indeed ψ_9 is an integral polynomial in x of degree forty with leading coefficient nine, and $x = t^{-2}(1 + t\mathbb{Z}_5[[t]])$. Both constants are units at five. Hence

$$(1105) \quad d(t) := \frac{\delta(Q)}{t^{81}} = \frac{t([9]Q)/t}{t^{80}\psi_9(Q)} \in 1 + t\mathbb{Z}_5[[t]].$$

This series is even, since δ and t both change sign under negation. This step uses the rational formal group, not an unproved integrality assertion for zero-slope sigma.

The exact original parameter of the second quotient is

$$\tau(z) = t(R_2) = \frac{2z(11z^2 - 3)}{3W(z)}.$$

The ordinate is an even integral series with unit constant. Thus τ is odd and τ/z is a unit with constant $-2/W_0$. The second quotient lies in the formal group throughout the disk, including O at its center. Consequently

$$\Xi(z) = \left(\frac{z}{\tau(z)}\right)^{81} \frac{\delta_1(R_1(z))}{d_2(\tau(z))}.$$

For the first quotient the center has $\psi_9 = 0, \phi = 4, \omega_9 = 2 \pmod{5}$. Its integral series therefore gives δ_1 a unit and $T_1 \in 5\mathcal{A}$, as in Lemma 288.1. The second has $\tau \in 5\mathcal{A}$, so formal multiplication gives $T_2 \in 5\mathcal{A}$. The displayed formula for Ξ is an even integral unit and gives (1104).

All composition bounds of Lemma 288.1 now apply to both quotients, giving $g \in \mathcal{A}$. Moreover $\Xi(z)/\Xi(0) \in 1 + z^2\mathbb{Z}_5[[z]]$, so after $z = 5s$ its logarithmic change belongs to $25s^2\mathcal{A}$. The logarithm converges in the Gauss norm with input in $25\mathcal{A}$. The R_0 terms have their established full tails. Finally, negating z fixes R_1 and negates R_2 ; the height and squared logarithms are unchanged. Thus g is even. $\square$

Theorem 288.8 (No zero in either zero-fiber disk). *On (1103),*

$$(1106) \quad g(s) = 2s^2 \pmod{5\mathcal{A}}, \quad g(0) = 20 \pmod{25}.$$

Consequently neither zero-fiber disk contains a zero of f .

Proof. The chosen central square root has Hensel residues $W_0 = 21 \pmod{25}$, $W_0 = 46 \pmod{125}$, with unit derivative. Evaluate $(-9/4, W_0/8)$ by (1096) and the cancelled formula (1104). The exact finite inputs are

	mod25	mod125
$\Xi(0)$	16	66
$\log_5 \Xi(0)$	15	15
$T_1(0)$	20	95
$\log_5 3$	20	95

They give $\ell_1(0)/5 = 1 \pmod{5}$ and $\ell_1(0)/5 = 16 \pmod{25}$. Every rational denominator is a unit. The logarithms use (1101); the entire error estimates of Lemma 288.5 permit omission of R_0 and replacement of ℓ_1 by $T_1/9$ at precision 125. At the center $R_2 = O$, so $T_2 = \ell_2 = 0$ exactly. With (1086), first precision gives

$$g(0) = -\frac{2}{81} 3 - 1^2 + \frac{4}{3} 4 = 0 \pmod{5}.$$

The pullback differentials (1097), integrated with their full coefficient bounds, give

$$\ell_1/5 = 1, \quad \ell_2/5 = (-2/W_0)s = 3s \pmod{5\mathcal{A}}.$$

The first differential has no constant term in z ; all terms after integration and substitution, apart from the stated linear term of the second logarithm, lie in $25\mathcal{A}$. The Ξ variation and R_0 tails contribute no nonconstant term modulo five to g , by Lemma 288.7. The squared-log contribution is therefore $3(3s)^2 = 2s^2$. This proves the first congruence in (1106).

At second precision the same table instead gives

$$g(0) = -\frac{2}{81} 3 - 16 \cdot 16^2 + \frac{4}{3} 19 = 20 \pmod{25}.$$

This is a direct value of the cancelled analytic formula; no rational-point identity has been applied to the algebraic center. Any zero would have $s \equiv 0 \pmod{5}$. Since g is even and integral restricted,

$$g(5t) = g(0) = 20 \pmod{25\mathbb{Z}_5\langle t \rangle}.$$

All nonconstant even terms are divisible by 25, and completeness makes this an entire-series congruence. It excludes the only possible root residue. The hyperelliptic involution gives the same exclusion on the second zero-fiber disk. $\square$

Finite evidence and proof boundary. The three exact input certificates are replayed, respectively, by `next_replay_unit_disk.py`, `next_replay_second_unit_disk.py`, and `next_replay_zero_fiber.py`. Their standard-library calculations include the normalized division recurrences, dual-number derivatives, Hensel square-root residues, exact unit logarithms and global constant residues. The canonical results, source hashes and check commands accompany the geometry inventory. The analytic proofs above separately establish the complete tails, unit denominators, derivative estimates, and exclusion or uniqueness on each whole disk. These are ordinary proofs with exact finite verification; no new Lean formalization of local heights or analytic zero counting is claimed. The separate infinity certificate supplies the last two disks, and the rational sieve decides the rationality of the additional first-disk orbit. Those subsequent conclusions do not change the individual local zero counts proved here.

289. THE TWO INFINITY DISKS AND THEIR RATIONAL DOUBLE ZEROS

We give the full ordinary certificate IF1–IF3, including the central height identity. Put $q = 1/z$, $v = Wq^3$, $v(0) = 1$. Then

$$\begin{aligned} v^2 &= 1 - 27q^2 + 99q^4 - 9q^6, \\ R_1 &= ((q^{-2} - 9)/4, v/(8q^3)), \quad R_2 = ((33 - 9q^2)/4, -9v/8). \end{aligned}$$

Set $q = 5s$ and $\mathcal{A} = \mathbb{Z}_5\langle s \rangle$, with its complete Gauss valuation.

IF1: all denominators and tails on the entire disk. The canceled expression $\Xi = q^{-81}\delta_1(R_1)/\delta_2(R_2)$ is an analytic unit, with

$$\Xi(q) = \Xi(0)(1 + q^2B(q^2)), \quad B \in \mathbb{Z}_5[[q^2]], \quad T_i \in 5\mathcal{A}.$$

Here is a formal-series proof. Hensel recursion gives $v \in 1 + q^2\mathbb{Z}_5[[q^2]]$ and $t_1 = -2q(1 - 9q^2)/v$. On E_1 , the zero-slope recursion gives $w = t^3U$, $U \in 1 + t^4\mathbb{Z}_5[[t^2]]$, $x = t^{-2}U^{-1}$ and $y = -t^{-3}U^{-1}$. The integral division polynomials have leading terms $\psi_9 = 9x^{40}$, $\phi_9 = x^{81}$ and $D = 9x^{120}$; see [149, Lemma 5.22] for the first two, and differentiate to obtain the last. Therefore

$$t^{80}\psi_9 \in 9 + t^2\mathbb{Z}_5[[t^2]], \quad t^{162}\phi_9 \in 1 + t^2\mathbb{Z}_5[[t^2]], \quad t^{240}D \in 9 + t^2\mathbb{Z}_5[[t^2]].$$

RD1, using the integral inverse of the last unit, yields

$$\delta_1(t) = t^{81}V(t^2), \quad V \in 1 + t^2\mathbb{Z}_5[[t^2]], \quad t([9]Q) \in t(9 + t^2\mathbb{Z}_5[[t^2]]).$$

Thus $q^{-81}\delta_1(R_1)$ is $(-2)^{81}$ times an even integral unit. The second quotient is an integral even point with center $(33/4, -9/8)$. Writing $e = q^2$, its actual first jets modulo five, and the full division recurrences applied to them, give

$$(x, y) = (2 + 4e, 2 + 3e), \quad \psi_9 = 3e, \quad \phi_9 = 4 + 3e, \quad \omega_9 = 2 + e \pmod{(5, e^2)}.$$

Consequently $\delta_2 = -\phi_9/\omega_9$ is an even integral unit, and $T_2 = -\phi_9\psi_9/\omega_9$ has constant in $5\mathbb{Z}_5$ and remaining terms in $q^2\mathbb{Z}_5[[q^2]]$. Substitution $q = 5s$ gives restricted series and proves the assertions. Moreover

$$\log_5 \Xi(q) - \log_5 \Xi(0) \in 25\mathcal{A},$$

because its j th logarithm term has valuation at least $2j - v_5(j)$, tending to infinity. This controls the entire tail.

IF2: the central value is exactly zero. The doubling law gives $2P_2 = (33/4, 9/8)$ for $P_2 = (6, 9)$, hence $R_2(0) = -2P_2$. As $q \rightarrow 0$, the singular terms combine as

$$\lambda_{01}(R_1) - 2\log_5 z = -2\log_5(t_1/q) - 2R_{01}(t_1) \rightarrow -2\log_5 2.$$

Also $\log_{E_1}(R_1) \rightarrow 0$. Evenness of local heights and the rank-one quadratic identity give

$$F_0(0) = -2\log_5 2 - \lambda_{02}(2P_2) + H_{02}(2P_2).$$

The height difference at this actual rational point can be computed at every place other than five. At two, $v_2(x) = -2$ and $3x^2 - 189$ has negative valuation, so Cremona case (a) in the previously fixed twice-Silverman normalization gives $2\log_5 2$. At three the exact tests are

$$v_3(x) = 1, \quad v_3(y) = 2, \quad A = 5, \quad B = 2, \quad C = 6, \quad v_3(c_4) = 4.$$

Thus case (c), $C \geq 3B$, gives $-(4/3)\log_5 3$. At all other primes different from five, the model has good reduction and the coordinates are integral, so the local height is zero. Changing the splitting affects only five. Therefore

$$H_{02}(2P_2) - \lambda_{02}(2P_2) = 2\log_5 2 - \frac{4}{3}\log_5 3,$$

which proves $F_0(0) = -(4/3)\log_5 3$ and $f(0) = 0$. No density of global rational points near infinity is assumed here.

IF3: an exact even factor. There is $U_\infty \in 1 + 5\mathcal{A}$ such that

$$f(5s) = 5s^2 U_\infty(s).$$

Indeed put $\ell_i = \log_{E_i}(R_i)$. The actual differential pullbacks give

$$\ell'_1 = -2/v, \quad \ell'_2 = 2q/v, \quad \ell_1(0) = 0, \quad \ell_2(0) = -2\log_{E_2}(P_2).$$

Integration of the integral even series $1/v$ and substitution $q = 5s$ give $\ell_1(5s)/5 \equiv -2s \pmod{5\mathcal{A}}$ and $\ell_2(5s) - \ell_2(0) \in 25\mathcal{A}$. At every degree j the integration denominator is controlled by $j - v_5(j) \rightarrow \infty$; the omitted first degrees are three and two respectively. ZS puts $R_{0i}(T_i)$ and their center differences in $25\mathcal{A}$. The logarithm change has this same bound by IF1. The second squared logarithm changes by $125\mathcal{A}$, since its center is in $5\mathbb{Z}_5$ and its change is in $25\mathcal{A}$; multiplication by α_2^0 of valuation -1 leaves $25\mathcal{A}$. Subtract the exact center value. The only surviving term in $g = f(5s)/5$ modulo five is

$$g \equiv -(5\alpha_1^0)(\ell_1(5s)/5)^2 \equiv -4s^2 = s^2 \pmod{5\mathcal{A}},$$

using the exact digit $5\alpha_1^0 \equiv 1$. DS proves exact evenness and IF2 proves zero constant term. Dividing the restricted series by s^2 therefore gives $U_\infty \in 1 + 5\mathcal{A}$. Its value is a unit at every $s \in \mathbb{Z}_5$. The only zero is the central rational infinity point and its multiplicity is exactly two. DS transports this statement to the other infinity disk.

The exact replay records the division jets, doubling coordinates and height valuation tests. The all-tail estimates, height identity and zero factorization above are ordinary proofs, not consequences of finite numerical root sampling or of a completed Lean height theory.

290. A FIVE-PRIMARY SIEVE AT THIRTEEN AND THE COMPLETE FIXED RATIONAL LOCUS

The ordinary arguments QL, HT, LH, ZS, DS, UD, SU, ZD and IF, followed by the exact sieve below, prove

$$C(\mathbb{Q}) = \{(1, 8), (1, -8), (-1, 8), (-1, -8), \infty_+, \infty_-\}.$$

Here C is the smooth projective completion of $W^2 = z^6 - 27z^4 + 99z^2 - 9$, with $W/z^3 = \pm 1$ at its two infinities. This is a theorem for one fixed curve. Its complete height and local analytic dependency chain has not been formalized in Lean.

SM1: transport through the global rational group. Let $P_1 = (-2, 1)$ on E_1 and $P_2 = (6, 9)$ on E_2 . QL2 proves that $E_i(\mathbb{Q})$ is torsion-free of rank one and $d_i = [E_i(\mathbb{Q}) : \mathbb{Z}P_i]$ is finite and prime to five. QL1 gives nonzero logarithms of valuation one. For any $Q \in E_i(\mathbb{Q})$ put $m_i(Q) = \log_{E_i}(Q)/\log_{E_i}(P_i)$. If the good reduction at thirteen has order fifteen, then

$$(SM1) \quad m_i(Q) \in \mathbb{Z}_5, \quad [3] \text{red}_{13}(Q) = [m_i(Q) \bmod 5] [3] \text{red}_{13}(P_i).$$

To prove this, the finite quotient of order d_i gives an integer a with $[d_i]Q = [a]P_i$. Apply the logarithm to get $m_i(Q) = a/d_i$. Reduce the integer group relation at thirteen and multiply by three. Both resulting points are killed by five, so multiplication by d_i is invertible on them; its inverse modulo five proves (SM1). The residue of a/d_i is exactly that of the five-adic coefficient. This does not reduce a value of $\mathbb{Q}_5$ modulo thirteen. It needs neither a global generator assertion $d_i = 1$ nor saturation at thirteen.

SM2: the complete obstruction over $\mathbb{F}_{13}$. There is no $X \in C(\mathbb{F}_{13})$ satisfying

$$(SM2) \quad [3]R_1(X) = O, \quad [3]R_2(X) \in \{(11, 2), (11, 11)\}.$$

Both reduced elliptic groups have order fifteen, and $[3] \text{red}_{13}(P_2) = (11, 2)$. Here is an exhaustive finite proof. The two discriminants are respectively 3 and 9 modulo thirteen. Their complete affine point lists, in addition to O , are

$$E_1 : (0, \pm 2), (1, \pm 3), (3, \pm 2), (6, \pm 6), (10, \pm 2), (11, \pm 1), (12, \pm 5),$$

$$E_2 : (3, \pm 2), (5, \pm 6), (6, \pm 4), (8, \pm 5), (9, \pm 1), (11, \pm 2), (12, \pm 2).$$

Testing the thirteen possible abscissas and ordinates proves exhaustion and hence order fifteen in each case. The reduced sextic is $F = z^6 + 12z^4 + 8z^2 + 4$, and the polynomial identity

$$(10 + 8z^2)F + (6z + 3z^3)F' = 1 \quad \text{in } \mathbb{F}_{13}[z]$$

proves squarefreeness even over the algebraic closure. The monic even-degree model is smooth at its two infinities too. The complete projective quotient table is

X on C	$[3]R_1(X)$	$[3]R_2(X)$
(0, 2)	(3, 11)	O
(0, 11)	(3, 2)	O
(1, 5)	(3, 11)	(11, 2)
(1, 8)	(3, 2)	(11, 11)
(3, 3)	(11, 1)	(11, 11)
(3, 10)	(11, 12)	(11, 2)
(6, 3)	O	O
(6, 10)	O	O
(7, 3)	O	O
(7, 10)	O	O
(10, 3)	(11, 1)	(11, 2)
(10, 10)	(11, 12)	(11, 11)
(12, 5)	(3, 11)	(11, 11)
(12, 8)	(3, 2)	(11, 2)
∞_+	O	(8, 8)
∞_-	O	(8, 5)

The fourteen affine points exhaust the same thirteen-by-thirteen enumeration. At $z = 0$ the projective quotient values are $R_2 = O$, $R_1 = (-9/4, W/8)$; at infinity of sign ϵ they are $R_1 = O$, $R_2 = (33/4, -9\epsilon/8)$. Applying chord and tangent addition gives the displayed triples and $[3]P_2$. The six rows with $[3]R_1 = O$ establish (SM2).

The reduction table applies to every rational point, including points outside the integral affine chart. If z is thirteen-adically integral, then W is integral. If z reduces to zero, R_2 has negative x -valuation and reduces to O , while R_1 is regular. If z has negative valuation, put $q = 1/z$ and $v = Wq^3$: then q reduces to zero and v to ± 1 , with the stated infinity quotient values. In the remaining cases all displayed affine denominators are units. Thus reduction of the rational quotient images agrees with the table.

Three independent finite implementations reproduce the full table, using chord and tangent arithmetic, division polynomials, and cubic line intersection. The primary canonical certificate also contains the smoothness identity, both entire elliptic point lists and both forbidden signs. Finite computation here proves only these finite inputs.

SM3: exclude the four extra local zeros. The complete ordinary disk certificates give exactly ten distinct $\mathbb{Q}_5$ zeros of f , counted with total multiplicity twelve:

disk orbit	distinct zeros	multiplicities
$z \equiv \pm 1$	8	all simple
$z \equiv \pm 2$	0	none
$z \equiv 0$	0	none
infinity	2	both double

DS covers all twelve residue disks. Four of the unit-disk zeros are the known rational points $(\pm 1, \pm 8)$. The other four are the orbit of the extra UD zero on $z = 1 + 5s$, $W(1) = 8$, with $s \equiv 4$. On that whole disk UD gives

$$\ell_1/5 \equiv 4 + 4s, \quad \ell_2/5 \equiv 3 + s \pmod{5},$$

while QL gives $\log_{E_1}(P_1)/5 \equiv 4$ and $\log_{E_2}(P_2)/5 \equiv 2$. The extra zero therefore has $(m_1(R_1), m_2(R_2)) \equiv (0, 1) \pmod{5}$. Under DS, either both quotient points change sign or only the second does. All four extra zeros thus have coefficients $(0, \pm 1)$ modulo five. If one were rational, (SM1) would force its reduced quotients to satisfy (SM2), a contradiction. None of the four is rational.

Every finite nonzero rational point satisfies $f = 0$ by LH and QL7. The zero fiber has no rational point because $W^2 = -9$; the infinity points were treated directly in IF2. The list is therefore exhaustive. Conversely the sextic takes value 64 at $z = \pm 1$ and its leading coefficient is one, so all six displayed rational points exist. This completes the fixed-curve ordinary proof. The subsequent exact inverse tests the second square at the same source and the real positive domain of the original problem. No varying-curve or varying-residual uniform height theorem follows from this classification.

291. THE UNIT-ZETA CUBIC LOCUS CONSISTS OF BOUNDARY POINTS

This section applies the complete fixed-curve classification of Section 290 to the original, common-source biquadratic curve. The definitions and projective inverses are those of Sections 248, 250, 256 and 262. Put $\zeta^2 - \zeta + 1 = 0$, with the same Eisenstein coordinate convention as those sections.

Proposition 291.1 (The six points in the original first quotient). *For the smooth projective curve*

$$H : y^2 = P(s) := s^6 + 3s^5 - 3s^4 - 11s^3 - 3s^2 + 3s + 1,$$

the complete rational locus consists of its two points over each of $s = 0, -1, \infty$.

Proof. Writing $S_C(z) = z^6 - 27z^4 + 99z^2 - 9$, the exact dense-chart maps are

$$(1107) \quad z = \frac{s-1}{s+1}, \quad W = \frac{8y}{(s+1)^3}, \quad s = \frac{1+z}{1-z}, \quad y = \frac{W}{(1-z)^3}.$$

They are the actual transformation of Theorem 256.1: its intermediate coordinates are $t = (s+1)/(s-1)$, $v = y(t-1)^3$, with $v^2 = -9t^6 + 99t^4 - 27t^2 + 1$. Substitution $z = 1/t$, $W = v/t^3$ gives (1107). Equivalently,

$$(s+1)^6 S_C\left(\frac{s-1}{s+1}\right) = 64P(s).$$

The inverse dense maps extend to an isomorphism of smooth projective curves, by [140, Lemmas 53.2.2 and 53.2.6].

The complete list $C(\mathbb{Q}) = \{(\pm 1, \pm 8), \infty_+, \infty_-\}$ therefore transfers as follows:

point of C	point of H
$z = -1, W = \pm 8$	$s = 0, y = \pm 1$
$z = 1, W = \pm 8$	$s = \infty, y/s^3 = \pm 1$
$z = \infty, W/z^3 = \epsilon$	$s = -1, y = -\epsilon$

The exceptional rows follow from the actual chart limits $y/s^3 = W/(1+z)^3$ and $y = (W/z^3)/(1/z - 1)^3$. Thus every apparent zero denominator is accounted for, and the six listed points exhaust $H(\mathbb{Q})$. $\square$

Theorem 291.2 (The complete common-source cubic locus). *Let*

$$(1108) \quad A_0 = S^3 - 3ST^2 - T^3, \quad B_0 = 3ST(S + T),$$

$$(1109) \quad A = -B_0, \quad B = A_0 + B_0, \quad r = A/B.$$

The smooth projective normalization $D = D_{3,\zeta}$ of

$$(1110) \quad v^2 = 1 + r, \quad w^2 = 1 - 3r$$

has exactly twelve rational points. Their source is $[S : T] = [0 : 1], [-1 : 1], [1 : 0]$, and at each source (v, w) takes all four values in $\{1, -1\}^2$. Their original projective parameter t is given by

$$(1111) \quad \begin{array}{cc|c} v & w & t \\ \hline 1 & 1 & \infty \\ 1 & -1 & 1 \\ -1 & 1 & -1 \\ -1 & -1 & -1/2 \end{array}$$

Proof. The first quotient of (1110) is exactly H , with $y_1 = Bv$ and $y_1^2 = A_0(A_0 + B_0)$. At the same source, the second quotient has $y_2 = Bw$ and

$$(1112) \quad y_2^2 = (A_0 + B_0)(A_0 + 4B_0).$$

Every rational point of D maps to a rational point of H under its projective quotient morphism. Proposition 291.1 therefore restricts its source to the following three values:

s	A_0	B_0	A	B
0	-1	0	0	-1
-1	1	0	0	1
∞	1	0	0	1

In particular $B \neq 0$ and $r = 0$ at each source. The second square (1112) is then B^2 , giving both rational choices $w = \pm 1$; the two first-quotient points give $v = \pm 1$. Points from different sources have not been paired. These are four distinct rational points at each source. Both square roots are nonzero, and $r = 0$ avoids the branch values $-1, 1/3, \infty$. The raw equations are smooth there, so normalization neither merges these points nor introduces another local branch. This proves the exact total of twelve.

For the two middle rows of (1111), the original inverse (910) is $t = (v + w + 2)/(v - w)$. The first and last rows require its genuine projective continuation; the last gives an apparent $0/0$. Use the original formulas instead:

$$v = \frac{t^2 + 2t}{t^2 + t + 1}, \quad w = \frac{t^2 - 2t - 2}{t^2 + t + 1}.$$

At infinity both ratios are one, and at $t = -1/2$ both are minus one. Also

$$r(t) = \frac{(t^2 - 1)(2t + 1)}{(t^2 + t + 1)^2}$$

has precisely the four simple projective zeros $\infty, 1, -1, -1/2$. These give all projective inverse values, including the removable $0/0$ case. $\square$

Corollary 291.3 (Exclusion of the actual positive unit-zeta branch). *The curve $D_{3,\zeta}$, and its Gaussian cover over the same positive domain, have no rational point with finite original parameter $t > 1$. In particular there are no positive coprime integers a, b such that*

$$M = a^2 + ab + b^2 = U^2, \quad F(a, b) = Q^3, \quad Q > 1,$$

whose actual oriented second Eisenstein decomposition has unit class ζ modulo cubes of units. More generally the same positive locus of $D_{g,\zeta}$ is empty for every positive odd integer g divisible by three.

Proof. The exact positive inverse of Theorem 250.3 requires finite $t > 1$, equivalently $0 < r < 1/3$ with $v > 0$. None of the points in (1111) has this property. For clarity, the homogeneous first-root output is

$$a_0 = T_0^2 - W_0^2, \quad b_0 = W_0(2T_0 + W_0).$$

At the four values of $t = T_0/W_0$ in the table, its primitive signed pairs are respectively $(1, 0), (0, 1), (0, -1), (-1, 0)$. The finite representatives at $t = 1, -1/2$ have content three, which must be divided out. Every primitive boundary pair has $ab = 0$ and $M = F = 1$; none is a positive seed or has $Q > 1$.

An actual seed in the asserted unit class has, after absorbing a minus sign into the root,

$$ab + M\zeta = \zeta w_1^3.$$

The exact projective inverse of Section 248 then supplies the common source and $t = (a+U)/b > 1$, contradicting Theorem 291.2. That inverse included primitive, ramified, unit and infinite root inputs before imposing the positive domain. Thus none of these input exceptions restores a positive point. The previously proved converse on that domain makes this an exclusion of the actual branch, not just a necessary auxiliary square.

For $g = 3k$ odd, the projective power maps satisfy the exact identity

$$L_{\zeta,g} = L_{\zeta,3} \circ L_{1,k}.$$

It gives a nonconstant rational curve map $(t, s) \mapsto (t, L_{1,k}(s))$ from $D_{g,\zeta}$ to $D_{3,\zeta}$, extending to the smooth projective models as in [140, Lemmas 53.2.2 and 53.2.6]. It retains t ; a rational point with finite $t > 1$ would map to the excluded positive locus. No enumeration of the boundary preimages for larger g is needed or asserted. $\square$

Scope. The identity cubic unit class already has its separate complete three-adic obstruction. The distinct class ζ^2 is not eliminated here: conjugation changes the target ratio r to $-1 - r$, which does not preserve $(0, 1/3)$. It cannot silently be substituted for a positive-domain isomorphism. General second residuals, the ramified first norm, other exponents and uniform point-height bounds remain outside this conclusion. The theorem uses the complete ordinary and exact finite fixed-curve proof, together with the already proved projective inverses; it is not a new Lean formalization of a curve, rational locus or ABC statement.

292. ACTUAL PRIME CHAINS AND A MEDIUM-PRIME CENTRAL BARRIER

For the actual endpoint $c = 2^e 3^f$, with integers $e, f \geq 2$, put

$$M = 2^e, \quad N = 3^f, \quad A = M/2, \quad B = N/3, \quad Q = \text{rad}(c - 1).$$

Retain both complete premises

$$(NG0) \quad \max(M, N) < 2 \min(M, N) - 1, \quad AB > Q.$$

The previously proved DP no-proper-face calculation identifies the original FCRT/SCRT/PBT optimum as $D + \Gamma$, where

$$D = \log(AB/Q) > 0, \quad \Gamma = \min_{H|Q} \text{dist}(\log H, [\log(Q/B), \log A]).$$

All divisors here are divisors of the complete actual radical. Each prime is assigned once; in the original endpoint this assigns its entire prime-power contribution. The following results concern that same optimum.

NG1: the exact largest divisor gap. Let $Q > 1$ be squarefree, with primes $p_1 < \dots < p_s$, and define

$$R_i = \prod_{j < i} p_j, \quad \Lambda(Q) = \max_{1 \leq i \leq s} \frac{p_i}{R_i}.$$

The largest difference of consecutive logarithms of divisors of Q is exactly $\log \Lambda(Q)$.

Proof. When p_i is added, the previous divisor-log set S is replaced by $S \cup (S + a)$, where $a = \log p_i$ and S has endpoints $0, T = \log R_i$. If $a > T$, the spans are disjoint and the largest gap is the maximum of the old gap and $a - T$. If $a \leq T$, the spans overlap. Every gap in the union is contained in a gap of at least one of the two sets, within that set's span, so it is at most the old maximum. Induction, starting with $\{0, \log p_1\}$, proves the upper bound. For the lower bound, if $p_i > R_i$, a divisor of the final Q smaller than p_i cannot contain p_i or any later prime. It divides R_i and is at most R_i . Thus R_i, p_i are consecutive divisors of the final integer. Since $p_1 > 1 = R_1$, the largest of these positive gaps attains $\log \Lambda(Q)$. $\square$

Once the full prime support is supplied, this is a test with one step per prime. It makes no assertion about the cost of factoring Q .

NG2: a sufficient arithmetic class. For any real $A, B \geq 1$ with $AB > Q$, the same definitions give

$$(NG1) \quad \Gamma \leq \frac{\max(0, \log \Lambda(Q) - D)}{2} \leq \frac{\max(0, \log P^+(Q) - D)}{2}.$$

In particular,

$$(NG2) \quad p_i \leq (AB/Q)R_i \quad \text{for every } i$$

implies $\Gamma = 0$.

Proof. If $\Gamma > 0$, the central closed interval contains neither 0 nor $\log Q$, since both are divisor logs. Because $A, B \geq 1$, it therefore lies strictly inside $[0, \log Q]$, in one open gap (u, v) of the full divisor-log set. Its length is D . The two endpoint distances sum to $v - u - D$, so their minimum is at most $(\log \Lambda(Q) - D)/2$. If $\Gamma = 0$, the bound is immediate. Finally $R_i \geq 1$ gives $\Lambda(Q) \leq P^+(Q)$, and (NG2) is exactly $\log \Lambda(Q) \leq D$. $\square$

This proves a full arithmetic subclass, but not its membership for all endpoints. A large gap near an end of the divisor set can also make (NG2) fail when the required central partition succeeds.

NG3: a certified nonsquare central success. Take $e = 41, f = 26$. The complete factorization is

$$(NG3) \quad c - 1 = 7^2 \cdot 439 \cdot 857 \cdot 2729 \cdot 292183 \cdot 380261663.$$

All six factors are prime, with the deterministic certificates described below. Both premises (NG0) hold. The actual divisors

$$H = 2729 \cdot 380261663 = 1037734078327, \quad J = 7 \cdot 439 \cdot 857 \cdot 292183$$

satisfy $HJ = Q$, $H \leq 2^{40} = A$ and $J \leq 3^{25} = B$ by integer comparison. Hence $\Gamma = 0$, while $D = \log(7c/(6(c-1))) > 0$. Since e is odd, this endpoint is outside the earlier DP square family. Its global value $\Lambda(Q) = 439/7$ is larger than AB/Q . Thus (NG2) does not certify every successful central partition. This is one certified nonsquare endpoint, not an assertion of infinitely many such endpoints.

NG4: a complete medium-prime bracket. Let $Q = qR$ be squarefree, with q prime and $p < r$ the two smallest prime factors of R . Suppose $A, B \geq 1$, $AB > Q$, and

$$(NG4) \quad prq \leq R < r^2q, \quad rq > \max(A, B).$$

Then $R/r, rq$ are consecutive divisors of the full Q , and

$$(NG5) \quad \Gamma = \log \frac{rq}{\max(A, B)} > 0, \quad D + \Gamma = \log \frac{\min(A, B)}{R/r}.$$

Proof. A divisor of R below r is either 1 or p , because R is squarefree. If $H \mid Q$ contains q and $H < rq$, write $H = qd$ with $d \mid R, d < r$. Then $H \leq pq \leq R/r$. If H omits q and $H > R/r$, then $R/H < r$, so $H = R$ or R/p ; both are at least rq . The endpoints themselves divide Q , are strictly ordered, and have product Q . Since $rq > A, B$, the entire central interval lies strictly

between their logarithms. Its two distances are $\log(rq/B)$ and $\log(rq/A)$, proving both formulas. $\square$

For an actual instance take $e = 123$, $f = 78$, the cube of the preceding endpoint. Put

$$q = 257077829249200341434761489003997119,$$

$$R = 7 \cdot 31 \cdot 439 \cdot 857 \cdot 919 \cdot 2729 \cdot 292183 \cdot 380261663 \cdot 4266021703.$$

The complete factorization is $c - 1 = 7qR$, hence $Q = qR$ without an unfactored remainder. Here $p = 7$, $r = 31$; both (NG0) and (NG4) hold. More strongly, every prime of Q is at most q and

$$(NG6) \quad 20q < \min(A, B),$$

where the exact integers are

$$A = 5316911983139663491615228241121378304,$$

$$B = 5474401089420219382077155933569751763,$$

$$R/31 = 3130566812026385226903752016561167657,$$

$$31q = 7969412706725210584477606159123910689.$$

Since $B > A$, the exact conclusions are

$$(NG7) \quad D = \log \frac{7c}{6(c-1)}, \quad \Gamma = \log \frac{31q}{B} > 2D > 0, \quad D < \log(6/5) < 1.$$

No decimal logarithm is needed: the two strict comparisons are

$$31q[6(c-1)]^2 > B(7c)^2, \quad 35c < 36(c-1).$$

The exact positive balance margin $2 \min(M, N) - 1 - \max(M, N)$ is

$$4844444664297995820229445163776257926.$$

The verifier checks these comparisons and all 1024 actual divisors. The proof of (NG5) separately establishes completeness of the bracket.

Consequently (NG0), nonsquareness, $0 < D < 1$, and even $P^+(Q) < \min(A, B)/20$ do not imply $\Gamma = 0$ or $\Gamma \leq D$. This excludes precisely those strengthened statements. It supplies neither a fixed- ϵ infinite counterexample nor a contradiction to a bound with an arbitrary additive C_ϵ . It does not disprove ABC or another certificate with different, proved rules. Within the original rules, (NG5) is the exact optimum.

NG5: deterministic primality and complete finite evidence. The standard-library verifier `next_replay_medium_partition.py` checks 47 prime nodes and 134 witnesses: base 2, 45 full predecessor nodes and one full Gaussian successor node. It uses no probable-prime predicate or factorization routine. For completeness, both criteria have elementary proofs.

For a predecessor node $m > 2$, factor $m - 1$ completely into previously proved primes. For each distinct $r \mid m - 1$, suppose an integer a_r satisfies

$$a_r^{m-1} \equiv 1 \pmod{m}, \quad \gcd(a_r^{(m-1)/r} - 1, m) = 1.$$

For every prime $\ell \mid m$, the order of a_r modulo ℓ divides $\ell - 1$ and $m - 1$ and contains the full r -part of $m - 1$. It follows that $m - 1 \mid \ell - 1$, hence $\ell \geq m$ and $\ell = m$. Thus m is prime. A different witness for each r is permitted.

For a successor node let $m > 2$ be odd and factor $m + 1$ completely into previously proved smaller primes. For every $r \mid m + 1$, suppose $z_r = a_r + b_r i$ in $(\mathbb{Z}/m\mathbb{Z})[i]$ satisfies

$$(NG8) \quad a_r^2 + b_r^2 \equiv 1 \pmod{m}, \quad z_r^{m+1} = 1, \quad \gcd(N(z_r^{(m+1)/r} - 1), m) = 1.$$

For any prime $\ell \mid m$, the algebra $\mathbb{F}_\ell[i]$ is either $\mathbb{F}_\ell \times \mathbb{F}_\ell$ or $\mathbb{F}_{\ell^2}$, since ℓ is odd. Its norm-one group has order $\ell - 1$ in the split case, where its elements are (u, u^{-1}) , and $\ell + 1$ in the field case, by the surjective finite-field norm. The last condition in (NG8) makes $z_r^{(m+1)/r} - 1$ a unit modulo ℓ , so the order of z_r contains the full r -part of $m + 1$. Therefore $m + 1$ divides that norm-one group order for every $\ell \mid m$, giving $\ell \geq m$ and hence primality. No Jacobi-symbol test is being substituted for this proof.

The sole successor node is

$$m = 13098839766085821942054493478243,$$

$$m + 1 = 2^2 \cdot 3^5 \cdot 7 \cdot 23 \cdot 1427 \cdot 10691 \cdot 40507 \cdot 135446661817793.$$

All factors are earlier proved nodes. The largest endpoint prime has $q - 1 = 2 \cdot 3 \cdot 3271 \cdot m$, a full predecessor node. The Gaussian operations are literal integer-pair operations modulo m , and every supplied witness is checked.

The verifier also checks both complete factorizations of $c - 1$, primitivity, balance, nonsquareness, $D > 0$, all 64 and 1024 divisors, the maximum-gap formula, the central memberships and the strict rational comparisons. Generation and read-only `--check` actually passed; independent root and peer executions also passed. The canonical output SHA256 is `26d04f39748d763f5139794465afb78058f18474567b629f6937abcae2ec9af2`. These finite certificates and the complete elementary proofs above are separate from any Lean theorem, prime-value infinitude, or uniform ABC estimate.

293. CONSTRUCTIVE PAIRED PACKETS AND RECIPROCAL ADJACENT DIVISORS

All factors in this section belong to the complete actual squarefree $Q = \text{rad}(c - 1)$. For $c = 2^e 3^f$, retain (NG0), $A = 2^{e-1}$, $B = 3^{f-1}$, $D = \log(AB/Q) > 0$, and the same original optimum $D + \Gamma$. The elementary divisor assertions also apply to any squarefree $Q > 1$ and real $A, B \geq 1$ with $AB > Q$.

PA1: a certificate with at most m exact updates. Suppose there is an exact integer factorization

$$Q = R \prod_{i=1}^m U_i V_i, \quad m \geq 1, \quad R \geq 1, \quad 1 < U_i < V_i.$$

All its nonunit factors are squarefree and pairwise coprime. They may be composite packets. Choose an actual positive divisor $K \mid R$ and define

$$h_{\min} = K \prod_i U_i, \quad h_{\max} = K \prod_i V_i, \quad \rho = \max_i (V_i/U_i) > 1.$$

If

$$(PA1) \quad h_{\min} \leq A, \quad h_{\max} \geq Q/B,$$

then

$$(PA2) \quad \Gamma \leq \frac{\max(0, \log \rho - D)}{2}, \quad D + \Gamma \leq \max\left(D, \frac{D + \log \rho}{2}\right).$$

In particular $\rho \leq AB/Q$ implies $\Gamma = 0$.

Proof. For each subset J of packet indices, the two integers

$$H_J = K \prod_{i \notin J} U_i \prod_{i \in J} V_i, \quad Q/H_J = (R/K) \prod_{i \notin J} V_i \prod_{i \in J} U_i$$

recover Q exactly. Every prime is used in exactly one output, so this is a legal subfamily of the original divisor certificate. Only $m + 1$ members are necessary. Fix the given order and put

$$H_j = K \prod_{i \leq j} V_i \prod_{i > j} U_i, \quad 0 \leq j \leq m.$$

This chain is strictly increasing from $h_{\min}$ to $h_{\max}$, with $H_j/H_{j-1} = V_j/U_j \leq \rho$. Let j be the first index with $H_j \geq Q/B$, which exists by (PA1). If $H_j \leq A$, a central divisor has been found. Otherwise $j \geq 1$, since $H_0 \leq A$, and

$$H_{j-1} < Q/B \leq A < H_j.$$

The distances from the central interval to these two genuine divisors sum to $\log(H_j/H_{j-1}) - D \leq \log \rho - D$. Their minimum bounds the optimum over all actual divisors, proving the first assertion of (PA2). Adding D proves the second. $\square$

After this factor certificate is supplied, the construction is an exact bounded algorithm. Start at H_0 and update $H_j = (H_{j-1}/U_j)V_j$; division is exact because that packet has not yet been

replaced. At most m updates find a central divisor or its two brackets. For integer endpoints, all tests are integer comparisons $BH_j \geq Q$ and $H_j \leq A$. In the bracket case compare the positive rational numbers $Q/(BH_{j-1})$ and H_j/A to choose the smaller logarithmic penalty. There is no 2^m search or floating-point comparison. This is not an efficient factorization algorithm or a universal theorem supplying suitable packets.

PA2: an actual instance and its conditional scope. For the completely certified $c = 2^{41}3^{26}$ of NG3, choose

$$\begin{aligned} U_1 &= 2729, & V_1 &= 3073 = 7 \cdot 439, \\ R &= 857 \cdot 292183 \cdot 380261663, & K &= 380261663. \end{aligned}$$

These factors recover the complete Q . The exact inequalities

$$K \cdot 2729 \leq 2^{40}, \quad K \cdot 3073 \cdot 3^{25} \geq Q, \quad 3073Q < 2729 \cdot 2^{40} \cdot 3^{25}$$

give (PA1) and $\rho < AB/Q$, hence $\Gamma = 0$. The endpoint is nonsquare, while its global gap parameter $439/7$ exceeds AB/Q . The local packet certificate is therefore a strictly sharper sufficient test in this actual example. Its added information is a pair of close composite packets in the required location.

For a sequence of actual endpoints with $Q \rightarrow \infty$, if $D/\log Q \rightarrow 0$ and certificates satisfying (PA1) can be supplied with $\log \rho/\log Q \rightarrow 0$, then (PA2) gives $(D + \Gamma)/\log Q \rightarrow 0$. Likewise the two inequalities

$$D \leq \epsilon \log Q + C, \quad \log \rho \leq \epsilon \log Q + C$$

give $D + \Gamma \leq \epsilon \log Q + C$ with the same constants. Neither hypothesis is established for all endpoints. In this stratum

$$D = \log c - \log \text{rad}(c(c-1)),$$

so the first hypothesis already contains the restricted scalar ABC problem. The paired-packet supply is a separate unproved arithmetic condition, not a consequence of one example or a moderate largest prime factor.

PA3: the general reciprocal obstruction. Let $Q = qR$ be squarefree with q prime, let $d > 1$ be a divisor of R , and set

$$t = \max\{u : u \mid R, u < d\}.$$

This maximum exists since $1 < d$. If

$$(PA3) \quad dtq \leq R < d^2q,$$

then $R/d, dq$ are consecutive actual divisors of Q . If also $dq > \max(A, B)$, then

$$(PA4) \quad \Gamma = \log \frac{dq}{\max(A, B)}, \quad D + \Gamma = \log \frac{\min(A, B)}{R/d}.$$

Proof. If a divisor H contains q and is below dq , write $H = qv$ with $v \mid R, v < d$; then $H \leq qt \leq R/d$. If it omits q and is above R/d , then $R/H < d$, hence $R/H \leq t$ and $H \geq R/t \geq dq$. Both endpoints divide Q , are strictly ordered, and have product Q . The central interval and its distances are therefore exactly as in NG4, proving (PA4). $\square$

There is a prime-prefix specialization without a search for t . If d is a prime factor of R and T is the product of every prime of R smaller than d , the additional condition $T < d$ gives $t = T$. Indeed, a divisor below d cannot contain a larger prime, and T itself is such a divisor. Thus $dTq \leq R < d^2q$ suffices; NG4 is the case that d is the second smallest prime of R .

This invariant uses the full actual $R = Q/q$ and a certified divisor threshold. Largest-prime size alone does not contain that information. An exact obstruction of this kind prevents a PA1 certificate with $\rho \leq AB/Q$ for the same central interval, since their proved conclusions would contradict one another. It does not exclude a different certificate with different proved rules, nor the parent ABC routes. All accounting above uses actual radicals and whole prime packets. No infinite fixed- ϵ counterexample or new Lean theorem is claimed.

294. THE ACTUAL SQUARE AND CUBE SELECTOR ARITHMETIC IN LEAN

The new module `NonlinearPowerSelectors.lean` contains 24 theorems for the actual root-parameter families used in NP1 and CU1. Its definitions multiply the integer coordinate pair $(3k, 1)$ literally. No output power representation or primitive Bezout witness is assumed for either concrete family.

Write $\zeta^2 = \zeta - 1$, and for pairs of integers set

$$(a, b)(c, d) = (ac - bd, ad + bc + bd), \quad N(a, b) = a^2 + ab + b^2, \quad T(a, b) = ab(a + b).$$

For every integer k , let $w = (3k, 1)$, $z_2 = w^2$, $z_3 = w^3$, and $N_k = 9k^2 + 3k + 1$. The implemented definitions of z_2, z_3 are respectively the product of w with itself and that product with w .

Exact coordinates and the complete boundaries. For all $k \in \mathbb{Z}$, the formal identities are

$$\begin{aligned} N(w) &= N_k, & N_k &\equiv 1 \pmod{3}, \\ z_2 &= (9k^2 - 1, 6k + 1), & (z_2)_1 + (z_2)_2 &= 3k(3k + 2), \\ z_3 &= ((3k)^3 - 3(3k) - 1, 3(3k)(3k + 1)), \\ (z_3)_1 + (z_3)_2 &= (3k)^3 + 3(3k)^2 - 1, \\ N(z_2) &= N_k^2, & N(z_3) &= N_k^3. \end{aligned}$$

The entire boundary products, with no removed prime factors, are

$$\begin{aligned} T(z_2) &= (3k - 1)(3k + 1)(6k + 1)(3k)(3k + 2), \\ T(z_3) &= 3(3k)(3k + 1)((3k)^3 - 3(3k) - 1) \\ &\quad \cdot ((3k)^3 + 3(3k)^2 - 1). \end{aligned}$$

These are obtained directly by expanding the coordinate products; the norm-power identities follow by substitution. The modulus-three claim follows from $N_k = 3(3k^2 + k) + 1$.

If $k \geq 1$, both coordinates of each output, their sum and their boundary are positive, and $N_k > 1$. For the cubic first coordinate, put $a = 3k \geq 3$; then $a^3 - 3a - 1 > 0$, while the second coordinate is $3a(a + 1) > 0$. The square and norm claims are immediate from their displayed formulas. Positivity alone uses $k \geq 1$; the polynomial identities and coprimality below hold for every integer k .

Actual primitivity without an assumed witness. For the square coordinates $A = 9k^2 - 1$, $C = 6k + 1$, the integer identity

$$(-12k^2 - 1)A + 3k^2(6k - 1)C = 1$$

proves coprimality. For the cube put $a = 3k$ and $F = a^3 - 3a - 1$. The three identities

$$\begin{aligned} -F + 3(9k^3 - 3k) &= 1, \\ -F + a(a^2 - 3) &= 1, \\ F - (a + 1)(a^2 - a - 2) &= 1 \end{aligned}$$

show that F is coprime to 3, a and $a + 1$, hence to the actual second coordinate $3a(a + 1)$. The Lean proofs construct these witnesses and use integer coprimality lemmas.

For any coprime integers A, C , if $uA + vC = 1$ then

$$(u - v)A + v(A + C) = 1, \quad (v - u)C + u(A + C) = 1.$$

Thus all three pairs of arms have gcd one. The module proves the corresponding actual equalities $\text{Int.gcd}(A, C) = \text{Int.gcd}(A, A + C) = \text{Int.gcd}(C, A + C) = 1$ and applies them to both concrete outputs.

Every boundary prime leaves the root norm a unit. Over any field, the simultaneous equations

$$AC(A + C) = 0, \quad A^2 + AC + C^2 = 0$$

force $A = C = 0$. If either coordinate is zero this follows from the square of the other; if $A + C = 0$, subtract $A(A + C)$ from the norm equation to get $C^2 = 0$ and then $A = 0$. For a

primitive integer pair, a prime divisor q of its full boundary therefore cannot divide its norm: reduction of a Bezout identity would give $0 = 1$ in $\mathbb{Z}/q\mathbb{Z}$.

Apply this proved helper to z_2 and z_3 , using their proved coprimality. Since their norms are N_k^2 and N_k^3 , a zero root norm modulo q would give a zero output norm, a contradiction. Consequently the final concrete theorems assert, for every $k \in \mathbb{Z}$ and natural prime q ,

$$q \mid T(z_i) \implies N_k \in (\mathbb{Z}/q\mathbb{Z})^\times, \quad q \nmid N_k, \quad i = 2, 3.$$

They do not assume a marked prime or a lower cutoff; primes two and three are included. The statements use the literal full boundary, not a selected or prefactored part of it.

Exact verification inventory. The 24 theorem declarations consist of three root-norm statements; six square and six cube statements (coordinates, sum, norm, boundary, positivity, coprimality in each family); three arm-gcd statements; two generic field/primitive-pair statements; and four concrete root-norm unit and nondivisibility consequences. Definitions are not counted as theorem declarations.

The source SHA256 is

```
38849679f2e456475bf9f9ef8dfaa0c5
5df2e0cba8d705d7ec579c2e111776e9
```

The independent fresh project compiled this single module with Lean 4.32.0 against the existing clean Mathlib cache at commit `81a5d257c8e410db227a6665ed08f64fea08e997`. Compilation used warnings as errors, one compiler thread and a fresh output file, without native C generation. No old project theorem module was imported or recompiled. All 24 explicit `#print axioms` queries completed, with union exactly `propext`, `Classical.choice`, `Quot.sound`. There is no `sorry`, `native_decide` or added axiom. The complete successful manifest and log are recorded separately; the manifest SHA256 is

```
b01e02f2482bef62985243547185acf3
c35b60fadd7cf088596a341fbeb22cc6
```

An independent full source, scope and evidence review passed. Root subsequently repeated the fresh compilation of the identical canonical release source; its separate `verification/mathlib_validation.json` also records all 24 declarations and queries as passing.

The formal result does not assert CRT packet selection, exact prescribed prime depths, cubic-factor irreducibility or separability, squarefree-value density, moving-data uniformity, infinite-tail control, or ABC. Those assertions were not inserted as hypotheses or axioms. The module proves the concrete finite arithmetic that the ordinary selector arguments use.

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